

Intelligent Security API

Developer Guide

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VISION AND SECURITY SYSTEM CO.,LTD.

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1 Reading Guide

Chapter	Description
Overview	Includes the ISAPI overview, applicable products, terms and definitions, abbreviations, and update history.
ISAPI Framework	Read the chapter to take a quick look at the ISAPI framework and basic functions.
Quick Start Guide	Read the chapter to quickly understand the programming process of basic functions such as authentication, message parsing, real-time live view, playback, and event uploading.
API Reference	Start programming according to API definitions.
How-To Video Guidance	How-to videos demonstrate detailed steps of different integration tasks.

2 Overview

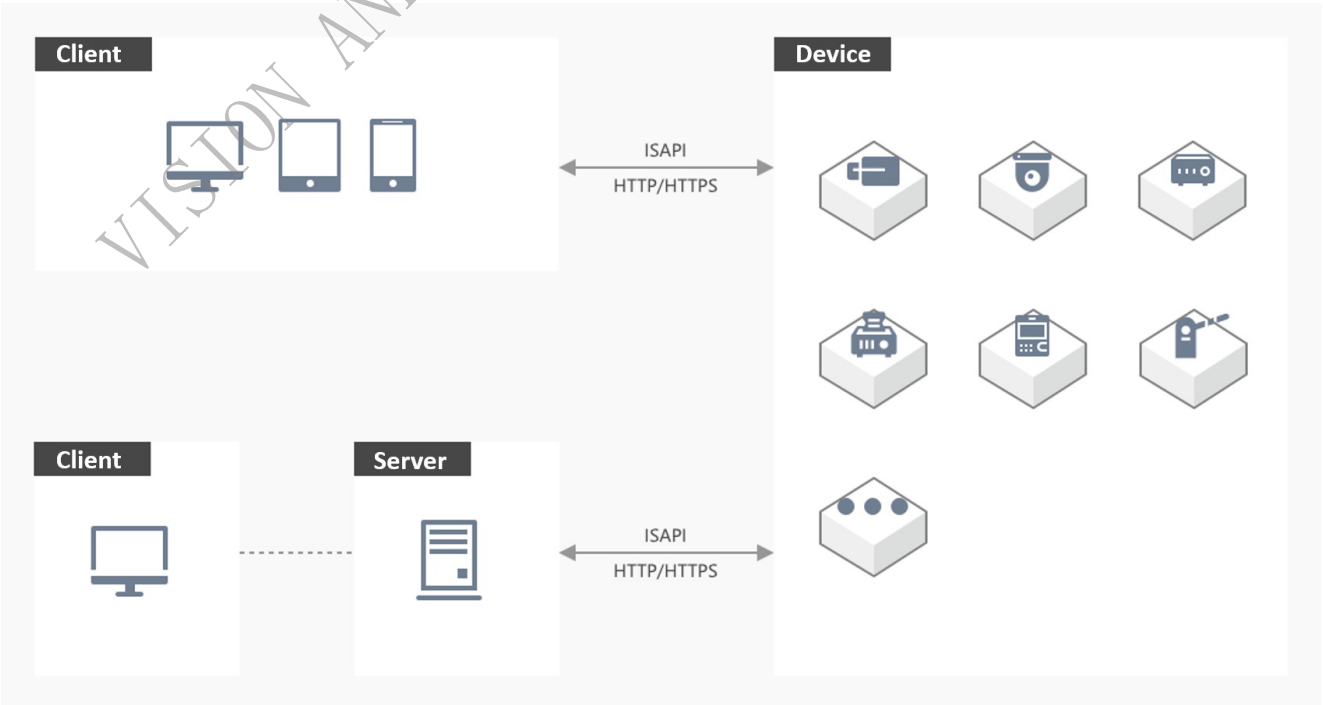
2.1 Introduction

Intelligent Security API (hereinafter referred to as ISAPI) is an application layer protocol based on HTTP (Hypertext Transfer Protocol) and adopts the REST (Representational State Transfer) architecture for communication between security devices (cameras, DVRs, NVRs, etc.) and the platform or client software.

Since established in 2013, ISAPI has included more than 11,000 APIs for different functions, including device management, vehicle recognition, parking lot management, intelligent facial application, access control management, interrogation management, and recording management. It is applicable to industries such as traffic, fire protection, education, and security inspection.

2.1.1 Application Scenario

When you integrate devices via ISAPI, the device acts as the server to listen on the fixed port and the user's application acts as the client to actively log in to the device for communication. To achieve the above goals, the device should be configured with a fixed IP address and the requests from the client can reach the server.



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2.1.2 Layers in the Network Model

ISAPI is an application layer protocol based on HTTP, thereby it inherits all specifications and properties from HTTP.

Protocols frequently used along with ISAPI include SADP (Search Active Device Protocol) based on multicast for discovering and activating devices, RTSP (Real-Time Streaming Protocol) based on TCP/UDP for live view and video playback of devices, etc.

Application Layer	Device Discovery SADP	Signaling Interaction ISAPI (HTTP)	Media Stream RTSP/RTP
Transport Layer	MCAST	TCP	TCP/UDP
Network Layer	IP		
Network Interface	Hardware Drive Interface		

2.2 Product Scope

- Network Cameras
  - DeepinView Series

    DS-2CD5026G0-(AP), DS-2CD5026G0/E-I(H), DS-2CD5028G0/E-HI, DS-2CD5046G0-(AP), DS-2CD5065G0-(AP), DS-2CD5085G0-(AP), DS-2CD50C5G0-(AP), DS-2CD5126G0-IZS, DS-2CD5146G0-IZS, DS-2CD5165G0-IZS, DS-2CD5185G0-IZS, DS-2CD51C5G0-IZS, DS-2CD5526G0-IZ(H)S, DS-2CD5526G0-IZ(H)SY, DS-2CD5526G1-IZ(H)S, DS-2CD5546G0-IZ(H)S, DS-2CD5546G0-IZ(H)SY, DS-2CD5546G1-IZ(H)S, DS-2CD5565G0-IZ(H)S, DS-2CD5585G0-IZ(H)S, DS-2CD5585G1-IZ(H)S, DS-2CD55C5G0-IZ(H)S, DS-2CD5A26G0-IZ(H)S, DS-2CD5A26G0-IZ(H)SY, DS-2CD5A26G1-IZ(H)S, DS-2CD5A46G0-IZ(H)S, DS-2CD5A46G0-IZ(H)SY, DS-2CD5A46G0-IZ/UH, DS-2CD5A46G1-IZ(H)S, DS-2CD5A65G0-IZ(H)S, DS-2CD5A85G0-IZ(H)S, DS-2CD5A85G1-IZ(H)S, DS-2CD5AC5G0-IZ(H)S, DS-2CD7026G0-(AP), DS-2CD7026G0/EP-I(H), DS-2CD7026G0/P-(AP), DS-2CD7046G0-(AP), DS-2CD7065G0-(AP), DS-2CD7085G0-(AP), DS-2CD7126G0-IZS, DS-2CD7126G0/L-IZS, DS-2CD7146G0-IZS, DS-2CD7165G0-IZS, DS-2CD7185G0-IZS, DS-2CD72205G0/E(70200mm), ~~DS-2CD72325G0/E(70~~200mm), DS-2CD7526G0-IZ(H)S, DS-2CD7526G0-IZ(H)SY, DS-2CD7546G0-IZ(H)S, DS-2CD7546G0-IZ(H)SY, DS-2CD7565G0-IZ(H)S, DS-2CD7585G0-IZ(H)S, DS-2CD7A26G0-IZ(H)S, DS-2CD7A26G0-IZ(H)SY, DS-2CD7A26G0/P-IZ(H)S, DS-2CD7A26G0/P-LZ(H)S, DS-2CD7A46G0-IZ(H)S, DS-2CD7A46G0-IZ(H)SY, DS-2CD7A65G0-IZ(H)S, DS-2CD7A85G0-IZ(H)S, IDS-2CD7A46G2/V-XZHS(Y), IDS-2CD7A86G2/V-XZHS(Y), iDS-2CD5347G2-XS, iDS-2CD5347G2/V-XS, iDS-2CD5387G2-XS, iDS-2CD5387G2/V-XS, iDS-2CD5546G2-XZ(H)S(Y), iDS-2CD5586G2-XZ(H)S(Y), iDS-2CD5A46G2-INZHSY, iDS-2CD5A46G2-XZ(H)S(Y), iDS-2CD5A46G2/V-XZ(H)S(Y), iDS-2CD5A86G2-XZ(H)S(Y), iDS-2CD5A86G2/V-XZ(H)S(Y), iDS-2CD5D47G2-XS, iDS-2CD5D47G2/V-XS, iDS-2CD5D87G2-XS, iDS-2CD5D87G2/V-XS, iDS-2CD5T47G2-X(H)S(Y), iDS-2CD5T47G2/V-X(H)S(Y), iDS-2CD5T47G2E/V-XS, iDS-2CD5T87G2-X(H)S(Y), iDS-2CD5T87G2/V-X(H)S(Y), iDS-2CD70166G2-AP, iDS-2CD70166G2/E-IHSYR, iDS-2CD7026G0(-AP), iDS-2CD7026G0/E-IHSY(R), iDS-2CD7026G0/EP-IHSY, iDS-2CD7026G0/P-AP, iDS-2CD7026G0/S-AP, iDS-2CD7026G2-AP, iDS-2CD7046G0(-AP)(/F11), iDS-2CD7046G0/E-IHSY(/F11)(R), iDS-2CD7046G0/EP-IHSY, iDS-2CD7046G0/H-AP, iDS-2CD7046G0/P-AP, iDS-2CD7046G0/S-AP, iDS-2CD7046G2-AP, iDS-2CD7046G2/E-IHSYR, iDS-2CD7046G2/EP-IHSYR, iDS-2CD7046G2/P-AP, iDS-2CD7086G0(-AP), iDS-2CD7086G0/E-IHSY(R), iDS-2CD7086G0/H-AP, iDS-2CD7086G2-AP, iDS-2CD7086G2/E-IHSYR, iDS-2CD7086G2/EP-IHSYR, iDS-2CD7086G2/P-AP, iDS-2CD70C5G0(-AP), iDS-2CD70C5G0/E-IHSY(R), iDS-2CD70C5G0/H-AP, iDS-2CD70C6G2-AP, iDS-2CD70C6G2/E-IHSYR, iDS-2CD71166G2-IZHSY, iDS-2CD7126G0-IZ(H)S(Y), iDS-2CD7126G0-IZS, iDS-2CD7126G2-IZ(H)S(Y), iDS-2CD7146G0-IZ(H)S(Y), iDS-2CD7146G0-IZS, iDS-2CD7146G0-PTR, iDS-2CD7146G0/C-IZS, iDS-2CD7146G0/H-IZ(H)S(Y), iDS-2CD7146G0/S-IZS, iDS-2CD7146G2-IZ(H)S(Y)(1T), iDS-

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2CD7146G2/LM-IZ(H)S(Y), iDS-2CD7186G0-IZ(H)S(Y), iDS-2CD7186G0-IZS, iDS-2CD7186G0/H-IZ(H)S(Y), iDS-2CD7186G0/S-IZS, iDS-2CD7186G2-IZ(H)S(Y), iDS-2CD7186G2/LM-IZ(H)S(Y), iDS-2CD71C5G0-IZS, iDS-2CD71C5G0/H-IZ(H)S(Y), iDS-2CD71C6G2-IZ(H)S(Y), iDS-2CD72265G0/E, iDS-2CD72325G0/E, iDS-2CD7347G0-XS, iDS-2CD7387G0-XS, iDS-2CD7526G0-IZHS(Y), iDS-2CD7546G0-IZHS(Y), iDS-2CD7546G0/S-IZHS(Y), iDS-2CD7546G2-XZHS(Y), iDS-2CD7546G2/P-XZHS(Y), iDS-2CD7547G0-XZHS(Y), iDS-2CD7547G0/P-XZHS(Y), iDS-2CD7547G2-XZHS(Y), iDS-2CD7547G2-ZHS/RC, iDS-2CD7547G2/P-XZHS(Y), iDS-2CD7586G0-IZHS(Y), iDS-2CD7586G0/S-IZHSY, iDS-2CD7586G2-XZHS(Y), iDS-2CD7586G2/P-XZHS(Y), iDS-2CD7587G0-XZHS(Y), iDS-2CD7587G2-XZHS(Y), iDS-2CD7587G2-ZHS/RC, iDS-2CD7587G2/P-XZHS(Y), iDS-2CD75C5G0-IZHS(Y)(R), iDS-2CD7A166G2-IZHSY, iDS-2CD7A26G0-IZHS(Y), iDS-2CD7A26G0/P-IZHS(Y), iDS-2CD7A26G2-IZHS(Y), iDS-2CD7A45G0-IZ(H)S(Y), iDS-2CD7A45G0/P-IZHS(Y), iDS-2CD7A45G2-IZHS(Y), iDS-2CD7A45G2/P-IZHS(Y), iDS-2CD7A46G0-IZHS(Y), iDS-2CD7A46G0/H-IZHS(Y), iDS-2CD7A46G0/P-IZHS(Y), iDS-2CD7A46G0/S-IZHS(Y), iDS-2CD7A46G2-IZHS(Y)(1T)(/5G), iDS-2CD7A46G2-IZHS(Y)/4G, iDS-2CD7A46G2/LM-IZHS(Y), iDS-2CD7A46G2/P-IZHS(Y)(/5G), iDS-2CD7A46G2/P-IZHS(Y)/4G, iDS-2CD7A47G0-XZHS(Y), iDS-2CD7A47G0/P-XZHS(Y), iDS-2CD7A47G2-XZHS(Y), iDS-2CD7A47G2/P-XZHS(Y), iDS-2CD7A86G0-IZHS(Y), iDS-2CD7A86G0/H-IZHS(Y), iDS-2CD7A86G0/S-IZHSY, iDS-2CD7A86G2-IZHS(Y)(/5G), iDS-2CD7A86G2-IZHS(Y)/4G, iDS-2CD7A86G2/LM-IZHS(Y), iDS-2CD7A86G2/P-IZHS(Y)(/5G), iDS-2CD7A86G2/P-IZHS(Y)/4G, iDS-2CD7A87G0-XZHS(Y), iDS-2CD7A87G2-XZHS(Y), iDS-2CD7A87G2/P-XZHS(Y), iDS-2CD7AC5G0-IZHS(Y)(R), iDS-2CD7AC5G0/H-IZHS(Y), iDS-2CD7AC6G2-IZHS(Y), iDS-2CD7D47G0-XS, iDS-2CD7D87G0-XS, iDS-2CD7T46G2/VX3-I(H)S(Y), iDS-2CD7T47G0-XHS(Y), iDS-2CD7T87G0-XHS(Y), iDS-2CD8146G0-IZS, iDS-2CD8426G0/B-I, iDS-2CD8426G0/F-I, iDS-2CD8447G0/B-RS, iDS-2CD8626G0/P-IZS, iDS-2CD8626G0/P-LZS, iDS-2CD8A46G0-IZ/UH, iDS-2CD8A46G0-IZHS, iDS-2CD8A46G0-XZ(H)S(Y), iDS-2CD8A46G2-XZHS(Y), iDS-2CD8A48G0-XZ(H)S(Y), iDS-2CD8A48G2-XZHS(Y), iDS-2CD8A86G0-XZ(H)S(Y), iDS-2CD8A86G2-XZHS(Y), iDS-2CD8C46G0-XZS, iDS-2CD8C46G0P-XZS, iDS-2CD8C46G0S-XZS, iDS-2CD8C48G0-XZS, iDS-2CD8C48G0-XZS/T2, iDS-2CD8V446G0/X2-XZHS(Y), iDS-2CD8V446G0/X4-XZHS(Y), iDS-2CD8V447G0E/X2-XZS, iDS-2CD8V886G0/X2-XZHS(Y), iDS-2CD8V886G0/X4-XZHS(Y)

2.3 Terms And Definitions

2.3.1 Event

Event refers to the information uploaded by the device actively. Event needs to be uploaded by the device in real time for the immediate response from the platform. If the device is offline, the event can be stored in the cache first and then be uploaded again when the connection is restored.

2.3.2 Arming

Arming means that the client establishes connection with the device so that events can be uploaded to the client via the connection. The client can subscribe to some event types, and the device will upload the specified events only, otherwise the device will upload all types of events to the client.

2.3.3 Motion Detection

It can realize image changes recognition and moving objects detection of a specified area by camera. For example, if a person passes by or a camera is moved, the motion will be recognized and then an alarm will be uploaded for notifications, and the event can trigger recording and alarm output.

2.3.4 Traffic Enforcement Event Detection

The traffic enforcement event detection supports the function of violation enforcement, which can automatically detect and capture the evidence of events, including illegal parking, wrong-way driving, illegal U-turning, driving on the lane line, illegal lane change, and motor vehicle on non-motor vehicle lane, on the roads with no more than four lanes. The enforcement results will be uploaded as alarms in real time. The difference between traffic incident detection and traffic enforcement event detection is that the former only detects events which are not always illegal (e.g., congestion, roadblock, etc.) and the latter detects events of violating the traffic rules. For the traffic enforcement event detection (e.g., illegal parking, wrong-way driving, illegal U-turning, driving on the lane line, illegal lane change, etc.), the detailed violation information needs to be recorded, including the background picture, close-up picture, vehicle picture, and

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license plate number.

2.3.5 Model Package

Model package is the combination of detection and classification models on the upper-level platform and distributed to device as a whole. There is no specific model data for device management (such as adding, deleting, editing and searching) and devices are managed uniformly according to model package.

2.3.6 Intelligent Event Analysis

Intelligent event analysis refers to a module whose logical rules (such as counting, alarms, etc.) is further designed based on the detection and recognition results of the AI Open Platform algorithms.

2.3.7 Video Analysis Task

Perform real-time detection on video streams. Applicable to scenarios with sufficient computing power and high real-time requirements. #

2.4 Symbols And Acronyms

HEOP: Embedded Open Platform

ISAPI: Intelligent Security API.

ISUP: Intelligent Security Uplink Protocol.

HTTP: Hypertext Transfer Protocol.

OSD: On-screen display

VMD: Motion Detection

PTZ: Pan-Tilt-Zoom

ANPR: Automatic Number Plate Recognition

TFS: Traffic Enforcement Event Detection (TFS is the abbreviation for Traffic Forensics System)

AIOP: AI Open Platform

GPU: Graphics Processing Unit

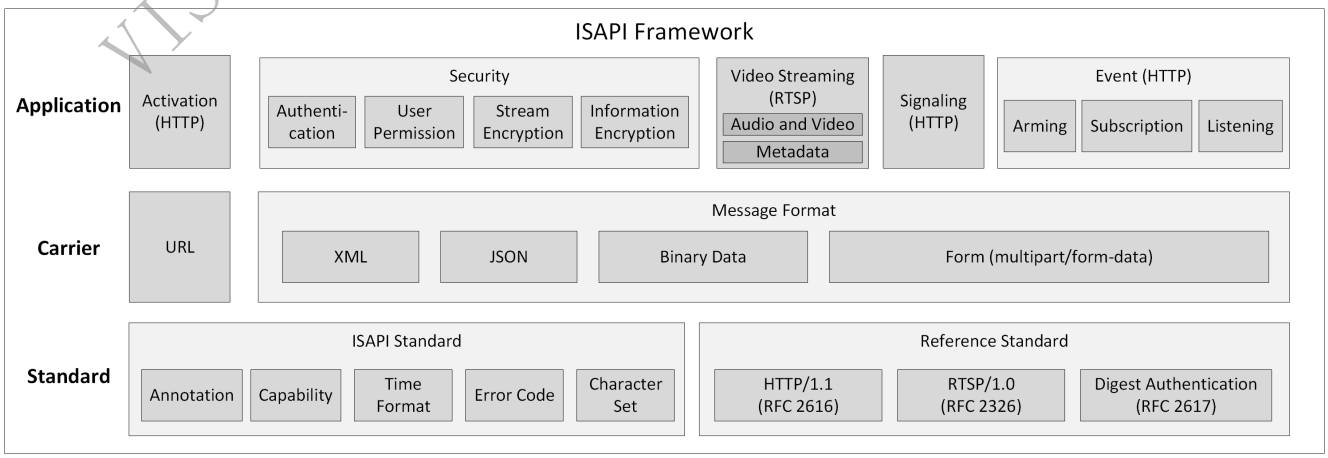
- GPU: Graphics Processing Unit

2.5 Update History

No update record

3 ISAPI Framework

3.1 Overview



Notes:

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In general, ISAPI refers to the communication protocol based on the HTTP standard. As ISAPI is usually used along with RTSP (Real-Time Streaming Protocol), the RTSP standard is brought into the ISAPI system.

The metadata scheme for transmitting additional information of the stream is extended based on the RTSP standard to transmit the video stream and the structured intelligent information of the stream simultaneously. It is compatible with the RTSP standard.

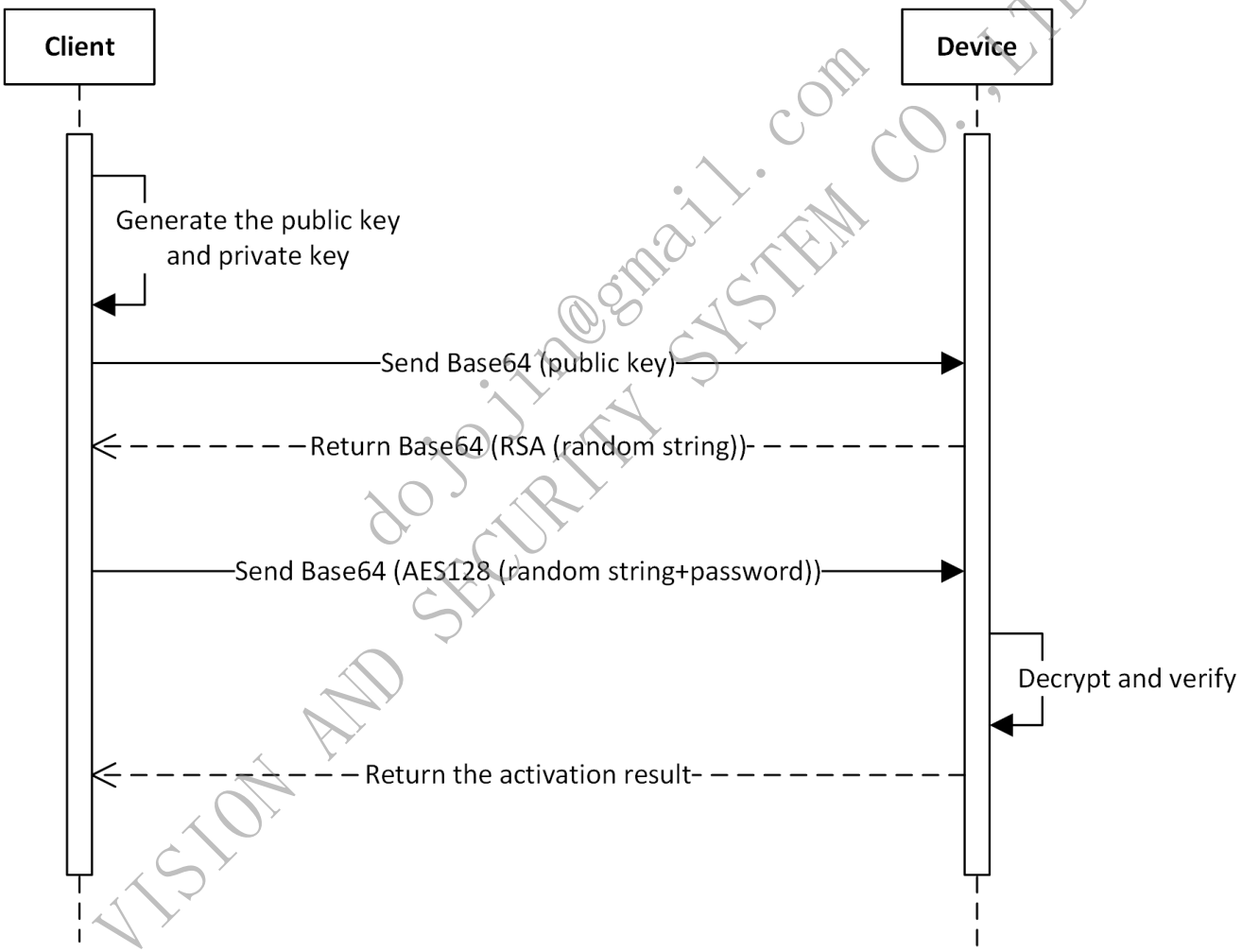
3.2 Activation

The purpose of activation is to ensure that the user can set the password for the device and the password meets the security requirement. After the device is activated, you can use the related functions.

ISAPI is a communication protocol running on the application layer. When activating the device via ISAPI, you should know the device's IP address and make sure that the device is connected to the client.

The web application built in the device supports activating the device via ISAPI. When you enter the device's IP address in the address bar of the web browser on the PC, you can activate the device according to the activation guide.

If you want to activate the device on your own application, you need to integrate the activation function via ISAPI. The API calling flow and related APIs are shown below.



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Firstly, two operations are defined:

- bytesToHexstring: it is used to convert a byte array (the length is N) to a hexadecimal string (the length is 2N). For example, `127,10,23` -> `7f0a17`
- hexStringToBytes: it is used to convert a hexadecimal string (the length is 2N) to a byte array (the length is N). For example, `7f0a17` -> `127,10,23`

1. The client generates a public and private key pair (1024 bits), and gets the 128-byte modulus in the public key (hereinafter referred to as public key modulus). If the length is longer than 128, the leading 0 needs to be removed.
2. The client converts the public key modulus to a 256-byte public key string via bytesToHexstring and sends the public key string to the device in XML message (related URI: `POST /ISAPI/Security/challenge`) after being encoded by Base64.

3. The device parses the request to obtain a 256-byte public key string decoded by Base64 and converts it to a 128-byte public key modulus via hexStringToBytes. The complete public key is the combination of obtained public key modulus and public exponent (the default value is `'010001'`).
4. The device generates a 32-byte hexadecimal random string, calls the RSA API to encrypt the random string with the private key, converts the encrypted data to a string via bytesToHexstring, encodes the string by Base64, and then sends it to the client.
5. The client decodes the string from the device by Base64, converts it via hexStringToBytes to get the encrypted data, decrypts the encrypted data with the private key via RSA to obtain a 32-byte hexadecimal random string, converts the obtained string via hexStringToBytes to get a 16-byte AES key. Then the client uses the AES key to encrypt the `"string consisting of the first 16 characters of the random string and the real password"` by AES128 ECB mode (with zero-padding method) to get a ciphertext, and converts the ciphertext via bytesToHexstring, encodes it by Base64, and sends it to the device in XML message (related URI: `PUT /ISAPI/System/activate`). Note: If the first 16 characters of the random string are `aaaabbbbccccdddd` and the real password is `Abc12345`, the data before encryption is `aaaabbbbccccddddAbc12345`. This can ensure that the client uses the random string as the key for encryption.
6. The device decodes the string by Base64, converts it via hexStringToBytes to get the ciphertext, uses the AES key to decrypt the ciphertext by AES128 ECB mode, and gets the real password via removing the first 16 characters.
7. The device verifies the password and returns the activation result.

Notes:

- You can get the device's activation status by calling the URI `GET /SDK/activateStatus` which requires no authentication.
- Devices also support to be activated via SADP (Search Active Device Protocol) which is based on the communication protocol of the data link layer. With SADP, you do not have to know the IP address of the device but need to ensure that the device and the application running SADP are connected to the same router. SADP also supports discovering devices in the LAN, changing the password of the devices, and so on. The HCSadpSDK is provided for SADP integration, including the developer guide, plug-in, and sample demo which can be used as a simple SADP tool.

3.3 Security Mechanism

3.3.1 Authentication

When the client applications send requests to devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Currently, all mainstream request class libraries of HTTP have encapsulated digest authentication. See details in Authentication of Quick Start Guide.

3.3.2 User Permission

There are three kinds of users with different permissions for access control and management.

Administrator: Has the permission to access all supported resources and should keep activated all the time. It is also known as "admin".

Operator: Has the permission to access general resources and a part of advanced resources.

Normal User: Only has the permission to access general resources.

3.3.3 Information Encryption

During ISAPI integration, the HTTPS service of devices is enabled by default. When the client applications communicate with devices via HTTPS, the information can be transmitted securely.

3.4 Video Streaming

3.4.1 Audio and Video Stream

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ISAPI supports getting and setting stream media parameters of the device, such as video resolution, encoding format, and stream.

Cameras support standard RTSP (Real-Time Streaming Protocol, see details in RFC 7826). Client applications can get the stream from devices via RTSP.

For details about real-time streaming and video playback, refer to **Real-Time Live View** and **Playback** in Quick Start Guide.

3.4.2 Metadata

The metadata is the structured intelligent information generated by intelligent devices. When the client applications get the audio and/or video stream from devices via RTSP, the metadata will be returned by the device at the same time. For example, to display the face target frame, face information, vehicle target frame, license plate number, vehicle information, and other information on the video stream, the client applications can overlay the above information on the video image.

Before using the metadata, you need to enable the metadata function of the device and then get the stream from the device via RTSP. Some devices support subscribing to the metadata by type. For details about the process of integrating the metadata function, refer to **Metadata Management**.

4 Quick Start Guide

4.1 Authentication

When the client applications send requests to the devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Client applications only need to call APIs of the class library to implement the digest authentication. The sample code is shown below.

4.1.1 C/C++ (libcurl)

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```
// #include <curl/curl.h>
// Callback Function
static size_t OnWriteData(void* buffer, size_t size, size_t nmemb, void* lpVoid)
{
    std::string* str = dynamic_cast<std::string*>((std::string *)lpVoid);
    if( NULL == str || NULL == buffer )
    {
        return -1;
    }

    char* pData = (char*)buffer;
    str->append(pData, size * nmemb);
    return nmemb;
}

std::string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
std::string strResponseData;
CURL *pCurlHandle = curl_easy_init();
curl_easy_setopt(pCurlHandle, CURLOPT_CUSTOMREQUEST, "GET");
curl_easy_setopt(pCurlHandle, CURLOPT_URL, strUrl.c_str());
// Set the user name and password
curl_easy_setopt(pCurlHandle, CURLOPT_USERPWD, "admin:admin12345");
// Set the authentication method to the digest authentication
curl_easy_setopt(pCurlHandle, CURLOPT_HTTPAUTH, CURLAUTH_DIGEST);
// Set the callback function
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEFUNCTION, OnWriteData);
// Set the parameters of the callback function to get the returned information
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEDATA, &strResponseData);
// Timeout settings for receiving the data. If receiving data is not completed within 5 seconds, the application will exit directly
curl_easy_setopt(pCurlHandle, CURLOPT_TIMEOUT, 5);
// Set the redirection times to avoid too many redirections
curl_easy_setopt(pCurlHandle, CURLOPT_MAXREDIRS, 1);
// Connection timeout duration. If the duration is too short, the client application will be disconnected before the data request sent by the application
reaches the device
curl_easy_setopt(pCurlHandle, CURLOPT_CONNECTTIMEOUT, 5);
CURLcode nRet = curl_easy_perform(pCurlHandle);
if (0 == nRet)
{
    // Output the received message
    std::cout << strResponseData << std::endl;
}
curl_easy_cleanup(pCurlHandle);
```

4.1.2 C# (WebClient)

```
// using System.Net;
// using System.Net.Security;
try
{
    string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
    WebClient client = new WebClient();
    // Set the user name and password
    client.Credentials = new NetworkCredential("admin", "admin12345");
    byte[] responseData = client.DownloadData(strUrl);
    string strResponseData = Encoding.UTF8.GetString(responseData);
    // Output received information
    Console.WriteLine(strResponseData);
}
catch (Exception ex)
{
    Console.WriteLine(ex.Message);
}
```

4.1.3 Java (HttpClient)

```
// import org.apache.commons.httpclient.HttpClient;
String url = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
HttpClient client = new HttpClient();
// Set the user name and password
UsernamePasswordCredentials creds = new UsernamePasswordCredentials("admin", "admin12345");
client.getState().setCredentials(AuthScope.ANY, creds);
GetMethod method = new GetMethod(url);
method.setDoAuthentication(true);
int statusCode = client.executeMethod(method);
byte[] responseData = method.getResponseBodyAsString().getBytes(method.getResponseCharSet());
String strResponseData = new String(responseData, "utf-8");
method.releaseConnection();
// Output received information
System.out.println(strResponseData);
```

4.1.4 Python (requests)

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```python
# - *- coding: utf-8 -*-

import requests
request_url = 'http://192.168.18.84:80/ISAPI/System/deviceInfo'
# Set the authentication information
auth = requests.auth.HTTPDigestAuth('admin', 'admin12345')
# Send the request and receive response
response = requests.get(request_url, auth=auth)
# Output response content
print(response.text)
```

4.2 Message Parsing

4.2.1 Message Format

During the process of communication and interaction via ISAPI, the request and response messages are often text data in XML or JSON format. Besides that, the data of firmware packages and configuration files is in binary format. A request can also be in form format with multiple formats of data (multipart/form-data).

4.2.1.1 XML

Generally, the `Content-Type` in the headers of the HTTP request is `application/xml; charset="UTF-8"`.

Request and response messages in XML format are all encoded with UTF-8 standards in ISAPI.

The namespace `http://www.isapi.org/ver20/XMLSchema` and ISAPI version number `2.0` of XML messages are configured by default, see the example below.

```xml
<?xml version="1.0" encoding="UTF-8"?>
<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <Node>
    <id>1</id>
    <enabled>true</enabled>
    <nodeName>nodeName</nodeName>
    <level>level1</level>
  </Node>
</NodeList>
```

4.2.1.2 JSON

The `Content-Type` in the headers of the HTTP request is often `application/json`.

To distinguish between APIs with XML messages and those with JSON messages, ISAPI adds the query parameter format=json to all request URLs with JSON messages, e.g., `http://192.168.1.1:80/ISAPI/System/Sensor/thermometrySensor?format=json`. Messages of request URLs without the query parameter format=json are usually in XML format. However, there may be some exceptions, and the message format is subject to the API definition.

Request and response messages in JSON format are all encoded by UTF-8 in ISAPI.

4.2.1.3 Binary Data

For the firmware and configuration files, the `Content-Type` in the header of an HTTP request is often `application/octet-stream`.

4.2.1.4 Form (multipart/form-data)

When multiple pieces of data are submitted at the same time in an ISAPI request (e.g., the person information and face picture need to be submitted at the same time when a face record is added to the face picture library), the `Content-Type` in the header of the corresponding HTTP request is usually `multipart/form-data, boundary=AaB03x`, where the boundary is a variable used to separate the entire HTTP body into multiple units and each unit is a piece of data with its own headers and body. In `Content-Disposition` of form unit headers, the `name` property refers to the form unit name, which is required for all form units; the `filename` property refers to the file name of form unit body, which is required only when the form unit body is a file. In headers of form units, `Content-Length` refers to the body length, which starts after CRLF(`\r\n`) and ends before two hyphens (`--`) of next form. There should be a CRLF used as the delimiter of two form units before two hyphens (`--`), and the `Content-Length` of previous form unit does not include the CRLF length.

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For the detailed format description, refer to RFC 1867 (Form-Based File Upload in HTML). Pay attention to two hyphens (`--`) before and after the boundary.

Notes

- In RFC specifications, it is strongly recommended to contain the field `Content-Length` in the entity header, and there is no requirement that the field `Content-Length` should be contained in the header of each form element. The absence of field `Content-Length` in the header should be considered when the client and device programs parse the form data.
- To avoid the conflict between message content and boundary value, it is recommended to use a longer and more complex string as the boundary value.

The example of ISAPI form data submitted by a client to a device is as follows.

```
POST /ISAPI/Intelligent/FDLib/pictureUpload
Content-Type: multipart/form-data; boundary=e5c2f8c5461142aea117791dade6414d
Content-Length: 56789

--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="PictureUploadData";
Content-Type: application/xml
Content-Length: 1234

<PictureUploadData/>
--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="face_picture"; filename="face_picture.jpg";
Content-Type: image/jpeg
Content-Length: 34567

Picture Data
--e5c2f8c5461142aea117791dade6414d--
```

The example of ISAPI form data responded from a device to a client is as follows.

In ISAPI messages, when there are multiple form units, three nodes (`pid`, `contentid`, and `filename`) are used for linking form units. The corresponding relations are as follows:

Node Name	Form Field	Description
pid	name	pid in XML/JSON messages corresponds to the name property of Content-Disposition in form headers.
contentid	Content-ID	contentid in XML/JSON messages corresponds to Content-ID in form headers.
filename	filename	filename in XML/JSON messages corresponds to filename property of Content-Disposition in form headers.

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```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=136a73438ecc4618834b999409d05bb9
Content-Length: 56789

--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="mixedTargetDetection";
Content-Type: application/json
Content-Length: 811

{
    "ipAddress": "172.6.64.7",
    "macAddress": "01:17:24:45:D9:F4",
    "channelID": 1,
    "dateTime": "2009-11-14T15:27+08:00",
    "eventType": "mixedTargetDetection",
    "eventDescription": "Mixed target detection",
    "deviceID": "123456789",
    "CaptureResult": [{
        "targetID": 1,
        "Human": {
            "Rect": {
                "height": 1.0,
                "width": 1.0,
                "x": 0.0,
                "y": 0.0
            },
            "contentID1": "humanImage", /*human body thumbnail*/
            "contentID2": "humanBackgroundImage", /*human body background picture*/
            "pId1": "9d48a26f7b8b4f2390c16808f93f3534", /*human body thumbnail ID */
            "pId2": "5EE7078E07BB47CF860DE8E4E9A85F28" /*ID of human body background picture*/
        }
    }]
}
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="9d48a26f7b8b4f2390c16808f93f3534"; filename="humanImage.jpg";
Content-Type: image/jpeg
Content-Length: 34567
Content-ID: humanImage

Picture Data
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="5EE7078E07BB47CF860DE8E4E9A85F28"; filename="humanBackgroundImage.jpg";
Content-Type: image/jpeg
Content-Length: 345678
Content-ID: humanBackgroundImage

Picture Data
--136a73438ecc4618834b999409d05bb9--
```

4.2.2 Annotation

The field descriptions of ISAPI request and response messages are marked as annotations in the example messages as shown below.

```
<?xml version="1.0" encoding="UTF-8"?>

<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, node list, attr:version{req, string, version No., range:[,]}-->
  <Node>
    <!--ro, opt, object, node information-->
    <id>
      <!--ro, req, int, node No., range:[,], step:, unit:, unitType:-->1
    </id>
    <enabled>
      <!--ro, opt, bool, whether to enable-->true
    </enabled>
    <nodeName>
      <!--ro, req, string, node name, range:[1,32]-->test
    </nodeName>
    <level>
      <!--ro, opt, enum, level, subType:string,
      [level1#Level 1,level2#Level 2,level3#Level 3]-->level1
    </level>
  </Node>
</NodeList>
```

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```
{
    "name":  "test",
    /*ro, req, string, name, range:[1,32]*/
    "type":  "type1",
    /*ro, req, enum, type, subType:string, [type1#type 1,type2#type 2]*/
    "enabled":  true,
    /*ro, opt, bool, enable or not, desc:xxxxxxx*/
    "NodeList": {
    /*opt, object, node list, dep:and,{$.enabled,eq,true}*/
        "scene":  1,
        /*req, enum, scene, subType:int, [1#scene 1; 2#scene 2; 3#scene 3]*/
        "ID":  1
        /*req, int, No., range:[1,8], step:, unit:, unitType:*/
    }
}
```

Key annotations are shown in the table below.

Annotation	Description	Remark
ro	Attribute: Read-Only	This field can only be obtained and cannot be edited.
wo	Attribute: Write-Only	This field can only be edited and cannot be obtained.
req	Attribute: Required	This field is required for request messages sent to the device and response messages returned from the device.
opt	Attribute: Optional	This field is optional for request messages sent to the device and response messages returned from the device.
dep	Attribute: Dependent	This field is valid and required when specific conditions are satisfied.
object	Field Type: Object	The field of type object contains multiple sub-fields.
list	Field Type: List	The subType following it refers to the data type of sub-items in the list.
subType	Field Type: String	The range following it refers to the maximum and the minimum string size of the field.
int	Field Type: Int	The range following it refers to the maximum and the minimum value of the field.
float	Field Type: Float	The range following it refers to the maximum and the minimum value of the field.
bool	Field Type: Boolean	The value can be true or false.
enum	Field Type: Enumeration	The subType following it indicates that the enumerators are of type string or int. The [] following the subType contains all enumerators.
subType	Sub-Type of Field	When the type of field is list or enum, the value of subType is the data type of each sub-object.
desc	Field Description	The detailed description of the field.

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4.2.3 Capability Set

ISAPI has designed capability sets for almost all functions, APIs, and fields. URLs for getting the capability set end with `/capabilities`. Some URLs may contain query parameters in the format: `/capabilities?format=json&type=xxx`.

There are two types of fields in the capability message of ISAPI: whether the device supports a function and the value range of a field in an API.

Whether the device supports a function: it is often in the format `isSupportXxxxxxxx`, which indicates that whether the device supports a function and a set of APIs for implementing this function.

The capability message example in JSON format is shown below:

```
{
    "isSupportMap":  true,
    /*ro, opt, bool, whether it supports the e-map function, desc:/ISAPI/SDT/Management/map/capabilities?format=json*/
    "isSupportAlgTrainResourceInfo":  true,
    /*ro, opt, bool, whether it supports only getting the resource information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/ResourceInfo?format=json*/
    "isSupportAlgTrainAuthInfo":  true,
    /*ro, opt, bool, whether it supports ony getting the authorization information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/SoftLock/AuthInfo?format=json*/
    "isSupportAlgTrainNodeList":  true,
    /*ro, opt, bool, whether it supports only getting the node information of the algorithm training platform, desc:/ISAPI/SDT/algorithmTraining/NodeList?
format=json*/
    "isSupportNAS":  true
    /*ro, opt, bool, whether it supports mounting and unmounting NAS, desc:/ISAPI/SDT/Management/NAS/capabilities?format=json*/
}
```

The capability message example in XML format is shown below:

```
<isSupportNetworkStatus>
    <!--ro, opt, bool, whether it supports searching the network status, desc: related API (/ISAPI/System/Network/status?format=json)-->true
</isSupportNetworkStatus>
```

The value range of the field: the maximum value, minimum value, the maximum size, the minimum size, options, and so on of each field of the API.

The example of JSON format is shown below:

```
{
    "boolType": {
        /*req, object, example of the capability of type bool*/
        "@opt": [true, false]
        /*req, array, options, subType: bool*/
    },
    "integerType": {
        /*req, object, example of the capability of type integer*/
        "@min": 0,
        /*ro, req, int, the minimum value*/
        "@max": 100
        /*ro, req, int, the maximum value*/
    },
    "stringType": {
        /*req, object, example of the capability of type string*/
        "@min": 0,
        /*ro, req, int, the minimum string size*/
        "@max": 32
        /*ro, req, int, the maximum string size*/
    },
    "enumType": {
        /*req, object, capability example of type enum*/
        "@opt": ["enum1", "enum2", "enum3"]
        /*req, array, options, subType: string*/
    }
}
```

The example of XML format is shown below:

```
<boolType opt="true,false" def="true">
    <!--ro, opt, bool, example of the capability of type bool-->true
</boolType>
<integerType min="0" max="100">
    <!--ro, opt, int, example of the capability of type int-->50
</integerType>
<stringType min="0" max="64">
    <!--ro, opt, string, example of the capability of type string-->test
</stringType>
<enumType opt="red,white,black" def="red">
    <!--ro, opt, string, example of the capability of type enum-->white
</enumType>
```

Note: For the same capability set, devices of different models and versions may return different results. The values shown in this document are only examples for reference. The capability set actually returned by the device takes precedence.

4.2.4 Time Format

ISAPI adopts ISO 8601 Standard Time Format, which is the same as W3C Standard Date and Time Formats.

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Format: `YYYY-MM-DDThh:mm:ss.sTZD`

```
YYYY = the year consisting of four decimal digits
MM = the month consisting of two decimal digits (01-January, 02-February, and so forth)
DD = the day consisting of two decimal digits (01 to 31)
hh = the hour consisting of two decimal digits (00 to 23, a.m. and p.m. are not allowed)
mm = the minute consisting of two decimal digits (00 to 59)
ss = the second consisting of two decimal digits (00 to 59)
s = one or more digits representing the fractional part of a second
TZD = time zone identifier (Z or +hh:mm or -hh:mm)
```

Example: 2017-08-16T20:17:06.123+08:00 refers to 20:17:06.123 on August 16, 2017 (local time which is 8 hours ahead of UTC). The plus sign (+) indicates that the local time is ahead of UTC, and the minus sign (-) means that the local time is behind UTC.

After the DST is enabled, the local time and time difference will change compared with UTC, and the values of related fields also need to be changed. Disabling the DST will bring into the opposite effect.

Example: In 1986, the DST was in effect from May 4 at 2:00 a.m. (GMT+8). During the DST period, the clocks were moved one hour ahead, which means that there was one less hour on that day. When the DST ends at 2:00 a.m. on September 14, 1986, the clocks were moved one hour back and there was an extra hour on that day. The changes of the time are as follows:

- DST Starts: 1986-05-04T02:00:00+08:00 --> 1986-05-04T03:00:00+09:00
- DST Ends: 1986-09-14T02:00:00+09:00 --> 1986-09-14T01:00:00+08:00

Notes:

- The time difference cannot be simply used to determine the time zone. Because when the DST starts, the time difference will change and it cannot represent the actual time zone.
- Both TZ (UTC time, e.g., 1986-05-03T18:00:00Z) and TD (local time and time difference, e.g., 1986-05-04T02:00:00+08:00) meet the time format standards of ISO 8601. In ISAPI, the TD format is recommended to be used in messages sent from the user applications and the devices.
- A few old-version devices will return the time in TZ format. For representing the time difference information and forward compatibility, an extra field `timeDiff` is added as shown in the example below. User applications need to support both TD format and TZ format when parsing the time in the messages returned by devices.

```
{
    "dateTime": "1986-05-03T18:00:00Z", /*device time. The value in TZ format is the UTC time and the value in TD format is the time difference between the device's local time and UTC*/
    "timeDiff": "+08:00" /*optional, time difference between the local time and UTC time. If this field does not exist, the user application will convert the dateTime into the local time for use*/
}
```

4.2.5 Character Set

To prevent characters not commonly used from resulting in exceptions in device programs and user applications, ISAPI limits the valid field values of type string to a specific range of characters. Character sets allowed to be used in the fields of type string in ISAPI are listed below.

- Single-byte character set: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and special characters (see details in the table below).
- Multi-byte character set: language characters based on Unicode and encoded by UTF-8 (UTF-8 encoding is a transformation format of Unicode character set. For details, refer to RFC 2044).

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No.	Name	Special Character	No.	Name	Special Character
1	Open Parenthesis	(	18	Dollar Sign	$
2	Close Parenthesis	)	19	Percent Sign	%
3	Plus Sign	+	20	Ampersand	&
4	Comma	,	21	Close Single Quotation Mark	'
5	Minus Sign	-	22	Asterisk	*
6	Period	.	23	Slash	/
7	Semicolon	;	24	Smaller Than	<
8	Equal Sign	=	25	Greater Than	>
9	At Sign	@	26	Question Mark	?
10	Open Square Bracket	[	27	Caret	^
11	Close Square Bracket	]	28	Open Single Quotation Mark	'
12	Underscore	_	29	Vertical Bar	|
13	Open Brace	{	30	Tilde	~
14	Close Brace	}	31	Double Quotation Marks	"
15	Space		32	Colon	:
16	Exclamation Mark	!	33	Backslash	\
17	Octothorpe	#			

The valid characters that can be used in some special fields are listed below.

- User name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 30 in the special character table.
- Password: User Name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 33 in the special character table.
- Names displayed on the UI (device name, person name, face picture library name, etc.): lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters.
- Normal fields of type string support lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters by default.

4.2.6 Error Processing

When requesting via ISAPI failed (the HTTP status code is not 200), the device will return the HTTP status code and ISAPI error code. For HTTP status codes, refer to 10 Status Code Definitions in RFC 2616. For ISAPI error codes, refer to Error Code Dictionary.

Message Example:

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```
HTTP/1.1 403 Forbidden
Content-Type: application/json; charset="UTF-8"
Date: Thu, 15 Jul 2021 20:43:30 GMT
Content-Length: 229
Connection: Keep-Alive

{
    "requestURL": "/ISAPI/Event/triggers/notifications/channels/whiteLightAlarm",
    "statusCode": 4,
    "statusString": "Invalid Operation",
    "subStatusCode": "notSupport",
    "errorCode": 1073741825,
    "errorMsg": "notSupport"
}
```

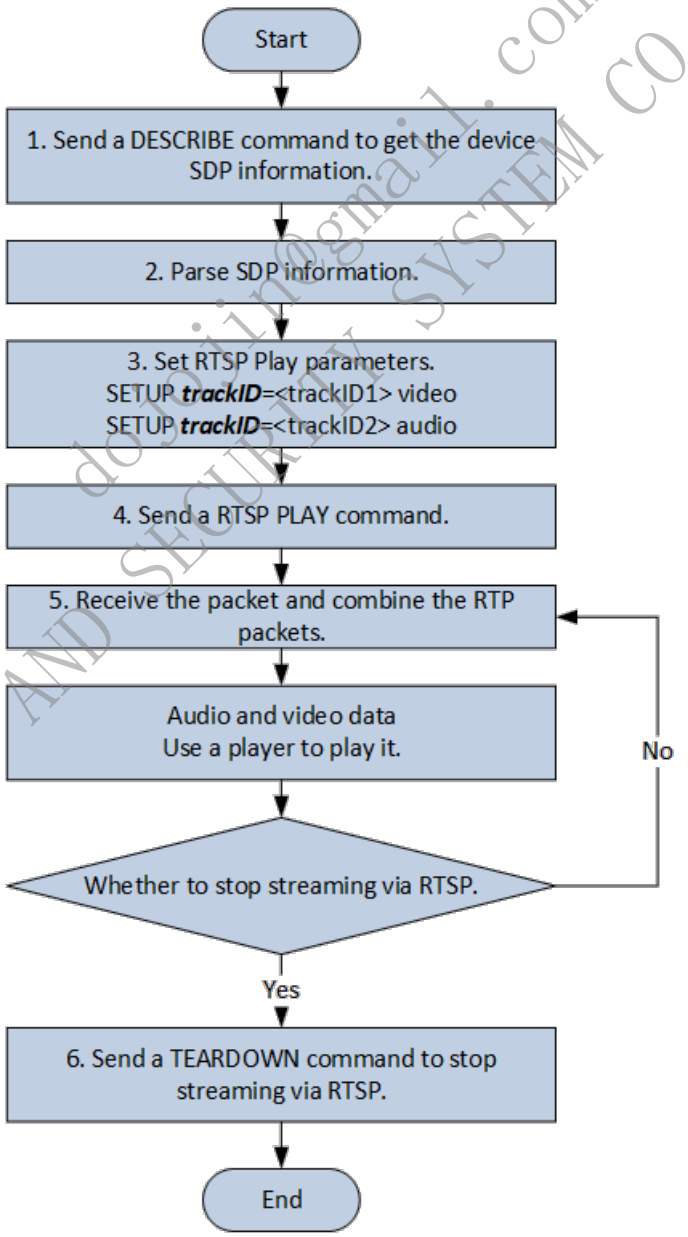
4.3 Real-Time Live View

4.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

4.3.2 API Calling Flow



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1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`. Digest authentication with devices is required before this step. 2. The client parses the media SDP information returned by the device. 3. Set RTSP play parameters, that is to set the track ID parsed from SDP information via SETUP commands. For

example, trackID=1 indicates videos while trackID=2 indicates audios. 4. The client sends an RTSP PLAY command, and the device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`. 5. The client receives the RTP packet sent by the device. Divided RTP packets should be assembled on the client before being parsed. 6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address; `[:port]` is optional, and 554 by default; `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, the IP address of the target device is `172.7.203.11`, and the streaming address of main stream for channel 17 will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

4.3.3 Example

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

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```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.22.126
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000020000011171011040lF000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP requests, and the server responds to them.

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

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```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J............X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

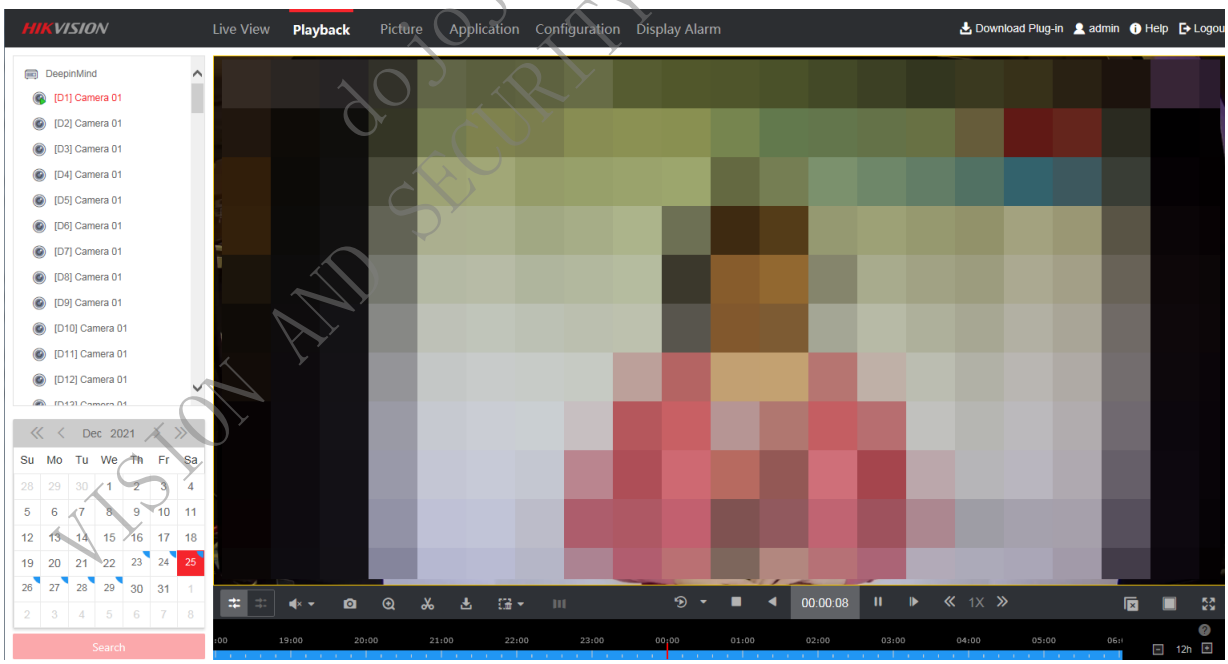
```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

4.4 Playback

4.4.1 Introduction to the Function

Devices such as cameras and NVRs can store videos. Storage mediums such as SD card, TF card and HDDs are needed before recording videos. The process of playback starts with searching for footages and then getting video stream via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

Below is an example of playback on a web client. The calendar in the bottom-left corner shows the results of searching for videos. Dates with videos are shown on the calendar and you can click a date to play back searched videos of the corresponding day.



4.4.2 API Calling Flow

1. (Optional) Check the calendar to find dates with stored videos: `POST /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/dailyDistribution`. `<trackStreamID>` = channel number * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, The rackStreamID of the main stream for channel 17 is 1701.
2. Searching for videos by parsing the `playbackURI` of the device to get the RTSP address `POST /ISAPI/ContentMgmt/search`.

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3. Play videos back via RTSP. Steps of playback via RTSP is the same as that of real-time live view. Refer to streaming via RTSP in real-time streaming. See details in RFC 7826 (Real Time Streaming Protocol).

Notes:

- Playback via RTSP needs digest authentication. The method is the same as the one of ISAPI digest authentication.
- Send PAUSE command to pause playback. Send PLAY command to continue. If you want to perform fast forward and slow forward, you can send a PLAY command and set Scale in headers. See details in 10.6 PAUSE and 12.34 Scale in RFC 7826 (Real Time Streaming Protocol).

4.4.3 Example

4.4.3.1 Search for Videos According to Calendar

Request from Client

```
POST /ISAPI/ContentMgmt/record/tracks/101/dailyDistribution HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 119
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<trackDailyParam>
    <year>2021</year>
    <monthOfYear>08</monthOfYear>
</trackDailyParam>
```

Response from Device

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```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:47:43 GMT
Content-Length: 2915
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<trackDailyDistribution version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <dayList>
        <day><id>1</id><dayOfMonth>1</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>2</id><dayOfMonth>2</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>3</id><dayOfMonth>3</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>4</id><dayOfMonth>4</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>5</id><dayOfMonth>5</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>6</id><dayOfMonth>6</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>7</id><dayOfMonth>7</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>8</id><dayOfMonth>8</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>9</id><dayOfMonth>9</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>10</id><dayOfMonth>10</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>11</id><dayOfMonth>11</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>12</id><dayOfMonth>12</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>13</id><dayOfMonth>13</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>14</id><dayOfMonth>14</dayOfMonth><record>false</record></day>
        <day><id>15</id><dayOfMonth>15</dayOfMonth><record>false</record></day>
        <day><id>16</id><dayOfMonth>16</dayOfMonth><record>false</record></day>
        <day><id>17</id><dayOfMonth>17</dayOfMonth><record>false</record></day>
        <day><id>18</id><dayOfMonth>18</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>19</id><dayOfMonth>19</dayOfMonth><record>false</record></day>
        <day><id>20</id><dayOfMonth>20</dayOfMonth><record>false</record></day>
        <day><id>21</id><dayOfMonth>21</dayOfMonth><record>false</record></day>
        <day><id>22</id><dayOfMonth>22</dayOfMonth><record>false</record></day>
        <day><id>23</id><dayOfMonth>23</dayOfMonth><record>false</record></day>
        <day><id>24</id><dayOfMonth>24</dayOfMonth><record>false</record></day>
        <day><id>25</id><dayOfMonth>25</dayOfMonth><record>false</record></day>
        <day><id>26</id><dayOfMonth>26</dayOfMonth><record>false</record></day>
        <day><id>27</id><dayOfMonth>27</dayOfMonth><record>false</record></day>
        <day><id>28</id><dayOfMonth>28</dayOfMonth><record>false</record></day>
        <day><id>29</id><dayOfMonth>29</dayOfMonth><record>false</record></day>
        <day><id>30</id><dayOfMonth>30</dayOfMonth><record>false</record></day>
        <day><id>31</id><dayOfMonth>31</dayOfMonth><record>false</record></day>
    </dayList>
</trackDailyDistribution>
```

4.4.3.2 Search for Videos

Request from Client

```
POST /ISAPI/ContentMgmt/search HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 486
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<CMSearchDescription>
    <searchID>88C2CD4D-D3FA-4AD4-BD80-555C18205DCC</searchID>
    <trackList>
        <trackID>101</trackID>
    </trackList>
    <timeSpanList>
        <timeSpan>
            <startTime>2021-08-16T00:00:00Z</startTime>
            <endTime>2021-08-18T23:59:59Z</endTime>
        </timeSpan>
    </timeSpanList>
    <maxResults>100</maxResults>
    <searchResultPostion>0</searchResultPostion>
    <metadataList>
        <metadataDescriptor>//recordType.meta.std-cgi.com</metadataDescriptor>
    </metadataList>
</CMSearchDescription>
```

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Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:19:13 GMT
Content-Length: 1021
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<CMSearchResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <searchID>{88c2cd4d-d3fa-4ad4-bd80-555c18205dcc}</searchID>
    <responseStatus>true</responseStatus>
    <responseStatusStrg>OK</responseStatusStrg>
    <numOfMatches>1</numOfMatches>
    <matchList>
        <searchMatchItem>
            <sourceID>{00000000-0000-0000-0000-000000000000}</sourceID>
            <trackID>101</trackID>
            <timeSpan>
                <startTime>2021-08-18T15:18:15Z</startTime>
                <endTime>2021-08-18T15:19:08Z</endTime>
            </timeSpan>
            <mediaSegmentDescriptor>
            <contentType>video</contentType>
            <codecType>H.264-BP</codecType>
            <playbackURI>rtsp://10.14.97.40/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&amp;endtime=20210818T151908Z&amp;name=00000004667000100&amp;size=1400788</playbackURI>
            <lockStatus>unlock</lockStatus>
            <name>00000004667000100</name>
            </mediaSegmentDescriptor>
            <metadataMatches>
                <metadataDescriptor>recordType.meta..com/timing</metadataDescriptor>
            </metadataMatches>
        </searchMatchItem>
    </matchList>
</CMSearchResult>
```

4.4.3.3 Playback via RTSP

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 4
Authorization: Digest username="", realm="323852ae0234c718f2d4198b", nonce="34e30476d", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="883546f4c19dd156fb3a490266c99715"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
```

2. The server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 5
WWW-Authenticate: Digest realm="323852ae0234c718f2d4198b", nonce="55e5895b9", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 7
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="cf33e4dc6b86a2fdd2e5b26d25e7b99d"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Accept: application/sdp
```

4. The server responds to SDP information.

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```
RTSP/1.0 200 OK
CSeq: 7
Content-Type: application/sdp
Content-Length: 546
Date: Wed, 18 Aug 2021 07:29:25 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.60.000
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:4106
a=range:clock=20210818T151815Z-20210818T151908Z
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H265/90000
a=control:trackID=video
b=AS:4096
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=audio
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000040005001171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP commands two times, and the server responds to them.

```
SETUP rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=video RTSP/1.0
CSeq: 8
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61048-61049
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61048-61049;server_port=62000-62001;ssrc=d6d3ed5
CSeq: 8
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

```
SETUP rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=audio RTSP/1.0
CSeq: 9
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61050-61051
Session: 225263317
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61050-61051;server_port=62002-62003;ssrc=d6d3ed6
CSeq: 9
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

6. The client sends an RTSP PLAY request, and the server starts to push streams for playback.

```
PLAY rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 10
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="4d8b7ceb301c06b40a9ae411189a909f"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
Range: npt=0.000-
```

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```
RTSP/1.0 200 OK
Range: clock=20210818T151815Z-20210818T151908Z
Session: 225263317
CSeq: 10
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

7. The client sends an RTSP TEARDOWN request to stop playback.

```
TEARDOWN rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 11
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="1d3f6f8d07d7087d341560b125445456"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
```

```
RTSP/1.0 200 OK
CSeq: 11
Date: Wed, 18 Aug 2021 07:29:38 GMT
Session: 225263317
Connection: close
```

4.5 Event Uploading

When the rules configured on the device are triggered, the device will generate event messages (e.g., motion detection, etc.) and actively upload them to the client. ISAPI supports three methods to receive event messages uploaded by the device, that is, in arming mode, in listening mode, and via subscription.

4.5.1 Arming

The client establishes a HTTP persistent connection with the device to receive event messages from the device.

There are two methods (arming with subscription and arming without subscription) to receive events from the device. The arming without subscription is to get all event messages from the device via HTTP GET method, while the arming with subscription is to get messages of subscribed events via HTTP POST method.

Notes

- ISAPI arming (with or without subscription) uses the HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send event messages continuously, while it's not supported for clients to send any message to the device via the connection.
- When the heartbeat timed out and no message is received from the device, you should terminate the arming connection and try establishing a new one.

4.5.1.1 Arming without Subscription

1. Establish the connection of arming without subscription: `GET /ISAPI/Event/notification/alertStream` and keep the connection alive via configuring `Connection: keep-alive` in HTTP headers on the client.
2. Receive events sent by the device. The event message will be separated and parsed by boundary. For parsing details, see **Event Message Parsing** below.
3. Terminate the arming connection when no event message needs to be received.

Event Message Parsing:

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```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

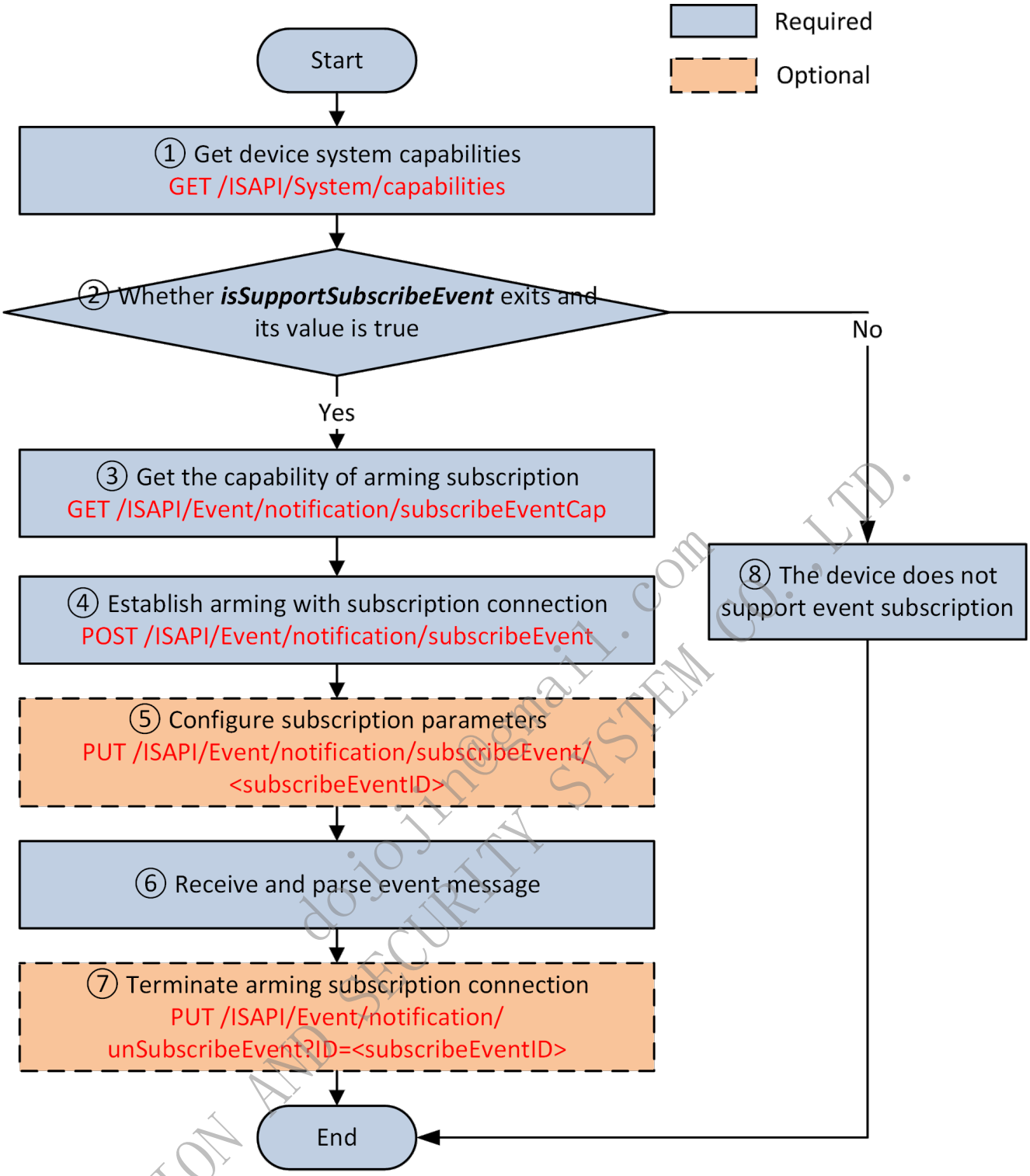
<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

Note: `<data_gateway_ip>` and `<frontier>` are variables, `[Picture Data]` indicates the raw data of a picture.

4.5.1.2 Arming with Subscription

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by the boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note：

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

Event Messages Parsing:

Client Creates Arming with Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

Server Responds to Request and Push Event Message

When the client sends a request to the device for establishing an arming connection, the device sends data in HTTP form format (multipart/form-data). In HTTP request headers, the `Content-Type` is `multipart/form-data, boundary=AaB03x`, of which the boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Please note two hyphens `--` before and after boundary. In normal conditions, devices will not close the arming link actively, and the end mark `--AaB03x--` will not be sent on the arming link.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Picture Data
--AaB03x--
```

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. A form message which starts with `--boundary` and ends with `--boundary--`.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description. `form-data` is a piece of form data.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next `\r\n` to the next `--boundary`.

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5 Device Management (General)

5.1 App Custom & Auto-Parsing Event(s)

5.1.1 Introduction to the Function

The HEOP's third-party app has the requirement of third-party platform communication. In order to improve the communication efficiency between third-party app and platforms, HEOP provides an event upload mode of reusing the existing communication link of HCNetSDK, ISAPI, ISUP, and OpenNetStream.

As the third-party apps are imported to the embedded devices, when apps reuse the existing communication link to upload events, the event content cannot be predicted by embedded devices. Therefore, it is necessary to constrain the events format uploaded by the third-party app to ensure the unified processing of embedded devices, and implement the auto-parsing and displaying of business platform.

This chapter mainly introduces the technical requirements and constraints on the content of events uploaded by third-party apps.

Two types of events are provided for uploading custom events of the third-party app.

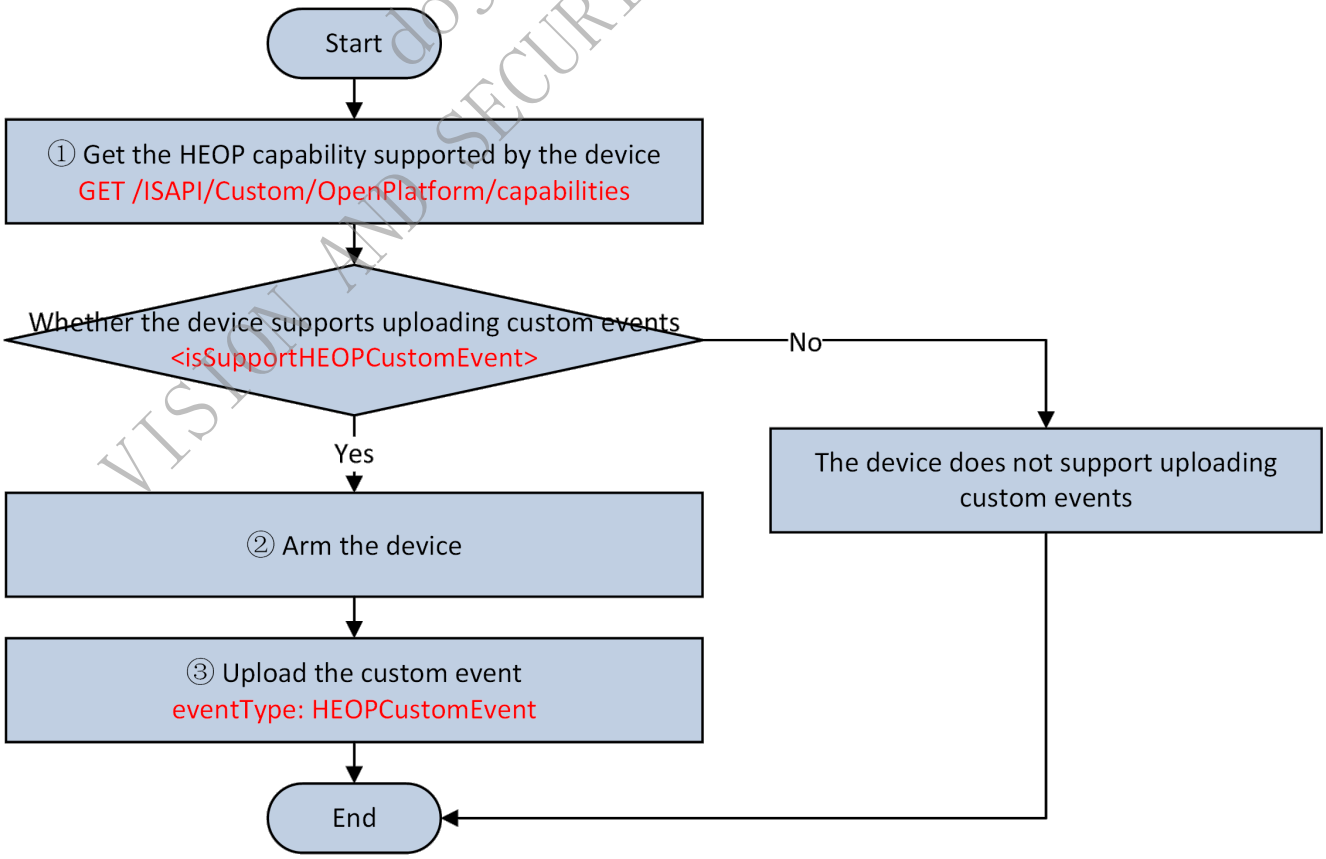
1. Uploading the custom event: the data is in JSON format, and the data content is agreed between the device and the platform. Embedded devices have no constraints on the uploaded content.
2. Uploading the auto-parsing event: the data is in JSON format, third-party apps need to comply with constraints on auto-parsing, and embedded devices verify the format of data uploaded by third-party apps. It realize the automatic parsing of platforms.

If the third-party app needs to use the existing communication link, you need to know the type of supported event uploading before programming.

5.1.2 API Calling Flow

5.1.2.1 Custom Event(s)

API Calling Flow of Uploading Custom Events



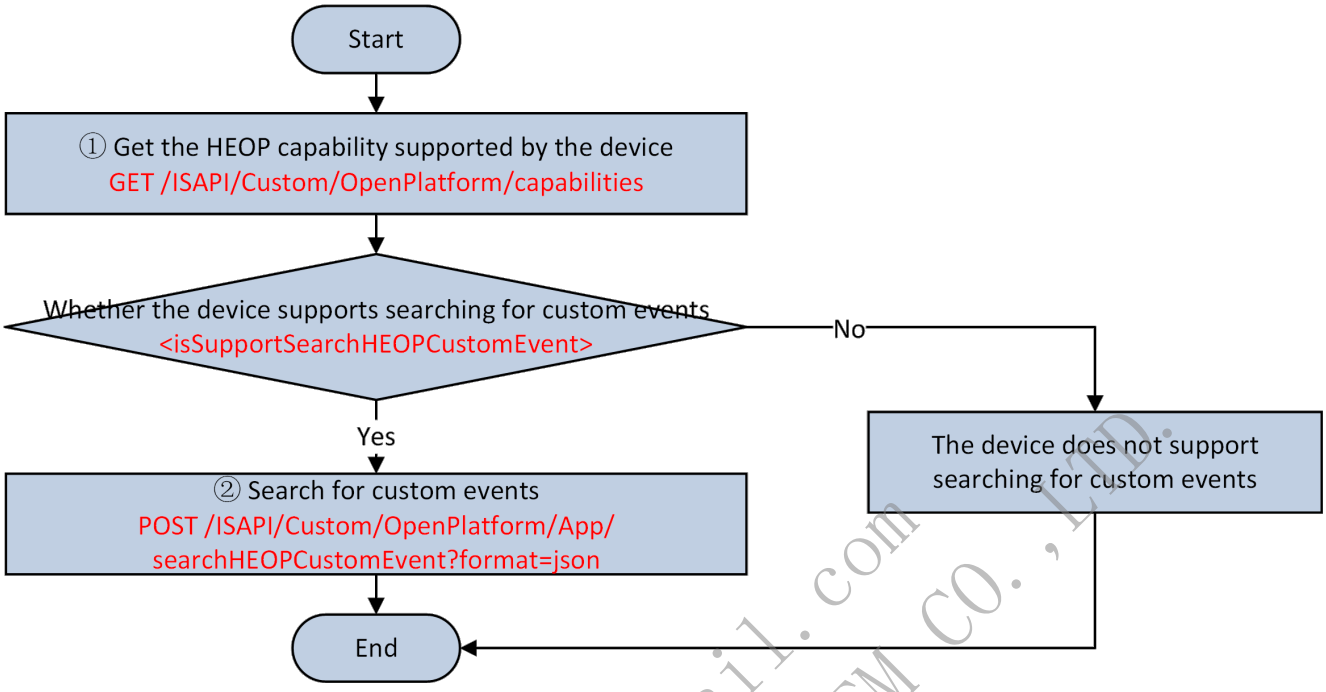
1. Get the HEOP capability supported by the device: `GET /ISAPI/Custom/OpenPlatform/capabilities`. If the node `<isSupportHEOPCustomEvent>` is returned in the capability and its value is true, it indicates that uploading custom

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events is supported by the device.

2. Arm the device, see "Event Arming and Subscription Management" for details.
3. Upload the custom event (eventType: `HEOPCustomEvent`).

API Calling Flow of Searching for Custom Events

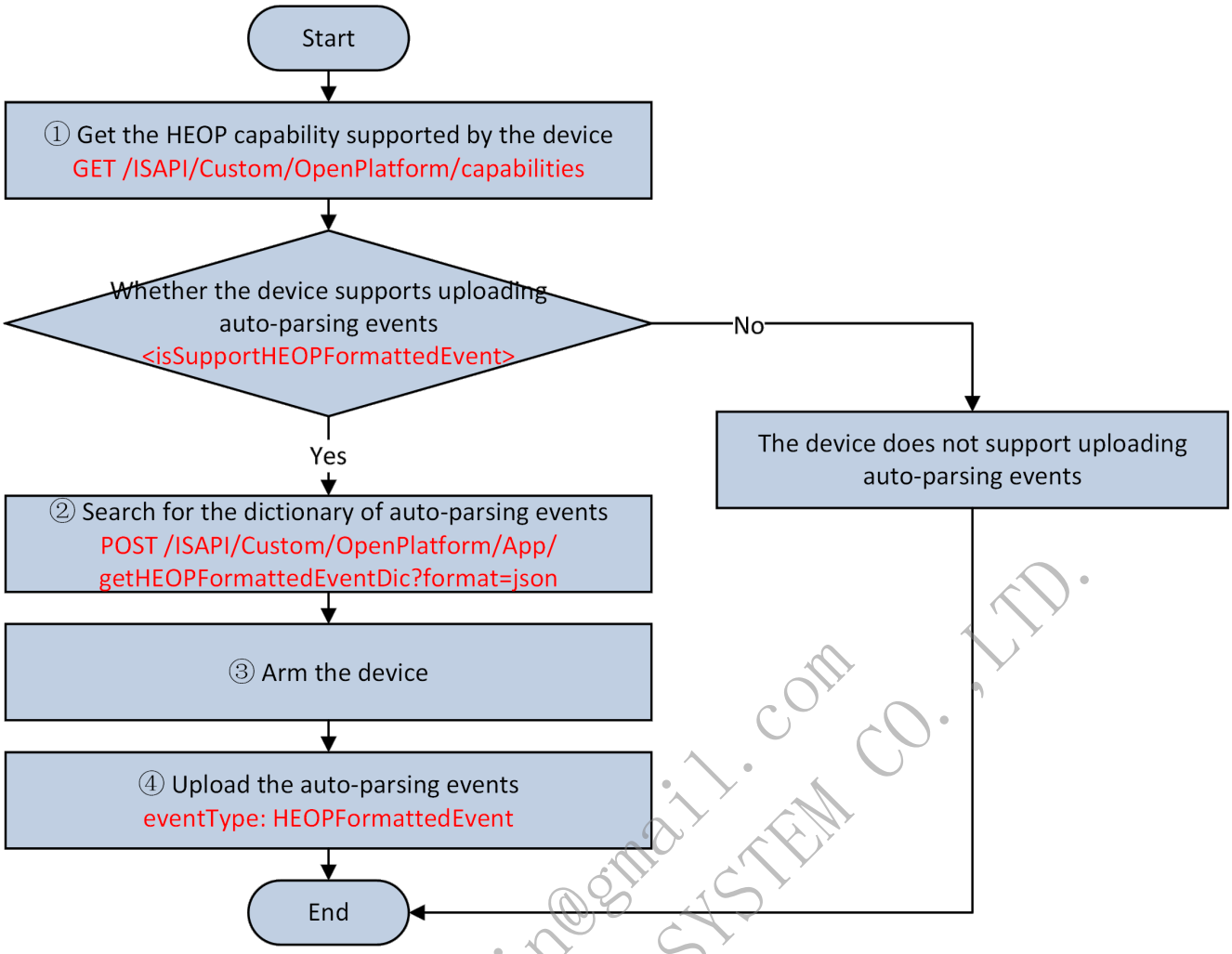


1. Get the HEOP capability supported by the device: `GET /ISAPI/Custom/OpenPlatform/capabilities`. If the node `<isSupportSearchHEOPCustomEvent>` is returned in the capability and its value is true, it indicates that searching for custom events is supported by the device.
2. Search for custom events: `POST /ISAPI/Custom/OpenPlatform/App/searchHEOPCustomEvent?format=json`.

5.1.2.2 Auto-Parsing Event(s)

API Calling Flow of Uploading Auto-Parsing Events

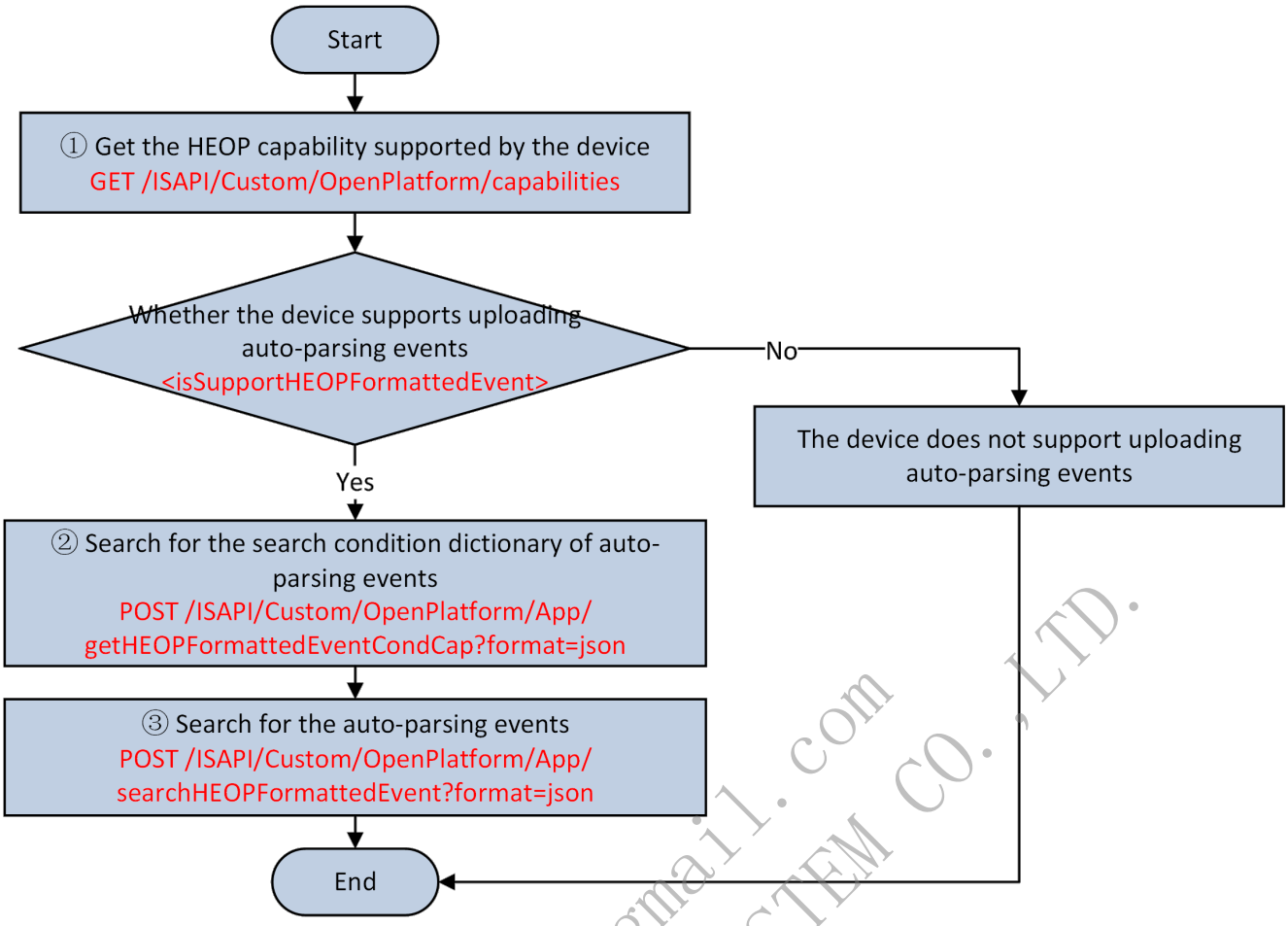
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dojojin@gmail.com



1. Get the HEOP capability supported by the device: `GET /ISAPI/Custom/OpenPlatform/capabilities`. If the node `<isSupportHEOPFormattedEvent>` is returned in the capability and its value is true, it indicates that uploading auto-parsing events is supported by the device.
2. Search for the dictionary of auto-parsing events: `POST /ISAPI/Custom/OpenPlatform/App/getHEOPFormattedEventDic?format=json`.
3. Arm the device, see "Event Arming and Subscription Management" for details.
4. Upload the auto-parsing event (event type: `HEOPFormattedEvent`).

API Calling Flow of Searching for Auto-Parsing Events

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1. Get the HEOP capability supported by the device: `GET /ISAPI/Custom/OpenPlatform/capabilities`. If the node `<isSupportSearchHEOPFormattedEvent>` is returned in the capability and its value is true, it indicates that searching for auto-parsing events is supported by the device.
2. Search for the search condition dictionary of auto-parsing events: `POST /ISAPI/Custom/OpenPlatform/App/getHEOPFormattedEventCondCap?format=json`.
3. Search for auto-parsing event: `POST /ISAPI/Custom/OpenPlatform/App/searchHEOPFormattedEvent?format=json`.

5.1.2.3 Message Example and UI Example

1. Example of Searching for Auto-Parsing Events Dictionary

Request Message Example

```
{
    "AppID": "12345"
}
```

Response Message Example

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```
{
    "HEOPFormattedEventDic": [
        {
            "subEventType": "faceInfo",
            "subEventTypeName": "Face Picture Information",
            "ParamsInfo": [
                {
                    "paramName": "gender",
                    "description": "Gender",
                    "type": "enumValue",
                    "enumValue": ["female", "male", "unknown"],
                    "enumValueDescription": ["Female", "Male", "Unknown"]
                },
                {
                    "paramName": "name",
                    "description": "Name",
                    "type": "int",
                },
                {
                    "paramName": "age",
                    "description": "Age",
                    "type": "int",
                },
                {
                    "paramName": "labels",
                    "description": "Labels",
                    "type": "stringArray",
                },
                {
                    "paramName": "faceRect",
                    "description": "Face Frame",
                    "type": "Region",
                }
            ]
        },
        {
            "subEventType": "vehicleInfo",
            "subEventTypeName": "Vehicle Information",
            "ParamsInfo": [
                {
                    "paramName": "plateNo",
                    "description": "License Plate No.",
                    "type": "string",
                    "enumValue": [],
                    "enumValueDescription": []
                }
            ]
        }
    ]
}
```


2. Example of Uploading Auto-Parsing Event

Complete Example

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```
POST /HEOPFormattedEvent HTTP/1.1
Content-Type: multipart/form-data; boundary=7e13971310878
Content-Length: 1343

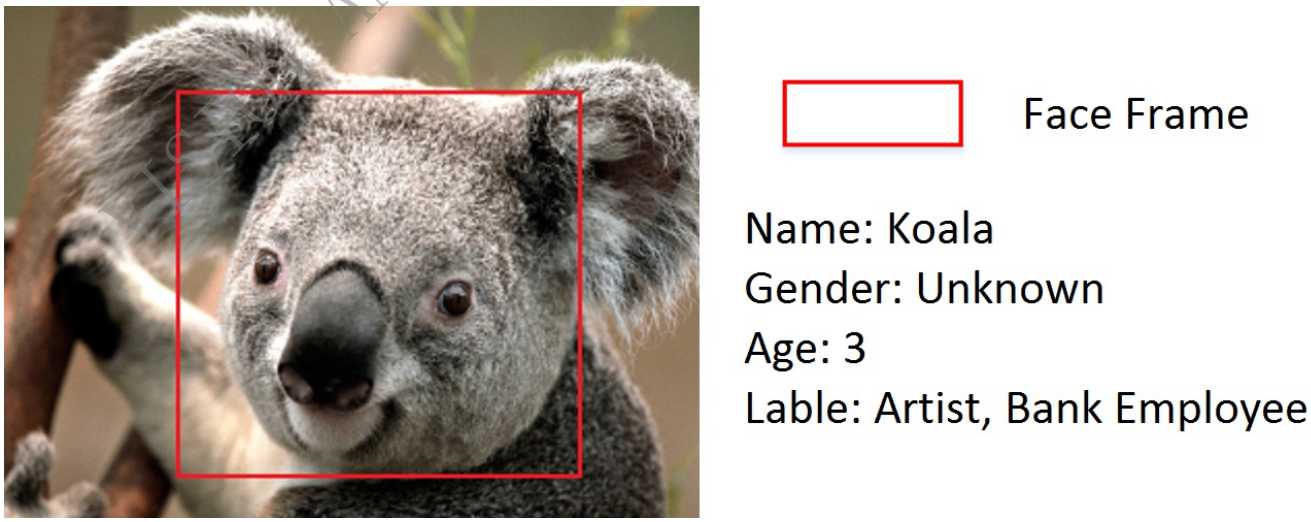
--7e13971310878
Content-Disposition: form-data; name="HEOPFormattedEvent.json";filename="HEOPFormattedEvent.json";
Content-Type: application/json
Content-Length: 997

{
    "ipAddress": "172.6.64.7",
    "portNo": 80,
    "protocol": "HTTP",
    "macAddress": "01:17:24:45:D9:F4",
    "channelID": 1,
    "dateTime": "2004-05-03T17:30:08+08:00",
    "activePostCount": 1,
    "eventType": "HEOPFormattedEvent",
    "eventState": "active",
    "eventDescription": "HEOP Custom Event",
    "channelName": "test",
    "deviceID": "test0123",
    "serialNumber": "test",
    "AppID": "12345",
    "subEventType": "faceInfo",
    "HEOPFormattedEvent": {
        "gender": "male",
        "name": "Tom",
        "age": 25,
        "labels": ["Young Artist", "Bank Staff"],
        "faceRect":[
            {
                "x": 0.234,
                "y": 0.234,
            },
            {
                "x": 0.234,
                "y": 0.456,
            },
            {
                "x": 0.456,
                "y": 0.456,
            },
            {
                "x": 0.456,
                "y": 0.234,
            },
        ]
    }
}
--7e13971310878
Content-Disposition: form-data; name="picture1";filename="picture1.jpg";
Content-Type: image/pjpeg
Content-Length: 516876

......JFIF.....
--7e13971310878--
```

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3. Example of Result for Parsing Uploaded Auto-Parsing Event



4. Example of Searching for the Search Condition Dictionary of Auto-Parsing Events

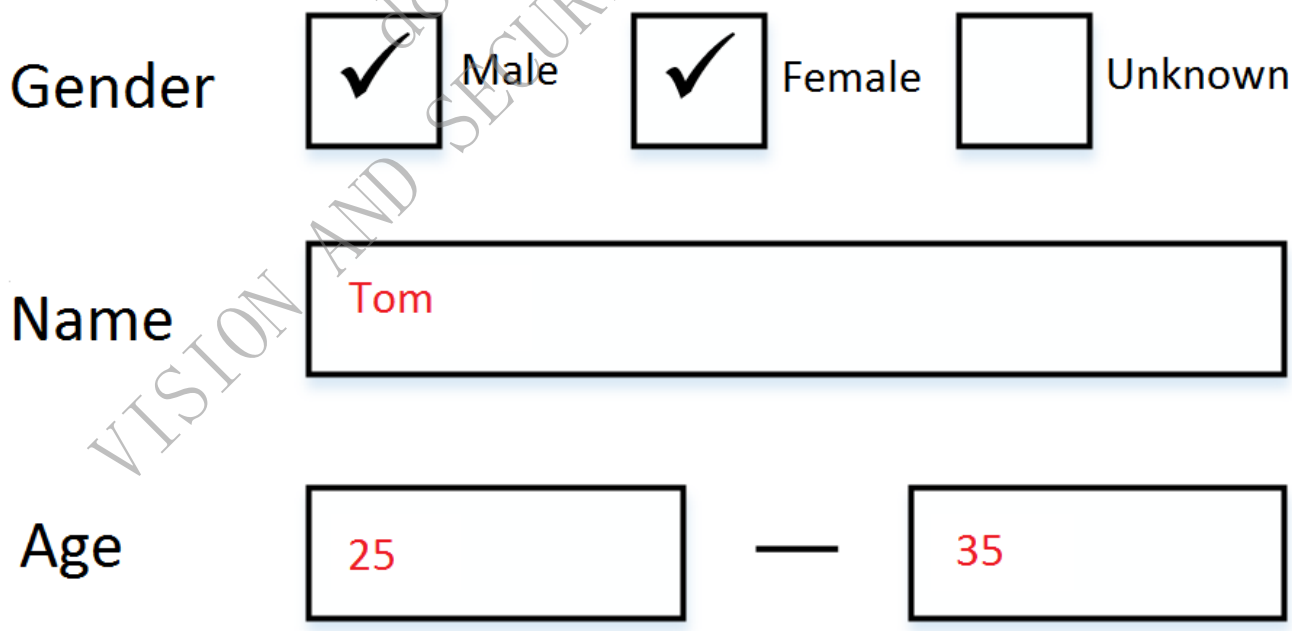
Request Message Example

```
{
    "AppID": "12345"
}
```

Response Message Example

```
{
    "HEOPFormattedEventCondCap": [
        {
            "subEventType": "faceInfo",
            "subEventTypeName": "Face Picture Information",
            "ParamsInfo": [
                {
                    "paramName": "gender",
                    "description": "Gender",
                    "type": "enumValueArray",
                    "enumValueArrayCap": {
                        "size": 3,
                        "enumValue": ["female","male","unknown"],
                        "enumValueDescription": ["Female","Male","Unknown"]
                    }
                },
                {
                    "paramName": "name",
                    "description": "Name",
                    "type": "string",
                    "stringCap": {
                        "size": 64
                    }
                },
                {
                    "paramName": "age",
                    "description": "Age",
                    "type": "intRange",
                    "IntRangeCap": {
                        "minRange": [0, 100],
                        "maxRange": [0, 100],
                    }
                }
            ]
        }
    ]
}
```

5. UI Example of Search Condition of Auto-Parsing Events



6. Example of Searching for Auto-Parsing Event

Request Message Example

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```
{
    "searchID": "123",
    "searchResultPostion": 1,
    "maxResults": 100,
    "startTime": "2020-01-01T08:00+08:00",
    "endTime": "2020-01-02T08:00+08:00",
    "AppID": "12345",
    "subEventType": "faceInfo",
    "HEOPFormattedEventCond": {
        "gender": ["female", "male"],
        "name": "Tom",
        "age": [25, 35]
    }
}
```

Response Message Example

```
{
    "searchID": "123",
    "responseStatusStrg": "OK",
    "numOfMatches": 1,
    "totalMatches": 1,
    "Results":[
        {
            "ipAddress": "172.6.64.7",
            "portNo": 80,
            "protocol": "HTTP",
            "macAddress": "01:17:24:45:D9:F4",
            "channelID": 1,
            "dateTime": "2004-05-03T17:30:08+08:00",
            "activePostCount": 1,
            "eventType": "HEOPFormattedEvent",
            "eventState": "active",
            "eventDescription": "HEOP Custom Event",
            "channelName": "test",
            "deviceID": "test0123",
            "serialNumber": "test",
            "AppID": "12345",
            "subEventType": "faceInfo",
            "HEOPFormattedEvent": {
                "gender": "male",
                "name": "Tom",
                "age": 25,
                "labels": ["Artist", "Bank Employee"],
                "faceRect":[
                    {
                        "x": 0.234,
                        "y": 0.234
                    },
                    {
                        "x": 0.234,
                        "y": 0.456
                    },
                    {
                        "x": 0.456,
                        "y": 0.456
                    },
                    {
                        "x": 0.456,
                        "y": 0.234
                    },
                ]
            }
            "picURL": "http://xxx.xxx.xxx/xxxx",
            "URLCertificationType": "digest"
        }
    ]
}
```

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5.2 Arming and Subscription

5.2.1 Introduction to the Function

With arming and subscription, the client can establish HTTP persistent connection with the device, and continuously receive the event messages from the device.

For ISAPI event arming, the client can receive all types of events by GET method, or receive the subscribed events only by POST method.

5.2.2 API Calling Flow

5.2.2.1 Without Subscription

1. Establish a connection for arming: `GET /ISAPI/Event/notification/alertStream`. You need to set `Connection: keep-alive` in HTTP Headers.
2. When receiving events sent by the device, the event messages can be separated and parsed by boundary. See "Parsing Event Messages" below for details.
3. Disable the arming connection when you do not need to receive event messages.

5.2.2.1.2 Syntax

```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/form-data; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!-- Some alarms are in JSON format, so the upper layer should parse based on the Content-Type field -->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/jpeg
Content-Length: image_length

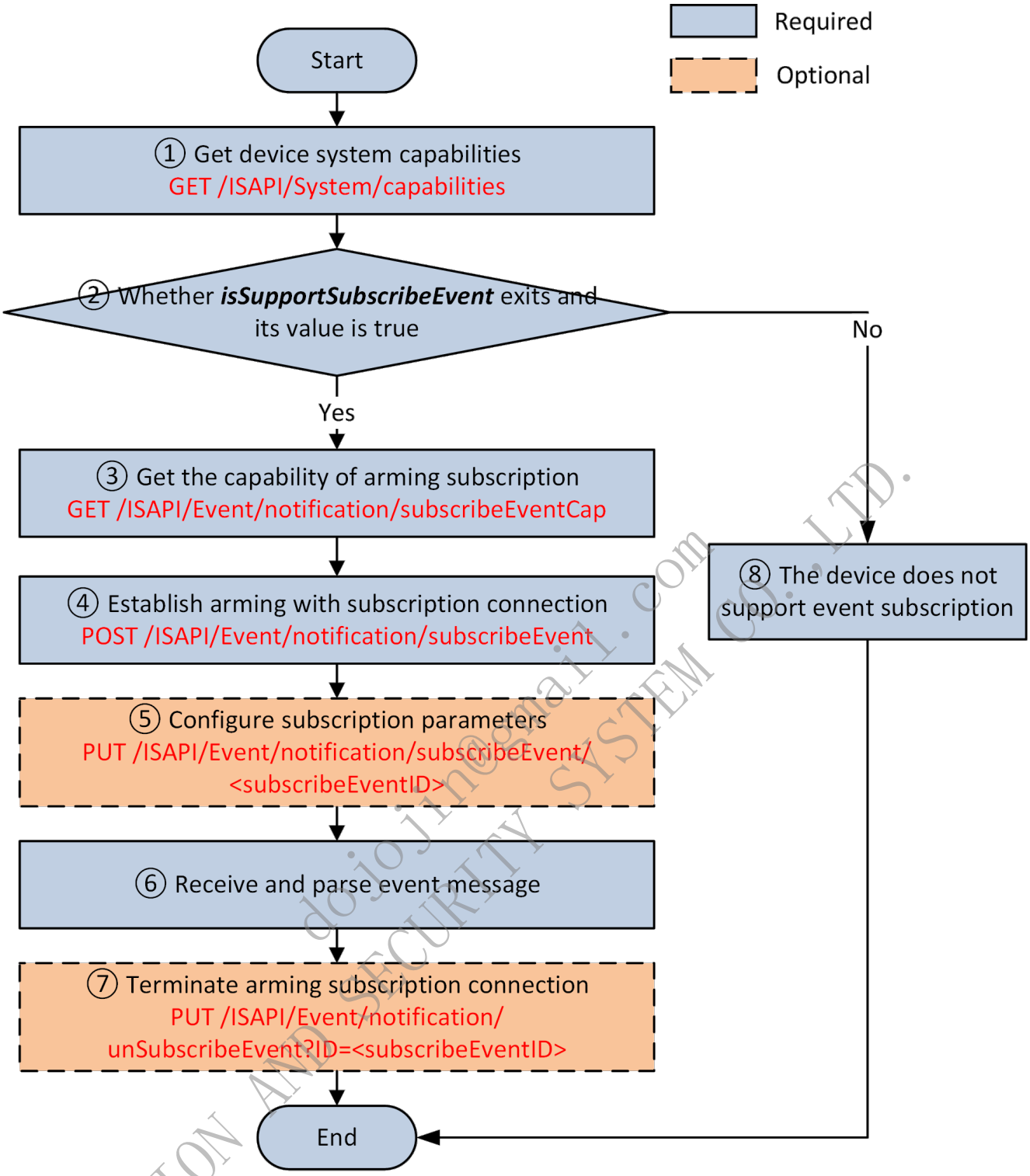
[Image Data]
--<frontier>
```

Note: `<data_gateway_ip>` and `<frontier>` are variables, and `[Image Data]` is an abbreviated representation indicating the raw data of an image at this location.

5.2.2.2 Subscription

5.2.2.2.1 API Calling Flow

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP Headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note:

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

5.2.2.2.2 Example

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Host: device_ip
Accept-Language: zh-cn
Date: YourDate
Content-Type: application/xml;
Content-Length: text_length
Connection: Keep-Alive

<SubscribeEvent/>
HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

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5.2.2.3 Event Messages Parsing

After the arming connection with the device is established, the data sent by the device is in HTTP form format (multipart/form-data). In an HTTP request, `Content-Type` in Headers is usually `multipart/form-data, boundary=AaB03x`, and boundary is a variable used to divide HTTP Body into multiple units, each being a set of data and has its own Headers and Body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Note the dash `--` before and after boundary. Under normal circumstances, the device will not actively close the arming connection, so the device will not send the form format end symbol `--AaB03x--` on the arming connection.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Image Data
--AaB03x--
```

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. --boundary is the start of a form. --boundary-- is the end of the whole HTTP form message.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next \r\n to the next --boundary.

5.2.3 Restriction Description

Note that ISAPI arming (with or without subscription) uses HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send out event messages continuously, while you cannot send any message to the device via the connection.

After the heartbeat time, if you do not receive any message from the device, you should disable the arming connection and try establishing a new one.

5.2.4 Sample Messages

5.2.4.1 Establish Arming Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

5.2.4.2 The Device Responses and Uploads an Event Message

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```
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <ipAddress>10.17.133.46</ipAddress>
    <portNo>80</portNo>
    <protocol>HTTP</protocol>
    <macAddress>44:19:b6:6d:24:85</macAddress>
    <channelID>1</channelID>
    <dateTime>2017-05-04T11:20:02+08:00</dateTime>
    <activePostCount>0</activePostCount>
    <eventType>heartBeat</eventType>
    <eventState>active</eventState>
    <eventDescription>heartBeat</eventDescription>
</EventNotificationAlert>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length
Content-ID: image_ID

[Picture Data]
--<frontier>
```

5.3 Calling Flow of Device Packet Capture

5.3.1 Function Introduction

When problems arise after a device is deployed on site, interaction message between the device and the external network is necessary to help developers for troubleshooting. Packet capture can be stored on the local device, and packet capture files can be exported after capture is complete. Also, packet capture files can be uploaded to cloud storage, and packet capture data can be obtained in real-time even if the device does not have the storage space.

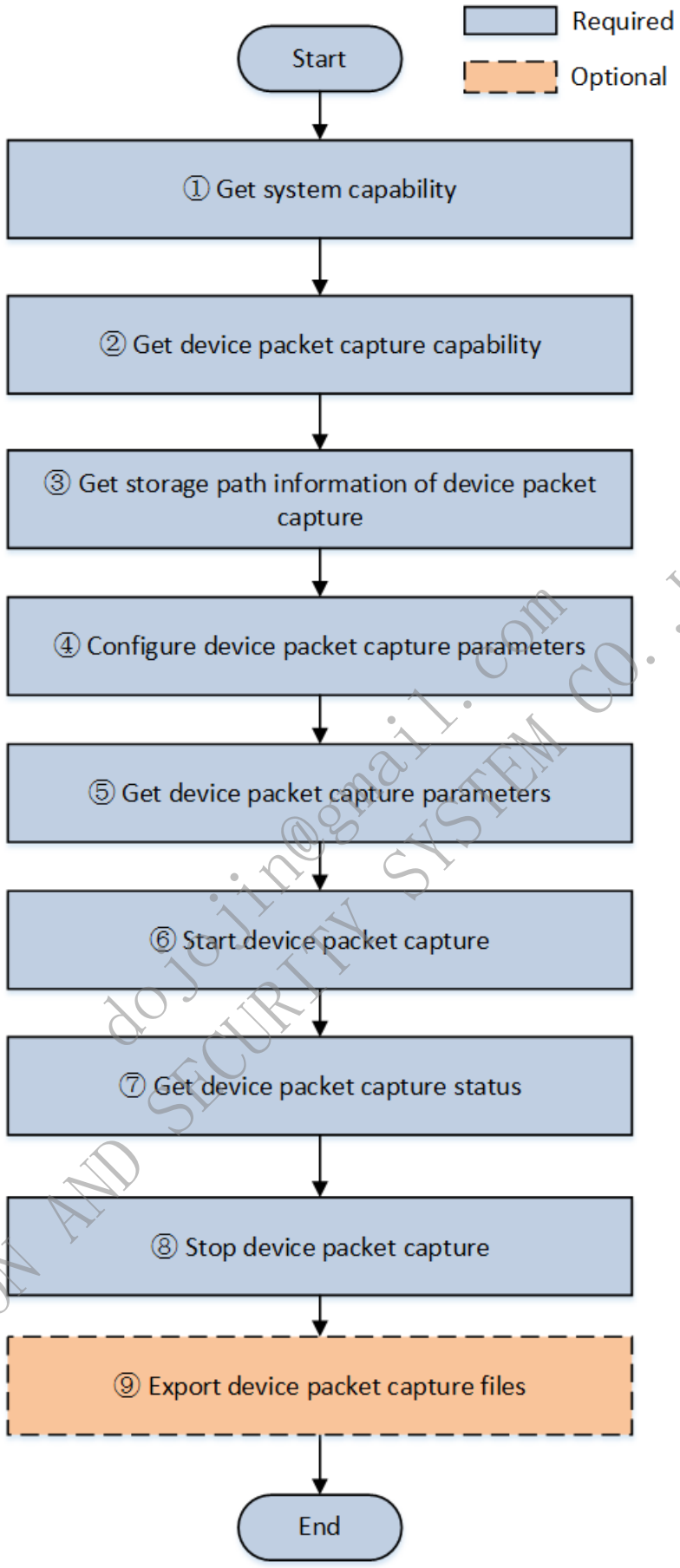
1. Device packet capture: save packet capture files on the local device, and export the files after capture is complete. Also, uploading packet capture files to cloud storage after capture is complete is supported. Then the client can obtain the storage URL and download packet capture files from the cloud storage.

2. Device real-time packet capture: after it is enabled, the device returns an URI for downloading packet capture data. The client can submit this URI to the browser to download the packet data. The device transmits packet data via HTTP Chunked, and users can store the packet data through the browser.

5.3.2 API Calling Flow

5.3.2.1 Device Packet Capture

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`. Get to know if the device supports packet capture by the field: `<isSupportNetworkCapture>true</isSupportNetworkCapture>`.

2. Check if the device supports packet captures: `GET /ISAPI/System/networkCapture/capabilities?format=json`. If `isSupportManualControl` is true, the device supports packet capture. If `isSupportManualControlAsyn` is true, the device supports asynchronous packet capture.

3. Get storage path information of device packet capture: `GET /ISAPI/System/networkCapture/StoragePathInfo?format=json`.

4. Configure device packet capture parameters such as capture duration, storage path, port, and address: `PUT /ISAPI/System/networkCapture/captureParams?format=json`.

5. Get device packet capture parameters such as capture duration, storage path, port, and address: `GET /ISAPI/System/networkCapture/captureParams?format=json`.

6. Start device packet capture: depending on the parameters, packet capture files can be saved on the local device for export after capture is complete, or packet capture files can be uploaded to cloud storage and downloaded via the storage URL, or the packet capture data can be returned in real-time.

Start device packet capture: `PUT /ISAPI/System/networkCapture/manualStart?format=json&asyn=<asyn>&realTime=<realTime>`.

Note:

- If the API does not contain URL parameters, the packet capture file is saved on the local device.
- If the API contains `asyn=true`, the packet capture file is automatically uploaded to cloud storage after capture is complete.

7. After starting capture, you can repeatedly get capture status, including whether the capture is ongoing, the size of the packet capture data, and the progress and storage URL for uploading the data to cloud storage.

Get status of device packet capture: `GET /ISAPI/System/networkCapture/manualStatus?format=json`.

8. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

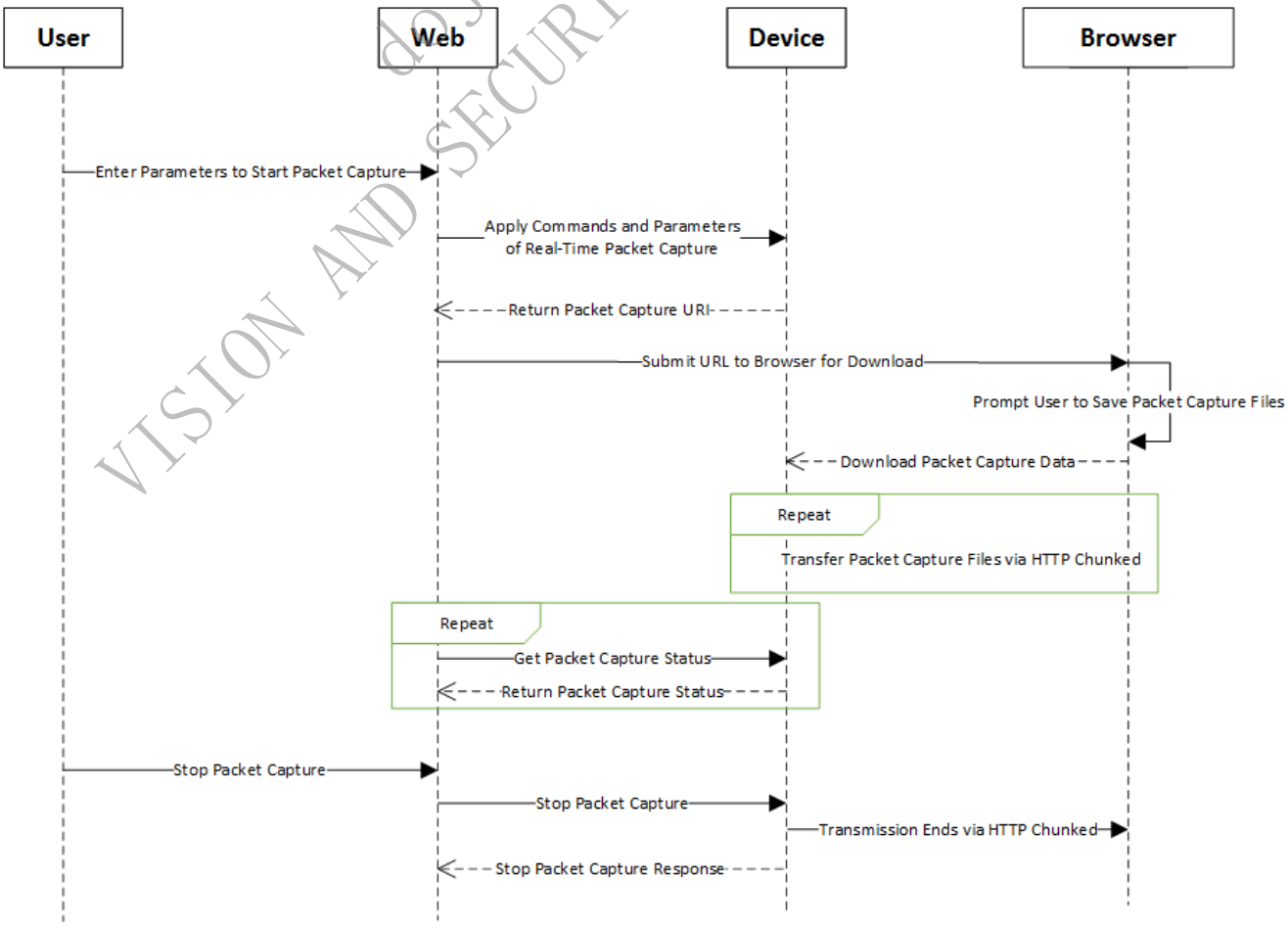
Stop device packet capture: `PUT /ISAPI/System/networkCapture/manualStop?format=json`.

9. (Optional) If packet capture data is stored on the local device, packet capture files need to be exported. If packet capture files are saved to cloud storage or captured in real-time, there is no need to export packet capture files.

Export device packet capture files: `GET /ISAPI/System/networkCapture/exportFile?format=json`.

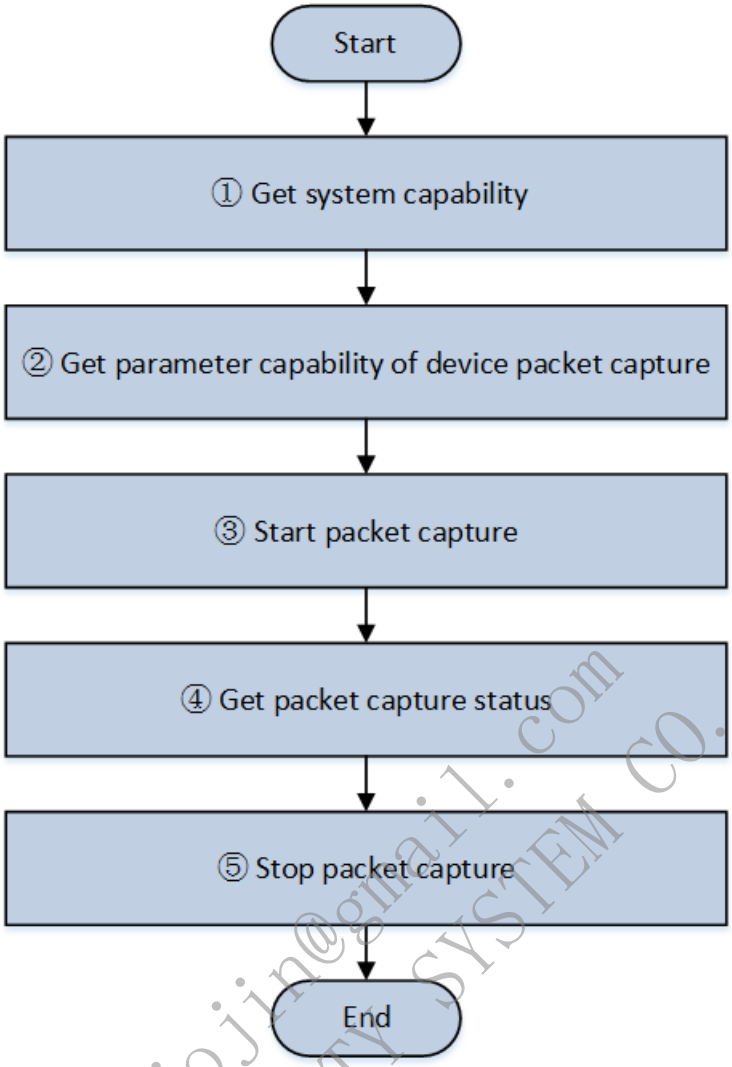
5.3.2.2 Device Real-Time Packet Capture

The process of real-time packet capture is shown as the following:



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See the following figure for the calling flow:



1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
`<isSupportStartNetworkCapture>true</isSupportStartNetworkCapture>` indicates the device supports starting packet capture. `<isSupportStopNetworkCapture>true</isSupportStopNetworkCapture>` indicates the device supports stopping packet capture. `<isSupportGetNetworkCaptureStatus>true</isSupportGetNetworkCaptureStatus>` indicates the device supports getting packet capture status.

2. Get capabilities of packet capture parameters: `GET /ISAPI/System/NetworkCaptureParams/capabilities?format=json`. The `realTimeEnabled` field indicates whether the device supports real-time packet capture.

3. Set the field realTimeEnabled as true in the parameters applied to the device to start device packet capture. The device returns an URI, and the client can download the real-time packet capture data from the device through a browser.

Start packet capture: `POST /ISAPI/System/StartNetworkCapture?format=json&security=<security>&iv=<iv>`.

Note:

- If `realTimeEnabled=true` is contained when starting device packet capture, it indicates packet capture data is uploaded in real-time by HTTP Chunked. The URL for downloading the returned packet capture data is valid for 30 seconds by default. If the download is attempted after this time, the device should return an HTTP 404 status code.

4. After starting packet capture, you can repeatedly get packet capture status, including whether packet capture is ongoing and the size of the packet capture data.

Get packet capture status: `GET /ISAPI/System/GetNetworkCaptureStatus?format=json`.

5. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

Stop device packet capture: `POST /ISAPI/System/StopNetworkCapture?format=json`.

5.4 Device Peripherals Upgrade

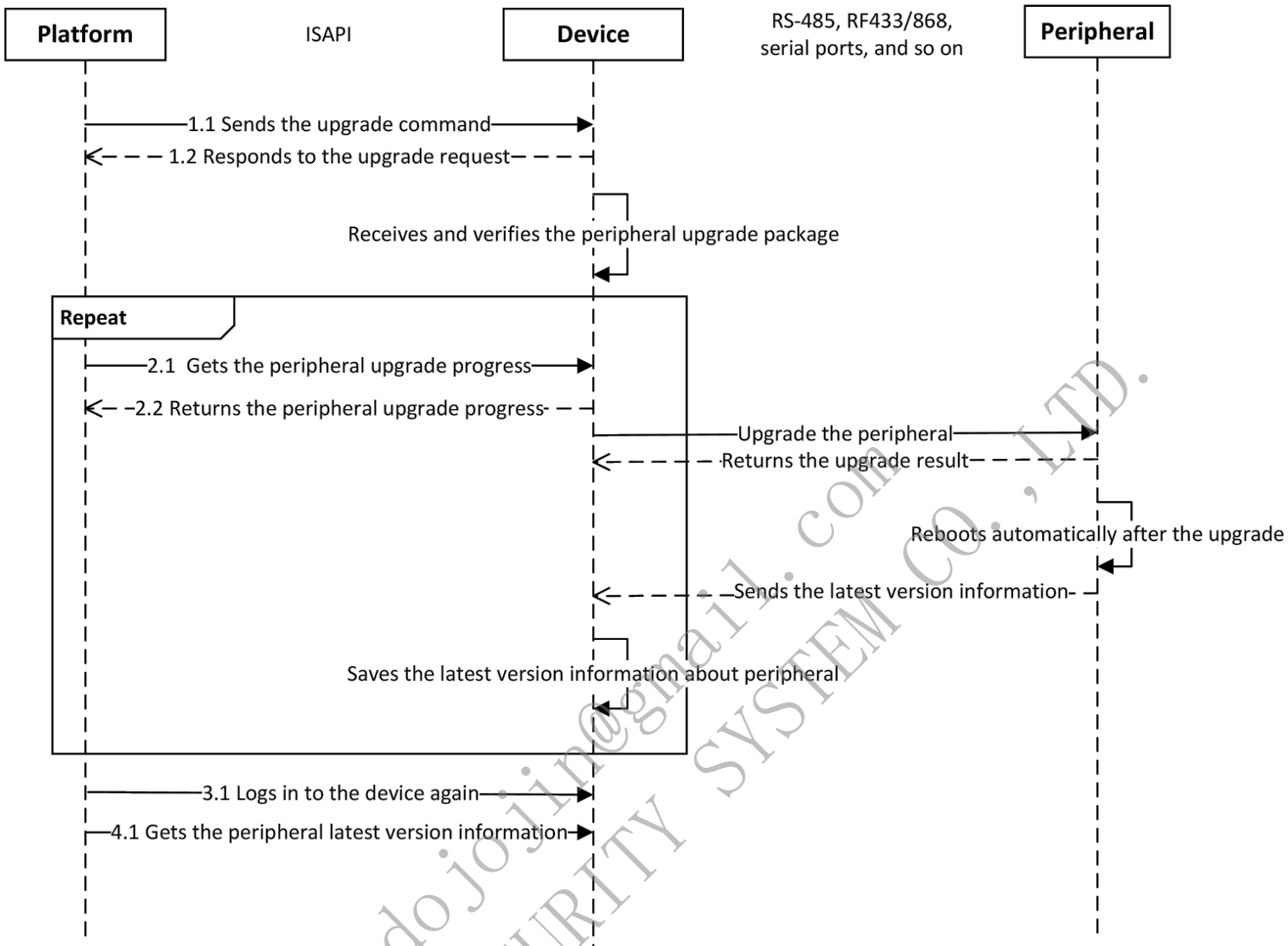
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5.4.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades device peripherals via ISAPI.

5.4.2 API Calling Flow

The sequence diagram of upgrading device peripherals by the platform is shown below.



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1. Get the device system capability `GET /ISAPI/System/capabilities` and check whether the device supports upgrading peripherals. If the field `isSupportAcsUpdate` is returned and its value is true, it indicates that the device supports this function, otherwise, the device does not support this function.
2. Get the capability of upgrading the peripherals module `GET /ISAPI/System/AcsUpdate/capabilities`, and get the types and IDs of peripherals that support upgrading.
3. The platform sends the upgrade command `POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>`. In the URL `type` refers to the peripheral type, `moduleAddress` refers to the peripheral module address, and `indexID` refers to the ID of peripheral to be upgraded. The platform will apply the upgrade peripheral package to the device.
4. Get the peripheral upgrade progress `GET /ISAPI/System/upgradeStatus?type=<Type>`.
5. Log in to the device again.
6. Get the peripheral latest version information.

5.5 Device Time Sync

5.5.1 Introduction to the Function

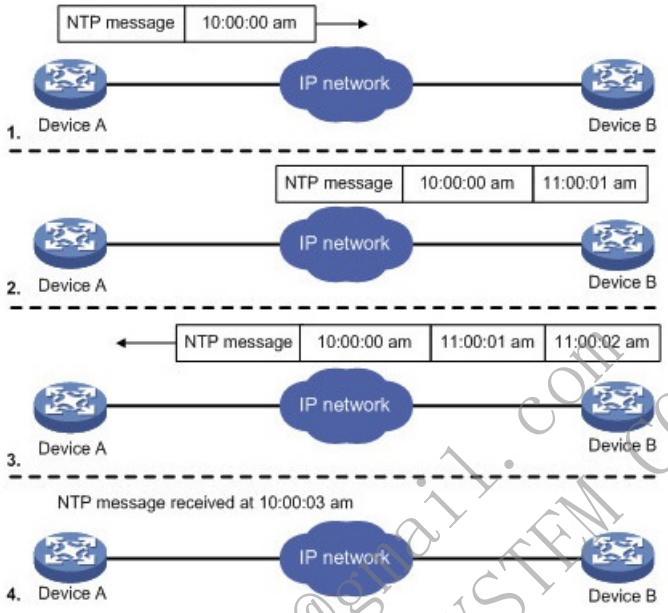
Time sync is a method to synchronize the time of all devices connecting to the NTP server, so that all devices can share the same clock time for providing related functions based on time. Supported time sync types: NTP time sync, manual sync, satellite time sync, platform time synchronization, etc. The following describes the method of NTP time sync.

5.5.1.1 NTP Time Sync

The local system of running NTP can receive sync from other clock sources (self as client), other clocks can sync from the local system (self as server), and sync with other devices.

The basic working principle of NTP is shown in the picture. Device A and Device B are connected via the network, and their systems follow their own independent system time. To auto sync their time, you can set device time auto sync via NTP. For example:

Before time sync between Device A and Device B, the time of Device A is 10:00:00 am, and that of Device B is 11:00:00 am. Device B is set as the server of NTP server, so that the time of Device A should be synchronized with that of Device B. The time of NTP message transmitted between Device A and Device B is 1 second.



The working process of system clock synchronization is as follows:

Device A sends an NTP message to Device B with a timestamp of 10:00:00 am (T1) that is when it leaves Device A.

When the NTP message reaches Device B. Device B will add its own timestamp, which is 11:00:01 am (T2).

Then the NTP message leaves Device B with Device B's timestamp, which is 11:00:02 am (T3).

Device A receives the response message, and the local time of Device A is 10:00:03 am (T4).

Above all, Device A can calculate two important parameters:

Round-trip delay of NTP message: Delay = (T4-T1) - (T3-T2) = 2 seconds.

Time difference between Device A and Device B: offset = ((T2-T1)+(T3-T4))/2=1 h.

Device A can sync its own time with that of Device B according to calculation results.

5.5.2 API Calling Flow

5.5.2.1 Time Sync Configuration

1. Get the Capability of Device Time synchronization Management

You can call this API to get the time sync types currently supported by the device, such as NTP time sync, manual time sync, satellite time sync, EZ platform time sync.

Get the capability: `GET /ISAPI/System/time/capabilities`.

2. Set device time synchronization management parameters

You can configure the time synchronization mode as follows：

Get device time synchronization management parameters: `GET /ISAPI/System/time`;

Set device time synchronization management parameters: `PUT /ISAPI/System/time`;

NTP time synchronization: See 4.2.2 NTP Time Sync (Client).

Manual time synchronization: Set the value of `timeMode` to `manual`, and set the device local time in nodes `localTime`、`timeZone`.

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Satellite time synchronization: Set the value of `timeMode` to `satellite`, and set the device local time in nodes `satelliteInterval`.

Platform time synchronization: Set the value of `timeMode` to `platform`.

Note： For manual time synchronization (time offset including time zone offset): Set manual time synchronization: localTime refers to the local time on device (time offset excluded, in format like 2019-02-28T10:50:44); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00); Get manual time synchronization: localTime refers to the local time on device (time offset included, in format like 2019-02-28T10:50:44+8:30); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00).

5.5.2.2 Time Zone Configuration

1. Get device time zone configuration capability

Call `GET /ISAPI/System/capabilities` to get the system capability. When `isSupportTimeZone` is returned, the time zone configuration is supported by the device.

2. Configure time zone parameters

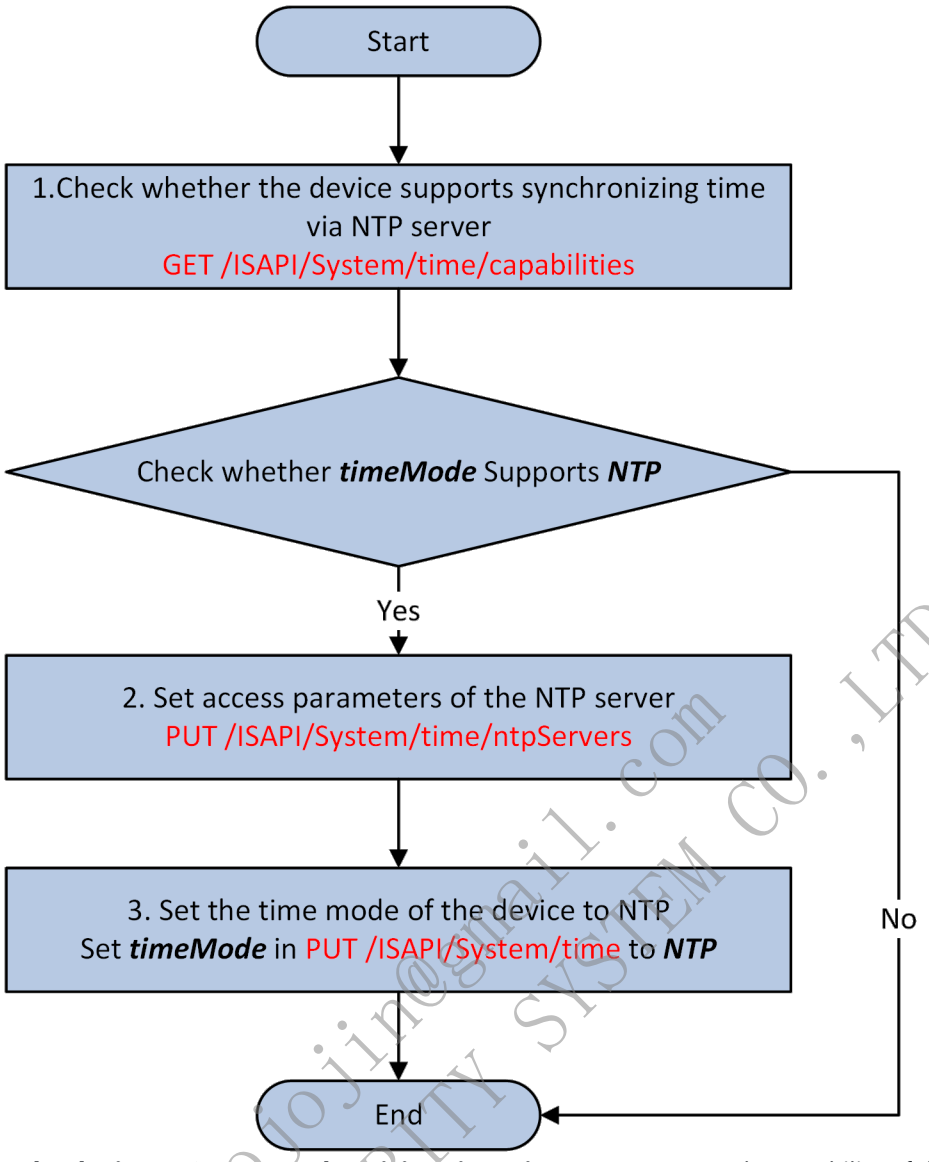
Get the device's time zone parameters: `GET /ISAPI/System/time/timeZone`. Set the device's time zone parameters: `PUT /ISAPI/System/time/timeZone`.

If DST (Daylight Saving Time) is disabled, the example of returned time zone parameters is: CST-8:00:00. It refers to UTC+8, and -8:00:00 is the UTC local time. If DST (Daylight Saving Time) is enabled, the example of returned time zone parameters is: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.2/02:00:00. It refers to UTC+8, the DST time is 30 minutes ahead of local time, the DST starts at 02:00:00 on the first Sunday of April and ends at 02:00:00 on the fifth Tuesday of October. MX.Y.Z: X is the month, Y is the week number in the month, Z is the day of a week (0-Sunday, 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday, 6-Saturday).

5.5.2.3 NTP Time Sync (Client)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as client):

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1. Check whether the device supports synchronizing time via NTP server Get the capability of the device: `GET /ISAPI/System/time/capabilities`, and check whether `timeMode` supports `NTP`.

2. Set access parameters of the NTP server

Supports accessing the NTP server by IP address to synchronize the device time.

Get the access parameter capability of the NTP server: `GET /ISAPI/System/time/ntpServers/capabilities`

Set access parameters of the NTP server: `PUT /ISAPI/System/time/ntpServers`

Get access parameters of the NTP server: `GET /ISAPI/System/time/ntpServers`

3. Set the time mode of the device to NTP

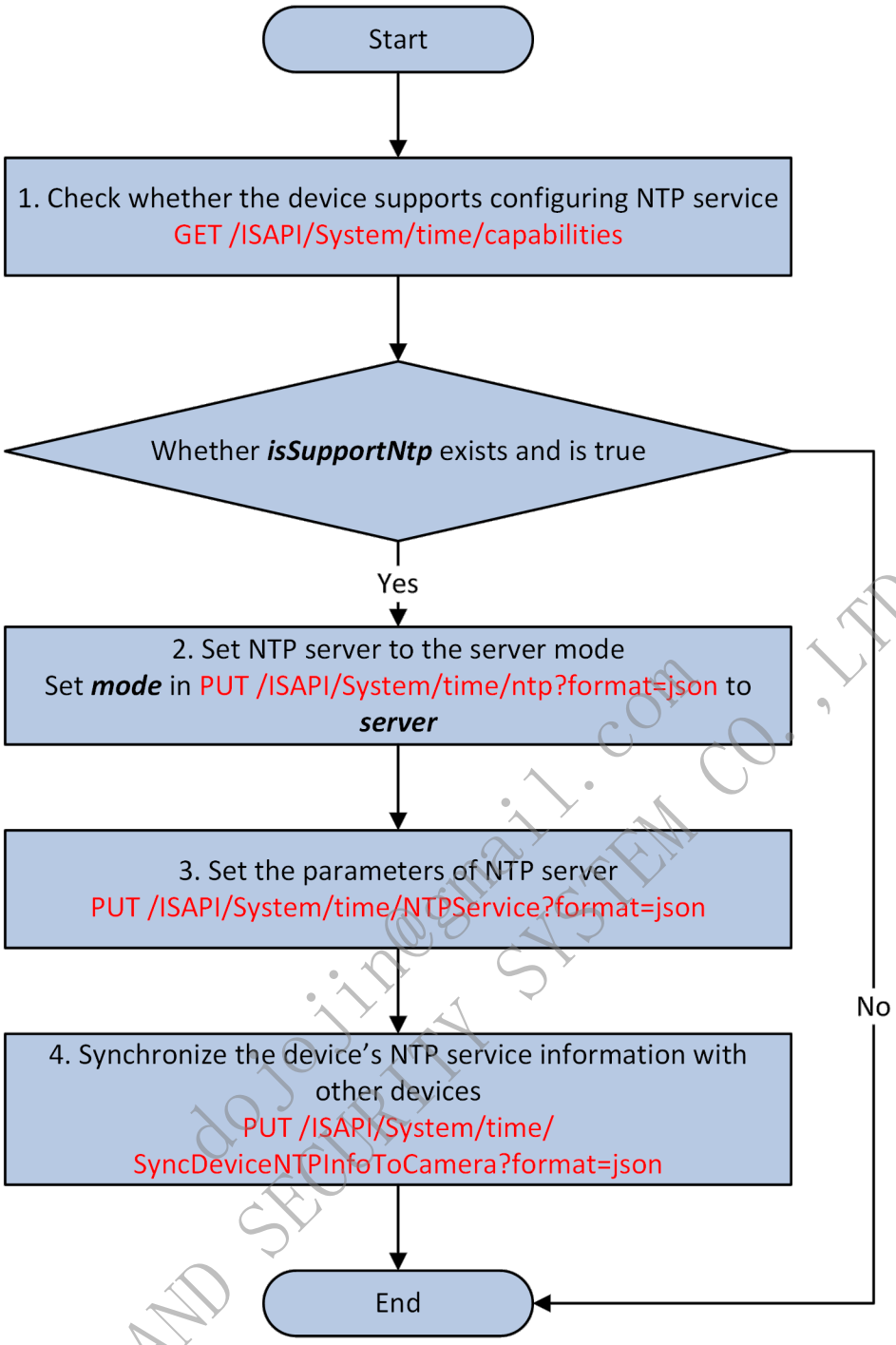
Supports setting the value of `timeMode` to `NTP`.

Get device time synchronization management parameters: `GET /ISAPI/System/time` Set device time synchronization management parameters: `PUT /ISAPI/System/time`

5.5.2.4 NTP Time Sync (Server Mode)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as server):

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1. Check whether the device supports configuring NTP service Get the capability of device time synchronization management: `GET /ISAPI/System/time/capabilities`; If `isSupportNtp` is returned, it indicates that the device supports time synchronization management.

2. Set NTP server to the server mode

Supports setting the value of `mode` to `server`.

Get the capability of server mode: `GET /ISAPI/System/time/ntp/capabilities?format=json`

Set NTP to server mode: `PUT /ISAPI/System/time/ntp?format=json`

Get parameters of NTP server mode: `GET /ISAPI/System/time/ntp?format=json`

3. Set the parameters of NTP server

Supports setting the IP address of the NTP server.

Get the capability of NTP server: `GET /ISAPI/System/time/NTPService/capabilitis?format=json`

Set the NTP server parameters: `PUT /ISAPI/System/time/NTPService?format=json`

Get the parameters of the NTP server: `GET /ISAPI/System/time/NTPService?format=json`

4. Synchronize the device's NTP service information with other devices

Supports synchronizing the time information to the camera.

Get the capability set of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json`

Synchronize device's NTP service information with the camera: `PUT /ISAPI/System/time/SyncDeviceNTPInfoToCamera?format=json`

Get the progress of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/Progress?format=json`

Search for the results of synchronizing device's NTP service information with the camera: `POST /ISAPI/System/time/SyncDeviceNTPInfoToCamera/SearchResult?format=json`

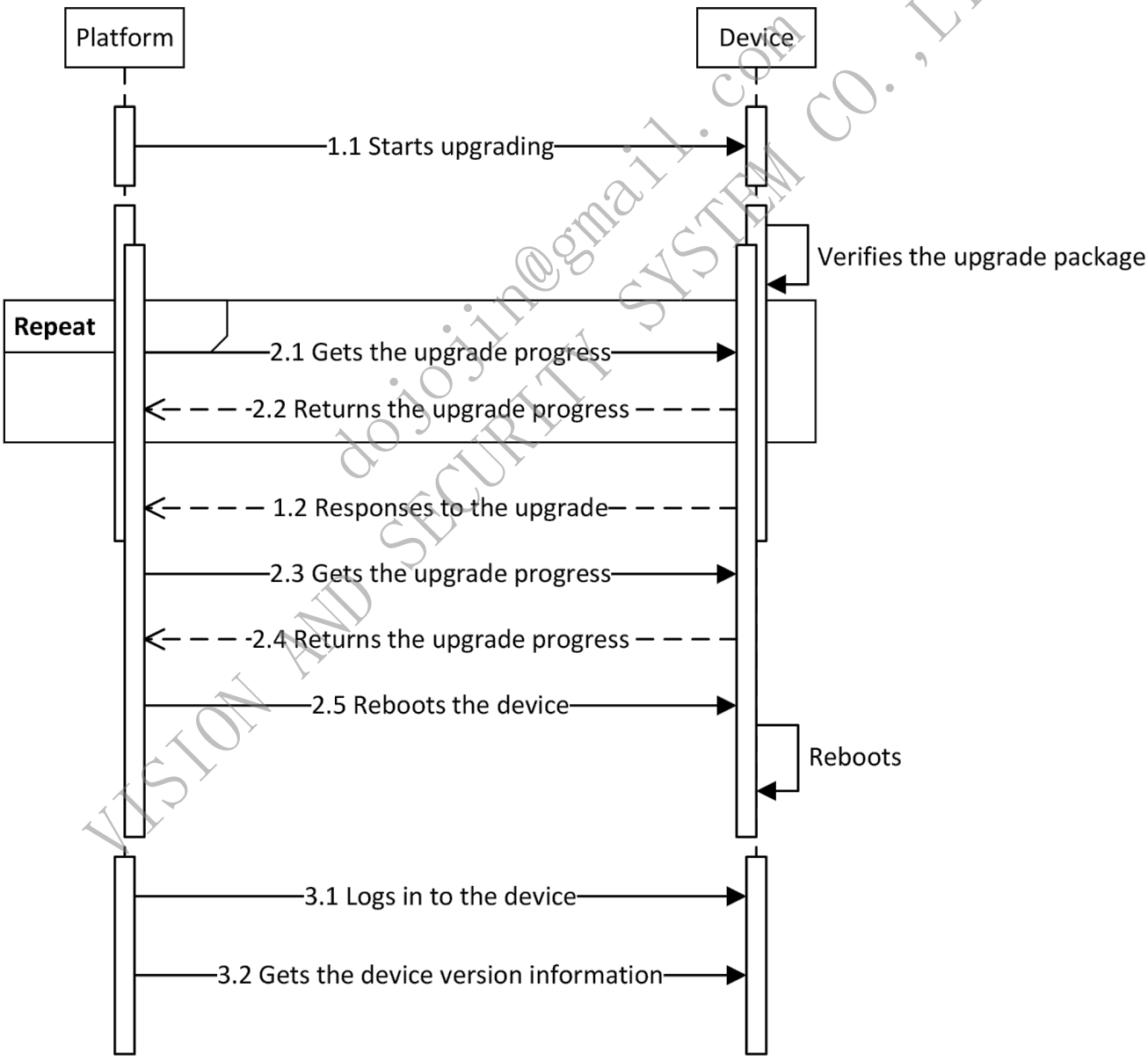
5.6 Device Upgrade

5.6.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades devices via ISAPI.

5.6.2 API Calling Flow

The sequence diagram of upgrading devices by the platform is shown below.



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1. **Upgrade devices.**

   Upgrade the device firmware: `POST /ISAPI/System/updateFirmware`.

2. **Get the device upgrade progress.**

Get the device upgrade progress: `GET /ISAPI/System/upgradeStatus`.

3. **Reboot devices.**

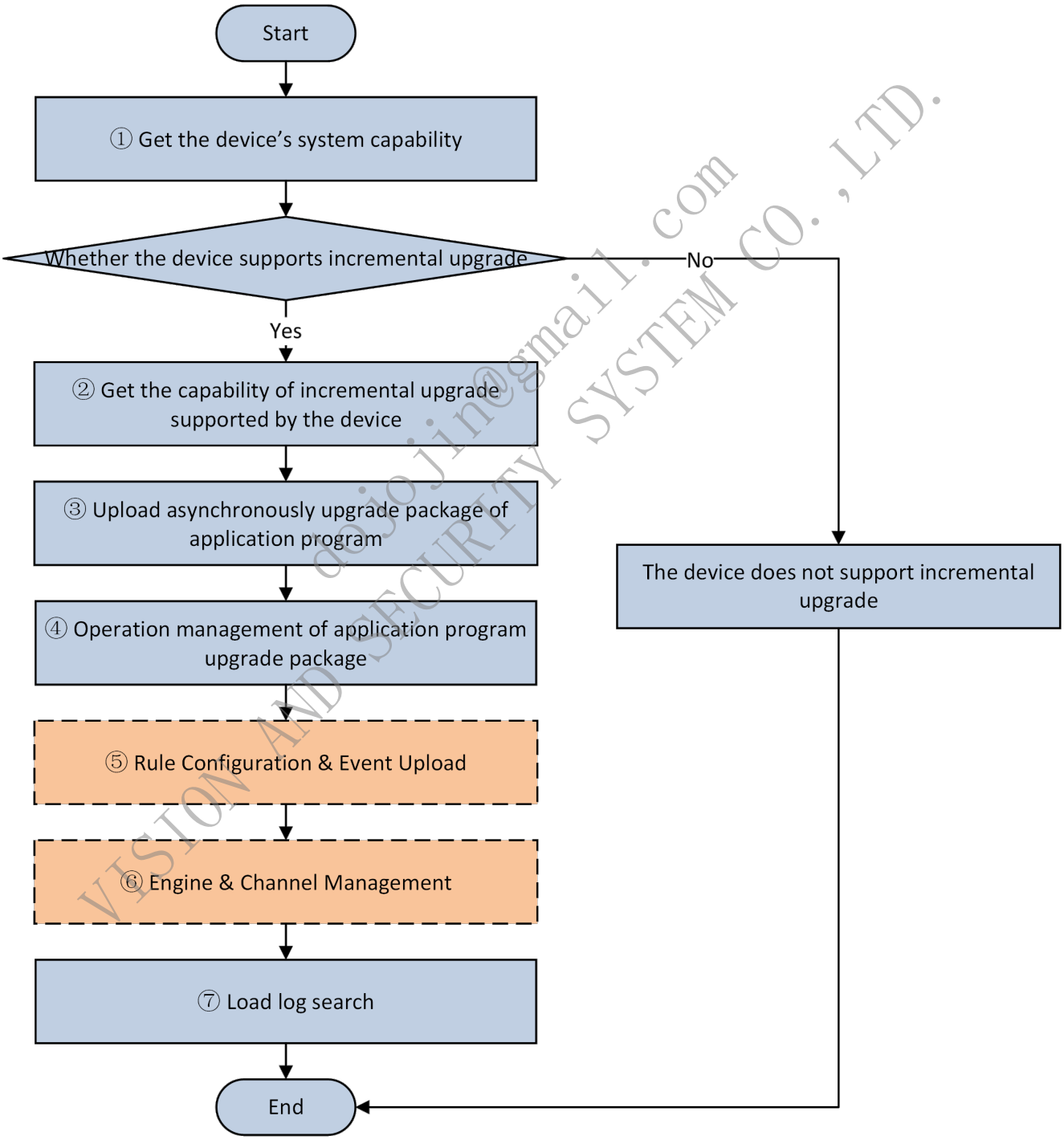
   Reboot devices: `PUT /ISAPI/System/reboot`.

5.7 HEOP Basics

5.7.1 Introduction to the Function

The HEOP integrated by Intelligent Security API (ISAPI) provides runtime environments and management APIs for third-party application programs to connect to and run on network cameras.

5.7.2 API Calling Flow



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1. Get the device system capability: `GET /ISAPI/System/capabilitie`. If the node `<OpenPlatformCap>` is returned in the capability message and and its value is true, it indicates that the HEOP function is supported by the device.
2. Get the HEOP capability supported by the device: `GET /ISAPI/OpenPlatform/capabilities`.
3. Search the HEOP remaining resources: `GET /ISAPI/Custom/OpenPlatform/resource`.

4. Upgrade synchronously the application program package:
    i. Configure synchronous upgrade of the application program full package: `PUT /ISAPI/Custom/OpenPlatform/uploadApp`.
    ii. Get the synchronous upgrading progress of the application program full package: `GET /ISAPI/Custom/OpenPlatform/uploadAppStatus`.
5. Application program information management:
    i. Get page information of application program: `GET /ISAPI/Custom/OpenPlatform/pageItem`.
    ii. Get the application program information list: `GET /ISAPI/Custom/OpenPlatform/App`.
    iii. Get the information of a single application program: `GET /ISAPI/Custom/OpenPlatform/App/<AppID>`.
    iv. Get the logs of the application program: `GET /ISAPI/Custom/OpenPlatform/App/<AppID>/syslog`.
6. Application program operation management:
    i. Run the application program: `PUT /ISAPI/Custom/OpenPlatform/App/<AppID>/start?engineID=<engineID>`.
    ii. Stop running the application program: `PUT /ISAPI/Custom/OpenPlatform/App/<AppID>/stop?engineID=<engineID>`.
    iii. Delete the application program: `DELETE /ISAPI/Custom/OpenPlatform/App/<AppID>/delete`.

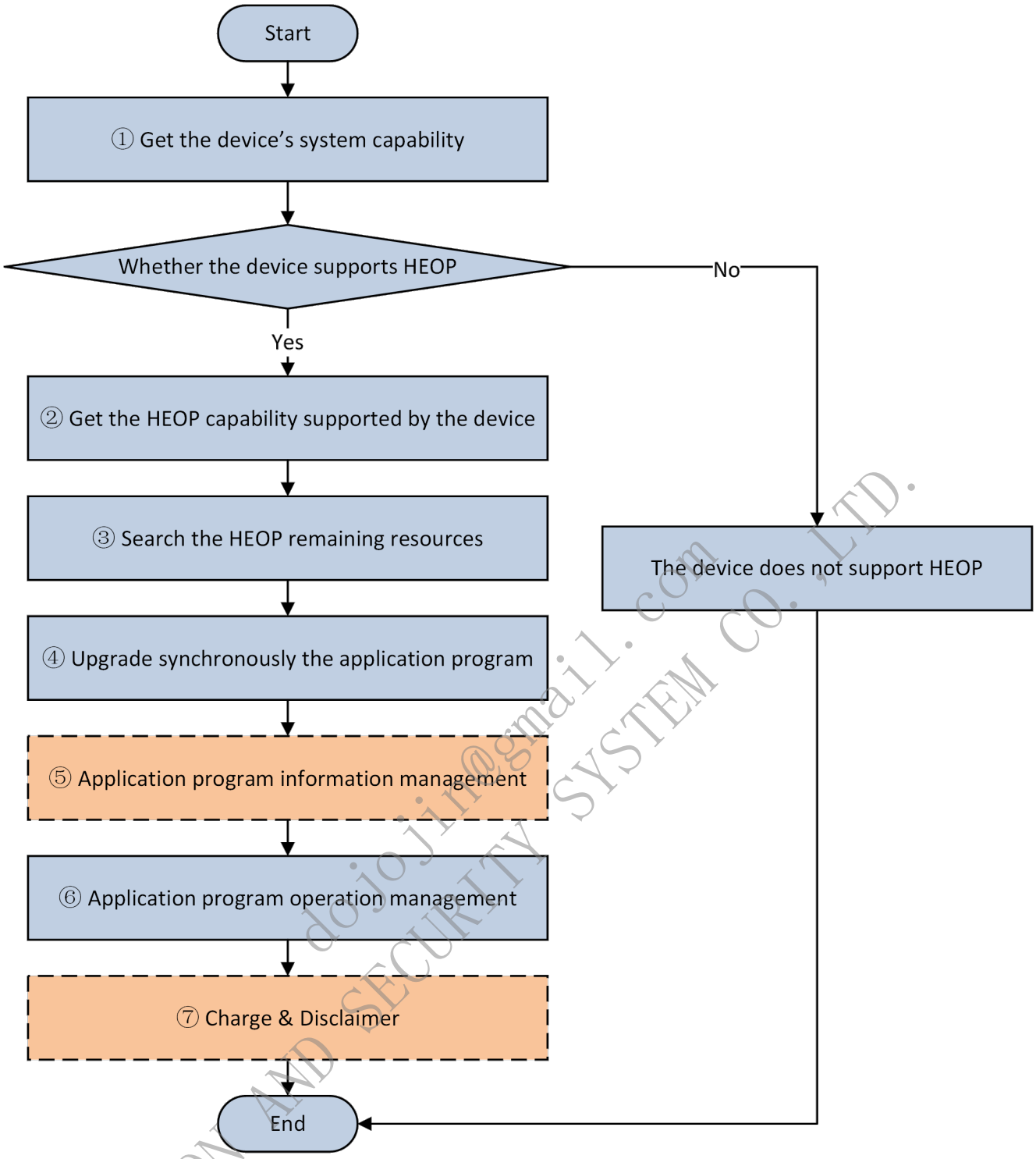
5.8 Incremental Upgrade

5.8.1 Introduction to the Function

If intelligent detection models of the front-end and back-end devices mismatch, the model data collected by the front-end device cannot be used for the comparison on back-end devices. At present, the device package is becoming larger, and the size some device upgrade packages reach level GB, which makes it difficult to upgrade the device (such as the upgrade package delivery, long upgrade time). The unified algorithm switch of front-end and back-end devices is designed to support HEOP framework, which supports patch upgrade of algorithm package.

5.8.2 API Calling Flow

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1. Get the device system capability: `GET /ISAPI/System/capabilitie`. If the node `<isSupportPatchUpgrade>` is returned in the capability message and its value is "true", it indicates that HEOP Synchronization is supported by the device.
2. Get the incremental upgrade capability supported by device: `GET /ISAPI/Custom/OpenPlatform/patchUpgrade/capabilities?format=json`.
3. Upload asynchronously the application program package:
   i. Upload asynchronously the application program package: `POST /ISAPI/Custom/OpenPlatform/uploadApp/async?format=json`.
   ii. Get the asynchronous uploading progress of the application program package: `GET /ISAPI/Custom/OpenPlatform/uploadAppStatus/async/<AppID>?format=json`.
4. Operation management of application program upgrade package:
   i. Run the specified application program upgrade package: `POST /ISAPI/Custom/OpenPlatform/appPackage/<AppUUID>/start`.
   ii. Stop running the application program upgrade package: `PUT /ISAPI/Custom/OpenPlatform/appPackage/<AppUUID>/stop`.

iii. Delete the specified application program upgrade package: `DELETE /ISAPI/Custom/OpenPlatform/appPackage/<AppUUID>`.

5. Rule Configuration & Event Upload: Reuse the existing protocol system (`HCNetSDK` and `ISAPI`), and the device selects whether to provide this function according to different intelligent types based on the existing intelligent rules.
6. Engine & Channel Management:
   i. Get the information about engine status: `GET /ISAPI/Custom/OpenPlatform/APP/engines?format=json`.
   ii. Get the information about the application program package that is bound to a specified engine: `GET /ISAPI/Custom/OpenPlatform/APP/loading/<engineID>?format=json`.
   iii. Set the information about the application program package that is bound to a specified engine: `PUT /ISAPI/Custom/OpenPlatform/APP/loading/<engineID>?format=json`.
   iv. Delete the binding between the application program package and the engine: `DELETE /ISAPI/Custom/OpenPlatform/APP/unloading/<engineID>?format=json`.
   v. Unbind the specified application program(s) from the engine: `PUT /ISAPI/Custom/OpenPlatform/APP/unloading?format=json`.

5.9 Mutually Exclusive Functions

5.9.1 Introduction to the Function

Some functions are mutually exclusive due to the device performance (for example, function A and function B cannot run at the same time, i.e, only one of them is allowed at one time).

5.9.2 API Calling Flow

The following three APIs are available for the integration of mutually exclusive functions:

1. Get the information of mutually exclusive functions: `GET /ISAPI/System/mutexFunction/capabilities?format=json`. Call this URL to get the list of existing mutually exclusive functions supported by the device. Note: NVR devices only support setting exlusive function "perimeter" (perimeter protection), and do not support "linedetection" (line crossing detection), "fielddetection" (intrusion detection), "regionEntrance" (region entrance), or "regionExiting" (region exiting).

2. Search for the functions that are mutually exclusive with a specified function: `POST /ISAPI/System/mutexFunction?format=json`. Based on the list of mutually exclusive functions returned by `GET /ISAPI/System/mutexFunction/capabilities?format=json`, you can search for the mutual exclusion status of a specified function and see whether to change the settings and disbale the mutually exclusive function.

3. Get the mutual exclusion information when device function exception occurs: `GET /ISAPI/System/mutexFunctionErrorMsg`. After getting the error code, you can call this API to get the current mutually exclusive functions.

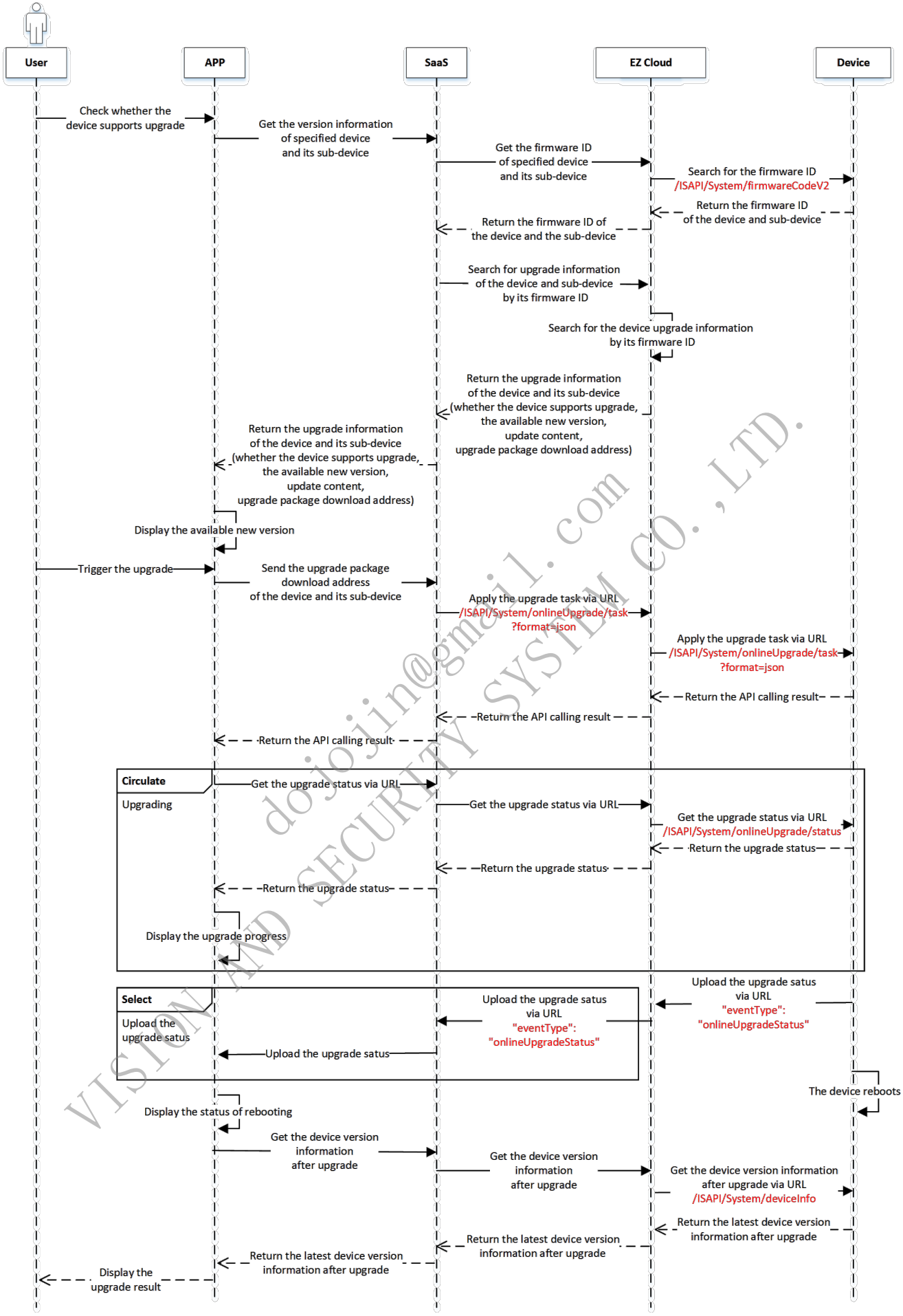
5.10 Remote Online Upgrade Based on The EZ2.0

5.10.1 Introduction to the Function

Considering that the device is connected to EZ via EZ 2.0 Protocol, EZ OTA upgrade solution is not supported. This solution applies to specified version upgrade.

5.10.2 API Calling Flow

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1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by its firmware ID. If there is an available new version, apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade devices (e.g., NVR) or sub-devices (e.g., IPC) via the node

device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.

3. During the device upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` and reboot automatically.
5. After both upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

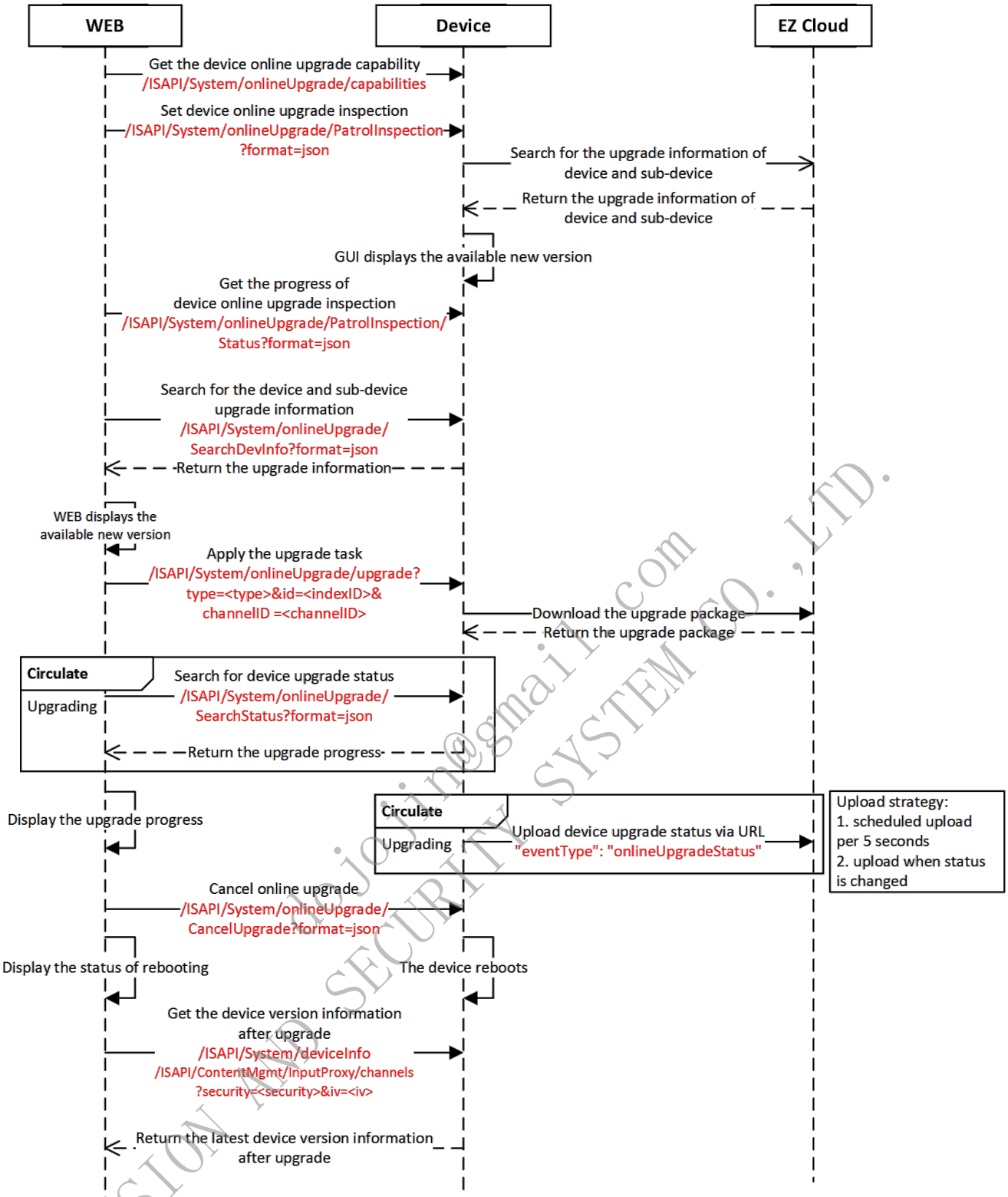
5.11 Remote Online Upgrade Based on the WEB Client

5.11.1 Introduction to the Function

This solution applies to remote online upgrade of the devices and sub-devices connected to WEB client.

5.11.2 API Calling Flow

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1. Get device online upgrade capability: `GET /ISAPI/System/onlineUpgrade/capabilities`.
2. Manually trigger the device online upgrade inspection: `POST /ISAPI/System/onlineUpgrade/PatrolInspection?format=json`. After it is enabled, the device will search for the available new version.
3. During the inspection, the WEB client can get the real-time inspection progress: `GET /ISAPI/System/onlineUpgrade/PatrolInspection/Status?format=json`.
4. After the device inspection is finished, search for the device and sub-device upgrade information: `GET /ISAPI/System/onlineUpgrade/SearchDevInfo?format=json`.
5. Apply the upgrade task: `PUT /ISAPI/System/onlineUpgrade/upgrade?type=<type>&id=<indexID>&channelID =<channelID>`. You can upgrade a specified device via the node `<channelID>` in the URL. Specify sub-device upgrade by configuring `<type>`.

6. You can search for online upgrade status during the upgrade: `GET /ISAPI/System/onlineUpgrade/SearchStatus?format=json`.
7. You can cancel the upgrade during the upgrade: `PUT /ISAPI/System/onlineUpgrade/CancelUpgrade?format=json`.
8. After the upgrade and reboot are finished, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

5.12 Serial Port Accessed External Device Management

5.12.1 Introduction to the Function

Information management of device accessed the serial are as follows: 1. Configure manufacturer, type, and model information of the specific serial port access device. 2. Search for the device type or model supported by the specific serial port.

5.12.2 API Calling Flow

1. Check whether the device supports information management of serial port. Get the capability of the device serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportDeviceInfo>` is returned, the device supports information configuration of devices access the serial port.

2. Set the information of serial port:

- Get the capability of device information parameters of a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Get device information parameters access single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Set device information parameter of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.

3. Check whether the device supports linking information of devices access the serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportSearchDeviceInfoRelations>` is returned, it indicates that the device supports searching for linked information od devices access the serial port.

4. Search for linked information of devices access the serial port.

- Get the capability of searching for linked parameters of information of devices access a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations/capabilities?format=json`.
- Search for linked parameters of information of devices access a single serial port:`POST /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations?format=json`.

5.13 Serial Port Data Transparent Transmission

5.13.1 Introduction to the Function

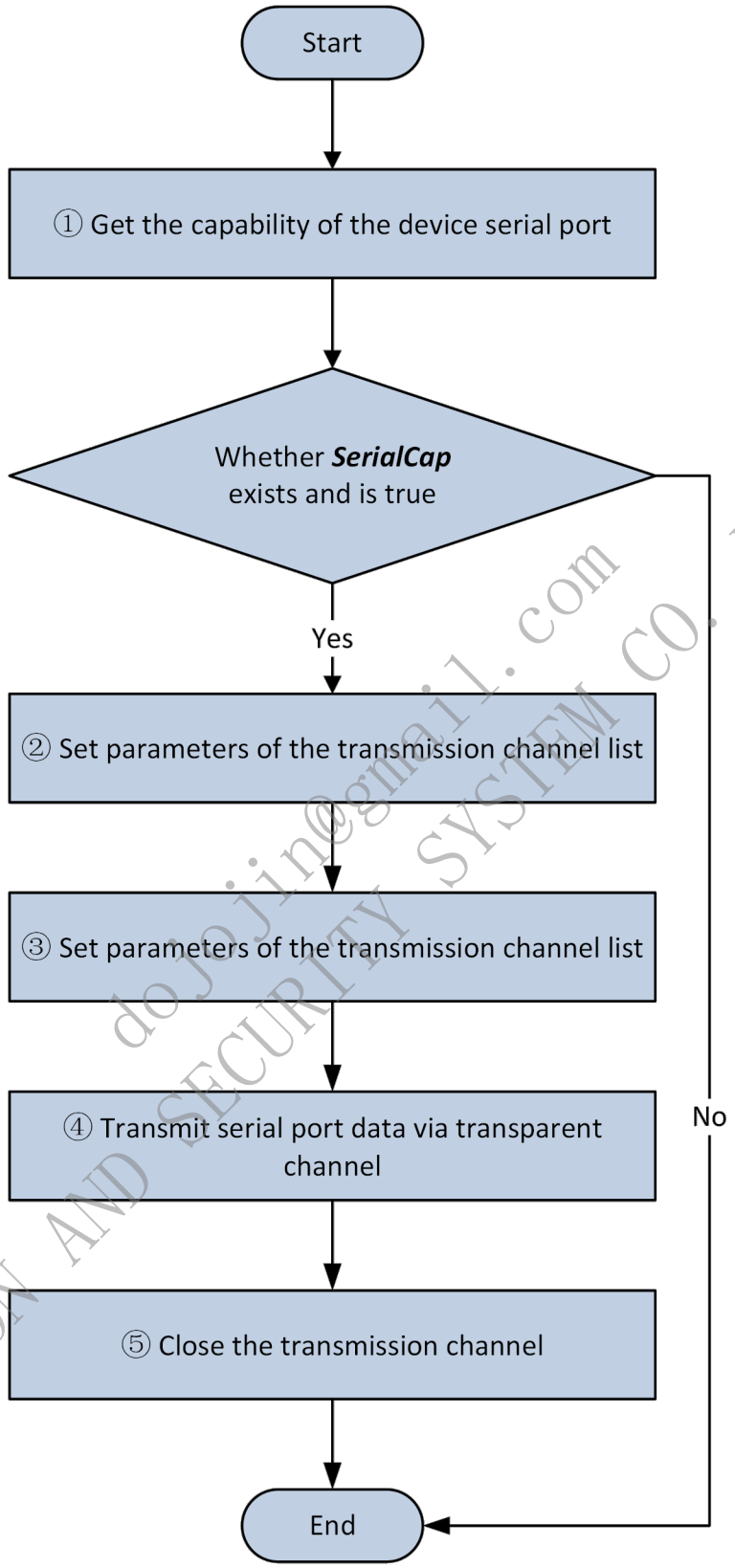
RS485, RS422 and RS232 serial ports external to the device are used as transparent channels to transmit serial port data. Supports the client sending serial data to the device, which then forwards the data to the serial port; conversely, when the serial port sends data to the device, which transparently transmits the data to the client. Note that the device's transparent transmission of serial data is half-duplex communication, meaning that bidirectional communication can be implemented between the device and the client, but the communication data is not necessarily in a question-and-answer format, and the client must match the request and response relationship on its own.

Since most standard HTTP request libraries do not support the persistent connection for receiving and sending data in real time, it is recommended that the client uses two TCP clients to implement the sending and receiving of

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transparently transmitted serial data.

5.13.2 API Calling Flow



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1. Check whether the device supports serial port data transmission.

Get the capability of the device serial port: `GET /ISAPI/System/capabilities`. If `SerialCap` is returned and the value is true, it indicates that the device supports the functions of the serial port.

2. Set parameters of the transmission channel list.

- Get parameters of the specific transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.
- Configure parameters of the specific transmission channel: `GET`

`/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.

3. Open the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/open`;

4. Transmit serial port data via transparent channel.

- Receive data uploaded by device serial port through transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.
- Send data to device serial port through transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.

5. Close the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/close`.

5.14 Serial Port Parameter Configuration

5.14.1 Introduction to the Function

Serial port parameter configuration.

5.14.2 API Calling Flow

1. Check whether the device supports configuring serial port parameters.

Get the capability of device serial port: `GET /ISAPI/System/capabilities. If is returned and its value is true, the device supports functions of serial port.

3. Get parameters of all serial ports.

- Get the capability of all serial ports: `GET /ISAPI/System/Serial/capabilities`.
- Get control parameters of all serial ports: `GET /ISAPI/System/Serial/ports?permissionController=<indexID>`.

4. Set control parameters of a single serial port.

- Get control parameters of single serial port: `GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.
- Configure control parameters of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.

5. Get the status of single serial port:`GET /ISAPI/System/Serial/ports/<portID>/status`.

5.15 Sub-device Batch Upgrade

5.15.1 Introduction to the Function

Sub-device Batch Upgrade (Single Task): applicable to situations of upgrading multiple sub-devices using one upgrade package. Application scenarios include: when the UWB positioning anchor connects to the web, upgrading multiple tags (sub-devices) through the positioning engine (gateway), with the same upgrade package.

Management of Sub-device Batch Upgrade Tasks: There are many types of sub-devices, such as LoRa nodes, which have a slow upgrade process and different models of LoRa nodes in the field also require different upgrade packages. Therefore, it is necessary to support the creation of multiple upgrade tasks at once, entrusting the upgrade to the devices, to enhance the practicality of batch upgrading sub-devices.

5.15.2 API Calling Flow

Sub-device Batch Upgrade (Single Task)

Get the device capability: `/ISAPI/System/capabilities`.

If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that the device support the function.

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If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that the device support getting the progress of batch upgraading sub-devices.

Sub-device batch upgrade: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.

Batch sub-device upgrade progress search: `POST /ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json`.

Sub-device Batch Upgrade Task Management

Get device system capabilities `/ISAPI/System/capabilities`, If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that batch upgrade of sub devices is supported,

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that progress of batch upgrading sub devices can be searched,

If the node `isSupportSearchBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading devices can be searched,

If the node `isSupportModifyBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading sub devices can be edited,

If the node `isSupportDeleteBulkUpgradeChildDeviceTask` exists and is true, it indicates that task of batch upgrading sub devices can be deleted.

Get the capability of batch upgrading sub-device: `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`.

Batch upgrade sub devices: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.****

Search for tasks of batch sub-device upgrade: `POST /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`.

Edit tasks of batch sub-device upgrade: **`PUT /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`**.****

Delete tasks of batch sub-device upgrade: `PUT /ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json`.

Note: 1. The two methods for batch upgrading child devices are the same API: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`. If the `BulkUpgradeChildDeviceListCap.taskID` is returned in `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`, it indicates that the task method is supported in the sub device upgrade. 2. To determine which sub devices can be upgraded, you can use the API `POST /ISAPI/IoTGateway/Childmanage/SearchChild?format=json`. Sub devices with the `tag` `isUpgradable` and `value` `true` indicate that the sub devices support upgrades.

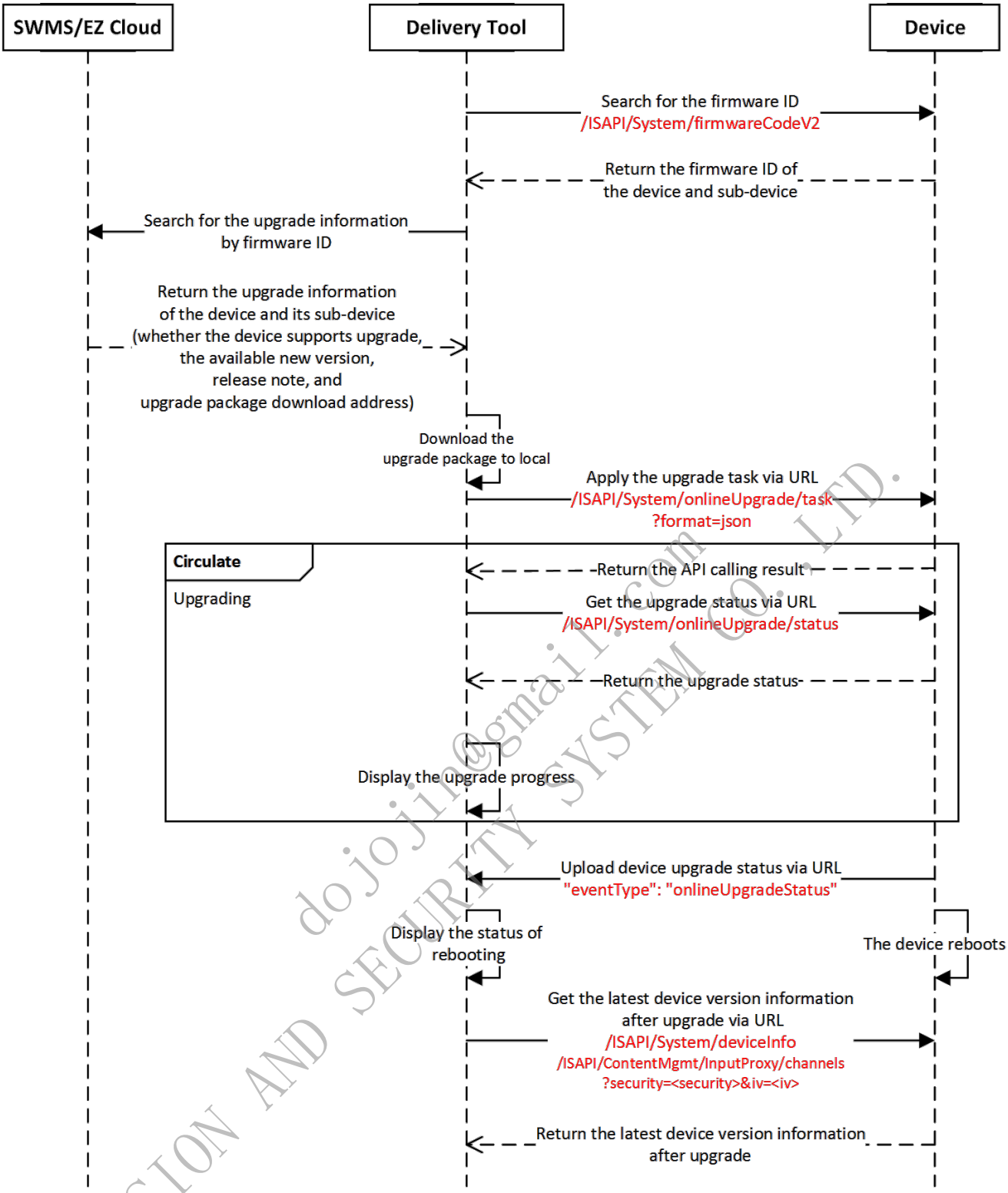
5.16 Upgrade Based on Interactive Tools Local Deployment

5.16.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI. If the device is connected via ISAPI protocols, it is required to set up HTTP(s) service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK, you can implement the device upgrade via transmitting ISAPI.

5.16.2 API Calling Flow

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1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by firmware ID on the SWMS/EZ Cloud, download upgrade package to local storage, and apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., camera) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress: `GET /ISAPI/System/onlineUpgrade/status`.
4. After upgrade is completed, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and reboot automatically.
5. After the device rebooting, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=`

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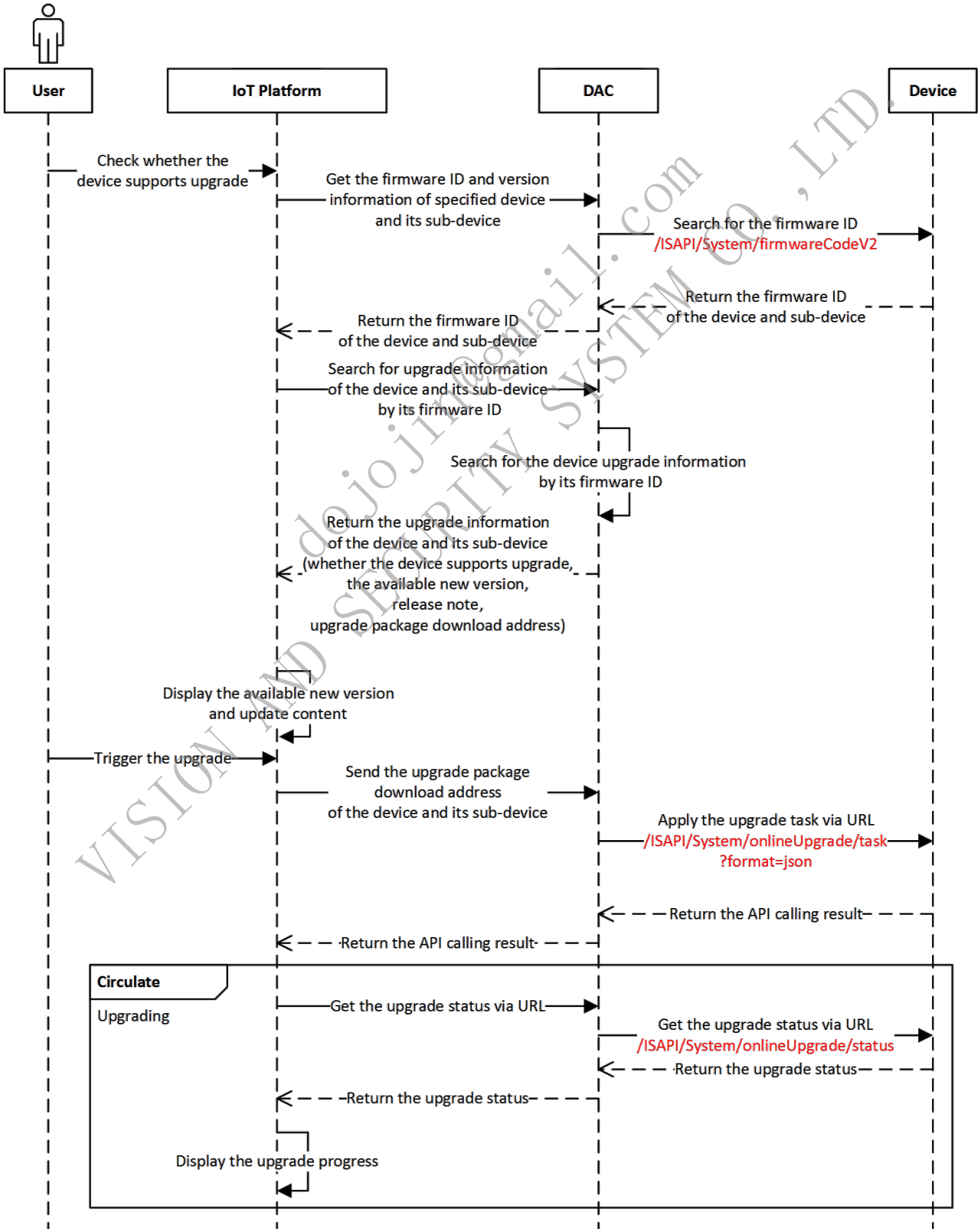
`<security>&iv=<iv>`.

5.17 Upgrade Based on Platform Local Deployment

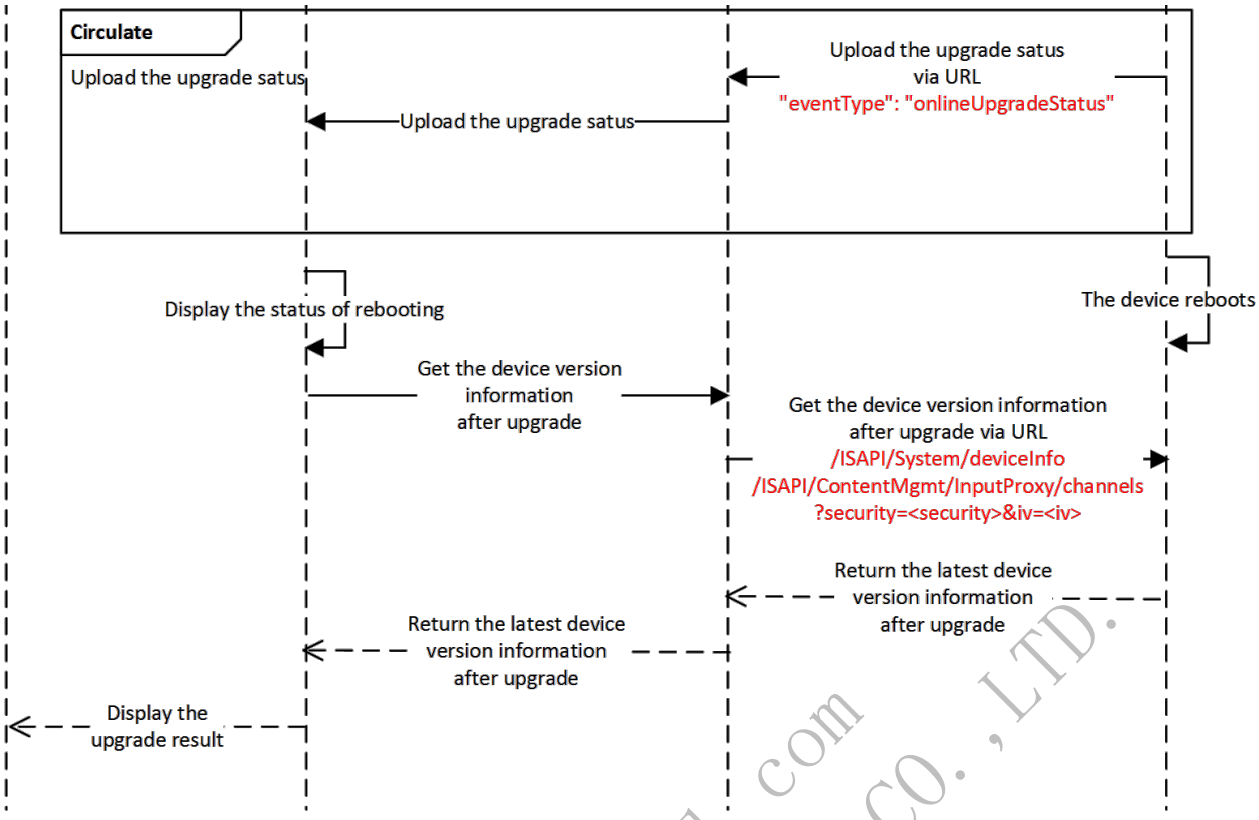
5.17.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI and ISUP. If via ISAPI, it is required to set up HTTP(s) service or transfer binary upgrade package files so as to store device upgrade package and generate URL for downloading upgrade package. If via ISUP, it is required to set up FTP service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK protocols, you can realize device upgrade via transmitting ISAPI.

5.17.2 API Calling Flow



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1. Search for device firmware ID on the platform: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information in DAC by device firmware ID on the platform. If there is an available new version, DAC will trigger an upgrade task and apply it to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and will reboot automatically.
5. After the upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

6 Video (General)

6.1 . OSD

6.1.1 Introduction to the Function

On-screen display is applied in the display interface of cameras, and is used to display some characters or graphics on the video to provide some information to the audience. After OSD configurations on the client software, the configured information text will be overlaid on the video stream of cameras and displayed on the screen.

The typical application scenarios of dynamic OSD: elevator floor number overlay for elevator monitoring, characters overlay for toll stations on highway/road, notification information overlay during live video programs. The typical application scenarios of static OSD: fixed information (such as camera position, date, time) overlay on the video signal.

6.1.2 API Calling Flow

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6.1.2.1 OSD

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`.
2. The client software configures OSD parameters, including whether to enable OSD, custom text overlay, time overlay, and overlay format, for a specified channel.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

   Set: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

   Remarks:

   - The configuration URL supports various parameters, including custom text overlay, channel name overlay, time overlay, overlay format, and so on. Some of these parameters can also be configured by other URLs, as shown below: `TextOverlayList` can also be configured via URL of custom text overlay.

     `channelNameOverlay` can also be configured via URL of channel name overlay.

     `DateTimeOverlay` can also be configured via URL of channel time overlay.

   - The overlaid information alignment mode (defined by `alignment`) refers to the alignment mode of custom text:

     `customize`: custom mode. The client software sends the overlay position coordinates of custom text to the device, and the device will overlay the characters on the stream for display according to the position coordinates.

     `alignRight`: right align. The device will display custom text on the right boundary of screen.

     `alignLeft`: left align. The device will display custom text on the left boundary of screen.

     `allRight`: all right align. The device will display all overlay contents, including custom text, the channel name, the time on the right boundary of screen.

     `allLeft`： all left align. The device will display all overlay contents, including custom text, the channel name, the time on the left boundary of screen.

6.1.2.2 Custom Text Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `TextOverlayList` is returned, it indicates that custom text overlay is supported and `size` defines the maximum number of custom texts.
2. The client software configures custom text OSD. `textID` refers to the custom text No., and the value range is: [1, `size`]:

   Create a custom text for a specified channel: `POST /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

   Get overlay parameters of a custom text for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Set overlay parameters of a custom text for a specified channel: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Delete a custom text: `DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Get overlay parameters of all custom texts for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

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Set overlay parameters of all custom texts for a specified channel: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

Delete all custom texts for a specified channel: `DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

6.1.2.3 Channel Name Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `channelNameOverlay` is returned, it indicates that channel name overlay is supported.
2. The client software configures the channel name overlay, including whether to enable channel name overlay, overlay position, and so on.

   Get channel name overlay parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay`.

   Set channel name overlay parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay`.

6.1.2.4 Channel Time Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `DateTimeOverlay` is returned, it indicates that channel time overlay is supported.
2. The client software configures the channel time overlay, including whether to enable channel time overlay, overlay position, date format, and so on.

   Get channel time overlay parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay`.

   Set channel time overlay parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay`.

6.2 Async Capturing Manually

6.2.1 Introduction to the Function

The client sends a capture command to the device, and the device immediately responds by returning the URL of the storage server where the captured picture is stored. The client can request image data from the storage server based on the received URL. It is suitable for the limited network bandwidth between the client and the device or the client centrally manages a large number of devices. The client can choose to use the image URL address carried in the alarm content to request pictures from the storage server when the network conditions improve or there is no concurrency of a large number of requests, so as to reduce the probability of image download failure and the risk of network congestion.

6.2.2 API Calling Flow

1. The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportSnapshotAsync` is returned and its value is `true`, async manual capturing shall be supported.
2. (Optional) The client gets the async manual capturing capability of a specified channel: `GET /ISAPI/Streaming/channels/<channelID>/picture/async/capabilities?format=json`.
3. The client controls the specified channel to manually and asynchronously capture pictures: `GET /ISAPI/Streaming/channels/<channelID>/picture/async?format=json&imageType=<imageType>&URLType=<URLType>`, and the device returns the URL of the storage server for capturing pictures.

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Remarks:

- If the step 2 returns a failure during the integration，it is recommended that you check whether the channel supports async manual capturing according to steps 1 and 3.

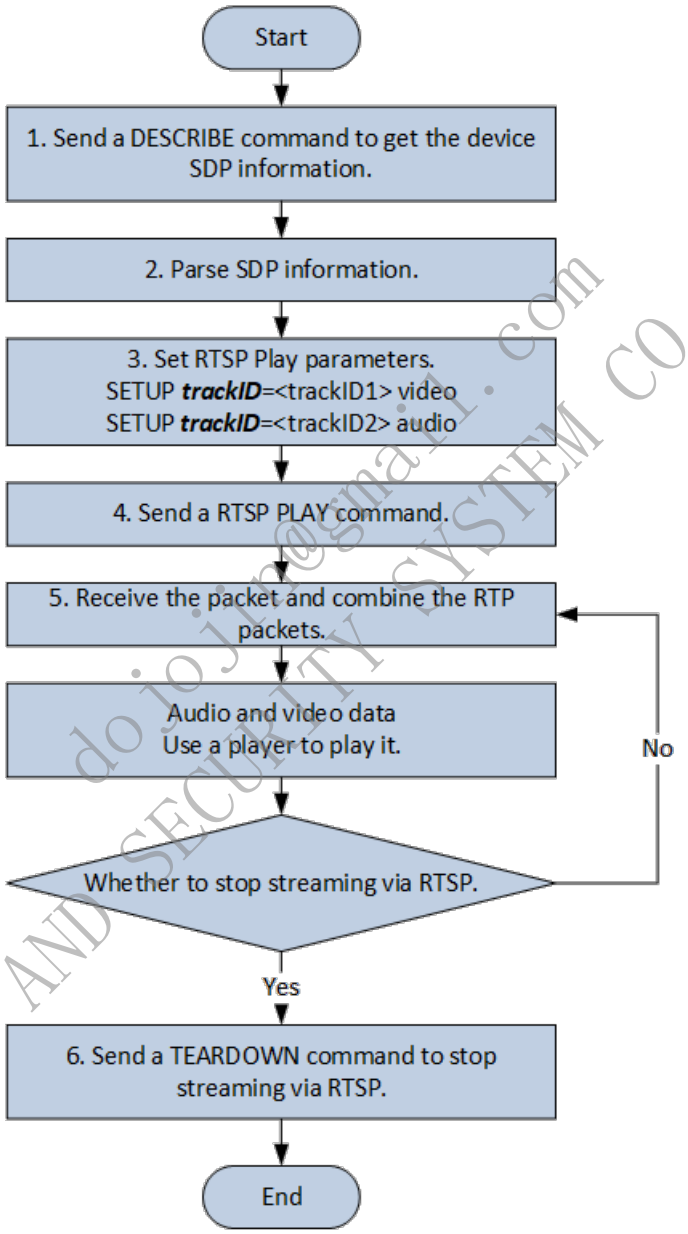
6.3 Live View

6.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

6.3.2 API Calling Flow



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1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`; Digest authentication with devices is required.
2. The client parses the media SDP information returned by the device.
3. Set RTSP play parameters, set the track ID parsed from SDP information via SETUP commands. For example, trackID=1 indicates videos while trackID=2 indicates audios.
4. The client sends an RTSP PLAY command, and device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`.
5. The client receives the RTP package sent by device. Divided RTP packets should be assembled on the client before being parsed.
6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address, `[:port]` is optional, and 554 by default, `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream); for example, the IP address of the target device is `172.7.203.11`, and if the channel number is 172, the streaming address will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

6.3.3. Example

1. A client sends an RTSP DESCRIBE request.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s=Media Server V4.22.126
i=Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000020000011171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP requests, and the server responds to them.

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```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J............X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

6.4 Manual Capturing Events

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6.4.1 Introduction to the Function

According to the preset information and capture interval configured by the client, the device manually captures images at the specified preset and reports the image information in the form of events. It is used for the client to identify emergency situations and perform an emergency patrol on some presets for analysis. It has a higher priority than the daily scheduled patrol capture. Common usage scenarios: according to the scheduled capturing event (eventType: imageCapture) rules, the store camera regularly audits in a number of key areas. If in the event of an emergency, and a worker wants to immediately re-inspect each area, you can set a manual capture rule to complete an independent audit. After the audit is completed, the camera will continue to audit according to the scheduled capturing rules.

6.4.2 API Calling Flow

1. The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportManualImageCapture` returns `true`, manual capturing shall be supported.
2. The client gets manual capturing capability for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture/capabilities?format=json`.
3. The client sets the parameters of manual capture events for a specified channel: Get parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture?format=json`. Set parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture?format=json`.
4. Receive the event uploaded by the device in arming or listening mode: `eventType:manualImageCapture`.

Remarks:

- Each time the client sets manual capture event parameters, the channel immediately takes a patrol capture of the relevant presets. When the last preset capture is finished, the manual capture event ends.

6.5 Motion Detection

6.5.1 Introduction to the Function

Commonly used in unattended camera recording and automatic alarm, motion detection can help to reduce the cost of manual monitoring for public institutions and companies. Besides, it can greatly improve monitoring efficiency and accuracy by avoiding the errors of manual monitoring due to fatigue after long-time work. For example, it can be applied in home and company theft proof, as well as monitoring and alarm in institutions like medical organizations and nursing homes.

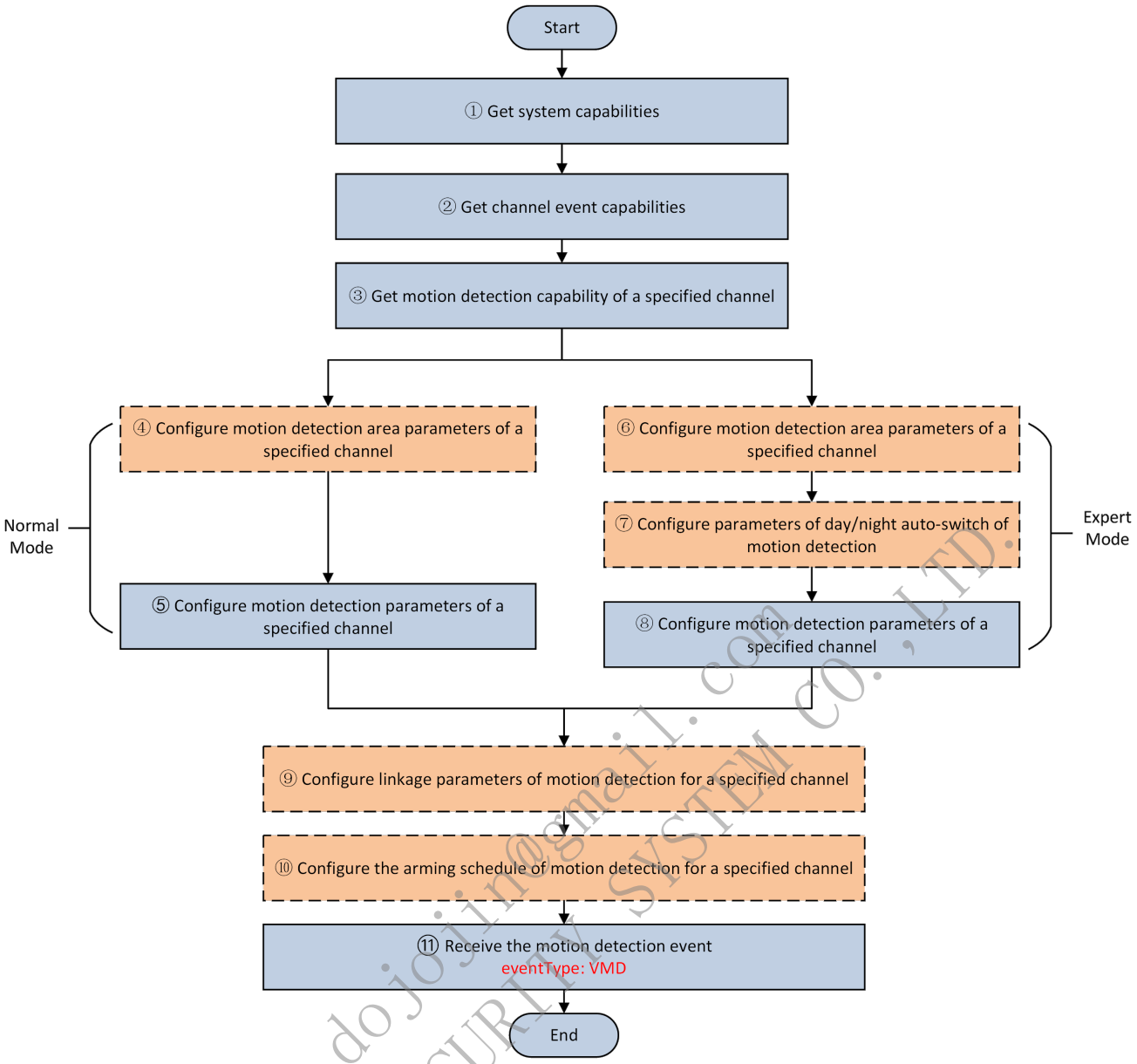
Motion detection has two modes: normal mode and expert mode.

Normal mode: the default mode of motion detection. In this mode, all detected areas only support a same sensitivity.

Expert mode: in this mode, every detected area can have its own detection sensitivity and percentage of objects in area. It supports configuring different sensitivities according to area types. For example, for the motion detection in the company hall, you can set higher detection sensitivity for the gate area to focus on people's entering and exiting.

6.5.2 API Calling Flow

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1. Get the system capability: `GET /ISAPI/System/capabilities`. If the node `isSupportMotionDetection` is returned and its value is `true`, it indicates that the device supports motion detection.

2. Get the device channel event capability. If the returned `eventType` contains `VMD`, it indicates that the corresponding channel supports motion detection.

   Get the event capability of all channels: `GET /ISAPI/Event/channels/capabilities`.

   Get the event capability of a specified channel: `GET /ISAPI/Event/channels/<channelID>/capabilities`.

3. Get the motion detection capability of a specified channel (normal mode): `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/capabilities`.

   Check the motion detection parameters (such as function enabling, detection sensitivity) supported by the device. You can check whether the device supports multiple areas via checking whether the node `regionType` contains `region`.

4. (Optional) Configure parameters of the motion detection area (normal mode).

   Capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout/capabilities`.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout?regionType=<regionType>`. **Note:** If the device supports multiple areas of motion detection, for compatibility, both `<gridMap>` and `<RegionList>` will be returned regardless the `regionType` value in URL.

Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout?regionType=<regionType>`. **Note:** If the device supports multiple areas of motion detection, to set multiple areas, you need to input `regionType=<region>` in URL.

5. Configure the motion detection parameters (such as motion detection enabling, detection sensitivity) (normal mode).

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection`. **Note:** If the device supports multiple areas of motion detection, for compatibility, both `<gridMap>` and `<RegionList>` will be returned regardless the `regionType` value in URL.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection`. **Note:** If the device supports multiple areas of motion detection, to set multiple areas, you need to input `regionType=<region>` in URL.

6. (Optional) Configure the parameters of motion detection area (expert mode). You can configure different detection conditions such as sensitivity for different areas.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/regions`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/regions`.

7. (Optional) Configure day/night auto switch of motion detection (expert mode). You can configure the time switch schedule of `day` and `night`.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/switch`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/switch`.

8. Configure motion detection conditions (expert mode).

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt`.

9. (Optional) Configure linkage parameters of the motion detection.

   Get: `GET /ISAPI/Event/triggers/VMD-<channelID>`.

   Configure: `PUT /ISAPI/Event/triggers/VMD-<channelID>`.

   Delete: `DELETE /ISAPI/Event/triggers/VMD-<channelID>`.

10. (Optional) Configure the arming schedule of motion detection.

    Get: `GET /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>`.

    Configure: `PUT /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>`.

11. Receive events reported by device in arming or listening mode: `eventType:VMD`.

Remarks:

1. For motion detection (normal mode), the uploaded `VMD` event includes device basic information only, such as channel No. and event time. For details, see 4.3.1 Motion Detection Event (Normal Mode).

2. For motion detection (expert mode), in addition to the device basic information, the uploaded `VMD` event includes the detection area ID, area coordinates and area detection sensitivity. For details, see 4.3.2 Motion Detection Event (Expert Mode).

3. In practical employment, you can select one mode as needed. There are no logical relations between the two modes.

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6.5.3 Message Example

6.5.3.1 Motion Detection Event (Normal Mode)

```
<?xml version="1.0" encoding="UTF-8"?>
<EventNotificationAlert version="1.0" xmlns="urn:psialliance-org">
<ipAddress>10.17.114.252</ipAddress>
<protocolType>HTTP</protocolType>
<macAddress>64:db:8b:6e:4f:08</macAddress>
<channelID>1</channelID>
<dateTime>2018-03-13T19:42:27+08:00</dateTime>
<activePostCount>4</activePostCount>
<eventType>VMD</eventType>
<eventState>active</eventState>
<eventDescription>Motion alarm</eventDescription>
<channelName>Camera 01</channelName>
<Extensions version="1.0" xmlns="urn:psialliance-org">
<serialNumber xmlns="urn:selfextension:psiaext-ver10-xsd">DS-2DF8236IX-AELW20170718CCWR799562898X</serialNumber>
<eventPush xmlns="urn:selfextension:psiaext-ver10-xsd">VMD&amp;amp;DS-2DF8236IX-AELW20170718CCWR799562898X,2018-01-21T12:50:39+08:00,1,1.0</eventPush>
</Extensions>
</EventNotificationAlert>
```

6.5.3.2 Motion Detection Event (Expert Mode)

```
<?xml version="1.0" encoding="UTF-8"?>
<EventNotificationAlert version="1.0" xmlns="urn:psialliance-org">
<ipAddress>10.17.114.252</ipAddress>
<protocolType>HTTP</protocolType>
<macAddress>64:db:8b:6e:4f:08</macAddress>
<channelID>1</channelID>
<dateTime>2018-03-13T20:36:34+08:00</dateTime>
<activePostCount>11</activePostCount>
<eventType>VMD</eventType>
<eventState>active</eventState>
<eventDescription>Motion alarm</eventDescription>
<DetectionRegionList>
<DetectionRegionEntry>
<regionID>1</regionID>
<sensitivityLevel>50</sensitivityLevel>
<RegionCoordinatesList>
<RegionCoordinates>
<positionX>216</positionX>
<positionY>216</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>756</positionX>
<positionY>756</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>756</positionX>
<positionY>756</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>216</positionX>
<positionY>216</positionY>
</RegionCoordinates>
</RegionCoordinatesList>
</DetectionRegionEntry>
</DetectionRegionList>
<channelName>Camera 01</channelName>
<Extensions version="1.0" xmlns="urn:psialliance-org">
<serialNumber xmlns="urn:selfextension:psiaext-ver10-xsd">DS-2DF8236IX-AELW20170718CCWR799562898X</serialNumber>
<eventPush xmlns="urn:selfextension:psiaext-ver10-xsd">VMD&amp;amp;DS-2DF8236IX-AELW20170718CCWR799562898X,2018-01-21T12:50:39+08:00,1,1.0</eventPush>
</Extensions>
</EventNotificationAlert>
```

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6.6 Preset Dedicated Focusing

6.6.1 Function Introduction

The preset dedicated focusing algorithm is designed for specific scenarios (such as instrument and gauge images) where the image is not clear at a certain position, to achieve clearer and faster focusing. The dedicated focusing algorithm includes foreground focusing, background focusing, and area focusing.

6.6.2 API Calling Flow

1. The client gets the device's PTZ capabilities via URL `GET /ISAPI/PTZCtrl/capabilities`. If `isSupportPresetsProFocus` is returned with `true` value, the device supports preset dedicated focusing.

2. The client gets the capability of preset dedicated focusing for the specified channel via URL `GET /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg/capabilities?format=json`.
3. The client batch configures presets' dedicated focusing parameters for the specified channel.
   - Get parameters: `GET /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg?format=json`
   - Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg?format=json`
4. The client configures the dedicated focusing parameters of a single preset for the specified channel. If a preset is not clear under the batch configuration in step 3, it can be configured separately.
   - Add dedicated focusing parameters of a preset for a specified channel: `POST /ISAPI/PTZCtrl/channels/<channelID>/AddPresetsProFocusCfg?format=json`
   - Get parameters: `POST /ISAPI/PTZCtrl/channels/<channelID>/GetPresetsProFocusCfg?format=json`
   - Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/ModifyPresetsProFocusCfg?format=json`
   - Delete: `POST /ISAPI/PTZCtrl/channels/<channelID>/DeletePresetsProFocusCfg?format=json`
   - Get parameters of multiple presets: `GET /ISAPI/PTZCtrl/channels/<channelID>/GetMultiPresetsProFocusCfg?format=json`

6.6.3 Exception Handling

Error Codes

statusCode	statusString	subStatusCode	errorCode	errorMsg	Chinese Description
4	Invalid Operation	numberReachedLlimit	0x40001119	The number reached limit.	The configured items have reached the limit and cannot be added further. Please delete unnecessary ones and try again.

6.7 Scene Change Detection

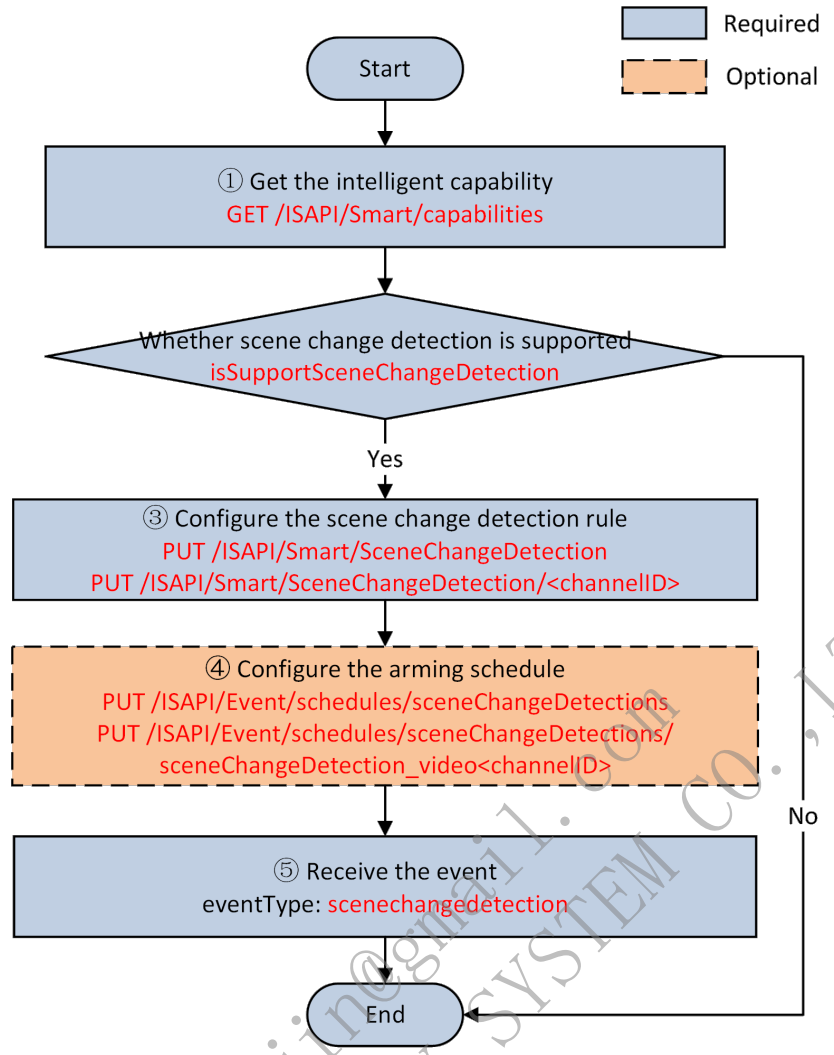
6.7.1 Function Introduction

The scene change detection function identifies whether a scene has changed based on configured rules. Once a scene change is detected, an alarm is triggered.

This function is commonly used to monitor whether external forces have caused changes in camera's viewing angle. When the viewing angle is changed by external interference, or the monitoring scene is tampered, affecting normal monitoring, relevant personnel can be notified promptly to take actions.

6.7.2 API Calling Flow

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1. Get the device's intelligent capability.

   `GET /ISAPI/Smart/capabilities`. If the response returns `isSupportSceneChangeDetection` as `true`, it indicates that the device supports the scene change detection.

2. Configure the detection rules.

   - Get the scene change detection capability: `GET /ISAPI/Smart/SceneChangeDetection/<channelID>/capabilities`.
   - Get scene change detection parameters: `GET /ISAPI/Smart/SceneChangeDetection`.
   - Set scene change detection parameters: `PUT /ISAPI/Smart/SceneChangeDetection`.
   - Get scene change detection parameters for a specific channel: `GET /ISAPI/Smart/SceneChangeDetection/<channelID>`.
   - Set scene change detection parameters for a specific channel: `PUT /ISAPI/Smart/SceneChangeDetection/<channelID>`.

   Note: A higher **sensitivityLevel** makes the camera more sensitive to scene changes. Adjust it based on actual environment.

3. (Optional) Configure the arming schedule.

   - Get schedule parameters of scene change detection for all channels: `GET /ISAPI/Event/schedules/sceneChangeDetections`.
   - Set schedule parameters of scene change detection for all channels: `PUT /ISAPI/Event/schedules/sceneChangeDetections`.

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- Get schedule parameters of scene change detection for a specific channel: `GET /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>`.
- Set schedule parameters of scene change detection for a specific channel: `PUT /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>`.

Note:

- Up to 8 time periods can be configured per day.
- Time periods must not overlap.

4. Receive the event (`"eventType":"scenechangedetection"`).

6.8 Sync Capturing Manually

6.8.1 Introduction to the Function

The client sends a capture command to the device, and the device immediately returns the captured picture data in respond. It is often used in scenarios with high network bandwidth between clients and devices to facilitate the rapid transmission of image data, such as a LAN built on a company campus.

6.8.2 API Calling Flow

1. (Optional) The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportManualSnapPicture` returns `true`, sync manual capturing shall be supported.
2. (Optional) The client gets sync manual capturing capability for a specified channel: `GET /ISAPI/Streaming/channels/<trackStreamID>/picture/capabilities?format=json`.
3. The client controls the specified channel to perform sync manual capturing: `GET /ISAPI/Streaming/channels/<trackStreamID>/picture?snapShotImageType=<snapShotImageType>&videoResolutionWidth=<videoResolutionWidth>&videoResolutionHeight=<videoResolutionHeight>&imageQuality=<imageQuality>&x=<x>&y=<y>&hight=<hight>&width=<width>`, and the device returns the captured binary picture data.

Remarks:

- If you only need to capture a JEPG picture, you can jump to step 3.
- If you need to capture pictures other than JEPG or capture in the designated area of the video, you need to perform steps 1 and 2 to determine whether the device supports it. If it does, go to step 3.
- In step 3, the value ranges of `videoResolutionWidth` and `videoResolutionHeight` are the same as `<videoResolutionWidth>` and `<videoResolutionHeight>` returned by `GET /ISAPI/Streaming/channels/<trackStreamID>/capabilities`.

6.9 Video Privacy Mask

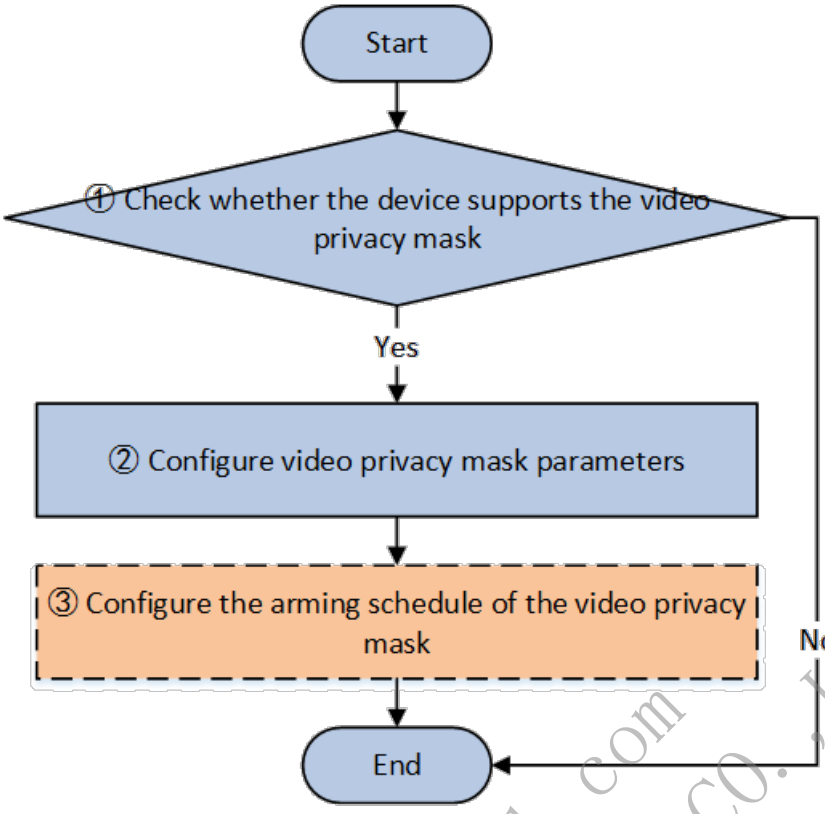
6.9.1 Introduction to the Function

Video privacy mask is to cover certain areas on the live image to protect personal privacy from being live viewed and recorded. You can use the masking method such as a mosaic or black frame as needed to make the area invisible. The device supporting privacy mask can specify the area(s) to be masked permanently on the live image, as a result of which the live view and recorded video during the live view will be influenced and the image cannot be restored. The following image shows the effect of masking an area with the mosaic.

6.9.2 API Calling Flow

6.9.2.1 Video Privacy Mask

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1. Check whether the device supports the video privacy mask. Get the device's system capability: `GET /ISAPI/System/capabilities`. If `isSupportPrivacyMask` is returned and its value is true, it indicates that the video privacy mask is supported by the device.

2. Configure the video privacy mask parameters. You can specify sensitive area(s) to be masked on the live image (related field: `PrivacyMaskRegionList`), and up to 8 areas are supported. You can also configure the masking type, such as mosaic, black frame, and red frame (related field: `maskType`). Get the video privacy mask capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/privacyMaskCap`. Configure the video privacy mask parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask`. Get the video privacy mask parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask`.

3. (Optional) Configure the arming schedule of the video privacy mask. You can configure the privacy mask schedule as needed. If not configured, the privacy mask is performed 24 hours a day by default. Get the video privacy mask schedule: `GET /ISAPI/Event/schedules/privacyMask/privacyMask-<channelID>`. Set the video privacy mask schedule:`PUT /ISAPI/Event/schedules/privacyMask/privacyMask-<channelID>`.

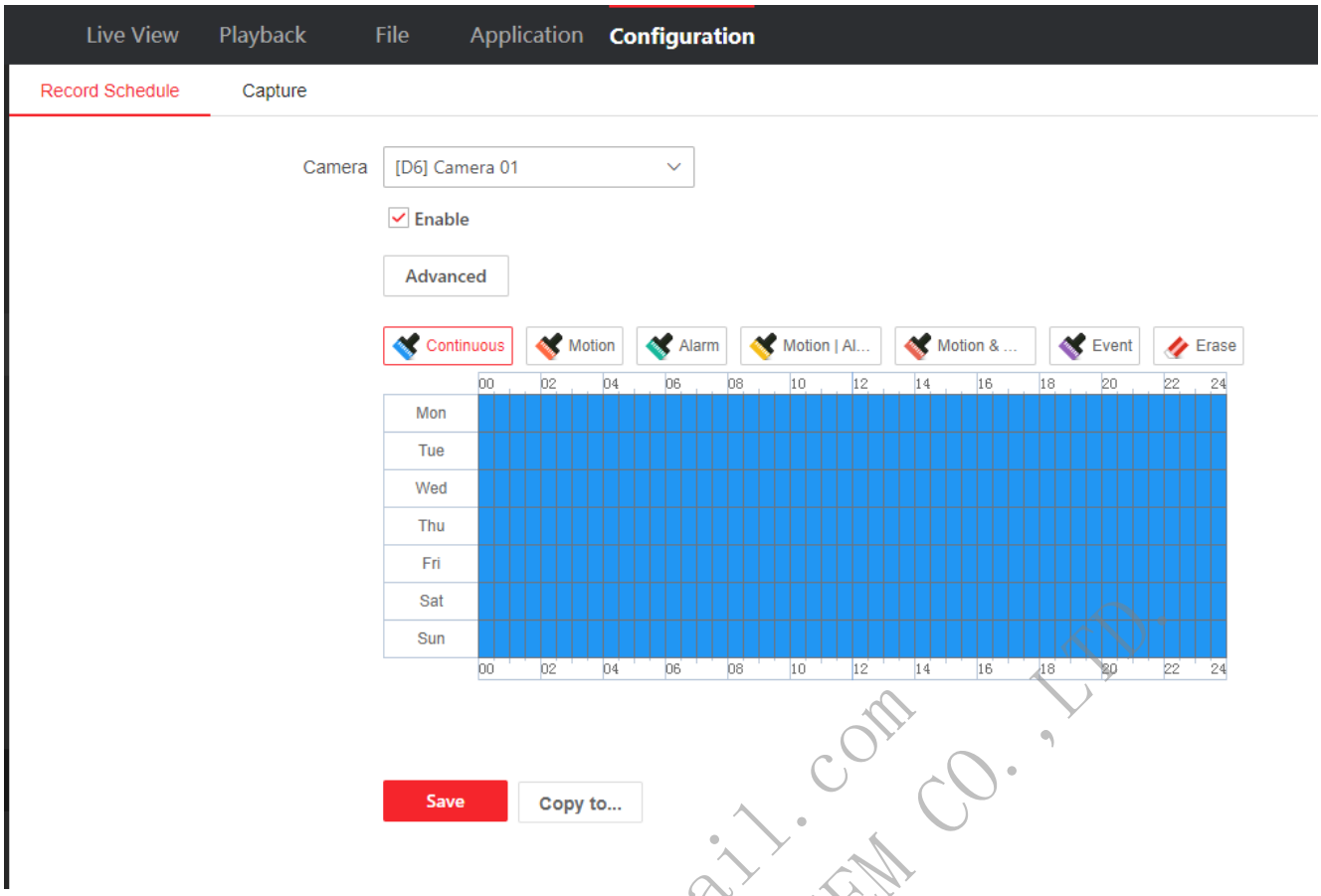
7 Video Management

7.1 . Scheduled Recording

7.1.1 Introduction to the Function

The web interface for reference is shown below.

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7.1.2 API Calling Flow

1. The client software gets the device system capability: `GET /ISAPI/System/capabilities`. When `isSupportScheduledRecordUpload` is returned and its value is `true`, it indicates that scheduled recording is supported.

2. The client software configure schedule parameters of scheduled recording. You can configure recording schedules for all channels or configure the recording schedule for a specified channel.

- Recording schedule management for all channels: Add a recording schedule: `POST /ISAPI/ContentMgmt/record/tracks`. Get all recording schedules parameters: `GET /ISAPI/ContentMgmt/record/tracks`. Set all recording schedules: `PUT /ISAPI/ContentMgmt/record/tracks`.

- Recording schedule management for a specified channel: Get the capability of configuring the recording schedule for a specified channel: `GET /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/capabilities?security=<security>&iv=<iv>`. Get the recording schedule parameters for a specified channel: `GET /ISAPI/ContentMgmt/record/tracks/<trackStreamID>?security=<security>&iv=<iv>`. Set the recording schedule for a specified channel: `PUT /ISAPI/ContentMgmt/record/tracks/<trackStreamID>?security=<security>&iv=<iv>`. Delete the recording schedule for a specified channel: `DELETE /ISAPI/ContentMgmt/record/tracks/<trackStreamID>?security=<security>&iv=<iv>`.

7.2 Search for Event Video

7.2.1 Introduction to the Function

This function supports searching for the videos of linked cameras when an event occurs.

7.2.2 API Calling Flow

1. Check whether the device supports searching for event videos.

Get the device sysgem capability: `GET /ISAPI/System/capabilities`. If the node `<SerialCap>` is returned and its value is true, the device supports searching for event videos.

2. Search for event videos: `POST /ISAPI/ContentMgmt/eventRecordSearch?format=json`.

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7.3 Search for Videos and Pictures

7.3.1 Introduction to the Function

This function supports searching for the video/picture data stored in the device. Video data source includes scheduled recording, manual recording and event recording, and picture data source includes scheduled capture and event linkage capture.

7.3.2 API Calling Flow

1. Check the device's supported time format. Get device's system capabilities: `GET /ISAPI/System/capabilities`.
   - If node `isSupportTimeTypeSTD` is returned with value `true`, it indicates that the device supports the ISO 8601 standard time, and URLs of functions related to ISO 8601 time parameters support additional query parameter `timeType=STD`.
   - If node `isSupportTimeTypeSTD` is not returned. Call `GET /ISAPI/System/time/timeType?format=json` to get the time type. If the `type` field returns `local`, it indicates that the device video search and playback use local time. If the `type` field returns `UTC`, it indicates that the device video search and playback use ISO 8601 time format.
   - If the device does not support API `GET /ISAPI/System/time/timeType?format=json`. Call `GET /ISAPI/System/deviceInfo` and check the time type of device via field `deviceType`. If `deviceType` (device type) is CVR, NVR, DVR, or ITCCAM, the device video search and playback use local time. Otherwise, follow the protocol specifications and use the TZ format, for example, `2024-05-13T16:00:00Z`.

2. Check whether the device supports video/picture search. Get the device's video search capability via `GET /ISAPI/ContentMgmt/capabilities`. If `<isSupportRecordSearch>` is returned, it indicates that the device supports video/picture search.

3. Search for videos and pictures:
   i. Get the capability of searching for videos and pictures: `GET /ISAPI/ContentMgmt/capabilities`. Get the picture search types that the device supports searching via the node `<pictureSearchType>` returned in the message, including pictures of event capture, scheduled capture and manual capture. Get the video types that the device supports searching for via the node `<recordSearchType>` returned in the message.
   ii. Get the parameters capability of searching for videos and pictures: `GET /ISAPI/ContentMgmt/search/capabilities`. Get the periods when the device supports searching for videos and pictures via the node `<timeSpanList>` returned in the message. Get the file tags that the device supports searching via the node `<labelSearch>` returned in the message.
   iii. Search for videos and pictures: `POST /ISAPI/ContentMgmt/search?security=<security>&iv=<iv>`.
   - The URL containing `timeType=STD` indicates that the time in the input and output messages are according to the ISO 8601 time format. The URL without `timeType=STD` indicates to use device's original time.
   - Check whether it is picture or video to be searched via the node `<trackID>` in the request message `<CMSearchDescription>`. "101" (search for the main stream videos of channel 1), "103" (search for the pictures of channel 1), and it is channel 1 by default .

   Note:

   - The value of `CMSearchDescription.trackIDList.trackID` is obtained from the `id` value returned in the response message of `GET /ISAPI/ContentMgmt/record/tracks`. The meaning of `id`: `101` is the main stream of channel 1, `102` is the sub-stream of channel 1, `103` is the capture schedule of channel 1, `201` is the main stream of channel 2, and so on. When the device saves only one copy of channel videos, it does not distinguish between the main and sub stream, and only supports setting `trackID` to `xx1`.
   - If `<trackIDList>` is returned in response message of `GET /ISAPI/ContentMgmt/search/capabilities`, it indicates that the device supports accurate search, and the returned value in `<trackIDList>` indicates the channels that support independent video search.

7.4 Search for Videos by Target Type

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7.4.1 Introduction to the Function

This function realizes persons, vehicles or mixed targets distinguishing when searching for motion detection event videos. Searching for videos by target type is newly added. The search conditions include target type information, and the returned result of the search contains the start and end time as well as the file name of the video.

7.4.2 API Calling Flow

1. Check whether the device supports searching by target type:

Get the device capability: `GET /ISAPI/System/capabilities`. If the node is returned and its value is true, it indicates that the device supports searching by target type. Currently, the device supports searching by person and by vehicle.

2. Get the capability of searching by target type: `GET /ISAPI/ContentMgmt/SearchByTargetType/capabilities?format=json`.

3. Search for videos by target type: `POST /ISAPI/ContentMgmt/SearchByTargetType?format=json`.

8 PTZ Control

8.1 . PTZ Lock

8.1.1 Introduction to the Function

The client can control a network camera's PTZ lock. During the locked period, the PTZ linked with alarm events will not work, but manual rotation is available, until the client unlocks the PTZ. This function can be used for long-term monitoring of key locations to prevent the lens from being moved. When configuring or modifying intelligent applications that involve patrol, the PTZ should be locked first and then unlocked after the configuration is completed to achieve internal control of the intelligent application's operation, while the actual PTZ operation remains functional.

8.1.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `ManualPTZLockCap` indicates the support for manual `PTZ` control locking;
2. (Client) Control PTZ manually:

   Manually lock the PTZ: `PUT /ISAPI/PTZCtrl/channels/<channelID>/manualPTZLock?format=json`;

   Manually unlock the PTZ: `PUT /ISAPI/PTZCtrl/channels/<channelID>/manualPTZUnlock?format=json`.
3. (Optional) The client configures the PTZ lock parameters to specify the lock duration:

   Capability: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `lockTime` indicates the maximum lock duration.

   Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/lockPTZ`.

   Get: `GET /ISAPI/PTZCtrl/channels/<channelID>/lockPTZ`.
4. The client gets the remaining time for PTZ unlock: `GET /ISAPI/PTZCtrl/channels/<channelID>/manualPTZLockCfg?format=json`.

8.2 Auxiliary Device Control

8.2.1 Introduction to the Function

The PTZ auxiliary devices help the device better display video content. The auxiliary devices includes lights, wipers, fans, and heaters. This function is used to control PTZ auxiliary devices.

8.2.2 API Calling Flow

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1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `PTZAuxType` indicates the support for PTZ auxiliary device control;
2. The client configures the PTZ auxiliary device parameters to control the auxiliary devices. It is recommended to first get all auxiliary device parameters before modifying parameter configurations:

   (Optional) Get all auxiliary device parameters: `GET /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols`;

   (Optional) Get a single auxiliary device parameter: `GET /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols/<auxCtrlID>`;

   (Optional) Configure auxiliary device parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols`;

   Configure a single auxiliary device parameter: `PUT /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols/<auxCtrlID>`.

Notes:

The `id` in the parameter configuration message corresponds to the `auxCtrlID` in the `url`, which is the auxiliary device index and starts from 1 by default.

8.3 PTZ 3D Positioning Control

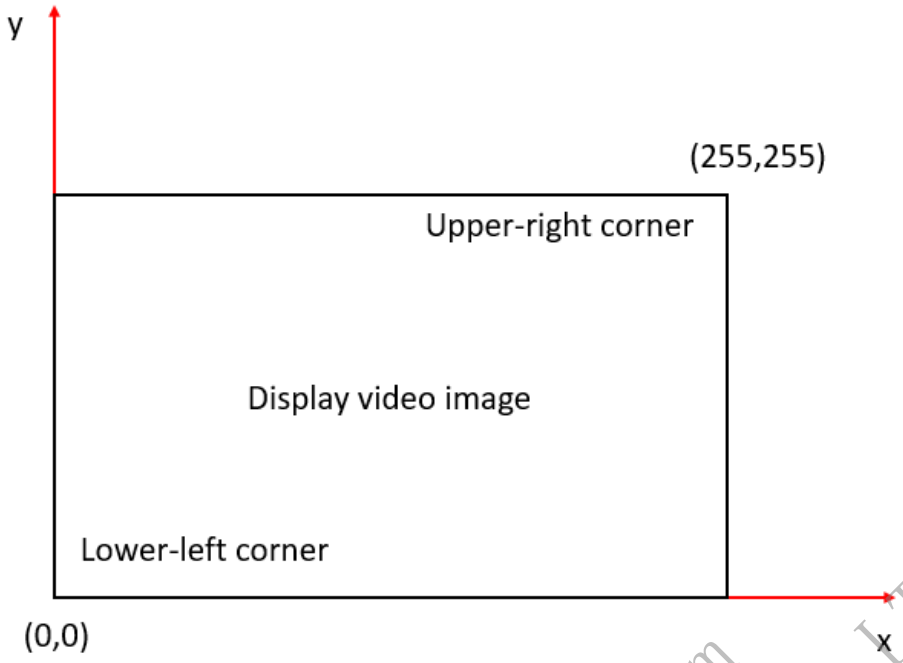
8.3.1 Introduction to the Function

Supports centering a point in the video image, and supports centering and zooming a rectangular area in the video image.

8.3.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportPosition3D` indicates the support for 3D positioning control;
2. (Client) Send a control command to the device: `PUT /ISAPI/PTZCtrl/channels/<channelID>/position3D`:
   - The point in the video image is centered by the device, and the coordinates of `StartPoint` and `EndPoint` are the same;
   - The device zooms in and centers a rectangular area in the video image, and the X coordinate value of `StartPoint` is smaller than that of `EndPoint`;
   - The device zooms out and centers a rectangular area in the video image, and the X coordinate value of `StartPoint` is greater than that of `EndPoint`.

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8.4 PTZ Absolute Position

8.4.1 Introduction to the Function

The PTZ absolute position is unique and remains the same. The client can record certain PTZ absolute positions in actual application scenarios to repeatedly inspect at specified positions. For example, in a supermarket, record the PTZ absolute position of the cash register and inspect the position periodically.

8.4.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. the returned `isSupportAbsoluteEx` indicates the support for high-precision absolute `PTZ` movement control;
2. (Client) Get the high-precision absolute `PTZ` movement control capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/RailwayRobot/capabilities?format=json`. The value range of control parameters will be returned;
3. The client specifies the PTZ to move to an absolute position: `PUT /ISAPI/PTZCtrl/channels/<channelID>/absoluteEx`. The `elevation` and `azimuth` represent the angles in the vertical and horizontal directions, `absoluteZoom` represents the zoom range, `focus` represents the focus range, and `absoluteRotate` represents the absolute rotation angle.
4. (Client) Get the PTZ absolute position: `GET /ISAPI/PTZCtrl/channels/<channelID>/absoluteEx`.

Notes:

- The zero coordinate position of `elevation` and `azimuth`, i.e., PT(0,0), can be specified by `PUT /ISAPI/PTZCtrl/channels/<channelID>/homeposition`.

8.5 PTZ Manual Control

8.5.1 Introduction to the Function

Adjust the horizontal and vertical positions of the PTZ, and control the zoom and focus of the lens at specified rotation/changing speeds. The client adjusts PTZ continuously through applying control commands, which is usually used for manual direction adjustment in remote live view.

8.5.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned

`ContinuousPanTiltSpace` in the message indicates the support for PTZ manual control;

2. The client sends a control command to the device: `PUT /ISAPI/PTZCtrl/channels/<channelID>/continuous`. The values of `pan`, `tilt`, `zoom`, and `rotate` represent the speed of rotation/change, and the characteristics (positive and negative) of the values represent the direction of rotation/change. If the control needs to be stopped, another command should be sent and values should be set as 0.

8.6 PTZ One-Touch Focus

8.6.1 Introduction to the Function

Help PTZ devices to focus. When defocus (blurry image) occurs , one-touch focus can make the image clear.

8.6.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportOnepushfoucsStart` indicates the support for one-touch focus;
2. (Client) Configure PTZ one-touch focus parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/onepushfoucs/start`;
3. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportOnepushfoucsReset` indicates the support for resetting focus parameters;
4. (Client) Reset the PTZ one-touch focus parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/onepushfoucs/reset`. The PTZ focus parameter setting will be restored.

8.7 Save PTZ Position

8.7.1 Introduction to the Function

The client controls the device to save the current PTZ position information. After the PTZ leaves the current position, the client can control the PTZ to return to the saved position, but only one position can be saved for each PTZ.

8.7.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportPTZSave` and `isSupportPTZSaveGet` indicates the support for keeping and getting `PTZ` parameters configurations;
2. (Client) Configure the device to save PTZ information: `PUT /ISAPI/PTZCtrl/channels/<channelID>/save?format=json`;
3. The client controls the PTZ to move to the saved position: `PUT /ISAPI/PTZCtrl/channels/<channelID>/gotoPosition?format=json`. The PTZ will rotates to the saved position;
4. The client gets the saved PTZ position information: `GET /ISAPI/PTZCtrl/channels/<channelID>/save`. The PTZ position information will be returned and PTZ does not move.

9 People Counting

9.1 People Counting Statistics Search

9.1.1 Introduction to the Function

For devices with local storage function, the people counting data will be stored in the local storage after the people counting function is enabled. You can search the local data to get multiple types of reports by setting different conditions.

9.1.2 API Calling Flow

9.1.2.1 Single Channel

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1. Get the capability of searching for people counting data of a single channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/search/capabilities`.
2. Search for people counting data of a single channel: `POST /ISAPI/System/Video/inputs/channels/<channelID>/counting/search`.

9.1.2.2 Multiple Channels

1. Get video capability: `GET /ISAPI/System/Video/capabilities`. If the node `isSupportMultiChannelCounting` is returend and its value is `true`, it indicates that searching for people counting data of multiple channels is supported.
2. Get the capability of searching for people counting data of multiple channels: `GET /ISAPI/System/Video/inputs/channels/counting/search/capabilities`.
3. Search for the people counting data of multiple channels: `POST /ISAPI/System/Video/inputs/channels/counting/search`.

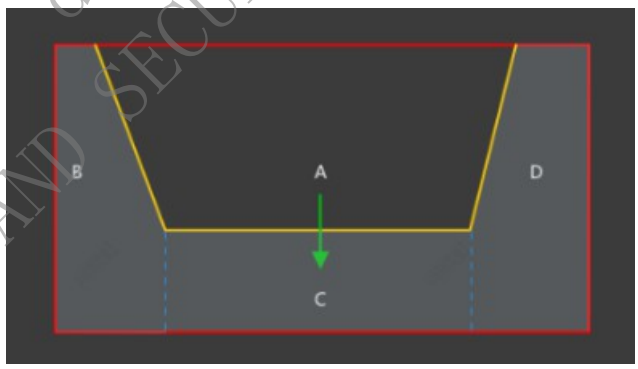
9.2 People Flow Direction Analysis

9.2.1 Introduction to the Function

People flow direction analysis is used for counting the people who crosses detection lines and enters the detection area. The flow analysis is valid only when the detection area is quadrilateral and the detection line and the detection area are overlapped only in the start and end points.

The requirements for drawing the rule/detection area of flow direction analysis are as follows:

1. The detection line (in yellow) and the detection area (in red) should have but two overlapped points.
2. The detection line should be perpendicular to the detection direction (along the direction of green arrow).
3. The four points on the detection line respectively are: point 1, 2, 3, 4.
4. Along the detection direction (along the green arrow), draw two parallel lines from point 2 and 3 to intersect the detection area (in red). The area between the two parallel lines is the counting region of area 2 (area C in the figure), and the rest two are area 1 (area B in the figure) and area 3 (area D in the figure).



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9.2.2 API Calling Flow

The process of people flow direction analysis is the same with that of the real-time people counting. You need to enable the flow direction analysis to receive and parse the flow statistics data.

1. Get the capability of people counting: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities`. If `flowDirectionAnalysis` is returned, it indicates that the device supports flow analysis.
2. Configure people counting parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting`. Configure detection lines parameters (`CountingLineItemList`) and rule parameters (`CountingRegionList.RuleRegionCoordinatesList`), select scene mode (`countingSceneMode`), enable the flow direction analysis (`flowDirectionAnalysis` is `true`).
3. Receive the event uploaded by devices in arming or listening mode: `PeopleCounting`. Parse

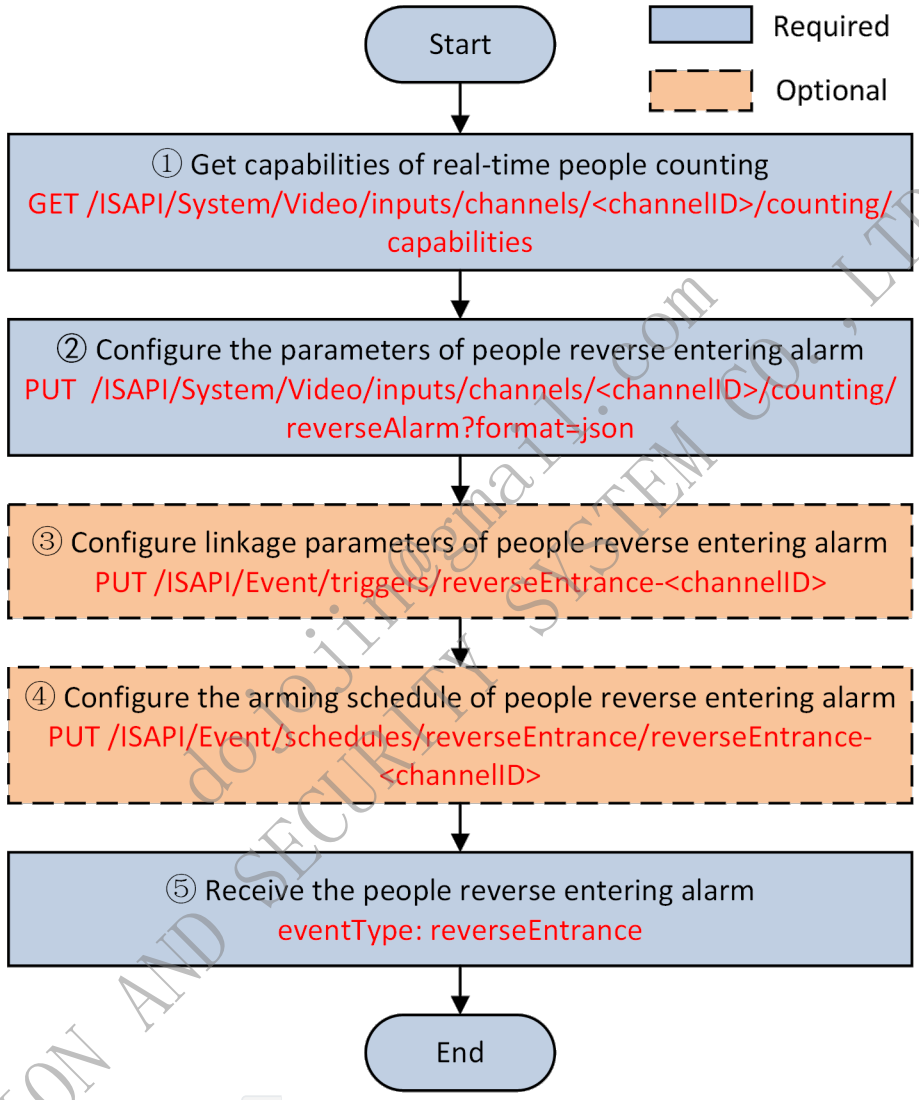
`peopleCounting.FlowStatisticsList` to get the flow data.

9.3 People Reverse Entering Alarm

9.3.1 Introduction to the Function

The people reverse entering alarm will be triggered if a person crosses the detection line from the opposite direction within a specified period. It applies to the circumstance when only single direction passing is allowed within a specified period. For example, if someone exits the workshop within working time, an alarm will be uploaded for notification.

9.3.2 API Calling Flow



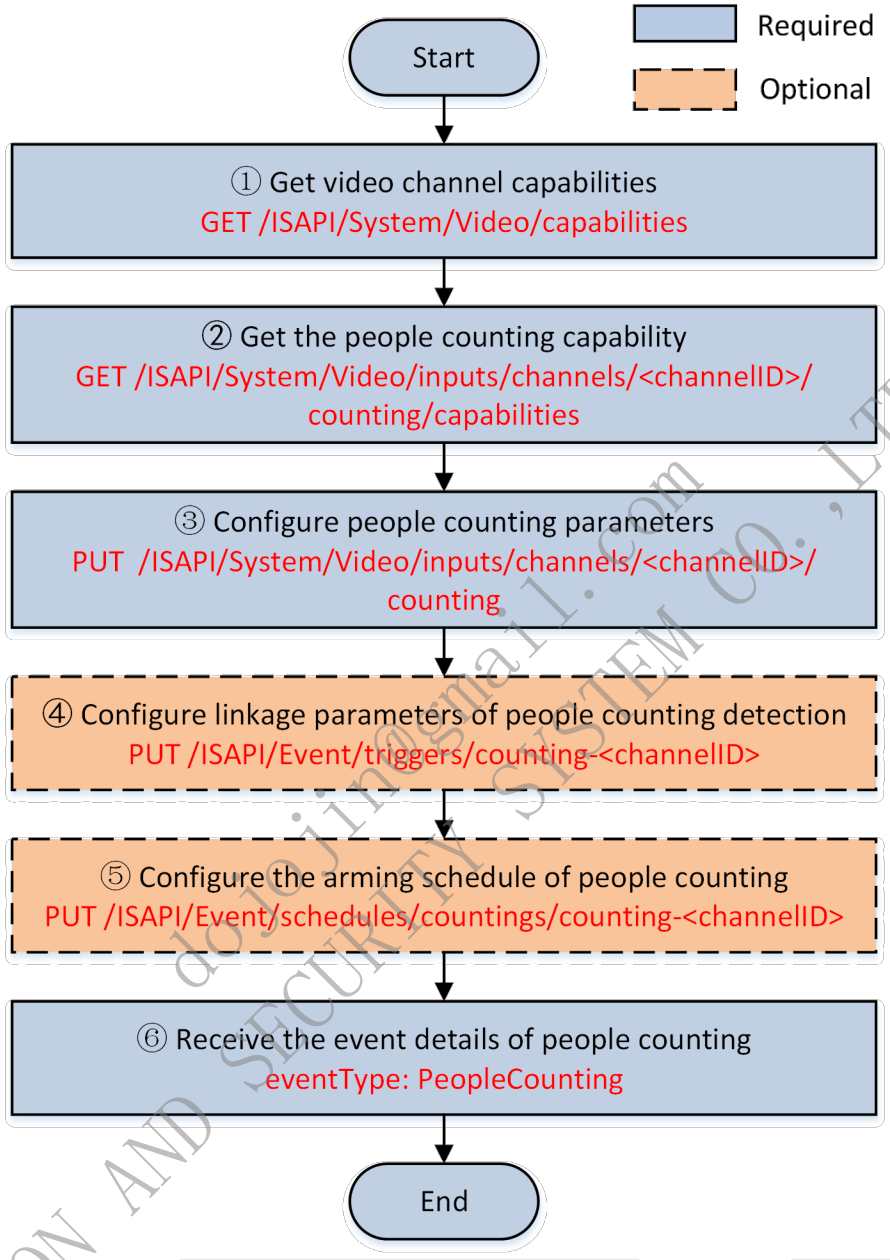
1. Get the people counting capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities`. If the node `isSupportReverseAlarm` is returned and its value is `true`, it indicates that people reverse entering alarm is supported.
2. Configure parameters of the people reverse entering alarm: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting/reverseAlarm?format=json`.
3. (Optional) Configure linkage parameters of the people reverse entering alarm: `PUT /ISAPI/Event/triggers/reverseEntrance-<channelID>`.
4. (Optional) Configure the arming schedule for the people reverse entering alarm: `PUT /ISAPI/Event/schedules/reverseEntrance/reverseEntrance-<channelID>`.
5. Receive events reported by device via arming or listening: `reverseEntrance`.

9.4 Real-time People Counting

9.4.1 Introduction to the Function

The function is to count line crossing persons or objects in a specified time period. It is commonly applied in the entrance and exit of the scenic spots, malls, stores, etc.

9.4.2 API Calling Flow



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1. Get video channel capabilities: `GET /ISAPI/System/Video/capabilities`. If the nodes `isSupportCounting` and `isSupportMultiChannelCounting` are returned and the value is true, it indicates that the device supports real-time people counting.
2. Get the people counting capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities`.
3. Configure people counting parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting`.
4. (Optional) Configure linkage rules for people counting detection: `PUT /ISAPI/Event/triggers/counting-<channelID>`.
5. (Optional) Configure the arming schedule for people counting: `PUT /ISAPI/Event/schedules/countings/counting-<channelID>`.
6. Receive events uploaded by device in listening or arming mode: `PeopleCounting`.

Notes:

- During people counting, considering the height difference between children and adults, you can enable the children detection mode so as to improve the accuracy. a. Get the capability of people counting. If `ChildFilter` is returned, it indicates that the device supports counting the number of children. b. Enable the children counting during the configuration of people counting parameters (set `enabled` of `ChildFilter` to true). c. Other operations are same as above steps.

- The people counting detection event (`PeopleCounting`) supports the following data types: a. Real-time statistics data (`statisticalMethods` is `realTime`): after enabled, persons' entry and exit will trigger the event data uploading, and the data is the real-time people counting statistics from 12:00 A.M. to the current clock on the day. b. Statistics data by period (`statisticalMethods` is `timeRange`): the device will upload the statistics data according to the scheduled uploading interval (`dataUploadCycle`) within a specified period. c. Statistics data triggered by alarm input (`statisticalMethods` is `signalTrigger`): the device is connected to the alarm input signal, and the statistics data will be accumulated and uploaded every time the signal is triggered.

- The dimensions of statistics data are as follows: the number of people entering (`enter`), the number of people exiting (`exit`), the number of people repeating passing (`duplicatePeople`, which is only valid for people entering, and deduplication is invalid for people exiting), the number of people according to different face attributes, including face expression (`FaceExpressionList`), gender (`GenderList`), mask wearing (`MaskList`), etc. See details in the event message.

- If the client software fails to receive the people counting event uploaded by devices due to exceptions, or the connected devices have operated for a while, you can get the history people counting data via URI (POST /ISAPI/System/Video/inputs/channels/counting/search).

10 Vehicle Recognition

10.1 Motor Vehicle Recognition

10.1.1 Introduction to the Function

This function is to detect whether there are vehicles appearing in the image for certain time periods. If a vehicle appears and the set recognition conditions are met, it will be captured, and its related information will be analyzed and uploaded as alarms. The integrator receives the event and analyzes the specific content as needed.

10.1.2. Device Event Upload

10.1.2.1 Check Whether Device Supports Baseline Vehicle Recognition Event

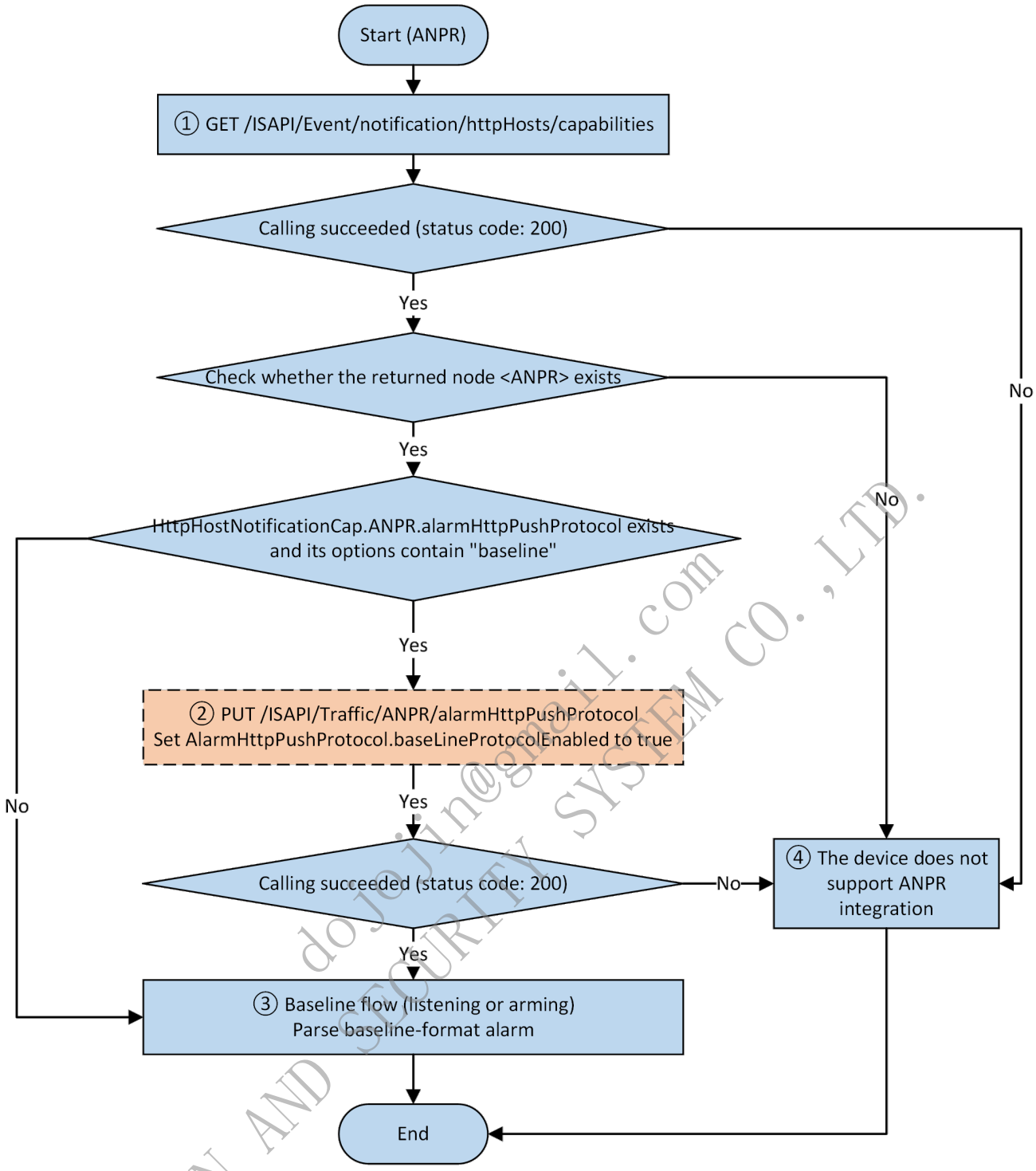
The integrator get the capability from device.

1. Get the system capability set via `GET /ISAPI/System/capabilities`: `true` is returned for `<ITCCap>.<isSupportVehicleDetection>` by device or `<ITCCap>.<isSupportHVTVehicleDetection>` is `true`.
2. Get the vehicle capture recognition capability via `GET /ISAPI/ITC/capabilities`: `true` is returned for `<isSupportVehicleDetection>` by device or `<isSupportHVTVehicleDetection>` is `true`.
3. Get the traffic service capability via `/ISAPI/Traffic/capabilities`: `<plateCap>` is returned by device.
4. Get the trigger mode capability set via `GET /ISAPI/ITC/TriggerMode/capabilities`: `<TriggerMode>` is returned.

When any one of the 4 conditions are met, it indicates the vehicle recognition event detection is supported. If none are met, the vehicle recognition event is not supported and please do not proceed to the following steps.

10.1.2.2 Check Whether Device Supports Baseline Flow

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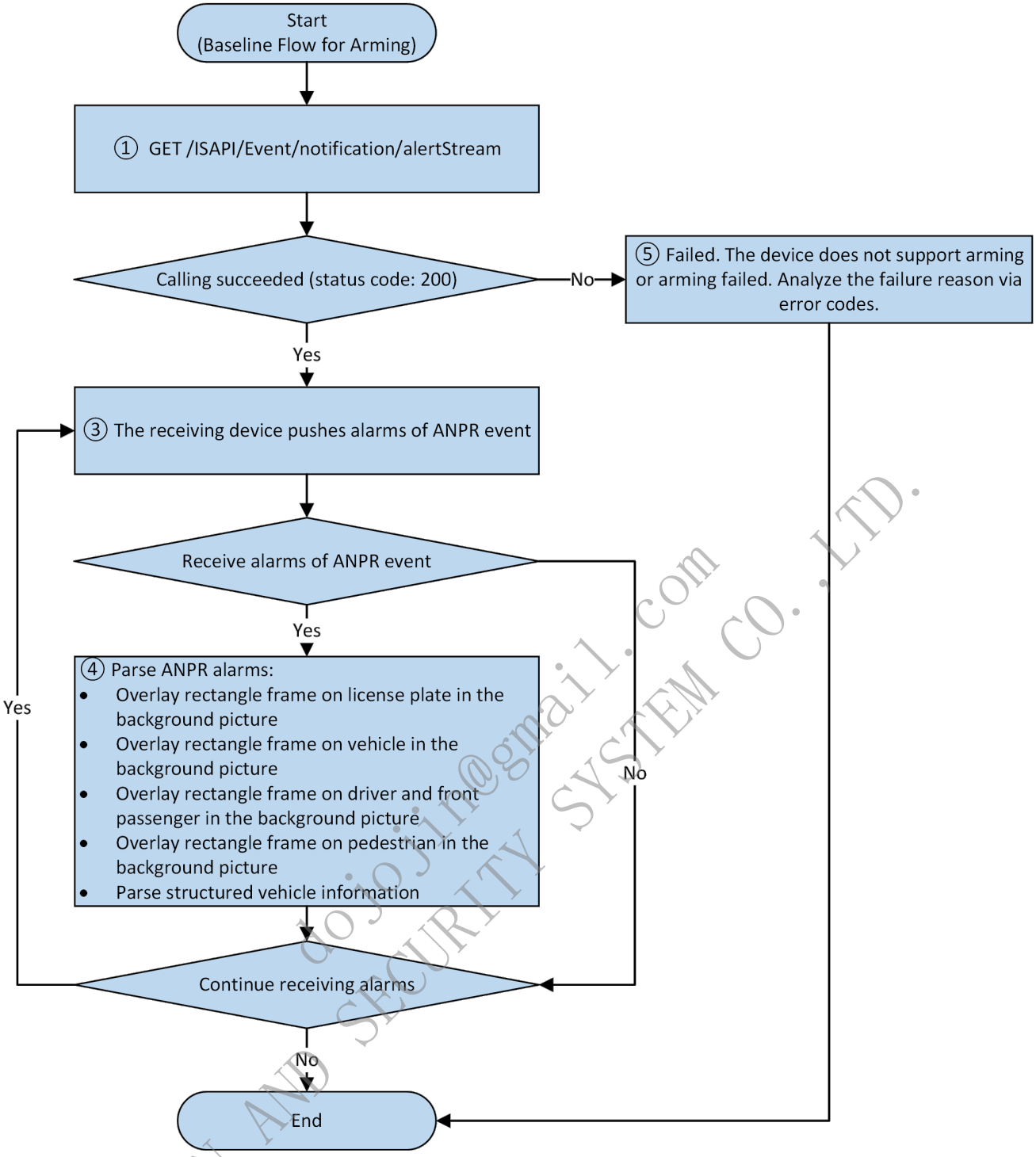


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1. (Client) Get the configuration capability of listening host via `GET /ISAPI/Event/notification/httpHosts/capabilities`.
2. (Client) Configure the alarm mode switch for device via `PUT /ISAPI/Traffic/ANPR/alarmHttpPushProtocol` and set `AlarmHttpPushProtocol.baseLineProtocolEnabled` to `true`.
3. Receive and parse alarms according to the baseline flow. See details in `4.3 Baseline Flow Event Upload` and `4.4 Baseline Flow Message Format and Example`.
4. The ANPR function integration is not supported by device.

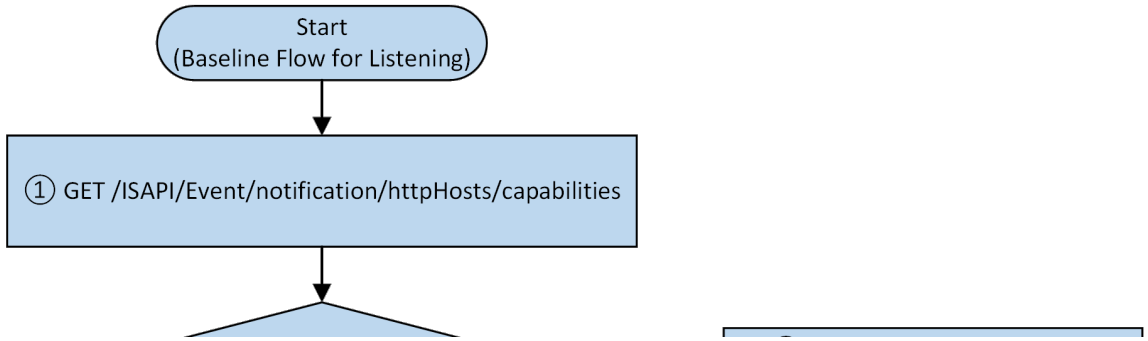
10.1.3 Baseline Flow Event Upload

10.1.3.1 Arming Flow

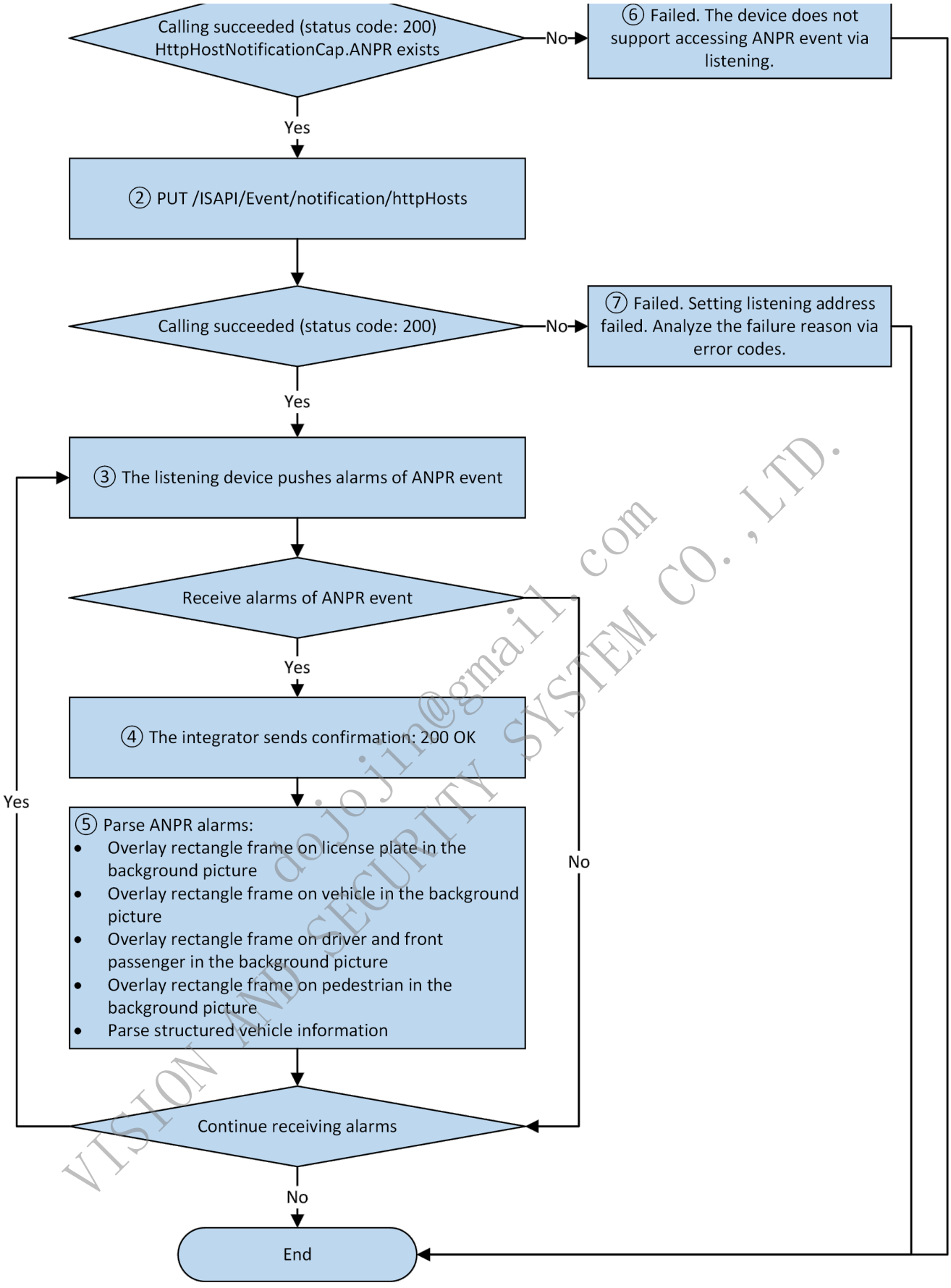


1. Arm the device.
2. Check whether arming succeeded.
3. Receive alarm notifications and check whether `eventType==ANPR`.
4. Parse ANPR alarm content.
5. Arming failed. Analyze the failure reason via error codes.

10.1.3.2 Listening Flow



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1. Get the capability set of device HTTP listening host location.
2. Change the HTTP listening host location, IP address, port configuration to the IP address of the server for receiving alarm notifications.
3. Receive alarm notifications and check whether `eventType==ANPR`.
4. The integrator sends confirmation (200 OK) to the device and notify the device of the successful alarm receiving; if the integrator does not reply, the device will consider the alarm notification lost and upload it again.
5. Parse ANPR alarm content.

6. The device does not support accessing ANPR event via listening.
7. Setting listening address failed. Analyze the failure reason via error codes.

10.1.4 Baseline Flow Message Format and Example

10.1.4.1 Baseline Flow Event Upload Method Configuration

Set `baseLineProtocolEnabled` to `true`,

URI: `/ISAPI/Traffic/ANPR/alarmHttpPushProtocol`

Method: `GET, PUT`

Body:

```
<AlarmHttpPushProtocol version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <baseLineProtocolEnabled>true</baseLineProtocolEnabled>
</AlarmHttpPushProtocol>
```

10.1.2.2 Baseline Flow Event Upload Example

Reported by device.

```
POST /test HTTP/1.1
Accept: text/html, application/xhtml+xml, */*
Accept-Language: zh-CN
Content-Type: multipart/form-data; boundary=7e13971310878
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)
Accept-Encoding: gzip, deflate
Host: 10.10.36.29:8080
Content-Length: 9907
Connection: Keep-Alive
Cache-Control: no-cache

--7e13971310878
Content-Disposition: form-data; name="anpr.xml";filename="anpr.xml";
Content-Type: text/xml
Content-Length: 150

<EventNotificationAlert/>
-----------------------------7e13971310878
Content-Disposition: form-data; name="licensePlatePicture.jpg";filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 200

......JFIF.....`.`.....C.......... .
..
.............. $.' ",#..(7),01444.'9=82<.342...C. ....

.2!.!222222222
```

Note: When there are multiple pictures of the same type, the following naming rule for will be applied: add suffix (e.g., "_1", "_2") after the extended name. For example, the names of three detection pictures can be detectionPicture_1.jpg, detectionPicture_2.jpg, and detectionPicture_3.jpg.

Attribute Value	Attribute Type Character String
ANPR Alarm XML	anpr.xml
Detection Picture (Background Picture)	detectionPicture.jpg
License Plate Thumbnail	licensePlatePicture.jpg
Driver's Face Matting	pilotPicture.jpg
Passenger's Face Matting	copilotPicture.jpg
Composite Picture	compositePicture.jpg
License Plate Binary Picture	plateBinaryPicture.jpg
Non-Motor Vehicle Matting	nonMotorPicture.jpg
Pedestrian Picture	pedestrianDetectionPicture.jpg
Pedestrian Matting	pedestrianPicture.jpg

11 Abnormal Event Detection

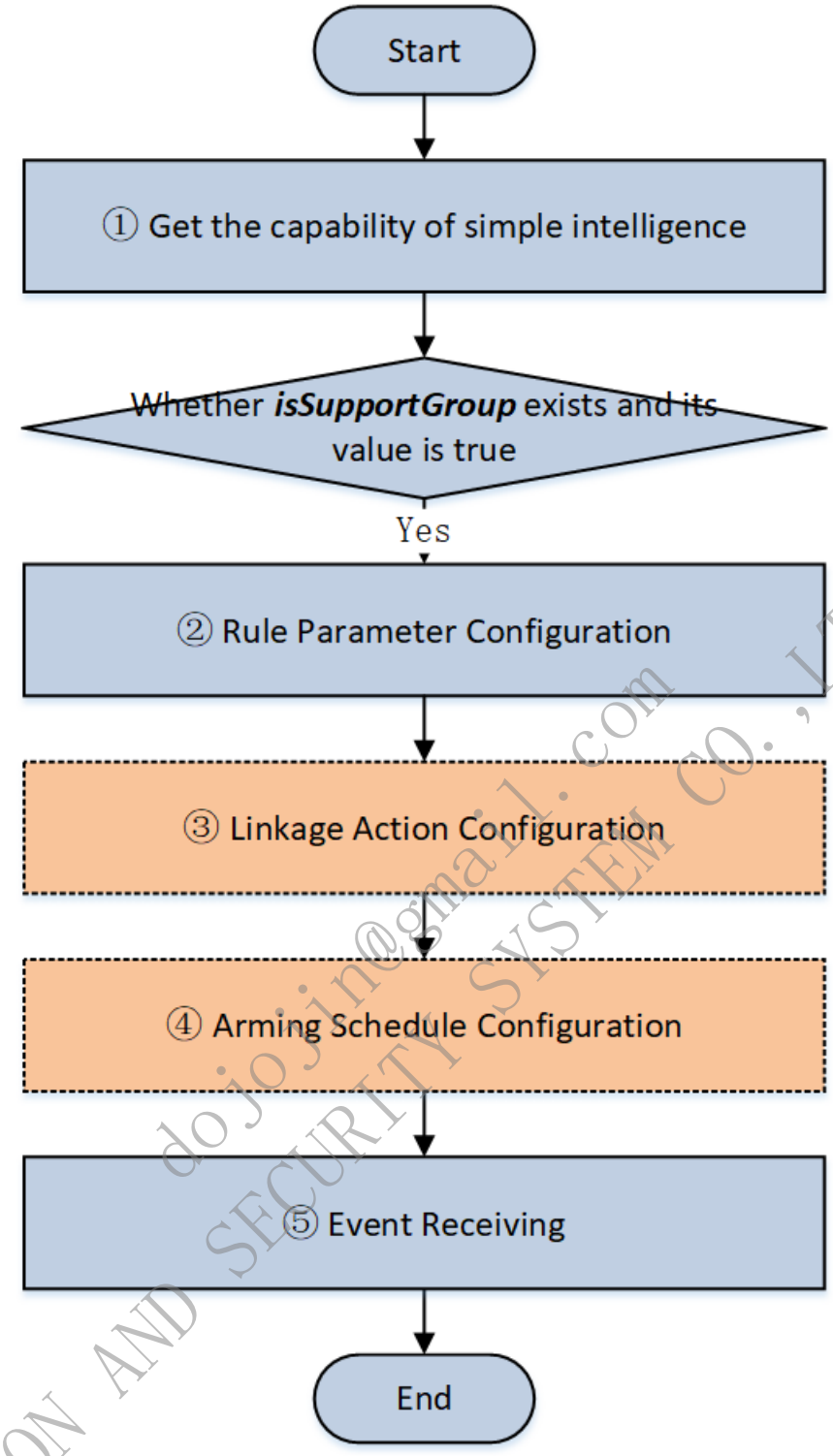
11.1 People Gathering

11.1.1 Introduction to the Function

The people gathering function is used for triggering the alarm in the fastest and best way when the people density in the set area exceeds the set threshold. The people gathering function is often used for performing the real-time monitoring of the abnormal behavior of people in the monitoring area, effectively preventing accidents caused by negligence, so as to avoid potential safety accidents and improve the supervision efficiency.

11.1.2 API Calling Flow

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1. Get the apability of AI intelligent search.

   Get the `isSupportGroup` in the capability of simple intelligence (related URI: `GET /ISAPI/Smart/capabilities`).

2. Configure rule parameters.

   Configure All Channels

- Get people gathering parameters (related URI: `GET /ISAPI/Smart/group`).
- Configure people gathering parameters (related URI: `PUT /ISAPI/Smart/group`).

  Configure Specified Channel

- Get the capability of people gathering parameters (related URI: `GET /ISAPI/Smart/group/<channelID>/capabilities`).

- Get the people gathering parameters of a specified channel (related URI: `GET /ISAPI/Smart/group/<channelID>`).
- Configure the people gathering parameters of a specified channel (related URI: `PUT /ISAPI/Smart/group/<channelID>`).

Configure All Areas of Specified Channel

- Get the capability of the parameters of all areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/capabilities`).
- Get the parameters of all areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions`).
- Configure the parameters of all areas for people gathering (related URI: `PUT /ISAPI/Smart/group/<channelID>/regions`).

Configure Specified Channel in Specified Channel

- Get the capability of the parameters of specified areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/<regionID>/capabilities`).
- Get the parameters of specified areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/<regionID>`).
- Configure the parameters of specified areas for people gathering (related URI: `PUT /ISAPI/Smart/group/<channelID>/regions/<regionID>`). Note: The percentage refers to the percentage of people number accounts for the alarming area. An alarm will be triggered in the system when the percentage is higher than the set percentage. Otherwise, no alarm will be triggered.

3. (Optional) Configure arming linkage parameters.

- Get the capability of event linkage configuration (related URI: `GET /ISAPI/Event/triggersCap`), whether the message `EventTriggersCap` contains the node `GroupDetectionTriggerCap`.
- Get the linkage parameters of people gathering (related URI: `GET /ISAPI/Event/triggers/group-<channelID>`).
- Configure linkage parameters of people gathering (related URI: `PUT /ISAPI/Event/triggers/group-<channelID>`).

4. (Optional) Configure arming schedule.

- Get the arming schedule of alarm outputs for all channels (related URI: `GET /ISAPI/Event/schedules/groups`).
- Configure the arming schedule of people gathering for all channels (related URI: `PUT /ISAPI/Event/schedules/groups`).
- Get the arming schedule of people gathering for a specified channel (related URI: `GET /ISAPI/Event/schedules/groups/group-<channelID>`).
- Configure the arming schedule of people gathering for a specified channel (related URI: `PUT /ISAPI/Event/schedules/groups/group-<channelID>`). Note: 1. Supports setting up to 8 time periods in one day on the timeline. 2. Overlapped time between any two time periods is not allowed when setting the arming schedule.

5. Receive event: `"eventType":"group"`, including the following. `detectionPictureTransType` refers to the transmission method of the pictures. The field value is bkgUrl for the picture URL or the picture encoding data via base64 when the value of detectionPictureTransType is "url" or "base64". The field value is the picture Content-ID when the value of detectionPictureTransType is "binary".

12 Facial Recognition

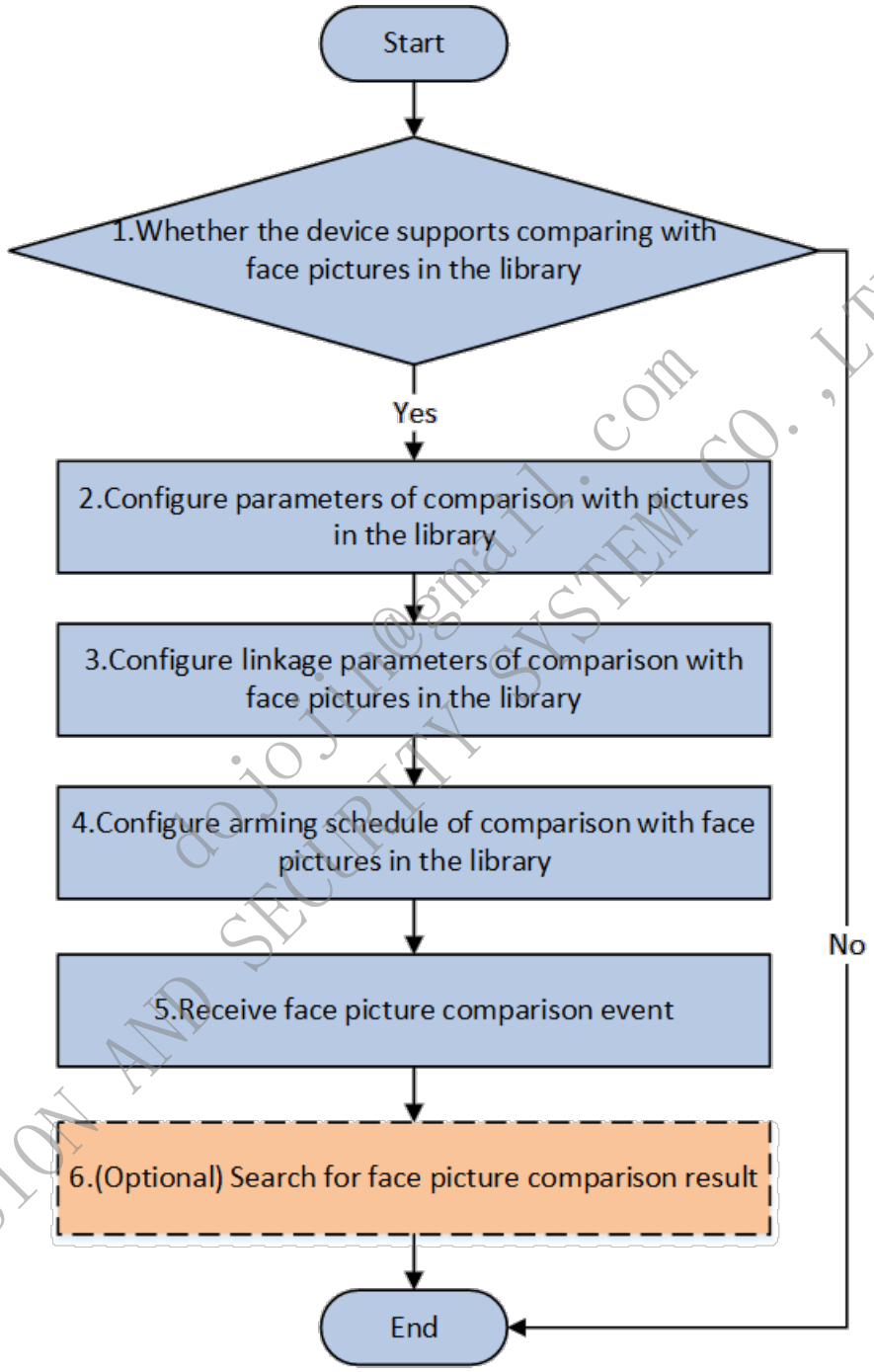
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12.1 . Comparison with Pictures in the Library

12.1.1 Introduction to the Function

Note: This feature depends on the face picture library management (refer to Face Picture Library Management Field for details). Analyze and model the face captured in the camera, then compare the face picture with pictures in the library, and lastly return the comparison similarity.

12.1.2 API Calling Flow



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1. **Whether the device supports comparing with face pictures in the library:**

   Get the capability node to evaluate whether the device supports comparing pictures with face pictures in the library, `GET /ISAPI/System/capabilities`; If the node `<isSupportFaceContrast>` is returned and its value is `true`, it indicates that the device supports comparing pictures with face pictures in the library.

2. **The parameters of comparison with pictures in the library including:**

   Get the capability of face picture comparison, `GET /ISAPI/Intelligent/channels/<channelID>/faceContrast/capabilities`. Configure the parameters of face

picture comparison, `PUT /ISAPI/Intelligent/channels/<channelID>/faceContrast`; Get the configuration of face picture comparison, `GET /ISAPI/Intelligent/channels/<channelID>/faceContrast`; **Note: The rule parameters support configuring the triggering threshold of face similarity event and specifying information including face picture library ID.**

3. **The linkage parameters of comparison with face pictures in the library:**

   Two sets of configurations linked to business systems are provided due to the front-end and back-end device differences:

   Linkage action of face picture library: for front-end devices (e.g., Deepinview), a business system of single channel and multiple face picture libraries. Therefore it supports configuring different arming linkage parameters via each compared face picture library. Call `GET /ISAPI/Event/triggersCap` to evaluate whether it supports the event linkage configuration capability, it does if `<FaceLibTriggerCap>` is returned, and the relevant configuration APIs are as follows: Configure the linkage action parameters, `PUT /ISAPI/Event/triggers/faceLib-<channelID>/<FDID>`; if users have added a customized face picture library ID when creating one, they can also call `PUT /ISAPI/Event/triggers/faceLib-<channelID>/<customFaceLibID>?FDType=custom` to configure via `customFaceLibID`; get the linkage action parameters, `GET /ISAPI/Event/triggers/faceLib-<channelID>/<FDID>`; Customizing the face picture library ID follows the same flow.

   Linkage action of channel: for back-end devices (e.g., DeepinMind), a business system of multiple channels and single face picture library. Therefore it supports configuring different arming linkage parameters via each video channel. Call `GET /ISAPI/Event/triggersCap` to evaluate whether it supports the event linkage configuration capability, it does if `<FaceContrastTriggerCap>` is returned, and the relevant configuration APIs are as follows: Configure the linkage action parameters of face picture comparison, `PUT /ISAPI/Event/triggers/faceContrast-<channelID>`; get the linkage action parameters of face picture comparison, `GET /ISAPI/Event/triggers/faceContrast-<channelID>`;

4. **Arming schedule of comparison with face pictures in the library**:

   Two sets of arming schedule configurations linked to business systems are provided due to the front-end and back-end device differences: Arming schedule of face picture library: Configure the single-channel arming schedule of face picture library, `PUT /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>`; if users have added a customized face picture library ID when creating one, they can also call `PUT /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>/<customFaceLibID>?FDType=custom` to configure via `customFaceLibID`; get the single-channel arming schedule of face picture library, `GET /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>`; Customizing the face picture library ID follows the same flow. Arming schedule of channel: Configure the arming schedule of face picture comparison, `PUT /ISAPI/Event/schedules/faceContrast-<channelID>`; get the arming schedule of face picture comparison, `GET /ISAPI/Event/schedules/faceContrast-<channelID>`;

5. **Face picture comparison event**:

   When the face captured on the camera is compared with pictures in the library, and the similarity has reached the rule parameters including the threshold, the device will upload the face picture comparison event; the event information mainly includes comparison similarity, captured face and its attributes, and the corresponding registered face picture and its attributes, `"eventType": "alarmResult"`.

6. **Search for face picture comparison result:**

   If supported by device, the face picture comparison result will be saved to the device local storage, and you can search for comparison history via the following API: Search for face picture comparison information, `GET /ISAPI/Intelligent/FDLib/FCSearch`.

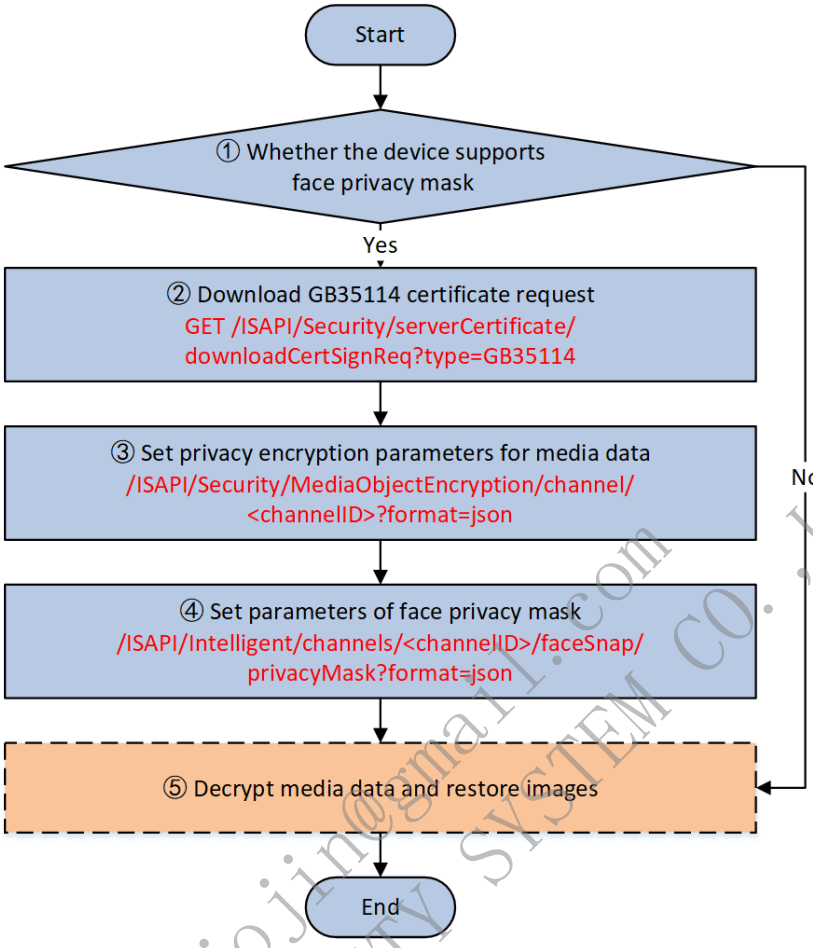
12.2 Face Capture

12.2.1 Introduction to the Function

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Face capture is to detect whether face pictures show up in images. If set conditions are met and there are faces showing up, the face pictures will be captured and uploaded as alarms.

12.2.2 API Calling Flow



1. **Whether the device supports face capture:**

   Get the system capabilities: `GET /ISAPI/System/capabilities`. If the node (its parent node is) is returned as true, it indicates that the device supports face capture.

2. **Whether the device supports multi-scene face capture:**

   Get the intelligent capabilities of device: `GET /ISAPI/Intelligent/channels/<channelID>/capabilities`. If the node is returned and its value is true, it indicates that the device supports face capture.

3. **Set face picture capturing rules of multiple scenes:**

   If multi-scene capture of a channel is required, you can set the face picture capturing rules for multiple scenes.

   Set the face capture rules of a specified scene: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/<SID>`;

   Get the face capture rules of a specified scene: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/<SID>`;

   Set the face picture capturing rules of multiple scenes for a specified channel: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene`;

   Get the face picture capturing rules of multiple scenes for a specified channel: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene`.

4. **Set face picture capturing rules of a specific scene.**

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If specific-scene capture of a channel is required, you can set the face picture capturing rules.

Get the capabilities of captured face picture rules `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/capabilities`;

Set the face capture rule parameters: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule`;

Get the face capture rule parameters: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule`;

Get the default parameters of face capture rule: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule?defualt=<defualt>`.

5. **Set linkage mode of face picture capture.**

   Set the linkage parameters of face capture: `PUT /ISAPI/Event/triggers/faceSnap-<channelID>`;

   Get the linkage parameters of face capture: `GET /ISAPI/Event/triggers/faceSnap-<channelID>`;

   Delete the linkage of face capture: `DELETE /ISAPI/Event/triggers/faceSnap-<channelID>`.

6. **Configure the arming schedule of face capture.**

   Set the arming schedule of face capture for a specified channel: `PUT /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>`;

   Get the arming schedule of face capture for a specified channel: `GET /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>`;

   Set the arming schedule of face capture: `PUT /ISAPI/Event/schedules/faceSnap`;

   Get the arming schedule of face capture: `GET /ISAPI/Event/schedules/faceSnap`.

7. **Receive face capture and recognition events.**

   If the wanted face shows up in the set area, face recognition event will be reported. The event information includes face pictures and attribute information, etc. `"eventType": "faceCapture"`.

12.3 Face Picture Library Management

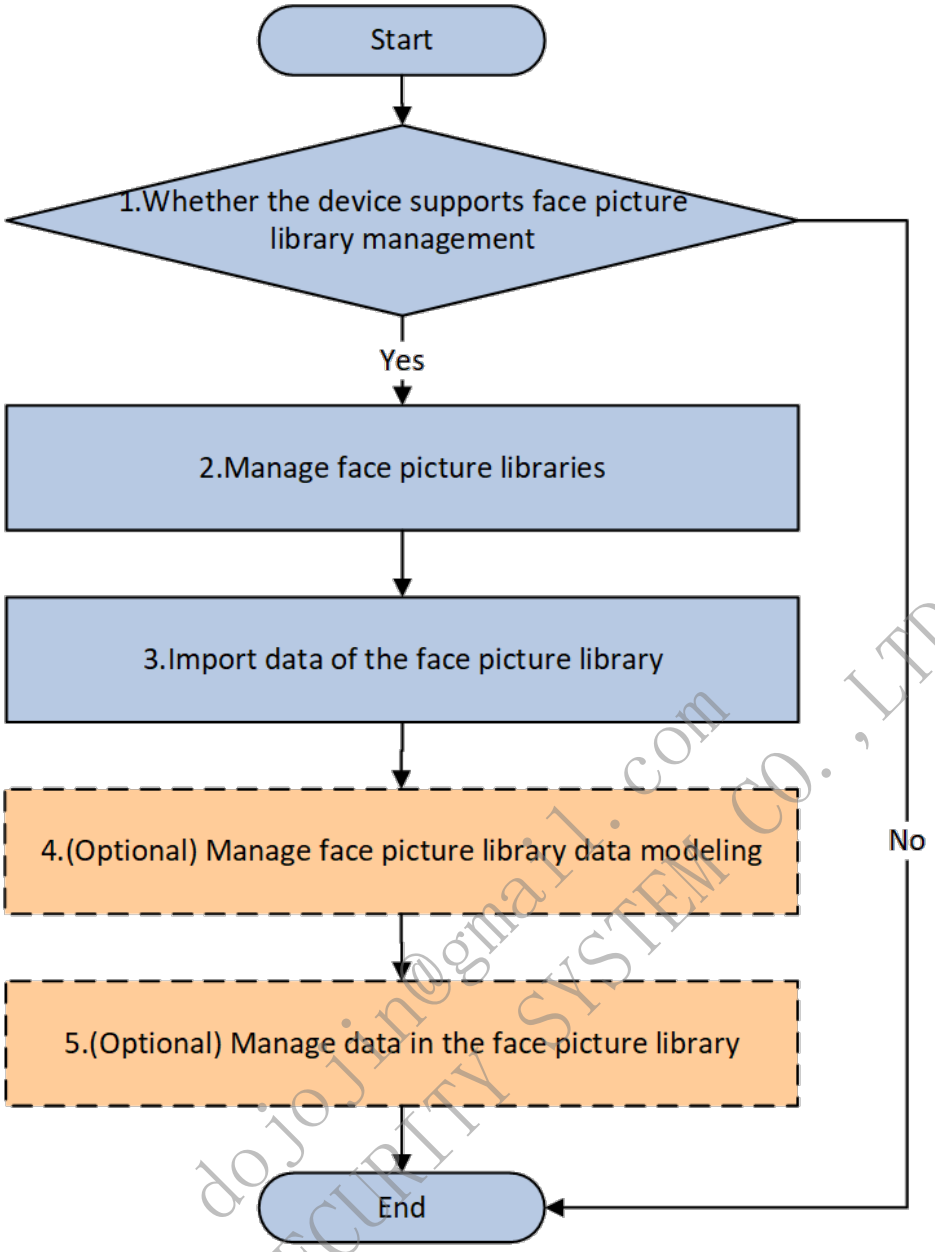
12.3.1 Face Picture Library Management

12.3.1.1 Introduction to the Function

This function is mainly used to manage operations on face picture libraries on the device, such as adding, deleting, editing, or searching for the face picture libraries, importing face pictures and additional person information to the libraries, editing, deleting, or searching for face pictures and additional person information in the libraries, and face picture modeling. This function is also the prerequisite of other face picture library functions such as face picture library comparison.

12.3.1.2 API Calling Flow

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1. Get the capability of managing face picture libraries for checking functions supported by the device: `GET /ISAPI/Intelligent/FDLib/capabilities`.
2. Manage face picture libraries.
   i. Create a face picture library: `POST /ISAPI/Intelligent/FDLib`. After the face picture library is created, the device will automatically generate the face picture library ID (that is, `FDID` is returned in the message `<FDLibInfoList>`). If you need to manage the face picture library ID by yourself, you can configure the custom face picture library ID (`customFaceLibID`) in the request message `<CreateFDLibList>` as needed. Afterwards, you can perform operations on the face picture library by the custom ID instead of `FDID`.
   ii. Edit parameters of the face picture library: `PUT /ISAPI/Intelligent/FDLib/<FDID>`. If you have added the custom face picture library ID in the above steps, you can also call `PUT /ISAPI/Intelligent/FDLib/<customFaceLibID>?FDType=custom` to configure parameters using `customFaceLibID`.
   iii. Get parameters of the face picture library: `GET /ISAPI/Intelligent/FDLib/<FDID>`. Operations involved with the custom face picture library ID are the same as those mentioned above.
   iv. Delete the face picture library: `DELETE /ISAPI/Intelligent/FDLib/<FDID>`. Operations involved with the custom face picture library ID are the same as those mentioned above.
3. Import data of the face picture library.

i. Import face picture data and automatically perform modeling, `POST /ISAPI/Intelligent/FDLib/pictureUpload`. By repeatedly applying the previous API, you can import the face picture data (including the face picture and additional face picture information `FaceAppendData`) to the corresponding face picture library; after it is successfully imported, the device will automatically model on the picture which could take some time, then the a face picture ID will be automatically generated (i.e., `PID` is returned in response parameters `MaskInfo`) by device after modeling succeeds; if you want to manually manage the face picture ID, you can add customized face picture ID (`customHumanID`) as needed when applying the request parameter `PictureUploadData`. `PID` can be replaced with customized face picture ID for subsequent face-picture-related operations.

ii. Import face picture library data in concurrent operation mode, `POST /ISAPI/Intelligent/FDLib/pictureUpload?type=concurrent`. If you want to improve the import efficiency, you can apply concurrent operation that the device will only store the picture not model on the picture. You will need to manually model on the picture (refer to Face Picture Library Data Modeling Management Flow for details). Note: Back-end DeepinMind devices will automatically perform modeling after data is imported even though under concurrent operation circumstances; if you demand to manually perform modeling, you can still apply modeling management.

4. (Optional) Manage face picture library data modeling. If you imported face picture library data in concurrent operation mode, you can select a time for device data modeling management after importing completed:
   i. Modeling on registered face pictures, `GET /ISAPI/Intelligent/FDLib/manualModeling`;
   ii. Search for modeling control results, `GET /ISAPI/Intelligent/FDLib/manualModelingRangeTask`;
   iii. Get modeling progress status, `GET /ISAPI/Intelligent/FDLib/manualModeling/progressState`.

5. (Optional) Manage data in the face picture library. You can search, edit, and delete data in the face picture library.
   i. Search for face picture data: `POST /ISAPI/Intelligent/FDLib/FDSearch`;
   ii. Edit face picture data: `PUT /ISAPI/Intelligent/FDLib/FDModify?format=json`, it is for editing data of imported pictures in binary format;
   iii. Edit face picture data in the library: `PUT /ISAPI/Intelligent/FDLib/FDSearch?format=json&FDID=<FDID>&FPID=<FPID>&faceLibType=<faceLibType>`, it is for editing data of imported pictures in URL format;
   iv. Delete face picture data in the library: `PUT /ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&FDID=<FDID>&faceLibType=<FDType>`.

12.3.1.3 Exception Handling

12.3.1.3.1 Error Code

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Rem
1	OK	ok	0x00000001			
2	Device Busy	noMemory	0x20000001		Not enough memory.	
4	Invalid Operation	notSupport	0x40000001		Not supported.	
4	Invalid Operation	faceLibDiskFull	0x400020F4		Face picture library storage media is full.	
4	Invalid Operation	faceLibraryIDNotExis	0x400020FD		The face picture library does not exist.	
					The face	

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4	Invalid Operation	faceLibInitializing	0x40002136		picture library is initializing. Please wait.	
4	Invalid Operation	notMatchFacelib	0x4000a004		Face picture library is not matched.	
4	Invalid Operation	existSameNameFacelib	0x4000a00a		Face picture library with duplicate name exists.	
4	Invalid Operation	delFacelibFailed	0x4000a00b		Deleting the face picture library failed.	
4	Invalid Operation	modifyFacelibFailed	0x4000a00c		Editing the face picture library failed.	
4	Invalid Operation	createFacelibFailed	0x4000a00d		Creating a face picture library failed.	
4	Invalid Operation	facelibAlreadyUsed	0x4000a010		The face picture library has been used.	
6	Invalid Content	faceLibraryDatabaseError	0x60000032		Database version of face picture library mismatch.	
6	Invalid Content	faceLibraryError	0x60000034		Face picture library error.	
6	Invalid Content	multipleFaceObjectError	0x60000035		Multiple faces are recognized in the pictures to be imported into the face picture library.	
6	Invalid Content	faceLibPicImportingError	0x6000003A		Importing picture into library...	
6	Invalid Content	faceLibraryNoData	0x60000069		No data found in face picture	

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					library.	
6	Invalid Content	NoDataForModelingError	0x6000006B		There is no data to be modeled in the face picture library.	
6	Invalid Content	customFaceLibIDRepeat	0x6000006C		Duplicate ID of custom face picture library.	
6	Invalid Content	customFaceDataDeleteing	0x600000A1		Deleting original customized face picture library data... The upper level cannot duplicate the customized face picture library of this ID.	
6	Invalid Content	analysisEnginesFacelibImporting	0x60001005		Importing face picture library to analysis engine...	
6	Invalid Content	facelibDataProcessing	0x6000100b		Processing face picture library data.	
6	Invalid Content	personInfoExtendValueIsTooLong	0x60001083		Customized tag for face picture library exceeded limit.	
6	Invalid Content	personInfoExtendValueContainsInvalidChar	0x60001084		Customized tag for face picture library contains invalid character(s).	

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12.3.1.4 Message Example

Import face pictures to face picture library in binary format and automatically perform modeling, `POST /ISAPI/Intelligent/FDLib/pictureUpload`:

```
POST http://10.16.96.29/ISAPI/Intelligent/FDLib/pictureUpload HTTP/1.1
If-Modified-Since: 0
Accept: */*
X-Requested-With: XMLHttpRequest
Content-Type: multipart/form-data; boundary=----------------------------7e27010d1dba
Referer: http://10.16.96.29/doc/page/config.asp
Accept-Language: zh-CN
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/5.0 (compatible; MSIE 10.0; Windows NT 6.1; WOW64; Trident/6.0)
Connection: Keep-Alive
Content-Length: 903330
DNT: 1
Host: 10.16.96.29
Pragma: no-cache
Cookie: language=zh; WebSession=48b693ca2b7301e3119d; sdMarkMenu=14%3AfaceSnapLib; sdMarkTab_1_0=0%3AsettingBasic

----------------------------7e27010d1dba
Content-Disposition: form-data; name="PictureUploadData"

<?xml version='1.0' encoding='UTF-8'?
><PictureUploadData><FDID>55869AFEA1E840958120C97054720372</FDID><FaceAppendData><name>20171222103007395_2</name></FaceAppendData></PictureUploadData>
----------------------------7e27010d1dba
Content-Disposition: form-data; name="importImage"; filename="20171222103007395_2.jpeg"
Content-Type: image/jpeg

【Picture data in binary format】
----------------------------7e27010d1dba--
```

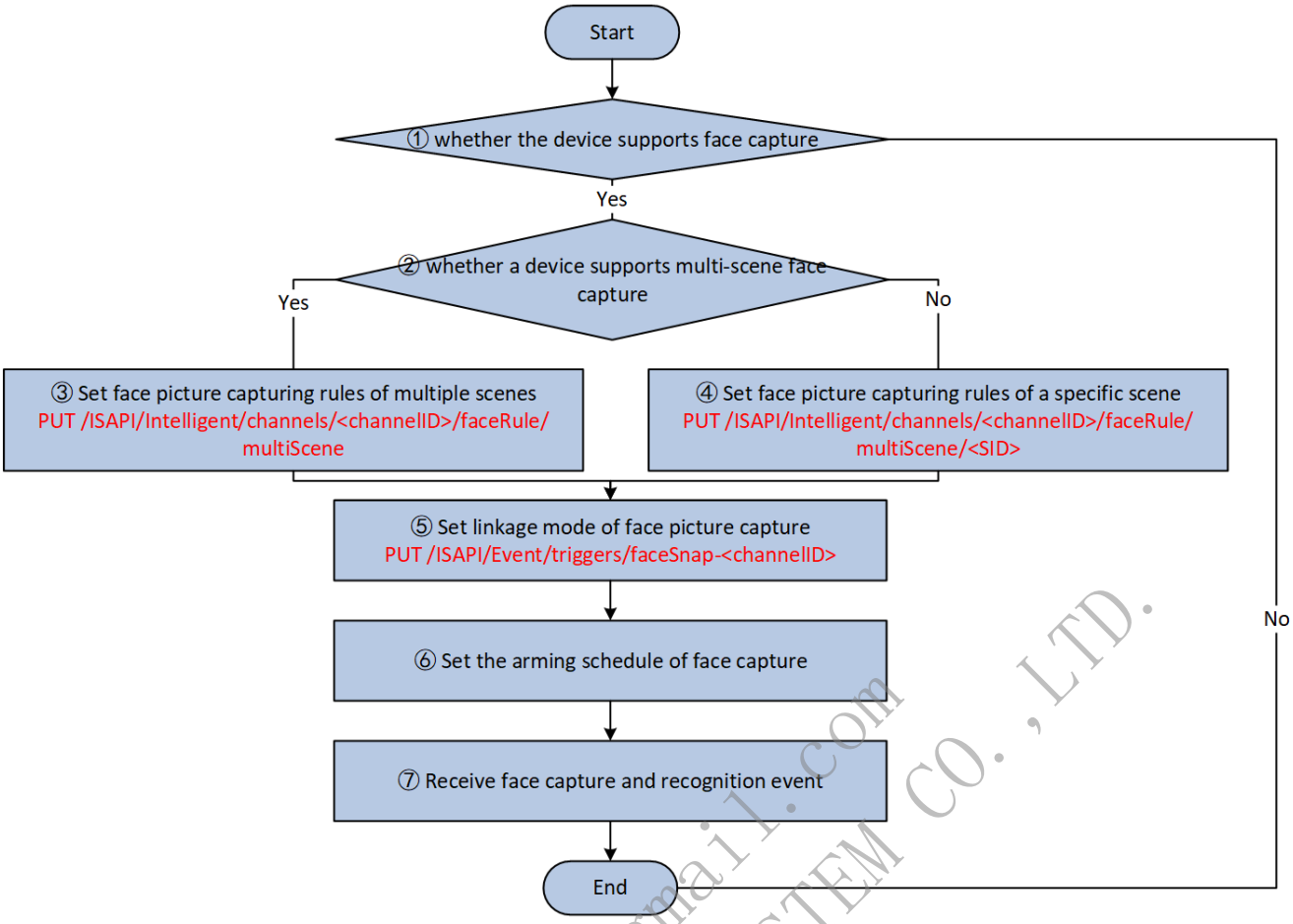
12.4 Privacy Mask on Faces

12.4.1 Introduction to the Function

Privacy mask on faces is to encrypt privacy contents for face capture and media data (see "Video Privacy Mask" API calling flow); it is the process of privacy masking for all faces of cameras and the encrypted images can be restored. First of all, all faces need to be identified, the original media data of faces in the images and masking block data will be replaced, the original media data of faces will be added to the private data of the live view stream and encrypted, then reported to the user synchronously; and the user can decrypt the private data to get the original media data and restore the face image. **Note: The current image restoration only supports private streams, which needs to be restored by integrating the Playback Library.** The following figure shows the effect of face privacy mask by mosaic processing.

12.4.2 API Calling Flow

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1. **Whether the video channel supports face privacy mask.** Get the channel intelligence capabilities: `GET /ISAPI/Intelligent/channels/<channelID>/capabilities`, if the node `<isSupportFacePrivacyMask>` is returned as true, it indicates that the device supports face privacy masking.

2. **Set privacy encryption parameters in media data.** Users apply the encryption parameters including secondary key `secretKey` and the update time of the primary key `secretKeyUpdateFrequency` to devices, etc. Get the encryption capabilities of media data privacy: `GET /ISAPI/Security/MediaObjectEncryption/channel/<channelID>/capabilities?format=json`. Set the encryption parameters of media data privacy `PUT /ISAPI/Security/MediaObjectEncryption/channel/<channelID>?format=json`.Get the encryption parameters of media data privacy: `GET /ISAPI/Security/MediaObjectEncryption/channel/<channelID>?format=json`.

3. **Set parameters of face privacy mask.** Users can enable face privacy mask for a channel `enabled`. Get the capability of face privacy masking: `GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json`;
Set the parameters of face privacy mask: `PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json`; Get the parameters of face privacy mask: `GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json`;

4. **(Optional) Decrypt media data and restore images**. The device generates the primary key and updates the primary key periodically according to the primary key update time in the media data privacy encryption parameters, then replaces the original media data with the masking block in the specified sensitive area in the live view screen, after which the images in the sensitive area are encrypted by the primary key and added to the private data of the live view stream, and at the same time the primary key is encrypted by the secondary key and added to the private data of the preview stream. The key is also encrypted by the secondary key and added to the private data of the live view stream. The user can parse the private data in the live view stream and decrypt it with the secondary key to get the primary key, then decrypt the private data with the primary key to get the original media data and restore the image.

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Note: At present, image restoration is only supported by private streams. You need to get the private data in the live view stream by integrating Playback Library, and get the primary key by decrypting the secondary key, then decrypt the private data by the primary key to get the original media data; and finally call the Playback Library API to restore the image. You can be refer to Playback library interface integration manual for playback library related APIs.

13 Traffic Control

13.1 Traffic Enforcement of Vehicles

13.1.1 Introduction to the Function

13.1.1.1 Smart Linkage System for Intelligent Transportation

Application Scenario: The smart linkage system of intelligent traffic consists of cameras and traffic speed domes, which support functions of remote monitoring via the network, video servers, and high definition, for traffic violation enforcement, traffic incident analysis, and traffic data statistics. Without people attending, the system can automatically capture and save key information for automatic enforcement, which helps save on labor costs, improve enforcement efficiency, and reduce disputes. The smart linkage system can be widely used in places that require large-scale high-definition monitoring, such as roads, airports, stations, docks, ports, tourist attractions, and streets.

Enforcement Function: The traffic enforcement event detection supports the function of violation enforcement, which can automatically detect and capture the evidence of events, including illegal parking, wrong-way driving, illegal U-turning, driving on the lane line, illegal lane change, and motor vehicle on non-motor vehicle lane, on the roads with no more than four lanes. The enforcement results will be uploaded as alarms in real time.

13.1.1.2 Smart Linkage System for Parking Management

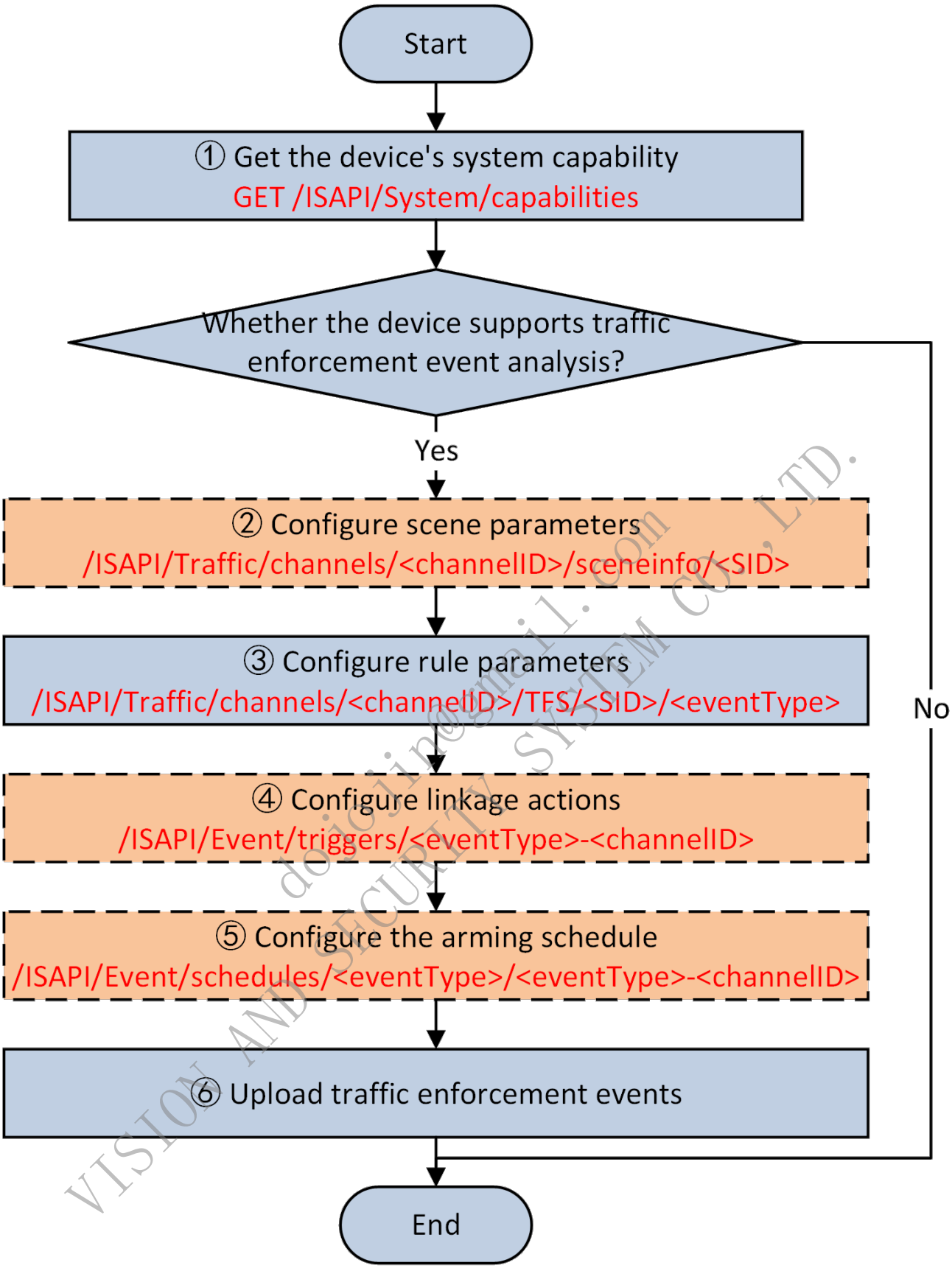
Application Scenario: The smart linkage system for parking management (hereinafter referred to as "the system") consists of parking speed domes and cameras (hereinafter referred to as "the devices"). The parking speed dome supports functions of remote monitoring via the network, video servers, and high definition. The system is easy to install and use without complex wiring. It focuses on the detection of the parallel parking event during enforcement. Without people attending, the system can automatically capture and save key information for automatic enforcement of the vehicle information in the parking space, which helps save on labor costs.

The devices are mainly used in places like the parking lot.

Parallel Parking Detection: The parallel parking detection can detect the parallel parking event of up to four parking spaces during enforcement and upload the vehicle information in the parking space in real time.

13.1.2 API Calling Flow

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1. **Get the device's system capability:**

   Get the device's system capability to check the traffic enforcement events supported by the device: `GET /ISAPI/System/capabilities`. The value of the node TFSEventSupport.opt can be illegalParking (illegal parking detection), wrongDirection (wrong-way driving detection), crossLane (driving on the lane line detection), laneChange (illegal lane change detection), vehicleExist (motor vehicle on non-motor vehicle lane detection), turnRound (illegal U-turning detection), parallelParking (parallel parking detection), notKeepDistance (not keeping vehicle distance detection), notSlowZebraCrossing (not slowing down at zebra crossing detection), overtakeRightSide (overtaking on the right detection), lowSpeed (driving in low speed detection), dragRacing

(street racing detection), changeLaneContinuously (continuous lane change detection), SSharpDriving (slalom driving detection), largeVehicleOccupyLine (lane occupation by large-sized vehicle detection), jamCrossLine (queue jumping detection), blackSmokeVehicle (vehicle emitting black smoke detection), or turnRightStop (large-sized vehicle not slowing down when turning right detection).

2. **Configure scenes:**

   Get the capability of configuring basic channel parameters to check whether the node `isEvidenceGettingSupport` exists and its value is true: `GET /ISAPI/Traffic/channels/<channelID>/capability`. If so, it indicates that the device supports configuring scene and rule parameters.

   Get the capability of configuring basic scene parameters of a specific channel: `GET /ISAPI/Traffic/channels/<channelID>/sceneinfo/capabilities`.

   Set the basic scene parameters for a specific channel: `PUT /ISAPI/Traffic/channels/<channelID>/sceneinfo/<SID>`. If you need to use the scene, the node SceneInfo.enabled should be set to true.

3. **Configure rules:**

   Get the capability of configuring the **event** rule parameters: `GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/XXX/capabilities`.

   Get the **event** rule parameters: `GET/ISAPI/Traffic/channels/<channelID>/TFS/<SID>/XXX`.

   Set the **event** rule parameters: `PUT/ISAPI/Traffic/channels/<channelID>/TFS/<SID>/XXX`.

   Notes:

   i. The value of `XXX` can be set to: illegalParking (illegal parking detection), wrongDirection (wrong-way driving detection), crossLane (driving on the lane line detection), laneChange (illegal lane change detection), vehicleExist (motor vehicle on non-motor vehicle lane detection), turnRound (illegal U-turning detection), parallelParking (parallel parking detection), notKeepDistance (not keeping vehicle distance detection), notSlowZebraCrossing (not slowing down at zebra crossing detection), overtakeRightSide (overtaking on the right detection), lowSpeed (driving in low speed detection), dragRacing (street racing detection), changeLaneContinuously (continuous lane change detection), SSharpDriving (slalom driving detection), largeVehicleOccupyLine (lane occupation by large-sized vehicle detection), jamCrossLine (queue jumping detection), blackSmokeVehicle (vehicle emitting black smoke detection), or turnRightStop (large-sized vehicle not slowing down when turning right detection).

   e.g.: Get the capability of configuring parameters of illegal parking detection: `GET /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/illegalParking/capabilities`.

   Get or set parameters of illegal parking detection: `GET/PUT /ISAPI/Traffic/channels/<channelID>/TFS/<SID>/illegalParking`.

   ii. For traffic speed domes, the rule parameters are required. For box cameras, the rule parameters are not required. The scene No. is 1 by default.

4. **Configure linkage actions:**

   Check whether the device supports configuring linkage actions of the illegal parking event by calling `GET /ISAPI/Event/triggersCap` and check whether the message `EventTriggersCap` contains the node `IllegalParkingTriggerCap`.

   Get the default or configured linkage action of the illegal parking event: `GET /ISAPI/Event/triggers/illegalParking-<channelID>`.

   Configure the linkage actions of the illegal parking event: `PUT /ISAPI/Event/triggers/illegalParking-<channelID>`.

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Notes: Only illegal parking detection supports linkage action configurations. You do not need to configure linkage actions for other events and they will be uploaded to the center by default.

5. **Configure arming schedules:**

   Check whether the device supports configuring the arming schedule for the illegal parking detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportillegalParking`.

   Configure the arming schedule for the illegal parking event: `GET/PUT /ISAPI/Event/schedules/illegalParkings/illegalParking-<channelID>`.

   Check whether the device supports configuring the arming schedule for the wrong-way driving detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportWrongDirection`.

   Configure the arming schedule for the wrong-way driving event: `GET/PUT /ISAPI/Event/schedules/wrongDirections/wrongDirection-<channelID>`.

   Check whether the device supports configuring the arming schedule for the illegal U-turning event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportTurnRound`.

   Configure the arming schedule for the illegal U-turning event: `GET/PUT /ISAPI/Event/schedules/turnRounds/turnRound-<channelID>`.

   Check whether the device supports configuring the arming schedule for the driving on the lane line event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportCrossLane`.

   Configure the arming schedule for the driving on the lane line event: `GET/PUT /ISAPI/Event/schedules/crossLanes/crossLane-<channelID>`.

   Check whether the device supports configuring the arming schedule for the illegal lane change detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportLaneChange`.

   Configure the arming schedule for the illegal lane change event: `GET/PUT /ISAPI/Event/schedules/laneChanges/laneChange-<channelID>`.

   Check whether the device supports configuring the arming schedule for the motor vehicle on non-motor vehicle lane event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportVehicleExist`.

   Configure the arming schedule for the motor vehicle on non-motor vehicle lane event: `GET/PUT /ISAPI/Event/schedules/vehicleExists/vehicleExist-<channelID>`.

   Check whether the device supports configuring the arming schedule for the parallel parking detection event by calling `GET /ISAPI/Event/capabilities` and checking whether the message `EventCap` contains the node `isSupportParallelParking`.

   Configure the arming schedule for the parallel parking event: `GET/PUT /ISAPI/Event/schedules/parallelParkings/parallelParking-<channelID>`.

   Notes:

   i. You can determine whether to configure the arming schedule or not. If the arming schedule is not configured, it is all-day arming by default.

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ii. Only the illegal parking detection, wrong-way driving detection, illegal U-turning detection, driving on the lane line detection, illegal lane change detection, motor vehicle on non-motor vehicle lane detection, and parallel parking detection support arming schedule configuration. For other events, configuring the arming schedule is not supported and the default arming schedule is all-day arming.

6. **Upload the traffic enforcement events** (`"eventType":"TFS"`). In the event message, TFS.illegalTrafficEvent is the specific traffic violation event, TFS.VehicleInfo is the vehicle information of the traffic violation, and TFS.PlateInfo is the license plate information of the vehicle violation. The detectionPicturesNumber refers to the number of detected violation pictures. The detectionPictureTransType refers to the transmission method of the pictures.

13.1.3 Exception Handling

13.1.3.1 Error Codes

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Rema
6	Invalid Content	intelligentTrafficMutexWithHighFrames	0x60008014		Disable all functions of traffic incident detection, traffic violation enforcement, and traffic data statistics, or adjust the video frame rate to 50 or below.	
6	Invalid Content	intelligentTrafficMutexWithHighFramesEx	0x60008018		Disable all functions of traffic incident detection, traffic violation enforcement, traffic data statistics, and vehicle detection, or adjust the video frame rate to 50 or below.	

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14 Crowd Trend Detection

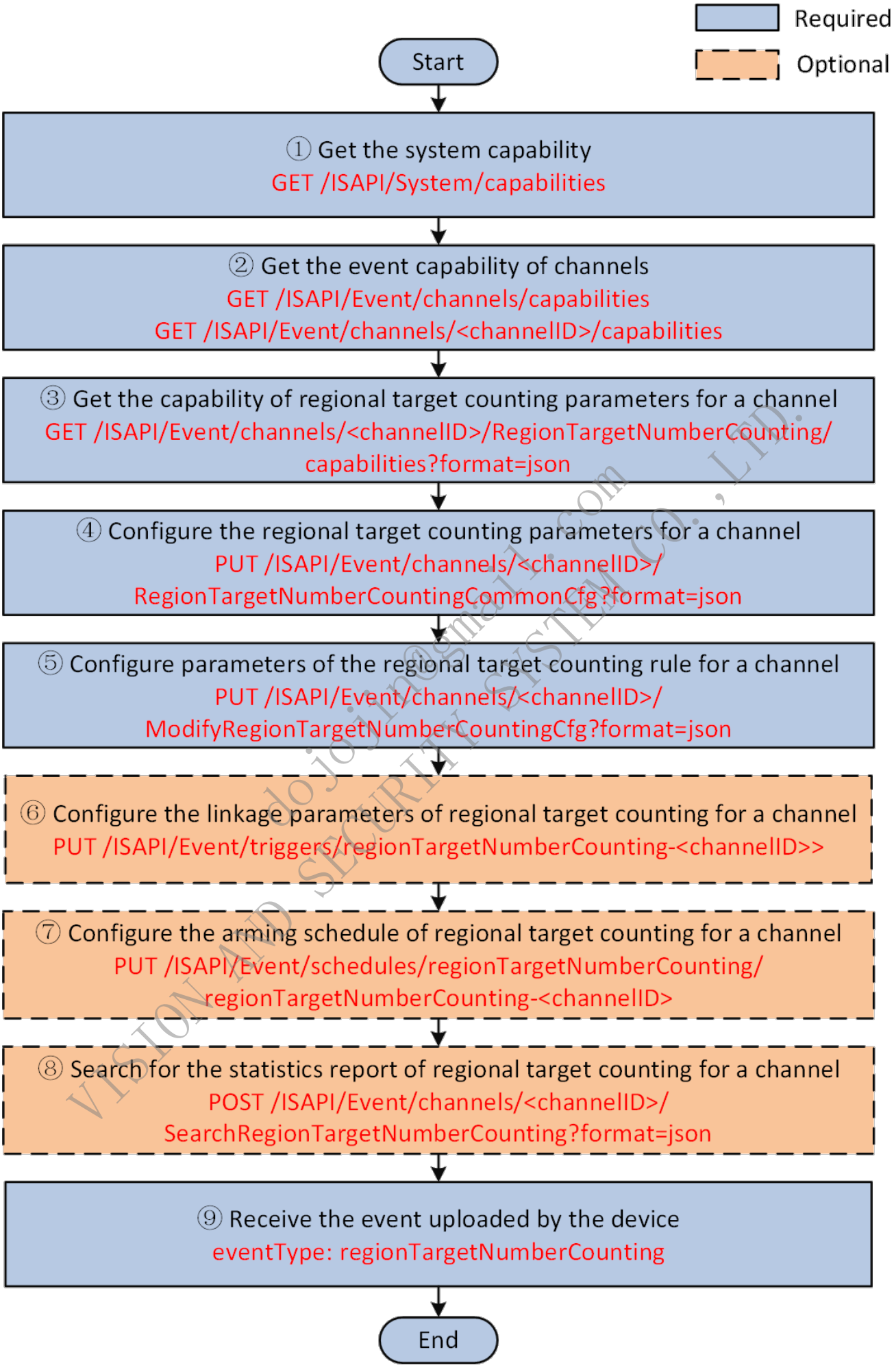
14.1 Regional Target Counting

14.1.1 Introduction to the Function

In the important area of monitoring, you can draw one or multiple detection lines, once targets (such as human, vehicles, and non-motor vehicles) cross the detection line, device will count the number of crossed targets according to the target type and upload the counting statistics to the client software in real time or regularly According to the statistics data, you can perform business deploy and analysis. For example, there is a billboard on the road, to analyze the advertising effect, you can draw detection lines around the billboard to get the number of crossed people, vehicles, and non-motor vehicles.

14.1.2 API Calling Flow

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Required
Optional
Start
① Get the system capability
GET /ISAPI/System/capabilities
② Get the event capability of channels
GET /ISAPI/Event/channels/capabilities
GET /ISAPI/Event/channels/<channelID>/capabilities
③ Get the capability of regional target counting parameters for a channel
GET /ISAPI/Event/channels/<channelID>/RegionTargetNumberCounting/capabilities?format=json
④ Configure the regional target counting parameters for a channel
PUT /ISAPI/Event/channels/<channelID>/RegionTargetNumberCountingCommonCfg?format=json
⑤ Configure parameters of the regional target counting rule for a channel
PUT /ISAPI/Event/channels/<channelID>/ModifyRegionTargetNumberCountingCfg?format=json
⑥ Configure the linkage parameters of regional target counting for a channel
PUT /ISAPI/Event/triggers/regionTargetNumberCounting-<channelID>>
⑦ Configure the arming schedule of regional target counting for a channel
PUT /ISAPI/Event/schedules/regionTargetNumberCounting/regionTargetNumberCounting-<channelID>
⑧ Search for the statistics report of regional target counting for a channel
POST /ISAPI/Event/channels/<channelID>/SearchRegionTargetNumberCounting?format=json
⑨ Receive the event uploaded by the device
eventType: regionTargetNumberCounting
End

1. Get the system capability: `GET /ISAPI/System/capabilities`. If `isSupportRegionTargetNumberCounting` is returned and its value is `true`, it indicates that the device supports regional target counting.
2. Get the event capability of channels. If the returned value of `eventType` contains `regionTargetNumberCounting`, it indicates that the channel supports regional target counting. Get the event capability of all channels: `GET /ISAPI/Event/channels/capabilities`. Get the event capability of a specified channel: `GET /ISAPI/Event/channels/<channelID>/capabilities`.
3. Get the capability of regional target counting parameters for a specified channel: `GET /ISAPI/Event/channels/<channelID>/RegionTargetNumberCounting/capabilities?format=json`.
4. Configure the regional target counting parameters (including enabling regional target counting, scheduled uploading, real-time uploading, and so on) for a specified channel. Get: `GET /ISAPI/Event/channels/<channelID>/RegionTargetNumberCountingCommonCfg?format=json`. Set: `PUT /ISAPI/Event/channels/<channelID>/RegionTargetNumberCountingCommonCfg?format=json`.
5. Configure parameters of the regional target counting rule (including detection line, detection line direction, statistics target, and so on) for a specified channel. Get a single rule: `GET /ISAPI/Event/channels/<channelID>/GetRegionTargetNumberCountingCfg?format=json`. Set a single rule: `PUT /ISAPI/Event/channels/<channelID>/ModifyRegionTargetNumberCountingCfg?format=json`. Get all rules: `GET /ISAPI/Event/channels/<channelID>/GetRegionTargetNumberCountingCfgList?format=json`.
6. (Optional) Configure the linkage parameters of regional target counting for a specified channel. Get: `GET /ISAPI/Event/triggers/regionTargetNumberCounting-<channelID>`. Set: `PUT /ISAPI/Event/triggers/regionTargetNumberCounting-<channelID>`.
7. (Optional) Configure the arming schedule of regional target counting for a specified channel. Get: `GET /ISAPI/Event/schedules/regionTargetNumberCounting/regionTargetNumberCounting-<channelID>`. Set: `PUT /ISAPI/Event/schedules/regionTargetNumberCounting/regionTargetNumberCounting-<channelID>`.
8. (Optional) Search for the statistics report of regional target counting for a specified channel. Get the capability: `GET /ISAPI/Event/channels/<channelID>/SearchRegionTargetNumberCounting/capabilities?format=json`. Search for the statistics report: `POST /ISAPI/Event/channels/<channelID>/SearchRegionTargetNumberCounting?format=json`.
9. Receive the event (`eventType:regionTargetNumberCounting`) uploaded by devices in listening or arming mode.

14.1.3 Exception Handling

Error Code

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Remarks
4	Invalid Operation	unopenMixedTargetDetectionError	0x4000216A	Operation failed, please start mixed target detection first.	Operation failed, please start mixed target detection first.	

15 Video Recognition

15.1 Professional Smart Abnormal Event Detection

15.1.1 Introduction to the Function

Professional Smart Abnormal Event Detection includes the following features:

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- Line Crossing Detection
- Intrusion Detection
- Region Entrance Detection
- Region Exiting Detection
- Loitering Detection
- People Gathering Detection
- Rapid Movement Detection
- Parking Detection
- Unattended Baggage Detection
- Object Removal Detection
- Running Detection
- Abnormal Number of People Detection
- Person Spacing Change Detection
- Violent Motion Detection
- On/Off Duty Detection
- Falling Down Detection
- Lingering Detection
- Person Density Detection
- Climbing Detection
- Trend Analysis
- In-Toilet Overtime Detection
- Sudden Change of Sound Intensity Detection
- Student Standing Up Detection
- Getting Up from Bed Detection

Professional smart abnormal event detection is designed for specific scenarios: workplace activity detection and pedestrian activity detection.

Workplace activity detection includes but not limited to:

- Using a Mobile Phone
- Number of People Changes
- On/Off Duty
- Person Lingering

Pedestrian activity detection includes but not limited to:

- Running Detection
- People Gathering Detection
- Violent Motion Detection
- Person Falling Detection

You can configure workplace activity detection and pedestrian activity detection rules and alarm linkage actions to suit your needs.

Abnormal event detection can be categorized into two types: Basic smart abnormal event detection and professional smart abnormal event detection. Their differences are as follows:

1. **Algorithms**: Traditional algorithms are used for basic smart abnormal event detection, while deep learning algorithms are used for professional smart abnormal event detection.
2. **Detection Effect**: Basic smart abnormal event detection can only analyze targets as people or vehicles. Professional smart abnormal event detection can not only distinguish between targets as people or vehicles but also identify specific behaviors, such as the number of people and human behaviors such as using a mobile phone, violent motion, and running.

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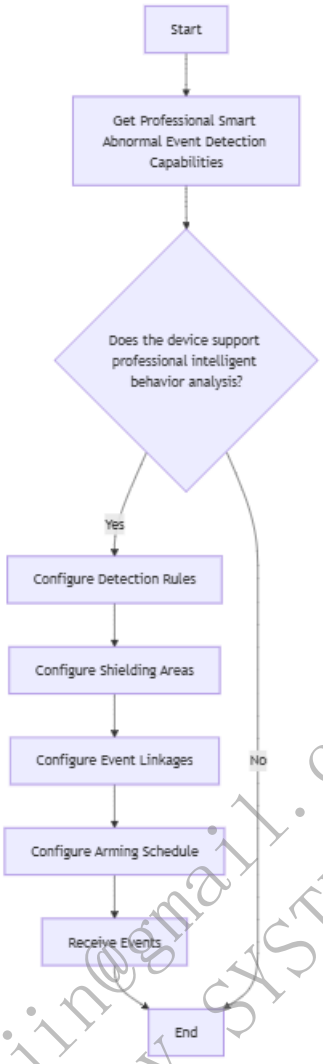
3. **Supported Events**: Smart Basic Intelligent Abnormal Behavior Detection supports events such as:
    - Line Crossing Detection
    - Intrusion Detection
    - Region Entrance Detection
    - Region Exiting Detection
    - Loitering Detection
    - People Gathering Detection
    - Rapid Movement Detection
    - Parking Detection
    - Unattended Baggage Detection
    - Object Removal Detection

Professional Smart Abnormal Event Detection supports all the events mentioned above and additional events including:

- Running Detection
- Abnormal Number of People Detection
- Person Spacing Change Detection
- Violent Motion Detection
- On/Off Duty Detection
- Falling Down Detection
- Lingering Detection
- Person Density Detection
- Climbing Detection
- Trend Analysis
- In-Toilet Overtime Detection
- Sudden Change of Sound Intensity Detection
- Student Standing Up Detection
- Getting Up from Bed Detection

15.1.2 API Calling Flow

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Before taking the following steps, ensure that this smart detection has been enabled for devices.

1. **Get Professional Smart Abnormal Event Detection Capabilities**
   - Call `GET /ISAPI/Intelligent/capabilities` to get professional smart abnormal event detetection capabilities and the field `isBehaviorSupport` is returned as `true` in the response message.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/capabilities` to get professional smart abnormal event detetection capabilities for a specified channel and the field `isBehaviorSupport` is returned as `true` in the response message.

   Note: `isBehaviorSupport`as `true` indicates support for professional smart abnormal event detetection. The fields of supported events are as follows:
   - `isLineDetectionSupport` indicates support for **Line Crossing Detection**.
   - `isFieldDetectionSupport` indicates support for **Intrusion Detection**.
   - `isRegionEntranceSupport` indicates support for **Region Entrance Detection**.
   - `isRegionExitingSupport` indicates support for **Region Exiting Detection**.
   - `isLoiteringSupport` indicates support for **Loitering Detection**.
   - `isGroupSupport` indicates support for **People Gathering Detection**.
   - `isRapidMoveSupport` indicates support for **Rapid Movement Detection**.
   - `isParkingSupport` indicates support for **Parking Detection**.
   - `isUnattendedBaggageSupport` indicates support for **Unattended Baggage Detection**.
   - `isAttendedBaggageSupport` indicates support for **Object Removal Detection**.
   - `isRunningSupport` indicates support for **Running Detection**.
   - `isPeopleNumChange` indicates support for **Abnormal Number of People Detection**.
   - `isSpacingChangeSupport` indicates support for **Person Spacing Change Detection**.

- `isViolentMotionSupport` indicates support for **Violent Movement Detection**.
- `isLeavePositionSupport` indicates support for **On/Off Duty Detection**.
- `isFailDownSupport` indicates support for ** Falling Down Detection**.
- `isRetentionParamSupport` indicates support for **Lingering Detection**.
- `isSupportPersonDensityDetection` indicates support for **Person Density Detection**.
- `isVcaEventAdvReachHeightSupport` indicates support for **Climbing Detection**.
- `isSupportSituationAnalysis` indicates support for **Trend Analysis**.
- `isSupportToiletTarry` indicates support for **In-Toilet Overtime Detection**.
- `isSupportAudioAbnormal` indicates support for **Sudden Change of Sound Intensity Detection**.
- `isSupportStandUp` indicates support for **Student Standing Up Detection**.
- `isSupportGetUp` indicates support for **Getting Up from Bed Detection**.

2. **Configure Detection Rules**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/capabilities` to get professional smart abnormal event detection capabilities.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to get detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to configure detection rules.
   - Call `DELETE /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to delete detection rules.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>` to get all detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>` to configure all detection rules.

   Note:
   - The `eventType` field in the rules indicates the corresponding event type.
   - The `ruleType` field indicates the event support drawing a detection line or a polygon detection area. Some events (sudden change of sound intensity) do not support drawing a box or line, some events (line crossing, climbing) support drawing a line, and some events (except sudden change of sound intensity, line crossing, and climbing) support for polygon detection areas.

3. **(Optional) Configure Shielding Areas**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/Shield/capabilities` to get professional smart area shielding capabilities.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/Shield` to get shielding areas for all events.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/Shield` to configure shielding areas for all events.

4. **(Optional) Configure Event Linkages**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/notifications` to get event linkages.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/notifications` to configure event linkages.

5. **(Optional) Configure Arming Schedule**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/schedules` to get the schedule to disable the professional smart abnormal event detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/schedules` to configure the schedule to disable the professional smart abnormal event detection rules.

   Note:
   - Up to eight time periods can be configured per day.

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- Each two time periods cannot overlap.

6. **Receive Events**
   - Line Crossing Detection: `"eventType":"linedetection"`
   - Intrusion Detection: `"eventType":"fielddetection"`
   - Region Entrance Detection: `"eventType":"regionEntrance"`
   - Region Exiting Detection: `"eventType":"regionExiting"`
   - Loitering Detection: `"eventType":"loitering"`
   - People Gathering Detection: `"eventType":"group"`
   - Rapid Movement Detection: `"eventType":"rapidMove"`
   - Parking Detection: `"eventType":"parking"`
   - Unattended Baggage Detection: `"eventType":"unattendedBaggage"`
   - Object Removal Detection: `"eventType":"attendedBaggage"`
   - Running Detection: `"eventType":"running"`
   - Abnormal Number of People Detection: `"eventType":"peopleNumChange"`
   - Person Spacing Change Detection: `"eventType":"spacingChange"`
   - Violent Movement Detection: `"eventType":"violentMotion"`
   - On/Off Duty Detection: `"eventType":"leavePosition"`
   - Falling Down Detection: `"eventType":"failDown"`
   - Lingering Detection: `"eventType":"retention"`
   - Person Density Event: `"eventType":"personDensityDetection"`
   - Climbing Detection: `"eventType":"advReachHeight"`
   - Trend Analysis: `"eventType":"situationAnalysis"`
   - In-Toilet Overtime Detection: `"eventType":"toiletTarry"`
   - Sudden Change of Sound Intensity Detection: `"eventType":"audioAbnormal"`
   - Student Standing Up Detection: `"eventType":"standUp"`
   - Getting Up from Bed Detection: `"eventType":"getUp"`
   - Sitting Still Detection: `"eventType":"sitQuietly"`
   - Key Target Standing Up Detection: `"eventType":"keyPersonGetUp"`
   - Using Mobile Phone Detection: `"eventType":"playCellphone"`
   - Yard Lingering: `"eventType":"yardTarry"`
   - Key Target Standing Up Detection: `"eventType":"keyPersonGetUp"`
   - Sitting Still Detection: `"eventType":"sitQuietly"`
   - Yard Lingering: `"eventType":"yardTarry"`
   - Using Mobile Phone Detection: `"eventType":"playCellphone"`

Note:

- The content of `targetAttrs` is in JSON format. When the event format is XML, use CDATA (Character Data) to include this data in the event and transmit it to the platform.
- CDATA (Character Data) is used to include large amounts of text data in XML documents. This text may contain special characters (such as `<`, `>`, `&`), which have special meanings in XML and typically need to be escaped. CDATA blocks allow these special characters to be included directly in the document without escaping, such as including JavaScript scripts or CSS styles in XML documents.

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15.2 Upgrading the Deep Learning Model for Perimeter Intrusion Detection

15.2.1 Introduction to the Function

You can upgrade the deep learning model for perimeter intrusion detection. You can also upgrade both the local training model for NVRs and the training model for network camera channels.

15.2.2 API Calling Flow

1. Determine if the device supports upgrading the deep learning model for perimeter intrusion detection: Call `GET /ISAPI/Intelligent/capabilities`. If the field `isSupportUpgradeDepthPerimeterModel` is returned as `true` in the response message, this function is supported.
2. Get the deep learning model upgrade capabilities: `GET /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/capabilities?format=json`
3. Set the model version: `PUT /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/VersionOperation?format=json`
    - You cannot specify a version to upgrade. Only rolling back to the previous version and restoring to the factory model version are currently supported.
4. Get the model upgrade progress: `GET /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/process?format=json`
5. Set the model version for a channel: `PUT /ISAPI/Intelligent/depthPerimeter/channels/<channelID>/upgradeTrainModel/VersionOperation?format=json`
6. Get the model upgrade progress for a channel: `GET /ISAPI/Intelligent/depthPerimeter/channels/<channelID>/upgradeTrainModel/process?format=json`

16 AIOP

16.1 AI Radar Rule Configuration

16.1.1 Introduction to the Function

Front-end devices rely on video images for AIOP task analysis, with relatively low accuracy, requiring the device's MCU to analyze radar data for secondary verification, when the camera detects the data of the radar that meets the configuration rules, it will trigger AI-related alarms; for example: "AIOP_Polling_Snap" (capture polling task detection event".

16.1.2 API Calling Flow

Get radar rule configuration:

Get: `GET /ISAPI/Intelligent/AIOpenPlatform/AIRadarRuleConfig?format=json`; Set: `PUT /ISAPI/Intelligent/AIOpenPlatform/AIRadarRuleConfig?format=json`; Capability: `GET /ISAPI/Intelligent/AIOpenPlatform/AIRadarRuleConfig/capabilities?format=json`.

16.2 AIOP Algorithm Model Management

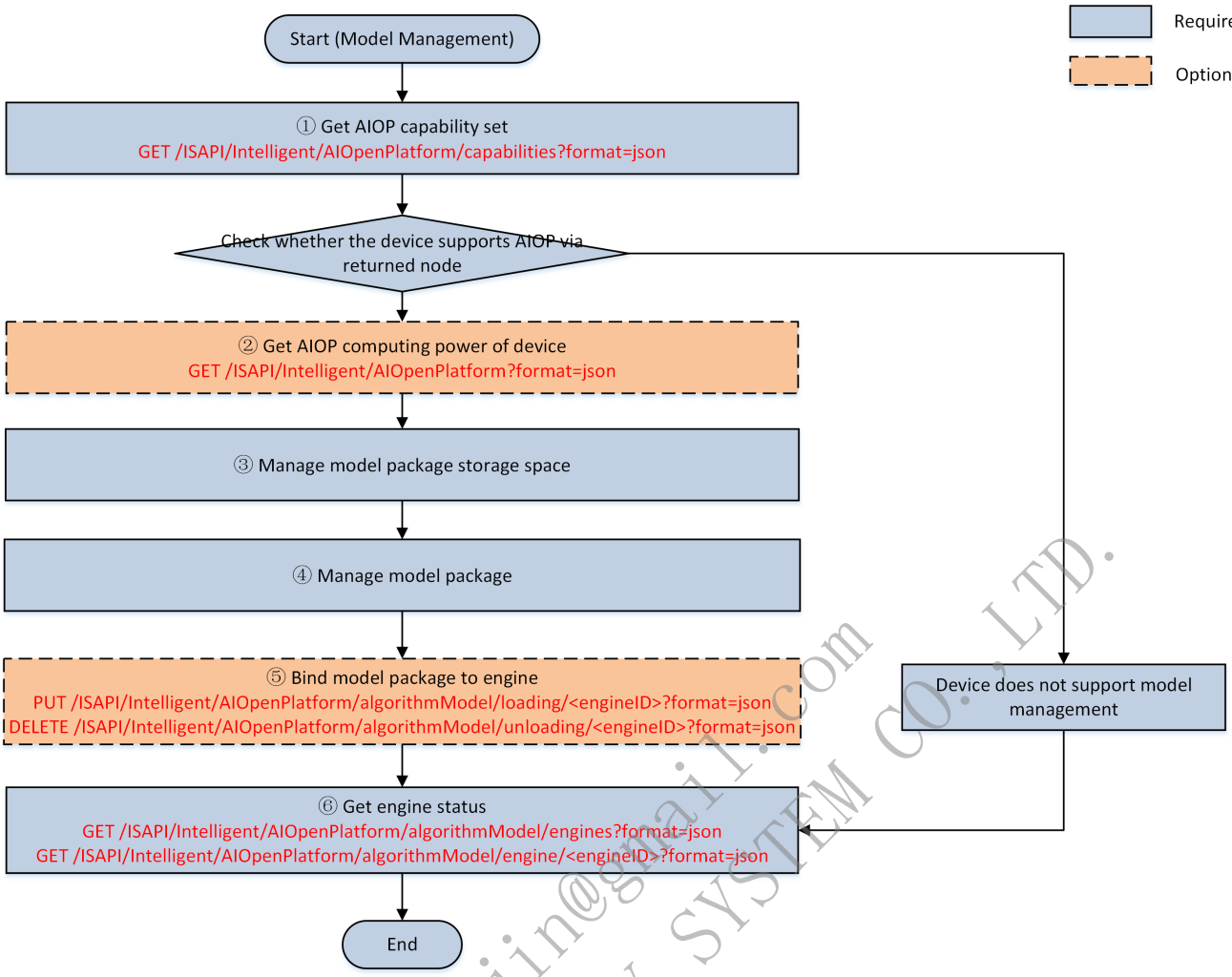
16.2.1 Model Management

16.2.1.1 Introduction to the Function

Model package is the basis for algorithm operations. Model management includes storage space management of model package, model package adding/deleting/editing/searching, binding/unbinding model package to the engine, and getting the engine status.

16.2.1.2 API Calling Flow

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1. Get the AIOP functions supported by device:

   Get the capability: `GET /ISAPI/System/capabilities`. If `isSupportAIOpenPlatform` is returned, it indicates that `AIOP` is supported.

   Get the `AIOP` capability: `GET /ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json`. If `200` is returned in the response message, it indicates that model management is supported.

2. Get the AIOP computing power supported by device: `GET /ISAPI/Intelligent/AIOpenPlatform?format=json`. Get the engine No. supported by the device.

3. Manage storage space of model package:

   - Search for storage space of device management model package: `GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/management/space?format=json`, to check whether the storage space is enough to save the model package.
   - Delete model package: `DELETE /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/management/space/<MPID>?format=json`.
   - Search for the model package imported to the device: `GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/management?format=json`. After loading and binding the model, you can get the model package containing an `array` object of `description`. You can parse the detection data according to the description.

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4. Model package management. Model output includes model package binary files, algorithm description files (applied in algorithm dictionary description and function definition), `License` files (required for model package needing authorization):
    - Preset model package (downloaded via `URL`): `POST /ISAPI/Intelligent/AIOpenPlatform/algorithmModel?format=json`. You should download the `URL` with authentication from the corresponding address. The obtained model package information contains an `array` object of `description`. You can parse the detection data according to the description.
    - Preset model package (pushed in `HTTP` form format): `POST /ISAPI/Intelligent/AIOpenPlatform/algorithmModelBinaryPackets?format=json`.
    - Search for upload progress of model package: `GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/progress/<MPID>?format=json`.
    - Restarting chip is generally for model update. For details, refer to the following model update flow.
    - (Optional) When presetting the model package, you can specify the engine for binding.
5. Bind the model package to an engine (the model package will start the analysis task only after it is bound to an engine). If the user has specified the engine when presetting the model package, this step can be skipped:
    - Bind: `PUT /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/loading/<engineID>?format=json`.
    - Unbind: `DELETE /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/unloading/<engineID>?format=json`.
6. Get the engine status:
    - Get status of all engines: `GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engines?format=json`.
    - Get status of specified engine: `GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engine/<engineID>?format=json`.

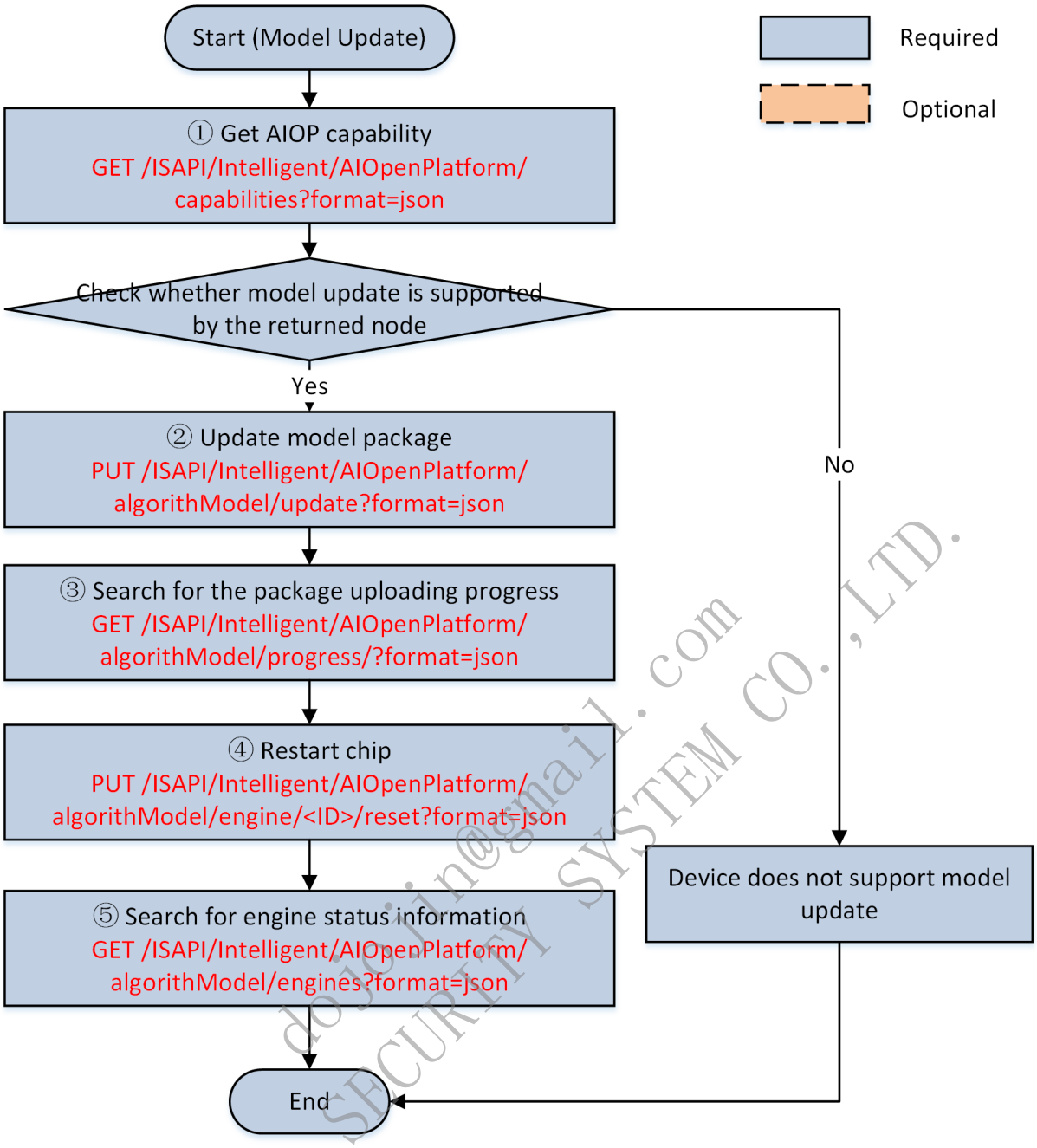
16.2.2 Model Update

16.2.2.1 Introduction to the Function

Supports updating the specified model package if no error occurs in the updated model package of the device. If the updated model package is not the same with the preset model package, the updated model package cannot be bound to the specified engine. Binding/unbinding is only allowed in the Engine Management API.

16.2.2.2 API Calling Flow

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1. Get the model updating capability supported by device: `GET /ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json`. By default, model update is supported when AIOP function is supported by device.
2. Model update: `PUT /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/update?format=json`. It supports `URL` and binary upgrade methods.
3. Get model package uploading progress: `GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/progress/<MPID>?format=json`.
4. Restart chip: `PUT /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engine/<ID>/reset?format=json`.
5. Get the engine status:
   - Get status of all engines: `GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engines?format=json`.
   - Get status of specified engine: `GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engine/<engineID>?format=json`.

16.2.3 Model Deployment

According to the deployment method of the algorithm model and the corresponding deployment area of cloud-edge-endpoint (CEE), the solution is divided into three deployment methods: edge device deployment, central deployment, and cloud deployment.

Edge device deployment: Deploy the algorithm model on the AI Open Platform device through the industry platform, and realize real-time response analysis through edge computing.

Central deployment: Algorithms and computing power are deployed in the center. The central unified intelligent analysis mode, through the algorithm scheduling platform, distributes intelligent algorithms from the algorithm library to the central computing resources (GPU servers) for recognition and analysis. Based on the fusion of scene algorithms, intelligent fusion of algorithms and computing power is realized.

Cloud deployment: Algorithm models are deployed on the cloud platform. Front-end devices upload the captured images to the cloud platform, where the cloud platform calls the algorithm interface for recognition and analysis, and presents the results. It also supports the third party to call algorithms through interfaces, and the cloud platform returns the recognition results to the third party.

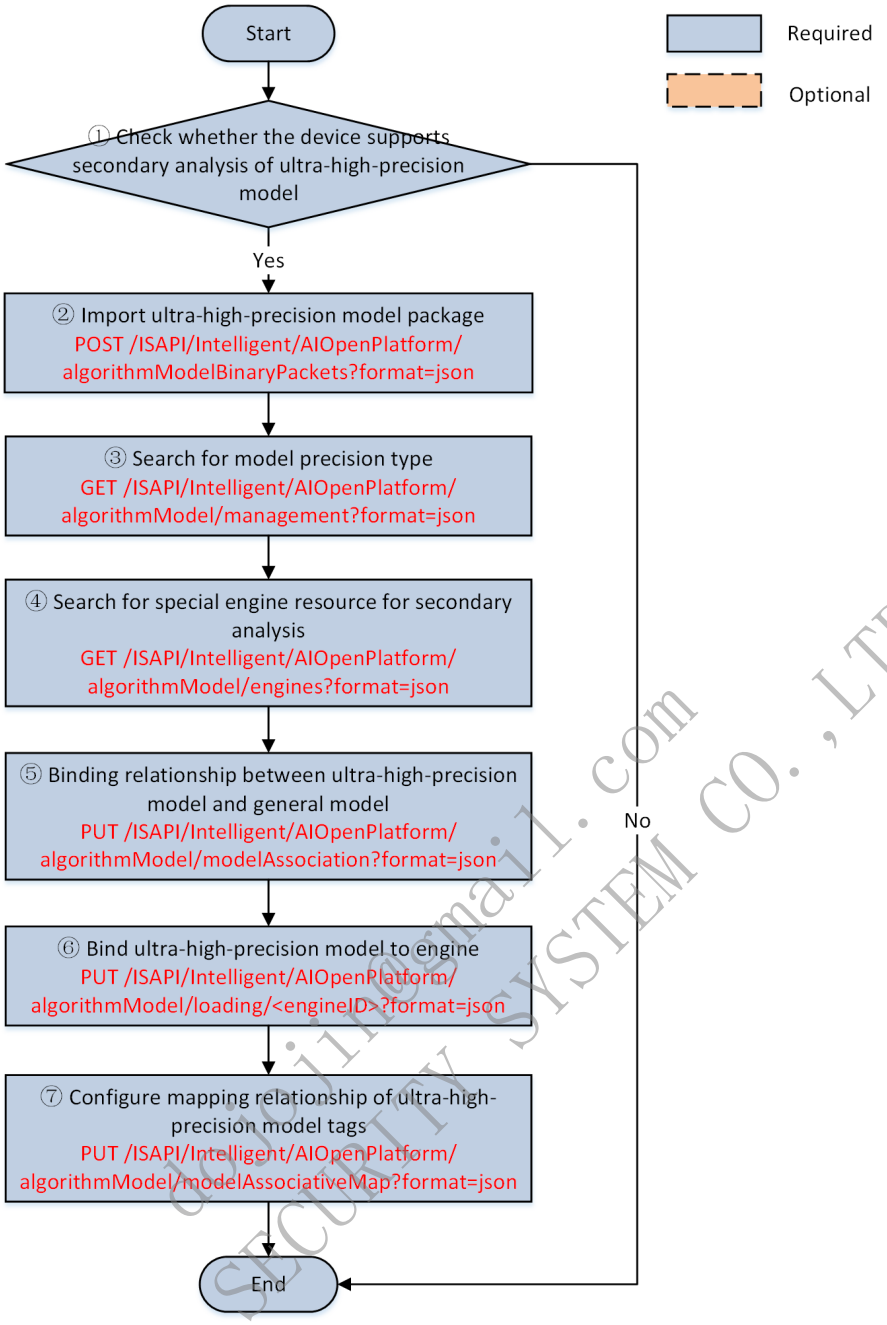
16.2.4 Secondary Analysis of Ultra-High-Precision Model

16.2.4.1 Introduction to the Function

Ultra-high-precision model: Generally refers to models with high detection accuracy, but they require a longer training time and larger model package size. The ultra-high-precision model is selected by the user during the model training on the training platform. The ultra-high-precision model supports the management mode of the ordinary model, importing, searching and updating models, binding an engine for independent detection and analysis, etc. Additionally, it supports linking with ordinary models and performing secondary analysis and verification of the analysis results generated by the ordinary models, improving the alarm accuracy.

16.2.4.2 API Calling Flow

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1. Get model type: `GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/management?format=json`. The `modelPrecision` represents the model precision: 0 (normal model), 2 (ultra-high-precision model).

2. Get secondary analysis engine resources: `GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engines?format=json`. The specialEngineList represents the information of special engines for secondary analysis, which are managed separately from normal engines.

- The engine in LoadingInfoList (engine status information list achieved by AI Super Brain is virtual engine ID and its parameter values is in the format of 2001, where 2 represents the physical engine No., and 1 represents the virtual engine No.; in specialEngineList, the engine represents the physical engine ID and its parameter value is in the format of 1.
- Ultra-high-precision models, due to the large model package and the high performance requirements for detection, only support binding to physical engines and do not support binding to virtual engines.

3. Bind ultra-high-precision model to normal model: `PUT /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/modelAssociation?format=json`.

- For a normal model to support binding to a super high-precision model for secondary analysis, the following conditions must be met The normal model is a detection, hybrid, or dual detection model; the IED rule is regional target detection, regional target counting, or cross-line target detection.

4. Bind ultra-high-precision model to physical engine: `PUT /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/loading/<engineID>?format=json`.

5. Configure ultra-high-precision model tag mapping relationship: `PUT /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/modelAssociativeMap?format=json`, where information such as detection model category, target classification attribute, and attribute value needs to be obtained from the model description file by the integrator and filled in.

16.2.Application of Collaborative Model Package

16.2.5.1 Introduction to the Function

Definition of collaborative model: When the AI training platform is training a model, it supports enabling model effect enhancement. After enabling it, the model trained is a mixed package of normal model and ultra-high-precision model, that is, collaborative package of large and small models. The small model refers to the normal model, and the large model refers to the ultra-high-precision model (hereinafter referred to as "collaborative package").

Collaborative package includes the large model package, the small model package, the collaborative model package, and the description files of the three model package types.

Application methods of collaborative package:

- Unified application: The collaborative package supports unified analysis after the engine resources are allocated. The collaborative package uses large and small models for collaborative analysis and requires the allocation of both general engine resources and dedicated engine resources for secondary analysis.
- Split application: When the secondary analysis resources of the device are insufficient, the collaborative package can be used independently as a general model or as an ultra-high-precision model.

  a. When applied independently as a general model, the integration process and the logic of the existing general model remain consistent. The general model within the collaborative package is associated with the engine by performanceMPID which is used to apply tasks.

  b. When applied independently as an ultra-high-precision model, the integration process and the logic of the existing ultra-high-precision model remain consistent. The ultra-high-precision model within the collaborative package is associated with the engine by superPrecisionMPID which is used to apply tasks.

16.2.5.2 API Calling Flow

1. Import collaborative model package: which is the same as that of the general model, refer to Chapter 5.1.2.

2. Get collaborative model package information: `GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/management?format=json`. The information defined in `coordinationModels` array object is the list of collaborative package information, including the MPID of the large and small models in the collaborative package, and the current application mode of the model package (indicating whether it is collaborative analysis of large and small models, or general small model analysis, or ultra-high-precision large model analysis). The performanceMPID of the small model in the collaborative package and the superPrecisionMPID of the large model support linking resources and apply task analysis independently.

2. Link collaborative model with engine: `PUT /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/loading/<engineID>?format=json`. In collaborative analysis mode, engine resources need to be allocated to large and small models. The dedicated engine resources for the large model for the secondary analysis are automatically allocated internally by the device, and the analysis resources for the small model are bound to a general engine by the user.

3. When the analysis resources of the device are insufficient, users can be guided to release analysis resources to ensure the running of the collaborative package; or the application mode of the collaborative package can be switched to the general model analysis mode or the ultra-high-precision model analysis mode. Switch the mode: `PUT /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/ApplicationParams?format=json`.

4. Collaborative package task analysis: `POST /ISAPI/Intelligent/AIOpenPlatform/videoTask?format=json`. When applying video task analysis, the MPID is the MPID of the collaborative package, and the device automatically adapts the model task analysis based on the current application mode of the collaborative package.

16.2.6 AIOP Multi-Chip Management

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16.2.6.1 Introduction to the Function

Due to the current limitations of chip performance, algorithm models need to run on different chips simultaneously. In this case, different chips have different functional capabilities. This function is mainly aimed at getting the capabilities supported by different chips when multiple chips coexist.

16.2.6.2 API Calling Flow

1. The client gets the computing power supported by the device on the AI Open Platform: `GET /ISAPI/Intelligent/AIOpenPlatform?format=json`. The `subPlatformInfo` list returns the firmware versions and basic information of other chips. 2. When multiple chips coexist:

a. First, check whether the output message contains the `engineList` field via `GET /ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json`. If it exists, it indicates that the device supports multiple chips. The `mainEngine` is the actual capability of the engine obtained by the current protocol.

b. The user can learn which engines the current device supports by checking engine ID returned in the `@opt` list in the `engineList` field, which includes physical engine ID and virtual engine ID. Finally, get the capabilities of other engines via `GET /ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json&engine=<engineID>`.

16.3 AIOP Image Analysis Task Management

16.3.1 Function Introduction

AIOP image analysis task management.

16.3.2 Integration Process

1. Obtain the device's support for AIOP function capability:
   - Obtain system capabilities: `GET /ISAPI/System/capabilities`, return `isSupportAIOpenPlatform` indicates support for `AIOP`;
   - Obtain device support capabilities: `GET /ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json`, if the protocol returns `isSupportPictureTask` as `true`, it indicates support for picture analysis task management;
   - Obtain picture task information capability: `GET /ISAPI/Intelligent/AIOpenPlatform/pictureTask/capabilities?format=json`;
2. Issue (create) an image task:
   - Create: `POST /ISAPI/Intelligent/AIOpenPlatform/pictureTask?format=json&security=<security>&iv=<iv>`;
3. (Optional) Manage image tasks
   - Query image task information: `GET /ISAPI/Intelligent/AIOpenPlatform/pictureTask?format=json`, the taskID (task ID) in the response message can be used to manage specific image tasks;
   - Modify image task: `PUT /ISAPI/Intelligent/AIOpenPlatform/pictureTask/<taskID>?format=json`;
   - Delete image task: `DELETE /ISAPI/Intelligent/AIOpenPlatform/pictureTask/<taskID>?format=json`;
   - Add analysis images: `POST /ISAPI/Intelligent/AIOpenPlatform/pictureTask/<taskID>?format=json&security=<security>&iv=<iv>`;
   - Import binary images: `POST /ISAPI/Intelligent/AIOpenPlatform/task/<taskID>/picture/<pictureID>?format=json`;

16.4 AIOP

16.4.1 Introduction to the Function

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AIOP is an all-in-one AI algorithm model development platform designed for the intelligent perception field, providing data management, data annotation, model automatic training, model verification, and model cloud-edge deployment services. It offers products and solutions which cover the full process from demand generation to AI application implementation.

Five major processes of AI application implementation: algorithm selection, data preparation, model training, model deployment, and model application.

Algorithm selection: select what type of algorithm template to implement.

Data preparation: Collecting suitable images or videos for algorithm training through cameras in specific application scenarios.

Model training: Upload the prepared data to the AI open platform, label the objects that need to be recognized by the algorithm via manual or automatic intelligent annotation, and select an appropriate application type to complete algorithm training. After the algorithm is trained, images can be directly uploaded to the AI open platform to verify the algorithm effect of the model, and the model can be downloaded to the local PC.

Model deployment: The downloaded algorithm model is applied to AI open platform devices with GPU chips through industry platforms or cloud view platform, or directly imported to cloud view platform. Then configure the IED intelligent analysis rule properties based on the specific application scenario.

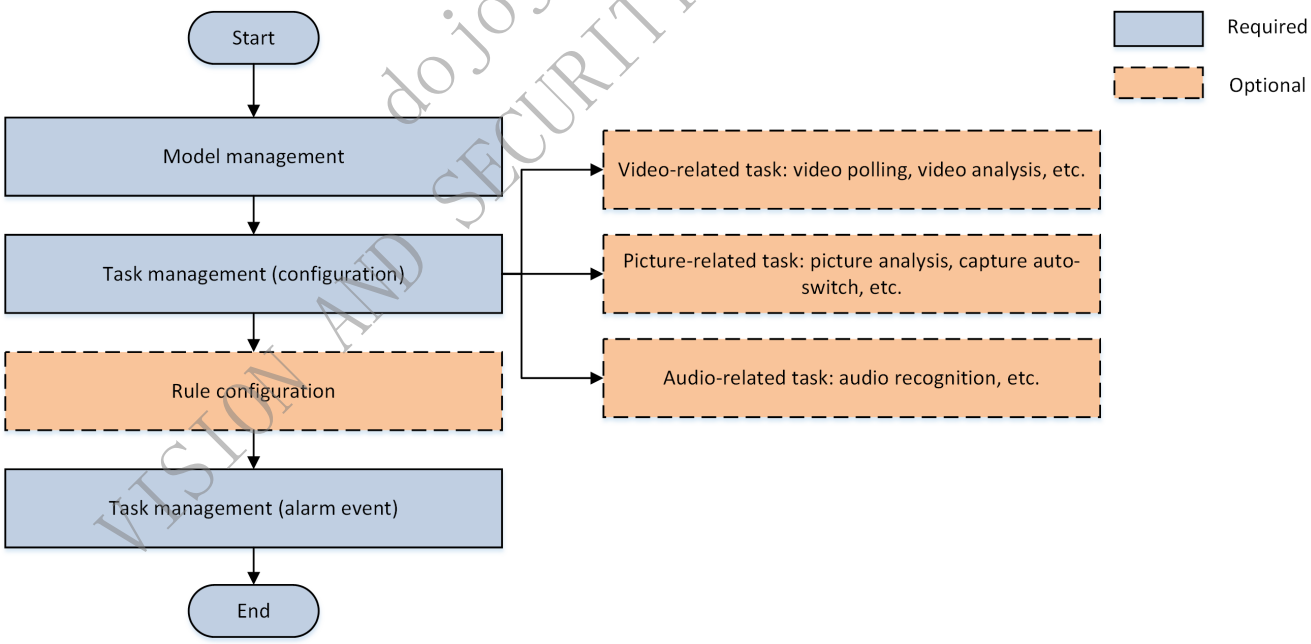
Model application: Alarm events generated by AI open platform devices are pushed to industry platforms or cloud view platform, where specific alarm event information, images, and videos can be viewed. At the same time, it can also be connected to devices (sound and light monitor, etc.) to remind the on-site person at the first time an event occurs. If users have their own application software, you can develop AI applications based on specific needs for users in the following methods: the API s of the device's protocol, the APIs of industry platform, or the APIs of cloud view platform.

The APIs of the device's protocol involved in model deployment are described in this document.

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16.4.2 API Calling Flow

The main processes involved in model deployment are shown in the following flowchart.



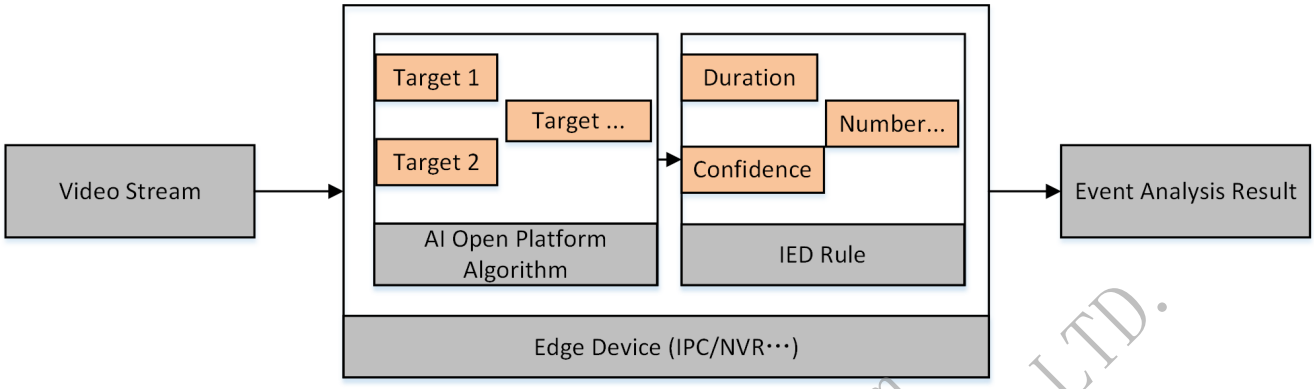
The rule configuration can be considered as additional parameters for task management, or the rules can be seen as optional post-operation for tasks:

- If AIOP runs tasks without linked task rules, the events reported by the device only contain target attributes, excluding contain alarm results.
- If AIOP runs tasks with linked task rules, the reported events will contain both target attributes and alarm results.

16.5 Analysis Task Rule Configuration

16.5.1 Introduction to the Function

Through the IED rules, flexible basic logic can be designed according to the actual business needs of users. The client sets the IED (analysis task rules) rules, and the device processes logical functions based on the algorithm analysis results, ultimately outputting the algorithm results processed by the IED rules.



IED application rule introduction:

For general analysis: Report all detection/classification results within the rule area; suitable for single detection scenarios with no strong business needs.

For statistics:

- Area target statistics: Output the number of targets (attributes) within the specified area; suitable for scenarios where the number of targets in a fixed area needs to be counted.
- Cross-line target statistics: Output the number of targets (attributes) crossing the line; suitable for scenarios where the number of objects on channels, conveyors, etc. needs to be counted.

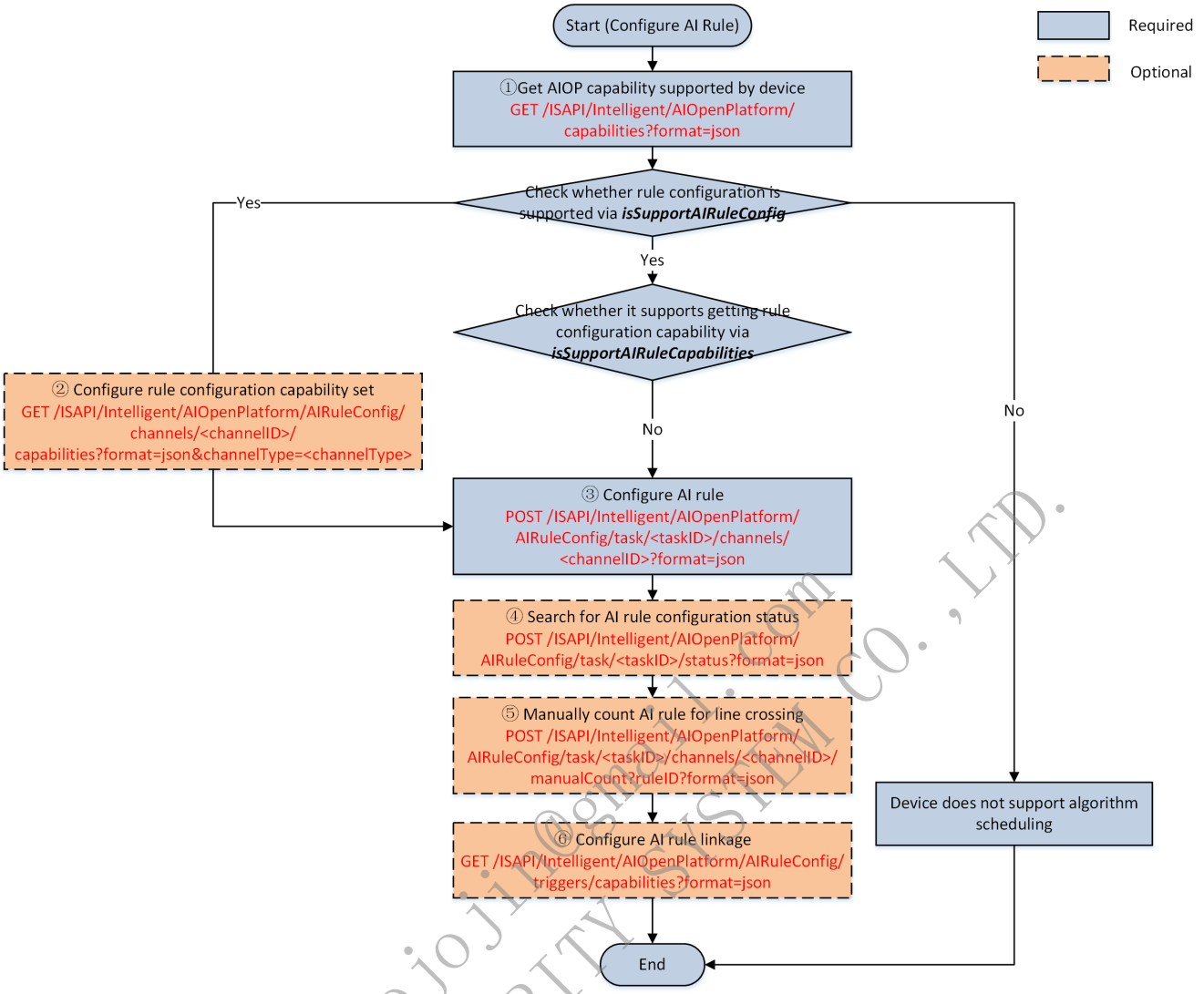
For detection:

- Area target exception detection: Extract alarm object information that meets the conditions of various attributes within the rule area; suitable for business scenarios where judgment needs to be made based on specific target attributes/quantity.
- Cross-line target detection: Extract alarm object information that meets the conditions of various attributes when crossing the line; suitable for scenarios where abnormal objects on channels, conveyors, etc. need to be detected.
- Area exception detection: Extract area status information that meets the alarm conditions; suitable for classification status detection scenarios of single classification model.

For composite rules: Combine multiple single rules according to the triggering logic to form composite alarm rules. The individual rule no longer trigger alarms separately, and the final alarm is reported after meeting the composite rule. For example, combining the vault door opening event with security personnel on duty for composite alarm. When security personnel leave and the vault door is open, a composite alarm is triggered.

16.5.2 API Calling Flow

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1. Get the device AIOP capability, `GET /ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json`;
   - `isSupportAIRuleConfig` indicates whether AI rule configuration is supported by device.
   - `isSupportOSDConfig` indicates whether OSD overlay configuration in rule configuration is supported by device.
   - `isSupportConfidenceConfig` indicates whether confidence configuration in rule configuration is supported by device.
   - `isSupportAIRuleCapabilities` indicates whether AI rule configuration capability is supported by device.
   - `isSupportAIRuleTrigger` indicates whether AI rule linkage configuration is supported by device.
2. Get the capability set of AI rule configuration: `GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/channels/<channelID>/capabilities?format=json&channelType=<channelType>`.

   Notes:
   - Get the capability set of AI rule configuration: `VCA_config`.
   - With AI rule configuration capability, the device can return the capability set according to the supported rule.
3. AI rule configuration is divided into two types according to different devices:

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- Perform rule analysis on the specified video channel:

  Get: `GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>?format=json&channelType=<channelType>`.

  Set: `PUT /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>?format=json&channelType=<channelType>`

  Notes:

  - If the value of `channelType` is `cameraIndexCode`, the data type of `cameraIndexCode` in `channelID` is string; If `channelType` does not exist, the data type of video channel No. in `channelID` is integer.
  - Rule name:

    (Client) Get the rule capability: `GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/channels/<channelID>/capabilities?format=json&channelType=<channelType>`. The returned `RuleNameInfo.ruleName` represents the custom rule name and `RuleNameInfo.ruleNameSpecCap` represents the supported special rule name.

    When configuring the rule, if `AIRuleConfig.RuleNameInfo.ruleName` is set to a non-empty string, the `ruleName` in the `AIOPData` detection data carried by the event is the same as `AIRuleConfig.RuleNameInfo.ruleName`. If `AIRuleConfig.RuleNameInfo.ruleName` is an empty character, the rule name in the `AIOPData` detection data carried by the event is the corresponding name of `ruleNameSpecCap`.
  - Default parameters for special rules: For special rules, the device will provide a set of default experience parameters:

    Get capabilities: `GET /ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json`. The returned `isSupportAIRuleDefaultParams` indicates that getting special AI intelligent rule default parameters is supported.

    Get: `GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/defaultParams?format=json`.

    If the default experience parameters do not meet the client's detection requirements, the client can adjust the rule parameters through the configuration interface.

- Analyze rules for a specified video channel's specified preset point (suitable for PTZ cameras):

  Get: `GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>/preset/<presetID>/?format=json`.

  Set: `PUT /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>/preset/<presetID>/?format=json`.

4. (Optional) Check whether the rule configuration for the specified task is successful: `GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/status?format=json`.
5. (Optional) Manual counting AI rule for line crossing: the manual counting should be set as the counting type in rule configuration:

   - Single rule counting:

     Get: `GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>/manualCount/<ruleID>?format=json`.

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Configure: `PUT /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>/manualCount/<ruleID>?format=json`.

- Batch rule counting:

  Get: `GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>/manualCount?format=json`.

  Configure: `PUT /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>/manualCount?format=json`.

6. (Optional) AI rule linkage configuration, supporting different linkage methods for individual rule configuration:
   i. Get the capability of AI rule linkage configuration: `/ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/channels/<channelID>/trigger/capabilities?format=json`.
   ii. Get AI rule linkage parameter: `GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>/trigger?format=json`.
   iii. Configure AI rule linkage parameter: `PUT /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>/trigger?format=json`.

Note:

- IED rule value description:

Object: The specific target category aimed by the target statistics rule (such as head and shoulder);

Attribute: The specific attribute of the specified object (such as wearing a hard hat / not wearing a hard hat);

Quantity and numerical value: The limit on the number of targets that trigger an alarm. It can be specified as >=, <=, >=, and <=, as well as not within the specified range;

Sensitivity: The standard for determining whether a target enters the specific area. For example, if it is set to 50, it means that 50% area of the target detection frame is within the specific area, and the target is considered to be within the specific area;

Alarm interval: The frequency of reporting analysis results, i.e., reporting the analysis results every n seconds;

Maximum alarm times: The maximum number of times an alarm can be triggered by the same target after the alarm state is triggered;

Duration: An alarm is triggered after the abnormal state lasts for a certain period of time;

Confidence filtering: The credibility of the target recognized by the algorithm. For example, if it is set to 800, it means that targets with a confidence level below 800 (80%) are filtered out.

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16.6 Capture Auto-Switch Analysis Task

16.6.1 Introduction to the Function

Analysis mode: According to the service requirement for real-time analysis and product computing power configuration, there are 4 types of analysis modes available:

Real-time video analysis: real-time inspection on channel video steam, with sufficient computing power and high demand for real-time nature.

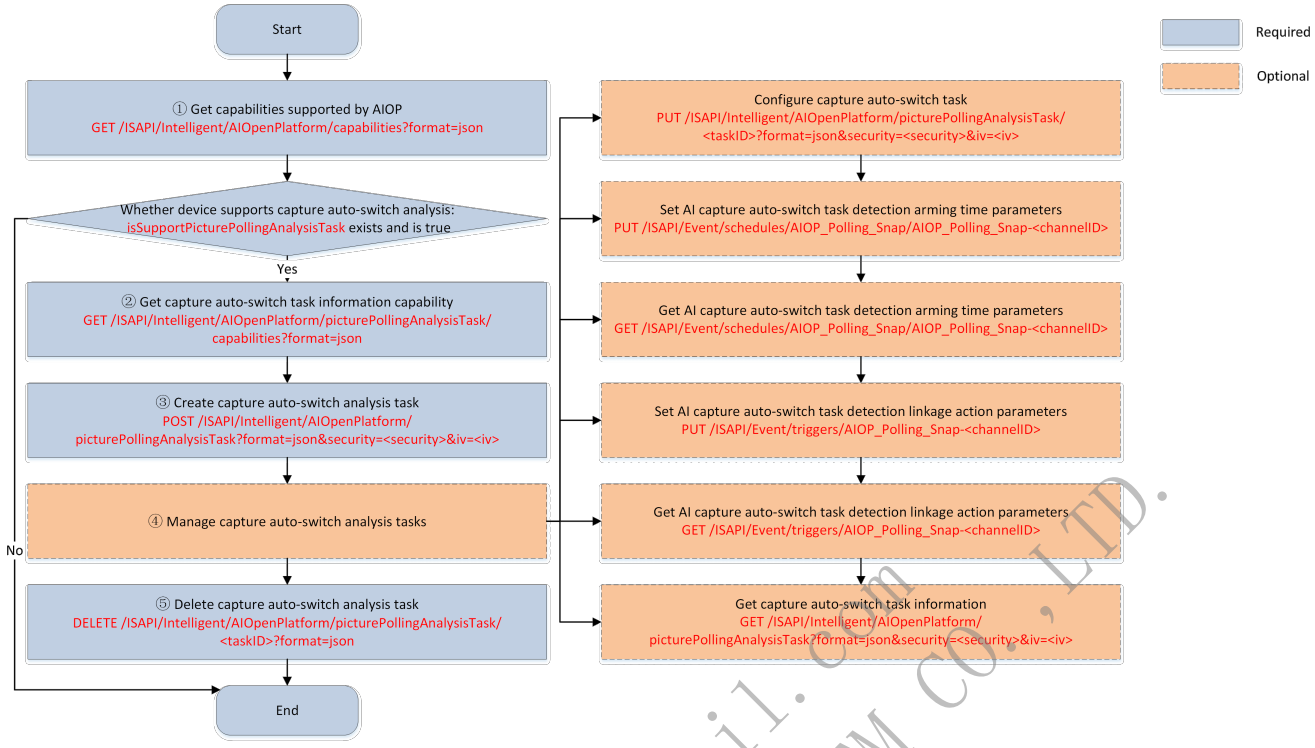
Auto-switch video analysis: real-time analysis on multiple channel videos in order, with insufficient GPU.

Scheduled capture analysis: analysis on channel video images captured by interval, with insufficient GPU.

Offline picture analysis: import offline pictures for algorithm recognition.

Import the model to edge device and bind it with engine, and then you can configure the analysis task.

16.6.2 API Calling Flow



1. (Client) Whether the device supports the capability of capture auto-switch task analysis:

System capability: `GET /ISAPI/System/capabilities`. If `isSupportAIOpenPlatform` is returned, it indicates that the device supports the AI Open Platform capability.

Task capability: `GET /ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json`. If `isSupportPicturePollingAnalysisTask` is `true`, it indicates that the device supports the capture auto-switch analysis task.

2. If the device returns not supported, the flow ends.

3. (Client) Get capture auto-switch task information capability:

Parameter capability: `GET /ISAPI/Intelligent/AIOpenPlatform/picturePollingAnalysisTask/capabilities?format=json`.

4. (Client) Create capture auto-switch analysis task:

Create task: `POST /ISAPI/Intelligent/AIOpenPlatform/picturePollingAnalysisTask?format=json&security=<security>&iv=<iv>`.

5. (Client) Get capture auto-switch analysis task:

Get task: `GET /ISAPI/Intelligent/AIOpenPlatform/picturePollingAnalysisTask?format=json&security=<security>&iv=<iv>`.

6. (Optional) (Client) Manage capture auto-switch analysis tasks:

Configure capture auto-switch task: `PUT /ISAPI/Intelligent/AIOpenPlatform/picturePollingAnalysisTask/<taskID>?format=json&security=<security>&iv=<iv>`;

Set AI capture auto-switch task detection arming time parameters: `PUT /ISAPI/Event/schedules/AIOP_Polling_Snap/AIOP_Polling_Snap-<channelID>`;

Get AI capture auto-switch task detection arming time parameters: `GET /ISAPI/Event/schedules/AIOP_Polling_Snap/AIOP_Polling_Snap-<channelID>`;

Set AI capture auto-switch task detection linkage action parameters: `PUT /ISAPI/Event/triggers/AIOP_Polling_Snap-<channelID>`;

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Get AI capture auto-switch task detection linkage action parameters: `GET /ISAPI/Event/triggers/AIOP_Polling_Snap-<channelID>`;

7. (Client) Delete capture auto-switch analysis task:

Delete task: `DELETE /ISAPI/Intelligent/AIOpenPlatform/picturePollingAnalysisTask/<taskID>?format=json`.

8. (Client) Receive AI capture auto-switch task detection event uploaded by device via arming or listening: `AIOP_Polling_Snap`.

16.7 Real-Time Video Analysis Task

16.7.1 Introduction to the Function

Analysis mode: According to the service requirements for real-time analysis and the computing power configuration of the product, there are four types of analysis modes available:

- Real-time video analysis: Real-time inspection of video streams with sufficient computing power and high real-time requirements.
- Auto-switch video analysis: Real-time analysis of multiple video streams in sequence with insufficient GPU.
- Scheduled capture analysis: Analysis of video images captured at intervals with insufficient GPU.
- Offline picture analysis: Import offline pictures for algorithm recognition.

Import the model to the edge device and bind it with the engine, and then you can configure the analysis task.

16.7.2 API Calling Flow

1. **Gets device's support for video task analysis capabilities:**
   - Device capability: `GET /ISAPI/System/capabilities`. If `isSupportAIOpenPlatform` is returned, it indicates that the device supports the AI Open Platform video analysis.
   - Task capability: `GET /ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json`, where `isSupportHistoryVideoTask` indicates support for historical video analysis.
   - Task capability: `GET /ISAPI/Intelligent/AIOpenPlatform/historyVideoTask/capabilities?format=json`.
2. **Manage video analysis tasks:**
   - Create task: `POST /ISAPI/Intelligent/AIOpenPlatform/videoTask?format=json`.
   - Modify task: `PUT /ISAPI/Intelligent/AIOpenPlatform/videoTask/<taskID>?format=json`.
   - Delete task: `DELETE /ISAPI/Intelligent/AIOpenPlatform/videoTask/<taskID>?format=json`.
   - Query task: `GET /ISAPI/Intelligent/AIOpenPlatform/videoTask?format=json`.
   - (Optional) Paginate task queries: `POST /ISAPI/Intelligent/AIOpenPlatform/SearchVideoTask?format=json`.

   Note: The task protocol includes arming time `TimeRange` and linkage actions `EventTrigger`:
   - Arming time: You can configure the analysis period as needed. For example, you can identify whether workers at a construction site are wearing safety helmets from 8 AM to 8 PM, which can help reduce false alarms due to poor lighting conditions at night.
   - Linkage action: You can configure alarm linkage actions as needed. For example, during the day at a construction site, if workers are not wearing safety helmets, an audible-visual alarm will be triggered.
3. **Receives video analysis events reported by the device: Refer to `AIOP_Video` for the event list**.

16.7.Historical Video Analysis Task

16.7.1 Introduction to the Function

Analyze video clips within a specified time period.

16.7.2 API Calling Flow

1. **Client gets device support for historical video task analysis capabilities:**

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- Device capability: `GET /ISAPI/System/capabilities`. If `isSupportAIOpenPlatform` is returned, it indicates that the device supports the AI Open Platform video analysis.
- Task capability: `GET /ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json`, where `isSupportVideoTask` indicates support for historical video analysis tasks.
- Task capability: `GET /ISAPI/Intelligent/AIOpenPlatform/videoTask/capabilities?format=json`.

2. **Client manages video analysis tasks:**
   - Create task: `POST /ISAPI/Intelligent/AIOpenPlatform/historyVideoTask?format=json`.
   - Modify task: `PUT /ISAPI/Intelligent/AIOpenPlatform/historyVideoTask/<taskID>?format=json`.
   - Delete task: `DELETE /ISAPI/Intelligent/AIOpenPlatform/historyVideoTask/<taskID>?format=json`.
   - Query task: `GET /ISAPI/Intelligent/AIOpenPlatform/historyVideoTask?format=json`.
3. **Receives video analysis events reported by the device: Refer to `AIOP_Video` for the event list**.

17 Two-Way Audio

17.1 Two-Way Audio

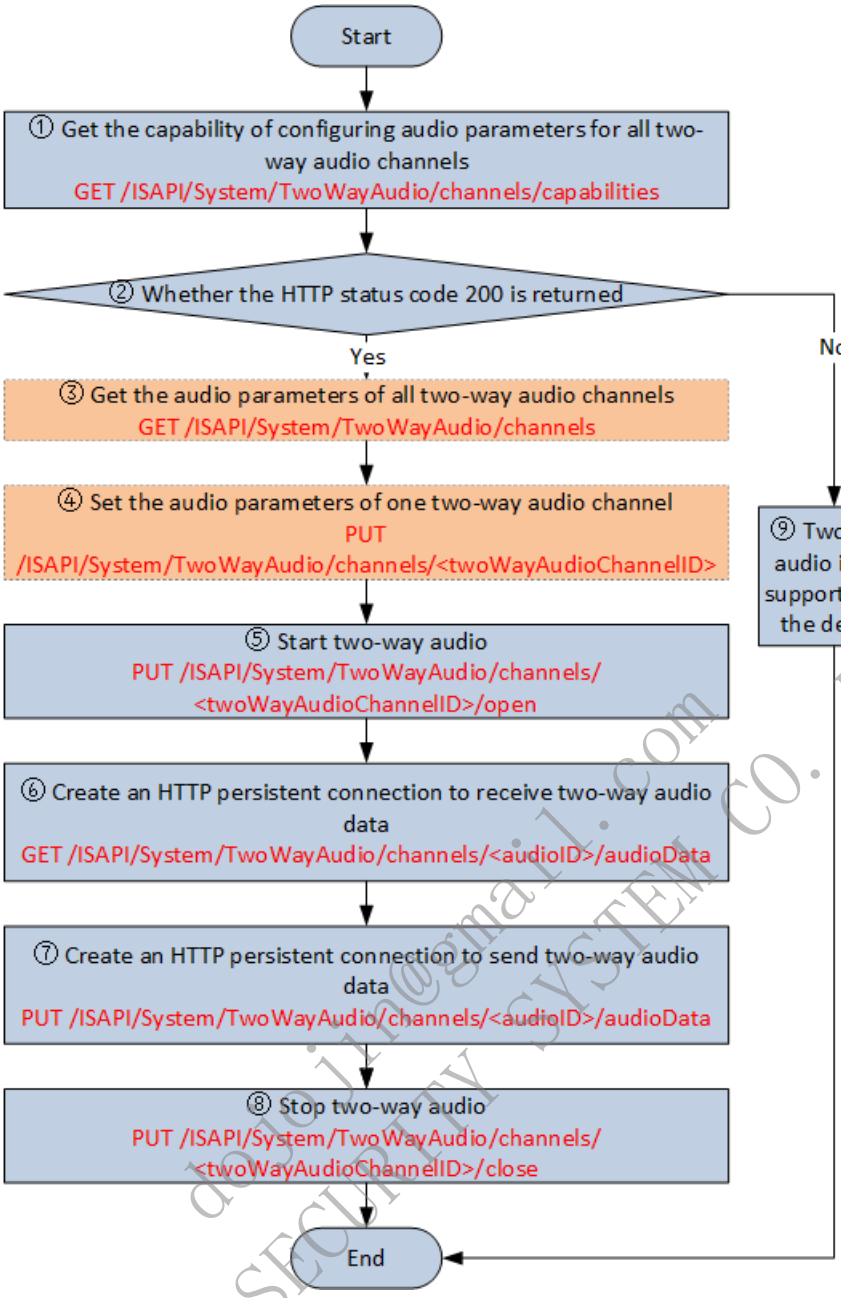
17.1.1 Introduction to the Function

This function is for the two-way audio between the client and device. When you call the API to implement the two-way audio, it requires the client to collect and encode the local audios and to decode the device audio data.

Note: Before starting two-way audio, check whether the way of collecting device audio data, audio encoding format, volume, etc., is correct or not. The audio encoding format of the client should be consistent with the device audio encoding format.

17.1.2 API Calling Flow

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1. Get the capability of configuring audio parameters for all two-way audio channels: `GET /ISAPI/System/TwoWayAudio/channels/capabilities`. Check whether the device supports two-way audio. If the HTTP status code `200 OK` is returned, then the device supports two-way audio.
2. Get the audio parameters of all two-way audio channels: `GET /ISAPI/System/TwoWayAudio/channels`. Get the No. of two-way audio channel for response and parsing `<id>`, audio type `<audioCompressionType>`, frame rate `<audioBitRate>`, sampling rate `<audioSamplingRate>` and so on, to ensure the consistent of audio format during the two-way audio process.
3. Set the audio parameters of one two-way audio channel: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>`. This step is optional, and the `<audioID>` is the channel No. `<id>` responded and parsed in step 2.
4. Start two-way audio: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/open`.
5. Create an HTTP persistent connection to receive the two-way audio data: `GET /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData`.
6. Create an HTTP persistent connection to send the two-way audio data: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData`.
7. When the two-way audio ends, the client disables the persistent connections in steps 6 and 7 and sends the message of stopping the two-way audio: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/close`.

Note:

- Two-way audio over ISAPI supports digest authentication.

- It supports the two-way audio mode or sending only mode (the platform only call `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData` to send the audio data to devices).
- The `<audioID>` in step 3, 4, 5, 6, and 7 is the two-way audio channel No. which starts from 1. It can be parsed from the `<id>` in the device response message of the step 2.
- The audio type `<audioCompressionType>`, frame rate `<audioBitRate>`, and sampling rate `<audioSamplingRate>` can be parsed from the device response message of the step 2 for audio playing and collection in the step 5 and 6.
- The step 6 and 7 use persistent connections. In the HTTP request headers, you don't need to set `Content-Legth`, but need to set the `Connection: keep-alive` and `Content-Type: application/octet-stream`.
- When the encoding format of audio data in the step 5 and 6 is AAC/MP2L2/MP3, the audio data length is variable, so the frame header should be lengthened by 4 bytes. When the encoding format is G.722.1/G.711alaw/G.711ulaw/G.726/G.729/G.729a/G.729b/PCM/ADPCM/Opus, the audio data length is fixed, there is no need to lengthen the frame header. For example, if the content length of a frame of MP3 audio data is 576 bytes and the header length is 4 bytes, the big end data is 0x00000240, the total length is 580 bytes.
- The damaged audio file in the AC3 format is usually incompatible when using. Currently, the audio algorithm library does not support this format.

Statistics of audio types with fixed lengths:

Audio Type	Fixed Length	Frame Interval
G.722.1	80	40ms
G.711alaw	160/320	20ms/40ms
G.711ulaw	160/320	20ms/40ms
G.726	80	40ms
G.729	10	10ms
G.729a	10	10ms
G.729b	10	10ms
PCM	1920	None
ADPCM	80	20ms
Opus	32/64	20ms/40ms

17.1.3 Exception Handling

17.1.3.1 Error Codes

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statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Re
4	Invalid Operation	twoWayAudioInProgressPleaseWait	0x40002068	Two-way audio in progress...Please wait.	Two-way audio in progress...Please wait.	Th op is all Tw au in pr Pl ag af tw au st

17.1.4 Message Format and Example

17.1.4.1 Receive Two-Way Audio Data

```
GET /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
Host: 10.17.115.128
Connection: keep-alive
Content-Type: application/octet-stream

HTTP/1.1 200 OK
Connection: keep-alive
Content-Type: application/octet-stream

//The following are the two-way audio data sent by the device.
```

17.1.4.2 Send Two-Way Audio Data

```
PUT /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
Host: 10.17.115.128
Connection: keep-alive
Content-Type: application/octet-stream

//The following are the two-way audio data sent by the client.
```

18 Audio Recognition

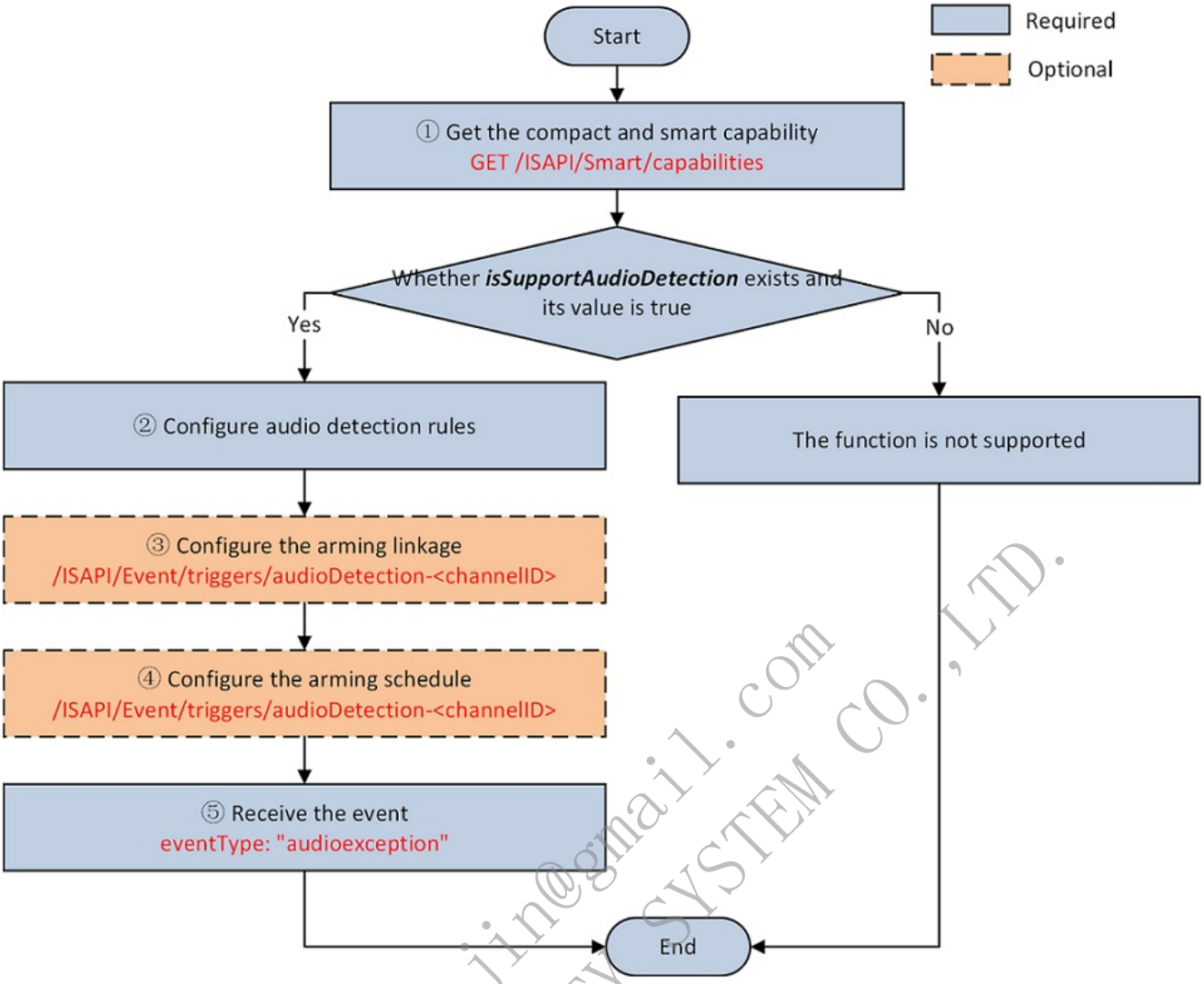
18.1 Audio Detection

18.1.1 Introduction to the Function

Detects the sound change, including the sudden increase or decrease of sound intensity and audio loss, of devices or the surrounding environment. Audio loss detection is used to timely detect audio loss situations, that are due to the audio configuration error or pickup damage.

18.1.2 API Calling Flow

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1. **Get the compact and smart capability.**

   Get the compact and smart capability: `GET /ISAPI/Smart/capabilities`. Check whether the `isSupportAudioDetection` is returned with true value.

2. **Configure rules.**

   Configure rules for all channels:

   Get audio detection parameters of all channels: `GET /ISAPI/Smart/AudioDetection/channels`.

   Set audio detection parameters of all channels: `PUT /ISAPI/Smart/AudioDetection/channels`.

   Configure rules for a specified channel:

   Get the capability of audio detection parameters of a specified channel: `GET /ISAPI/Smart/AudioDetection/channels/<channelID>/capabilities`.

   Get audio detection parameters of a specified channel: `GET /ISAPI/Smart/AudioDetection/channels/<channelID>`.

   Set audio detection parameters of a specified channel: `PUT /ISAPI/Smart/AudioDetection/channels/<channelID>`.

3. **(Optional) Configure arming linkages.**

   Get the event linkage configuration capability: `GET /ISAPI/Event/triggersCap`. Check if `AudioExceptionCap` is contained in `EventTriggersCap`.

Get the audio detection linkage parameters: `GET /ISAPI/Event/triggers/audioDetection-<channelID>`.

Set the audio detection linkage parameters: `PUT /ISAPI/Event/triggers/audioDetection-<channelID>`.

4. **(Optional) Configure the arming schedule.**

   Get the audio detection arming schedule for all channels: `GET /ISAPI/Event/schedules/audioDetections`.

   Set the audio detection arming schedule for all channels: `PUT /ISAPI/Event/schedules/audioDetections`.

   Get the audio detection arming schedule for a specified channel: `GET /ISAPI/Event/schedules/audioDetections/audioDetection-<channelID>`.

   Set the audio detection arming schedule for a specified channel: `PUT /ISAPI/Event/schedules/audioDetections/audioDetection-<channelID>`.

   Notes:

   i. Up to eight time periods can be configured on the timeline for one day.
   ii. Time of each two time periods cannot be overlapped.

5. **Receive the event**.

   The event type (`eventType`) is `audioexception`, and it contains `alarmType` (the alarm type, including audio loss, sudden decrease of sound intensity, and sudden increase of sound intensity).

19 API Reference

19.1 Device Management (General)

19.1.1 Event Subscription Management

19.1.1.1 Establish arming connection

Request URL

GET /ISAPI/Event/notification/alertStream?deployID=<deployID>

Query Parameter

Parameter Name	Parameter Type	Description
deployID	string	0 (client-side arming), 1 (real-time arming). Defaults to the client-side arming when delpoyID does not exist.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, event massage, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPV4 address of the device that triggers an alarm-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPV6 address of the device that triggers an alarm-->test
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm, range:[0,65535]-->0
  </portNo>
  <protocol>
    <!--ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address, range:[0,17]-->00-16-EA-AE-3C-40
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers an alarm, range:[0,65535]-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm triggering time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, the uploading times of the same alarm, range:[0,65535]-->0
  </activePostCount>
  <eventType>
    <!--ro, req, string, heartbeat type-->test
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:"active" (triggered), "inactive" (not triggered, the heartbeat data)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, range:[0,128]-->test
  </eventDescription>
</EventNotificationAlert>
```

19.1.1.2 Subscribe to events/alarms in arming mode

Request URL

POST /ISAPI/Event/notification/subscribeEvent?deployID=<deployID>

Query Parameter

Parameter Name	Parameter Type	Description
deployID	enum	0 (device local arming), 1 (real-time arming by platform). If the deployID is absent, it defaults to device local arming.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel
list)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)--
>list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--opt, object, subscription of a specified alarm/event-->
      <type>
        <!--req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
```

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```
"faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "CertificateCaptureEvent"(person ID
capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event),
"TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status),
"certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison), "regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, minor operation type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, minor event type, desc:this node is required when type is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel check-in.
3. When this node is not applied, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
        <childDevID>
          <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--opt, string, arm by radar channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </audioChannels>
      <childDeviceFormat>
        <!--opt, bool, upload by sub device, desc:upload by sub device-->true
      </childDeviceFormat>
      <unitChannels>
        <!--opt, string, arm by optical fiber channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </unitChannels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is all, this node should not be applied-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "localURL" (device local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
```

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```
  <ChangedUploadSub>
    <!--opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client
software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--opt, object, status information-->
      <all>
        <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--opt, bool, status of subscribed channels, desc:it is not required if the value of <all> is true-->true
      </channel>
      <hd>
        <!--opt, bool, subscribe to HDD status, desc:it is not required if the value of <all> is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:it is not required if the value of <all> is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When
this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform
focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-
->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will
be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm--
>true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no
related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <subscribeEventID>
    <!--opt, string, custom subscription ID. The platform should guarantee the uniqueness of ID when applying it. The automatic network replenishment of
events will be enabled by default after the node is applied, range:[1,128], desc:when an event/alarm is subscribed (related URL: POST
/ISAPI/Event/notification/subscribeEvent), if this node is applied, it can be used to replace <id> (subscription ID) returned in message
SubscribeEventResponse-->test
  </subscribeEventID>
  <level>
    <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
    <childDevID>
      <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
    </childDevID>
  </childDevIDList>
  <SubscribeISAPIMessage>
    <!--opt, object-->
    <EventTypeList>
      <!--opt, array, event list, subType:object, desc:event list-->
      <eventType>
        <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
      </eventType>
      <uploadPicEnabled>
        <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
      </uploadPicEnabled>
    </EventTypeList>
    <EventList>
      <!--req, array, event list, subType:object, range:[0,200]-->
      <Event>
        <!--opt, object, a single event-->
        <eventType>
          <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
        </eventType>
        <uploadPicEnabled>
          <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
```

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```
        </uploadPicEnabled>
      </Event>
    </EventList>
  </SubscribeISAPIMessage>
  <radarChannels>
    <!--opt, string, batch arming by radar channel, desc:if this field is applied, radarChannels in Event will not take effect. To arm all radar channels, this field should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object, subscription events qos level, desc:subscription events qos level-->
    <defaultLevel>
      <!--ro, req, enum, default qos level, subType:int, desc:default qos level-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, custom event qos level list, subType:object-->
      <event>
        <!--ro, opt, object, custom event qos level-->
        <eventType>
          <!--ro, req, string, event type, range:[0,64], desc:its values refer to event types of different domains-->test
        </eventType>
        <level>
          <!--ro, req, enum, qos level of this event type, subType:int, desc:qos level of this event type-->1
        </level>
        <channels>
          <!--ro, opt, string, video resource channel No., range:[0,256], desc:no such node means it applies to all channels; otherwise it applies to the channels listed in the array, and channels not listed use default qos level-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
  <audioChannels>
    <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
  </audioChannels>
  <unitChannels>
    <!--opt, string, batch arming by optical fiber channel, desc:if this field is applied, unitChannels in Event will not take effect. To arm all fiber channels, this node should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
  </unitChannels>
  <retransmissionID>
    <!--opt, string, ANR arming ID, range:[1,64], desc:UUID (an unique ID generated by the client). If this field is applied during establishing the event arming/subscription connection, the ANR of subscribed event will be enabled by default; when the network reconnected, the platform can apply the same UUID to tell the device to re-upload saved event information during network disconnection period-->123456
  </retransmissionID>
  <retransmissionTimeoutDuration>
    <!--opt, int, data transmission timeout for ANR, range:[1,168], unit:h, desc:data transmission timeout for ANR-->1
  </retransmissionTimeoutDuration>
  <subscribeWithoutPic>
    <!--opt, bool-->false
  </subscribeWithoutPic>
</SubscribeEvent>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEventResponse xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, subscription result, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, subscription ID, which is returned by the device. The device should guarantee the uniqueness of ID, range:[1,64]-->1
  </id>
  <FailedEventList>
    <!--ro, opt, array, list of events, of which subscription failed, subType:object, desc:when subscription of one or multiple events failed, this node
should be returned. The client software can check if all events are subscribed to by checking whether the node <FailedEventList> is returned-->
    <Event>
      <!--ro, opt, object, event, of which the subscription failed. The event/alarm type name can be obtained via the capability set-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:event type-->ADAS
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL).-->binary
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel No., desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <subStatusCode>
        <!--ro, req, enum, error code, subType:string, desc:refer to the error code dictionary-->activeNumMax
      </subStatusCode>
      <minorType>
        <!--ro, opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel
check-in.
3. When this node is not returned, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
    </Event>
  </FailedEventList>
</SubscribeEventResponse>
```

19.1.1.3 Get the alarm/event subscription capability

Request URL

GET /ISAPI/Event/notification/subscribeEventCap

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<SubscribeEventCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <format opt="xml,json">
    <!--ro, opt, string, supported message data format, attr:opt{req, string}-->xml
  </format>
  <heartbeat min="1" max="180">
    <!--ro, opt, int, heartbeat interval time, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->1
  </heartbeat>
  <channelMode opt="all,list">
    <!--ro, opt, enum, channel subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all channels), "list" (subscribe to
channels according to channel list)-->list
  </channelMode>
  <eventMode opt="all,list,allAndChild">
    <!--ro, req, enum, event subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all alarms/events), "list" (subscribe to
specified alarms/events)-->list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--ro, opt, object, subscription of a specified alarm/event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm), "emergencyAlarm",
"faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent",
"IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison
event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event),
"TransactionRecordEvent", "HealthInfoSyncQuery" (health information search event), "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"
(searching the consumption status), "certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison),
"regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
      </type>
      <channels>
        <!--ro, opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
    </Event>
  </EventList>
  <isSupportUnSubscribeEvent>
    <!--ro, opt, bool, whether the device supports canceling arming subscription-->true
  </isSupportUnSubscribeEvent>
</SubscribeEventCap>
```

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19.1.1.4 Cancel subscribing alarm/event

Request URL

PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	Subscription event ID, which should match the id field in the SubscribeEventResponse of the response message from the POST /ISAPI/Event/notification/subscribeEvent?deployID=.This field is optional. When the protocol is /ISAPI/Event/notification/unSubscribeEvent, it indicates that all subscription links are to be unsubscribed. When the protocol is /ISAPI/Event/notification/unSubscribeEvent?ID=XXX, it indicates that a specific subscription link is to be unsubscribed.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.1.1.5 Event subscription heartbeat

EventType:heartBeat

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, alarm message, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, transmission communication protocol type, range:[0,64], desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, string, event type-->heartBeat
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->heartBeat
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
</EventNotificationAlert>
```

19.1.2 Exception Detection

19.1.2.1 Get the linkage parameters of HDD error

Request URL

GET /ISAPI/Event/triggers/diskerror

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->diskerror
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->diskerror
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.1.2.2 Set the linkage parameters of HDD error

Request URL

PUT /ISAPI/Event/triggers/diskerror

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->diskerror
  </id>
  <eventType>
    <!--req, enum, subType:string-->diskerror
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.2.3 Delete the linkage of HDD error

Request URL

DELETE /ISAPI/Event/triggers/diskerror

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.2.4 Get the linkage parameters of full HDD

Request URL

GET /ISAPI/Event/triggers/diskfull

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->diskfull
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->diskfull
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.1.2.5 Set the linkage parameters of full HDD

Request URL

PUT /ISAPI/Event/triggers/diskfull

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->diskfull
  </id>
  <eventType>
    <!--req, enum, subType:string-->diskfull
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.2.6 Delete the linkage of full HDD

Request URL

DELETE /ISAPI/Event/triggers/diskfull

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.2.7 Delete the linkage of illegal login

Request URL

DELETE /ISAPI/Event/triggers/illaccess

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.2.8 Set the linkage parameters of illegal login

Request URL

PUT /ISAPI/Event/triggers/illaccess

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->illaccess
  </id>
  <eventType>
    <!--req, enum, subType:string-->illaccess
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.2.9 Get the linkage parameters of illegal login

Request URL

GET /ISAPI/Event/triggers/illaccess

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->illaccess
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->illaccess
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.1.2.10 Get the linkage parameters of IP address conflict event

Request URL

GET /ISAPI/Event/triggers/ipconflict

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->ipconflict
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->ipconflict
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.1.2.11 Delete the linkage of IP address conflict event

Request URL

DELETE /ISAPI/Event/triggers/ipconflict

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.2.12 Set the linkage parameters of IP address conflict event

Request URL

PUT /ISAPI/Event/triggers/ipconflict

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->ipconflict
  </id>
  <eventType>
    <!--req, enum, subType:string-->ipconflict
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.2.13 Get the linkage parameters of network disconnection

Request URL

GET /ISAPI/Event/triggers/nicbroken

Query Parameter

None

Request Message

None

Response Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->nicbroken
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->nicbroken
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.1.2.14 Set the linkage parameters of network disconnection

Request URL

PUT /ISAPI/Event/triggers/nicbroken

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->nicbroken
  </id>
  <eventType>
    <!--req, enum, subType:string-->nicbroken
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.2.15 Delete the linkage of network disconnection

Request URL

DELETE /ISAPI/Event/triggers/nicbroken

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.2.16 HDD error

EventType:diskerror

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"diskerror" (HDD Error)-->diskerror
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:HDD Error-->diskerror
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <diskNo>
    <!--ro, opt, int, HDD number-->1
  </diskNo>
  <HDDList>
    <!--ro, opt, array, HDD list, subType:object-->
    <HDD>
      <!--ro, opt, object, HDD-->
      <id>
        <!--ro, req, int, index, desc:it starts from 1-->1
      </id>
      <diskNumber>
        <!--ro, req, int, HDD No.-->1
      </diskNumber>
    </HDD>
  </HDDList>
</EventNotificationAlert>
```

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19.1.2.17 Invalid access

EventType:illaccess

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event parameters, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the alarm device-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the alarm device-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the alarm device-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, transfer protocol type, range:[0,64], desc:transfer protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZVIZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, No, of the channel that triggers an alarm, desc:No. of the video channel that triggers an alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, the uploading times of the same alarm, desc:event frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"illaccess" (invalid access)-->illaccess
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for persistent events: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:illaccess-invalid access-->illaccess
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <username>
    <!--ro, opt, string, locked user name, range:[0,32], desc:the number of attempts by current user name has reached limit. The user name and IPv4 address may be locked at the same time-->test
  </username>
  <ipV4Address>
    <!--ro, opt, string, locked IPv4 address, desc:the number of attempts by current IPv4 address has reached limit. The user name and Ipv4 address may be locked at the same time-->test
  </ipV4Address>
  <lockStatus>
    <!--ro, opt, enum, locking status, subType:string, desc:; “unlocked”, “locked”-->unlocked
  </lockStatus>
  <retryLoginTime>
    <!--ro, opt, int, remaining attempts, range:[0,65535], desc:it is valid when the value of lockStatus is “unlocked”-->0
  </retryLoginTime>
  <lockedTime>
    <!--ro, opt, int, locking duration (minutes), range:[0,65535], unit:秒, desc:it is valid when the value of lockStatus is “locked”-->0
  </lockedTime>
  <accessInfoList>
    <!--ro, opt, array, invalid access information list, subType:object-->
    <accessInfo>
      <!--ro, opt, object, invalid access information-->
      <ipV4Address>
        <!--ro, opt, string, Invalid access IP address-->test
      </ipV4Address>
      <dateTime>
        <!--ro, opt, datetime, invalid access time-->2004-05-03T17:30:08+08:00
      </dateTime>
    </accessInfo>
  </accessInfoList>
</EventNotificationAlert>
```

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19.1.2.18 Network disconnected

EventType:nicbroken

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers an alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers an alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, communication protocol types, range:[0,64], desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome" (the value
should be "HTTP" when ISAPI protocol is transmitted via EZ protocol, and the value should be "EHome" when ISAPI protocol is transmitted via ISUP)-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers an alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, the uploading times of the same alarm, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, string, event type, desc:"nicbroken" (network disconnected)-->nicbroken
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:"nicbroken" (network disconnected)-->nicbroken
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DevicePositionInfo>
    <!--ro, opt, object, device location, desc:the current location of the device when triggering the event, which is mainly for on-board device-->
    <divisionEW>
      <!--ro, opt, enum, eastern/western hemisphere, subType:string, desc:"E" (eastern hemisphere), "W" (western hemisphere)-->E
    </divisionEW>
    <longitude>
      <!--ro, opt, int, longitude, desc:degree*3600*100 + minute*60*100 + second*100-->100
    </longitude>
    <divisionNS>
      <!--ro, opt, enum, southern/northern hemisphere, subType:string, desc:"S" (southern hemisphere), "N" (northern hemisphere)-->N
    </divisionNS>
    <latitude>
      <!--ro, opt, int, latitude, desc:degree*3600*100 + minute*60*100 + second*100-->100
    </latitude>
    <direction>
      <!--ro, opt, int, vehicle direction, desc:actual direction (degree) (due north as 0 degree, calculated clockwise)-->100
    </direction>
    <speed>
      <!--ro, opt, int, speed, desc:cm/h-->100
    </speed>
    <height>
      <!--ro, opt, int, height, unit:cm, desc:cm-->1
    </height>
  </DevicePositionInfo>
</EventNotificationAlert>
```

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19.1.3 HEOP

19.1.3.1 Get Version Information of HEOP Software Library

Request URL

GET /ISAPI/Custom/OpenPlatform/algInfo?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "HEOPVersion":  "1.0.1",
    /*ro, opt, string, HEOP SDK version information, range:[1,64], desc:format: 1.0.1*/
    "BASEVersion":  "1.0.1",
    /*ro, opt, string, BASE SDK version information, range:[1,64], desc:format: 1.0.1*/
    "MEDIAVersion":  "1.0.1"
    /*ro, opt, string, MEDIA SDK version information, range:[1,64], desc:format: 1.0.1*/
}
```

19.1.3.2 Get the application program information list

Request URL

GET /ISAPI/Custom/OpenPlatform/App

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<AppList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, application program information list, subType:object, attr:version{req, string, protocolVersion}-->
  <App>
    <!--ro, opt, object, application program information-->
    <id>
      <!--ro, req, int, index (it is not AppID), range:[0,128]-->1
    </id>
    <AppID>
      <!--ro, opt, string, application program ID-->test
    </AppID>
    <packageName>
      <!--ro, req, string, software package name to be displayed, range:[0,128]-->test
    </packageName>
    <vendor>
      <!--ro, req, string, manufacturer name of the third-party application program, range:[0,128]-->test
    </vendor>
    <softVersion>
      <!--ro, req, string, software version number, range:[0,128]-->test
    </softVersion>
    <runStatus>
      <!--ro, req, enum, program running status, subType:string, desc:"true" (running), "false" (stopped)-->true
    </runStatus>
    <licenseInstalled>
      <!--ro, opt, enum, License status, subType:string, desc:"none" (free), "no" (inactivity), "yes" (activated), "expired"-->none
    </licenseInstalled>
    <expirationDateLeft>
      <!--ro, opt, int, days remaining until expiration from today, range:[-1,365], step:1, unit:d, desc:-1 (permanent valid)-->1
    </expirationDateLeft>
    <memory>
      <!--ro, opt, int, memory size occupied by the application program, range:[0,1024], step:1, unit:MB-->1
    </memory>
    <flash>
      <!--ro, opt, int, flash size occupied by the application program, range:[0,1024], step:1, unit:MB-->1
    </flash>
    <bStartedOnce>
      <!--ro, opt, bool, whether the application program has been started, desc:"true" (yes), "false" (no)-->true
    </bStartedOnce>
    <appPackageUUID>
      <!--ro, opt, string, the UUID of the application program package, range:[0,128]-->test
    </appPackageUUID>
    <buildTime>
      <!--ro, opt, string, compilation date, desc:format: buildxxxxxxxx-->test
    </buildTime>
    <status>
      <!--ro, opt, enum, status, subType:string, desc:"notDownloaded" (it is not downloaded), "installed" (it is installed), "uninstalled" (it is not
installed)-->notDownloaded
    </status>
    <CPU>
      <!--ro, opt, float, CPU occupation rate percentage-->0.000
    </CPU>
    <caculation>
      <!--ro, opt, float, computing power used by the application program. It should be accurate to one decimal place, desc:unit: TOPS-->0.000
    </caculation>
    <intelligentMemory>
      <!--ro, opt, int, intelligent memory size occupied by the application program, unit:Byte-->0
    </intelligentMemory>
    <GFLOPS>
      <!--ro, opt, float, computing power required for the application program, range:[0.1,100.0]-->0.1
    </GFLOPS>
    <extendRunStatus>
      <!--ro, opt, enum, extension module running status, subType:string, dep:and,{$.AppList[*].App.runStatus,eq,false}, desc:"abnormal"-->abnormal
    </extendRunStatus>
    <totalEffectiveDays>
      <!--ro, opt, int, the validity period of application program, step:1, desc:-1 (no limit, i.e., permanent valid)-->90
    </totalEffectiveDays>
    <surplusEffectiveDays>
      <!--ro, opt, int, the remaining days in validity period of application program, step:1, desc:-1 (no limit, i.e., permanent valid)-->30
    </surplusEffectiveDays>
    <totalAnalysisNum>
      <!--ro, opt, int, the total number of video channels for the application program, step:1-->5
    </totalAnalysisNum>
    <surplusAnalysisNum>
      <!--ro, opt, int, the number of available video channels for the application program, step:1-->1
    </surplusAnalysisNum>
  </App>
</AppList>
```

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19.1.3.3 Get Information of an Application Program

Request URL

GET /ISAPI/Custom/OpenPlatform/App/<AppID>

Query Parameter

Parameter Name	Parameter Type	Description
AppID	string	App ID, which is unique and composed of strings containing numbers and letters. For details, see the App ID description in Terminology and Definitions in the HEOP API calling flow

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<App xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, application program information list, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, application program ID-->1
  </id>
  <packageName>
    <!--ro, req, string, packet name for display-->test
  </packageName>
  <vendor>
    <!--ro, req, string, vendor name of third-party application program-->test
  </vendor>
  <softVersion>
    <!--ro, req, string, software version number-->test
  </softVersion>
  <runStatus>
    <!--ro, req, enum, program running status, subType:string, desc:“true” (running), “false” (stopped)-->true
  </runStatus>
  <licenseInstalled>
    <!--ro, opt, enum, whether the License is installed, subType:string, desc:"none” (free), "no” (inactivity), "yes” (activated), "expired”-->none
  </licenseInstalled>
  <expirationDateLeft>
    <!--ro, opt, int, remaining days of expiry date, range:[-1,365], step:1, desc:remaining days of expiry date-->1
  </expirationDateLeft>
  <memory>
    <!--ro, opt, int, memory size occupied by the application program, unit:MB-->1
  </memory>
  <flash>
    <!--ro, opt, int, flash size occupied by the application program, unit:MB-->1
  </flash>
  <bStartedOnce>
    <!--ro, opt, bool, whether the application program has been started:, desc:"true” (yes), "false” (no)-->true
  </bStartedOnce>
  <appPackageUUID>
    <!--ro, opt, string, the UUID of the application program package, range:[0,128]-->test
  </appPackageUUID>
  <buildTime>
    <!--ro, opt, string, compilation date, desc:e.g., buildxxxxxxxx-->test
  </buildTime>
  <status>
    <!--ro, opt, enum, status, subType:string, desc:“notDownloaded”, “installed”, “uninstalled”-->notDownloaded
  </status>
  <CPU>
    <!--ro, opt, float, CPU occupation rate percentage-->0.000
  </CPU>
  <caculation>
    <!--ro, opt, float, computing power size occupied by the application program, desc:The number is accurate to one decimal place (e.g., 0.1)-->0.1
  </caculation>
  <intelligentMemory>
    <!--ro, opt, int, intelligent memory size occupied by the application program-->0
  </intelligentMemory>
  <isMaxIntelligentMemory>
    <!--ro, opt, bool-->true
  </isMaxIntelligentMemory>
  <extendRunStatus>
    <!--ro, opt, enum, subType:string, dep:and,{$.App.runStatus,eq,false}-->abnormal
  </extendRunStatus>
  <totalEffectiveDays>
    <!--ro, opt, int, step:1-->90
  </totalEffectiveDays>
  <surplusEffectiveDays>
    <!--ro, opt, int, step:1-->30
  </surplusEffectiveDays>
  <totalAnalysisNum>
    <!--ro, opt, int, step:1-->5
  </totalAnalysisNum>
  <surplusAnalysisNum>
    <!--ro, opt, int, step:1-->1
  </surplusAnalysisNum>
  <isPreInstalled>
    <!--ro, opt, bool-->true
  </isPreInstalled>
  <licenseInstalledDate>
```

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```
    <!--ro, opt, date-->2023-08-18
  </licenseInstalledDate>
  <moduleLicenseInfoList>
    <!--ro, opt, array, subType:object-->
    <moduleLicenseInfo>
      <!--ro, opt, object-->
      <moduleID>
        <!--ro, req, int, range:[1,16]-->0
      </moduleID>
      <moduleName>
        <!--ro, opt, string, range:[0,64]-->0
      </moduleName>
      <licenseInstalled>
        <!--ro, opt, enum, subType:string-->none
      </licenseInstalled>
      <expirationDateLeft>
        <!--ro, opt, int, range:[-1,365], step:1, unit:d-->1
      </expirationDateLeft>
      <licenseInstalledDate>
        <!--ro, opt, date-->2023-08-18
      </licenseInstalledDate>
    </moduleLicenseInfo>
  </moduleLicenseInfoList>
  <defaultIconURL>
    <!--ro, opt, string-->test
  </defaultIconURL>
  <uploadInfo>
    <!--ro, opt, object-->
    <uploadVersion>
      <!--ro, opt, string-->V1.0.0 build YYMMDD
    </uploadVersion>
    <uploadSize>
      <!--ro, opt, int, range:[0,200], unit:MB-->49
    </uploadSize>
    <uploadDescription>
      <!--ro, opt, string, range:[0,400]-->test
    </uploadDescription>
    <URL>
      <!--ro, opt, string-->test
    </URL>
    <appPackageUUID>
      <!--ro, opt, string, range:[0,128]-->abcdefg1234567890
    </appPackageUUID>
  </uploadInfo>
  <packageDescription>
    <!--ro, opt, string, range:[0,256]-->test
  </packageDescription>
  <instanceType>
    <!--ro, opt, enum, subType:string-->multiInstanc
  </instanceType>
  <envVariables>
    <!--ro, opt, string, range:[0,256]-->test
  </envVariables>
  <abnormalDetail>
    <!--ro, opt, enum, subType:string, dep:and,{$.App.extendRunStatus,eq,abnormal}-->insufficientContinuousMMZ
  </abnormalDetail>
  <engineID>
    <!--ro, opt, string-->1
  </engineID>
  <engineRunningList>
    <!--ro, opt, array, subType:object-->
    <engineRunning>
      <!--ro, opt, object-->
      <engineID>
        <!--ro, req, string-->1
      </engineID>
      <vendor>
        <!--ro, req, string, range:[0,128]-->test
      </vendor>
      <softVersion>
        <!--ro, req, string, range:[0,128]-->test
      </softVersion>
      <status>
        <!--ro, opt, enum, subType:string-->notDownloaded
      </status>
      <runStatus>
        <!--ro, req, enum, subType:string-->true
      </runStatus>
      <licenseInstalled>
        <!--ro, opt, enum, subType:string-->none
      </licenseInstalled>
      <expirationDateLeft>
        <!--ro, opt, int, range:[-1,365], step:1, unit:d-->1
      </expirationDateLeft>
      <memory>
        <!--ro, opt, int, range:[0,1024], step:1, unit:MB-->1
      </memory>
      <flash>
        <!--ro, opt, int, range:[0,1024], step:1, unit:MB-->1
      </flash>
      <bStartedOnce>
        <!--ro, opt, bool-->true
      </bStartedOnce>
      <appPackageUUID>
```

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```
        <!--ro, opt, string, range:[0,128]-->test
      </appPackageUUID>
      <caculation>
        <!--ro, opt, float-->0.1
      </caculation>
      <intelligentMemory>
        <!--ro, opt, int, unit:Byte-->0
      </intelligentMemory>
      <isMaxIntelligentMemory>
        <!--ro, opt, bool-->true
      </isMaxIntelligentMemory>
      <extendRunStatus>
        <!--ro, opt, enum, subType:string, dep:and,{$.App.engineRunningList[*].engineRunning.runStatus,eq,false}-->abnormal
      </extendRunStatus>
      <uploadInfo>
        <!--ro, opt, object-->
        <uploadVersion>
          <!--ro, opt, string-->V1.0.0 build YYMMDD
        </uploadVersion>
        <uploadSize>
          <!--ro, opt, int, range:[0,200], unit:MB-->49
        </uploadSize>
        <uploadDescription>
          <!--ro, opt, string, range:[0,400]-->test
        </uploadDescription>
        <URL>
          <!--ro, opt, string-->test
        </URL>
        <appPackageUUID>
          <!--ro, opt, string, range:[0,128]-->abcdefg1234567890
        </appPackageUUID>
      </uploadInfo>
      <abnormalDetail>
        <!--ro, opt, enum, subType:string, dep:and,{$.App.engineRunningList.extendRunStatus,eq,abnormal}-->insufficientContinuousMMZ
      </abnormalDetail>
    </engineRunning>
  </engineRunningList>
  <displaySeq>
    <!--ro, opt, int, range:[1,64]-->1
  </displaySeq>
</App>
```

19.1.3.4 Set permission of the third-party application program

Request URL

PUT /ISAPI/Custom/OpenPlatform/App/<AppID>/permission?format=json&engineID=<engineID>

Query Parameter

Parameter Name	Parameter Type	Description
AppID	string	App ID, and the length is 1 to 28 bytes. It is unique and composed of strings containing numbers and letters. For details, see the App ID description in Terminology and Definitions in the HEOP API calling flow.
engineID	string	Engine ID.

Request Message

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```
{
    "PermissionList": {
    /*req, object, list of permissions*/
        "Stream": {
        /*opt, object*/
            "enabled":  true
            /*req, bool*/
        },
        "configAuth": {
        /*opt, object, configure the permission for free authentication*/
            "enabled":  true
            /*req, bool, enable the permission to get streams*/
        },
        "Storage": {
        /*opt, object*/
            "enabled":  true
            /*req, bool*/
        },
        "Network": {
        /*opt, object*/
            "enabled":  true
            /*req, bool*/
        },
        "OSD": {
        /*opt, object*/
            "enabled":  true
            /*req, bool*/
        }
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

19.1.3.5 Get permissions of the third-party application program

Request URL

GET /ISAPI/Custom/OpenPlatform/App/<AppID>/permission?format=json&engineID=<engineID>

Query Parameter

Parameter Name	Parameter Type	Description
AppID	string	App ID, and the length is 1 to 28 bytes. It is unique and composed of strings containing numbers and letters. For details, see the App ID description in Terminology and Definitions in the HEOP API calling flow.
engineID	string	Engine ID

Request Message

None

Response Message

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```
{
    "PermissionList": {
    /*ro, req, object, permission information*/
        "Stream": {
        /*ro, opt, object, streaming permission*/
            "enabled":  true
            /*ro, req, bool, enable the permission to get streams*/
        },
        "configAuth": {
        /*ro, opt, object, configure the permission for free authentication*/
            "enabled":  true
            /*ro, req, bool, enable the permission for free authentication*/
        },
        "Storage": {
        /*ro, opt, object, storage accessing permission*/
            "enabled":  true
            /*ro, req, bool, enable the permission to access storages, desc:enable the permission to get streams*/
        },
        "Network": {
        /*ro, opt, object, network accessing permission*/
            "enabled":  true
            /*ro, req, bool, enable the permission to access the network, desc:enable the permission to get streams*/
        },
        "OSD": {
        /*ro, opt, object*/
            "enabled":  true
            /*ro, req, bool*/
        }
    }
}
```

19.1.3.6 Run Specified Application Program

Request URL

PUT /ISAPI/Custom/OpenPlatform/App/<AppID>/start?engineID=<engineID>&channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
AppID	string	App ID, and the length is 1 to 28 bytes. It is unique and composed of strings containing numbers and letters. For details, see the App ID description in Terminology and Definitions in the HEOP API calling flow.
engineID	string	Engine ID
channelID	string	Video channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:(0) OK, (1) OK, (2) Device Busy, (3) Device Error, (4) Invalid Operation, (5) Invalid XML Format, (6)
Invalid XML Content, (7) Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:“OK" (succeeded), "Device Busy", "Device Error", “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, detailed description of the error code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
  <AttachInfo>
    <!--ro, opt, array, subType:object-->
    <StatusList>
      <!--ro, req, object-->
      <Status>
        <!--ro, req, object-->
        <index>
          <!--ro, req, int-->0
        </index>
        <statusCode>
          <!--ro, req, enum, subType:int-->0
        </statusCode>
        <statusString>
          <!--ro, req, enum, subType:string-->OK
        </statusString>
        <subStatusCode>
          <!--ro, req, string, range:[1,64]-->ok
        </subStatusCode>
        <description>
          <!--ro, opt, string, range:[0,1024]-->test
        </description>
      </Status>
    </StatusList>
  </AttachInfo>
</ResponseStatus>
```

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19.1.3.7 Stop Running Specified Application Program

Request URL

PUT /ISAPI/Custom/OpenPlatform/App/<AppID>/stop?engineID=<engineID>&channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
AppID	string	App ID, length is 1 to 28 bytes, which is unique and composed of strings containing numbers and letters. For details, see the App ID description in Terminology and Definitions in the HEOP API calling flow.
engineID	string	Engine ID
channelID	string	Video channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
  <AttachInfo>
    <!--ro, opt, array, subType:object-->
    <StatusList>
      <!--ro, req, object-->
      <Status>
        <!--ro, req, object-->
        <index>
          <!--ro, req, int-->0
        </index>
        <statusCode>
          <!--ro, req, enum, subType:int-->0
        </statusCode>
        <statusString>
          <!--ro, req, enum, subType:string-->OK
        </statusString>
        <subStatusCode>
          <!--ro, req, string, range:[1,64]-->ok
        </subStatusCode>
        <description>
          <!--ro, opt, string, range:[0,1024]-->test
        </description>
      </Status>
    </StatusList>
  </AttachInfo>
</ResponseStatus>
```

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19.1.3.8 Get Specified Application Program Logs

Request URL

GET /ISAPI/Custom/OpenPlatform/App/<AppID>/syslog?engineID=<engineID>

Query Parameter

Parameter Name	Parameter Type	Description
AppID	string	App ID, and the length is 1 to 28 bytes. It is unique and composed of strings containing numbers and letters. For details, see the App ID description in Terminology and Definitions in the HEOP API calling flow.
engineID	string	Engine ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<logInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, log information, attr:version{req, string, protocolVersion}-->
  <logPath>
    <!--ro, opt, string, saving path of the log files, range:[0,256], desc:log files are generated from the compressed text file with the suffix tar.gz--
>test
  </logPath>
</logInfo>
```

19.1.3.9 Get Device Engine Status

Request URL

GET /ISAPI/Custom/OpenPlatform/APP/engines?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "LoadingInfoList": [
    /*ro, req, array, the list of the loading information of the engine, subType:object*/
        {
            "engineID": 1,
            /*ro, req, int, engine ID, desc:used to distinguish different engine No.*/
            "engineStatus": 1,
            /*ro, req, enum, engine status, subType:int, desc:0 (not loaded), 1 (loaded & runs well), 2 (running exception), 10 (initializing), 11 (engine
exception), 12 (initialization exception), 13 (algorithm exception)*/
            "platform": 1,
            /*ro, req, enum, the platform where the algorithm runs, subType:int, desc:1 (TX1), 2 (P4), 3 (3559), 4 (3519), 5 (3516)*/
            "platformName": "K81",
            /*ro, req, string, the platform where the algorithm runs, desc:which is product-defined and used for display*/
            "PackageInfo": [
            /*ro, opt, array, upgrade information of the application program package, subType:object*/
                {
                    "packageName": "test",
                    /*ro, opt, string, the name of the application program package, range:[0,128], desc:which is used to distinguish the application program
information (fill in the name after loading)*/
                    "appPackageUUID": "abcdefg1234567890",
                    /*ro, opt, string, the UUID of the application program package, range:[0,128]*/
                    "runStatus": "true"
                    /*ro, opt, enum, the running status of the application program package:, subType:string, desc:“true” (running), “false” (stopped)*/
                }
            ],
            "AbilityInfos": {
            /*ro, opt, object, the capability supported by device*/
                "cardType": "DSP",
                /*ro, req, enum, computer power unit type, subType:string, desc:“DSP”, “NNIE”*/
                "totalGFLOPS": 0.1,
                /*ro, req, float, total computer power value, range:[0.1,100.0]*/
                "surplusGFLOPS": 0.1
                /*ro, req, float, remaining computer power value, range:[0.1,100.0]*/
            },
            "BaseInfo": {
            /*ro, opt, object*/
                "totalMemory": 1000,
                /*ro, opt, int, step:1, unit:MB*/
                "surplusMemory": 500,
                /*ro, opt, int, step:1, unit:MB*/
                "totalFlash": 50,
                /*ro, opt, int, step:1, unit:MB*/
                "surplusFlash": 25,
                /*ro, opt, int, step:1, unit:MB*/
                "CPUUtility": 80
                /*ro, opt, int, range:[0,100], step:1*/
            }
        }
    ]
}
```

19.1.3.10 Get the HEOP capability supported by device

Request URL

GET /ISAPI/Custom/OpenPlatform/capabilities

Query Parameter

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None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OpenPlatformCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, HEOP capability set supported by device, attr:version{req, string, protocolVersion}-->
  <isSupportOpenPlatform>
    <!--ro, req, bool, whether the device supports HEOP (Hikvision Embedded Open Platform)-->true
  </isSupportOpenPlatform>
  <maxAPPNum>
    <!--ro, req, int, the maximum number of third-party application programs allowed to be installed-->0
  </maxAPPNum>
  <AppInstallConfig>
    <!--ro, opt, string, the configuration capability set of installing application programs, desc:If this node is returned, it indicates that the device
supports configuring installation rules, otherwise it does not support-->test
    <downloadUrl min="0" max="128">
      <!--ro, opt, string, the URI for downloading the application program, range:[0,128], attr:min{req, int},max{req, int}, desc:the complete URI-->test
    </downloadUrl>
    <installationTime>
      <!--ro, opt, datetime, installation time, dep:and,{$.OpenPlatformCap.AppInstallConfig.method,eq,timing}-->1970-01-01T00:00:00+08:00
    </installationTime>
    <method opt="immediate,timing">
      <!--ro, opt, enum, installation method, subType:string, attr:opt{req, string},
desc:“immediate”(instant downloading and installation),“timing” (scheduled
downloading and installation)-->immediate
    </method>
  </AppInstallConfig>
  <AppManage>
    <!--ro, opt, object, the capability set of managing application programs, desc:If this node is returned, it indicates that the device supports managing
application programs, otherwise it does not support-->
    <manualCmd opt="download,clear">
      <!--ro, opt, enum, manual command, subType:string, attr:opt{req, string}, desc:“download” (download), “clear” (clear)-->download
    </manualCmd>
  </AppManage>
  <isNotSupportAppUpload>
    <!--ro, opt, bool, whether it does not support uploading the application program to the device-->true
  </isNotSupportAppUpload>
  <isNotSupportAppUploadStatus>
    <!--ro, opt, bool, whether it does not support getting the application program uploading progress-->true
  </isNotSupportAppUploadStatus>
  <isNotSupportAppDelete>
    <!--ro, opt, bool, whether it does not support deleting the application program-->true
  </isNotSupportAppDelete>
  <isSupportAppUserDataDelete>
    <!--ro, opt, bool, whether it supports deleting the user data of the application program, desc:If supports, userData can be loaded in
/ISAPI/Custom/OpenPlatform/App/<AppID>/delete, such as /ISAPI/Custom/OpenPlatform/App/<AppID>/delete?userData=true, which indicates that deleting the user
data of the application program while deleting the application program-->true
  </isSupportAppUserDataDelete>
  <isNotSupportAppStart>
    <!--ro, opt, bool, whether it does not support starting the application program-->true
  </isNotSupportAppStart>
  <isNotSupportAppList>
    <!--ro, opt, bool, whether it does not support getting the list of application programs-->true
  </isNotSupportAppList>
  <isNotSupportUploadLicenseKey>
    <!--ro, opt, bool-->true
  </isNotSupportUploadLicenseKey>
  <isNotSupportPermission>
    <!--ro, opt, bool-->true
  </isNotSupportPermission>
  <isNotSupportDisclaimer>
    <!--ro, opt, bool-->true
  </isNotSupportDisclaimer>
  <isNotSupportPageItem>
    <!--ro, opt, bool-->true
  </isNotSupportPageItem>
  <isNotSupportSyslog>
    <!--ro, opt, bool-->true
  </isNotSupportSyslog>
  <isSupportServerAddress>
    <!--ro, opt, bool-->true
  </isSupportServerAddress>
  <isSupportAlgInfo>
    <!--ro, opt, bool, whether it supports getting the information about the version of the HEOP software library-->true
  </isSupportAlgInfo>
  <isSupportAppBindMultiEngines>
    <!--ro, opt, bool, whether it supports binding one application program with multiple engines, desc:returning "true" indicates that it supports this
function, returning "false" or not returning this node indicates that it does not support this function-->true
  </isSupportAppBindMultiEngines>
  <AppChannels>
    <!--ro, opt, object, the capability of binding the application program with channel(s), desc:If this node appears in the message, it indicates that the
device supports the function: : /ISAPI/Custom/OpenPlatform/searchAppChannels?format=json, /ISAPI/Custom/OpenPlatform/modifyAppChannels?format=json-->
    <channel min="0" max="128">
      <!--ro, opt, int, the capability of binding the application with video channel(s), attr:min{req, int, range:[0,256]},max{req, int, range:[0,256]}-->1
    </channel>
```

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```
      <maxChannelNum>
        <!--ro, opt, int, the maximum number of channels allowed to be bound with the application program, range:[0,256]-->1
      </maxChannelNum>
    </AppChannels>
    <isSupportEngineBindMultiApp>
      <!--ro, opt, bool, whether it supports binding one engine with multiple application programs, desc:returning "true" indicates that it supports this
function, returning "false" or not returning this node indicates that it does not support this function-->true
    </isSupportEngineBindMultiApp>
    <isSupportReboot>
      <!--ro, opt, bool, whether it supports restarting the specified application program-->true
    </isSupportReboot>
    <isSupportUploadLicenseKey>
      <!--ro, opt, bool, whether it supports uploading License file, desc:corresponding URL: /ISAPI/Custom/OpenPlatform/uploadLicenseKey-->true
    </isSupportUploadLicenseKey>
    <isSupportHEOPCustomEvent>
      <!--ro, opt, bool-->true
    </isSupportHEOPCustomEvent>
    <isSupportHEOPFormattedEvent>
      <!--ro, opt, bool-->true
    </isSupportHEOPFormattedEvent>
    <isSupportSearchHEOPCustomEvent>
      <!--ro, opt, bool-->true
    </isSupportSearchHEOPCustomEvent>
    <isSupportSearchHEOPFormattedEvent>
      <!--ro, opt, bool-->true
    </isSupportSearchHEOPFormattedEvent>
    <isSupportEngineSyslog>
      <!--ro, opt, bool-->true
    </isSupportEngineSyslog>
    <isSupportAPPPortMap>
      <!--ro, opt, bool-->true
    </isSupportAPPPortMap>
    <isSupportAPPAutoReboot>
      <!--ro, opt, bool-->true
    </isSupportAPPAutoReboot>
  </OpenPlatformCap>
```

19.1.3.11 Get selection status of open platform disclaimer

Request URL

GET /ISAPI/Custom/OpenPlatform/disclaimer?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "Disclaimer": {
    /*ro, req, object*/
        "enabled":  true
        /*ro, req, bool, whether the disclaimer is selected*/
    }
}
```

19.1.3.12 Select the disclaimer

Request URL

PUT /ISAPI/Custom/OpenPlatform/disclaimer?format=json

Query Parameter

None

Request Message

```
{
    "Disclaimer": {
    /*req, object*/
        "enabled":  true
        /*req, bool, whether the disclaimer is selected*/
    }
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

19.1.3.13 Get Application Program Page Information

Request URL

GET /ISAPI/Custom/OpenPlatform/pageItem

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<pageItemList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, page information list of application program, subType:object, attr:version{req, string, protocolVersion}-->
  <pageItem>
    <!--ro, opt, object, page information of application program-->
    <id>
      <!--ro, req, int, page No. of application program, range:[0,128], step:1-->1
    </id>
    <appName>
      <!--ro, req, string, the third-party application program name, range:[0,128]-->test
    </appName>
    <htmlPath>
      <!--ro, opt, string, html page path of the third-party application program, range:[0,128]-->test
    </htmlPath>
    <introducePath>
      <!--ro, req, string, the introduction path of the third-party application program name, range:[0,128]-->test
    </introducePath>
    <logoPath>
      <!--ro, opt, string, picture path of the third-party application program logo, range:[0,128]-->test
    </logoPath>
    <copyRight>
      <!--ro, opt, string, copyright statement of the third-party application program, range:[0,128]-->test
    </copyRight>
    <hostname>
      <!--ro, req, string, home page domain of the third-party application program, range:[0,128]-->test
    </hostname>
  </pageItem>
</pageItemList>
```

19.1.3.14 Get the details of incremental upgrade capability

Request URL

GET /ISAPI/Custom/OpenPlatform/patchUpgrade/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "PatchUpgradeCap": {
    /*ro, req, object, incremental upgrade capability*/
        "isSupportAsyncUpload":  true,
        /*ro, opt, bool, whether it supports managing the asynchronous upgrade of application program, desc:corresponding URI:
/ISAPI/Custom/OpenPlatform/uploadApp/async?format=json; /ISAPI/Custom/OpenPlatform/uploadAppStatus/async?format=json*/
        "isSupportManagement":  true,
        /*ro, opt, bool, whether it supports managing the running of application program, desc:corresponding URI:
/ISAPI/Custom/OpenPlatform/appPackage/<UUID>/start; /ISAPI/Custom/OpenPlatform/appPackage/<UUID>/stop; /ISAPI/Custom/OpenPlatform/appPackage/<UUID>*/
        "isSupportEngine":  true
        /*ro, opt, bool, whether it supports the engine that manages the running of application program, desc:corresponding URI:
/ISAPI/Custom/OpenPlatform/APP/engines?format=json; /ISAPI/Custom/OpenPlatform/APP/loading/<engineID>?format=json;
/ISAPI/Custom/OpenPlatform/APP/unloading/<engineID>?format=json*/
    }
}
```

19.1.3.15 Get HEOP Remaining Resources

Request URL

GET /ISAPI/Custom/OpenPlatform/resource

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<resource xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--ro, req, object, remaining resource information, attr:version{req, string, protocolVersion}-->
  <totalMemory>
    <!--ro, opt, int, the size of all available memory for open platform, unit:MB-->0
  </totalMemory>
  <restMemory>
    <!--ro, opt, int, the size of remaining available memory for open platform, unit:MB-->0
  </restMemory>
  <totalFlash>
    <!--ro, opt, int, the size of all available memory for open platform, unit:MB-->0
  </totalFlash>
  <restFlash>
    <!--ro, opt, int, the size of remaining available flash for open platform, unit:MB-->0
  </restFlash>
  <totalCalculation>
    <!--ro, opt, float, the size of all available computing power for open platform, desc:The number is accurate to one decimal place (e.g.,0.1).-->0.000
  </totalCalculation>
  <restCalculation>
    <!--ro, opt, float, the size of remaining available computing power for open platform, desc:The number is accurate to one decimal place (e.g.,0.1).--
>0.000
  </restCalculation>
  <totalIntelligentMemory>
    <!--ro, opt, int, the size of all available intelligent memory for open platform, unit:MB, desc:The physical address of intelligent memory is
continuous, which is used to speed up processing.-->0
  </totalIntelligentMemory>
  <restIntelligentMemory>
    <!--ro, opt, int, the size of remaining available intelligent memory for open platform, unit:MB-->0
  </restIntelligentMemory>
</resource>
```

19.1.3.16 Search for the list of channels that are bound with the application program

Request URL

POST /ISAPI/Custom/OpenPlatform/searchAppChannels?format=json

Query Parameter

None

Request Message

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dojojin@gmail.com

```
{
    "appID":  "1",
    /*req, string, the unique ID of an application program, range:[1,64]*/
    "engineID":  "1"
    /*opt, string, range:[1,64]*/
}
```

Response Message

```
{
    "channelList": [1, 2, 3, 4]
    /*ro, req, array, list of channels, subType:int, range:[0,128], desc:list of channels*/
}
```

19.1.3.17 Get Asynchronous Uploading Progress of Application Program Package

Request URL

GET /ISAPI/Custom/OpenPlatform/uploadAppStatus/async/<AppID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
AppID	string	UUID of the App package, with a maximum length of 128. This ID can be used to determine the uniqueness of the upgrade package, and it is consistent with the appPackageUUID during asynchronous loading.

Request Message

None

Response Message

```
{
    "UploadAppStatus": {
    /*ro, req, object, the upload status of the application program*/
        "percent":  100,
        /*ro, req, int, the upload progress of the application program, range:[0,100], desc:in the format of percentage*/
        "status":  1,
        /*ro, req, enum, upload status, subType:int, desc:0 (exception occurred), 1 (uploading), 2 (uploaded)*/
        "statusExceptionDetails":  "appFileCorruption"
        /*ro, opt, string, dep:and,{$.UploadAppStatus.status,ue,0}*/
    }
}
```

19.1.3.18 Get Synchronous Upgrading Progress of Application Program Full Package

Request URL

GET /ISAPI/Custom/OpenPlatform/uploadAppStatus?taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID: When the taskID is specified in POST /ISAPI/Custom/OpenPlatform/uploadAppBinaryPackets?format=json, it indicates asynchronous uploading of the App. The specified taskID is used to search for the import progress of the App.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<uploadAppStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--ro, req, object, uploading status, attr:version{req, string, protocolVersion}-->
  <uploading>
    <!--ro, req, bool, whether the application program is uploading-->true
  </uploading>
  <percent>
    <!--ro, req, int, uploading progress of the application program, range:[0,100], step:1-->1
  </percent>
  <uploadStatus>
    <!--ro, opt, enum, subType:string-->nonexecution
  </uploadStatus>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, opt, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, opt, string-->OK
  </subStatusCode>
</uploadAppStatus>
```

19.1.3.19 Delete the third-party server information

Request URL

DELETE /ISAPI/OrgPreview/Channels/<channelID>/Server

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.1.3.20 Set address and port parameters of third-party server

Request URL

PUT /ISAPI/OrgPreview/Channels/<channelID>/Server

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Server xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--req, object, the address and port parameters of third-party server, attr:version{req, string, protocolVersion}-->
  <addressingFormatType>
    <!--req, enum, address type, subType:string, desc:ipaddress; hostname;-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--opt, string, domain name, range:[1,64]-->time.windows.com
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,32]-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--opt, int, port No., range:[1,65535], desc:port No.-->74
  </portNo>
</Server>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.3.21 Get the address and port No. of the third-party server

Request URL

GET /ISAPI/OrgPreview/Channels/<channelID>/Server

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Server xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--ro, req, object, the address and port number of the third-party server, attr:version{req, string, protocolVersion}-->
  <addressingFormatType>
    <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, host name, range:[1,64]-->time.windows.com
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,32]-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128], desc:IPv6 address (not supported)-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, Port No., range:[1,65535], desc:Port No.-->74
  </portNo>
</Server>
```

19.1.4 Online Upgrading Management

19.1.4.1 Get the device online upgrade capability

Request URL

GET /ISAPI/System/onlineUpgrade/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, online upgrade, attr:version{req, string, protocolVersion}-->
  <firmwareNum max="10">
    <!--ro, req, int, number of supported online upgrade packages, attr:max{req, int}-->1
  </firmwareNum>
  <firmwareCode max="256">
    <!--ro, req, int, maximum length of firmware ID, attr:max{req, int}-->1
  </firmwareCode>
  <firmwareVersion max="64">
    <!--ro, req, int, maximum length of version, attr:max{req, int}-->1
  </firmwareVersion>
  <firmwareCodeNumOnce max="10">
    <!--ro, req, int, maximum number of obtained firmware IDs, attr:max{req, int}-->1
  </firmwareCodeNumOnce>
  <upgradePercent min="0" max="100">
    <!--ro, req, int, upgrade progress capability in percentage, attr:min{req, int},max{req, int}-->1
  </upgradePercent>
  <Version>
    <!--ro, opt, object, upgrade package version information-->
    <newVersion max="64">
      <!--ro, req, string, version information of the new upgrade package, attr:max{req, int}-->test
    </newVersion>
    <changeLog max="64">
      <!--ro, req, string, log of the new upgrade package, attr:max{req, int}-->test
    </changeLog>
  </Version>
  <rebootAfterUpgrade>
    <!--ro, opt, enum, upgrade options after device reboot, subType:string, desc:"auto" (auto upgrade), "manual" (manual upgrade)-->auto
  </rebootAfterUpgrade>
  <DeviceParameter>
    <!--ro, opt, object, online upgrade parameters-->
    <isSupportAutoDownloadPackage>
      <!--ro, opt, bool, whether it supports automatic download of upgrade package-->true
    </isSupportAutoDownloadPackage>
    <notSupportAutoUpgrade>
      <!--ro, opt, bool, whether it no longer supports automatic download of upgrade package and automatic upgrade, desc:corresponding URL: PUT
/ISAPI/System/onlineUpgrade/upgrade-->true
    </notSupportAutoUpgrade>
    <isSupportTimingUpgrade>
      <!--ro, opt, bool, whether it supports scheduled upgrade-->true
    </isSupportTimingUpgrade>
  </DeviceParameter>
  <ManualDownloadPackage>
    <!--ro, opt, object, manual download of upgrade package-->
    <supportOperation opt="start,cancel,pause,resume">
```

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```
      <!--ro, opt, enum, supported operations, subType:string, attr:opt{req, string}, desc:"start", "cancel", "pause", "resume"-->start
    </supportOperation>
  </ManualDownloadPackage>
  <isSupportIgnoreCurrentVersion>
    <!--ro, opt, bool, whether it supports ignoring the current version-->true
  </isSupportIgnoreCurrentVersion>
  <UpgradePackageTask>
    <!--ro, opt, object, whether the device supports applying upgrade tasks, desc:if this node is returned, it indicates that the following URLs are also
supported: /ISAPI/System/onlineUpgrade/task?format=json, /ISAPI/System/onlineUpgrade/status/<ID>-->
    <isSupportUpgradeModules>
      <!--ro, opt, bool, whether the device supports upgrading sub modules-->true
    </isSupportUpgradeModules>
    <isSupportUpgradeChannels>
      <!--ro, opt, bool, whether the device supports upgrading sub channels-->true
    </isSupportUpgradeChannels>
    <isSupportDownloadStrategy>
      <!--ro, opt, bool, whether the device supports strategy parameters of downloading upgrade package, desc:corresponding filed is "downloadStrategy" of
URL (/ISAPI/System/onlineUpgrade/task?format=json)-->true
    </isSupportDownloadStrategy>
    <isSupportDelayedUpgradeByDateTime>
      <!--ro, opt, bool, whether the device supports delaying the upgrade by time, desc:corresponding filed is "delayedUpgradeDateTime" of URL
(/ISAPI/System/onlineUpgrade/task?format=json)-->true
    </isSupportDelayedUpgradeByDateTime>
    <upgradePackageAddressType opt="URL, ftpIPAddress, ftpDomainAddress">
      <!--ro, opt, string, attr:opt{req, string}-->URL
    </upgradePackageAddressType>
  </UpgradePackageTask>
  <isSupportPatrolInspection>
    <!--ro, opt, bool, whether the device supports patrol inspection-->true
  </isSupportPatrolInspection>
  <isSupportBatchUpgrade>
    <!--ro, opt, bool, whether the device supports batch online upgrades-->true
  </isSupportBatchUpgrade>
  <isSupportCancelUpgrade>
    <!--ro, opt, bool, whether the device supports canceling online upgrade-->true
  </isSupportCancelUpgrade>
  <isSupportPackagePause>
    <!--ro, opt, bool, whether the device supports pausing upgrade package download, desc:/ISAPI/System/onlineUpgrade/PackagePause?format=json-->true
  </isSupportPackagePause>
  <isSupportPackageResume>
    <!--ro, opt, bool, whether the device supports resuming upgrade package download, desc:/ISAPI/System/onlineUpgrade/PackageResume?format=json-->true
  </isSupportPackageResume>
  <isSupportCancelUpgradeByTaskID>
    <!--ro, opt, bool, whether the device supports canceling online upgrade by task ID, desc:corresponding to the field assignByTaskID of operationType in
URL PUT  /ISAPI/System/onlineUpgrade/CancelUpgrade?format=json-->true
  </isSupportCancelUpgradeByTaskID>
</OnlineUpgradeCap>
```

19.1.4.2 Get the online upgrade progress of device

Request URL

GET /ISAPI/System/onlineUpgrade/status

Query Parameter

None

Request Message

None

Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, online upgrade status, attr:version{req, string, protocolVersion}-->
  <status>
    <!--ro, req, enum, online upgrade status, subType:string, desc:"notUpgrade" (not upgraded), "upgrading", "successful" (upgraded), "LanguageMismatch"
(language mismatch), "writeFlashError" (failed to write to flash), "packageTypeMismatch" (upgrade package type mismatch), "packageVersionMismatch" (upgrade
version mismatch), "netUnreachable" (network disconnected), "unknownError" (unknown error), "engineVersionMismatch" (mismatch of upgrade package engine
version)-->notUpgrade
  </status>
  <percent>
    <!--ro, req, int, current upgrade progress in percentage, range:[0,100]-->1
  </percent>
  <onlineUpdateDevInfoList>
    <!--ro, opt, object, online upgrade device information, desc:containing the online upgrade status of sub devices, sub modules, or sub channels-->
    <onlineUpdateDevInfo>
      <!--ro, opt, object, online upgrade device information-->
      <moduleType>
        <!--ro, req, enum, module type, subType:string, desc:"keypad", "detector", "cardReader" (card reader), "siren" (sounder), "relay", "transmitter"
(transmitter), "remoteCtrl" (remote control),"repeater" (repeater), "SSD", "extensionModule" (extension module)-->keypad
      </moduleType>
      <moduleID>
        <!--ro, req, int, module ID, range:[1,128], step:1-->1
      </moduleID>
      <model>
        <!--ro, opt, string, model, range:[1,32]-->test
      </model>
      <status>
        <!--ro, opt, enum, upgrade status, subType:string, desc:"allowance" (upgradeable, the upgrade command is not triggered), "ready" (waiting for
upgrade, the upgrade command has been triggered), "upgriding" (upgrading), "success" (upgraded), "failed" (upgrade failed)-->allowance
      </status>
      <percent>
        <!--ro, opt, int, current upgrade percentage, range:[0,100]-->1
      </percent>
      <updateTime>
        <!--ro, opt, datetime, upgrade completion time-->2004-05-03T17:30:08+08:00
      </updateTime>
    </onlineUpdateDevInfo>
  </onlineUpdateDevInfoList>
</OnlineUpgradeStatus>
```

19.1.4.3 Get all of the online upgrade task progress

Request URL

GET /ISAPI/System/onlineUpgrade/status/task?format=json

Query Parameter

None

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO., LTD.
dojojin@gmail.com

```
{
    "onlineUpgradeStatusList": [
    /*ro, req, array, online upgrade status list, subType:object*/
        {
            "status":  "notUpgrade",
            /*ro, req, enum, upgrade status, subType:string, desc:“notUpgrade”, “upgrading”, “successful” (upgrading succeeded), “languageMismatch”,
“writeFlashError”, “packageTypeMismatch”, “packageVersionMismatch”, “netUnreachable” (network disconnected), “unknownError”, “waitManualTrigger”*/
            "percent":  0,
            /*ro, req, int, upgrade progress percentage, range:[0,100]*/
            "taskID":  "test",
            /*ro, opt, string, task ID, range:[1,64], desc:generated when creating upgrade tasks, related URI: /ISAPI/System/onlineUpgrade/task?
format=json*/
            "moudleType":  "stationHostAndScreen",
            /*ro, opt, enum, component type, subType:string, desc:"stationHostAndScreen" (dock station upgrade package, including host and screen),
"stationHost" (dock station host), "spellingControllerMainAndSubBoard" (main board and sub board of splicing controller), "spellingControllerMainBoard"
(main board of splicing controller), "spellingControllerSubBoard" (sub board of splicing controller), "bodyCamera" (body camera)*/
            "packageName":  "ZCS-H1_target_V1.0.1_build201123.zip",
            /*ro, opt, string, upgrade package name, range:[1,128]*/
            "taskTime":  "1970-01-01T00:00:00+08:00",
            /*ro, opt, datetime, applying upgrade task time*/
            "taskParam": {
            /*ro, opt, object, task parameters, desc:corresponding URL: POST /ISAPI/System/onlineUpgrade/task?format=json*/
                "URL":  "test",
                /*ro, req, string, upgrade package address, range:[1,1024]*/
                "delayedUpgrade":  true,
                /*ro, opt, bool, whether to delay the upgrade, desc:if this node is not applied, it indicates that the upgrade will be started immediately;
if this node is applied and the value is false, it indicates that the upgrade will be delayed*/
                "delayedUpgradeDateTime":  "1970-01-01T00:00:00+08:00"
                /*ro, opt, datetime, time of starting the delayed upgrade, dep:or,{$.PackageDescription.delayedUpgrade,eq,false}*/
            }
        }
    ]
}
```

19.1.4.4 Apply the task of online upgrade

Request URL

POST /ISAPI/System/onlineUpgrade/task?format=json&confirmEnabled=<confirmEnabled>

Query Parameter

Parameter Name	Parameter Type	Description
confirmEnabled	string	--

Request Message

dojojin@mail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
{
    "PackageDescription": {
    /*req, object, description of the upgrade package*/
        "name":  "DSJ_N3_GML_DEBUG_STD_1.2.0_190524.zip",
        /*opt, string, upgrade package name, range:[1,256]*/
        "URL":  "test",
        /*req, string, URL of the upgrade package, range:[1,1024], desc:the task ID is contained in the URL. The uploading and downloading of the upgrade package is based on the object storage protocol (version 2.0). If this node is null, the FTP address will take effect. If both the URL and FTP address are configured, the URL has priority over FTP address*/
        "MD5":  "test",
        /*opt, string, range:[0,256]*/
        "moduleType":  "cardReader",
        /*opt, enum, module type, subType:string, desc:“cardReader” (card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm programma of card reader), "dockStationHost" (dock station host), "dockStationScreen" (dock station screen), "stationHostAndScreen" (dock station host and screen)*/
        "moduleID":  1,
        /*opt, int, module ID*/
        "spellingControllerSubBoardList": [
        /*opt, array, splicing controller sub board list, subType:object, range:[1,1024], dep:or,{$.PackageDescription.moduleType,eq,spellingControllerMainAndSubBoard},{$.PackageDescription.moduleType,eq,spellingControllerSubBoard}, desc:it is valid when the value of the node moduleType is "spellingControllerMainAndSubBoard" or "spellingControllerSubBoard". If it is not configured, it indicates that all the sub boards will be updated*/
            {
                "subBoardID":  1,
                /*opt, int, sub board ID, range:[1,1024]*/
                "ipV4Address":  "test"
                /*ro, opt, string, IPv4, range:[1,16], desc:when an exception occurs on the main board, the sub board can be updated through the IP address. The default sub board interface will be used*/
            }
        ],
        "channel":  1,
        /*opt, int, channel No., desc:this node is used for upgrading the IP channel of the NVR*/
        "childDevID":  "3434645611",
        /*opt, string, range:[9,32]*/
        "customTaskID":  "test",
        /*opt, string, user-defined task ID, range:[1,128], desc:if this node exists, it will be returned in the result*/
        "delayedUpgrade":  true,
        /*opt, bool, whether to delay upgrading the package, desc:if this node is not applied, it indicates that the package will be upgraded immediately; if this node is applied and the value is false, it indicates that the upgrade will be delayed. User can decide the time to upgrade the package of the dock station or body camera if the upgrade is delayed*/
        "delayedUpgradeDateTime":  "1970-01-01T00:00:00+08:00",
        /*opt, datetime, time of starting the delayed upgrade, dep:or,{$.PackageDescription.delayedUpgrade,eq,false}, desc:the related capability is "isSupportDelayedUpgradeByDateTime" in GET  /ISAPI/System/onlineUpgrade/capabilities*/
        "ftpAddress":  "test",
        /*opt, string, FTP address of the upgrade package, desc:this node will take effect when URL is null and the format is: ftp://[username]:[password]@[IP address]:[port No.]*/
        "downloadStrategy": {
        /*opt, object, device's upgrade package downloading strategy parameters*/
            "sequelEnabled":  true,
            /*opt, bool, whether to enable ANR (Automatic Network Replenishment), desc:if enabled, when downloading is interrupted, the device will try downloading the package again; if disabled, the default strategy of the device will be adopted*/
            "sequelTimeout":  3000,
            /*opt, int, timeout of a single ANR attempt, step:1, unit:ms, desc:when a single ANR attempt exceeds the timeout period but the download does not continue, the ANR attempt will be recognized as failed*/
            "maxRetryTimes":  5
            /*opt, int, ANR maximum attempt times, desc:when the device ANR exceeds the maximum attempt times but the download does not continue, the download and the upgrade will be recognized as failed*/
        },
        "upgradePackageURLType":  "full"
        /*opt, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1-succeeded, other value-error occurred. This field is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"OK" (succeeded), other value (error occurred). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded), other value (error occurred). This field is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error message, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "test"
    /*ro, opt, string, ID of the upgrade task, desc:it can be used to get the status and progress of the task*/
}
```

19.1.4.5 Upload online upgrade status event

EventType:onlineUpgradeStatus

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:the event triggering frequency. The pulse event is an event which is triggered continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object recognition occurs, and the recognition of each object will trigger the event once, e.g., the face capture. When the pulse events are triggered, this node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously according to the device detection frequency; if it changes to another source, then the counting will start again. It can be regarded as the event triggering frequency*/
    "eventType":  "onlineUpgradeStatus",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid event or event starts), "inactive" (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "Online Upgrade Status",
    /*ro, req, string, event description, desc:online upgrade status event*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID, desc:PUID. This field must be returned when ISAPI event information is transmitted via ISUP. It is the same as deviceID of URI (/ISAPI/System/Network/Ehome)*/
    "serialNumber":  "test",
    /*ro, opt, string, device short serial No., range:[1,9]*/
    "status":  "failure",
    /*ro, req, enum, upgrade status, subType:string, desc:"failure" (upgrade failed), "success" (upgrade succeeded), "upgrading"*/
    "failedReason":  "languageMismatch",
    /*ro, opt, enum, cause of failure, subType:string, desc:"languageMismatch" (language mismatch), "writeFlashError" (failed to write to flash), "packageTypeMismatch" (upgrade package type mismatch), "packageVersionMismatch" (upgrade version mismatch), "netUnreachable" (network disconnected), "unknownError" (unknown error)*/
    "moduleType":  "cardReader",
    /*ro, opt, enum, device component type, subType:string, desc:"cardReader" (card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm programma of card reader), "dockStationHost" (dock station host), "dockStationScreen" (dock station screen)*/
    "moduleID":  1,
    /*ro, opt, int, device component No., desc:it depends on the node moduleType*/
    "channel":  1,
    /*ro, opt, int, channel No., desc:it is used for IP channel (to upgrade digital channel managed by NVR)*/
    "taskID":  "test",
    /*ro, opt, string, task ID, range:[1,64], desc:generated when creating upgrade tasks, related URI: /ISAPI/System/onlineUpgrade/task?format=json*/
    "process":  1
    /*ro, opt, int, upgrade progress, range:[1,100], step:1, unit:%, dep:or,{$.status,eq,upgrading}, desc:progress percentage*/
}
```


19.1.5 Serial Port Management

19.1.5.1 Get the serial port capability of the device

Request URL

GET /ISAPI/System/Serial/capabilities

Query Parameter

None

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<SerialCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, range of RS-485 serial port numbers supported by the device, attr:version{req, string, protocolVersion}-->
  <rs485PortNums>
    <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->0
  </rs485PortNums>
  <supportRS232Config>
    <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
  </supportRS232Config>
  <rs422PortNums>
    <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->0
  </rs422PortNums>
  <rs232PortNums>
    <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->0
  </rs232PortNums>
  <rs485WorkMode opt="Led, CaptureTrigger,transparent">
    <!--ro, opt, string, range of RS-485 serial port No. supported by the device, attr:opt{req, string}-->test
  </rs485WorkMode>
  <rs232SerialNumber opt="1,2,3">
    <!--ro, opt, int, range of RS-232 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs232SerialNumber>
  <rs485SerialNumber opt="4,5,6">
    <!--ro, opt, int, range of RS-485 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs485SerialNumber>
  <isSupportAuthenticationService>
    <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
  </isSupportAuthenticationService>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports configuring the information about the serial port device-->true
  </isSupportDeviceInfo>
  <isSupportSearchDeviceInfoRelations>
    <!--ro, opt, bool, whether the device supports searching the linkage information of the serial port device-->true
  </isSupportSearchDeviceInfoRelations>
</SerialCap>
```

19.1.5.2 Get the capability of a specific serial port

Request URL

GET /ISAPI/System/Serial/ports/<portID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--

Request Message

None

Response Message

dejojin@gmail.com VISION AND SECURITY SYSTEM CO., LTD.

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port No., attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID-->0
  </id>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <serialPortType opt="RS485,RS422,RS232">
    <!--ro, opt, string, serial port type, attr:opt{req, string}-->RS485
  </serialPortType>
  <duplexMode opt="half,full">
    <!--ro, opt, string, duplex mode of the serial port, attr:opt{req, string}-->half
  </duplexMode>
  <baudRate opt="1200,2400,4800,9600,19200,38400,57600,115200">
    <!--ro, opt, int, attr:opt{req, string}-->1200
  </baudRate>
  <dataBits opt="6,7,8">
    <!--ro, opt, int, attr:opt{req, string}-->6
  </dataBits>
  <parityType opt="none,even,odd,mark,space">
    <!--ro, opt, string, attr:opt{req, string}-->none
  </parityType>
  <stopBits opt="1,1.5,2">
    <!--ro, opt, string, stop bit, attr:opt{req, string}-->1
  </stopBits>
  <flowCtrl opt="none,software,hardware">
    <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
  </flowCtrl>
</SerialPort>
```

19.1.5.3 Close the transmission channel

Request URL

PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/close?type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
channelID	string	--
type	string	Data (purpose) type "general" (general data transparent transmission, the device sends the transparent transmission data via serial port, regardless of whether there is an external component connected to the serial port), "sensorCtrl" (sensor data transparent transmission control, when the device sends the transparent transmission data via serial port, it will also stop collecting sensor data, and compared with general type, the device will perform specific control and management on the sensors externally connected to the serial port)

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.1.5.4 Open the transmission channel

Request URL

PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/open?type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
channelID	string	Camera ID
type	string	Data (purpose) type. "general" (general data transparent transmission, the device sends the transparent transmission data via serial port, regardless of whether there is an external component connected to the serial port), "sensorCtrl" (sensor data transparent transmission control, when the device sends the transparent transmission data via serial port, it will also stop collecting sensor data, and compared with general type, the device will perform specific control and management on the sensors externally connected to the serial port)

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO., LTD. dongjin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.1.5.5 Set the parameters of a specific serial port supported by the device

Request URL

PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	--
childDevID	string	--
deviceIndex	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, serial port ID-->0
  </id>
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
  <serialPortType>
    <!--opt, enum, serial port type: "RS485","RS422","RS232", subType:string, desc:serial port type: "RS485","RS422","RS232"-->RS485
  </serialPortType>
  <duplexMode>
    <!--opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"-->half
  </duplexMode>
  <baudRate>
    <!--opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--opt, int-->6
  </dataBits>
  <parityType>
    <!--opt, enum, parity type, subType:string, desc:"none, even, odd, mark, space"-->none
  </parityType>
  <stopBits>
    <!--opt, string, stop bit: "1,1.5,2"-->1
  </stopBits>
  <flowCtrl>
    <!--opt, enum, "none,software,hardware", subType:string, desc:"none, software, hardware"-->none
  </flowCtrl>
</SerialPort>
```

Response Message

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error reason description in detail, desc:error reason description in detail-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.5.6 Get the parameters of a specific port supported by the device

Request URL

GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	0 (main permission controller), 1 (sub permission controller) When this parameter does not exist, it indicates getting the serial port information of the main permission controller (i.e., the device serial port information).
childDevID	string	--
deviceIndex	string	For C60S devices connected to multiple LED sending cards, the URL needs to include deviceIndex= to distinguish which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, serial port ID-->0
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable the serial port-->true
  </enabled>
  <serialPortType>
    <!--ro, opt, enum, serial port type, subType:string, desc:"RS485", "RS422", "RS232"-->RS485
  </serialPortType>
  <duplexMode>
    <!--ro, opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"-->half
  </duplexMode>
  <baudRate>
    <!--ro, opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--ro, opt, int-->6
  </dataBits>
  <parityType>
    <!--ro, opt, enum, parity type, subType:string, desc:parity type: "none,even,odd,mark,space"-->none
  </parityType>
  <stopBits>
    <!--ro, opt, enum, stop bit, subType:string, desc:stop bit-->1
  </stopBits>
  <flowCtrl>
    <!--ro, opt, enum, "none,software,hardware", subType:string, desc:"none", "software", "hardware"-->none
  </flowCtrl>
  <accessProtocolType>
    <!--ro, opt, enum, subType:string-->PELCO
  </accessProtocolType>
</SerialPort>
```

19.1.5.7 Get the list of serial ports supported by the device

Request URL

GET /ISAPI/System/Serial/ports?permissionController=<indexID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	0 (main permission controller), 1 (sub permission controller) If this parameter does not exist, it indicates getting the serial port information of main permission controller, that is, the serial port information of the device.
deviceIndex	string	For C60S device connecting multiple LED sending cards, the URL needs to include the parameter deviceIndex= to specify which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPortList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <SerialPort>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->0
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <serialPortType>
      <!--ro, opt, enum, subType:string-->RS485
    </serialPortType>
    <duplexMode>
      <!--ro, opt, enum, subType:string-->half
    </duplexMode>
    <baudRate>
      <!--ro, opt, enum, subType:int-->2400
    </baudRate>
    <dataBits>
      <!--ro, opt, int-->6
    </dataBits>
    <parityType>
      <!--ro, opt, enum, subType:string-->none
    </parityType>
    <stopBits>
      <!--ro, opt, enum, subType:string-->1
    </stopBits>
    <workMode>
      <!--ro, opt, enum, subType:string-->console
    </workMode>
    <flowCtrl>
      <!--ro, opt, enum, subType:string-->none
    </flowCtrl>
    <accessProtocolType>
      <!--ro, opt, enum, subType:string-->PELCO
    </accessProtocolType>
  </SerialPort>
</SerialPortList>
```

19.1.6 System Maintenance

19.1.6.1 Get the storage capability of the device

Request URL

GET /ISAPI/ContentMgmt/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<RacmCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device storage capability, attr:version{req, string, protocolVersion}-->
  <nasNums>
    <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->0
  </nasNums>
  <pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
personQueueCounting,personQueueTime,violentMotion,leavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,r
oadblock,
abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminite,OPTEX,securityControlPanel,ge
tUp,toiletTarry,
playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePh
one,fatigueDriving,
notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringW
arning,
PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideS
hop,stallOccupyingRoad,     illegalHeap,illegalParkingofNon-
MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel,
ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,temperatureIntervalMeasurement,AIOP,tail,abnormalFace,normalFace,multiFace,callFace,sunglasses
Face,humanEnter,operateOverTime,stickUp,installScanner,objectPlacedOrTaken,fakeCard,securityChecking,pothole,crack,roadMaint,imageCapture,methaneConcentrati
onException,methaneLightIntensityException,bannerDetection,variableDataEvent,shipChannelAbnormal,radarVideoDetection,smokeAndFireDetection,smokeDetection,se
nsorAlarm,alarmResult,AIOP_Video,AIOP_Polling_Snap,swipePatrol,BatteryCarInfoEvent,indoorPasswayBlock,detectorTemperatureAlarm,fireNoRegulation,detectorMoti
onAlarm,detectorShelterAlarm,detectorRunningStatus,faceCapture,UVPhotonDetection,DevPowerStatusReport,HFPD,audioAnalysis,IndoorPasswayRecovery,BatteryCarRec
overy,DetectorTemperatureRecovery,RadarFallenObjectsOnTracks,VegetationSpectrumDetection,personDensityDetection,ChannelClearEvent,channelOccupy,ChannelPassi
```

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```
ngEvent,FallenObjectsOnTracksEvent,audioexception,CarWindow,shipsDetection,InsulatorAbnormalAlarm,PackingSpaceRecognition,smokeAlarm,SmokeAlarmRecovery,fire
SmartFireDetect,FirePointAlarmRecovery,offDuty,backDuty,sleepDuty,batteryCarDisobey,SmokingDetectionRecoverEvent,noFireExtinguisherRecog,FireExtinguisherRec
overyEvent,TelephoningBehaviorAlarmEvent,TelephoningBehaviorRecoveryEvent,GasTankMonitoringEvent,TemperatureMeasurementPreviewRecovery,TMPA,TMA,TemperatureM
easurementPreviewAlarmRecovery,TemperatureMeasurementAlarmRecovery,DetectorShelterAlarmRecovery,TemperatureUp,TemperatureDown,TemperatureEvent,TemperatureUp
Recover,TemperatureDownRecover,TemperatureEventRecover,ElectricFenceAlarmEvent,cityManagement">
    <!--ro, opt, enum, picture search type, subType:string, attr:opt{req, string}, desc:"AllEvent", "CMR", "MOTION", "ALARM", "EDR", "ALARMANDMOTION",
"pir:PIR", "wlsensor",  "callhelp", "facedetection", "FieldDetection", "scenechangedetection", "LineDetection", "regionEntrance", "regionExiting",
"loitering", "group", "rapidMove", "parking", "unattendedBaggage", "attendedBaggage", "vehicleDetection", "manual", "manualSnapShot",  "playSnapShot",
"safetyHelmet", "personQueueCounting", "personQueueTime", "violentMotion", "leavePosition", "advReachHeight", "peopleNumChange", "mixedTargetDetection",
"illegalParking", "pedestrian", "construction", "roadblock", "evidence", "crosslane", "vehicleexist", "lanechange", "wrongdirection", "congestion",
"turnround", "parallelParking", "abandonedObject", "trafficAccident", "fogDetection", "smokeDetection", "personDensity", "failDown", "accessController",
"videoInterCom", "GJD", "Luminite", "OPTEX", "securityControlPanel", "getUp", "toiletTarry", "playCellphone", "vehicleMonitor", "running", "retention",
"spacingChange", "yardTarry", "keyPersonGetUp", "sitQuietly", "standup", "plateRecognition", "smoking", "onThePhone", "fatigueDriving",
"notLookingStraightAhead", "LeaveThePost", "seatbelt", "sunglasses", "identityAbnormal", "ForwardCollisionWarning", "LaneDepartureWarning",
"BlindSpotDetection", "HeadwayMonitoringWarning", "PedestrianCollisionWarning", "SuddenSpeedUp", "SuddenSpeedDown", "SuddenLeftTurn", "SuddenRightTurn",
"Turnover", "NotYieldToPedestrian;allPic", "ATMPanel", "ATMSurround", "ATMFace", "ATMSafetyCabin", "smokeDetectAlarm", "temperatureIntervalMeasurement",
"AIOP", "tail", "abnormalFace", "normalFace;callFace", "sunglassesFace", "humanEnter", "operateOverTime", "stickup", "installScanner",
"objectPlacedOrTaken", "fakeCard", "securityChecking", "pothole", "crack", "roadMaint", "imageCapture", "ReID", "thermalVehicleDetection",
"methaneConcentrationException", "methaneLightIntensityException", "bannerDetection", "waterQualityDetection", "waterLevelDetection", "parallelParking",
"methaneConcentrationException", "variableDataEvent", "VOCsGasDetection", "shipChannelAbnormal", "radarVideoDetection", "AIDfogDetection:AID",
"smokeAndFireDetection", "smokeDetection", "sensorAlarm", "GasTankDetection", "alarmResult", "AIOP_Video", "AIOP_Polling_Snap", "swipePatrol",
"BatteryCarInfoEvent", "indoorPasswayBlock", "detectorTemperatureAlarm", "fireNoRegulation", "detectorMotionAlarm", "detectorShelterAlarm",
"detectorRunningStatus", "faceCapture", "UVPhotonDetection", "DevPowerStatusReport", 'HFPD"-->AllEvent
  </pictureSearchType>
  <recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofN
on-MotorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
,temperatureIntervalMeasurement,AIOP,securityChecking,scheduledRecordUpload,variableDataEvent,smokeAndFireDetection,smokeDetection,AIOP_Video,AIOP_Polling_S
nap,BatteryCarInfoEvent,indoorPasswayBlock,detectorTemperatureAlarm,fireNoRegulation,detectorMotionAlarm,detectorShelterAlarm,detectorRunningStatus,faceCapt
ure,UVPhotonDetection,DevPowerStatusReport,ChannelClearEvent,channelOccupy,ChannelPassingEvent,personQueueCounting,personQueueTime,leavePosition,localButton
Record,callRecord">
    <!--ro, opt, enum, video search type, subType:string, attr:opt{req, string}, desc:"AllEvent" (all events), "CMR" (capture by schedule), "MOTION" (motion
detection), "ALARM", "EDR", "ALARMANDMOTION", "pir", "wlsensor", "callhelp", "facedetection", "FieldDetection", "scenechangedetection", "LineDetection",
"regionEntrance", "regionExiting", "loitering", "group", "rapidMove", "parking", "unattendedBagggage", "attendedBaggage", "vehicleDetection", "manual",
"manualSnapShot", "playSnapShot", "safetyHelmet",  "personDensityDetection", "smokeDetection", "ATMPanel", "ATMSurround", "ATMFace", "ATMSafetyCabin",
"temperatureIntervalMeasurement", "AIOP", "securityChecking", "scheduledRecordUpload", "thermalVehicleDetection", "waterQualityDetection",
"waterLevelDetection", "methaneConcentrationException", "variableDataEvent"-->AllEvent
  </recordSearchType>
  <isSupportRacmChannelsCap opt="true,false">
    <!--ro, opt, bool, whether the device supports the Rcam capability by channel, attr:opt{opt, string}, desc:related URI:
/ISAPI/ContentMgmt/channels/<ID>/capabilities-->true
  </isSupportRacmChannelsCap>
  <SecurityLog>
    <!--ro, opt, object, security log-->
    <isSupportSecurityLog>
      <!--ro, opt, bool, whether the device supports the security log-->true
    </isSupportSecurityLog>
    <isSupportLogServer>
      <!--ro, opt, bool, whether it supports security log server configuration-->true
    </isSupportLogServer>
    <isSupportLogServerTest>
      <!--ro, opt, bool, whether it supports security log server test-->true
    </isSupportLogServerTest>
    <SecurityLogTypeList>
      <!--ro, req, array, list of security log types, subType:object-->
      <SecurityLogType>
        <!--ro, opt, object, list of log types-->
        <primaryType>
          <!--ro, req, enum, main log type, subType:string, desc:"Event", "Operation", "Other" (other types), "All" (all types)-->Event
        </primaryType>
        <secondaryType opt="test">
          <!--ro, req, enum, sub type, subType:string, attr:opt{req, string}, desc:see other types on the appendix. "all" (all types)-->all
        </secondaryType>
      </SecurityLogType>
    </SecurityLogTypeList>
  </SecurityLog>
  <PictureSearchSubTypeList>
    <!--ro, opt, object, list of picture searching sub type-->
    <PictureSearchSubType>
      <!--ro, opt, object, the sub type of picture search-->
      <mainType>
        <!--ro, req, enum, main event type of the searched picture, subType:string, desc:"vehicleDetection", "HVTVehicleDetection" (mixed-traffic
detection)-->vehicleDetection
      </mainType>
      <subType opt="all,motorVehicle,nonMotorVehicle,pedestrian">
        <!--ro, req, enum, sub type of each main event type, subType:string, attr:opt{req, string}, desc:"all", "motorVehicle", "nonMotorVehicle",
"pedestrian"-->all
      </subType>
    </PictureSearchSubType>
  </PictureSearchSubTypeList>
</RacmCap>
```

19.1.6.2 Get device storage capability by channel

Request URL

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GET /ISAPI/ContentMgmt/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RacmChannelsCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, storage capability of a device, attr:version{req, string, protocolVersion}-->
  <pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,regionEntran
ce,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,allPic,methaneConce
ntrationException,methaneLightIntensityException,bannerDetectio,AIOpenPlatform,alarmResult,UVPhotonDetection,DevPowerStatusReport,FallenObjectsOnTracksEvent
,ChannelClearEvent,channelOccupy,ChannelPassingEvent,audioexception,fielddetection, linedetection, regionEntrance,
regionExiting,smokeDetectAlarm,audioAnalysis,smokeAlarm,SmokeAlarmRecovery,fireSmartFireDetect,FirePointAlarmRecovery,indoorPasswayBlock,IndoorPasswayRecove
ry,offDuty,backDuty,sleepDuty,sleepDutyRecoveryEvent,batteryCarDisobey,BatteryCarRecovery,SmokingDetectionRecoverEvent,noFireExtinguisherRecog,FireExtinguis
herRecoveryEvent,TelephoningBehaviorAlarmEvent,TelephoningBehaviorRecoveryEvent,GasTankMonitoringEvent,GasTankMonitoringRecoveryEvent,TemperatureMeasurement
PreviewRecovery,TemperatureMeasurementRecovery,playCellphone,PlayCellphoneRecoveryEvent,TMPA,TMA,TemperatureMeasurementPreviewAlarmRecovery,TemperatureMeasu
rementAlarmRecovery,detectorShelterAlarm,DetectorShelterAlarmRecovery,,TemperatureUp,TemperatureDown,TemperatureEvent,TemperatureUpRecover,TemperatureDownRe
cover,TemperatureEventRecover,localButtonRecord,flowVelocity,volumeFlowRate,rainfall,detectorTemperatureAlarm,AlarmTriggeredByPanicButton,detectorSmokeAlarm
,shipsDetection">
    <!--ro, opt, enum, picture search type, subType:string, attr:opt{req, string}, desc:“AllEvent”, “CMR”, “MOTION”, “ALARM”, “EDR”, “ALARMANDMOTION”,
“Command”, “pir”, “wlsensor”, “callhelp”, “facedetection”, “FieldDetection”, “scenechangedetection”, “LineDetection”, “regionEntrance”, “regionExiting”,
“loitering”, “group”, “rapidMove”, “parking”, “unattendedBaggage”, “attendedBaggage”, “vehicleDetection”, “manual”, “manualSnapShot”, “playSnapShot”,
“allPic”, “methaneConcentrationException”, “methaneLightIntensityException”, “bannerDetection”, “waterQualityDetection”, “waterLevelDetection”,
“parallelParking”, “waterLevelAndRainfall”, “AIOpenPlatform”, “VOCsGasDetection”, “angleDeviationDetection”-->AllEvent
  </pictureSearchType>
</RacmChannelsCap>
```

19.1.6.3 Get the system security capability

Request URL

GET /ISAPI/Security/capabilities?username=<userName>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
userName	string	user name
deviceIndex	string	For centralized splicing controllers connecting to LED sending cards, the URL needs to include the parameter deviceIndex= to specify LED sending card The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SecurityCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, system security capability, attr:version{req, string, protocolVersion}-->
  <supportUserNums>
    <!--ro, opt, int, supported maximum number of users-->0
  </supportUserNums>
  <userBondIpNums>
    <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->0
  </userBondIpNums>
  <userBondMacNums>
    <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->0
  </userBondMacNums>
  <isSupCertificate>
    <!--ro, opt, bool, whether the device supports authentication-->true
  </isSupCertificate>
  <issupIllegalLoginLock>
    <!--ro, opt, bool, whether the device supports locking login-->true
```

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```
  </issupIllegalLoginLock>
  <isSupportOnlineUser>
    <!--ro, opt, bool, whether the device supports the online user configuration-->true
  </isSupportOnlineUser>
  <isSupportAnonymous>
    <!--ro, opt, bool, whether the device supports anonymous login-->true
  </isSupportAnonymous>
  <isSupportStreamEncryption>
    <!--ro, opt, bool, whether the device supports stream encryption-->true
  </isSupportStreamEncryption>
  <securityVersion opt="1,2,3,4">
    <!--ro, opt, string, encryption capability set, attr:opt{req, string}, desc:the encryption capability of each version consists of two parts: encryption
algorithm and the range of encrypted nodes currently 1 refers to AES128 encryption and 2 refers to AES256 encryption, the range of encrypted nodes is
described in each protocol-->1
  </securityVersion>
  <keyIterateNum>
    <!--ro, opt, int, secret key iteration times, dep:or,{$.SecurityCap.securityVersion,eq,1},{$.SecurityCap.securityVersion,eq,2}, desc:this node depends
on the node securityVersion, the range is between 100 and 1000-->100
  </keyIterateNum>
  <isSupportUserCheck>
    <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, dep:or,
{$.SecurityCap.securityVersion,eq,0},{$.SecurityCap.securityVersion,eq,1}, desc:it is an added capability, which indicates that whether supporting the login
password verification for editing/adding/deleting user parameters, this node depends on the node securityVersion, which means that it is only valid for the
versions that support encrypting the sensitive information-->true
  </isSupportUserCheck>
  <isSupportSecurityQuestionConfig>
    <!--ro, opt, bool, whether the device supports answering security questions, desc:related URI: /ISAPI/Security/questionConfiguration-->true
  </isSupportSecurityQuestionConfig>
  <supportSecurityNode opt="wirelessServer,wirelessDial">
    <!--ro, opt, string, additional node for supporting sensitive information encryption, attr:opt{req, string},
desc:in the earlier version of the solution for encrypting the sensitive information via ISAPI (i.e., the solution for transmitting plaintext via ISAPI),
some URIs containing sensitive information are left out now these URIs need to be encrypted for transmission, this capability node is added to be compatible
with old devices _x000D_
          if wirelessServer in this node is returned, it indicates that the following sensitive information should be encrypted _x000D_
          /ISAPI/System/Network/interfaces/<ID>/wirelessServer/capabilities, /ISAPI/System/Network/interfaces/<ID>/wirelessServer, if wirelessDial in this
node is returned, it indicates that the following sensitive information should be encrypted _x000D_
          /ISAPI/System/Network/WirelessDial/Interfaces,_x000D_
    /ISAPI/System/Network/WirelessDial/Interfaces/<ID>-->wirelessServer
  </supportSecurityNode>
  <SecurityLimits>
    <!--ro, opt, object, capability of configuring security limit parameters-->
    <LoginPasswordLenLimit min="1" max="16">
      <!--ro, opt, string, length limit of the user's login password, attr:min{req, int},max{req, int}-->1
    </LoginPasswordLenLimit>
    <SecurityAnswerLenLimit min="1" max="128">
      <!--ro, opt, string, length limit of the security question's answer, attr:min{req, int},max{req, int}-->1
    </SecurityAnswerLenLimit>
  </SecurityLimits>
  <RSAKeyLength opt="512,1024,2048" def="2048">
    <!--ro, opt, enum, HTTPS certificate length, subType:int, attr:opt{req, string},def{req, string}, desc:512, 1024, 2048-->2048
  </RSAKeyLength>
  <isSupportONVIFUserManagement>
    <!--ro, opt, bool, whether the device supports user management of Open Network Video Interface Protocol-->true
  </isSupportONVIFUserManagement>
  <WebCertificateCap>
    <!--ro, opt, object, HTTP authentication capability, desc:if this node is not returned, it indicates that device supports basic and digest
authentication-->
    <CertificateType opt="basic,digest,digest/basic">
      <!--ro, req, enum, certificate type, subType:string, attr:opt{req, string}, desc:“basic” (basic authentication), “digest” (digest authentication),
“digest/basic” (digest/basic authentication)-->basic
    </CertificateType>
    <SecurityAlgorithm>
      <!--ro, opt, object, security algorithm, dep:or,{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest},
{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
      <algorithmType opt="MD5,SHA256,MD5/SHA256">
        <!--ro, opt, enum, algorithm type, subType:string, attr:opt{req, string}, desc:“MD5”, “SHA256”, “MD5/SHA256”-->MD5
      </algorithmType>
    </SecurityAlgorithm>
  </WebCertificateCap>
  <isSupportConfigFileImport>
    <!--ro, opt, bool, whether the device supports importing the configuration file, desc:true (support), this node is not returned (not support)-->true
  </isSupportConfigFileImport>
  <isSupportConfigFileExport>
    <!--ro, opt, bool, whether the device supports exporting the configuration file, desc:true (support), this node is not returned (not support)-->true
  </isSupportConfigFileExport>
  <cfgFileSecretKeyLenLimit min="0" max="16">
    <!--ro, opt, string, length limit of the configuration file's verification key, attr:min{req, int},max{req, int}-->1
  </cfgFileSecretKeyLenLimit>
  <isIrreversible>
    <!--ro, opt, bool, whether the device supports irreversible password storage, desc:If this function is not supported, the plaintext password of the user
information will be stored in the device; otherwise, the password will be hashed for storage in the device.-->true
  </isIrreversible>
  <salt>
    <!--ro, opt, string, the specific salt used by the user to log in-->test
  </salt>
  <isSupportOnvifInfo>
    <!--ro, opt, bool, whether the device supports getting the Open Network Video Interface Protocol, desc:true (support), this node is not returned (not
support)-->true
  </isSupportOnvifInfo>
  <isSupportGB35114Certificate>
    <!--ro, opt, bool, whether supports GB35114 protocol and CA/SIP authentication certificate-->true
  </isSupportGB35114Certificate>
  <isSupportDeviceCertificatesManagement>
    <!--ro, opt, bool, whether the device supports device certificate management, desc:if this node is not returned, it indicates that device does not
```

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```
    <!--ro, opt, bool, whether the device supports device certificate management, desc:if this node is not returned, it indicates that device does not
support this function-->true
  </isSupportDeviceCertificatesManagement>
  <isSupportSIPCertificatesManagement>
    <!--ro, opt, bool, whether the device supports SIP certificate management, desc:this node is returned if the device supports this function and not
returned if device does not support this function, related URI: /ISAPI/Security/SIPCertificate, additional node: <SIPCertificatesManagementCap>-->true
  </isSupportSIPCertificatesManagement>
  <isSupportDeviceSelfSignCertExport>
    <!--ro, opt, bool, whether the device supports exporting self-signed certificate-->true
  </isSupportDeviceSelfSignCertExport>
  <isSupportSecurityEmail>
    <!--ro, opt, bool, whether the device supports configuring the security email, desc:if this node is not returned, it indicates that device does not
support this function-->true
  </isSupportSecurityEmail>
  <maxIllegalLoginTimes min="3" max="20" def="5">
    <!--ro, opt, int, maximum failed illegal login attempts, range:[3,20], attr:min{req, int},max{req, int},def{req, int}, desc:this node is valid when the
value of the node issupportIllegalLoginLock is true and will be returned when the maximum failed attempts of illegally logging in to the device reach the
limit-->3
  </maxIllegalLoginTimes>
  <SecurityAdvanced>
    <!--ro, opt, object, security advanced parameters-->
    <securityEnhanced>
      <!--ro, opt, bool, whether the device supports security reinforcement-->true
    </securityEnhanced>
    <noOperationEnabled>
      <!--ro, req, bool, whether to enable the control timeout when there is no operation-->true
    </noOperationEnabled>
    <noOperationTime min="1" max="60" def="15">
      <!--ro, req, int, control timeout when there is no operation, range:[1,60], unit:min, attr:min{req, int},max{req, int},def{req, int}-->15
    </noOperationTime>
  </SecurityAdvanced>
  <LoginLinkNum>
    <!--ro, opt, object, number of controls logging in at the same time-->
    <maxLinkNum min="1" max="128" def="50">
      <!--ro, req, int, maximum number of controls logging in at the same time, range:[1,128], attr:min{req, int},max{req, int},def{req, int}-->50
    </maxLinkNum>
  </LoginLinkNum>
  <isSupportEncryptCertificate>
    <!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
  </isSupportEncryptCertificate>
  <maxIllegalLoginLockTime min="1" max="120" def="30.00">
    <!--ro, opt, int, lock time range, range:[1,120], unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
  </maxIllegalLoginLockTime>
  <isSupportCertificateCustomID>
    <!--ro, opt, bool, whether the device supports using the user's custom ID to configure the certificate-->true
  </isSupportCertificateCustomID>
  <isSupportSoftwareLicense>
    <!--ro, opt, bool, whether support configuring the product open source statement, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/softwareLicense?format=json-->true
  </isSupportSoftwareLicense>
  <isSupportPasswordProtection>
    <!--ro, opt, bool, whether it supports configuring SDK protocol authentication mode parameters, desc:PUT/GET
/ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
  </isSupportPasswordProtection>
</SecurityCap>
```

19.1.6.4 Get device system capability

Request URL

GET /ISAPI/System/capabilities?type=<all>

Query Parameter

Parameter Name	Parameter Type	Description
all	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device system capability, attr:version{opt, string, protocolVersion}-->
  <SysCap>
    <!--ro, opt, object, system capability-->
    <isSupportDst>
      <!--ro, opt, bool, whether the device supports DST (Daylight Saving Time)-->true
    </isSupportDst>
    <NetworkCap>
      <!--ro, opt, object, network capability, desc:related URI: /ISAPI/System/Network/capabilities-->
      <isSupportWireless>
        <!--ro, req, bool, whether the device supports wireless network-->true
      </isSupportWireless>
```

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```
<isSupportPPPoE>
  <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
</isSupportPPPoE>
<isSupportBond>
  <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
</isSupportBond>
<isSupport802_1x>
  <!--ro, req, bool, whether the device supports 802.1x protocol-->true
</isSupport802_1x>
<isSupportNtp>
  <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
</isSupportNtp>
<isSupportFtp>
  <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
</isSupportFtp>
<isSupportUpnp>
  <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play ) protocol-->true
</isSupportUpnp>
<isSupportDdns>
  <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
</isSupportDdns>
<isSupportHttps>
  <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
</isSupportHttps>
<SnmpCap>
  <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
  <isSupport>
    <!--ro, req, bool, whether the device supports SNMP-->true
  </isSupport>
</SnmpCap>
<isSupportExtNetCfg>
  <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
</isSupportExtNetCfg>
<isSupportIPFilter>
  <!--ro, opt, bool, whether the device supports IP filtering-->true
</isSupportIPFilter>
<isSupportSSH>
  <!--ro, opt, bool, whether the device supports SSH-->true
</isSupportSSH>
<isSupportEZVIZ>
  <!--ro, opt, bool, whether the device supports EZ protocol-->true
</isSupportEZVIZ>
<isSupportEhome>
  <!--ro, opt, bool, whether the device supports ISUP server configuration-->true
</isSupportEhome>
<isSupportWirelessDial>
  <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
</isSupportWirelessDial>
<isSupportWirelessServer>
  <!--ro, opt, bool, whether the device supports wireless server-->true
</isSupportWirelessServer>
<isSupportWPS>
  <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
</isSupportWPS>
<isWirelessMutexWithWirelessServer>
  <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
</isWirelessMutexWithWirelessServer>
<isSupportMACFilter>
  <!--ro, opt, bool, whether the device supports physical address filtering-->true
</isSupportMACFilter>
<isSupportRFIDData>
  <!--ro, opt, bool, whether the device supports RFID collection configuration-->true
</isSupportRFIDData>
<isSupportwifiProbeSSID>
  <!--ro, opt, bool, whether the device supports configuration of Wi-Fi signal detection-->true
</isSupportwifiProbeSSID>
<isSupportWifiProbe>
  <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
</isSupportWifiProbe>
<verificationCode min="1" max="1">
  <!--ro, opt, string, verification code, attr:min{opt, string},max{opt, string}-->test
</verificationCode>
<VerificationCodeModification>
  <!--ro, opt, object, device verification code configuration-->
  <verificationCodeType opt="normal,empty">
    <!--ro, opt, string, verification code type, attr:opt{opt, string}-->empty
  </verificationCodeType>
  <isSupportDeclarationURL>
    <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
  </isSupportDeclarationURL>
  <isSupportPrivacyPolicyURL>
    <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
  </isSupportPrivacyPolicyURL>
  <verificationCodeModify opt="true,false">
    <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{opt, string}-->true
  </verificationCodeModify>
  <Hyperlinks>
    <!--ro, opt, object, hyperlink configuration-->
    <declarationURL>
      <!--ro, opt, string, hyperlink to the Terms of Service-->test
    </declarationURL>
    <privacyPolicyURL>
      <!--ro, opt, string, hyperlink to the Privacy Policy-->test
    </privacyPolicyURL>
```

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```
        </Hyperlinks>
      </VerificationCodeModification>
      <isSupportIntegrate>
        <!--ro, opt, bool, whether the device supports access protocol configuration-->true
      </isSupportIntegrate>
      <isSupportPlatformAccess>
        <!--ro, opt, bool, whether the device supports platform access, desc:GET /ISAPI/System/Network/platformAccessFilter/capabilities (platform access IP
address filtering configuration (Ehome and GB28181 allowlist configuration))-->true
      </isSupportPlatformAccess>
      <isSupportGetLinkSocketIP>
        <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
      </isSupportGetLinkSocketIP>
      <isSupportWebSocket>
        <!--ro, opt, bool, whether the device supports WebSocket-->true
      </isSupportWebSocket>
      <isSupportWebSocketS>
        <!--ro, opt, bool, whether the device supports WebSocketS-->true
      </isSupportWebSocketS>
      <Adaption>
        <!--ro, opt, object, network self-adaption configuration-->
        <streamType opt="0,1,2,3,4,5,7,8,9,10">
          <!--ro, opt, string, stream type that supports network self-adaption during live view, attr:opt{opt, string}, desc:0 (main stream), 1 (sub-
stream), 2 (third stream), 3 (fourth stream/virtual stream), 4 (fifth stream), 5 (sixth stream), 7 (seventh stream), 8 (eighth stream), 9 (ninth stream), 10
(tenth stream), and so forth. The index value here is the same as that in the live view structure NET_DVR_PREVIEWINFO of Device Network SDK. Note that 6 is
an invalid value, and from the seventh stream (index value: 7), the index value is the same as that for live view via RTSP-->test
        </streamType>
      </Adaption>
      <isSupportVideoImgDB>
        <!--ro, opt, bool, whether the device supports image and video library configuration-->true
      </isSupportVideoImgDB>
      <isSupportEventDataOverWebSocket opt="true,false">
        <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{opt, string}-->true
      </isSupportEventDataOverWebSocket>
      <isSupportDynamicHostName>
        <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) configuration-->true
      </isSupportDynamicHostName>
      <isSupportTrafficMonitor>
        <!--ro, opt, bool, whether the device supports traffic monitoring, desc:related URI:
/ISAPI/System/Network/WirelessDial/Interfaces/<ID>/trafficMonitor?format=json-->true
      </isSupportTrafficMonitor>
      <isSupportWiredandWirelessTrafficMonitor>
        <!--ro, opt, bool, whether the device supports traffic monitoring (wireless NIC and wired NIC), desc:related URI:
/ISAPI/System/Network/Interfaces/<ID>/trafficMonitor?format=json-->true
      </isSupportWiredandWirelessTrafficMonitor>
      <isSupportEmailEncrypt>
        <!--ro, opt, bool, whether the device supports email encryption configuration-->true
      </isSupportEmailEncrypt>
      <isSupportEZVIZUnbind>
        <!--ro, opt, bool, whether the device supports unbinding from EZ, desc:/ISAPI/System/Network/EZVIZ/unbind?format=json-->true
      </isSupportEZVIZUnbind>
      <isSupportIntranetSegment>
        <!--ro, opt, bool, whether the device supports LAN segment configuration, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will be returned and its value is true or not be returned. Related URI:
/ISAPI/System/Network/intranetSegment/capabilities?format=json-->true
      </isSupportIntranetSegment>
      <isSupportBasebandCapture>
        <!--ro, opt, bool, whether the device supports capturing packets of the baseband, desc:you can capture packets transmitted between the 4G module and
the base station using the ping command. Related URI: /ISAPI/System/Network/cellularNetwork/basebandCapture/capabilities?format=json-->true
      </isSupportBasebandCapture>
      <isSupportCellularNetworkCellInfo>
        <!--ro, opt, bool, whether the device supports getting a cell's information of the cellular network, desc:you can get the information of the
cellular network where the device is. Related URI: /ISAPI/System/Network/cellularNetwork/cellInfo?format=json&ecurity=<security>&iv=<iv>-->true
      </isSupportCellularNetworkCellInfo>
      <isSupportCellularNetworkLockBand>
        <!--ro, opt, bool, whether the device accessing the locked frequency band of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/lockBand/capabilities?format=json-->true
      </isSupportCellularNetworkLockBand>
      <isSupportRadioFrequencyFluctuation>
        <!--ro, opt, bool, whether the device supports configuring RF parameters of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/radioFrequencyFluctuation/capabilities?format=json-->true
      </isSupportRadioFrequencyFluctuation>
      <isSupportNetworkModuleUpgrade>
        <!--ro, opt, bool, whether it supports upgrading network module, desc:including importing upgrade package
(/ISAPI/System/Network/WirelessDial/ImportUpgradePackage?format=json), getting importing progress
(/ISAPI/System/Network/WirelessDial/GetUpgradePackageImportStatus?format=json), starting upgrade
(/ISAPI/System/Network/WirelessDial/StartUpgradeNetworkModule?format=json)-->true
      </isSupportNetworkModuleUpgrade>
      <isSupportNetworkModuleVersionInfo>
        <!--ro, opt, bool, whether it supports getting the version information of network module, desc:GET
/ISAPI/System/Network/WirelessDial/NetworkModuleVersionInfo?format=json-->true
      </isSupportNetworkModuleVersionInfo>
      <isSupportDSTPServerCfg>
        <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET or PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
format=json-->true
      </isSupportDSTPServerCfg>
      <isSupportDSTPProtocolsCap>
        <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
      </isSupportDSTPProtocolsCap>
    </NetworkCap>
    <IOCap>
      <!--ro, opt, object, I/O capability, desc:related URI: /ISAPI/System/IO/capabilities-->
      <IOInputPortNums>
        <!--ro, opt, int, number of local input ports of the device-->true
```

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```
      </IOInputPortNums>
      <IOOutputPortNums>
        <!--ro, opt, int, number of local output ports of the device-->true
      </IOOutputPortNums>
      <isSupportStrobeLamp>
        <!--ro, opt, bool, whether the device supports configuring the output of the gate's light, desc:related URI:
/ISAPI/System/IO/outputs/strobelampConf-->true
      </isSupportStrobeLamp>
    </IOCap>
    <SerialCap>
      <!--ro, opt, object, range of RS-485 serial port numbers supported by the device-->
      <rs485PortNums>
        <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->true
      </rs485PortNums>
      <supportRS232Config>
        <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
      </supportRS232Config>
      <rs422PortNums>
        <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->true
      </rs422PortNums>
      <rs232PortNums>
        <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->true
      </rs232PortNums>
    </SerialCap>
    <VideoCap>
      <!--ro, opt, object, video encoding capability, desc:related URI: /ISAPI/System/Video/capabilities-->
      <videoInputPortNums>
        <!--ro, opt, int, number of video input ports-->true
      </videoInputPortNums>
      <videoOutputPortNums>
        <!--ro, opt, int, number of video output ports-->true
      </videoOutputPortNums>
      <isSupportHeatmap>
        <!--ro, opt, bool, whether the device supports heat map-->true
      </isSupportHeatmap>
      <isSupportCounting>
        <!--ro, opt, bool, whether the device supports people counting-->true
      </isSupportCounting>
      <countingType>
        <!--ro, opt, string, statistics type, desc:human (people counting); object (counting)-->test
      </countingType>
      <isSupportPicture>
        <!--ro, opt, bool, whether the device supports OSD picture overlay-->true
      </isSupportPicture>
      <isSupportCountingCollection>
        <!--ro, opt, bool, whether the device supports people counting ANR-->true
      </isSupportCountingCollection>
      <isSupportPrivacyMask>
        <!--ro, opt, bool, whether the device supports privacy mask-->true
      </isSupportPrivacyMask>
      <isSupportBinocularPreviewSwitch>
        <!--ro, opt, bool, whether the device supports switching live view of the dual-lens camera-->true
      </isSupportBinocularPreviewSwitch>
      <isSupportCalibCheck>
        <!--ro, opt, bool, whether the device supports calibration verification-->true
      </isSupportCalibCheck>
      <isSupportPIP>
        <!--ro, opt, bool, whether the device supports PIP-->true
      </isSupportPIP>
      <isSupportVideoOutputMode>
        <!--ro, opt, bool, whether the device supports video output pattern parameters-->true
      </isSupportVideoOutputMode>
      <pictureChannelList>
        <!--ro, opt, array, subType:object-->
        <channelID>
          <!--ro, opt, int-->0
        </channelID>
      </pictureChannelList>
    </VideoCap>
    <AudioCap>
      <!--ro, opt, object, audio encoding capability, desc:related URL: /ISAPI/System/Audio/capabilities-->
      <audioInputNums>
        <!--ro, req, int, number of audio inputs-->1
      </audioInputNums>
      <audioOutputNums>
        <!--ro, req, int, number of audio outputs-->1
      </audioOutputNums>
      <muteDuringPanTilt opt="MicIn">
        <!--ro, opt, string, whether the device supports muting during motion, attr:opt{opt, string}, desc:muteDuringPanTilt in
/ISAPI/System/TwoWayAudio/channels/ID-->true
      </muteDuringPanTilt>
      <isSupportDeviceAudioCapture>
        <!--ro, opt, bool, whether the device supports audio collection, desc:GET/PUT /ISAPI/System/Audio/DeviceAudioCapture?format=json-->true
      </isSupportDeviceAudioCapture>
    </AudioCap>
    <isSupportExternalDevice>
      <!--ro, opt, bool, whether the device supports connecting to peripherals: true (support), desc:related URI: /ISAPI/System/externalDevice/capabilities-
->true
    </isSupportExternalDevice>
    <isSupportSubscribeEvent>
      <!--ro, opt, bool, whether the device supports subscribing to events, desc:related URI: /ISAPI/Event/notification/subscribeEventCap-->true
    </isSupportSubscribeEvent>
    <isSupportDiagnosedData>
      <!--ro, opt, bool, whether the device supports exporting the device diagnosis data, desc:true (support), this node is not returned (not support)
```

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```
      <!--ro, opt, bool, whether the device supports exporting the device diagnosis data, desc:true (support), this node is not returned (not support). Related URI: /ISAPI/System/diagnosedData-->true
    </isSupportDiagnosedData>
    <isSupportPTZChannels>
      <!--ro, opt, bool, whether the device supports returning the PTZ channel (it is distinguished from the video channel)-->true
    </isSupportPTZChannels>
    <isSupportMetadata>
      <!--ro, opt, bool, whether the device supports metadata, desc:related URI: /ISAPI/Streaming/channels/<ID>/metadata/capabilities-->true
    </isSupportMetadata>
    <isSupportManualPowerConsumption>
      <!--ro, opt, bool, whether the device supports controlling the device consumption manually-->true
    </isSupportManualPowerConsumption>
    <supportSmartOverlapChannles opt="1">
      <!--ro, opt, bool, whether the device supports configuring the smart event stream, attr:opt{opt, string}, desc:if this function is supported, this node and the corresponding channel ID will be returned; otherwise, this node will not be returned-->true
    </supportSmartOverlapChannles>
  </SysCap>
  <voicetalkNums>
    <!--ro, opt, int, number of two-way audio channels-->2
  </voicetalkNums>
  <isSupportSnapshot>
    <!--ro, opt, bool, whether the device supports capturing pictures-->true
  </isSupportSnapshot>
  <SecurityCap>
    <!--ro, opt, object, encryption capability set-->
    <supportUserNums>
      <!--ro, opt, int, supported maximum number of users-->1
    </supportUserNums>
    <userBondIpNums>
      <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->1
    </userBondIpNums>
    <userBondMacNums>
      <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->1
    </userBondMacNums>
    <isSupCertificate>
      <!--ro, opt, bool, whether the device supports authentication-->true
    </isSupCertificate>
    <issupIllegalLoginLock>
      <!--ro, opt, bool, whether the device supports locking login-->true
    </issupIllegalLoginLock>
    <isSupportOnlineUser>
      <!--ro, opt, bool, whether the device supports the online user configuration-->true
    </isSupportOnlineUser>
    <isSupportAnonymous>
      <!--ro, opt, bool, whether the device supports anonymous login-->true
    </isSupportAnonymous>
    <isSupportStreamEncryption>
      <!--ro, opt, bool, whether the device supports stream encryption-->true
    </isSupportStreamEncryption>
    <securityVersion opt="1,2,3,4,7">
      <!--ro, opt, string, encryption capability, attr:opt{opt, string}, desc:the encryption capability of each version consists of two parts: encryption algorithm and the range of encrypted nodes. Currently 1 refers to AES128 encryption and 2 refers to AES256 encryption. The range of encrypted nodes is described in each protocol-->test
    </securityVersion>
    <keyIterateNum>
      <!--ro, opt, int, iteration times, desc:the value is usually between 100 and 1000-->100
    </keyIterateNum>
    <isSupportUserCheck>
      <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, desc:this node depends on the node <securityVersion>, which means that it is only valid for the versions that support encrypting the sensitive information-->true
    </isSupportUserCheck>
    <isSupportSecurityQuestionConfig>
      <!--ro, opt, bool, whether the device supports answering security questions-->true
    </isSupportSecurityQuestionConfig>
    <supportSecurityNode opt="wirelessServer,wirelessDial">
      <!--ro, opt, string, additional node for supporting sensitive information encryption, attr:opt{opt, string}, desc:in the earlier version of the solution for encrypting the sensitive information via ISAPI (i.e., the solution for transmitting plaintext via ISAPI), some URIs containing sensitive information are left out. Now these URIs need to be encrypted for transmission. This capability node is added to be compatible with old devices-->test
    </supportSecurityNode>
    <SecurityLimits>
      <!--ro, opt, object, capability of configuring security limit parameters-->
      <LoginPasswordLenLimit min="1" max="16">
        <!--ro, opt, string, length limit of the user's login password, attr:min{opt, string},max{opt, string}-->test
      </LoginPasswordLenLimit>
      <SecurityAnswerLenLimit min="1" max="128">
        <!--ro, opt, string, length limit of the security question's answer, attr:min{opt, string},max{opt, string}-->test
      </SecurityAnswerLenLimit>
    </SecurityLimits>
    <RSAKeyLength opt="512,1024,2048,3072" def="3072">
      <!--ro, opt, enum, HTTPS certificate length, subType:string, attr:opt{opt, string},def{opt, string}, desc:512, 1024, 2048-->3072
    </RSAKeyLength>
    <isSupportONVIFUserManagement>
      <!--ro, opt, bool, whether the device supports ONVIF user management-->true
    </isSupportONVIFUserManagement>
    <WebCertificateCap>
      <!--ro, opt, object, HTTP authentication capability-->
      <CertificateType opt="basic,digest,digest/basic">
        <!--ro, req, string, certificate type: basic (authentication), attr:opt{opt, string}-->test
      </CertificateType>
      <SecurityAlgorithm>
        <!--ro, opt, object, encryption algorithm, dep:or,{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,degist},{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
        <algorithmType opt="MD5,SHA256,MD5/SHA256">
          <!--ro, opt, string, algorithm type, attr:opt{req, string}-->test
```

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```
        </algorithmType>
      </SecurityAlgorithm>
    </WebCertificateCap>
    <isSupportConfigFileImport>
      <!--ro, opt, bool, whether the device supports importing the configuration file-->true
    </isSupportConfigFileImport>
    <isSupportConfigFileExport>
      <!--ro, opt, bool, whether the device supports exporting the configuration file-->true
    </isSupportConfigFileExport>
    <cfgFileSecretKeyLenLimit min="0" max="16">
      <!--ro, opt, string, length limit of the configuration file's verification key, attr:min{opt, string},max{opt, string}-->0
    </cfgFileSecretKeyLenLimit>
    <isIrreversible>
      <!--ro, opt, bool, whether the device supports irreversible password storage, desc:if this function is not supported, the plaintext password of the
user information will be stored in the device; otherwise, the password will be hashed for storage in the device-->true
    </isIrreversible>
    <salt>
      <!--ro, opt, string, the specific salt used by the user to log in-->test
    </salt>
    <isSupportOnvifInfo>
      <!--ro, opt, bool, whether the device supports getting the ONVIF protocol information-->true
    </isSupportOnvifInfo>
    <isSupportDeviceCertificatesManagement>
      <!--ro, opt, bool, whether the device supports certificate management-->true
    </isSupportDeviceCertificatesManagement>
    <isSupportSIPCertificatesManagement>
      <!--ro, opt, bool, whether the device supports SIP certificate management-->true
    </isSupportSIPCertificatesManagement>
    <isSupportDeviceSelfSignCertExport>
      <!--ro, opt, bool, whether the device supports exporting the self-signed certificate-->true
    </isSupportDeviceSelfSignCertExport>
    <isSupportSecurityEmail>
      <!--ro, opt, bool, whether the device supports configuring the security email-->true
    </isSupportSecurityEmail>
    <isSupportGB35114Certificate>
      <!--ro, opt, bool, N/A-->true
    </isSupportGB35114Certificate>
    <maxIllegalLoginTimes min="3" max="20" def="5">
      <!--ro, opt, string, maximum failed illegal login attempts, attr:min{opt, string},max{opt, string},def{opt, string}, desc:this node is valid when the
value of the node <issupportIllegalLoginLock> is true and will be returned when the maximum failed attempts of illegally logging in to the device reach the
limit-->1
    </maxIllegalLoginTimes>
    <SecurityAdvanced>
      <!--ro, opt, object, advanced security configuration-->
      <securityEnhanced>
        <!--ro, opt, bool, whether the device supports security reinforcement-->true
      </securityEnhanced>
      <noOperationEnabled>
        <!--ro, req, bool, whether to enable the control timeout when there is no operation-->true
      </noOperationEnabled>
      <noOperationTime min="1" max="60" def="15">
        <!--ro, req, int, control timeout when there is no operation, unit:分钟, attr:min{opt, string},max{opt, string},def{opt, string}-->1
      </noOperationTime>
    </SecurityAdvanced>
    <LoginLinkNum>
      <!--ro, opt, object, number of users logging in at the same time-->
      <maxLinkNum min="1" max="128" def="50">
        <!--ro, req, int, number of controls logging in at the same time, attr:min{opt, string},max{opt, string},def{opt, string}-->64
      </maxLinkNum>
    </LoginLinkNum>
    <isSupportEncryptCertificate>
      <!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
    </isSupportEncryptCertificate>
    <maxIllegalLoginLockTime min="1" max="120" def="30">
      <!--ro, opt, int, lock time range, range:[1,120], unit:min, attr:min{req, int},max{req, int},def{req, int}-->1
    </maxIllegalLoginLockTime>
    <isSupportCertificateCustomID>
      <!--ro, opt, bool, whether the device supports using the user's custom ID to configure the certificate-->true
    </isSupportCertificateCustomID>
    <isSupportSoftwareLicense>
      <!--ro, opt, bool, whether support configuring the product open source statement, desc:if this function is supported, this node will be returned and
its value is true, otherwise, this node will not be returned. Related URI: /ISAPI/Security/softwareLicense?format=json-->true
    </isSupportSoftwareLicense>
    <isSupportPasswordProtection>
      <!--ro, opt, bool, whether it supports parameter configuration of SDK protocol authentication mode, desc:PUT/GET
/ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
    </isSupportPasswordProtection>
  </SecurityCap>
  <EventCap>
    <!--ro, opt, object, event capability-->
    <isSupportHDFull>
      <!--ro, opt, bool, whether the device supports HDD full alarm-->true
    </isSupportHDFull>
    <isSupportHDError>
      <!--ro, opt, bool, whether the device supports HDD error alarm-->true
    </isSupportHDError>
    <isSupportNicBroken>
      <!--ro, opt, bool, whether the device supports the network disconnected event-->true
    </isSupportNicBroken>
    <isSupportIpConflict>
      <!--ro, opt, bool, whether the device supports the IP address conflict event-->true
    </isSupportIpConflict>
    <isSupportIllAccess>
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    <!--ro, opt, bool, whether the device supports the illegal login event-->true
  </isSupportIllAccess>
  <isSupportViException>
    <!--ro, opt, bool, whether the device supports video input exception (only for analog channels)-->true
  </isSupportViException>
  <isSupportViMismatch>
    <!--ro, opt, bool, whether the device supports the event of video standard mismatch-->true
  </isSupportViMismatch>
  <isSupportRecordException>
    <!--ro, opt, bool, whether the device supports the recording exception event-->true
  </isSupportRecordException>
  <isSupportTriggerFocus>
    <!--ro, opt, bool, whether the device supports defocus exception-->true
  </isSupportTriggerFocus>
  <isSupportMotionDetection>
    <!--ro, opt, bool, whether the device supports motion detection-->true
  </isSupportMotionDetection>
  <isSupportVideoLoss>
    <!--ro, opt, bool, whether the device supports video loss events-->true
  </isSupportVideoLoss>
  <isSupportTamperDetection>
    <!--ro, opt, bool, whether the device supports tampering detection-->true
  </isSupportTamperDetection>
  <isSupportFireDetection>
    <!--ro, opt, bool, whether the device supports fire detection events-->true
  </isSupportFireDetection>
  <isSupportFaceContrast>
    <!--ro, opt, bool, whether the device supports face picture comparison events-->true
  </isSupportFaceContrast>
  <isSupportFaceLib>
    <!--ro, opt, bool, Front-end devices support linkage by face picture library., desc:related URI: /ISAPI/Event/schedules/faceLib/<ID>/<FDID>-->true
  </isSupportFaceLib>
  <isSupportStudentsStoodUp>
    <!--ro, opt, bool, whether the device supports the student standing up detection event-->true
  </isSupportStudentsStoodUp>
  <isSupportFramesPeopleCounting>
    <!--ro, opt, bool, whether the device supports the regional people counting event-->true
  </isSupportFramesPeopleCounting>
  <isSupportFaceSnap>
    <!--ro, opt, bool, whether the device supports face capture events-->true
  </isSupportFaceSnap>
  <isSupportPersonDensityDetection>
    <!--ro, opt, bool, whether the device supports people density detection-->true
  </isSupportPersonDensityDetection>
  <isSupportPersonQueueDetection>
    <!--ro, opt, bool, whether the device supports queue detection events-->true
  </isSupportPersonQueueDetection>
  <isSupportTeacherBehaviorDetect>
    <!--ro, opt, bool, whether the device supports teacher's behavior detection-->true
  </isSupportTeacherBehaviorDetect>
  <isSupportCityManagement>
    <!--ro, opt, bool, whether the device supports the city management function-->true
  </isSupportCityManagement>
  <isSupportMixedTargetDetection>
    <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
  </isSupportMixedTargetDetection>
  <isSupportFaceSnapModeling>
    <!--ro, opt, bool, whether the device supports face modeling events-->true
  </isSupportFaceSnapModeling>
  <isSupportIntersectionAnalysis>
    <!--ro, opt, bool, whether the device supports the intersection analysis-->true
  </isSupportIntersectionAnalysis>
  <isSupportLuma>
    <!--ro, opt, bool, VQD brightness exception-->true
  </isSupportLuma>
  <isSupportChroma>
    <!--ro, opt, bool, VQD image color cast-->true
  </isSupportChroma>
  <isSupportSnow>
    <!--ro, opt, bool, VQD snow noise-->true
  </isSupportSnow>
  <isSupportStreak>
    <!--ro, opt, bool, VQD stripe noise-->true
  </isSupportStreak>
  <isSupportFreeze>
    <!--ro, opt, bool, VQD image freeze-->true
  </isSupportFreeze>
  <isSupportSigLose>
    <!--ro, opt, bool, VQD signal loss-->true
  </isSupportSigLose>
  <isSupportClarity>
    <!--ro, opt, bool, VQD resolution exception-->true
  </isSupportClarity>
  <isSupportJitter>
    <!--ro, opt, bool, VQD image flutter-->true
  </isSupportJitter>
  <isSupportBlock>
    <!--ro, opt, bool, VQD video tampering-->true
  </isSupportBlock>
  <isSupportFlowers>
    <!--ro, opt, bool, VQD blurred image alarm-->true
  </isSupportFlowers>
  <isSupportNoise>
    <!--ro, opt, bool, VQD blurred screen-->true
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    </isSupportNoise>
    <isSupportGhost>
      <!--ro, opt, bool, VQD image noise-->true
    </isSupportGhost>
    <isSupportPurple>
      <!--ro, opt, bool, VQD purple fringing-->true
    </isSupportPurple>
    <isSupportICR>
      <!--ro, opt, bool, VQD ICP-->true
    </isSupportICR>
    <isSupportProtectiveFilm>
      <!--ro, opt, bool, VQD protective film unremoved-->true
    </isSupportProtectiveFilm>
    <isSupportAppAbnormal>
      <!--ro, opt, bool, whether the device supports the App exception alarm or event-->true
    </isSupportAppAbnormal>
  </EventCap>
  <ITCCap>
    <!--ro, opt, object, related node and URIs: the node <isSupportVehicleDetection> in /ISAPI/ITC/capability and /ISAPI/ITC/capabilities-->
    <isSupportVehicleDetection>
      <!--ro, opt, bool, whether the device supports vehicle detection-->true
    </isSupportVehicleDetection>
  </ITCCap>
  <RacmCap>
    <!--ro, opt, object, UI before picture search-->
    <nasNums>
      <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->1
    </nasNums>
    <pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
personQueueCounting,personQueueTime,violentMotion,leavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,r
oadblock,
abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminite,OPTEX,securityControlPanel,ge
tUp,toiletTarry,
playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePh
one,fatigueDriving,
notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringW
arning,
PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideS
hop,stallOccupyingRoad,     illegalHeap,illegalParkingofNon-
MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel,
ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,alarmResult">
      <!--ro, opt, string, picture search type, attr:opt{opt, string}-->test
    </pictureSearchType>
    <recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofN
on-MotorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
">
      <!--ro, opt, string, video search type, attr:opt{opt, string}, desc:"llEvent" (all events), "CMR" (capture by schedule), "MOTION" (capture motion
detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures
and motion detection pictures), "Command", "pir" (pir alarm), "wlsensor" (wireless alarm), "callhelp" (panic alarm), "facedetection" (face detection),
"FieldDetection" (intrusion detection), "scenechangedetection" (scene change detection), "LineDetection" (line crossing detection)-->test
    </recordSearchType>
    <isSupportRacmChannelsCap opt="true,false">
      <!--ro, opt, bool, whether the device supports the Rcam capability by channel, attr:opt{opt, string}, desc:related URI:
/ISAPI/ContentMgmt/channels/<ID>/capabilities-->true
    </isSupportRacmChannelsCap>
    <SecurityLog>
      <!--ro, opt, object, device's capability of security log-->
      <isSupportSecurityLog>
        <!--ro, opt, bool, whether the device supports the security log-->true
      </isSupportSecurityLog>
      <isSupportLogServer>
        <!--ro, opt, bool, whether it supports security log server configuration-->true
      </isSupportLogServer>
      <isSupportLogServerTest>
        <!--ro, opt, bool, whether it supports security log server test-->true
      </isSupportLogServerTest>
      <SecurityLogTypeList>
        <!--ro, req, array, list of security log types, subType:object-->
        <SecurityLogType>
          <!--ro, opt, object, security log type-->
          <primaryType>
            <!--ro, req, string, main log type, desc:"Event", "Operation", "Other", "All"-->test
          </primaryType>
          <secondaryType opt="all">
            <!--ro, req, string, minor log type, attr:opt{opt, string}, desc:for other types, refer to the appendix-->test
          </secondaryType>
        </SecurityLogType>
      </SecurityLogTypeList>
    </SecurityLog>
    <PictureSearchSubTypeList>
      <!--ro, opt, array, list of picture search sub type, subType:object-->
      <PictureSearchSubType>
        <!--ro, opt, object, the sub type of picture search-->
        <mainType>
          <!--ro, req, string, main event type of the searched picture-->test
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          </mainType>
          <subType opt="all,motorVehicle,nonMotorVehicle,pedestrian">
            <!--ro, req, string, sub type of each main event type, attr:opt{opt, string}-->test
          </subType>
        </PictureSearchSubType>
      </PictureSearchSubTypeList>
    </RacmCap>
    <PTZCtrlCap>
      <!--ro, opt, object, device's PTZ capability-->
      <isSupportPatrols>
        <!--ro, opt, bool, whether the device supports patrol-->true
      </isSupportPatrols>
    </PTZCtrlCap>
    <SmartCap>
      <!--ro, opt, object, compact and smart capability, desc:/ISAPI/Smart/capabilities-->
      <isSupportROI>
        <!--ro, opt, bool, whether the device supports ROI encoding configuration-->true
      </isSupportROI>
      <isSupportFaceDetect>
        <!--ro, opt, bool, whether the device supports face detection-->true
      </isSupportFaceDetect>
      <isSupportIntelliTrace>
        <!--ro, opt, bool, whether the device supports smart tracking-->true
      </isSupportIntelliTrace>
      <isSupportFieldDetection>
        <!--ro, opt, bool, whether the device supports intrusion detection-->true
      </isSupportFieldDetection>
      <isSupportDefocusDetection>
        <!--ro, opt, bool, whether the device supports defocus detection-->true
      </isSupportDefocusDetection>
      <isSupportAudioDetection>
        <!--ro, opt, bool, whether the device supports audio detection-->true
      </isSupportAudioDetection>
      <isSupportSceneChangeDetection>
        <!--ro, opt, bool, whether the device supports scene change detection-->true
      </isSupportSceneChangeDetection>
      <isSupportLineDetection>
        <!--ro, opt, bool, whether the device supports line crossing detection-->true
      </isSupportLineDetection>
      <isSupportRegionEntrance>
        <!--ro, opt, bool, whether the device supports region entrance detection-->true
      </isSupportRegionEntrance>
      <isSupportRegionExiting>
        <!--ro, opt, bool, whether the device supports region exiting detection-->true
      </isSupportRegionExiting>
      <isSupportLoitering>
        <!--ro, opt, bool, whether the device supports loitering detection-->true
      </isSupportLoitering>
      <isSupportGroup>
        <!--ro, opt, bool, whether the device supports people gathering detection-->true
      </isSupportGroup>
      <isSupportRapidMove>
        <!--ro, opt, bool, whether the device supports fast moving detection-->true
      </isSupportRapidMove>
      <isSupportParking>
        <!--ro, opt, bool, whether the device supports parking detection-->true
      </isSupportParking>
      <isSupportUnattendedBaggage>
        <!--ro, opt, bool, whether the device supports unattended baggage detection-->true
      </isSupportUnattendedBaggage>
      <isSupportAttendedBaggage>
        <!--ro, opt, bool, whether the device supports object removal detection-->true
      </isSupportAttendedBaggage>
      <isSupportPeopleDetection>
        <!--ro, opt, bool, whether the device supports human body detection-->true
      </isSupportPeopleDetection>
      <isSupportSmartCalibration>
        <!--ro, opt, bool, whether the device supports calibration filter size-->true
      </isSupportSmartCalibration>
      <isSupportStorageDetection>
        <!--ro, opt, bool, whether the device supports health monitoring storage-->true
      </isSupportStorageDetection>
      <isSupportChannelResource>
        <!--ro, opt, bool, whether the device supports channel resources-->true
      </isSupportChannelResource>
      <isSupportSmartOverlapParams>
        <!--ro, opt, bool, whether it supports smart overlay parameters, desc:related URI: /ISAPI/Smart/channels/<channelID>/SmartOverlapParams?format=json--
>true
      </isSupportSmartOverlapParams>
    </SmartCap>
    <TestCap>
      <!--ro, opt, object, test capability-->
      <isSupportFTPTest>
        <!--ro, opt, bool, whether it supports FTP test-->true
      </isSupportFTPTest>
      <isSupportPingTest>
        <!--ro, opt, bool, whether it supports PING test-->true
      </isSupportPingTest>
      <isSupportNTPTest>
        <!--ro, opt, bool, whether it supports NTP test-->true
      </isSupportNTPTest>
      <isSupportNASTest>
        <!--ro, opt, bool, whether it supports NAS test-->true
      </isSupportNASTest>
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    <isSupportEmailTest>
      <!--ro, opt, bool, whether the device supports email test-->true
    </isSupportEmailTest>
  </TestCap>
  <PanoramaCap>
    <!--ro, opt, object, capability of panoramic channels, desc:the object cannot be extended. You can extend it in /ISAPI/Panorama/capabilities-->
    <isSupportGeneratePanorama>
      <!--ro, opt, bool, whether the device supports generating panoramic images-->true
    </isSupportGeneratePanorama>
    <isSupportPanoramaPosition3D>
      <!--ro, opt, bool, whether the device supports panoramic 3D position-->true
    </isSupportPanoramaPosition3D>
    <isSupportPreset>
      <!--ro, opt, bool, whether the device supports configuring panoramic preset-->true
    </isSupportPreset>
    <isSupportGetPanoramicSplicingStatus>
      <!--ro, opt, bool-->true
    </isSupportGetPanoramicSplicingStatus>
    <isSupportStopPanoramicSplicing>
      <!--ro, opt, bool-->true
    </isSupportStopPanoramicSplicing>
    <isSupportSplicingChannelMotionControl>
      <!--ro, opt, bool-->true
    </isSupportSplicingChannelMotionControl>
    <isSupportGetPanoramaSplicingPictures>
      <!--ro, opt, bool-->true
    </isSupportGetPanoramaSplicingPictures>
    <isSupportImportPanoramaSplicingParamFile>
      <!--ro, opt, bool-->true
    </isSupportImportPanoramaSplicingParamFile>
    <isSupportExportPanoramaSplicingParamFile>
      <!--ro, opt, bool-->true
    </isSupportExportPanoramaSplicingParamFile>
    <isSupportImportPanoramicSplicingPicture>
      <!--ro, opt, bool-->true
    </isSupportImportPanoramicSplicingPicture>
    <isSupportExportPanoramicSplicingPicture>
      <!--ro, opt, bool-->true
    </isSupportExportPanoramicSplicingPicture>
  </PanoramaCap>
  <WLAlarmCap>
    <!--ro, opt, object, wireless alarm capability-->
    <isSupportTeleControl>
      <!--ro, opt, bool, whether the device supports remote control-->true
    </isSupportTeleControl>
    <isSupportPIR>
      <!--ro, opt, bool, whether the device supports PIR configuration-->true
    </isSupportPIR>
    <isSupportWLSensors>
      <!--ro, opt, bool, N/A-->true
    </isSupportWLSensors>
    <isSupportCallHelp>
      <!--ro, opt, bool, whether the device supports calling for help-->true
    </isSupportCallHelp>
    <WLSensorsNum>
      <!--ro, opt, int, N/A-->1
    </WLSensorsNum>
  </WLAlarmCap>
  <isSupportGIS>
    <!--ro, opt, bool, whether the device supports GIS, desc:related URI: /ISAPI/GIS/channels-->true
  </isSupportGIS>
  <isSupportCompass>
    <!--ro, opt, bool, whether the device supports compass configuration, desc:related URI: /ISAPI/Compass/channels/<ID>/capabilities-->true
  </isSupportCompass>
  <isSupportRoadInfoOverlays>
    <!--ro, opt, bool, whether the device supports overlaying the lane information-->true
  </isSupportRoadInfoOverlays>
  <isSupportFaceCaptureStatistics>
    <!--ro, opt, bool, whether the device supports face capture statistics, desc:related URI: /ISAPI/Intelligent/channels/<ID>/faceCaptureStatistics/search-
->true
  </isSupportFaceCaptureStatistics>
  <isSupportElectronicsEnlarge>
    <!--ro, opt, bool, whether the device supports electronics enlarge-->true
  </isSupportElectronicsEnlarge>
  <isSupportRemoveStorage>
    <!--ro, opt, bool, whether the device supports remote storage-->true
  </isSupportRemoveStorage>
  <isSupportViewshedOverlay>
    <!--ro, opt, bool, whether the device supports overlaying viewshed camera target, desc:related URI:
/ISAPI/System/Video/inputs/channels/<ID>/viewshedOverlay-->true
  </isSupportViewshedOverlay>
  <isSupportGPSLabelTracking>
    <!--ro, opt, bool, whether the device supports GPS tag tracking, desc:related URI:/ISAPI/GIS/channels/<ID>/GPSLabelTracking?format=json-->true
  </isSupportGPSLabelTracking>
  <isSupportFirmwareVersionInfo>
    <!--ro, opt, bool, whether the device supports displaying the firmware version information-->true
  </isSupportFirmwareVersionInfo>
  <isSupportSetupCalibration>
    <!--ro, opt, bool, whether the device supports configuring calibration-->true
  </isSupportSetupCalibration>
  <isSupportStreamDualVCA>
    <!--ro, opt, bool, whether the device supports overlaying intelligent private data on the stream, desc:related URI: in the
/ISAPI/Intelligent/channels/<ID>/intelliResource; corresponding node: <isOverlapIntelli>-->true
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    </isSupportStreamDualVCA>
    <isSupportLaserSpotManual>
      <!--ro, opt, bool, whether the device supports configuring laser light spot-->true
    </isSupportLaserSpotManual>
    <isSupportRTMP>
      <!--ro, opt, bool, whether the device supports RTMP configuration, desc:related URI: /ISAPI/Streaming/channels/<ID>/RTMPCfg/capabilities-->true
    </isSupportRTMP>
    <isSupportTraffic>
      <!--ro, opt, bool, traffic service capability-->true
    </isSupportTraffic>
    <isSupportPicInfoOverlap>
      <!--ro, opt, bool, whether the device supports overlaying pictures and texts-->true
    </isSupportPicInfoOverlap>
    <isSupportChannelEventCap>
      <!--ro, opt, bool, whether the device supports getting the event capability by channel, desc:related URI: /ISAPI/Event/channels/<ID>/capabilities-->true
    </isSupportChannelEventCap>
    <isSupportMixedTargetDetection>
      <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
    </isSupportMixedTargetDetection>
    <isSupportElevatorFloorCalibration>
      <!--ro, opt, bool, whether the device supports configuring floor display-->true
    </isSupportElevatorFloorCalibration>
    <isSupportGPSCalibratation>
      <!--ro, opt, bool, whether the device supports the GPS calibration function-->true
    </isSupportGPSCalibratation>
    <isSupportBVCorrect>
      <!--ro, opt, bool, whether the device supports configuring the parameters of camera correction-->true
    </isSupportBVCorrect>
    <isSupportSteamingCodingModeSwitch>
      <!--ro, opt, bool, encoding output mode (true-support), desc:related URI: /ISAPI/System/streamingCoding/mode/capabilities?format=json-->true
    </isSupportSteamingCodingModeSwitch>
    <supportSnapshotChannel opt="1">
      <!--ro, opt, string, the capability supported by the capture schedule channel, attr:opt{opt, string}-->test
    </supportSnapshotChannel>
    <isSupportActiveMulticast>
      <!--ro, opt, bool, whether the device supports multicast configuration (/ISAPI/Streaming/channels/<ID>/capabilities)-->true
    </isSupportActiveMulticast>
    <isSupportChannelEventListCap>
      <!--ro, opt, bool, it indicates that whether the device supports getting the event capability of all channels of the device-->true
    </isSupportChannelEventListCap>
    <VCAResourceChannelsCap>
      <!--ro, opt, array, subType:object-->
      <ChannelsList>
        <!--ro, opt, object-->
        <channelsID>
          <!--ro, req, int-->1
        </channelsID>
      </ChannelsList>
    </VCAResourceChannelsCap>
    <isSupportTimerVCAResource>
      <!--ro, opt, bool, whether the device supports scheduled switch of intelligent resources-->true
    </isSupportTimerVCAResource>
    <MixedTargetDetection>
      <!--ro, opt, object, the capability of recognizing specified target among mixed targets supported by the device-->
      <isSupportFaceRecognition>
        <!--ro, opt, bool, whether the device supports facial recognition-->true
      </isSupportFaceRecognition>
      <isSupportHumanRecognition>
        <!--ro, opt, bool, whether the device supports human body recognition-->true
      </isSupportHumanRecognition>
      <isSupportVehicleRecognition>
        <!--ro, opt, bool, whether the device supports vehicle recognition-->true
      </isSupportVehicleRecognition>
    </MixedTargetDetection>
    <isSupportDiscoveryMode>
      <!--ro, opt, bool, whether the device supports discovery mode, desc:related URI: /ISAPI/System/discoveryMode/capabilities-->true
    </isSupportDiscoveryMode>
    <isSupportAutoMaintenance>
      <!--ro, opt, bool, whether the device supports automatic maintenance, desc:related URI: /ISAPI/System/autoMaintenance/capabilities?format=json-->true
    </isSupportAutoMaintenance>
    <isSupportRtspOverHTTPS>
      <!--ro, opt, bool, supports tspOverHTTPS-->true
    </isSupportRtspOverHTTPS>
    <isSupportAUXInfoCap>
      <!--ro, opt, bool, it indicates that whether the device supports getting the capability of getting the attributes of device's all channels-->true
    </isSupportAUXInfoCap>
    <regionClip>
      <!--ro, opt, object, the channels and streams that supports area cropping-->
      <channels opt="101">
        <!--ro, opt, string, the channels and streams that supports area cropping, attr:opt{opt, string}, desc:101: channel No.*100+stream type-->test
      </channels>
    </regionClip>
    <isVehicleStatisticsSupport>
      <!--ro, opt, bool, whether the device supports configuring the parameters for traffic data statistics-->true
    </isVehicleStatisticsSupport>
    <isSupportElevatorFloorName>
      <!--ro, opt, bool, whether the device supports configuring floor custom name-->true
    </isSupportElevatorFloorName>
    <VQDEventSupport opt="luma,chroma,snow,streak,freeze,sigLose,clarity,jitter,block,flowers,noise,ghost,purple,ICR,protectiveFilm,blur,contrast">
      <!--ro, opt, string, VQD event list, attr:opt{opt, string}, desc:"Luma" (brightness exception); "Chroma" (image color cast); "Snow" (snow noise);
"Streak" (strip noise); "Freeze" (frozen screen); "SigLose" (signal loss); "Clarity" (resolution exception);         "JItter" (image flutter); "Block"
(video tampering); "Flowers" (blurred screen); "Noise" (image noise); "Ghost" (ghost image); "Purple" (purple fringing); "ICR" (ICP); "protectiveFilm"
(protective film unremoved)-->test
    </VQDEventSupport>
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  </vQuEventSupport>
  <isSupportDisplayTrajectory>
    <!--ro, opt, bool, whether the device supports target track overlay (playing library)-->true
  </isSupportDisplayTrajectory>
  <isSupportEmmcInfo>
    <!--ro, opt, bool, the capability of supporting the EMMC configuration, desc:related URI: /ISAPI/System/emmcInfo/capabilities?format=json-->true
  </isSupportEmmcInfo>
  <DevInfoSearch opt="tiltAngle">
    <!--ro, opt, string, whether the device supports searching for the device information, attr:opt{opt, string}, desc:related URI:
/ISAPI/System/workingstatus/devInfoSearch?format=json-->test
  </DevInfoSearch>
  <isSupportAIOpenPlatform>
    <!--ro, opt, bool, whether the device supports AI open platform device capability, desc:related URI: /ISAPI/Intelligent/AIOpenPlatform/capabilities?
format=json-->true
  </isSupportAIOpenPlatform>
  <FaceCounting>
    <!--ro, opt, object, whether the device supports the customer traffic (face)-->
    <ChannelsList>
      <!--ro, opt, array, channel list, subType:object-->
      <channelsID>
        <!--ro, req, int, No. of channels supported by the device-->1
      </channelsID>
    </ChannelsList>
  </FaceCounting>
  <isSupportBVCorrectPassthrough>
    <!--ro, opt, bool, whether the device supports transparent transmission of camera correction data, desc:related URI:
/ISAPI/System/BVCorrect/Passthrough/channels/<ID>/capabilities?format=json-->true
  </isSupportBVCorrectPassthrough>
  <isSupportISUPHttpPassthrough>
    <!--ro, opt, bool, whether the device supports HTTP transparent transmission via ISUP V5.0-->true
  </isSupportISUPHttpPassthrough>
  <isSupportDataPrealarm>
    <!--ro, opt, bool, whether the device supports network traffic pre-alarm event, desc:eventType: dataPrealarm-->true
  </isSupportDataPrealarm>
  <isSupportPatchUpgrade>
    <!--ro, opt, bool, whether it supports incremental upgrade, desc:related URI: /ISAPI/Custom/OpenPlatform/patchUpgrade/capabilities?format=json-->true
  </isSupportPatchUpgrade>
  <isSupportSnapshotAsync>
    <!--ro, opt, bool, whether the device supports capturing pictures asynchronously (the picture URL will be returned), desc:related URI:
/ISAPI/Streaming/channels/<ID>/picture/async?format=json-->true
  </isSupportSnapshotAsync>
  <isSupportVehicleMonitor>
    <!--ro, opt, bool, whether the device supports intelligent arming and tracking of vehicles, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/capabilities?format=json-->true
  </isSupportVehicleMonitor>
  <isSupportManualVehicleMonitor>
    <!--ro, opt, bool, whether the device supports manual arming and tracking of vehicles, desc:related URI:
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/manual/capabilities?format=json-->true
  </isSupportManualVehicleMonitor>
  <isSupportCalibrationStream>
    <!--ro, opt, bool, supported capability of calibration stream, desc:related URI: /ISAPI/Streaming/calibStream/capabilities?format=json-->true
  </isSupportCalibrationStream>
  <isSupportTimeZone>
    <!--ro, opt, bool, whether the device supports configuring the DST (Daylight Saving Time), desc:related URI: /ISAPI/System/time/timeZone-->true
  </isSupportTimeZone>
  <isSupportVibrationDetection>
    <!--ro, opt, bool, whether the device supports the vibration detection, desc:related URI: /ISAPI/System/channels/<ID>/vibrationDetection?format=json--
>true
  </isSupportVibrationDetection>
  <isSupportOnlineUpgradeTask>
    <!--ro, opt, bool, whether device online upgrading is supported-->true
  </isSupportOnlineUpgradeTask>
  <isSupportICCID>
    <!--ro, opt, bool, whether the device supports getting the ICCID information, desc:related URI: /ISAPI/System/deviceInfo/ICCID?format=json-->true
  </isSupportICCID>
  <isSupportDataAware>
    <!--ro, opt, bool, the data aware capability, desc:related URI: /ISAPI/System/dataAware/capabilities?format=json-->true
  </isSupportDataAware>
  <isSupportAbnormalReboot>
    <!--ro, opt, bool, whether the device supports abnormal reboot event, desc:eventType: abnormalReboot-->true
  </isSupportAbnormalReboot>
  <isSupportVCAResourceList>
    <!--ro, opt, bool, whether the device supports configuring intelligent resources in a batch, desc:related URI:
/ISAPI/System/Video/inputs/channels/VCAResourceList?format=json-->true
  </isSupportVCAResourceList>
  <isSupportDebuggingDataExport>
    <!--ro, opt, bool, whether the device supports importing the parameters (in-out parameters, rule configuration parameters), desc:related URI:
/ISAPI/System/debuggingData/parameters?format=json; this node is not returned if not supported-->true
  </isSupportDebuggingDataExport>
  <isSupportInternalParaAdjust>
    <!--ro, opt, bool, whether the device supports trinocular device; whether the device supports editing in-parameters, desc:related URI:
/ISAPI/System/internalParameters/adjustment?format=json; this node is not returned if not supported-->true
  </isSupportInternalParaAdjust>
  <characteristicCode min="1" max="128">
    <!--ro, opt, string, device attribute code, range:[1,128], attr:min{opt, string},max{opt, string}, desc:related URI:
/ISAPI/System/deviceInfo/characteristicCode?format=json-->test
  </characteristicCode>
  <isSupportSIMCardStatus>
    <!--ro, opt, bool, whether the device supports getting the information of SIM card status, desc:related URI: /ISAPI/System/SIMCardStatus?format=json--
>true
  </isSupportSIMCardStatus>
  <OpenPlatformCap>
    <!--ro, opt, object, detailed HEOP capability supported by the device, desc:related URI: /ISAPI/Custom/OpenPlatform/capabilities-->
    <isSupportOpenPlatform>
```

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```
      <!--ro, req, bool, whether the device supports HEOP (Embedded Open Platform)-->true
    </isSupportOpenPlatform>
    <maxAPPNum>
      <!--ro, opt, int, the maximum number of third-party application packages allowed to be installed-->0
    </maxAPPNum>
  </OpenPlatformCap>
  <manualEvent
opt="fielddetection,linedetection,ANPR,faceCapture,mixedTargetDetection,studentsStoodUp,headRiseRate,teacherBehavior,ElevatorRunningDetectionEvent">
    <!--ro, opt, string, whether the device supports URI for mannual triggering event, attr:opt{req, string}, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/manualEvent/<eventType>?format=json-->test
  </manualEvent>
  <isSupportWiegand>
    <!--ro, opt, bool, whether the device supports the Wiegand protocol, desc:related URI: /ISAPI/System/Wiegand/<wiegandID>/capabilities?format=json-->true
  </isSupportWiegand>
  <zoomFocusWebDisplay
opt="ROI,roadTrafficDetection,SMD,mixedTargetDetection,faceCapture,personCounting,multidimensionalPeopleCounting,AIOP,heelPDC,packingSpaceRecognition,tunnel
Detection,cityManagement">
    <!--ro, opt, string, whether the web supports displaying the zoomfocus page, attr:opt{req, string},
desc:ROI page path: Configuration—Vedio/Audio—ROI
roadTrafficDetection:Configuration—Road Traffic—Detection Configuration; SMD; mixedTargetDetection; faceCapture-->ROI
  </zoomFocusWebDisplay>
  <isSupportDeviceStatusUploadStrategy>
    <!--ro, opt, bool, whether the device supports configuring the parameters of device status uploading strategy, desc:related URL:
/ISAPI/System/status/uploadStrategy/capabilities?format=json-->true
  </isSupportDeviceStatusUploadStrategy>
  <isSupportTOFShelterAlarm>
    <!--ro, opt, bool, whether the device supports TOF tampering alarm, desc:/ISAPI/Event/channels/<channelID>/TOFShelterAlarm/capabilities?format=json--
>true
  </isSupportTOFShelterAlarm>
  <isSupportT1Test>
    <!--ro, opt, bool, whether the device supports the configuration of controlling T1 tool-->true
  </isSupportT1Test>
  <OTAP>
    <!--ro, opt, object, the capability of Open Things Access Protocol, desc:OTAP capability-->
    <isSupportOTAPModel>
      <!--ro, opt, bool, whether the device supports getting the virtual No. and device No. of the OTAP, desc:GET /ISAPI/System/Network/OTAP/Model?
format=json-->true
    </isSupportOTAPModel>
  </OTAP>
  <isSupportTimeTypeSTD>
    <!--ro, opt, bool, whether the device supports the time type of ISO8601, desc:if "true" is returned, it indicates that all the devices containing
ISO8601 parameters support adding the timeType=STD, such as /ISAPI/ContentMgmt/search?timeType=STD, the time will be implemented according to the standard--
>true
  </isSupportTimeTypeSTD>
  <isSupportRegionTargetNumberCounting>
    <!--ro, opt, bool, whether the device supports the region target counting, desc:related URI:
/ISAPI/Event/channels/<channelID>/RegionTargetNumberCounting/Capabilities?format=json-->true
  </isSupportRegionTargetNumberCounting>
  <isSupportSearchVoltageAndPowerConsumptionReport>
    <!--ro, opt, bool, whether the device supports searching for voltage and consumption report, desc:related URI:
/ISAPI/System/SearchVoltageAndPowerConsumptionReport/capabilities?format=json-->true
  </isSupportSearchVoltageAndPowerConsumptionReport>
  <isSupportWebPTZCfg>
    <!--ro, opt, bool, whether it supports displaying the web page for PTZ configuration, desc:this node is to determine whether to display the web page for
PTZ configuration and is irrelevant to whether PTZ is supported; default process logic: when this node does not exist, whether to display the web page for
PTZ configuration is subject to the device type-->true
  </isSupportWebPTZCfg>
  <isSupportRealTimeStreamInfo>
    <!--ro, opt, bool, whether the device supports getting the real-time stream information, desc:GET /ISAPI/Streaming/RealTimeStreamInfo?format=json-->true
  </isSupportRealTimeStreamInfo>
  <isSupportGetPlaybackStreamInfo>
    <!--ro, opt, bool, whether the device supports getting video playback stream information, desc:GET /ISAPI/Streaming/GetPlaybackStreamInfo?format=json--
>true
  </isSupportGetPlaybackStreamInfo>
  <isSupportSearchVideoCodeMealParams>
    <!--ro, opt, bool, whether the device supports searching for video encoding plan parameters, desc:POST
/ISAPI/Streaming/channels/<channelID>/SearchVideoCodeMealParams?format=json-->true
  </isSupportSearchVideoCodeMealParams>
  <isSupportVideoCodeMealParams>
    <!--ro, opt, bool, whether the device supports video encoding plan parameters, desc:GET/PUT /ISAPI/Streaming/channels/<channelID>/VideoCodeMealParams?
format=json-->true
  </isSupportVideoCodeMealParams>
  <isSupportRestoreDefaultParam>
    <!--ro, opt, bool, whether the device supports restoring default parameters, desc:PUT /ISAPI/Streaming/RestoreDefaultParam?format=json-->true
  </isSupportRestoreDefaultParam>
  <isSupportElevatorRunningDetectionEvent>
    <!--ro, opt, bool, whether the device supports elevator running event, desc:eventType: ElevatorRunningDetectionEvent-->true
  </isSupportElevatorRunningDetectionEvent>
  <isSupportElevatorOperationStatistics>
    <!--ro, opt, bool, whether the device supports elevator running statistics event, desc:eventType: ElevatorOperationStatistics-->true
  </isSupportElevatorOperationStatistics>
  <isSupportAuthorizeMicAcquisitionParam>
    <!--ro, opt, bool, whether the device supports audio authorization information, desc:GET/PUT
/ISAPI/System/InformedConsentMgr/AuthorizeMicAcquisitionParam?format=json-->true
  </isSupportAuthorizeMicAcquisitionParam>
  <isSupportCombinationRuleParams>
    <!--ro, opt, bool, whether the device supports combination rule parameters of perimeter detection, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterCombinationDetection/CombinationRuleParams?format=json-->true
  </isSupportCombinationRuleParams>
  <DualVCA>
    <!--ro, opt, object, intelligent information overlay-->
    <overlapRangeType opt="ruleRegion,fullScreen" def="ruleRegion">
      <!--ro, opt, string, type of overlay area, attr:opt{req, string},def{req, string}-->ruleRegion
```

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```
      </overlapRangeType>
    </DualVCA>
    <isSupportGetAllPIPCofigParams>
      <!--ro, opt, bool, whether it supports getting all picture-in-picture configuration parameters, desc:GET /ISAPI/System/Video/GetAllPIPCofigParams?
format=json-->true
    </isSupportGetAllPIPCofigParams>
    <isSupportPosOverlayTest>
      <!--ro, opt, bool, whether it supports the test of enabling POS information overlay configuration, desc:POST
/ISAPI/System/Video/inputs/channels/<channelID>/PosOverlayTest?format=json-->true
    </isSupportPosOverlayTest>
    <isSupportFullScreenPosOverlay>
      <!--ro, opt, bool, whether it supports full-screen POS information overlay parameters, desc:GET/PUT
/ISAPI/System/Video/inputs/channels/<channelID>/FullScreenPosOverlay?format=json-->true
    </isSupportFullScreenPosOverlay>
    <isSupportFullScreenPosOverlayCap>
      <!--ro, opt, bool, whether to get the dynamic capability of full-screen POS information overlay parameters, desc:GET
/ISAPI/System/Video/inputs/channels/<channelID>/FullScreenPosOverlayCap?format=json-->true
    </isSupportFullScreenPosOverlayCap>
    <isSupportGetFlashLifespanInfo>
      <!--ro, opt, bool, whether it supports getting the lifespan information of the built-in disk storage medium, desc:GET
/ISAPI/System/GetFlashLifespanInfo?format=json-->true
    </isSupportGetFlashLifespanInfo>
    <isSupportTOFCalibrate>
      <!--ro, opt, bool, whether it supports configuring TOF detection calibration, desc:PUT /ISAPI/System/Video/inputs/tamperDetection/TOFCalibrate?
format=json-->true
    </isSupportTOFCalibrate>
    <isSupportPerimeterAlgoTargetFilterParams>
      <!--ro, opt, bool, whether it supports target filtering parameters of perimeter algorithm, desc:GET/PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterAlgoMgr/PerimeterAlgoTargetFilterParams?format=json-->true
    </isSupportPerimeterAlgoTargetFilterParams>
    <videoEncodeChannel opt="1,2">
      <!--ro, opt, string, channel supporting video parameter display on the web, attr:opt{req, string}, desc:N/A-->1,2
    </videoEncodeChannel>
    <isSupportRestorePerimeterAlgoDefaultParam>
      <!--ro, opt, bool, whether it supports restoring parameters of perimeter algorithm, desc:PUT
/ISAPI/Intelligent/channels/<channelID>/PerimeterAlgoMgr/RestorePerimeterAlgoDefaultParam?format=json-->true
    </isSupportRestorePerimeterAlgoDefaultParam>
    <isSupportFlashLifespanProtectParams>
      <!--ro, opt, bool, whether it supports flash life protection parameter, desc:GET/PUT /ISAPI/System/FLASHMgr/FlashLifespanProtectParams?format=json--
>true
    </isSupportFlashLifespanProtectParams>
    <isSupportHVTVehicleDetectUploadChannel>
      <!--ro, opt, bool, whether it supports configuring the channel of uploading mixed-traffic detection information to the FTP server, desc:GET/PUT
/ISAPI/System/Network/ftp/HVTVehicleDetectUpload?format=json&channelID=<channelID>-->true
    </isSupportHVTVehicleDetectUploadChannel>
    <isSupportSynchronizationRulesBetweenChannels>
      <!--ro, opt, bool, POST /ISAPI/System/IntelligentRulesMgr/channels/<channelID>/SynchronizationRulesBetweenChannels?format=json, desc:POST
/ISAPI/System/IntelligentRulesMgr/channels/<channelID>/SynchronizationRulesBetweenChannels?format=json-->true
    </isSupportSynchronizationRulesBetweenChannels>
    <isSupportVideoPipList>
      <!--ro, opt, bool-->true
    </isSupportVideoPipList>
    <isOnlySupportAnalogChannel>
      <!--ro, opt, bool-->true
    </isOnlySupportAnalogChannel>
    <isSupportSoundClassificationDetection>
      <!--ro, opt, bool-->true
    </isSupportSoundClassificationDetection>
    <isSupportManualTriggerDeicing>
      <!--ro, opt, bool-->true
    </isSupportManualTriggerDeicing>
    <isSupportGetChallengeV3>
      <!--ro, opt, bool-->true
    </isSupportGetChallengeV3>
    <isSupportStartActivateV3>
      <!--ro, opt, bool-->true
    </isSupportStartActivateV3>
  </DeviceCap>
```

19.1.6.5 Export device configuration file

Request URL

GET /ISAPI/System/configurationData?secretkey=<secretkey>

Query Parameter

Parameter Name	Parameter Type	Description
secretkey	string	The verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.

Request Message

None

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Response Message

Binary Data

19.1.6.6 Get the device attribute code

Request URL

GET /ISAPI/System/deviceInfo/characteristicCode?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "Characteristic": {
    /*ro, req, object, attribute code, desc:attribute code*/
        "code":  "+/"
        /*ro, opt, string, device attribute code, range:[1,128], desc:device attribute code*/
    }
}
```

19.1.6.7 Set restoring devices' default parameters

Request URL

PUT /ISAPI/System/factoryReset?mode=<mode>&childDevID=<devIndex>&loginPassword=<loginPassword>&module=<module>

Query Parameter

Parameter Name	Parameter Type	Description
mode	enum	--
devIndex	string	--
loginPassword	string	--
module	enum	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.6.8 Get the network service capability

Request URL

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GET /ISAPI/System/Network/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<NetworkCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, get the network service capability, attr:version{opt, string, protocolVersion}-->
  <isSupportWireless>
    <!--ro, req, bool, whether the device supports wireless network-->true
  </isSupportWireless>
  <isSupportPPPoE>
    <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
  </isSupportPPPoE>
  <isSupportBond>
    <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
  </isSupportBond>
  <isSupport802_1x>
    <!--ro, req, bool, whether the device supports 802.1x protocol-->true
  </isSupport802_1x>
  <isSupportNtp>
    <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
  </isSupportNtp>
  <isSupportFtp>
    <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
  </isSupportFtp>
  <isSupportUpnp>
    <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play) protocol-->true
  </isSupportUpnp>
  <isSupportDdns>
    <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
  </isSupportDdns>
  <isSupportHttps>
    <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
  </isSupportHttps>
  <SnmpCap>
    <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
    <isSupport>
      <!--ro, req, bool, whether the device supports SNMP-->true
    </isSupport>
  </SnmpCap>
  <isSupportExtNetCfg>
    <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
  </isSupportExtNetCfg>
  <isSupportIPFilter>
    <!--ro, opt, bool, whether the device supports IP filtering-->true
  </isSupportIPFilter>
  <isSupportSSH>
    <!--ro, opt, bool, whether the device supports SSH-->true
  </isSupportSSH>
  <isSupportEZVIZ>
    <!--ro, opt, bool, whether the device supports EZ protocol-->true
  </isSupportEZVIZ>
  <isSupportEhome>
    <!--ro, opt, bool, whether the device supports EHome (ISUP) protocol-->true
  </isSupportEhome>
  <isSupportWirelessDial>
    <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
  </isSupportWirelessDial>
  <isSupportWirelessServer>
    <!--ro, opt, bool, whether the device supports wireless server-->true
  </isSupportWirelessServer>
  <isSupportWPS>
    <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
  </isSupportWPS>
  <isWirelessMutexWithWirelessServer>
    <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
  </isWirelessMutexWithWirelessServer>
  <isSupportMACFilter>
    <!--ro, opt, bool, whether the device supports physical address filtering-->true
  </isSupportMACFilter>
  <isSupportRFIDData>
    <!--ro, opt, bool, whether the device supports RFID collection configuration-->true
  </isSupportRFIDData>
  <isSupportwifiProbeSSID>
    <!--ro, opt, bool, whether the device supports configuration of Wi-Fi signal detection-->true
  </isSupportwifiProbeSSID>
  <isSupportWifiProbe>
    <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
  </isSupportWifiProbe>
  <verificationCode min="1" max="10">
    <!--ro, opt, string, verification code, attr:min{req, int},max{req, int}-->test
  </verificationCode>
  <VerificationCodeModification>
```

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```
    <!--ro, opt, object, verification code editing-->
    <verificationCodeType opt="normal,empty">
      <!--ro, opt, string, verification code type, attr:opt{req, string}-->test
    </verificationCodeType>
    <isSupportDeclarationURL>
      <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
    </isSupportDeclarationURL>
    <isSupportPrivacyPolicyURL>
      <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
    </isSupportPrivacyPolicyURL>
    <verificationCodeModify opt="true,false">
      <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{req, string}-->true
    </verificationCodeModify>
    <Hyperlinks>
      <!--ro, opt, object, hyperlink-->
      <declarationURL>
        <!--ro, opt, string, hyperlink to the Terms of Service-->test
      </declarationURL>
      <privacyPolicyURL>
        <!--ro, opt, string, hyperlink to the Privacy Policy-->test
      </privacyPolicyURL>
    </Hyperlinks>
  </VerificationCodeModification>
  <isSupportIntegrate>
    <!--ro, opt, bool, whether the device supports access protocol configuration-->true
  </isSupportIntegrate>
  <isSupportPlatformAccess>
    <!--ro, opt, bool, whether the device supports platform access, desc:GET /ISAPI/System/Network/platformAccessFilter/capabilities (platform access IP
address filtering configuration (Ehome allowlist configuration))-->true
  </isSupportPlatformAccess>
  <isSupportGetLinkSocketIP>
    <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
  </isSupportGetLinkSocketIP>
  <isSupportWebSocket>
    <!--ro, opt, bool, whether the device supports WebSocket-->true
  </isSupportWebSocket>
  <isSupportWebSocketS>
    <!--ro, opt, bool, whether the device supports WebSocketS-->true
  </isSupportWebSocketS>
  <Adaption>
    <!--ro, opt, object, network self-adaption-->
    <streamType opt="0,1,2,3,4,5,7,8,9,10">
      <!--ro, opt, enum, stream type that supports network self-adaption during live view, subType:int, attr:opt{req, string}, desc:stream types that
support network self-adaption during live view: 0 (main stream), 1 (sub-stream), 2 (third stream), 3 (fourth stream / virtual stream), 4 (fifth stream), 5
(sixth stream), 7 (seventh stream), 8 (eighth stream), 9 (ninth stream), 10 (tenth stream), and so on-->0
    </streamType>
  </Adaption>
  <isSupportVideoImgDB>
    <!--ro, opt, bool, whether the device supports image and video library configuration-->true
  </isSupportVideoImgDB>
  <isSupportEventDataOverWebSocket opt="true,false">
    <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{req, string}-->true
  </isSupportEventDataOverWebSocket>
  <isSupportDynamicHostName>
    <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) configuration-->true
  </isSupportDynamicHostName>
  <isSupportTrafficMonitor>
    <!--ro, opt, bool, whether the device supports traffic monitoring-->true
  </isSupportTrafficMonitor>
  <isSupportWiredandWirelessTrafficMonitor>
    <!--ro, opt, bool, whether the device supports traffic monitoring (wireless NIC and wired NIC)-->true
  </isSupportWiredandWirelessTrafficMonitor>
  <isSupportEmailEncrypt>
    <!--ro, opt, bool, whether the device supports email encryption configuration-->true
  </isSupportEmailEncrypt>
  <isSupportEZVIZUnbind>
    <!--ro, opt, bool, whether the device supports unbinding from EZ, desc:/ISAPI/System/Network/EZVIZ/unbind?format=json-->true
  </isSupportEZVIZUnbind>
  <isSupportIntranetSegment>
    <!--ro, opt, bool, whether the device supports LAN segment configuration, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is false or not be returned. Related URI: /ISAPI/System/Network/intranetSegment/capabilities?
format=json-->true
  </isSupportIntranetSegment>
  <isSupportBasebandCapture>
    <!--ro, opt, bool, whether the device supports capturing packets of the baseband, desc:you can capture packets transmitted between the 4G module and the
base station using the ping command. Related URI: /ISAPI/System/Network/cellularNetwork/basebandCapture/capabilities?format=json-->true
  </isSupportBasebandCapture>
  <isSupportCellularNetworkCellInfo>
    <!--ro, opt, bool, whether the device supports getting a cell's information of the cellular network, desc:you can get the information of the cellular
network where the device is. Related URI: /ISAPI/System/Network/cellularNetwork/cellInfo?format=json&ecurity=<security>&iv=<iv>-->true
  </isSupportCellularNetworkCellInfo>
  <isSupportCellularNetworkLockBand>
    <!--ro, opt, bool, whether the device supports accessing the locked frequency band of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/lockBand/capabilities?format=json-->true
  </isSupportCellularNetworkLockBand>
  <isSupportRadioFrequencyFluctuation>
    <!--ro, opt, bool, whether the device supports configuring RF parameters of the cellular network, desc:related URI:
/ISAPI/System/Network/cellularNetwork/radioFrequencyFluctuation/capabilities?format=json-->true
  </isSupportRadioFrequencyFluctuation>
  <isSupportNetworkModuleUpgrade>
    <!--ro, opt, bool, whether the device supports upgrading network module, desc:including importing upgrade package
(/ISAPI/System/Network/WirelessDial/ImportUpgradePackage?format=json), getting importing progress
(/ISAPI/System/Network/WirelessDial/GetUpgradePackageImportStatus?format=json), starting upgrading
(/ISAPI/System/Network/WirelessDial/StartUpgradeNetworkModule?format=json), get upgrade status
```

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```
(/ISAPI/System/Network/WirelessDial/StartUpgradeNetworkModule?format=json), get upgrade status
(/ISAPI/System/Network/WirelessDial/GetUpgradeNetworkModuleStatus?format=json)-->true
  </isSupportNetworkModuleUpgrade>
  <isSupportNetworkModuleVersionInfo>
    <!--ro, opt, bool, whether the device supports getting the version information of network module, desc:GET
/ISAPI/System/Network/WirelessDial/NetworkModuleVersionInfo?format=json-->true
  </isSupportNetworkModuleVersionInfo>
  <isSupportAutoRecampParams>
    <!--ro, opt, bool, whether the device supports parameters of auto re-camping on a regular basis, desc:GET/PUT
/ISAPI/System/Network/WirelessDial/AutoRecampParams?format=json-->true
  </isSupportAutoRecampParams>
  <isSupportDSTPServerCfg>
    <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET/PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
format=json-->true
  </isSupportDSTPServerCfg>
  <isSupportDSTPProtocolsCap>
    <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
  </isSupportDSTPProtocolsCap>
  <isSupportMQTTAccessCfg>
    <!--ro, opt, bool, whether the device supports MQTT access configuration, desc:GET/PUT /ISAPI/System/Network/MQTTAccess/MQTTAccessCfg?format=json-->true
  </isSupportMQTTAccessCfg>
</NetworkCap>
```

19.1.6.9 Set SSH parameters

Request URL

PUT /ISAPI/System/Network/ssh?readerID=<readerID>&type=<type>&SOCChipID=<SOCChipID>&controllerID=<controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	--
type	enum	--
SOCChipID	string	--
controllerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SSH xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, opt, object, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--wo, req, bool, whether to enable the function-->true
  </enabled>
  <port>
    <!--wo, opt, int-->22
  </port>
</SSH>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, read-only,status description: OK,Device Busy,Device Error,Invalid Operation,Invalid XML Format,Invalid XML Content,Reboot,Additional
Error, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot"
(reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.1.6.10 Reboot device

Request URL

PUT /ISAPI/System/reboot?childDevID=<devIndex>&module=<module>&loginPassword=<loginPassword>

Query Parameter

Parameter Name	Parameter Type	Description
devIndex	string	Reboot a sub-device via specifying sub-device ID. The ID of the sub-device connected to the current device can be searched via /ISAPI/IoTGateway/Childmanage/SearchChild?format=json.
module	enum	The "module" specifies a module to restart, without requiring a device reboot, corresponding to the capability (/ISAPI/System/capabilities/deviceReboot->module). "algorithmProgram (traffic terminal radar-assisted algorithm program, supports independent algorithm program startup), "radar" (restart radar module of the radar-assisted camera).
loginPassword	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.1.6.11 Get the capability of configuring the upload strategy of device status

Request URL

GET /ISAPI/System/status/uploadStrategy/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "enabled": {
    /*ro, req, object, whether to enable the function, desc:whether to enable the function*/
        "@opt": [true, false]
        /*ro, req, array, options, subType:bool*/
    },
    "uploadInterval": {
    /*ro, opt, object, uploading interval*/
        "@min":  5,
        /*ro, req, int, the minimum value, range:[5,300], unit:s*/
        "@max":  300
        /*ro, req, int, the maximum value, range:[5,300], unit:s*/
    }
}
```

19.1.6.12 Get device status

Request URL

GET /ISAPI/System/status?controllerID=<controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
controllerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device status, attr:version{opt, string, protocolVersion}-->
  <currentDeviceTime>
    <!--ro, opt, datetime, device current time-->1970-01-01T00:00:00+08:00
  </currentDeviceTime>
  <deviceUpTime>
    <!--ro, opt, int, device running time, unit:s-->1200
  </deviceUpTime>
  <CPUList>
    <!--ro, opt, array, CPU data list, subType:object-->
    <CPU>
      <!--ro, opt, object, CPU data-->
      <cpuDescription>
        <!--ro, req, string, description-->test
      </cpuDescription>
      <cpuUtilization>
        <!--ro, req, int, CPU usage, range:[0,100], desc:CPU usage-->1
      </cpuUtilization>
    </CPU>
  </CPUList>
  <MemoryList>
    <!--ro, opt, array, memory list, subType:object-->
    <Memory>
      <!--ro, opt, object, memory-->
      <memoryDescription>
        <!--ro, req, string, description-->test
      </memoryDescription>
      <memoryUsage>
        <!--ro, req, float, used memory size, unit:MB-->1.0
      </memoryUsage>
      <memoryAvailable>
        <!--ro, req, float, available memory size, unit:MB-->1.0
      </memoryAvailable>
    </Memory>
  </MemoryList>
  <CameraList>
    <!--ro, opt, array, camera list, subType:object, desc:camera list-->
    <Camera>
      <!--ro, opt, object, camera-->
      <zoomReverseTimes>
        <!--ro, req, int, zoom reverse times-->0
      </zoomReverseTimes>
      <zoomTotalSteps>
        <!--ro, req, int, zoom total steps-->0
      </zoomTotalSteps>
      <focusReverseTimes>
        <!--ro, req, int, focus reverse times-->0
      </focusReverseTimes>
      <focusTotalSteps>
        <!--ro, req, int, focus total steps-->0
      </focusTotalSteps>
```

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```
      <irisShiftTimes>
        <!--ro, req, int, iris shift times-->0
      </irisShiftTimes>
      <irisTotalSteps>
        <!--ro, req, int, iris total steps-->0
      </irisTotalSteps>
      <icrShiftTimes>
        <!--ro, req, int, ICR shift times-->0
      </icrShiftTimes>
      <icrTotalSteps>
        <!--ro, req, int, ICR total steps-->0
      </icrTotalSteps>
      <lensIntirTimes>
        <!--ro, req, int, lens Initialization times-->0
      </lensIntirTimes>
      <cameraRunTotalTime>
        <!--ro, req, int, camera total running time-->0
      </cameraRunTotalTime>
    </Camera>
  </CameraList>
  <DomeInfoList>
    <!--ro, opt, array, PTZ information list, subType:object-->
    <DomeInfo>
      <!--ro, opt, object, PTZ information-->
      <domeRunTotalTime>
        <!--ro, opt, int, the combination of running time at -20° below/40° above/-20°- 40°-->1
      </domeRunTotalTime>
      <runTimeUnderNegativetwenty>
        <!--ro, opt, int, running time at -20° below-->1
      </runTimeUnderNegativetwenty>
      <runTimeBetweenNtwentyPforty>
        <!--ro, opt, int, running time at -20°- 40°-->1
      </runTimeBetweenNtwentyPforty>
      <runtimeOverPositiveforty>
        <!--ro, opt, int, running time at 40° above-->1
      </runtimeOverPositiveforty>
      <panTotalRounds>
        <!--ro, opt, int, pan rounds-->1
      </panTotalRounds>
      <tiltTotalRounds>
        <!--ro, opt, int, tilt rounds-->1
      </tiltTotalRounds>
      <heatState>
        <!--ro, opt, enum, heater status, subType:int, desc:0 (temperature control), 1 (remain open), 2 (remain closed)-->1
      </heatState>
      <fanState>
        <!--ro, opt, enum, fan status, subType:int, desc:0 (temperature control), 1 (remain open), 2 (remain closed)-->1
      </fanState>
      <panFrecRecord>
        <!--ro, opt, int, pan calling times, range:[0,2147483647]-->1
      </panFrecRecord>
      <tiltFrecRecord>
        <!--ro, opt, int, tilt calling times, range:[0,2147483647]-->1
      </tiltFrecRecord>
    </DomeInfo>
  </DomeInfoList>
  <totalRebootCount>
    <!--ro, opt, int, device reboot times-->1
  </totalRebootCount>
  <batteryAllowance>
    <!--ro, opt, int, remaining battery, range:[0,100], desc:remaining battery, "0~100"%-->50
  </batteryAllowance>
  <AlarmInfoList>
    <!--ro, opt, array, alarm information list, subType:object, desc:add-->
    <AlarmInfo>
      <!--ro, opt, object, alarm information-->
      <channelId>
        <!--ro, req, int, channel No.-->1
      </channelId>
      <FaceAlarm>
        <!--ro, opt, object, face picture alarm-->
        <faceSnapRecvAlarm>
          <!--ro, req, int, alarm times of DSP received via face picture-->1
        </faceSnapRecvAlarm>
        <faceSnapSendAlarm>
          <!--ro, req, int, alarm times of face picture sent to the platform-->1
        </faceSnapSendAlarm>
      </FaceAlarm>
      <MixTargetAlarm>
        <!--ro, opt, object, multi-target-type detection alarm-->
        <mixTargetRecvAlarm>
          <!--ro, req, int, alarm times of DSP received via multi-target-type detection-->1
        </mixTargetRecvAlarm>
        <mixTargetSendAlarm>
          <!--ro, req, int, times of face picture alarm sent to the platform-->1
        </mixTargetSendAlarm>
      </MixTargetAlarm>
    </AlarmInfo>
  </AlarmInfoList>
  <SDCardStatusInfo>
    <!--ro, opt, object, SD card status information-->
    <videoRewritingTimes>
      <!--ro, opt, int, range:[0,2147483647]-->0
    </videoRewritingTimes>
```

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```
  </SDCardStatusInfo>
</DeviceStatus>
```

19.1.6.13 Get the video analysis task capability

Request URL

GET /ISAPI/System/Video/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <videoInputPortNums>
    <!--ro, opt, int, number of video input ports-->0
  </videoInputPortNums>
  <videoOutputPortNums>
    <!--ro, opt, int, number of video output ports-->0
  </videoOutputPortNums>
  <menuNums>
    <!--ro, opt, int, number of local menus that can be displayed-->0
  </menuNums>
  <externalChannelNum>
    <!--ro, opt, bool, number of extended analog channels-->true
  </externalChannelNum>
  <isSupportHeatmap>
    <!--ro, opt, bool, whether the device supports heat map-->true
  </isSupportHeatmap>
  <isSupportCounting>
    <!--ro, opt, bool, whether the device supports people counting-->true
  </isSupportCounting>
  <countingType>
    <!--ro, opt, enum, statistics type, subType:string, desc:statistics type-->human
  </countingType>
  <isSupportPicture>
    <!--ro, opt, bool, whether the device supports OSD picture overlay, desc:related URI: /ISAPI/System/Video/inputs/channels/<channelID>/image/picture--
>true
  </isSupportPicture>
  <isSupportPreviewSwitch>
    <!--ro, opt, bool, whether the device supports switching live view-->true
  </isSupportPreviewSwitch>
  <isSupportRecodStatus>
    <!--ro, opt, bool, whether the device supports searching for the recording status-->true
  </isSupportRecodStatus>
  <isSupportPrivacyMask>
    <!--ro, opt, bool, whether the device supports privacy mask-->true
  </isSupportPrivacyMask>
  <isSupportBinocularPreviewSwitch>
    <!--ro, opt, bool, whether the device supports switching live view of the dual-lens camera-->true
  </isSupportBinocularPreviewSwitch>
  <isSupportCalibCheck>
    <!--ro, opt, bool, whether the device supports calibration verification-->true
  </isSupportCalibCheck>
  <isSupportPIP>
    <!--ro, opt, bool-->true
  </isSupportPIP>
  <isSupportFocusVideoMode>
    <!--ro, opt, bool-->true
  </isSupportFocusVideoMode>
  <isSupportExternalChannel>
    <!--ro, opt, bool, whether the device supports extended analog channels-->true
  </isSupportExternalChannel>
  <isSupportMultiChannelCounting>
    <!--ro, opt, bool-->true
  </isSupportMultiChannelCounting>
  <isSupportCountingCollection>
    <!--ro, opt, bool, whether the device supports people counting ANR-->true
  </isSupportCountingCollection>
  <isSupportHeatmapCollection>
    <!--ro, opt, bool, whether the device supports heat map data ANR-->true
  </isSupportHeatmapCollection>
  <channelFlexible opt="name,enable,online,linknum">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->name
  </channelFlexible>
  <isSupportOutputsResource>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/resource?format=json, desc:related URI: /ISAPI/System/Video/outputs/resource?format=json--
>true
  </isSupportOutputsResource>
  <OSDLanguage opt="GBK,EUC-KR,Hebrew" def="GBK">
```

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```
  <OSDLanguage opt="GBK,EUC-KR,Hebrew" def="GBK">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->GBK
  </OSDLanguage>
  <isSupportMixedChannel>
    <!--ro, opt, bool-->true
  </isSupportMixedChannel>
  <isSupportMixedChannelStatus>
    <!--ro, opt, bool-->true
  </isSupportMixedChannelStatus>
  <isSupportOutputCourseware>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/courseware/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/courseware/capabilities?format=json-->true
  </isSupportOutputCourseware>
  <isSupportVideoInputMode>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/inputs/mode/capabilities, desc:related URI: /ISAPI/System/Video/inputs/mode/capabilities-->true
  </isSupportVideoInputMode>
  <isSupportVideoOutputMode>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/mode/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/mode/capabilities?format=json-->true
  </isSupportVideoOutputMode>
  <PreviewMode>
    <!--ro, opt, object-->
    <PIPType>
      <!--ro, opt, object-->
      <mainScreenChannelID>
        <!--ro, opt, int-->0
      </mainScreenChannelID>
      <subScreenChannelID>
        <!--ro, opt, int-->0
      </subScreenChannelID>
    </PIPType>
  </PreviewMode>
  <isSupportViewTag>
    <!--ro, opt, bool-->true
  </isSupportViewTag>
  <isSupportMenuStatus>
    <!--ro, opt, bool-->true
  </isSupportMenuStatus>
</VideoCap>
```

19.1.6.14 Get the wireless alarm capabilities

Request URL

GET /ISAPI/WLAlarm/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WLAlarmCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <isSupportTeleControl>
    <!--ro, opt, bool-->true
  </isSupportTeleControl>
  <isSupportPIR>
    <!--ro, opt, bool-->true
  </isSupportPIR>
  <isSupportWLSensors>
    <!--ro, opt, bool-->true
  </isSupportWLSensors>
  <isSupportCallHelp>
    <!--ro, opt, bool-->true
  </isSupportCallHelp>
  <WLSensorsNum>
    <!--ro, opt, int-->0
  </WLSensorsNum>
</WLAlarmCap>
```

19.1.7 Time Management

19.1.7.1 Get device time synchronization management parameters

Request URL

GET /ISAPI/System/time

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Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--ro, req, enum, time synchronization mode, subType:string, desc:"NTP" (NTP time synchronization), "manual" (manual time synchronization), "satellite" (satellite time synchronization), "platform" (platform time synchronization), "NONE" (time synchronization is not allowed or no time synchronization source), "GB28181" (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--ro, opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}-->2019-02-28T10:50:44+08:30
  </localTime>
  <timeZone>
    <!--ro, opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:daylight saving time configuration in time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the device time returned currently is in DST (Daylight Saving Time) system, desc:whether the device time returned currently is in DST (Daylight Saving Time) system-->true
  </isSummerTime>
  <platformType>
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform-->EZVIZ
  </platformType>
  <platformNo>
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--ro, opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--ro, opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--ro, opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

19.1.7.2 Set device time synchronization management parameters

Request URL

PUT /ISAPI/System/time

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}, desc:the time difference between local time and coordinated universal time (UTC) is configured via the timeZone field-->2019-02-28T10:50:44
  </localTime>
  <timeZone>
    <!--opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the time returned by the current device is that in the DST (daylight saving time), desc:whether the time returned by the current device is that in the DST (daylight saving time)-->true
  </isSummerTime>
  <platformType>
    <!--opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform, related URI: /ISAPI/System/Network/EZVIZ-->EZVIZ
  </platformType>
  <platformNo>
    <!--opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <FailedNodeInfoList>
    <!--ro, opt, object, information list of failed nodes, desc:for the manual time synchronization of central analysis cluster, this field is returned if time synchronization failed-->
    <FailedNodeInfo>
      <!--ro, opt, object, information of failed nodes-->
      <nodeID>
        <!--ro, req, string, node ID, range:[0,64]-->test
      </nodeID>
      <nodeIP>
        <!--ro, req, string, node IP, range:[0,20]-->test
      </nodeIP>
      <reason>
        <!--ro, opt, string, reason why the node failed to synchronize time, range:[0,128]-->test
      </reason>
    </FailedNodeInfo>
  </FailedNodeInfoList>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>
```

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19.1.7.3 Get the capability of device time synchronization management

Request URL

GET /ISAPI/System/time/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management capability set, attr:version{opt, string, protocolVersion}-->
  <timeMode opt="NTP,manual,satellite,SDK,28181,ONVIF,ALL(任何支持的校时方式都允许校时),NONE(不允校时或无校时源),platform,PTP">
    <!--ro, req, enum, time synchronization mode, subType:string, attr:opt{opt, string}, desc:“NTP” (NTP time synchronization), “manual” (manual time
synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or
no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime min="0" max="256">
    <!--ro, opt, string, local time, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTime>
  <timeZone min="0" max="256">
    <!--ro, opt, string, time zone, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </timeZone>
  <satelliteInterval min="0" max="3600">
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, attr:min{opt, string},max{opt, string}, desc:unit: minute-->60
  </satelliteInterval>
  <timeType opt="local,UTC">
    <!--ro, opt, enum, time type, subType:string, attr:opt{opt, string}, desc:“local” (local time), “UTC” (UTC time)-->local
  </timeType>
  <platformType opt="EZVIZ">
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, attr:opt{opt, string}, desc:platform type-->EZVIZ
  </platformType>
  <platformNo min="1" max="2">
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, attr:min{req, int},max{req, int}, desc:it is the only ID, which is
configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <isSupportHistoryTime>
    <!--ro, opt, bool, supported capability of the historical time synchronization list, desc:related URI: /ISAPI/System/time/historyInfo?format=json-->true
  </isSupportHistoryTime>
  <isSupportTimeFilter>
    <!--ro, opt, bool, supported capability of filtering time synchronization, desc:related URI: /ISAPI/System/time/filter/capabilities?format=json-->true
  </isSupportTimeFilter>
  <displayFormat opt="MM/dd/yyyy hh:mm,mm,dd-MM-yyyy hh,MM-dd-yyyy hh:mm,yyyy-MM-dd hh:mm,MM/dd/yyyy hh:mm:ss,dd-MM-yyyy hh:mm:ss,MM-dd-yyyy hh:mm:ss,yyyy-
MM-dd hh:mm:ss,yyyy-MM-dd,dd-MM-yyyy,MM-dd-yyyy">
    <!--ro, opt, enum, time display format, subType:string, attr:opt{req, string}, desc:if this node is returned, it indicates that the device supports
configuring time display format, related URI: /ISAPI/System/time/timeType?format=json-->MM/dd/yyyy hh:mm
  </displayFormat>
  <isSupportSyncDeviceNTPInfoToCamera>
    <!--ro, opt, bool, the capability of synchronizing device’s NTP service information with the camera, desc:related URI:
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json-->true
  </isSupportSyncDeviceNTPInfoToCamera>
  <isSupportNTPService>
    <!--ro, opt, bool-->true
  </isSupportNTPService>
  <isSupportTimeZoneListInfo>
    <!--ro, opt, bool-->true
  </isSupportTimeZoneListInfo>
  <isSupportAutoEnterSummerTime>
    <!--ro, opt, bool-->true
  </isSupportAutoEnterSummerTime>
  <ethernetPort min="0" max="255">
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}, attr:min{req, int},max{req, int}-->1
  </ethernetPort>
  <IANA
opt="Africa/Abidjan,Africa/Accra,Africa/Addis_Ababa,Africa/Algiers,Africa/Blantyre,Africa/Brazzaville,Africa/Cairo,Africa/Casablanca,Africa/Conakry,Africa/D
akar,Africa/Dar_es_Salaam,Africa/Djibouti,Africa/Freetown,Africa/Gaborone,Africa/Harare,Africa/Johannesburg,Africa/Kampala,Africa/Khartoum,Africa/Kigali,Afr
ica/Kinshasa,Africa/Lagos,Africa/Maseru,Africa/Mogadishu,Africa/Nairobi,Africa/Sao_Tome,Africa/Timbuktu,Africa/Tripoli,Africa/Tunis,America/Anchorage,Americ
a/Aruba,America/Asuncion,America/Barbados,America/Belize,America/Bogota,America/Buenos_Aires,America/Cancun,America/Caracas,America/Cayman,America/Chicago,A
merica/Curacao,America/Dawson_Creek,America/Denver,America/Detroit,America/Dominica,America/Edmonton,America/El_Salvador,America/Fortaleza,America/Grand_Tur
k,America/Grenada,America/Guatemala,America/Guyana,America/Halifax,America/Havana,America/Indiana/Indianapolis,America/Indiana/Knox,America/Indiana/Marengo,
America/Indiana/Petersburg,America/Indiana/Tell_City,America/Indiana/Vevay,America/Indiana/Vincennes,America/Indiana/Winamac,America/Jamaica,America/La_Paz,
America/Lima,America/Los_Angeles,America/Louisville,America/Managua,America/Martinique,America/Mendoza,America/Metlakatla,America/Mexico_City,America/Monter
rey,America/Montevideo,America/Montreal,America/Nassau,America/New_York,America/North_Dakota/Beulah,America/North_Dakota/Center,America/North_Dakota/New_Sal
em,America/Panama,America/Phoenix,America/Port_of_Spain,America/Port-au-
Prince,America/Puerto_Rico,America/Santo_Domingo,America/Sao_Paulo,America/St_Johns,America/St_Kitts,America/St_Lucia,America/St_Thomas,America/Tijuana,Amer
ica/Toronto,America/Vancouver,America/Winnipeg,Antarctica/South_Pole,Arctic/Longyearbyen,Asia/Almaty,Asia/Amman,Asia/Anadyr,Asia/Aqtau,Asia/Baghdad,Asia/Bah
rain,Asia/Baku,Asia/Bangkok,Asia/Beirut,Asia/Calcutta,Asia/Damascus,Asia/Dhaka,Asia/Dubai,Asia/Gaza,Asia/Hebron,Asia/Ho_Chi_Minh,Asia/Hong_Kong,Asia/Jakarta
,Asia/Jerusalem,Asia/Kabul,Asia/Karachi,Asia/Kathmandu,Asia/Kuala_Lumpur,Asia/Kuwait,Asia/Macau(China),Asia/Manila,Asia/Muscat,Asia/Phnom_Penh,Asia/Pyongyan
g,Asia/Rangoon,Asia/Riyadh,Asia/Seoul,Asia/Shanghai,Asia/Singapore,Asia/Taipei,Asia/Tehran,Asia/Tel_Aviv,Asia/Tokyo,Asia/Ulaanbaatar,Atlantic/Bermuda,Atlant
ic/Canary,Atlantic/Cape_Verde,Atlantic/Reykjavik,Atlantic/Stanley,Australia/Adelaide,Australia/Brisbane,Australia/Canberra,Australia/Darwin,Australia/Melbou
rne,Australia/NSW,Australia/Perth,Australia/Queensland,Australia/Sydney,Australia/Victoria,Canada/Newfoundland,Canada/Saskatchewan,Chile/EasterIsland,Europe
/Amsterdam,Europe/Andorra,Europe/Athens,Europe/Belfast,Europe/Belgrade,Europe/Berlin,Europe/Bratislava,Europe/Brussels,Europe/Bucharest,Europe/Budapest,Euro
pe/Chisinau,Europe/Copenhagen,Europe/Dublin,Europe/Gibraltar,Europe/Helsinki,Europe/Isle_of_Man,Europe/Istanbul,Europe/Kiev,Europe/Lisbon,Europe/London,Euro
```

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```
pe/Luxembourg,Europe/Madrid,Europe/Malta,Europe/Minsk,Europe/Monaco,Europe/Moscow,Europe/Oslo,Europe/Paris,Europe/Prague,Europe/Riga,Europe/Rome,Europe/San_
Marino,Europe/Sarajevo,Europe/Simferopol,Europe/Skopje,Europe/Sofia,Europe/Stockholm,Europe/Tallinn,Europe/Vatican,Europe/Vienna,Europe/Vilnius,Europe/Warsa
w,Europe/Zagreb,Europe/Zurich,Indian/Antananarivo,Indian/Maldives,Indian/Mauritius,Pacific/Auckland,Pacific/Fiji,Pacific/Guam,Pacific/Honolulu,Pacific/Kirit
imati,Pacific/Noumea,America/Antigua,MST,Asia/Kolkata,America/Santiago,Africa/Monrovia,Asia/Colombo,America/Chihuahua,America/Tegucigalpa,America/Paramaribo
,Europe/Ljubljana,Asia/Ashgabat,Africa/Asmara,Asia/Brunei,Africa/Bangui,Africa/Banjul,Asia/Bishkek,Africa/Bissau,Africa/Bujumbura,Asia/Dili,Asia/Dushanbe,Pa
cific/Funafuti,Pacific/Guadalcanal,Africa/Juba,America/St_Vincent,Africa/Libreville,Africa/Lome,Africa/Luanda,Africa/Lusaka,Pacific/Majuro,Africa/Malabo,Afr
ica/Maputo,Africa/Mbabane,Indian/Comoro,Africa/Ndjamena,Pacific/Palau,Africa/Niamey,Europe/Nicosia,Africa/Nouakchott,Pacific/Tongatapu,Africa/Ouagadougou,Am
erica/Lower_Princes,Europe/Podgorica,Pacific/Efate,Africa/Porto-
Novo,Asia/Tashkent,Asia/Tbilisi,Asia/Thimphu,Europe/Tirane,Europe/Vaduz,Asia/Vientiane,Africa/Windhoek,Africa/Douala,Pacific/Nauru,Asia/Yerevan,Etc/GMT+12,E
tc/GMT+11,America/Adak,Pacific/Marquesas,Etc/GMT+9,Etc/GMT+8,America/Whitehorse,Pacific/Easter,America/Regina,America/Indianapolis,America/Cuiaba,America/Ar
aguaina,America/Cayenne,America/Godthab,America/Punta_Arenas,America/Miquelon,America/Bahia,Etc/GMT+2,Atlantic/Azores,Etc/UTC,Europe/Kaliningrad,Europe/Astr
akhan,Europe/Samara,Europe/Saratov,Europe/Volgograd,Asia/Yekaterinburg,Asia/Qyzylorda,Asia/Katmandu,Asia/Omsk,Asia/Barnaul,Asia/Hovd,Asia/Krasnoyarsk,Asia/N
ovosibirsk,Asia/Tomsk,Asia/Irkutsk,Australia/Eucla,Asia/Chita,Asia/Yakutsk,Pacific/Port_Moresby,Australia/Hobart,Asia/Vladivostok,Australia/Lord_Howe,Pacifi
c/Bougainville,Asia/Srednekolymsk,Asia/Magadan,Pacific/Norfolk,Asia/Sakhalin,Asia/Kamchatka,Etc/GMT-12,Pacific/Chatham,Etc/GMT-13,Pacific/Apia">
    <!--ro, opt, enum, IANA synchronization, subType:string, attr:opt{opt, string}-->Africa/Abidjan
  </IANA>
  <isSupportSearchDeviceIANAList>
    <!--ro, opt, bool, whether the device supports searching for supported IANA list, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
  </isSupportSearchDeviceIANAList>
  <windowsZone opt="Dateline Standard Time,UTC-11,Aleutian Standard Time,Hawaiian Standard Time,Marquesas Standard Time,Alaskan Standard Time,UTC-09,Pacific
Standard Time (Mexico),UTC-08,Pacific Standard Time,US Mountain Standard Time,Mountain Standard Time (Mexico),Mountain Standard Time,Yukon Standard
Time,Central America Standard Time,Central Standard Time,Easter Island Standard Time,Central Standard Time (Mexico),Canada Central Standard Time,SA Pacific
Standard Time,Eastern Standard Time (Mexico),Eastern Standard Time,Haiti Standard Time,Cuba Standard Time,US Eastern Standard Time,Turks And Caicos Standard
Time,Paraguay Standard Time,Atlantic Standard Time,Venezuela Standard Time,Central Brazilian Standard Time,SA Western Standard Time,Pacific SA Standard
Time,Newfoundland Standard Time,Tocantins Standard Time,E. South America Standard Time,SA Eastern Standard Time,Argentina Standard Time,Greenland Standard
Time,Montevideo Standard Time,Magallanes Standard Time,Saint Pierre Standard Time,Bahia Standard Time,UTC-02,Azores Standard Time,Cape Verde Standard
Time,UTC,GMT Standard Time,Greenwich Standard Time,Sao Tome Standard Time,Morocco Standard Time,W. Europe Standard Time,Central Europe Standard Time,Romance
Standard Time,Central European Standard Time,W. Central Africa Standard Time,Jordan Standard Time,GTB Standard Time,Middle East Standard Time,Egypt Standard
Time,E. Europe Standard Time,Syria Standard Time,West Bank Standard Time,South Africa Standard Time,FLE Standard Time,Israel Standard Time,South Sudan
Standard Time,Kaliningrad Standard Time,Sudan Standard Time,Libya Standard Time,Namibia Standard Time,Arabic Standard Time,Turkey Standard Time,Arab
Standard Time,Belarus Standard Time,Russian Standard Time,E. Africa Standard Time,Iran Standard Time,Arabian Standard Time,Astrakhan Standard
Time,Azerbaijan Standard Time,Russia Time Zone 3,Mauritius Standard Time,Saratov Standard Time,Georgian Standard Time,Volgograd Standard Time,Caucasus
Standard Time,Afghanistan Standard Time,West Asia Standard Time,Ekaterinburg Standard Time,Pakistan Standard Time,Qyzylorda Standard Time,India Standard
Time,Sri Lanka Standard Time,Nepal Standard Time,Central Asia Standard Time,Bangladesh Standard Time,Omsk Standard Time,Myanmar Standard Time,SE Asia
Standard Time,Altai Standard Time,W. Mongolia Standard Time,North Asia Standard Time,N. Central Asia Standard Time,Tomsk Standard Time,China Standard
Time,North Asia East Standard Time,Singapore Standard Time,W. Australia Standard Time,Taipei Standard Time,Ulaanbaatar Standard Time,Aus Central W. Standard
Time,Transbaikal Standard Time,Tokyo Standard Time,North Korea Standard Time,Korea Standard Time,Yakutsk Standard Time,Cen. Australia Standard Time,AUS
Central Standard Time,E. Australia Standard Time,AUS Eastern Standard Time,West Pacific Standard Time,Tasmania Standard Time,Vladivostok Standard Time,Lord
Howe Standard Time,Bougainville Standard Time,Russia Time Zone 10,Magadan Standard Time,Norfolk Standard Time,Sakhalin Standard Time,Central Pacific
Standard Time,Russia Time Zone 11,New Zealand Standard Time,UTC+12,Fiji Standard Time,Chatham Islands Standard Time,UTC+13,Tonga Standard Time,Samoa
Standard Time,Line Islands Standard Time">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string}-->Dateline Standard Time
  </windowsZone>
  <isSupportTimeStateParam>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeStateParam?format=json-->true
  </isSupportTimeStateParam>
  <localTimeDST min="0" max="256">
    <!--ro, opt, string, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTimeDST>
  <isSupportStandard>
    <!--ro, opt, bool-->true
  </isSupportStandard>
  <isSupportPTPParams>
    <!--ro, opt, bool, desc:GET/PUT /ISAPI/System/time/PTPParams?format=json-->true
  </isSupportPTPParams>
  <isSupportTimeZoneLock>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeZoneLock?format=json-->true
  </isSupportTimeZoneLock>
</Time>
```


19.1.7.4 Set device local time

Request URL

PUT /ISAPI/System/time/localTime

Query Parameter

None

Request Message

Binary Data

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.1.7.5 Set parameters of all NTP servers

Request URL

PUT /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--opt, object, NTP server information-->
    <id>
      <!--req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

Response Message

dongjin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.7.6 Get the parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{req, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--ro, req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--ro, opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--ro, opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

19.1.7.7 Set the parameters of a NTP server

Request URL

PUT /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

dojo jin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, NTP server domain name地址去掉, range:[1,64]-->12345
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,32], desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--opt, bool-->false
  </enabled>
</NTPServer>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.1.7.8 Get the parameters of a NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	NTP server No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--ro, req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, NTP server domain name, range:[1,64], dep:and,{$.NTPServer.addressingFormatType,eq,hostname}-->xxx12345
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,32], dep:and,{$.NTPServer.addressingFormatType,eq,ipAddress}-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--ro, opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--ro, opt, bool-->false
  </enabled>
  <devicePortNoValid>
    <!--ro, opt, enum, subType:string-->yes
  </devicePortNoValid>
  <portType>
    <!--ro, opt, enum, subType:string-->auto
  </portType>
  <customPortNo>
    <!--ro, opt, int, range:[1,65535], step:1, dep:and,{$.NTPServer.portType,eq,custom}-->1
  </customPortNo>
  <hostNameExampleList>
    <!--ro, opt, array, subType:object-->
    <hostNameExample>
      <!--ro, opt, object-->
      <hostName>
        <!--ro, opt, string-->xxx12345
      </hostName>
    </hostNameExample>
  </hostNameExampleList>
</NTPServer>
```

19.1.7.9 Get the configuration capability of parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers/capabilities

Query Parameter

None

Request Message

None

Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, NTP server address type, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName min="1" max="64">
      <!--ro, opt, string, NTP server domain name, range:[1,64], attr:min{req, int},max{req, int}-->12345
    </hostName>
    <ipAddress min="1" max="32">
      <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
    </ipAddress>
    <ipv6Address min="1" max="128">
      <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo min="1" max="65535">
      <!--ro, opt, int, port No., range:[1,65535], attr:min{req, int},max{req, int}, desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval min="1" max="10800">
      <!--ro, opt, int, time synchronization interval, unit:min, attr:min{req, int},max{req, int}-->1440
    </synchronizeInterval>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable, attr:opt{req, string}, desc:true (enable), false (disable)-->true
    </enabled>
  </NTPServer>
</NTPServerList>
```

19.1.7.10 Execute a test to check if the NTP server is available

Request URL

POST /ISAPI/System/time/ntpServers/test

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPTestDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <addressingFormatType>
    <!--req, enum, "ipaddress,hostname", subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--req, int, port No.-->1
  </portNo>
</NTPTestDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPTestResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <errorDescription>
    <!--ro, req, string, error description-->test
  </errorDescription>
</NTPTestResult>
```

19.1.7.11 Set the device time type

Request URL

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PUT /ISAPI/System/time/timeType?format=json

Query Parameter

None

Request Message

```
{
    "TimeType": {
    /*req, object, time type information, desc:time type information*/
        "type":  "UTC",
        /*req, enum, time type, subType:string, desc:"UTC, local "*/
        "displayFormat":  "MM/dd/yyyy hh:mm"
        /*opt, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

19.1.7.12 Get the device time type

Request URL

GET /ISAPI/System/time/timeType?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "TimeType": {
    /*ro, req, object*/
        "type":  "UTC",
        /*ro, req, enum, value: local,UTC, subType:string, desc:"UTC,local"*/
        "displayFormat":  "MM/dd/yyyy hh:mm"
        /*ro, opt, enum, subType:string*/
    }
}
```

19.1.7.13 Get time zone

Request URL

GET /ISAPI/System/time/timeZone

Query Parameter

None

Request Message

None

Response Message

None

19.1.7.14 Set time zone

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Request URL

PUT /ISAPI/System/time/timeZone

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.1.8 Upgrade Management

19.1.8.1 Get the device upgrading status and progress

Request URL

GET /ISAPI/System/upgradeStatus?type=<type>&childDevID=<childDevID>&id=<id>

Query Parameter

Parameter Name	Parameter Type	Description
type	string	"usbmic" (array microphone module)
childDevID	string	--
id	string	The id indicates the peripheral No. to be upgraded.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<upgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, upgrade status and result, attr:version{req, string, protocolVersion}-->
  <upgrading>
    <!--ro, req, bool, upgrade status-->true
  </upgrading>
  <percent>
    <!--ro, req, int, upgrade progress (% complete), range:[0,100]-->1
  </percent>
  <reboot>
    <!--ro, opt, bool, dep:and,{$.upgradeStatus.percent,eq,100}-->false
  </reboot>
  <upgradeStage>
    <!--ro, opt, enum, subType:string-->software
  </upgradeStage>
</upgradeStatus>
```

19.2 Video (General)

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19.2.1 Exposure Parameters

19.2.1.1 Set the exposure mode for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/exposure

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Exposure xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, exposure mode, attr:version{req, string, protocolVersion}-->
  <ExposureType>
    <!--req, enum, exposure type, subType:string, desc:exposure type-->auto
  </ExposureType>
  <autoIrisLevel>
    <!--opt, int, automatic aperture level-->1
  </autoIrisLevel>
  <OverexposeSuppress>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <Type>
      <!--opt, enum, "AUTO", "MANUAL", subType:string, desc:"AUTO", "MANUAL"-->AUTO
    </Type>
    <DistanceLevel>
      <!--opt, int, focal length-->1
    </DistanceLevel>
    <shortIRDistanceLevel>
      <!--opt, int, this node depends on  <Type>-->1
    </shortIRDistanceLevel>
    <longIRDistanceLevel>
      <!--opt, int, this node depends on  <Type>-->1
    </longIRDistanceLevel>
    <supplementLightIntensity1>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity1>
    <supplementLightIntensity2>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity2>
    <supplementLightIntensity3>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity3>
    <supplementLightIntensity4>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity4>
  </OverexposeSuppress>
  <pIris>
    <!--opt, object, aperture-->
    <pIrisType>
      <!--opt, enum, aperture type, subType:string, desc:aperture type-->AUTO
    </pIrisType>
    <IrisLevel>
      <!--opt, int, aperture level-->1
    </IrisLevel>
  </pIris>
  <faceExposure>
    <!--opt, object, face exposure-->
    <enabled>
      <!--opt, bool, whether to enable face exposure-->true
    </enabled>
    <sensitivity>
      <!--opt, int, sensitivity of face exposure-->1
    </sensitivity>
  </faceExposure>
  <PIrisGeneral>
    <!--opt, object, it is available when <ExposureType> is set to "pIris-General", desc:it is available when ExposureType is set to "pIris-General"-->
    <irisLevel>
      <!--opt, int, aperture level of general lens, range:[1,100]-->1
    </irisLevel>
  </PIrisGeneral>
</Exposure>
```

Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.1.2 Get the exposure mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/exposure?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Camera ID
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<Exposure xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, exposure mode, attr:version{req, string, protocolVersion}-->
  <ExposureType>
    <!--ro, req, enum, lens type, subType:string, desc:lens type-->auto
  </ExposureType>
  <autoIrisLevel>
    <!--ro, opt, int-->1
  </autoIrisLevel>
  <OverexposeSuppress>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <Type>
      <!--ro, opt, enum, this node depends on <enabled>,"AUTO,MANUAL", subType:string, desc:this node depends on <enabled>,"AUTO,MANUAL"-->AUTO
    </Type>
    <DistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </DistanceLevel>
    <shortIRDistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </shortIRDistanceLevel>
    <longIRDistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </longIRDistanceLevel>
    <supplementLightIntensity1>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity1>
    <supplementLightIntensity2>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity2>
    <supplementLightIntensity3>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity3>
    <supplementLightIntensity4>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity4>
    <hightLightDistanceLevel>
      <!--ro, opt, int, range:[1,100]-->50
    </hightLightDistanceLevel>
    <lowLightDistanceLevel>
      <!--ro, opt, int, range:[1,100]-->50
    </lowLightDistanceLevel>
  </OverexposeSuppress>
  <pIris>
    <!--ro, opt, object, this node depends on <ExposureType>,"AUTO,MANUAL"-->
    <pIrisType>
      <!--ro, opt, enum, subType:string-->AUTO
    </pIrisType>
    <IrisLevel>
      <!--ro, opt, int, this node depends on <pIrisType>-->1
    </IrisLevel>
  </pIris>
  <faceExposure>
    <!--ro, opt, object, face exposure-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <sensitivity>
      <!--ro, opt, int, sensitivity-->1
    </sensitivity>
  </faceExposure>
  <PIrisGeneral>
    <!--ro, opt, object, it is available when <ExposureType> is set to "pIris-General", desc:it is valid when <ExposureType> is set to "pIris-General"-->
    <irisLevel>
      <!--ro, opt, int, iris level of general lens, range:[1,100]-->1
    </irisLevel>
  </PIrisGeneral>
  <combinationParams>
    <!--ro, opt, int, range:[0,100]-->50
  </combinationParams>
  <opticalFilter>
    <!--ro, opt, enum, subType:string-->none
  </opticalFilter>
  <lowGhostingExposure>
    <!--ro, opt, object-->
    <lowGhostingShutterLevel>
      <!--ro, opt, enum, subType:string-->1/100
    </lowGhostingShutterLevel>
  </lowGhostingExposure>
  <exposureCompensateLevel>
    <!--ro, opt, int, range:[-100,100]-->0
  </exposureCompensateLevel>
  <exposureControl>
    <!--ro, opt, enum, subType:string-->center
  </exposureControl>
</Exposure>
```

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19.2.1.3 Set the exposure time parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/shutter

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Shutter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <ShutterLevel>
    <!--opt, enum, exposure level, subType:string, desc:"1/1", "1/2", "1/3", "1/6", "1/12", "1/25", "1/50", "1/75", "1/100", "1/120", "1/125", "1/150",
"1/175", this node is valid when exposure type is "shutter first"-->1/1
  </ShutterLevel>
  <scheduleEnabled>
    <!--opt, bool-->true
  </scheduleEnabled>
  <scheduleShutterLevel>
    <!--opt, int-->1
  </scheduleShutterLevel>
  <ScheduleList>
    <!--opt, array, subType:object-->
    <Schedule>
      <!--opt, object-->
      <startHour>
        <!--opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--opt, int, end time (hour)-->1
      </endHour>
      <endMinute>
        <!--opt, int, end time (minute)-->1
      </endMinute>
    </Schedule>
  </ScheduleList>
  <redShutterLevel>
    <!--opt, int, range:[0,20000]-->1
  </redShutterLevel>
</Shutter>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.1.4 Get the exposure time parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/shutter?parameterType=<parameterType>

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Shutter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <ShutterLevel>
    <!--ro, opt, enum, exposure level, subType:string, desc:"1/1", "1/2", "1/3", "1/6", "1/12", "1/25", "1/50", "1/75", "1/100", "1/120", "1/125", "1/150",
"1/175", this node is valid when exposure type is "shutter first"-->1/1
  </ShutterLevel>
  <scheduleEnabled>
    <!--ro, opt, bool-->true
  </scheduleEnabled>
  <scheduleShutterLevel>
    <!--ro, opt, int-->1
  </scheduleShutterLevel>
  <ScheduleList>
    <!--ro, opt, array, subType:object-->
    <Schedule>
      <!--ro, opt, object-->
      <startHour>
        <!--ro, opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--ro, opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--ro, opt, int, end time (hour)-->1
      </endHour>
      <endMinute>
        <!--ro, opt, int, end time (minute)-->1
      </endMinute>
    </Schedule>
  </ScheduleList>
  <redShutterLevel>
    <!--ro, opt, int, range:[0,20000]-->1
  </redShutterLevel>
  <maxShutterLevelLimit>
    <!--ro, opt, enum, subType:string-->1/1
  </maxShutterLevelLimit>
  <minShutterLevelLimit>
    <!--ro, opt, enum, subType:string-->1/1
  </minShutterLevelLimit>
</Shutter>
```

19.2.2 Focus Parameters

19.2.2.1 Get the focus parameters of the specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/focusConfiguration

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FocusConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, focus configuration, attr:version{req, string, protocolVersion}-->
  <focusStyle>
    <!--ro, req, enum, focus mode, subType:string, desc:"AUTO" (automatic focus), "MANUAL" (manual focus), "SEMIAUTOMATIC" (semi-automatic focus)-->AUTO
  </focusStyle>
  <focusLimited>
    <!--ro, opt, enum, focus limits, subType:int, desc:focus limits-->10
  </focusLimited>
  <focusPosition>
    <!--ro, opt, int, focus point position-->1
  </focusPosition>
  <focusSpeed>
    <!--ro, opt, int, focus speed-->1
  </focusSpeed>
  <focusSensitivity>
    <!--ro, opt, int, focus sensitivity-->1
  </focusSensitivity>
  <temperatureChangeAdaptEnabled>
    <!--ro, opt, bool, whether to enable temperature change adaption-->true
  </temperatureChangeAdaptEnabled>
  <relativeFocusPos>
    <!--ro, opt, int, relative focus point position-->1
  </relativeFocusPos>
  <lensType>
    <!--ro, opt, enum, lens type, subType:string, desc:lens type-->1140
  </lensType>
  <highTemperaturePriority>
    <!--ro, opt, bool, whether to enable high temperature priority mode-->true
  </highTemperaturePriority>
  <focusRange>
    <!--ro, opt, enum, focus adjusting range, subType:string, desc:focus adjusting range-->max
  </focusRange>
  <focusDistanceMode>
    <!--ro, opt, enum, focus distance mode, subType:string, desc:focus distance mode-->compatible
  </focusDistanceMode>
  <zoomFocusFollowEnabled>
    <!--ro, opt, bool-->true
  </zoomFocusFollowEnabled>
  <focusRegion>
    <!--ro, opt, enum, subType:string, dep:or,{$.FocusConfiguration.focusStyle,eq,AUTO},{$.FocusConfiguration.focusStyle,eq,SEMIAUTOMATIC}-->center
  </focusRegion>
  <focusScene>
    <!--ro, opt, enum, subType:string-->normal
  </focusScene>
  <timeFocusEnabled>
    <!--ro, opt, bool, dep:or,{$.FocusConfiguration.focusStyle,eq,AUTO},{$.FocusConfiguration.focusStyle,eq,SEMIAUTOMATIC}-->true
  </timeFocusEnabled>
  <timeFocus>
    <!--ro, opt, int, range:[1,1440], unit:min, dep:and,{$.FocusConfiguration.timeFocusEnabled,eq,true}-->1
  </timeFocus>
  <focusDistance>
    <!--ro, opt, float, range:[0.30,99.99], step:0.01, unit:m, dep:and,{$.FocusConfiguration.focusStyle,eq,distanceCalculate}-->0.35
  </focusDistance>
  <focusStrategy>
    <!--ro, opt, enum, subType:string-->DTOF
  </focusStrategy>
</FocusConfiguration>
```

19.2.2.2 Set the focus parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/focusConfiguration

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FocusConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, focus configuration, attr:version{req, string, protocolVersion}-->
  <focusStyle>
    <!--req, enum, focus mode, subType:string, desc:"auto" (automatic focus), "manual" (manual focus), "semiautomatic" (semiautomatic focus)-->AUTO
  </focusStyle>
  <focusLimited>
    <!--opt, enum, subType:int-->10
  </focusLimited>
  <focusPosition>
    <!--opt, int, focus point position-->1
  </focusPosition>
  <focusSpeed>
    <!--opt, int, focus speed-->1
  </focusSpeed>
  <focusSensitivity>
    <!--opt, int, focus sensitivity-->1
  </focusSensitivity>
  <temperatureChangeAdaptEnabled>
    <!--opt, bool, whether to enable temperature change adaption-->true
  </temperatureChangeAdaptEnabled>
  <relativeFocusPos>
    <!--opt, int, relative focus point position-->1
  </relativeFocusPos>
  <lensType>
    <!--opt, enum, lens type, subType:string, desc:"1140", "1050"-->1140
  </lensType>
  <highTemperaturePriority>
    <!--opt, bool, whether to enable high temperature priority mode-->true
  </highTemperaturePriority>
  <focusRange>
    <!--opt, enum, focus adjustment range, subType:string, desc:"min" (the minimum), "medium" (the medium), "max" (the maximum)-->max
  </focusRange>
  <focusDistanceMode>
    <!--opt, enum, focus distance mode, subType:string, desc:focus distance mode-->compatible
  </focusDistanceMode>
</FocusConfiguration>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.2.3 Get the image capability of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/imageCap

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ImageCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <isSupportRegionalExposure>
    <!--ro, opt, bool-->true
  </isSupportRegionalExposure>
  <isSupportRegionalFocus>
    <!--ro, opt, bool-->true
  </isSupportRegionalFocus>
  <isSupportManualBackGroundCorrect>
    <!--ro, opt, bool-->true
  </isSupportManualBackGroundCorrect>
  <isSupportManualShutterCorrect>
    <!--ro, opt, bool-->true
  </isSupportManualShutterCorrect>
  <RegionalExposure>
    <!--ro, opt, object-->
    <enabled opt="true,false" def="false">
      <!--ro, opt, bool, attr:opt{req, string},def{req, string}-->true
    </enabled>
    <showRlueEnabled opt="true,false" def="false">
      <!--ro, opt, bool, attr:opt{req, string},def{req, string}-->true
    </showRlueEnabled>
  </RegionalExposure>
</ImageCap>
```

19.2.3 Image Adjustment

19.2.3.1 Set image parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <videoInputID>
    <!--req, int-->1
  </videoInputID>
  <ImageFlip>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
  </ImageFlip>
  <WDR>
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->open
    </mode>
    <WDRLevel>
      <!--opt, int, range:[1,100]-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <BLCRegionList>
      <!--opt, array, subType:object-->
      <BLCRegion>
        <!--opt, object-->
        <id>
          <!--req, int, area No., range:[1,10]-->1
        </id>
        <RegionCoordinatesList>
          <!--req, array, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
```

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```
          <RegionCoordinates>
            <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,704]-->1
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,576]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </BLCRegion>
    </BLCRegionList>
  </BLC>
  <NoiseReduce>
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->close
    </mode>
    <GeneralMode>
      <!--opt, object-->
      <generalLevel>
        <!--req, int, range:[0,100]-->1
      </generalLevel>
    </GeneralMode>
  </NoiseReduce>
  <WhiteBalance>
    <!--opt, object, white balance-->
    <WhiteBalanceStyle>
      <!--opt, enum, subType:string-->auto
    </WhiteBalanceStyle>
    <WhiteBalanceRed>
      <!--opt, int, range:[0,100]-->1
    </WhiteBalanceRed>
    <WhiteBalanceBlue>
      <!--opt, int, range:[0,100]-->1
    </WhiteBalanceBlue>
  </WhiteBalance>
  <Exposure>
    <!--opt, object-->
    <ExposureType>
      <!--req, enum, subType:string-->auto
    </ExposureType>
    <autoIrisLevel>
      <!--opt, int, range:[0,100]-->1
    </autoIrisLevel>
    <OverexposeSuppress>
      <!--opt, object-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <Type>
        <!--opt, enum, subType:string-->AUTO
      </Type>
      <DistanceLevel>
        <!--opt, int, focal length, range:[1,100]-->1
      </DistanceLevel>
    </OverexposeSuppress>
    <pIris>
      <!--opt, object, aperture-->
      <pIrisType>
        <!--opt, enum, subType:string-->MANUAL
      </pIrisType>
      <IrisLevel>
        <!--opt, int, range:[1,100]-->1
      </IrisLevel>
    </pIris>
  </Exposure>
  <Sharpness>
    <!--opt, object-->
    <SharpnessLevel>
      <!--req, int, range:[0,100]-->1
    </SharpnessLevel>
  </Sharpness>
  <Shutter>
    <!--opt, object-->
    <ShutterLevel>
      <!--opt, enum, subType:string-->1/1
    </ShutterLevel>
  </Shutter>
  <powerLineFrequency>
    <!--opt, object-->
    <powerLineFrequencyMode>
      <!--opt, enum, subType:string-->50hz
    </powerLineFrequencyMode>
  </powerLineFrequency>
  <Gain>
    <!--opt, object, gain-->
    <GainLevel>
      <!--opt, string, range:[0,10]-->test
    </GainLevel>
  </Gain>
  <Color>
    <!--opt, object-->
    <brightnessLevel>
```

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```
        <!--opt, int, brightness, range:[0,100]-->0
      </brightnessLevel>
      <contrastLevel>
        <!--opt, int, contrast, range:[0,100]-->0
      </contrastLevel>
      <saturationLevel>
        <!--opt, int, saturation, range:[0,100]-->0
      </saturationLevel>
      <grayScale>
        <!--opt, object-->
        <grayScaleMode>
          <!--opt, enum, subType:string-->indoor
        </grayScaleMode>
      </grayScale>
    </Color>
    <IrcutFilter>
      <!--opt, object-->
      <IrcutFilterType>
        <!--opt, enum, subType:string-->auto
      </IrcutFilterType>
      <nightToDayFilterLevel>
        <!--opt, enum, subType:string-->low
      </nightToDayFilterLevel>
      <nightToDayFilterTime>
        <!--opt, int, range:[0,10]-->1
      </nightToDayFilterTime>
    </IrcutFilter>
    <Scene>
      <!--opt, object-->
      <mode>
        <!--req, enum, subType:string-->indoor
      </mode>
    </Scene>
    <PTZ>
      <!--opt, object, PTZ-->
      <enabled>
        <!--opt, bool, whether to enable the function-->true
      </enabled>
    </PTZ>
    <EIS>
      <!--opt, object-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
    </EIS>
    <HLC>
      <!--opt, object-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <HLCLevel>
        <!--opt, int, range:[0,10]-->0
      </HLCLevel>
    </HLC>
    <Dehaze>
      <!--opt, object-->
      <DehazeMode>
        <!--opt, enum, subType:string-->open
      </DehazeMode>
      <DehazeLevel>
        <!--opt, int, range:[0,10]-->0
      </DehazeLevel>
    </Dehaze>
    <CaptureMode>
      <!--opt, object, capture mode-->
      <mode>
        <!--req, enum, subType:string-->close
      </mode>
    </CaptureMode>
    <LensDistortionCorrection>
      <!--opt, object-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
    </LensDistortionCorrection>
    <LaserLight>
      <!--opt, object-->
      <mode>
        <!--opt, enum, subType:string-->auto
      </mode>
    </LaserLight>
  </ImageChannel>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.3.2 Get image configuration capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, temperature range capability, see XML_Cap_tempRange for details, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <videoInputID>
    <!--ro, req, int-->1
  </videoInputID>
  <FocusConfiguration>
    <!--ro, opt, object-->
    <focusStyle opt="AUTO,MANUAL,SEMIAUTOMATIC,distanceCalculate">
      <!--ro, req, string, attr:opt{req, string}-->test
    </focusStyle>
    <focusLimited opt="10,50,100,150,300,600,1000,2000,5000,20000,50000,65535">
      <!--ro, opt, int, attr:opt{req, string}-->1
    </focusLimited>
  </FocusConfiguration>
  <ImageFlip>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <ImageFlipStyle opt="LEFTRIGHT,UPDOWN,CENTER,AUTO">
      <!--ro, opt, string, type, attr:opt{req, string}-->LEFTRIGHT
    </ImageFlipStyle>
  </ImageFlip>
  <WDR>
    <!--ro, opt, object-->
    <mode opt="open,close,auto,SDR,time">
      <!--ro, req, enum, WDR mode: "open,close,auto", subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->open
    </mode>
    <WDRLevel min="1" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <BLCMode opt="UP,DOWN,LEFT,RIGHT,CENTER,MULTI-AREA,Region,AUTO">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->UP
    </BLCMode>
```

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AND

```
      <BLCRegionList>
        <!--ro, opt, array, subType:object-->
        <BLCRegion>
          <!--ro, opt, object-->
          <id>
            <!--ro, req, int, area No.-->1
          </id>
          <RegionCoordinatesList>
            <!--ro, req, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
            <RegionCoordinates>
              <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--ro, req, int, X-coordinate, range:[0,704]-->1
              </positionX>
              <positionY>
                <!--ro, req, int, Y-coordinate, range:[0,576]-->1
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </BLCRegion>
      </BLCRegionList>
    </BLC>
    <NoiseReduce>
      <!--ro, opt, object-->
      <mode opt="close,general,advanced">
        <!--ro, req, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->close
      </mode>
      <GeneralMode>
        <!--ro, opt, object, mode-->
        <generalLevel min="0" max="10">
          <!--ro, req, int, attr:min{req, int},max{req, int}-->1
        </generalLevel>
      </GeneralMode>
      <AdvancedMode>
        <!--ro, opt, object-->
        <FrameNoiseReduceLevel min="0" max="10">
          <!--ro, req, int, attr:min{req, int},max{req, int}-->1
        </FrameNoiseReduceLevel>
        <InterFrameNoiseReduceLevel min="0" max="10">
          <!--ro, req, int, attr:min{req, int},max{req, int}-->1
        </InterFrameNoiseReduceLevel>
      </AdvancedMode>
    </NoiseReduce>
    <WhiteBalance>
      <!--ro, opt, object, white balance-->
      <WhiteBalanceStyle opt="manual、auto1、auto2、locked、outdoor、indoor、daylightLamp、sodiumlight、autotrace、onece、autooutdoor、autosodiumlight、
mercurylight、auto、incandescentlight、warmlight、naturallight,customColorTemperature,highTemperature">
        <!--ro, req, string, attr:opt{req, string}-->test
      </WhiteBalanceStyle>
      <WhiteBalanceRed min="-127" max="127">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
      </WhiteBalanceRed>
      <WhiteBalanceBlue min="-127" max="127">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
      </WhiteBalanceBlue>
    </WhiteBalance>
    <Exposure>
      <!--ro, opt, object-->
      <ExposureType opt="auto,IrisFirst,ShutterFirst,gainFirst,manual,pIris,T5280-PQ1,T5289-PQ1,T1140-PQ1,T2712-PQ1,HV1250P-MPIR,pc,pIris-General,HZ1140P-
8MPIRA,HZ1135P-8MPSGA,MV1550P-12MPIR,MZ1545P-12MP,HZ0618P-4MPSG,MZ1050P-12MP,SF5014M-12MP,SF3018M-10MP,MZ1140M,MZ2514M-10MP,MZ7018M-8MP,MZ2516M-
10MPIR,indoor50HZ,indoor60HZ,lowGhosting,customize">
        <!--ro, req, string, attr:opt{req, string}-->test
      </ExposureType>
      <autoIrisLevel min="0" max="100">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
      </autoIrisLevel>
      <OverexposeSuppress>
        <!--ro, opt, object-->
        <enabled opt="true,false">
          <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
        </enabled>
        <Type opt="AUTO,MANUAL">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->AUTO
        </Type>
        <DistanceLevel min="1" max="100">
          <!--ro, opt, int, Focal length, attr:min{req, int},max{req, int}-->1
        </DistanceLevel>
        <hightLightDistanceLevel min="1" max="100">
          <!--ro, opt, int, range:[1,100], attr:min{opt, int},max{opt, int}-->50
        </hightLightDistanceLevel>
        <lowLightDistanceLevel min="1" max="100">
          <!--ro, opt, int, range:[1,100], attr:min{opt, int},max{opt, int}-->50
        </lowLightDistanceLevel>
      </OverexposeSuppress>
      <pIris>
        <!--ro, opt, object, aperture-->
        <pIrisType opt="AUTO,MANUAL">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->MANUAL
        </pIrisType>
        <IrisLevel min="1" max="100">
          <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
        </IrisLevel>
      </pIris>
    </Exposure>
```

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```
  <Sharpness>
    <!--ro, opt, object-->
    <SharpnessLevel min="0" max="255">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </SharpnessLevel>
  </Sharpness>
  <Shutter>
    <!--ro, opt, object-->
    <ShutterLevel
opt="1/1,1/2,1/3,1/6,1/12,1/25,1/50,1/60,1/75,1/100,1/120,1/125,1/150,1/175,1/15,1/20,1/30,1/60,1/180,1/200,1/250,1/500,1/1000,1/2000,1/4000,1/8000,1/5000,1
/10000,1/225,1/300,1/400,,1/425,1/600,1/750,1/1250,1/1750,1/2500,1/3500,1/6000,1/30000,1/100000">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </ShutterLevel>
  </Shutter>
  <powerLineFrequency>
    <!--ro, opt, object-->
    <powerLineFrequencyMode opt="50hz,60hz">
      <!--ro, opt, string, mode, attr:opt{req, string}-->test
    </powerLineFrequencyMode>
  </powerLineFrequency>
  <Gain>
    <!--ro, opt, object, gain-->
    <GainLevel min="0" max="10">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </GainLevel>
  </Gain>
  <Color>
    <!--ro, opt, object, image adjustment capability-->
    <brightnessLevel min="0" max="255">
      <!--ro, opt, int, brightness, attr:min{req, int},max{req, int}-->0
    </brightnessLevel>
    <contrastLevel min="0" max="255">
      <!--ro, opt, int, contrast, attr:min{req, int},max{req, int}-->0
    </contrastLevel>
    <saturationLevel min="0" max="255">
      <!--ro, opt, int, saturation, attr:min{req, int},max{req, int}-->0
    </saturationLevel>
    <grayScale>
      <!--ro, opt, object, gray scale-->
      <grayScaleMode opt="indoor,outdoor">
        <!--ro, opt, string, gray scale mode, attr:opt{req, string}-->test
      </grayScaleMode>
    </grayScale>
  </Color>
  <IrcutFilter>
    <!--ro, opt, object-->
    <IrcutFilterType opt="auto,day,night,schedule,eventTrigger,videoAuto,IrcutFilterType,darkFighterX,darkFighterXAuto,darkFighterXSchedule,colorVuAuto">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
    </IrcutFilterType>
    <nightToDayFilterLevel min="0" max="7">
      <!--ro, opt, int, range:[0,7], attr:min{req, int},max{req, int}-->4
    </nightToDayFilterLevel>
    <nightToDayFilterTime min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </nightToDayFilterTime>
    <Schedule>
      <!--ro, opt, object-->
      <scheduleType opt="day,night">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->day
      </scheduleType>
      <TimeRange>
        <!--ro, req, object, time period-->
        <beginTime>
          <!--ro, req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </Schedule>
  </IrcutFilter>
  <Scene>
    <!--ro, opt, object-->
    <mode opt="indoor,outdoor,standard,dimLight,riverSeaProtection,perimeterProtection,forestFirePrevention">
      <!--ro, req, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->indoor
    </mode>
  </Scene>
  <PTZ>
    <!--ro, opt, object, PTZ-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
  </PTZ>
  <EPTZ>
    <!--ro, opt, object, EPTZ-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <mode opt="cruise,autoTrack">
      <!--ro, opt, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->cruise
    </mode>
  </EPTZ>
  <EIS>
```

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```
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
  </EIS>
  <HLC>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <HLCLevel min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </HLCLevel>
  </HLC>
  <Dehaze>
    <!--ro, opt, object-->
    <DehazeMode opt="open,close,auto,advancedMode">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->open
    </DehazeMode>
    <DehazeLevel min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </DehazeLevel>
  </Dehaze>
  <LensDistortionCorrection>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <accurateLevel min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
    </accurateLevel>
  </LensDistortionCorrection>
  <SupplementLight>
    <!--ro, opt, object-->
    <supplementLightMode
opt="mixed,whiteLight,colorVuWhiteLight,irLight,dualLight,close,eventIntelligence,multiAzimuthMixed,multiAzimuthWhiteLight,multiAzimuthIrLight,panoramicWhit
eLight,panoramicIrLight">
      <!--ro, opt, enum, mode of supplement light, subType:string, attr:opt{req, string}, desc:mode of supplement light-->mixed
    </supplementLightMode>
    <mixedLightBrightnessRegulatMode opt="manual,auto">
      <!--ro, opt, enum, brightness adjustment mode of mixed supplement light, subType:string, attr:opt{req, string}, desc:"manual" (in this mode, you need
to set the parameters irLightBrightness and brightnessLimit), "auto" (brightness-related parameters are not required). This node is valid when
supplementLightMode is set to "mixed"-->auto
    </mixedLightBrightnessRegulatMode>
    <highIrLightBrightness min="0" max="100">
      <!--ro, opt, int, brightness of far IR supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different supplement
light modes, including "mixed", "whiteLight", "colorVuWhiteLight", "irLight", and "dualLight"-->80
    </highIrLightBrightness>
    <highWhiteLightBrightness min="0" max="100">
      <!--ro, opt, int, brightness of far white supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different supplement
light modes, including "mixed", "whiteLight", "colorVuWhiteLight", "irLight", and "dualLight"-->80
    </highWhiteLightBrightness>
    <lowIrLightBrightness min="0" max="100">
      <!--ro, opt, int, brightness of near IR supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different supplement
light modes, including "mixed", "whiteLight", "colorVuWhiteLight", "irLight", and "dualLight"-->80
    </lowIrLightBrightness>
    <lowWhiteLightBrightness min="0" max="100">
      <!--ro, opt, int, brightness of near white supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different
supplement light modes, including "mixed", "whiteLight", "colorVuWhiteLight", "irLight", and "dualLight"-->80
    </lowWhiteLightBrightness>
    <whiteLightBrightness min="0" max="100">
      <!--ro, opt, int, light brightness (regardless of far or near light), range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different
supplement light modes, including "mixed", "whiteLight", "colorVuWhiteLight", "irLight", and "dualLight", when the adjustment mode of the supplement light
is "auto"-->80
    </whiteLightBrightness>
    <WhiteLightModeCfg>
      <!--ro, opt, object-->
      <highWhiteLightBrightness min="0" max="100">
        <!--ro, opt, int, brightness of far white supplement light, range:[0,100], attr:min{req, int},max{req, int}-->80
      </highWhiteLightBrightness>
      <lowWhiteLightBrightness min="0" max="100">
        <!--ro, opt, int, brightness of near white supplement light, range:[0,100], attr:min{req, int},max{req, int}-->80
      </lowWhiteLightBrightness>
    </WhiteLightModeCfg>
    <isAutoModeBrightnessCfg>
      <!--ro, opt, bool-->true
    </isAutoModeBrightnessCfg>
    <EventIntelligenceModeCfg>
      <!--ro, opt, object-->
      <brightnessRegulatMode opt="manual,auto">
        <!--ro, opt, enum, supplement light adjustment mode, subType:string, attr:opt{req, string}, desc:supplement light adjustment mode-->auto
      </brightnessRegulatMode>
      <BrightnessCtrlList size="4">
        <!--ro, opt, array, subType:object, attr:size{req, int}-->
        <BrightnessCtrl>
          <!--ro, opt, object-->
          <lightType opt="highWhite,lowWhite,highIrLight,lowIrLight">
            <!--ro, opt, enum, subType:string, attr:opt{req, string}-->highWhite
          </lightType>
          <brightness min="0" max="100">
            <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->80
          </brightness>
        </BrightnessCtrl>
      </BrightnessCtrlList>
```

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```
      </EventIntelligenceModeCfg>
      <IrLightModeCfg>
        <!--ro, opt, object-->
      </IrLightModeCfg>
    </SupplementLight>
    <isSupportLaserSpotManual>
      <!--ro, opt, bool-->true
    </isSupportLaserSpotManual>
    <isSupportDOFAdjust>
      <!--ro, opt, bool-->true
    </isSupportDOFAdjust>
    <isSupportAntiBandingParams>
      <!--ro, opt, bool-->true
    </isSupportAntiBandingParams>
  </ImageChannel>
```

19.2.3.3 Set the image adjustment parameters in auto mode of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/color

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--opt, object, gray scale-->
    <grayScaleMode>
      <!--opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--opt, bool, whether to enable night mode, when its value is "true", the saturation can be adjusted, otherwise, the saturation cannot be adjusted--
>true
  </nightMode>
  <redBrightnessLevel>
    <!--opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--opt, int, range:[0,100]-->0
  </sharpnessLevel>
</Color>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.3.4 Get the image adjustment parameters in auto mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/color?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--ro, opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--ro, opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--ro, opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--ro, opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--ro, opt, object-->
    <grayScaleMode>
      <!--ro, opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--ro, opt, bool, whether to enable night mode-->true
  </nightMode>
  <redBrightnessLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--ro, opt, int, sharpness, range:[0,100]-->0
  </sharpnessLevel>
  <hueMode>
    <!--ro, opt, enum, subType:string-->warm
  </hueMode>
</Color>
```

19.2.3.5 Set the e-PTZ parameters for a specified channel

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Request URL

PUT /ISAPI/Image/channels/<channelID>/EPTZ

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EPTZ xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, e-PTZ, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <streamType>
    <!--opt, enum, stream type, subType:int, desc:1 (main stream), 2 (sub-stream), 3 (three-stream), 4 (four-stream)-->1
  </streamType>
</EPTZ>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.3.6 Get the e-PTZ parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/EPTZ

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EPTZ xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <streamType>
    <!--ro, opt, enum, stream type, subType:int, desc:stream type-->1
  </streamType>
</EPTZ>
```

19.2.3.7 Get the capability of switching e-PTZ mode

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Request URL

GET /ISAPI/Image/channels/<channelID>/EPTZ/mode/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "EPTZModeCap": {
    /*ro, req, object, modes supported by e-PTZ*/
        "mode": {
        /*ro, opt, object, modes supported by e-PTZ: "cruise"-patrol (PTZ supports preset,patrol,zoom,and eight directions),"autoTrack"-automatic tracking*/
            "@opt":  ["cruise", "autoTrack"]
            /*ro, opt, enumarray, "cruise” (patrol (PTZ supports preset,patrol,zoom,and eight directions)), "autoTrack” (automatic tracking),
subType:string, desc:"cruise” (patrol (PTZ supports preset,patrol,zoom,and eight directions)), "autoTrack” (automatic tracking)*/
        },
        "AutoTrackCap": {
        /*ro, opt, object, this node is returned when mode contains "autoTrack", dep:and,{$.EPTZModeCap.mode,eq,autoTrack}*/
            "sensitivity": {
            /*ro, opt, object, sensitivity*/
                "@min":  0,
                /*ro, opt, int*/
                "@max":  100
                /*ro, opt, int*/
            },
            "detectionTarget": {
            /*ro, opt, object, detection target*/
                "@opt":  ["human", "vehicle", "all"]
                /*ro, opt, enumarray, subType:string*/
            },
            "ReginonList": {
            /*ro, req, object*/
                "id": {
                /*ro, opt, object, rule ID*/
                    "@min":  1,
                    /*ro, opt, int*/
                    "@max":  4
                    /*ro, req, int*/
                },
                "Region": {
                /*ro, req, object, rule area*/
                    "number": {
                    /*ro, opt, object, supported number of edges*/
                        "@min":  3,
                        /*ro, opt, int*/
                        "@max":  10
                        /*ro, req, int*/
                    },
                    "x": {
                    /*ro, opt, object, x-coordinate, desc:the origin is the upper-left corner of the screen*/
                        "@min":  0.000,
                        /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
                        /*ro, req, float, range:[0.000,1.000]*/
                    },
                    "y": {
                    /*ro, opt, object, y-coordinate, desc:the origin is the upper-left corner of the screen*/
                        "@min":  0.000,
                        /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
                        /*ro, req, float, range:[0.000,1.000]*/
                    }
                }
            },
            "FilterSize": {
            /*ro, opt, object, size filter of rectangle*/
                "MaxTargetRect": {
                /*ro, req, object*/
                    "height": {
                    /*ro, opt, object, height, desc:the origin is the upper-left corner of the screen*/
                        "@min":  0.000,
                        /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
                        /*ro, req, float, range:[0.000,1.000]*/
                    },
                    "width": {
                    /*ro, opt, object, width, desc:the origin is the upper-left corner of the screen*/
                        "@min":  0.000,
                        /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
```

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```
                    /*ro, req, float, range:[0.000,1.000]*/
                },
                "x": {
                /*ro, opt, object, x-coordinate, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, opt, object, y-coordinate, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, range:[0.000,1.000]*/
                }
            },
            "MinTargetRect": {
            /*ro, req, object*/
                "height": {
                /*ro, opt, object, height, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                },
                "width": {
                /*ro, opt, object, width,range, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, range:[0.000,1.000]*/
                },
                "x": {
                /*ro, opt, object, x-coordinate, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, opt, object, y-coordinate, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, range:[0.000,1.000]*/
                }
            }
        }
    }
}
```

19.2.3.8 Get the parameters of switching e-PTZ mode

Request URL

GET /ISAPI/Image/channels/<channelID>/EPTZ/mode?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
{
    "EPTZMode": {
    /*ro, req, object, modes supported by e-PTZ*/
        "mode":  "autoTrack",
        /*ro, opt, enum, mode, subType:string, desc:"cruise" (patrol), "autoTrack" (automatic tracking)*/
        "AutoTrack": {
        /*ro, opt, object, auto-tracking mode configuration, dep:and,{$.EPTZMode.mode,eq,autoTrack}*/
            "sensitivity":  10,
            /*ro, opt, int, sensitivity*/
            "detectionTarget": ["human", "vehicle"],
            /*ro, opt, array, subType:string*/
            "ReginonList": [
            /*ro, req, array, region list, subType:object*/
                {
                    "id":  1,
                    /*ro, req, int, rule ID*/
                    "Region": [
                    /*ro, req, array, rule region, subType:object, range:[3,10], desc:the origin is the upper-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                }
            ],
            "FilterSize": {
            /*ro, opt, object, size filter*/
                "MaxTargetRect": {
                /*ro, req, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.0,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.0,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.0,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.0
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "MinTargetRect": {
                /*ro, req, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.0,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "width":  1.0,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.0,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.0
                    /*ro, opt, float, Y-coordinate*/
                }
            }
        }
    }
}
```

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19.2.3.9 Set parameters of switching the e-PTZ mode for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/EPTZ/mode?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "EPTZMode": {
    /*req, object, modes supported by e-PTZ*/
        "mode":  "autoTrack",
        /*opt, enum, e-PTZ mode, subType:string, desc:"cruise" (patrol), "autoTrack" (automatic tracking)*/
        "AutoTrack": {
        /*opt, object, auto-tracking mode configurations, dep:and,{$.EPTZMode.mode,eq,autoTrack}*/
            "sensitivity":  10,
            /*opt, int, sensitivity*/
            "detectionTarget": ["human", "vehicle"],
            /*opt, array, detection target, subType:string, desc:"human", "vehicle", "all" (human and vehicle)*/
            "ReginonList": [
            /*req, array, area list, subType:object*/
                {
                    "id":  1,
                    /*req, int, rule ID*/
                    "Region": [
                    /*req, array, rule area, subType:object, range:[3,10], desc:the origin is the upper-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*opt, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                }
            ],
            "FilterSize": {
            /*opt, object, size filter*/
                "MaxTargetRect": {
                /*req, object, the maximum target area, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.0,
                    /*req, float, height, range:[0.000,1.000]*/
                    "width":  1.0,
                    /*req, float, width, range:[0.000,1.000]*/
                    "x":  0.0,
                    /*req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.0
                    /*opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "MinTargetRect": {
                /*req, object, the minimum target area, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.0,
                    /*req, float, width, range:[0.000,1.000]*/
                    "width":  1.0,
                    /*req, float, width, range:[0.000,1.000]*/
                    "x":  0.0,
                    /*req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.0
                    /*opt, float, Y-coordinate*/
                }
            }
        }
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

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19.2.3.10 Get day/night auto switch parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/ircutFilter

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IrcutFilter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <IrcutFilterType>
    <!--ro, opt, enum, day/night auto switch mode, subType:string, desc:"auto, day, night, schedule, eventTrigger, darkFighterX, darkFighterXAuto,
darkFighterXSchedule"-->auto
  </IrcutFilterType>
  <dayToNightFilterLevel>
    <!--ro, opt, int, level of switching day to night, range:[0,7]-->4
  </dayToNightFilterLevel>
  <dayToNightFilterTime>
    <!--ro, opt, int, time interval of switching day to night-->1
  </dayToNightFilterTime>
  <nightToDayFilterLevel>
    <!--ro, opt, int, level of switching night to day, range:[0,7]-->4
  </nightToDayFilterLevel>
  <nightToDayFilterTime>
    <!--ro, opt, int, time interval of switching night to day-->1
  </nightToDayFilterTime>
  <Schedule>
    <!--ro, opt, object-->
    <scheduleType>
      <!--ro, req, enum, schedule type, subType:string, desc:"day", "night"-->day
    </scheduleType>
    <TimeRange>
      <!--ro, req, object, time period on schedule-->
      <beginTime>
        <!--ro, req, time, start time-->00:00:00+08:00
      </beginTime>
      <endTime>
        <!--ro, req, time, end time-->00:00:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <EventTrigger>
    <!--ro, opt, object-->
    <eventType>
      <!--ro, req, enum, event type, subType:string, desc:event type-->IO
    </eventType>
    <IrcutFilterAction>
      <!--ro, req, enum, "day, night", subType:string, desc:"day", "night"-->day
    </IrcutFilterAction>
  </EventTrigger>
  <dayNightStatus>
    <!--ro, opt, enum, subType:string-->day
  </dayNightStatus>
</IrcutFilter>
```

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19.2.3.11 Set day/night auto switch parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/ircutFilter

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IrcutFilter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, IrcutFilter, attr:version{req, string, protocolVersion}-->
  <IrcutFilterType>
    <!--opt, enum, IrcutFilter type, subType:string, desc:"auto", "day", "night", "schedule" , "eventTrigger", "videoAuto", "darkFighterX",
"darkFighterXAuto", "darkFighterXSchedule", "colorVuAuto"-->auto
  </IrcutFilterType>
  <dayToNightFilterLevel>
    <!--opt, enum, level of switching day to night, subType:string, desc:"low", "normal", "high"-->low
  </dayToNightFilterLevel>
  <dayToNightFilterTime>
    <!--opt, int, time interval of switching day to night-->1
  </dayToNightFilterTime>
  <nightToDayFilterLevel>
    <!--opt, enum, level of switching night to day, subType:string, desc:"low", "normal", "high"-->low
  </nightToDayFilterLevel>
  <nightToDayFilterTime>
    <!--opt, int, time interval of switching night to day-->1
  </nightToDayFilterTime>
  <Schedule>
    <!--opt, object, schedule-->
    <scheduleType>
      <!--req, enum, schedule type, subType:string, desc:"day", "night"-->day
    </scheduleType>
    <TimeRange>
      <!--req, object, time period on schedule-->
      <beginTime>
        <!--req, time, start time-->00:00:00+08:00
      </beginTime>
      <endTime>
        <!--req, time, end time-->00:00:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <EventTrigger>
    <!--opt, object, event trigger-->
    <eventType>
      <!--req, enum, event type, subType:string, desc:"IO", "VMD"-->IO
    </eventType>
    <IrcutFilterAction>
      <!--req, enum, "day,night", subType:string, desc:"day", "night"-->day
    </IrcutFilterAction>
  </EventTrigger>
</IrcutFilter>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.3.12 Reset the image parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/reset

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.2.3.13 Restore the image parameters of a specific channel to default settings

Request URL

PUT /ISAPI/Image/channels/<channelID>/restore

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.2.3.14 Get the scene parameters of a specified channel

Request URL

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GET /ISAPI/Image/channels/<channelID>/scene

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Scene xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, subType:string-->indoor
  </mode>
</Scene>
```

19.2.3.15 Set the scene parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/scene

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Scene xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->indoor
  </mode>
</Scene>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.3.16 Set the sharpness control parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/sharpness

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.3.17 Get the sharpness control parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/sharpness?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--ro, opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--ro, req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```

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19.2.3.18 Get the image processing parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <videoInputID>
    <!--ro, req, int-->1
  </videoInputID>
  <FocusConfiguration>
    <!--ro, opt, object-->
    <focusStyle>
      <!--ro, req, enum, subType:string-->AUTO
    </focusStyle>
    <focusLimited>
      <!--ro, opt, enum, subType:int-->10
    </focusLimited>
  </FocusConfiguration>
  <ImageFlip>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </ImageFlip>
  <ImageFreeze>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </ImageFreeze>
  <proportionalpan>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </proportionalpan>
  <WDR>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->open
    </mode>
    <WDRLevel>
      <!--ro, opt, int, range:[1,100]-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <BLCRegionList>
      <!--ro, opt, array, subType:object-->
      <BLCRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, range:[1,10]-->1
        </id>
        <RegionCoordinatesList>
          <!--ro, req, array, subType:object, range:[0,4]-->
          <RegionCoordinates>
```

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```
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,704]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,576]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </BLCRegion>
    </BLCRegionList>
  </BLC>
  <NoiseReduce>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->close
    </mode>
    <GeneralMode>
      <!--ro, opt, object-->
      <generalLevel>
        <!--ro, req, int, range:[0,100]-->1
      </generalLevel>
    </GeneralMode>
  </NoiseReduce>
  <WhiteBalance>
    <!--ro, opt, object-->
    <WhiteBalanceStyle>
      <!--ro, opt, enum, subType:string-->auto
    </WhiteBalanceStyle>
    <WhiteBalanceRed>
      <!--ro, opt, int, range:[-127,127]-->1
    </WhiteBalanceRed>
    <WhiteBalanceBlue>
      <!--ro, opt, int, range:[-127,127]-->1
    </WhiteBalanceBlue>
  </WhiteBalance>
  <Exposure>
    <!--ro, opt, object-->
    <ExposureType>
      <!--ro, req, enum, subType:string-->auto
    </ExposureType>
    <autoIrisLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </autoIrisLevel>
    <OverexposeSuppress>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <Type>
        <!--ro, opt, enum, subType:string-->AUTO
      </Type>
      <DistanceLevel>
        <!--ro, opt, int, range:[1,100]-->1
      </DistanceLevel>
    </OverexposeSuppress>
    <pIris>
      <!--ro, opt, object-->
      <pIrisType>
        <!--ro, opt, enum, subType:string-->MANUAL
      </pIrisType>
      <IrisLevel>
        <!--ro, opt, int, range:[1,100]-->1
      </IrisLevel>
    </pIris>
  </Exposure>
  <Sharpness>
    <!--ro, opt, object-->
    <SharpnessLevel>
      <!--ro, req, int, range:[0,100]-->1
    </SharpnessLevel>
  </Sharpness>
  <Shutter>
    <!--ro, opt, object-->
    <ShutterLevel>
      <!--ro, opt, enum, subType:string-->1/1
    </ShutterLevel>
  </Shutter>
  <powerLineFrequency>
    <!--ro, opt, object-->
    <powerLineFrequencyMode>
      <!--ro, opt, enum, subType:string-->50hz
    </powerLineFrequencyMode>
  </powerLineFrequency>
  <Gain>
    <!--ro, opt, object-->
    <GainLevel>
      <!--ro, opt, string, range:[0,10]-->test
    </GainLevel>
  </Gain>
  <Color>
    <!--ro, opt, object-->
    <brightnessLevel>
```

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```
      <!--ro, opt, int, range:[0,255]-->0
    </brightnessLevel>
    <contrastLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </contrastLevel>
    <saturationLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </saturationLevel>
    <grayScale>
      <!--ro, opt, object-->
      <grayScaleMode>
        <!--ro, opt, enum, subType:string-->indoor
      </grayScaleMode>
    </grayScale>
  </Color>
  <IrcutFilter>
    <!--ro, opt, object-->
    <IrcutFilterType>
      <!--ro, opt, enum, subType:string-->auto
    </IrcutFilterType>
    <nightToDayFilterLevel>
      <!--ro, opt, int, range:[0,7]-->4
    </nightToDayFilterLevel>
    <nightToDayFilterTime>
      <!--ro, opt, int, range:[0,10]-->1
    </nightToDayFilterTime>
    <EventTrigger>
      <!--ro, opt, object-->
      <eventType>
        <!--ro, req, enum, subType:string-->IO
      </eventType>
      <IrcutFilterAction>
        <!--ro, req, enum, subType:string-->day
      </IrcutFilterAction>
    </EventTrigger>
  </IrcutFilter>
  <Scene>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->indoor
    </mode>
  </Scene>
  <PTZ>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
  </PTZ>
  <EIS>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </EIS>
  <HLC>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <HLCLevel>
      <!--ro, opt, int, range:[0,10]-->0
    </HLCLevel>
  </HLC>
  <Dehaze>
    <!--ro, opt, object-->
    <DehazeMode>
      <!--ro, opt, enum, subType:string-->open
    </DehazeMode>
    <DehazeLevel>
      <!--ro, opt, int, range:[0,10]-->0
    </DehazeLevel>
  </Dehaze>
  <CaptureMode>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->close
    </mode>
  </CaptureMode>
  <LensDistortionCorrection>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </LensDistortionCorrection>
  <SupplementLight>
    <!--ro, opt, object-->
    <supplementLightMode>
      <!--ro, opt, enum, subType:string-->mixed
    </supplementLightMode>
    <mixedLightBrightnessRegulatMode>
      <!--ro, opt, enum, subType:string-->auto
    </mixedLightBrightnessRegulatMode>
    <highIrLightBrightness>
      <!--ro, opt, int, range:[0,100]-->80
```

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```
      </highIrLightBrightness>
      <highWhiteLightBrightness>
        <!--ro, opt, int, range:[0,100]-->80
      </highWhiteLightBrightness>
      <lowIrLightBrightness>
        <!--ro, opt, int, range:[0,100]-->80
      </lowIrLightBrightness>
      <lowWhiteLightBrightness>
        <!--ro, opt, int, range:[0,100]-->80
      </lowWhiteLightBrightness>
      <IrLightModeCfg>
        <!--ro, opt, object-->
      </IrLightModeCfg>
    </SupplementLight>
    <LaserLight>
      <!--ro, opt, object-->
      <mode>
        <!--ro, opt, enum, subType:string-->auto
      </mode>
    </LaserLight>
  </ImageChannel>
```

19.2.4 Image Adjustment

19.2.4.1 Set image parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <videoInputID>
    <!--req, int-->1
  </videoInputID>
  <ImageFlip>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
  </ImageFlip>
  <WDR>
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->open
    </mode>
    <WDRLevel>
      <!--opt, int, range:[1,100]-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <BLCRegionList>
      <!--opt, array, subType:object-->
      <BLCRegion>
        <!--opt, object-->
        <id>
          <!--req, int, area No., range:[1,10]-->1
        </id>
        <RegionCoordinatesList>
          <!--req, array, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,704]-->1
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,576]-->1
```

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```
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </BLCRegion>
    </BLCRegionList>
  </BLC>
  <NoiseReduce>
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->close
    </mode>
    <GeneralMode>
      <!--opt, object-->
      <generalLevel>
        <!--req, int, range:[0,100]-->1
      </generalLevel>
    </GeneralMode>
  </NoiseReduce>
  <WhiteBalance>
    <!--opt, object, white balance-->
    <WhiteBalanceStyle>
      <!--opt, enum, subType:string-->auto
    </WhiteBalanceStyle>
    <WhiteBalanceRed>
      <!--opt, int, range:[0,100]-->1
    </WhiteBalanceRed>
    <WhiteBalanceBlue>
      <!--opt, int, range:[0,100]-->1
    </WhiteBalanceBlue>
  </WhiteBalance>
  <Exposure>
    <!--opt, object-->
    <ExposureType>
      <!--req, enum, subType:string-->auto
    </ExposureType>
    <autoIrisLevel>
      <!--opt, int, range:[0,100]-->1
    </autoIrisLevel>
    <OverexposeSuppress>
      <!--opt, object-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <Type>
        <!--opt, enum, subType:string-->AUTO
      </Type>
      <DistanceLevel>
        <!--opt, int, focal length, range:[1,100]-->1
      </DistanceLevel>
    </OverexposeSuppress>
    <pIris>
      <!--opt, object, aperture-->
      <pIrisType>
        <!--opt, enum, subType:string-->MANUAL
      </pIrisType>
      <IrisLevel>
        <!--opt, int, range:[1,100]-->1
      </IrisLevel>
    </pIris>
  </Exposure>
  <Sharpness>
    <!--opt, object-->
    <SharpnessLevel>
      <!--req, int, range:[0,100]-->1
    </SharpnessLevel>
  </Sharpness>
  <Shutter>
    <!--opt, object-->
    <ShutterLevel>
      <!--opt, enum, subType:string-->1/1
    </ShutterLevel>
  </Shutter>
  <powerLineFrequency>
    <!--opt, object-->
    <powerLineFrequencyMode>
      <!--opt, enum, subType:string-->50hz
    </powerLineFrequencyMode>
  </powerLineFrequency>
  <Gain>
    <!--opt, object, gain-->
    <GainLevel>
      <!--opt, string, range:[0,10]-->test
    </GainLevel>
  </Gain>
  <Color>
    <!--opt, object-->
    <brightnessLevel>
      <!--opt, int, brightness, range:[0,100]-->0
    </brightnessLevel>
    <contrastLevel>
      <!--opt, int, contrast, range:[0,100]-->0
    </contrastLevel>
    <saturationLevel>
```

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```
      <!--opt, int, saturation, range:[0,100]-->0
    </saturationLevel>
    <grayScale>
      <!--opt, object-->
      <grayScaleMode>
        <!--opt, enum, subType:string-->indoor
      </grayScaleMode>
    </grayScale>
  </Color>
  <IrcutFilter>
    <!--opt, object-->
    <IrcutFilterType>
      <!--opt, enum, subType:string-->auto
    </IrcutFilterType>
    <nightToDayFilterLevel>
      <!--opt, enum, subType:string-->low
    </nightToDayFilterLevel>
    <nightToDayFilterTime>
      <!--opt, int, range:[0,10]-->1
    </nightToDayFilterTime>
  </IrcutFilter>
  <Scene>
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->indoor
    </mode>
  </Scene>
  <PTZ>
    <!--opt, object, PTZ-->
    <enabled>
      <!--opt, bool, whether to enable the function-->true
    </enabled>
  </PTZ>
  <EIS>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
  </EIS>
  <HLC>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <HLCLevel>
      <!--opt, int, range:[0,10]-->0
    </HLCLevel>
  </HLC>
  <Dehaze>
    <!--opt, object-->
    <DehazeMode>
      <!--opt, enum, subType:string-->open
    </DehazeMode>
    <DehazeLevel>
      <!--opt, int, range:[0,10]-->0
    </DehazeLevel>
  </Dehaze>
  <CaptureMode>
    <!--opt, object, capture mode-->
    <mode>
      <!--req, enum, subType:string-->close
    </mode>
  </CaptureMode>
  <LensDistortionCorrection>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
  </LensDistortionCorrection>
  <LaserLight>
    <!--opt, object-->
    <mode>
      <!--opt, enum, subType:string-->auto
    </mode>
  </LaserLight>
</ImageChannel>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.4.2 Get image configuration capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, temperature range capability, see XML_Cap_tempRange for details, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <videoInputID>
    <!--ro, req, int-->1
  </videoInputID>
  <FocusConfiguration>
    <!--ro, opt, object-->
    <focusStyle opt="AUTO,MANUAL,SEMIAUTOMATIC,distanceCalculate">
      <!--ro, req, string, attr:opt{req, string}-->test
    </focusStyle>
    <focusLimited opt="10,50,100,150,300,600,1000,2000,5000,20000,50000,65535">
      <!--ro, opt, int, attr:opt{req, string}-->1
    </focusLimited>
  </FocusConfiguration>
  <ImageFlip>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <ImageFlipStyle opt="LEFTRIGHT,UPDOWN,CENTER,AUTO">
      <!--ro, opt, string, type, attr:opt{req, string}-->LEFTRIGHT
    </ImageFlipStyle>
  </ImageFlip>
  <WDR>
    <!--ro, opt, object-->
    <mode opt="open,close,auto,SDR,time">
      <!--ro, req, enum, WDR mode: "open,close,auto", subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->open
    </mode>
    <WDRLevel min="1" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <BLCMode opt="UP,DOWN,LEFT,RIGHT,CENTER,MULTI-AREA,Region,AUTO">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->UP
    </BLCMode>
```

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```
      <BLCRegionList>
        <!--ro, opt, array, subType:object-->
        <BLCRegion>
          <!--ro, opt, object-->
          <id>
            <!--ro, req, int, area No.-->1
          </id>
          <RegionCoordinatesList>
            <!--ro, req, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
            <RegionCoordinates>
              <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--ro, req, int, X-coordinate, range:[0,704]-->1
              </positionX>
              <positionY>
                <!--ro, req, int, Y-coordinate, range:[0,576]-->1
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </BLCRegion>
      </BLCRegionList>
    </BLC>
    <NoiseReduce>
      <!--ro, opt, object-->
      <mode opt="close,general,advanced">
        <!--ro, req, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->close
      </mode>
      <GeneralMode>
        <!--ro, opt, object, mode-->
        <generalLevel min="0" max="10">
          <!--ro, req, int, attr:min{req, int},max{req, int}-->1
        </generalLevel>
      </GeneralMode>
      <AdvancedMode>
        <!--ro, opt, object-->
        <FrameNoiseReduceLevel min="0" max="10">
          <!--ro, req, int, attr:min{req, int},max{req, int}-->1
        </FrameNoiseReduceLevel>
        <InterFrameNoiseReduceLevel min="0" max="10">
          <!--ro, req, int, attr:min{req, int},max{req, int}-->1
        </InterFrameNoiseReduceLevel>
      </AdvancedMode>
    </NoiseReduce>
    <WhiteBalance>
      <!--ro, opt, object, white balance-->
      <WhiteBalanceStyle opt="manual、auto1、auto2、locked、outdoor、indoor、daylightLamp、sodiumlight、autotrace、onece、autooutdoor、autosodiumlight、
mercurylight、auto、incandescentlight、warmlight、naturallight,customColorTemperature,highTemperature">
        <!--ro, req, string, attr:opt{req, string}-->test
      </WhiteBalanceStyle>
      <WhiteBalanceRed min="-127" max="127">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
      </WhiteBalanceRed>
      <WhiteBalanceBlue min="-127" max="127">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
      </WhiteBalanceBlue>
    </WhiteBalance>
    <Exposure>
      <!--ro, opt, object-->
      <ExposureType opt="auto,IrisFirst,ShutterFirst,gainFirst,manual,pIris,T5280-PQ1,T5289-PQ1,T1140-PQ1,T2712-PQ1,HV1250P-MPIR,pc,pIris-General,HZ1140P-
8MPIRA,HZ1135P-8MPSGA,MV1550P-12MPIR,MZ1545P-12MP,HZ0618P-4MPSG,MZ1050P-12MP,SF5014M-12MP,SF3018M-10MP,MZ1140M,MZ2514M-10MP,MZ7018M-8MP,MZ2516M-
10MPIR,indoor50HZ,indoor60HZ,lowGhosting,customize">
        <!--ro, req, string, attr:opt{req, string}-->test
      </ExposureType>
      <autoIrisLevel min="0" max="100">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
      </autoIrisLevel>
      <OverexposeSuppress>
        <!--ro, opt, object-->
        <enabled opt="true,false">
          <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
        </enabled>
        <Type opt="AUTO,MANUAL">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->AUTO
        </Type>
        <DistanceLevel min="1" max="100">
          <!--ro, opt, int, Focal length, attr:min{req, int},max{req, int}-->1
        </DistanceLevel>
        <hightLightDistanceLevel min="1" max="100">
          <!--ro, opt, int, range:[1,100], attr:min{opt, int},max{opt, int}-->50
        </hightLightDistanceLevel>
        <lowLightDistanceLevel min="1" max="100">
          <!--ro, opt, int, range:[1,100], attr:min{opt, int},max{opt, int}-->50
        </lowLightDistanceLevel>
      </OverexposeSuppress>
      <pIris>
        <!--ro, opt, object, aperture-->
        <pIrisType opt="AUTO,MANUAL">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->MANUAL
        </pIrisType>
        <IrisLevel min="1" max="100">
          <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
        </IrisLevel>
      </pIris>
    </Exposure>
```

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```
  <Sharpness>
    <!--ro, opt, object-->
    <SharpnessLevel min="0" max="255">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </SharpnessLevel>
  </Sharpness>
  <Shutter>
    <!--ro, opt, object-->
    <ShutterLevel
opt="1/1,1/2,1/3,1/6,1/12,1/25,1/50,1/60,1/75,1/100,1/120,1/125,1/150,1/175,1/15,1/20,1/30,1/60,1/180,1/200,1/250,1/500,1/1000,1/2000,1/4000,1/8000,1/5000,1
/10000,1/225,1/300,1/400,,1/425,1/600,1/750,1/1250,1/1750,1/2500,1/3500,1/6000,1/30000,1/100000">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </ShutterLevel>
  </Shutter>
  <powerLineFrequency>
    <!--ro, opt, object-->
    <powerLineFrequencyMode opt="50hz,60hz">
      <!--ro, opt, string, mode, attr:opt{req, string}-->test
    </powerLineFrequencyMode>
  </powerLineFrequency>
  <Gain>
    <!--ro, opt, object, gain-->
    <GainLevel min="0" max="10">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </GainLevel>
  </Gain>
  <Color>
    <!--ro, opt, object, image adjustment capability-->
    <brightnessLevel min="0" max="255">
      <!--ro, opt, int, brightness, attr:min{req, int},max{req, int}-->0
    </brightnessLevel>
    <contrastLevel min="0" max="255">
      <!--ro, opt, int, contrast, attr:min{req, int},max{req, int}-->0
    </contrastLevel>
    <saturationLevel min="0" max="255">
      <!--ro, opt, int, saturation, attr:min{req, int},max{req, int}-->0
    </saturationLevel>
    <grayScale>
      <!--ro, opt, object, gray scale-->
      <grayScaleMode opt="indoor,outdoor">
        <!--ro, opt, string, gray scale mode, attr:opt{req, string}-->test
      </grayScaleMode>
    </grayScale>
  </Color>
  <IrcutFilter>
    <!--ro, opt, object-->
    <IrcutFilterType opt="auto,day,night,schedule,eventTrigger,videoAuto,IrcutFilterType,darkFighterX,darkFighterXAuto,darkFighterXSchedule,colorVuAuto">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
    </IrcutFilterType>
    <nightToDayFilterLevel min="0" max="7">
      <!--ro, opt, int, range:[0,7], attr:min{req, int},max{req, int}-->4
    </nightToDayFilterLevel>
    <nightToDayFilterTime min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </nightToDayFilterTime>
    <Schedule>
      <!--ro, opt, object-->
      <scheduleType opt="day,night">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->day
      </scheduleType>
      <TimeRange>
        <!--ro, req, object, time period-->
        <beginTime>
          <!--ro, req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </Schedule>
  </IrcutFilter>
  <Scene>
    <!--ro, opt, object-->
    <mode opt="indoor,outdoor,standard,dimLight,riverSeaProtection,perimeterProtection,forestFirePrevention">
      <!--ro, req, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->indoor
    </mode>
  </Scene>
  <PTZ>
    <!--ro, opt, object, PTZ-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
  </PTZ>
  <EPTZ>
    <!--ro, opt, object, EPTZ-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <mode opt="cruise,autoTrack">
      <!--ro, opt, enum, mode, subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->cruise
    </mode>
  </EPTZ>
  <EIS>
```

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```
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
      </enabled>
    </EIS>
    <HLC>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
      </enabled>
      <HLCLevel min="0" max="10">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
      </HLCLevel>
    </HLC>
    <Dehaze>
      <!--ro, opt, object-->
      <DehazeMode opt="open,close,auto,advancedMode">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->open
      </DehazeMode>
      <DehazeLevel min="0" max="10">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
      </DehazeLevel>
    </Dehaze>
    <LensDistortionCorrection>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
      </enabled>
      <accurateLevel min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
      </accurateLevel>
    </LensDistortionCorrection>
    <SupplementLight>
      <!--ro, opt, object-->
      <supplementLightMode
opt="mixed,whiteLight,colorVuWhiteLight,irLight,dualLight,close,eventIntelligence,multiAzimuthMixed,multiAzimuthWhiteLight,multiAzimuthIrLight,panoramicWhit
eLight,panoramicIrLight">
        <!--ro, opt, enum, mode of supplement light, subType:string, attr:opt{req, string}, desc:mode of supplement light-->mixed
      </supplementLightMode>
      <mixedLightBrightnessRegulatMode opt="manual,auto">
        <!--ro, opt, enum, brightness adjustment mode of mixed supplement light, subType:string, attr:opt{req, string}, desc:"manual" (in this mode, you need
to set the parameters irLightBrightness and brightnessLimit), "auto" (brightness-related parameters are not required). This node is valid when
supplementLightMode is set to "mixed"-->auto
      </mixedLightBrightnessRegulatMode>
      <highIrLightBrightness min="0" max="100">
        <!--ro, opt, int, brightness of far IR supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different supplement
light modes, including "mixed", "whiteLight", "colorVuWhiteLight", "irLight", and "dualLight"-->80
      </highIrLightBrightness>
      <highWhiteLightBrightness min="0" max="100">
        <!--ro, opt, int, brightness of far white supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different supplement
light modes, including "mixed", "whiteLight", "colorVuWhiteLight", "irLight", and "dualLight"-->80
      </highWhiteLightBrightness>
      <lowIrLightBrightness min="0" max="100">
        <!--ro, opt, int, brightness of near IR supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different supplement
light modes, including "mixed", "whiteLight", "colorVuWhiteLight", "irLight", and "dualLight"-->80
      </lowIrLightBrightness>
      <lowWhiteLightBrightness min="0" max="100">
        <!--ro, opt, int, brightness of near white supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different
supplement light modes, including "mixed", "whiteLight", "colorVuWhiteLight", "irLight", and "dualLight"-->80
      </lowWhiteLightBrightness>
      <whiteLightBrightness min="0" max="100">
        <!--ro, opt, int, light brightness (regardless of far or near light), range:[0,100], attr:min{req, int},max{req, int}, desc:it is shared by different
supplement light modes, including "mixed", "whiteLight", "colorVuWhiteLight", "irLight", and "dualLight", when the adjustment mode of the supplement light
is "auto"-->80
      </whiteLightBrightness>
      <WhiteLightModeCfg>
        <!--ro, opt, object-->
        <highWhiteLightBrightness min="0" max="100">
          <!--ro, opt, int, brightness of far white supplement light, range:[0,100], attr:min{req, int},max{req, int}-->80
        </highWhiteLightBrightness>
        <lowWhiteLightBrightness min="0" max="100">
          <!--ro, opt, int, brightness of near white supplement light, range:[0,100], attr:min{req, int},max{req, int}-->80
        </lowWhiteLightBrightness>
      </WhiteLightModeCfg>
      <isAutoModeBrightnessCfg>
        <!--ro, opt, bool-->true
      </isAutoModeBrightnessCfg>
      <EventIntelligenceModeCfg>
        <!--ro, opt, object-->
        <brightnessRegulatMode opt="manual,auto">
          <!--ro, opt, enum, supplement light adjustment mode, subType:string, attr:opt{req, string}, desc:supplement light adjustment mode-->auto
        </brightnessRegulatMode>
        <BrightnessCtrlList size="4">
          <!--ro, opt, array, subType:object, attr:size{req, int}-->
          <BrightnessCtrl>
            <!--ro, opt, object-->
            <lightType opt="highWhite,lowWhite,highIrLight,lowIrLight">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->highWhite
            </lightType>
            <brightness min="0" max="100">
              <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->80
            </brightness>
          </BrightnessCtrl>
        </BrightnessCtrlList>
```

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```
      </EventIntelligenceModeCfg>
      <IrLightModeCfg>
        <!--ro, opt, object-->
      </IrLightModeCfg>
    </SupplementLight>
    <isSupportLaserSpotManual>
      <!--ro, opt, bool-->true
    </isSupportLaserSpotManual>
    <isSupportDOFAdjust>
      <!--ro, opt, bool-->true
    </isSupportDOFAdjust>
    <isSupportAntiBandingParams>
      <!--ro, opt, bool-->true
    </isSupportAntiBandingParams>
  </ImageChannel>
```

19.2.4.3 Set the image adjustment parameters in auto mode of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/color

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--opt, object, gray scale-->
    <grayScaleMode>
      <!--opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--opt, bool, whether to enable night mode, when its value is "true", the saturation can be adjusted, otherwise, the saturation cannot be adjusted--
>true
  </nightMode>
  <redBrightnessLevel>
    <!--opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--opt, int, range:[0,100]-->0
  </sharpnessLevel>
</Color>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.4.4 Get the image adjustment parameters in auto mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/color?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--ro, opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--ro, opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--ro, opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--ro, opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--ro, opt, object-->
    <grayScaleMode>
      <!--ro, opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--ro, opt, bool, whether to enable night mode-->true
  </nightMode>
  <redBrightnessLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--ro, opt, int, sharpness, range:[0,100]-->0
  </sharpnessLevel>
  <hueMode>
    <!--ro, opt, enum, subType:string-->warm
  </hueMode>
</Color>
```

19.2.4.5 Set the e-PTZ parameters for a specified channel

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Request URL

PUT /ISAPI/Image/channels/<channelID>/EPTZ

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EPTZ xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, e-PTZ, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <streamType>
    <!--opt, enum, stream type, subType:int, desc:1 (main stream), 2 (sub-stream), 3 (three-stream), 4 (four-stream)-->1
  </streamType>
</EPTZ>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.4.6 Get the e-PTZ parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/EPTZ

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EPTZ xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <streamType>
    <!--ro, opt, enum, stream type, subType:int, desc:stream type-->1
  </streamType>
</EPTZ>
```

19.2.4.7 Get the capability of switching e-PTZ mode

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Request URL

GET /ISAPI/Image/channels/<channelID>/EPTZ/mode/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "EPTZModeCap": {
    /*ro, req, object, modes supported by e-PTZ*/
        "mode": {
        /*ro, opt, object, modes supported by e-PTZ: "cruise"-patrol (PTZ supports preset,patrol,zoom,and eight directions),"autoTrack"-automatic tracking*/
            "@opt":  ["cruise", "autoTrack"]
            /*ro, opt, enumarray, "cruise” (patrol (PTZ supports preset,patrol,zoom,and eight directions)), "autoTrack” (automatic tracking),
subType:string, desc:"cruise” (patrol (PTZ supports preset,patrol,zoom,and eight directions)), "autoTrack” (automatic tracking)*/
        },
        "AutoTrackCap": {
        /*ro, opt, object, this node is returned when mode contains "autoTrack", dep:and,{$.EPTZModeCap.mode,eq,autoTrack}*/
            "sensitivity": {
            /*ro, opt, object, sensitivity*/
                "@min":  0,
                /*ro, opt, int*/
                "@max":  100
                /*ro, opt, int*/
            },
            "detectionTarget": {
            /*ro, opt, object, detection target*/
                "@opt":  ["human", "vehicle", "all"]
                /*ro, opt, enumarray, subType:string*/
            },
            "ReginonList": {
            /*ro, req, object*/
                "id": {
                /*ro, opt, object, rule ID*/
                    "@min":  1,
                    /*ro, opt, int*/
                    "@max":  4
                    /*ro, req, int*/
                },
                "Region": {
                /*ro, req, object, rule area*/
                    "number": {
                    /*ro, opt, object, supported number of edges*/
                        "@min":  3,
                        /*ro, opt, int*/
                        "@max":  10
                        /*ro, req, int*/
                    },
                    "x": {
                    /*ro, opt, object, x-coordinate, desc:the origin is the upper-left corner of the screen*/
                        "@min":  0.000,
                        /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
                        /*ro, req, float, range:[0.000,1.000]*/
                    },
                    "y": {
                    /*ro, opt, object, y-coordinate, desc:the origin is the upper-left corner of the screen*/
                        "@min":  0.000,
                        /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
                        /*ro, req, float, range:[0.000,1.000]*/
                    }
                }
            },
            "FilterSize": {
            /*ro, opt, object, size filter of rectangle*/
                "MaxTargetRect": {
                /*ro, req, object*/
                    "height": {
                    /*ro, opt, object, height, desc:the origin is the upper-left corner of the screen*/
                        "@min":  0.000,
                        /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
                        /*ro, req, float, range:[0.000,1.000]*/
                    },
                    "width": {
                    /*ro, opt, object, width, desc:the origin is the upper-left corner of the screen*/
                        "@min":  0.000,
                        /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
```

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```
                    /*ro, req, float, range:[0.000,1.000]*/
                },
                "x": {
                /*ro, opt, object, x-coordinate, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, opt, object, y-coordinate, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, range:[0.000,1.000]*/
                }
            },
            "MinTargetRect": {
            /*ro, req, object*/
                "height": {
                /*ro, opt, object, height, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                },
                "width": {
                /*ro, opt, object, width,range, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, range:[0.000,1.000]*/
                },
                "x": {
                /*ro, opt, object, x-coordinate, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, opt, object, y-coordinate, desc:the origin is the upper-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, range:[0.000,1.000]*/
                }
            }
        }
    }
}
```


19.2.4.8 Get the parameters of switching e-PTZ mode

Request URL

GET /ISAPI/Image/channels/<channelID>/EPTZ/mode?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
{
    "EPTZMode": {
    /*ro, req, object, modes supported by e-PTZ*/
        "mode":  "autoTrack",
        /*ro, opt, enum, mode, subType:string, desc:"cruise" (patrol), "autoTrack" (automatic tracking)*/
        "AutoTrack": {
        /*ro, opt, object, auto-tracking mode configuration, dep:and,{$.EPTZMode.mode,eq,autoTrack}*/
            "sensitivity":  10,
            /*ro, opt, int, sensitivity*/
            "detectionTarget": ["human", "vehicle"],
            /*ro, opt, array, subType:string*/
            "ReginonList": [
            /*ro, req, array, region list, subType:object*/
                {
                    "id":  1,
                    /*ro, req, int, rule ID*/
                    "Region": [
                    /*ro, req, array, rule region, subType:object, range:[3,10], desc:the origin is the upper-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                }
            ],
            "FilterSize": {
            /*ro, opt, object, size filter*/
                "MaxTargetRect": {
                /*ro, req, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.0,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.0,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.0,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.0
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "MinTargetRect": {
                /*ro, req, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.0,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "width":  1.0,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.0,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.0
                    /*ro, opt, float, Y-coordinate*/
                }
            }
        }
    }
}
```

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19.2.4.9 Set parameters of switching the e-PTZ mode for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/EPTZ/mode?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "EPTZMode": {
    /*req, object, modes supported by e-PTZ*/
        "mode":  "autoTrack",
        /*opt, enum, e-PTZ mode, subType:string, desc:"cruise" (patrol), "autoTrack" (automatic tracking)*/
        "AutoTrack": {
        /*opt, object, auto-tracking mode configurations, dep:and,{$.EPTZMode.mode,eq,autoTrack}*/
            "sensitivity":  10,
            /*opt, int, sensitivity*/
            "detectionTarget": ["human", "vehicle"],
            /*opt, array, detection target, subType:string, desc:"human", "vehicle", "all" (human and vehicle)*/
            "ReginonList": [
            /*req, array, area list, subType:object*/
                {
                    "id":  1,
                    /*req, int, rule ID*/
                    "Region": [
                    /*req, array, rule area, subType:object, range:[3,10], desc:the origin is the upper-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*opt, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                }
            ],
            "FilterSize": {
            /*opt, object, size filter*/
                "MaxTargetRect": {
                /*req, object, the maximum target area, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.0,
                    /*req, float, height, range:[0.000,1.000]*/
                    "width":  1.0,
                    /*req, float, width, range:[0.000,1.000]*/
                    "x":  0.0,
                    /*req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.0
                    /*opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "MinTargetRect": {
                /*req, object, the minimum target area, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.0,
                    /*req, float, width, range:[0.000,1.000]*/
                    "width":  1.0,
                    /*req, float, width, range:[0.000,1.000]*/
                    "x":  0.0,
                    /*req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.0
                    /*opt, float, Y-coordinate*/
                }
            }
        }
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

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19.2.4.10 Get day/night auto switch parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/ircutFilter

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IrcutFilter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <IrcutFilterType>
    <!--ro, opt, enum, day/night auto switch mode, subType:string, desc:"auto, day, night, schedule, eventTrigger, darkFighterX, darkFighterXAuto,
darkFighterXSchedule"-->auto
  </IrcutFilterType>
  <dayToNightFilterLevel>
    <!--ro, opt, int, level of switching day to night, range:[0,7]-->4
  </dayToNightFilterLevel>
  <dayToNightFilterTime>
    <!--ro, opt, int, time interval of switching day to night-->1
  </dayToNightFilterTime>
  <nightToDayFilterLevel>
    <!--ro, opt, int, level of switching night to day, range:[0,7]-->4
  </nightToDayFilterLevel>
  <nightToDayFilterTime>
    <!--ro, opt, int, time interval of switching night to day-->1
  </nightToDayFilterTime>
  <Schedule>
    <!--ro, opt, object-->
    <scheduleType>
      <!--ro, req, enum, schedule type, subType:string, desc:"day", "night"-->day
    </scheduleType>
    <TimeRange>
      <!--ro, req, object, time period on schedule-->
      <beginTime>
        <!--ro, req, time, start time-->00:00:00+08:00
      </beginTime>
      <endTime>
        <!--ro, req, time, end time-->00:00:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <EventTrigger>
    <!--ro, opt, object-->
    <eventType>
      <!--ro, req, enum, event type, subType:string, desc:event type-->IO
    </eventType>
    <IrcutFilterAction>
      <!--ro, req, enum, "day, night", subType:string, desc:"day", "night"-->day
    </IrcutFilterAction>
  </EventTrigger>
  <dayNightStatus>
    <!--ro, opt, enum, subType:string-->day
  </dayNightStatus>
</IrcutFilter>
```

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19.2.4.11 Set day/night auto switch parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/ircutFilter

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IrcutFilter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, IrcutFilter, attr:version{req, string, protocolVersion}-->
  <IrcutFilterType>
    <!--opt, enum, IrcutFilter type, subType:string, desc:“auto”, “day”, “night”, “schedule” , “eventTrigger”, “videoAuto”, “darkFighterX”,
“darkFighterXAuto”, “darkFighterXSchedule”, “colorVuAuto”-->auto
  </IrcutFilterType>
  <dayToNightFilterLevel>
    <!--opt, enum, level of switching day to night, subType:string, desc:“low”, “normal”, “high”-->low
  </dayToNightFilterLevel>
  <dayToNightFilterTime>
    <!--opt, int, time interval of switching day to night-->1
  </dayToNightFilterTime>
  <nightToDayFilterLevel>
    <!--opt, enum, level of switching night to day, subType:string, desc:“low”, “normal”, “high”-->low
  </nightToDayFilterLevel>
  <nightToDayFilterTime>
    <!--opt, int, time interval of switching night to day-->1
  </nightToDayFilterTime>
  <Schedule>
    <!--opt, object, schedule-->
    <scheduleType>
      <!--req, enum, schedule type, subType:string, desc:“day”, “night”-->day
    </scheduleType>
    <TimeRange>
      <!--req, object, time period on schedule-->
      <beginTime>
        <!--req, time, start time-->00:00:00+08:00
      </beginTime>
      <endTime>
        <!--req, time, end time-->00:00:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <EventTrigger>
    <!--opt, object, event trigger-->
    <eventType>
      <!--req, enum, event type, subType:string, desc:“IO”, “VMD”-->IO
    </eventType>
    <IrcutFilterAction>
      <!--req, enum, "day,night", subType:string, desc:“day”, “night”-->day
    </IrcutFilterAction>
  </EventTrigger>
</IrcutFilter>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.4.12 Reset the image parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/reset

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.2.4.13 Restore the image parameters of a specific channel to default settings

Request URL

PUT /ISAPI/Image/channels/<channelID>/restore

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.2.4.14 Get the scene parameters of a specified channel

Request URL

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GET /ISAPI/Image/channels/<channelID>/scene

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Scene xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, subType:string-->indoor
  </mode>
</Scene>
```

19.2.4.15 Set the scene parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/scene

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Scene xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->indoor
  </mode>
</Scene>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.4.16 Set the sharpness control parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/sharpness

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.4.17 Get the sharpness control parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/sharpness?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--ro, opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--ro, req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```

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19.2.4.18 Get the image processing parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <videoInputID>
    <!--ro, req, int-->1
  </videoInputID>
  <FocusConfiguration>
    <!--ro, opt, object-->
    <focusStyle>
      <!--ro, req, enum, subType:string-->AUTO
    </focusStyle>
    <focusLimited>
      <!--ro, opt, enum, subType:int-->10
    </focusLimited>
  </FocusConfiguration>
  <ImageFlip>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </ImageFlip>
  <ImageFreeze>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </ImageFreeze>
  <proportionalpan>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </proportionalpan>
  <WDR>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->open
    </mode>
    <WDRLevel>
      <!--ro, opt, int, range:[1,100]-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <BLCRegionList>
      <!--ro, opt, array, subType:object-->
      <BLCRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, range:[1,10]-->1
        </id>
        <RegionCoordinatesList>
          <!--ro, req, array, subType:object, range:[0,4]-->
          <RegionCoordinates>
```

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```
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,704]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,576]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </BLCRegion>
    </BLCRegionList>
  </BLC>
  <NoiseReduce>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->close
    </mode>
    <GeneralMode>
      <!--ro, opt, object-->
      <generalLevel>
        <!--ro, req, int, range:[0,100]-->1
      </generalLevel>
    </GeneralMode>
  </NoiseReduce>
  <WhiteBalance>
    <!--ro, opt, object-->
    <WhiteBalanceStyle>
      <!--ro, opt, enum, subType:string-->auto
    </WhiteBalanceStyle>
    <WhiteBalanceRed>
      <!--ro, opt, int, range:[-127,127]-->1
    </WhiteBalanceRed>
    <WhiteBalanceBlue>
      <!--ro, opt, int, range:[-127,127]-->1
    </WhiteBalanceBlue>
  </WhiteBalance>
  <Exposure>
    <!--ro, opt, object-->
    <ExposureType>
      <!--ro, req, enum, subType:string-->auto
    </ExposureType>
    <autoIrisLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </autoIrisLevel>
    <OverexposeSuppress>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <Type>
        <!--ro, opt, enum, subType:string-->AUTO
      </Type>
      <DistanceLevel>
        <!--ro, opt, int, range:[1,100]-->1
      </DistanceLevel>
    </OverexposeSuppress>
    <pIris>
      <!--ro, opt, object-->
      <pIrisType>
        <!--ro, opt, enum, subType:string-->MANUAL
      </pIrisType>
      <IrisLevel>
        <!--ro, opt, int, range:[1,100]-->1
      </IrisLevel>
    </pIris>
  </Exposure>
  <Sharpness>
    <!--ro, opt, object-->
    <SharpnessLevel>
      <!--ro, req, int, range:[0,100]-->1
    </SharpnessLevel>
  </Sharpness>
  <Shutter>
    <!--ro, opt, object-->
    <ShutterLevel>
      <!--ro, opt, enum, subType:string-->1/1
    </ShutterLevel>
  </Shutter>
  <powerLineFrequency>
    <!--ro, opt, object-->
    <powerLineFrequencyMode>
      <!--ro, opt, enum, subType:string-->50hz
    </powerLineFrequencyMode>
  </powerLineFrequency>
  <Gain>
    <!--ro, opt, object-->
    <GainLevel>
      <!--ro, opt, string, range:[0,10]-->test
    </GainLevel>
  </Gain>
  <Color>
    <!--ro, opt, object-->
    <brightnessLevel>
```

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```
      <!--ro, opt, int, range:[0,255]-->0
    </brightnessLevel>
    <contrastLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </contrastLevel>
    <saturationLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </saturationLevel>
    <grayScale>
      <!--ro, opt, object-->
      <grayScaleMode>
        <!--ro, opt, enum, subType:string-->indoor
      </grayScaleMode>
    </grayScale>
  </Color>
  <IrcutFilter>
    <!--ro, opt, object-->
    <IrcutFilterType>
      <!--ro, opt, enum, subType:string-->auto
    </IrcutFilterType>
    <nightToDayFilterLevel>
      <!--ro, opt, int, range:[0,7]-->4
    </nightToDayFilterLevel>
    <nightToDayFilterTime>
      <!--ro, opt, int, range:[0,10]-->1
    </nightToDayFilterTime>
    <EventTrigger>
      <!--ro, opt, object-->
      <eventType>
        <!--ro, req, enum, subType:string-->IO
      </eventType>
      <IrcutFilterAction>
        <!--ro, req, enum, subType:string-->day
      </IrcutFilterAction>
    </EventTrigger>
  </IrcutFilter>
  <Scene>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->indoor
    </mode>
  </Scene>
  <PTZ>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
  </PTZ>
  <EIS>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </EIS>
  <HLC>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <HLCLevel>
      <!--ro, opt, int, range:[0,10]-->0
    </HLCLevel>
  </HLC>
  <Dehaze>
    <!--ro, opt, object-->
    <DehazeMode>
      <!--ro, opt, enum, subType:string-->open
    </DehazeMode>
    <DehazeLevel>
      <!--ro, opt, int, range:[0,10]-->0
    </DehazeLevel>
  </Dehaze>
  <CaptureMode>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->close
    </mode>
  </CaptureMode>
  <LensDistortionCorrection>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </LensDistortionCorrection>
  <SupplementLight>
    <!--ro, opt, object-->
    <supplementLightMode>
      <!--ro, opt, enum, subType:string-->mixed
    </supplementLightMode>
    <mixedLightBrightnessRegulatMode>
      <!--ro, opt, enum, subType:string-->auto
    </mixedLightBrightnessRegulatMode>
    <highIrLightBrightness>
      <!--ro, opt, int, range:[0,100]-->80
```

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```
      </highIrLightBrightness>
      <highWhiteLightBrightness>
        <!--ro, opt, int, range:[0,100]-->80
      </highWhiteLightBrightness>
      <lowIrLightBrightness>
        <!--ro, opt, int, range:[0,100]-->80
      </lowIrLightBrightness>
      <lowWhiteLightBrightness>
        <!--ro, opt, int, range:[0,100]-->80
      </lowWhiteLightBrightness>
      <IrLightModeCfg>
        <!--ro, opt, object-->
      </IrLightModeCfg>
    </SupplementLight>
    <LaserLight>
      <!--ro, opt, object-->
      <mode>
        <!--ro, opt, enum, subType:string-->auto
      </mode>
    </LaserLight>
  </ImageChannel>
```

19.2.5 Image Enhancement

19.2.5.1 Get defog mode parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/dehaze

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Dehaze xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, "open,close,auto", attr:version{req, string, protocolVersion}-->
  <DehazeMode>
    <!--ro, opt, enum, defog mode, subType:string, desc:"open", "close", "auto"-->open
  </DehazeMode>
  <DehazeLevel>
    <!--ro, opt, int, defog level-->1
  </DehazeLevel>
  <dehazeIntensity>
    <!--ro, opt, int, range:[0,100], dep:and,{$.Dehaze.DehazeMode,eq,advancedMode}-->0
  </dehazeIntensity>
  <smoothIntensity>
    <!--ro, opt, int, range:[0,100], dep:and,{$.Dehaze.DehazeMode,eq,advancedMode}-->0
  </smoothIntensity>
</Dehaze>
```

19.2.5.2 Set defog mode parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/dehaze

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Dehaze xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, "open,close,auto", attr:version{req, string, protocolVersion}-->
  <DehazeMode>
    <!--opt, enum, defog mode, subType:string, desc:"open", "close", "auto"-->open
  </DehazeMode>
  <DehazeLevel>
    <!--opt, int, defog level-->1
  </DehazeLevel>
</Dehaze>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.5.3 Get the electronic image stabilization (in auto mode) parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/EIS

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EIS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <EISLevel>
    <!--ro, opt, int-->1
  </EISLevel>
</EIS>
```

19.2.5.4 Set the electronic image stabilization (in auto mode) parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/EIS

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EIS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <EISLevel>
    <!--opt, int-->1
  </EISLevel>
</EIS>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.5.5 Get the electronic image stabilization (in day mode) parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/EIS/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EIS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <EISLevel>
    <!--ro, opt, int-->1
  </EISLevel>
</EIS>
```

19.2.5.6 Set the electronic image stabilization (in day mode) parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/EIS/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EIS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <EISLevel>
    <!--opt, int-->1
  </EISLevel>
</EIS>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.5.7 Get the electronic image stabilization (in night mode) parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/EIS/night

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EIS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <EISLevel>
    <!--opt, int-->1
  </EISLevel>
</EIS>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.5.8 Get the electronic image stabilization (in night mode) parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/EIS/night

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EIS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <EISLevel>
    <!--ro, opt, int-->1
  </EISLevel>
</EIS>
```

19.2.5.9 Set the picture gain parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/gain

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Gain xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, gain, attr:version{req, string, protocolVersion}-->
  <GainLevel>
    <!--opt, int, gain level-->1
  </GainLevel>
  <GainWindow>
    <!--opt, object-->
    <RegionCoordinatesList>
      <!--opt, array, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GainWindow>
  <scheduleEnabled>
    <!--opt, bool-->true
  </scheduleEnabled>
  <scheduleGainLevel>
    <!--opt, int, dep:and,{$.Gain.scheduleEnabled,eq,true}-->1
  </scheduleGainLevel>
  <Schedule>
    <!--opt, object, dep:and,{$.Gain.scheduleEnabled,eq,true}-->
    <startHour min="0" max="23">
      <!--opt, int, start time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </startHour>
    <startMinute min="0" max="59">
      <!--opt, int, start time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </startMinute>
    <endHour min="0" max="23">
      <!--opt, int, end time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </endHour>
    <endMinute min="0" max="59">
      <!--opt, int, end time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </endMinute>
  </Schedule>
  <redGainLevel>
    <!--opt, int, range:[0,100]-->1
  </redGainLevel>
</Gain>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.5.10 Get the picture gain parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/gain?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Gain xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, gain, attr:version{req, string, protocolVersion}-->
  <GainLevel>
    <!--ro, opt, int, gain level-->1
  </GainLevel>
  <GainWindow>
    <!--ro, opt, object-->
    <RegionCoordinatesList>
      <!--ro, opt, array, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, opt, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GainWindow>
  <scheduleEnabled>
    <!--ro, opt, bool-->true
  </scheduleEnabled>
  <scheduleGainLevel>
    <!--ro, opt, int, dep:and,{$.Gain.scheduleEnabled,eq,true}-->1
  </scheduleGainLevel>
  <Schedule>
    <!--ro, opt, object, time range, dep:and,{$.Gain.scheduleEnabled,eq,true}-->
    <startHour min="0" max="23">
      <!--ro, opt, int, start time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </startHour>
    <startMinute min="0" max="59">
      <!--ro, opt, int, start time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </startMinute>
    <endHour min="0" max="23">
      <!--ro, opt, int, end time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </endHour>
    <endMinute min="0" max="59">
      <!--ro, opt, int, end time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </endMinute>
  </Schedule>
  <redGainLevel>
    <!--ro, opt, int, range:[0,100]-->1
  </redGainLevel>
  <minExposureGainLevel>
    <!--ro, opt, int, range:[0,10]-->0
  </minExposureGainLevel>
  <maxExposureGainLevel>
    <!--ro, opt, int, range:[0,10]-->0
  </maxExposureGainLevel>
</Gain>
```

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19.2.5.11 Set the image distortion correction parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/lensDistortionCorrection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LensDistortionCorrection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, parameters of image distortion correction, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable image distortion correction-->true
  </enabled>
  <level>
    <!--opt, enum, distortion correction level, subType:string, desc:"Level 1", "Level 2", "Level 3", "custom"-->level1
  </level>
  <accurateLevel>
    <!--opt, int, correction strength, range:[0,100]-->0
  </accurateLevel>
  <zoomedInDistantViewLevel>
    <!--opt, int, zoomed-in distant view, range:[0,100]-->0
  </zoomedInDistantViewLevel>
  <horizontalFOV>
    <!--opt, int, horizontal field of view, range:[0,100]-->0
  </horizontalFOV>
  <verticalFOV>
    <!--opt, int, vertical field of view, range:[0,100]-->0
  </verticalFOV>
  <correctionFOV>
    <!--opt, int, correction field of view, range:[0,100], desc:correction field of view-->0
  </correctionFOV>
</LensDistortionCorrection>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.5.12 Get the capability of image distortion correction of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/lensDistortionCorrection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LensDistortionCorrection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <level>
    <!--ro, opt, enum, subType:string-->level1
  </level>
  <accurateLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </accurateLevel>
  <zoomedInDistantViewLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </zoomedInDistantViewLevel>
  <horizontalFOV>
    <!--ro, opt, int, range:[0,100]-->0
  </horizontalFOV>
  <verticalFOV>
    <!--ro, opt, int, range:[0,100]-->0
  </verticalFOV>
  <correctionFOV>
    <!--ro, opt, int, range:[0,100]-->0
  </correctionFOV>
  <mode>
    <!--ro, opt, enum, subType:string-->custom
  </mode>
</LensDistortionCorrection>
```

19.2.5.13 Get the capability of image distortion correction of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/lensDistortionCorrection/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LensDistortionCorrection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the capability of image distortion correction, attr:version{req, string, protocolVersion}-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <level opt="level1,level2,level3,custom,visualEffect,viewingAngle">
    <!--ro, opt, enum, distortion correction level, subType:string, attr:opt{req, string}, desc:"Level 1", "Level 2", "Level 3", "custom"-->level1
  </level>
  <accurateLevel min="0" max="100">
    <!--ro, opt, int, correction strength, range:[0,100], attr:min{req, int},max{req, int}-->0
  </accurateLevel>
  <zoomedInDistantViewLevel min="0" max="100">
    <!--ro, opt, int, zoomed-in distant view, range:[0,100], attr:min{req, int},max{req, int}-->0
  </zoomedInDistantViewLevel>
  <horizontalFOV min="0" max="100">
    <!--ro, opt, int, horizontal field of view, range:[0,100], attr:min{req, int},max{req, int}-->0
  </horizontalFOV>
  <verticalFOV min="0" max="100">
    <!--ro, opt, int, vertical field of view, range:[0,100], attr:min{req, int},max{req, int}-->0
  </verticalFOV>
  <correctionFOV min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
  </correctionFOV>
  <isSupportLevelTypeAccurate>
    <!--ro, opt, bool-->true
  </isSupportLevelTypeAccurate>
  <supportPrompt opt="levelPrompt">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->levelPrompt
  </supportPrompt>
  <isLDCModifyResolution>
    <!--ro, opt, bool-->true
  </isLDCModifyResolution>
  <mode opt="custom,visualEffect,viewingAngle" def="viewingAngle">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{opt, string}-->custom
  </mode>
</LensDistortionCorrection>
```

19.2.5.14 Set the DNR (Digital Noise Reduction) parameters in auto mode of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/noiseReduce

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, mode, subType:string, desc:"close", "general", "advanced"-->close
  </mode>
  <GeneralMode>
    <!--opt, object, mode-->
    <generalLevel>
      <!--req, int, level-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--opt, object, advanced mode-->
    <FrameNoiseReduceLevel>
      <!--req, int-->1
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--req, int-->1
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.5.15 Get the DNR (Digital Noise Reduction) parameters in auto mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/noiseReduce

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, mode, subType:string, desc:"close", "general", "advanced"-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object, this node is valid only when <mode> is "general"-->
    <generalLevel>
      <!--ro, req, int-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object, advanced mode-->
    <FrameNoiseReduceLevel>
      <!--ro, req, int-->1
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--ro, req, int-->1
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

19.2.5.16 Set the digital noise reduction (DNR) parameters in day mode of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/noiseReduce/day

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->close
  </mode>
  <GeneralMode>
    <!--opt, object, dep:and,{$.NoiseReduce.mode,eq,general}-->
    <generalLevel>
      <!--req, int, range:[0,100]-->0
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--opt, object, dep:and,{$.NoiseReduce.mode,eq,advanced}-->
    <FrameNoiseReduceLevel>
      <!--req, int, range:[0,100]-->0
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--req, int, range:[0,100]-->0
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.5.17 Get the digital noise reduction (DNR) parameters in day mode of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/noiseReduce/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, subType:string-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object, dep:and,{$.NoiseReduce.mode,eq,general}-->
    <generalLevel>
      <!--ro, req, int, range:[0,100]-->0
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object, dep:and,{$.NoiseReduce.mode,eq,advanced}-->
    <FrameNoiseReduceLevel>
      <!--ro, req, int, range:[0,100]-->0
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--ro, req, int, range:[0,100]-->0
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

19.2.5.18 Set the digital noise reduction (DNR) parameters in night mode a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/noiseReduce/night

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->close
  </mode>
  <GeneralMode>
    <!--opt, object, dep:and,{$.NoiseReduce.mode,eq,general}-->
    <generalLevel>
      <!--req, int, range:[0,100]-->0
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--opt, object, dep:and,{$.NoiseReduce.mode,eq,advanced}-->
    <FrameNoiseReduceLevel>
      <!--req, int, range:[0,100]-->0
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--req, int, range:[0,100]-->0
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.5.19 Get the digital noise reduction (DNR) parameters in day mode of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/noiseReduce/night

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, subType:string-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object, dep:and,{$.NoiseReduce.mode,eq,general}-->
    <generalLevel>
      <!--ro, req, int, range:[0,100]-->0
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object, dep:and,{$.NoiseReduce.mode,eq,advanced}-->
    <FrameNoiseReduceLevel>
      <!--ro, req, int, range:[0,100]-->0
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--ro, req, int, range:[0,100]-->0
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

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19.2.6 Image Light Supplement

19.2.6.1 Set the laser supplement light parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/laserLight

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LaserLight xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->auto
  </mode>
  <sensitivityLevel>
    <!--opt, int, range:[0,100], dep:and,{$.LaserLight.mode,eq,manual}-->0
  </sensitivityLevel>
  <brightnessLevel>
    <!--opt, int, range:[0,255], dep:and,{$.LaserLight.mode,eq,manual}-->0
  </brightnessLevel>
  <laserangle>
    <!--opt, int, range:[1,36], dep:and,{$.LaserLight.mode,eq,manual}-->1
  </laserangle>
  <triggerMode>
    <!--opt, enum, subType:string, dep:and,{$.LaserLight.mode,eq,manual}-->camera
  </triggerMode>
  <brightnessLimit>
    <!--opt, int, range:[0,255], dep:and,{$.LaserLight.mode,eq,auto}-->0
  </brightnessLimit>
  <enabled>
    <!--opt, bool, dep:and,{$.LaserLight.mode,eq,manual}-->true
  </enabled>
  <illumination>
    <!--opt, int, range:[0,100], dep:and,{$.LaserLight.mode,eq,manual}-->0
  </illumination>
  <lightAngle>
    <!--opt, int, range:[0,100], dep:and,{$.LaserLight.mode,eq,manual}-->0
  </lightAngle>
  <LaserSpotManual>
    <!--opt, object-->
    <normalizedScreenSize>
      <!--req, object-->
      <normalizedScreenWidth>
        <!--req, int, range:[0,704]-->0
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, range:[0,576]-->0
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <laserSpotLevel>
      <!--req, int, range:[1,3]-->1
    </laserSpotLevel>
    <LaserWarning>
      <!--opt, object-->
      <enabled>
        <!--opt, bool-->true
      </enabled>
      <warningWord>
        <!--opt, string, range:[0,44]-->test
      </warningWord>
      <TopLeftPoint>
        <!--opt, object-->
        <positionX>
          <!--req, int, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--req, int, range:[0,1000]-->0
        </positionY>
      </TopLeftPoint>
    </LaserWarning>
  </LaserSpotManual>
  <compensationRegionMode>
    <!--opt, enum, subType:string-->center
  </compensationRegionMode>
  <compensationLevel>
    <!--opt, enum, subType:string-->low
  </compensationLevel>
  <LaserColor>
    <!--opt, object-->
    <R>
      <!--req, int, range:[0,100]-->0
    </R>
    <G>
      <!--req, int, range:[0,100]-->0
    </G>
    <B>
      <!--req, int, range:[0,100]-->0
    </B>
  </LaserColor>
</LaserLight>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.6.2 Get the laser supplement light parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/laserLight

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LaserLight xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, subType:string-->auto
  </mode>
  <sensitivityLevel>
    <!--ro, opt, int, range:[0,100], dep:and,{$.LaserLight.mode,eq,manual}-->0
  </sensitivityLevel>
  <brightnessLevel>
    <!--ro, opt, int, range:[0,255], dep:and,{$.LaserLight.mode,eq,manual}-->0
  </brightnessLevel>
  <laserangle>
    <!--ro, opt, int, range:[1,36], dep:and,{$.LaserLight.mode,eq,manual}-->1
  </laserangle>
  <triggerMode>
    <!--ro, opt, enum, subType:string, dep:and,{$.LaserLight.mode,eq,manual}-->camera
  </triggerMode>
  <brightnessLimit>
    <!--ro, opt, int, range:[0,255], dep:and,{$.LaserLight.mode,eq,auto}-->0
  </brightnessLimit>
  <enabled>
    <!--ro, opt, bool, dep:and,{$.LaserLight.mode,eq,manual}-->true
  </enabled>
  <illumination>
    <!--ro, opt, int, range:[0,100], dep:and,{$.LaserLight.mode,eq,manual}-->0
  </illumination>
  <lightAngle>
    <!--ro, opt, int, range:[0,100], dep:and,{$.LaserLight.mode,eq,manual}-->0
  </lightAngle>
  <LaserSpotManual>
    <!--ro, opt, object-->
    <normalizedScreenSize>
      <!--ro, req, object-->
      <normalizedScreenWidth>
        <!--ro, req, int, range:[0,704]-->0
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, range:[0,576]-->0
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <laserSpotLevel>
      <!--ro, req, int, range:[1,3]-->1
    </laserSpotLevel>
    <LaserWarning>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, opt, bool-->true
      </enabled>
      <warningWord>
        <!--ro, opt, string, range:[0,44]-->test
      </warningWord>
      <TopLeftPoint>
        <!--ro, opt, object-->
        <positionX>
          <!--ro, req, int, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, range:[0,1000]-->0
        </positionY>
      </TopLeftPoint>
    </LaserWarning>
  </LaserSpotManual>
  <compensationRegionMode>
    <!--ro, opt, enum, subType:string-->center
  </compensationRegionMode>
  <compensationLevel>
    <!--ro, opt, enum, subType:string-->low
  </compensationLevel>
  <LaserColor>
    <!--ro, opt, object-->
    <R>
      <!--ro, req, int, range:[0,100]-->0
    </R>
    <G>
      <!--ro, req, int, range:[0,100]-->0
    </G>
    <B>
      <!--ro, req, int, range:[0,100]-->0
    </B>
  </LaserColor>
  <beamAngle>
    <!--ro, opt, int, range:[0,180], unit:°-->15
  </beamAngle>
</LaserLight>
```

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19.2.6.3 Set the mounting scenario mode of a camera

Request URL

PUT /ISAPI/Image/channels/<channelID>/mountingScenario

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, "indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight","lowIllumination"-low
illumination,"backlight"-back light,"frontlight"-front light,"faceSnap"-face picture capture, subType:string, desc:“indoor“, “outdoor”, “day”, “night”,
“morning”, “nightfall”, “mode1”, “mode2”, “mode3”, “mode4”, “street”, “lowIllumination”-low illumination, “custom1”, “custom2”, “normal”, “road”,
“faceSnap”-face picture capture, “highway”, “road2”, “backlight”-back light, “frontlight”-front light, “facenight”, “fastRoadAndLight”, “fastRoadNoLihgt”,
“normalRoadAndLight”, “normalRoadNoLight”, “cityRoadAndBright”-->indoor
  </mode>
</MountingScenario>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.6.4 Get the capability of configurations of mounting scenario mode

Request URL

GET /ISAPI/Image/channels/<channelID>/mountingScenario/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode
opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowIllumination,custom1,custom2,normal,road,faceSnap,highway,road2,backlight,
frontlight,facenight,fastRoadAndLight,fastRoadNoLight,normalRoadAndLight,normalRoadNolight,riverSeaProtection,perimeterProtection,forestFirePrevention,foggy
,faceFirst,plateFirst,moveFirst,autoWDR,sportSnap,dustMoisture,rampWay">
    <!--ro, req, enum, Mode, subType:string, attr:opt{req, string}, desc:"lowIllumination"-Low illumination, "backlight"-back light, "frontlight"-front
light, "faceSnap"-face picture capture-->indoor
  </mode>
  <isSupportRecommendation>
    <!--ro, opt, bool-->true
  </isSupportRecommendation>
  <modeNameList size="2">
    <!--ro, opt, object, attr:size{req, int}-->
    <modeNameItem>
      <!--ro, opt, object-->
      <mode
opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowIllumination,custom1,custom2,normal,road,faceSnap,highway,road2,backlight,
frontlight,facenight,fastRoadAndLight,fastRoadNoLight,normalRoadAndLight,normalRoadNolight,riverSeaProtection,perimeterProtection,forestFirePrevention,foggy
,faceFirst,plateFirst,moveFirst,autoWDR">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->indoor
      </mode>
      <modeName>
        <!--ro, req, string-->test
      </modeName>
    </modeNameItem>
  </modeNameList>
</MountingScenario>
```

19.2.6.5 Get the mounting scenario mode of a camera

Request URL

GET /ISAPI/Image/channels/<channelID>/mountingScenario?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	Parameter type. When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured, it will be the value of the parameter configured on the device.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, "indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight","lowIllumination"-
Low illumination,"backlight"-back light,"frontlight"-front light,"faceSnap"-face picture capture, subType:string, desc:“indoor”, “outdoor”, “day”, “night”,
“morning”, “nightfall”, “mode1”, “mode2”,“ mode3”, “mode4”, “highway”, “road2”, “faceSnap”, “backlight”, “frontlight”, “lowIllumination”-Low illumination,
“backlight”-back light, “frontlight”-front light, “faceSnap”-face picture capture-->indoor
  </mode>
  <modeNameList>
    <!--ro, opt, array, subType:object-->
    <modeNameItem>
      <!--ro, opt, object-->
      <mode>
        <!--ro, req, enum, subType:string-->indoor
      </mode>
      <modeName>
        <!--ro, req, string-->test
      </modeName>
    </modeNameItem>
  </modeNameList>
</MountingScenario>
```

19.2.7 Image White Balance

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

19.2.7.1 Set the white balance parameters of s specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/whiteBalance

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, white balance, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--opt, enum, white balance type, subType:string, desc:“auto”, “atuo1”, “auto2” “manual”, “indoor”, “outdoor”, “autotrace”, “onece”, “sodiumlight”,
“mercurylight”-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--req, int, this node depends on <WhiteBalanceStyle>, desc:this node depends on <WhiteBalanceStyle>-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--req, int, this node depends on <WhiteBalanceStyle>, desc:this node depends on <WhiteBalanceStyle>-->1
  </WhiteBalanceBlue>
</WhiteBalance>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.7.2 Get the white balance (in day mode) parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/whiteBalance/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dongjijin@mail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--ro, opt, enum, subType:string-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--ro, req, int-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--ro, req, int-->1
  </WhiteBalanceBlue>
  <whiteBalanceLevel>
    <!--ro, opt, int-->1
  </whiteBalanceLevel>
  <customColorTemperatureLevel>
    <!--ro, opt, int, range:[2700,7500], dep:and,{$.WhiteBalance,eq,customColorTemperature}-->1
  </customColorTemperatureLevel>
</WhiteBalance>
```

19.2.7.3 Set the white balance (in day mode) parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/whiteBalance/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--opt, enum, subType:string-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--req, int-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--req, int-->1
  </WhiteBalanceBlue>
  <customColorTemperatureLevel>
    <!--opt, int, range:[2700,7500], dep:and,{$.WhiteBalance,eq,customColorTemperature}-->1
  </customColorTemperatureLevel>
</WhiteBalance>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.7.4 Get the white balance parameters of s specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/whiteBalance?parameterType=<parameterType>

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, white balance, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--ro, opt, enum, "auto,manual,indoor,outdoor,autotrace,onece,sodiumlight,mercurylight,auto0,atuo1,fluorescent,natural,warm,incandescent",
subType:string, desc:"auto", "auto1", "atuo2", "manual", "indoor", "outdoor", "autotrace", "onece", "sodiumlight", "mercurylight"-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--ro, req, int, white balance, range:[0,100], desc:white balance-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--ro, req, int, white balance, range:[0,100], desc:white balance-->1
  </WhiteBalanceBlue>
  <whiteBalanceLevel>
    <!--ro, opt, int, range:[0,100], dep:or,{$.WhiteBalance.WhiteBalanceStyle,eq,auto},{$.WhiteBalance.WhiteBalanceStyle,eq,auto1},
{$.WhiteBalance.WhiteBalanceStyle,eq,auto2}-->1
  </whiteBalanceLevel>
  <customColorTemperatureLevel>
    <!--ro, opt, int, range:[1800,10000]-->1
  </customColorTemperatureLevel>
  <WhiteBalanceRedFineTuning>
    <!--ro, opt, int, range:[0,100]-->1
  </WhiteBalanceRedFineTuning>
  <WhiteBalanceBlueFineTuning>
    <!--ro, opt, int, range:[0,100]-->1
  </WhiteBalanceBlueFineTuning>
  <autoMode>
    <!--ro, opt, enum, subType:string, dep:and,{$.WhiteBalance.WhiteBalanceStyle,eq,auto}-->video
  </autoMode>
  <manualMode>
    <!--ro, opt, enum, subType:string, dep:and,{$.WhiteBalance.WhiteBalanceStyle,eq,manual}-->colorFineTuning
  </manualMode>
</WhiteBalance>
```

19.2.8 Live Streaming

19.2.8.1 Get the Real-Time Transport Control Protocol (RTCP) parameters of a specified channel

Request URL

GET /ISAPI/Streaming/channels/<trackStreamID>/RTCP

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	Channel No. * 100 + stream type (-1: main stream, -2: sub-stream, and so forth).

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<RTCP xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
</RTCP>
```

19.2.8.2 Set the Real-Time Transport Control Protocol (RTCP) parameters of a specified channel

Request URL

PUT /ISAPI/Streaming/channels/<trackStreamID>/RTCP

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RTCP xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
</RTCP>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.9 Manual Capture

19.2.9.1 Get the manual capture result information of a specified channel

Request URL

GET /ISAPI/Streaming/channels/<trackStreamID>/picture?snapShotImageType=<snapShotImageType>&videoResolutionWidth=<videoResolutionWidth>&videoResolutionHeight=<videoResolutionHeight>&imageQuality=<imageQuality>&x=<x>&y=<y>&hight=<hight>&width=<width>&maxSnapResolution=<maxSnapResolution>&livingBodyDetect=<livingBodyDetect>

Query Parameter

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

Parameter Name	Parameter Type	Description
trackStreamID	string	Channel No. * 100 + stream type (-1: main stream, -2: sub-stream).
snapShotImageType	string	Image formats: "bayer" (image not processed by ISP, which can be used to test whether the image clarity of the factory equipment meets the standard), "JPEGWITHRULE" (this type of image is specially for HIKMICRO devices, indicating a JPEG image captured with temperature rule information overlaid), "JPEGWITHOUTRULE" (when this type is used by HIKMICRO devices, it indicates a JPEG image captured without temperature rule information overlaid), "JPEG" (when this type is used by HIKMICRO devices, whether the captured JPEG image includes rule information depends on the enabling status of overlay temperature information on stream function. If it is enabled, the image returned will have rule information overlaid; otherwise, it will not. If the HIKMICRO device calls the URL without using this parameter, the logic for the returned JPEG image is consistent with the above JPEG type, depending on the enabling status of overlay temperature information on stream function).
videoResolutionWidth	string	Resolution width, and if not configured, it is 704 by default (D1 resolution width).
videoResolutionHeight	string	Resolution height, and if not configured, it is 576 by default (D1 resolution height).
imageQuality	string	Image quality: "best", "better", "general"
x	string	x-coordinate of the specified area, the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen
y	string	y-coordinate of the specified area, the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen
hight	string	Height of the specified area. The value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen.
width	string	Width of the specified area. The value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen.
maxSnapResolution	enum	Capture images using the maximum image capture resolution. For certain devices, such as low-power devices, the actual maximum image capture resolution differs from the video encoding resolution. During certain intelligent detections, images are captured at the maximum image capture resolution. If not configured, the default value is 1.
livingBodyDetect	enum	Whether to perform face anti-spoofing detection when capturing images.

Request Message

None

Response Message

Binary Data

19.2.10 Motion Detection

19.2.10.1 Get arming schedule of motion detection of all channels

Request URL

GET /ISAPI/Event/schedules/motionDetections

Query Parameter

VISION AND SECURITY SYSTEM CO., LTD. doc.jc.jl@gmail.com

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, arming time list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, ID, desc:ID-->VMD_video1
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string, desc:event type-->VMD
    </eventType>
    <inputIOPortID>
      <!--ro, opt, string-->test
    </inputIOPortID>
    <outputIOPortID>
      <!--ro, opt, string-->test
    </outputIOPortID>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, arming time list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <dayOfWeek>
          <!--ro, opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-
->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</MotionDetectionScheduleList>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

19.2.10.2 Set arming schedule of motion detection of all channels

Request URL

PUT /ISAPI/Event/schedules/motionDetections

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, arming time list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string, No., desc:No.-->VMD_video1
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string, desc:"VMD" (motion detection)-->VMD
    </eventType>
    <inputIOPortID>
      <!--opt, string-->test
    </inputIOPortID>
    <outputIOPortID>
      <!--opt, string-->test
    </outputIOPortID>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming time list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</MotionDetectionScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.10.3 Get the arming schedule of motion detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>

Query Parameter

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->VMD_video1
  </id>
  <eventType>
    <!--ro, opt, string-->VMD
  </eventType>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <outputIOPortID>
    <!--ro, opt, string-->test
  </outputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

19.2.10.4 Set the arming schedule of motion detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->VMD_video1
  </id>
  <eventType>
    <!--opt, string-->VMD
  </eventType>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <outputIOPortID>
    <!--opt, string-->test
  </outputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

19.2.10.5 Delete the linkage of motion detection for a specified channel

Request URL

DELETE /ISAPI/Event/triggers/VMD-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.2.10.6 Set the linkage triggering parameters of motion detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/VMD-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <notificationMethod>
    <!--req, enum, subType:string-->[email
  </notificationMethod>
  <notificationRecurrence>
    <!--opt, enum, subType:string-->beginning
  </notificationRecurrence>
  <notificationInterval>
    <!--opt, int, unit:ms-->1
  </notificationInterval>
  <outputIOPortID>
    <!--opt, string--> l
  </outputIOPortID>
  <dynOutputIOPortID>
    <!--opt, string--> l
  </dynOutputIOPortID>
  <videoInputID>
    <!--opt, string-->1
  </videoInputID>
  <dynVideoInputID>
    <!--opt, string-->1
  </dynVideoInputID>
  <ptzAction>
    <!--opt, object-->
    <ptzChannelID>
      <!--req, string-->1
    </ptzChannelID>
    <actionName>
      <!--req, enum, subType:string-->preset
    </actionName>
    <actionNum>
      <!--opt, int-->1
    </actionNum>
    <presetDurationTime>
      <!--opt, int, unit:s-->1
    </presetDurationTime>
  </ptzAction>
  <cellphoneNumber>
    <!--opt, string, range:[1,11]-->test
  </cellphoneNumber>
  <targetNotificationInterval>
    <!--opt, int, range:[0,1000], unit:s-->30
  </targetNotificationInterval>
  <SupplementLightAlarm>
    <!--opt, object-->
    <durationTime>
      <!--opt, int, range:[0,90], unit:s-->5
    </durationTime>
  </SupplementLightAlarm>
</EventTriggerNotification>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.10.7 Get the linkage triggering parameters of motion detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/VMD-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, float, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, req, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
      <presetDurationTime>
        <!--ro, opt, int, unit:s-->1
      </presetDurationTime>
    </ptzAction>
    <cellphoneNumber>
      <!--ro, opt, string, range:[1,11]-->test
    </cellphoneNumber>
    <targetNotificationInterval>
      <!--ro, opt, int, range:[0,1000], unit:s-->30
    </targetNotificationInterval>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

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19.2.10.8 Get the linkage parameters of motion detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/VMD-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupporttriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, string-->VMD
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->[email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

19.2.10.9 Set the linkage parameters of motion detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/VMD-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->VMD
```

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```
    </eventType>
    <eventDescription>
      <!--opt, string-->test
    </eventDescription>
    <inputIOPortID>
      <!--opt, string-->1
    </inputIOPortID>
    <dynInputIOPortID>
      <!--opt, string-->1
    </dynInputIOPortID>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <dynVideoInputChannelID>
      <!--opt, string-->1
    </dynVideoInputChannelID>
    <intervalBetweenEvents>
      <!--opt, int, unit:s-->1
    </intervalBetweenEvents>
    <WLSensorID>
      <!--opt, string-->1
    </WLSensorID>
    <EventTriggerNotificationList>
      <!--opt, array, subType:object-->
      <EventTriggerNotification>
        <!--opt, object-->
        <id>
          <!--req, string-->test
        </id>
        <notificationMethod>
          <!--req, enum, subType:string-->email
        </notificationMethod>
        <notificationRecurrence>
          <!--opt, enum, subType:string-->beginning
        </notificationRecurrence>
        <notificationInterval>
          <!--opt, int, unit:ms-->1
        </notificationInterval>
        <outputIOPortID>
          <!--opt, string-->1
        </outputIOPortID>
        <dynOutputIOPortID>
          <!--opt, string-->1
        </dynOutputIOPortID>
        <videoInputID>
          <!--opt, string-->1
        </videoInputID>
        <dynVideoInputID>
          <!--opt, string-->1
        </dynVideoInputID>
        <ptzAction>
          <!--opt, object-->
          <ptzChannelID>
            <!--req, string-->1
          </ptzChannelID>
          <actionName>
            <!--req, enum, subType:string-->preset
          </actionName>
          <actionNum>
            <!--opt, int-->1
          </actionNum>
          <presetDurationTime>
            <!--opt, int, unit:s-->1
          </presetDurationTime>
        </ptzAction>
        <cellphoneNumber>
          <!--opt, string, range:[1,11]-->test
        </cellphoneNumber>
        <LightAudioByIPAction>
          <!--opt, object-->
          <notificationType>
            <!--req, enum, subType:string-->audio
          </notificationType>
          <notificationName>
            <!--opt, string-->test
          </notificationName>
        </LightAudioByIPAction>
        <SupplementLightAlarm>
          <!--opt, object-->
          <durationTime>
            <!--opt, int, range:[0,90], unit:s-->5
          </durationTime>
        </SupplementLightAlarm>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
  </EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.10.10 Set the motion detection parameters of a specific channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, motion detection parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable motion detection-->true
  </enabled>
  <enableHighlight>
    <!--opt, bool, whether to enable dynamic analysis-->true
  </enableHighlight>
  <samplingInterval>
    <!--opt, int, frame interval, range:[0,2]-->0
  </samplingInterval>
  <startTriggerTime>
    <!--opt, int, start time, range:[0,500], unit:ms-->0
  </startTriggerTime>
  <endTriggerTime>
    <!--opt, int, end time, range:[0,500], unit:ms-->0
  </endTriggerTime>
  <regionType>
    <!--req, enum, region type, subType:string, desc:"grid", "roi" (region of interest), "none", "region" (multiple regions)-->grid
  </regionType>
  <Grid>
    <!--opt, object, grid-->
    <rowGranularity>
      <!--req, int, total number of rows, range:[0,18]-->0
    </rowGranularity>
    <columnGranularity>
      <!--req, int, total number of columns, range:[0,22]-->0
    </columnGranularity>
  </Grid>
  <ROI>
    <!--opt, object, region of interest (ROI)-->
    <normalizedScreenWidth>
      <!--req, int, normalized width, range:[0,1000]-->0
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height, range:[0,1000]-->0
    </normalizedScreenHeight>
  </ROI>
  <MotionDetectionLayout>
    <!--opt, object, layout configuration-->
    <sensitivityLevel>
      <!--req, int, sensitivity, range:[0,100]-->0
    </sensitivityLevel>
    <layout>
      <!--opt, object, layout information-->
      <gridMap>
        <!--opt, string, grid map, desc:the image is divided in to N × N grids, and motion detection can be enabled (value: 1) or disabled (value 0) for
each grid. Each four horizontal grids consist of a hexadecimal number, e.g., the corresponding hexadecimal number of "0011" is 3-->test
      </gridMap>
      <roiMap>
        <!--opt, string, ROI map-->test
      </roiMap>
      <RegionList>
        <!--opt, array, region list, subType:object-->
        <Region>
          <!--opt, object, region information-->
          <id>
            <!--req, string, region ID, range:[1,4]-->test
          </id>
          <RegionCoordinatesList>
            <!--opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
            <RegionCoordinates>
              <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--req, int, X-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--req, int, Y-coordinate, range:[0,1000]-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </Region>
      </RegionList>
    </layout>
    <targetType>
      <!--opt, enum, target classification, subType:string, desc:"human", "vehicle"-->human
    </targetType>
  </MotionDetectionLayout>
  <enableWithMoving>
    <!--opt, bool, whether to enable motion detection-->true
  </enableWithMoving>
</MotionDetection>
```

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dongjin@gmail.com

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

19.2.10.11 Get the motion detection parameters of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, motion detection parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable motion detection-->true
  </enabled>
  <enableHighlight>
    <!--ro, opt, bool, whether to enable dynamic analysis-->true
  </enableHighlight>
  <samplingInterval>
    <!--ro, opt, int, frame interval, range:[0,2]-->0
  </samplingInterval>
  <startTriggerTime>
    <!--ro, opt, int, start time, range:[0,500], unit:ms-->0
  </startTriggerTime>
  <endTriggerTime>
    <!--ro, opt, int, end time, range:[0,500], unit:ms-->0
  </endTriggerTime>
  <regionType>
    <!--ro, req, enum, region type, subType:string, desc:"grid", "roi" (region of interest), "none", "region" (multiple regions)-->grid
  </regionType>
  <Grid>
    <!--ro, opt, object, grid-->
    <rowGranularity>
      <!--ro, req, int, total number of rows, range:[0,18]-->0
    </rowGranularity>
    <columnGranularity>
      <!--ro, req, int, total number of columns, range:[0,22]-->0
    </columnGranularity>
  </Grid>
  <ROI>
    <!--ro, opt, object, region of interest (ROI)-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[0,1000]-->0
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height, range:[0,1000]-->0
    </normalizedScreenHeight>
  </ROI>
  <MotionDetectionLayout>
    <!--ro, opt, object, layout configuration-->
    <sensitivityLevel>
      <!--ro, req, int, sensitivity, range:[0,100]-->0
    </sensitivityLevel>
    <layout>
      <!--ro, opt, object, layout information-->
      <gridMap>
        <!--ro, opt, string, grid map, desc:the image is divided in to N × N grids, and motion detection can be enabled (value: 1) or disabled (value: 0)
for each grid. Each four horizontal grids consist of a hexadecimal number, e.g., the corresponding hexadecimal number of "0011" is 3-->test
      </gridMap>
      <roiMap>
        <!--ro, opt, string, ROI map-->test
      </roiMap>
      <RegionList>
        <!--ro, opt, array, this node is available when the regionType is set to "region", subType:object-->
        <Region>
          <!--ro, opt, object, list-->
          <id>
            <!--ro, req, string, region ID, range:[1,4]-->test
          </id>
          <RegionCoordinatesList>
            <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
            <RegionCoordinates>
              <!--ro, opt, object, list, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--ro, req, int, X-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </Region>
      </RegionList>
    </layout>
    <targetType>
      <!--ro, opt, enum, target classification, subType:string, desc:"human", "vehicle"-->human
    </targetType>
  </MotionDetectionLayout>
  <enableWithMoving>
    <!--ro, opt, bool, whether to enable motion detection-->true
  </enableWithMoving>
</MotionDetection>
```

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19.2.10.12 Get the motion detection capability of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, motion detection parameters, attr:version{req, string, protocolVersion}-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable motion detection, attr:opt{req, string}-->true
  </enabled>
  <enableHighlight opt="true,false">
    <!--ro, opt, bool, whether to enable dynamic analysis, attr:opt{req, string}-->true
  </enableHighlight>
  <samplingInterval>
    <!--ro, opt, int, frame interval-->0
  </samplingInterval>
  <startTriggerTime>
    <!--ro, opt, int, start time, unit:ms-->0
  </startTriggerTime>
  <endTriggerTime>
    <!--ro, opt, int, end time, unit:ms-->0
  </endTriggerTime>
  <regionType opt="grid,roi,none,region">
    <!--ro, req, enum, region type, subType:string, attr:opt{req, string}, desc:"grid", "roi" (region of interest), "none", "region" (multiple regions)--
>grid
  </regionType>
  <Grid>
    <!--ro, opt, object, grid-->
    <rowGranularity>
      <!--ro, req, int, total number of rows-->0
    </rowGranularity>
    <columnGranularity>
      <!--ro, req, int, total number of columns-->0
    </columnGranularity>
  </Grid>
  <ROI>
    <!--ro, opt, object, region of interest (ROI)-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->0
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->0
    </normalizedScreenHeight>
  </ROI>
  <MotionDetectionLayout>
    <!--ro, req, object, layout configuration capability-->
    <sensitivityLevel min="0" max="100" step="20">
      <!--ro, req, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int},step{opt, int}-->0
    </sensitivityLevel>
    <layout>
      <!--ro, opt, object, layout information-->
      <gridMap>
        <!--ro, opt, string, grid map, desc:the image is divided in to N × N grids, and motion detection can be enabled (value: 1) or disabled (value: 0)
for each grid. Each four horizontal grids consist of a hexadecimal number, e.g., the corresponding hexadecimal number of "0011" is 3-->test
      </gridMap>
      <roiMap>
        <!--ro, opt, string, ROI map-->test
      </roiMap>
      <RegionList size="4">
        <!--ro, opt, array, this node is available when the regionType is set to "region", subType:object, attr:size{req, int}-->
        <Region>
          <!--ro, opt, object, list-->
          <id>
            <!--ro, req, string, region ID-->test
          </id>
          <RegionCoordinatesList>
            <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
            <RegionCoordinates>
              <!--ro, opt, object, list, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--ro, req, int, X-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </Region>
```

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```
      </RegionList>
    </layout>
    <targetType opt="human,vehicle">
      <!--ro, opt, enum, target classification, subType:string, attr:opt{req, string}, desc:"human", "vehicle"-->human
    </targetType>
    <regionEdgesNum min="4" max="10">
      <!--ro, opt, int, number of detection area sides, range:[4,10], attr:min{req, int},max{req, int}, desc:if this node is not returned, the default
detection area is quadrilateral-->0
    </regionEdgesNum>
  </MotionDetectionLayout>
  <enableWithMoving opt="true,false">
    <!--ro, opt, bool, whether to enable motion detection, attr:opt{req, string}-->true
  </enableWithMoving>
  <isSupportMotionDetectionResolution>
    <!--ro, opt, bool-->true
  </isSupportMotionDetectionResolution>
</MotionDetection>
```

19.2.10.13 Set grid layout parameters of motion detection of a specific channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout/gridLayout?regionType=<regionType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionType	enum	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetectionGridLayout xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, grid layout parameters of motion detection, attr:version{req, string, protocolVersion}-->
  <sensitivityLevel>
    <!--req, int, sensitivity, range:[0,100], desc:sensitivity-->1
  </sensitivityLevel>
  <layout>
    <!--opt, object-->
    <gridMap>
      <!--opt, string, grid map, desc:the image is divided in to N × N grids, and motion detection can be enabled (value: 1) or disabled (value 0) for each
grid. Each four horizontal grids consist of a hexadecimal number-->0011
    </gridMap>
    <roiMap>
      <!--opt, string-->test
    </roiMap>
    <RegionList size="10">
      <!--opt, array, subType:object, attr:size{req, int}-->
      <Region>
        <!--opt, object-->
        <id>
          <!--req, string, region ID-->test
        </id>
        <RegionCoordinatesList>
          <!--opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType>
    <!--opt, enum, target classification, subType:string, desc:target classification-->human
  </targetType>
</MotionDetectionGridLayout>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.10.14 Get grid layout parameters of motion detection of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout/gridLayout?regionType=<regionType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionType	enum	Configure region: /ISAPI/System/Video/inputs/channels//motionDetection/layout/gridLayout?regionType=region; configure grid: /ISAPI/System/Video/inputs/channels//motionDetection/layout/gridLayout?regionType=grid.

Request Message

None

Response Message

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do jo.jin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetectionGridLayout xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <sensitivityLevel>
    <!--ro, req, int, sensitivity, range:[0,100], desc:sensitivity-->1
  </sensitivityLevel>
  <layout>
    <!--ro, opt, object-->
    <gridMap>
      <!--ro, opt, string, grid map, desc:the image is divided in to N × N grids, and motion detection can be enabled (value: 1) or disabled (value 0) for
each grid. Each four horizontal grids consist of a hexadecimal number-->0011
    </gridMap>
    <roiMap>
      <!--ro, opt, string-->test
    </roiMap>
    <RegionList size="10">
      <!--ro, opt, array, this node is available when the regionType is set to "region", subType:object, attr:size{req, int}-->
      <Region>
        <!--ro, opt, object, list-->
        <id>
          <!--ro, req, string-->test
        </id>
        <RegionCoordinatesList>
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, opt, object, coordinate, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType>
    <!--ro, opt, enum, target classification, subType:string, desc:target classification-->human
  </targetType>
</MotionDetectionGridLayout>
```

19.2.10.15 Get the region parameters of motion detection of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout?regionType=<regionType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionType	enum	Configure region: /ISAPI/System/Video/inputs/channels//motionDetection/layout?regionType=region; configure grid: /ISAPI/System/Video/inputs/channels//motionDetection/layout?regionType=grid.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetectionLayout xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <sensitivityLevel>
    <!--ro, req, int, sensitivity-->1
  </sensitivityLevel>
  <layout>
    <!--ro, opt, object-->
    <gridMap>
      <!--ro, opt, string, grid map, desc:the image is divided in to N × N grids, and motion detection can be enabled (value: 1) or disabled (value 0) for
each grid. Each four horizontal grids consist of a hexadecimal number-->0011
    </gridMap>
    <roiMap>
      <!--ro, opt, string-->test
    </roiMap>
    <RegionList>
      <!--ro, opt, array, this node is available when the regionType is set to "region", subType:object-->
      <Region>
        <!--ro, opt, object, list-->
        <id>
          <!--ro, req, string-->test
        </id>
        <RegionCoordinatesList>
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, opt, object, coordinate, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType>
    <!--ro, opt, enum, target classification, subType:string, desc:"human", "vehicle"-->human
  </targetType>
</MotionDetectionLayout>
```

19.2.10.16 Set the region parameters of motion detection of a specific channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout?regionType=<regionType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionType	enum	--

Request Message

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dongjin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetectionLayout xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <sensitivityLevel>
    <!--req, int, sensitivity-->1
  </sensitivityLevel>
  <layout>
    <!--opt, object-->
    <gridMap>
      <!--opt, string, grid map, desc:the image is divided in to N × N grids, and motion detection can be enabled (value: 1) or disabled (value 0) for each
grid. Each four horizontal grids consist of a hexadecimal number-->0011
    </gridMap>
    <roiMap>
      <!--opt, string-->test
    </roiMap>
    <RegionList>
      <!--opt, array, this node is available when the regionType is set to "region", subType:object-->
      <Region>
        <!--opt, object, list-->
        <id>
          <!--req, string, region ID-->test
        </id>
        <RegionCoordinatesList>
          <!--opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </Region>
    </RegionList>
  </layout>
  <targetType>
    <!--opt, enum, target classification, subType:string, desc:"human", "vehicle"-->human
  </targetType>
</MotionDetectionLayout>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.10.17 Get motion detection parameters of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetectionExt xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, motion detection parameters of a specified video input channel, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <minObjectSize>
    <!--ro, opt, int, minimum target resolution-->0
  </minObjectSize>
  <maxObjectSize>
    <!--ro, opt, int, maximum target resolution-->0
  </maxObjectSize>
  <ROI>
    <!--ro, opt, object, region of interest (ROI)-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->0
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->0
    </normalizedScreenHeight>
  </ROI>
  <enableHighlight>
    <!--ro, opt, bool, whether to enable highlight-->true
  </enableHighlight>
  <MotionDetectionSwitch>
    <!--ro, opt, object, day/night settings switch-->
    <type>
      <!--ro, opt, enum, type, subType:string, desc:"off", "auto", "schedule" (scheduled)-->off
    </type>
    <Schedule>
      <!--ro, opt, object, schedule-->
      <scheduleType>
        <!--ro, req, enum, schedule type, subType:string, desc:"day" (daytime), "night" (night)-->day
      </scheduleType>
      <TimeRange>
        <!--ro, req, object, time period-->
        <beginTime>
          <!--ro, req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </Schedule>
  </MotionDetectionSwitch>
  <activeMode>
    <!--ro, opt, enum, effective mode, subType:string, desc:"normal", "expert"-->normal
  </activeMode>
  <MotionDetectionRegionList size="10">
    <!--ro, opt, array, detection area list, subType:object, attr:size{req, int}-->
    <MotionDetectionRegion>
      <!--ro, opt, object, detection area-->
      <id>
        <!--ro, req, string, ID-->test
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable the function-->true
      </enabled>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->0
      </sensitivityLevel>
      <daySensitivityLevel>
        <!--ro, opt, int, sensitivity (during day time), range:[0,100]-->0
      </daySensitivityLevel>
      <nightSensitivityLevel>
        <!--ro, opt, int, sensitivity (during night), range:[0,100]-->0
      </nightSensitivityLevel>
      <objectSize>
        <!--ro, opt, int, percentage, range:[0,100]-->0
      </objectSize>
      <dayObjectSize>
        <!--ro, opt, int, percentage (during day time), range:[0,100]-->0
      </dayObjectSize>
      <nightObjectSize>
        <!--ro, opt, int, percentage (during night), range:[0,100]-->0
      </nightObjectSize>
      <RegionCoordinatesList>
        <!--ro, req, array, rule region list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </MotionDetectionRegion>
  </MotionDetectionRegionList>
  <enableWithMoving>
    <!--ro, opt, bool, whether to enable dynamic analysis-->true
```

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```
    </enableWithMoving>
  </MotionDetectionExt>
```

19.2.10.18 Set motion detection parameters of a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetectionExt xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, motion detection parameters of a specified video input channel, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <minObjectSize>
    <!--opt, int, minimum target resolution-->0
  </minObjectSize>
  <maxObjectSize>
    <!--opt, int, maximum target resolution-->0
  </maxObjectSize>
  <ROI>
    <!--opt, object, region of interest (ROI)-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->0
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->0
    </normalizedScreenHeight>
  </ROI>
  <enableHighlight>
    <!--opt, bool, whether to enable highlight-->true
  </enableHighlight>
  <MotionDetectionSwitch>
    <!--opt, object, day/night settings switch-->
    <type>
      <!--opt, enum, type, subType:string, desc:"off", "auto", "schedule" (scheduled)-->off
    </type>
    <Schedule>
      <!--opt, object, schedule-->
      <scheduleType>
        <!--req, enum, schedule type, subType:string, desc:"day" (daytime), "night" (night)-->day
      </scheduleType>
      <TimeRange>
        <!--req, object, time period-->
        <beginTime>
          <!--req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </Schedule>
  </MotionDetectionSwitch>
  <activeMode>
    <!--opt, enum, effective mode, subType:string, desc:"normal", "expert"-->normal
  </activeMode>
  <MotionDetectionRegionList size="10">
    <!--opt, array, detection area list, subType:object, attr:size{req, int}-->
    <MotionDetectionRegion>
      <!--opt, object, detection area-->
      <id>
        <!--req, string, ID-->test
      </id>
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <sensitivityLevel>
        <!--opt, int, sensitivity, range:[0,100]-->0
      </sensitivityLevel>
      <daySensitivityLevel>
        <!--opt, int, sensitivity (during day time), range:[0,100]-->0
      </daySensitivityLevel>
      <nightSensitivityLevel>
        <!--opt, int, sensitivity (during night), range:[0,100]-->0
      </nightSensitivityLevel>
      <objectSize>
        <!--opt, int, percentage, range:[0,100]-->0
```

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```
        </objectSize>
        <dayObjectSize>
          <!--opt, int, percentage (during day time), range:[0,100]-->0
        </dayObjectSize>
        <nightObjectSize>
          <!--opt, int, percentage (during night), range:[0,100]-->0
        </nightObjectSize>
        <RegionCoordinatesList>
          <!--req, array, rule region list, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--opt, object, area, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, x-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </MotionDetectionRegion>
    </MotionDetectionRegionList>
    <enableWithMoving>
      <!--opt, bool, whether to enable dynamic analysis-->true
    </enableWithMoving>
  </MotionDetectionExt>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

19.2.10.19 Set the motion detection parameters of a single region of a specific channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, detection area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->test
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <sensitivityLevel>
    <!--opt, int, sensitivity, range:[0,100]-->0
  </sensitivityLevel>
  <daySensitivityLevel>
    <!--opt, int, sensitivity (during day time), range:[0,100]-->0
  </daySensitivityLevel>
  <nightSensitivityLevel>
    <!--opt, int, sensitivity (during night), range:[0,100]-->0
  </nightSensitivityLevel>
  <objectSize>
    <!--opt, int, percentage, range:[0,100]-->0
  </objectSize>
  <dayObjectSize>
    <!--opt, int, percentage (during day time), range:[0,100]-->0
  </dayObjectSize>
  <nightObjectSize>
    <!--opt, int, percentage (during night), range:[0,100]-->0
  </nightObjectSize>
  <RegionCoordinatesList>
    <!--req, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</MotionDetectionRegion>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.10.20 Motion detection

EventType:VMD

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, EventNotificationAlert, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPv4 address-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 Address-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, device's communication port No.-->1
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol, subType:string, desc:"HTTP", "HTTPS", "EHome"-->HTTP
  </protocol>
  <macAddress>
```

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```
    <!--ro, opt, string, MAC address-->test
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->1
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No.-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, event occurred time-->2017-04-22T15:39:01+08:0
  </dateTime>
  <activePostCount>
    <!--ro, req, int, event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"VMD" (motion detection)-->VMD
  </eventType>
  <eventState>
    <!--ro, req, enum, event status (for persistent event), subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <DetectionRegionList>
    <!--ro, opt, array, detection area parameter list (for expert mode only), subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, parameters of a detection area (for expert mode only), desc:parameters of a detection area (for expert mode only)-->
      <regionID>
        <!--ro, req, string, detection area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, req, array, detection area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, detection area origin coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID-->1
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, req, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, req, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <targetType>
    <!--ro, opt, string, target classification, desc:"human", "vehicle", "human,vehicle" (human and vehicle). If detection target classification is not
supported, this field is not returned-->human
  </targetType>
  <bkgUrl>
    <!--ro, opt, string, background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (it is for the cloud storage protocol), "digest" (digest authentication,
it is for the device local storage)-->no
  </URLCertificationType>
</EventNotificationAlert>
```

19.2.11 Privacy Mask

19.2.11.1 Set the arming schedule of video tampering detection for all channels

Request URL

PUT /ISAPI/Event/schedules/tamperDetections

Query Parameter

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None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TamperDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--req, object-->
    <id>
      <!--req, string-->tamperdetection_video1
    </id>
    <eventType>
      <!--opt, string-->tamperDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</TamperDetectionScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.11.2 Get the arming schedule of video tampering detection for all channels

Request URL

GET /ISAPI/Event/schedules/tamperDetections

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TamperDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, req, object-->
    <id>
      <!--ro, req, string-->tamperdetection_video1
    </id>
    <eventType>
      <!--ro, opt, string-->tamperDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</TamperDetectionScheduleList>
```

19.2.11.3 Get the arming schedule of video tampering detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/tamperDetections/tamperdetection_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->tamperdetection_video1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->tamperDetection
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.2.11.4 Set the arming schedule of video tampering detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/tamperDetections/tamperdetection_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->tamperdetection_video1
  </id>
  <eventType>
    <!--opt, string-->tamperDetection
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

19.2.11.5 Get the linkage parameters of video tampering detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/tamper-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->tamper-1
  </id>
  <eventType>
    <!--ro, req, string-->tamper
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s-->0
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
      <modbus>
        <!--ro, opt, object-->
        <outputPortID>
          <!--ro, opt, int, range:[0,15]-->1
        </outputPortID>
        <delayTime>
          <!--ro, opt, int, range:[1,10], unit:s-->1
        </delayTime>
        <alarmStatus>
          <!--ro, opt, enum, subType:string-->open
        </alarmStatus>
      </modbus>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.2.11.6 Delete the linkage of video tampering detection for a specified channel

Request URL

DELETE /ISAPI/Event/triggers/tamper-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.2.11.7 Set the linkage parameters of video tampering detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/tamper-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->tamper-1
  </id>
  <eventType>
    <!--req, string-->tamper
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s-->0
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
      <modbus>
        <!--opt, object-->
        <outputPortID>
          <!--opt, int, range:[0,15]-->1
        </outputPortID>
        <delayTime>
          <!--opt, int, range:[1,10], unit:s-->1
        </delayTime>
        <alarmStatus>
          <!--opt, enum, subType:string-->open
        </alarmStatus>
      </modbus>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.11.8 Set the linkage triggering parameters of video tampering detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/tamper-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <notificationMethod>
    <!--req, enum, subType:string-->FTP
  </notificationMethod>
  <notificationRecurrence>
    <!--opt, enum, subType:string-->beginning
  </notificationRecurrence>
  <notificationInterval>
    <!--opt, int, unit:ms-->1
  </notificationInterval>
  <outputIOPortID>
    <!--opt, string-->test
  </outputIOPortID>
  <dynOutputIOPortID>
    <!--opt, string-->test
  </dynOutputIOPortID>
  <videoInputID>
    <!--opt, string-->1
  </videoInputID>
  <dynVideoInputID>
    <!--opt, string-->1
  </dynVideoInputID>
  <ptzAction>
    <!--opt, object-->
    <ptzChannelID>
      <!--req, string-->1
    </ptzChannelID>
    <actionName>
      <!--req, enum, subType:string-->preset
    </actionName>
    <actionNum>
      <!--opt, int-->1
    </actionNum>
    <presetDurationTime>
      <!--opt, int, unit:s-->0
    </presetDurationTime>
  </ptzAction>
  <cellphoneNumber>
    <!--opt, string, range:[0,11]-->test
  </cellphoneNumber>
  <modbus>
    <!--opt, object-->
    <outputPortID>
      <!--opt, int, range:[0,15]-->1
    </outputPortID>
    <delayTime>
      <!--opt, int, range:[1,10], unit:s-->1
    </delayTime>
    <alarmStatus>
      <!--opt, enum, subType:string-->open
    </alarmStatus>
  </modbus>
</EventTriggerNotification>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.11.9 Get the linkage triggering parameters of video tampering detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/tamper-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <notificationMethod>
    <!--ro, req, enum, subType:string-->FTP
  </notificationMethod>
  <notificationRecurrence>
    <!--ro, opt, enum, subType:string-->beginning
  </notificationRecurrence>
  <notificationInterval>
    <!--ro, opt, int, unit:ms-->1
  </notificationInterval>
  <outputIOPortID>
    <!--ro, opt, string-->test
  </outputIOPortID>
  <dynOutputIOPortID>
    <!--ro, opt, string-->test
  </dynOutputIOPortID>
  <videoInputID>
    <!--ro, opt, string-->1
  </videoInputID>
  <dynVideoInputID>
    <!--ro, opt, string-->1
  </dynVideoInputID>
  <ptzAction>
    <!--ro, opt, object-->
    <ptzChannelID>
      <!--ro, req, string-->1
    </ptzChannelID>
    <actionName>
      <!--ro, req, enum, subType:string-->preset
    </actionName>
    <actionNum>
      <!--ro, opt, int-->1
    </actionNum>
    <presetDurationTime>
      <!--ro, opt, int, unit:s-->0
    </presetDurationTime>
  </ptzAction>
  <cellphoneNumber>
    <!--ro, opt, string, range:[0,11]-->test
  </cellphoneNumber>
  <modbus>
    <!--ro, opt, object-->
    <outputPortID>
      <!--ro, opt, int, range:[0,15]-->1
    </outputPortID>
    <delayTime>
      <!--ro, opt, int, range:[1,10], unit:s-->1
    </delayTime>
    <alarmStatus>
      <!--ro, opt, enum, subType:string-->open
    </alarmStatus>
  </modbus>
</EventTriggerNotification>
```

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19.2.11.10 Get the video tampering parameters of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/tamperDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TamperDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video tampering parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable video tampering-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, req, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <tampersensitivityLevel>
    <!--ro, req, int, video tampering sensitivity, range:[0,100]-->1
  </tampersensitivityLevel>
  <TamperDetectionRegionList>
    <!--ro, req, array, video tampering detection area list, subType:object-->
    <TamperDetectionRegion>
      <!--ro, opt, object, video tampering detection area-->
      <id>
        <!--ro, req, string, area ID-->test
      </id>
      <enabled>
        <!--ro, opt, bool, whether to enable the area-->true
      </enabled>
      <sensitivityLevel>
        <!--ro, req, int, area sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, req, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </TamperDetectionRegion>
  </TamperDetectionRegionList>
  <onlyRecordEnabled>
    <!--ro, opt, bool-->false
  </onlyRecordEnabled>
  <alarmTriggerDelay>
    <!--ro, opt, int, range:[2,30], unit:s-->3
  </alarmTriggerDelay>
</TamperDetection>
```

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19.2.11.11 Set the video tampering parameters of a specific channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/tamperDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TamperDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, video tampering parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable video tampering-->true
  </enabled>
  <normalizedScreenSize>
    <!--req, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <tampersensitivityLevel>
    <!--req, int, video tampering sensitivity, range:[0,100]-->1
  </tampersensitivityLevel>
  <TamperDetectionRegionList>
    <!--req, array, video tampering detection area list, subType:object-->
    <TamperDetectionRegion>
      <!--opt, object, video tampering detection area-->
      <id>
        <!--req, string, area ID-->test
      </id>
      <enabled>
        <!--opt, bool, whether to enable the area-->true
      </enabled>
      <sensitivityLevel>
        <!--req, int, area sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--req, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, x-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </TamperDetectionRegion>
  </TamperDetectionRegionList>
  <onlyRecordEnabled>
    <!--opt, bool-->false
  </onlyRecordEnabled>
  <alarmTriggerDelay>
    <!--opt, int, range:[2,30], unit:s-->3
  </alarmTriggerDelay>
</TamperDetection>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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19.2.11.12 Set the video tampering parameters of a single detection area for a specified channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/tamperDetection/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TamperDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, video tampering detection areas, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, area ID-->test
  </id>
  <enabled>
    <!--opt, bool, whether to enable area detection-->true
  </enabled>
  <sensitivityLevel>
    <!--req, int, area detection sensitivity, range:[0,100]-->1
  </sensitivityLevel>
  <RegionCoordinatesList>
    <!--req, array, list of area coordinates, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,704]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,576]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</TamperDetectionRegion>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.11.13 Video tampering alarm

EventType:shelteralarm

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event details, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPv4 Address-->10.10.10.10
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 Address-->ABCD:EF01:2345:6789:ABCD:EF01:2345:6789
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, device's communication port No.-->1
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:"HTTP, HTTPS, EHome". The value should be "HTTP" when ISAPI protocol
is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP.-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->test
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No.-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, event occurred time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type: shelteralarm, subType:string, desc:event type: shelteralarm-->shelteralarm
  </eventType>
  <eventState>
    <!--ro, req, enum, event status (for persistent event), subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID-->test
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, req, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, req, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <Image>
    <!--ro, opt, object, event related pictures-->
    <resourcesContentType>
      <!--ro, opt, enum, resource transmission type, subType:string, desc:"url" (transmitted via URL), "binary" (transmitted in binary format). When the
returned alarm message contains additional resource (e.g., pictures), this field should be returned.-->binary
    </resourcesContentType>
    <resourcesContent>
      <!--ro, opt, string, resource ID, range:[1,128], desc:when the returned alarm message contains additional resources (e.g., pictures), this field
should be returned. When the value of resourcesContentType is "binary", it must be the same as Content-ID of the resource. When the value of
resourcesContentType is "url", its value is the resource URL-->image
    </resourcesContent>
  </Image>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
EventNotificationAlert	[Message content]	application/xml	--	--	--
image	[Binary picture data]	image/jpeg	image	image.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.2.12 Scene Change

19.2.12.1 Set the arming schedule of scene change detection for all channels

Request URL

PUT /ISAPI/Event/schedules/sceneChangeDetections

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SceneChangeDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->sceneChangeDetection_video1
    </id>
    <eventType>
      <!--opt, enum, subType:string-->sceneChangeDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</SceneChangeDetectionScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.12.2 Get the arming schedule of scene change detection for all channels

Request URL

GET /ISAPI/Event/schedules/sceneChangeDetections

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<SceneChangeDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->sceneChangeDetection_video1
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->sceneChangeDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</SceneChangeDetectionScheduleList>
```

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19.2.12.3 Get the arming schedule of scene change detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->sceneChangeDetection_video1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->sceneChangeDetection
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.2.12.4 Set the arming schedule of scene change detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->sceneChangeDetection_video1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->sceneChangeDetection
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.12.5 Sudden scene change detection

EventType:scenechangedetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"scenechangedetection" (sudden scene change detection)-->scenechangedetection
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
</EventNotificationAlert>
```

19.2.13 Video Metadata

19.2.13.1 Get all metadata parameters of the specified channel

Request URL

GET /ISAPI/Streaming/channels/<channelID>/metadata

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<MetadataCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, metadata configurations, attr:version{req, string, protocolVersion}-->
  <MetadataList>
    <!--ro, req, array, metadata information list, subType:object-->
    <Metadata>
      <!--ro, req, object, metadata information-->
      <type>
        <!--ro, req, enum, event types that support metadata, subType:string, desc:"thermometry" (real-time temperature measurement), "fireDetection" (fire
detection), "shipsDetection" (ship detection), "behaviorAnalysis" (abnormal event detection), "ANPR" (automatic number plate recognition), "personQueue"
(queue management, including people queuing-up detection and waiting time detection), "faceSnap" (face capture), "mixedTargetDetection" (multi-target-
detection), "AIOP" (AIOP detection), "radarDetection" (radar detection), "personalTrack" (personal tracking)-->thermometry
      </type>
      <enable>
        <!--ro, req, bool, whether to enable uploading metadata-->true
      </enable>
    </Metadata>
  </MetadataList>
  <uploadDataFormat>
    <!--ro, opt, enum, uploaded message format, subType:string, desc:"xml" (XML format), "json" (JSON format)-->xml
  </uploadDataFormat>
</MetadataCfg>
```

19.2.13.2 Set all metadata parameters for a specified channel

Request URL

PUT /ISAPI/Streaming/channels/<channelID>/metadata

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MetadataCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, metadata configurations, attr:version{req, string, protocolVersion}-->
  <MetadataList>
    <!--req, array, metadata information list, subType:object-->
    <Metadata>
      <!--req, object, metadata information-->
      <type>
        <!--req, enum, event types that support metadata, subType:string, desc:"thermometry" (real-time temperature measurement), "fireDetection" (fire
detection), "shipsDetection" (ship detection), "behaviorAnalysis" (abnormal event detection), "ANPR" (automatic number plate recognition), "personQueue"
(queue management, including people queuing-up detection and waiting time detection), "faceSnap" (face capture), "mixedTargetDetection" (multi-target-
detection), "AIOP" (AIOP detection)-->thermometry
      </type>
      <enable>
        <!--req, bool, whether to enable uploading metadata-->true
      </enable>
    </Metadata>
  </MetadataList>
  <uploadDataFormat>
    <!--opt, enum, uploaded message format, subType:string, desc:"xml" (XML format), "json" (JSON format)-->xml
  </uploadDataFormat>
</MetadataCfg>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

19.2.13.3 Get metadata configuration capabilities of the specified channel

Request URL

GET /ISAPI/Streaming/channels/<channelID>/metadata/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MetadataCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, metadata configurations, attr:version{req, string, protocolVersion}-->
  <MetadataList size="10">
    <!--ro, req, array, metadata information list, subType:object, attr:size{req, int}-->
    <Metadata>
      <!--ro, req, object, metadata information-->
      <type>
        <!--ro, req, enum, event types that support metadata, subType:string, desc:"thermometry" (real-time temperature measurement), "fireDetection" (fire
detection), "shipsDetection" (ship detection), "behaviorAnalysis" (abnormal event detection), "ANPR" (automatic number plate recognition), "personQueue"
(queue management, including people queuing-up detection and waiting time detection), "faceSnap" (face capture), "mixedTargetDetection" (multi-target-
detection), "AIOP" (AIOP detection)-->thermometry
      </type>
      <enable>
        <!--ro, req, bool, whether to enable uploading metadata-->true
      </enable>
    </Metadata>
  </MetadataList>
  <uploadDataFormat opt="xml,json">
    <!--ro, opt, string, message format, attr:opt{req, string}-->test
  </uploadDataFormat>
  <isSupportSubscribeType>
    <!--ro, opt, bool, whether the device supports subscribing to the metadata, desc:this field is returned when the device supports subscribing to the
metadata. By default, the subscription of metadata is according to channels-->true
  </isSupportSubscribeType>
</MetadataCfg>
```

19.2.13.4 Subscribe to metadata RTSP signaling of a specified channel

Request URL

POST /ISAPI/Streaming/channels/<channelID>/metadata/subscribeType?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

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Request Message

```
{
    "type": ["thermometry", "fireDetection", "mixedTargetDetection"]
    /*req, array, event types that support metadata, subType:string, desc:"thermometry"-real-time temperature measurement, "fireDetection"-fire source detection, "shipsDetection"-ship detection, "behaviorAnalysis"-abnormal event detection, "ANPR", "personQueue"-queue management, "faceSnap"-face capture, “mixedTargetDetection”, “AIOP”, “AIOPModelDescription”;*/
}
```

Response Message

```
{
    "rtspURI":  "/ISAPI/Streaming/channels/101?type=personalTrack"
    /*ro, req, string, RTSP URI, range:[1,128], desc:RTSP URI*/
}
```

19.2.14 Video OSD Information Overlay

19.2.14.1 Set the parameters of a specific video input channel of the device

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/image/<pictureID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
pictureID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, picture overlay, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, picture ID, range:[0,64]-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <imageName>
    <!--req, string, picture name, range:[0,63]-->test
  </imageName>
  <positionX>
    <!--opt, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--opt, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <transparentColorEnabled>
    <!--opt, bool, whether to enable transparent color-->true
  </transparentColorEnabled>
  <transparentColor>
    <!--opt, string, transparent color, desc:transparent color-->ffffff
  </transparentColor>
  <imageWidth>
    <!--opt, int, picture width, range:[0,1000]-->1
  </imageWidth>
  <imageHeight>
    <!--opt, int, picture height, range:[0,1000]-->1
  </imageHeight>
</ImageOverlay>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK”, “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML
Content”, “Reboot”-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.14.2 Get the OSD parameters of a specified channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, text overlay, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, req, object, screen size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--ro, opt, object, attributes-->
    <transparent>
      <!--ro, req, bool, whether it is transparent-->true
    </transparent>
    <flashing>
      <!--ro, req, bool, whether it is flickering-->true
    </flashing>
  </attribute>
  <TextOverlayList>
    <!--ro, opt, array, text content list, subType:object-->
    <TextOverlay>
      <!--ro, opt, object, text content-->
      <id>
        <!--ro, req, string, text No.-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionY>
      <displayText>
        <!--ro, req, string, overlay text-->test
      </displayText>
      <isPersistentText>
        <!--ro, opt, bool, whether the overlaid text remains the previous settings after device rebooting, desc:if the value is true or the node is not
transmitted, it indicates to write OSD to flash, and it is not suitable for frequent or fast callings. If the value is false, it indicates not to writer OSD
to flash and the OSD settings will not restore-->true
      </isPersistentText>
```

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```
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object, channel time information overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
    <dateStyle>
      <!--ro, opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY",
"CHR-YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
    </dateStyle>
    <timeStyle>
      <!--ro, opt, enum, time format, subType:string, desc:"12hour", "24hour"-->12hour
    </timeStyle>
    <displayWeek>
      <!--ro, opt, bool, whether to display week information-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object, channel name overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable channel name overlay-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </channelNameOverlay>
  <HumitureOverlay>
    <!--ro, opt, object, temperature and humidity-->
    <enabled>
      <!--ro, req, bool, whether to display temperature and humidity-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </HumitureOverlay>
  <fontSize>
    <!--ro, opt, enum, font size, subType:string, desc:"adaptive", "16*16", "32*32", "48*48", "64*64", "96*96", "128*128"-->96*96
  </fontSize>
  <frontColorMode>
    <!--ro, opt, enum, font color mode, subType:string, desc:"auto", "customize", "outline"-->auto
  </frontColorMode>
  <frontColor>
    <!--ro, opt, string, font color, desc:hexadecimal digit-->000000
  </frontColor>
  <alignment>
    <!--ro, opt, enum, alignment mode, subType:string, desc:"customize" (custom), "alignRight" (right align), "alignLeft" (left align), "GB", "adsorption"
(adsorption mode)-->alignRight
  </alignment>
  <boundary>
    <!--ro, opt, int, the minimum boundary, range:[0,2], dep:and,{$.VideoOverlay.alignment,eq,GB}, desc:unit: number of characters-->1
  </boundary>
  <FloorNumOverlay>
    <!--ro, opt, object, elevator floor No.-->
    <enabled>
      <!--ro, req, bool, whether to display the elevator floor No.-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </FloorNumOverlay>
  <upDownboundary>
    <!--ro, opt, int, range:[0,2]-->1
  </upDownboundary>
  <leftRightboundary>
    <!--ro, opt, int, range:[0,2]-->1
  </leftRightboundary>
  <Angle>
    <!--ro, opt, object, device angle-->
    <TiltAngle>
      <!--ro, opt, object, pitch angle-->
```

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```
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionY>
    </TiltAngle>
  </Angle>
  <GPSOverlay>
    <!--ro, opt, object, GPS information overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </GPSOverlay>
  <WirelessDialOverlay>
    <!--ro, opt, object, wireless dial-up status overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable. If it is enabled, the information such as wireless network type (5G/4G), upstream speed, and downstream speed
will be overlaid-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </WirelessDialOverlay>
  <WirelessDataOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </WirelessDataOverlay>
  <elevatorRunningOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </elevatorRunningOverlay>
</VideoOverlay>
```

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19.2.14.3 Set OSD parameters of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, text overlay, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--req, object, screen size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--opt, object, attributes-->
    <transparent>
      <!--req, bool, whether it is transparent-->true
    </transparent>
    <flashing>
      <!--req, bool, whether it is flickering-->true
    </flashing>
  </attribute>
  <TextOverlayList>
    <!--opt, array, text content list, subType:object-->
    <TextOverlay>
      <!--opt, object, text content-->
      <id>
        <!--req, string, text No.-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionY>
      <displayText>
        <!--req, string, overlay text-->test
      </displayText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--opt, object, channel time information overlay-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
    <dateStyle>
      <!--opt, enum, date, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-
YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
    </dateStyle>
    <timeStyle>
      <!--opt, enum, time format, subType:string, desc:time format-->12hour
    </timeStyle>
    <displayWeek>
      <!--opt, bool, whether to display week information-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--opt, object, channel name overlay-->
    <enabled>
      <!--req, bool, whether to enable channel name overlay-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </channelNameOverlay>
  <fontSize>
    <!--opt, enum, font size, subType:string, desc:"adaptive", "16*16", "32*32", "48*48", "64*64", "96*96", "128*128"-->96*96
  </fontSize>
  <frontColorMode>
    <!--opt, enum, font color mode, subType:string, desc:"auto", "customize", "outline"-->auto
  </frontColorMode>
  <frontColor>
    <!--opt, string, font color, desc:hexadecimal digit-->000000
  </frontColor>
  <alignment>
    <!--opt, enum, alignment mode, subType:string, desc:"customize" (custom), "alignRight" (right align), "alignLeft" (left align), "GB", "adsorption"
(adsorption mode)-->alignRight
```

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```
  </alignment>
  <boundary>
    <!--opt, int, range:[0,2], dep:and,{$.VideoOverlay.alignment,eq,GB}, desc:unit: number of characters-->000000
  </boundary>
  <upDownboundary>
    <!--opt, int, range:[0,2]-->1
  </upDownboundary>
  <leftRightboundary>
    <!--opt, int, range:[0,2]-->1
  </leftRightboundary>
</VideoOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, it is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

19.2.14.4 Get the OSD capability of a specific video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, see details in the message of XML_BatteryPowerOverlay, attr:version{req, string, protocolVersion}-->
  <TextOverlayList size="8">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <TextOverlay>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, listening host ID-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--ro, req, int, range:[0,704]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,576]-->1
      </positionY>
      <displayText min="0" max="64">
        <!--ro, req, string, attr:min{req, int},max{req, int}-->test
      </displayText>
      <isPersistentText>
        <!--ro, opt, bool-->true
      </isPersistentText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
```

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```
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <positionX>
      <!--ro, req, int, range:[0,704]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,576]-->1
    </positionY>
    <dateStyle opt="YYYY-MM-DD,MM-DD-YYYY,DD-MM-YYYY,CHR-YYYY-MM-DD,CHR-MM-DD-YYYY,CHR-DD-MM-YYYY,CHR-YYYY/MM/DD,CHR-MM/DD/YYYY,CHR-DD/MM/YYYY">
      <!--ro, opt, string, attr:opt{req, string}-->YYYY-MM-DD
    </dateStyle>
    <timeStyle opt="12hour,24hour">
      <!--ro, opt, string, time format, attr:opt{req, string}-->12hour
    </timeStyle>
    <displayWeek opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <positionX>
      <!--ro, opt, int, range:[0,1920]-->1
    </positionX>
    <positionY>
      <!--ro, opt, int, range:[0,1920]-->1
    </positionY>
    <name min="0" max="128">
      <!--ro, opt, string, range:[0,128], attr:min{req, int},max{req, int}-->Camera 01
    </name>
  </channelNameOverlay>
  <HumitureOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <positionX min="0" max="1920">
      <!--ro, opt, int, range:[0,1920], attr:min{req, int},max{req, int}-->1
    </positionX>
    <positionY min="0" max="1920">
      <!--ro, opt, int, range:[0,1920], attr:min{req, int},max{req, int}-->1
    </positionY>
  </HumitureOverlay>
  <fontSize opt="16*16,32*32,48*48,64*64,80*80,96*96,128*128,adaptive">
    <!--ro, opt, string, font size, attr:opt{req, string}-->adaptive
  </fontSize>
  <frontColorMode opt="auto,customize,outline" def="auto">
    <!--ro, opt, string, font color mode, attr:opt{req, string},def{opt, string}-->outline
  </frontColorMode>
  <frontColor>
    <!--ro, opt, string, font color, desc:font color-->000000
  </frontColor>
  <alignment opt="customize,alignRight,alignLeft,GB,GB2025,allRight,allLeft,adsorption">
    <!--ro, opt, string, alignment mode, attr:opt{req, string}-->customize
  </alignment>
  <lowRightTextNumber min="1" max="6">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </lowRightTextNumber>
  <lowLeftTextNumber min="7" max="8">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->7
  </lowLeftTextNumber>
  <boundary min="0" max="2">
    <!--ro, opt, int, dep:and,{$.VideoOverlay.alignment,eq,GB}, attr:min{req, int},max{req, int}-->1
  </boundary>
  <TextOverlaySize min="0" max="44">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </TextOverlaySize>
  <FloorNumOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <positionX>
      <!--ro, req, int, range:[0,704]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,576]-->1
    </positionY>
  </FloorNumOverlay>
  <upDownboundary min="0" max="5">
    <!--ro, opt, int, attr:min{opt, int},max{opt, int}-->1
  </upDownboundary>
  <leftRightboundary min="0" max="5">
    <!--ro, opt, int, attr:min{opt, int},max{opt, int}-->1
  </leftRightboundary>
  <Angle>
    <!--ro, opt, object-->
    <TiltAngle>
      <!--ro, opt, object-->
      <enabled opt="true,false">
```

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```
        <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
      </enabled>
      <positionX min="0" max="704">
        <!--ro, req, int, range:[0,704], attr:min{req, int},max{req, int}-->1
      </positionX>
      <positionY min="0" max="576">
        <!--ro, req, int, range:[0,576], attr:min{req, int},max{req, int}-->1
      </positionY>
    </TiltAngle>
  </Angle>
  <GPSOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <positionX min="0" max="704">
      <!--ro, req, int, range:[0,704], attr:min{req, int},max{req, int}-->1
    </positionX>
    <positionY min="0" max="576">
      <!--ro, req, int, range:[0,576], attr:min{req, int},max{req, int}-->1
    </positionY>
  </GPSOverlay>
  <WirelessDialOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <positionX min="0" max="704">
      <!--ro, req, int, range:[0,704], attr:min{req, int},max{req, int}-->1
    </positionX>
    <positionY min="0" max="576">
      <!--ro, req, int, range:[0,576], attr:min{req, int},max{req, int}-->1
    </positionY>
  </WirelessDialOverlay>
  <WirelessDataOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <positionX min="0" max="704">
      <!--ro, req, int, range:[0,704], attr:min{req, int},max{req, int}-->1
    </positionX>
    <positionY min="0" max="576">
      <!--ro, req, int, range:[0,576], attr:min{req, int},max{req, int}-->1
    </positionY>
  </WirelessDataOverlay>
  <elevatorRunningOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <positionX min="0" max="704">
      <!--ro, req, int, range:[0,704], attr:min{req, int},max{req, int}-->1
    </positionX>
    <positionY min="0" max="576">
      <!--ro, req, int, range:[0,576], attr:min{req, int},max{req, int}-->1
    </positionY>
  </elevatorRunningOverlay>
  <elevatorOpenCloseDoorOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <positionX>
      <!--ro, req, int, range:[0,704]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,576]-->1
    </positionY>
  </elevatorOpenCloseDoorOverlay>
  <isSupportLensMultipleDisplayCfg>
    <!--ro, opt, bool-->true
  </isSupportLensMultipleDisplayCfg>
</VideoOverlay>
```

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

19.2.14.5 Get the channel name overlay parameters of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, channel name overlay parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable channel name overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate-->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate-->1
  </positionY>
  <name>
    <!--ro, opt, string, channel name, range:[0,128]-->Camera 01
  </name>
  <OverlayRegion>
    <!--ro, opt, object, coordinates of overlaid string frame (the origin is the is the upper-left corner of the screen)-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</channelNameOverlay>
```

19.2.14.6 Set the channel name overlay parameters for a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable channel name overlay-->true
  </enabled>
  <positionX>
    <!--req, int, X-coordinate-->1
  </positionX>
  <positionY>
    <!--req, int, Y-coordinate-->1
  </positionY>
  <name>
    <!--opt, string, channel name, range:[0,32]-->Camera 01
  </name>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</channelNameOverlay>
```

Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.14.7 Set the parameters of date and time information overlay for a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, configurations of date and time information overlay of video input channels, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable date and time information overlay-->true
  </enabled>
  <positionX>
    <!--req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <dateStyle>
    <!--opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-
YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--opt, enum, time format, subType:string, desc:"12hour" (12-hour system), "24hour" (24-hour system)-->12hour
  </timeStyle>
  <displayWeek>
    <!--opt, bool, whether to display week information-->true
  </displayWeek>
  <displayDate>
    <!--opt, bool, whether to display date-->true
  </displayDate>
  <displayTime>
    <!--opt, bool, whether to display time-->true
  </displayTime>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</DateTimeOverlay>
```

Response Message

VERSION AND SECURITY SYSTEM CO.,LTD.
dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.14.8 Get the parameters of date and time information overlay of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable date and time overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate of the upper-left corner of the overlay area, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate of the upper-left corner of the overlay area, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <dateStyle>
    <!--ro, opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY"--
>YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--ro, opt, enum, time format, subType:string, desc:time format-->12hour
  </timeStyle>
  <displayWeek>
    <!--ro, opt, bool, whether to display week information-->true
  </displayWeek>
  <displayDate>
    <!--ro, opt, bool-->true
  </displayDate>
  <displayTime>
    <!--ro, opt, bool-->true
  </displayTime>
  <OverlayRegion>
    <!--ro, opt, object-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</DateTimeOverlay>
```

19.2.14.9 Delete the text overlay parameters of a specific video input channel

Request URL

VISION AND SECURITY SYSTEM CO.,LTD. dongjin@gmail.com

DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.2.14.10 Set the text overlay parameters of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, see details in the message of XML_TextOverlay, subType:object, attr:version{opt, string, protocolVersion}-->
  <TextOverlay>
    <!--opt, object-->
    <id>
      <!--req, string, text No.-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <positionX>
      <!--req, int, X-coordinate-->1
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate-->1
    </positionY>
    <displayText>
      <!--req, string, overlay character-->test
    </displayText>
    <OverlayRegion>
      <!--opt, object-->
      <width>
        <!--req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </TextOverlay>
</TextOverlayList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.14.11 Get parameters of text information overlay of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dongjin@mail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, text information overlay, subType:object, attr:version{req, string, protocolVersion}-->
  <TextOverlay>
    <!--ro, opt, object, text overlay configuration-->
    <id>
      <!--ro, req, string, text No.-->test
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the
screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the
screen-->1
    </positionY>
    <displayText>
      <!--ro, req, string, overlaid text-->test
    </displayText>
    <OverlayRegion>
      <!--ro, opt, object-->
      <width>
        <!--ro, req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </TextOverlay>
</TextOverlayList>
```

19.2.14.12 Add a text to overlay on the video for a specific video input channel

Request URL

POST /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, text overlay settings, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <positionX>
    <!--req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--req, int, the origin is the lower-left corner of the screen, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, text overlay-->test
  </displayText>
</TextOverlay>
```

Response Message

doc.jiin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.2.14.13 Set the parameters of a specific piece of text overlay information of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, text No.-->test
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <positionX>
    <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, overlay character-->test
  </displayText>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</TextOverlay>
```

Response Message

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.14.14 Get the text overlay parameters of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
textID	string	Text overlay ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, text overlay configuration, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, text No.-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the screen-
->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the screen-
->1
  </positionY>
  <displayText>
    <!--ro, req, string, overlaid text-->test
  </displayText>
  <OverlayRegion>
    <!--ro, opt, object-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</TextOverlay>
```

19.2.14.15 Delete a specified piece of text overlay of a video input channel

Request URL

DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.
textID	string	Text overlay ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.2.14.16 Get the OSD language

Request URL

GET /ISAPI/System/Video/inputs/OSDLanguage

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Language xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, OSD language, attr:version{req, string, protocolVersion}-->
  <type>
    <!--ro, req, enum, language type, subType:string, desc:"GBK", "EUC-KR"-->GBK
  </type>
</Language>
```

19.2.14.17 Set OSD language parameters

Request URL

PUT /ISAPI/System/Video/inputs/OSDLanguage

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Language xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <type>
    <!--req, enum, "GBK,EUC-KR",def="GBK", subType:string, desc:"GBK", "EUC-KR"-->GBK
  </type>
</Language>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.15 Video Privacy Mask

19.2.15.1 Set arming schedules of privacy mask for all channels

Request URL

PUT /ISAPI/Event/schedules/privacyMask

Query Parameter

None

Request Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, arming schedule list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming schedule configurations-->
    <id>
      <!--req, string, No.-->test
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string, desc:"privacyMask" (privacy mask)-->privacyMask
    </eventType>
    <inputIOPortID>
      <!--opt, string, input port No.-->1
    </inputIOPortID>
    <outputIOPortID>
      <!--opt, string, output port No.-->1
    </outputIOPortID>
    <videoInputChannelID>
      <!--opt, string, video input channel No.-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, list of time periods, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, time period-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, start and end time-->
          <beginTime>
            <!--req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, schedule list of holidays, subType:object-->
      <TimeBlock>
        <!--opt, object, time period-->
        <TimeRange>
          <!--req, object, start and end time-->
          <beginTime>
            <!--req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</PrivacyMaskScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.15.2 Get arming schedules of privacy mask for all channels

Request URL

GET /ISAPI/Event/schedules/privacyMask

Query Parameter

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, arming schedule list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule configurations-->
    <id>
      <!--ro, req, string, No.-->test
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string, desc:"privacyMask" (privacy mask)-->privacyMask
    </eventType>
    <inputIOPortID>
      <!--ro, opt, string, input port No.-->1
    </inputIOPortID>
    <outputIOPortID>
      <!--ro, opt, string, output port No.-->1
    </outputIOPortID>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel No.-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, list of time periods, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, time period-->
        <dayOfWeek>
          <!--ro, opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)--
>1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, start and end time-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, schedule list of holidays, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, time period-->
        <TimeRange>
          <!--ro, req, object, start and end time-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</PrivacyMaskScheduleList>
```

19.2.15.3 Get the arming schedule of privacy mask for the specified channel

Request URL

GET /ISAPI/Event/schedules/privacyMask/privacyMask-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, arming schedule configurations, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, No.-->test
  </id>
  <eventType>
    <!--ro, opt, enum, event type, subType:string, desc:"privacyMask" (privacy mask)-->privacyMask
  </eventType>
  <inputIOPortID>
    <!--ro, opt, string, input port No.-->1
  </inputIOPortID>
  <outputIOPortID>
    <!--ro, opt, string, output port No.-->1
  </outputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel No.-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, list of time periods, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object, time period-->
      <dayOfWeek>
        <!--ro, opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object, start and end time-->
        <beginTime>
          <!--ro, req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, schedule list of holidays, subType:object-->
    <TimeBlock>
      <!--ro, opt, object, time period-->
      <TimeRange>
        <!--ro, req, object, start and end time-->
        <beginTime>
          <!--ro, req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.2.15.4 Set the arming schedule of privacy mask for the specified channel

Request URL

PUT /ISAPI/Event/schedules/privacyMask/privacyMask-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

VISION AND SECURITY SYSTEM CO.,LTD.
dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, arming schedule configurations, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, No.-->test
  </id>
  <eventType>
    <!--opt, enum, event type, subType:string, desc:"privacyMask" (privacy mask)-->privacyMask
  </eventType>
  <inputIOPortID>
    <!--opt, string, input port No.-->1
  </inputIOPortID>
  <outputIOPortID>
    <!--opt, string, output port No.-->1
  </outputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel No.-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, list of time periods, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object, time period-->
      <dayOfWeek>
        <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object, start and end time-->
        <beginTime>
          <!--req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, schedule list of holidays, subType:object-->
    <TimeBlock>
      <!--opt, object, time period-->
      <TimeRange>
        <!--req, object, start and end time-->
        <beginTime>
          <!--req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.15.5 Get the privacy mask configuration capability of a specified channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/privacyMaskCap

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the privacy mask configuration capability, attr:version{req, string, protocolVersion}-->
  <maskTypeDscriptor opt="gray,red,yellow,blue,orange,green,transparent,half-transparent,mosaic,black,matchman,groundGlass,custom">
    <!--ro, opt, enum, mask type, subType:string, attr:opt{req, string}, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-
transparent", "mosaic", "black"-->gray
  </maskTypeDscriptor>
  <privacymaskName min="0" max="16">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </privacymaskName>
  <regionNum min="0" max="8">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->8
  </regionNum>
  <minZoomdoorlimit>
    <!--ro, opt, int, the minimum zooming in limit-->1
  </minZoomdoorlimit>
  <maxZoomdoorlimit>
    <!--ro, opt, int, the maximum zooming in limit-->1
  </maxZoomdoorlimit>
  <videoPrivacyType opt="privacyMask,privacyCover">
    <!--ro, opt, enum, privacy type, subType:string, attr:opt{req, string}, desc:"privacyMask"-privacy mask, "privacyCover"-video tampering-->privacyMask
  </videoPrivacyType>
  <supportRegionType>
    <!--ro, opt, enum, the default type of area drawing, subType:string, desc:"rectangle". If this node is not supported, the area is rectangular for
privacy mask, and is polygon for video tampering by default-->rectangle
  </supportRegionType>
  <mediaObjectEncryption>
    <!--ro, opt, object, encryption of the private target in media data, desc:if it is enabled, the main steam of corresponding area will be covered, but
the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?format=json). You can use the key to decrypt the
encrypted area if this node is not returned, it indicates that device does not support this function-->
    <enabled opt="true,false" def="false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string},def{req, string}-->true
    </enabled>
    <maskType opt="mosaic" def="mosaic">
      <!--ro, opt, string, mask type, attr:opt{req, string},def{req, string}, desc:"mosaic"-->mosaic
    </maskType>
  </mediaObjectEncryption>
  <polygonCoordinatesNum>
    <!--ro, opt, int, dep:and,{$.PrivacyMaskCap.supportRegionType,eq,polygon}-->1
  </polygonCoordinatesNum>
  <regionRangeType opt="fullScreen,region">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->region
  </regionRangeType>
  <privacyMaskRegionListNum>
    <!--ro, opt, int, range:[0,8]-->4
  </privacyMaskRegionListNum>
  <switch opt="local,remote">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->local
  </switch>
  <isSupportUpdateBackgroundPicture>
    <!--ro, opt, bool-->true
  </isSupportUpdateBackgroundPicture>
  <isSupportHumanBodyPrivacyMaskParams>
    <!--ro, opt, bool-->true
  </isSupportHumanBodyPrivacyMaskParams>
  <isSupportHeadShoulderPrivacyMaskParams>
    <!--ro, opt, bool-->true
  </isSupportHeadShoulderPrivacyMaskParams>
  <isSupportPlatePrivacyMaskParams>
    <!--ro, opt, bool-->true
  </isSupportPlatePrivacyMaskParams>
  <isSupportImportPrivacyMaskPicture>
    <!--ro, opt, bool-->true
  </isSupportImportPrivacyMaskPicture>
</PrivacyMaskCap>
```

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19.2.15.6 Get the parameters of all privacy mask areas of a specified channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, privacy mask area configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <PrivacyMaskRegion>
    <!--ro, opt, object, configurations of a privacy mask area-->
    <id>
      <!--ro, req, int, area ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the area-->true
    </enabled>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,576]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <privacymaskName>
      <!--ro, opt, string, privacy mask name-->test
    </privacymaskName>
    <maskType>
      <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
    </maskType>
    <zoomdoorlimit>
      <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private region will be displayed, range:[10,1000]-->1
    </zoomdoorlimit>
    <mediaObjectEncryption>
      <!--ro, opt, object, encryption of the private target in media data, desc:it is disabled when this field is not returned. If it is enabled, the main
steam of corresponding area will be covered, but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?
format=json). You can use the key to decrypt the encrypted area-->
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <maskType>
        <!--ro, opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
      </maskType>
    </mediaObjectEncryption>
    <regionRangeType>
      <!--ro, opt, enum, subType:string-->region
    </regionRangeType>
  </PrivacyMaskRegion>
</PrivacyMaskRegionList>
```

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19.2.15.7 Set the parameters of all privacy mask areas of a specified channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, privacy mask area configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <PrivacyMaskRegion>
    <!--opt, object, configurations of a privacy mask area-->
    <id>
      <!--req, int, area ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the area-->true
    </enabled>
    <RegionCoordinatesList>
      <!--req, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <privacymaskName>
      <!--opt, string, privacy mask name-->test
    </privacymaskName>
    <maskType>
      <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
    </maskType>
    <zoomdoorlimit>
      <!--opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
    </zoomdoorlimit>
    <mediaObjectEncryption>
      <!--opt, object, encryption of the private target in media data, desc:it is disabled when this field is not configured. If it is enabled, the main
steam of corresponding area will be covered, but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?
format=json). You can use the key to decrypt the encrypted area-->
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <maskType>
        <!--opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
      </maskType>
    </mediaObjectEncryption>
  </PrivacyMaskRegion>
</PrivacyMaskRegionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.2.15.8 Delete parameters of all privacy mask areas of the specified channel

Request URL

DELETE /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.2.15.9 Get parameters of a privacy mask area of a specified channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD.
dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, configurations of a privacy mask area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, region ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the area-->true
  </enabled>
  <RegionCoordinatesList>
    <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,704]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,576]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <privacymaskName>
    <!--ro, opt, string, privacy mask name-->test
  </privacymaskName>
  <maskType>
    <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
  </maskType>
  <zoomdoorlimit>
    <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private region will be displayed, range:[10,1000]-->1
  </zoomdoorlimit>
  <mediaObjectEncryption>
    <!--ro, opt, object, encryption of the private target in media data, desc:If it is enabled, the main steam of corresponding region will be covered, but
the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?format=json). You can use the key to decrypt the
encrypted region.-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <maskType>
      <!--ro, opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
    </maskType>
  </mediaObjectEncryption>
  <regionRangeType>
    <!--ro, opt, enum, subType:string-->region
  </regionRangeType>
</PrivacyMaskRegion>
```

19.2.15.10 Delete parameters of a privacy mask area of the specified channel

Request URL

DELETE /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

None

Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.2.15.11 Set parameters of a privacy mask area of a specified channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, configurations of a privacy mask area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, area ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the area-->true
  </enabled>
  <RegionCoordinatesList>
    <!--req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,704]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,576]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <privacymaskName>
    <!--opt, string, privacy mask name-->test
  </privacymaskName>
  <maskType>
    <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic", "black"-
->gray
  </maskType>
  <zoomdoorlimit>
    <!--opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
  </zoomdoorlimit>
  <mediaObjectEncryption>
    <!--opt, object, encryption of the private target in media data, desc:it is disabled when this parameter is not configured if it is enabled, the main
steam of corresponding region will be covered, but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?
format=json). You can use the key to decrypt the encrypted region-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <maskType>
      <!--opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
    </maskType>
  </mediaObjectEncryption>
</PrivacyMaskRegion>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

VISION AND SECURITY SYSTEM CO., LTD. dojojin@gmail.com

19.2.15.12 Get the privacy mask parameters of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask?switch=<switchType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
switchType	string	Analysis unit switch: "local" (analysis by local device, e.g., executed locally by NVR), "remote" (the device transmits data to sub-device for analysis, e.g., from the NVR to the network camera).

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, privacy mask configurations, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the privacy mask-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--ro, opt, array, list of video tampering region(s), subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--ro, opt, object, configurations of a privacy mask region-->
      <id>
        <!--ro, req, int, region ID-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable the region-->true
      </enabled>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--ro, opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private region will be displayed, range:[10,1000]-->1
      </zoomdoorlimit>
      <mediaObjectEncryption>
        <!--ro, opt, object, encryption of the private target in media data, desc:If it is enabled, the main steam of corresponding region will be covered,
but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?format=json). You can use the key to decrypt the
encrypted region.-->
        <enabled>
          <!--ro, req, bool, whether to enable-->true
        </enabled>
        <maskType>
          <!--ro, opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
        </maskType>
      </mediaObjectEncryption>
      <regionRangeType>
        <!--ro, opt, enum, subType:string-->region
      </regionRangeType>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--ro, opt, enum, region type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

19.2.15.13 Set the privacy mask parameters of a specified channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask?switch=<switchType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
switchType	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, privacy mask configurations, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable the privacy mask-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--opt, array, list of video privacy mask area(s), subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--opt, object, configurations of a privacy mask area-->
      <id>
        <!--req, int, area ID-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable the area-->true
      </enabled>
      <RegionCoordinatesList>
        <!--req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--opt, int, zooming value limit range, range:[10,1000]-->1
      </zoomdoorlimit>
      <mediaObjectEncryption>
        <!--opt, object, encryption of the private target in media data, desc:it is disabled when this field is not configured. If it is enabled, the main
steam of corresponding area will be covered, but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?
format=json). You can use the key to decrypt the encrypted area-->
        <enabled>
          <!--req, bool, whether to enable-->true
        </enabled>
        <maskType>
          <!--opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
        </maskType>
      </mediaObjectEncryption>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--opt, enum, area type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.16 Wide Dynamic Range / Backlight Compensation / Highlight Compensation

19.2.16.1 Set the BLC (Backlight Compensation) parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/BLC

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, BLC area, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable BLC-->true
  </enabled>
  <BLCMode>
    <!--opt, enum, BLC mode, subType:string, desc:"UP", "DOWN", "LEFT", "RIGHT", "CENTER", "MULTI-AREA", "Region", "AUTO"-->UP
  </BLCMode>
  <BLCLevel>
    <!--opt, int, BLC level-->1
  </BLCLevel>
  <BLCRegionList>
    <!--opt, array, List of BLC areas, subType:object-->
    <BLCRegion>
      <!--opt, object, BLC area-->
      <id>
        <!--req, int, area No.-->1
      </id>
      <RegionCoordinatesList>
        <!--req, array, coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.16.2 Get the BLC (Backlight Compensation) parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/BLC

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, BLC mode: "UP, DOWN, LEFT, RIGHT, CENTER, MULTI-AREA, Region", attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <BLCMode>
    <!--ro, opt, enum, BLC mode, subType:string, desc:"UP", "DOWN", "LEFT", "RIGHT", "CENTER", "MULTI-AREA", "Region", "AUTO"-->UP
  </BLCMode>
  <BLCLevel>
    <!--ro, opt, int-->1
  </BLCLevel>
  <BLCRegionList>
    <!--ro, opt, array, subType:object-->
    <BLCRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, req, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
  <switchMode>
    <!--ro, opt, enum, subType:string-->time
  </switchMode>
  <timeMode>
    <!--ro, opt, object, dep:and,{$.BLC.switchMode,eq,time}-->
    <timeRangeList>
      <!--ro, req, array, subType:object, range:[0,1]-->
      <timeRange>
        <!--ro, req, object-->
        <startTime>
          <!--ro, req, time-->08:00:00
        </startTime>
        <endTime>
          <!--ro, req, time-->18:00:00
        </endTime>
      </timeRange>
    </timeRangeList>
  </timeMode>
</BLC>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

19.2.16.3 Get the backlight compensation (in day mode) parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/BLC/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <BLCMode>
    <!--ro, opt, enum, subType:string-->UP
  </BLCMode>
  <BLCLevel>
    <!--ro, opt, int-->1
  </BLCLevel>
  <BLCRegionList>
    <!--ro, opt, array, subType:object-->
    <BLCRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, req, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

19.2.16.4 Set the backlight compensation (in day mode) parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/BLC/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <BLCMode>
    <!--opt, enum, subType:string-->UP
  </BLCMode>
  <BLCLevel>
    <!--opt, int-->1
  </BLCLevel>
  <BLCRegionList>
    <!--opt, array, subType:object-->
    <BLCRegion>
      <!--opt, object-->
      <id>
        <!--req, int-->1
      </id>
      <RegionCoordinatesList>
        <!--req, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--opt, object-->
          <positionX>
            <!--req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.16.5 Get the backlight compensation (in night mode) parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/BLC/night

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojo_jia@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <BLCMode>
    <!--ro, opt, enum, subType:string-->UP
  </BLCMode>
  <BLCLevel>
    <!--ro, opt, int-->1
  </BLCLevel>
  <BLCRegionList>
    <!--ro, opt, array, subType:object-->
    <BLCRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, req, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

19.2.16.6 Set the backlight compensation (in night mode) parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/BLC/night

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

dejojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <BLCMode>
    <!--opt, enum, subType:string-->UP
  </BLCMode>
  <BLCLevel>
    <!--opt, int-->1
  </BLCLevel>
  <BLCRegionList>
    <!--opt, array, subType:object-->
    <BLCRegion>
      <!--opt, object-->
      <id>
        <!--req, int-->1
      </id>
      <RegionCoordinatesList>
        <!--req, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--opt, object-->
          <positionX>
            <!--req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.16.7 Get the highlight compensation (in auto mode) parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/HLC

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojo jia@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<HLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <HLCMode>
    <!--ro, opt, enum, subType:string-->close
  </HLCMode>
  <HLCLevel>
    <!--ro, opt, int-->1
  </HLCLevel>
  <timePlanList>
    <!--ro, opt, array, subType:object, dep:and,{$.HLC.HLCMode,eq,timing}-->
    <startTime>
      <!--ro, req, time-->08:00:00
    </startTime>
    <endTime>
      <!--ro, req, time-->10:00:00
    </endTime>
  </timePlanList>
</HLC>
```

19.2.16.8 Set the highlight compensation (in auto mode) parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/HLC

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <HLCMode>
    <!--opt, enum, subType:string-->[close
  </HLCMode>
  <HLCLevel>
    <!--opt, int-->1
  </HLCLevel>
</HLC>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.2.16.9 Set the wide dynamic range (WDR) parameters in auto mode of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/WDR

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Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WDR xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->auto
  </mode>
  <WDRLevel>
    <!--opt, int, range:[1,100]-->50
  </WDRLevel>
  <WDRContrastLevel>
    <!--opt, int-->1
  </WDRContrastLevel>
  <WDRLevel1>
    <!--opt, int-->1
  </WDRLevel1>
</WDR>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

19.2.16.10 Get the WDR (Wide Dynamic Range) parameters in auto mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/WDR

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WDR xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, WDR mode, subType:string, desc:WDR mode-->auto
  </mode>
  <WDRLevel>
    <!--ro, opt, int, WDR level, range:[0,100], desc:WDR level-->50
  </WDRLevel>
  <WDRContrastLevel>
    <!--ro, opt, int-->1
  </WDRContrastLevel>
  <WDRLevel1>
    <!--ro, opt, int-->1
  </WDRLevel1>
  <timeMode>
    <!--ro, opt, object, dep:and,{$.WDR.mode,eq,time}-->
    <timeRangeList>
      <!--ro, req, array, subType:object, range:[0,1]-->
      <timeRange>
        <!--ro, req, object-->
        <startTime>
          <!--ro, req, time-->08:00:00
        </startTime>
        <endTime>
          <!--ro, req, time-->18:00:00
        </endTime>
      </timeRange>
    </timeRangeList>
  </timeMode>
  <WDRStreakInhibition>
    <!--ro, opt, bool-->false
  </WDRStreakInhibition>
  <WDRCloseToOpenFilterLevel>
    <!--ro, opt, int, range:[1,9]-->5
  </WDRCloseToOpenFilterLevel>
  <WDROpenToCloseFilterLevel>
    <!--ro, opt, int, range:[1,9]-->5
  </WDROpenToCloseFilterLevel>
</WDR>
```

19.3 Video Management

19.3.1 Manual Recording

19.3.1.1 Start the manual recording

Request URL

PUT /ISAPI/ContentMgmt/record/control/manual/start/tracks/<trackStreamID>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.3.1.2 Stop the manual recording

Request URL

PUT /ISAPI/ContentMgmt/record/control/manual/stop/tracks/<trackStreamID>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.3.2 Scheduled Recording

19.3.2.1 Add a recording schedule

Request URL

POST /ISAPI/ContentMgmt/record/tracks

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Track xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, search all recording schedules, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, channel corresponding to the stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main stream
of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type should
be subject to that of SrcUrl-->201
  </id>
  <Channel>
    <!--req, int, channel No.-->1
  </Channel>
  <Enable>
    <!--req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
  </Enable>
  <Description>
    <!--req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20
fps,bitrate=6000 kbps
  </Description>
  <TrackGUID>
    <!--req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
  </TrackGUID>
  <Size>
    <!--opt, int, recording size-->1
  </Size>
  <Duration min="0" max="750">
    <!--opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15 hours; 0
indicates the video is valid until it is recurrently overwritten-->P10DT15H
  </Duration>
  <DefaultRecordingMode>
    <!--req, enum, default record type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture
alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection pictures),
"AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording), "temperatureIntervalMeasurement"
```

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```
(temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm" (panic alarm),
"zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from vehicl entering to
vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed
detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
  </DefaultRecordingMode>
  <LoopEnable>
    <!--opt, bool, whether to support recurrently overwriting-->true
  </LoopEnable>
  <SrcDescriptor>
    <!--opt, object, video source description-->
    <SrcGUID>
      <!--req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
    </SrcGUID>
    <SrcChannel>
      <!--req, int, source channel-->1
    </SrcChannel>
    <StreamHint>
      <!--req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
    </StreamHint>
    <SrcDriver>
      <!--req, string, stream executable driver name-->RTP/RTSP
    </SrcDriver>
    <SrcType>
      <!--opt, string, source type-->mp4 video
    </SrcType>
    <SrcUrl>
      <!--req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID> refers
to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main stream
of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
    </SrcUrl>
    <SrcUrlMethods>
      <!--req, enum, methods supported by the source, subType:string, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
    </SrcUrlMethods>
    <SrcLogin>
      <!--req, string, login source password-->admin:admin
    </SrcLogin>
  </SrcDescriptor>
  <TrackSchedule>
    <!--opt, object, recording schedule-->
    <ScheduleBlockList>
      <!--opt, array, recording schedule list, subType:object-->
      <ScheduleBlock>
        <!--opt, object, recording schedule module-->
        <ScheduleBlockGUID>
          <!--opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
        </ScheduleBlockGUID>
        <ScheduleBlockType>
          <!--req, string, schedule type-->/ISAPI/recording/schedule/default
        </ScheduleBlockType>
        <ScheduleAction>
          <!--opt, array, schedule, subType:object, desc:N/A-->
          <id>
            <!--req, int, ID-->1
          </id>
          <ScheduleActionStartTime>
            <!--opt, object, scheduled start time-->
            <DayOfWeek>
              <!--req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionStartTime>
          <ScheduleActionEndTime>
            <!--opt, object, scheduled end time-->
            <DayOfWeek>
              <!--req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionEndTime>
          <ScheduleDSTEnable>
            <!--req, bool, whether to enable Daylight Saving Time (DST)-->true
          </ScheduleDSTEnable>
          <Description>
            <!--req, string, Description-->PreMorning (Midnight to 8am,local time)
          </Description>
          <Actions>
            <!--opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
            <Record>
              <!--opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Record>
            <Log>
              <!--opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Log>
            <SaveImg>
              <!--opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </SaveImg>
            <ActionRecordingMode>
              <!--req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures),
"ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "COMMAND" (recording by commands), "SMART" (smart recording), "thermalVehicleDetection" (thermal imaging vehicle detection),
```

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```
"fishingShipDetection" (fishing ship detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"AllEvent" (all events), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "temperatureIntervalMeasurement" (temperature range
detection), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay
detection), "failDown" (getting up or falling down detection)-->CMR
            </ActionRecordingMode>
            <PreRecordTimeSeconds>
              <!--opt, int, pre-record duration, unit:s, desc:from 0 to 10 seconds; default: 5 seconds-->5
            </PreRecordTimeSeconds>
            <PostRecordTimeSeconds>
              <!--opt, int, post-record duration, unit:s, desc:from 0 to 20 seconds; default: 5 seconds-->5
            </PostRecordTimeSeconds>
          </Actions>
        </ScheduleAction>
      </ScheduleBlock>
    </ScheduleBlockList>
  </TrackSchedule>
  <CustomExtensionList>
    <!--opt, array, custom extension list, subType:object-->
    <CustomExtension>
      <!--opt, object, custom extension-->
      <CustomExtensionName>
        <!--opt, string, custom extension name-->www.xxx.com/RaCM/trackExt/ver10
      </CustomExtensionName>
      <enableSchedule>
        <!--opt, bool, whether to enable recording schedule configuration-->true
      </enableSchedule>
      <SaveAudio>
        <!--opt, bool, whether to enable audio recording-->true
      </SaveAudio>
      <PreRecordTimeSeconds>
        <!--opt, int, pre-record, unit:s-->1
      </PreRecordTimeSeconds>
      <PostRecordTimeSeconds>
        <!--opt, int, post-record, unit:s-->1
      </PostRecordTimeSeconds>
      <HolidaySchedule>
        <!--opt, object, holiday schedule-->
        <ScheduleBlock>
          <!--opt, object, schedule module-->
          <ScheduleBlockGUID>
            <!--req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
          </ScheduleBlockGUID>
          <ScheduleBlockType>
            <!--opt, string, schedule module type-->www.xxx.com/racm/schedule/ver10
          </ScheduleBlockType>
        </ScheduleBlock>
      </HolidaySchedule>
    </CustomExtension>
  </CustomExtensionList>
  <IntelligentRecord>
    <!--opt, bool, whether to enable VCA recording function-->false
  </IntelligentRecord>
  <delayTime>
    <!--opt, int, capture delay time, unit:s-->1
  </delayTime>
</Track>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

19.3.2.2 Get all parameters of a recording schedule

Request URL

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GET /ISAPI/ContentMgmt/record/tracks

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TrackList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, get all parameters of a recording schedule, subType:object, attr:version{req, string, protocolVersion}-->
  <Track>
    <!--ro, opt, object, recording schedule configuration-->
    <id>
      <!--ro, opt, int, channel number corresponding to stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main stream of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type should be subject to that of SrcUrl-->201
    </id>
    <Channel>
      <!--ro, req, int, channel No.-->1
    </Channel>
    <Enable>
      <!--ro, req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
    </Enable>
    <Description>
      <!--ro, req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20 fps,bitrate=6000 kbps
    </Description>
    <TrackGUID>
      <!--ro, req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
    </TrackGUID>
    <Size>
      <!--ro, opt, int, recording size-->1
    </Size>
    <Duration min="0" max="750">
      <!--ro, opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15 hours; 0 indicates the video is valid until it is recurrently overwritten-->P10DT15H
    </Duration>
    <DefaultRecordingMode>
      <!--ro, req, enum, default record type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection pictures), "AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording), "temperatureIntervalMeasurement" (temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from vehicl entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay detection), "fallDown" (getting up or falling down detection)-->CMR
    </DefaultRecordingMode>
    <LoopEnable>
      <!--ro, opt, bool, whether to support recurrently overwriting-->true
    </LoopEnable>
    <SrcDescriptor>
      <!--ro, opt, object, video source description-->
      <SrcGUID>
        <!--ro, req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
      </SrcGUID>
      <SrcChannel>
        <!--ro, req, int, source channel-->1
      </SrcChannel>
      <StreamHint>
        <!--ro, req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
      </StreamHint>
      <SrcDriver>
        <!--ro, req, string, stream executable driver name-->RTP/RTSP
      </SrcDriver>
      <SrcType>
        <!--ro, opt, string, source type-->mp4 video
      </SrcType>
      <SrcUrl>
        <!--ro, req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID> refers to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main stream of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
      </SrcUrl>
      <SrcUrlMethods>
        <!--ro, req, enum, methods supported by the source, subType:string, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
      </SrcUrlMethods>
      <SrcLogin>
        <!--ro, req, string, login source password-->admin:admin
      </SrcLogin>
    </SrcDescriptor>
    <TrackSchedule>
      <!--ro, opt, object, recording schedule-->
      <ScheduleBlockList>
        <!--ro, opt, array, recording schedule list, subType:object-->
        <ScheduleBlock>
          <!--ro, opt, object, recording schedule module-->
```

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```
            <ScheduleBlockGUID>
              <!--ro, opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
            </ScheduleBlockGUID>
            <ScheduleBlockType>
              <!--ro, req, string, schedule type-->/ISAPI/recording/schedule/default
            </ScheduleBlockType>
            <ScheduleAction>
              <!--ro, opt, array, schedule, subType:object, desc:N/A-->
              <id>
                <!--ro, req, int, ID-->1
              </id>
              <ScheduleActionStartTime>
                <!--ro, opt, object, scheduled start time-->
                <DayOfWeek>
                  <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
                </DayOfWeek>
                <TimeOfDay>
                  <!--ro, opt, time, time-->08:00:00
                </TimeOfDay>
              </ScheduleActionStartTime>
              <ScheduleActionEndTime>
                <!--ro, opt, object, scheduled end time-->
                <DayOfWeek>
                  <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
                </DayOfWeek>
                <TimeOfDay>
                  <!--ro, opt, time, time-->08:00:00
                </TimeOfDay>
              </ScheduleActionEndTime>
              <ScheduleDSTEnable>
                <!--ro, req, bool, whether to enable Daylight Saving Time (DST)-->true
              </ScheduleDSTEnable>
              <Description>
                <!--ro, req, string, description-->PreMorning (Midnight to 8am,local time)
              </Description>
              <Actions>
                <!--ro, opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
                <Record>
                  <!--ro, opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
                </Record>
                <Log>
                  <!--ro, opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
                </Log>
                <SaveImg>
                  <!--ro, opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
                </SaveImg>
                <ActionRecordingMode>
                  <!--ro, req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures),
"ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "COMMAND" (recording by commands), "SMART" (smart recording), "thermalVehicleDetection" (thermal imaging vehicle detection),
"fishingShipDetection" (fishing ship detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"AllEvent" (all events), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "temperatureIntervalMeasurement" (temperature range
detection), "Linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay
detection), "failDown" (getting up or falling down detection)-->CMR
                </ActionRecordingMode>
                <PreRecordTimeSeconds>
                  <!--ro, opt, int, pre-record duration, unit:s, desc:from 0 to 10 seconds; default: 5 seconds-->5
                </PreRecordTimeSeconds>
                <PostRecordTimeSeconds>
                  <!--ro, opt, int, post-record duration, unit:s, desc:from 0 to 20 seconds; default: 5 seconds-->5
                </PostRecordTimeSeconds>
              </Actions>
            </ScheduleAction>
          </ScheduleBlock>
        </ScheduleBlockList>
      </TrackSchedule>
      <CustomExtensionList>
        <!--ro, opt, array, custom extension list, subType:object-->
        <CustomExtension>
          <!--ro, opt, object, custom extension-->
          <CustomExtensionName>
            <!--ro, opt, string, custom extension name-->www.xxxx.com/RaCM/trackExt/ver10
          </CustomExtensionName>
          <enableSchedule>
            <!--ro, opt, bool, whether to enable recording schedule configuration-->true
          </enableSchedule>
          <SaveAudio>
            <!--ro, opt, bool, whether to enable audio recording-->true
          </SaveAudio>
          <PreRecordTimeSeconds>
            <!--ro, opt, int, pre-record, unit:s-->1
          </PreRecordTimeSeconds>
          <PostRecordTimeSeconds>
            <!--ro, opt, int, post-record, unit:s-->1
          </PostRecordTimeSeconds>
          <HolidaySchedule>
            <!--ro, opt, object, holiday schedule-->
            <ScheduleBlock>
              <!--ro, opt, object, schedule module-->
              <ScheduleBlockGUID>
                <!--ro, req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
              </ScheduleBlockGUID>
              <ScheduleBlockType>
                <!--ro, opt, string, schedule module type-->www.xxxx.com/racm/schedule/ver10
```

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```
            <!--ro, opt, string, schedule module type-->www.xxxx.com/ISAPI/Schedule/ver10
          </ScheduleBlockType>
        </ScheduleBlock>
      </HolidaySchedule>
    </CustomExtension>
  </CustomExtensionList>
  <IntelligentRecord>
    <!--ro, opt, bool, whether to enable VCA recording function-->false
  </IntelligentRecord>
  <delayTime>
    <!--ro, opt, int, capture delay time, unit:s-->1
  </delayTime>
  <durationEnabled>
    <!--ro, opt, bool, whether to enable video expiry time, desc:this node is not returned when it is not supported; you can configure video expiry time
via the node "Duration" when it is supported-->true
  </durationEnabled>
  <redundancyRec>
    <!--ro, opt, bool, whether it is redundant recording, desc:it is false when this node does not exist-->false
  </redundancyRec>
  <passbackRecord>
    <!--ro, opt, bool, whether to copy back recording, desc:it is false when this node does not exist-->false
  </passbackRecord>
  <lockDuration>
    <!--ro, opt, int, recording locking duration, unit:h, desc:0-unlocked or this node does not exist, 0xffff-locked permanently; if the total recording
is longer than the locked part, it will not be locked-->0
  </lockDuration>
  <recordBackup>
    <!--ro, opt, bool, whether to enable video archiving, desc:it is false when this node does not exist-->false
  </recordBackup>
  <SVCLevel>
    <!--ro, opt, enum, SVC frame extracting type, subType:int, desc:0-not extracted or this node does not exist, 1-half extracted, 2-three quarters
extracted-->0
  </SVCLevel>
  <recordManage>
    <!--ro, opt, bool, whether to enable video scheduling, desc:true (enabled scheduled recording), false (no scheduled recording but the recording
schedule is still in use, e.g., motion detection and copy-back); it is false when this node does not exist-->false
  </recordManage>
  <extraSaveAudio>
    <!--ro, opt, bool, whether to enable separate audio storage, desc:it is false when this node does not exist-->false
  </extraSaveAudio>
 </Track>
</TrackList>
```

19.3.2.3 Set all recording schedules

Request URL

PUT /ISAPI/ContentMgmt/record/tracks

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TrackList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
 <!--req, array, all recording schedules parameters, subType:object, attr:version{req, string, protocolVersion}-->
 <Track>
  <!--opt, object, recording schedule configuration-->
  <id>
   <!--opt, int, channel corresponding to the stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main stream
of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type should
be subject to that of SrcUrl-->201
  </id>
  <Channel>
   <!--req, int, channel No.-->1
  </Channel>
  <Enable>
   <!--req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
  </Enable>
  <Description>
   <!--req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20
fps,bitrate=6000 kbps
  </Description>
  <TrackGUID>
   <!--req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
  </TrackGUID>
  <Duration min="0" max="750">
   <!--opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15 hours;
0 indicates the video is valid until it is recurrently overwritten-->P10DT15H
  </Duration>
  <DefaultRecordingMode>
   <!--req, enum, default recording type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM"
(capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "ALLEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording),
"temperatureIntervalMeasurement" (temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by
default), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm"
(panic alarm), "zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from
vehicl entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp"
```

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```
vehicle entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toilethuman" (in-toilet overtime detection), "getup"
(get-off-bed detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
    </DefaultRecordingMode>
    <LoopEnable>
      <!--opt, bool, whether to support recurrently overwriting-->true
    </LoopEnable>
    <SrcDescriptor>
      <!--opt, object, video source description-->
      <SrcGUID>
        <!--req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
      </SrcGUID>
      <SrcChannel>
        <!--req, int, source channel-->1
      </SrcChannel>
      <StreamHint>
        <!--req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
      </StreamHint>
      <SrcDriver>
        <!--req, string, stream executable driver name-->RTP/RTSP
      </SrcDriver>
      <SrcType>
        <!--opt, string, source type-->mp4 video
      </SrcType>
      <SrcUrl>
        <!--req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID> refers
to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main stream
of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
      </SrcUrl>
      <SrcLogin>
        <!--req, string, login source password-->admin:admin
      </SrcLogin>
    </SrcDescriptor>
    <TrackSchedule>
      <!--opt, object, recording schedule-->
      <ScheduleBlockList>
        <!--opt, array, recording schedule list, subType:object-->
        <ScheduleBlock>
          <!--opt, object, recording schedule module-->
          <ScheduleBlockGUID>
            <!--opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
          </ScheduleBlockGUID>
          <ScheduleBlockType>
            <!--req, string, schedule type-->/ISAPI/recording/schedule/default
          </ScheduleBlockType>
        </ScheduleBlock>
      </ScheduleBlockList>
    </TrackSchedule>
    <CustomExtensionList>
      <!--opt, array, custom extension list, subType:object-->
      <CustomExtension>
        <!--opt, object, custom extension-->
        <CustomExtensionName>
          <!--opt, string, custom extension name-->www.xxx.com/RaCM/trackExt/ver10
        </CustomExtensionName>
        <enableSchedule>
          <!--opt, bool, whether to enable recording schedule configuration-->true
        </enableSchedule>
        <PreRecordTimeSeconds>
          <!--opt, int, pre-record, unit:s-->1
        </PreRecordTimeSeconds>
        <PostRecordTimeSeconds>
          <!--opt, int, post-record, unit:s-->1
        </PostRecordTimeSeconds>
      </CustomExtension>
    </CustomExtensionList>
    <durationEnabled>
      <!--opt, bool, whether to enable video expiry time, desc:this node is not returned when it is not supported; you can configure video expiry time via
the node Duration when it is supported-->true
    </durationEnabled>
  </Track>
</TrackList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

19.3.2.4 Get the configuration capability of the recording schedule

Request URL

GET /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	Channel number corresponding to the stream type, for example, 101 indicates main stream of channel 1.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Track xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, configuration capability of the recording schedule, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, channel corresponding to the stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main
stream of channel 1, 202 refers to the sub-stream of channel 2, and so on-->201
  </id>
  <Channel>
    <!--ro, req, int, channel No.-->1
  </Channel>
  <Enable>
    <!--ro, req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
  </Enable>
  <Description>
    <!--ro, req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20
fps,bitrate=6000 kbps
  </Description>
  <TrackGUID>
    <!--ro, req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
  </TrackGUID>
  <Size>
    <!--ro, opt, int, recording size-->1
  </Size>
  <Duration min="0" max="750">
    <!--ro, opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15
hours-->P10DT15H
  </Duration>
  <DefaultRecordingMode
opt="CMR,MOTION,ALARM,EDR,ALARMANDMOTION,AllEvent,POS,PECCANCY,SMART,temperatureIntervalMeasurement,methaneConcentrationException,thermalVehicleDetection,fi
shingShipDetection,consultation,panicAlarm,zoneAlarm,parallelParking,parallelParkingContinuousRecord,linedetection,toiletTarry,getUp,retention,failDown">
    <!--ro, req, enum, default recording type, subType:string, attr:opt{req, string}, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection
pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and
motion detection pictures), "AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording),
"temperatureIntervalMeasurement" (temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by
default), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm"
(panic alarm), "zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from
vehicl entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp"
(get-off-bed detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
  </DefaultRecordingMode>
```

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```
  <LoopEnable>
    <!--ro, opt, bool, whether to support recurrent overwriting-->true
  </LoopEnable>
  <SrcDescriptor>
    <!--ro, opt, object, video source description-->
    <SrcGUID>
      <!--ro, req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
    </SrcGUID>
    <SrcChannel>
      <!--ro, req, int, source channel-->1
    </SrcChannel>
    <StreamHint>
      <!--ro, req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
    </StreamHint>
    <SrcDriver>
      <!--ro, req, string, stream executable driver name-->RTP/RTSP
    </SrcDriver>
    <SrcType>
      <!--ro, opt, string, source type-->mp4 video
    </SrcType>
    <SrcUrl
opt="rtsp://localhost/PSIA/Streaming/channels/101,rtsp://localhost/PSIA/Streaming/channels/102,rtsp://localhost/PSIA/Streaming/channels/103,rtsp://localhost
/PSIA/Streaming/channels/107">
      <!--ro, req, string, source URL, attr:opt{req, string}-->rtsp://10.3.2.26/mpeg4/media.amp
    </SrcUrl>
    <SrcUrlMethods>
      <!--ro, req, string, methods supported by the source, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
    </SrcUrlMethods>
    <SrcLogin>
      <!--ro, req, string, login source password, desc:user name for source channel RTSP authentication returned fby device; encryption required for
sensitive information-->admin
    </SrcLogin>
  </SrcDescriptor>
  <TrackSchedule>
    <!--ro, opt, object, recording schedule-->
    <ScheduleBlockList>
      <!--ro, opt, array, recording schedule list, subType:object-->
      <ScheduleBlock>
        <!--ro, opt, object, recording schedule module-->
        <ScheduleBlockGUID>
          <!--ro, opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
        </ScheduleBlockGUID>
        <ScheduleBlockType>
          <!--ro, req, string, schedule type-->/ISAPI/recording/schedule/default
        </ScheduleBlockType>
        <ScheduleAction>
          <!--ro, opt, array, schedule, subType:object, desc:N/A-->
          <id>
            <!--ro, req, int, ID-->1
          </id>
          <ScheduleActionStartTime>
            <!--ro, opt, object, scheduled start time-->
            <DayOfWeek>
              <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--ro, opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionStartTime>
          <ScheduleActionEndTime>
            <!--ro, opt, object, scheduled end time-->
            <DayOfWeek>
              <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--ro, opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionEndTime>
          <ScheduleDSTEnable>
            <!--ro, req, bool, whether to enable Daylight Saving Time (DST)-->true
          </ScheduleDSTEnable>
          <Description>
            <!--ro, req, string, description-->PreMorning (Midnight to 8am,local time)
          </Description>
          <Actions>
            <!--ro, opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
            <Record>
              <!--ro, opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Record>
            <Log>
              <!--ro, opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Log>
            <SaveImg>
              <!--ro, opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </SaveImg>
            <ActionRecordingMode>
              <!--ro, req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures),
"ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "Command" (recording by commands), "SMART" (smart recording), "AllEvent" (all events), "temperatureIntervalMeasurement" (temperature range
detection), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "methaneConcentrationException"
(dangerous gas detection: methane concentration exception by default), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "linedetection"
(line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay detection), "failDown"
(getting up or falling down detection)-->CMR
```

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```
            </ActionRecordingMode>
            <PreRecordTimeSeconds min="0" max="10" def="5">
              <!--ro, opt, int, pre-record duration, unit:s, attr:min{req, int},max{req, int},def{req, int}, desc:from 0 to 10 seconds; default: 5 seconds-->5
            </PreRecordTimeSeconds>
            <PostRecordTimeSeconds min="30" max="600" def="30">
              <!--ro, opt, int, post-record duration, unit:s, attr:min{req, int},max{req, int},def{req, int}, desc:from 30 to 600 seconds; default: 30 seconds-->30
            </PostRecordTimeSeconds>
          </Actions>
        </ScheduleAction>
      </ScheduleBlock>
    </ScheduleBlockList>
  </TrackSchedule>
  <CustomExtensionList>
    <!--ro, opt, array, custom extension list, subType:object-->
    <CustomExtension>
      <!--ro, opt, object, custom extension-->
      <CustomExtensionName>
        <!--ro, opt, string, custom extension name-->www.xxx.com/RaCM/trackExt/ver10
      </CustomExtensionName>
      <enableSchedule>
        <!--ro, opt, bool, whether to enable recording schedule configuration-->true
      </enableSchedule>
      <SaveAudio>
        <!--ro, opt, bool, whether to enable audio recording-->true
      </SaveAudio>
      <PreRecordTimeSeconds>
        <!--ro, opt, int, pre-record, unit:s-->1
      </PreRecordTimeSeconds>
      <PostRecordTimeSeconds>
        <!--ro, opt, int, post-record, unit:s-->1
      </PostRecordTimeSeconds>
      <HolidaySchedule>
        <!--ro, opt, object, holiday schedule-->
        <ScheduleBlock>
          <!--ro, opt, object, schedule module-->
          <ScheduleBlockGUID>
            <!--ro, req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
          </ScheduleBlockGUID>
          <ScheduleBlockType>
            <!--ro, opt, string, schedule module type-->www.xxx.com/racm/schedule/ver10
          </ScheduleBlockType>
        </ScheduleBlock>
      </HolidaySchedule>
    </CustomExtension>
  </CustomExtensionList>
  <IntelligentRecord>
    <!--ro, opt, enum, whether to enable VCA recording, subType:string, desc:0-no, 1-yes.-->0
  </IntelligentRecord>
  <delayTime opt="0,3,4,5,10,30,60,120,300">
    <!--ro, opt, int, capture delay time, unit:s, attr:opt{req, string}-->1
  </delayTime>
  <durationEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable video expiry date, attr:opt{req, string}, desc:1. front-end device: this node is not returned when this function is not supported; the video expiry date will be set by the node <Duration> when this function is supported and <Duration> is not 0
2. back-end device: the video expiry date will be set by the node <Duration> when this node is not returned; this node is "false" when <Duration> is 0-->true
  </durationEnabled>
  <redundancyRec opt="true,false">
    <!--ro, opt, bool, whether it is redundant recording, attr:opt{req, string}-->false
  </redundancyRec>
  <passbackRecord opt="true,false">
    <!--ro, opt, bool, whether to copy back recording, attr:opt{req, string}-->false
  </passbackRecord>
  <lockDuration min="0" max="65535">
    <!--ro, opt, int, recording locking duration, unit:h, attr:min{req, int},max{req, int}, desc:0-unlocked or this node does not exist, 0xffff-locked permanently; if the total recording is longer than the locked part, it will not be locked-->0
  </lockDuration>
  <recordBackup opt="true,false">
    <!--ro, opt, bool, whether to enable video archiving, attr:opt{req, string}-->false
  </recordBackup>
  <SVCLevel opt="0,1,2,3">
    <!--ro, opt, enum, SVC frame extracting type, subType:int, attr:opt{req, string}, desc:0-not extracted, 1-half extracted, 2-three quarters extracted-->0
  </SVCLevel>
  <recordManage opt="true,false">
    <!--ro, opt, bool, whether to enable video scheduling, attr:opt{req, string}, desc:enabled-scheduled recording; disabled-no scheduled recording but the recording schedule still in use for, e.g., motion detection and copy-back-->false
  </recordManage>
  <extraSaveAudio opt="true,false">
    <!--ro, opt, bool, whether to enable separate audio storage, attr:opt{req, string}-->false
  </extraSaveAudio>
</Track>
```

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19.3.2.5 Get the recording schedule parameters of a specified channel

Request URL

GET /ISAPI/ContentMgmt/record/tracks/<trackStreamID>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	Channel number corresponding to the stream type, for example, 101 indicates main stream of channel 1.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Track xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parameters of a recording schedule, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, channel corresponding to the stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main stream of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type should be subject to that of SrcUrl-->201
  </id>
  <Channel>
    <!--ro, req, int, channel No.-->1
  </Channel>
  <Enable>
    <!--ro, req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
  </Enable>
  <Description>
    <!--ro, req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20 fps,bitrate=6000 kbps
  </Description>
  <TrackGUID>
    <!--ro, req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
  </TrackGUID>
  <Size>
    <!--ro, opt, int, video size-->1
  </Size>
  <Duration min="0" max="750">
    <!--ro, opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15 hours-->P10DT15H
  </Duration>
  <DefaultRecordingMode>
    <!--ro, req, enum, default recording type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection pictures), "AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording), "temperatureIntervalMeasurement" (temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from vehicl entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
  </DefaultRecordingMode>
  <LoopEnable>
    <!--ro, opt, bool, whether to support recurrent overwriting-->true
  </LoopEnable>
  <SrcDescriptor>
    <!--ro, opt, object, video source description-->
    <SrcGUID>
      <!--ro, req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
    </SrcGUID>
    <SrcChannel>
      <!--ro, req, int, source channel-->1
    </SrcChannel>
    <StreamHint>
      <!--ro, req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
    </StreamHint>
    <SrcDriver>
      <!--ro, req, string, stream executable driver name-->RTP/RTSP
    </SrcDriver>
    <SrcType>
      <!--ro, opt, string, source type-->mp4 video
    </SrcType>
    <SrcUrl>
      <!--ro, req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID> refers to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main stream of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
    </SrcUrl>
    <SrcUrlMethods>
      <!--ro, req, enum, methods supported by the source, subType:string, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
    </SrcUrlMethods>
    <SrcLogin>
      <!--ro, req, string, login source password, desc:user name for source channel RTSP authentication returned by device; encryption required for sensitive information-->admin
    </SrcLogin>
  </SrcDescriptor>
  <TrackSchedule>
    <!--ro, opt, object, recording schedule-->
```

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```
    <ScheduleBlockList>
      <!--ro, opt, array, recording schedule list, subType:object-->
      <ScheduleBlock>
        <!--ro, opt, object, recording schedule module-->
        <ScheduleBlockGUID>
          <!--ro, opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
        </ScheduleBlockGUID>
        <ScheduleBlockType>
          <!--ro, req, string, schedule type-->/ISAPI/recording/schedule/default
        </ScheduleBlockType>
        <ScheduleAction>
          <!--ro, opt, array, schedule, subType:object, desc:N/A-->
          <id>
            <!--ro, req, int, ID-->1
          </id>
          <ScheduleActionStartTime>
            <!--ro, opt, object, scheduled start time-->
            <DayOfWeek>
              <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--ro, opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionStartTime>
          <ScheduleActionEndTime>
            <!--ro, opt, object, scheduled end time-->
            <DayOfWeek>
              <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
            </DayOfWeek>
            <TimeOfDay>
              <!--ro, opt, time, time-->08:00:00
            </TimeOfDay>
          </ScheduleActionEndTime>
          <ScheduleDSTEnable>
            <!--ro, req, bool, whether to enable Daylight Saving Time (DST)-->true
          </ScheduleDSTEnable>
          <Description>
            <!--ro, req, string, description-->PreMorning (Midnight to 8am,local time)
          </Description>
          <Actions>
            <!--ro, opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
            <Record>
              <!--ro, opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Record>
            <Log>
              <!--ro, opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </Log>
            <SaveImg>
              <!--ro, opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
            </SaveImg>
            <ActionRecordingMode>
              <!--ro, req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures),
"ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "COMMAND" (recording by commands), "SMART" (smart recording), "thermalVehicleDetection" (thermal imaging vehicle detection),
"fishingShipDetection" (fishing ship detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"AllEvent" (all events), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "temperatureIntervalMeasurement" (temperature range
detection), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay
detection), "failDown" (getting up or falling down detection)-->CMR
            </ActionRecordingMode>
            <PreRecordTimeSeconds>
              <!--ro, opt, int, pre-record duration, unit:s, desc:from 0 to 10 seconds; default: 5 seconds-->5
            </PreRecordTimeSeconds>
            <PostRecordTimeSeconds>
              <!--ro, opt, int, post-record duration, unit:s, desc:from 0 to 20 seconds; default: 5 seconds-->5
            </PostRecordTimeSeconds>
          </Actions>
        </ScheduleAction>
      </ScheduleBlock>
    </ScheduleBlockList>
  </TrackSchedule>
  <CustomExtensionList>
    <!--ro, opt, array, custom extension list, subType:object-->
    <CustomExtension>
      <!--ro, opt, object, custom extension-->
      <CustomExtensionName>
        <!--ro, opt, string, custom extension name-->www.xxx.com/RaCM/trackExt/ver10
      </CustomExtensionName>
      <enableSchedule>
        <!--ro, opt, bool, whether to enable recording schedule configuration-->true
      </enableSchedule>
      <SaveAudio>
        <!--ro, opt, bool, whether to enable audio recording-->true
      </SaveAudio>
      <PreRecordTimeSeconds>
        <!--ro, opt, int, pre-record, unit:s-->1
      </PreRecordTimeSeconds>
      <PostRecordTimeSeconds>
        <!--ro, opt, int, post-record, unit:s-->1
      </PostRecordTimeSeconds>
      <HolidaySchedule>
        <!--ro, opt, object, holiday schedule-->
        <ScheduleBlock>
          <!--ro, opt, object, schedule module-->
```

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```
            <ScheduleBlockGUID>
              <!--ro, req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
            </ScheduleBlockGUID>
            <ScheduleBlockType>
              <!--ro, opt, string, schedule module type-->www.xxx.com/racm/schedule/ver10
            </ScheduleBlockType>
          </ScheduleBlock>
        </HolidaySchedule>
      </CustomExtension>
    </CustomExtensionList>
    <IntelligentRecord>
      <!--ro, opt, bool, whether to enable VCA recording-->false
    </IntelligentRecord>
    <delayTime>
      <!--ro, opt, int, capture delay time, unit:s-->1
    </delayTime>
    <durationEnabled>
      <!--ro, opt, bool, whether to enable video expiry time, desc:this node is not returned when it is not supported; you can configure video expiry time via the node Duration when it is supported-->true
    </durationEnabled>
    <redundancyRec>
      <!--ro, opt, bool, whether it is redundant recording-->false
    </redundancyRec>
    <passbackRecord>
      <!--ro, opt, bool, whether to copy back recording-->false
    </passbackRecord>
    <lockDuration>
      <!--ro, opt, int, recording locking duration, unit:h, desc:0-unlocked or this node does not exist, 0xffff-locked permanently; if the total recording is longer than the locked part, it will not be locked-->0
    </lockDuration>
    <recordBackup>
      <!--ro, opt, bool, whether to enable video archiving-->false
    </recordBackup>
    <SVCLevel>
      <!--ro, opt, enum, SVC frame extracting type, subType:int, desc:0-not extracted, 1-half extracted, 2-three quarters extracted-->0
    </SVCLevel>
    <recordManage>
      <!--ro, opt, bool, whether to enable video scheduling, desc:true (enabled scheduled recording), false (no scheduled recording but the recording schedule is still in use, e.g., motion detection and copy-back)-->false
    </recordManage>
    <extraSaveAudio>
      <!--ro, opt, bool, whether to enable separate audio storage-->false
    </extraSaveAudio>
  </Track>
```

19.3.2.6 Set the recording schedule for a specified channel

Request URL

PUT /ISAPI/ContentMgmt/record/tracks/<trackStreamID>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	Channel number corresponding to the stream type, for example, 101 indicates main stream of channel 1.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<Track xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, get the parameters of a recording schedule, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, channel corresponding to the stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main stream of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type should be subject to that of SrcUrl-->201
  </id>
  <Channel>
    <!--req, int, channel No.-->1
  </Channel>
  <Enable>
    <!--req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
  </Enable>
  <Description>
    <!--req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20 fps,bitrate=6000 kbps
  </Description>
  <TrackGUID>
    <!--req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
  </TrackGUID>
  <Duration min="0" max="750">
    <!--opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15 hours-->P10DT15H
  </Duration>
```

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```
  <DefaultRecordingMode>
    <!--req, enum, default record type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture
alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection pictures),
"AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording), "temperatureIntervalMeasurement"
(temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm" (panic alarm),
"zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from vehicl entering to
vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed
detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
  </DefaultRecordingMode>
  <LoopEnable>
    <!--opt, bool, whether to support recurrently overwriting-->true
  </LoopEnable>
  <SrcDescriptor>
    <!--opt, object, video source description-->
    <SrcGUID>
      <!--req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
    </SrcGUID>
    <SrcChannel>
      <!--req, int, source channel-->1
    </SrcChannel>
    <StreamHint>
      <!--req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
    </StreamHint>
    <SrcDriver>
      <!--req, string, stream executable driver name-->RTP/RTSP
    </SrcDriver>
    <SrcType>
      <!--opt, string, source type-->mp4 video
    </SrcType>
    <SrcUrl>
      <!--req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID> refers
to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main stream
of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
    </SrcUrl>
    <SrcLogin>
      <!--req, string, login source password, desc:user name for source channel RTSP authentication; encryption required for sensitive information--
>admin:admin
    </SrcLogin>
  </SrcDescriptor>
  <TrackSchedule>
    <!--opt, object, recording schedule-->
    <ScheduleBlockList>
      <!--opt, array, recording schedule list, subType:object-->
      <ScheduleBlock>
        <!--opt, object, recording schedule module-->
        <ScheduleBlockGUID>
          <!--opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
        </ScheduleBlockGUID>
        <ScheduleBlockType>
          <!--req, string, schedule type-->/ISAPI/recording/schedule/default
        </ScheduleBlockType>
      </ScheduleBlock>
    </ScheduleBlockList>
  </TrackSchedule>
  <CustomExtensionList>
    <!--opt, array, custom extension list, subType:object-->
    <CustomExtension>
      <!--opt, object, custom extension-->
      <CustomExtensionName>
        <!--opt, string, custom extension name-->www.xxx.com/RaCM/trackExt/ver10
      </CustomExtensionName>
      <enableSchedule>
        <!--opt, bool, whether to enable recording schedule configuration-->true
      </enableSchedule>
      <PreRecordTimeSeconds>
        <!--opt, int, pre-record, unit:s-->1
      </PreRecordTimeSeconds>
      <PostRecordTimeSeconds>
        <!--opt, int, post-record, unit:s-->1
      </PostRecordTimeSeconds>
    </CustomExtension>
  </CustomExtensionList>
  <durationEnabled>
    <!--opt, bool, whether to enable video expiry time, desc:this node is not returned when it is not supported; you can configure video expiry time via the
node Duration when it is supported-->true
  </durationEnabled>
</Track>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

19.3.3 Video Search

19.3.3.1 Refresh the video index manually before playback

Request URL

PUT /ISAPI/ContentMgmt/record/control/manualRefresh/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>
```

19.3.3.2 Get the capability of searching for video and picture files

Request URL

GET /ISAPI/ContentMgmt/search/capabilities

Query Parameter

None

Request Message

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None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search conditions, attr:version{opt, string, protocolVersion}-->
  <timeSpanList>
    <!--ro, opt, array, list of time periods, subType:object-->
    <timeSpan>
      <!--ro, opt, object, time period-->
      <startTime>
        <!--ro, req, datetime, start time-->1970-01-01T00:00:00+08:00
      </startTime>
      <endTime>
        <!--ro, req, datetime, end time-->1970-01-01T00:00:00+08:00
      </endTime>
    </timeSpan>
  </timeSpanList>
</CMSearchDescription>
```

19.3.3.3 Search for video and picture files

Request URL

POST /ISAPI/ContentMgmt/search?timeType=<timeType>

Query Parameter

Parameter Name	Parameter Type	Description
timeType	enum	1. If "isSupportTimeTypeSTD" is true in GET /ISAPI/System/capabilities, it indicates that the device supports parsing the optional parameter timeType=STD in the URL.2. If the parameter timeType=STD is included in the URL, time-related parameters (e.g., "datetime") is processed in ISO8601 time format.3. If the parameter timeType=STD is not included in the URL, time-related parameters is processed according to the original time format set by the device.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, search conditions, attr:version{opt, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two
searches, the search history will be saved in the memory to speed up next search-->test
  </searchID>
  <trackIDList>
    <!--opt, array, channel list, subType:object-->
    <trackID>
      <!--opt, int, channel stream ID, desc:"XX01" (main stream videos of XX channel), "XX03" (search for XX channel pictures), 1 by default-->101
    </trackID>
  </trackIDList>
  <timeSpanList>
    <!--opt, array, list of time periods, subType:object-->
    <timeSpan>
      <!--opt, object, time period-->
      <startTime>
        <!--req, datetime, start time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T00:00:00Z
      </startTime>
      <endTime>
        <!--req, datetime, end time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T01:00:00Z
      </endTime>
      <laneNumber>
        <!--opt, int, lane No.-->1
      </laneNumber>
      <carType>
        <!--opt, enum, vehicle type, subType:string, desc:"coach", "truck", "car", "minibus", "smalltruck" (small-sized truck), "bigcar" (large-sized
vehicle), "smallcar" (small-sized vehicle), "human", "tumbrel" (two wheeler), "trike" (tricycle), "suv", "midcar" (middle-sized bus), "all", "unknown"--
>coach
      </carType>
      <illegalType>
        <!--opt, enum, violation type, subType:string, desc:"all", "normal", "LowSpeed" (driving in low speed), "highSpeed" (driving in high speed),
"reverse" (wrong-way driving), "redLight" (red light running), "driveLine" (driving on the lane line), "direction" (driving against direction guidance),
"intersectionCongest" (overstay at intersection), "nonDriveWay" (motor vehicle on non-motor vehicle lane), "changeLane" (illegal lane change), "wrongLine"
(occupy dedicated lane), "badSign" (prohibition violation), "intersectionStop" (stop vehicle at intersection), "greenLightStop (stop vehicle when green
light), "overYello" (over yellow line), "illegalParking" (illegal parking), "unTurn" (illegal U-turn), "emergencyLane" (occupy emergency lane), "banRight"
(no right turn), "banLeft" (no left turn), "failSafeBelt" (seatbelt unbuckled), "pedRedLight" (pedestrian red light running), "failYieldPed" (not yield to
pedestrian), "overStop" (over stop line), "bigCarOccupy" (lane occupation by large-sized vehicle), "leftNotYieldStraight" (left turn not yield to straight),
"rightNotYieldLeft" (right turn not yield to left turn), "uTurnNotYieldGoingStaight", "leftSideDrivingAfterLeftTurn", "gasser", "threeWheel", "Passenger",
```

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```
"uphone", "congestion", "accident", "runGreeLight", "nonMotorExist", "failHelmet", "smoke", "playPhone", "failSafeBeltVice", "roadBlock", "construction",
"notKeepDistance", "overtakeRightSide", "dragRacing", "changeLaneContinuously", "SSharpDriving", "abandonedObject", "pedestrian", "notSlowZebraCrossing",
"overThermal", "turnRightStop", "occupyDedicatedLane", "notDriveInDedicatedLane", "smog", "conflagration", "animalsOccupy", "irregularSafeBeltVic",
"notYieldToRightRoad", "rampNotYieldToMainRoad", "enterNotYieldToInRoundabout", "TruckOccupyOvertakingLane", "noAlterTraffic", "banStraight", "followClose"-
->all
      </illegalType>
      <vehicleTypeByWeight>
        <!--opt, enum, vehicle type according to the vehicle weight, subType:int, desc:1 (7 and fewer seats passenger vehicle / 2 tons or fewer truck), 2 (8
to 19 seats passenger vehicle / 2 to 5 tons truck), 3 (20 to 39 seats passenger vehicle/5 to 10 tons truck), 4 (40 and more seats passenger vehicle/10 to 15
tons truck), 5 (15 and more tons truck)-->4
      </vehicleTypeByWeight>
      <mediaID>
        <!--opt, int, file ID, desc:file ID (media ID), which is used to return video file information after segmented by startTime and endTime-->0
      </mediaID>
      <listType>
        <!--ro, opt, enum, name list type, subType:string, desc:"black" (blocklist). If this node is empty, it indicates that the name list type is not
limited, corresponding to the listType in ANPR-->black
      </listType>
    </timeSpan>
  </timeSpanList>
  <sourceIDList>
    <!--opt, array, stream ID list, subType:object-->
    <sourceID>
      <!--req, string, GUID, range:[0,64]-->test
    </sourceID>
  </sourceIDList>
  <searchTypeList>
    <!--opt, array, search type list, subType:object, range:[0,20]-->
    <searchKeyword>
      <!--opt, string, keyword-->test
    </searchKeyword>
    <keywordType>
      <!--opt, enum, keyword type, subType:string, desc:“policeCode” (police No.), “shortSerialNumber” (device serial No.); when searching for videos stored
in dock stations via the platform, videos can be filtered according to the serial No. of body camera; if the dock station does not send this parameter, the
policeCode will be used by default-->shortSerialNumber
    </keywordType>
  </searchTypeList>
  <contentTypeList>
    <!--opt, array, file type list for search, subType:object-->
    <contentType>
      <!--opt, enum, file type for search, subType:string, desc:"video", "audio", "metadata", "text", "mixed", "other". Note: "metadata" is used when
searching for pictures-->video
    </contentType>
  </contentTypeList>
  <uploadState>
    <!--opt, enum, uploading status, subType:string, desc:"all", "uploaded", "nonUploaded" (not uploaded), "uploadFailed” (uploading failed)-->all
  </uploadState>
  <lockState>
    <!--opt, enum, locking status, subType:string, desc:"lock” (locked), "unlock” (unlocked), "all"-->lock
  </lockState>
  <maxResults>
    <!--opt, int, the maximum number of results returned for one time, range:[1,500]-->500
  </maxResults>
  <searchResultPostion>
    <!--opt, int, start position for search, desc:the node of body camera is in front of <maxResults>-->1
  </searchResultPostion>
  <metadataList>
    <!--req, array, search conditions, subType:object, desc:only supports a single description list-->
    <metadataDescriptor>
      <!--req, enum, video or file type for search, subType:string, desc:1. the value of CMSearchDescription->contentTypeList->contentType indicates the
type (video or picture) for search; 2. the video and picture file type supported by the device will be returned by the <recordSearchType> node and the
<pictureSearchType> node in the RacmCap message returned by calling /ISAPI/ContentMgmt/capabilities; 3. transmit the data in the format of
“recordType.meta..com/[type]” to search for videos or pictures; if /[type] is not contained, it indicates that all types of video files will be searched; if
the [type] is set to “allPic”, it indicates that all types of picture files will be searched; the [type] refers to the followings: “imageCapture” (scheduled
capture by channel), “pothole”, “crack”, “roadMaint” (road maintenance), “scheduledRecordUpload” (scheduled recording by channel), "ReID" (target pattern
detection), “thermalVehicleDetection” (thermal-based vehicle detection), “objectsThrownDetection”, “methaneConcentrationException”,
“methaneLightIntensityException” (hazardous gas detection), “bannerDetection”, “waterQualityDetection”, “waterLevelDetection”, “parallelParking”,
“waterLevelAndRainfall” (water level and rainfall), “AIOpenPlatform” (events on AI open platform), “fishingShipDetection”, “variableDataEvent”,
“VOCsGasDetection”, “angleDeviationDetection” (tilt detection), “pir”, “shipChannelAbnormal” (ship deviation), “radarVideoDetection” (radar-assisted target
detection), “AIDfogDetection”, “consultation”, “panicAlarm”, “zoneAlarm”, “smokeAndFireDetection”, “smokeDetection”, “facedetection”, “FieldDetection”
(intrusion detection), “scenechangedetection”, “LineDetection” (line crossing detection), “regionEntrance”, “regionExiting”, “loitering”, “group” (people
gathering), “rapidMove” (fast moving”, “parking”, “unattendedBaggage”, “attendedBaggage”, “vehicleDetection”, “manual”, “manualSnapShot” (manual capture),
“playSnapShot” (capture in playback), “safetyHelmet” (hard hat detection), “personQueueCounting” (people queuing-up detection), “personQueueTime” (waiting
time in queue detection), “violentMotion”, “leavePosition” (absent detection), “advReachHeight” (climing detection), “peopleNumChange” (people count
change), “mixedTargetDetection” (multi-target-type detection), “illegalParking” (parking violation), “pedestrian” (pedestrian detection), “construction”,
“roadblock”, “evidence” (manual evidence capture), “crosslane” (driving on the lane line), “vehicleexist” (motor vehicle on non-motor vehicle lane),
“Lanechange”, “wrongdirection” (wron-way driving), “congestion”, “turnround”, “abandonedObject”, “trafficAccident”, “fogDetection”, “sensorAlarm”,
“parallelParkingContinuousRecord” (contimuous recording for parallel parking), “GasTankDetection”, “swipePatrol” (patrolling by card), “alarmResult” (face
picture comparison), “UVPhotonDetection”, “DevPowerStatusReport” (device power status uploading), “playCellphone” (using mobile phone detection), “HFPD”,
“audioAnalysis”, “PeopleFallingDownDetection”, “PeopleSedentary”, “RadarFallenObjectsOnTracks”, “VegetationSpectrumDetection”, “personDensityDetection”,
“ChannelClearEvent”, “channelOccupy”, ChannelPassingEvent”, and “FallenObjectsOnTracksEvent”-->recordType.meta..com/imageCapture
    </metadataDescriptor>
    <SearchProperity>
      <!--opt, object, search by features-->
      <plateSearchMask>
        <!--opt, enum, license plate-->test
      </plateSearchMask>
      <stateOrProvince>
        <!--opt, int, province, desc:"0xffff" (undefined)-->0xffff
      </stateOrProvince>
      <country>
        <!--opt, enum, country/region, subType:string, desc:0: Algorithm library does not support the national identification card; 1: CZ - Czech Republic;
2: FRA - France; 3: DE - Germany; 4: E - Spain; 5: IT - Italy; 6: NL - Netherlands; 7: PL - Poland; 8: SVK - Slovakia; 9: BY - Belorussia; 10: MDA -
Moldova; 11: RU - Russia; 12: UA - Ukraine; 169: Canada; 172: US; 227: Australia; 0xff: All-->0
```


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```
Moldova, 11: RU - Russia, 12: UA - Ukraine, 169: Canada, 172: US, 227: Australia, 0xff: All-->0
      </country>
      <subType>
        <!--opt, enum, sub types, subType:string, desc:only supported by vehicleDetection and HVTVehicleDetection. "all", "motorVehicle", "nonMotorVehicle",
"pedestrian"-->all
      </subType>
      <capTypeDescriptor>
        <!--opt, enum, types of captured pictures, subType:string, desc:"event", "evidence". Events include parkEvent, crosslane, vehicleexist, lanechange,
wrongdirection, congestion, turnaround, edfManual. Evidences include parkEvent, crosslane, vehicleexist, lanechange, wrongdirection, and anyType-->anyType
      </capTypeDescriptor>
      <nation>
        <!--opt, enum, area, subType:string, desc:EU: Europe Region; ER: Middle East; AP: Asia Pacific; AfricaAndAmerica: Africa and America-->EU
      </nation>
      <vehicleType>
        <!--opt, enum, vehicle type, subType:string, desc:"unknown", "bus", "truck", "vehicle", "van", "buggy", "SUVMPV": SUV/MPV, "mediumBus"-->bus
      </vehicleType>
      <absoulteParkingNum>
        <!--opt, string, parking space No., desc:the parking space No. is the same as the number on the actual ground surface, which can be captured by the
camera-->test
      </absoulteParkingNum>
    </SearchProperity>
  </metadataList>
  <cardNumber>
    <!--opt, string, card No., range:[0,32], desc:if this node is empty or does not exist, it indicates that the card No. is not restricted-->test
  </cardNumber>
  <drawFrame>
    <!--opt, bool, whether to enable frame extraction, desc:the value of this node is false by default-->false
  </drawFrame>
  <AlarmInNoList>
    <!--opt, array, triggered alarm input channel list, subType:object-->
    <alarmInNo>
      <!--req, int, triggered alarm input channel No.-->0
    </alarmInNo>
  </AlarmInNoList>
  <ruleID>
    <!--opt, string, rule ID, desc:if this node is not configured, the search result will not be restricted by any rule-->test
  </ruleID>
  <roomIndex>
    <!--opt, int, interrogation room No., desc:it starts from 1-->0
  </roomIndex>
  <StreamIDParam>
    <!--opt, object, stream ID parameters-->
    <streamID>
      <!--req, string, stream ID, range:[0,72]-->test
    </streamID>
    <channel>
      <!--opt, int, channel No., range:[1,256]-->1
    </channel>
    <cmdType>
      <!--opt, int, trigger type, range:[0,255], desc:connect NVR with cloud storage to use it-->0
    </cmdType>
    <backupVolumeNum>
      <!--opt, int, archive No., range:[0,255], desc:applicable to CVR-->0
    </backupVolumeNum>
    <archiveLabel>
      <!--opt, string, archive No.-->test
    </archiveLabel>
  </StreamIDParam>
  <PosAlarm>
    <!--opt, object, pos video-->
    <caseSensitive>
      <!--req, bool, whether it is case-sensitive-->true
    </caseSensitive>
    <combinateMode>
      <!--req, enum, keyword combination, subType:string, desc:"or", "and"-->or
    </combinateMode>
    <KeyWordList>
      <!--opt, array, keyword list, subType:object, desc:supports up to three key words-->
      <keyWord>
        <!--req, string, keyword, range:[0,128]-->test
      </keyWord>
    </KeyWordList>
  </PosAlarm>
  <TrialParam>
    <!--opt, object, search conditions of trail information-->
    <caseNo>
      <!--opt, string, case No., range:[0,56]-->test
    </caseNo>
    <caseName>
      <!--opt, string, case name, range:[0,56]-->test
    </caseName>
    <litigant1>
      <!--opt, string, party 1, range:[0,32]-->test
    </litigant1>
    <litigant2>
      <!--opt, string, party 2, range:[0,32]-->test
    </litigant2>
    <chiefJudge>
      <!--opt, string, jury, range:[0,32]-->test
    </chiefJudge>
    <caseType>
      <!--opt, enum, case type, subType:int, desc:0 (all), 1 (penal case), 2 (civil case)-->0
    </caseType>
  </TrialParam>
```

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```
<ConferenceParam>
  <!--opt, object, video conference parameters-->
  <conferenceID>
    <!--opt, string, conference ID, range:[1,64]-->test
  </conferenceID>
</ConferenceParam>
<ATMParam>
  <!--opt, object, ATM search-->
  <transactionType>
    <!--opt, enum, transaction type, subType:int, desc:0 (all), 1(search), 2(withdrawal), 3 (deposit), 4(change password), 5(transfer), 6(search with no card), 7(deposit with no card), 8 (cash swallowed), 9(card swallowed) 10(undefined)-->0
  </transactionType>
  <transationAmount>
    <!--opt, int, amount, step:1, desc:unit: yuan-->0
  </transationAmount>
</ATMParam>
<labelSearch>
  <!--opt, object, search by tag type list of video file, desc:if the node is not returned, it means search for all videos. Related URI: /ISAPI/ContentMgmt/record/control/remark?format=json&security=<security>&iv=<iv>-->
  <labelSearchType>
    <!--opt, enum, type of searching by video file tag, subType:string, desc:"recordLabelTypeList", "allLabel", "noLabel"-->recordLabelTypeList
  </labelSearchType>
  <labelTypeList>
    <!--opt, object, by tag type list of video file, dep:and,{$.CMSearchDescription.recordLabelSearch.recordLabelSearchType,eq,recordLabelTypeList}-->
    <labelTypes>
      <!--opt, object, tag type of video file-->
      <labelType>
        <!--opt, enum, tag type of video file, subType:string, desc:"importance" (important), "unimportance" (not important)-->importance
      </labelType>
    </labelTypes>
  </labelTypeList>
</labelSearch>
<dockStationAudioFile>
  <!--opt, object, dock station audio file-->
  <analysisState>
    <!--ro, opt, enum, analysis status, subType:string, desc:"complete" (completed), "running", "notStarted" (not started)-->complete
  </analysisState>
  <keyWordList>
    <!--opt, array, dock station audio file label, subType:object, range:[0,100]-->
    <keyWord>
      <!--opt, string, label, range:[0,256]-->test
    </keyWord>
  </keyWordList>
</dockStationAudioFile>
<objectsThrownDetectionSearch>
  <!--opt, object, object thrown from building search condition-->
  <alarmLevel>
    <!--opt, int, alarm priority, range:[1,6], desc:1 (low), 2 (medium), 3 (high), 4 (low and medium), 5 (low and high), 6 (medium and high). If this node is empty, alarms of all levels will be displayed-->3
  </alarmLevel>
  <lowFloorID>
    <!--opt, int, low floor, range:[-100,100]-->0
  </lowFloorID>
  <highFloorID>
    <!--opt, int, high floor, range:[-100,100]-->0
  </highFloorID>
</objectsThrownDetectionSearch>
<alarmResultSearch>
  <!--opt, object, face picture comparison search condition-->
  <faceContrastType>
    <!--req, enum, face picture comparison type, subType:string, desc:"all", "faceContrast" (face picture comparison succeeded), "whiteFaceContrast" (allowlist face picture comparison), "faceContrastFailure" (face picture comparison failed)-->faceContrast
  </faceContrastType>
</alarmResultSearch>
<radarID>
  <!--opt, int, radar channel No.-->1
</radarID>
<synthesisType>
  <!--opt, enum, subType:string-->all
</synthesisType>
</CMSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search results, attr:version{opt, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searches, the search history will be saved in the memory to speed up next search-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search status, subType:string, desc:true (succeeded), false (failed)-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"OK", "NO MATCHES", "MORE"-->OK
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of returned records-->500
  </numOfMatches>
  <matchList>
```

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```
<matchList>
  <!--ro, opt, array, result list, subType:object-->
  <searchMatchItem>
    <!--ro, opt, object, results-->
    <sourceID>
      <!--ro, req, string, source ID, desc:source ID-->test
    </sourceID>
    <trackID>
      <!--ro, opt, int, channel stream ID, desc:"XX01" (main stream videos of XX channel), "XX03" (search for XX channel pictures), 1 by default-->101
    </trackID>
    <timeSpan>
      <!--ro, opt, object, time range-->
      <startTime>
        <!--ro, req, datetime, video start time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T00:10:00Z
      </startTime>
      <endTime>
        <!--ro, req, datetime, video end time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T00:20:00Z
      </endTime>
    </timeSpan>
    <mediaSegmentDescriptor>
      <!--ro, opt, object, data descrition-->
      <contentType>
        <!--ro, req, string, file type, desc:corresponding to conditions-->test
      </contentType>
      <codecType>
        <!--ro, req, string, video encoding type-->test
      </codecType>
      <rateType>
        <!--ro, opt, string, video bit rate-->test
      </rateType>
      <playbackURI>
        <!--ro, opt, string, video URL,
desc:"name=" is followed by the video file name, and "size=" is followed by the size of the video file. The "starttime" and "endtime" are UTC time, but as the time returned by front-end devices is UTC time, while the time returned by DVR/NVR devices is the devices' local time, there are different mechanisms to deal with the time:
1. For DVR/NVR devices that return the local time:
    1) When the request URL does not include "timeType=STD" (/ISAPI/ContentMgmt/search), then the returned playbackURI will not include "timeType=STD". The "starttime" and "endtime" will refer to the devices' local time.
    2) When the request URL includes "timeType=STD" (/ISAPI/ContentMgmt/search&timeType=STD), and the devices support parsing "timeType" ("isSupportTimeTypeSTD" is true in GET /ISAPI/System/capabilities), then the returned playbackURI will include "timeType=STD". The "starttime" and "endtime" will refer to the UTC time in ISO 8601 format.
2. For IPC devices that return UTC time, whether or not the request URL includes "timeType=STD", the devices do not need to parse "timeType". The returned playbackURI will not include "timeType=STD", and the "starttime" and "endtime" will refer to the UTC time in ISO 8601 format.
3. To play back the video files, synchronize the playbackURI to the video playback API: /ISAPI/Streaming/tracks/<trackStreamID>?starttime=<startTime>&endtime=<endTimeime>&name=<fileName>&size=<fileSize>&timeType=<timeType>.-->rtsp://10.65.130.168/Streaming/tracks/201/?starttime=20190213T091134Z&amp;endtime=20190213T092116Z&amp;name=01010000093000100&amp;size=732335288;timeType=STD
      </playbackURI>
      <lockStatus>
        <!--ro, opt, enum, whether video file is locked, subType:string, desc:"lock", "unlock"-->lock
      </lockStatus>
      <remainLockTime>
        <!--ro, opt, int, remaining lock time, unit:s, desc:only returned when "lockStatus" is "lock". If 0 is returned, it indicates that it is locked permanently-->1
      </remainLockTime>
      <name>
        <!--ro, opt, string, file name-->test
      </name>
      <size>
        <!--ro, opt, int, file size-->0
      </size>
      <mediaID>
        <!--ro, opt, int, file ID, desc:unique identifier of file-->0
      </mediaID>
      <remark>
        <!--ro, opt, string, file remarks-->test
      </remark>
      <GPSInfo>
        <!--ro, opt, string, file GPS, desc:required, if no GPS information, a blank is returned-->test
      </GPSInfo>
      <recorderCode>
        <!--ro, opt, string, video recorder No.-->test
      </recorderCode>
      <uploadState>
        <!--ro, opt, enum, upload status, subType:string, desc:"uploaded", "nonUploaded", "uploadFailed", "uploading"-->uploaded
      </uploadState>
      <uploadTime>
        <!--ro, opt, datetime, video uploading time, desc:ISO8601 time (TD format, local time and time difference)-->1970-01-01T00:00:00+08:00
      </uploadTime>
      <policeCode>
        <!--ro, opt, string, police No., desc:sensitive information is encrypted-->test
      </policeCode>
      <policeName>
        <!--ro, opt, string, police name, desc:sensitive information is encrypted-->test
      </policeName>
      <shootingTime>
        <!--ro, opt, datetime, time, desc:ISO8601 time (TD format, local time and time difference)-->1970-01-01T00:00:00+08:00
      </shootingTime>
      <FileTime>
        <!--ro, opt, int, the file is too long, unit:min-->1
      </FileTime>
      <remarks>
        <!--ro, opt, string, file remarks-->test
      </remarks>
      <cloudStorageType>
```

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```
        <!--ro, opt, enum, storage server type, subType:string, desc:0 (cloud storage), 1 (CVR), 2 (pstor)-->0
      </cloudStorageType>
      <fileUrl>
        <!--ro, opt, string, file URL-->test
      </fileUrl>
      <thumbnailUrl>
        <!--ro, opt, string, thumbnail URL of picture or video-->test
      </thumbnailUrl>
      <encryption>
        <!--ro, opt, object, encryption-->
        <enabled>
          <!--ro, opt, bool, whether to encrypt the file-->true
        </enabled>
        <encryptionkey>
          <!--ro, opt, string, encryption key, desc:sensitive information is encrypted-->test
        </encryptionkey>
        <encryptionLevel>
          <!--ro, opt, enum, encryption level, subType:int, desc:1 (AES128), 2 (AES256)-->1
        </encryptionLevel>
      </encryption>
      <videoTargetTypeList>
        <!--ro, opt, object, list of target types, desc:this node is only applicable for HEOP protocol of intercom products. Several target types are included in the time range. Every type is for a stream. Playback is according to video of corresponding time-->
        <videoTargetType>
          <!--ro, opt, enum, list of target types, subType:string, desc:this node is only applicable for HEOP protocol of intercom products. Several target types are included in the time range. Every type is for a stream. Playback is according to video of corresponding time. "local", "peer"-->local
        </videoTargetType>
      </videoTargetTypeList>
      <TargetRegionList size="8">
        <!--ro, opt, array, search for the target region list of the picture, subType:object, attr:size{req, int}, desc:related URL: /ISAPI/ContentMgmt/capabilities  targetRegionEventPicSearch-->
        <TargetRegion>
          <!--ro, opt, object, target area-->
          <regionID>
            <!--ro, req, string, detection area ID-->1
          </regionID>
          <RegionCoordinatesList>
            <!--ro, req, array, list of point coordinates, subType:object, range:[3,10], desc:rectangle-->
            <RegionCoordinates>
              <!--ro, opt, object, point coordinates, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--ro, req, float, X-coordinate, range:[0,1]-->1
              </positionX>
              <positionY>
                <!--ro, req, float, Y-coordinate, range:[0,1]-->1
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </TargetRegion>
      </TargetRegionList>
      <objectsThrownDetectionResult>
        <!--ro, opt, object, object thrown from building search result-->
        <alarmLevel>
          <!--ro, opt, int, alarm level, desc:it is categorized into three levels according to the target track and the actual event credibility. The higher the level, the greater the credibility of the alarm being triggered by an actual object thrown from building event.-->0
        </alarmLevel>
        <floorID>
          <!--ro, opt, int, floor No., range:[-100,100]-->0
        </floorID>
      </objectsThrownDetectionResult>
      <synthesisType>
        <!--ro, opt, enum, subType:string-->auto
      </synthesisType>
      <bodyCameraShortSN>
        <!--ro, opt, string, range:[0,9]-->P20151543
      </bodyCameraShortSN>
      <shipsDetection>
        <!--ro, opt, object-->
        <shipName>
          <!--ro, opt, string, range:[1,64]-->浙嘉兴资03181
        </shipName>
        <shipsSpeed>
          <!--ro, opt, float, range:[0.0,300.0], unit:km/h-->23.4
        </shipsSpeed>
        <shipsLength>
          <!--ro, opt, float, range:[0.000,500.000], unit:m-->50.000
        </shipsLength>
        <shipLocation>
          <!--ro, opt, object-->
          <longitudeType>
            <!--ro, opt, enum, subType:string-->E
          </longitudeType>
          <latitudeType>
            <!--ro, opt, enum, subType:string-->S
          </latitudeType>
          <longitude>
            <!--ro, opt, object-->
            <degree>
              <!--ro, opt, int, range:[0,180]-->0
            </degree>
            <minute>
              <!--ro, opt, int, range:[0,60]-->0
            </minute>
```

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```
            <sec>
              <!--ro, opt, float, range:[0,60.000000]-->0.000
            </sec>
          </longitude>
          <latitude>
            <!--ro, opt, object-->
            <degree>
              <!--ro, opt, int, range:[0,180]-->0
            </degree>
            <minute>
              <!--ro, opt, int, range:[0,60]-->0
            </minute>
            <sec>
              <!--ro, opt, float, range:[0,60.000000]-->0.000
            </sec>
          </latitude>
        </shipLocation>
        <abnormalType>
          <!--ro, opt, enum, subType:int-->shipCourseDeviation
        </abnormalType>
        <shipChannelID>
          <!--ro, opt, int, range:[1,4]-->4
        </shipChannelID>
      </shipsDetection>
    </mediaSegmentDescriptor>
    <metadataMatches>
      <!--ro, opt, object, metadata match information-->
      <metadataDescriptor>
        <!--ro, req, string, metadata description, desc:eg: recordType.meta..com/MOTION. The supported video file type by a specific device can be
returned by opt in RacmCap->recordSearchType of /ISAPI/ContentMgmt/capabilities-->test
      </metadataDescriptor>
    </metadataMatches>
    <face>
      <!--ro, opt, object, face search result-->
      <name>
        <!--ro, opt, string, name-->test
      </name>
      <gender>
        <!--ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
      </gender>
      <certificateType>
        <!--ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport" (passport), "other" (the
other certificate)-->officerID
      </certificateType>
      <certificateNumber>
        <!--ro, opt, string, certificate No.-->test
      </certificateNumber>
      <currTemperature>
        <!--ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃-->36.5
      </currTemperature>
      <mask>
        <!--ro, opt, enum, whether the person wears a mask, subType:string, desc:"unknown", "no", "yes"-->unknown
      </mask>
      <phoneNumber>
        <!--ro, opt, string, phone number, range:[1,64], desc:it is valid when event type is face picture comparison-->test
      </phoneNumber>
      <FDID>
        <!--ro, opt, string, face picture library ID, range:[1,64], desc:it is valid when event type is face picture comparison-->test
      </FDID>
      <FDLibName>
        <!--ro, opt, string, face picture library name, range:[1,64], desc:it is valid when event type is face picture comparison-->test
      </FDLibName>
      <FDPicURL>
        <!--ro, opt, string, face picture library picture URL, desc:it is valid when event type is face picture comparison-->test
      </FDPicURL>
      <similarity>
        <!--ro, opt, float, comparison similarity, range:[0,1], step:0.001, desc:it is valid when event type is face picture comparison-->test
      </similarity>
    </face>
    <AlarmInNoList>
      <!--ro, opt, array, triggered alarm input channel list, subType:object-->
      <alarmInNo>
        <!--ro, req, int, triggered alarm input channel No.-->0
      </alarmInNo>
    </AlarmInNoList>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:if this node is not returned, the search result is not restricted by any rule-->test
    </ruleID>
    <roomIndex>
      <!--ro, opt, int, interrogation room No., desc:it starts from 1-->0
    </roomIndex>
    <StreamIDParam>
      <!--ro, opt, object, stream ID parameters-->
      <streamID>
        <!--ro, req, string, stream ID, range:[0,72]-->test
      </streamID>
      <cmdType>
        <!--ro, opt, int, trigger type, range:[0,255], desc:connect NVR with cloud storage to use it-->0
      </cmdType>
      <backupVolumeNum>
        <!--ro, opt, int, archive No., range:[0,255], desc:applicable to CVR-->0
      </backupVolumeNum>
      <archiveLabel>
        <!--ro, opt, string, archive No.-->test
```

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```
        </archiveLabel>
      </StreamIDParam>
      <PosAlarm>
        <!--ro, opt, object, pos video-->
        <caseSensitive>
          <!--ro, req, bool, whether it is case-sensitive-->true
        </caseSensitive>
        <combinateMode>
          <!--ro, req, enum, keyword combination, subType:string, desc:"or", "and"-->or
        </combinateMode>
        <KeyWordList>
          <!--ro, opt, array, keyword list, subType:object, desc:supports a maximum of three words-->
          <keyWord>
            <!--ro, req, string, keyword, range:[0,128]-->test
          </keyWord>
        </KeyWordList>
      </PosAlarm>
      <TrialParam>
        <!--ro, opt, object, search conditions of trail information-->
        <caseNo>
          <!--ro, opt, string, case No., range:[0,56]-->test
        </caseNo>
        <caseName>
          <!--ro, opt, string, case name, range:[0,56]-->test
        </caseName>
        <litigant1>
          <!--ro, opt, string, party 1, range:[0,32]-->test
        </litigant1>
        <litigant2>
          <!--ro, opt, string, party 2, range:[0,32]-->test
        </litigant2>
        <chiefJudge>
          <!--ro, opt, string, jury, range:[0,32]-->test
        </chiefJudge>
        <caseType>
          <!--ro, opt, enum, case type, subType:int, desc:0 (all), 1 (penal case), 2 (civil case)-->0
        </caseType>
      </TrialParam>
      <plateNo>
        <!--ro, opt, string, license plate number, range:[1,64], desc:noPlate (vehicle without license plate), unknown (no license plate recognized),
XXXXXXXX (license plate number. For motor vehicles, it is a 16-byte string; for non-motor vehicles, it is a 48-byte string)-->蓝浙AX2P12
      </plateNo>
      <ConferenceParam>
        <!--ro, opt, object, video conference parameters-->
        <conferenceName>
          <!--ro, opt, string, conference name, range:[1,64], desc:from the "conferenceName" of POST /ISAPI/VCS/conferences-->test
        </conferenceName>
      </ConferenceParam>
      <ATMParam>
        <!--ro, opt, object, ATM search-->
        <transactionType>
          <!--ro, opt, enum, transaction type, subType:int, desc:0 (all), 1(search), 2(withdrawal), 3 (deposit), 4 (change password), 5(transfer), 6 (search
with no card), 7 (deposit with no card), 8 (cash swallowed), 9 (card swallowed) 10 (undefined)-->0
        </transactionType>
        <transationAmount>
          <!--ro, opt, int, amount, step:1, desc:unit: yuan-->0
        </transationAmount>
      </ATMParam>
      <labelTypeList>
        <!--ro, opt, object, tag type list of video file, desc:URL for configuring video tags: /ISAPI/ContentMgmt/record/control/remark?
format=json&security=<security>&iv=<iv>-->
        <labelTypes>
          <!--ro, opt, object, tag type of video file-->
          <labelType>
            <!--ro, opt, enum, tag type of video file, subType:string, desc:"importance", "unimportance"-->importance
          </labelType>
        </labelTypes>
      </labelTypeList>
      <isInFalseAlarmMaterialLib>
        <!--ro, opt, bool, whether imported to false alarm library, desc:not imported if this node is not returned or false is returned-->true
      </isInFalseAlarmMaterialLib>
      <vegetationSpectrumDetection>
        <!--ro, opt, object-->
        <vegetationIndex>
          <!--ro, opt, enum, subType:string-->NDVI
        </vegetationIndex>
        <vegetationIndexInitData>
          <!--ro, opt, enum, subType:string-->NDVI
        </vegetationIndexInitData>
        <reflectivityType>
          <!--ro, opt, enum, subType:string-->blue
        </reflectivityType>
        <isVisibleImage>
          <!--ro, opt, bool-->true
        </isVisibleImage>
      </vegetationSpectrumDetection>
    </searchMatchItem>
    <dockStationAudioFile>
      <!--ro, opt, object, dock station audio file-->
      <analysisState>
        <!--ro, opt, enum, analysis status, subType:string, desc:"complete" (completed), "running", "notStarted" (not started)-->complete
      </analysisState>
      <keyWordList>
```

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```
        <!--ro, opt, array, dock station audio file label, subType:object, range:[0,100]-->
        <keyWord>
          <!--ro, opt, string, label, range:[0,256]-->test
        </keyWord>
      </keyWordList>
    </dockStationAudioFile>
    <pictureType>
      <!--ro, opt, enum, subType:string-->licensePlatePicture
    </pictureType>
  </matchList>
  <totalMatches>
    <!--ro, opt, int, total number of matched results-->12
  </totalMatches>
</CMSearchResult>
```

19.4 PTZ Control

19.4.1 Preset Management

19.4.1.1 Delete all patrols by channel.

Request URL

DELETE /ISAPI/PTZCtrl/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, read-only,describe the error reason in detail, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.4.1.2 Set parameters of all patrols for a specified channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, list of patrol parameters, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPatrol>
    <!--opt, object, patrol parameters-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <id>
      <!--req, string, ID-->test
    </id>
    <patrolName>
      <!--req, string, patrol name-->test
    </patrolName>
    <resumeType>
      <!--opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
    </resumeType>
    <PatrolSequenceList>
      <!--req, array, list of patrol results, subType:object-->
      <PatrolSequence>
        <!--req, object, patrol result-->
        <presetID>
          <!--req, string, preset ID-->1
        </presetID>
        <seqSpeed>
          <!--req, string, patrol speed-->1
        </seqSpeed>
        <delay>
          <!--req, string, delay-->1
        </delay>
      </PatrolSequence>
    </PatrolSequenceList>
  </PTZPatrol>
</PTZPatrolList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.4.1.3 Get the parameters of all patrols of a specific channel

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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dojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPatrol>
    <!--ro, opt, object, patrol parameters-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <patrolName>
      <!--ro, req, string, patrol name-->test
    </patrolName>
    <resumeType>
      <!--ro, opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
    </resumeType>
    <PatrolSequenceList>
      <!--ro, req, array, subType:object-->
      <PatrolSequence>
        <!--ro, req, object, list-->
        <presetID>
          <!--ro, req, string, preset ID-->1
        </presetID>
        <seqSpeed>
          <!--ro, req, string, patrol speed-->1
        </seqSpeed>
        <delay>
          <!--ro, req, int, delay, unit:ms-->1
        </delay>
      </PatrolSequence>
    </PatrolSequenceList>
  </PTZPatrol>
</PTZPatrolList>
```

19.4.1.4 Get patrol parameters of a specified channel

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <patrolName>
    <!--ro, req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--ro, opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--ro, req, array, patrol result list, subType:object-->
    <PatrolSequence>
      <!--ro, req, object, patrol result-->
      <presetID>
        <!--ro, req, string, preset ID-->1
      </presetID>
      <seqSpeed>
        <!--ro, req, string, patrol speed-->1
      </seqSpeed>
      <delay>
        <!--ro, req, int, delay, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

19.4.1.5 Delete a patrol by channel

Request URL

DELETE /ISAPI/PTZCtrl/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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19.4.1.6 Set path parameters of a single patrol for a specified channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, string, ID-->test
  </id>
  <patrolName>
    <!--req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--req, array, patrol result list, subType:object-->
    <PatrolSequence>
      <!--req, object, patrol result-->
      <presetID>
        <!--req, string, preset ID-->1
      </presetID>
      <seqSpeed>
        <!--req, string, patrol speed-->1
      </seqSpeed>
      <delay>
        <!--req, int, delay, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.4.1.7 Start calling a patrol by channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/patrols/<patrolID>/start

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.4.1.8 Stop calling a patrol by channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/patrols/<patrolID>/stop

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.4.1.9 Add a patrol by channel

Request URL

POST /ISAPI/PTZCtrl/channels/<channelID>/patrols?delayUnitType=<delayUnitType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
delayUnitType	string	If the ?delayUnitType=ms is not contained in the configuration URI, the unit of the delay field will follow the device's default behavior (milliseconds for back-end, front-end, and on-board vehicle, seconds for front-end and thermal device).

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, string, ID-->test
  </id>
  <patrolName>
    <!--req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--req, array, subType:object-->
    <PatrolSequence>
      <!--req, object, list-->
      <presetID>
        <!--req, string, preset ID-->1
      </presetID>
      <seqSpeed>
        <!--req, string, patrol speed-->1
      </seqSpeed>
      <delay>
        <!--req, int, unit: millisecond, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.4.1.10 Set all presets parameters

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/presets

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPresetList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, presets list, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPreset>
    <!--opt, object, PTZ preset-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <id>
      <!--req, string, ID-->test
    </id>
    <presetName>
      <!--req, string, preset name-->test
    </presetName>
    <AbsoluteHigh>
      <!--opt, object, absolute height-->
      <elevation>
        <!--opt, int, altitude, range:[-900,2700]-->1
      </elevation>
      <azimuth>
        <!--opt, int, panning angle, range:[0,3600]-->1
      </azimuth>
      <absoluteZoom>
        <!--opt, int, absolute zoom, range:[0,1000]-->1
      </absoluteZoom>
    </AbsoluteHigh>
  </PTZPreset>
</PTZPresetList>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.4.1.11 Delete all presets parameters

Request URL

DELETE /ISAPI/PTZCtrl/channels/<channelID>/presets

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.4.1.12 Get all presets' parameters

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/presets

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPresetList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPreset>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <id>
      <!--ro, req, string-->test
    </id>
    <presetName>
      <!--ro, req, string-->test
    </presetName>
    <AbsoluteHigh>
      <!--ro, opt, object-->
      <elevation>
        <!--ro, opt, int, range:[-900,2700]-->1
      </elevation>
      <azimuth>
        <!--ro, opt, int, range:[0,3600]-->1
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, int, range:[0,1000]-->1
      </absoluteZoom>
      <focus>
        <!--ro, opt, int, range:[1,100000]-->1
      </focus>
    </AbsoluteHigh>
  </PTZPreset>
</PTZPresetList>
```

19.4.1.13 Get the single PTZ control parameters

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/presets/<presetID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

None

Response Message

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dongjin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPreset xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--ro, req, string, listening host ID-->test
  </id>
  <presetName>
    <!--ro, req, string, preset name-->test
  </presetName>
  <AbsoluteHigh>
    <!--ro, opt, object, absolute height-->
    <elevation>
      <!--ro, opt, int, altitude, range:[-900,2700], desc:altitude-->1
    </elevation>
    <azimuth>
      <!--ro, opt, int, panning angle, range:[0,3600], desc:panning angle-->1
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, int, range:[0,1000]-->1
    </absoluteZoom>
  </AbsoluteHigh>
</PTZPreset>
```

19.4.1.14 Set parameters of a preset

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/presets/<presetID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPreset xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, preset parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, string, ID-->test
  </id>
  <presetName>
    <!--req, string, preset name-->test
  </presetName>
  <AbsoluteHigh>
    <!--opt, object, absolute height-->
    <elevation>
      <!--opt, int, altitude, range:[-900,2700]-->1
    </elevation>
    <azimuth>
      <!--opt, int, range:[0,3600]-->1
    </azimuth>
    <absoluteZoom>
      <!--opt, int, range:[0,1000]-->1
    </absoluteZoom>
  </AbsoluteHigh>
</PTZPreset>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.4.1.15 Delete a preset

Request URL

DELETE /ISAPI/PTZCtrl/channels/<channelID>/presets/<presetID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel ID
presetID	string	Preset ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.4.1.16 Call a preset

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/presets/<presetID>/goto

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.4.2 PTZ Initial Position

19.4.2.1 Set the initial position of the PTZ of a specific channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/homeposition

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.4.2.2 Delete the initial position

Request URL

DELETE /ISAPI/PTZCtrl/channels/<channelID>/homeposition

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Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.4.3 PTZ Maintenance

19.4.3.1 Get the PTZ capability supported by the device

Request URL

GET /ISAPI/PTZCtrl/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<PTZCtrlCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, PTZ control capability, attr:version{req, string, protocolVersion}-->
  <maxPTZChannleNum min="1" max="10">
    <!--ro, opt, int, number of PTZ channels, attr:min{req, int},max{req, int}-->1
  </maxPTZChannleNum>
  <isSupportPTZWithPreview opt="true,false">
    <!--ro, opt, bool, whether to support live view, attr:opt{req, string}-->true
  </isSupportPTZWithPreview>
  <isSupportZoomLinkage opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isSupportZoomLinkage>
  <isSupportManualWiper opt="true,false">
    <!--ro, opt, bool, whether to support wiper control interface manually, attr:opt{req, string}-->true
  </isSupportManualWiper>
  <isSupportTiltSync opt="true,false">
    <!--ro, opt, bool, whether to support tilt sync control of sound and bullet camera, attr:opt{req, string}-->true
  </isSupportTiltSync>
  <isSupportPTZProtocolList>
    <!--ro, opt, bool, whether to support getting PTZ protocol supported by device, desc:/ISAPI/PTZCtrl/PTZProtocolList?format=json-->true
  </isSupportPTZProtocolList>
  <isSupportRailwayRobot>
    <!--ro, opt, bool-->true
  </isSupportRailwayRobot>
  <isSupportPTZViewPreset>
    <!--ro, opt, bool-->true
  </isSupportPTZViewPreset>
  <isSupportPresetsProFocus>
    <!--ro, opt, bool-->true
  </isSupportPresetsProFocus>
  <isSupportGetCurrentPresetInfo>
    <!--ro, opt, bool-->true
  </isSupportGetCurrentPresetInfo>
</PTZCtrlCap>
```

19.4.3.2 Set control parameter of all PTZ.

Request URL

PUT /ISAPI/PTZCtrl/channels

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<PTZChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, PTZ channels' configuration list, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZChannel>
    <!--req, object, PTZ channel configuration-->
    <id>
      <!--req, int-->1
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <serialNumber>
      <!--req, int-->1
    </serialNumber>
    <videoInputID>
      <!--req, int-->1
    </videoInputID>
    <panMaxSpeed>
      <!--opt, int-->1
    </panMaxSpeed>
    <tiltMaxSpeed>
      <!--opt, int-->1
    </tiltMaxSpeed>
    <presetSpeed>
      <!--opt, int, range:[1,8]-->1
    </presetSpeed>
    <autoPatrolSpeed>
      <!--opt, int, range:[0,100]-->1
    </autoPatrolSpeed>
    <keyBoardControlSpeed>
      <!--opt, int, range:[0,100]-->1
    </keyBoardControlSpeed>
    <controlProtocol>
      <!--opt, enum, control protocol type, subType:string, desc:control protocol type-->pelco-d
    </controlProtocol>
    <controlAddress>
      <!--req, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <Address>
        <!--opt, string-->1
      </Address>
    </controlAddress>
    <defaultPresetID>
      <!--opt, string-->test
    </defaultPresetID>
    <PTZRs485Para>
      <!--req, object-->
      <baudRate>
        <!--req, int-->9600
      </baudRate>
      <dataBits>
        <!--req, int, data bit-->1
      </dataBits>
      <parityType>
        <!--req, enum, parity type, subType:string, desc:parity type-->space
      </parityType>
      <stopBits>
        <!--req, enum, stop bit, subType:string, desc:1, 1.5, 2-->1
      </stopBits>
      <flowCtrl>
        <!--req, enum, flow control type, subType:string, desc:flow control type-->hardware
      </flowCtrl>
    </PTZRs485Para>
    <manualControlSpeed>
      <!--opt, enum, manual control speed type, subType:string, desc:"pedestrian", "nonMotorVehicle" (non-motor vehicles), "motorVehicle" (vehicle),
"selfadaptive" (self-adaptive), "compatible"-->pedestrian
    </manualControlSpeed>
    <PTQueryFrequency>
      <!--opt, enum, subType:int-->30
    </PTQueryFrequency>
  </PTZChannel>
</PTZChannelList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.4.3.3 Add control parameter of single PTZ

Request URL

POST /ISAPI/PTZCtrl/channels

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<PTZChannel xmlns="http://www.isapi.org/ver20/XMLSchema">
  <!--req, object, PTZ channel configuration-->
  <id>
    <!--req, int, ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <serialNumber>
    <!--req, int, serial port No.-->1
  </serialNumber>
  <videoInputID>
    <!--req, int-->1
  </videoInputID>
  <panMaxSpeed>
    <!--opt, int, ro,degrees/sec-->1
  </panMaxSpeed>
  <tiltMaxSpeed>
    <!--opt, int, ro,degrees/sec-->1
  </tiltMaxSpeed>
  <presetSpeed>
    <!--opt, int, 1..8, range:[1,8]-->1
  </presetSpeed>
  <autoPatrolSpeed>
    <!--opt, int, 0..100, range:[0,100]-->1
  </autoPatrolSpeed>
  <keyBoardControlSpeed>
    <!--opt, int, 0..100, range:[0,100]-->1
  </keyBoardControlSpeed>
  <controlProtocol>
    <!--opt, enum, control protocol type, subType:string, desc:control protocol type-->pelco-d
  </controlProtocol>
  <controlAddress>
    <!--req, object-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <Address>
      <!--opt, string, address-->1
    </Address>
  </controlAddress>
  <defaultPresetID>
    <!--opt, string, id-->test
  </defaultPresetID>
  <PTZRs485Para>
    <!--req, object-->
    <baudRate>
      <!--req, int, baud rate-->9600
    </baudRate>
    <dataBits>
      <!--req, int, data bit-->1
    </dataBits>
    <parityType>
      <!--req, enum, parity type, subType:string, desc:"none" (no parity bit), "even" (even parity), "odd" (odd parity), "mark" (total number of parity bits
is 1), "space" (total number of parity bits is 0)-->space
    </parityType>
    <stopBits>
      <!--req, enum, stop bit, subType:string, desc:"1", "1.5", "2"-->1
    </stopBits>
    <flowCtrl>
      <!--req, enum, "none,software,hardware", subType:string, desc:"none", "software", "hardware"-->hardware
    </flowCtrl>
  </PTZRs485Para>
  <manualControlSpeed>
    <!--opt, enum, manual control speed type, subType:string, desc:"pedestrian", "nonMotorVehicle" (non-motor vehicles), "motorVehicle" (vehicle),
"selfadaptive" (self-adaptive), "compatible"-->pedestrian
  </manualControlSpeed>
  <PTQueryFrequency>
    <!--opt, enum, subType:int-->30
  </PTQueryFrequency>
</PTZChannel>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.4.3.4 Set a specified PTZ channel's parameters

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, PTZ channel's parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <serialNumber>
    <!--req, int, serial port No., desc:this field should be supported by both GET and PUT methods. If the value is 0, it indicates that no serial port is
linked-->1
  </serialNumber>
  <videoInputID>
    <!--req, int, video input ID-->1
  </videoInputID>
  <panMaxSpeed>
    <!--opt, int, the maximum panning speed, unit: degrees/sec-->1
  </panMaxSpeed>
  <tiltMaxSpeed>
    <!--opt, int, the maximum titling speed, unit: degrees/sec-->1
  </tiltMaxSpeed>
  <presetSpeed>
    <!--opt, int, preset speed-->1
  </presetSpeed>
  <autoPatrolSpeed>
    <!--opt, int, automatic patrol speed-->1
  </autoPatrolSpeed>
  <keyBoardControlSpeed>
    <!--opt, int, keypad control speed, range:[0,100]-->1
  </keyBoardControlSpeed>
  <controlProtocol>
    <!--opt, enum, control protocol, subType:string, desc:"pelco-d", "modbus-RTU", "modbus-ASCII"-->pelco-d
  </controlProtocol>
  <controlAddress>
    <!--req, object, control address-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <Address>
      <!--opt, string, address, range:[1,255]-->test
    </Address>
  </controlAddress>
  <defaultPresetID>
    <!--opt, string, preset ID-->test
  </defaultPresetID>
  <PTZRs485Para>
    <!--opt, object-->
    <baudRate>
      <!--req, int, baud rate-->9600
    </baudRate>
```

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```
    <dataBits>
      <!--req, int, data bit-->1
    </dataBits>
    <parityType>
      <!--req, enum, parity type, subType:string, desc:"none" (no parity bit), "even" (even parity), "odd" (odd parity), "mark" (total number of parity bits is 1), "space" (total number of parity bits is 0)-->space
    </parityType>
    <stopBits>
      <!--req, enum, stop bit, subType:string, desc:"1", "1.5", "2"-->1
    </stopBits>
    <flowCtrl>
      <!--req, enum, flow control type, subType:string, desc:"none", "software", "hardware"-->hardware
    </flowCtrl>
  </PTZRs485Para>
  <manualControlSpeed>
    <!--opt, enum, manual control speed type, subType:string, desc:"pedestrian", "nonMotorVehicle" (non-motor vehicles), "motorVehicle" (vehicle), "selfadaptive" (self-adaptive), "compatible"-->pedestrian
  </manualControlSpeed>
  <panSupport>
    <!--opt, bool, whether it supports panning-->true
  </panSupport>
  <tiltSupport>
    <!--opt, bool, whether it supports tilting-->true
  </tiltSupport>
  <zoomSupport>
    <!--opt, bool, whether it supports zooming-->true
  </zoomSupport>
  <PTPositiveDirection>
    <!--opt, object, panning and tilting positive direction, desc:panning and tilting positive direction-->
    <pan>
      <!--opt, enum, panning positive direction, subType:string, desc:"left", "right"-->left
    </pan>
    <tile>
      <!--opt, enum, tilting positive direction, subType:string, desc:"up", "down"-->up
    </tile>
  </PTPositiveDirection>
  <PTQueryFrequency>
    <!--opt, enum, subType:int-->30
  </PTQueryFrequency>
</PTZChannel>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.4.3.5 Get parameters of a specified PTZ channel

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
```

<PTZChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
<!--ro, req, object, attr:version{req, string, protocolVersion}-->
<id>
<!--ro, req, int-->1
</id>
<enabled>
<!--ro, req, bool, ro-->true
</enabled>
<serialNumber>
<!--ro, req, int-->1
</serialNumber>
<videoInputID>
<!--ro, req, int-->1
</videoInputID>
<panMaxSpeed>
<!--ro, opt, int, the maximum speed for panning-->1
</panMaxSpeed>
<tiltMaxSpeed>
<!--ro, opt, int, the maximum speed for tilting-->1
</tiltMaxSpeed>
<presetSpeed>
<!--ro, opt, int, preset speed-->1
</presetSpeed>
<autoPatrolSpeed>
<!--ro, opt, int, automatic patrol speed-->1
</autoPatrolSpeed>
<keyBoardControlSpeed>
<!--ro, opt, int, keyboard control speed, range:[0,100]-->1
</keyBoardControlSpeed>
<controlProtocol>
<!--ro, opt, enum, "pelco-d,modbus-RTU,modbus-ASCII", subType:string, desc:"pelco-d,modbus-RTU,modbus-ASCII"-->pelco-d
</controlProtocol>
<controlAddress>
<!--ro, opt, object-->
<enabled>
<!--ro, req, bool, ro-->true
</enabled>
<Address>
<!--ro, opt, string, 1-255, range:[1,255]-->test
</Address>
</controlAddress>
<defaultPresetID>
<!--ro, opt, string, default preset ID-->test
</defaultPresetID>
<PTZRs485Para>
<!--ro, opt, object-->
<baudRate>
<!--ro, req, int, baud rate-->9600
</baudRate>
<dataBits>
<!--ro, req, int, data bit-->1
</dataBits>
<parityType>
<!--ro, req, enum, "none,even,odd,mark,space", subType:string, desc:"none,even,odd,mark,space"-->space
</parityType>
<stopBits>
<!--ro, req, enum, stop bit, subType:string, desc:stop bit-->1
</stopBits>
<flowCtrl>
<!--ro, req, enum, flow control mode, subType:string, desc:flow control mode-->hardware
</flowCtrl>
</PTZRs485Para>
<manualControlSpeed>
<!--ro, opt, enum, manual control speed, subType:string, desc:manual control speed-->pedestrian
</manualControlSpeed>
<panSupport>
<!--ro, opt, bool, whether it supports panning-->true
</panSupport>
<tiltSupport>
<!--ro, opt, bool, whether it supports tilting-->true
</tiltSupport>
<zoomSupport>
<!--ro, opt, bool, whether it supports zooming-->true
</zoomSupport>
<PTPositiveDirection>
<!--ro, opt, object, panning and tilting positive direction, desc:panning and tilting positive direction-->
<pan>
<!--ro, opt, enum, panning positive direction (horizontal), subType:string, desc:"left", "right"-->left
</pan>
<tile>
<!--ro, opt, enum, tilting positive direction (vertical), subType:string, desc:"up", "down"-->up
</tile>
</PTPositiveDirection>
<PTQueryFrequency>
<!--ro, opt, enum, subType:int-->30
</PTQueryFrequency>
<tiltNegationEnabled>
<!--ro, opt, bool-->false
</tiltNegationEnabled>
<presetAccuracy>
<!--ro, opt, float, range:[0.00,0.50]-->0.01
</presetAccuracy>
<autoFocusCompletionWaitEnabled>
<!--ro, opt, bool-->false

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```
    </autoFocusCompletionWaitEnabled>
</PTZChannel>
```

19.4.3.6 Get the PTZ component control capability of a specified channel

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/componentCtrl/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "ComponentCtrlCap": {
    /*ro, req, object, PTZ component control capability*/
        "locked": {
        /*ro, req, object, whether the PTZ component control is locked*/
            "@opt": [true, false]
            /*ro, opt, array, true (PTZ component control is disabled), false (PTZ component control is enabled; enabled by default), subType:bool,
desc:true (PTZ component control is disabled), false (PTZ component control is enabled; enabled by default)*/
        }
    }
}
```

19.4.3.7 Set PTZ component control parameters of a specific channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/componentCtrl?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "ComponentCtrl": {
    /*req, object, PTZ component control parameters*/
        "locked":  false
        /*req, bool, whether to disable PTZ component control, desc:true (disable), false (enable), the default value is false*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

19.4.3.8 Get PTZ component control parameters of a specified channel

Request URL

VISION AND SECURITY SYSTEM CO ,LTD. dcjojin@gmail.com

GET /ISAPI/PTZCtrl/channels/<channelID>/componentCtrl?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "ComponentCtrl": {
    /*ro, req, object, PTZ component control*/
        "locked":  false
        /*ro, req, bool, whether the PTZ component control is locked, desc:true (PTZ component control is disabled), false (PTZ component control is
enabled; enabled by default)*/
    }
}
```

19.4.3.9 Get the PTZ self-checking and control capability

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/initializationCtrl/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "InitializationCtrlCap": {
    /*ro, req, object, PTZ self-checking and control capability*/
        "status": {
        /*ro, opt, object, status*/
            "@opt": ["initialized", "noInitialization", "close", "success", "building", "false"]
            /*ro, req, array, status value, subType:string, desc:initialized-initialized already, noInitialization-not initialized yet*/
        },
        "initializationPercent": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int*/
            "@max":  100
            /*ro, req, int*/
        }
    }
}
```

19.4.3.10 Get PTZ self-checking and control status

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/initializationCtrl/status?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
{
    "InitializationCtrl": {
    /*ro, opt, object, PTZ self-checking and control status*/
        "status":  "initialized",
        /*ro, req, enum, status, subType:string, desc:"initialized" (initialized already), "noInitialization" (not initialized yet)*/
        "initializationPercent":  0
        /*ro, opt, int, range:[0,100], unit:%*/
    }
}
```

19.4.3.11 Set the parameters for PTZ self-checking and control of a specific channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/initializationCtrl?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  "test",
    /*ro, opt, string, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "test",
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

19.4.3.12 Get the power-off memory parameters of a specific channel

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/saveptzpoweroff

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<savePtzPoweroff xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, power-off memory parameters, attr:version{req, string, protocolVersion}-->
  <savePtzPoweroffType>
    <!--ro, req, enum, "disable,30sec,60sec,300sec,600sec", subType:string, desc:"disable", "30sec", "60sec", "300sec", "600sec"-->disable
  </savePtzPoweroffType>
</savePtzPoweroff>
```

19.4.3.13 Set the power-off memory parameters of a specific channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/saveptzpoweroff

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dojoji1@gmail.com

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<savePtzPoweroff xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, power-off memory parameters, attr:version{req, string, protocolVersion}-->
  <savePtzPoweroffType>
    <!--req, enum, mode, subType:string, desc:"disable", "30sec", "60sec", "300sec", "600sec"-->disable
  </savePtzPoweroffType>
</savePtzPoweroff>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.4.4 PTZ Manual Control

19.4.4.1 Get the PTZ status of a specific channel of the device

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/status

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <AbsoluteHigh>
    <!--ro, opt, object, absolute height-->
    <elevation>
      <!--ro, opt, int, altitude, range:[-900,2700], desc:altitude-->1
    </elevation>
    <azimuth>
      <!--ro, opt, int, panning angle, range:[0,3600], desc:panning angle-->1
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, int, zooming parameter, range:[0,1000], desc:zooming parameter-->1
    </absoluteZoom>
  </AbsoluteHigh>
</PTZStatus>
```

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19.4.5 PTZ Scan

19.4.5.1 Get a pattern scan path parameters of a specified channel

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/patterns/<patternID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patternID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPattern xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, PTZ scanning pattern, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable, desc:whether to enable-->true
  </enabled>
  <id>
    <!--ro, req, int, ID-->1
  </id>
  <space>
    <!--ro, req, int, read-only,range:[0,100], range:[0,100], step:1-->1
  </space>
</PTZPattern>
```

19.4.5.2 Delete a pattern by channel

Request URL

DELETE /ISAPI/PTZCtrl/channels/<channelID>/patterns/<patternID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patternID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.4.5.3 Set path parameters of a pattern scan for a specified channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/patterns/<patternID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patternID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPattern xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, PTZ scanning pattern, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, int, ID-->1
  </id>
  <space>
    <!--req, int, read-only,range:[0,100], range:[0,100], step:1-->1
  </space>
</PTZPattern>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.5 People Counting

19.5.1 Real-Time People Counting

19.5.1.1 Clear the people counting data in the flash memory

Request URL

PUT /ISAPI/ContentMgmt/FlashStorage/remove/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FlashStorageRemove xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{opt, string, protocolVersion}-->
  <coutingRemove>
    <!--opt, bool-->true
  </coutingRemove>
</FlashStorageRemove>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.5.1.2 Get the arming schedule of people counting for all channels

Request URL

GET /ISAPI/Event/schedules/countings

Query Parameter

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None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CountingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{opt, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->1
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->counting
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, string}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</CountingScheduleList>
```

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19.5.1.3 Set arming schedules of people counting of all channels

Request URL

PUT /ISAPI/Event/schedules/countings

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CountingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{opt, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->1
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->counting
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming time, subType:object, attr:size{opt, string}-->
      <TimeBlock>
        <!--opt, object, time-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:days of the week-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range, desc:ISO8601-->
          <beginTime>
            <!--req, time, start time, desc:ISO8601-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time, desc:ISO8601-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--opt, object, time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time, desc:ISO8601-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time, desc:ISO8601-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</CountingScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.5.1.4 Set the arming schedule of people counting for a specified channel

Request URL

PUT /ISAPI/Event/schedules/countings/counting-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

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Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->counting
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{req, string}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00[+08:00]
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00[+08:00]
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00[+08:00]
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00[+08:00]
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.5.1.5 Get the arming schedule of people counting for a specified channel

Request URL

GET /ISAPI/Event/schedules/countings/counting-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->counting
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, string}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.5.1.6 Set the linkage parameters of people counting

Request URL

PUT /ISAPI/Event/triggers/counting-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->counting-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->counting
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <AudioAction>
        <!--opt, object-->
        <audioID>
          <!--req, int-->0
        </audioID>
        <alarmTimes>
          <!--req, int, range:[0,255]-->0
        </alarmTimes>
      </AudioAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.5.1.7 Get the linkage parameters of people counting

Request URL

GET /ISAPI/Event/triggers/counting-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->counting-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->counting
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <AudioAction>
        <!--ro, opt, object-->
        <audioID>
          <!--ro, req, int-->0
        </audioID>
        <alarmTimes>
          <!--ro, req, int, range:[0,255]-->0
        </alarmTimes>
      </AudioAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.5.1.8 Get the shielded area parameters of people counting event

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/Shield/peopleCounting

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Shield xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, enum, subType:string-->true
  </enabled>
  <RegionList>
    <!--ro, opt, array, subType:object-->
    <Region>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->1
      </id>
      <enabled>
        <!--ro, req, enum, subType:string-->true
      </enabled>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, req, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </RegionList>
</Shield>
```

19.5.1.9 Set the shielded area parameters of people counting event

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/Shield/peopleCounting

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Shield xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, enum, subType:string-->true
  </enabled>
  <RegionList>
    <!--opt, array, subType:object-->
    <Region>
      <!--opt, object-->
      <id>
        <!--req, string-->1
      </id>
      <enabled>
        <!--req, enum, subType:string-->true
      </enabled>
      <RegionCoordinatesList>
        <!--opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </RegionList>
</Shield>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.5.1.10 Get the people counting parameters of one video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/counting

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<Counting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
```

```
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1000
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <OverlayConfiguration>
    <!--ro, opt, object, counting OSD parameters-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
    <OSDType>
      <!--ro, opt, enum, OSD type, subType:string, desc:it is valid when enabled is "true","enter,leave,entreLeave,peoplePassing"-->none
    </OSDType>
    <child>
      <!--ro, opt, bool, whether to enable children counting for OSD-->true
    </child>
  </OverlayConfiguration>
  <CountingRegionType>
    <!--ro, req, enum, detection type, subType:string, desc:read-only,detection type: "region,line"-->region
  </CountingRegionType>
  <CountingRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <CountingRegion>
      <!--ro, opt, object, detection area-->
      <id>
        <!--ro, req, int, detection area ID-->1
      </id>
      <direction>
        <!--ro, opt, object-->
        <startPoint>
          <!--ro, req, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </startPoint>
        <endPoint>
          <!--ro, req, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </endPoint>
      </direction>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[3,10]-->
      </RegionCoordinatesList>
      <RuleRegionCoordinatesList>
        <!--ro, opt, array, rule frame, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RuleRegionCoordinatesList>
    </CountingRegion>
  </CountingRegionList>
  <CountingLineItemList>
    <!--ro, opt, array, list of detection lines, subType:object-->
    <CountingLineItem>
      <!--ro, opt, object, detection line-->
      <id>
        <!--ro, req, int, detection line ID-->1
      </id>
      <direction>
        <!--ro, req, object, detection line direction-->
        <startPoint>
          <!--ro, req, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </startPoint>
        <endPoint>
          <!--ro, req, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
```

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```
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </endPoint>
      </direction>
      <sensitivityLevel>
        <!--ro, opt, int, target detection sensitivity, range:[1,100], desc:the default value is 50-->50
      </sensitivityLevel>
      <LineCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,4]-->
        <Coordinates>
          <!--ro, req, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </Coordinates>
      </LineCoordinatesList>
    </CountingLineItem>
  </CountingLineItemList>
  <dataUploadCycle>
    <!--ro, opt, int, interval for uploading data, unit:min-->1
  </dataUploadCycle>
  <SECUploadEnabled>
    <!--ro, opt, bool, uploading mechanism for each second-->true
  </SECUploadEnabled>
  <EmailReport>
    <!--ro, opt, object-->
    <DayReport>
      <!--ro, opt, bool, daily report-->true
    </DayReport>
    <WeekReport>
      <!--ro, opt, bool, weekly report-->true
    </WeekReport>
    <MonthReport>
      <!--ro, opt, bool, monthly report-->true
    </MonthReport>
    <YearReport>
      <!--ro, opt, bool, yearly report-->true
    </YearReport>
    <reportFormat>
      <!--ro, opt, string, email report format, desc:"excel" (default), "csv", "txt", "xml"-->excel
    </reportFormat>
  </EmailReport>
  <CountingCalibrate>
    <!--ro, opt, object-->
    <calibrateType>
      <!--ro, opt, enum, calibration type, subType:string, desc:"automatic,manual,no",calibration types,including Auto,Manual,and None-->automatic
    </calibrateType>
    <SetupParam>
      <!--ro, req, object, installation parameters-->
      <height>
        <!--ro, opt, float, height, range:[200.0,500.0], unit:cm, desc:the default value is 300-->300.0
      </height>
      <tiltAngle>
        <!--ro, opt, float, angle of pitch, range:[-90.0,90.0], unit:°, desc:the default value is 0-->0.0
      </tiltAngle>
      <heelAngle>
        <!--ro, opt, float, angle of inclination, range:[0.0,180.0], unit:°, desc:the default value is 0-->0.0
      </heelAngle>
    </SetupParam>
    <CountingArea>
      <!--ro, req, object, counting area (in red frame)-->
      <RegionCoordinatesList>
        <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </CountingArea>
    <AutomaticCalib>
      <!--ro, opt, object, it is valid when calibrateType is "automaticCalib"-->
      <CalibRegion>
        <!--ro, opt, object, calibration area (in green frame)-->
        <RegionCoordinatesList>
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
```

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```
        </CalibRegion>
      </AutomaticCalib>
    </CountingCalibrate>
    <HeightFilterOverlay>
      <!--ro, opt, object, height filter-->
      <enabled>
        <!--ro, opt, bool-->true
      </enabled>
      <heightFilter>
        <!--ro, opt, int, height filter, range:[0,200], unit:cm-->0
      </heightFilter>
    </HeightFilterOverlay>
    <countingType>
      <!--ro, opt, enum, method to trigger counting, subType:string, desc:"none", "alarmInputTrigger", "videoTrigger"-->none
    </countingType>
    <signalType>
      <!--ro, opt, enum, signal type, subType:string, desc:"level", "pulses"-->level
    </signalType>
    <RS485TransmissionEnabled>
      <!--ro, opt, bool, "false,true"-->true
    </RS485TransmissionEnabled>
    <dailyResetTime>
      <!--ro, opt, time, e.g.,"22:00:00+08:00", desc:e.g.,"22:00:00+08:00"-->22:00:00+08:00
    </dailyResetTime>
    <streamOverlayRuleInfos>
      <!--ro, opt, bool, whether to enable overlaying rule information in stream-->true
    </streamOverlayRuleInfos>
    <detectThreshold>
      <!--ro, opt, int, global detection threshold, range:[0,100], desc:the default value is 90-->90
    </detectThreshold>
    <ChildFilter>
      <!--ro, opt, object, counting children-->
      <enabled>
        <!--ro, req, bool, whether to enable the function-->true
      </enabled>
      <heightThreshold>
        <!--ro, opt, int, height threshold of children,the counting will start when the height of children is higher than configured height threshold,unit:
cm,value range: [100,200], range:[100,200], unit:cm, desc:height threshold of children,the counting will start when the height of children is higher than
configured height threshold,unit: cm,value range: [100,200]-->150
      </heightThreshold>
    </ChildFilter>
    <MisinfoFilter>
      <!--ro, opt, object, false alarm filtering-->
      <enabled>
        <!--ro, req, bool, whether to enable the function-->true
      </enabled>
      <threshold>
        <!--ro, opt, int, threshold of false alarm filtering, range:[0,100], desc:the value ranges from 0 to 100 and the default value is 15. Larger threshold
value corresponds to higher precision-->15
      </threshold>
      <timesLimit>
        <!--ro, opt, int, judge times of false alarm, range:[1,5], desc:value range: [1,5], default value: 3, more attempts corresponds to higher precision--
>3
      </timesLimit>
    </MisinfoFilter>
    <detectionMode>
      <!--ro, opt, enum, target detection mode, subType:string, desc:detection mode: single-detect based on tracking algorithm,double-detect based on depth
map,double_single-detect based on depth map mainly and tracking algorithm secondarily,single_double-detect based on tracking algorithm mainly and depth map
secondarily-->single
    </detectionMode>
    <TrajectoryCountFilter>
      <!--ro, opt, object, pattern counting filtering-->
      <enabled>
        <!--ro, req, bool, whether to enable the function, desc:whether to enable the function-->true
      </enabled>
      <movementDisplacement>
        <!--ro, opt, int, displacement range, range:[0,200], unit:cm-->40
      </movementDisplacement>
      <residenceTime>
        <!--ro, opt, float, dwell time duration, range:[0.0,10.0], unit:s-->0.1
      </residenceTime>
    </TrajectoryCountFilter>
    <RegionsDirectionList>
      <!--ro, opt, array, counting direction, subType:object-->
    </RegionsDirectionList>
    <maintenanceModeEnabled>
      <!--ro, opt, bool, whether to enable maintenance mode-->true
    </maintenanceModeEnabled>
  </Counting>
```

19.5.1.11 Set the people counting parameters for a specific channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<Counting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <OverlayConfiguration>
    <!--opt, object, counting OSD parameters-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
    <OSDType>
      <!--opt, enum, OSD type, subType:string, desc:it is valid when enabled is "true","enter,Leave,entreLeave,peoplePassing"-->none
    </OSDType>
    <child>
      <!--opt, bool, whether to enable children counting for OSD-->true
    </child>
  </OverlayConfiguration>
  <CountingRegionType>
    <!--req, enum, detection type, subType:string, desc:"region", "line"-->region
  </CountingRegionType>
  <CountingRegionList>
    <!--opt, array, detection area list, subType:object-->
    <CountingRegion>
      <!--opt, object, detection area-->
      <id>
        <!--req, int, detection area ID-->1
      </id>
      <direction>
        <!--req, object-->
        <startPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </startPoint>
        <endPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </endPoint>
      </direction>
      <RuleRegionCoordinatesList>
        <!--opt, array, rule frame, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RuleRegionCoordinatesList>
    </CountingRegion>
  </CountingRegionList>
  <CountingLineItemList>
    <!--opt, array, list of detection lines, subType:object-->
    <CountingLineItem>
      <!--opt, object, detection line-->
      <id>
        <!--req, int, detection line ID-->1
```

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```
      </id>
      <direction>
        <!--req, object, detection line direction-->
        <startPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </startPoint>
        <endPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </endPoint>
      </direction>
      <LineCoordinatesList>
        <!--opt, array, subType:object, range:[0,2]-->
        <Coordinates>
          <!--req, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </Coordinates>
      </LineCoordinatesList>
    </CountingLineItem>
  </CountingLineItemList>
  <dataUploadCycle>
    <!--opt, int, interval for uploading data, unit:min-->1
  </dataUploadCycle>
  <SECUploadEnabled>
    <!--opt, bool, uploading data for each second-->true
  </SECUploadEnabled>
  <EmailReport>
    <!--opt, object-->
    <DayReport>
      <!--opt, bool, daily report-->true
    </DayReport>
    <WeekReport>
      <!--opt, bool, weekly report-->true
    </WeekReport>
    <MonthReport>
      <!--opt, bool, monthly report-->true
    </MonthReport>
    <YearReport>
      <!--opt, bool, yearly report-->true
    </YearReport>
    <reportFormat>
      <!--opt, string, email report format, desc:"excel" (default), "csv", "txt", "xml"-->excel
    </reportFormat>
  </EmailReport>
  <CountingCalibrate>
    <!--opt, object-->
    <calibrateType>
      <!--opt, enum, calibration type, subType:string, desc:"automatic,manual,no",calibration types,including Auto,Manual,and None-->automatic
    </calibrateType>
    <SetupParam>
      <!--req, object, installation parameters-->
      <height>
        <!--opt, float, height, range:[200.0,500.0], unit:cm, desc:the default value is 300-->300.0
      </height>
      <tiltAngle>
        <!--opt, float, angle of pitch, range:[-90.0,90.0], unit:°, desc:the default value is 0-->0.0
      </tiltAngle>
      <heelAngle>
        <!--opt, float, angle of inclination, range:[0.0,180.0], unit:°, desc:the default value is 0-->0.0
      </heelAngle>
    </SetupParam>
    <CountingArea>
      <!--req, object, counting area (in red frame)-->
      <RegionCoordinatesList>
        <!--opt, array, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </CountingArea>
    <AutomaticCalib>
```

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```
      <!--opt, object, auto calibration-->
      <CalibRegion>
        <!--opt, object, calibration area (in green frame)-->
        <RegionCoordinatesList>
          <!--opt, array, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </CalibRegion>
    </AutomaticCalib>
  </CountingCalibrate>
  <HeightFilterOverlay>
    <!--opt, object-->
    <enabled>
      <!--opt, bool-->true
    </enabled>
    <heightFilter>
      <!--opt, int, filter height, range:[0,200], unit:cm-->0
    </heightFilter>
  </HeightFilterOverlay>
  <countingType>
    <!--opt, enum, method to trigger counting, subType:string, desc:method to trigger counting: "none,alarmInputTrigger,videoTrigger"-->none
  </countingType>
  <dailyResetTime>
    <!--opt, time, e.g.,"22:00:00+08:00", desc:e.g.,"22:00:00+08:00"-->22:00:00+08:00
  </dailyResetTime>
  <streamOverlayRuleInfos>
    <!--opt, bool, whether to enable overlaying rule information in stream-->true
  </streamOverlayRuleInfos>
  <detectThreshold>
    <!--opt, int, global detection threshold, range:[0,100], desc:the default value is 90-->90
  </detectThreshold>
  <ChildFilter>
    <!--opt, object, counting children-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <heightThreshold>
      <!--opt, int, threshold of children's height, range:[100,200], unit:cm, desc:the counting will start when the height of children is higher than the
configured height threshold-->150
    </heightThreshold>
  </ChildFilter>
  <MisinfoFilter>
    <!--opt, object, whether to filter false alarms-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <threshold>
      <!--opt, int, threshold of false alarm filtering, range:[0,100], desc:the value ranges from 0 to 100 and the default value is 15. Larger threshold
value corresponds to higher precision-->15
    </threshold>
    <timesLimit>
      <!--opt, int, limit of times for checking false alarms, range:[1,5], desc:the value ranges from 1 to 5 and the default value is 3. Larger value
corresponds to higher precision-->3
    </timesLimit>
  </MisinfoFilter>
  <detectionMode>
    <!--opt, enum, target detection mode, subType:string, desc:"single" (detect based on tracking algorithm), "double" (detect based on depth map),
"double_single" (detect based on depth map mainly and tracking algorithm secondarily), "single_double" (detect based on tracking algorithm mainly and depth
map secondarily)-->single
  </detectionMode>
  <TrajectoryCountFilter>
    <!--opt, object, track counting filtering-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <movementDisplacement>
      <!--opt, int, displacement range, range:[0,200], unit:cm-->40
    </movementDisplacement>
    <residenceTime>
      <!--opt, float, dwell time duration, range:[0.0,10.0], unit:s-->0.1
    </residenceTime>
  </TrajectoryCountFilter>
  <RegionsDirectionList>
    <!--opt, array, counting direction, subType:object-->
  </RegionsDirectionList>
  <maintenanceModeEnabled>
    <!--opt, bool, whether to enable maintenance mode-->true
  </maintenanceModeEnabled>
</Counting>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.5.1.12 Get capability of getting people counting statistics

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CountingCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <CountingRegionType opt="line,region,lineAndRegion">
    <!--ro, opt, string, area type, attr:opt{opt, string}, desc:"line"-->line
  </CountingRegionType>
  <dataUploadCycle opt="1,5,10,15,20,30,60">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
  </dataUploadCycle>
  <isSupportSECUpload>
    <!--ro, opt, bool-->true
  </isSupportSECUpload>
  <isSupportFlashRemoveCouting>
    <!--ro, opt, bool, "true"-->true
  </isSupportFlashRemoveCouting>
  <OSDType opt="enter,leave,enterLeave,
none,enterPeoplePassing,pepolePassing,enterDuplicatePeople,enterLeaveDuplicatePeople,enterLeaveDuplicatePassByShop,enterLeaveDuplicatePassing" def="none">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{opt, string}-->enter
  </OSDType>
  <EmailReport>
    <!--ro, opt, object-->
    <DayReport opt="true,false">
      <!--ro, opt, bool, daily report, attr:opt{req, bool}-->true
    </DayReport>
    <WeekReport opt="true,false">
      <!--ro, opt, bool, weekly report, attr:opt{req, bool}-->true
    </WeekReport>
    <MonthReport opt="true,false">
      <!--ro, opt, bool, monthly report, attr:opt{req, bool}-->true
    </MonthReport>
    <YearReport opt="true,false">
      <!--ro, opt, bool, yearly report, attr:opt{req, bool}-->true
    </YearReport>
    <reportFormat opt="excel,csv,txt,xml">
      <!--ro, opt, string, email report format, attr:opt{req, string}-->excel
    </reportFormat>
  </EmailReport>
  <CountingCalibrate>
    <!--ro, opt, object-->
    <calibrateType opt="automatic,manual,no,smart">
      <!--ro, opt, enum, calibration type, subType:string, attr:opt{req, string}, desc:calibration type-->automatic
    </calibrateType>
    <SetupParam>
      <!--ro, req, object-->
      <height min="200.0" max="500.0">
        <!--ro, opt, float, height, range:[200.0,500.0], unit:cm, attr:min{req, float},max{req, float}-->200.0
      </height>
```

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```
      <tiltAngle min="-90.0" max="90.0">
        <!--ro, opt, float, range:[-90,90], unit:°, attr:min{req, float},max{req, float}-->-90
      </tiltAngle>
      <heelAngle min="0.0" max="180.0">
        <!--ro, opt, float, range:[0,180], unit:°, attr:min{req, float},max{req, float}-->0
      </heelAngle>
    </SetupParam>
    <CountingArea>
      <!--ro, req, object, counting area (red frame)-->
      <RegionCoordinatesList size="4">
        <!--ro, opt, array, rectangle, subType:object, range:[0,4], attr:size{req, string}, desc:rectangle-->
        <RegionCoordinates>
          <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </CountingArea>
    <AutomaticCalib>
      <!--ro, opt, object, automatic calibration-->
      <CalibRegion>
        <!--ro, opt, object, calibration area (green frame)-->
        <RegionCoordinatesList size="4">
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], attr:size{req, string}, desc:rectangle-->
          <RegionCoordinates>
            <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </CalibRegion>
    </AutomaticCalib>
  </CountingCalibrate>
  <HeightFilterOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false" default="false">
      <!--ro, opt, bool, attr:opt{req, bool},default{req, bool}-->true
    </enabled>
    <heightFilter min="0" max="250" default="120">
      <!--ro, opt, int, range:[0,250], unit:cm, attr:min{req, int},max{req, int},default{req, int}-->0
    </heightFilter>
  </HeightFilterOverlay>
  <isSupportCalibrate opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isSupportCalibrate>
  <isSupportPosInfoOverlay opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isSupportPosInfoOverlay>
  <countingType opt="none,alarmInputTrigger,videoTrigger">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->none
  </countingType>
  <alarmInputTrigger_signalType opt="level,pulses">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->level
  </alarmInputTrigger_signalType>
  <videoTrigger_RS485Transmission opt="true,false">
    <!--ro, opt, bool, attr:opt{req, bool}-->true
  </videoTrigger_RS485Transmission>
  <isSupportCountingStatus>
    <!--ro, opt, bool-->true
  </isSupportCountingStatus>
  <isSupportTriggerPeopleCountingDataSearch>
    <!--ro, opt, bool-->true
  </isSupportTriggerPeopleCountingDataSearch>
  <isSupportDailyResetTime>
    <!--ro, opt, bool, whether it supports resetting and clearing by schedule-->true
  </isSupportDailyResetTime>
  <Polyline>
    <!--ro, opt, object, whether it supports polyline-->
    <maxPointNumber>
      <!--ro, req, int-->1
    </maxPointNumber>
  </Polyline>
  <isSupportBusDetectionArea>
    <!--ro, opt, bool-->true
  </isSupportBusDetectionArea>
  <isSupportStreamOverlayRuleInfos>
    <!--ro, opt, bool, whether it supports overlaying intelligent information in stream-->true
  </isSupportStreamOverlayRuleInfos>
  <isSupportMobileDescribeChange>
    <!--ro, opt, bool, this node is for marking the editing of the displayed parameter name of people counting camera-->true
  </isSupportMobileDescribeChange>
  <isSupportSearchDoorStates>
    <!--ro, opt, bool, this node if for supporting the detection of door status triggered by signal of vehicle-mounted device-->true
  </isSupportSearchDoorStates>
  <isSupportRuleRegion>
    <!--ro, opt, bool, whether it supports setting counting area-->true
```

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```
  </isSupportRuleRegion>
  <notSupportStatisticType>
    <!--ro, opt, enum, this node is valid when <isSupportTriggerPeopleCountingDataSearch> is set to "true", subType:string, desc:this node is valid when
<isSupportTriggerPeopleCountingDataSearch> is set to "true"-->enternum
  </notSupportStatisticType>
  <detectThreshold min="0" max="100" def="90">
    <!--ro, opt, int, detection threshold, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->90
  </detectThreshold>
  <ruleRegionEdgesNum min="4" max="10">
    <!--ro, opt, int, capability of edge quantity of rule frame, range:[4,10], attr:min{req, int},max{req, int}-->4
  </ruleRegionEdgesNum>
  <ChildFilter>
    <!--ro, opt, object, children counting-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <heightThreshold min="100" max="200" def="150">
      <!--ro, opt, int, threshold of the height of children, range:[100,200], unit:cm, attr:min{req, int},max{req, int},def{req, int}-->150
    </heightThreshold>
  </ChildFilter>
  <MisinfoFilter>
    <!--ro, opt, object, false alarm filtering-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <threshold min="0" max="100" def="15">
      <!--ro, opt, int, filtering threshold, range:[0,100], attr:min{req, int},max{req, int},def{req, int}, desc:the larger the threshold value, the higher
the precision-->15
    </threshold>
    <timesLimit min="1" max="5" def="3">
      <!--ro, opt, int, prevention attempts of false alarm, range:[1,5], attr:min{req, int},max{req, int},def{req, int}, desc:the larger the number of
attempts, the higher the precision-->3
    </timesLimit>
  </MisinfoFilter>
  <isSupportReverseAlarm>
    <!--ro, opt, bool, whether it supports setting reverse counting alarm-->true
  </isSupportReverseAlarm>
  <detectionMode opt="single,double,double_single,single_double" def="double_single">
    <!--ro, opt, enum, detection mode, subType:string, attr:opt{req, string},def{req, string}, desc:"single" (detect based on tracking algorithm), "double"
(detect based on depth map), "double_single" (detect based on depth map mainly and tracking algorithm secondarily), "single_double" (detect based on
tracking algorithm mainly and depth map secondarily)-->double_single
  </detectionMode>
  <TrajectoryCountFilter>
    <!--ro, opt, object, pattern counting filtering-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <movementDisplacement min="0" max="200" def="40">
      <!--ro, opt, int, motion displacement range, range:[0,200], attr:min{req, int},max{req, int},def{req, int}-->40
    </movementDisplacement>
    <residenceTime min="0.0" max="10.0" def="0.1">
      <!--ro, opt, float, dwell time duration, range:[0.0,10.0], unit:s, attr:min{req, float, unit:s},max{req, float, unit:s},def{req, float, unit:s}-->0.1
    </residenceTime>
  </TrajectoryCountFilter>
  <isSupportParameterExport>
    <!--ro, opt, bool, whether it supports one-touch exporting people counting parameters-->true
  </isSupportParameterExport>
  <MultiRegions min="1" max="8">
    <!--ro, opt, int, number of counting regions, attr:min{req, int},max{req, int}, desc:this node will not be returned if people counting in multiple
regions is not supported-->1
  </MultiRegions>
  <isSupportRegionsDirection>
    <!--ro, opt, bool, whether it supports configuring counting direction-->true
  </isSupportRegionsDirection>
  <maintenanceModeEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable maintenance mode, attr:opt{req, bool}-->true
  </maintenanceModeEnabled>
  <flowDirectionAnalysis opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->false
  </flowDirectionAnalysis>
  <isSupportRemoveDuplicate>
    <!--ro, opt, bool-->true
  </isSupportRemoveDuplicate>
  <isSupportCaptureOverlap>
    <!--ro, opt, bool-->true
  </isSupportCaptureOverlap>
  <InterPupileRegion>
    <!--ro, opt, object-->
    <positionX>
      <!--ro, req, int, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,1000]-->50
    </positionY>
    <width min="0" max="1000">
      <!--ro, req, int, range:[0,1000], attr:min{opt, string},max{opt, string}-->50
    </width>
    <height>
      <!--ro, req, int, range:[0,1000]-->50
    </height>
  </InterPupileRegion>
  <MaxInterPupilRegion>
    <!--ro, opt, object-->
```

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```
      <positionX>
        <!--ro, req, int, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,1000]-->50
      </positionY>
      <width min="50" max="50">
        <!--ro, req, int, range:[0,1000], attr:min{opt, string},max{opt, string}-->50
      </width>
      <height>
        <!--ro, req, int, range:[0,1000]-->50
      </height>
    </MaxInterPupilRegion>
  </CountingCap>
```

19.5.1.13 Set the parameters of a specific video input channel of the device

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting/lineItem/<lineRuleID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
lineRuleID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<CountingLineItem xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, int, detection line ID-->1
  </id>
  <direction>
    <!--req, object, detection line direction-->
    <StartPoint>
      <!--req, object, start point-->
      <positionX>
        <!--req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate-->1
      </positionY>
    </StartPoint>
    <EndPoint>
      <!--req, object, end point-->
      <positionX>
        <!--req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate-->1
      </positionY>
    </EndPoint>
  </direction>
  <sensitivityLevel>
    <!--req, int, target detection sensitivity, range:[1,100], desc:the default value is 50-->50
  </sensitivityLevel>
  <spaceGenerationSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </spaceGenerationSpeed>
  <timeGenerationSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </timeGenerationSpeed>
  <countingSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </countingSpeed>
  <detectionType>
    <!--opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
  </detectionType>
  <objectSizeCorrection>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </objectSizeCorrection>
  <LineCoordinatesList>
    <!--opt, array, subType:object-->
    <Coordinates>
      <!--req, object, coordinates-->
      <positionX>
        <!--req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate-->1
      </positionY>
    </Coordinates>
  </LineCoordinatesList>
  <polyline>
    <!--opt, bool-->true
  </polyline>
</CountingLineItem>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.5.1.14 Export the people counting parameters

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Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/parameterExport

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

Binary Data

19.5.1.15 Set parameters for overlaying people counting POS information on the stream of a video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting/posInfoOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PosInfoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters for overlaying people counting POS information on the stream of a video input channel, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--opt, bool, whether to enable-->true
  </enabled>
  <type>
    <!--opt, enum, type, subType:string, desc:"inOut" (entered and exited information), "inOutPassIdHeight" (entered, exited, passed, person/object No., and height information).  "height"-->inOut
  </type>
</PosInfoOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.5.1.16 Get parameters for overlaying people counting POS information on the stream of a video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/posInfoOverlay

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PosInfoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters for overlaying people counting POS information on the stream of a video input channel, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--ro, opt, bool, whether to enable-->true
  </enabled>
  <type opt="inOut,inOutPassIdHeight,height">
    <!--ro, req, enum, type, subType:string, attr:opt{req, string}, desc:"inOut" (entered and exited information), "inOutPassIdHeight" (entered, exited, passed, person/object No., and height information),"height"-->inOut
  </type>
</PosInfoOverlay>
```

19.5.1.17 Get the capability of overlaying people counting POS information on the stream of a video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/posInfoOverlay/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PosInfoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the capability of overlaying people counting POS information on the stream of a video input channel, attr:version{opt, string, protocolVersion}-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable, attr:opt{opt, string}-->true
  </enabled>
  <type opt="inOut,inOutPassIdHeight,height">
    <!--ro, req, enum, type, subType:string, attr:opt{req, string}, desc:"inOut", "inOutPassIdHeight", "height"-->inOut
  </type>
</PosInfoOverlay>
```

19.5.1.18 Set the parameters of a specific video input channel of the device

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<CountingRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, area ID-->1
  </id>
  <Direction>
    <!--req, object, direction-->
    <StartPoint>
      <!--req, object, start point, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </StartPoint>
    <EndPoint>
      <!--req, object, end point, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </EndPoint>
  </Direction>
  <sensitivityLevel>
    <!--req, int, target detection sensitivity, range:[1,100], desc:the default value is 50-->50
  </sensitivityLevel>
  <spaceGenerationSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </spaceGenerationSpeed>
  <timeGenerationSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </timeGenerationSpeed>
  <countingSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </countingSpeed>
  <detectionType>
    <!--opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
  </detectionType>
  <objectSizeCorrection>
    <!--opt, int, range:[1,100]-->50
  </objectSizeCorrection>
  <RegionCoordinatesList>
    <!--req, array, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--req, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <RuleRegionCoordinatesList>
    <!--opt, array, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--req, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RuleRegionCoordinatesList>
  <RuleRegionLimit>
    <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->1
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->1
    </positionY>
    <height>
      <!--req, int, height, range:[0,1000]-->1
    </height>
    <width>
      <!--req, int, width, range:[0,1000]-->1
    </width>
  </RuleRegionLimit>
  <enabled>
    <!--opt, bool, whether to enable the function, desc:whether to enable the function-->true
  </enabled>
</CountingRegion>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.5.1.19 Reset the people counting statistics

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting/resetCount

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.5.1.20 Set parameters of reverse entering alarm

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting/reverseAlarm?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
{
    "reverseAlarm": [
    /*opt, array, subType:object*/
        {
            "id":  1,
            /*req, int, armed region ID*/
            "enabled":  true
            /*req, bool, whether to enable reverse entering alarm*/
        }
    ]
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

19.5.1.21 Get parameters of people reverse entering alarm

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/reverseAlarm?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "reverseAlarm": [
    /*ro, opt, array, configuration of people reverse entering alarm, subType:object*/
        {
            "id":  1,
            /*ro, req, int, armed region ID*/
            "enabled":  true
            /*ro, req, bool, whether to enable reverse entering alarm*/
        }
    ]
}
```

19.5.1.22 Search people counting report data

Request URL

POST /ISAPI/System/Video/inputs/channels/<channelID>/counting/search

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<CountingStatisticsDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, search people counting report data, attr:version{opt, string, protocolVersion}-->
  <statisticType>
    <!--req, enum, data type, subType:string, desc:"enternum,exitnum,peoplePassing,all"-->enternum
  </statisticType>
  <reportType>
    <!--req, enum, report type, subType:string, desc:"daily,weekly,monthly,yearly"-->daily
  </reportType>
  <timeSpanList>
    <!--opt, array, subType:object-->
    <timeSpan>
      <!--opt, object, time range-->
      <startTime>
        <!--req, datetime, start time-->1970-01-01T00:00:00+08:00
      </startTime>
      <endTime>
        <!--req, datetime, end time-->1970-01-01T00:00:00+08:00
      </endTime>
    </timeSpan>
  </timeSpanList>
  <triggerPeopleCountingData>
    <!--opt, bool, whether to enable people counting-->true
  </triggerPeopleCountingData>
  <PagingSearchCond>
    <!--opt, object, this node is valid when <triggerPeopleCountingData> is set to true-->
    <searchID>
      <!--req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two
searching, the search history will be saved in the memory to speed up next searching-->test
    </searchID>
    <searchResultPosition>
      <!--req, int-->1
    </searchResultPosition>
    <maxResults>
      <!--req, int-->1
    </maxResults>
  </PagingSearchCond>
  <MinTimeInterval>
    <!--opt, enum, minimum time interval, subType:string, desc:minimum time interval: quarter,half,hour,day,month,week-->quarter
  </MinTimeInterval>
  <child>
    <!--opt, bool, whether supports counting children only, desc:true (yes), false (counting both adults and children)-->true
  </child>
  <FaceAttribute>
    <!--opt, object, facial features, desc:facial features-->
    <faceExpression>
      <!--opt, enum, facial expression, subType:string, desc:facial expression-->unknown
    </faceExpression>
    <gender>
      <!--opt, enum, gender, subType:string, desc:gender-->unknown
    </gender>
    <mask>
      <!--opt, enum, whether the person wears a mask, subType:string, desc:whether the person wears a mask-->unknown
    </mask>
    <ageGroup>
      <!--opt, enum, age group, subType:string, desc:age group-->unknown
    </ageGroup>
    <glass>
      <!--opt, enum, subType:string-->yes
    </glass>
  </FaceAttribute>
  <regionID>
    <!--opt, string, region ID, desc:region ID-->1
  </regionID>
  <HumanBodyAttribute>
    <!--opt, object, human body attributes, desc:human body attributes-->
    <things>
      <!--opt, enum, "unknown", "yes", "no", subType:string, desc:"unknown", "yes", "no"-->unknown
    </things>
  </HumanBodyAttribute>
  <statisticTypeList>
    <!--opt, array, subType:object, range:[1,13]-->
    <statisticType>
      <!--opt, enum, subType:string-->enternum
    </statisticType>
  </statisticTypeList>
</CountingStatisticsDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CountingStatisticsResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, search people counting report data, attr:version{opt, string, protocolVersion}-->
  <responseStatus>
    <!--ro, req, bool-->true
  </responseStatus>
  <responseStatusStrg>
```

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```
    <!--ro, req, string, "MORE"-more items to be searched,"OK"-search completed-->test
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int-->1
  </numOfMatches>
  <matchList>
    <!--ro, opt, array, subType:object-->
    <matchElement>
      <!--ro, opt, object-->
      <timeSpan>
        <!--ro, opt, object, time range-->
        <startTime>
          <!--ro, req, datetime, start time-->1970-01-01T00:00:00+08:00
        </startTime>
        <endTime>
          <!--ro, req, datetime, end time-->1970-01-01T00:00:00+08:00
        </endTime>
      </timeSpan>
      <enterCount>
        <!--ro, opt, int-->1
      </enterCount>
      <exitCount>
        <!--ro, opt, int-->1
      </exitCount>
      <peoplePassingCount>
        <!--ro, opt, int-->1
      </peoplePassingCount>
      <duplicatePeopleCount>
        <!--ro, opt, int-->1
      </duplicatePeopleCount>
      <duplicateEnter>
        <!--ro, opt, int-->1
      </duplicateEnter>
      <duplicateExit>
        <!--ro, opt, int-->1
      </duplicateExit>
      <duplicatePass>
        <!--ro, opt, int-->1
      </duplicatePass>
      <FaceExpressionList>
        <!--ro, opt, array, subType:object-->
        <FaceExpression>
          <!--ro, opt, object, facial expression-->
          <value>
            <!--ro, req, enum, subType:string-->unknown
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </FaceExpression>
      </FaceExpressionList>
      <GenderList>
        <!--ro, opt, array, subType:object-->
        <Gender>
          <!--ro, opt, object, gender-->
          <value>
            <!--ro, req, enum, subType:string-->unknown
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </Gender>
      </GenderList>
      <MaskList>
        <!--ro, opt, array, subType:object-->
        <Mask>
          <!--ro, opt, object, whether the person wears a mask-->
          <value>
            <!--ro, req, enum, subType:string-->unknown
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </Mask>
      </MaskList>
      <AgeGroupList>
        <!--ro, opt, array, subType:object-->
        <AgeGroup>
          <!--ro, opt, object, age group-->
          <value>
            <!--ro, req, enum, subType:string-->unknown
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </AgeGroup>
      </AgeGroupList>
      <GlassList>
        <!--ro, opt, array, subType:object-->
        <Glass>
          <!--ro, opt, object-->
          <value>
            <!--ro, req, enum, subType:string-->unknown
          </value>
```

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```
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </Glass>
      </GlassList>
      <RegionList>
        <!--ro, opt, array, subType:object-->
        <Region>
          <!--ro, opt, object, region-->
          <id>
            <!--ro, req, int-->1
          </id>
          <enterCount>
            <!--ro, opt, int-->1
          </enterCount>
          <exitCount>
            <!--ro, opt, int-->1
          </exitCount>
          <passingCount>
            <!--ro, opt, int-->1
          </passingCount>
          <duplicateEnter>
            <!--ro, opt, int-->1
          </duplicateEnter>
          <duplicateExit>
            <!--ro, opt, int-->1
          </duplicateExit>
        </Region>
      </RegionList>
      <ThingsList>
        <!--ro, opt, array, subType:object-->
        <Things>
          <!--ro, opt, object-->
          <value>
            <!--ro, req, enum, "unknown", "yes", "no", subType:string, desc:"unknown", "yes", "no"-->unknown
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </Things>
      </ThingsList>
      <enterRate>
        <!--ro, opt, float, range:[0.00,100.00], step:0.01-->1
      </enterRate>
    </matchElement>
  </matchList>
</CountingStatisticsResult>
```

19.5.1.23 Get people counting status by channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/status

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CountingStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <status>
    <!--ro, req, enum, "counting,stopped,paused", subType:string, desc:"counting", "stopped", "paused"-->counting
  </status>
  <time>
    <!--ro, req, time, operation time (local time + time zone)-->22:00:00+08:00
  </time>
  <doorStatus>
    <!--ro, opt, enum, door triggering status, subType:string, desc:"open", "close", "N_A"-->open
  </doorStatus>
</CountingStatus>
```

19.5.1.24 Get the capability of people counting data replenishment

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Request URL

GET /ISAPI/System/Video/inputs/channels/counting/collection/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CollectionDescriptionCap": {
    /*ro, opt, object*/
        "timeRange":  15,
        /*ro, req, int, supported time range for people counting replenishment*/
        "timeSpanMaxNum":  1,
        /*ro, req, int, the maximum number of time buckets supported for one time search integer*/
        "resultMaxNum":  100,
        /*ro, req, int, the maximum number of items can be searched*/
        "channelMaxNum":  1
        /*ro, req, int, the maximum number of channels supported for one time search*/
    }
}
```

19.5.1.25 Search for people counting data replenishment

Request URL

POST /ISAPI/System/Video/inputs/channels/counting/collection?format=json

Query Parameter

None

Request Message

```
{
    "CollectionDescription": {
    /*opt, object, search for people counting data replenishment*/
        "searchID":  "test",
        /*req, string, search ID, desc:this node is used to check whether searches are the same one. Due to device performance limitation, you can search
asynchronously by repeatedly applying the search conditions and getting the search progress (the searchID should be the same)*/
        "searchResultPosition":  0,
        /*req, int, initial position of search result list, desc:in a single search, if you cannot get all the records in the result list, you can mark the
end position and get the following records after the marked position in the next search*/
        "maxResults":  100,
        /*req, int, the maximum number of search results this time*/
        "timeSpanList": [
        /*opt, array, time period list, subType:object*/
            {
                "startTime":  "2004-05-03T17:30:08+08:00",
                /*req, string, start time*/
                "endTime":  "2004-05-03T17:30:08+08:00"
                /*req, string, end time*/
            }
        ],
        "channels": [1, 2, 3]
        /*opt, array, array, subType:int, desc:sub type in the array is integer32, linked channel No., if the channel information is not specified, it
indicates searching for all channels*/
    }
}
```

Response Message

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```
{
    "CollectionResult": {
    /*ro, opt, object, search for people counting data replenishment*/
        "responseStatusStrg":  "OK",
        /*ro, req, string, search status description, desc:search status description OK- Searching ended,NO MATCHES-No matched data found,MORE-Search again
for more results,the max. length is 32,{dep if errcode == 1 && errMsg == ok}*/
        "numOfMatches":  1,
        /*ro, req, int, returned number of results for current search*/
        "totalMatches":  1,
        /*ro, req, int, total number of matched results*/
        "targets": [
        /*ro, opt, array, statistics results, subType:object*/
            {
                "startTime":  "2017-08-25T11:34:59+08:00",
                /*ro, req, string, start time of statistics*/
                "endTime":  "2017-08-25T11:34:59+08:00",
                /*ro, req, string, end time of statistics*/
                "channel":  1,
                /*ro, req, int, channel No.*/
                "enter":  34,
                /*ro, req, int, the number of people entered*/
                "exit":  39,
                /*ro, req, int, the number of people exited*/
                "pass":  0
                /*ro, req, int, the number of people passed by*/
            }
        ],
        "collectionStatus":  "syncing"
        /*ro, opt, enum, the status of people counting data replenishment, subType:string, desc:syncing indicates that the people counting data
synchronization is not completed yet; the platform will not parse the targets statistics result and will continue searching until the "finished" is returned
if this node does not exist, it indicates that the replenishment result will be returned and the platform will directly parse the targets no matter the
synchronization is completed or not; "syncing" (synchronizing the IPC data), "finished" (synchronization completed)*/
    }
}
```

19.5.1.26 People counting event

EventType:PeopleCounting

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, people counting event details, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPv4 address-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 Address-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, device's communication port No.-->1
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol, subType:string, desc:"HTTP, HTTPS, EHome". The value should be "EHome" when ISAPI protocol is
transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->test
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No.-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, event occurred time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"PeopleCounting" (people counting)-->PeopleCounting
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:"active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <deviceID>
```

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```
    <!--ro, opt, string, PUID, range:[1,64], desc:required in Ehome alarm, corresponding to deviceID in /ISAPI/System/Network/Ehome-->test
  </deviceID>
  <peopleCounting>
    <!--ro, opt, object, people counting-->
    <statisticalMethods>
      <!--ro, opt, enum, people counting method, subType:string, desc:"realTime" (real-time people counting), "timeRange" (scheduled people counting),
"signalTrigger" (alarm input triggered people counting)-->realTime
    </statisticalMethods>
    <RealTime>
      <!--ro, opt, object, real-time, desc:it is required when statisticalMethods is "realTime"-->
      <time>
        <!--ro, req, datetime, people counting time-->1970-01-01T00:00:00+08:00
      </time>
    </RealTime>
    <TimeRange>
      <!--ro, opt, object, time period for real-time people counting, desc:it is required when statisticalMethods is "timeRange" or "signalTrigger".-->
      <startTime>
        <!--ro, req, datetime, start time of people counting-->1970-01-01T00:00:00+08:00
      </startTime>
      <endTime>
        <!--ro, req, datetime, end time of people counting-->1970-01-01T00:00:00+08:00
      </endTime>
    </TimeRange>
    <enter>
      <!--ro, opt, int, number of entered people-->1
    </enter>
    <exit>
      <!--ro, opt, int, number of exited people-->1
    </exit>
    <pass>
      <!--ro, opt, int, passers-by number, desc:passers-by number-->1
    </pass>
    <duplicatePeople>
      <!--ro, opt, int, number of duplicated people, desc:number of duplicated people-->1
    </duplicatePeople>
    <FlowStatisticsList>
      <!--ro, opt, array, flow direction counting list, subType:object-->
      <FlowStatistics>
        <!--ro, opt, object, flow direction counting, desc:for deduplicated people counting statistics, the flow direction analysis is only valid in the
following circumstances: the rule area is a quadrilateral, the counting broken line is formed with four points, and the counting line has only two points
that overlap with the rule area line-->
        <countingLineID>
          <!--ro, opt, int, counting line ID-->1
        </countingLineID>
        <pass>
          <!--ro, opt, int, passers-by number, desc:people who enter the counting area but do not cross the counting line-->1
        </pass>
        <movingDirectionList>
          <!--ro, opt, array, people moving direction list, subType:object, desc:count the moving direction of people who enter the counting area and cross
the counting line-->
          <MovingDirection>
            <!--ro, opt, object, people moving direction-->
            <countingLineSegmentID>
              <!--ro, opt, int, area ID, range:[1,3], desc:the four points of the broken line are marked as 1, 2, 3, and 4 respectively in counterclockwise
direction from the counting direction. Draw two lines parallel to the counting direction respectively starting from point 2 and point 3 to intersect the
counting rule area. The formed three areas are marked as 1, 2, and 3 in counterclockwise direction from the counting direction-->1
            </countingLineSegmentID>
            <count>
              <!--ro, opt, int, the number of people-->0
            </count>
          </MovingDirection>
        </movingDirectionList>
      </FlowStatistics>
    </FlowStatisticsList>
    <countingSceneMode>
      <!--ro, opt, enum, people counting scene mode, subType:string, desc:"entrance", "exit". The default mode is entrance-->entrance
    </countingSceneMode>
    <deDuplicateEnter>
      <!--ro, opt, int, deduplicated number of entered people-->0
    </deDuplicateEnter>
    <deDuplicateExit>
      <!--ro, opt, int, deduplicated number of exited people-->0
    </deDuplicateExit>
    <faceLibDuplicatePeopleList>
      <!--ro, opt, array, list of duplicated people in the face picture library, subType:object, desc:count the number of duplicated people in every face
picture library-->
      <faceLibDuplicatePeople>
        <!--ro, opt, object, number of duplicated people in the face picture library-->
        <FDID>
          <!--ro, req, string, face picture library ID, desc:check details of the library with /ISAPI/Intelligent/FDLib/search?format=json-->1
        </FDID>
        <name>
          <!--ro, req, string, face picture library name-->11
        </name>
        <duplicate>
          <!--ro, opt, int, number of duplicated people in the library-->1
        </duplicate>
      </faceLibDuplicatePeople>
    </faceLibDuplicatePeopleList>
  </peopleCounting>
  <childCounting>
    <!--ro, opt, object, children counting-->
    <enter>
      <!--ro, opt, int, number of entered people-->1
```

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```
      <!--ro, opt, int, number of entered people-->1
    </enter>
    <exit>
      <!--ro, opt, int, number of exited people-->1
    </exit>
  </childCounting>
  <FaceExpressionList>
    <!--ro, opt, array, facial expression statistics, subType:object-->
    <FaceExpression>
      <!--ro, opt, object, facial expression-->
      <value>
        <!--ro, req, enum, facial attributes, subType:string, desc:"unknown", "poker-faced" (no facial expression), "happy", "surprised", "disgusted",
"sad", "angry", "contemptuous", "panic", "all"-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people corresponding to the face expression-->1
      </count>
    </FaceExpression>
  </FaceExpressionList>
  <GenderList>
    <!--ro, opt, array, gender statistics, subType:object-->
    <Gender>
      <!--ro, opt, object, gender-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "female", "male", "all"-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people corresponding to the gender-->1
      </count>
    </Gender>
  </GenderList>
  <MaskList>
    <!--ro, opt, array, mask statistics, subType:object-->
    <Mask>
      <!--ro, opt, object, mask wearing status-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "yes" (wearing a mask), "no" (not wearing a mask)-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people corresponding to the masks wearing status-->1
      </count>
    </Mask>
  </MaskList>
  <AgeGroupList>
    <!--ro, opt, array, age group statistics, subType:object-->
    <AgeGroup>
      <!--ro, opt, object, age group-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "child", "young", "middle", "old", "infant", "kid", "teenager", "prime", "middleAged"
(the middle age), "all"-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people corresponding to the age group-->1
      </count>
    </AgeGroup>
  </AgeGroupList>
  <GlassList>
    <!--ro, opt, array, glasses statistics, subType:object-->
    <Glass>
      <!--ro, opt, object, glasses wearing status-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "yes" (wearing glasses), "no" (not wearing glasses), "sunglasses" (wearing sunglasses),
"all"-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people corresponding to the glass wearing attribute-->1
      </count>
    </Glass>
  </GlassList>
  <RegionList>
    <!--ro, opt, array, people counting information in the area, subType:object-->
    <Region>
      <!--ro, opt, object, area-->
      <id>
        <!--ro, req, int, area ID-->1
      </id>
      <name>
        <!--ro, opt, string, area name-->test
      </name>
      <enter>
        <!--ro, opt, int, number of entered people-->1
      </enter>
      <exit>
        <!--ro, opt, int, number of exited people-->1
      </exit>
      <totalEnter>
        <!--ro, opt, int, total number of entered people, desc:before resetting on the current day, the number of people will accumulate the number will be
initialized only after manual resetting on the current day or auto resetting on the next day (if the device supports duplicated people counting, the number
will be the deduplicated number of people of the current day)-->1
      </totalEnter>
      <totalExit>
        <!--ro, opt, int, total number of exited people, desc:before resetting on the current day, the number of people will accumulate the number will be
initialized only after manual resetting on the current day or auto resetting on the next day (if the device supports duplicated people counting, the number
will be the deduplicated number of people of the current day)-->1
```

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```
      </totalExit>
      <totalPass>
        <!--ro, opt, int-->1
      </totalPass>
    </Region>
  </RegionList>
  <targetAttrs>
    <!--ro, opt, string, target attributes, range:[1,1024],
desc:transparent transmission field, which corresponds to targetAttrs in applying task, format:
    <![CDATA[{
    /*opt, transparent transmission field, the following three fileds are recommended*/
    "deviceId": "1",
    /*opt, device ID, string, maximum length 64*/
    "deviceChannel": 1,
    /*opt, device channel, integer32*/
    "deviceName": "shebei"
    /*opt, device name, string, maximum length 128*/
    }]]>, example: <![CDATA[{ "deviceId": "1", "deviceChannel": 1, "deviceName": "shebei"}]]>-->test
  </targetAttrs>
  <targetInfo>
    <!--ro, opt, object, target information, desc:corresponding to targetInfo in heat analysis task /ISAPI/SDT/Management/HeatAnalysis/video/AddTask?
format=json. Either targetInfo or targetAttrs is required-->
    <cameraInfo>
      <!--ro, opt, string, camera information-->test
      <cameraIndexCode>
        <!--ro, req, string, camera ID-->test
      </cameraIndexCode>
      <cameraName>
        <!--ro, opt, string, camera name-->test
      </cameraName>
    </cameraInfo>
    <mapFileID>
      <!--ro, opt, string, map file ID-->test
    </mapFileID>
    <storeID>
      <!--ro, opt, string, store ID-->test
    </storeID>
  </targetInfo>
  <customInfo>
    <!--ro, opt, string, custom information-->test
  </customInfo>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle), 2-"hightPerformanceVehicle" (high-performance
vehicle), 3-"personStructModel" (person structural modeling), 4-"vehicleAndPersonStructModel" (person and vehicle structural modeling), 5-"videoFaceDetect"
(video face detection), 6-"videoFaceDetectModel" (video face detection and modeling), 7-"smart" (face detection + human body detection + behavior
detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime", "historyvideo", "Localvideo"-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to
128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
```

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```
    </TaskInfo>
    <bkgUrl>
      <!--ro, opt, string, background picture URL, desc:background picture URL-->test
    </bkgUrl>
    <timeStamp>
      <!--ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format, e.g., 2019-12-30T15:00:00.23+08:00-->test
    </timeStamp>
    <DetectionRegionList>
      <!--ro, opt, array, detection area, subType:object-->
      <DetectionRegionEntry>
        <!--ro, opt, object, detection area-->
        <regionID>
          <!--ro, req, string, area ID-->test
        </regionID>
        <sensitivityLevel>
          <!--ro, opt, int, sensitivity level, range:[0,100]-->1
        </sensitivityLevel>
        <RegionCoordinatesList>
          <!--ro, opt, array, target area, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </DetectionRegionEntry>
    </DetectionRegionList>
    <HatList>
      <!--ro, opt, array, wearing hat statistics, subType:object-->
      <Hat>
        <!--ro, opt, object, whether the person wears a hat-->
        <value>
          <!--ro, req, enum, options, subType:string, desc:"unknown", "yes" (wearing a hat), "no" (not wearing a hat), "helmet" (wearing a helmet)-->unknown
        </value>
        <count>
          <!--ro, req, int, number of people wearing hats-->1
        </count>
      </Hat>
    </HatList>
    <ThingsList>
      <!--ro, opt, array, carrying objects statistics, subType:object-->
      <Things>
        <!--ro, opt, object, carrying objects-->
        <value>
          <!--ro, req, enum, options, subType:string, desc:"unknown", "yes", "no"-->unknown
        </value>
        <count>
          <!--ro, req, int, number of people carrying objects-->1
        </count>
      </Things>
    </ThingsList>
    <isDataRetransmission>
      <!--ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be
uploaded again after the device recovers-->true
    </isDataRetransmission>
  </EventNotificationAlert>
```

19.5.1.27 People reverse entering alarm

EventType:reverseEntrance

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```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "test",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, string, protocol type, desc:"HTTP", "HTTPS"*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2009-11-14T15:27Z",
    /*ro, req, string, alarm trigger time*/
    "activePostCount":  1,
    /*ro, req, int, times that the same alarm has been uploaded*/
    "eventType":  "reverseEntrance",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, string, event status, desc:"active" (triggered), "inactive" (not triggered, the heartbeat data)*/
    "eventDescription":  "reverseEntrance alarm",
    /*ro, req, string, event description*/
    "url":  "test"
    /*ro, opt, string, URL of reverse entering alarm picture, desc:this field is not returned when the data is transmitted in binary format*/
}
```

19.6 Vehicle Recognition

19.6.1 Motor Vehicle Detection

19.6.1.1 Get arming schedule of blocklist ANPR of all channels

Request URL

GET /ISAPI/Event/schedules/blackList

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<BlackListScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, see details in the message of XML_Schedule, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, ID, desc:ID-->blackList-1
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string, desc:event type-->blackList
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming the task by day each week-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming the task by day each week-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</BlackListScheduleList>
```

19.6.1.2 Set ANPR arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/vehicledetects

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<VehicledetectScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, arming time list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming schedule-->
    <id>
      <!--req, string, index, desc:the format is event type-channel No.-->vehicledetect-1
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->vehicledetect
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming time list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</VehicledetectScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.6.1.3 Set arming schedule of allowlist ANPR of all channels

Request URL

PUT /ISAPI/Event/schedules/whiteList

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteListScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, arming time list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, Configure arming schedules:-->
    <id>
      <!--req, string, index, desc:the format is event type-channel No.-->whiteList-1
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->whiteList
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming time list, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</WhiteListScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.6.1.4 Set the linkage parameters of vehicle detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/vehicledetection-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->vehicledetection-1
  </id>
  <eventType>
    <!--req, string-->vehicledetection
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <direction>
    <!--opt, enum, subType:string-->both
  </direction>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,IO}-->1
      </outputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <wiegandOutputID>
        <!--opt, int, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,wiegand}-->1
      </wiegandOutputID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.6.1.5 Get the linkage parameters of vehicle detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/vehicledetection-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->vehicledetection-1
  </id>
  <eventType>
    <!--ro, req, string-->vehicledetection
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <direction>
    <!--ro, opt, enum, subType:string-->both
  </direction>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,IO}-->1
      </outputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <wiegandOutputID>
        <!--ro, opt, int, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,wiegand}-->1
      </wiegandOutputID>
      <triggerDirection>
        <!--ro, opt, enum, subType:string-->forward
      </triggerDirection>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.6.1.6 Get the total capability of vehicle capture and recognition

Request URL

GET /ISAPI/ITC/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ITCCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
</ITCCap>
```

19.6.1.7 Get the parameters of traffic rule violation dictionary

Request URL

GET /ISAPI/ITC/illegalDictionary

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IllegalDictionary xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, traffic rule violation dictionary, attr:version{req, string, protocolVersion}-->
  <IllegalCodeList>
    <!--ro, req, array, violation code list, subType:object-->
    <IllegalCodeItem>
      <!--ro, req, object, violation code-->
      <idx>
        <!--ro, req, int, ID, range:[0,65535]-->0
      </idx>
      <illegalCode>
        <!--ro, req, string, violation code, range:[0,64]-->test
      </illegalCode>
      <illegalName>
        <!--ro, req, string, violation name, range:[0,256]-->test
      </illegalName>
      <illegalStringCode>
        <!--ro, opt, string, violation code in string format, range:[0,128]-->test
      </illegalStringCode>
      <illegalDescription>
        <!--ro, opt, string, violation description, range:[0,128]-->test
      </illegalDescription>
    </IllegalCodeItem>
  </IllegalCodeList>
</IllegalDictionary>
```

19.6.1.8 Get the capability of traffic rule violation dictionary

Request URL

GET /ISAPI/ITC/illegalDictionary/capabilities

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<IllegalDictionary xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, traffic rule violation dictionary, attr:version{req, string, protocolVersion}-->
  <IllegalCodeList>
    <!--ro, req, array, violation code list, subType:object-->
    <IllegalCodeItem>
      <!--ro, req, object, violation code-->
      <idx min="0" max="64">
        <!--ro, req, int, index, attr:min{req, int},max{req, int}-->0
      </idx>
      <illegalCode min="0" max="64">
        <!--ro, req, string, violation code, attr:min{req, int},max{req, int}-->test
      </illegalCode>
      <illegalName min="0" max="256">
        <!--ro, req, string, violation name, attr:min{req, int},max{req, int}-->test
      </illegalName>
      <illegalStringCode min="0" max="10">
        <!--ro, opt, string, violation code (string), attr:min{req, int},max{req, int}-->test
      </illegalStringCode>
      <illegalCodeLetterNum min="0" max="10">
        <!--ro, opt, int, supported number of letters in violation code, attr:min{req, int},max{req, int}-->0
      </illegalCodeLetterNum>
      <illegalDescription size="256">
        <!--ro, opt, string, violation description, attr:size{req, int}-->test
      </illegalDescription>
    </IllegalCodeItem>
  </IllegalCodeList>
  <isSupportReset>
    <!--ro, opt, bool, whether it supports restoring to default setting, desc:/ISAPI/ITC/illegalDictionary/reset-->true
  </isSupportReset>
  <isSupportOverspeedAdvancedParam>
    <!--ro, opt, bool, whether it supports overspeed advanced configuration of traffic rule violation dictionary-->true
  </isSupportOverspeedAdvancedParam>
</IllegalDictionary>
```

19.6.1.9 Restore the traffic violation dictionary to default settings

Request URL

PUT /ISAPI/ITC/illegalDictionary/reset

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

19.6.1.10 Get the capability of arming parameters for traffic violations capture

Request URL

GET /ISAPI/ITC/illegalSchedules/capabilities?format=json&channelID=<channelID>

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
{
    "eventPriorityEnable": {
    /*ro, opt, object, whether to enable violation priority*/
        "@opt": [true, false]
        /*ro, opt, array, options, subType:bool*/
    },
    "IllegalScheduleListCap": {
    /*ro, req, object, list of arming parameters for traffic violations capture, desc:this node is only available for ANPR violation information, related node is illegalInfo*/
        "@size":  1,
        /*ro, req, int, the maximum number supported in a single capture*/
        "Schedule": {
        /*ro, opt, object, arming parameters*/
            "eventType":  "post,driveLine",
            /*ro, req, string, event type, desc:"post" (checkpoint), "driveLine" (driving on the lane line), "reverse" (wrong-way driving), "banSign" (prohibition violation), "overSpeed" (driving in overspeed), "lowSpeed" (driving in low speed), "changeLane" (illegal lane change), "safeBelt", "uphone" (making a phone call when driving), "gasser" (queue jumping), "congestion", "helmet" (driver is wearing a helmet), "twoWheelPassenger" (manned non-motor vehicle), "nonMotorExist" (non-motor vehicle on motor vehicle lane), "intersectionCongest" (overstay at intersection), "nonDriveWay" (motor vehicle on non-motor vehicle lane), "redLight" (red light running), "intersectionStop" (stop vehicle at intersection), "direction" (driving against direction guidance), "greenLightStop (stop vehicle when green light), "illegalTurn", "accident", "runGreeLight" (running green light in congestion), "leftNotYieldStraight" (left turn not yield to straight), "rightNotYieldLeft" (right turn not yield to left turn), "uTurnNotYieldGoingStaight", "leftSideDrivingAfterLeftTurn", "failYieldPed" (not yield to pedestrian), "illegalParking" (illegal parking), "nonMotorRedLight" (non-motor vehicle red light running), "nonMotorCrossing" (non-motor vehicle crossing line), "nonMotorOccupyZebraCrossing" (non-motor vehicle crossing zebra line), "roadBlock", "construction", "notKeepDistance" (not keeping a safe distance), "overtakeRightSide" (overtaking on the right), "dragRacing" (street racing), "changeLaneContinuously" (continuous lane change), "SSharpDriving" (slalom driving), "abandonedObject" (thrown object), "pedestrian" (pedestrian pre-alarm), "notSlowZebraCrossing", "overThermal", "turnRightStop" (large-sized vehicle not slowing down when turning right), "enterNotYieldToInRoundabout" (not yield at roundabout intersection), "rampNotYieldToMainRoad" (vehicle from ramp not yielding to vehicle on main road), "notYieldToRightRoad" (not yielding to vehicle from right side), "TruckOccupyOvertakingLane" (large truck occupying overtaking lane), "TruckTurnRightStop" (large truck not slowing down when turning right), "banStraight" (going straight prohibited), "emergencyLane" (occupy emergency lane), "smoke", "conflagration", "VehicleSpeedDrop", "animalsOccupy", "suonaDet" (whistle), "roarDet"*/
            "noPlateCaptureEnable": {
            /*ro, opt, object, whether to enable no plate vehicle capture*/
                "@opt": [true, false]
                /*ro, opt, array, options, subType:bool*/
            },
            "eventPriority": {
            /*ro, opt, object, event capture priority, desc:value range 0 to 10; the higher the value, the higher the priority; when the value is 0, this node will not have priority*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  10
                /*ro, req, int, the maximum value*/
            },
            "TimeBlockList": {
            /*ro, opt, object, list of time periods*/
                "@size":  7,
                /*ro, req, int, the maximum number supported in a single capture*/
                "TimeBlock": {
                /*ro, opt, object, time period*/
                    "dayOfWeek": {
                    /*ro, req, object, day of a week, desc:Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday*/
                        "@min":  1,
                        /*ro, req, int, the minimum value*/
                        "@max":  7
                        /*ro, req, int, the maximum value*/
                    },
                    "TimeRangeList": {
                    /*ro, opt, object, list of upload intervals*/
                        "@size":  4,
                        /*ro, req, int, the maximum number supported in a single capture*/
                        "TimeRange": {
                        /*ro, opt, object, time interval*/
                            "beginTime":  "00:00:00+08:00",
                            /*ro, req, time, start time*/
                            "endTime":  "00:00:00+08:00"
                            /*ro, req, time, end time*/
                        }
                    }
                }
            },
            "RelatedLanList": {
            /*ro, opt, object, list of linked lane information*/
                "@size":  6,
                /*ro, opt, int, size*/
                "RelatedLan": {
                /*ro, opt, object, linked lane information*/
                    "lanID": {
                    /*ro, opt, object, whether to enable linked lane No.*/
                        "@min":  1,
                        /*ro, req, int, the minimum value*/
                        "@max":  6
                        /*ro, req, int, the maximum value*/
                    }
                }
            }
        }
    }
}
```

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19.6.1.11 Get the arming parameters for traffic violation capture

Request URL

GET /ISAPI/ITC/illegalSchedules?format=json&channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "eventPriorityEnable":  true,
    /*ro, req, bool, whether to enable violation priority*/
    "IllegalScheduleList": [
    /*ro, req, array, arming time of traffic violation capture, subType:object, desc:this node is only available for ANPR violation information, which is related to the node illegalInfo*/
        {
            "Schedule": {
            /*ro, opt, object, arming schedules*/
                "eventType":  "post",
                /*ro, req, enum, event type, subType:string, desc:"post" (checkpoint), "driveLine" (driving on the lane line), "reverse" (wrong-way driving), "banSign" (prohibition violation), "overSpeed" (driving in overspeed), "lowSpeed" (driving in low speed), "changeLane" (illegal lane change), "safeBelt", "uphone" (making a phone call when driving), "gasser" (queue jumping), "congestion", "helmet" (driver is wearing a helmet), "twoWheelPassenger" (manned non-motor vehicle), "nonMotorExist" (non-motor vehicle on motor vehicle lane), "intersectionCongest" (overstay at intersection), "nonDriveWay" (motor vehicle on non-motor vehicle lane), "redLight" (red light running), "intersectionStop" (stop vehicle at intersection), "direction" (driving against direction guidance), "greenLightStop (stop vehicle when green light), "illegalTurn", "accident", "runGreeLight" (running green light in congestion), "leftNotYieldStraight" (left turn not yield to straight), "rightNotYieldLeft" (right turn not yield to left turn), "uTurnNotYieldGoingStaight", "leftSideDrivingAfterLeftTurn", "failYieldPed" (not yield to pedestrian), "illegalParking" (illegal parking), "nonMotorRedLight" (non-motor vehicle red light running), "nonMotorCrossing" (non-motor vehicle crossing line), "nonMotorOccupyZebraCrossing" (non-motor vehicle crossing zebra line), "roadBlock", "construction", "notKeepDistance" (not keeping a safe distance), "overtakeRightSide" (overtaking on the right), "dragRacing" (street racing), "changeLaneContinuously" (continuous lane change), "SSharpDriving" (slalom driving), "abandonedObject" (thrown object), "pedestrian" (pedestrian pre-alarm), "notSlowZebraCrossing", "overThermal", "turnRightStop" (large-sized vehicle not slowing down when turning right), "enterNotYieldToInRoundabout" (not yield at roundabout intersection), "rampNotYieldToMainRoad" (vehicle from ramp not yielding to vehicle on main road), "notYieldToRightRoad" (not yielding to vehicle from right side), "TruckOccupyOvertakingLane" (large truck occupying overtaking lane), "TruckTurnRightStop" (large truck not slowing down when turning right), "banStraight" (going straight prohibited), "emergencyLane" (occupy emergency lane), "smoke", "conflagration", "VehicleSpeedDrop", "animalsOccupy", "suonaDet" (whistle), "roarDet"*/
                "noPlateCaptureEnable":  true,
                /*ro, opt, bool, whether to enable no plate vehicle capture*/
                "eventPriority":  0,
                /*ro, opt, int, event capture priority, range:[0,10], desc:value ranges from 0 to 10; the higher the value, the higher the priority; when the value is 0, this event will not have priority*/
                "TimeBlockList": [
                /*ro, opt, array, list of time periods, subType:object, range:[0,7]*/
                    {
                        "TimeBlock": {
                        /*ro, opt, object, time period*/
                            "dayOfWeek":  1,
                            /*ro, req, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)*/
                            "TimeRangeList": [
                            /*ro, opt, array, list of time intervals, subType:object, range:[0,4], desc:4 time periods can be configured for a day*/
                                {
                                    "TimeRange": {
                                    /*ro, opt, object, time interval*/
                                        "beginTime":  "00:00:00+08:00",
                                        /*ro, req, time, start time*/
                                        "endTime":  "00:00:00+08:00"
                                        /*ro, req, time, end time*/
                                    }
                                }
                            ]
                        }
                    }
                ],
                "RelatedLanList": [
                /*ro, opt, array, list of linked lane information, subType:object, range:[0,6]*/
                    {
                        "RelatedLan": {
                        /*ro, opt, object, linked lane information*/
                            "lanID":  1
                            /*ro, opt, int, whether to enable linked lane No.*/
                        }
                    }
                ]
            }
        }
    ]
}
```

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19.6.1.12 Get trigger mode capability of traffic cameras

Request URL

GET /ISAPI/ITC/TriggerMode/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TriggerMode xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, trigger mode, attr:version{req, string, protocolVersion}-->
  <usedTriggerMode
opt="postIOSpeed,postSingleIO,postRS485,postRadar,postVTCoil,postHVT,postMPR,epoliceRS485,postEpoliceRS485,videoEpolice,TPS,videoMonitor,postNoComityPed,pos
tRedLightPed,postSideRadarSpeed,parkingIncident">
    <!--ro, req, enum, trigger mode supported by the device, different devices support different trigger modes, and a device may support multiple trigger
modes, subType:string, attr:opt{req, string}, desc:"postMobile", "postDetect", "TPS", "trafficIncident", "trafficIncidentTPS", "roadWarning", "postIOSpeed",
"postSingleIO", "postRS485", "postRadar", "postVTCoil", "postHVT", "postMPR", "epoliceRS485", "postEpoliceRS485", "videoEpolice", "videoMonitor",
"postNoComityPed", "postRedLightPed", "postSideRadarSpeed", "parkingIncident", "vehicleDetect", "boxNumDetection", "parkSpeedDetection"-->postMobile
  </usedTriggerMode>
  <isSupportChannelConfig>
    <!--ro, opt, bool, whether it supports channel configuration, desc:related URL: /ISAPI/ITC/channels/<ID>/TriggerMode/capabilities-->true
  </isSupportChannelConfig>
  <isSupportChannelSwitchStatus>
    <!--ro, opt, bool, whether it supports switching trigger mode status by channel No., desc:true (the optional node <channelID> in
/ISAPI/ITC/TriggerMode/switchStatus?format=json&channelID=<channelID> is supported)-->true
  </isSupportChannelSwitchStatus>
</TriggerMode>
```

19.6.1.13 Get the status of video detection traffic lights

Request URL

GET /ISAPI/ITC/TriggerMode/lightStatus?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Camera ID (this optional parameter is supported when isSupportChannelLightStatus is "true" returned from /ISAPI/ITC/TriggerMode/capabilities).

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VepLightStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, status of video detection traffic lights, attr:version{req, string, protocolVersion}-->
  <lightStatus opt="normal,abnomal">
    <!--ro, req, enum, traffic light status, subType:string, attr:opt{req, string}, desc:"normal", "abnormal"-->normal
  </lightStatus>
</VepLightStatus>
```

19.6.1.14 Set the multiple-frame recognition mode parameters of smart video security

Request URL

PUT /ISAPI/ITC/TriggerMode/videoMonitor

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, trigger mode parameters of smart video security, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--req, int, number of supported lanes, range:[1,3]-->0
  </relatedLaneCount>
  <capType>
    <!--req, enum, captured target type, subType:string, desc:"vehicle" (motor vehicle), "vehWithHum" (non-motor vehicle)-->vehicle
  </capType>
  <ViolationDetectLineList>
    <!--req, array, list of violation detection line on the single lane, subType:object-->
    <ViolationDetectLine>
      <!--opt, object, violation detection line on the single lane-->
      <lineName>
        <!--req, enum, detection line name, subType:string, desc:"laneLine" (for a single lane only)-->laneLine
      </lineName>
      <relatedDriveWay>
        <!--req, int, linked lane No., range:[1,99]-->1
      </relatedDriveWay>
      <laneDirectionType>
        <!--req, enum, lane direction type No., subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from
north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <carDriveDirect>
        <!--opt, enum, subType:string-->up2down
      </carDriveDirect>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--opt, object, only two coordinates are required, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <signSpeed>
        <!--opt, int, marked speed limit, range:[0,255]-->0
      </signSpeed>
      <speedLimit>
        <!--opt, int, speed limit, range:[0,255]-->0
      </speedLimit>
      <lowSpeedLimit>
        <!--opt, int, low speed limit for small-sized vehicle, range:[0,255]-->0
      </lowSpeedLimit>
      <bigCarSignSpeed>
        <!--opt, int, marked speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarSignSpeed>
      <bigCarSpeedLimit>
        <!--opt, int, speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarSpeedLimit>
      <bigCarLowSpeedLimit>
        <!--opt, int, low speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarLowSpeedLimit>
    </ViolationDetectLine>
  </ViolationDetectLineList>
  <VirtualLane>
    <!--req, object, draw violation detection line on all lanes-->
    <lineName>
      <!--req, enum, detection line name, subType:string, desc:"laneRightBoundaryLine" (parking space boundary line)-->laneRightBoundaryLine
    </lineName>
    <RegionCoordinatesList>
      <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </VirtualLane>
  <enabled>
    <!--opt, bool, whether to enable smart video security, desc:if this node is not returned, smart video security of the device is enabled by default--
>true
  </enabled>
  <TriggerLine>
    <!--opt, object, triggering line of smart video security capture-->
    <RegionCoordinatesList>
      <!--opt, array, area coordinates list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
```

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```
      </RegionCoordinatesList>
    </TriggerLine>
    <capNo>
      <!--opt, int, captured pictures of smart video security linked cameras, range:[1,2]-->1
    </capNo>
    <interval>
      <!--opt, int, capture interval of smart video security, range:[1,3000], unit:ms-->1
    </interval>
    <carDriveDirect>
      <!--opt, enum, vehicle detection type, subType:string, desc:"up2down" (up to down), "down2up" (down to up)-->up2down
    </carDriveDirect>
    <sceneMode>
      <!--opt, enum, scene mode, subType:string, desc:"standardScene" (standard scene), "lowPoleScene" (low pole scene), "entrance" (entrance and exit scene),
"other" (other scenes)-->standardScene
    </sceneMode>
    <overSpeed>
      <!--opt, bool, whether to enable overspeed trigger mode, desc:whether to enable overspeed trigger mode-->true
    </overSpeed>
    <overSpeedCapNo>
      <!--opt, int, capture times of overspeed, range:[1,3]-->1
    </overSpeedCapNo>
    <lowSpeed>
      <!--opt, bool, whether to enable low speed trigger mode-->true
    </lowSpeed>
    <lowSpeedCapNo>
      <!--opt, int, capture times of low speed, range:[1,3]-->1
    </lowSpeedCapNo>
    <carHighSpeed>
      <!--opt, int, abnormal overspeed, range:[0,255]-->0
    </carHighSpeed>
    <carLowSpeed>
      <!--opt, int, abnormal low speed, range:[0,255]-->0
    </carLowSpeed>
    <bigCarHighSpeed>
      <!--opt, int, abnormal overspeed of large-sized vehicle, range:[0,255]-->0
    </bigCarHighSpeed>
    <bigCarLowSpeed>
      <!--opt, int, abnormal low speed of large-sized vehicle, range:[0,255]-->0
    </bigCarLowSpeed>
    <nonmotorOverSpeed>
      <!--ro, opt, bool, whether to enable capturing non-motor vehicle-->true
    </nonmotorOverSpeed>
    <nonmotorHighSpeed>
      <!--ro, opt, int, speed limit for non-motor vehicle (unit: km/h), range:[0,255], unit:km/h-->0
    </nonmotorHighSpeed>
  </VideoMonitor>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

19.6.1.15 Get the multiple-frame recognition mode capability of smart video security

Request URL

GET /ISAPI/ITC/TriggerMode/videoMonitor/capabilities

Query Parameter

None

Request Message

None

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, multiple-frame recognition mode capability of smart video security, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount min="1" max="16">
    <!--ro, req, int, total number of linked lanes, attr:min{req, int},max{req, int}-->0
  </relatedLaneCount>
  <relatedDriveWay min="1" max="99">
    <!--ro, req, int, linked lane No., attr:min{req, int},max{req, int}-->0
  </relatedDriveWay>
  <laneDirectionType opt="0,1,2,3,4,5,6,7,8">
    <!--ro, req, enum, lane direction, subType:int, attr:opt{req, string}, desc:0-unknown, 1-from east to west, 2-from west to east, 3-from south to north,
4-from north to south, 5-from southeast to northwest, 6-from northwest to southeast, 7-from northeast to southwest, 8-from southwest to northeast-->0
  </laneDirectionType>
  <capType opt="vehWithHum,vehicle,nonmotor,human,motorWithHum,nonmotorWithHum,motorWithNonmotor">
    <!--ro, req, enum, captured target type, subType:string, attr:opt{req, string}, desc:"vehWithHum" (motor vehicle, non-motor vehicle, and pedestrian),
"vehicle" (motor vehicle), "nonmotor" (non-motor vehicle), "human" (pedestrian), "motorWithHum" (motor vehicle and pedestrian), "nonmotorWithHum" (non-motor
vehicle and pedestrian), "motorNonmotor" (motor vehicle and non-motor vehicle)-->vehWithHum
  </capType>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable smart video security, attr:opt{req, string}-->true
  </enabled>
  <TriggerLine>
    <!--ro, opt, object, triggering line of smart video security capture-->
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinate list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </TriggerLine>
  <capNo opt="1,2">
    <!--ro, opt, int, number of captured pictures, attr:opt{req, string}-->0
  </capNo>
  <interval min="1" max="3000">
    <!--ro, opt, int, capture time, unit:ms, attr:min{req, int},max{req, int}-->0
  </interval>
  <carDriveDirect opt="up2down,down2up">
    <!--ro, req, enum, vehicle detection type, subType:string, attr:opt{req, string}, desc:"up2down" (up to down), "down2up" (down to up)-->up2down
  </carDriveDirect>
  <sceneMode opt="standardScene ,lowPoleScene">
    <!--ro, opt, enum, scene mode, subType:string, attr:opt{req, string}, desc:"standardScene" (standard scene), "lowPoleScene" (low pole scene), "entrance"
(entrance and exit scene), "other" (other scenes)-->standardScene
  </sceneMode>
  <overSpeed>
    <!--ro, opt, bool, whether to enable overspeed trigger mode-->true
  </overSpeed>
  <overSpeedCapNo opt="1,2,3">
    <!--ro, opt, int, capture times of overspeed, attr:opt{req, string}-->0
  </overSpeedCapNo>
  <lowSpeed>
    <!--ro, opt, bool, whether to enable low speed trigger mode-->true
  </lowSpeed>
  <lowSpeedCapNo opt="1,2,3">
    <!--ro, opt, int, capture times of low speed, attr:opt{req, string}-->0
  </lowSpeedCapNo>
  <signSpeed min="0" max="255">
    <!--ro, opt, int, marked speed limit, attr:min{req, int},max{req, int}-->0
  </signSpeed>
  <speedLimit min="0" max="255">
    <!--ro, opt, int, speed limit, attr:min{req, int},max{req, int}-->0
  </speedLimit>
  <bigCarSignSpeed min="0" max="255">
    <!--ro, opt, int, marked speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarSignSpeed>
  <bigCarHighSpeedLimit min="0" max="255">
    <!--ro, opt, int, maximum speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarHighSpeedLimit>
  <bigCarLowSpeedLimit min="0" max="255">
    <!--ro, opt, int, low speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarLowSpeedLimit>
  <lowSpeedLimit min="0" max="60">
    <!--ro, opt, int, low speed limit for small-sized vehicle, attr:min{req, int},max{req, int}-->0
  </lowSpeedLimit>
  <bigCarSpeedLimit min="0" max="60">
    <!--ro, opt, int, speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarSpeedLimit>
  <carHighSpeed min="0" max="255">
    <!--ro, opt, int, abnormal overspeed, attr:min{req, int},max{req, int}-->0
  </carHighSpeed>
  <carLowSpeed min="0" max="255">
    <!--ro, opt, int, abnormal low speed, attr:min{req, int},max{req, int}-->0
  </carLowSpeed>
  <bigCarHighSpeed min="0" max="255">
```

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```
    <!--ro, opt, int, abnormal overspeed of large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarHighSpeed>
  <bigCarLowSpeed min="0" max="255">
    <!--ro, opt, int, abnormal low speed of large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarLowSpeed>
  <nonmotorHighSpeedLimit min="0" max="255">
    <!--ro, opt, int, speed limit for non-motor vehicle, attr:min{req, int},max{req, int}-->0
  </nonmotorHighSpeedLimit>
</VideoMonitor>
```

19.6.1.16 Set the configuration parameters for uploading vehicle detection information to the FTP server

Request URL

PUT /ISAPI/System/Network/ftp/uploadInfo?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FtpUpload xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, the configuration parameters for uploading vehicle detection information to the FTP server, attr:version{req, string, protocolVersion}-->
  <vehiclePicName>
    <!--opt, object, picture name-->
    <mode>
      <!--req, enum, mode, subType:string, desc:"default", "custom"-->default
    </mode>
    <NameRuleType>
      <!--opt, object, naming rule, dep:and,{$.FtpUpload.vehiclePicName.mode,eq,custom}-->
      <RuleTypeItemList>
        <!--opt, array, rule list, subType:object-->
        <RuleTypeItem>
          <!--opt, object, rule information-->
          <id>
            <!--req, int, index-->1
          </id>
          <item>
            <!--opt, enum, rule items, subType:string, desc:"capture_time"-capture time, "plate_No"-license plate number, "alarm_type"-alarm type,
"camera_name"-camera name, "countryOrRegion"-country/region, "carDirectionType"-vehicle driving direction (upward,downward)-->capture_time
          </item>
          <cameraName>
            <!--opt, string, camera name, range:[1,32], dep:and,{$.FtpUpload.vehiclePicName.NameRuleType.RuleTypeItemList.RuleTypeItem.item,eq,camera_name}-
->test
          </cameraName>
        </RuleTypeItem>
      </RuleTypeItemList>
    </NameRuleType>
  </vehiclePicName>
</FtpUpload>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.6.1.17 Get the configuration parameters for uploading vehicle detection information to the FTP server

Request URL

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GET /ISAPI/System/Network/ftp/uploadInfo?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Camera ID. If . is true in /ISAPI/System/Network/ftp/capabilities, linking the camera is supported.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FtpUpload xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the configuration parameters for uploading vehicle detection information to the FTP server, attr:version{req, string,
protocolVersion}-->
  <vehiclePicName>
    <!--ro, opt, object, picture name-->
    <mode>
      <!--ro, req, enum, mode, subType:string, desc:default; custom;-->default
    </mode>
    <NameRuleType>
      <!--ro, opt, object, naming rule, dep:and,{$.FtpUpload.vehiclePicName.mode,eq,custom}-->
      <RuleTypeItemList>
        <!--ro, opt, array, rule list, subType:object-->
        <RuleTypeItem>
          <!--ro, opt, object, rule-->
          <id>
            <!--ro, req, int, rule ID-->1
          </id>
          <item>
            <!--ro, opt, enum, rule item, subType:string, desc:capture_time: capture time; plate_No: license plate number; alarm_type: alarm type;
camera_name: camera name; countryOrRegion: country/region; carDirectionType: vehicle driving direction (upward,downward);-->capture_time
          </item>
          <cameraName>
            <!--ro, opt, string, camera name, range:[1,32], dep:and,
{$.FtpUpload.vehiclePicName.NameRuleType.RuleTypeItemList.RuleTypeItem.item,eq,camera_name}-->test
          </cameraName>
        </RuleTypeItem>
      </RuleTypeItemList>
    </NameRuleType>
  </vehiclePicName>
</FtpUpload>
```

19.6.1.18 Set ANR parameters

Request URL

PUT /ISAPI/Traffic/ANR

Query Parameter

None

Request Message

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dojo.jin@mail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ANRControl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, ANR parameters, attr:version{req, string, protocolVersion}-->
  <uploadInterval>
    <!--req, int, upload interval, unit:ms-->0
  </uploadInterval>
  <uploadTimeout>
    <!--opt, int, uploading timed out, unit:ms-->0
  </uploadTimeout>
  <DeviceIpAddress>
    <!--opt, object, IP address of ANR arming server-->
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host server name, range:[0,64], dep:and,{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[0,32], dep:and,{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[0,128], dep:and,{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
  </DeviceIpAddress>
</ANRControl>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

19.6.1.19 Get the ANR parameters

Request URL

GET /ISAPI/Traffic/ANR

Query Parameter

None

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD.
hojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ANRControl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ANR (Automatic Network Replenishment) parameters, attr:version{req, string, protocolVersion}-->
  <uploadInterval>
    <!--ro, req, int, uploading interval, unit:ms-->0
  </uploadInterval>
  <uploadTimeout>
    <!--ro, opt, int, unit: ms, unit:ms-->0
  </uploadTimeout>
  <DeviceIpAddress>
    <!--ro, opt, object, IP address of ANR arming server-->
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, range:[0,64], dep:and,{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, this node is valid only when addressingFormatType is "ipaddress", range:[0,32], dep:and,
{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, this node is valid only when addressingFormatType is "ipaddress", range:[0,128], dep:and,
{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
  </DeviceIpAddress>
</ANRControl>
```

19.6.1.20 Set the text overlay configuration parameters

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/overlap

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Overlap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of text overlay on the picture, attr:version{req, string, protocolVersion}-->
  <isOverlap>
    <!--req, bool-->true
  </isOverlap>
  <overlapMergeType>
    <!--req, enum, subType:string-->outside
  </overlapMergeType>
  <linePercent>
    <!--req, int, range:[1,100]-->0
  </linePercent>
  <startPosTop>
    <!--req, int, range:[1,100]-->0
  </startPosTop>
  <startPosLeft>
    <!--req, int, range:[1,100]-->0
  </startPosLeft>
  <spaceNum>
    <!--opt, int, range:[0,10]-->0
  </spaceNum>
  <charSize>
    <!--req, enum, subType:string-->16
  </charSize>
  <charInterval>
    <!--req, int, range:[0,10]-->0
  </charInterval>
  <frontColor>
    <!--req, string-->ffffff
  </frontColor>
  <backColor>
    <!--req, string-->ffffff
  </backColor>
  <OverlapItemList>
    <!--opt, array, subType:object-->
    <OverlapItem>
      <!--opt, object-->
      <id>
        <!--req, string, index, range:[0,10]-->test
      </id>
      <item>
        <!--req, enum, subType:string-->captureTimeFormatType
```

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```
        <!--req, enum, subType:string-->captureTimeFormatType
      </item>
      <itemOverlayEnabled>
        <!--opt, bool-->true
      </itemOverlayEnabled>
      <changeLineNum>
        <!--req, int-->0
      </changeLineNum>
      <spaceNum>
        <!--req, int-->0
      </spaceNum>
      <overlayInfo>
        <!--req, string-->test
      </overlayInfo>
      <captureTimeFormatType>
        <!--opt, enum, subType:string-->0
      </captureTimeFormatType>
      <parkingTimeFormatType>
        <!--opt, enum, subType:string-->0
      </parkingTimeFormatType>
    </OverlapItem>
  </OverlapItemList>
  <generalOverlapEnabled>
    <!--opt, bool-->true
  </generalOverlapEnabled>
  <ImageMergeOverlap>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <overlapMergeType>
      <!--req, enum, subType:string-->outside
    </overlapMergeType>
    <linePercent>
      <!--req, int, range:[1,100]-->0
    </linePercent>
    <startPosTop>
      <!--req, int, range:[1,100]-->0
    </startPosTop>
    <startPosLeft>
      <!--req, int, range:[1,100]-->0
    </startPosLeft>
    <spaceNum>
      <!--opt, int, range:[0,10]-->0
    </spaceNum>
    <charSize>
      <!--req, enum, subType:string-->16
    </charSize>
    <charInterval>
      <!--req, int, range:[0,10]-->0
    </charInterval>
    <frontColor>
      <!--req, int-->0
    </frontColor>
    <backColor>
      <!--req, int-->0
    </backColor>
    <OverlapItemList>
      <!--opt, array, subType:object-->
      <OverlapItem>
        <!--opt, object-->
        <id>
          <!--req, string, index-->test
        </id>
        <item>
          <!--req, enum, subType:string-->captureTimeFormatType
        </item>
        <itemOverlayEnabled>
          <!--opt, bool-->true
        </itemOverlayEnabled>
        <changeLineNum>
          <!--req, int-->0
        </changeLineNum>
        <spaceNum>
          <!--req, int-->0
        </spaceNum>
        <overlayInfo>
          <!--req, string-->test
        </overlayInfo>
        <captureTimeFormatType>
          <!--opt, enum, subType:string-->0
        </captureTimeFormatType>
        <parkingTimeFormatType>
          <!--opt, enum, subType:string-->0
        </parkingTimeFormatType>
      </OverlapItem>
    </OverlapItemList>
  </ImageMergeOverlap>
</Overlap>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.6.1.21 Set the parameters of vehicle detection captured pictures

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/picParam

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<PicParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of vehicle detection captured pictures, attr:version{opt, string, protocolVersion}-->
  <PictureCfg>
    <!--opt, object, picture parameters-->
    <mode>
      <!--req, enum, capture mode, subType:string, desc:"quality" (by quality), "size" (by size)-->quality
    </mode>
    <pictureQuality>
      <!--opt, int, picture quality, range:[1,100]-->50
    </pictureQuality>
    <pictureSize>
      <!--opt, int, picture size, unit:KB-->100
    </pictureSize>
  </PictureCfg>
  <Overlap>
    <!--opt, object, picture overlay-->
    <enabled>
      <!--req, bool, whether to enable text overlay-->true
    </enabled>
    <OverlapItemList>
      <!--opt, array, overlay item list, subType:object-->
      <OverlapItem>
        <!--opt, object, overlay items-->
        <id>
          <!--req, int, index-->1
        </id>
        <item opt="positionNo">
          <!--req, string, overlay content, attr:opt{opt, enum, subType:string}-->test
        </item>
      </OverlapItem>
    </OverlapItemList>
    <fontColor>
      <!--opt, string, font color, desc:the previous version of protocol; hexBinary-->test
    </fontColor>
    <backColor>
      <!--opt, string, background color, desc:the previous version of protocol; hexBinary-->test
    </backColor>
  </Overlap>
  <CapturePicOverlays>
    <!--opt, object, Capture Overlay, desc:professional OSD configuration for picture capture, which is only supported by capture cameras now-->
  </CapturePicOverlays>
  <MergePicOverlays>
    <!--opt, object, Composite Picture Overlay, desc:professional OSD configuration for picture composite, which is only supported by capture cameras now-->
  </MergePicOverlays>
  <PlateEnhancement>
    <!--opt, object, License Plate Enhancement-->
    <enabled>
      <!--opt, bool, whether to enable license plate enhancement on captured picture-->true
    </enabled>
    <level>
      <!--opt, int, set the level of license plate enhancement from 0 to 100, range:[0,100]-->1
    </level>
  </PlateEnhancement>
  <CapturePicTypeList>
    <!--opt, array, the list of the types of captured pictures, subType:object, range:[1,3], desc:all types of picture shall be captured if this node is not
returned-->
    <capturePicType>
      <!--opt, enum, types of captured pictures, subType:string, desc:"licensePlatePicture" (license plate picture), "vehiclePicture" (vehicle picture),
"detectionPicture" (detection picture (background picture))-->licensePlatePicture
    </capturePicType>
  </CapturePicTypeList>
  <resolution>
    <!--opt, string, picture resolution, desc:subject to the returned range-->1920*1080
  </resolution>
  <capturePicIntervalEnabled>
    <!--ro, opt, bool, whether to enable picture capture interval-->true
  </capturePicIntervalEnabled>
  <capturePicIntervalTime>
    <!--ro, opt, int, capture interval, range:[1,30], unit:s, dep:and,{$.PicParam.capturePicIntervalEnabled,eq,true}-->1
  </capturePicIntervalTime>
  <syncCaptureBackPanoramaEnabled>
    <!--opt, bool, whether to enable background picture of panoramic channel, desc:a scene can be monitored by both channels of the front-end dual-channel
device. The alarm from the detailed-image channel should be linked with the capture of panoramic background pictures by the panoramic-image channel-->true
  </syncCaptureBackPanoramaEnabled>
</PicParam>
```

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dojojin@gmail.com

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

19.6.1.22 Get parameters of vehicle detection captured pictures

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

None

Response Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<PicParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of vehicle detection captured pictures, attr:version{opt, string, protocolVersion}-->
  <PictureCfg>
    <!--ro, opt, object, picture parameters-->
    <mode>
      <!--ro, req, enum, capture mode, subType:string, desc:"quality" (by quality), "size" (by size)-->quality
    </mode>
    <pictureQuality>
      <!--ro, opt, int, picture quality, range:[1,100]-->50
    </pictureQuality>
    <pictureSize>
      <!--ro, opt, int, picture size, unit:KB-->100
    </pictureSize>
  </PictureCfg>
  <Overlap>
    <!--ro, opt, object, picture overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable text overlay-->true
    </enabled>
    <OverlapItemList>
      <!--ro, opt, array, overlay item list, subType:object-->
      <OverlapItem>
        <!--ro, opt, object, overlay items-->
        <id>
          <!--ro, req, int, ID-->1
        </id>
        <item opt="positionNo">
          <!--ro, req, string, overlay content, attr:opt{opt, enum, subType:string}-->test
        </item>
      </OverlapItem>
    </OverlapItemList>
    <fontColor>
      <!--ro, opt, string, font color, desc:the previous version of protocol; hexBinary-->test
    </fontColor>
    <backColor>
      <!--ro, opt, string, background color, desc:the previous version of protocol; hexBinary-->test
    </backColor>
  </Overlap>
  <CapturePicOverlays>
    <!--ro, opt, object, capture overlay, desc:professional OSD configuration for picture capture, which is only supported by capture cameras now-->
  </CapturePicOverlays>
  <MergePicOverlays>
    <!--ro, opt, object, composite picture overlay, desc:professional OSD configuration for picture composite, which is only supported by capture cameras
now-->
  </MergePicOverlays>
  <PlateEnhancement>
    <!--ro, opt, object, license plate enhancement-->
    <enabled>
      <!--ro, opt, bool, whether to enable license plate enhancement on captured picture-->true
    </enabled>
    <level>
      <!--ro, opt, int, set the level of license plate enhancement from 0 to 100, range:[0,100]-->1
    </level>
  </PlateEnhancement>
  <CapturePicTypeList>
    <!--ro, opt, array, the list of the types of captured pictures, subType:object, range:[1,3], desc:all types of picture shall be captured if this node is
not returned-->
    <capturePicType>
      <!--ro, opt, enum, types of captured pictures, subType:string, desc:"licensePlatePicture" (license plate picture), "vehiclePicture" (vehicle picture),
"detectionPicture" (detection picture (background picture))-->licensePlatePicture
    </capturePicType>
  </CapturePicTypeList>
  <resolution>
    <!--ro, opt, string, picture resolution, desc:subject to the returned range-->1920*1080
  </resolution>
  <capturePicIntervalEnabled>
    <!--ro, opt, bool, whether to enable picture capture interval-->true
  </capturePicIntervalEnabled>
  <capturePicIntervalTime>
    <!--ro, opt, int, capture interval, range:[1,30], unit:s, dep:and,{$.PicParam.capturePicIntervalEnabled,eq,true}-->1
  </capturePicIntervalTime>
  <syncCaptureBackPanoramaEnabled>
    <!--ro, opt, bool, whether to enable background picture of panoramic channel, desc:a scene can be monitored by both channels of the front-end dual-
channel device. The alarm from the detailed-image channel should be linked with the capture of panoramic background pictures by the panoramic-image channel-
->true
  </syncCaptureBackPanoramaEnabled>
</PicParam>
```

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19.6.1.23 Get captured picture configuration capability

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PicParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture capture configuration capability, attr:version{opt, string, protocolVersion}-->
  <PictureCfg>
    <!--ro, req, object, picture parameters-->
    <mode opt="quality,size">
      <!--ro, req, enum, capture mode, subType:string, attr:opt{opt, enum, subType:string}, desc:"quality" (by quality), "size" (by size)-->quality
    </mode>
    <pictureQuality min="1" max="100">
      <!--ro, opt, int, picture quality, range:[1,100], attr:min{opt, string},max{opt, string}-->1
    </pictureQuality>
    <pictureSize min="1" max="6553500000">
      <!--ro, opt, int, picture size, unit:KB, attr:min{req, int},max{req, int}-->1
    </pictureSize>
  </PictureCfg>
  <Overlap>
    <!--ro, opt, object, picture overlay-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable text overlay, attr:opt{req, string}-->true
    </enabled>
    <OverlapItem opt="positionNo,positionInfo,cameraNo,captureTime,plateNo,vehicleColor,sceneName,carType,vehicleLogo,sceneNo,GPS,vehicleSpeed">
      <!--ro, req, enum, overlay items, subType:string, attr:opt{opt, enum}, desc:"positionNo" (detection point No.), "positionInfo" (detection point),
"cameraNo" (camera No.), "captureTime" (capture time), "plateNo" (license plate No.), "vehicleColor" (vehicle color), "sceneName" (scene name), "carType"
(vehicle type), "vehicleLogo" (main vehicle brand) , "sceneNo" (scene No.), "GPS" (GPS device)-->positionNo
    </OverlapItem>
    <fontColor>
      <!--ro, opt, string, font color, desc:the previous version of protocol; hexBinary-->test
    </fontColor>
    <backColor>
      <!--ro, opt, string, background color, desc:the previous version of protocol; hexBinary-->test
    </backColor>
  </Overlap>
  <CapturePicOverlays>
    <!--ro, opt, object, professional OSD configuration for captured pictures,currently this configuration is only supported by capture cameras,
desc:professional OSD configuration for captured pictures,currently this configuration is only supported by capture cameras-->
  </CapturePicOverlays>
  <MergePicOverlays>
    <!--ro, opt, object, professional OSD configuration for combined pictures,currently this configuration is only supported by capture cameras,
desc:professional OSD configuration for combined pictures,currently this configuration is only supported by capture cameras-->
  </MergePicOverlays>
  <PlateEnhancement>
    <!--ro, opt, object, license plate enhancement on captured picture-->
    <enabled opt="true,false" def="false">
      <!--ro, opt, bool, whether to enable license plate enhancement on captured picture, attr:opt{opt, string},def{opt, string}-->true
    </enabled>
    <level min="0" max="100" def="50">
      <!--ro, opt, int, license plate enhancement level range, range:[0,100], attr:min{opt, string},max{opt, string},def{opt, string}-->1
    </level>
  </PlateEnhancement>
  <CapturePicTypeList size="3">
    <!--ro, opt, array, the list of the types of captured pictures, subType:object, attr:size{req, int}, desc:all types of picture shall be captured if this
node is not returned-->
    <capturePicType opt="licensePlatePicture,vehiclePicture,detectionPicture">
      <!--ro, opt, string, types of captured pictures, attr:opt{req, string}, desc:"licensePlatePicture" (license plate picture), "vehiclePicture" (vehicle
picture), "detectionPicture" (detection picture (background picture))-->licensePlatePicture
    </capturePicType>
  </CapturePicTypeList>
  <resolution opt="1920*1080,2688*1520,1280*720,704*576,352*288">
    <!--ro, opt, string, picture resolution, attr:opt{req, string}-->1920*1080
  </resolution>
  <capturePicIntervalEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable picture capture interval, attr:opt{req, string}-->true
  </capturePicIntervalEnabled>
  <capturePicIntervalTime min="1" max="30" def="1">
    <!--ro, opt, int, capture interval, range:[1,30], unit:s, dep:and,{$.PicParam.capturePicIntervalEnabled,eq,true}, attr:min{req, int},max{req,
int},def{req, int}-->1
  </capturePicIntervalTime>
  <syncCaptureBackPanoramaEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable background picture of panoramic channel, attr:opt{req, string}, desc:a scene can be monitored by both channels of
the front-end dual-channel device. The alarm from the detailed-image channel should be linked with the capture of panoramic background pictures by the
panoramic-image channel-->true
  </syncCaptureBackPanoramaEnabled>
</PicParam>
```

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19.6.1.24 Get the capability of OSD overlay on a captured picture

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CapturePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
</CapturePicOverlays>
```

19.6.1.25 Set the configuration parameters for overlaying all OSD text information on a captured picture

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/info

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, see details in the message of XML_OverlayInfo, subType:object, attr:version{opt, string, protocolVersion}-->
  <OverlayInfo>
    <!--opt, object, overlay information-->
    <overlayInfoID>
      <!--req, int, No., range:[1,50]-->1
    </overlayInfoID>
    <itemType>
      <!--req, enum, the type of overlaid information on a picture, subType:int, desc:the type of overlaid information on a picture-->70
    </itemType>
    <itemOverlayEnabled>
      <!--req, bool-->true
    </itemOverlayEnabled>
    <customName>
      <!--req, string, self-defined name of the overlay information, range:[1,32]-->test
    </customName>
    <changeLineNum>
      <!--req, int, range:[0,100]-->2
    </changeLineNum>
    <spaceNum>
      <!--req, int, range:[0,255]-->1
    </spaceNum>
    <startPosEnable>
      <!--opt, bool, whether to enable coordinate configuration-->true
    </startPosEnable>
    <startPosTop>
      <!--req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
height, and the default value is 0-->0
    </startPosTop>
    <startPosLeft>
      <!--req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
width, and the default value is 0-->0
    </startPosLeft>
    <overlayInfoText>
      <!--opt, string, overlay character string, range:[0,128], desc:the maximum string length of place is 128 bytes, the maximum string length of
intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and
the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
    </overlayInfoText>
    <overlayInfoText2>
      <!--opt, string, overlay character string, range:[1,32], desc:overlay character string-->test
    </overlayInfoText2>
  </OverlayInfo>
</OverlayInfoList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.6.1.26 Set the configuration parameters for overlaying a

piece of OSD text information on a captured picture **Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/info/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, overlay information, attr:version{opt, string, protocolVersion}-->
  <overlayInfoID>
    <!--req, int, overlayed information ID, range:[1,50]-->1
  </overlayInfoID>
  <itemType>
    <!--req, enum, overlayed information type, subType:int, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line,etc.), 18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30/31/32-reserved, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-congestion-->70
  </itemType>
  <itemOverlayEnabled>
    <!--req, bool, whether to overlay the item-->true
  </itemOverlayEnabled>
  <customName>
    <!--req, string, custom overlaying name, range:[1,32]-->test
  </customName>
  <changeLineNum>
    <!--req, int, number of line feeds, range:[0,100]-->2
  </changeLineNum>
  <spaceNum>
    <!--req, int, number of spaces, range:[0,255]-->1
  </spaceNum>
  <startPosEnable>
    <!--opt, bool, whether to enable coordinate configuration-->true
  </startPosEnable>
  <startPosTop>
    <!--req, int, start top coordinate, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
  </startPosTop>
  <startPosLeft>
    <!--req, int, start left coordinate, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
  </startPosLeft>
  <overlayInfoText>
    <!--opt, string, overlay character string, range:[0,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
  </overlayInfoText>
  <overlayInfoText2>
    <!--opt, string, overlay character string for camera 2, range:[1,32], desc:overlay character string for camera 2-->test
  </overlayInfoText2>
</OverlayInfo>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.6.1.27 Get overlay parameters of captured traffic pictures by channel

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays?multiCharPosition=<multiCharPosition>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
multiCharPosition	enum	--

Request Message

None

Response Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>
<CapturePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, see details in the message of XML_OverlayInfoList, attr:version{opt, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--ro, req, bool, whether to enable text overlay, desc:whether to enable text overlay-->true
  </overlayInfoEnabled>
  <OverlayInfoList>
    <!--ro, opt, array, subType:object-->
    <OverlayInfo>
      <!--ro, opt, object-->
      <overlayInfoID>
        <!--ro, req, int-->1
      </overlayInfoID>
      <itemType>
        <!--ro, req, enum, overlay mode, subType:int, desc:overlay mode-->70
      </itemType>
      <itemOverlayEnabled>
        <!--ro, req, bool-->true
      </itemOverlayEnabled>
      <customName>
        <!--ro, req, string, self-defined name of the overlay information-->test
      </customName>
      <changeLineNum>
        <!--ro, req, int, range:[0,100]-->2
      </changeLineNum>
      <spaceNum>
        <!--ro, req, int, space number, range:[0,255]-->1
      </spaceNum>
      <startPosEnable>
        <!--ro, opt, bool, whether to enable coordinate configuration-->true
      </startPosEnable>
      <startPosTop>
        <!--ro, req, int, top coordinate of the start point-->0
      </startPosTop>
      <startPosLeft>
        <!--ro, req, int, left coordinate of the start point-->0
      </startPosLeft>
      <overlayInfoText>
        <!--ro, opt, string, overlay character string, range:[0,128]-->test
      </overlayInfoText>
      <overlayInfoText2>
        <!--ro, opt, string, overlay character string, range:[0,32]-->test
      </overlayInfoText2>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent>
    <!--ro, req, int, percentage of overlaying lines, range:[0,100]-->1
  </linePercent>
  <itemsStlye>
    <!--ro, req, enum, overlay mode, subType:string, desc:"horizontal", "vertical"-->horizontal
  </itemsStlye>
  <charStyle>
    <!--ro, req, enum, font type, subType:string, desc:font type: 0-SimSun (default),1-Wei-->song_type
  </charStyle>
  <charSize>
    <!--ro, req, enum, character size, subType:int, desc:0 (8*16), 1 (16*32), 2 (48*48), 3 (32*64), 4 (64*128)-->0
  </charSize>
  <charPosition>
    <!--ro, req, enum, text position overlaid on the picture, subType:int, desc:0 (in the picture), 1 (outside the top edge of the picture), 2 (outside the
bottom edge of the picture)-->1
  </charPosition>
  <charInterval>
    <!--ro, req, int, character separation distance, range:[0,16]-->0
  </charInterval>
  <colorAdapt>
    <!--ro, opt, bool, whether to enable color self-adaption-->true
  </colorAdapt>
  <zeroizeEnable>
    <!--ro, opt, bool, whether to enable zero filling for OSD overlay-->true
  </zeroizeEnable>
  <platePicOverlay>
    <!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
  </platePicOverlay>
  <platePicPosTop>
    <!--ro, req, int, start top coordinate-->0
  </platePicPosTop>
  <platePicPosLeft>
    <!--ro, req, int, start left coordinate-->0
  </platePicPosLeft>
  <featurePicOSDEnable>
    <!--ro, opt, bool-->true
  </featurePicOSDEnable>
</CapturePicOverlays>
```

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19.6.1.28 Get the overlay capability in the composite captured pictures

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<MergePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, composite picture overlay, attr:version{req, string, protocolVersion}-->
</MergePicOverlays>
```

19.6.1.29 Get parameters of overlaying characters on composite pictures

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/info

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, overlay information list, subType:object, attr:version{req, string, protocolVersion}-->
  <OverlayInfo>
    <!--ro, req, object, overlay information-->
    <overlayInfoID>
      <!--ro, req, int, overlay ID-->0
    </overlayInfoID>
    <itemType>
      <!--ro, req, enum, overlaid information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for parking space patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask-->0
    </itemType>
    <itemOverlayEnabled>
      <!--ro, req, bool, whether to overlay this item-->true
    </itemOverlayEnabled>
    <customName>
      <!--ro, req, string, custom overlaying name, range:[0,32], desc:customize name of the overlay information-->test
    </customName>
    <changeLineNum>
      <!--ro, req, int, the default value is 0, range:[0,100], desc:the default value is 0-->0
    </changeLineNum>
    <spaceNum>
      <!--ro, req, int, the number of spaces, range:[0,255], desc:the default value is 0-->0
    </spaceNum>
    <startPosEnable>
      <!--ro, opt, bool, whether to enable coordinate configuration, desc:whether to enable coordinate configuration-->true
    </startPosEnable>
    <startPosTop>
      <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
    </startPosTop>
    <startPosLeft>
      <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
    </startPosLeft>
    <overlayInfoText>
      <!--ro, opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
    </overlayInfoText>
    <overlayInfoText2>
      <!--ro, opt, string, overlay characters, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
    </overlayInfoText2>
  </OverlayInfo>
</OverlayInfoList>
```

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19.6.1.30 Get the configuration parameters of overlaying a piece of OSD text information on the composite captured picture

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/info/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, overlay information, attr:version{req, string, protocolVersion}-->
  <overlayInfoID>
    <!--ro, req, int, overlayed information ID, which is between 1 and 50-->0
  </overlayInfoID>
  <itemType>
    <!--ro, req, enum, overlayed information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask-->0
  </itemType>
  <itemOverlayEnabled>
    <!--ro, req, bool, whether to overlay the item-->true
  </itemOverlayEnabled>
  <customName>
    <!--ro, req, string, custom overlaying name, range:[0,32], desc:self-defined name of the overlay information-->test
  </customName>
  <changeLineNum>
    <!--ro, req, int, number of line feeds, and the default value is 0, range:[0,100], desc:the default value is 0-->0
  </changeLineNum>
  <spaceNum>
    <!--ro, req, int, number of spaces, and the default value is 0, range:[0,255], desc:the default value is 0-->0
  </spaceNum>
  <startPosEnable>
    <!--ro, opt, bool, whether to enable coordinate configuration, desc:whether to enable coordinate configuration-->true
  </startPosEnable>
  <startPosTop>
    <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
  </startPosTop>
  <startPosLeft>
    <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
  </startPosLeft>
  <overlayInfoText>
    <!--ro, opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
  </overlayInfoText>
  <overlayInfoText2>
    <!--ro, opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
  </overlayInfoText2>
</OverlayInfo>
```

19.6.1.31 Get the parameters of OSD overlay on the composite captured picture

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays?multiCharPosition=<multiCharPosition>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
multiCharPosition	enum	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<MergePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, see details in the message of XML_OverlayInfoList, attr:version{req, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--ro, req, bool, whether to enable the function-->true
  </overlayInfoEnabled>
```

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```
  <OverlayInfoList>
    <!--ro, req, array, list of overlay information, subType:object-->
    <OverlayInfo>
      <!--ro, req, object, overlay information-->
      <overlayInfoID>
        <!--ro, req, int, overlay ID-->0
      </overlayInfoID>
      <itemType>
        <!--ro, req, enum, overlay information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask, 83-vehicle noise decible detection by device (in inverse proportion to the distance between device and vehicle), 85-reflective stripe-->0
      </itemType>
      <itemOverlayEnabled>
        <!--ro, req, bool, whether to overlay the information-->true
      </itemOverlayEnabled>
      <customName>
        <!--ro, req, string, custom overlaying name, range:[0,32], desc:max. 32 bytes; if this node is not returned or empty, the default name will be overlaid-->test
      </customName>
      <changeLineNum>
        <!--ro, req, int, number of line feeds, range:[0,100], desc:the default value is 0-->0
      </changeLineNum>
      <spaceNum>
        <!--ro, req, int, number of spaces, range:[0,255], desc:the default value is 0-->0
      </spaceNum>
      <startPosEnable>
        <!--ro, opt, bool, whether to enable coordinate configuration, desc:it is valid for overlay within the picture; line breaking and space function will be unavailable after it is enabled; it is recommended to be disabled-->true
      </startPosEnable>
      <startPosTop>
        <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
      </startPosTop>
      <startPosLeft>
        <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
      </startPosLeft>
      <overlayInfoText>
        <!--ro, opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
      </overlayInfoText>
      <overlayInfoText2>
        <!--ro, opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
      </overlayInfoText2>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent>
    <!--ro, req, int, percentage of overlaying lines, range:[0,100], desc:the default value is 100-->1
  </linePercent>
  <itemsStlye>
    <!--ro, req, enum, overlay mode, subType:string, desc:"horizontal" (default), "vertical"-->horizontal
  </itemsStlye>
  <charStyle>
    <!--ro, req, enum, font type, subType:string, desc:“song_type”-SimSun (default), “wei_type”-Wei-->song_type
  </charStyle>
  <charSize>
    <!--ro, req, enum, font size, subType:string, desc:“0”-16*16(Chinese)/8*16(English), “1”-32*32(Chinese)/16*32(English), “2”-48*48, “3”-64*64(Chinese)/32*64(English)-->0
  </charSize>
  <charPosition>
    <!--ro, req, enum, position of the text overlaid in the picture, subType:int, desc:“0”-overlaid on the picture, "1”-overlaid outside the top edge of the picture, “2”-overlaid outside the bottom edge of the picture-->0
  </charPosition>
  <charInterval>
    <!--ro, req, int, character pitch, range:[0,16], desc:the default value is 0, unit: pixel-->0
  </charInterval>
  <colorAdapt>
    <!--ro, opt, bool, whether to enable color self-adaption-->true
  </colorAdapt>
  <zeroizeEnable>
    <!--ro, opt, bool, whether to enable zero filling for OSD overlay, desc:it is used to enable zero filling for vehicle speed, speed limit, overspeed ratio, and lane No. Zero filling is enabled by default-->true
  </zeroizeEnable>
  <platePicOverlay>
    <!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
  </platePicOverlay>
  <platePicPosTop>
```

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```
    <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
height, and the default value is 0-->0
  </platePicPosTop>
  <platePicPosLeft>
    <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
width, and the default value is 0-->0
  </platePicPosLeft>
</MergePicOverlays>
```

19.6.1.32 Set vehicle detection parameters

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/vehicleDetect

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VehicleDetectCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, vehicle detection parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <nation>
    <!--req, enum, region, subType:string, desc:"ER"-CIS Region,"EU"-Europe Region,"ME"-Middle East,"AP"-Asia Pacific,"AfricaAndAmerica"-Africa and
America,"All"-All Region-->All
  </nation>
  <bestDetectionSize>
    <!--opt, int, range:[0,1000]-->0
  </bestDetectionSize>
  <VehicleDetectSceneList size="10">
    <!--opt, array, subType:object, attr:size{req, int}-->
    <VehicleDetectScene>
      <!--opt, object, Vehicle Detection Scene-->
      <id>
        <!--req, int, ID, range:[1,10]-->0
      </id>
      <sceneName>
        <!--opt, string, scene name, range:[0,64]-->test
      </sceneName>
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <LaneConfig>
        <!--opt, object, lane line-->
        <LaneList size="10">
          <!--opt, object, lane list, attr:size{req, int}-->
          <Lane>
            <!--opt, object, lane-->
            <laneId>
              <!--req, int, lane No., range:[1,10]-->1
            </laneId>
            <RegionCoordinatesList>
              <!--opt, array, area coordinates list, subType:object-->
              <RegionCoordinates>
                <!--opt, object, area coordinates-->
                <positionX>
                  <!--req, int, X-coordinate, range:[0,1000]-->0
                </positionX>
                <positionY>
                  <!--req, int, Y-coordinate, range:[0,1000]-->0
                </positionY>
              </RegionCoordinates>
            </RegionCoordinatesList>
          </Lane>
        </LaneList>
      </LaneConfig>
      <AboveRoadCalib>
        <!--opt, object-->
        <RegionCoordinatesList>
          <!--opt, array, area coordinates list, subType:object-->
          <RegionCoordinates>
            <!--opt, object, area coordinates-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
```

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```
      </AboveRoadCalib>
      <RelatedLaneParam>
        <!--opt, object, linked lane parameters-->
        <RelatedLaneNoList>
          <!--req, array, linked lane number, subType:object-->
        </RelatedLaneNoList>
      </RelatedLaneParam>
    </VehicleDetectScene>
  </VehicleDetectSceneList>
  <autoBuildRecogArea>
    <!--opt, enum, subType:string-->line
  </autoBuildRecogArea>
  <RodeType>
    <!--opt, object-->
    <type>
      <!--opt, enum, transmitter type, subType:string, desc:"entrance,city,custom,alarmInput"-->entrance
    </type>
  </RodeType>
  <countryIndex>
    <!--opt, int, country/region No., range:[1,255]-->0
  </countryIndex>
  <CRIndex>
    <!--req, enum, country/region, subType:int, desc:0 (Unsupported. It indicates that the algorithm library does not support recognizing license plates of this country), 1 (Czech Republic), 10 (Moldova), 100 (Reserved), 101 (Reserved), 102 (Reserved), 103 (Reserved), 104 (Egypt), 105 (Libya), 106 (Sudan), 107 (Tunisia), 108 (Algeria), 109 (Morocco), 11 (Russia), 110 (Ethiopia), 111 (Eritrea), 112 (Somalia Democratic), 113 (Djibouti), 114 (Kenya), 115 (Tanzania), 116 (Uganda), 117 (Rwanda), 118 (Burundi), 119 (Seychelles), 12 (Ukraine), 120 (Chad), 121 (Central African), 122 (Cameroon), 123 (Equatorial Guinea), 124 (Gabon), 125 (Republic of the Congo (Congo-Brazzaville)), 126 (Democratic Republic of the Congo (Congo-Kinshasa)), 127 (São Tomé and Príncipe), 128 (Mauritania), 129 (Western Sahara (Saharaui)), 13 (Belgium), 130 (Senegal), 131 (Gambia), 132 (Mali), 133 (Burkina Faso), 134 (Guinea), 135 (Guinea-Bissau), 136 (Cape Verde), 137 (Sierra Leone), 138 (Liberia), 139 (Ivory Coast), 14 (Bulgaria), 140 (Ghana), 141 (Togo), 142 (Benin), 143 (Niger), 144 (Zambia), 145 (Angola), 146 (Zimbabwe), 147 (Malawi), 148 (Mozambique), 149 (Botswana), 15 (Denmark), 150 (Namibia), 151 (South Africa), 152 (Swaziland), 153 (Lesotho), 154 (Madagascar), 155 (Comoros), 156 (Mauritius), 157 (Nigeria), 158 (South Sudan), 159 (Saint Helena), 16 (Finland), 160 (Mayotte), 161 (Reunion), 162 (Canary Islands), 163 (AZORES), 164 (Madeira), 165 (Reserved), 166 (Reserved), 167 (Reserved), 168 (Reserved), 169 (Canada), 17 (United Kingdom Great Britain), 170 (Greenland), 171 (St. Pierre and Miquelon), 172 (United State), 173 (Bermuda), 174 (Mexico), 175 (Guatemala), 176 (Belize), 177 (El Salvador), 178 (Honduras), 179 (Nicaragua), 18 (Greece), 180 (Costa Rica), 181 (Panama), 182 (Bahamas), 183 (Turks and  Caicos Islands), 184 (Cuba), 185 (Jamaica), 186 (Cayman Islands), 187 (Haiti), 188 (Dominican), 189 (Puerto Rico), 19 (Croatia), 190 (United States Virgin Islands), 191 (British Virgin Islands), 192 (Anguilla), 193 (Antigua and Barbuda), 194 (Collectivité de Saint-Martin), 195 (Autonomous country), 196 (Saint-Barthélemy), 197 (Saint Kitts and Nevis), 198 (Montserrat), 199 (Guadeloupe), 2 (France), 20 (Hungary), 200 (Dominica), 201 (Martinique), 202 (St. Lucia), 203 (Saint Vincent and the Grenadines), 204 (Grenada), 205 (Barbados), 206 (Trinidad and Tobago), 207 (Curaçao), 208 (Aruba), 209 (Netherlands Antilles), 21 (Israel), 210 (Colombia), 211 (Venezuela), 212 (Guyana), 213 (Suriname), 214 (French Guiana), 215 (Ecuador), 216 (Peru), 217 (Bolivia), 218 (Paraguay), 219 (Chile), 22 (Luxembourg), 220 (Brazil), 221 (Uruguay), 222 (Argentina), 223 (Reserved), 224 (Reserved), 225 (Reserved), 226 (Reserved), 227 (Australia), 228 (New Zealand ), 229 (Papua New Guinea), 23 (Macedonia, North Macedonia after 2018), 230 (Solomon Islands), 231 (Vanuatu), 232 (New Caledonia), 233 (Palau), 234 (Federated States of Micronesia), 235 (Marshall Island), 236 (Northern Mariana Islands), 237 (Guam), 238 (Nauru), 239 (Kiribati), 24 (Norway), 240 (Fiji Islands), 241 (Tonga), 242 (Tuvalu), 243 (Wallis et Futuna), 244 (Samoa), 245 (Eastern Samoa), 246 (Tokelau), 247 (Niue), 248 (Cook Islands), 249 (French Polynesia), 25 (Portugal), 250 (Pitcairn Islands), 251 (Hawaii State), 252 (Reserved), 253 (invalid), 254 (Unrecognized), 255 (ALL), 26 (Romania), 27 (Serbia), 28 (Azerbaijan), 29 (Georgia), 3 (Germany), 30 (Kazakhstan), 31 (Lithuania), 32 (Turkmenistan), 33 (Uzbekistan), 34 (Latvia), 35 (Estonia), 36 (Albania), 37 (Austria), 38 (Bosnia and Herzegovina), 39 (Ireland), 4 (Spain), 40 (Iceland), 41 (Vatican), 42 (Malta), 43 (Sweden), 44 (Switzerland), 45 (Cyprus), 46 (Turkey), 47 (Slovenia), 48 (Montenegro), 49 (Kosovo), 5 (Italy), 50 (Andorra), 51 (Armenia), 52 (Monaco), 53 (Liechtenstein), 54 (San Marino), 55 (Reserved), 56 (Reserved), 57 (Reserved), 58 (Reserved), 59 (China), 6 (Netherlands), 60 (Bahrain), 61 (South Korea), 62 (Lebanon), 63 (Nepal), 64 (Thailand), 65 (Pakistan), 66 (United Arab Emirates), 67 (Bhutan), 68 (Oman), 69 (North Korea), 7 (Poland), 70 (Philippines), 71 (Cambodia), 72 (Qatar), 73 (Kyrgyzstan), 74 (Maldives), 75 (Malaysia), 76 (Mongolia), 77 (Saudi Arabia), 78 (Brunei), 79 (Laos), 8 (Slovakia), 80 (Japan), 81 (Turkey), 82 (Palestinian), 83 (Tajikistan), 84 (Kuwait), 85 (Syria), 86 (India), 87 (Indonesia), 88 (Afghanistan), 89 (Sri Lanka), 9 (Belarus), 90 (Iraq), 91 (Vietnam), 92 (Iran), 93 (Yemen), 94 (Jordan), 95 (Burma/Myanmar), 96 (Sikkim), 97 (Bangladesh), 98 (Singapore), 99 (Democratic Republic of Timor-Leste)-->0
  </CRIndex>
  <detectionMode>
    <!--opt, enum, detection mode, subType:string-->vehiclePriority
  </detectionMode>
  <Filtration>
    <!--opt, object-->
    <enabled>
      <!--opt, bool-->true
    </enabled>
    <filtrationTime>
      <!--opt, int, filtering time, range:[1,5], unit:min-->1
    </filtrationTime>
  </Filtration>
</VehicleDetectCfg>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.6.1.33 Get vehicle detection parameters

Request URL

GET /ISAPI/Traffic/channels/<channelID>/vehicleDetect

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VehicleDetectCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, vehicle detection parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <nation>
    <!--ro, req, enum, region, subType:string, desc:"ER"-CIS Region,"EU"-Europe Region,"ME"-Middle East,"AP"-Asia Pacific,"AfricaAndAmerica"-Africa and
America,"All"-All Region-->All
  </nation>
  <bestDetectionSize>
    <!--ro, opt, int, range:[0,1000]-->0
  </bestDetectionSize>
  <VehicleDetectSceneList size="10">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <VehicleDetectScene>
      <!--ro, opt, object, Vehicle Detection Scene-->
      <id>
        <!--ro, req, int, ID, range:[1,10]-->0
      </id>
      <sceneName>
        <!--ro, opt, string, scene name, range:[0,64]-->test
      </sceneName>
      <enabled>
        <!--ro, req, bool, whether to enable the function-->true
      </enabled>
      <LaneConfig>
        <!--ro, opt, object, lane line-->
        <LaneList size="10">
          <!--ro, opt, array, subType:object, attr:size{opt, int}-->
          <Lane>
            <!--ro, opt, object-->
            <laneId>
              <!--ro, req, int, range:[1,10]-->1
            </laneId>
            <RegionCoordinatesList>
              <!--ro, opt, array, subType:object, range:[0,2]-->
              <RegionCoordinates>
                <!--ro, opt, object-->
                <positionX>
                  <!--ro, req, int, range:[0,1000]-->0
                </positionX>
                <positionY>
                  <!--ro, req, int, range:[0,1000]-->0
                </positionY>
              </RegionCoordinates>
            </RegionCoordinatesList>
          </Lane>
        </LaneList>
      </LaneConfig>
      <AboveRoadCalib>
        <!--ro, opt, object-->
        <RegionCoordinatesList>
          <!--ro, opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </AboveRoadCalib>
      <RelatedLaneParam>
        <!--ro, opt, object, linked lane parameters-->
        <RelatedLaneNoList>
          <!--ro, req, array, linked lane number, subType:object, range:[0,6]-->
        </RelatedLaneNoList>
      </RelatedLaneParam>
```

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```
    </VehicleDetectScene>
  </VehicleDetectSceneList>
  <autoBuildRecogArea>
    <!--ro, opt, enum, subType:string-->line
  </autoBuildRecogArea>
  <RodeType>
    <!--ro, opt, object-->
    <type>
      <!--ro, opt, enum, transmitter type, subType:string, desc:"entrance,city,custom,alarmInput"-->entrance
    </type>
    <quickPoleLift>
      <!--ro, opt, bool, dep:and,{$.VehicleDetectCfg.RodeType.type,eq,entrance}-->true
    </quickPoleLift>
  </RodeType>
  <countryIndex>
    <!--ro, opt, int, country/region No., range:[0,255]-->0
  </countryIndex>
  <CRIndex>
    <!--ro, req, enum, country/region, subType:int, desc:0 (Unsupported. It indicates that the algorithm library does not support recognizing license plates
of this country), 1 (Czech Republic), 10 (Moldova), 100 (Reserved), 101 (Reserved), 102 (Reserved), 103 (Reserved), 104 (Egypt), 105 (Libya), 106 (Sudan),
107 (Tunisia), 108 (Algeria), 109 (Morocco), 11 (Russia), 110 (Ethiopia), 111 (Eritrea), 112 (Somalia Democratic), 113 (Djibouti), 114 (Kenya), 115
(Tanzania), 116 (Uganda), 117 (Rwanda), 118 (Burundi), 119 (Seychelles), 12 (Ukraine), 120 (Chad), 121 (Central African), 122 (Cameroon), 123 (Equatorial
Guinea), 124 (Gabon), 125 (Republic of the Congo (Congo-Brazzaville)), 126 (Democratic Republic of the Congo (Congo-Kinshasa)), 127 (São Tomé and Príncipe),
128 (Mauritania), 129 (Western Sahara (Saharaui)), 13 (Belgium), 130 (Senegal), 131 (Gambia), 132 (Mali), 133 (Burkina Faso), 134 (Guinea), 135 (Guinea-
Bissau), 136 (Cape Verde), 137 (Sierra Leone), 138 (Liberia), 139 (Ivory Coast), 14 (Bulgaria), 140 (Ghana), 141 (Togo), 142 (Benin), 143 (Niger), 144
(Zambia), 145 (Angola), 146 (Zimbabwe), 147 (Malawi), 148 (Mozambique), 149 (Botswana), 15 (Denmark), 150 (Namibia), 151 (South Africa), 152 (Swaziland),
153 (Lesotho), 154 (Madagascar), 155 (Comoros), 156 (Mauritius), 157 (Nigeria), 158 (South Sudan), 159 (Saint Helena), 16 (Finland), 160 (Mayotte), 161
(Reunion), 162 (Canary Islands), 163 (AZORES), 164 (Madeira), 165 (Reserved), 166 (Reserved), 167 (Reserved), 168 (Reserved), 169 (Canada), 17 (United
Kingdom Great Britain), 170 (Greenland), 171 (St. Pierre and Miquelon), 172 (United State), 173 (Bermuda), 174 (Mexico), 175 (Guatemala), 176 (Belize), 177
(El Salvador), 178 (Honduras), 179 (Nicaragua), 18 (Greece), 180 (Costa Rica), 181 (Panama), 182 (Bahamas), 183 (Turks and  Caicos Islands), 184 (Cuba), 185
(Jamaica), 186 (Cayman Islands), 187 (Haiti), 188 (Dominican), 189 (Puerto Rico), 19 (Croatia), 190 (United States Virgin Islands), 191 (British Virgin
Islands), 192 (Anguilla), 193 (Antigua and Barbuda), 194 (Collectivité de Saint-Martin), 195 (Autonomous country), 196 (Saint-Barthélemy), 197 (Saint Kitts
and Nevis), 198 (Montserrat), 199 (Guadeloupe), 2 (France), 20 (Hungary), 200 (Dominica), 201 (Martinique), 202 (St. Lucia), 203 (Saint Vincent and the
Grenadines), 204 (Grenada), 205 (Barbados), 206 (Trinidad and Tobago), 207 (Curaçao), 208 (Aruba), 209 (Netherlands Antilles), 21 (Israel), 210 (Colombia),
211 (Venezuela), 212 (Guyana), 213 (Suriname), 214 (French Guiana), 215 (Ecuador), 216 (Peru), 217 (Bolivia), 218 (Paraguay), 219 (Chile), 22 (Luxembourg),
220 (Brazil), 221 (Uruguay), 222 (Argentina), 223 (Reserved), 224 (Reserved), 225 (Reserved), 226 (Reserved), 227 (Australia), 228 (New Zealand ), 229
(Papua New Guinea), 23 (Macedonia, North Macedonia after 2018), 230 (Solomon Islands), 231 (Vanuatu), 232 (New Caledonia), 233 (Palau), 234 (Federated
States of Micronesia), 235 (Marshall Island), 236 (Northern Mariana Islands), 237 (Guam), 238 (Nauru), 239 (Kiribati), 24 (Norway), 240 (Fiji Islands), 241
(Tonga), 242 (Tuvalu), 243 (Wallis et Futuna), 244 (Samoa), 245 (Eastern Samoa), 246 (Tokelau), 247 (Niue), 248 (Cook Islands), 249 (French Polynesia), 25
(Portugal), 250 (Pitcairn Islands), 251 (Hawaii State), 252 (Reserved), 253 (invalid), 254 (Unrecognized), 255 (All), 26 (Romania), 27 (Serbia), 28
(Azerbaijan), 29 (Georgia), 3 (Germany), 30 (Kazakhstan), 31 (Lithuania), 32 (Turkmenistan), 33 (Uzbekistan), 34 (Latvia), 35 (Estonia), 36 (Albania), 37
(Austria), 38 (Bosnia and Herzegovina), 39 (Ireland), 4 (Spain), 40 (Iceland), 41 (Vatican), 42 (Malta), 43 (Sweden), 44 (Switzerland), 45 (Cyprus), 46
(Turkey), 47 (Slovenia), 48 (Montenegro), 49 (Kosovo), 5 (Italy), 50 (Andorra), 51 (Armenia), 52 (Monaco), 53 (Liechtenstein), 54 (San Marino), 55
(Reserved), 56 (Reserved), 57 (Reserved), 58 (Reserved), 59 (China), 6 (Netherlands), 60 (Bahrain), 61 (South Korea), 62 (Lebanon), 63 (Nepal), 64
(Thailand), 65 (Pakistan), 66 (United Arab Emirates), 67 (Bhutan), 68 (Oman), 69 (North Korea), 7 (Poland), 70 (Philippines), 71 (Cambodia), 72 (Qatar), 73
(Kyrgyzstan), 74 (Maldives), 75 (Malaysia), 76 (Mongolia), 77 (Saudi Arabia), 78 (Brunei), 79 (Laos), 8 (Slovakia), 80 (Japan), 81 (Turkey), 82
(Palestinian), 83 (Tajikistan), 84 (Kuwait), 85 (Syria), 86 (India), 87 (Indonesia), 88 (Afghanistan), 89 (Sri Lanka), 9 (Belarus), 90 (Iraq), 91 (Vietnam),
92 (Iran), 93 (Yemen), 94 (Jordan), 95 (Burma/Myanmar), 96 (Sikkim), 97 (Bangladesh), 98 (Singapore), 99 (Democratic Republic of Timor-Leste)-->0
  </CRIndex>
  <detectionMode>
    <!--ro, opt, enum, detection mode, subType:string, desc:detection mode-->vehiclePriority
  </detectionMode>
  <Filtration>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <filtrationTime>
      <!--ro, opt, int, filtering time, range:[1,5], unit:min, desc:filtering time-->1
    </filtrationTime>
  </Filtration>
  <zeroAngleInstallationAlgoEnabled>
    <!--ro, opt, bool-->true
  </zeroAngleInstallationAlgoEnabled>
  <plateSubSectionShowEnabled>
    <!--ro, opt, bool-->true
  </plateSubSectionShowEnabled>
  <motorbikeReportEnable>
    <!--ro, opt, bool-->true
  </motorbikeReportEnable>
  <doubleLicensePlateReportEnable>
    <!--ro, opt, bool-->true
  </doubleLicensePlateReportEnable>
</VehicleDetectCfg>
```

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19.6.1.34 Set vehicle detection parameters at a specific scene

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/vehicleDetect/<SID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
SID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VehicleDetectScene xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, vehicle detection scene, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, index, range:[1,10]-->0
  </id>
  <sceneName>
    <!--opt, string, scene name, range:[0,64]-->test
  </sceneName>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <PlateRecogParam>
    <!--opt, object-->
    <PlateRecogRegionList>
      <!--opt, array, subType:object-->
      <PlateRecogRegion>
        <!--opt, object, ANPR region-->
        <id>
          <!--req, string, index, range:[1,10]-->test
        </id>
        <RegionCoordinatesList>
          <!--opt, array, area coordinates list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </PlateRecogRegion>
    </PlateRecogRegionList>
  </PlateRecogParam>
  <LaneConfig>
    <!--opt, object, lane line-->
    <LaneList size="10">
      <!--opt, object, lane list, attr:size{req, int}-->
      <Lane>
        <!--opt, object, lane-->
        <laneId>
          <!--req, int, lane No., range:[1,10]-->1
        </laneId>
        <RegionCoordinatesList>
          <!--opt, array, area coordinates list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <lineType>
          <!--req, enum, line type, subType:string, desc:laneBoundaryLine; aneLine;-->laneBoundaryLine
        </lineType>
        <carDriveDirect>
          <!--req, enum, vehicle driving direction, subType:string, desc:down2up: from down to up; up2down: from up to down;-->down_to_up
        </carDriveDirect>
        <RightLaneLine>
          <!--opt, object, right lane line, desc:right lane line coordinate. When this node is returned,it indicates that both left and right lane lines are
required to be configured for each lane line-->
          <RegionCoordinatesList size="2">
            <!--req, array, area coordinates list, subType:object, range:[0,2], attr:size{req, int}-->
            <RegionCoordinates>
              <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--req, int, X-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--req, int, Y-coordinate, range:[0,1000]-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </RightLaneLine>
      </Lane>
```

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```
      </LaneList>
    </LaneConfig>
    <AtRoadsideCalib>
      <!--opt, object-->
      <RegionCoordinatesList>
        <!--opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:The origin is the lower-left corner of the screen.-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </AtRoadsideCalib>
    <ParkingDetection>
      <!--opt, object, vehicle parking detection-->
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <duration>
        <!--req, int, detection time interval, range:[5,60], unit:s-->0
      </duration>
      <alarmIntervalTime>
        <!--req, int, uploading interval, range:[1,30], unit:min-->0
      </alarmIntervalTime>
    </ParkingDetection>
    <AboveRoadCalib>
      <!--opt, object-->
      <RegionCoordinatesList>
        <!--opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </AboveRoadCalib>
    <RelatedLaneParam>
      <!--opt, object, linked lane parameters-->
      <RelatedLaneNoList>
        <!--req, array, linked lane No., subType:object-->
        <relatedLaneNo>
          <!--req, int, linked lane No., range:[1,6]-->0
        </relatedLaneNo>
      </RelatedLaneNoList>
    </RelatedLaneParam>
    <relatedPlateColor>
      <!--opt, string-->other
    </relatedPlateColor>
    <relatedVehicleType>
      <!--opt, string-->vehicle
    </relatedVehicleType>
  </VehicleDetectScene>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.6.1.35 Get vehicle detection parameters at a specific scene

Request URL

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GET /ISAPI/Traffic/channels/<channelID>/vehicleDetect/<SID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
SID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VehicleDetectScene xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, vehicle detection scene, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID, range:[1,10]-->0
  </id>
  <sceneName>
    <!--ro, opt, string, scene name, range:[0,64]-->test
  </sceneName>
  <enabled>
    <!--ro, req, bool, whether to enable the scene-->true
  </enabled>
  <PlateRecogParam>
    <!--ro, opt, object-->
    <PlateRecogRegionList>
      <!--ro, opt, array, subType:object-->
      <PlateRecogRegion>
        <!--ro, opt, object, list-->
        <id>
          <!--ro, req, string, ID, range:[1,10]-->test
        </id>
        <RegionCoordinatesList>
          <!--ro, opt, array, area coordinate list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, x-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </PlateRecogRegion>
    </PlateRecogRegionList>
  </PlateRecogParam>
  <LaneConfig>
    <!--ro, opt, object, lane line-->
    <LaneList size="10">
      <!--ro, opt, object, lane list, attr:size{req, int}-->
      <Lane>
        <!--ro, opt, object, lane-->
        <laneId>
          <!--ro, req, int, lane No., range:[1,10]-->1
        </laneId>
        <RegionCoordinatesList>
          <!--ro, opt, array, area coordinate list, subType:object, range:[0,2], desc:area coordinate list-->
          <RegionCoordinates>
            <!--ro, opt, object, coordinates of the license plate recognition region, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, x-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <lineType>
          <!--ro, req, enum, lane line type, subType:string, desc:line type: "LaneBoundaryLine" (lane boundary line),"LaneLine" (lane line)--
>laneBoundaryLine
        </lineType>
        <carDriveDirect>
          <!--ro, req, enum, driving direction, subType:string, desc:vehicle driving direction,this node is supported when the value of lineType is
"LaneLine": "unknow","up_to_down","down_to_up"-->down_to_up
        </carDriveDirect>
        <RightLaneLine>
          <!--ro, opt, object, right lane line, desc:right lane line coordinate. When this node is returned,it indicates that both left and right lane lines
are required to be configured for each lane line-->
          <RegionCoordinatesList size="2">
            <!--ro, req, array, area coordinate list, subType:object, range:[0,2], attr:size{req, int}-->
            <RegionCoordinates>
              <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
```

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```
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, x-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </RightLaneLine>
    </Lane>
  </LaneList>
</LaneConfig>
<AtRoadsideCalib>
  <!--ro, opt, object-->
  <RegionCoordinatesList>
    <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</AtRoadsideCalib>
<ParkingDetection>
  <!--ro, opt, object, vehicle parking detection-->
  <enabled>
    <!--ro, req, bool, whether to enable the scene-->true
  </enabled>
  <duration>
    <!--ro, req, int, detection interval, range:[5,60], unit:s-->0
  </duration>
  <alarmIntervalTime>
    <!--ro, req, int, uploading interval, range:[1,30], unit:min-->0
  </alarmIntervalTime>
</ParkingDetection>
<AboveRoadCalib>
  <!--ro, opt, object-->
  <RegionCoordinatesList>
    <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</AboveRoadCalib>
<RelatedLaneParam>
  <!--ro, opt, object, linked lane parameters-->
  <RelatedLaneNoList>
    <!--ro, req, array, list of related lane No.s,the maximum number can be 6, subType:object-->
    <relatedLaneNo>
      <!--ro, req, int, related lane No., range:[1,6]-->0
    </relatedLaneNo>
  </RelatedLaneNoList>
</RelatedLaneParam>
<relatedPlateColor>
  <!--ro, opt, string-->other
</relatedPlateColor>
<relatedVehicleType>
  <!--ro, opt, string-->vehicle
</relatedVehicleType>
</VehicleDetectScene>
```

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19.6.1.36 Get the capability of vehicle detection parameters

Request URL

GET /ISAPI/Traffic/channels/<channelID>/vehicleDetect/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VehicleDetectCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <nation opt="EU,ER,EUandCIS,ME,APAC,AfricaAndAmerica,All">
    <!--ro, req, enum, region, subType:string, attr:opt{req, string}, desc:region-->All
  </nation>
  <bestDetectionSize>
    <!--ro, opt, int, range:[0,1000]-->0
  </bestDetectionSize>
  <VehicleDetectSceneList size="10">
    <!--ro, opt, array, list of vehicle detection scenes, subType:object, attr:size{req, int}-->
    <VehicleDetectScene>
      <!--ro, opt, object, vehicle detection scene-->
      <id>
        <!--ro, req, int, ID, range:[1,10]-->0
      </id>
      <sceneName min="0" max="64">
        <!--ro, opt, string, scene name, range:[0,64], attr:min{req, int},max{req, int}-->test
      </sceneName>
      <enabled>
        <!--ro, req, bool, whether to enable the function-->true
      </enabled>
      <LaneConfig>
        <!--ro, opt, object, lane line-->
        <LaneList size="4">
          <!--ro, opt, array, lane list, subType:object, attr:size{req, int}-->
          <Lane>
            <!--ro, opt, object, lane-->
            <laneId min="1" max="4">
              <!--ro, req, int, lane No., range:[1,4], attr:min{req, int},max{req, int}-->1
            </laneId>
            <RegionCoordinatesList size="10">
              <!--ro, opt, array, area coordinates list, subType:object, range:[0,2], attr:size{req, int}-->
              <RegionCoordinates>
                <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
                <positionX>
                  <!--ro, req, int, X-coordinate, range:[0,1000]-->0
                </positionX>
                <positionY>
                  <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
                </positionY>
              </RegionCoordinates>
            </RegionCoordinatesList>
          </Lane>
        </LaneList>
      </LaneConfig>
    </VehicleDetectScene>
  </VehicleDetectSceneList>
  <autoBuildRecogArea opt="line,region">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->line
  </autoBuildRecogArea>
  <RodeType>
    <!--ro, opt, object, road type-->
    <type opt="entrance,city,custom,alarmInput,post,publicSecurity">
      <!--ro, opt, enum, type, subType:string, attr:opt{req, string}, desc:type-->entrance
    </type>
    <LaneLineNum>
      <!--ro, opt, object-->
      <entrance min="0" max="10">
        <!--ro, opt, int, range:[0,10], attr:min{req, int},max{req, int}-->0
      </entrance>
      <city min="0" max="10">
        <!--ro, opt, int, range:[0,10], attr:min{req, int},max{req, int}-->0
      </city>
      <alarmInput min="0" max="10">
        <!--ro, opt, int, range:[0,10], attr:min{req, int},max{req, int}-->0
      </alarmInput>
    </LaneLineNum>
    <quickPoleLift opt="true,false">
      <!--ro, opt, bool, dep:and,{$.VehicleDetectCfg.RodeType.type,eq,entrance}, attr:opt{req, string}-->true
    </quickPoleLift>
  </RodeType>
  <countryIndex opt="1,2,3,4,5,6,7,8,12,14,17,18,19,20,23,39,44,55">
    <!--ro, opt, int, attr:opt{req, string}-->0
  </countryIndex>
  <RelatedLaneParam>
    <!--ro, opt, object, linked lane parameters-->
    <RelatedLaneNoList size="6">
      <!--ro, req, array, linked lane number, subType:object, attr:size{req, int}, desc:linked lane number-->
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <CRIndex
opt="0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,25,26,28,29,30,31,33,34,35,36,37,38,39,40,42,43,44,45,46,47,49,51,53,55,59,60,62,65,68,70
,71,72,73,76,77,84,87,89,90,94,95,91,104,107,108,110,114,115,130,137,139,140,144,145,146,151,156,157,169,174,175,177,180,181,188,206,210,215,216,217,218,219
,220,221,222,227,228,256">
    <!--ro, req, enum, country/region, subType:int, attr:opt{req, string}, desc:country/region-->0
  </CRIndex>
```

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```
  <detectionMode opt="vehiclePriority,licensePlateOrVehicle">
    <!--ro, opt, enum, detection mode, subType:string, attr:opt{req, string}, desc:detection mode-->vehiclePriority
  </detectionMode>
  <Filtration>
    <!--ro, opt, object-->
    <enabled opt="true,false" def="false">
      <!--ro, opt, bool, whether to enable filtering the same license plate, attr:opt{req, string},def{req, string}, desc:whether to enable filtering the
same license plate-->true
    </enabled>
    <filtrationTime min="1" max="5" def="4">
      <!--ro, opt, int, filtering time, range:[1,5], unit:min, attr:min{req, int},max{req, int},def{req, int}, desc:filtering time-->1
    </filtrationTime>
  </Filtration>
  <RelatedNationAndCRListCap>
    <!--ro, opt, array, subType:object-->
    <RelatedNationAndCR>
      <!--ro, opt, object-->
      <nation>
        <!--ro, opt, enum, region, subType:string, desc:region-->EU
      </nation>
      <CRIndex opt="1,2,3,4,5,6,7">
        <!--ro, opt, int, attr:opt{req, string}-->1
      </CRIndex>
    </RelatedNationAndCR>
  </RelatedNationAndCRListCap>
  <zeroAngleInstallationAlgoEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </zeroAngleInstallationAlgoEnabled>
  <plateSubSectionShowEnabled opt="true,false" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </plateSubSectionShowEnabled>
  <motorbikeReportEnable opt="true,false" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </motorbikeReportEnable>
  <doubleLicensePlateReportEnable opt="true,false" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </doubleLicensePlateReportEnable>
  <fuzzyMatchEnabled opt="true,false" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->false
  </fuzzyMatchEnabled>
</VehicleDetectCfg>
```

19.6.1.37 Set vehicle detection configuration parameters

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/vehicleDetect/config

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Configuration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, vehicle detection parameters, attr:version{req, string, protocolVersion}-->
  <needAuth>
    <!--req, bool, whether to enable certification-->true
  </needAuth>
  <dir>
    <!--opt, string, path-->test
  </dir>
  <picTime>
    <!--opt, string-->test
  </picTime>
  <picDisplay>
    <!--req, bool-->true
  </picDisplay>
</Configuration>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.6.1.38 Get vehicle detection parameters

Request URL

GET /ISAPI/Traffic/channels/<channelID>/vehicleDetect/config

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Configuration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, vehicle detection parameters, attr:version{req, string, protocolVersion}-->
  <needAuth>
    <!--ro, req, bool-->true
  </needAuth>
  <dir>
    <!--ro, opt, string-->test
  </dir>
  <picTime>
    <!--ro, opt, string-->test
  </picTime>
  <picDisplay>
    <!--ro, req, bool-->true
  </picDisplay>
</Configuration>
```

19.6.1.39 License plate recognition

EventType:ANPR

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, license plate recognition result, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, req, enum, protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP"
when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
```

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```
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"ANPR" (license plate recognition)-->ANPR
  </eventType>
  <eventState>
    <!--ro, req, enum, continuous event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, enum, event description, subType:string, desc:"ANPR" (license plate recognition)-->ANPR
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,128]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <ANPR>
    <!--ro, opt, object, information about license plate recognition alarm-->
    <region>
      <!--ro, opt, enum, region, subType:string, desc:"ER" (Russian-speaking region), "EU" (Europe), "EUandCIS" (Europe and Russia), "ME" (Middle East),
"other" (other regions), "APAC" (Asia-Pacific region), "AFandAM" (Africa and America), "THAandLA" (Thailand and Laos), "HKandMO" (Hong Kong China and Macao
China), "All" (all regions),-->ER
    </region>
    <country>
      <!--ro, opt, enum, country or region, subType:int, desc:253-invalid, 70-Philippines, 17-United Kingdom (previously Great Britain), 23-Macedonia (North
Macedonia since 2018), 188-Dominican, 227-Australia, 60-Bahrain, 95-Burma/Myanmar, 228-New Zealand, 28-Azerbaijan, 131-Gambia, 132-Mali, 159-Saint Helena,
133-Burkina Faso, 134-Guinea, 135-Guinea-Bissau, 136-Cape Verde, 137-Sierra Leone, 138-Liberia, 139-Ivory Coast, 140-Ghana, 141-Togo, 142-Benin, 143-Niger,
144-Zambia, 145-Angola, 146-Zimbabwe, 147-Malawi, 148-Mozambique, 149-Botswana, 150-Namibia, 151-South Africa, 152-Swaziland, 153-Lesotho, 154-Madagasca,
155-Comoros, 156-Mauritius, 157-Nigeria, 158-South Sudan, 160-Mayotte, 161-Reunion, 162-Canary Islands, 163-AZORES, 164-Madeira, 165-reserved, 166-reserved,
167-reserved, 168-reserved, 169-Canada, 170-Greenland Nuuk, 171-Pierre and Miquelon, 172-United States, 173-Bermuda, 174-Mexico, 175-Guatemala, 176-Belize,
177-El Salvador, 178-Honduras, 179-Nicaragua, 180-Costa Rica, 181-Panama, 182-Bahamas, 183-Turks and  Caicos Islands, 184-Cuba, 185-Jamaica, 186-Cayman
Islands, 187-Haiti, 189-Puerto Rico, 190-United States Virgin Islands, 191-British Virgin Islands, 192-Anguilla, 193-Antigua and Barbuda, 194-Collectivit de
Saint-Martin, 195-Autonomous country, 196-Saint-Barthlemy, 197-Saint Kitts and Nevis, 198-Montserrat, 199-Guadeloupe, 200-Dominica, 201-Martinique, 202-St.
Lucia, 203-Saint Vincent and the Grenadines, 204-Grenada, 205-Barbados, 206-Trinidad and Tobago, 207-Cura?ao, 0-Unsupported, 1-Czech Republic, 2-France, 3-
Germany, 4-Spain, 5-Italy, 6-Netherlands, 7-Poland, 8-Slovakia, 9-Belarus, 10-Moldova, 11-Russia, 12-Ukraine, 13-Belgium, 14-Bulgaria, 15-Denmark, 16-
Finland, 18-Greece, 19-Croatia, 20-Hungary, 21-Israel(), 22-Luxembourg, 24-Norway, 25-Portuga, 26-Romania, 27-Serbia, 29-Georgia(), 30-Kazakhstan(), 31-
Lithuania, 32-Turkmenistan(), 33-Uzbekistan(), 34-Latvia, 35-Estonia, 36-Albania, 37-Austria, 38-Bosnia and Herzegovina, 39-Ireland, 40-Iceland, 41-Vatican,
42-Malta, 43-Sweden, 44-Switzerland, 45-Cyprus, 46-Turkey, 47-Slovenia, 48-Montenegro, 49-Kosovo, 50-Andorra, 51-Armenia(), 52-Monaco, 53-Liechtenstein, 54-
San Marino, 55-reserved, 56-reserved, 90-Iraq, 57-reserved, 58-reserved, 59-China, 91-Vietnam, 61-South Korea, 62-Lebanon, 63-Nepal, 64-Thailand, 65-
Pakistan, 66-United Arab Emirates, 67-Bhutan, 68-Oman, 69-North Korea, 71-Cambodia, 72-Qatar, 73-Kyrgyzstan, 74-Maldives, 75-Malaysia, 76-Mongolia, 77-Saudi
Arabia, 78-Brunei, 79-Laos, 80-Japan, 81-Turkey, 82-Palestinian, 83-Tajikistan, 84-Kuwait, 85-Syria, 86-India, 87-Indonesia, 88-Afghanistan, 89-Sri Lanka,
92-Iran, 93-Yemen, 94-Jordan, 96-Sikkim, 97-Bangladesh, 98-Singapore, 99-Democratic Republic of Timor-Leste, 100-reserved, 101-reserved, 102-reserved, 103-
reserved, 104-Egypt, 105-Libya, 106-Sudan, 107-Tunisia, 108-Algeria, 109-Morocco, 110-Ethiopia, 111-Eritrea, 112-Somalia Democratic, 113-Djibouti, 114-
Kenya, 115-Tanzania, 116-Uganda, 117-Rwanda, 118-Burundi, 119-Seychelles, 120-Chad, 121-Central African, 122-Cameroon, 123-Equatorial Guinea, 124-Gabon,
125-Congo, 126-Democratic Republic of the Congo, 127-Sao Tome and Principe, 128-Mauritania, 129-Western Sahara, 130-Senegal, 208-Aruba, 209-Netherlands
Antilles, 210-Colombia, 211-Venezuela, 212-Guyana, 213-Suriname, 214-Guyane Francaise, 215-Ecuador, 216-Peru, 217-Bolivia, 218-Paraguay, 219-Chile, 220-
Brazil, 221-Uruguay, 222-Argentina, 223-reserved, 224-reserved, 225-reserved, 226-reserved, 229-Papua New Guinea, 230-Salomonen, 231-Vanuatu, 232-New
Caledonia, 233-Palau, 234-Federated States of Micronesia, 235-Marshall Island, 236-Northern Mariana Islands, 237-Guam, 238-Nauru, 239-Kiribati, 240-Fidschi,
241-Tonga, 242-Tuvalu, 243-Wallis et Futuna, 244-Samoa, 245-Eastern Samoa, 246-Tokelau, 247-Niue, 248-Cook Islands, 249-French Polynesia, 250-Pitcairn
Islands, 251-Hawaii State, 252-reserved, 254-unrecognized, 255-all-->253
    </country>
    <area>
      <!--ro, opt, enum, United Arab Emirates, subType:string, desc:"FJR" (Fujairah), "AD" (Abu Dhabi), "unknown", "UMW" (Umm al-Qaiwain), "other", "AM"
(Ajman), "RAK" (Ras al-Khaimah), "DB" (Dubai), "SJ" (Sharjah)-->FJR
    </area>
    <licensePlate>
      <!--ro, req, string, license plate number, range:[1,32], desc:noPlate (vehicle without license plate), unknown (no license plate recognized), XXXXXXXX
(specific license plate number; color information required by Chinese license plate; for motor vehicles, it is a 16-byte string; for non-motor vehicles, it
is a 48-byte string)-->A283KY77
    </licensePlate>
    <cameraNo>
      <!--ro, opt, string, device No., range:[1,48], desc:corresponds to <cameraNum> from /ISAPI/System/Video/inputs/channels/<ID>/cameraInfo (recommended
for new devices) and /ISAPI/Traffic/channels/<channelID>/cameraInfo applied by some traffic monitoring device-->test
    </cameraNo>
    <line>
      <!--ro, req, int, recognized lane No.-->1
    </line>
    <direction>
      <!--ro, opt, enum, license plate recognition direction, subType:string, desc:"reverse", "forward", "unknown"-->reverse
    </direction>
    <confidenceLevel>
      <!--ro, req, int, confidence level, range:[0,100]-->50
    </confidenceLevel>
    <plateType>
      <!--ro, opt, enum, license plate type, subType:string, desc:"unknown", "arm" (police vehicle), "92TypeArm" (type 92 armed police vehicle),
"02TypePersonalized" (type 02 custom vehicle),  "yellowTwoLine" (yellow two-line rear license plate), "embassy" (embassy vehicle), "oneLineArm"(new armed
police vehicle, one-line), "twoLineArm" (new armed police vehicle, two-line), "yellow1225FarmVehicle" (agricultural vehicle, yellow 1225),
"green1325FarmVehicle" (agricultural vehicle, green 1325), "yellow1325FarmVehicle" (agricultural vehicle, yellow 1325), "motorola" (motorcycle), "coach"
(driver-training vehicle), "tempTravl" (vehicle with temporary license plate), "trailer" (trailer), "consulate" (consular vehicle), "hongKongMacao" (Hong
Kong China/Macao China entrance vehicle), "tempEntry" (temporary vehicle), "civilAviation" (civil aviation license plate), "newEnergy" (new energy vehicle),
```

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```
"92FarmVehicle" (two-line license plate civil vehicle), "emergency" (emergency license plate), "oneLineArmHeadquarters" (armed police headquarter license plate, one-line), "twoLineArmHeadquarters" (armed police headquarter license plate, two-line), "twoWheelVehicle" (two wheeler)-->unknown
    </plateType>
    <plateColor>
      <!--ro, opt, enum, license plate color, subType:string, desc:"black", "blue","golden", "orange", "red", "yellow", "white", "unknown", "other", "newEnergyYellowGreen"-new energy yellow-green,"civilAviationBlack", "civilAviationGreen" (civil aviation green), "green", "mixedColor", "newEnergyGreen" (new energy green), "brown"-->black
    </plateColor>
    <licenseBright>
      <!--ro, opt, int, license plate brightness, which ranges from 0 to 255, range:[0,255]-->50
    </licenseBright>
    <Rect>
      <!--ro, opt, object, coordinates of the license plate thumbnail in the matched picture, desc:used by back-end DeepinMind; the origin is the upper-left corner of the screen-->
      <height>
        <!--ro, req, float, height, range:[0.000,1.000]-->1.000
      </height>
      <width>
        <!--ro, req, float, width, range:[0.000,1.000]-->1.000
      </width>
      <x>
        <!--ro, req, float, the reference origin is the upper left corner of image, range:[0.000,1.000]-->1.000
      </x>
      <y>
        <!--ro, req, float, the reference origin is the upper left corner of image, range:[0.000,1.000]-->1.000
      </y>
    </Rect>
    <pilotsafebelt>
      <!--ro, opt, enum, whether the driver is wearing a safety belt, subType:string, desc:"unknown", "yes", "no"-->unknown
    </pilotsafebelt>
    <vicepilotsafebelt>
      <!--ro, opt, enum, whether the co-driver is wearing a safety belt, subType:string, desc:"unknown", "yes", "no"-->unknown
    </vicepilotsafebelt>
    <pilotsunvisor>
      <!--ro, opt, enum, whether the driver room's sun visor is open, subType:string, desc:"unknown", "yes", "no"-->unknown
    </pilotsunvisor>
    <vicepilotsunvisor>
      <!--ro, opt, enum, whether the co-driver room's sun visor is open, subType:string, desc:"unknown", "yes", "no"-->unknown
    </vicepilotsunvisor>
    <envprosign>
      <!--ro, opt, enum, whether it is a yellow-label vehicle, subType:string, desc:"unknown", "green" (green-label), "yellow" (yellow-lable), "yes", "no"-->unknown
    </envprosign>
    <dangmark>
      <!--ro, opt, enum, whether it is a dangerous goods vehicle, subType:string, desc:"unknown", "yes", "no"-->unknown
    </dangmark>
    <uphone>
      <!--ro, opt, enum, whether the driver is making a phone call, subType:string, desc:"unknown", "yes", "no"-->unknown
    </uphone>
    <pendant>
      <!--ro, opt, enum, whether there are window hangings detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </pendant>
    <tissueBox>
      <!--ro, opt, enum, whether there is a tissue box detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </tissueBox>
    <frontChild>
      <!--ro, opt, enum, whether the co-driver is with a baby in arm, subType:string, desc:"unknown", "yes", "no"-->unknown
    </frontChild>
    <label>
      <!--ro, opt, enum, whether there are stickers detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </label>
    <decoration>
      <!--ro, opt, enum, whether there are decorations detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </decoration>
    <smoking>
      <!--ro, opt, enum, whether anyone is smoking, subType:string, desc:"unknown", "yes", "no"-->yes
    </smoking>
    <perfumeBox>
      <!--ro, opt, enum, whether there is perfume box detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </perfumeBox>
    <pdvs>
      <!--ro, opt, enum, whether there is a person sticking out of sunroof, subType:string, desc:"unknown", "yes", "no"-->unknown
    </pdvs>
    <helmet>
      <!--ro, opt, enum, whether there is helmet detected, subType:string, desc:"unknown", "yes", "no"-->no
    </helmet>
    <twoWheelVehicle>
      <!--ro, opt, enum, whether there is a two-wheel vehicle detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </twoWheelVehicle>
    <threeWheelVehicle>
      <!--ro, opt, enum, whether there is a three-wheel vehicle detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </threeWheelVehicle>
    <blackness>
      <!--ro, opt, int, Ringelmann emittance, desc:it is used for black smoke detection-->2
    </blackness>
    <plateCharBelieve>
      <!--ro, opt, string, confidence of each character in the recognized license plate, desc:if "A12345" is detected as a license plate number and the confidence is 10, 20, 30, 40, 50, 60, it indicates that the accuracy of "A" is 20%-->test
    </plateCharBelieve>
    <speedLimit>
      <!--ro, opt, int, maximum speed limit, desc:it is valid only when overspeeding occurred-->50
    </speedLimit>
    <illegalInfo>
```

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```
      <!--ro, opt, object, vehicle traffic violation information, desc:illegal action information-->
      <illegalCode>
        <!--ro, req, string, illegal action code, range:[0,64]-->1301
      </illegalCode>
      <illegalName>
        <!--ro, req, string, illegal action name, range:[0,128]-->逆行
      </illegalName>
      <illegalDescription>
        <!--ro, opt, string, illegal action description, range:[0,256]-->车辆逆行
      </illegalDescription>
    </illegalInfo>
    <vehicleType>
      <!--ro, req, enum, vehicle type, subType:string, desc:vehicle type:
"unknown,largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickup
Truck"-->nonmotorVehicle
    </vehicleType>
    <postPicFileName>
      <!--ro, opt, string, name of the picture selected as the checkpoint picture when illegal action occurs, desc:"none" (no picture selected)-->test
    </postPicFileName>
    <featurePicFileName>
      <!--ro, opt, string, name of the picture selected as the close-up picture when running the red light in the intersection violation system is detected,
desc:"none" (no picture selected)-->test
    </featurePicFileName>
    <detectDir>
      <!--ro, opt, enum, detection direction, subType:int, desc:1 (upward), 2 (downward), 3 (bidirectional), 4 (westward), 5 (northward), 6 (eastward), 7
(southward), 8 (other)-->1
    </detectDir>
    <detectType>
      <!--ro, opt, enum, detection type, subType:int, desc:0 (vehicle detection), 1 (inductive loop trigger), 2 (video trigger), 3 (multiple-frame
recognition), 4 (radar trigger), 5 (mixed-traffic detection)-->1
    </detectType>
    <barrierGateCtrlType>
      <!--ro, opt, enum, whether to enable elapsed time, subType:int, desc:0 (enabled), 1 (disabled)-->0
    </barrierGateCtrlType>
    <alarmDataType>
      <!--ro, opt, enum, alarm data type (real-time data or history data), subType:int, desc:0 (real-time data), 1 (history data)-->0
    </alarmDataType>
    <dwIllegalTime>
      <!--ro, opt, int, illegal action duration, unit:ms, desc:it is the difference between the capture time of the last picture and the capture time of the
first picture-->100
    </dwIllegalTime>
    <vehicleInfo>
      <!--ro, opt, object, vehicle information-->
      <index>
        <!--ro, req, int, vehicle No.-->1
      </index>
      <vehicleType>
        <!--ro, opt, enum, vehicle type, subType:int, desc:vehicle type:
"unknown,largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickup
Truck"-->0
      </vehicleType>
      <colorDepth>
        <!--ro, req, enum, shade of the vehicle color, subType:int, desc:0 (deep color), 1 (light color), 2 (unknown)-->0
      </colorDepth>
      <color>
        <!--ro, req, enum, vehicle color, subType:string, desc:"unknown", "green", "brown", "pink", "purple", "deepGray" (dark gray), "cyan", "orange",
"white", "silver" (silvery), "gray", "blacks", "red", "deepBlue" (dark blue), "blue", "yellow"-->green
      </color>
      <speed>
        <!--ro, opt, int, vehicle speed, unit:km/h-->1
      </speed>
      <length>
        <!--ro, req, int, length of the former vehicle, unit:dm-->10
      </length>
      <vehicleLogoRecog>
        <!--ro, req, int, vehicle main brand-->1
      </vehicleLogoRecog>
      <vehileSubLogoRecog>
        <!--ro, opt, int, vehicle sub-brand-->1
      </vehileSubLogoRecog>
      <vehileModel>
        <!--ro, opt, int, model year of vehicle sub-brand-->1
      </vehileModel>
      <vehicleTypeByWeight>
        <!--ro, opt, enum, vehicle type according to the vehicle weight, subType:int, desc:1 (class 1: buses with seven or less seats, trucks with capacity
of 2 tons or less), 2 (class 2: buses with 8 to 19 seats, trucks with capacity of 2 to 5 (included) tons), 3 (class 3: buses with 20 to 39 seats, trucks
with capacity of 5 to 10 (included) tons),4 (class 4: buses with 40 or more seats, trucks with capacity of 10 to 15 (included) tons), 5 (class 5: trucks
with capacity of more than 15 tons), 6 (class 6: special vehicle such as ambulance, fire truck, and garbage truck)-->4
      </vehicleTypeByWeight>
      <tollRoadVehicleSeries>
        <!--ro, opt, enum, vehicles on tollway, subType:int, desc:N/A-->2
      </tollRoadVehicleSeries>
      <tollRoadVehicleType>
        <!--ro, opt, enum, vehicle types on tollway, subType:int, desc:N/A-->4
      </tollRoadVehicleType>
      <CarWindowFeature>
        <!--ro, opt, object, vehicle window feature, desc:configured via <CarWindowFeature> from /ISAPI/ITC/carFeatureParam-->
        <tempPlate>
          <!--ro, opt, enum, whether there is temporary license plate, subType:string, desc:detect whether there is temporary license plate on the window:
"unknown", "yes", "no"-->unknown
        </tempPlate>
        <passCard>
          <!--ro, opt, enum, whether there is entry & exit pass, subType:string, desc:detect whether there is entry & exit pass on the window: "unknown",
"yes", "no"-->unknown
```

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        </passCard>
        <carCard>
          <!--ro, opt, enum, whether there are cards detected, subType:string, desc:whether there is any card (business cards, flyers, etc.) attached to the window: "unknown", "yes", "no"-->unknown
        </carCard>
      </CarWindowFeature>
      <CarBodyFeature>
        <!--ro, opt, object, bodywork feature, desc:configured via <CarBodyFeature> from /ISAPI/ITC/carFeatureParam-->
        <sparetire>
          <!--ro, opt, enum, whether it is with a spare tire, subType:string, desc:"unknown", "yes", "no"-->unknown
        </sparetire>
        <rack>
          <!--ro, opt, enum, whether it is with a luggage rack, subType:string, desc:"unknown", "yes", "no"-->unknown
        </rack>
        <sunRoof>
          <!--ro, opt, enum, whether there is a sunroof, subType:string, desc:"unknown", "yes", "no"-->unknown
        </sunRoof>
        <words>
          <!--ro, opt, enum, whether the vehicle is painted, subType:string, desc:"unknown", "yes", "no"-->unknown
        </words>
        <slagTruckCoverPlate>
          <!--ro, opt, enum, whether the back cover of the dump truck is covered, subType:string, desc:"unknown", "yes", "no"-->unknown
        </slagTruckCoverPlate>
        <reflectiveTape>
          <!--ro, opt, enum, subType:string-->unknown
        </reflectiveTape>
        <closureWindow>
          <!--ro, opt, enum, subType:string-->unknown
        </closureWindow>
      </CarBodyFeature>
      <vehicleUseType>
        <!--ro, opt, enum, vehicle purpose type, subType:string, desc:configured via vehicleUseEnable from /ISAPI/ITC/carFeatureParam: "taxi", "ambulance", "bus", "schoolbus", "coach", "unknown"-->taxi
      </vehicleUseType>
      <carWindowInfo>
        <!--ro, opt, object, vehicle window information, desc:configured via /ISAPI/ITC/TriggerMode/HOVLaneDetection?format=json-->
        <carWindowNum>
          <!--ro, opt, int, number of vehicle windows-->4
        </carWindowNum>
        <carWindowStatusList>
          <!--ro, opt, object, vehicle window status list-->
          <carWindowStatus>
            <!--ro, opt, object, vehicle window status-->
            <passengers>
              <!--ro, opt, int, number of persons behind the window, range:[0,10]-->1
            </passengers>
            <windowStatus>
              <!--ro, opt, enum, whether the vehicle windows are transparent, subType:string, desc:"unknown", "yes", "no"-->yes
            </windowStatus>
          </carWindowStatus>
        </carWindowStatusList>
      </carWindowInfo>
      <actualPassengers>
        <!--ro, opt, int, number of passengers in the vehicle, range:[0,100]-->1
      </actualPassengers>
      <vehiclePosition>
        <!--ro, opt, enum, detected vehicle body part, subType:string, desc:"leftSide", "rightSide", "headStock", "tailStock", "roof"-->leftSide
      </vehiclePosition>
      <containerInfo>
        <!--ro, opt, object, container information-->
        <containerNum>
          <!--ro, opt, int, number of transferred containers, range:[1,10]-->2
        </containerNum>
        <containerList>
          <!--ro, opt, array, list of container information, subType:object, range:[1,10]-->
          <container>
            <!--ro, opt, object, container information-->
            <containerMainNum>
              <!--ro, opt, string, major No. of captured container, range:[1,32]-->abcd1234
            </containerMainNum>
            <containerSubNum>
              <!--ro, opt, string, minor No. of captured container, range:[1,32]-->abcd1234
            </containerSubNum>
            <containerISONum>
              <!--ro, opt, string, ISO No. of captured container, range:[1,32]-->abcd1234
            </containerISONum>
            <containerNumConfidence>
              <!--ro, opt, int, confidence of captured container No., range:[0,100]-->0
            </containerNumConfidence>
          </container>
        </containerList>
      </containerInfo>
      <RFID>
        <!--ro, opt, string, link RFID, range:[1,64], dep:and,{$.EventNotificationAlert.ANPR.vehicleType,eq,nonmotorVehicle}-->12345
      </RFID>
      <vehicleCarryFeature>
        <!--ro, opt, object-->
        <isCarry>
          <!--ro, opt, enum, subType:string-->unknown
        </isCarry>
        <isTarpaulin>
          <!--ro, opt, enum, subType:string-->unknown
        </isTarpaulin>
```

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```
        <isCoverplate>
          <!--ro, opt, enum, subType:string-->unknown
        </isCoverplate>
      </vehicleCarryFeature>
    </vehicleInfo>
    <EntranceInfo>
      <!--ro, opt, object, entrance and exit information-->
      <parkingID>
        <!--ro, opt, string, parking space No.-->test
      </parkingID>
      <gateID>
        <!--ro, opt, string, entrance and exit No.-->test
      </gateID>
      <direction>
        <!--ro, opt, enum, target direction, subType:string, desc:"enter", "leave"-->enter
      </direction>
      <cardNo>
        <!--ro, opt, string, card No.-->test
      </cardNo>
      <parkType>
        <!--ro, opt, enum, parking type, subType:string, desc:"temporary", "permanent"-->temporary
      </parkType>
    </EntranceInfo>
    <pictureInfoList>
      <!--ro, req, array, picture list, subType:object, range:[0,8]-->
      <pictureInfo>
        <!--ro, req, object, picture information-->
        <fileName>
          <!--ro, req, enum, picture name, subType:string, desc:"detectionPicture.jpg" (background picture), "licensePlatePicture.jpg" (license plate
picture), "pilotPicture.jpg" (driver's picture matting), "copilotPicture.jpg" (co-driver's picture matting), "compositePicture.jpg" (composite picture),
"plateBinaryPicture.jpg" (license plate binary picture), "nonMotorPicture.jpg" (non-motor vehicle picture matting), "pedestrianDetectionPicture.jpg"
(pedestrian picture), "pedestrianPicture.jpg" (pedestrian's picture matting), "vehiclePicture.jpg" (vehicle picture). Picture name, which must correspond to
the picture name transmitted with the alarm message-->detectionPicture.jpg
        </fileName>
        <type>
          <!--ro, req, enum, picture type, subType:string, desc:"detectionPicture" (background picture), "licensePlatePicture" (license plate picture),
"pilotPicture" (driver's picture matting), "copilotPicture" (co-driver's picture matting), "compositePicture" (composite picture), "plateBinaryPicture"
(license plate binary picture), "nonMotorPicture" (non-motor vehicle picture matting), "pedestrianDetectionPicture" (pedestrian picture),
"pedestrianPicture" (pedestrian's picture matting), "vehiclePicture" (vehicle picture)-->vehiclePicture
        </type>
        <dataType>
          <!--ro, req, enum, data type, subType:int, desc:0 (binary), 1 (URL)-->0
        </dataType>
        <picRecogMode>
          <!--ro, opt, enum, license plate recognition mode, subType:int, desc:0 (front license plate recognition), 1 (rear license plate recognition)-->0
        </picRecogMode>
        <redLightTime>
          <!--ro, opt, int, red light time elapsed, unit:s-->60
        </redLightTime>
        <vehicleHead>
          <!--ro, opt, enum, license plate recognition direction, subType:string, desc:"unknown", "forward" (front license plate recognition), "back" (rear
license plate recognition)-->unknown
        </vehicleHead>
        <absTime>
          <!--ro, opt, string, absolute time, desc:format: yyyyMMddHHmmssxxx, e.g.: 20090810235959999, the last three digits are time in millisecond--
>20090810235959999
        </absTime>
        <plateRect>
          <!--ro, opt, object, license plate area coordinates, desc:this field is valid when the value of type is "detectionPicture". The normalized value
is the current image size in percentage multiplying 1000 and it is accurate to three decimal places. The origin is the upper-left corner of screen-->
          <X>
            <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </X>
          <Y>
            <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </Y>
          <width>
            <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
          </height>
        </plateRect>
        <vehicelRect>
          <!--ro, opt, object, the vehicle area coordinates, desc:this field is valid when the value of type is "detectionPicture". The normalized value is
the current image size in percentage multiplying 1000. The origin is the upper-left corner of screen-->
          <X>
            <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </X>
          <Y>
            <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </Y>
          <width>
            <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
          </height>
        </vehicelRect>
        <pictureURL>
          <!--ro, opt, string, picture URL, desc:it is valid when the value of dataType is "URL"-->test
        </pictureURL>
        <pId>
          <!--ro, opt, string, strLen.max=32, range:[1,32], desc:strLen.max=32-->test
```

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```
        </pId>
        <province>
          <!--ro, opt, enum, province index, subType:int, desc:N/A-->0
        </province>
        <PilotRect>
          <!--ro, opt, object-->
          <x>
            <!--ro, req, int, range:[0,1000]-->1000
          </x>
          <y>
            <!--ro, req, int, range:[0,1000]-->1000
          </y>
          <width>
            <!--ro, req, int, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, range:[0,1000]-->1000
          </height>
        </PilotRect>
        <VicepilotRect>
          <!--ro, opt, object-->
          <x>
            <!--ro, req, int, range:[0,1000]-->1000
          </x>
          <y>
            <!--ro, req, int, range:[0,1000]-->1000
          </y>
          <width>
            <!--ro, req, int, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, range:[0,1000]-->1000
          </height>
        </VicepilotRect>
        <VehicelWindowRect>
          <!--ro, opt, object-->
          <x>
            <!--ro, req, int, range:[0,1000]-->1000
          </x>
          <y>
            <!--ro, req, int, range:[0,1000]-->1000
          </y>
          <width>
            <!--ro, req, int, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, range:[0,1000]-->1000
          </height>
        </VehicelWindowRect>
        <extraPlateRect>
          <!--ro, opt, object-->
          <x>
            <!--ro, req, int, range:[0,1000]-->1000
          </x>
          <y>
            <!--ro, req, int, range:[0,1000]-->1000
          </y>
          <width>
            <!--ro, req, int, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, range:[0,1000]-->1000
          </height>
        </extraPlateRect>
      </pictureInfo>
    </pictureInfoList>
    <hasMoreData>
      <!--ro, opt, bool, whether there is more data, desc:this field is used to report the license plate information first, and then report XML message with picture data; the XM message with picture data and license plate information are linked by UUID-->true
    </hasMoreData>
    <listType>
      <!--ro, opt, enum, List Property, subType:string, desc:"white" (allowlist), "black" (blocklist), "temporary" (temporary list)-->white
    </listType>
    <originalLicensePlate>
      <!--ro, opt, string, original license plate number, desc:when the original license plate number contains non-English characters, the original license plate number can be returned-->test
    </originalLicensePlate>
    <CRIndex>
      <!--ro, opt, enum, country or region index, subType:int, desc:country or region index-->258
    </CRIndex>
    <VehicelGPSInfo>
      <!--ro, opt, object, GPS information of the vehicle-->
      <longitudeType>
        <!--ro, req, enum, longitude, subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)-->E
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, latitude, subType:string, desc:"S" (Southern Hemisphere), "N" (North Hemisphere)-->S
      </latitudeType>
      <Longitude>
        <!--ro, req, object, longitude information-->
        <degree>
          <!--ro, req, int, degree(s)-->60
        </degree>
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```
          <minute>
            <!--ro, req, int, minute(s), range:[0,59]-->59
          </minute>
          <sec>
            <!--ro, req, float, second(s), range:[0,59.999999]-->59.000000
          </sec>
        </Longitude>
        <Latitude>
          <!--ro, req, object, latitude information-->
          <degree>
            <!--ro, req, int, degree(s)-->60
          </degree>
          <minute>
            <!--ro, req, int, minute(s), range:[0,59]-->59
          </minute>
          <sec>
            <!--ro, req, float, second(s), range:[0,59.999999], desc:the value is accurate to six decimal places-->59.000000
          </sec>
        </Latitude>
      </VehicleGPSInfo>
      <vehiclePositionControl>
        <!--ro, opt, enum, vehicle arming type, subType:string, desc:"vehicleMonitor" (intelligent arming of vehicle, related URI: PUT
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/<taskID>/startTask), "manualVehicleMonitor" (manual arming of vehicle, related URI: PUT
/ISAPI/Traffic/channels/<ID>/manualVehicleMonitor), "dailyVehicleMonitor" (daily arming of vehicle)-->dailyVehicleMonitor
      </vehiclePositionControl>
      <vehicleMonitorTaskID>
        <!--ro, opt, string, task ID of intelligent arming of vehicle, range:[1,64], desc:the task ID is sent to the device by the upper layer, which should
guarantee the uniqueness of ID. It's valid when the value of vehiclePositionControl is "vehicleMonitor"-->test
      </vehicleMonitorTaskID>
      <vehicleListName>
        <!--ro, opt, string, name of the list that the vehicle belongs to, range:[0,128], desc:name of the list that the vehicle belongs to,the maximum size
is 128 bytes-->test
      </vehicleListName>
      <vehicleThermometryEnabled>
        <!--ro, opt, bool, whether to enable vehicle temperature measurement-->true
      </vehicleThermometryEnabled>
      <currTemperature>
        <!--ro, opt, float, temperature-->36.5
      </currTemperature>
      <thermometryUnit>
        <!--ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
      </thermometryUnit>
      <plateCategory>
        <!--ro, opt, string, license plate additional information, range:[0,8], desc:license plate additional information-->test
      </plateCategory>
      <plateSize>
        <!--ro, opt, enum, license plate size, subType:int, desc:license plate size-->1
      </plateSize>
      <isNeedVerification>
        <!--ro, opt, bool, whether it requires verification, desc:false-do not verify-->true
      </isNeedVerification>
      <ISO-CR>
        <!--ro, opt, string, ISO 3166-1 standard country/region code, desc:refer to protocol dictionary-->CN
      </ISO-CR>
      <vehicelRectName>
        <!--ro, opt, string, country/region where the vehicle triggering events belongs, range:[0,64]-->11
      </vehicelRectName>
      <rodeType>
        <!--ro, opt, enum, road type, subType:string, desc:"entrance" (entrance and exit), "city", "custom", "alarmInput" (alarm input triggered by coil),
"post" (checkpoint), "publicSecurity" (general video security mode, in which the scene with mixed vehicles and without lane lines can be detected)--
>entrance
      </rodeType>
      <Custom>
        <!--ro, opt, object, custom parameters, dep:and,{$.EventNotificationAlert.ANPR.rodeType,eq,custom}-->
        <direction>
          <!--ro, opt, enum, target direction, subType:string, desc:"enter", "parkind", "leave"-->enter
        </direction>
      </Custom>
      <accelerationNoise>
        <!--ro, opt, object, vehicle revving behavior that causes huge noise-->
        <startTime>
          <!--ro, req, datetime, start time of the behavior-->1970-01-01T00:00:00+08:00
        </startTime>
        <endTime>
          <!--ro, req, datetime, end time of the behavior-->1970-01-01T00:00:00+08:00
        </endTime>
        <noiseDecibel>
          <!--ro, req, float, noise decibel-->36.5
        </noiseDecibel>
      </accelerationNoise>
      <ADRNo>
        <!--ro, opt, string, hazardous material ID, range:[1,8]-->231965
      </ADRNo>
      <pilotmask>
        <!--ro, opt, enum, subType:string-->unknown
      </pilotmask>
      <vicepilotMask>
        <!--ro, opt, enum, subType:string-->unknown
      </vicepilotMask>
      <nonMotorMask>
        <!--ro, opt, enum, subType:string-->unknown
      </nonMotorMask>
      <pedestrianMask>
        <!--ro, opt, enum, subType:string-->unknown
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```
      <!--ro, opt, enum, subType:string-->unknown
    </pedestrianMask>
    <vehicleTemperature>
      <!--ro, opt, object-->
      <vehicleHeadTemperature>
        <!--ro, opt, float, range:[-20.0,550.0]-->50.2
      </vehicleHeadTemperature>
      <vehicleBodyTemperature>
        <!--ro, opt, float, range:[-20.0,550.0]-->35.2
      </vehicleBodyTemperature>
      <leftWheelHubList>
        <!--ro, opt, array, subType:object, range:[0,6]-->
        <leftWheelHub>
          <!--ro, opt, object-->
          <ID>
            <!--ro, opt, int, range:[1,6]-->1
          </ID>
          <wheelHubTemperature>
            <!--ro, opt, float, range:[-20.0,550.0]-->38.2
          </wheelHubTemperature>
        </leftWheelHub>
      </leftWheelHubList>
      <rightWheelHubList>
        <!--ro, opt, array, subType:object, range:[0,6]-->
        <rightWheelHub>
          <!--ro, opt, object-->
          <ID>
            <!--ro, opt, int, range:[1,6]-->1
          </ID>
          <wheelHubTemperature>
            <!--ro, opt, float, range:[-20.0,550.0]-->38.2
          </wheelHubTemperature>
        </rightWheelHub>
      </rightWheelHubList>
    </vehicleTemperature>
    <deliveryVan>
      <!--ro, opt, enum, subType:string-->unknown
    </deliveryVan>
    <nonMotorShedUmbrella>
      <!--ro, opt, enum, subType:string-->unknown
    </nonMotorShedUmbrella>
    <nonMotorManned>
      <!--ro, opt, enum, subType:string-->unknown
    </nonMotorManned>
  </ANPR>
  <UUID>
    <!--ro, opt, string, unique common ID, which is used to link the same capture across multiple servers, desc:if the vehicle is captured many times, the
UUID is the same-->test
  </UUID>
  <picNum>
    <!--ro, opt, int, number of pictures-->2
  </picNum>
  <monitoringSiteID>
    <!--ro, opt, string, camera No., desc:corresponds to <positionNum> from /ISAPI/System/Video/inputs/channels/<channelID>/cameraInfo (recommended for new
devices) and /ISAPI/Traffic/channels/<channelID>/cameraInfo applied by some traffic monitoring device-->test
  </monitoringSiteID>
  <ePlateUUID>
    <!--ro, opt, string, electronic license plate ID, desc:if this field has a value, it indicates that an electronic license plate is linked-->test
  </ePlateUUID>
  <isDataRetransmission>
    <!--ro, opt, bool, whether the data is replenished-->true
  </isDataRetransmission>
  <SceneInfo>
    <!--ro, opt, object, scene information-->
    <scenesID>
      <!--ro, opt, string, scene ID, which is between 1 and 16-->test
    </scenesID>
    <sceneName>
      <!--ro, opt, string, scene name, range:[0,128]-->test
    </sceneName>
    <PTZPos>
      <!--ro, opt, object, PTZ information-->
      <elevation>
        <!--ro, opt, int, tilting angle, range:[-900,2700]-->0
      </elevation>
      <azimuth>
        <!--ro, opt, int, panning angle, range:[0,3600]-->0
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, int, zooming ratio, range:[0,1000]-->1
      </absoluteZoom>
    </PTZPos>
  </SceneInfo>
  <monitorDescription>
    <!--ro, opt, string, camera information, range:[0,256], desc:corresponds to <positionInfo> from /ISAPI/System/Video/inputs/channels/<ID>/cameraInfo
(recommended for new devices) and /ISAPI/Traffic/channels/<channelID>/cameraInfo applied by some traffic monitoring device-->test
  </monitorDescription>
  <DeviceGPSInfo>
    <!--ro, opt, object, device GPS information-->
    <longitudeType>
      <!--ro, req, enum, longitude, subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)-->E
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, latitude, subType:string, desc:"S" (Southern Hemisphere), "N" (North Hemisphere)-->S
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```
    </latitudeType>
    <Longitude>
      <!--ro, req, object, longitude information-->
      <degree>
        <!--ro, req, int, degree(s)-->60
      </degree>
      <minute>
        <!--ro, req, int, minute(s), range:[0,59]-->59
      </minute>
      <sec>
        <!--ro, req, float, second(s), range:[0,59.999999]-->59.000000
      </sec>
    </Longitude>
    <Latitude>
      <!--ro, req, object, latitude information-->
      <degree>
        <!--ro, req, int, degree(s)-->60
      </degree>
      <minute>
        <!--ro, req, int, minute(s), range:[0,59]-->59
      </minute>
      <sec>
        <!--ro, req, float, second(s), range:[0,59.999999], desc:the value is accurate to six decimal places-->59.000000
      </sec>
    </Latitude>
  </DeviceGPSInfo>
  <pilotStandardSafebelt>
    <!--ro, opt, enum, whether the driver wears the seat belt properly, subType:string, desc:"unknown", "yes", "no"-->yes
  </pilotStandardSafebelt>
  <vicepilotStandardSafebelt>
    <!--ro, opt, enum, whether the front passenger wears the seat belt properly, subType:string, desc:"unknown", "yes", "no"-->yes
  </vicepilotStandardSafebelt>
  <trafficLightSnap>
    <!--ro, opt, enum, whether it is captured at the traffic lights, subType:string, desc:"yes", "no"-->yes
  </trafficLightSnap>
  <sequence>
    <!--ro, opt, int, No. of vehicle capture from network-triggered burst, range:[1,4294967295], desc:command applied by network-triggered burst control;
related API: /ISAPI/Traffic/startRecognition; SDK interface: NET_DVR_ContinuousShoot-->0
  </sequence>
  <relaLaneDirectionType>
    <!--ro, opt, enum, linked lane direction, subType:int, desc:a camera can be linked with multiple lane directions, which is independent of the camera
setup angle <detectDir>; 0 (others),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to south), 5 (from southeast to
northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
  </relaLaneDirectionType>
  <carDirectionType>
    <!--ro, opt, enum, vehicle running direction on the lane, subType:int, desc:independent of the camera setup angle <detectDir>; 0-from up to down, 1-from
down to up-->1
  </carDirectionType>
  <targetID>
    <!--ro, opt, string, target ID, range:[1,64], desc:UUID, of which the uniqueness is guaranteed by device, and corresponds to <dwMatchNo> in the private
protocol-->test
  </targetID>
  <isSecondCamera>
    <!--ro, opt, bool, whether to enable capture by the second camera, desc:the second camera: e.g., the perspective camera from distant/close-view capture
cameras, the back camera from front/back capture camera, etc.-->false
  </isSecondCamera>
  <dataAnalysisType>
    <!--ro, opt, enum, data analysis type, subType:int, desc:0-data not analyzed, 1-data analyzed-->1
  </dataAnalysisType>
  <RecordInfo>
    <!--ro, opt, object, vehicle video information-->
    <fileName>
      <!--ro, req, enum, video name, subType:string, desc:the video name should match <Content-ID> for binary transmission: "record.mp4" (vehicle video)--
>record.mp4
    </fileName>
    <dataType>
      <!--ro, req, enum, data type, subType:int, desc:0 (binary), 1 (URL)-->0
    </dataType>
    <URL>
      <!--ro, opt, string, URL, desc:it is valid when the value of dataType is "URL"-->test
    </URL>
  </RecordInfo>
  <VehicleWeightInfo>
    <!--ro, opt, object, vehicle weighing information, desc:this node is currently only supported by weighing data management server-->
    <isOverWeight>
      <!--ro, opt, bool, whether it is overweight, desc:set overweight reporting or overrun reporting as priority via <illegalPriorityType> from
/ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->false
    </isOverWeight>
    <axleNum>
      <!--ro, opt, int, number of axles, range:[1,10]-->4
    </axleNum>
    <axleModel>
      <!--ro, opt, enum, vehicle type by axle, subType:int, desc:0 (unknown), 512 (2-axle truck), 756 (3-axle centre-axle trailer combination), 766 (3-axle
articulated vehicle), 767 (3-axle truck), 768 (3-axle truck), 1024 (type-125 4-axle centre-axle trailer combination), 1025 (type-152 4-axle centre-axle
trailer combination), 1026 (4-axle articulated vehicle), 1027 (4-axle full trailer combination), 1028 (4-axle truck), 1280 (type-155 5-axle centre-axle
trailer combination), 1281 (type-1125 5-axle centre-axle trailer combination), 1282 (type-155 5-axle articulated vehicle), 1283 (type-1125 5-axle
articulated vehicle), 1284 (type-129 5-axle articulated vehicle), 1285 (type-1522 5-axle full trailer combination), 1286 (type-11222 5-axle full trailer
combination), 1536 (type-159 6-axle centre-axle trailer combination), 1537 (type-159-2 6-axle centre-axle trailer combination), 1538 (type-1155-1 6-axle
centre-axle trailer combination), 1539 (type-1155-2 6-axle centre-axle trailer combination), 1540 (type-159-3 6-axle articulated vehicle), 1541 (type-159-4
6-axle articulated vehicle), 1542 (type-1129 6-axle articulated vehicle), 1543 (type-11522-1 6-axle full trailer combination), 1544 (type-11522-2 6-axle
full trailer combination)-->0
    </axleModel>
    <overWeight>
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```
      <!--ro, opt, float, over-limit weight (unit: ton), range:[0.000,100.000], unit:吨, desc:accurate to three decimal places-->4.502
    </overWeight>
    <weight>
      <!--ro, opt, float, vehicle weight (unit: ton), range:[0.000,100.000], unit:吨, desc:accurate to three decimal places-->4.502
    </weight>
    <limitWeight>
      <!--ro, opt, float, weight limit (unit: ton), range:[0.000,100.000], unit:吨, desc:accurate to three decimal places-->4.502
    </limitWeight>
    <axleLen>
      <!--ro, opt, float, distance between tyres, range:[0.00,100.00], unit:吨, desc:accurate to two decimal places-->4.502
    </axleLen>
    <devDescInfo>
      <!--ro, opt, string, device description, range:[0,64]-->test
    </devDescInfo>
    <AxleInfoList>
      <!--ro, opt, object, axle information list, desc:<axleNum>, number of axles-->
      <AxleInfo>
        <!--ro, opt, object, information of a single axle-->
        <axleWeight>
          <!--ro, opt, float, axle weight, range:[0.00,10000.00], unit:kg-->100.00
        </axleWeight>
        <axleDistance>
          <!--ro, opt, int, distance between axles, range:[0,100000], unit:mm, desc:the distance between the current axle and the next axle-->30000
        </axleDistance>
        <wheelNum>
          <!--ro, opt, int, range:[0,8]-->2
        </wheelNum>
      </AxleInfo>
    </AxleInfoList>
    <length>
      <!--ro, opt, int, vehicle length, range:[1,1000000], unit:cm-->4000
    </length>
    <width>
      <!--ro, opt, int, vehicle width, range:[1,1000000], unit:cm-->4000
    </width>
    <height>
      <!--ro, opt, int, vehicle height, range:[1,1000000], unit:cm-->4000
    </height>
    <tollwayVehicleType>
      <!--ro, opt, enum, vehicle types on tollway, subType:int, desc:0 (unknown), 1 (type-1 minibus), 2 (type-1 small-sized bus), 3 (type-2 medium-sized bus), 4 (type-2 bus and passenger trailer combination), 5 (type-3 large-sized bus), 6 (type-4 large-sized bus), 7 (type-1 2-axle truck), 8 (type-2 2-axle truck), 9 (type-3 3-axle truck), 10 (type-4 4-axle truck), 11 (type-5 5-axle truck), 12 (type-6 6-axle truck), 13 (type-1 2-axle operation vehicle), 14 (type-2 2-axle operation vehicle), 15 (type-3 3-axle operation vehicle), 16 (type-4 4-axle operation vehicle), 17 (type-5 5-axle operation vehicle), 18 (type-6 operation vehicle with 6 or more axles)-->1
    </tollwayVehicleType>
    <tiresNum>
      <!--ro, opt, int, number of tyres, range:[1,20]-->4
    </tiresNum>
    <approvedPassengers>
      <!--ro, opt, int, authorized passenger capacity, range:[1,100]-->7
    </approvedPassengers>
    <limitLength>
      <!--ro, opt, int, vehicle length limit, range:[1,1000000], unit:cm, desc:configured via /ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->4000
    </limitLength>
    <limitWidth>
      <!--ro, opt, int, vehicle width limit, range:[1,1000000], unit:cm, desc:configured via /ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->4000
    </limitWidth>
    <limitHeight>
      <!--ro, opt, int, vehicle height limit, range:[1,1000000], unit:cm, desc:configured via /ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->4000
    </limitHeight>
    <overLength>
      <!--ro, opt, int, over-limit length, range:[1,1000000], unit:cm, desc:the over-limit part equals <length> minus <limitLength>-->4000
    </overLength>
    <overWidth>
      <!--ro, opt, int, over-limit width, range:[1,1000000], unit:cm, desc:the over-limit part equals <width> minus <limitWidth>-->4000
    </overWidth>
    <overHeight>
      <!--ro, opt, int, over-limit height, range:[1,1000000], unit:cm, desc:the over-limit part equals <height> minus <limitHeight>-->4000
    </overHeight>
    <isOverLimit>
      <!--ro, opt, bool, whether it is over-limit, desc:set overweight reporting or overrun reporting as priority via <illegalPriorityType> from /ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->false
    </isOverLimit>
    <axleNumConfidence>
      <!--ro, opt, int, confidence of captured number of axles, range:[0,100]-->0
    </axleNumConfidence>
    <standardAxleTypeInfo>
      <!--ro, opt, object-->
      <axleType>
        <!--ro, req, enum, subType:int-->12
      </axleType>
      <axleDescribeInfo>
        <!--ro, req, string, range:[0,64]-->2轴载货汽车
      </axleDescribeInfo>
    </standardAxleTypeInfo>
  </VehicleWeightInfo>
  <isNotSlowZebraCrossing>
    <!--ro, opt, bool, whether the vehicle did not slow down at zebra crossing-->false
  </isNotSlowZebraCrossing>
  <isTurnRightStop>
    <!--ro, opt, bool, whether the vehicle did not stop at a right turn-->false
```

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```
  </isTurnRightStop>
  <PlateInfoList>
    <!--ro, opt, object, this node is returned when license plates from Mainland, Hongkong, and Maokao are all supported-->
    <PlateInfo>
      <!--ro, opt, object, information of a single license plate-->
      <plateRect>
        <!--ro, opt, object, license plate area coordinates, desc:this field is valid when the value of type is "detectionPicture". The normalized value is
the current image size in percentage multiplying 1000 and it is accurate to three decimal places. The origin is the upper-left corner of screen-->
        <X>
          <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
        </X>
        <Y>
          <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
        </Y>
        <width>
          <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
        </width>
        <height>
          <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
        </height>
      </plateRect>
      <plateColor>
        <!--ro, req, enum, license plate color, subType:string, desc:license plate color: "white","yellow","blue","black","green","newEnergyGreen"-new
energy green,"newEnergyYellowGreen"-new energy flavogreen,"other"-other color-->black
      </plateColor>
      <licensePlate>
        <!--ro, req, string, license plate number, range:[1,32], desc:license plate number,e.g.,"123456"-->A283KY77
      </licensePlate>
      <confidenceLevel>
        <!--ro, req, int, confidence level, range:[0,100]-->50
      </confidenceLevel>
      <province>
        <!--ro, opt, enum, province index, subType:int, desc:N/A-->0
      </province>
      <isExtraPlate>
        <!--ro, opt, bool-->true
      </isExtraPlate>
    </PlateInfo>
  </PlateInfoList>
  <deviceUUID>
    <!--ro, opt, string, device No., range:[1,32], desc:the device number is the device UUID which is the device's serial number and can be edited by the
node <deviceID> in the message of /ISAPI/System/deviceInfo-->12345
  </deviceUUID>
  <VehicleGATInfo>
    <!--ro, opt, object, N/A, desc:N/A-->
    <palteTypeByGAT>
      <!--ro, opt, enum, N/A, subType:int, desc:N/A-->1
    </palteTypeByGAT>
    <plateColorByGAT>
      <!--ro, opt, enum, N/A, subType:int, desc:0-white, 1-yellow, 2-blue, 3-black, 4-green, 5-not recognized, 9, other colors-->0
    </plateColorByGAT>
    <vehicleTypeByGAT>
      <!--ro, opt, enum, N/A, subType:string, desc:GA/T 16.4 complied: "K10" (large-sized bus), "K20" (medium-sized bus), "K30" (small-sized bus), "K33"
(small-sized car), "H10" (heavy truck), "H20" (medium-sized truck), "H30" (light truck), "M10" (three-wheel motorcycle), "M20" (two-wheel motorcycle), "X99"
(others)-->K10
    </vehicleTypeByGAT>
    <colorByGAT>
      <!--ro, opt, enum, N/A, subType:string, desc:GA/T 16.8 complied: A-white, B-gray, C-yellow, D-pink, E-red, F-purple, G-green, H-blue, I-brown, J-
black, K-not recognized, Z-other colors-->A
    </colorByGAT>
  </VehicleGATInfo>
  <EPlateInfo>
    <!--ro, opt, object, RFID detection result-->
    <EPlateChipNo>
      <!--ro, opt, string, chip No. of electronic license plate (detected by RFID), range:[0,16]-->E855000011F9AAA5
    </EPlateChipNo>
    <EPlateCardNo>
      <!--ro, opt, string, electronic license plate No. (detected by RFID), range:[0,12], desc:electronic license plate No. (12 digits; the first two
represent the province code-->E855000011F9AAA5
    </EPlateCardNo>
    <EPlateNo>
      <!--ro, opt, string, license plate No. (detected by RFID), range:[0,32], desc:"noPlate" (vehicle without license plate detected), "unknown" (license
plate not detected; temporary license plate detected)-->浙A12345
    </EPlateNo>
    <EPlateManufacturingDate>
      <!--ro, opt, string, production date (detected by RFID), range:[0,7], desc:format: yyyy-MM-->2014-01
    </EPlateManufacturingDate>
    <EPlateInspectionValidity>
      <!--ro, opt, string, inspection validity (detected by RFID), range:[0,7], desc:accurate to month; format: yyyy-MM-->2014-01
    </EPlateInspectionValidity>
    <EPlateMandaScrapPeriod>
      <!--ro, opt, int, compulsory scrapping period (detected by RFID), range:[0,100], unit:年, desc:from the start of inspection validity to the compulsory
scrapping time (unit: year)-->10
    </EPlateMandaScrapPeriod>
    <EPlatePassengersCapacity>
      <!--ro, opt, int, authorized passenger capacity (detected by RFID), range:[0,100], desc:e.g. 1-->1
    </EPlatePassengersCapacity>
    <EPlateCarryingCapacity>
      <!--ro, opt, float, tractive tonnage, range:[0,100], unit:吨, desc:tractive tonnage (unit: ton)-->2.0
    </EPlateCarryingCapacity>
    <EPlateVehicleColor>
      <!--ro, opt, enum, vehicle color (detected by RFID), subType:string, desc:"unknown", "white", "silver", "gray", "black", "red", "deepBlue", "blue",
"yellow", "green", "brown", "pink", "purple", "deepGray", "cyan", "orange"-->black
```

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```
    </EPlateVehicleColor>
    <EPlateNonMotorType>
      <!--ro, opt, enum, non-motor vehicle type (detected by RFID), subType:string, desc:"unknown", "threeWheelVehicle" (tricycle), "twoWheelVehicle" (two wheeler)-->threeWheelVehicle
    </EPlateNonMotorType>
    <EPlateNonMotorPlateType>
      <!--ro, opt, enum, non-motor vehicle license plate type (detected by RFID), subType:string, desc:"unknown", "temporary", "normal"-->unknown
    </EPlateNonMotorPlateType>
    <EPlateColor>
      <!--ro, opt, enum, vehicle color (detected by RFID), subType:string, desc:"black", "blue","golden", "orange", "red", "yellow", "white", "unknown", "other", "newEnergyYellowGreen"-new energy yellow-green,"civilAviationBlack", "civilAviationGreen" (civil aviation green), "green", "mixedColor", "newEnergyGreen" (new energy green), "brown"-->black
    </EPlateColor>
    <EPlateUseType>
      <!--ro, opt, enum, non-motor vehicle purpose (detected by RFID), subType:string, desc:"unknown", "nonOperating" (non-operating), "takeOut" (take-out delivery), "express" (express delivery)-->nonOperating
    </EPlateUseType>
    <EPlateTriggerTime>
      <!--ro, req, datetime, device detection trigger time of electronic license plate (detected by RFID)-->1970-01-01T00:00:00+08:00
    </EPlateTriggerTime>
    <EPlateListenAuthCode>
      <!--ro, opt, string, licensing authority code of electronic license plate (detected by RFID)-->粤B
    </EPlateListenAuthCode>
  </EPlateInfo>
  <accelerationNoise>
    <!--ro, opt, object, vehicle revving behavior that causes huge noise-->
    <isExistsAccelerationNoise>
      <!--ro, req, bool, whether there is vehicle revving behavior that causes huge noise-->true
    </isExistsAccelerationNoise>
    <startTime>
      <!--ro, req, datetime, start time of the behavior, desc:accurate to millisecond-->1970-01-01T00:00:00.000+08:00
    </startTime>
    <endTime>
      <!--ro, req, datetime, end time of the behavior, desc:accurate to millisecond-->1970-01-01T00:00:00.000+08:00
    </endTime>
  </accelerationNoise>
  <noiseDecibel>
    <!--ro, opt, float, noise decibel of vehicle running, range:[0,300], unit:分贝, desc:accurate to one decimal place-->65.5
  </noiseDecibel>
  <exhaustPipeNum>
    <!--ro, opt, int, number of exhaust pipes, range:[0,4]-->2
  </exhaustPipeNum>
  <exhaustPipePositionType>
    <!--ro, opt, enum, exhaust pipe installation location, subType:string, desc:"unilateral", "bilateral", "middle"-->unilateral
  </exhaustPipePositionType>
  <openGateType>
    <!--ro, opt, enum, reasons for opening the barrier gate, subType:string, dep:and,{$.EventNotificationAlert.ANPR.barrierGateCtrlType,eq,0}, desc:"white" (auto-open barrier gate via device according to the allowList), "manual" (manually open barrier gate via platform), "abnormal" (auto-open barrier gate via device for exceptional circumstances)-->white
  </openGateType>
  <plateStandardStatus>
    <!--ro, opt, enum, whether the license plate is valid, subType:string, desc:"yes", "no"-->yes
  </plateStandardStatus>
  <limitSurfaceSpeed>
    <!--ro, opt, int, cross-sectional device-detected speed limit (required for overspeed and violation), range:[0,255], unit:km/h-->1
  </limitSurfaceSpeed>
  <startCaptureSpeed>
    <!--ro, opt, int, device-detected capture triggered speed (required for overspeed and violation), range:[0,255], unit:km/h-->1
  </startCaptureSpeed>
  <redlightIllegalCode>
    <!--ro, opt, string, traffic lights and lane matching code for red light running, range:[0,4], desc:0-all lanes, 1-turn left, 2-go straight, 3-turn right, 4-go straight & turn left, 5-go straight & turn right-->1
  </redlightIllegalCode>
  <redlightSignalStartTime>
    <!--ro, opt, string, start time of the red light (required for red light running), desc:ISO 8601 format yyyy-MM-ddTHH:mm:ss+current zone, e.g., 2017-08-01T17:30:08+08:00-->"2004-05-03T17:30:08+08:00"
  </redlightSignalStartTime>
  <redlightSignalEndTime>
    <!--ro, opt, string, end time of the red light (required for red light running), desc:ISO 8601 format yyyy-MM-ddTHH:mm:ss+current zone, e.g., 2017-08-01T17:30:08+08:00-->"2004-05-03T17:30:08+08:00"
  </redlightSignalEndTime>
  <trafficControlTimeSpanList>
    <!--ro, opt, array, time period for traffic restriction, subType:object-->
    <TimeRange>
      <!--ro, req, object, time period-->
      <beginTime>
        <!--ro, req, time, start time-->10:00:00
      </beginTime>
      <endTime>
        <!--ro, req, time, end time-->10:00:00
      </endTime>
    </TimeRange>
  </trafficControlTimeSpanList>
  <detectionBackgroundImageResolution>
    <!--ro, opt, object, -ro, desc:corresponds to the picture represented by <pictureURL> and <pId> when <fileName> is "detectionPicture.jpg" and <type> is "detectionPicture" from <pictureInfoList> and <pictureInfo> in ANPR when the normalized coordinates of <Rect>, <pictureInfoList>, <pictureInfo>, <plateRect>, <vehicelRect>, <PlateInfoList>, <PlateInfo>, and <plateRect> are converted to the resolution coordinates, this node is required-->
    <height>
      <!--ro, req, int, resolution (height), range:[1,65535], unit:px-->1
    </height>
    <width>
      <!--ro, req, int, resolution (width), range:[1,65535], unit:px-->1
    </width>
  </detectionBackgroundImageResolution>
```

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```
    <trafficSurveyVehicleType>
      <!--ro, opt, enum, subType:int-->1
    </trafficSurveyVehicleType>
    <doorTextInfoList>
      <!--ro, opt, array, subType:object, range:[0,2]-->
      <doorTextInfo>
        <!--ro, opt, object-->
        <doorID>
          <!--ro, opt, int, range:[0,2]-->1
        </doorID>
        <doorTextList>
          <!--ro, opt, array, subType:object, range:[0,4]-->
          <doorText>
            <!--ro, opt, string, range:[0,128]-->载荷50人
          </doorText>
        </doorTextList>
        <approvedPassengers>
          <!--ro, opt, int, range:[1,100]-->7
        </approvedPassengers>
        <doorTextConfidence>
          <!--ro, opt, int, range:[0,100]-->0
        </doorTextConfidence>
      </doorTextInfo>
    </doorTextInfoList>
    <splicingDetectInfo>
      <!--ro, opt, object-->
      <startSplicingTime>
        <!--ro, req, datetime-->1970-01-01T00:00:00 000+08:00
      </startSplicingTime>
      <endSplicingTime>
        <!--ro, req, datetime-->1970-01-01T00:00:00 000+08:00
      </endSplicingTime>
    </splicingDetectInfo>
  </EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
anpr.xml	[Message content]	application/xml	--	--	--
.value	[Binary picture data]	image/jpeg	detectionPicture	detectionPicture.jpg	--
.value	[Binary picture data]	image/jpeg	licensePlatePicture	licensePlatePicture.jpg	--
.value	[Binary picture data]	image/jpeg	pilotPicture	pilotPicture.jpg	--
.value	[Binary picture data]	image/jpeg	copilotPicture	copilotPicture.jpg	--
.value	[Binary picture data]	image/jpeg	compositePicture	compositePicture.jpg	--
.value	[Binary picture data]	image/jpeg	plateBinaryPicture	plateBinaryPicture.jpg	--
.value	[Binary picture data]	image/jpeg	nonMotorPicture	nonMotorPicture.jpg	--
.value	[Binary picture data]	image/jpeg	pedestrianDetectionPicture	pedestrianDetectionPicture.jpg	--
.value	[Binary picture data]	image/jpeg	vehiclePicture	vehiclePicture.jpg	--
.value	[Binary picture data]	image/jpeg	featurePicture	featurePicture.jpg	--
.value	[Binary picture data]	image/jpeg	panoramaPicture	panoramaPicture.jpg	--
.value	[Binary picture data]	image/jpeg	extraLicensePlatePicture	extraLicensePlatePicture.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.7 Abnormal Event Detection

19.7.1 Fast Moving

19.7.1.1 Get the arming schedule of fast moving detection for all channels

Request URL

GET /ISAPI/Event/schedules/rapidMoves

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<RapidMoveScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->rapidMove
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RapidMoveScheduleList>
```

19.7.1.2 Set the arming schedule of fast moving detection for all channels

Request URL

PUT /ISAPI/Event/schedules/rapidMoves

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<RapidMoveScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, enum, subType:string-->rapidMove
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RapidMoveScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.7.1.3 Set the arming schedule of fast moving detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/rapidMoves/rapidMove-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--opt, string-->rapidMove
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.7.1.4 Get the arming schedule of fast moving detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/rapidMoves/rapidMove-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->rapidMove
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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19.7.1.5 Set the linkage parameters of fast moving detection

Request URL

PUT /ISAPI/Event/triggers/rapidMove-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--req, enum, subType:string-->rapidMove
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.1.6 Get the linkage parameters of fast moving detection

Request URL

GET /ISAPI/Event/triggers/rapidMove-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->rapidMove
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

19.7.1.7 Fast moving

EventType:rapidMove

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, transmitted parameters, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
```

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```
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"rapidMove" (fast moving detection)-->rapidMove
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->0
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
```

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```
    <targetAttrs>
      <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
    </targetAttrs>
    <TaskInfo>
      <!--ro, opt, object, task information submitted by data center products-->
      <taskID>
        <!--ro, req, string, task ID, range:[0,64]-->test
      </taskID>
      <algorithmType>
        <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
      </algorithmType>
      <taskName>
        <!--ro, opt, string, task name-->test
      </taskName>
      <algorithmID>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmID>
      <monitorPointID>
        <!--ro, opt, string, camera No.-->test
      </monitorPointID>
      <monitorPointName>
        <!--ro, opt, string, camera name-->test
      </monitorPointName>
      <streamType>
        <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
      </streamType>
      <ruleID>
        <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
      </ruleID>
      <ruleCustomName>
        <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
      </ruleCustomName>
      <algorithmInfo>
        <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
        <algorithmId>
          <!--ro, opt, string, algorithm package ID-->test
        </algorithmId>
        <algoVendorId>
          <!--ro, opt, string, manufacturer ID-->test
        </algoVendorId>
        <algoVersionId>
          <!--ro, opt, string, algorithm version No.-->test
        </algoVersionId>
        <algoChipType>
          <!--ro, opt, string, chip type-->test
        </algoChipType>
        <algoModelVersion>
          <!--ro, opt, string, model version-->test
        </algoModelVersion>
      </algorithmInfo>
    </TaskInfo>
    <timeStamp>
      <!--ro, opt, datetime, timestamp of the frame, which triggers the alarm-->1970-01-01T00:00:00+08:00
    </timeStamp>
    <isDataRetransmission>
      <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
    </isDataRetransmission>
    <ruleID>
      <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
    </ruleID>
    <ruleName>
      <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
    </ruleName>
  </EventNotificationAlert>
```

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19.7.2 Loitering Detection

19.7.2.1 Set arming schedules of loitering detection of all channels

Request URL

PUT /ISAPI/Event/schedules/loiterings

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, arming schedule list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming schedule-->
    <id>
      <!--req, string, ID, desc:ID-->loitering-1
    </id>
    <eventType>
      <!--opt, string, event type-->loitering
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming schedule list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1: Monday; 2: Tuesday; 3: Wednesday; 4: Thursday; 5: Friday; 6: Saturday; 7: Sunday;-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming schedule, subType:object-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LoiteringScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.2.2 Get arming schedules of the loitering detection of all channels

Request URL

GET /ISAPI/Event/schedules/loiterings

Query Parameter

None

Request Message

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None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, arming schedule list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule information-->
    <id>
      <!--ro, req, string, ID-->loitering-1
    </id>
    <eventType>
      <!--ro, opt, string, event type-->loitering
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, arming schedule list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1: Monday), 2: Tuesday, 3: Wednesday, 4: Thursday, 5: Friday, 6: Saturday, 7: Sunday-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time period-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time period-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LoiteringScheduleList>
```

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19.7.2.3 Get the arming schedule of loitering detection for all channels

Request URL

GET /ISAPI/Event/schedules/loiterings/loitering

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <eventType>
      <!--ro, opt, string-->loitering
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LoiteringScheduleList>
```

19.7.2.4 Set the arming schedule of loitering detection for all channels

Request URL

PUT /ISAPI/Event/schedules/loiterings/loitering

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, string-->loitering
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LoiteringScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.7.2.5 Set arming schedule of loitering detection of a channel

Request URL

PUT /ISAPI/Event/schedules/loiterings/loitering-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID, desc:ID-->test
  </id>
  <eventType>
    <!--opt, enum, event type, subType:string-->loitering
  </eventType>
  <videoInputChannelID>
    <!--opt, string, video input channel ID-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, arming schedule list, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, day of the week, subType:int, desc:1(Monday); 2(Tuesday); 3(Wednesday); 4(Thursday); 5(Friday); 6(Saturday); 7(Sunday)-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time, start time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object, holiday arming schedule-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <TimeRange>
        <!--req, object, time period-->
        <beginTime>
          <!--req, time, start time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.7.2.6 Get the arming schedule of loitering detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/loiterings/loitering-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->loitering
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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19.7.2.7 Get the linkage parameters of loitering detection

Request URL

GET /ISAPI/Event/triggers/loitering-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->loitering
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.7.2.8 Set the linkage parameters of loitering detection

Request URL

PUT /ISAPI/Event/triggers/loitering-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--req, enum, subType:string-->loitering
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.2.9 Set loitering detection parameters of all channels

Request URL

PUT /ISAPI/Smart/loitering

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Loitering>
    <!--opt, object, loitering detection parameters configuration of a channel-->
    <id>
      <!--req, string, channel No.-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <normalizedScreenSize>
      <!--opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <LoiteringRegionList>
      <!--opt, array, information list of loitering detection areas, subType:object-->
      <LoiteringRegion>
        <!--opt, object, loitering detection area-->
        <id>
          <!--req, string-->1
        </id>
        <sensitivityLevel>
          <!--opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
        </sensitivityLevel>
        <timeThreshold>
          <!--opt, int, time threshold-->1
        </timeThreshold>
        <RegionCoordinatesList>
          <!--opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, x-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </LoiteringRegion>
    </LoiteringRegionList>
    <mutexAbility opt="PDC">
      <!--opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
  </Loitering>
</LoiteringList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.2.10 Get loitering detection parameters of all channels

Request URL

GET /ISAPI/Smart/loitering

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Loitering>
    <!--ro, opt, object, loitering detection parameters configuration of a channel-->
    <id>
      <!--ro, req, string, channel No.-->test
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <normalizedScreenSize>
      <!--ro, opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <LoiteringRegionList>
      <!--ro, opt, array, information list of loitering detection areas, subType:object-->
      <LoiteringRegion>
        <!--ro, opt, object, loitering detection area-->
        <id>
          <!--ro, req, string-->1
        </id>
        <sensitivityLevel>
          <!--ro, opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
        </sensitivityLevel>
        <timeThreshold>
          <!--ro, opt, int, time threshold-->1
        </timeThreshold>
        <RegionCoordinatesList>
          <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </LoiteringRegion>
    </LoiteringRegionList>
    <mutexAbility opt="PDC">
      <!--ro, opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--ro, opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
  </Loitering>
</LoiteringList>
```

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19.7.2.11 Get loitering detection parameters of a channel

Request URL

GET /ISAPI/Smart/loitering/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Loitering xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, loitering detection parameters configuration of a channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <LoiteringRegionList>
    <!--ro, opt, array, information list of loitering detection areas, subType:object-->
    <LoiteringRegion>
      <!--ro, opt, object, loitering detection area-->
      <id>
        <!--ro, req, string, area ID-->1
      </id>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
      </sensitivityLevel>
      <timeThreshold>
        <!--ro, opt, int, time threshold-->1
      </timeThreshold>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </LoiteringRegion>
  </LoiteringRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, ro,"PDC", attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool-->true
  </isSupportMultiScene>
</Loitering>
```

19.7.2.12 Get the loitering detection capability

Request URL

GET /ISAPI/Smart/loitering/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Loitering xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the loitering detection capability, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, req, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <LoiteringRegionList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <LoiteringRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->1
      </id>
      <sensitivityLevel min="1" max="100">
        <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->1
      </sensitivityLevel>
      <timeThreshold min="1" max="10" def="1">
        <!--ro, req, int, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
      </timeThreshold>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,4]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </LoiteringRegion>
  </LoiteringRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool-->true
  </isSupportMultiScene>
  <isSupportOverlapParams>
    <!--ro, opt, bool-->true
  </isSupportOverlapParams>
</Loitering>
```

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19.7.2.13 Batch get parameters of multiple areas

Request URL

GET /ISAPI/Smart/loitering/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, information list of loitering detection areas, subType:object, attr:version{req, string, protocolVersion}-->
  <LoiteringRegion>
    <!--ro, opt, object, loitering detection area-->
    <id>
      <!--ro, req, string, detection area ID-->1
    </id>
    <sensitivityLevel>
      <!--ro, opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
    </sensitivityLevel>
    <timeThreshold>
      <!--ro, opt, int, time threshold-->1
    </timeThreshold>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </LoiteringRegion>
</LoiteringRegionList>
```

19.7.2.14 Batch set parameters of multiple areas

Request URL

PUT /ISAPI/Smart/loitering/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, information list of loitering detection areas, subType:object, attr:version{req, string, protocolVersion}-->
  <LoiteringRegion>
    <!--opt, object, loitering detection area-->
    <id>
      <!--req, string, area ID-->1
    </id>
    <sensitivityLevel>
      <!--opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
    </sensitivityLevel>
    <timeThreshold>
      <!--opt, int, time threshold-->1
    </timeThreshold>
    <RegionCoordinatesList>
      <!--opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </LoiteringRegion>
</LoiteringRegionList>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:detailed description of the error code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.2.15 Set parameters of a single area

Request URL

PUT /ISAPI/Smart/loitering/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, loitering detection area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, area ID-->1
  </id>
  <sensitivityLevel>
    <!--opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
  </sensitivityLevel>
  <timeThreshold>
    <!--opt, int, time threshold-->1
  </timeThreshold>
  <RegionCoordinatesList>
    <!--opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</LoiteringRegion>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:detailed description of the error code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.2.16 Get loitering detection parameters of a single area

Request URL

GET /ISAPI/Smart/loitering/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, loitering detection area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, area ID-->1
  </id>
  <sensitivityLevel>
    <!--ro, opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
  </sensitivityLevel>
  <timeThreshold>
    <!--ro, opt, int, time threshold-->1
  </timeThreshold>
  <RegionCoordinatesList>
    <!--ro, opt, array, list of area coordinates, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</LoiteringRegion>
```

19.7.2.17 Loitering detection

EventType:loitering

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
```

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```
<portNo>
  <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
</portNo>
<protocol>
  <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
</protocol>
<macAddress>
  <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
</macAddress>
<dynChannelID>
  <!--ro, opt, string, digital channel No.-->test
</dynChannelID>
<channelID>
  <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
</channelID>
<releatedChannelList>
  <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
</releatedChannelList>
<dateTime>
  <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
</dateTime>
<activePostCount>
  <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
</activePostCount>
<eventType>
  <!--ro, req, enum, event type, subType:string, desc:"loitering" (loitering detection)-->loitering
</eventType>
<eventState>
  <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
</eventState>
<eventDescription>
  <!--ro, req, string, event description-->test
</eventDescription>
<deviceID>
  <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
</deviceID>
<DetectionRegionList>
  <!--ro, opt, array, detection area list, subType:object-->
  <DetectionRegionEntry>
    <!--ro, opt, object, detection area-->
    <regionID>
      <!--ro, req, string, detection area ID-->test
    </regionID>
    <sensitivityLevel>
      <!--ro, opt, int, sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, target type, subType:string, desc:"human"-->human
    </detectionTarget>
  </DetectionRegionEntry>
</DetectionRegionList>
<channelName>
  <!--ro, opt, string, channel name, range:[1,64]-->test
</channelName>
<Extensions>
  <!--ro, opt, object, extended information-->
  <serialNumber>
    <!--ro, opt, string, serial No.-->test
  </serialNumber>
  <eventPush>
    <!--ro, opt, string, event notification-->test
  </eventPush>
</Extensions>
<detectionPictureTransType>
  <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
</detectionPictureTransType>
<detectionPicturesNumber>
  <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
</detectionPicturesNumber>
<bkgUrl>
  <!--ro, opt, string, background picture URL, desc:background picture URL-->test
</bkgUrl>
<URLCertificationType>
  <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
</URLCertificationType>
<pId>
```

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```
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

19.7.3 Object Removal

19.7.3.1 Get the arming schedule of object removal detection for all channels

Request URL

GET /ISAPI/Event/schedules/attendedBaggages

Query Parameter

None

Request Message

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None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AttendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->attendedBaggage-1
    </id>
    <eventType>
      <!--ro, opt, string-->attendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</AttendBaggageScheduleList>
```

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19.7.3.2 Set the arming schedule of object removal detection for all channels

Request URL

PUT /ISAPI/Event/schedules/attendedBaggages

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AttendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->attendedBaggage-1
    </id>
    <eventType>
      <!--opt, string-->attendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</AttendBaggageScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.7.3.3 Get the arming schedule of object removal detection for all channels

Request URL

GET /ISAPI/Event/schedules/attendedBaggages/attendedBaggage

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AttendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule version="2.0">
    <!--ro, opt, object, attr:version{req, string}-->
    <id>
      <!--ro, req, string-->attendedBaggage-1
    </id>
    <eventType>
      <!--ro, opt, string-->attendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</AttendBaggageScheduleList>
```

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19.7.3.4 Set the arming schedule of object removal detection for all channels

Request URL

PUT /ISAPI/Event/schedules/attendedBaggages/attendedBaggage

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AttendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule version="2.0">
    <!--opt, object, attr:version{req, string}-->
    <id>
      <!--req, string-->attendedBaggage-1
    </id>
    <eventType>
      <!--opt, string-->attendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</AttendBaggageScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.7.3.5 Set the arming schedule of object removal detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/attendedBaggages/attendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->attendedBaggage-1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->attendedBaggage
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.7.3.6 Get the arming schedule of object removal detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/attendedBaggages/attendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->attendedBaggage-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->attendedBaggage
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.7.3.7 Get the linkage parameters of object removal detection

Request URL

GET /ISAPI/Event/triggers/attendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->attendedBaggage-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->attendedBaggage
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.7.3.8 Set the linkage parameters of object removal detection

Request URL

PUT /ISAPI/Event/triggers/attendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->attendedBaggage-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->attendedBaggage
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.3.9 Object removal

EventType:attendedBaggage

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, object removal alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, event triggering frequency, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"attendedBaggage" (object removal)-->attendedBaggage
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
```

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```
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"others"-->others
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
```

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```
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format, e.g., 2019-12-30T15:00:00.23+08:00-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

19.7.4 People Gathering Detection

19.7.4.1 Get the arming time parameters of people gathering for all channels

Request URL

GET /ISAPI/Event/schedules/groups

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<GroupScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, people gathering arming time configuration list, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, people gathering arming time configuration-->
    <id>
      <!--ro, req, string, index, desc:the format is event type-channel No.-->test
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string, desc:;group:group-->group
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <dayOfWeek>
          <!--ro, opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-
->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming time-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start Time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</GroupScheduleList>
```

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19.7.4.2 Set the arming time parameters of people gathering for all channels

Request URL

PUT /ISAPI/Event/schedules/groups

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, people gathering arming time configuration list, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, people gathering arming time configuration-->
    <id>
      <!--req, string, index, desc:the format is event type-channel No.-->test
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string, desc:;group:group-->group
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start Time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start Time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</GroupScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.7.4.3 Get the arming schedule parameters of people gathering for a specified channel

Request URL

GET /ISAPI/Event/schedules/groups/group-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, arming time, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index, desc:the format is event type-channel No.-->test
  </id>
  <eventType>
    <!--ro, opt, enum, event type, subType:string, desc:;group:group-->group
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel ID-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, arming time list, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object, arming time-->
      <dayOfWeek>
        <!--ro, opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)--
>1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object, time range-->
        <beginTime>
          <!--ro, req, time, Start Time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object, holiday arming time-->
    <TimeBlock>
      <!--ro, opt, object, arming time-->
      <TimeRange>
        <!--ro, req, object, time range-->
        <beginTime>
          <!--ro, req, time, start time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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19.7.4.4 Set the arming time parameters of people gathering for all channels

Request URL

PUT /ISAPI/Event/schedules/groups/group-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, people gathering arming time configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, desc:the format is event type-channel No.-->test
  </id>
  <eventType>
    <!--opt, enum, event type, subType:string, desc:;group:group-->group
  </eventType>
  <videoInputChannelID>
    <!--opt, string, video input channel ID-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, arming time list, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <dayOfWeek>
        <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object, time range-->
        <beginTime>
          <!--req, time, start time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object, holiday arming time-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <TimeRange>
        <!--req, object, time range-->
        <beginTime>
          <!--req, time, start time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.7.4.5 Set the linkage parameters of people gathering

Request URL

PUT /ISAPI/Event/triggers/group-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, people gathering linkage configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, desc:the format is event type-channel No.-->test
  </id>
  <eventType>
    <!--req, enum, event type, subType:string, desc:;group:group-->group
  </eventType>
  <eventDescription>
    <!--opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string, video input channel ID, desc:if <eventType> is
"VMD,videoloss,tamperdetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage"-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, event capture interval, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string, wireless ID-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, linkage action list, subType:object-->
    <EventTriggerNotification>
      <!--opt, object, linkage action-->
      <id>
        <!--req, string, index-->test
      </id>
      <notificationMethod>
        <!--req, enum, linkage action, subType:string, desc:"email", "IM" (instant messaging), "IO" (I/O output), "syslog" (system log), "HTTP" (uploading
via HTTP), "FTP" (uploading via FTP), "beep" (buzzer), "ptz" (PTZ), "record" (recording), "monitorAlarm" (alarm on monitor), "center" (uploading to the
center), "LightAudioAlarm" (light and audio alarm), "focus" (focus), "trace" (tracking), "whiteLight" (white light), "audio" (audio)-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, linkage cycle, subType:string, desc:"beginning", "beginningandend", "recurring"-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, linkage time interval, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string, video channel ID-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string, dynamic video input channel ID-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object, PTZ linkage-->
        <ptzChannelID>
          <!--req, string, PTZ linked channel No.-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
        </actionName>
        <actionNum>
          <!--opt, int, linked No., desc:depend on the content of <actionName>, it can be preset No., patrol No., and pattern No.-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object, linkage parameters of the white light, desc:it is valid when the value of notificationMethod is "whiteLight"-->
        <whiteLightDurationTime>
          <!--opt, int, light flashing duration, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.4.6 Get linkage parameters of people gathering

Request URL

GET /ISAPI/Event/triggers/group-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, people gathering linkage configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index, desc:the format is event type-channel No.-->test
  </id>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:group:group-->group
  </eventType>
  <eventDescription>
    <!--ro, opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel ID, desc:if <eventType> is
"VMD,videoloss,tamperdetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage"-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, event capture interval, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string, wireless ID-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, linkage action list, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object, linkage action-->
      <id>
        <!--ro, req, string, index-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, linkage action, subType:string, desc:"email", "IM" (instant messaging), "IO" (I/O output), "syslog" (system log), "HTTP"
(uploading via HTTP), "FTP" (uploading via FTP), "beep" (buzzer), "ptz" (PTZ), "record" (recording), "monitorAlarm" (alarm on monitor), "center" (uploading
to the center), "LightAudioAlarm" (light and audio alarm), "focus" (focus), "trace" (tracking), "whiteLight" (white light), "audio" (audio)-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, linkage cycle, subType:string, desc:“beginning”, “beginningandend”, “recurring”-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, linkage time interval, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string, video channel ID-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string, dynamic video input channel ID-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object, PTZ linkage-->
        <ptzChannelID>
          <!--ro, req, string, PTZ linked channel No.-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int, linked No., desc:depend on the content of <actionName>, it can be preset No., patrol No., and pattern No.-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object, linkage parameters of the white light, desc:it is valid when the value of notificationMethod is "whiteLight"-->
        <whiteLightDurationTime>
          <!--ro, opt, int, light flashing duration, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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dojojing@mail.com

19.7.4.7 Get parameters of people gathering for a specified channel

Request URL

PUT /ISAPI/Smart/group/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Group xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters configuration of people gathering for a channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <GroupRegionList>
    <!--opt, array, people gathering area list, subType:object-->
    <GroupRegion>
      <!--opt, object, people gathering area-->
      <id>
        <!--req, string, area ID-->1
      </id>
      <objectOccupation>
        <!--opt, int, percentage-->1
      </objectOccupation>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
  </GroupRegionList>
  <mutexAbility opt="PDC">
    <!--opt, string, mutual exclusion ability, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--opt, bool, whether supports multiple scene detection-->true
  </isSupportMultiScene>
</Group>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.4.8 Get parameters of people gathering for a specified channel

Request URL

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GET /ISAPI/Smart/group/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<Group xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters configuration of people gathering for a channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <GroupRegionList>
    <!--ro, opt, array, people gathering area list, subType:object-->
    <GroupRegion>
      <!--ro, opt, object, people gathering area-->
      <id>
        <!--ro, req, string, area ID-->1
      </id>
      <objectOccupation>
        <!--ro, opt, int, percentage-->1
      </objectOccupation>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
    <headCount>
      <!--ro, opt, int-->5
    </headCount>
    <duration>
      <!--ro, opt, int-->10
    </duration>
  </GroupRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, mutual exclusion ability, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool, whether supports multiple scene detection-->true
  </isSupportMultiScene>
  <shieldedRegionList>
    <!--ro, opt, array, subType:object-->
    <GroupRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,4]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
  </shieldedRegionList>
  <sensitivityLevel>
    <!--ro, opt, int, range:[0,100]-->1
  </sensitivityLevel>
</Group>
```

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VISION AND SECURITY SYSTEM CO.,LTD.

19.7.4.9 Get the capability of people gathering

Request URL

GET /ISAPI/Smart/group/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<Group xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, get the capability of people gathering, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, req, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <GroupRegionList size="4">
    <!--ro, opt, object, people gathering area list, attr:size{req, int}-->
    <GroupRegion>
      <!--ro, opt, object, people gathering area-->
      <id>
        <!--ro, req, string, area ID-->1
      </id>
      <objectOccupation min="1" max="100">
        <!--ro, req, int, percentage, range:[1,100], attr:min{req, int},max{req, int}-->1
      </objectOccupation>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
    <headCount min="2" max="20">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->5
    </headCount>
    <duration min="1" max="60">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->10
    </duration>
  </GroupRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, mutual exclusion ability, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool, whether supports multiple scene detection-->true
  </isSupportMultiScene>
  <isSupportOverlapParams>
    <!--ro, opt, bool, whether supports the parameters for overlaying intelligent information of people gathering, desc:GET/PUT
/ISAPI/Smart/group/channels/<channelID>/GroupOverlapParams?format=json-->true
  </isSupportOverlapParams>
  <shieldedRegionList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <GroupRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,4]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
  </shieldedRegionList>
  <sensitivityLevel min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
  </sensitivityLevel>
</Group>
```

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19.7.4.10 Set parameters of people gathering for all areas

Request URL

PUT /ISAPI/Smart/group/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, batch configure the parameters of people gathering area, subType:object, attr:version{req, string, protocolVersion}-->
  <GroupRegion>
    <!--opt, object, configure the parameter of people gathering area-->
    <id>
      <!--req, string, area ID-->1
    </id>
    <objectOccupation>
      <!--opt, int, percentage-->1
    </objectOccupation>
    <RegionCoordinatesList>
      <!--opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, x-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GroupRegion>
</GroupRegionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.4.11 Get the parameters of people gathering in all areas

Request URL

GET /ISAPI/Smart/group/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<GroupRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, batch set the parameters of people gathering area, subType:object, attr:version{req, string, protocolVersion}-->
  <GroupRegion>
    <!--ro, opt, object, set a specified parameter of people gathering area-->
    <id>
      <!--ro, req, string, area ID-->1
    </id>
    <objectOccupation>
      <!--ro, opt, int, percentage-->1
    </objectOccupation>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GroupRegion>
</GroupRegionList>
```

19.7.4.12 Set parameters of people gathering for a specified area

Request URL

PUT /ISAPI/Smart/group/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, configure the parameter of people gathering area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, area ID-->1
  </id>
  <objectOccupation>
    <!--opt, int, percentage-->1
  </objectOccupation>
  <RegionCoordinatesList>
    <!--opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, x-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</GroupRegion>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.4.13 Get parameters of a specified area for people gathering

Request URL

GET /ISAPI/Smart/group/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, a specified parameter of people gathering area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, area ID-->1
  </id>
  <objectOccupation>
    <!--ro, opt, int, percentage-->1
  </objectOccupation>
  <RegionCoordinatesList>
    <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</GroupRegion>
```

19.7.4.14 People gathering

EventType:group

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
```

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```
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"group" (people gathering)-->group
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <objectOccupation>
        <!--ro, opt, int, percentage, range:[1,100]-->1
      </objectOccupation>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
```

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```
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
group	[Message content]	application/xml	--	--	--
.value	[Binary picture data]	image/jpeg	随机值	pic.jpeg	--
video	--	video/mp4	随机值	video.mp4	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.7.5 Unattended Baggage

19.7.5.1 Set the arming schedule of unattended baggage detection for all channels

Request URL

PUT /ISAPI/Event/schedules/unattendedBaggages

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<UnattendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->unattendedBaggage-1
    </id>
    <eventType>
      <!--opt, enum, subType:string-->unattendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, range:[0,56], attr:size{req, int, range:[0,8]}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object, range:[0,8]-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</UnattendBaggageScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.5.2 Get the arming schedule of unattended baggage detection for all channels

Request URL

GET /ISAPI/Event/schedules/unattendedBaggages

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<UnattendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->unattendedBaggage-1
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->unattendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, range:[0,56], attr:size{req, int, range:[0,8]}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object, range:[0,8]-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</UnattendBaggageScheduleList>
```

19.7.5.3 Get the arming schedule of unattended baggage detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/unattendedBaggages/unattendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->unattendedBaggage-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->unattendedBaggage
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, range:[0,56], attr:size{req, int, range:[0,8]}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object, range:[0,8]-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.7.5.4 Set the arming schedule of unattended baggage detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/unattendedBaggages/unattendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--opt, enum, subType:string-->unattendedBaggage
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, range:[0,56], attr:size{req, int, range:[0,8]}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object, range:[0,8]-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.7.5.5 Get the linkage parameters of unattended baggage detection

Request URL

GET /ISAPI/Event/triggers/unattendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->unattendedBaggage-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->unattendedBaggage
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.7.5.6 Set the linkage parameters of unattended baggage detection

Request URL

PUT /ISAPI/Event/triggers/unattendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->unattendedBaggage-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->unattendedBaggage
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.7.5.7 Unattended baggage detection

EventType:unattendedBaggage

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, unattended baggage alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, event triggering frequency, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"unattendedBaggage" (unattended baggage)-->unattendedBaggage
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
```

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```
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"others"-->others
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
```

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```
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format, e.g., 2019-12-30T15:00:00.23+08:00-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

19.8 Facial Recognition

19.8.1 Face Capture

19.8.1.1 Get the arming schedule of face capture

Request URL

GET /ISAPI/Event/schedules/faceSnap

Query Parameter

None

Request Message

None

Response Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, arming schedule of face capture for all channels, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule of face capture for a specified channel-->
    <id>
      <!--ro, req, string, channel No.-->test
    </id>
    <eventType>
      <!--ro, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpLoad",
"HFP", "IO", "IOTD", "LES", 'LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fieldetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "leavePosition",
"linedetection", "listSyncException", "loitering", "LowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollover", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->ADAS
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel No.-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, arming time list, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--ro, opt, object, arming advanced configurations-->
          <vehicleDetectSceneID>
            <!--ro, opt, int, vehicle detection scene ID-->1
          </vehicleDetectSceneID>
          <sceneID>
            <!--ro, opt, int, scene ID-->1
          </sceneID>
          <SceneList>
            <!--ro, opt, array, switching rules of multiple scenes, subType:object, desc:mutually exclusive with "sceneID"-->
            <Scene>
              <!--ro, opt, object, multiple scene list-->
              <sceneID>
                <!--ro, req, string, scene ID-->1
              </sceneID>
              <durationTime>
                <!--ro, req, int, stayed time of a single scene-->1
              </durationTime>
            </Scene>
          </SceneList>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapScheduleList>
```

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19.8.1.2 Set the arming schedule of face capture

Request URL

PUT /ISAPI/Event/schedules/faceSnap

Query Parameter

None

Request Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, req, array, arming schedule of face capture for all channels, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming schedule of face capture for a specified channel-->
    <id>
      <!--wo, req, string, channel No.-->test
    </id>
    <eventType>
      <!--wo, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpLoad",
"HFP", "IO", "IOTD", "LES", 'LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fielddetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "leavePosition",
"linedetection", "listSyncException", "loitering", "LowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollover", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->faceSnap
    </eventType>
    <videoInputChannelID>
      <!--wo, opt, string, video input channel No.-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--wo, req, array, arming time list, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--wo, opt, object, arming slots-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:days of the week-->1
        </dayOfWeek>
        <TimeRange>
          <!--wo, req, object, time range-->
          <beginTime>
            <!--wo, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--wo, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--wo, opt, object, arming advanced configurations-->
          <vehicleDetectSceneID>
            <!--wo, opt, int, vehicle detection scene-->1
          </vehicleDetectSceneID>
          <sceneID>
            <!--wo, opt, int, scene ID-->1
          </sceneID>
          <SceneList>
            <!--wo, opt, array, switching rules of multiple scenes, subType:object, desc:mutually exclusive with "sceneID"-->
            <Scene>
              <!--wo, opt, object, multiple scene list-->
              <sceneID>
                <!--wo, req, string, scene ID-->1
              </sceneID>
              <durationTime>
                <!--wo, req, int, stayed time of a single scene-->1
              </durationTime>
            </Scene>
          </SceneList>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--wo, opt, array, holiday arming time list, subType:object-->
      <TimeBlock>
        <!--wo, opt, object, arming time-->
        <TimeRange>
          <!--wo, req, object, time range-->
          <beginTime>
            <!--wo, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--wo, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.1.3 Get the arming schedule of face capture for a specified channel

Request URL

GET /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.

Request Message

None

Response Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, arming schedule of face capture for a specified channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpload", "HFP",
"IO", "IOTD", "LES", 'LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fielddetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "leavePosition",
"linedetection", "listSyncException", "loitering", "LowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollover", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->faceSnap
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel No.-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00+08:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--ro, opt, object-->
        <vehicleDetectSceneID>
          <!--ro, opt, int-->1
        </vehicleDetectSceneID>
        <sceneID>
          <!--ro, opt, int-->1
        </sceneID>
        <SceneList>
          <!--ro, opt, array, subType:object-->
          <Scene>
            <!--ro, opt, object-->
            <sceneID>
              <!--ro, req, string-->1
            </sceneID>
            <durationTime>
              <!--ro, req, int-->1
            </durationTime>
          </Scene>
        </SceneList>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
  <backToSceneTime>
    <!--ro, opt, int, range:[5,100], step:1, unit:s-->15
  </backToSceneTime>
</Schedule>
```

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19.8.1.4 Set the arming schedule of face capture for a specified channel

Request URL

PUT /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, opt, object, arming schedule of face capture for a specified channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--wo, req, string, channel No.-->test
  </id>
  <eventType>
    <!--wo, req, enum, event type, subType:string, desc:“ADAS”, “ADASAlarm”, AID, ANPR, AccessControllerEvent, CDsStatus, DBD, GPSUpload, HFP, IO, IOTD,
LES, LFPD, PALMismatch, PIR, PeopleCounting, PeopleNumChange, Standup, TMA, TMPAM, VMD, abnormalAcceleration, abnormalDriving, advReachHeight, alarmResult,
attendance, attendedBaggage, audioAbnormal, audioexception, behaviorResult, blindSpotDetection, cardMatch, changedStatus, collision, containerDetection,
crowdSituationAnalysis, databaseException, defocus, diskUnformat, diskerror, diskfull, driverConditionMonitor, emergencyAlarm, faceCapture,
faceSnapModeling, facedetection, failDown, faultAlarm, fielddetection, fireDetection, fireEscapeDetection, flowOverrun, framesPeopleCounting, getUp, group,
hdBadBlock, hdImpact, heatmap, highHDTemperature, highTempAlarm, hotSpare, illaccess, ipcTransferAbnorma, ipconflict, keyPersonGetUp, leavePosition,
linedetection , listSyncException, loitering, lowHDTemperature, mixedTargetDetection, modelError, nicbroken, nodeOffline, nonPoliceIntrusio, overSpeed,
overtimeTarry, parking, peopleNumChange, peopleNumCounting, personAbnormalAlarm:, personDensityDetection, personQueueCounting, personQueueDetection,
personQueueRealTime, personQueueTime, playCellphone, pocException, poe, policeAbsent, radarAlarm, radarFieldDetection, radarLineDetection,
radarPerimeterRule, radarTargetDetection, radarVideoDetection, raidException, rapidMove, reachHeight, recordCycleAbnormal, recordException, regionEntrance,
regionExiting, retention, rollover, running, safetyHelmetDetection, scenechangedetection, sensorAlarm, severeHDFailure, shelteralarm, shipsDetection,
sitQuietly, smokeAndFireDetection, smokeDetection, softIO, spacingChange, sysStorFull, takingElevatorDetection, targetCapture, temperature, thermometry,
thirdPartyException, toiletTarry, tollCodeInfo, tossing, unattendedBaggage, vehicleMatchResult, vehicleRcogResult, versionAbnormal, videoException,
videoloss, violationAlarm, violentMotion, yardTarry, faceSnap-->faceSnap
  </eventType>
  <videoInputChannelID>
    <!--wo, opt, string, video input channel No.-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--wo, req, array, arming time list, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--wo, opt, object, arming time-->
      <dayOfWeek>
        <!--wo, opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)--
>1
      </dayOfWeek>
      <TimeRange>
        <!--wo, req, object, time range-->
        <beginTime>
          <!--wo, req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--wo, req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--wo, opt, object, arming advanced configurations-->
        <vehicleDetectSceneID>
          <!--wo, opt, int, vehicle detection scene-->1
        </vehicleDetectSceneID>
        <sceneID>
          <!--wo, opt, int, scene ID-->1
        </sceneID>
        <SceneList>
          <!--wo, opt, array, switching rules of multiple scenes, subType:object, desc:mutually exclusive with “sceneID”-->
          <Scene>
            <!--wo, opt, object, multiple scene list-->
            <sceneID>
              <!--wo, req, string, scene ID-->1
            </sceneID>
            <durationTime>
              <!--req, int, stayed time of a single scene-->1
            </durationTime>
          </Scene>
        </SceneList>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, holiday arming time list, subType:object-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <TimeRange>
        <!--req, object, time range-->
        <beginTime>
          <!--req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.1.5 Set the arming schedule of face modeling for all channels

Request URL

PUT /ISAPI/Event/schedules/faceSnapModeling

Query Parameter

None

Request Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapModelingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--req, object-->
    <id>
      <!--req, string-->faceSnapModeling-1
    </id>
    <eventType>
      <!--opt, enum, subType:string-->faceSnapModeling
    </eventType>
    <inputIOPortID>
      <!--opt, string-->test
    </inputIOPortID>
    <outputIOPortID>
      <!--opt, string-->test
    </outputIOPortID>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--opt, object-->
          <vehicleDetectSceneID>
            <!--opt, int-->0
          </vehicleDetectSceneID>
          <sceneID>
            <!--opt, int-->0
          </sceneID>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapModelingScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.8.1.6 Get the arming schedule of face modeling for all channels

Request URL

GET /ISAPI/Event/schedules/faceSnapModeling

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapModelingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, req, object-->
    <id>
      <!--ro, req, string-->faceSnapModeling-1
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->faceSnapModeling
    </eventType>
    <inputIOPortID>
      <!--ro, opt, string-->test
    </inputIOPortID>
    <outputIOPortID>
      <!--ro, opt, string-->test
    </outputIOPortID>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--ro, opt, object-->
          <vehicleDetectSceneID>
            <!--ro, opt, int-->0
          </vehicleDetectSceneID>
          <sceneID>
            <!--ro, opt, int-->0
          </sceneID>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapModelingScheduleList>
```

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19.8.1.7 Set the arming schedule of face modeling for a specified channel

Request URL

PUT /ISAPI/Event/schedules/faceSnapModeling/faceSnapModeling-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->faceSnapModeling-1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->faceSnapModeling
  </eventType>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <outputIOPortID>
    <!--ro, opt, string-->test
  </outputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--opt, object-->
        <vehicleDetectSceneID>
          <!--opt, int-->0
        </vehicleDetectSceneID>
        <sceneID>
          <!--opt, int-->0
        </sceneID>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.1.8 Get the arming schedule of face modeling for a specified channel

Request URL

GET /ISAPI/Event/schedules/faceSnapModeling/faceSnapModeling-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceSnapModeling-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->faceSnapModeling
  </eventType>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <outputIOPortID>
    <!--ro, opt, string-->test
  </outputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--ro, opt, object-->
        <vehicleDetectSceneID>
          <!--ro, opt, int-->0
        </vehicleDetectSceneID>
        <sceneID>
          <!--ro, opt, int-->0
        </sceneID>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.8.1.9 Delete the linkage of face capture

Request URL

DELETE /ISAPI/Event/triggers/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.8.1.10 Set the linkage parameters of face capture

Request URL

PUT /ISAPI/Event/triggers/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, req, object, Configure linkage actions:, attr:version{req, string, protocolVersion}-->
  <id>
    <!--wo, req, string, channel No.-->test
  </id>
  <eventType>
    <!--wo, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpload", "HFP",
"IO", "IOTD", "LES", "LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fielddetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "leavePosition",
"linedetection", "listSyncException", "loitering", "lowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollOver", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->faceSnap
  </eventType>
  <eventDescription>
    <!--wo, opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--wo, opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--wo, opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--wo, opt, string, video input channel ID, desc:only valid when event types are VMD, videoloss, shelteralarm, regionEntrance, regionExiting,
loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, thermal, zoneTriggeredAlarm, criticalAlerts, businessCounselingLinkage-->test
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--wo, opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--wo, opt, int, event capture interval, step:1, unit:s-->1
```

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```
    </intervalBetweenEvents>
    <WLSensorID>
      <!--wo, opt, string, wireless ID-->test
    </WLSensorID>
    <EventTriggerNotificationList>
      <!--wo, opt, array, configuration list of event linkage, subType:object-->
      <EventTriggerNotification>
        <!--wo, opt, object, list-->
        <id>
          <!--wo, req, string, linkage configuration ID-->test
        </id>
        <notificationMethod>
          <!--wo, req, enum, linkage action, subType:string, desc:"FTP" (uploading via FTP), "HTTP" (uploading via HTTP), "IM" (instant messaging), "IO" (I/O
output), "lightAudioAlarm" (light and audio alarm), "audio" (audio), "beep" (buzzer), "center" (uploading to the center), "cloud" (cloud storage), "email",
"focus" (focus), "monitorAlarm" (alarm on monitor), "ptz" (PTZ), "record" (recording), "syslog" (system log), "trace" (tracking), "whiteLight" (white
light)-->FTP
        </notificationMethod>
        <notificationRecurrence>
          <!--wo, opt, enum, linkage cycle, subType:string, desc:“beginning”, “beginningandend”, “recurring”-->beginning
        </notificationRecurrence>
        <notificationInterval>
          <!--wo, opt, int, linkage time interval, step:1, unit:ms-->1
        </notificationInterval>
        <videoInputID>
          <!--wo, opt, string, video input ID-->test
        </videoInputID>
        <dynVideoInputID>
          <!--wo, opt, string, dynamic video input ID-->test
        </dynVideoInputID>
        <ptzAction>
          <!--wo, opt, object, PTZ linkage action-->
          <ptzChannelID>
            <!--wo, req, string, PTZ linked channel No.-->test
          </ptzChannelID>
          <actionName>
            <!--wo, req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
          </actionName>
          <actionNum>
            <!--wo, opt, int, linkage value-->1
          </actionNum>
        </ptzAction>
        <WhiteLightAction>
          <!--wo, opt, object, white light linkage-->
          <whiteLightDurationTime>
            <!--wo, req, int, light flashing duration, range:[1,60], step:1, unit:s-->1
          </whiteLightDurationTime>
        </WhiteLightAction>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
  </EventTrigger>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.8.1.11 Get the linkage parameters of face capture

Request URL

GET /ISAPI/Event/triggers/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, linkage configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpload", "HFP",
"IO", "IOTD", "LES", 'LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fielddetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "leavePosition",
"linedetection", "listSyncException", "loitering", "lowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollover", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->faceSnap
  </eventType>
  <eventDescription>
    <!--ro, opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel ID, desc:only valid when event types are VMD, videoloss, shelteralarm, regionEntrance, regionExiting,
loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, thermal, zoneTriggeredAlarm, criticalAlerts, businessCounselingLinkage-->test
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, event capture interval, step:1, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string, wireless ID-->test
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, step:1, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->test
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->test
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, req, int, range:[1,60], step:1, unit:s-->1
        </whiteLightDurationTime>
```

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dojojin@gmail.com

```
        </WhiteLightDurationTime>
      </WhiteLightAction>
      <childDevID>
        <!--ro, opt, string-->1
      </childDevID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

19.8.1.12 Set the face capture rule parameters of a device

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceRule

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, face capture rule capability, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--req, enum, whether to enable the function, subType:string, desc:“true”, “false”-->true
  </enabled>
  <FaceDetectionRegion>
    <!--opt, object, detection area-->
    <RegionCoordinatesList>
      <!--opt, array, list of region coordinates, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--req, object, area point coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </FaceDetectionRegion>
  <InterPupileRegion>
    <!--opt, object, rectangle frame of minimum pupil distance, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--req, int, pupil distance, range:[0,1000]-->50
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->50
    </height>
  </InterPupileRegion>
  <MaxInterPupilRegion>
    <!--opt, object, the maximum pupil distance, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--req, int, pupil distance, range:[0,1000]-->50
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->50
    </height>
  </MaxInterPupilRegion>
  <SizeFilter>
    <!--opt, object, size filter-->
    <enabled>
      <!--req, bool, whether to enable size filter-->true
    </enabled>
    <ratio>
      <!--opt, float, ratio of head-shoulder size filter to minimum pupil distance, range:[4.0,6.0]-->5.0
    </ratio>
    <MaxTargetSize>
      <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000], desc:X-coordinate-->0
      </positionX>
```

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```
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
        <width>
          <!--req, int, width, range:[0,1000]-->0
        </width>
        <height>
          <!--req, int, height, range:[0,1000]-->0
        </height>
      </MaxTargetSize>
      <MinTargetSize>
        <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
        <width>
          <!--req, int, width, range:[0,1000]-->0
        </width>
        <height>
          <!--req, int, height, range:[0,1000]-->0
        </height>
      </MinTargetSize>
    </SizeFilter>
    <detectionType>
      <!--opt, string, face type for detection, desc:"abnormalFace" (abnormal face),"normalFace" (normal face),"multiFace" (multiple faces),"sunglassesFace"
(face,on which wearing sunglasses),"callFace" (face of a person who is making a phone call). Multiple values are separated by commas--
>abnormalFace,normalFace
    </detectionType>
    <autoROIEnable>
      <!--opt, bool, whether to enable face automatic ROI, desc:the value of this node is false by default-->false
    </autoROIEnable>
  </FaceRule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.1.13 Get the capability of configuring parameters of face capture rules.

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceRule/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--ro, req, object, capability of configuring parameters of face capture rules, attr:version{opt, string, protocolVersion}-->
  <enabled opt="true,false">
    <!--ro, req, enum, whether to enable, subType:string, attr:opt{opt, string}, desc:whether to enable-->true
  </enabled>
  <FaceDetectionRegion>
```

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```
    <FaceDetectionRegion>
      <!--ro, opt, object, detection area-->
      <RegionCoordinatesList>
        <!--ro, opt, array, coordinate list of the area, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, req, object, coordinates of the area, desc:The origin is the lower-left corner of the screen.-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </FaceDetectionRegion>
    <InterPupileRegion>
      <!--ro, opt, object, the rectangle frame on the minimum pupillary distance, desc:The origin is the lower-left corner of the screen.-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
      </positionY>
      <width min="0" max="1000">
        <!--ro, req, int, pupillary distance, range:[0,1000], attr:min{opt, string},max{opt, string}-->50
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->50
      </height>
    </InterPupileRegion>
    <MaxInterPupilRegion>
      <!--ro, opt, object, the maximum pupillary distance, desc:The origin is the lower-left corner of the screen.-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
      </positionY>
      <width min="50" max="50">
        <!--ro, req, int, pupillary distance, range:[0,1000], attr:min{opt, string},max{opt, string}-->50
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->50
      </height>
    </MaxInterPupilRegion>
    <SizeFilter>
      <!--ro, opt, object, size filter-->
      <enabled opt="true,false" def="false">
        <!--ro, req, bool, whether to enable size filter, attr:opt{opt, string},def{opt, string}-->true
      </enabled>
      <ratio min="4.0" max="6.0" def="5.0">
        <!--ro, opt, float, ratio of the size filter of the head and shoulders to the minimum pupillary distance, range:[4.0,6.0], attr:min{opt,
string},max{opt, string},def{opt, string}-->5.0
      </ratio>
      <MaxTargetSize>
        <!--ro, opt, object, target frame on the largest face, desc:The origin is the lower-left corner of the screen.-->
        <positionX min="0" max="1000">
          <!--ro, req, int, X-coordinate, attr:min{req, int},max{req, int}-->0
        </positionX>
        <positionY min="0" max="1000">
          <!--ro, req, int, Y-coordinate, attr:min{req, int},max{req, int}-->0
        </positionY>
        <width min="0" max="1000">
          <!--ro, req, int, width, attr:min{req, int},max{req, int}-->0
        </width>
        <height min="0" max="1000">
          <!--ro, req, int, height, attr:min{req, int},max{req, int}-->0
        </height>
      </MaxTargetSize>
      <MinTargetSize>
        <!--ro, opt, object, target frame on the smallest face, desc:The origin is the lower-left corner of the screen.-->
        <positionX min="0" max="1000">
          <!--ro, req, int, X-coordinate, attr:min{req, int},max{req, int}-->0
        </positionX>
        <positionY min="0" max="1000">
          <!--ro, req, int, Y-coordinate, attr:min{req, int},max{req, int}-->0
        </positionY>
        <width min="0" max="1000">
          <!--ro, req, int, width, attr:min{req, int},max{req, int}-->0
        </width>
        <height min="0" max="1000">
          <!--ro, req, int, height, attr:min{req, int},max{req, int}-->0
        </height>
      </MinTargetSize>
    </SizeFilter>
    <isSupportTrigger>
      <!--ro, opt, bool, whether it supports trigger mode, desc:Related URI: /ISAPI/Intelligent/channels/<ID>/faceRule/trigger?format=json-->true
    </isSupportTrigger>
    <detectionType opt="abnormalFace,normalFace,multiFace,sunglassesFace,callFace" def="normalFace,multiFace,sunglassesFace,callFace">
      <!--ro, opt, string, face type for detection, attr:opt{req, string},def{req, string}, desc:abnormalFace (abnormal face), normalFace (normal face),
multiFace (multiple faces), sunglassesFace (face with sunglasses), callFace (face of a person who is making a phone call, which is used to detect faces in
ATM mode). Multiple types are separated by commas.-->test
    </detectionType>
    <defualt opt="true">
```

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```
    <!--ro, opt, bool, whether it supports getting the default parameters, attr:opt{req, string}, desc:If the value of this node is true, it indicates that
  the query parameter default can be added to the URI GET /ISAPI/Intelligent/channels/<ID>/faceRule for getting the default parameters, i.e., GET
  /ISAPI/Intelligent/channels/<ID>/faceRule?defualt=true.-->true
  </defualt>
  <autoROIEnable opt="true,false">
    <!--ro, opt, bool, whether to enable automatic face ROI, attr:opt{req, string}, desc:By default it is false.-->false
  </autoROIEnable>
  <MEF>
    <!--ro, opt, object, mobile electric fence, desc:Related URI: /ISAPI/Intelligent/channels/<ID>/faceRule/MEF?format=json-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable uploading the electric fence data linked with the event, attr:opt{req, string}-->true
    </enabled>
  </MEF>
  <isSupportFaceSnapPosOverlayParams>
    <!--ro, opt, bool-->true
  </isSupportFaceSnapPosOverlayParams>
  <backToSceneTime min="5" max="100">
    <!--ro, opt, int, range:[5,100], step:1, unit:s, attr:min{req, int},max{req, int}-->15
  </backToSceneTime>
  <minInterPupileDistance min="13" max="666">
    <!--ro, opt, int, range:[13,666], step:1, unit:px, attr:min{req, int},max{req, int}-->13
  </minInterPupileDistance>
  <isSupportSearchFaceSnapAlarm>
    <!--ro, opt, bool-->true
  </isSupportSearchFaceSnapAlarm>
  <isSupportFaceApproachParams>
    <!--ro, opt, bool-->true
  </isSupportFaceApproachParams>
  <isSupportChannelPrivacyMaskList>
    <!--ro, opt, bool-->true
  </isSupportChannelPrivacyMaskList>
</FaceRule>
```

19.8.1.14 Get the face picture capturing rules of multiple scenes for a specified channel

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<MultiSceneFaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, face picture capturing rules of multiple scenes, attr:version{req, string, protocolVersion}-->
  <FaceRuleList>
    <!--ro, req, array, list of face picture rules, subType:object-->
    <FaceRule>
      <!--ro, req, object, face capture rule-->
      <enabled>
        <!--ro, req, enum, whether to enable, subType:string, desc:“true”, “false”-->true
      </enabled>
      <FaceDetectionRegion>
        <!--ro, opt, object, detection area-->
        <RegionCoordinatesList>
          <!--ro, opt, array, list of region coordinates, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--ro, req, object, area point coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->50
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </FaceDetectionRegion>
      <InterPupileRegion>
        <!--ro, opt, object, rectangle frame of minimum pupil distance, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
        <width>
          <!--ro, req, int, pupil distance, range:[0,1000]-->50
        </width>
        <height>
          <!--ro, req, int, height, range:[0,1000]-->50
        </height>
      </InterPupileRegion>
      <MaxInterPupilRegion>
        <!--ro, opt, object, the maximum pupil distance, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
        <width>
          <!--ro, req, int, pupil distance, range:[0,1000]-->50
        </width>
        <height>
          <!--ro, req, int, height, range:[0,1000]-->50
        </height>
      </MaxInterPupilRegion>
      <SizeFilter>
        <!--ro, opt, object, size filter-->
        <enabled>
          <!--ro, req, bool, whether to enable size filter-->true
        </enabled>
        <ratio>
          <!--ro, opt, float, ratio of head-shoulder size filter to minimum pupil distance, range:[4.0,6.0]-->5.0
        </ratio>
      </SizeFilter>
    </FaceRule>
  </FaceRuleList>
</MultiSceneFaceRule>
```

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19.8.1.15 Set the face capture rules of a specified scene

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/<SID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
SID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SingleSceneFaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, face capture rules of a specified scene, attr:version{req, string, protocolVersion}-->
  <sid>
    <!--req, string, scene ID-->test
  </sid>
  <FaceRule>
    <!--req, object, face capture rule-->
    <enabled>
      <!--req, enum, whether to enable, subType:string, desc:“true”, “false”-->test
    </enabled>
    <FaceDetectionRegion>
      <!--opt, object, detection area-->
      <RegionCoordinatesList>
        <!--opt, array, list of region coordinates, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object, area point coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </FaceDetectionRegion>
    <InterPupileRegion>
      <!--opt, object, rectangle frame of minimum pupil distance, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->50
      </positionY>
      <width>
        <!--req, int, pupil distance, range:[0,1000]-->50
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->50
      </height>
    </InterPupileRegion>
    <MaxInterPupilRegion>
      <!--opt, object, the maximum pupil distance, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->50
      </positionY>
      <width>
        <!--req, int, pupil distance, range:[0,1000]-->50
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->50
      </height>
    </MaxInterPupilRegion>
    <SizeFilter>
      <!--opt, object, size filter-->
      <enabled>
        <!--req, bool, whether to enable size filter-->true
      </enabled>
      <ratio>
        <!--opt, float, ratio of head-shoulder size filter to minimum pupil distance, range:[4.0,6.0]-->5.0
      </ratio>
    </SizeFilter>
  </FaceRule>
</SingleSceneFaceRule>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.1.16 Set the scene mode parameters of multiple scenes for face capture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/sceneMode?format=json&mode=<mode>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
mode	string	--

Request Message

```
{
    "FaceSnapMultiSceneMode": {
    /*req, object*/
        "sceneMode":  "normal"
        /*req, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

19.8.1.17 Get the trigger mode parameters capability of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceRule/trigger/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "TriggerCap": {
    /*ro, req, object*/
        "type": {
        /*ro, opt, object*/
            "@opt": ["IO", "video"]
            /*ro, opt, array, subType:string*/
        },
        "IOSignalType": {
        /*ro, opt, object*/
            "@opt": ["levelSignal", "pulseSignal"]
            /*ro, opt, array, subType:string*/
        },
        "doorStatus": {
        /*ro, opt, object*/
            "@opt": ["open", "close", "N_A"]
            /*ro, opt, array, subType:string*/
        }
    }
}
```

19.8.1.18 Set the trigger mode parameters of face capture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/trigger?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "Trigger": {
    /*req, object*/
        "type":  "video",
        /*req, enum, subType:string*/
        "IOSignalType":  "levelSignal"
        /*req, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

19.8.1.19 Get the trigger mode parameters of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceRule/trigger?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "Trigger": {
    /*ro, req, object*/
        "type":  "video",
        /*ro, req, enum, subType:string*/
        "IOSignalType":  "levelSignal",
        /*ro, req, enum, subType:string*/
        "doorStatus":  "open"
        /*ro, req, enum, subType:string*/
    }
}
```

19.8.1.20 Get the face capture rule parameters of a device

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceRule?defualt=<defualt>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
defualt	string	Get default parameters via FaceRule.defualt in /ISAPI/Intelligent/channels//faceRule/capabilities.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, face capture rule capability, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--ro, req, enum, whether to enable, subType:string, desc:"true", "false"-->true
  </enabled>
  <FaceDetectionRegion>
    <!--ro, opt, object, detection area-->
    <RegionCoordinatesList>
      <!--ro, opt, array, list of region coordinates, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, req, object, area point coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </FaceDetectionRegion>
  <InterPupileRegion>
    <!--ro, opt, object, rectangle frame of minimum pupil distance, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--ro, req, int, pupillary distance, range:[0,1000]-->50
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->50
    </height>
  </InterPupileRegion>
  <MaxInterPupilRegion>
    <!--ro, opt, object, the maximum pupillary distance, desc:The origin is the lower-left corner of the screen.-->
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--ro, req, int, pupillary distance, range:[0,1000]-->50
    </width>
    <height>
```

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```
      <!--ro, req, int, height, range:[0,1000]-->50
    </height>
  </MaxInterPupilRegion>
  <SizeFilter>
    <!--ro, opt, object, size filter-->
    <enabled>
      <!--ro, req, bool, whether to enable size filter-->true
    </enabled>
    <ratio>
      <!--ro, opt, float, ratio of head-shoulder size filter to minimum pupil distance, range:[4.0,6.0]-->5.0
    </ratio>
    <MaxTargetSize>
      <!--ro, opt, object, target frame on the largest face, desc:The origin is the lower-left corner of the screen.-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000], desc:Value range: [0,1000]-->0
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
      <width>
        <!--ro, req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->0
      </height>
    </MaxTargetSize>
    <MinTargetSize>
      <!--ro, opt, object, target frame on the smallest face, desc:The origin is the lower-left corner of the screen.-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
      <width>
        <!--ro, req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->0
      </height>
    </MinTargetSize>
  </SizeFilter>
  <detectionType>
    <!--ro, opt, string, face type for detection, desc:abnormalFace (abnormal face), normalFace (normal face), multiFace (multiple faces), sunglassesFace (face with sunglasses), callFace (face of a person who is making a phone call, which is used to detect faces in ATM mode). This node supports to be set to multiple values and they should be separated by commas.-->abnormalFace,normalFace
  </detectionType>
  <autoROIEnable>
    <!--ro, opt, bool, whether to enable automatic face ROI, desc:By default it is false.-->false
  </autoROIEnable>
  <minInterPupileDistance>
    <!--ro, opt, int, range:[13,666], step:1, unit:px-->58
  </minInterPupileDistance>
</FaceRule>
```

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19.8.1.21 Get the capability of configuring advanced parameters of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/advancedParam/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "AdvancedParamCap": {
    /*ro, req, object*/
        "snapMode": {
        /*ro, opt, object*/
            "@opt": ["fast", "best"],
            /*ro, opt, array, subType:string*/
        },
        "quicksnapThreshold": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  100,
            /*ro, opt, int*/
```

```
            "@def":  70
            /*ro, opt, int*/
        },
        "quicksnapMaxtime": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  300,
            /*ro, opt, int*/
            "@def":  5
            /*ro, opt, int*/
        },
        "quicksnapCount": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  100,
            /*ro, opt, int*/
            "@def":  1
            /*ro, opt, int*/
        },
        "snapTime": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  3,
            /*ro, opt, int*/
            "@def":  1
            /*ro, opt, int*/
        },
        "snapThreshold": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  100,
            /*ro, opt, int*/
            "@def":  20
            /*ro, opt, int*/
        },
        "faceExposurEnable": {
        /*ro, opt, object*/
            "@opt": [true, false],
            /*ro, opt, array, subType:bool*/
            "@def":  false
            /*ro, opt, bool*/
        },
        "referenceBright": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  100,
            /*ro, opt, int*/
            "@def":  50
            /*ro, opt, int*/
        },
        "minContinueTime": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  60,
            /*ro, opt, int*/
            "@def":  5
            /*ro, opt, int*/
        },
        "faceFilterTime": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  100000,
            /*ro, opt, int*/
            "@def":  0
            /*ro, opt, int*/
        },
        "faceAttributEnable": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "facesnapFilterEnable": {
        /*ro, opt, object*/
            "@opt": [true, false],
            /*ro, opt, array, subType:bool*/
            "@def":  false
            /*ro, opt, bool*/
        },
        "swingAngleLeft": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  70
            /*ro, opt, int*/
        },
        "swingAngleRight": {
        /*ro, opt, object*/
```

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```
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  70
            /*ro, opt, int*/
        },
        "tiltAngleUp": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  50
            /*ro, opt, int*/
        },
        "tiltAngleDown": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  50
            /*ro, opt, int*/
        },
        "RemoveDuplicate": {
        /*ro, opt, object, dep:and,{$.AdvancedParam.snapMode,eq,best}*/
            "enabled": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, opt, array, subType:bool*/
            },
            "threshold": {
            /*ro, opt, object*/
                "@min":  70,
                /*ro, opt, int*/
                "@max":  100,
                /*ro, opt, int*/
                "@def":  88
                /*ro, opt, int*/
            },
            "faceScore": {
            /*ro, opt, object*/
                "@min":  20,
                /*ro, opt, int*/
                "@max":  60,
                /*ro, opt, int*/
                "@def":  20
                /*ro, opt, int*/
            },
            "updateTime": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, opt, int*/
                "@max":  300,
                /*ro, opt, int*/
                "@def":  60
                /*ro, opt, int*/
            }
        },
        "quickContinueSnapEnable": {
        /*ro, opt, object, dep:and,{$.AdvancedParamCap.snapMode,eq,fast}*/
            "@opt": [true, false],
            /*ro, opt, array, subType:bool*/
            "@def":  false
            /*ro, opt, bool*/
        },
    }
}
```

19.8.1.22 Set the advanced parameters of face capture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/advancedParam?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
{
    "AdvancedParam": {
    /*wo, req, object*/
        "tarGenerateRate": 3,
        /*wo, opt, int, range:[1,5]*/
        "sensitivity": 3,
        /*wo, opt, int, range:[1,5]*/
        "snapMode": "fast",
        /*wo, req, enum, subType:string*/
        "quicksnapThreshold": 70,
        /*wo, opt, int, range:[0,100], unit:s*/
        "quicksnapMaxtime": 5,
        /*wo, opt, int, range:[0,300], step:1, unit:s*/
        "quicksnapCount": 1,
        /*wo, opt, int, range:[1,100]*/
        "snapTime": 1,
        /*wo, opt, int, range:[1,3]*/
        "snapThreshold": 20,
        /*wo, opt, int, range:[0,100]*/
        "snapInterval": 1,
        /*wo, opt, int*/
        "faceExposurEnable": true,
        /*wo, opt, bool*/
        "referenceBright": 50,
        /*wo, opt, int, range:[0,100]*/
        "minContinueTime": 5,
        /*wo, opt, int, range:[0,60], step:1, unit:min*/
        "faceFilterTime": 0,
        /*wo, opt, int, range:[0,100000], step:1, unit:ms*/
        "faceAttributEnable": true,
        /*wo, opt, bool*/
        "facesnapFilterEnable": false,
        /*wo, opt, bool*/
        "swingAngleLeft": 10,
        /*wo, opt, int, range:[0,70], step:1, unit:°*/
        "swingAngleRight": 1,
        /*wo, opt, int, range:[0,70], step:1, unit:°*/
        "tiltAngleUp": 10,
        /*wo, opt, int, range:[0,50], step:1, unit:°*/
        "tiltAngleDown": 10,
        /*wo, opt, int, range:[0,50], step:1, unit:°*/
        "instantLightMidEnable": true,
        /*wo, opt, bool*/
        "fixedLightMidEnable": true,
        /*wo, opt, bool*/
        "severeEnable": true,
        /*wo, opt, bool*/
        "OSDEnable": true,
        /*wo, opt, bool*/
        "RemoveDuplicate": {
        /*wo, opt, object, dep:and,{$.AdvancedParam.snapMode,eq,best}*/
            "enabled": true,
            /*wo, opt, bool*/
            "threshold": 88,
            /*wo, opt, int*/
            "faceScore": 20,
            /*wo, opt, int*/
            "updateTime": 60
            /*wo, opt, int*/
        },
        "quickContinueSnapEnable": false
        /*wo, opt, bool, dep:and,{$.AdvancedParam.snapMode,eq,fast}*/
    }
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int*/
    "statusString": "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode": "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode": 1,
    /*ro, opt, int*/
    "errorMsg": "ok"
    /*ro, opt, string*/
}
```

19.8.1.23 Get the capability of configuring parameters for capturing face pictures and overlaying information on the captured face pictures.

Request URL

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GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/overlapPic/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "OverlapPicCap": {
    /*ro, opt, object, capability of configuring parameters for overlaying information on the picture*/
        "AddIntelInfo": {
        /*ro, opt, object, overlay the VCA information*/
            "streamWithIntelInfo": {
            /*ro, opt, object, overlay the VCA information on the stream*/
                "@opt": [true, false]
                /*ro, req, array, value range, subType:bool*/
            },
            "alarmWithTargetInfo": {
            /*ro, opt, object, overlay the target frame on the alarm picture*/
                "@opt": [true, false]
                /*ro, req, array, value range, subType:bool*/
            }
        },
        "TargetPicParam": {
        /*ro, opt, object, target picture parameters*/
            "targetPicMode": {
            /*ro, opt, object, target picture mode*/
                "@opt": ["custom", "head", "headShoulders", "human", "features", "IDCard", "faceCloseup"]
                /*ro, req, array, value range, subType:string, desc:"custom", "head" (head shot), "headShoulders" (head and shoulders picture), "human"
(full-body shot)*/
            },
            "CustomImgSizeCap": {
            /*ro, opt, object, picture size in custom mode*/
                "targetPicWidth": {
                /*ro, req, object, width*/
                    "@min":  0.0,
                    /*ro, req, float, the minimum value*/
                    "@max":  100.0,
                    /*ro, req, float, the maximum value*/
                    "@def":  3.0
                    /*ro, req, float, the default value*/
                },
                "headHeight": {
                /*ro, req, object, head height*/
                    "@min":  0.0,
                    /*ro, req, float, the minimum value*/
                    "@max":  100.0,
                    /*ro, req, float, the maximum value*/
                    "@def":  2.0
                    /*ro, req, float, the default value*/
                },
                "bodyHeight": {
                /*ro, req, object, body height*/
                    "@min":  1.0,
                    /*ro, req, float, the minimum value*/
                    "@max":  100.0,
                    /*ro, req, float, the maximum value*/
                    "@def":  1.0
                    /*ro, req, float, the default value*/
                }
            },
            "HeadImgSizeCap": {
            /*ro, opt, object, picture size in head shot mode*/
                "targetPicWidth": {
                /*ro, req, object, width*/
                    "@min":  1.0,
                    /*ro, req, float, the minimum value*/
                    "@max":  100.0,
                    /*ro, req, float, the maximum value*/
                    "@def":  1.5
                    /*ro, req, float, the default value*/
                },
                "headHeight": {
                /*ro, req, object, head height*/
                    "@min":  1.0,
                    /*ro, req, float, the minimum value*/
                    "@max":  100.0,
                    /*ro, req, float, the maximum value*/
                    "@def":  1.5
                    /*ro, req, float, the default value*/
                },
                "bodyHeight": {
```

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```
            /*ro, req, object, body height*/
                "@min":  1.0,
                /*ro, req, float, the minimum value*/
                "@max":  100.0,
                /*ro, req, float, the maximum value*/
                "@def":  0.5
                /*ro, req, float, the default value*/
            }
        },
        "HeadShouldersImgSizeCap": {
        /*ro, opt, object, picture size in head and shoulders picture mode*/
            "targetPicWidth": {
            /*ro, req, object, width*/
                "@min":  1.0,
                /*ro, req, float, the minimum value*/
                "@max":  100.0,
                /*ro, req, float, the maximum value*/
                "@def":  3.0
                /*ro, req, float, the default value*/
            },
            "headHeight": {
            /*ro, req, object, head height*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  100,
                /*ro, req, int, the maximum value*/
                "@def":  2
                /*ro, req, int, the default value*/
            },
            "bodyHeight": {
            /*ro, req, object, body height*/
                "@min":  1.0,
                /*ro, req, float, the minimum value*/
                "@max":  100.0,
                /*ro, req, float, the maximum value*/
                "@def":  3.5
                /*ro, req, float, the default value*/
            }
        },
        "HumanImgSizeCap": {
        /*ro, opt, object, picture size in full-body shot mode*/
            "targetPicWidth": {
            /*ro, req, object, width*/
                "@min":  1.0,
                /*ro, req, float, the minimum value*/
                "@max":  100.0,
                /*ro, req, float, the maximum value*/
                "@def":  3.0
                /*ro, req, float, the default value*/
            },
            "headHeight": {
            /*ro, req, object, head height*/
                "@min":  1.0,
                /*ro, req, float, the minimum value*/
                "@max":  100.0,
                /*ro, req, float, the maximum value*/
                "@def":  2.0
                /*ro, req, float, the default value*/
            },
            "bodyHeight": {
            /*ro, req, object, body height*/
                "@min":  1.100,
                /*ro, req, float, the minimum value*/
                "@max":  100.0,
                /*ro, req, float, the maximum value*/
                "@def":  7.0
                /*ro, req, float, the default value*/
            }
        },
        "TargetPicHeight": {
        /*ro, opt, object, target picture with fixed pixels*/
            "enable": {
            /*ro, req, object, whether to enable*/
                "@opt": [true, false]
                /*ro, req, array, value range, subType:bool*/
            },
            "height": {
            /*ro, req, object, height*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  100,
                /*ro, req, int, the maximum value*/
                "@def":  1
                /*ro, req, int, the default value*/
            }
        },
        "faceEnhancement": {
        /*ro, opt, object, whether to enable face enhancement*/
            "@opt": [true, false]
            /*ro, req, array, value range, subType:bool*/
        },
        "faceEnhancementLevel": {
        /*ro, opt, object*/
            "@min":  0,
```

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```
                /*ro, req, int*/
                "@max":  100,
                /*ro, req, int*/
                "@def":  50
                /*ro, req, int*/
            },
        },
        "AlarmPicParam": {
        /*ro, opt, object, alarm picture parameters*/
            "backgroundPicUpload": {
            /*ro, opt, object, whether to upload the background picture*/
                "@opt": [true, false]
                /*ro, req, array, value range, subType:bool*/
            },
            "picQuality": {
            /*ro, opt, object, picture quality*/
                "@opt": ["best", "better", "general"]
                /*ro, req, array, value range, subType:string, desc:"best", "better", "general"*/
            },
            "PicSize": {
            /*ro, opt, object, picture size*/
                "width": {
                /*ro, req, object, width*/
                    "@opt": [1280, 1280, 1920, 2560],
                    /*ro, req, array, value range, subType:int*/
                },
                "height": {
                /*ro, req, object, height*/
                    "@opt": [720, 960, 1080, 1440],
                    /*ro, req, array, value range, subType:int*/
                }
            },
            "facePictureUpload": {
            /*ro, opt, object, whether to upload the close-up face picture (head and shoulders picture), desc:If this field does not exist, the close-up
face picture will be uploaded by default.*/
                "@opt": [true, false]
                /*ro, req, array, value range, subType:bool*/
            }
        },
        "AlarmOsdParam": {
        /*ro, opt, object, parameters of overlaying OSD on the alarm picture*/
            "@size":  1,
            /*ro, opt, int, the maximum number of overlaid items*/
            "osdIndex": {
            /*ro, opt, object, overlaid items*/
                "@opt": ["cameraNo", "positionInfo", "snapTimeOsd", "facingPupil", "GPS"]
                /*ro, req, array, value range, subType:string, desc:"cameraNo" (channel No.), "positionInfo" (position information), "snapTimeOsd" (capture
time), "facingPupil" (pupillary distance)*/
            },
            "enabled": {
            /*ro, opt, object, whether to enable overlaying the item*/
                "@opt": [true, false]
                /*ro, req, array, value range, subType:bool*/
            }
        },
        "backgroundCompress": {
        /*ro, opt, object*/
            "enabled": {
            /*ro, req, object*/
                "@opt": [true, false],
                /*ro, req, array, subType:bool*/
                "@def":  true
                /*ro, opt, bool*/
            },
            "backgroundCompressLevel": {
            /*ro, opt, object*/
                "@opt":  "high,middle,low",
                /*ro, req, string*/
                "@def":  "middle"
                /*ro, opt, string*/
            }
        }
    }
}
```

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19.8.1.24 Get the parameters of capturing face pictures and overlaying information on the captured face pictures.

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/overlapPic?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "OverlapPic": {
    /*ro, opt, object, parameters of overlaying information on the picture*/
        "AddIntelInfo": {
        /*ro, opt, object, overlay the VCA information*/
            "streamWithIntelInfo":  true,
            /*ro, opt, bool, overlay the VCA information on the stream*/
            "alarmWithTargetInfo":  false
            /*ro, opt, bool, overlay the target frame on the alarm picture*/
        },
        "TargetPicParam": {
        /*ro, opt, object, target picture parameters*/
            "targetPicMode":  "custom",
            /*ro, opt, enum, target picture mode, subType:string, desc:"custom", "head" (head shot), "headShoulders" (head and shoulders picture), "human"
(full-body shot)*/
            "targetPicWidth":  3.0,
            /*ro, opt, float, width*/
            "headHeight":  2.0,
            /*ro, opt, float, head height*/
            "bodyHeight":  7.0,
            /*ro, opt, float, body height*/
            "TargetPicHeight": {
            /*ro, opt, object, target picture with fixed pixels*/
                "enable":  true,
                /*ro, req, bool, whether to enable*/
                "height":  100
                /*ro, req, int, height*/
            },
            "FaceBeautification": {
            /*ro, opt, object, face beautification*/
                "enable":  true,
                /*ro, req, bool, whether to enable*/
                "level":  100
                /*ro, req, int, beautification level*/
            },
            "faceSharpeningLevel":  50,
            /*ro, opt, int, range:[0,100]*/
            "faceEnhancement":  true,
            /*ro, opt, bool, whether to enable face enhancement*/
            "faceEnhancementLevel":  50,
            /*ro, opt, int, range:[0,100]*/
            "faceCloseupRatio": {
            /*ro, opt, object*/
                "upRatio":  20,
                /*ro, opt, int*/
                "downRatio":  20,
                /*ro, opt, int*/
                "leftRatio":  20,
                /*ro, opt, int*/
                "rightRatio":  20
                /*ro, opt, int*/
            },
            "humanEyeLighteningEnabled":  false,
            /*ro, opt, bool*/
            "colorReproduction":  true
            /*ro, opt, bool*/
        },
        "AlarmPicParam": {
        /*ro, opt, object, alarm picture parameters*/
            "backgroundPicUpload":  true,
            /*ro, opt, bool, whether to upload the background picture*/
            "picQuality":  "general",
            /*ro, opt, enum, picture quality, subType:string, desc:"best", "better", "general"*/
            "PicSize": {
            /*ro, req, object, picture size*/
                "width":  1920,
                /*ro, req, int, width*/
                "height":  1080
                /*ro, req, int, height*/
            },
            "facePictureUpload":  true
            /*ro, opt, bool, whether to upload the close-up face picture (head and shoulders picture), desc:If this field does not exist, the close-up face
picture will be uploaded by default.*/
        },
        "AlarmOsdParam": [
        /*ro, opt, array, parameters of overlaying OSD on the alarm picture, subType:object*/
            {
                "osdIndex":  "cameraNo",
                /*ro, opt, enum, overlaid items, subType:string, desc:"cameraNo" (channel No.), "positionInfo" (position information), "snapTimeOsd"
(capture time), "facingPupil" (pupillary distance)*/
                "osdValue":  "test",
                /*ro, opt, string, overlaid content, desc:If the value of this field is empty, it indicates that the content needs to be overlaid by DSP.*/
                "enabled":  true
                /*ro, opt, bool, whether to enable overlaying the item*/
            }
        ]
```

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```
        ],
        "faceNoiseReduction":  true,
        /*ro, opt, bool*/
        "faceMosaic":  false,
        /*ro, opt, bool*/
        "backgroundCompress": {
        /*ro, opt, object*/
            "enabled":  true,
            /*ro, req, bool*/
            "backgroundCompressLevel":  "middle"
            /*ro, opt, enum, subType:string*/
        }
    }
}
```

19.8.1.25 Set the overlay and capture parameters for face capture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/overlapPic?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
{
    "OverlapPic": {
    /*opt, object, picture overlay*/
        "AddIntelInfo": {
        /*opt, object, overlay intelligent information*/
            "streamWithIntelInfo":  true,
            /*opt, bool, overlay intelligent information with the stream*/
            "alarmWithTargetInfo":  false
            /*opt, bool, overlay the alarm picture with target information*/
        },
        "TargetPicParam": {
        /*opt, object, target picture parameters*/
            "targetPicMode":  "custom",
            /*ro, opt, enum, target picture mode, subType:string, desc:"custom", "head" (head shot), "headShoulders" (half-body shot), "human" (full-body
shot)*/
            "targetPicWidth":  3.0,
            /*opt, float, target picture width*/
            "headHeight":  2.0,
            /*opt, float, head height*/
            "bodyHeight":  7.0,
            /*opt, float, body height*/
            "TargetPicHeight": {
            /*opt, object, target picture fixed pixels*/
                "enable":  true,
                /*req, bool, whether to enable*/
                "height":  100
                /*req, int, height*/
            },
            "FaceBeautification": {
            /*opt, object, face picture beauty mode*/
                "enable":  true,
                /*req, bool, whether to enable*/
                "level":  100
                /*req, int, beauty level*/
            },
            "faceSharpeningLevel":  50,
            /*opt, int, range:[0,100]*/
            "faceEnhancement":  true,
            /*opt, bool, whether to enable face enhancement*/
            "faceEnhancementLevel":  50,
            /*opt, int, range:[0,100]*/
            "faceCloseupRatio": {
            /*ro, opt, object*/
                "upRatio":  20,
                /*ro, opt, int*/
                "downRatio":  20,
                /*ro, opt, int*/
                "leftRatio":  20,
                /*ro, opt, int*/
                "rightRatio":  20
                /*ro, opt, int*/
            },
            "humanEyeLighteningEnabled":  false,
            /*opt, bool*/
            "colorReproduction":  true
            /*opt, bool*/
        },
        "AlarmPicParam": {
        /*opt, object, alarm picture parameters*/
```

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```
            "backgroundPicUpload": true,
            /*opt, bool, whether to upload the background picture*/
            "picQuality": "general",
            /*opt, enum, picture quality, subType:string, desc:"best", "better", "general"*/
            "PicSize": {
            /*req, object, picture size*/
                "width": 1920,
                /*req, int, width*/
                "height": 1080
                /*req, int, height*/
            },
            "facePictureUpload": true
            /*opt, bool*/
        },
        "AlarmOsdParam": [
        /*opt, array, text overlay parameters on captured pictures, subType:object*/
            {
                "osdIndex": "cameraNo",
                /*opt, enum, overlaid item, subType:string, desc:"cameraNo" (channel number), positionInfo (position information), "snapTimeOsd" (capture
time)*/
                "osdValue": "test",
                /*opt, string, overlay text, desc:when this field does not exists, it indicates DSP overlay*/
                "enabled": true
                /*opt, bool, whether to enable text overlay*/
            }
        ],
        "faceNoiseReduction": true,
        /*opt, bool*/
        "faceMosaic": false,
        /*opt, bool*/
        "backgroundCompress": {
        /*opt, object*/
            "enabled": true,
            /*req, bool*/
            "backgroundCompressLevel": "middle"
            /*opt, enum, subType:string*/
        }
    }
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode": "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx": "0x00000000"
    /*ro, opt, string*/
}
```

19.8.1.26 Get the capability of captured face picture display

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/picParam/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <startTime>
    <!--ro, req, datetime-->1970-01-01T00:00:00+08:00
  </startTime>
</FaceSnapDescription>
```

19.8.1.27 Get the face picture privacy mask configuration capability of a specified channel

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "enabled": {
    /*ro, req, object, whether to enable*/
        "@opt": [true, false],
        /*ro, req, array, range, subType:bool*/
        "@def":  false
        /*ro, opt, bool, default value*/
    },
    "privacyMaskLevel": {
    /*ro, opt, object*/
        "@opt": ["high", "mid", "low", "auto"]
        /*ro, opt, array, subType:string*/
    },
    "modeType": {
    /*ro, opt, object*/
        "@opt": [1, 2, 3]
        /*ro, opt, array, subType:int*/
    },
    "nonMosaicFaceProperty": {
    /*ro, opt, object*/
        "@size":  7,
        /*ro, opt, int*/
        "@opt": ["glass", "mask", "hat", "glassAndMask", "glassAndHat", "maskAndHat", "glassAndMaskAdnHat"]
        /*ro, opt, array, subType:string*/
    },
    "nonMosaicFaceLib": {
    /*ro, req, object*/
        "@opt": [true, false],
        /*ro, req, array, subType:bool*/
        "@def":  false
        /*ro, opt, bool*/
    }
}
```

19.8.1.28 Set the privacy mask parameters of a specified channel

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
{
    "enabled":  true
    /*req, bool, whether to enable*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

19.8.1.29 Get the capability of privacy mask parameters of a specific channel

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "enabled":  true,
    /*ro, req, bool, whether to enable, desc:whether to enable*/
    "privacyMaskLevel":  "high",
    /*ro, opt, enum, subType:string*/
    "modeType":  1,
    /*ro, opt, enum, subType:int, dep:or,{$.enabled,eq,true}*/
    "nonMosaicFaceProperty": ["glass", "mask", "hat", "glassAndMask", "glassAndHat", "maskAndHat", "glassAndMaskAdnHat"],
    /*ro, opt, array, subType:string*/
    "nonMosaicFaceLib":  true
    /*ro, opt, bool*/
}
```

19.8.1.30 Restore the face capture algorithm to defaults

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/restoreAlgParam?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

19.8.1.31 Get the capability of configuring shielded area of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/shieldRegion/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "ShieldRegionCap": {
    /*ro, opt, object*/
        "@size":  8,
        /*ro, req, int*/
        "enable": {
        /*ro, req, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "PointF": {
        /*ro, opt, object*/
            "@minSize":  0,
            /*ro, req, int*/
            "@maxSize":  4,
            /*ro, req, int*/
            "x": {
            /*ro, req, object*/
                "@min":  0.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@def":  0.001
                /*ro, req, float, range:[0.000,1.000]*/
            },
            "y": {
            /*ro, req, object*/
                "@min":  0.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@def":  0.001
                /*ro, req, float, range:[0.000,1.000]*/
            }
        }
    }
}
```

19.8.1.32 Get the shielded area parameters of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/shieldRegion?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

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Request Message

None

Response Message

```
{
    "ShieldRegion": [
    /*ro, opt, array, subType:object*/
        {
            "ruleID":  1,
            /*ro, req, int*/
            "enable":  true,
            /*ro, req, bool*/
            "PointF": [
            /*ro, opt, array, subType:object, range:[3,10]*/
                {
                    "x":  0.120,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "y":  0.120
                    /*ro, req, float, range:[0.000,1.000]*/
                }
            ]
        }
    ]
}
```

19.8.1.33 Set the shielded area parameters of face capture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/shieldRegion?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "ShieldRegion": [
    /*opt, array, subType:object*/
        {
            "ruleID":  1,
            /*req, int*/
            "enable":  true,
            /*req, bool*/
            "PointF": [
            /*opt, array, subType:object, range:[3,10]*/
                {
                    "x":  0.120,
                    /*req, float, range:[0.000,1.000]*/
                    "y":  0.120
                    /*req, float, range:[0.000,1.000]*/
                }
            ]
        }
]
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

19.8.1.34 Get the VCA resource parameters of face capture and recognition channel

Request URL

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GET /ISAPI/Intelligent/channels/<channelID>/intelliResource

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliResource xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <BehaviorInfo>
    <!--ro, opt, object-->
    <IntelliImage>
      <!--ro, opt, object-->
      <isUpload>
        <!--ro, req, enum, subType:string-->true
      </isUpload>
      <imgQuality>
        <!--ro, opt, enum, subType:string-->best
      </imgQuality>
      <imgResolutionWidth>
        <!--ro, req, int-->1
      </imgResolutionWidth>
      <imgResolutionHeight>
        <!--ro, req, int-->1
      </imgResolutionHeight>
      <isImgTargetOverlap>
        <!--ro, req, bool-->true
      </isImgTargetOverlap>
      <isImgRuleOverlap>
        <!--ro, req, bool-->true
      </isImgRuleOverlap>
      <isImgSizeOverlap>
        <!--ro, opt, bool-->bool
      </isImgSizeOverlap>
    </IntelliImage>
    <VideoOverlapInfo>
      <!--ro, opt, object-->
      <isOverlapIntelli>
        <!--ro, req, bool-->true
      </isOverlapIntelli>
      <isOverlapTarget>
        <!--ro, req, bool-->true
      </isOverlapTarget>
      <isOverlapRule>
        <!--ro, req, bool-->true
      </isOverlapRule>
    </VideoOverlapInfo>
    <IntelliAnalysisEnable>
      <!--ro, req, bool-->true
    </IntelliAnalysisEnable>
    <sceneType>
      <!--ro, opt, enum, subType:string-->perimeterProtection
    </sceneType>
    <DisplayTrajectory>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <durationLimit>
        <!--ro, opt, enum, subType:string-->5s
      </durationLimit>
    </DisplayTrajectory>
    <targetMarkingColorEnabled>
      <!--ro, req, bool-->true
    </targetMarkingColorEnabled>
    <picRecordEnable>
      <!--ro, opt, bool-->true
    </picRecordEnable>
    <autoTrack>
      <!--ro, opt, bool-->false
    </autoTrack>
    <NotificationPTZRecheck>
      <!--ro, opt, bool-->false
    </NotificationPTZRecheck>
  </BehaviorInfo>
  <FaceCaptureInfo>
    <!--ro, opt, object-->
    <imgQuality>
      <!--ro, opt, enum, subType:string-->best
    </imgQuality>
```

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```
    <isImgTargetOverlap>
      <!--ro, req, bool-->true
    </isImgTargetOverlap>
    <VideoOverlapInfo>
      <!--ro, opt, object-->
      <isOverlapIntelli>
        <!--ro, req, bool-->true
      </isOverlapIntelli>
      <isOverlapTarget>
        <!--ro, opt, bool-->true
      </isOverlapTarget>
      <isOverlapRule>
        <!--ro, opt, bool-->true
      </isOverlapRule>
    </VideoOverlapInfo>
    <backgroundUpload>
      <!--ro, opt, bool-->true
    </backgroundUpload>
    <imgResolutionWidth>
      <!--ro, req, int-->1
    </imgResolutionWidth>
    <imgResolutionHeight>
      <!--ro, req, int-->1
    </imgResolutionHeight>
    <TargetImgInfo>
      <!--ro, opt, object-->
      <targetImgType>
        <!--ro, opt, enum, subType:string-->custom
      </targetImgType>
      <TargetImgSize>
        <!--ro, opt, object-->
        <widthRatio>
          <!--ro, req, float-->0.000
        </widthRatio>
        <upperPartHeightRatio>
          <!--ro, req, float-->0.000
        </upperPartHeightRatio>
        <underPartHeightRatio>
          <!--ro, req, float-->0.000
        </underPartHeightRatio>
      </TargetImgSize>
      <fixedPixelEnable>
        <!--ro, opt, bool-->true
      </fixedPixelEnable>
      <heightPixel>
        <!--ro, opt, int-->1
      </heightPixel>
      <FaceBeautification>
        <!--ro, opt, object-->
        <enabled>
          <!--ro, opt, bool-->true
        </enabled>
        <level>
          <!--ro, opt, int, range:[0,100]-->1
        </level>
      </FaceBeautification>
      <faceEnhancement>
        <!--ro, opt, bool-->true
      </faceEnhancement>
    </TargetImgInfo>
    <backgroundIntervalUpload>
      <!--ro, opt, int, range:[0,65534], unit:s-->1
    </backgroundIntervalUpload>
    <picRecordEnable>
      <!--ro, opt, bool-->true
    </picRecordEnable>
  </FaceCaptureInfo>
  <AlgVersionInfo>
    <!--ro, opt, object-->
    <AlgItem>
      <!--ro, opt, object-->
      <id>
        <!--ro, opt, string-->test
      </id>
      <algName>
        <!--ro, opt, string-->test
      </algName>
    </AlgItem>
  </AlgVersionInfo>
  <JudiciaryInfo>
    <!--ro, opt, object-->
    <IntelliImage>
      <!--ro, req, object-->
      <isUpload>
        <!--ro, req, enum, subType:string-->true
      </isUpload>
      <imgQuality>
        <!--ro, opt, enum, subType:string-->best
      </imgQuality>
      <imgResolutionWidth>
        <!--ro, req, int-->1
      </imgResolutionWidth>
      <imgResolutionHeight>
        <!--ro, req, int-->1
```

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```
      </imgResolutionHeight>
    </IntelliImage>
    <VideoOverlapInfo>
      <!--ro, opt, object-->
      <isOverlapIntelli>
        <!--ro, req, bool-->true
      </isOverlapIntelli>
    </VideoOverlapInfo>
  </JudiciaryInfo>
  <QRCodeUrl>
    <!--ro, opt, object-->
    <vcaVideoQRCodeUrl>
      <!--ro, opt, string, range:[0,256]-->http://127.0.0.1
    </vcaVideoQRCodeUrl>
  </QRCodeUrl>
</IntelliResource>
```

19.8.1.35 Set the intelligent resource parameters of face capture channel

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/intelliResource

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliResource xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <BehaviorInfo>
    <!--opt, object-->
    <IntelliImage>
      <!--opt, object-->
      <isUpload>
        <!--req, enum, subType:string-->true
      </isUpload>
      <imgQuality>
        <!--opt, enum, subType:string-->best
      </imgQuality>
      <imgResolutionWidth>
        <!--opt, int-->1
      </imgResolutionWidth>
      <imgResolutionHeight>
        <!--opt, int-->1
      </imgResolutionHeight>
      <isImgTargetOverlap>
        <!--req, bool-->true
      </isImgTargetOverlap>
      <isImgRuleOverlap>
        <!--req, bool-->true
      </isImgRuleOverlap>
      <isImgSizeOverlap>
        <!--opt, bool-->bool
      </isImgSizeOverlap>
    </IntelliImage>
    <VideoOverlapInfo>
      <!--opt, object-->
      <isOverlapIntelli>
        <!--req, bool-->true
      </isOverlapIntelli>
      <isOverlapTarget>
        <!--req, bool-->true
      </isOverlapTarget>
      <isOverlapRule>
        <!--req, bool-->true
      </isOverlapRule>
    </VideoOverlapInfo>
    <IntelliAnalysisEnable>
      <!--req, bool-->true
    </IntelliAnalysisEnable>
    <DisplayTrajectory>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <durationLimit>
        <!--opt, enum, subType:string-->5s
      </durationLimit>
    </DisplayTrajectory>
    <targetMarkingColorEnabled>
      <!--req, bool-->true
```

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```
    </targetMarkingColorEnabled>
    <picRecordEnable>
      <!--opt, bool-->true
    </picRecordEnable>
  </BehaviorInfo>
  <FaceCaptureInfo>
    <!--opt, object-->
    <imgQuality>
      <!--opt, enum, subType:string-->best
    </imgQuality>
    <isImgTargetOverlap>
      <!--req, bool-->true
    </isImgTargetOverlap>
    <VideoOverlapInfo>
      <!--opt, object-->
      <isOverlapIntelli>
        <!--req, bool-->true
      </isOverlapIntelli>
      <isOverlapTarget>
        <!--opt, bool-->true
      </isOverlapTarget>
      <isOverlapRule>
        <!--opt, bool-->true
      </isOverlapRule>
    </VideoOverlapInfo>
    <backgroundUpload>
      <!--opt, bool-->true
    </backgroundUpload>
    <imgResolutionWidth>
      <!--req, int-->1
    </imgResolutionWidth>
    <imgResolutionHeight>
      <!--req, int-->1
    </imgResolutionHeight>
    <TargetImgInfo>
      <!--opt, object-->
      <targetImgType>
        <!--opt, enum, subType:string-->custom
      </targetImgType>
      <TargetImgSize>
        <!--opt, object-->
        <widthRatio>
          <!--req, float-->0.000
        </widthRatio>
        <upperPartHeightRatio>
          <!--req, float-->0.000
        </upperPartHeightRatio>
        <underPartHeightRatio>
          <!--req, float-->0.000
        </underPartHeightRatio>
      </TargetImgSize>
      <fixedPixelEnable>
        <!--opt, bool-->true
      </fixedPixelEnable>
      <heightPixel>
        <!--opt, int-->1
      </heightPixel>
      <FaceBeautification>
        <!--opt, object-->
        <enabled>
          <!--opt, bool-->true
        </enabled>
        <level>
          <!--opt, int, range:[0,100]-->1
        </level>
      </FaceBeautification>
      <faceEnhancement>
        <!--opt, bool-->true
      </faceEnhancement>
    </TargetImgInfo>
    <backgroundIntervalUpload>
      <!--opt, int, range:[0,65534], unit:s-->1
    </backgroundIntervalUpload>
    <picRecordEnable>
      <!--opt, bool-->true
    </picRecordEnable>
  </FaceCaptureInfo>
  <AlgVersionInfo>
    <!--opt, object-->
    <AlgItem>
      <!--opt, object-->
      <id>
        <!--opt, string, index-->test
      </id>
      <algName>
        <!--opt, string-->test
      </algName>
    </AlgItem>
  </AlgVersionInfo>
  <JudiciaryInfo>
    <!--opt, object-->
    <IntelliImage>
      <!--req, object-->
```

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```
        <isUpload>
          <!--req, enum, subType:string-->true
        </isUpload>
        <imgQuality>
          <!--opt, enum, subType:string-->best
        </imgQuality>
        <imgResolutionWidth>
          <!--req, int-->1
        </imgResolutionWidth>
        <imgResolutionHeight>
          <!--req, int-->1
        </imgResolutionHeight>
      </IntelliImage>
      <VideoOverlapInfo>
        <!--opt, object-->
        <isOverlapIntelli>
          <!--req, bool-->true
        </isOverlapIntelli>
      </VideoOverlapInfo>
    </JudiciaryInfo>
    <QRCodeUrl>
      <!--opt, object-->
      <vcaVideoQRCodeUrl>
        <!--opt, string, range:[0,256]-->http://127.0.0.1
      </vcaVideoQRCodeUrl>
    </QRCodeUrl>
  </IntelliResource>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.1.36 Get the configuration capability of intelligent

resources by channel **Request URL**

GET /ISAPI/Intelligent/channels/<channelID>/intelliResource/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IntelliResourceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
</IntelliResourceCap>
```

19.8.1.37 Set the VCA type parameters

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/intelliType

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliType xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <type>
    <!--req, enum, subType:string-->behavior
  </type>
</IntelliType>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.1.38 Get the VCA type parameters

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/intelliType

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliType xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <type>
    <!--ro, req, enum, subType:string-->behavior
  </type>
</IntelliType>
```

19.8.1.39 Get the configuration capability of manual face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/manualFaceSnap/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

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Request Message

None

Response Message

```
{
    "ManualFaceSnapCap": {
    /*ro, req, object*/
        "enabled": {
        /*ro, req, object*/
            "@opt":  "true,false",
            /*ro, req, string*/
            "#text":  true
            /*ro, opt, bool*/
        }
    }
}
```

19.8.1.40 Get the parameters of manual face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/manualFaceSnap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "ManualFaceSnap": {
    /*ro, req, object*/
        "enabled":  true
        /*ro, req, bool*/
    }
}
```

19.8.1.41 Get the face modeling configuration capability

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/modeling/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "ModelingCap": {
    /*ro, opt, object*/
        "FaceSnapModeling": {
        /*ro, req, object*/
            "enabled":  "true,false",
            /*ro, req, string*/
            "QuickModeling": {
            /*ro, opt, object*/
                "enabled":  "true,false",
                /*ro, opt, string*/
                "threshold": {
                /*ro, opt, object*/
                    "@min":  20,
                    /*ro, req, int*/
                    "@max":  80,
                    /*ro, req, int*/
                    "@def":  60
                    /*ro, req, int*/
                },
                "snapTime": {
                /*ro, opt, object*/
                    "@min":  0.0,
                    /*ro, req, float*/
                    "@max":  300.0,
                    /*ro, req, float*/
                    "@def":  2.0
                    /*ro, req, float*/
                }
            },
            "LivenessDetection": {
            /*ro, req, object*/
                "enabled":  "true,false"
                /*ro, req, string*/
            },
            "alarmStorageEnable":  "true,false",
            /*ro, opt, string*/
            "mixedTargetDetectionWithFaceSnapModeling":  "true,false"
            /*ro, opt, string*/
        },
        "uploadPictureEnabled":  "true,false"
        /*ro, opt, string*/
    }
}
```

19.8.1.42 Get the face modeling parameters

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/modeling?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "Modeling": {
    /*ro, opt, object*/
        "FaceSnapModeling": {
        /*ro, req, object*/
            "enabled":  true,
            /*ro, opt, bool*/
            "QuickModeling": {
            /*ro, req, object*/
                "enabled":  true,
                /*ro, opt, bool*/
                "threshold":  60,
                /*ro, opt, int*/
                "snapTime":  2.0
                /*ro, opt, float, unit:s*/
            },
            "LivenessDetection": {
            /*ro, req, object*/
                "enabled":  true
                /*ro, opt, bool*/
            },
            "alarmStorageEnable":  true,
            /*ro, opt, bool*/
            "mixedTargetDetectionWithFaceSnapModeling":  true
            /*ro, opt, bool*/
        },
        "uploadPictureEnabled":  false
        /*ro, opt, bool*/
    }
}
```

19.8.1.43 Set the face modeling parameters

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/modeling?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "Modeling": {
    /*opt, object*/
        "FaceSnapModeling": {
        /*req, object*/
            "enabled":  true,
            /*opt, bool*/
            "QuickModeling": {
            /*req, object*/
                "enabled":  true,
                /*opt, bool*/
                "threshold":  60,
                /*opt, int*/
                "snapTime":  2.0
                /*opt, float, unit:s*/
            },
            "LivenessDetection": {
            /*req, object*/
                "enabled":  true
                /*opt, bool*/
            },
            "alarmStorageEnable":  true,
            /*opt, bool*/
            "mixedTargetDetectionWithFaceSnapModeling":  true
            /*opt, bool*/
        },
        "uploadPictureEnabled":  false
        /*opt, bool*/
    }
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

19.8.1.44 Set the overlay parameters of captured face picture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/picInfoOverlap/faceSnap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "PicInfoOverlap": {
    /*req, object*/
        "enabled":  true,
        /*req, bool*/
        "OverlapItem": [
        /*req, array, subType:object*/
            {
                "overlapItemId":  "1",
                /*req, string*/
                "itemType":  "cameraNo"
                /*req, enum, subType:string*/
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

19.8.1.45 Get the overlay parameters of captured face picture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/picInfoOverlap/faceSnap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "PicInfoOverlap": {
    /*ro, req, object*/
        "enabled":  true,
        /*ro, req, bool*/
        "OverlapItem": [
        /*ro, req, array, subType:object*/
            {
                "overlapItemId":  "1",
                /*ro, req, string*/
                "itemType":  "cameraNo"
                /*ro, req, enum, subType:string*/
            }
        ]
    }
}
```

19.8.1.46 Set the shielded area parameters of face capture event

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/Shield/faceCapture

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Shield xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, enum, subType:string-->true
  </enabled>
  <RegionList>
    <!--opt, array, subType:object-->
    <Region>
      <!--opt, object-->
      <id>
        <!--req, string-->1
      </id>
      <enabled>
        <!--req, enum, subType:string-->true
      </enabled>
      <RegionCoordinatesList>
        <!--opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </RegionList>
</Shield>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.1.47 Set the picture information overlay configurations of a device

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/picInfoOverlap

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PicInfoOverlap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, opt, object, attr:version{req, string, protocolVersion}-->
  <Overlap>
    <!--wo, opt, object-->
    <enabled>
      <!--wo, opt, bool-->true
    </enabled>
    <OverlapItemList>
      <!--wo, opt, array, list of overlay information, subType:object-->
      <OverlapItem>
        <!--wo, opt, object, overlay information-->
        <id>
          <!--wo, req, int-->1
        </id>
        <item>
          <!--wo, req, enum, subType:string-->positionInfo
        </item>
      </OverlapItem>
    </OverlapItemList>
  </Overlap>
</PicInfoOverlap>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.1.48 Face capture and recognition event

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EventType:faceCapture

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, req, int, times that the same alarm has been uploaded*/
    "eventType":  "faceCapture",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)*/
    "eventDescription":  "faceCapture",
    /*ro, req, string, event description, desc:face capture data upload*/
    "channelName":  "ABC",
    /*ro, req, string, channel name (camera name)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID (PUID)*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished*/
    "faceCapture": [
    /*ro, opt, array, face capture and recognition results, one result can contain multiple pictures and one picture can contains multiple faces,
subType:object*/
        {
            "errorCode":  1,
            /*ro, opt, int, error code*/
            "errorMsg":  "ok",
            /*ro, opt, string, error code description*/
            "taskid":  "temp_dfasdffsadfdsadada",
            /*ro, opt, string, task ID*/
            "faceTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, capture time*/
            "traceUuid":  "sddddddddff",
            /*ro, opt, string, target ID, the same person in multiple capture results corresponds to same target ID*/
            "traceIdx":  1,
            /*ro, opt, int, No. of picture with highest comparison similarity*/
            "targetAttrs": {
            /*ro, opt, object, face attributes*/
                "deviceId":  "ee3c4000-a22d-11b4-8334-5803fb77894c",
                /*ro, opt, string, device UUID*/
                "deviceChannel":  1,
                /*ro, opt, int, device channel No.*/
                "deviceName":  "shebei",
                /*ro, opt, string, device name*/
                "faceTime":  "2004-05-03T17:30:08+08:00",
                /*ro, opt, datetime, capture time*/
                "bkgUrl":  "test",
                /*ro, opt, string, background picture URL*/
                "thermalURL":  "test",
                /*ro, opt, string, thermal picture URL*/
                "contentID":  "backgroundImage",
                /*ro, opt, string, background picture (binary)*/
                "pId":  "F4F665D6A18E41308CE9934DCDDD1111"
                /*ro, opt, string*/
            },
            "faces": [
            /*ro, opt, array, face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on,
subType:object*/
                {
                    "faceId":  1,
                    /*ro, opt, int, face ID*/
                    "faceRect": {
                    /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "age": {
                    /*ro, opt, object, age*/
                        "range":  5,
                        /*ro, opt, int, age deviation*/
                        "value":  23,
                        /*ro, opt, int, age*/
```

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```
                    "ageGroup": "old"
                    /*ro, opt, string, age group*/
                },
                "gender": {
                /*ro, opt, object, gender*/
                    "confidence": 99.9999,
                    /*ro, opt, float, confidence*/
                    "value": "male"
                    /*ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                },
                "glass": {
                /*ro, opt, object, whether the person wears glasses*/
                    "confidence": 99.945,
                    /*ro, opt, float, confidence*/
                    "value": "yes"
                    /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no", "yes"*/
                },
                "mask": {
                /*ro, opt, object, whether the person wears a mask*/
                    "confidence": 99.945,
                    /*ro, opt, float, confidence*/
                    "value": "yes"
                    /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "yes"*/
                },
                "smile": {
                /*ro, opt, object, whether the person is smiling*/
                    "confidence": 99.945,
                    /*ro, opt, float, confidence*/
                    "value": "no"
                    /*ro, opt, enum, whether the person is smiling, subType:string, desc:"no", "yes"*/
                },
                "faceExpression": {
                /*ro, opt, object, facial expression*/
                    "value": "unknown"
                    /*ro, opt, enum, facial expression, subType:string, desc:"unknown", "surprised", "panic", "disgusted", "happy", "sad", "angry",
"poker-faced"*/
                },
                "hat": {
                /*ro, opt, object, whether the person wears a hat*/
                    "value": "no"
                    /*ro, opt, enum, whether the person wears a hat, subType:string, desc:"no", "yes", "unknown"*/
                },
                "URL": "xxxxxxx",
                /*ro, opt, string, face picture thumbnail URL*/
                "contentID": "faceImage",
                /*ro, opt, string, face picture thumbnail (binary)*/
                "stayDuration": 20,
                /*ro, opt, int, dwell time duration in the image, unit:ms*/
                "faceScore": 30,
                /*ro, req, int, face score (face picture quality), range:[0,100]*/
                "captureEndMark": true,
                /*ro, req, bool, end flag of capture, desc:true (ended), false (not ended)*/
                "targetID": "xxxxxxx",
                /*ro, opt, string, face and human body linkage ID*/
                "FacePictureRect": {
                /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                    "height": 1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width": 1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x": 0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y": 0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "blockingState": "noBlocking",
                /*ro, opt, enum, target face shielding status, subType:string, desc:"noBlocking" (unshielded), "instantaneousInsignificantBlocking"
(lightly shielded), "continuousInsignificantBlocking" (keep shielding), "significantBlocking" (seriously shielded), "OSD", "unknown"*/
                "swingAngle": -90,
                /*ro, opt, int, panning angle, range:[-90,90]*/
                "tiltAngle": -90,
                /*ro, opt, int, tilting angle, range:[-90,90]*/
                "pupilDistance": 10,
                /*ro, opt, int, pupil distance of target face, desc:the maximum value is 10 bytes, and the maximum value is (resolution width/1.6),
unit: pixel*/
                "livenessDetectionStatus": "liveFace",
                /*ro, opt, enum, face anti-spoofing detection status, subType:string, desc:"liveFace" (live face), "notLiveFace" (the face is not real),
"detectionFailed" (detection failed), "notEnabled" (disabled).*/
                "pId": "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string*/
                "enterTime": "2004-05-03T17:30:08+08:00",
                /*ro, opt, datetime*/
                "exitTime": "2004-05-03T17:30:08+08:00",
                /*ro, opt, datetime*/
                "perClassify": 0,
                /*ro, opt, int, face data presorting*/
                "faceSnapThermometryEnabled": true,
                /*ro, opt, bool, whether to enable uploading captured face picture with temperature information*/
                "currTemperature": 36.8,
                /*ro, opt, float, face temperature*/
                "isAbnomalTemperature": true,
                /*ro, opt, bool, whether the temperature is abnormal or not: true, false*/
                "thermometryUnit": "celsius",
```


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```
                    /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
                    "alarmTemperature":  36.7,
                    /*ro, opt, float, temperature threshold to trigger alarm*/
                    "interiorPosition":  "driver"
                    /*ro, opt, enum, subType:string*/
                }
            ],
            "uid":  "20190520103922026O0Kb4s3iQPzLignc1sRUbFT0hR039Y0YzuLt0VLfl99yZi",
            /*ro, opt, string, the unique ID of event uploading*/
            "OperatorInfo": {
            /*ro, opt, object, operator information, desc:operator information*/
                "personUUID":  "test",
                /*ro, opt, string, range:[1,32]*/
                "personName":  "test"
                /*ro, opt, string, range:[1,64]*/
            },
            "MEFData": [
            /*ro, opt, array, subType:object*/
                {
                    "time":  "2004-05-03T17:30:08.000+08:00",
                    /*ro, req, datetime, it should be accurate to the millisecond, desc:it should be accurate to the millisecond*/
                    "IMSI":  "460001357924680",
                    /*ro, req, string, range:[1,15]*/
                    "bandIdx":  11
                    /*ro, opt, enum, subType:int*/
                }
            ],
            "isNeedVerification":  true
            /*ro, opt, bool*/
        }
    ],
    "GPS": {
    /*ro, opt, object*/
        "divisionEW":  "E",
        /*ro, req, enum, "E" (Eastern Hemisphere), W" (West Hemisphere), subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)*/
        "longitude":  100,
        /*ro, req, int, longitude, desc:longitude*/
        "divisionNS":  "N",
        /*ro, req, enum, "S" (Southern Hemisphere), "N" (North Hemisphere), subType:string, desc:"S" (Southern Hemisphere), "N" (North Hemisphere)*/
        "latitude":  100
        /*ro, req, int, latitude, desc:latitude*/
    },
    "URLCertificationType":  "digest"
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (no authentication, it is used for the cloud protocol), "digest" (digest
authentication, it is used for local picture URL returned by NVR or DVR)*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceCapture	[Message content]	application/json	--	--	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.8.2 Face Picture Comparison

19.8.2.1 Set the arming schedule of multiple face picture comparison alarms

Request URL

PUT /ISAPI/Event/schedules/faceContrast

Query Parameter

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None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceContrastScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, schedule information list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--req, object, schedule information-->
    <id>
      <!--req, string, index, desc:the format is event type-channel No.-->faceContrast-1
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->faceContrast
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming time list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceContrastScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.2.2 Set the arming schedule parameters of a specified face picture library by channel

Request URL

PUT /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>?FDType=<FDType>

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--
FDID	string	--
FDType	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, arming schedule information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, desc:the format is event type-channel No.-->faceLib-1
  </id>
  <eventType>
    <!--opt, enum, event type, subType:string-->faceLib
  </eventType>
  <videoInputChannelID>
    <!--opt, string, video input channel ID-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, arming time list, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <dayOfWeek>
        <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object, time period-->
        <beginTime>
          <!--req, time, start time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, holiday arming time, subType:object-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <TimeRange>
        <!--req, object, time period-->
        <beginTime>
          <!--req, time, start time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.2.3 Get the linkage parameters of face picture comparison

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Request URL

GET /ISAPI/Event/triggers/faceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceContrast-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->faceContrast
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.8.2.4 Get the linkage parameters of event when face picture comparison failed

Request URL

GET /ISAPI/Event/triggers/faceContrastFailure-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceContrastFailure-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->faceContrastFailure
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <AudioAction>
        <!--ro, opt, object, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,audio}-->
        <audioID>
          <!--ro, opt, int-->1
        </audioID>
        <alarmTimes>
```

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```
          <alarmTimes>
            <!--ro, opt, int, range:[0,255]-->1
          </alarmTimes>
          <childDevID>
            <!--ro, opt, string, range:[9,32]-->3434645611
          </childDevID>
          <audioContent>
            <!--ro, opt, string, range:[0,256]-->test
          </audioContent>
        </AudioAction>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
  </EventTrigger>
```

19.8.2.5 Get the linkage parameters of event when face picture comparison completed

Request URL

GET /ISAPI/Event/triggers/faceContrastSuccess-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceContrastSuccess-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->faceContrastSuccess
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
```

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```
        </dynVideoInputID>
        <ptzAction>
          <!--ro, opt, object-->
          <ptzChannelID>
            <!--ro, req, string-->1
          </ptzChannelID>
          <actionName>
            <!--ro, req, enum, subType:string-->preset
          </actionName>
          <actionNum>
            <!--ro, opt, int-->1
          </actionNum>
        </ptzAction>
        <WhiteLightAction>
          <!--ro, opt, object-->
          <whiteLightDurationTime>
            <!--ro, opt, int, range:[1,60], unit:s-->1
          </whiteLightDurationTime>
        </WhiteLightAction>
        <AudioAction>
          <!--ro, opt, object, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,audio}-->
          <audioID>
            <!--ro, opt, int-->1
          </audioID>
          <alarmTimes>
            <!--ro, opt, int, range:[0,255]-->1
          </alarmTimes>
          <childDevID>
            <!--ro, opt, string, range:[9,32]-->3434645611
          </childDevID>
          <audioContent>
            <!--ro, opt, string, range:[0,256]-->test
          </audioContent>
        </AudioAction>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
  </EventTrigger>
```

19.8.2.6 Set the linkage action parameters of a specified face picture library

Request URL

PUT /ISAPI/Event/triggers/faceLib-<channelID>/<FDID>?FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
FDID	string	--
FDType	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, linkage action parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, desc:the format is event type-channel No.-->faceLib-1
  </id>
  <eventType>
    <!--req, enum, event type, subType:string-->faceLib
  </eventType>
  <eventDescription>
    <!--opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string, I/O input channel ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string, dynamic I/O input channel ID-->1
  </dynInputIOPortID>
  <intervalBetweenEvents>
    <!--opt, int, event capture interval, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string, wireless sensor ID-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, linkage action list, subType:object-->
    <EventTriggerNotification>
      <!--opt, object, linkage action-->
      <id>
        <!--req, string, index-->test
      </id>
      <notificationMethod>
        <!--req, enum, linkage action, subType:string, desc:"FTP" (uploading via FTP), "HTTP" (uploading via HTTP), "IM" (instant messaging), "IO" (I/O
output), "LightAudioAlarm" (light and audio alarm), "audio" (audio), "beep" (buzzer), "center" (uploading to the center), "cloud" (cloud storage), "email",
"focus" (focus), "monitorAlarm" (alarm on monitor), "ptz" (PTZ), "record" (recording), "syslog" (system log), "trace" (tracking), "whiteLight" (white
light)-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, linkage time interval, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string, I/O output port ID-->1
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string, dynamic I/O output port ID-->1
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string, video channel ID-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string, dynamic video input channel ID-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object, PTZ linkage, desc:it is valid when the value of notificationMethod is "PTZ"-->
        <ptzChannelID>
          <!--req, string, PTZ linked channel No.-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
        </actionName>
        <actionNum>
          <!--opt, int, operation No., desc:operation No.-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object, white light linkage, desc:it is valid when the value of notificationMethod is "whiteLight"-->
        <whiteLightDurationTime>
          <!--opt, int, light flashing duration, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.2.7 Set the linkage parameters of allowlist face picture comparison

Request URL

PUT /ISAPI/Event/triggers/whiteListFaceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->whiteListFaceContrast-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->whiteListFaceContrast
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.2.8 Set the face picture comparison parameters

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceContrast

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceContrastList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FaceContrast>
    <!--opt, object-->
    <id>
      <!--req, int-->1
    </id>
    <enable>
      <!--opt, bool-->true
    </enable>
    <faceContrastType>
      <!--req, enum, face picture comparison type, subType:string-->faceContrast
    </faceContrastType>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->1
    </thresholdValue>
    <FDLibList>
      <!--opt, array, subType:object-->
      <FDLib>
        <!--opt, object, face picture library-->
        <id>
          <!--req, int-->1
        </id>
        <FDID>
          <!--req, string, a unique ID of face picture library-->test
        </FDID>
        <thresholdValue>
          <!--opt, int, range:[0,100]-->1
        </thresholdValue>
        <customFaceLibID>
          <!--opt, string, custom face picture library ID-->test
        </customFaceLibID>
      </FDLib>
    </FDLibList>
    <faceSnapDataUpload>
      <!--opt, bool, upload captured face image, desc:true by default-->true
    </faceSnapDataUpload>
    <AttendanceSaveEnable>
      <!--opt, bool, desc:true by default-->true
    </AttendanceSaveEnable>
    <contrastFailureAlarmUpload>
      <!--opt, bool, desc:true by default-->true
    </contrastFailureAlarmUpload>
    <alarmOutType>
      <!--opt, enum, subType:string-->off
    </alarmOutType>
    <plusesTime>
      <!--opt, int, unit:ms-->1
    </plusesTime>
    <contrastSuccessAlarmUpload>
      <!--opt, bool-->true
    </contrastSuccessAlarmUpload>
```

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```
<LivenessDetection>
  <!--opt, object-->
  <enable>
    <!--req, bool-->true
  </enable>
</LivenessDetection>
<voiceEnabled>
  <!--opt, bool-->true
</voiceEnabled>
<faceSnapDataSave>
  <!--opt, bool-->true
</faceSnapDataSave>
<QuickContrast>
  <!--opt, object-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <snapTime>
    <!--req, float, unit:s-->0.000
  </snapTime>
  <threshold>
    <!--opt, int-->1
  </threshold>
  <quickConfigMode>
    <!--opt, enum, subType:string-->custom
  </quickConfigMode>
  <Custom>
    <!--opt, object-->
    <timeOutMode>
      <!--opt, enum, subType:string-->infinite
    </timeOutMode>
    <timeOut>
      <!--opt, int, range:[1,99], unit:s-->1
    </timeOut>
    <duplicateContrastMode>
      <!--opt, enum, subType:string-->success
    </duplicateContrastMode>
    <duplicateContrastTimes>
      <!--opt, int, range:[1,99]-->1
    </duplicateContrastTimes>
  </Custom>
</QuickContrast>
<mixedTargetDetectionWithFaceContrast>
  <!--opt, bool-->true
</mixedTargetDetectionWithFaceContrast>
<brokenNetHttp>
  <!--opt, bool-->true
</brokenNetHttp>
<relationId>
  <!--opt, string, arming ID-->test
</relationId>
<alarmStorageEnable>
  <!--opt, bool-->true
</alarmStorageEnable>
<realTimeContrast>
  <!--opt, bool-->true
</realTimeContrast>
<analysisTime>
  <!--opt, datetime-->1970-01-01T00:00:00+08:00
</analysisTime>
<contrastSuccessAlarmout>
  <!--opt, bool, desc:true by default-->true
</contrastSuccessAlarmout>
<contrastFailureAlarmout>
  <!--opt, bool, desc:true by default-->true
</contrastFailureAlarmout>
<removeDuplicateEnabled>
  <!--opt, bool-->true
</removeDuplicateEnabled>
<removeDuplicateTime>
  <!--opt, float, unit:s-->0.000
</removeDuplicateTime>
<repetitionTime>
  <!--opt, int-->1
</repetitionTime>
<topOverlayNameEnabled>
  <!--opt, bool-->true
</topOverlayNameEnabled>
<personListShowEnabled>
  <!--opt, bool-->true
</personListShowEnabled>
<refreshListTime>
  <!--opt, int, unit:s-->1
</refreshListTime>
<personCountShowEnabled>
  <!--opt, bool-->true
</personCountShowEnabled>
<FaceROI>
  <!--opt, object-->
  <enabled>
    <!--opt, bool-->true
  </enabled>
  <level>
```

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```
        <!--opt, int-->1
      </level>
    </FaceROI>
    <uploadResultCount>
      <!--opt, int-->1
    </uploadResultCount>
    <normalTemperatureAlarmout>
      <!--opt, bool-->true
    </normalTemperatureAlarmout>
    <abnormalTemperatureAlarmout>
      <!--opt, bool-->true
    </abnormalTemperatureAlarmout>
    <matchType>
      <!--opt, enum, subType:int-->0
    </matchType>
    <alarmSourceType>
      <!--opt, enum, subType:string-->timing
    </alarmSourceType>
    <timingInterval>
      <!--opt, enum, subType:int, dep:and,{$.FaceContrastList[*].FaceContrast.alarmSourceType,eq,timing}-->5
    </timingInterval>
  </FaceContrast>
</FaceContrastList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.2.9 Get the face picture comparison parameters.

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceContrast

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<FaceContrastList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FaceContrast>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int, ID-->1
    </id>
    <enable>
      <!--ro, opt, bool-->true
    </enable>
    <faceContrastType>
      <!--ro, req, enum, subType:string-->faceContrast
    </faceContrastType>
    <contrastFailureAlarmUpload>
      <!--ro, opt, bool, true by default, desc:true by default-->true
    </contrastFailureAlarmUpload>
    <QuickContrast>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <snapTime>
        <!--ro, req, float, unit:s-->0.000
      </snapTime>
      <threshold>
        <!--ro, opt, int-->1
      </threshold>
      <quickConfigMode>
        <!--ro, opt, enum, subType:string-->custom
      </quickConfigMode>
      <Custom>
        <!--ro, opt, object-->
        <timeOutMode>
          <!--ro, opt, enum, subType:string-->infinite
        </timeOutMode>
        <duplicateContrastMode>
          <!--ro, opt, enum, subType:string-->success
        </duplicateContrastMode>
      </Custom>
    </QuickContrast>
    <mixedTargetDetectionWithFaceContrast>
      <!--ro, opt, bool-->true
    </mixedTargetDetectionWithFaceContrast>
    <contrastSuccessAlarmout>
      <!--ro, opt, bool, true by default, desc:true by default-->true
    </contrastSuccessAlarmout>
    <contrastFailureAlarmout>
      <!--ro, opt, bool, true by default, desc:true by default-->true
    </contrastFailureAlarmout>
  </FaceContrast>
</FaceContrastList>
```

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19.8.2.10 Get the face picture comparison capability

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceContrast/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<FaceContrastCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enable>
  <faceContrastType opt="faceContrast,blacklistFaceContrast,whitelistFaceContrast">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->faceContrast
  </faceContrastType>
  <contrastFailureAlarmUpload opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </contrastFailureAlarmUpload>
  <QuickContrast>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </enabled>
    <threshold min="20" max="80" def="60">
      <!--ro, opt, int, attr:min{req, int},max{req, int},def{req, int}-->1
    </threshold>
    <snapTime min="0.0" max="300.0" def="2.0">
      <!--ro, opt, float, unit:s, attr:min{req, int},max{req, int},def{req, int}-->0.000
    </snapTime>
    <quickConfigMode opt="custom,faceSignIn,faceRecognition">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->custom
    </quickConfigMode>
    <CustomCap>
      <!--ro, opt, object-->
      <timeOutMode opt="infinite,finite" def="finite">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->infinite
      </timeOutMode>
      <timeOut min="1" max="99" default="1">
        <!--ro, opt, int, range:[1,99], unit:s, attr:min{req, int},max{req, int},default{opt, int}-->1
      </timeOut>
      <duplicateContrastMode opt="success,infinite,finite" def="success">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->success
      </duplicateContrastMode>
      <duplicateContrastTimes min="1" max="99" default="1">
        <!--ro, opt, int, range:[1,99], attr:min{req, int},max{req, int},default{opt, int}-->1
      </duplicateContrastTimes>
    </CustomCap>
    <FaceSignInCap>
      <!--ro, opt, object-->
      <timeOutMode opt="infinite,finite" def="finite">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->infinite
      </timeOutMode>
      <duplicateContrastMode opt="success,infinite,finite" def="success">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->success
      </duplicateContrastMode>
    </FaceSignInCap>
    <FaceRecognitionCap>
      <!--ro, opt, object-->
      <timeOutMode opt="infinite,finite" def="finite">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->infinite
      </timeOutMode>
      <duplicateContrastMode opt="success,infinite,finite" def="success">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->success
      </duplicateContrastMode>
    </FaceRecognitionCap>
  </QuickContrast>
  <mixedTargetDetectionWithFaceContrast>
    <!--ro, opt, bool-->true
  </mixedTargetDetectionWithFaceContrast>
  <brokenNetHttp opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </brokenNetHttp>
  <relationId min="1" max="64">
    <!--ro, opt, string, range:[1,64], attr:min{req, int},max{req, int}-->test
  </relationId>
  <contrastSuccessAlarmout opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </contrastSuccessAlarmout>
  <contrastFailureAlarmout opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </contrastFailureAlarmout>
</FaceContrastCap>
```

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19.8.2.11 Search for face picture comparison results of a single channel

Request URL

POST /ISAPI/Intelligent/FDLib/FCSearch/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string-->test
  </searchID>
  <searchResultPosition>
    <!--req, int-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int-->1
  </maxResults>
  <snapStartTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </snapStartTime>
  <snapEndTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </snapEndTime>
  <faceMatchInfoEnable>
    <!--opt, bool-->true
  </faceMatchInfoEnable>
  <monitorName>
    <!--opt, string-->test
  </monitorName>
  <similarity>
    <!--opt, float, range:[0.0,100.0]-->0.0
  </similarity>
  <eventType>
    <!--opt, enum, subType:string-->unlimit
  </eventType>
  <FaceModeList>
    <!--opt, array, subType:object-->
    <FaceMode>
      <!--opt, object-->
      <ModeInfo>
        <!--opt, object-->
        <similarity>
          <!--opt, float, range:[0.0,100.0]-->0.0
        </similarity>
        <modeData>
          <!--opt, string-->test
        </modeData>
      </ModeInfo>
    </FaceMode>
  </FaceModeList>
  <PictureInfoList>
    <!--opt, array, subType:object-->
    <PictureInfo>
      <!--opt, object-->
      <FDID>
        <!--req, string-->test
      </FDID>
      <PID>
        <!--req, string-->test
      </PID>
    </PictureInfo>
  </PictureInfoList>
  <totalSearchResults>
    <!--opt, int-->1
  </totalSearchResults>
  <FDIDList>
    <!--opt, array, subType:object-->
    <FDID>
      <!--opt, string-->test
    </FDID>
  </FDIDList>
  <sortord>
    <!--opt, enum, subType:string-->similarity
  </sortord>
  <sex>
    <!--opt, enum, subType:string-->male
  </sex>
  <glass>
    <!--opt, enum, subType:string-->yes
  </glass>
  <smile>
    <!--opt, enum, subType:string-->yes
  </smile>
  <high_risk_group>
    <!--opt, enum, subType:string-->yes
  </high_risk_group>
  <importLib>
    <!--opt, enum, subType:string-->yes
```

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```
  </importLib>
  <CustomFaceLibIDList>
    <!--opt, array, subType:object-->
    <customFaceLibID>
      <!--opt, string-->test
    </customFaceLibID>
  </CustomFaceLibIDList>
</FCSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string-->test
  </searchID>
  <responseStatus>
    <!--ro, req, bool-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, subType:string-->MORE
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int-->1
  </totalMatches>
  <progress>
    <!--ro, opt, int, range:[0,100]-->1
  </progress>
  <MatchList>
    <!--ro, opt, array, subType:object-->
    <MatchElement>
      <!--ro, opt, object-->
      <snapPicURL>
        <!--ro, opt, string-->test
      </snapPicURL>
      <snapTime>
        <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
      </snapTime>
      <facePicURL>
        <!--ro, opt, string-->test
      </facePicURL>
      <FaceMatchInfoList>
        <!--ro, opt, array, subType:object-->
        <FaceMatchInfo>
          <!--ro, opt, object-->
          <FDID>
            <!--ro, req, string-->test
          </FDID>
          <FDname>
            <!--ro, opt, string-->test
          </FDname>
          <thresholdValue>
            <!--ro, opt, int, range:[0,100]-->1
          </thresholdValue>
          <bornTime>
            <!--ro, opt, date-->1970-01-01+08:00
          </bornTime>
          <name>
            <!--ro, opt, string-->test
          </name>
          <sex>
            <!--ro, opt, enum, subType:string-->male
          </sex>
          <province>
            <!--ro, req, string-->0
          </province>
          <city>
            <!--ro, opt, string-->11
          </city>
          <certificateType>
            <!--ro, req, enum, subType:string-->ID
          </certificateType>
          <certificateNumber>
            <!--ro, opt, string, range:[8,32]-->test
          </certificateNumber>
          <picURL>
            <!--ro, req, string-->test
          </picURL>
          <PID>
            <!--ro, req, string-->test
          </PID>
          <PersonInfoExtendList>
            <!--ro, opt, array, subType:object-->
            <PersonInfoExtend>
              <!--ro, opt, object-->
```

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```
            <id>
              <!--ro, req, int-->1
            </id>
            <enable>
              <!--ro, req, bool-->true
            </enable>
            <name>
              <!--ro, opt, string-->test
            </name>
            <value>
              <!--ro, opt, string-->test
            </value>
          </PersonInfoExtend>
        </PersonInfoExtendList>
        <phoneNumber>
          <!--ro, opt, string-->test
        </phoneNumber>
        <caseInfo>
          <!--ro, opt, string-->test
        </caseInfo>
        <similarity>
          <!--ro, opt, float, range:[0.0,100.0]-->0.0
        </similarity>
        <modelStatus>
          <!--ro, opt, enum, subType:string-->modeling
        </modelStatus>
        <ModelingStatus>
          <!--ro, opt, object-->
          <FDID>
            <!--ro, opt, string-->test
          </FDID>
          <PID>
            <!--ro, opt, string-->test
          </PID>
          <name>
            <!--ro, req, string-->test
          </name>
          <status>
            <!--ro, req, enum, subType:string-->abnormal
          </status>
          <reason>
            <!--ro, req, enum, subType:string-->analysisFailed
          </reason>
          <customHumanID>
            <!--ro, opt, string-->123456789012345678901234567890123456789012
          </customHumanID>
          <customFaceLibID>
            <!--ro, opt, string-->123456789012345678901234567890123456789012
          </customFaceLibID>
          <modeData>
            <!--ro, opt, string-->test
          </modeData>
        </ModelingStatus>
      </FaceMatchInfo>
    </FaceMatchInfoList>
    <monitorName>
      <!--ro, opt, string-->test
    </monitorName>
    <glass>
      <!--ro, req, enum, subType:string-->any
    </glass>
    <smile>
      <!--ro, req, enum, subType:string-->no
    </smile>
    <high_risk_group>
      <!--ro, opt, enum, subType:string-->yes
    </high_risk_group>
    <similarity>
      <!--ro, opt, float, range:[0.0,100.0]-->0.0
    </similarity>
    <Rect>
      <!--ro, opt, object-->
      <height>
        <!--ro, req, float, range:[0.000,1.000]-->0.000
      </height>
      <width>
        <!--ro, req, float, range:[0.000,1.000]-->0.000
      </width>
      <x>
        <!--ro, req, float, range:[0.000,1.000]-->0.000
      </x>
      <y>
        <!--ro, req, float, range:[0.000,1.000]-->0.000
      </y>
    </Rect>
    <channelID>
      <!--ro, opt, int-->1
    </channelID>
    <name>
      <!--ro, opt, string-->test
    </name>
    <sex>
      <!--ro, opt, enum, subType:string-->male
    </sex>
```

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```
      <ageGroup>
        <!--ro, req, enum, subType:string-->any
      </ageGroup>
      <mask>
        <!--ro, req, enum, subType:string-->no
      </mask>
      <imageQuality>
        <!--ro, opt, int-->1
      </imageQuality>
      <faceExpression>
        <!--ro, req, enum, subType:string-->angry
      </faceExpression>
      <contrastStatus>
        <!--ro, opt, bool-->true
      </contrastStatus>
      <age>
        <!--ro, opt, int, range:[1,255]-->1
      </age>
      <targetID>
        <!--ro, opt, string, range:[1,48]-->test
      </targetID>
      <MEFData>
        <!--ro, opt, array, subType:object-->
        <MEF>
          <!--ro, opt, object-->
          <time>
            <!--ro, req, datetime-->2004-05-03T17:30:08+08:00.000
          </time>
          <IMSI>
            <!--ro, req, string, range:[1,15]-->test
          </IMSI>
        </MEF>
      </MEFData>
    </MatchElement>
  </MatchList>
</FCSearchResult>
```

19.8.2.12 Search for the face picture comparison alarm information

Request URL

POST /ISAPI/Intelligent/FDLib/FCSearch

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of search results this time, desc:the maximum number of search results this time-->1
  </maxResults>
  <snapStartTime>
    <!--opt, datetime, capture start time with ISO8601 time format-->1970-01-01T00:00:00+08:00
  </snapStartTime>
  <snapEndTime>
    <!--opt, datetime, capture end time with ISO8601 time format-->1970-01-01T00:00:00+08:00
  </snapEndTime>
  <faceMatchInfoEnable>
    <!--opt, bool, whether to only upload data with matched information-->true
  </faceMatchInfoEnable>
  <monitorName>
    <!--opt, string, camera name-->test
  </monitorName>
  <similarity>
    <!--opt, float, similarity, range:[0.0,100.0]-->0.0
  </similarity>
  <eventType>
    <!--opt, enum, event type, subType:string, desc:event type-->unlimit
  </eventType>
  <FaceModeList>
    <!--opt, array, face picture information list, subType:object-->
    <FaceMode>
      <!--opt, object, face picture information-->
      <ModeInfo>
        <!--opt, object, model information-->
        <similarity>
```

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```
          <!--opt, float, similarity, range:[0.0,100.0]-->0.0
        </similarity>
        <modeData>
          <!--opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method-
->test
        </modeData>
      </ModeInfo>
    </FaceMode>
  </FaceModeList>
  <PictureInfoList>
    <!--opt, array, list of the picture information, subType:object, desc:list of the picture information-->
    <PictureInfo>
      <!--opt, object, picture information-->
      <FDID>
        <!--req, string-->test
      </FDID>
      <PID>
        <!--req, string-->test
      </PID>
    </PictureInfo>
  </PictureInfoList>
  <ChannelList>
    <!--opt, array, search channel list, subType:object-->
    <Channel>
      <!--opt, object-->
      <channelID>
        <!--req, int, channel No.-->1
      </channelID>
    </Channel>
  </ChannelList>
  <totalSearchResults>
    <!--opt, int, the total number of results in a single search, desc:the total number of results in a single search-->1
  </totalSearchResults>
  <FDIDList>
    <!--opt, array, face picture list, subType:object-->
    <FDID>
      <!--opt, string, face picture library ID-->test
    </FDID>
  </FDIDList>
  <sortord>
    <!--opt, enum, subType:string-->similarity
  </sortord>
  <sex>
    <!--opt, enum, gender, subType:string, desc:gender-->male
  </sex>
  <glass>
    <!--opt, enum, whether the person wears glasses, subType:string, desc:whether the person wears glasses-->yes
  </glass>
  <smile>
    <!--opt, enum, whether the person wears a smile, subType:string, desc:whether the person wears a smile-->yes
  </smile>
  <high_risk_group>
    <!--opt, enum, subType:string-->yes
  </high_risk_group>
  <importLib>
    <!--opt, enum, subType:string-->yes
  </importLib>
  <CustomFaceLibIDList>
    <!--opt, array, custom list of face picture library ID, subType:object-->
    <customFaceLibID>
      <!--opt, string, custom face picture library ID-->test
    </customFaceLibID>
  </CustomFaceLibIDList>
  <thermometryUnit>
    <!--opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
  </thermometryUnit>
  <currTemperature>
    <!--opt, float, face temperature, range:[-20.0,105.0], unit:℃, desc:face temperature-->36.7
  </currTemperature>
  <temperatureCondition>
    <!--opt, enum, face temperature status, subType:string, desc:face temperature status-->normal
  </temperatureCondition>
  <temperatureValue>
    <!--opt, float, temperature value, range:[-20,105], unit:℃, dep:and,{$.FCSearchDescription.temperatureCondition,eq,above}-->37.2
  </temperatureValue>
  <humanIDList>
    <!--opt, array, subType:object-->
    <humanID>
      <!--req, string-->test
    </humanID>
  </humanIDList>
  <collectEnabled>
    <!--opt, enum, subType:string-->yes
  </collectEnabled>
</FCSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

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```
<FCSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search results, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search result, subType:string, desc:search result-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"MORE" (searching for more data), "NO MATCHES" (no matched data),"OK" (searching completed)-->MORE
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <progress>
    <!--ro, opt, int, range:[0,100]-->1
  </progress>
  <MatchList>
    <!--ro, opt, array, matched results list, subType:object-->
    <MatchElement>
      <!--ro, opt, object, matched results-->
      <snapPicURL>
        <!--ro, opt, string, URL of captured background picture-->test
      </snapPicURL>
      <snapTime>
        <!--ro, opt, datetime, capture time-->1970-01-01T00:00:00+08:00
      </snapTime>
      <facePicURL>
        <!--ro, opt, string, face sub-picture URL-->test
      </facePicURL>
      <FaceMatchInfoList>
        <!--ro, opt, array, subType:object-->
        <FaceMatchInfo>
          <!--ro, opt, object-->
          <FDID>
            <!--ro, req, string-->test
          </FDID>
          <FDname>
            <!--ro, opt, string, face picture library name-->test
          </FDname>
          <thresholdValue>
            <!--ro, opt, int, detection threshold, range:[0,100], desc:detection threshold-->1
          </thresholdValue>
          <bornTime>
            <!--ro, opt, date, date of birth-->1970-01-01+08:00
          </bornTime>
          <name>
            <!--ro, opt, string, name-->test
          </name>
          <sex>
            <!--ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
          </sex>
          <province>
            <!--ro, req, string-->0
          </province>
          <city>
            <!--ro, opt, string, city, desc:see the data dictionary for details-->11
          </city>
          <certificateType>
            <!--ro, req, enum, certificate type, subType:string, desc:certificate type-->ID
          </certificateType>
          <certificateNumber>
            <!--ro, opt, string, certificate No., range:[8,32], desc:certificate No.-->test
          </certificateNumber>
          <picURL>
            <!--ro, req, string-->test
          </picURL>
          <PID>
            <!--ro, req, string, picture ID-->test
          </PID>
          <PersonInfoExtendList>
            <!--ro, opt, array, person extended information list, subType:object-->
            <PersonInfoExtend>
              <!--ro, opt, object, person extended information-->
              <id>
                <!--ro, req, int, serial No. of person custom extension information-->1
              </id>
              <enable>
                <!--ro, req, bool, enable person extended information or not-->true
              </enable>
              <name>
                <!--ro, opt, string, person tag extended information name-->test
              </name>
              <value>
                <!--ro, opt, string, person tag extended information-->test
              </value>
            </PersonInfoExtend>
          </PersonInfoExtendList>
          <phoneNumber>
```

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```
            <!--ro, opt, string, phone No.-->test
          </phoneNumber>
          <caseInfo>
            <!--ro, opt, string, case description-->test
          </caseInfo>
          <similarity>
            <!--ro, opt, float, similarity, range:[0.0,100.0]-->0.0
          </similarity>
          <modelStatus>
            <!--ro, opt, enum, subType:string-->modeling
          </modelStatus>
          <ModelingStatus>
            <!--ro, opt, object-->
            <FDID>
              <!--ro, opt, string, face picture library ID-->test
            </FDID>
            <PID>
              <!--ro, opt, string, picture ID-->test
            </PID>
            <name>
              <!--ro, req, string, picture name-->test
            </name>
            <status>
              <!--ro, req, enum, subType:string-->abnormal
            </status>
            <reason>
              <!--ro, req, enum, subType:string-->analysisFailed
            </reason>
            <customHumanID>
              <!--ro, opt, string-->123456789012345678901234567890123456789012
            </customHumanID>
            <customFaceLibID>
              <!--ro, opt, string, custom face picture library ID, desc:custom face picture library ID-->123456789012345678901234567890123456789012
            </customFaceLibID>
            <modeData>
              <!--ro, opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64
method-->test
            </modeData>
          </ModelingStatus>
        </FaceMatchInfo>
      </FaceMatchInfoList>
      <monitorName>
        <!--ro, opt, string, camera information-->test
      </monitorName>
      <glass>
        <!--ro, req, enum, whether the person wears glasses, subType:string, desc:"any" (unlimited condition which is used in search), "no" (not wear
glasses), "sunglasses" (wears sunglasses), "unknown"-->any
      </glass>
      <smile>
        <!--ro, req, enum, whether the person wears a smile, subType:string, desc:whether the person wears a smile-->no
      </smile>
      <high_risk_group>
        <!--ro, opt, enum, subType:string-->yes
      </high_risk_group>
      <similarity>
        <!--ro, opt, float, similarity, range:[0.0,100.0], desc:similarity-->0.0
      </similarity>
      <Rect>
        <!--ro, opt, object-->
        <height>
          <!--ro, req, float, width, range:[0.000,1.000]-->0.000
        </height>
        <width>
          <!--ro, req, float, height, range:[0.000,1.000]-->0.000
        </width>
        <x>
          <!--ro, req, float, X-coordinate, range:[0.000,1.000], desc:the reference origin is the upper left corner of image-->0.000
        </x>
        <y>
          <!--ro, req, float, Y-coordinate, range:[0.000,1.000], desc:the reference origin is the upper left corner of image-->0.000
        </y>
      </Rect>
      <channelID>
        <!--ro, opt, int, channel No., desc:channel No.-->1
      </channelID>
      <name>
        <!--ro, opt, string-->test
      </name>
      <sex>
        <!--ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
      </sex>
      <ageGroup>
        <!--ro, req, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime",
"teenager", "unknown", "young"-->any
      </ageGroup>
      <mask>
        <!--ro, req, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"-->no
      </mask>
      <imageQuality>
        <!--ro, opt, int, picture quality-->1
      </imageQuality>
      <faceExpression>
        <!--ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-faced", "sad",
```

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```
"surprised", "unknown"-->angry
      </faceExpression>
      <contrastStatus>
        <!--ro, opt, bool, comparison result-->true
      </contrastStatus>
      <age>
        <!--ro, opt, int, age, range:[1,255]-->1
      </age>
      <targetID>
        <!--ro, opt, string, target ID, range:[1,48]-->test
      </targetID>
      <thermometryUnit>
        <!--ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
      </thermometryUnit>
      <currTemperature>
        <!--ro, opt, float, face temperature, range:[-20.0,105.0], unit:℃, desc:face temperature-->36.7
      </currTemperature>
      <temperatureCondition>
        <!--ro, opt, enum, face temperature status, subType:string, desc:face temperature status-->normal
      </temperatureCondition>
      <temperatureValue>
        <!--ro, opt, float, temperature value, range:[-20,105], unit:℃, dep:and,{$.FCSearchResult.MatchList[*].MatchElement.temperatureCondition,eq,above}-
->37.2
      </temperatureValue>
      <MEFData>
        <!--ro, opt, array, subType:object-->
        <MEF>
          <!--ro, opt, object, electronic fence data-->
          <time>
            <!--ro, req, datetime-->2004-05-03T17:30:08+08:00.000
          </time>
          <IMSI>
            <!--ro, req, string, range:[1,15]-->test
          </IMSI>
        </MEF>
      </MEFData>
      <alarmTemperature>
        <!--ro, opt, float, alarm temperature threshold, range:[-20.0,60.0], unit:℃-->37.6
      </alarmTemperature>
      <collectEnabled>
        <!--ro, opt, bool-->false
      </collectEnabled>
    </MatchElement>
  </MatchList>
</FCSearchResult>
```

19.8.2.13 Get the capabilities of configuring prompt for face picture comparison

Request URL

GET /ISAPI/Intelligent/FDLib/prompt/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "faceContrastPromptLen": {
    /*ro, opt, object*/
        "@min":  "0",
        /*ro, req, string, range:[0,16]*/
        "@max":  "16"
        /*ro, req, string, range:[0,16]*/
    },
    "whiteListFaceContrastPromptLen": {
    /*ro, opt, object*/
        "@min":  "0",
        /*ro, req, string, range:[0,16]*/
        "@max":  "16"
        /*ro, req, string, range:[0,16]*/
    }
}
```

19.8.2.14 Face picture comparison event

EventType:alarmResult

```
{
    "ipAddress":  "172.6.64.7",
```

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```
    /*ro, req, string, IPv4 address of the device that triggers the alarm, range:[1,32]*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm, range:[1,128]*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm, range:[1,65535]*/
    "protocol":  "HTTP",
    /*ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address, range:[1,32]*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, req, int, times that the same alarm has been uploaded*/
    "eventType":  "alarmResult",
    /*ro, req, enum, event type, subType:string, desc:"alarmResult" (face picture comparison)*/
    "eventState":  "active",
    /*ro, req, enum, continuous event status, subType:string, desc:active (valid event); inactive (invalid event)*/
    "eventDescription":  "alarmResult",
    /*ro, req, string, event description, range:[1,128], desc:event description*/
    "channelName":  "ABC",
    /*ro, req, string, channel name (camera name), range:[1,128]*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID (PUID), range:[1,64], desc:this field must be returned when ISAPI event information is transmitted via ISUP*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be
uploaded again after the device recovers*/
    "alarmResult": [
    /*ro, opt, array, face picture comparison results, subType:object, desc:face picture comparison results*/
        {
            "errorCode":  1,
            /*ro, opt, int, when analyzing face picture failed, the corresponding error code will be returned*/
            "errorMsg":  "ok",
            /*ro, opt, string, when analyzing face picture failed, the corresponding error information will be returned, desc:when analyzing face picture
failed, the corresponding error information will be returned*/
            "image":  " http://10.7.52.174:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/011_163957_23006&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, opt, string, picture URL*/
            "modelData":  "test",
            /*ro, opt, string, modeling data of background picture, which is encoded by Base64, range:[1,2048]*/
            "traceUuid":  "sddddddddff",
            /*ro, opt, string, target ID, desc:one person corresponds to one target ID in multiple alarms (related URI:
/ISAPI/SDT/Management/Task/facePicture); the maximum size*/
            "traceIdx":  1,
            /*ro, opt, int, picture index, desc:one person can correspond to multiple pictures*/
            "targetAttrs": {
            /*ro, opt, object, target attributes*/
                "deviceId":  "ee3c4000-a22d-11b4-8334-5803fb77894c",
                /*ro, opt, string, device UUID, range:[1,64]*/
                "deviceChannel":  1,
                /*ro, opt, int, device channel No.*/
                "deviceName":  "shebei",
                /*ro, opt, string, device name*/
                "faceTime":  "2016-09-10 00:00:00.000",
                /*ro, opt, string, capture time, range:[1,64]*/
                "rect": {
                /*ro, opt, object, background picture frame information of face, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "contentID":  "backgroundImage",
                /*ro, opt, string, contentID of the picture data binary data, range:[1,128], desc:the bkgURL cannot be returned when this field is
returned*/
                "bkgUrl":  "http://absddsdfsdfsdk.jpg",
                /*ro, opt, string, background picture URL*/
                "pId":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, captured background picture ID, range:[1,32], desc:captured background picture ID*/
                "ThermalImage": {
                /*ro, opt, object, thermal imaging picture information*/
                    "resourcesContentType":  "binary",
                    /*ro, opt, string, resource transmission type, desc:"url" (transmitted via URL), "binary" (transmitted in binary format). When the
returned alarm message contains additional resource (e.g., pictures), this field should be returned*/
                    "resourcesContent":  "test"
                    /*ro, opt, string, resource ID, desc:when the returned alarm message contains additional resources (e.g., pictures), this field should
be returned. When the value of resourcesContentType is "binary", it must be the same as Content-ID of the resource. When the value of resourcesContentType
is "url", its value is the resource URL*/
                }
            },
            "faces": [
            /*ro, opt, array, face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on,
subType:object, desc:face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on*/
                {
                    "faceId":  1,
                    /*ro, opt, int, face ID, desc:the face ID cannot be duplicated in the same picture*/
```

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```
                    /*ro, opt, int, face ID, desc:the face ID cannot be duplicated in the same picture*/
                    "faceRect": {
                    /*ro, opt, object, face thumbnail frame, desc:it is used for modeling. The origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "faceMark": {
                    /*ro, opt, object, face features in the picture*/
                        "leftEye": {
                        /*ro, opt, object, left eye coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "rightEye": {
                        /*ro, opt, object, right eye's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "noseTip": {
                        /*ro, opt, object, nose tip's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "leftMouth": {
                        /*ro, opt, object, left mouth corner's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "rightMouth": {
                        /*ro, opt, object, right mouth corner's position, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        }
                    },
                    "recommendFaceRect": {
                    /*ro, opt, object, recommended face frame coordinates, desc:recommended face frame coordinates*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "facePose": {
                    /*ro, opt, object, face angle*/
                        "pitch":  0.998,
                        /*ro, opt, float, face pitch angle, face pitching up is regarded as positive, range:[-90.000,90.000], desc:this field does not
exists if it is invalid. It is accurate to three decimal places*/
                        "yaw":  0.876,
                        /*ro, opt, float, face yaw angle, face yawing left is regarded as positive, range:[-90.000,90.000], desc:this field does not exists
if it is invalid. It is accurate to three decimal places*/
                        "roll":  0.623
                        /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive, range:[-90.000,90.000], desc:this field does not
exists if it is invalid. It is accurate to three decimal places*/
                    },
                    "age": {
                    /*ro, opt, object, age*/
                        "range":  5,
                        /*ro, opt, int, age deviation; if the age deviation already exists, the age is required*/
                        "value":  23,
                        /*ro, opt, int, age*/
                        "ageGroup":  "any"
                        /*ro, opt, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old",
"prime", "teenager", "unknown", "young"*/
                    },
                    "gender": {
                    /*ro, opt, object, gender*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "male"
                        /*ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                    },
                    "glass": {
                    /*ro, opt, object, whether the person wears glasses*/
                        "confidence":  99.94499999999999,
                        /*ro, opt, float, confidence*/
```


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```
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no", "yes", "sunglasses"*/
                    },
                    "smile": {
                    /*ro, opt, object, whether the person is smiling*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person is smiling, subType:string, desc:"no", "yes"*/
                    },
                    "mask": {
                    /*ro, opt, object, whether the person wears a mask*/
                        "confidence":  99.94499999999999,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "yes"*/
                    },
                    "faceExpression": {
                    /*ro, opt, object, facial expression*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "angry"
                        /*ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-
faced", "sad", "surprised", "unknown"*/
                    },
                    "hat": {
                    /*ro, opt, object, whether the person wears a hat*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "unknown"
                        /*ro, opt, enum, whether the person wears a hat, subType:string, desc:"no", "yes", "unknown"*/
                    },
                    "score": {
                    /*ro, opt, object, face score (face picture quality)*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  65
                        /*ro, opt, int, score value, range:[1,100]*/
                    },
                    "modeldata":  "test",
                    /*ro, opt, string, face model data (encoded by base64)*/
                    "AbsoluteHigh": {
                    /*ro, opt, object, PTZ position*/
                        "elevation":  -900,
                        /*ro, opt, int, tilting angle, range:[-900,2700]*/
                        "azimuth":  22,
                        /*ro, opt, int, azimuth, range:[0,3600]*/
                        "absoluteZoom":  1
                        /*ro, opt, int, zooming ratio, range:[1,1000]*/
                    },
                    "identify": [
                    /*ro, opt, array, arming linkage information, subType:object*/
                        {
                            "relationId":  "9887",
                            /*ro, opt, string, arming linkage ID, range:[0,64], desc:arming linkage ID*/
                            "maxsimilarity":  0.780,
                            /*ro, opt, float, the maximum similarity of face comparison results, range:[0.000,1.000], step:0.001*/
                            "similarityRange":  1,
                            /*ro, opt, enum, similarity range, subType:int, desc:0 (smaller than the minimum value), 1 (between the minimum value and
maximum value), 2 (larger than the maximum value). If the similarity is smaller than the minimum value, the person will be marked*/
                            "candidate": [
                            /*ro, opt, array, matched face picture information, subType:object*/
                                {
                                "alarmId":  1,
                                /*ro, opt, int, alarm ID, desc:it cannot be duplicated of one arming linkage*/
                                "blacklist_id":  "2323",
                                /*ro, opt, string, face picture library ID, range:[1,64], desc:it corresponds to the FDID of face picture library. For
cameras, only the default FDID will be returned, the custom face picture library ID will be returned in customFaceLibID*/
                                "human_data": [
                                /*ro, opt, array, person information of matched face pictures, subType:object*/
                                    {
                                        "face_id":  "123445",
                                        /*ro, opt, string, face ID, range:[1,32]*/
                                        "face_picurl":  "http://127.0.0.1:6120/face_picurl",
                                        /*ro, opt, string, face picture URL*/
                                        "pId":  "F4F665D6A18E41308CE9934DCDDD766F",
                                        /*ro, opt, string, face thumbnail ID, range:[1,64], desc:face thumbnail ID*/
                                        "face_rect": {
                                        /*ro, opt, object, face thumbnail coordinates, desc:the origin is the upper-left corner of the screen*/
                                            "height":  1.000,
                                            /*ro, opt, float, height, range:[0.000,1.000]*/
                                            "width":  1.000,
                                            /*ro, opt, float, width, range:[0.000,1.000]*/
                                            "x":  0.000,
                                            /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                                            "y":  0.000
                                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                                        },
                                        "similarity":  0.997000,
                                        /*ro, opt, float, face picture similarity*/
                                        "isNoSaveFDPicture":  true,
                                        /*ro, opt, bool, whether to not save the picture of face picture library, desc:true (do not save); if this node
is not returned, it indicates that saving the picture*/
```


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```
                                        "contentID":  "faceLibImage",
                                        /*ro, opt, string, picture in the face picture library, desc:picture in the face picture library*/
                                        "bkg_picurl":  "http://10639103:80/FDLib?
FDID=529FB9EAC2A04041A603E18F300CD3A5&pId=F4F665D6A18E41308CE9934DCDDD766F&size=22332"
                                        /*ro, opt, string, background picture URL*/
                                    }
                                ],
                                "human_id":  "123",
                                /*ro, opt, string, person ID, desc:person ID in the list, which corresponds to FPID in the face picture library*/
                                "reserve_field": {
                                /*ro, opt, object, reserved (transmitted data)*/
                                    "name":  "张三",
                                    /*ro, req, string, name, range:[1,256]*/
                                    "gender":  "any",
                                    /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition which is used for search), "male",
"female", "unknown"*/
                                    "bornTime":  "2004-05-03",
                                    /*ro, req, string, date of birth*/
                                    "city":  "130100",
                                    /*ro, opt, string, city code of person birth, range:[1,32], desc:see the data dictionary for details*/
                                    "certificateType":  "officerID",
                                    /*ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport"
(passport), "other" (the other certificate)*/
                                    "certificateNumber":  "test",
                                    /*ro, opt, string, certificate No., range:[0,32]*/
                                    "ageGroup":  "any",
                                    /*ro, opt, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle
age), "old", "prime", "teenager", "unknown", "young"*/
                                    "phoneNumber":  "test",
                                    /*ro, opt, string, phone number, range:[1,64]*/
                                    "cityName":  "石家庄市",
                                    /*ro, opt, string, city (corresponding to the field "city"), range:[0,32], desc:city (corresponding to the field
"city")*/
                                    "provinceName":  "河北省"
                                    /*ro, opt, string, province, range:[0,32], desc:province*/
                                },
                                "similarity":  0.780,
                                /*ro, opt, float, range:[0.000,1.000]*/
                                "listType":  "blackList",
                                /*ro, opt, enum, list type, subType:string, desc:"blacklist" (blocklist), "whitelist" (allowlist)*/
                                "extendData": [
                                /*ro, opt, array, person extended information, subType:object*/
                                    {
                                        "extendID":  1,
                                        /*ro, opt, int, person extended information ID, desc:it starts from 1*/
                                        "enable":  true,
                                        /*ro, opt, bool, enable person extended information or not*/
                                        "name":  "test",
                                        /*ro, opt, string, name of person tag extended information, range:[1,96]*/
                                        "value":  "test"
                                        /*ro, opt, string, person tag extended information, range:[0,64]*/
                                    }
                                ],
                                "FDLibName":  "faceLib1",
                                /*ro, opt, string, face picture library name, desc:face picture library name*/
                                "FDLibThreshold":  50,
                                /*ro, opt, int, detection threshold of face picture library, range:[0,100]*/
                                "customFaceLibID":  "test",
                                /*ro, opt, string, custom face picture library ID*/
                                "customHumanID":  "test"
                                /*ro, opt, string, custom person ID*/
                            }
                        ],
                        "alarmRecordID":  "test",
                        /*ro, opt, string, alarm record ID*/
                        "libAttribute":  "general"
                        /*ro, opt, enum, library type, subType:string, desc:"general" (normal library), "blackList" (blocklist library), "VIP" (VIP
library), "passerby" (allowlist library)*/
                    }
                ],
                "contentID":  "faceImage",
                /*ro, opt, string, background picture ID, desc:background picture ID*/
                "URL":  "test",
                /*ro, opt, string, face picture thumbnail URL, range:[1,256], desc:face picture thumbnail URL*/
                "pId":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, picture ID in face picture library, range:[1,32]*/
                "stayDuration":  20,
                /*ro, opt, int, dwell time duration in the image, unit:ms*/
                "prompt":  "test",
                /*ro, opt, string, prompt contents of face picture comparison, range:[1,16]*/
                "contrastStatus":  true,
                /*ro, opt, bool, comparison result*/
                "realTimeContrast":  true,
                /*ro, opt, bool, whether to enable real-time face picture comparison*/
                "remainReportNum":  0,
                /*ro, opt, int, the remaining times of uploading the comparison alarm of the current target, desc:the remaining times of uploading the
comparison alarm of the current target*/
                "UID":  "test",
                /*ro, opt, string*/
                "UUPID":  "test",
                /*ro, opt, string, background picture ID of face picture library, range:[1,64], desc:it consists of device serial No., imported time,
and random digits*/
                "frontMatchType":  1,
                /*ro, opt, enum, face picture library type, subType:int, desc:1 (cluster library), 0 (normal library)*/
```


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```
                    "isImportFD":  true,
                    /*ro, opt, bool, whether to import the captured face data to the library, desc:true (yes), false (no). When face picture comparison
failed, the face picture quality meets the requirements, and is imported to the library, true will be returned. When face picture comparison failed, and the
face picture quality does not meet the requirements, false will be returned*/
                    "faceSnapThermometryEnabled":  true,
                    /*ro, opt, bool, whether to enable temperature screening when capturing face pictures*/
                    "currTemperature":  36.7,
                    /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃, desc:it should be used with thermometryUnit together*/
                    "isAbnomalTemperature":  true,
                    /*ro, opt, bool, whether the temperature is abnormal, desc:true (yes), false (no)*/
                    "thermometryUnit":  "celsius",
                    /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
                    "alarmTemperature":  36.7,
                    /*ro, opt, float, range:[-20.0,150.0], unit:℃*/
                    "facePicQuality":  "low"
                    /*ro, opt, enum, subType:string*/
                }
            ],
            "alarmRecordID":  "test",
            /*ro, opt, string, alarm record ID*/
            "uid":  "test",
            /*ro, opt, string, the unique ID of event uploading, range:[1,64], desc:it can be composed of time (accurate to the millisecond) and random
digits*/
            "remainReportNum":  0,
            /*ro, opt, int, the remaining times of uploading the comparison alarm of the current target*/
            "algorithmInfo": {
            /*ro, opt, object, algorithm version information, desc:this field is required for secondary analysis*/
                "algorithmId":  "test",
                /*ro, opt, string, algorithm package ID*/
                "algoVendorId":  "test",
                /*ro, opt, string, manufacturer ID*/
                "algoVersionId":  "test",
                /*ro, opt, string, algorithm version No.*/
                "algoChipType":  "test",
                /*ro, opt, string, chip type*/
                "algoModelVersion":  "test"
                /*ro, opt, string, model version*/
            }
        }
    ],
    "URLCertificationType":  "digest",
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no"-no authentication, "digest"-digest authentication (it is used for local
picture URL returned by NVR or DVR)*/
    "GPS": {
    /*ro, opt, object, device GPS information*/
        "divisionEW":  "E",
        /*ro, req, enum, "E" (Eastern Hemisphere), W" (West Hemisphere), subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)*/
        "longitude":  100,
        /*ro, req, int, longitude, desc:longitude*/
        "divisionNS":  "N",
        /*ro, req, enum, subType:string*/
        "latitude":  100
        /*ro, req, int, latitude, desc:latitude*/
    }
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
alarmResult	[Message content]	application/json	--	--	--
..value	[Binary picture data]	image/jpeg	backgroundImage	backgroundImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
faceLibImage	[Binary picture data]	image/jpeg	faceLibImage	faceLibImage.jpg	--
...value	[Binary picture data]	image/jpeg			--
..value	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.8.3 Face Picture Library Management

19.8.3.1 Get the parameters of person custom information

Request URL

GET /ISAPI/Intelligent/faceContrast/personInfoExtend

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PersonInfoExtendList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PersonInfoExtend>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->1
    </id>
    <enable>
      <!--ro, req, bool-->true
    </enable>
    <name>
      <!--ro, opt, string-->test
    </name>
    <value>
      <!--ro, opt, string-->test
    </value>
  </PersonInfoExtend>
</PersonInfoExtendList>
```

19.8.3.2 Set the parameters of person custom information

Request URL

PUT /ISAPI/Intelligent/faceContrast/personInfoExtend

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<PersonInfoExtendList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PersonInfoExtend>
    <!--opt, object-->
    <id>
      <!--req, int-->1
    </id>
    <enable>
      <!--req, bool-->true
    </enable>
    <name>
      <!--opt, string-->test
    </name>
    <value>
      <!--opt, string-->test
    </value>
  </PersonInfoExtend>
</PersonInfoExtendList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.3.3 Get the capability of person custom information

Request URL

GET /ISAPI/Intelligent/faceContrast/personInfoExtend/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PersonInfoExtendList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PersonInfoExtend>
    <!--ro, opt, object-->
    <id min="1" max="10">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </id>
    <enable>
      <!--ro, req, bool-->true
    </enable>
    <name min="1" max="10">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </name>
    <value min="1" max="10">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </value>
  </PersonInfoExtend>
</PersonInfoExtendList>
```

19.8.3.4 Delete all face picture libraries

Request URL

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DELETE /ISAPI/Intelligent/FDLib

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.3.5 Import the face pictures to a specific face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/<FDID>/data?checkCode=<checkCode>&FDType=<FDType>&mode=<mode>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
checkCode	string	Verification code, and the maximum length is 16. When "security" is 1, it indicates that "checkCode" in the URL is encrypted in AES128 CBC mode; when "security" is 2, it indicates that "checkCode" in the URL is encrypted in AES256 CBC mode.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.
mode	string	When importing a face library, an error "faceLibraryNumError" occurs to prompt the user whether to overwrite. If the overwrite import mode is selected, and the face library has reached its maximum capacity, the subsequently imported faces will overwrite the earliest imported face information.

Request Message

Binary Data

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.8.3.6 Update the specified face picture of a face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>/imageData?identityKey=<identityKey>&FDType=<FDType>&modeling=<modeling>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	face picture library ID
pictureID	string	picture ID
identityKey	string	The maximum length is 64. This parameter stands for the identity key of the operation platform. The device can know the operator's identity by checking this parameter in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.
modeling	enum	whether it is synchronous modeling true (synchronous modeling), false or not applied (asynchronous modeling).

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ImageData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, picture parameters, attr:version{req, string, protocolVersion}-->
  <picType>
    <!--req, enum, picture type, subType:string, desc:"binary" (binary picture in form format), "url" (picture transmitted in URL)-->binary
  </picType>
  <picUrl>
    <!--opt, picurl, picture URL-->http://127.0.0.1
  </picUrl>
  <Rect>
    <!--opt, object, target frame, desc:the origin is the upper-left corner of the screen. When there are multiple face targets in one picture, you need to
specify the modeling target-->
    <height>
      <!--req, float, height, range:[0.000,1.000]-->0.000
    </height>
    <width>
      <!--req, float, width, range:[0.000,1.000]-->0.000
    </width>
    <x>
      <!--req, float, X-coordinate, range:[0.000,1.000]-->0.000
    </x>
    <y>
      <!--req, float, Y-coordinate, range:[0.000,1.000]-->0.000
    </y>
  </Rect>
</ImageData>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
ImageData	[Message content]	application/xml	--	--	--
faceImage	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

19.8.3.7 Delete the linked information of a picture in the face picture

library **Request URL**

DELETE /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
pictureID	string	Picture ID
identityKey	string	The maximum length is 64. This parameter stands for the identity key of the operation platform. The device can know the operator's identity by checking this parameter in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.8.3.8 Get the number of remaining face pictures in a face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/<FDID>/picture/surplusCapacity?FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SurplusCapacity xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, undefined, attr:version{req, string, protocolVersion}-->
  <FDPicturesNumber>
    <!--ro, opt, int, supported number of remaining face pictures in the face picture library-->1
  </FDPicturesNumber>
</SurplusCapacity>
```

19.8.3.9 Get the basic parameters of a face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/<FDID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

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Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID.
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, face picture library basic information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, index-->1
  </id>
  <FDID>
    <!--ro, opt, string, face picture library ID-->test
  </FDID>
  <name>
    <!--ro, opt, string, face picture library name-->test
  </name>
  <thresholdValue>
    <!--ro, opt, int, detection threshold, range:[0,100]-->1
  </thresholdValue>
  <customInfo>
    <!--ro, opt, string, custom information-->test
  </customInfo>
  <customFaceLibID>
    <!--ro, opt, string, custom face picture library ID-->test
  </customFaceLibID>
  <faceLibType>
    <!--ro, opt, enum, face picture library type, subType:string, desc:"ordinary" (normal library), "private" (private library), "stranger" (stranger
library (clustering library)), "allowable" (allowlist library), "privacyMask" (privacy mask library)-->ordinary
  </faceLibType>
  <autoUpdata>
    <!--ro, opt, bool, whether to enable auto update of background picture in face picture library, desc:if autoUpdata in the capability set only supports
"true", the auto update feature is enabled by default; if not, it is disabled.-->true
  </autoUpdata>
  <qualityThreshold>
    <!--ro, opt, int, picture quality threshold, range:[0,100], desc:this node is valid when <autoUpdata> is "true"-->1
  </qualityThreshold>
  <importFDThreshold>
    <!--ro, opt, int, put-in-storage similarity threshold, range:[0,100], desc:this node is valid when <autoUpdata> is "true"-->1
  </importFDThreshold>
  <Age>
    <!--ro, opt, object, evaluation cycle, desc:this node is valid when <autoUpdata> is "true"-->
    <judgementCycleType>
      <!--ro, req, enum, evaluation cycle type, subType:string, desc:"day" (evaluate at 0 o'clock every day), "week" (evaluate at 0 o'clock every Sunday),
"month" (default; evaluate at 0 o'clock on the first day of every month), "duration" (evaluate by duration), "quarter" (evaluate at 0 o'clock on the first
day of every quarter), "halfYear" (evaluate at 0 o'clock on the first day of every six months)-->day
    </judgementCycleType>
    <statisticalCycleType>
      <!--ro, opt, enum, statistics cycle type of face picture matches, subType:string, desc:"day" (daily), "week" (weekly), "month" (monthly)-->day
    </statisticalCycleType>
    <statisticalCycle>
      <!--ro, opt, int, statistics cycle of face picture matches, range:[0,99999]-->1
    </statisticalCycle>
    <minFMCount>
      <!--ro, opt, int, min. number of face picture matches within statistics cycle, range:[0,99999], desc:the default value is 30-->1
    </minFMCount>
    <FMCountEnabled>
      <!--ro, opt, bool, whether to enable total number of face picture matches-->true
    </FMCountEnabled>
    <FMCount>
      <!--ro, opt, int, total number of face picture matches, range:[0,99999]-->100
```

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```
      </FMCount>
      <backgroundPicThreshold>
        <!--ro, opt, int, background picture replacement similarity threshold, range:[1,100]-->90
      </backgroundPicThreshold>
      <judgementCycle>
        <!--ro, opt, int, evaluation cycle of clustering data aging in face picture library, range:[1,365], unit:d, dep:and,
{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
      </judgementCycle>
      <agingExecDataTime>
        <!--ro, opt, time, clustering library aging execution time-->00:00:00+08:00
      </agingExecDataTime>
      <automaticClearEnabled>
        <!--ro, opt, bool, whether to enable auto clear-up when the clustering library is full-->true
      </automaticClearEnabled>
    </Age>
    <FaceCounting>
      <!--ro, opt, object, face picture statistics-->
      <enabled>
        <!--ro, req, bool, whether to enable face-picture people counting deduplication-->true
      </enabled>
      <uploadType>
        <!--ro, req, enum, reporting mode after face-picture people counting deduplication is enabled, subType:string, desc:"out_repetition" (report outside
deduplication time), "noUpload" (do not report)-->out_repetition
      </uploadType>
      <channelList>
        <!--ro, opt, object, video channel list, desc:this node is used on normal cameras accessed to NVR/CVR to specify which video channels should use the
face picture library to perform people counting-->
        <channelID>
          <!--ro, req, int, video channel No., range:[1,256]-->true
        </channelID>
      </channelList>
    </FaceCounting>
    <faceMonitor>
      <!--ro, opt, bool, whether it is the face picture library linked with intelligent face picture arming-->true
    </faceMonitor>
    <monitorFaceNum>
      <!--ro, opt, bool, max. face pictures supported by face picture arming-->true
    </monitorFaceNum>
    <customNoConfig>
      <!--ro, opt, bool, this node is not valid for Web client, desc:it is only valid in person arming mode-->true
    </customNoConfig>
</FDLibBaseCfg>
```

19.8.3.10 Get the status of asynchronously importing base picture data to face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/asyncImportDatasTasks?format=json&identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	If the URL does not contain the identityKey parameter, it indicates searching for tasks of the normal custom library; if the identityKey parameter is contained, it indicates searching for tasks related to the private library only.

Request Message

None

Response Message

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```
{
    "AsyncImportDatasTasks": [
    /*ro, opt, array, subType:object*/
        {
            "customFaceLibID":  "test",
            /*ro, req, string, range:[1,64]*/
            "taskID":  "test",
            /*ro, req, string, range:[1,64]*/
            "URL":  "test",
            /*ro, req, string, range:[1,255]*/
            "type":  0,
            /*ro, opt, enum, subType:int*/
            "URLCertificationType":  "AWS2_0",
            /*ro, req, enum, subType:string*/
            "status":  0,
            /*ro, req, enum, subType:int*/
            "state":  "URLError",
            /*ro, opt, enum, subType:string*/
            "progress":  1
            /*ro, req, int, range:[1,100], step:1*/
        }
    ]
}
```

19.8.3.11 Get face picture library capability

Request URL

GET /ISAPI/Intelligent/FDLib/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<FDLibCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, face picture library capability object, attr:version{opt, string, protocolVersion}-->
  <CreateFDLibList size="15">
    <!--ro, req, array, face picture library capability object array, subType:object, attr:size{opt, string}-->
    <CreateFDLib>
      <!--ro, req, object, face picture library capability object-->
      <id max="15" min="1">
        <!--ro, req, int, number range capability of face picture library, range:[1,15], attr:max{opt, string},min{opt, string}-->10
      </id>
      <name max="64" min="1">
        <!--ro, opt, string, name length of face picture library, range:[1,64], attr:max{opt, string},min{opt, string}-->64
      </name>
      <thresholdValue max="100" min="0" def="60">
        <!--ro, opt, int, detection threshold range capability of face picture library, range:[0,100], attr:max{opt, string},min{opt, string},def{req, int},
desc:the higher the threshold, the lower the detection accuracy-->50
      </thresholdValue>
      <customInfo max="128" min="1">
        <!--ro, opt, string, custom description length in the face picture library, range:[1,128], attr:max{opt, string},min{opt, string}-->128
      </customInfo>
      <faceDataMax max="150000" min="1">
        <!--ro, opt, int, the maximum number of human face base picture allowed in the face picture library, range:[1,150000], attr:max{opt,
string},min{opt, string}-->30000
      </faceDataMax>
      <customFaceLibID max="64" min="1">
        <!--ro, opt, string, the custom ID length in the face picture library, range:[1,64], attr:max{opt, string},min{opt, string}-->64
      </customFaceLibID>
    </CreateFDLib>
  </CreateFDLibList>
  <FaceAppendData>
    <!--ro, opt, object, the object capability of background picture data extension of face picture library-->
    <bornTime>
      <!--ro, opt, date, birth date of the person in the face picture-->1970-01-01+08:00
    </bornTime>
    <name max="64" min="1">
      <!--ro, opt, string, the description length of person name, range:[1,64], attr:max{opt, string},min{opt, string}-->64
    </name>
    <sex opt="male,female">
      <!--ro, opt, enum, gender, subType:string, attr:opt{opt, string}, desc:male; female;-->male
    </sex>
    <province>
      <!--ro, opt, string, province of birth-->01
    </province>
    <city>
      <!--ro, opt, string, city-->01
    </city>
    <certificateType opt="officerID,ID,passportID,other">
```

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```
      <!--ro, opt, enum, certificate type capability, subType:string, attr:opt{opt, string}, desc:"officerID"-officer ID,"ID"-identify card,"passportID"-passport,"other"-other certificate-->officerID
    </certificateType>
    <certificateNumber max="32" min="1">
      <!--ro, opt, string, capability of certificate No., range:[1,128], attr:max{opt, string},min{opt, string}-->32
    </certificateNumber>
    <PersonInfoExtendList size="128">
      <!--ro, opt, array, object array of base picture's extended information in the face picture library, subType:object, attr:size{opt, string}, desc:if the node is returned, it indicates that the device supports. Corresponding URL: /ISAPI/Intelligent/FDLib/pictureUpload-->
      <PersonInfoExtend>
        <!--ro, opt, object, the capability of base picture's extended information in the face picture library-->
        <id max="15000" min="1">
          <!--ro, req, int, No. range of extended information, range:[1,15000], attr:max{opt, string},min{opt, string}-->30000
        </id>
        <enable opt="true,false">
          <!--ro, req, bool, whether to enable the extended information, attr:opt{opt, string}-->true
        </enable>
        <name max="128" min="1">
          <!--ro, opt, string, the name length of extended information, range:[1,128], attr:max{opt, string},min{opt, string}-->128
        </name>
        <value max="256" min="1">
          <!--ro, opt, string, the content length of extended information, range:[1,256], attr:max{opt, string},min{opt, string}-->256
        </value>
      </PersonInfoExtend>
    </PersonInfoExtendList>
    <customHumanID max="128" min="1">
      <!--ro, opt, string, the capability of the person ID of custom face picture, range:[1,128], attr:max{opt, string},min{opt, string}-->64
    </customHumanID>
  </FaceAppendData>
  <FDSearchDescription>
    <!--ro, opt, object, the search capability of face picture library-->
    <searchID max="63" min="1">
      <!--ro, opt, string, search ID, attr:max{opt, string},min{opt, string}, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searches, the search history will be saved in the memory to speed up next search; min/max will be returned according to the device actual capability, if not returned: min=1 max=63 (default)-->63
    </searchID>
    <searchResultPosition>
      <!--ro, opt, int, start position, desc:if this node is not returned, confirm <isSupportStandardSearch> exists and the node "true" is returned; it will start from 0 by default; if not returned, and it will start from 1 by default-->1
    </searchResultPosition>
    <maxResults>
      <!--ro, opt, int, the number of search results this time, desc:the number of search results in a single search is between 1 and 500; if exceeding the max number of search results supported by the device, the device will return the max number of results in a single search by default; if this node is not returned, the number of search results is between 1 and 500 by default-->30
    </maxResults>
    <FDID>
      <!--ro, opt, string, face picture library ID-->1
    </FDID>
    <startTime>
      <!--ro, opt, datetime, the birth start time of the persons in the face picture library-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--ro, opt, datetime, the birth end time of the persons in the face picture library-->1970-01-01T00:00:00+08:00
    </endTime>
    <name max="64" min="1">
      <!--ro, opt, string, person name, attr:max{opt, string},min{opt, string}-->64
    </name>
    <sex opt="male,female">
      <!--ro, opt, enum, gender, subType:string, attr:opt{opt, string}, desc:male; female;-->male
    </sex>
    <province>
      <!--ro, opt, string, province of birth-->01
    </province>
    <city>
      <!--ro, opt, string, city-->01
    </city>
    <certificateType opt="officerID,ID,passportID,other">
      <!--ro, opt, enum, certificate type capability, subType:string, attr:opt{opt, string}, desc:"officerID" (officer ID), "ID" (identify card), "passportID" (passport), "other" (other certificate)-->officerID
    </certificateType>
    <certificateNumber max="32" min="1">
      <!--ro, opt, string, capability of certificate No. length, range:[1,128], attr:max{opt, string},min{opt, string}-->32
    </certificateNumber>
    <customFaceLibID max="64" min="0">
      <!--ro, opt, string, the custom ID length in the face picture library, range:[1,64], attr:max{opt, string},min{opt, string}-->64
    </customFaceLibID>
  </FDSearchDescription>
  <isSuportFDLibImpEmp opt="true,false">
    <!--ro, opt, bool, whether to support the import and export of the face picture library, attr:opt{opt, string}-->true
  </isSuportFDLibImpEmp>
  <isSuportFCSearch opt="true,false">
    <!--ro, opt, bool, whether supports searching captured face picture library, attr:opt{opt, string}-->true
  </isSuportFCSearch>
  <FCSearchDescription>
    <!--ro, opt, object, the object capability of searching captured face picture library-->
    <searchID max="63" min="1">
      <!--ro, opt, string, search ID, attr:max{opt, string},min{opt, string}-->63
    </searchID>
    <searchResultPosition>
      <!--ro, opt, int, start position; this node is not returned-->1
    </searchResultPosition>
    <maxResults>
      <!--ro, opt, int, the number of search results this time-->30
    </maxResults>
```

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```
    <snapStartTime>
      <!--ro, opt, datetime, ISO 8601 time-->1970-01-01T00:00:00+08:00
    </snapStartTime>
    <snapEndTime>
      <!--ro, opt, datetime, ISO 8601 time-->1970-01-01T00:00:00+08:00
    </snapEndTime>
    <eventType opt="unlimit,faceContrast,whiteFaceContrast,faceCapture,faceContrastFailure">
      <!--ro, opt, enum, type of face capture event, subType:string, attr:opt{opt, string}, desc:"unlimit", "faceContrast" (face picture comparison),
"whiteFaceContrast" (allowlist face picture comparison (strangers alarm)), "faceCapture" (face capture), "faceContrastFailure" (face picture comparison
failed)-->unlimit
    </eventType>
    <isSupportMultiChannelSearch>
      <!--ro, opt, bool, whether the device supports multichannel search-->true
    </isSupportMultiChannelSearch>
    <isSupportTotalSearchResult>
      <!--ro, opt, bool, whether the device supports the device result limit for each search-->true
    </isSupportTotalSearchResult>
  </FCSearchDescription>
  <isSupportFDExecuteControl>
    <!--ro, opt, bool, whether the device supports creating relation between face picture libraries and cameras-->true
  </isSupportFDExecuteControl>
  <isSupportManualModeling>
    <!--ro, opt, bool, whether the device supports modeling control-->true
  </isSupportManualModeling>
  <isSupportSingleManualModeling>
    <!--ro, opt, bool, whether the device supports single modeling-->true
  </isSupportSingleManualModeling>
  <isSupportRangeManualModeling>
    <!--ro, opt, bool, whether the device supports all modeling-->true
  </isSupportRangeManualModeling>
  <isSupportSingleManualModelingSearch>
    <!--ro, opt, bool, whether the device supports searching for single modeling-->true
  </isSupportSingleManualModelingSearch>
  <isSupportRangeManualModelingSearch>
    <!--ro, opt, bool, whether the device supports all modeling control-->true
  </isSupportRangeManualModelingSearch>
  <isSupportModelingStatusSearch>
    <!--ro, opt, bool, whether the device supports searching for the condition for modeling status-->true
  </isSupportModelingStatusSearch>
  <Reboot>
    <!--ro, opt, object, the specific parameters of prompt for device reboot to take effect-->
    <thresholdValue>
      <!--ro, opt, bool, after editing the threshold for the face picture library, you need to reboot the device to take effect-->true
    </thresholdValue>
    <delFaceLib>
      <!--ro, opt, bool, after deleting the face picture library, you need to reboot the device to take effect-->true
    </delFaceLib>
  </Reboot>
  <isSupportCoverImportpictures>
    <!--ro, opt, bool, whether to support importing and overwriting the existing pictures-->true
  </isSupportCoverImportpictures>
  <isSupportCoverImportFaceLib>
    <!--ro, opt, bool, whether to support importing and overwriting the existing face picture library-->true
  </isSupportCoverImportFaceLib>
  <isSupportConcurrentImportpictures>
    <!--ro, opt, bool, whether to support concurrent import of pictures-->true
  </isSupportConcurrentImportpictures>
  <isSupportSurplusCapacity>
    <!--ro, opt, bool, whether to support searching for the number of remaining face pictures of specified face picture library-->true
  </isSupportSurplusCapacity>
  <isSupportFDlibProgress>
    <!--ro, opt, bool, whether to support searching for the import and output status of the face picture library-->true
  </isSupportFDlibProgress>
  <isSupportFDlibPhotoWall>
    <!--ro, opt, bool, whether to support the picture display capability-->true
  </isSupportFDlibPhotoWall>
  <isSupportNewlyPictureUpload>
    <!--ro, opt, bool, whether to support importing pictures into the face picture library, desc:it is used for the device to be compatible with large-sized
picture import-->true
  </isSupportNewlyPictureUpload>
  <isSupportCustomHumanID>
    <!--ro, opt, bool, whether to support custom person ID for developer-->true
  </isSupportCustomHumanID>
  <isSupportCustomFaceLibID>
    <!--ro, opt, bool, whether to support custom face picture library ID for developer-->true
  </isSupportCustomFaceLibID>
  <isSupportFDCapacity>
    <!--ro, opt, bool, whether to support getting the capacity of face picture library-->true
  </isSupportFDCapacity>
  <isSupportNoSaveUploadPicture>
    <!--ro, opt, bool, whether to support not uploading the original pictures-->true
  </isSupportNoSaveUploadPicture>
  <isSupportUploadModeling>
    <!--ro, opt, bool, whether to support directly uploading the picture modeling data-->true
  </isSupportUploadModeling>
  <importImageSize max="300" min="1">
    <!--ro, opt, int, the size of supported JPG/JPEG pictures, unit:KB, attr:max{opt, string},min{opt, string}-->30
  </importImageSize>
  <FDModelingStatusSearchDescription>
    <!--ro, opt, object, task ID-->
    <searchID max="63" min="1">
      <!--ro, opt, string, search ID, attr:max{opt, string, range:[0,64]},min{opt, string}, desc:it is used to confirm the upper-level platform or system.
If the platform or the system is the same one during two searches, the search history will be saved in the memory to speed up next search; min/max will be
```

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```
returned according to the device actual capability, if not returned: min=1 max=63 (default)-->searchIDtest
    </searchID>
    <searchResultPosition>
      <!--ro, opt, int, start position; this node is not returned, desc:this node is not returned, confirm <isSupportStandardSearch> exists and the node
"true" is returned, it will start from 0 by default; if not returned, it will start from 1 by default-->30
    </searchResultPosition>
    <maxResults>
      <!--ro, opt, int, the number of search results this time, desc:the number of search results in a single search is between 1 and 500; if exceeding the
max number of search results supported by the device, the device will return the max number of results in a single search by default; if this node is not
returned, the number of search results is between 1 and 500 by default-->30
    </maxResults>
    <FDIDList>
      <!--ro, opt, string, ID of face picture library supported by the device to search for face modeling data-->test
    </FDIDList>
    <taskID max="64" min="1">
      <!--ro, opt, string, taskID, attr:max{opt, string},min{opt, string}, desc:the taskID has priority, corresponding URL
(/ISAPI/Intelligent/FDLib/manualModelingRangeTask or /ISAPI/Intelligent/FDLib/manualModeling?FDID=&&PID=), this node will be returned-->64
    </taskID>
  </FDModelingStatusSearchDescription>
</FDLibCap>
```

19.8.3.12 Search face picture data

Request URL

POST /ISAPI/Intelligent/FDLib/FDSearch?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, desc:it is used to check whether the current search requester is the same as the previous one. If they are the same, the
search record will be stored in the device to speed up the next search-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, initial position of search result list, desc:if it cannot get all records in one time searching,you can search the records followed
specified position for next search. For video intercom devices,this field can only be set to 0 as the picture will be returned along with the message-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, maximum number of records for single searching-->1
  </maxResults>
  <FDID>
    <!--opt, string, face picture library ID-->test
  </FDID>
  <startTime>
    <!--opt, date, start time-->1970-01-01+08:00
  </startTime>
  <endTime>
    <!--opt, date, end time-->1970-01-01+08:00
  </endTime>
  <name>
    <!--opt, string, name, range:[1,32], desc:name-->test
  </name>
  <sex>
    <!--opt, enum, gender, subType:string, desc:gender-->male
  </sex>
  <province>
    <!--opt, string-->11
  </province>
  <city>
    <!--opt, string, city, desc:city code of birth for the person in the face picture,the maximum size is 32 bytes-->11
  </city>
  <certificateType>
    <!--opt, enum, certificate type, subType:string, desc:the maximum size is 10 bytes,certificate type: "officerID"-officer ID,"ID"-identify
card,passport,other-->officerID
  </certificateType>
```

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```
    <certificateNumber>
      <!--opt, string, certificate No., range:[8,32], desc:certificate No.,the maximum size is 32 bytes-->test
    </certificateNumber>
    <phoneNumber>
      <!--opt, string, range:[1,64]-->test
    </phoneNumber>
    <FaceModeList>
      <!--opt, array, subType:object-->
      <FaceMode>
        <!--opt, object-->
        <ModeInfo>
          <!--opt, object-->
          <similarity>
            <!--opt, float, similarity, range:[0.0,100.0]-->0.0
          </similarity>
          <modeData>
            <!--opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method-
->test
          </modeData>
        </ModeInfo>
      </FaceMode>
    </FaceModeList>
    <modelStatus>
      <!--opt, enum, subType:string-->modeling
    </modelStatus>
    <modelingStatus>
      <!--opt, enum, subType:string-->success
    </modelingStatus>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID, desc:custom face picture library ID-->test
    </customFaceLibID>
    <sortord>
      <!--opt, enum, subType:string-->time
    </sortord>
    <order>
      <!--opt, enum, subType:string-->DESC
    </order>
    <StrangerLibFaceAppendData>
      <!--opt, object-->
      <PersonInfoExtendList size="4">
        <!--opt, array, subType:object, attr:size{req, int}-->
        <PersonInfoExtend>
          <!--opt, object, person extended information-->
          <id>
            <!--req, int, serial No. of person custom extension information, desc:serial No. of person custom extension information-->1
          </id>
          <enable>
            <!--req, bool, enable person extended information or not-->true
          </enable>
          <name>
            <!--opt, string, person tag extended information name, desc:person tag extended information name-->test
          </name>
          <value>
            <!--opt, string, person tag extended information, desc:person tag extended information-->test
          </value>
        </PersonInfoExtend>
      </PersonInfoExtendList>
    </StrangerLibFaceAppendData>
    <OccurrencesInfo>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <occurrences>
        <!--opt, int-->1
      </occurrences>
      <occurrencesSearchType>
        <!--opt, enum, subType:string-->greaterThanOrEqual
      </occurrencesSearchType>
    </OccurrencesInfo>
    <faceScore>
      <!--opt, int, face score (face picture quality), range:[-1,100], desc:face score (face picture quality)-->1
    </faceScore>
    <FDIDList>
      <!--opt, object-->
      <FDID>
        <!--req, string, range:[1,64]-->test
      </FDID>
    </FDIDList>
    <faceScoreMax>
      <!--opt, int, range:[-1,100]-->0
    </faceScoreMax>
    <pictureStatus>
      <!--opt, enum, subType:string-->normal
    </pictureStatus>
    <importStartTime>
      <!--opt, datetime-->1970-01-01T00:00:00+08:00
    </importStartTime>
    <importEndTime>
      <!--opt, datetime-->1970-01-01T00:00:00+08:00
    </importEndTime>
    <faceContrastSuccessStartTime>
      <!--opt, datetime-->1970-01-01T00:00:00+08:00
    </faceContrastSuccessStartTime>
```

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```
  <faceContrastSuccessEndTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </faceContrastSuccessEndTime>
  <userType>
    <!--opt, enum, subType:string-->normal
  </userType>
  <associatedPlateType>
    <!--opt, enum, subType:string-->greenChannel
  </associatedPlateType>
</FDSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, search result, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, search ID, desc:it is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device to speed up the next search-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search result, subType:string, desc:search result-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data).-->OK
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <progress>
    <!--ro, opt, int, range:[0,100]-->1
  </progress>
  <MatchList>
    <!--ro, opt, array, searched matched data list, subType:object-->
    <MatchElement>
      <!--ro, opt, object-->
      <FDID>
        <!--ro, opt, string, face picture library ID-->test
      </FDID>
      <thresholdValue>
        <!--ro, opt, int, range:[0,100]-->1
      </thresholdValue>
      <bornTime>
        <!--ro, opt, date, birthday of person in the face picture-->1970-01-01+08:00
      </bornTime>
      <name>
        <!--ro, req, string, name, range:[1,32], desc:name-->test
      </name>
      <sex>
        <!--ro, opt, enum, gender, subType:string, desc:male; female;-->male
      </sex>
      <province>
        <!--ro, opt, string-->11
      </province>
      <city>
        <!--ro, opt, string, city, desc:city code of birth for the person in the face picture,the maximum size is 32 bytes-->11
      </city>
      <certificateType>
        <!--ro, opt, enum, certificate type, subType:string, desc:the max. size is 10 bytes,certificate type: "officerID"-officer ID,"ID"-identify card,passport,other-->officerID
      </certificateType>
      <certificateNumber>
        <!--ro, opt, string, certificate No., range:[8,32], desc:certificate No.,the max. size is 32 bytes-->test
      </certificateNumber>
      <picURL>
        <!--ro, req, string-->test
      </picURL>
      <PID>
        <!--ro, req, string, picture ID-->test
      </PID>
      <PersonInfoExtendList>
        <!--ro, opt, array, subType:object-->
        <PersonInfoExtend>
          <!--ro, opt, object, person extended information-->
          <id>
            <!--ro, req, int, serial No. of person custom extension information-->1
          </id>
          <enable>
            <!--ro, req, bool, enable person extended information or not-->true
          </enable>
          <name>
            <!--ro, opt, string, person tag extended information name, desc:person tag extended information name-->test
          </name>
          <value>
            <!--ro, opt, string, person tag extended information, desc:person tag extended information-->test
```

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```
          </value>
        </PersonInfoExtend>
      </PersonInfoExtendList>
      <similarity>
        <!--ro, opt, float, similarity, range:[0.0,100.0]-->0.0
      </similarity>
      <ModelingStatus>
        <!--ro, opt, object-->
        <FDID>
          <!--ro, opt, string, face picture library ID-->test
        </FDID>
        <PID>
          <!--ro, opt, string, picture ID-->test
        </PID>
        <name>
          <!--ro, req, string, picture name-->test
        </name>
        <status>
          <!--ro, req, enum, subType:string-->success
        </status>
        <reason>
          <!--ro, opt, enum, subType:string-->noface
        </reason>
        <customHumanID>
          <!--ro, opt, string-->123456789012345678901234567890l2
        </customHumanID>
        <customFaceLibID>
          <!--ro, opt, string-->123456789012345678901234567890l2
        </customFaceLibID>
        <modeData>
          <!--ro, opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64
method-->test
        </modeData>
      </ModelingStatus>
      <isNoSaveFDPicture>
        <!--ro, opt, bool-->true
      </isNoSaveFDPicture>
      <humanId>
        <!--ro, opt, string-->test
      </humanId>
      <modelStatus>
        <!--ro, opt, enum, subType:string-->modeling
      </modelStatus>
      <FMCount>
        <!--ro, opt, int-->1
      </FMCount>
      <importTime>
        <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
      </importTime>
      <UUPID>
        <!--ro, opt, string-->test
      </UUPID>
      <occurrences>
        <!--ro, opt, int-->1
      </occurrences>
      <IMSIList>
        <!--ro, opt, array, subType:object-->
        <IMSIItem>
          <!--ro, opt, object-->
          <IMSI>
            <!--ro, opt, string-->123456789012345
          </IMSI>
          <count>
            <!--ro, opt, int-->10
          </count>
          <lastFMTime>
            <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
          </lastFMTime>
        </IMSIItem>
      </IMSIList>
      <lastFMTime>
        <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
      </lastFMTime>
      <faceScore>
        <!--ro, opt, int, range:[-1,100]-->1
      </faceScore>
      <pictureStatus>
        <!--ro, opt, enum, subType:string-->normal
      </pictureStatus>
      <userType>
        <!--ro, opt, enum, subType:string-->normal
      </userType>
      <associatedPlateType>
        <!--ro, opt, enum, subType:string-->greenChannel
      </associatedPlateType>
      <contrastStatus>
        <!--ro, opt, enum, subType:string-->contrast
      </contrastStatus>
    </MatchElement>
  </MatchList>
</FDSearchResult>
```

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19.8.3.13 Specify a face picture library to start model the face pictures in a batch

Request URL

GET /ISAPI/Intelligent/FDLib/manualModeling?range=<range>&FDID=<FDID>&FDType=<FDType>&identityKey=<identityKey>&pictureSaveMode=<pictureSaveMode>

Query Parameter

Parameter Name	Parameter Type	Description
range	string	Control data range, and when the range is unmodeled, it indicates that unmodeled image data in the face database should be uniformly modeled; when the range is all, it indicates that all images stored in the device (regardless of whether they have modeling data) need to be remodeled, for example, after a device upgrade (algorithm library upgrade), the modeling data needs to be updated.
FDID	string	Face picture library ID
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.
identityKey	string	When importing a face library, an error "faceLibraryNumError" occurs to prompt the user whether to overwrite. If the overwrite import mode is selected, and the face library has reached its maximum capacity, the subsequently imported faces will overwrite the earliest imported face information.
pictureSaveMode	string	When pictureSaveMode is noSave, it indicates that the uploaded original image will not be saved. When isSupportNoSaveUploadPicture is returned in /ISAPI/Intelligent/channels//faceContrast/capabilities, it represents the operation of not saving the uploaded original image. After the device modeling is successful, the uploaded original image will be deleted.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <id>
    <!--ro, opt, string, task ID, desc:task ID-->test
  </id>
</ResponseStatus>
```

19.8.3.14 Search control results of discrete modeling

Request URL

POST /ISAPI/Intelligent/FDLib/manualModelingRangeTask?identityKey=<identityKey>&pictureSaveMode=

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<pictureSaveMode>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
pictureSaveMode	string	When pictureSaveMode is noSave, it indicates that the uploaded original image will not be saved. When isSupportNoSaveUploadPicture is returned in /ISAPI/Intelligent/channels//faceContrast/capabilities, it represents the operation of not saving the uploaded original image. After the device modeling is successful, the uploaded original image will be deleted.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ModelingRangeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <ModelingRange>
    <!--opt, object-->
    <FDID>
      <!--req, string, face picture library ID-->test
    </FDID>
    <PID>
      <!--req, string, picture ID-->test
    </PID>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID, desc:custom face picture library ID-->test
    </customFaceLibID>
  </ModelingRange>
</ModelingRangeList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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19.8.3.15 Get the modeling status of a face picture in a specific face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/modelingStatus?FDID=<FDID>&PID=<pictureID>&identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
pictureID	string	Picture ID
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ModelingStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <FDID>
    <!--ro, opt, string-->test
  </FDID>
  <PID>
    <!--ro, opt, string-->test
  </PID>
  <name>
    <!--ro, req, string-->test
  </name>
  <status>
    <!--ro, req, enum, subType:string-->success
  </status>
  <reason>
    <!--ro, opt, enum, subType:string-->noface
  </reason>
  <customHumanID>
    <!--ro, opt, string-->123456789012345678901234567890l2
  </customHumanID>
  <customFaceLibID>
    <!--ro, opt, string-->123456789012345678901234567890l2
  </customFaceLibID>
  <modeData>
    <!--ro, opt, string-->test
  </modeData>
</ModelingStatus>
```

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19.8.3.16 Search for face pictures by modeling status

Request URL

POST /ISAPI/Intelligent/FDLib/modelingStatus?status=<status>&identityKey=<identityKey>&FDType=<FDType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
status	string	Modeling status: "success" (search for successfully modeled data), "failed" (search for failed modeling data), "none" (search for unmodeled data).
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.
terminalNo	string	Terminal No., which is used for information release terminal and can be obtained from the TerminalInfo->id field in the response message of POST /ISAPI/Publish/TerminalMgr/terminalSearch.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDModelingStatusSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, undefined, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of search results this time-->1
  </maxResults>
  <CondList>
    <!--opt, array, subType:object-->
    <Cond>
      <!--opt, object-->
      <FDID>
        <!--opt, string-->test
      </FDID>
    </Cond>
  </CondList>
  <taskID>
    <!--opt, string, task ID, desc:taskID and CondList cannot exist at the same time, and taskID is in prior-->test
  </taskID>
</FDModelingStatusSearchDescription>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FDModelingStatusSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, undefined, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <responseStatus>
    <!--ro, req, bool, search result-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data)-->OK
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <ModelingStatusList>
    <!--ro, opt, array, subType:object-->
    <ModelingStatus>
      <!--ro, opt, object-->
      <FDID>
        <!--ro, opt, string-->test
      </FDID>
      <PID>
        <!--ro, opt, string-->test
      </PID>
      <name>
        <!--ro, req, string-->test
      </name>
      <status>
        <!--ro, req, enum, subType:string-->success
      </status>
      <reason>
        <!--ro, opt, enum, subType:string-->noface
      </reason>
      <customHumanID>
        <!--ro, opt, string-->123456789012345678901234567890 12
      </customHumanID>
      <customFaceLibID>
        <!--ro, opt, string-->123456789012345678901234567890 12
      </customFaceLibID>
      <modeData>
        <!--ro, opt, string-->test
      </modeData>
    </ModelingStatus>
  </ModelingStatusList>
</FDModelingStatusSearchResult>
```

19.8.3.17 Search for all modeling statuses of the device

Request URL

POST /ISAPI/Intelligent/FDLib/modelingStatusRange?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ModelingRangeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <ModelingRange>
    <!--opt, object-->
    <FDID>
      <!--req, string-->test
    </FDID>
    <PID>
      <!--req, string-->test
    </PID>
    <customFaceLibID>
      <!--opt, string-->test
    </customFaceLibID>
  </ModelingRange>
</ModelingRangeList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ModelingStatusList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <ModelingStatus>
    <!--ro, opt, object-->
    <FDID>
      <!--ro, opt, string-->test
    </FDID>
    <PID>
      <!--ro, opt, string-->test
    </PID>
    <name>
      <!--ro, req, string-->test
    </name>
    <status>
      <!--ro, req, enum, subType:string-->success
    </status>
    <reason>
      <!--ro, opt, enum, subType:string-->noface
    </reason>
    <customHumanID>
      <!--ro, opt, string-->123456789012345678901234567890123
    </customHumanID>
    <customFaceLibID>
      <!--ro, opt, string-->123456789012345678901234567890123
    </customFaceLibID>
    <modeData>
      <!--ro, opt, string-->test
    </modeData>
  </ModelingStatus>
</ModelingStatusList>
```

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19.8.3.18 Import face pictures and the additional information to a specific face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/pictureUpload?identityKey=<identityKey>&type=<type>&dataType=<dataType>&pictureSaveMode=<pictureSaveMode>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	--
type	string	--
dataType	string	--
pictureSaveMode	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PictureUploadData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, picture information, attr:version{req, string, protocolVersion}-->
  <FDID>
    <!--req, string, a unique ID of face comparison picture library-->test
```

```
  </FDID>
  <customFaceLibID>
    <!--opt, string, custom face picture library ID, desc:custom face picture library ID-->test
  </customFaceLibID>
  <FaceAppendData>
    <!--opt, object, face picture information-->
    <bornTime>
      <!--opt, string, date of birth-->test
    </bornTime>
    <name>
      <!--req, string, name, range:[1,32]-->test
    </name>
    <sex>
      <!--opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
    </sex>
    <province>
      <!--opt, string, province, desc:see the data dictionary for details-->11
    </province>
    <city>
      <!--opt, string, city, desc:see the data dictionary for details-->11
    </city>
    <certificateType>
      <!--opt, enum, certificate type, subType:string, desc:"officerID"-officer ID,"ID"-identify card,"passportID"-passport,"other"-other certificate--
>officerID
    </certificateType>
    <certificateNumber>
      <!--opt, string, certificate No., range:[8,32], desc:certificate No.-->test
    </certificateNumber>
    <PersonInfoExtendList>
      <!--opt, array, custom extension information of person, subType:object-->
      <PersonInfoExtend>
        <!--opt, object, person information-->
        <id>
          <!--req, int, serial No. of person custom extension information, desc:serial No. of person custom extension information-->1
        </id>
        <enable>
          <!--req, bool, enable person extended information or not-->true
        </enable>
        <name>
          <!--opt, string, person tag extended information name-->test
        </name>
        <value>
          <!--opt, string, person tag extended information-->test
        </value>
      </PersonInfoExtend>
    </PersonInfoExtendList>
    <caseInfo>
      <!--opt, string, case information, range:[1,64]-->test
    </caseInfo>
    <customHumanID>
      <!--opt, string, custom person ID, range:[1,64]-->test
    </customHumanID>
    <phoneNumber>
      <!--opt, string, phone number, range:[1,64]-->test
    </phoneNumber>
    <RegionCoordinatesList>
      <!--opt, array, face picture target, subType:object, range:[0,4], desc:face picture target-->
      <RegionCoordinates>
        <!--opt, object, coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <modeldata>
      <!--opt, string, modeling data, desc:modeling data-->test
    </modeldata>
    <IMSIList>
      <!--opt, array, subType:object-->
      <IMSI>
        <!--opt, string, range:[1,15]-->123456789012345
      </IMSI>
    </IMSIList>
    <userType>
      <!--opt, enum, subType:string-->normal
    </userType>
    <associatedPlateType>
      <!--opt, enum, subType:string-->greenChannel
    </associatedPlateType>
  </FaceAppendData>
  <picURL>
    <!--opt, string, picture URL-->test
  </picURL>
</PictureUploadData>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
PictureUploadData	[Message content]	application/xml	--	--	--
importImage	[Binary picture data]	image/jpeg		file.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<MaskInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, information, attr:version{req, string, protocolVersion}, desc:information-->
  <PID>
    <!--ro, opt, string, picture ID, desc:picture ID-->test
  </PID>
  <picURL>
    <!--ro, opt, picurl, picture URL generated by device after importing pictures, desc:picture URL generated by device after importing pictures--
>http://127.0.0.1
  </picURL>
  <isNoSavePDPicture>
    <!--ro, opt, bool-->true
  </isNoSavePDPicture>
  <FaceGradingDetails>
    <!--ro, opt, object, face score (face picture quality), desc:face score (face picture quality)-->
    <featureConfidence>
      <!--ro, opt, float, [0,1], range:[0.000,1.000], desc:[0,1]-->0.000
    </featureConfidence>
    <faceRect>
      <!--ro, opt, enum, rectangle frame for human face, subType:int, desc:rectangle frame for human face-->1
    </faceRect>
    <pupillaryDistance>
      <!--ro, opt, int, pupil distance, desc:pupil distance-->1
    </pupillaryDistance>
    <colorConfidence>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </colorConfidence>
    <grayScale>
      <!--ro, opt, int, grayscale value, range:[0,255]-->1
    </grayScale>
    <grayMean>
      <!--ro, opt, int, range:[0,255]-->1
    </grayMean>
    <grayVariance>
      <!--ro, opt, int, range:[0,128]-->1
    </grayVariance>
    <clearityScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </clearityScore>
    <pitchAngle>
      <!--ro, opt, float, range:[-90.000,90.000]-->0.000
    </pitchAngle>
    <yawAngle>
      <!--ro, opt, float, range:[-90.000,90.000]-->0.000
    </yawAngle>
    <rollAngle>
      <!--ro, opt, float, face roll angle, range:[-90.000,90.000], desc:face roll angle-->0.000
    </rollAngle>
    <poseConfidence>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </poseConfidence>
    <frontalScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </frontalScore>
    <visibleScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </visibleScore>
    <totalScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </totalScore>
    <detectConfidence>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </detectConfidence>
    <invisibleClass>
      <!--ro, opt, array, subType:object, range:[1,8]-->
    </invisibleClass>
    <faceClass>
      <!--ro, opt, enum, face picture type, subType:int, desc:face picture type-->0
    </faceClass>
    <illuminationClass>
      <!--ro, opt, enum, subType:int-->0
    </illuminationClass>
    <leftEyeOpen>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </leftEyeOpen>
    <rightEyeOpen>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </rightEyeOpen>
    <mouthOpen>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </mouthOpen>
    <pictureStatus>
      <!--ro, opt, enum, subType:string-->normal
    </pictureStatus>
  </FaceGradingDetails>
</MaskInfo>
```

VISION AND SECURITY SYSTEM CO.,LTD. dejijin@gmail.com

19.8.3.19 Get the capacity of face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/surplusCapacity?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "SurplusCapacity": {
    /*ro, req, object*/
        "FDPicturesNumber":  100,
        /*ro, req, int*/
        "capacity":  100
        /*ro, req, int*/
    }
}
```

19.8.3.20 Set the face picture library parameters

Request URL

PUT /ISAPI/Intelligent/FDLib?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfgList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, face picture library information list, subType:object, attr:version{req, string, protocolVersion}-->
  <FDLibBaseCfg>
    <!--opt, object, face picture library information-->
    <id>
      <!--opt, int, range:[1,15]-->1
    </id>
    <FDID>
      <!--opt, string, unique ID of face picture library-->test
    </FDID>
    <name>
      <!--req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->50
    </thresholdValue>
    <customInfo>
      <!--opt, string, additional information of face picture library, range:[1,128]-->test
    </customInfo>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID, range:[1,64]-->test
    </customFaceLibID>
    <faceLibType>
      <!--ro, opt, enum, face picture library type, subType:string, desc:"ordinary" (normal library), "private" (private library), "stranger" (stranger
library), "allowable" (allowlist library), "privacyMask" (privacy mask library)-->allowable
    </faceLibType>
    <autoUpdata>
      <!--opt, bool-->true
    </autoUpdata>
    <qualityThreshold>
      <!--opt, int, range:[0,100]-->50
    </qualityThreshold>
    <importFDThreshold>
      <!--opt, int, range:[0,100]-->40
    </importFDThreshold>
    <Age>
      <!--opt, object-->
      <judgementCycleType>
        <!--req, enum, subType:string-->day
      </judgementCycleType>
      <statisticalCycleType>
        <!--ro, req, enum, subType:string-->month
      </statisticalCycleType>
      <statisticalCycle>
        <!--req, int, range:[0,99999]-->1
      </statisticalCycle>
      <minFMCount>
        <!--req, int, range:[0,99999]-->1
      </minFMCount>
      <FMCountEnabled>
        <!--opt, bool-->true
      </FMCountEnabled>
      <FMCount>
        <!--opt, int, range:[0,99999]-->1
      </FMCount>
      <backgoundPicThreshold>
        <!--opt, int, range:[1,100]-->90
      </backgoundPicThreshold>
      <judgementCycle>
        <!--opt, int, range:[1,365], unit:d, dep:and,{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
      </judgementCycle>
    </Age>
    <FaceCounting>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <uploadType>
        <!--ro, req, enum, subType:string-->noUpload
      </uploadType>
    </FaceCounting>
    <faceMonitor>
      <!--ro, opt, bool-->true
    </faceMonitor>
    <monitorFaceNum>
      <!--ro, opt, int-->1
    </monitorFaceNum>
    <customNoConfig>
      <!--ro, opt, bool-->true
    </customNoConfig>
    <uploadID>
      <!--opt, string, range:[1,128], dep:and,{$.FDLibBaseCfglist.FDLibBaseCfg.autoUpdata,eq,true}-->test
    </uploadID>
  </FDLibBaseCfg>
</FDLibBaseCfgList>
```

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VISION AND SECURITY SYSTEM CO.,LTD.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.8.3.21 Create a face picture library

Request URL

POST /ISAPI/Intelligent/FDLib?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64 characters. This parameter stands for the identity key of the operation platform. The device can know the operator's identity by checking this parameter in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<CreateFDLibList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, create the list of face picture library, subType:object, attr:version{req, string, protocolVersion}-->
  <CreateFDLib>
    <!--opt, object, create a face picture library-->
    <id>
      <!--opt, int, sub item No. in the list; it increases from 1, range:[1,15]-->1
    </id>
    <name>
      <!--req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->50
    </thresholdValue>
    <customInfo>
      <!--opt, string, custom information, range:[1,128]-->test
    </customInfo>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID, range:[1,64]-->test
    </customFaceLibID>
    <autoUpdata>
      <!--opt, bool, whether to enable auto update of background picture in face picture library-->true
    </autoUpdata>
    <qualityThreshold>
      <!--opt, int, picture quality threshold, range:[0,100]-->50
    </qualityThreshold>
    <importFDThreshold>
      <!--opt, int, inbound similarity threshold, range:[0,100]-->40
    </importFDThreshold>
    <Age>
      <!--opt, object, the aggregate aging parameters configuration of face picture library; it is valid when autoUpdata is true-->
      <judgementCycleType>
        <!--req, enum, the time type for aging judgement cycle of the aggregate data, subType:string, desc:"day", "week", "month", "duration" (judge
according to actual duration)-->day
      </judgementCycleType>
      <statisticalCycleType>
        <!--req, enum, statistics cycle type of clustering matches in face picture library, subType:string, desc:"day", "week", "month"-->month
      </statisticalCycleType>
      <statisticalCycle>
        <!--req, int, statistics cycle of face picture matches, range:[0,99999]-->1
      </statisticalCycle>
      <minFMCount>
        <!--req, int, min. number of face picture matches within statistics cycle, range:[0,99999]-->1
      </minFMCount>
      <FMCountEnabled>
        <!--opt, bool, whether to enable total number of face picture matches-->true
      </FMCountEnabled>
      <FMCount>
        <!--opt, int, total number of face picture matches, range:[0,99999]-->1
      </FMCount>
      <backgroundPicThreshold>
        <!--opt, int, range:[1,100]-->90
      </backgroundPicThreshold>
      <judgementCycle>
        <!--opt, int, evaluation cycle range of clustering data aging in face picture library, range:[1,365], unit:d, dep:and,
{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->60
      </judgementCycle>
      <agingExecDataTime>
        <!--opt, time-->00:00:00+08:00
      </agingExecDataTime>
      <automaticClearEnabled>
        <!--opt, bool-->true
      </automaticClearEnabled>
    </Age>
    <FaceCounting>
      <!--opt, object, the configuration for customer traffic (face) deduplication of face picture comparison-->
      <enabled>
        <!--req, bool, whether to enable customer traffic (face) deduplication-->true
      </enabled>
      <uploadType>
        <!--req, enum, uploading mode after the customer traffic (face) deduplication is enabled, subType:string, desc:"out_repetition" (uploading outside
the deduplication period); "noUpload" (not uploading outside the deduplication period)-->noUpload
      </uploadType>
    </FaceCounting>
    <faceLibType>
      <!--opt, enum, face picture library type, subType:string, desc:if this node is not returned, a normal library is created by default; "ordinary"
(normal library); "privacyMask" (privacy mask library)-->ordinary
    </faceLibType>
  </CreateFDLib>
</CreateFDLibList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, create the result list of face picture library, subType:object, attr:version{req, string, protocolVersion}-->
  <FDLibInfo>
    <!--ro, opt, object, face picture library information-->
    <id>
      <!--ro, req, int, sub item No. in the list; it increases from 1 and corresponds to the id of CreateFDLibList-->1
    </id>
    <name>
      <!--ro, req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <FDID>
      <!--ro, req, string, returned face picture library ID when it created, the library ID of the same type is unique, range:[1,64]-->test
    </FDID>
    <statusCode>
      <!--ro, opt, enum, status code; it is returned when creating a face picture library failed, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3
(Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->1
    </statusCode>
    <subStatusCode>
      <!--ro, opt, string, sub status code; it is returned when creating a face picture library failed-->existSameNameFacelib
    </subStatusCode>
  </FDLibInfo>
</FDLibInfoList>
```

19.8.3.22 Get the information, including library ID, library type, name, and custom information, of all face picture libraries

Request URL

GET /ISAPI/Intelligent/FDLib?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64 characters.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfgList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FDLibBaseCfg>
    <!--ro, opt, object, face picture library information-->
    <id>
      <!--ro, opt, int, range:[1,15]-->1
    </id>
    <FDID>
      <!--ro, opt, string, face picture library ID-->test
    </FDID>
    <name>
      <!--ro, req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <thresholdValue>
      <!--ro, opt, int, detection threshold, range:[0,100]-->50
    </thresholdValue>
    <customInfo>
      <!--ro, opt, string, custom information, range:[1,128]-->test
    </customInfo>
    <customFaceLibID>
      <!--ro, opt, string, custom face picture library ID, range:[1,64]-->test
    </customFaceLibID>
    <faceLibType>
      <!--ro, opt, enum, face picture library type, subType:string, desc:face picture library type-->ordinary
    </faceLibType>
    <autoUpdata>
      <!--ro, opt, bool-->true
    </autoUpdata>
    <qualityThreshold>
      <!--ro, opt, int, range:[0,100]-->50
    </qualityThreshold>
    <importFDThreshold>
      <!--ro, opt, int, range:[0,100]-->40
    </importFDThreshold>
    <Age>
      <!--ro, opt, object-->
      <judgementCycleType>
        <!--ro, req, enum, subType:string-->day
      </judgementCycleType>
      <statisticalCycleType>
        <!--ro, req, enum, "day", "week", "month", subType:string, desc:"day", "week", "month"-->month
      </statisticalCycleType>
      <statisticalCycle>
        <!--ro, req, int, range:[0,99999]-->1
      </statisticalCycle>
      <minFMCount>
        <!--ro, req, int, range:[0,99999]-->1
      </minFMCount>
      <FMCountEnabled>
        <!--ro, opt, bool-->true
      </FMCountEnabled>
      <FMCount>
        <!--ro, opt, int, range:[0,99999]-->1
      </FMCount>
      <backgroundPicThreshold>
        <!--ro, opt, int, range:[1,100]-->90
      </backgroundPicThreshold>
      <judgementCycle>
        <!--ro, opt, int, range:[1,365], unit:d, dep:and,{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
      </judgementCycle>
    </Age>
    <FaceCounting>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <uploadType>
        <!--ro, req, enum, subType:string-->noUpload
      </uploadType>
      <faceSimilarity>
        <!--ro, opt, int, range:[0,100]-->40
      </faceSimilarity>
    </FaceCounting>
    <faceMonitor>
      <!--ro, opt, bool-->true
    </faceMonitor>
    <monitorFaceNum>
      <!--ro, opt, int-->1
    </monitorFaceNum>
    <customNoConfig>
      <!--ro, opt, bool-->true
    </customNoConfig>
    <uploadID>
      <!--ro, opt, string, range:[1,128], dep:and,{$.FDLibBaseCfglist.FDLibBaseCfg.autoUpdata,eq,true}-->test
    </uploadID>
  </FDLibBaseCfg>
</FDLibBaseCfgList>
```

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19.8.3.23 Export the package of face capture library attributes

Request URL

POST /ISAPI/SDT/Face/captureSearch/dataPackage

Query Parameter

None

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int*/
    "maxResults":  100,
    /*req, int*/
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*opt, datetime*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*opt, datetime*/
    "choiceChannel": [
    /*opt, array, subType:object*/
        {
            "channelID":  "test"
            /*opt, string*/
        }
    ],
    "ageGroup":  "any",
    /*opt, enum, subType:string*/
    "gender":  "male",
    /*opt, enum, subType:string*/
    "glasses":  "yes",
    /*opt, enum, subType:string*/
    "smile":  "yes",
    /*opt, enum, subType:string*/
    "isInLibrary":  "yes",
    /*opt, enum, subType:string*/
    "thermometryUnit":  "celsius",
    /*opt, enum, subType:string*/
    "currTemperature":  38.0,
    /*opt, float, range:[-20.0,150.0]*/
    "temperatureCondition":  "normal",
    /*opt, enum, subType:string*/
    "FPID":  " "
    /*opt, string*/
}
```

Response Message

Binary Data

19.9 Traffic Control

19.9.1 Traffic Enforcement of Vehicles

19.9.1.1 Get the traffic service capability

Request URL

GET /ISAPI/Traffic/capabilities

Query Parameter

None

Request Message

None

Response Message

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dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>
<TrafficCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{opt, string, protocolVersion}-->
  <plateCap>
    <!--ro, opt, object, capability of license plate recognition-->
    <isSupportPlateList>
      <!--ro, opt, bool, whether it supports importing and exporting the license plate blocklist and allowlist, desc:whether it supports importing and exporting the license plate blocklist and allowlist-->true
    </isSupportPlateList>
    <plateListNum>
      <!--ro, opt, int, the capacity of license plate blocklists, desc:maximum number of imported blocklists and allowlists supported by the device-->64
    </plateListNum>
    <plateMaskLen>
      <!--ro, opt, int, the length of license plate characters, desc:maximum length of license plate characters supported by the device-->64
    </plateMaskLen>
    <isSupportLicenseModify>
      <!--ro, opt, bool-->true
    </isSupportLicenseModify>
    <isSupportLPAuditDataDelete>
      <!--ro, opt, bool-->true
    </isSupportLPAuditDataDelete>
  </plateCap>
  <isNotSupportOverlaps>
    <!--ro, opt, bool, whether it supports text overlay, desc:whether it does not support text overlay-->true
  </isNotSupportOverlaps>
  <isSupportVehicleDetection>
    <!--ro, opt, bool, whether it supports vehicle detection, desc:whether it supports vehicle recognition-->true
  </isSupportVehicleDetection>
  <isSupportHVTVehicleDetection>
    <!--ro, opt, bool, whether it supports HVT vehicle detection, desc:whether it supports mixed traffic detection-->true
  </isSupportHVTVehicleDetection>
</TrafficCap>
```

19.9.1.2 Search for the status of manual traffic law enforcement

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/manualItsCapStatus

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ManualItsCapStatus xmlns="http://www.isapi.org/ver20/XMLSchema">
  <!--ro, opt, object-->
  <manualItsCapID>
    <!--ro, req, int-->0
  </manualItsCapID>
</ManualItsCapStatus>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ManualItsCapStatusResult xmlns="http://www.isapi.org/ver20/XMLSchema">
  <!--ro, opt, object-->
  <errorCode>
    <!--ro, opt, int-->1
  </errorCode>
  <errorStr>
    <!--ro, opt, string-->test
  </errorStr>
  <manualItsCapID>
    <!--ro, req, int-->0
  </manualItsCapID>
</ManualItsCapStatusResult>
```

19.10 Perimeter Intrusion Detection

19.10.1 Intrusion Detection

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19.10.1.1 Set intrusion detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/fieldDetections

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->fieldDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FieldDetectionScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.10.1.2 Get arming schedules of all channels for intrusion detection

Request URL

GET /ISAPI/Event/schedules/fieldDetections

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, configuration of intrusion detection arming schedules-->
    <id>
      <!--ro, req, string, ID, desc:format: event type-channel No.-->test
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->fieldDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, list of arming schedules, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming schedules-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FieldDetectionScheduleList>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

19.10.1.3 Set the arming schedule of intrusion detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/fieldDetections/fieldDetection-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->格式：事件类型-通道号
  </id>
  <eventType>
    <!--opt, string-->fieldDetection
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

19.10.1.4 Get the arming schedule of intrusion detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/fieldDetections/fieldDetection-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, opt, string-->fieldDetection
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.10.1.5 Get arming schedules of all radar channels for intrusion detection

Request URL

GET /ISAPI/Event/schedules/radarFieldDetections

Query Parameter

None

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<RadarFieldDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, list of arming schedules, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule information-->
    <id>
      <!--ro, req, string, ID, desc:format: event type-channel No.-->radarFieldDetection-1
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->radarFieldDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, list of arming schedules, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, holiday arming schedules, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RadarFieldDetectionScheduleList>
```

19.10.1.6 Set arming schedules of all radar channels for intrusion detection

Request URL

PUT /ISAPI/Event/schedules/radarFieldDetections

Query Parameter

None

Request Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<RadarFieldDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, refer to the message XML_Schedule for details, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->radarFieldDetection-1
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->radarFieldDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming time list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RadarFieldDetectionScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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VISION AND SECURITY SYSTEM CO.,LTD.

19.10.1.7 Set the arming schedule of radar intrusion detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/radarFieldDetections/radarFieldDetection-<radarChannelID>

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->radarFieldDetection-1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->radarFieldDetection
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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VISION AND SECURITY SYSTEM CO.,LTD.

19.10.1.8 Get the arming schedule of radar intrusion detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/radarFieldDetections/radarFieldDetection-<radarChannelID>

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->radarFieldDetection-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->radarFieldDetection
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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19.10.1.9 Set the linkage triggering parameters of intrusion detection event for a specified channel

Request URL

PUT /ISAPI/Event/triggers/FieldDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <notificationMethod>
      <!--req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object-->
      <ptzChannelID>
        <!--req, string-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object-->
      <whiteLightDurationTime>
        <!--opt, int, range:[1,60]-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <AudioAction>
      <!--opt, object-->
      <audioID>
        <!--req, int-->1
      </audioID>
      <alarmTimes>
        <!--req, int, range:[0,9]-->1
      </alarmTimes>
    </AudioAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

19.10.1.10 Get the linkage triggering parameters of intrusion detection event for a specified channel

Request URL

GET /ISAPI/Event/triggers/FieldDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60]-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <AudioAction>
      <!--ro, opt, object-->
      <audioID>
        <!--ro, req, int-->1
      </audioID>
      <alarmTimes>
        <!--ro, req, int, range:[0,9]-->1
      </alarmTimes>
    </AudioAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

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19.10.1.11 Set the linkage parameters of intrusion detection event for a specified channel

Request URL

PUT /ISAPI/Event/triggers/FieldDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->fielddetection
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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VISION AND SECURITY SYSTEM CO.,LTD.

19.10.1.12 Get the linkage parameters of intrusion detection event for a specified channel

Request URL

GET /ISAPI/Event/triggers/FieldDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->fielddetection
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <SupplementLightAlarm>
        <!--ro, opt, object-->
        <durationTime>
          <!--ro, opt, int, range:[0,90], unit:s-->5
        </durationTime>
      </SupplementLightAlarm>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.10.1.13 Get the linkage parameters of radar intrusion detection event for a specified channel

Request URL

GET /ISAPI/Event/triggers/radarFieldDetection-<radarChannelID>

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->radarFieldDetection-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->radarFieldDetection
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.10.1.14 Set intrusion detection parameters of all channels

Request URL

PUT /ISAPI/Smart/FieldDetection

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<?xml version="1.0" encoding="UTF-8" ?>

<FieldDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FieldDetection>
    <!--opt, object-->
    <id>
      <!--req, string, channel No.-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <intelliBackSearch>
      <!--opt, bool, whether the device supports video dual-VCA-->true
    </intelliBackSearch>
    <startTriggerTime>
      <!--req, int, unit:ms-->1
    </startTriggerTime>
    <endTriggerTime>
      <!--req, int, unit:ms-->1
    </endTriggerTime>
    <normalizedScreenSize>
      <!--opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <minObjectSize>
      <!--opt, int, minimum pixels of each object-->1
    </minObjectSize>
    <maxObjectSize>
      <!--opt, int, maximum pixels of each object-->1
    </maxObjectSize>
    <FieldDetectionRegionList size="4">
      <!--opt, array, subType:object, attr:size{req, int}-->
      <FieldDetectionRegion>
        <!--opt, object, intrusion region-->
        <id>
          <!--req, string-->1
        </id>
        <enabled>
          <!--req, bool, whether to enable the function-->true
        </enabled>
        <sensitivityLevel>
          <!--req, int, sensitivity, range:[0,100]-->1
        </sensitivityLevel>
        <timeThreshold>
          <!--req, int-->1
        </timeThreshold>
        <objectOccupation>
          <!--req, int, percentage-->1
        </objectOccupation>
        <detectionTarget>
          <!--opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <RegionCoordinatesList>
          <!--opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, x-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <humanMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (human)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--opt, enum, subType:string-->low
        </recordConfidence>
      </FieldDetectionRegion>
    </FieldDetectionRegionList>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </FieldDetection>
```


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```
</FieldDetectionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.1.15 Get intrusion detection parameters of all channels

Request URL

GET /ISAPI/Smart/FieldDetection

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, intrusion detection parameter list, subType:object, attr:version{req, string, protocolVersion}-->
  <FieldDetection>
    <!--ro, opt, object, intrusion detection parameters-->
    <id>
      <!--ro, req, string, channel No.-->test
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <intelliBackSearch>
      <!--ro, opt, bool, whether the device supports video dual-VCA-->true
    </intelliBackSearch>
    <startTriggerTime>
      <!--ro, req, int, unit:ms-->1
    </startTriggerTime>
    <endTriggerTime>
      <!--ro, req, int, unit:ms-->1
    </endTriggerTime>
    <normalizedScreenSize>
      <!--ro, opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <minObjectSize>
      <!--ro, opt, int, minimum pixels of each object-->1
    </minObjectSize>
    <maxObjectSize>
      <!--ro, opt, int, maximum pixels of each object-->1
    </maxObjectSize>
    <FieldDetectionRegionList size="4">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <FieldDetectionRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, string-->1
        </id>
        <enabled>
          <!--ro, req, bool, whether to enable the function-->true
        </enabled>
```

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```
        <sensitivityLevel>
          <!--ro, req, int, sensitivity-->1
        </sensitivityLevel>
        <timeThreshold>
          <!--ro, req, int, time threshold-->1
        </timeThreshold>
        <objectOccupation>
          <!--ro, req, int, percentage-->1
        </objectOccupation>
        <detectionTarget>
          <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <RegionCoordinatesList>
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <humanMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--ro, opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--ro, opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--ro, opt, enum, subType:string-->low
        </recordConfidence>
      </FieldDetectionRegion>
    </FieldDetectionRegionList>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </FieldDetection>
</FieldDetectionList>
```

19.10.1.16 Get intrusion detection parameters by video input channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, intrusion detection, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <intelliBackSearch>
    <!--ro, opt, bool, whether the device supports video dual-VCA-->true
  </intelliBackSearch>
  <startTriggerTime>
    <!--ro, req, int, unit: milliseconds, unit:ms-->1
  </startTriggerTime>
  <endTriggerTime>
    <!--ro, req, int, unit: milliseconds, unit:ms-->1
  </endTriggerTime>
```

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```
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <minObjectSize>
    <!--ro, opt, int, minimum pixels of each object-->1
  </minObjectSize>
  <maxObjectSize>
    <!--ro, opt, int, maximum pixels of each object-->1
  </maxObjectSize>
  <FieldDetectionRegionList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <FieldDetectionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, ID-->1
      </id>
      <enabled>
        <!--ro, opt, bool, whether to enable the function, desc:whether to enable the function-->true
      </enabled>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity-->1
      </sensitivityLevel>
      <timeThreshold>
        <!--ro, req, int, time threshold-->1
      </timeThreshold>
      <objectOccupation>
        <!--ro, req, int, percentage-->1
      </objectOccupation>
      <detectionTarget>
        <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
      </detectionTarget>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <humanMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
      </vehicleMisinfoFilterEnabled>
      <priority>
        <!--ro, opt, enum, priority, subType:string, desc:priority-->low
      </priority>
      <alarmConfidence>
        <!--ro, opt, enum, subType:string-->low
      </alarmConfidence>
      <recordConfidence>
        <!--ro, opt, enum, subType:string-->low
      </recordConfidence>
      <ruleName>
        <!--ro, opt, string, range:[1,64]-->test
      </ruleName>
      <detectionTargetProperty>
        <!--ro, opt, object-->
        <human>
          <!--ro, opt, object, dep:or,{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,all},
{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,human}-->
          <propertyList>
            <!--ro, opt, array, subType:object-->
            <property>
              <!--ro, opt, enum, subType:string-->nonMotor
            </property>
          </propertyList>
        </human>
        <animal>
          <!--ro, opt, object, dep:or,{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,all},
{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,animal}-->
          <propertyList>
            <!--ro, opt, array, subType:object-->
            <property>
              <!--ro, opt, enum, subType:string-->large
            </property>
          </propertyList>
        </animal>
      </detectionTargetProperty>
    </FieldDetectionRegion>
  </FieldDetectionRegionList>
  <humanMisinfoFilterEnabled>
```

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```
    <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
  </vehicleMisinfoFilterEnabled>
  <contAlarmForStaticTargetEnabled>
    <!--ro, opt, bool-->true
  </contAlarmForStaticTargetEnabled>
  <targetAlarmInterval>
    <!--ro, opt, int, range:[1,5], step:1, unit:s, dep:and,{$.FieldDetection.targetAlarmInterval,eq,true}-->1
  </targetAlarmInterval>
  <stackTarget>
    <!--ro, opt, bool-->true
  </stackTarget>
  <targetSensitivityLevel>
    <!--ro, opt, int, range:[0,100]-->50
  </targetSensitivityLevel>
</FieldDetection>
```

19.10.1.17 Set intrusion detection parameters by channel

Request URL

PUT /ISAPI/Smart/FieldDetection/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, intrusion detection parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <intelliBackSearch>
    <!--opt, bool, whether the device supports video dual-VCA-->true
  </intelliBackSearch>
  <startTriggerTime>
    <!--req, int, trigger start time, unit:ms-->1
  </startTriggerTime>
  <endTriggerTime>
    <!--req, int, trigger end time, unit:ms-->1
  </endTriggerTime>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <minObjectSize>
    <!--opt, int, minimum pixels of each object-->1
  </minObjectSize>
  <maxObjectSize>
    <!--opt, int, maximum pixels of each object-->1
  </maxObjectSize>
  <FieldDetectionRegionList size="4">
    <!--opt, array, intrusion detection region list, subType:object, attr:size{req, int}-->
    <FieldDetectionRegion>
      <!--opt, object, intrusion detection region-->
      <id>
        <!--req, string, region ID-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <sensitivityLevel>
        <!--req, int, sensitivity-->1
      </sensitivityLevel>
      <timeThreshold>
        <!--req, int, time threshold-->1
      </timeThreshold>
      <objectOccupation>
        <!--req, int, percentage-->1
      </objectOccupation>
      <detectionTarget>
```

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```
        <!--opt, enum, detection target, subType:string-->test
      </detectionTarget>
      <RegionCoordinatesList>
        <!--opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <humanMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
      </vehicleMisinfoFilterEnabled>
      <priority>
        <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
      </priority>
      <alarmConfidence>
        <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </alarmConfidence>
      <recordConfidence>
        <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </recordConfidence>
    </FieldDetectionRegion>
  </FieldDetectionRegionList>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
</FieldDetection>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.10.1.18 Get configuration capability of intrusion detection by channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<FieldDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <FieldDetectionRegionList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <FieldDetectionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, region ID-->1
      </id>
      <minRegionCoordinatesNum>
        <!--ro, opt, int-->1
      </minRegionCoordinatesNum>
      <maxRegionCoordinatesNum>
        <!--ro, opt, int-->1
      </maxRegionCoordinatesNum>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity-->1
      </sensitivityLevel>
      <timeThreshold min="0" max="60" def="0">
        <!--ro, req, int, time threshold, attr:min{req, int},max{req, int},def{req, int}-->1
      </timeThreshold>
      <detectionTarget opt="all,human,vehicle,animal,others" def="human">
        <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string},def{req, string}, desc:detection target-->all
      </detectionTarget>
      <alarmConfidence opt="low,mediumLow,mediumHigh,high" def="low">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
      </alarmConfidence>
    </FieldDetectionRegion>
  </FieldDetectionRegionList>
  <isSupportMultiScene opt="true,false">
    <!--ro, req, bool, whether it supports multiple scene detection, attr:opt{req, string}-->true
  </isSupportMultiScene>
  <isSupportHumanMisinfoFilter>
    <!--ro, opt, bool, whether it supports false human body alarm prevention, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will not be returned-->true
  </isSupportHumanMisinfoFilter>
  <isSupportVehicleMisinfoFilter>
    <!--ro, opt, bool, whether it supports false vehicle alarm prevention, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned-->true
  </isSupportVehicleMisinfoFilter>
  <isSupportTargetMultiSelect>
    <!--ro, opt, bool, whether it supports selecting multiple detection targets, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned-->true
  </isSupportTargetMultiSelect>
  <isSupportAllDayUpload>
    <!--ro, opt, bool, whether it supports uploading all-day event-->true
  </isSupportAllDayUpload>
  <contAlarmForStaticTargetEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </contAlarmForStaticTargetEnabled>
  <targetAlarmInterval min="1" max="5">
    <!--ro, opt, int, range:[1,5], step:1, unit:s, dep:and,{$.FieldDetection.targetAlarmInterval,eq,true}, attr:min{req, int},max{req, int}-->1
  </targetAlarmInterval>
</FieldDetection>
```

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19.10.1.19 Get parameters of all intrusion detection regions of a specific channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <channelEnabled>
    <!--ro, opt, bool-->true
  </channelEnabled>
  <FieldDetectionRegion>
    <!--ro, opt, object, intrusion detection-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, req, int, sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <timeThreshold>
      <!--ro, req, int, time threshold-->1
    </timeThreshold>
    <objectOccupation>
      <!--ro, req, int-->1
    </objectOccupation>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
    </detectionTarget>
    <RegionCoordinatesList>
      <!--ro, opt, array, list of area coordinates, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:confidence of alarm notification-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, confidence of video recording, subType:string, desc:confidence of video recording-->low
    </recordConfidence>
  </FieldDetectionRegion>
</FieldDetectionRegionList>
```

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19.10.1.20 Set parameters of all intrusion detection regions of a specified channel

Request URL

PUT /ISAPI/Smart/FieldDetection/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameters of all intrusion detection regions, subType:object, attr:version{req, string, protocolVersion}-->
  <FieldDetectionRegion>
    <!--opt, object, intrusion detection regions-->
    <id>
      <!--req, string, No.-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, sensitivity-->1
    </sensitivityLevel>
    <timeThreshold>
      <!--req, int, time threshold-->1
    </timeThreshold>
    <objectOccupation>
      <!--req, int, percentage-->1
    </objectOccupation>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <RegionCoordinatesList>
      <!--opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </FieldDetectionRegion>
</FieldDetectionRegionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.10.1.21 Get parameters of an intrusion detection region of a specific channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <sensitivityLevel>
    <!--ro, req, int, sensitivity, range:[0,100]-->1
  </sensitivityLevel>
  <timeThreshold>
    <!--ro, req, int, time threshold-->1
  </timeThreshold>
  <objectOccupation>
    <!--ro, req, int-->1
  </objectOccupation>
  <detectionTarget>
    <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
  </detectionTarget>
  <RegionCoordinatesList>
    <!--ro, opt, object, list of area coordinates-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates-->
      <positionX>
        <!--ro, req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:confidence of alarm notification: "low","mediumLow","mediumHigh","high","low"-
->low
  </alarmConfidence>
  <recordConfidence>
    <!--ro, opt, enum, confidence of video recording, subType:string, desc:confidence of recording video: "low","mediumLow","mediumHigh","high","low"-->low
  </recordConfidence>
  <ruleName>
    <!--ro, opt, string, range:[1,64]-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object, dep:or,{$.FieldDetectionRegion.detectionTarget,eq,all},{$.FieldDetectionRegion.detectionTarget,eq,human}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object, dep:or,{$.FieldDetectionRegion.detectionTarget,eq,all},{$.FieldDetectionRegion.detectionTarget,eq,animal}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</FieldDetectionRegion>
```

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19.10.1.22 Set parameters of an intrusion detection region of a specified channel

Request URL

PUT /ISAPI/Smart/FieldDetection/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of an intrusion detection region of a specified channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, No.-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <sensitivityLevel>
    <!--req, int, sensitivity-->1
  </sensitivityLevel>
  <timeThreshold>
    <!--req, int, time threshold-->1
  </timeThreshold>
  <objectOccupation>
    <!--req, int, percentage-->1
  </objectOccupation>
  <detectionTarget>
    <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all" is
mutually exclusive with "others"-->all
  </detectionTarget>
  <RegionCoordinatesList>
    <!--opt, object, region coordinate list-->
    <RegionCoordinates>
      <!--opt, object, region coordinates-->
      <positionX>
        <!--req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </alarmConfidence>
  <recordConfidence>
    <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </recordConfidence>
</FieldDetectionRegion>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.1.23 Get configuration capability of intrusion detection by channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/regions/<regionID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region ID

Request Message

None

Response Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, detection area ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <sensitivityLevel min="1" max="100">
    <!--ro, req, int, sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->1
  </sensitivityLevel>
  <timeThreshold min="1" max="100" def="1">
    <!--ro, req, int, range:[1,100], attr:min{req, int},max{req, int},def{req, int}-->1
  </timeThreshold>
  <objectOccupation min="1" max="100">
    <!--ro, req, int, range:[1,100], attr:min{req, int},max{req, int}-->1
  </objectOccupation>
  <maxRegionCoordinatesNum>
    <!--ro, req, int, the maximum number of coordinates-->1
  </maxRegionCoordinatesNum>
  <minRegionCoordinatesNum>
    <!--ro, req, int, the minimum number of coordinates-->1
  </minRegionCoordinatesNum>
  <detectionTarget opt="all,human,vehicle,animal,others">
    <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string}, desc:detection target-->all
  </detectionTarget>
  <RegionCoordinatesList>
    <!--ro, opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <humanMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, false alarm reduction of human body, attr:opt{req, string}-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, false alarm reduction of vehicle, attr:opt{req, string}-->true
  </vehicleMisinfoFilterEnabled>
  <priority opt="low,middle,high">
    <!--ro, opt, enum, priority, subType:string, attr:opt{req, string}, desc:priority: "low,middle,high"-->low
  </priority>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
  </isSupportAllTargetDetectionWebDisplay>
  <ruleName min="1" max="64">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object-->
      <propertyList size="1">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="nonMotor">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object-->
      <propertyList size="3">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="large,small,poultry">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</FieldDetectionRegion>
```

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19.10.1.24 Get radar intrusion detection capability

Request URL

GET /ISAPI/Smart/RadarChannels/<radarChannelID>/radarFieldDetection/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

None

Response Message

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```
{
    "RadarFieldDetectionCap": {
    /*ro, req, object, radar intrusion detection capability*/
        "enabled": [true, false],
        /*ro, req, array, whether to enable detection, subType:bool*/
        "intelliBackSearch":  true,
        /*ro, req, bool, whether it supports dual-VCA search*/
        "startTriggerTime": {
        /*ro, req, object, triggering start time*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "endTriggerTime": {
        /*ro, req, object, triggering end time*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  0
            /*ro, req, int, the maximum value*/
        },
        "minObjectSize": {
        /*ro, opt, object, the minimum number of pixels*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  0
            /*ro, req, int, the maximum value*/
        },
        "maxObjectSize": {
        /*ro, opt, object, the maximum number of pixels*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  0
            /*ro, req, int, the maximum value*/
        },
        "RegionListCap": {
        /*ro, req, object, detection region parameters*/
            "maxSize":  4,
            /*ro, req, int, the maximum number of detection regions*/
            "enabled": [true, false],
            /*ro, req, array, whether to enable detection, subType:bool*/
            "sensitivityLevel": {
            /*ro, req, object, detection sensitivity*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  100
                /*ro, req, int, the maximum value*/
            },
            "objectOccupation": {
            /*ro, req, object*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "RegionCap": {
            /*ro, opt, object*/
                "minSize":  3,
                /*ro, req, int, the minimum number of detection region edges*/
                "maxSize":  10,
                /*ro, req, int, the maximum number of detection regions*/
                "x": {
                /*ro, req, object, x-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, req, object, y-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                }
            }
        },
        "isSupportMultiScene":  true,
        /*ro, opt, bool, whether it supports multi-scene mode*/
        "humanMisinfoFilter": [true, false],
        /*ro, opt, array, whether it supports preventing false human body detection alarm, subType:bool*/
        "vehicleMisinfoFilter": [true, false],
        /*ro, opt, array, whether it supports preventing false vehicle detection alarm, subType:bool*/
        "isSupportTargetMultiSelect":  true,
        /*ro, opt, bool, whether it supports multi-target detection*/
        "isSupportAllDayUpload":  true
        /*ro, opt, bool, whether it supports uploading all-day events*/
    }
}
```

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19.10.1.25 Get parameters of a radar detection region

Request URL

GET /ISAPI/Smart/RadarChannels/<radarChannelID>/radarFieldDetection/regions/<regionID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--
regionID	string	--

Request Message

None

Response Message

```
{
    "RegionInfo": {
    /*ro, opt, object, region information*/
        "regionID":  1,
        /*ro, req, int, region ID*/
        "enabled":  true,
        /*ro, req, bool, whether to enable the function*/
        "sensitivityLevel":  50,
        /*ro, req, int, region detection sensitivity, range:[1,100]*/
        "objectOccupation":  1,
        /*ro, req, int, object occupation rate*/
        "Region": [
        /*ro, opt, array, area, subType:object, range:[3,10], desc:the origin is the lower-left corner of the screen*/
            {
                "x":  0.120,
                /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                "y":  0.120
                /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
            }
        ]
    }
}
```

19.10.1.26 Set parameters of all radar detection regions

Request URL

PUT /ISAPI/Smart/RadarChannels/<radarChannelID>/radarFieldDetection/regions?format=json

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

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```
{
    "RegionList": [
    /*opt, array, region information list, subType:object*/
        {
            "RegionInfo": {
            /*opt, object, region information*/
                "regionID": 1,
                /*req, int, region ID*/
                "enabled": true,
                /*req, bool, whether to enable the function*/
                "sensitivityLevel": 50,
                /*req, int, region detection sensitivity, range:[1,100]*/
                "objectOccupation": 1,
                /*req, int, object occupation rate*/
                "Region": [
                /*opt, array, region coordinates, subType:object, range:[3,10], desc:the origin is the lower-left corner of the screen*/
                    {
                        "x": 0.120,
                        /*req, float, X-coordinate, range:[0.000,1.000]*/
                        "y": 0.120
                        /*req, float, Y-coordinate, range:[0.000,1.000]*/
                    }
                ]
            }
        }
    ]
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

19.10.1.27 Get parameters of radar intrusion detection

Request URL

GET /ISAPI/Smart/RadarChannels/<radarChannelID>/radarFieldDetection?format=json

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

None

Response Message

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```
{
    "RadarFieldDetection": {
    /*ro, req, object, radar intrusion detection parameters*/
        "enabled":  true,
        /*ro, req, bool, whether to enable the function*/
        "intelliBackSearch":  true,
        /*ro, opt, bool, whether the device supports video dual-VCA*/
        "startTriggerTime":  50,
        /*ro, req, int, triggering start time, range:[1,100], unit:ms*/
        "endTriggerTime":  50,
        /*ro, req, int, triggering end time, range:[1,100], unit:ms*/
        "minObjectSize":  0,
        /*ro, opt, int, minimum number of pixels*/
        "maxObjectSize":  0,
        /*ro, opt, int, maximum number of pixels*/
        "RegionList": [
        /*ro, opt, array, area information list, subType:object*/
            {
                "RegionInfo": {
                /*ro, opt, object, area information*/
                    "regionID":  1,
                    /*ro, req, int, area ID*/
                    "enabled":  true,
                    /*ro, req, bool, whether to enable detection*/
                    "sensitivityLevel":  50,
                    /*ro, req, int, detection sensitivity, range:[1,100]*/
                    "objectOccupation":  1,
                    /*ro, req, int*/
                    "Region": [
                    /*ro, opt, array, area, subType:object, range:[3,10], desc:the origin is the lower-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                }
            }
        ],
        "humanMisinfoFilter":  true,
        /*ro, opt, bool, whether it supports preventing false human body detection alarm; if it does not support, this node will not be returned*/
        "vehicleMisinfoFilter":  true
        /*ro, opt, bool, whether it supports preventing false vehicle detection alarm; if it does not support, this node will not be returned*/
    }
}
```

19.10.1.28 Intrusion detection

EventType:fielddetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
```

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```
<eventType>
  <!--ro, req, enum, event type, subType:string, desc:"fielddetection" (intrusion detection)-->fielddetection
</eventType>
<eventState>
  <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
</eventState>
<eventDescription>
  <!--ro, req, string, event description-->test
</eventDescription>
<deviceID>
  <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
</deviceID>
<DetectionRegionList>
  <!--ro, opt, array, detection area list, subType:object-->
  <DetectionRegionEntry>
    <!--ro, opt, object, detection area-->
    <regionID>
      <!--ro, req, string, area ID-->test
    </regionID>
    <sensitivityLevel>
      <!--ro, opt, int, sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
    </detectionTarget>
    <TargetRect>
      <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
      <X>
        <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
      </X>
      <Y>
        <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
      </Y>
      <width>
        <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
      </width>
      <height>
        <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
      </height>
    </TargetRect>
    <sceneID>
      <!--ro, opt, int, scene ID-->1
    </sceneID>
  </DetectionRegionEntry>
</DetectionRegionList>
<channelName>
  <!--ro, opt, string, channel name, range:[1,64]-->test
</channelName>
<Extensions>
  <!--ro, opt, object, extended information-->
  <serialNumber>
    <!--ro, opt, string, serial No.-->test
  </serialNumber>
  <eventPush>
    <!--ro, opt, string, event notification-->test
  </eventPush>
</Extensions>
<detectionPictureTransType>
  <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
</detectionPictureTransType>
<detectionPicturesNumber>
  <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
</detectionPicturesNumber>
<bkgUrl>
  <!--ro, opt, string, background picture URL, desc:background picture URL-->test
</bkgUrl>
<URLCertificationType>
  <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
</URLCertificationType>
<pId>
  <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
</pId>
<visibleLightURL>
  <!--ro, opt, string, visible light picture URL-->test
</visibleLightURL>
<thermalURL>
  <!--ro, opt, string, thermal imaging picture URL-->test
</thermalURL>
<targetAttrs>
  <!--ro, opt, object, transmitted parameters-->{ /*可选，透传字段，推荐使用以下三个字段*/ "deviceId": "1",  /*可选，设备ID，string类型，最大长度为64*/
```

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<!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
</targetAttrs>
<TaskInfo>
<!--ro, opt, object, task information submitted by data center products-->
<taskID>
<!--ro, req, string, task ID, range:[0,64]-->test
</taskID>
<algorithmType>
<!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
</algorithmType>
<taskName>
<!--ro, opt, string, task name-->test
</taskName>
<algorithmID>
<!--ro, opt, string, algorithm package ID-->test
</algorithmID>
<monitorPointID>
<!--ro, opt, string, camera No.-->test
</monitorPointID>
<monitorPointName>
<!--ro, opt, string, camera name-->test
</monitorPointName>
<streamType>
<!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
</streamType>
<ruleID>
<!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
</ruleID>
<ruleCustomName>
<!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
</ruleCustomName>
<algorithmInfo>
<!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
<algorithmId>
<!--ro, opt, string, algorithm package ID-->test
</algorithmId>
<algoVendorId>
<!--ro, opt, string, manufacturer ID-->test
</algoVendorId>
<algoVersionId>
<!--ro, opt, string, algorithm version No.-->test
</algoVersionId>
<algoChipType>
<!--ro, opt, string, chip type-->test
</algoChipType>
<algoModelVersion>
<!--ro, opt, string, model version-->test
</algoModelVersion>
</algorithmInfo>
</TaskInfo>
<timeStamp>
<!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
</timeStamp>
<targetSpeed>
<!--ro, opt, int-->0
</targetSpeed>
<targetDistance>
<!--ro, opt, int, unit:m-->2
</targetDistance>
<triggerType>
<!--ro, opt, enum, subType:string-->video
</triggerType>
<uid>
<!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits-->test
</uid>
<isDataRetransmission>
<!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers.)-->true
</isDataRetransmission>
<ruleID>
<!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
</ruleID>
<ruleName>
<!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
</ruleName>
<nonTargetRate>
<!--ro, opt, int, range:[1,100], unit:%-->1
</nonTargetRate>
<visibleLightAbsoluteHigh>
<!--ro, opt, object-->
<elevation>
<!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
</elevation>
<azimuth>
<!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
</azimuth>

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```
      <absoluteZoom>
        <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
      </absoluteZoom>
      <focus>
        <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
      </focus>
    </visibleLightAbsoluteHigh>
    <thermalAbsoluteHigh>
      <!--ro, opt, object-->
      <elevation>
        <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
      </elevation>
      <azimuth>
        <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
      </absoluteZoom>
      <focus>
        <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
      </focus>
    </thermalAbsoluteHigh>
    <deviceLocation>
      <!--ro, opt, object-->
      <longitudeType>
        <!--ro, req, enum, subType:string-->W
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
      </latitudeType>
      <longitude>
        <!--ro, req, object-->
        <degree>
          <!--ro, req, int, degree(s), range:[0,180]-->1
        </degree>
        <minute>
          <!--ro, req, float, minute(s), range:[0,60]-->0.0
        </minute>
        <sec>
          <!--ro, req, float, range:[0,60]-->0.0
        </sec>
      </longitude>
      <latitude>
        <!--ro, req, object-->
        <degree>
          <!--ro, req, int, range:[0,90]-->1
        </degree>
        <minute>
          <!--ro, req, float, minute(s), range:[0,60]-->0.0
        </minute>
        <sec>
          <!--ro, req, float, range:[0,60]-->0.0
        </sec>
      </latitude>
    </deviceLocation>
    <laserRanging>
      <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
    </laserRanging>
  </EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
fielddetection	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.10.2 Line Crossing Detection

19.10.2.1 Get the arming schedules of line crossing detection of all channels

Request URL

GET /ISAPI/Event/schedules/lineDetections

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the arming schedules of line crossing detection of all channels, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule information-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->lineDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming time-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LineDetectionScheduleList>
```

19.10.2.2 Set line crossing detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/lineDetections

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming information-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->lineDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LineDetectionScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.10.2.3 Get the arming schedule of line crossing detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/lineDetections/lineDetections_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->lineDetection
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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19.10.2.4 Set the arming schedule of line crossing detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/lineDetections/lineDetections_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--opt, enum, subType:string-->lineDetection
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.10.2.5 Set the arming schedule of radar intrusion detection for all channels

Request URL

PUT /ISAPI/Event/schedules/radarLineDetections

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RadarLineDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->radarLineDetection-1
    </id>
    <eventType>
      <!--opt, enum, subType:string-->radarLineDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RadarLineDetectionScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.10.2.6 Get arming schedules of all radar channels for line crossing detection

Request URL

GET /ISAPI/Event/schedules/radarLineDetections

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RadarLineDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, list of arming schedules, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule information-->
    <id>
      <!--ro, req, string, ID, desc:format: event type-channel No.-->radarLineDetection-1
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->radarLineDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, list of arming schedules, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, holiday arming schedules, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RadarLineDetectionScheduleList>
```

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19.10.2.7 Get the arming schedule of radar line crossing detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/radarLineDetections/radarLineDetection-<radarChannelID>

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->radarLineDetection-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->radarLineDetection
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.10.2.8 Set the arming schedule of radar line crossing detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/radarLineDetections/radarLineDetection-<radarChannelID>

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->radarLineDetection-1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->radarLineDetection
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

19.10.2.9 Set alarm linkage parameters of line crossing detection by channel

Request URL

PUT /ISAPI/Event/triggers/LineDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, linkage action list, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object, linkage action-->
    <id>
      <!--req, string, No.-->test
    </id>
    <notificationMethod>
      <!--req, enum, linkage action, subType:string, desc:"email", "IM" (instant messaging), "IO" (I/O output), "syslog" (system log), "HTTP" (uploading via
HTTP), "FTP" (uploading via FTP), "beep" (buzzer), "ptz" (PTZ), "record" (recording), "monitorAlarm" (alarm on monitor), "center" (uploading to the center),
"LightAudioAlarm" (light and audio alarm), "focus" (focus), "trace" (tracking), "whiteLight" (white light), "audio" (audio), "supplementLight" (supplement
light)-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, linkage time interval, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string, video channel ID-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string, dynamic video input channel ID-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object, PTZ linkage-->
      <ptzChannelID>
        <!--req, string, PTZ linked channel No.-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object, linkage parameters of the white light, desc:it is valid when the value of notificationMethod is "whiteLight"-->
      <whiteLightDurationTime>
        <!--opt, int, light flashing duration, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, supplement light alarm duration, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.2.10 Get alarm linkage parameters of a line crossing detection channel

Request URL

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GET /ISAPI/Event/triggers/LineDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object, linkage type-->
    <id>
      <!--ro, req, string-->test
    </id>
    <notificationMethod>
      <!--ro, req, enum, linkage type, subType:string, desc:linkage type-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

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19.10.2.11 Get the linkage parameters of line crossing detection event for a specified channel

Request URL

GET /ISAPI/Event/triggers/LineDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->linedetection
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <SupplementLightAlarm>
        <!--ro, opt, object-->
        <durationTime>
          <!--ro, opt, int, range:[0,90], unit:s-->5
        </durationTime>
      </SupplementLightAlarm>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.10.2.12 Set the linkage parameters of line crossing detection event for a specified channel

Request URL

PUT /ISAPI/Event/triggers/LineDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->LineDetection
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.2.13 Get the linkage parameters of radar line crossing detection event for a specified channel

Request URL

GET /ISAPI/Event/triggers/radarLineDetection-<radarChannelID>

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->radarLineDetection-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->radarLineDetection
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

19.10.2.14 Set line crossing detection parameters of all channels

Request URL

PUT /ISAPI/Smart/LineDetection

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

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```
<?xml version="1.0" encoding="UTF-8" ?>

<LineDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LineDetection>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <intelliBackSearch>
      <!--opt, bool-->true
    </intelliBackSearch>
    <duration>
      <!--opt, int-->1
    </duration>
    <startTriggerTime>
      <!--opt, int-->1
    </startTriggerTime>
    <endTriggerTime>
      <!--opt, int-->1
    </endTriggerTime>
    <normalizedScreenSize>
      <!--opt, object-->
      <normalizedScreenWidth>
        <!--req, int-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <minObjectSize>
      <!--opt, int-->1
    </minObjectSize>
    <maxObjectSize>
      <!--opt, int-->1
    </maxObjectSize>
    <LineItemList size="4">
      <!--opt, array, subType:object, attr:size{req, int}-->
      <LineItem>
        <!--opt, object-->
        <id>
          <!--req, string-->1
        </id>
        <enabled>
          <!--req, bool, whether to enable the function-->true
        </enabled>
        <sensitivityLevel>
          <!--req, int, detection sensitivity, range:[0,100]-->1
        </sensitivityLevel>
        <directionSensitivity>
          <!--opt, enum, detection direction, subType:string, desc:detection direction-->any
        </directionSensitivity>
        <CoordinatesList>
          <!--opt, array, subType:object, range:[0,2]-->
          <Coordinates>
            <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, x-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->1
            </positionY>
          </Coordinates>
        </CoordinatesList>
        <detectionTarget>
          <!--opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <humanMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (human body)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--opt, enum, subType:string-->low
        </recordConfidence>
      </LineItem>
    </LineItemList>
    <recogRuleType>
      <!--opt, enum, subType:string-->vectorMode
    </recogRuleType>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
```

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </LineDetection>
</LineDetectionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.2.15 Get line crossing detection parameters of all channels

Request URL

GET /ISAPI/Smart/LineDetection

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LineDetection>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, channel No.-->test
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <intelliBackSearch>
      <!--ro, opt, bool-->true
    </intelliBackSearch>
    <duration>
      <!--ro, opt, int, duration-->1
    </duration>
    <startTriggerTime>
      <!--ro, opt, int-->1
    </startTriggerTime>
    <endTriggerTime>
      <!--ro, opt, int-->1
    </endTriggerTime>
    <normalizedScreenSize>
      <!--ro, opt, object-->
      <normalizedScreenWidth>
        <!--ro, req, int-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <minObjectSize>
      <!--ro, opt, int-->1
    </minObjectSize>
    <maxObjectSize>
      <!--ro, opt, int-->1
    </maxObjectSize>
    <LineItemList size="4">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <LineItem>
        <!--ro, opt, object-->
```

dojojin@gmail.com
PROTECTION AND SECURITY SYSTEM CO.,LTD.

```
        <id>
          <!--ro, req, string, ID-->1
        </id>
        <enabled>
          <!--ro, req, bool, whether to enable the function-->true
        </enabled>
        <sensitivityLevel>
          <!--ro, req, int, detection sensitivity, range:[0,100]-->1
        </sensitivityLevel>
        <directionSensitivity>
          <!--ro, opt, enum, detection direction, subType:string, desc:detection direction-->any
        </directionSensitivity>
        <CoordinatesList>
          <!--ro, opt, array, rectangle, subType:object, range:[0,2], desc:rectangle-->
          <Coordinates>
            <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </Coordinates>
        </CoordinatesList>
        <detectionTarget>
          <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <humanMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--ro, opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--ro, opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--ro, opt, enum, subType:string-->low
        </recordConfidence>
      </LineItem>
    </LineItemList>
    <recogRuleType>
      <!--ro, opt, enum, subType:string-->vectorMode
    </recogRuleType>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </LineDetection>
</LineDetectionList>
```

19.10.2.16 Set line crossing detection parameters by channel

Request URL

PUT /ISAPI/Smart/LineDetection/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, line crossing detection parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <intelliBackSearch>
    <!--opt, bool, whether to enable dual-VCA-->true
  </intelliBackSearch>
  <duration>
    <!--opt, int, duration-->1
  </duration>
  <startTriggerTime>
```

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```
    <!--opt, int, trigger start time-->1
  </startTriggerTime>
  <endTriggerTime>
    <!--opt, int, trigger end time-->1
  </endTriggerTime>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <minObjectSize>
    <!--opt, int, minimum number of pixels of each object-->1
  </minObjectSize>
  <maxObjectSize>
    <!--opt, int, maximum number of pixels of each object-->1
  </maxObjectSize>
  <LineItemList size="4">
    <!--opt, array, subType:object, attr:size{req, int}-->
    <id>
      <!--req, string, No.-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, detection sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <directionSensitivity>
      <!--opt, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any" (all directions)-
->any
    </directionSensitivity>
    <CoordinatesList>
      <!--opt, array, coordinates list, subType:object, range:[0,2], desc:line-->
      <Coordinates>
        <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </Coordinates>
    </CoordinatesList>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </LineItemList>
  <recogRuleType>
    <!--opt, enum, direction recognition mode, subType:string, desc:"vectorMode" (vector mode), "slopeMode" (slope mode)-->vectorMode
  </recogRuleType>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
</LineDetection>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.2.17 Get configuration capability of line crossing detection by channel

Request URL

GET /ISAPI/Smart/LineDetection/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <LineItemList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <LineItem>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
      </enabled>
      <sensitivityLevel min="1" max="100">
        <!--ro, req, int, detection sensitivity, attr:min{req, int},max{req, int}-->1
      </sensitivityLevel>
      <directionSensitivity opt="left-right,right-left,any">
        <!--ro, opt, enum, detection direction, subType:string, attr:opt{req, string}, desc:detection direction-->any
      </directionSensitivity>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,2], desc:line-->
        <RegionCoordinates>
          <!--ro, req, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
```

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```
        <detectionTarget opt="all,human,vehicle,animal,others" def="human">
          <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string},def{req, string}, desc:detection target-->all
        </detectionTarget>
        <alarmConfidence opt="low,mediumLow,mediumHigh,high" def="low">
          <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
        </alarmConfidence>
        <recordConfidence opt="low,mediumLow,mediumHigh,high" def="low">
          <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
        </recordConfidence>
        <ruleName min="1" max="64">
          <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
        </ruleName>
        <detectionTargetProperty>
          <!--ro, opt, object-->
          <human>
            <!--ro, opt, object-->
            <propertyList size="1">
              <!--ro, opt, object, attr:size{req, int}-->
              <property opt="nonMotor">
                <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
              </property>
            </propertyList>
          </human>
          <animal>
            <!--ro, opt, object-->
            <propertyList size="3">
              <!--ro, opt, object, attr:size{req, int}-->
              <property opt="large,small,poultry">
                <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
              </property>
            </propertyList>
          </animal>
        </detectionTargetProperty>
      </LineItem>
    </LineItemList>
    <isSupportPerimeterDetection opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </isSupportPerimeterDetection>
    <isSupportMultiScene opt="true,false">
      <!--ro, opt, bool, whether supports multiple scene detection, attr:opt{req, string}-->true
    </isSupportMultiScene>
    <recogRuleType>
      <!--ro, opt, enum, subType:string-->vectorMode
    </recogRuleType>
    <isSupportHumanMisinfoFilter>
      <!--ro, opt, bool, whether supports false human body alarm prevention, desc:If this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned.-->true
    </isSupportHumanMisinfoFilter>
    <isSupportVehicleMisinfoFilter>
      <!--ro, opt, bool, whether supports false vehicle alarm prevention, desc:If this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned.-->true
    </isSupportVehicleMisinfoFilter>
    <isSupportTargetMultiSelect>
      <!--ro, opt, bool, whether supports selecting multiple detection targets, desc:If this function is supported, this node will be returned and its value
is true; otherwise, this node will not be returned.-->true
    </isSupportTargetMultiSelect>
    <isSupportAllDayUpload>
      <!--ro, opt, bool, whether supports uploading all-day event-->true
    </isSupportAllDayUpload>
    <isSupportOverlapParams>
      <!--ro, opt, bool-->true
    </isSupportOverlapParams>
    <warningSystemEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </warningSystemEnabled>
    <isSupportStackTarget>
      <!--ro, opt, bool-->true
    </isSupportStackTarget>
    <isSupportAllTargetDetectionWebDisplay>
      <!--ro, opt, bool-->false
    </isSupportAllTargetDetectionWebDisplay>
    <targetSensitivityLevel min="0" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
    </targetSensitivityLevel>
  </LineDetection>
```

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19.10.2.18 Add detection lines for a specific channel

Request URL

POST /ISAPI/Smart/LineDetection/<channelID>/lineItem

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineItemList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LineItem>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, req, int, detection sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <directionSensitivity>
      <!--ro, opt, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any"-->any
    </directionSensitivity>
    <CoordinatesList>
      <!--ro, opt, array, list of coordinates, subType:object, range:[0,2], desc:line-->
      <Coordinates>
        <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </Coordinates>
    </CoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:"Low", "middle" "high"-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, confidence of alarm notification, subType:string-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, confidence of video recording, subType:string-->low
    </recordConfidence>
  </LineItem>
</LineItemList>
```

19.10.2.19 Get parameters of a detection line of a specific channel

Request URL

GET /ISAPI/Smart/LineDetection/<channelID>/lineItem/<ruleID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
ruleID	string	Rule ID

Request Message

None

Response Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<LineItem xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable the function, desc:whether to enable the function-->true
  </enabled>
  <sensitivityLevel>
    <!--ro, req, int, detection sensitivity-->1
  </sensitivityLevel>
  <directionSensitivity>
    <!--ro, opt, enum, detection direction, subType:string, desc:"left-right,right-left,any"-->any
  </directionSensitivity>
  <CoordinatesList>
    <!--ro, opt, array, line, subType:object, range:[0,2], desc:line-->
    <Coordinates>
      <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </Coordinates>
  </CoordinatesList>
  <detectionTarget>
    <!--ro, opt, enum, detection target, subType:string, desc:"all,human,vehicle,others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm reduction of vehicle-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--ro, opt, enum, priority, subType:string, desc:“low”, “middle”, “high”-->low
  </priority>
  <alarmConfidence>
    <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:"low", "mediumLow", "mediumHigh", "high", "low”-->low
  </alarmConfidence>
  <recordConfidence>
    <!--ro, opt, enum, confidence of recording video, subType:string, desc:"low", "mediumLow", "mediumHigh", "high", "low”-->low
  </recordConfidence>
  <ruleName>
    <!--ro, opt, string, range:[1,64]-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object, dep:or,{$.LineItem.detectionTarget,eq,all},{$.LineItem.detectionTarget,eq,human}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object, dep:or,{$.LineItem.detectionTarget,eq,all},{$.LineItem.detectionTarget,eq,animal}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</LineItem>
```

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19.10.2.20 Set parameters of a detection line of a specified channel

Request URL

PUT /ISAPI/Smart/LineDetection/<channelID>/lineItem/<ruleID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
ruleID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineItem xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of a detection line, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, No.-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <sensitivityLevel>
    <!--req, int, detection sensitivity-->1
  </sensitivityLevel>
  <directionSensitivity>
    <!--opt, enum, detection direction, subType:string, desc:"Left-right" (from left to right), "right-left" (from right to left), "any" (all directions)--
>any
  </directionSensitivity>
  <CoordinatesList>
    <!--opt, array, coordinates list, subType:object, range:[0,2], desc:line-->
    <Coordinates>
      <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </Coordinates>
  </CoordinatesList>
  <detectionTarget>
    <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all" is
mutually exclusive with "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </alarmConfidence>
  <recordConfidence>
    <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </recordConfidence>
</LineItem>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.2.21 Get configuration capability of line crossing detection by channel

Request URL

GET /ISAPI/Smart/LineDetection/<channelID>/lineItem/<ruleID>/capabilities

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.
ruleID	string	Rule ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineItem xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <sensitivityLevel min="1" max="100">
    <!--ro, req, int, sensitivity, attr:min{req, int},max{req, int}-->1
  </sensitivityLevel>
  <directionSensitivity opt="left-right,right-left,any">
    <!--ro, opt, enum, direction, subType:string, attr:opt{req, string}, desc:direction-->any
  </directionSensitivity>
  <CoordinatesList>
    <!--ro, opt, array, line, subType:object, range:[0,2], desc:line-->
    <Coordinates>
      <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,1000]-->1
      </positionY>
    </Coordinates>
  </CoordinatesList>
  <detectionTarget opt="all,human,vehicle,animal,others">
    <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string}, desc:detection target-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable false alarm filtering (human body), attr:opt{req, string}-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle), attr:opt{req, string}-->true
  </vehicleMisinfoFilterEnabled>
  <priority opt="low,middle,high">
    <!--ro, opt, enum, priority, subType:string, attr:opt{req, string}, desc:priority-->low
  </priority>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
  </isSupportAllTargetDetectionWebDisplay>
  <ruleName min="1" max="64">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object-->
      <propertyList size="1">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="nonMotor">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object-->
      <propertyList size="3">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="large,small,poultry">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</LineItem>
```

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19.10.2.22 Get radar line crossing detection capability

Request URL

GET /ISAPI/Smart/RadarChannels/<radarChannelID>/radarLineDetection/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

None

Response Message

```
{
    "RadarLineDetectionCap": {
    /*ro, req, object, radar line crossing detection capability*/
        "enabled": [true, false],
        /*ro, req, array, whether to enable the function, subType:bool*/
        "intelliBackSearch":  "true,false",
        /*ro, req, string, whether it supports dual-VCA search*/
        "duration": {
        /*ro, req, object, duration*/
            "@min":  0,
            /*ro, req, int, minimum value*/
            "@max":  0
            /*ro, req, int, maximum value*/
        },
        "startTriggerTime": {
        /*ro, req, object, triggering start time*/
            "@min":  0,
            /*ro, req, int, minimum value*/
            "@max":  0
            /*ro, req, int, maximum value*/
        },
        "endTriggerTime": {
        /*ro, req, object, triggering end time*/
            "@min":  0,
            /*ro, req, int, minimum value*/
            "@max":  0
            /*ro, req, int, maximum value*/
        },
        "minObjectSize": {
        /*ro, opt, object, optional*/
            "@min":  0,
            /*ro, req, int, minimum value*/
            "@max":  0
            /*ro, req, int, maximum value*/
        },
        "maxObjectSize": {
        /*ro, opt, object, optional*/
            "@min":  0,
            /*ro, req, int, minimum value*/
            "@max":  0
            /*ro, req, int, maximum value*/
        },
        "LineItemListCap": {
        /*ro, req, object, detection line list*/
            "maxSize":  4,
            /*ro, req, int, maximum number of detection lines*/
            "enabled": [true, false],
            /*ro, req, array, whether to enable the detection, subType:bool*/
            "sensitivityLevel":{
            /*ro, req, object, detection sensitivity*/
                "@min": 1,
                /*ro, req, int, minimum value*/
                "@max":  100
                /*ro, req, int, maximum value*/
            },
            "directionSensitivity": {
            /*ro, req, object, detection direction, desc:detection direction: "left-right" (from left to right),"right-left" (from right to left),"any" (all
directions)*/
                "@opt": ["left-right", "right-left", "any"],
                /*ro, req, array, range, subType:string*/
                "@def":  "any"
                /*ro, req, string, the default value*/
            },
            "LineItemCap": {
            /*ro, req, object, parameters of a detection line*/
                "maxSize":  2,
                /*ro, req, int, maximum number of detection lines*/
                "x": {
                /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, maximum value, range:[0.000,1.000]*/
                },
                "y": {
```

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```
                /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, maximum value, range:[0.000,1.000]*/
                }
            }
        },
        "isSupportMultiScene":  true,
        /*ro, opt, bool, whether it supports multi-scene mode*/
        "recogRuleType": {
        /*ro, req, object, direction recognition mode, desc:"vectorMode" (vector mode), "slopeMode" (slope mode)*/
            "@opt":  "vectorMode,slopeMode"
            /*ro, req, string, range*/
        },
        "humanMisinfoFilter": [true, false],
        /*ro, opt, array, whether it supports preventing false human body detection alarm, subType:bool*/
        "vehicleMisinfoFilter": [true, false],
        /*ro, opt, array, whether it supports preventing false vehicle detection alarm, subType:bool*/
        "isSupportTargetMultiSelect":  true,
        /*ro, opt, bool, whether it supports multi-target detection*/
        "isSupportAllDayUpload":  true
        /*ro, opt, bool, whether it supports uploading all-day events*/
    }
}
```

19.10.2.23 Set parameters of a radar detection line

Request URL

PUT /ISAPI/Smart/RadarChannels/<radarChannelID>/radarLineDetection/lineItems/<lineRuleID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--
lineRuleID	string	--

Request Message

```
{
    "LineItemInfo": {
    /*opt, object, parameters of a radar detection line*/
        "lineID":  4,
        /*req, int, detection line ID*/
        "enabled":  true,
        /*req, bool, whether to enable detection*/
        "sensitivityLevel":  50,
        /*req, int, detection sensitivity, range:[0,100]*/
        "directionSensitivity":  "any",
        /*req, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any" (all
directions)*/
        "LineItem": [
        /*req, array, detection line, subType:object, range:[0,2], desc:the origin is the lower-left corner of the screen*/
            {
                "x":  0.120,
                /*req, float, X-coordinate, range:[0.000,1.000]*/
                "y":  0.120
                /*req, float, Y-coordinate, range:[0.000,1.000]*/
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

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19.10.2.24 Get parameters of all radar detection lines

Request URL

GET /ISAPI/Smart/RadarChannels/<radarChannelID>/radarLineDetection/lineItems?format=json

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

None

Response Message

```
{
    "LineItemList": [
    /*ro, opt, array, subType:object*/
        {
            "LineItemInfo": {
            /*ro, opt, object*/
                "lineID": 4,
                /*ro, req, int, detection line ID*/
                "enabled": true,
                /*ro, req, bool, whether to enable the detection*/
                "sensitivityLevel": 50,
                /*ro, req, int, detection sensitivity, range:[0,100]*/
                "directionSensitivity": "any",
                /*ro, req, enum, detection direction, subType:string, desc:detection direction: "left-right" (from left to right),"right-left" (from right
to left),"any" (all directions)*/
                "LineItem": [
                /*ro, req, array, parameters of a detection line, subType:object, range:[0,2], desc:the origin is the lower-left corner of the screen*/
                    {
                        "x": 0.120,
                        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                        "y": 0.120
                        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    }
                ]
            }
        }
    ]
}
```

19.10.2.25 Get parameters of radar's line crossing detection rule

Request URL

GET /ISAPI/Smart/RadarChannels/<radarChannelID>/radarLineDetection?format=json

Query Parameter

Parameter Name	Parameter Type	Description
radarChannelID	string	--

Request Message

None

Response Message

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```
{
    "RadarLineDetection": {
    /*ro, req, object, parameters of radar's line crossing detection rule*/
        "enabled":  true,
        /*ro, req, bool, whether to enable detection*/
        "intelliBackSearch":  true,
        /*ro, req, bool, dual-VCA search*/
        "duration":  500,
        /*ro, req, int, duration*/
        "startTriggerTime":  500,
        /*ro, req, int, triggering start time, range:[0,100], unit:ms*/
        "endTriggerTime":  500,
        /*ro, req, int, triggering end time, range:[0,100], unit:ms*/
        "minObjectSize":  0,
        /*ro, opt, int, minimum pixels of each object*/
        "maxObjectSize":  0,
        /*ro, opt, int, maximum pixels of each object*/
        "LineItemList": [
        /*ro, opt, array, detection line list, subType:object*/
            {
                "LineItemInfo": {
                /*ro, opt, object*/
                    "lineID":  4,
                    /*ro, req, int, detection line ID*/
                    "enabled":  true,
                    /*ro, req, bool, whether to enable detection*/
                    "sensitivityLevel":  50,
                    /*ro, req, int, detection sensitivity, range:[0,100]*/
                    "directionSensitivity":  "any",
                    /*ro, req, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any"
(all directions)*/
                    "LineItem": [
                    /*ro, req, array, parameters of a detection line, subType:object, range:[0,2], desc:the origin is the lower-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                }
            }
        ],
        "recogRuleType":  "vectorMode",
        /*ro, opt, enum, direction recognition mode, subType:string, desc:"vectorMode" (vector mode), "slopeMode" (slope mode)*/
        "humanMisinfoFilter":  true,
        /*ro, opt, bool, whether to enable false alarm filtering of human body detection*/
        "vehicleMisinfoFilter":  true
        /*ro, opt, bool, whether to enable false alarm filtering of vehicle detection*/
    }
}
```

19.10.2.26 Line crossing detection

EventType:linedetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
```

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```
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"linedetection" (Line Crossing Detection)-->linedetection
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
        </height>
      </TargetRect>
      <sceneID>
        <!--ro, opt, int, scene ID-->1
      </sceneID>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
```

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```
  <visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "Localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <targetSpeed>
    <!--ro, opt, datetime-->2019-12-30T15:29:23+08:00
  </targetSpeed>
  <targetDistance>
    <!--ro, opt, int, unit:m-->2
  </targetDistance>
  <triggerType>
    <!--ro, opt, enum, subType:string-->video
  </triggerType>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits--
>test
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
  <applicationScene>
    <!--ro, opt, enum, Application Scene, subType:string-->indoor
  </applicationScene>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
```

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```
        <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
      </elevation>
      <azimuth>
        <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
      </absoluteZoom>
      <focus>
        <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
      </focus>
    </visibleLightAbsoluteHigh>
    <thermalAbsoluteHigh>
      <!--ro, opt, object-->
      <elevation>
        <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
      </elevation>
      <azimuth>
        <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
      </absoluteZoom>
      <focus>
        <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
      </focus>
    </thermalAbsoluteHigh>
    <deviceLocation>
      <!--ro, opt, object-->
      <longitudeType>
        <!--ro, req, enum, subType:string-->W
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
      </latitudeType>
      <longitude>
        <!--ro, req, object-->
        <degree>
          <!--ro, req, int, degree(s), range:[0,180]-->1
        </degree>
        <minute>
          <!--ro, req, float, minute(s), range:[0,60]-->0.0
        </minute>
        <sec>
          <!--ro, req, float, range:[0,60]-->0.0
        </sec>
      </longitude>
      <latitude>
        <!--ro, req, object-->
        <degree>
          <!--ro, req, int, range:[0,90]-->1
        </degree>
        <minute>
          <!--ro, req, float, minute(s), range:[0,60]-->0.0
        </minute>
        <sec>
          <!--ro, req, float, range:[0,60]-->0.0
        </sec>
      </latitude>
    </deviceLocation>
    <laserRanging>
      <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
    </laserRanging>
  </EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
linedetection	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

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Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.10.3 Region Entrance

19.10.3.1 Set region entrance detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/regionEntrances

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, string, event type-->regionEntrance
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, day of a week, subType:int, desc:day of a week-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionEntranceScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.3.2 Get arming schedules of all channels for region entrance detection

Request URL

GET /ISAPI/Event/schedules/regionEntrances

Query Parameter

None

Request Message

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None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, configuration of region entrance detection arming schedules-->
    <id>
      <!--ro, req, string, ID, desc:format: event type-channel No.-->1
    </id>
    <eventType>
      <!--ro, opt, string, event type-->regionEntrance
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, list of arming schedules, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming schedule-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionEntranceScheduleList>
```

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19.10.3.3 Set the arming schedule of region entrance detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/regionEntrances/regionEntrance-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--opt, string-->regionEntrance
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.10.3.4 Get the arming schedule of region entrance detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/regionEntrances/regionEntrance-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, opt, string-->regionEntrance
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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19.10.3.5 Get the linkage triggering parameters of region entrance for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionEntrance-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

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19.10.3.6 Set the linkage triggering parameters of region entrance for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionEntrance-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <notificationMethod>
      <!--req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object-->
      <ptzChannelID>
        <!--req, string-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object-->
      <whiteLightDurationTime>
        <!--opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.10.3.7 Get the linkage parameters of region entrance for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionEntrance-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->regionEntrance
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <SupplementLightAlarm>
        <!--ro, opt, object-->
        <durationTime>
          <!--ro, opt, int, range:[0,90], unit:s-->5
        </durationTime>
      </SupplementLightAlarm>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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19.10.3.8 Set the linkage parameters of region entrance for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionEntrance-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->regionEntrance
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string--> 1
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string--> 1
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <SupplementLightAlarm>
        <!--opt, object-->
        <durationTime>
          <!--opt, int, range:[0,90], unit:s-->5
        </durationTime>
      </SupplementLightAlarm>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.3.9 Set region entrance detection parameters of all channels

Request URL

PUT /ISAPI/Smart/regionEntrance

Query Parameter

None

Request Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_RegionEntrance for details, attr:version{req, string, protocolVersion}-->
  <RegionEntrance>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <normalizedScreenSize>
      <!--opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <RegionEntranceRegionList>
      <!--opt, object, area list-->
      <RegionEntranceRegion>
        <!--opt, object, intrusion region-->
        <id>
          <!--req, string, detection area ID-->1
        </id>
        <sensitivityLevel>
          <!--opt, int, sensitivity, range:[0,100]-->1
        </sensitivityLevel>
        <RegionCoordinatesList>
          <!--opt, array, region coordinates list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, x-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <detectionTarget>
          <!--opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <humanMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (human body)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--opt, enum, subType:string-->low
        </recordConfidence>
      </RegionEntranceRegion>
    </RegionEntranceRegionList>
    <mutexAbility opt="PDC">
      <!--opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </RegionEntrance>
</RegionEntranceList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.3.10 Get region entrance detection parameters of all channels

Request URL

GET /ISAPI/Smart/regionEntrance

Query Parameter

None

Request Message

None

Response Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, refer to the message XML_RegionEntrance for details, attr:version{req, string, protocolVersion}-->
  <RegionEntrance>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, channel No.-->test
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <normalizedScreenSize>
      <!--ro, opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <RegionEntranceRegionList>
      <!--ro, opt, array, area list, subType:object-->
      <RegionEntranceRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, string-->1
        </id>
        <sensitivityLevel>
          <!--ro, opt, int, sensitivity, range:[1,100]-->1
        </sensitivityLevel>
        <RegionCoordinatesList>
          <!--ro, opt, array, region coordinates list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <detectionTarget>
          <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <humanMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--ro, opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--ro, opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--ro, opt, enum, subType:string-->low
        </recordConfidence>
      </RegionEntranceRegion>
    </RegionEntranceRegionList>
    <mutexAbility opt="PDC">
      <!--ro, opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--ro, opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </RegionEntrance>
</RegionEntranceList>
```

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19.10.3.11 Set parameters of the region entrance detection rule by channel

Request URL

PUT /ISAPI/Smart/regionEntrance/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntrance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, region entrance detection configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <RegionEntranceRegionList>
    <!--opt, object, region list-->
    <RegionEntranceRegion>
      <!--opt, object, region entrance detection region-->
      <id>
        <!--req, string, region ID-->1
      </id>
      <sensitivityLevel>
        <!--opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--opt, object, region coordinate list-->
        <RegionCoordinates>
          <!--opt, object, region coordinates-->
          <positionX>
            <!--req, int, X-coordinate-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value
"all" is mutually exclusive with "others"-->all
      </detectionTarget>
      <humanMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
      </vehicleMisinfoFilterEnabled>
      <priority>
        <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
      </priority>
      <alarmConfidence>
        <!--opt, enum, alarm confidence, subType:string, desc:"Low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </alarmConfidence>
      <recordConfidence>
        <!--opt, enum, video confidence, subType:string, desc:"Low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </recordConfidence>
    </RegionEntranceRegion>
  </RegionEntranceRegionList>
  <mutexAbility opt="PDC">
    <!--opt, string, mutually exclusive capability: "PDC" (people counting statistics, which is not supported during region entrance detection),
attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--opt, bool, whether the device supports multiple scenes detection-->true
  </isSupportMultiScene>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
</RegionEntrance>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.3.12 Get region entrance detection parameters by video input channel

Request URL

GET /ISAPI/Smart/regionEntrance/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntrance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, read-only, "PDC"-people counting statistics, which is not supported during region entrance detection, attr:version{req, string,
protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <RegionEntranceRegionList>
    <!--ro, opt, object-->
    <RegionEntranceRegion>
      <!--ro, opt, object, Intrusion Region-->
      <id>
        <!--ro, req, string-->1
      </id>
      <sensitivityLevel>
        <!--ro, opt, int, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
      </detectionTarget>
      <humanMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
```

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```
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--ro, opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--ro, opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--ro, opt, enum, subType:string-->low
        </recordConfidence>
        <ruleName>
          <!--ro, opt, string, range:[1,64]-->test
        </ruleName>
        <detectionTargetProperty>
          <!--ro, opt, object-->
          <human>
            <!--ro, opt, object, dep:or,{$.RegionEntrance.RegionEntranceRegionList.RegionEntranceRegion.detectionTarget,eq,all},
{$.RegionEntrance.RegionEntranceRegionList.RegionEntranceRegion.detectionTarget,eq,human}-->
            <propertyList>
              <!--ro, opt, array, subType:object-->
              <property>
                <!--ro, opt, enum, subType:string-->nonMotor
              </property>
            </propertyList>
          </human>
          <animal>
            <!--ro, opt, object, dep:or,{$.RegionEntrance.RegionEntranceRegionList.RegionEntranceRegion.detectionTarget,eq,all},
{$.RegionEntrance.RegionEntranceRegionList.RegionEntranceRegion.detectionTarget,eq,animal}-->
            <propertyList>
              <!--ro, opt, array, subType:object-->
              <property>
                <!--ro, opt, enum, subType:string-->large
              </property>
            </propertyList>
          </animal>
        </detectionTargetProperty>
      </RegionEntranceRegion>
    </RegionEntranceRegionList>
    <mutexAbility opt="PDC">
      <!--ro, opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--ro, opt, bool, whether supports multiple-scene detection-->true
    </isSupportMultiScene>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <stackTarget>
      <!--ro, opt, bool-->true
    </stackTarget>
    <targetSensitivityLevel>
      <!--ro, opt, int, range:[0,100]-->50
    </targetSensitivityLevel>
  </RegionEntrance>
```

19.10.3.13 Get configuration capability of region entrance detection by channel

Request URL

GET /ISAPI/Smart/regionEntrance/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntrance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ro, mutex capability, it indicates that which function is mutually exclusive with the region entrance detection function:
"PDC,videoFrameRate50,videoFrameRate60,19-1920*1080@50fps,20-1920*1080@60fps,scheduled recording", attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
```

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```
<enabled opt="true,false">
  <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
</enabled>
<normalizedScreenSize>
  <!--ro, opt, object, normalized screen size-->
  <normalizedScreenWidth>
    <!--ro, req, int, normalized screen width-->1
  </normalizedScreenWidth>
  <normalizedScreenHeight>
    <!--ro, req, int, normalized screen height-->1
  </normalizedScreenHeight>
</normalizedScreenSize>
<RegionEntranceRegionList size="4">
  <!--ro, opt, array, subType:object, attr:size{req, int}-->
  <RegionEntranceRegion>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->1
    </id>
    <sensitivityLevel min="1" max="100">
      <!--ro, req, int, sensitivity, attr:min{req, int},max{req, int}-->1
    </sensitivityLevel>
    <RegionCoordinatesList size="10">
      <!--ro, opt, array, region coordinate list, subType:object, range:[3,10], attr:size{req, int}, desc:region coordinate list-->
      <RegionCoordinates>
        <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <maxRegionCoordinatesNum>
      <!--ro, opt, int-->1
    </maxRegionCoordinatesNum>
    <minRegionCoordinatesNum>
      <!--ro, opt, int-->1
    </minRegionCoordinatesNum>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
    </detectionTarget>
    <alarmConfidence opt="low,mediumLow,mediumHigh,high" def="low">
      <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
    </alarmConfidence>
    <recordConfidence opt="low,mediumLow,mediumHigh,high" def="low">
      <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
    </recordConfidence>
    <ruleName min="1" max="64">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </ruleName>
    <detectionTargetProperty>
      <!--ro, opt, object-->
      <human>
        <!--ro, opt, object-->
        <propertyList size="1">
          <!--ro, opt, object, attr:size{req, int}-->
          <property opt="nonMotor">
            <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
          </property>
        </propertyList>
      </human>
      <animal>
        <!--ro, opt, object-->
        <propertyList size="3">
          <!--ro, opt, object, attr:size{req, int}-->
          <property opt="large,small,poultry">
            <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
          </property>
        </propertyList>
      </animal>
    </detectionTargetProperty>
  </RegionEntranceRegion>
</RegionEntranceRegionList>
<mutexAbility opt="PDC">
  <!--ro, opt, string, attr:opt{req, string}-->PDC
</mutexAbility>
<isSupportMultiScene opt="true,false">
  <!--ro, req, bool, whether supports multiple scenes detection, attr:opt{req, string}-->true
</isSupportMultiScene>
<isSupportHumanMisinfoFilter>
  <!--ro, opt, bool, whether supports preventing false human body alarm, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned-->true
</isSupportHumanMisinfoFilter>
<isSupportVehicleMisinfoFilter>
  <!--ro, opt, bool, whether supports preventing false vehicle alarm, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned-->true
</isSupportVehicleMisinfoFilter>
<isSupportTargetMultiSelect>
  <!--ro, opt, bool, whether supports selecting multiple detection targets, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned-->true
</isSupportTargetMultiSelect>
<isSupportAllDayUpload>
```

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HIKVISION AND SECURITY SYSTEM CO.,LTD.

```
    <isSupportAllDayUpload>
      <!--ro, opt, bool, whether supports uploading all-day event-->true
    </isSupportAllDayUpload>
    <isSupportOverlapParams>
      <!--ro, opt, bool-->true
    </isSupportOverlapParams>
    <isSupportStackTarget>
      <!--ro, opt, bool-->true
    </isSupportStackTarget>
    <isSupportAllTargetDetectionWebDisplay>
      <!--ro, opt, bool-->false
    </isSupportAllTargetDetectionWebDisplay>
    <targetSensitivityLevel min="0" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
    </targetSensitivityLevel>
  </RegionEntrance>
```

19.10.3.14 Get parameters of all region entrance detection regions by channel

Request URL

GET /ISAPI/Smart/regionEntrance/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, refer to the message XML_RegionEntranceRegion for details, attr:version{req, string, protocolVersion}-->
  <RegionEntranceRegion>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, region ID-->1
    </id>
    <sensitivityLevel>
      <!--ro, req, int, sensitivity-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--ro, opt, array, region coordinate list, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:priority-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, subType:string-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, subType:string-->low
    </recordConfidence>
  </RegionEntranceRegion>
</RegionEntranceRegionList>
```

19.10.3.15 Set parameters of all region entrance detection regions by channel

Request URL

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PUT /ISAPI/Smart/regionEntrance/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, region parameters list, attr:version{req, string, protocolVersion}-->
  <RegionEntranceRegion>
    <!--opt, object, region parameters-->
    <id>
      <!--req, string, region ID-->1
    </id>
    <sensitivityLevel>
      <!--req, int, sensitivity-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--opt, array, region coordinate list, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </RegionEntranceRegion>
</RegionEntranceRegionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.10.3.16 Set parameters of a region entrance detection region by channel

Request URL

PUT /ISAPI/Smart/regionEntrance/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, region configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, region ID-->1
  </id>
  <sensitivityLevel>
    <!--req, int, sensitivity-->1
  </sensitivityLevel>
  <RegionCoordinatesList>
    <!--opt, array, region coordinate list, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <detectionTarget>
    <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all" is
mutually exclusive with "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--opt, enum, alarm confidence, subType:string, desc:"Low", "mediumLow", "mediumHigh", "high"-->low
  </alarmConfidence>
  <recordConfidence>
    <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </recordConfidence>
</RegionEntranceRegion>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.3.17 Get parameters of a region entrance detection region of a specific channel

Request URL

GET /ISAPI/Smart/regionEntrance/<channelID>/regions/<regionID>

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, region ID-->1
  </id>
  <sensitivityLevel>
    <!--ro, req, int, sensitivity-->1
  </sensitivityLevel>
  <RegionCoordinatesList>
    <!--ro, opt, array, list of area coordinates, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <detectionTarget>
    <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:confidence of alarm notification: "low","mediumLow","mediumHigh","high","low"-
->low
  </alarmConfidence>
  <recordConfidence>
    <!--ro, opt, enum, confidence of video recording, subType:string, desc:confidence of recording video: "low","mediumLow","mediumHigh","high","low"-->low
  </recordConfidence>
  <ruleName>
    <!--ro, opt, string, range:[1,64]-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object, dep:or,{$.RegionEntranceRegion.detectionTarget,eq,all},{$.RegionEntranceRegion.detectionTarget,eq,human}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object, dep:or,{$.RegionEntranceRegion.detectionTarget,eq,all},{$.RegionEntranceRegion.detectionTarget,eq,animal}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</RegionEntranceRegion>
```

19.10.3.18 Region entrance

EventType:regionEntrance

```
<?xml version="1.0" encoding="UTF-8"?>
```

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```
<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"regionEntrance" (region entrance)-->regionEntrance
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area points coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
        </height>
      </TargetRect>
      <sceneID>
        <!--ro, opt, int, scene ID-->1
      </sceneID>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
```

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```
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transparent transmission field-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为
64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
```

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```
  </timeStamp>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits-->test
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:Rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:Rule Name-->test
  </ruleName>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </visibleLightAbsoluteHigh>
  <thermalAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </thermalAbsoluteHigh>
  <deviceLocation>
    <!--ro, opt, object-->
    <longitudeType>
      <!--ro, req, enum, subType:string-->W
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
    </latitudeType>
    <longitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, degree(s), range:[0,180]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </longitude>
    <latitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, range:[0,90]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </latitude>
  </deviceLocation>
  <laserRanging>
    <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
  </laserRanging>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
regionEntrance	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.10.4 Region Exiting

19.10.4.1 Get arming schedules of all channels for region exiting detection

Request URL

GET /ISAPI/Event/schedules/regionExitings

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, list of arming schedules, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, ID, desc:format: event type-channel No.-->test
    </id>
    <eventType>
      <!--ro, opt, string, event type-->regionExiting
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, list of arming schedules, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming schedule-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionExitingScheduleList>
```

19.10.4.2 Set region exiting detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/regionExitings

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, string, event type-->regionExiting
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionExitingScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.10.4.3 Get the arming schedule of region exiting detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/regionExitings/regionExiting-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, opt, string-->regionExiting
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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19.10.4.4 Set the arming schedule of region exiting detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/regionExitings/regionExiting-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--opt, string-->regionExiting
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.10.4.5 Get the linkage triggering parameters of region exiting for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionExiting-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, enum, subType:string-->regionExiting
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, boot-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

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19.10.4.6 Set the linkage triggering parameters of region exiting for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionExiting-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object-->
    <id>
      <!--req, enum, subType:string-->regionExiting
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object-->
      <ptzChannelID>
        <!--req, string-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object-->
      <whiteLightDurationTime>
        <!--opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

19.10.4.7 Set the linkage parameters of region exiting for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionExiting-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, string-->regionExiting
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
      <LightAudioByIPAction>
        <!--opt, object-->
        <notificationType>
          <!--req, enum, subType:string-->audio
        </notificationType>
        <notificationName>
          <!--opt, string-->10.16.99.98:8000:光
        </notificationName>
      </LightAudioByIPAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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VISION AND SECURITY SYSTEM CO.,LTD.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.4.8 Get the linkage parameters of region exiting for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionExiting-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

dojo jin@gmail.com VISION AND SECURITY SYSTEM CO., LTD.

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, string-->regionEntrance
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <SupplementLightAlarm>
        <!--ro, opt, object-->
        <durationTime>
          <!--ro, opt, int, range:[0,90], unit:s-->5
        </durationTime>
      </SupplementLightAlarm>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

19.10.4.9 Get region exiting detection parameters of all channels

Request URL

GET /ISAPI/Smart/regionExiting

Query Parameter

None

Request Message

None

Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <RegionExiting>
    <!--ro, opt, object, region exiting-->
    <id>
      <!--ro, req, string, channel No.-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <normalizedScreenSize>
      <!--ro, opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <RegionExitingRegionList>
      <!--ro, opt, array, subType:object-->
      <RegionExitingRegion>
        <!--ro, opt, object, region exiting-->
        <id>
          <!--ro, req, string, detection area ID-->1
        </id>
        <sensitivityLevel>
          <!--ro, req, int, sensitivity, range:[0,100]-->1
        </sensitivityLevel>
        <RegionCoordinatesList>
          <!--ro, opt, array, region coordinates list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <detectionTarget>
          <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <humanMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--ro, opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--ro, opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--ro, opt, enum, subType:string-->low
        </recordConfidence>
      </RegionExitingRegion>
    </RegionExitingRegionList>
    <mutexAbility opt="PDC">
      <!--ro, opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--ro, opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </RegionExiting>
</RegionExitingList>
```

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19.10.4.10 Set region exiting detection parameters of all channels

Request URL

PUT /ISAPI/Smart/regionExiting

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <RegionExiting>
    <!--opt, object-->
    <id>
      <!--req, string-->1
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <normalizedScreenSize>
      <!--opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <RegionExitingRegionList>
      <!--opt, array, subType:object-->
      <RegionExitingRegion>
        <!--opt, object-->
        <id>
          <!--req, string-->1
        </id>
        <sensitivityLevel>
          <!--req, int, sensitivity, range:[0,100]-->1
        </sensitivityLevel>
        <RegionCoordinatesList>
          <!--opt, array, region coordinates list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, x-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <detectionTarget>
          <!--opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <humanMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (human body)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--opt, enum, subType:string-->low
        </recordConfidence>
      </RegionExitingRegion>
    </RegionExitingRegionList>
    <mutexAbility opt="PDC">
      <!--opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </RegionExiting>
</RegionExitingList>
```

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.4.11 Set parameters of region exiting detection rule by channel

Request URL

PUT /ISAPI/Smart/regionExiting/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExiting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of region exiting detection rule, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <RegionExitingRegionList>
    <!--opt, array, region list of region exiting detection, subType:object-->
    <RegionExitingRegion>
      <!--opt, object, region exiting detection region-->
      <id>
        <!--req, string, region ID-->1
      </id>
      <sensitivityLevel>
        <!--req, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--opt, array, region coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value
"all" is mutually exclusive with "others"-->all
      </detectionTarget>
      <humanMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
      </vehicleMisinfoFilterEnabled>
      <priority>
        <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
      </priority>
      <alarmConfidence>
        <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </alarmConfidence>
      <recordConfidence>
        <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </recordConfidence>
    </RegionExitingRegion>
  </RegionExitingRegionList>
  <mutexAbility opt="PDC">
    <!--opt, string, mutually exclusive capability: "PDC" (people counting statistics, which is not supported during region entrance detection),
attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--opt, bool, whether supports multi-scene detection-->true
  </isSupportMultiScene>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
</RegionExiting>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.4.12 Get parameters of region exiting detection rule by channel

Request URL

GET /ISAPI/Smart/regionExiting/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExiting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, read-only, "PDC"-people counting statistics, which is not supported during region exiting detection, attr:version{req, string,
protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <RegionExitingRegionList>
    <!--ro, opt, array, list of region exiting, subType:object-->
    <RegionExitingRegion>
      <!--ro, opt, object, region exiting-->
      <id>
        <!--ro, req, string, region ID-->1
      </id>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value
"all" is mutually exclusive with "others"-->all
      </detectionTarget>
      <humanMisinfoFilterEnabled>
```

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```
        <!--ro, opt, bool, whether to enable false alarm filtering of human body detection-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering of vehicle detection-->true
      </vehicleMisinfoFilterEnabled>
      <priority>
        <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
      </priority>
      <alarmConfidence>
        <!--ro, opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </alarmConfidence>
      <recordConfidence>
        <!--ro, opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </recordConfidence>
      <ruleName>
        <!--ro, opt, string, range:[1,64]-->test
      </ruleName>
      <detectionTargetProperty>
        <!--ro, opt, object-->
        <human>
          <!--ro, opt, object, dep:or,{$.RegionExiting.RegionExitingRegionList.RegionExitingRegion.detectionTarget,eq,all},
{$.RegionExiting.RegionExitingRegionList.RegionExitingRegion.detectionTarget,eq,human}-->
          <propertyList>
            <!--ro, opt, array, subType:object-->
            <property>
              <!--ro, opt, enum, subType:string-->nonMotor
            </property>
          </propertyList>
        </human>
        <animal>
          <!--ro, opt, object, dep:or,{$.RegionExiting.RegionExitingRegionList.RegionExitingRegion.detectionTarget,eq,all},
{$.RegionExiting.RegionExitingRegionList.RegionExitingRegion.detectionTarget,eq,animal}-->
          <propertyList>
            <!--ro, opt, array, subType:object-->
            <property>
              <!--ro, opt, enum, subType:string-->large
            </property>
          </propertyList>
        </animal>
      </detectionTargetProperty>
    </RegionExitingRegion>
  </RegionExitingRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool, whether supports multi-scene detection-->true
  </isSupportMultiScene>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <stackTarget>
    <!--ro, opt, bool-->true
  </stackTarget>
  <targetSensitivityLevel>
    <!--ro, opt, int, range:[0,100]-->50
  </targetSensitivityLevel>
</RegionExiting>
```

19.10.4.13 Get configuration capability of region exiting detection by channel

Request URL

GET /ISAPI/Smart/regionExiting/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExiting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, ro, mutex capability, it indicates that which function is mutually exclusive with the region exiting detection:
"PDC,videoFrameRate50,videoFrameRate60,19-1920*1080@50fps,20-1920*1080@60fps,scheduled recording", attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
```

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```
    </id>
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <normalizedScreenSize>
      <!--ro, opt, object, normalized screen size-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized screen width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized screen height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <RegionExitingRegionList size="4">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <RegionExitingRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, string-->1
        </id>
        <sensitivityLevel min="1" max="100">
          <!--ro, req, int, sensitivity, attr:min{req, int},max{req, int}-->1
        </sensitivityLevel>
        <RegionCoordinatesList>
          <!--ro, opt, array, region coordinate list, subType:object, range:[3,10], desc:region coordinate list-->
          <RegionCoordinates>
            <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <maxRegionCoordinatesNum>
          <!--ro, opt, int-->1
        </maxRegionCoordinatesNum>
        <minRegionCoordinatesNum>
          <!--ro, opt, int-->1
        </minRegionCoordinatesNum>
        <detectionTarget opt="all,human,vehicle,animal,others">
          <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string}, desc:detection target-->all
        </detectionTarget>
        <alarmConfidence opt="low,mediumLow,mediumHigh,high" def="low">
          <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
        </alarmConfidence>
        <recordConfidence opt="low,mediumLow,mediumHigh,high" def="low">
          <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
        </recordConfidence>
        <ruleName min="1" max="64">
          <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
        </ruleName>
        <detectionTargetProperty>
          <!--ro, opt, object-->
          <human>
            <!--ro, opt, object-->
            <propertyList size="1">
              <!--ro, opt, object, attr:size{req, int}-->
              <property opt="nonMotor">
                <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
              </property>
            </propertyList>
          </human>
          <animal>
            <!--ro, opt, object-->
            <propertyList size="3">
              <!--ro, opt, object, attr:size{req, int}-->
              <property opt="large,small,poultry">
                <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
              </property>
            </propertyList>
          </animal>
        </detectionTargetProperty>
      </RegionExitingRegion>
    </RegionExitingRegionList>
    <mutexAbility opt="PDC">
      <!--ro, opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--ro, req, bool, whether supports multiple scenes.-->true
    </isSupportMultiScene>
    <isSupportHumanMisinfoFilter>
      <!--ro, opt, bool, whether supports preventing false human body alarm, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned-->true
    </isSupportHumanMisinfoFilter>
    <isSupportVehicleMisinfoFilter>
      <!--ro, opt, bool, whether supports preventing false vehicle alarm, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned-->true
    </isSupportVehicleMisinfoFilter>
    <isSupportTargetMultiSelect>
      <!--ro, opt, bool, whether supports selecting multiple detection targets, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will not be returned-->true
    </isSupportTargetMultiSelect>
```

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```
  </isSupportTargetMultiSelect>
  <isSupportAllDayUpload>
    <!--ro, opt, bool, whether supports uploading all-day event-->true
  </isSupportAllDayUpload>
  <isSupportOverlapParams>
    <!--ro, opt, bool-->true
  </isSupportOverlapParams>
  <isSupportStackTarget>
    <!--ro, opt, bool-->true
  </isSupportStackTarget>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
  </isSupportAllTargetDetectionWebDisplay>
  <targetSensitivityLevel min="0" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
  </targetSensitivityLevel>
</RegionExiting>
```

19.10.4.14 Set parameters of all region exiting detection regions by channel

Request URL

PUT /ISAPI/Smart/regionExiting/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameter list of all region exiting detection regions, subType:object, attr:version{req, string, protocolVersion}-->
  <RegionExitingRegion>
    <!--opt, object, parameters of region exiting detection region-->
    <id>
      <!--req, string, region ID-->1
    </id>
    <sensitivityLevel>
      <!--req, int, sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--opt, array, region coordinate list, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </RegionExitingRegion>
</RegionExitingRegionList>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.4.15 Get parameters of all region exiting detection regions by channel

Request URL

GET /ISAPI/Smart/regionExiting/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <RegionExitingRegion>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->1
    </id>
    <sensitivityLevel>
      <!--ro, req, int, range:[0,100]-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinate list, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:priority-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, subType:string-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, subType:string-->low
    </recordConfidence>
  </RegionExitingRegion>
</RegionExitingRegionList>
```

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19.10.4.16 Get parameters of a region exiting detection region of a specific channel

Request URL

GET /ISAPI/Smart/regionExiting/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, region ID-->1
  </id>
  <sensitivityLevel>
    <!--ro, req, int, sensitivity, range:[0,100]-->1
  </sensitivityLevel>
  <RegionCoordinatesList>
    <!--ro, opt, array, list of area coordinates, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <detectionTarget>
    <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:confidence of alarm notification: "low","mediumLow","mediumHigh","high","low"-
->low
  </alarmConfidence>
  <recordConfidence>
    <!--ro, opt, enum, confidence of video recording, subType:string, desc:confidence of recording video: "low","mediumLow","mediumHigh","high","low"-->low
  </recordConfidence>
  <ruleName>
    <!--ro, opt, string, range:[1,64]-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object, dep:or,{$.RegionExitingRegion.detectionTarget,eq,all},{$.RegionExitingRegion.detectionTarget,eq,human}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object, dep:or,{$.RegionExitingRegion.detectionTarget,eq,all},{$.RegionExitingRegion.detectionTarget,eq,animal}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</RegionExitingRegion>
```

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19.10.4.17 Set parameters of a region exiting detection region by channel

Request URL

PUT /ISAPI/Smart/regionExiting/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of a region exiting detection region, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, region ID-->1
  </id>
  <sensitivityLevel>
    <!--req, int, sensitivity, range:[0,100]-->1
  </sensitivityLevel>
  <RegionCoordinatesList>
    <!--opt, array, region coordinate list, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <detectionTarget>
    <!--opt, enum, detection target type, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value
"all" is mutually exclusive with "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </alarmConfidence>
  <recordConfidence>
    <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </recordConfidence>
</RegionExitingRegion>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.10.4.18 Region exiting

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EventType:regionExiting

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome".
The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP--
>HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"regionExiting" (region exiting)-->regionExiting
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height-->1.000
        </height>
      </TargetRect>
      <sceneID>
```

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```
        <!--ro, opt, int, scene ID-->1
      </sceneID>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transparent transmission field-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选，设备ID，string类型，最大长度为
64*/ "deviceChannel": 1, /*可选，设备通道号，integer32类型*/ "deviceName": "shebei", /*可选，设备名称，string类型，最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
```

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```
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits--
>test
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </visibleLightAbsoluteHigh>
  <thermalAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </thermalAbsoluteHigh>
  <deviceLocation>
    <!--ro, opt, object-->
    <longitudeType>
      <!--ro, req, enum, subType:string-->W
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
    </latitudeType>
    <longitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, degree(s), range:[0,180]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </longitude>
    <latitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, range:[0,90]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </latitude>
  </deviceLocation>
  <laserRanging>
    <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
  </laserRanging>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
regionExiting	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.11 Crowd Trend Detection

19.11.1 Regional Target Counting

19.11.1.1 Get parameters of a single rule of regional target counting for a specified channel

Request URL

POST /ISAPI/Event/channels/<channelID>/GetRegionTargetNumberCountingCfg?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "sceneID":  1,
    /*opt, int, scene ID, range:[1,64], desc:this node is not required for certain types of devices, such as network cameras*/
    "ruleID":  1
    /*req, int, rule ID, range:[1,4]*/
}
```

Response Message

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```
{
    "ruleID":  1,
    /*ro, req, int, rule ID, range:[1,4]*/
    "enabled":  true,
    /*ro, req, bool, whether to enable the function*/
    "ruleName":  "test",
    /*ro, opt, string, rule name, range:[0,64]*/
    "ruleDraw": {
    /*ro, opt, object, rule area drawing parameters*/
        "ruleLinelist": [
        /*ro, req, array, line coordinates, subType:object, range:[0,2], desc:the origin is the upper-left corner of the screen*/
            {
                "x":  0.000,
                /*ro, req, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*ro, req, float, y-coordinate, range:[0.000,1.000]*/
            }
        ],
        "ruleLineDirection": {
        /*ro, req, object, detection line positive direction, desc:the origin is the upper-left corner of the screen*/
            "startPoint": {
            /*ro, req, object, start point*/
                "x":  0.000,
                /*ro, req, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*ro, req, float, y-coordinate, range:[0.000,1.000]*/
            },
            "endPoint": {
            /*ro, req, object, end point*/
                "x":  0.000,
                /*ro, req, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*ro, req, float, y-coordinate, range:[0.000,1.000]*/
            }
        }
    },
    "ruleParams": {
    /*ro, opt, object, rule parameters*/
        "statisticalObjectives": ["human", "nonMotor", "vehicle"],
        /*ro, opt, array, statistics target, subType:string, desc:"human", "nonMotor" (non-motor vehicles), "vehicle"*/
        "vehicleStatistical": ["bus", "van", "smallCar", "truck"],
        /*ro, opt, array, subType:string, dep:and,{$.ruleParams.statisticalObjectives[*],eq,vehicle}*/
        "interPupileRegion": {
        /*ro, opt, object, the area of minimum pupil distance, desc:the origin is the upper-left corner of the screen*/
            "x":  0.120,
            /*ro, req, float, x-coordinate of the upper-left corner, range:[0.000,1.000]*/
            "y":  0.120,
            /*ro, req, float, y-coordinate of the upper-left corner, range:[0.000,1.000]*/
            "width":  0.120,
            /*ro, req, float, rectangle width, range:[0.000,1.000]*/
            "height":  0.120
            /*ro, req, float, rectangle height, range:[0.000,1.000]*/
        },
        "dataOverlap": {
        /*ro, opt, object, OSD*/
            "overlapEnabled":  true,
            /*ro, req, bool, whether to enable the OSD*/
            "overlapDirection":  "bothway",
            /*ro, opt, enum, overlay direction, subType:string, desc:"bothway" (bidirectional); "forward", "back" (reverse)*/
            "forwardOverlapName":  "test",
            /*ro, opt, string, range:[0,48]*/
            "backOverlapName":  "test",
            /*ro, opt, string, range:[0,48]*/
            "overlapPosition": {
            /*ro, opt, object*/
                "x":  0.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "y":  0.000
                /*ro, req, float, range:[0.000,1.000]*/
            },
            "dailyResetTime":  "00:00:00+08:00",
            /*ro, opt, time, Daily Reset Time*/
            "fontSize":  "adaptive"
            /*ro, opt, enum, subType:string*/
        }
    }
}
```

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19.11.1.2 Get all rules parameters of regional target counting for a specified channel

Request URL

POST /ISAPI/Event/channels/<channelID>/GetRegionTargetNumberCountingCfgList?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "sceneID":  1
    /*opt, int, scene ID (unique), range:[1,64], desc:if this node is not configured, it indicates to get rules of all scenes*/
}
```

Response Message

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```
{
    "ruleList": [
    /*ro, opt, array, rule list, subType:object*/
        {
            "sceneID":  1,
            /*ro, opt, int, scene ID, range:[1,64], desc:this node is not required for certain types of devices, such as network cameras*/
            "ruleID":  1,
            /*ro, req, int, rule ID, range:[1,4]*/
            "enabled":  true,
            /*ro, req, bool, whether to enable the function*/
            "ruleName":  "test",
            /*ro, opt, string, rule name, range:[0,64]*/
            "ruleDraw": {
            /*ro, opt, object, rule area drawing parameters*/
                "ruleLinelist": [
                /*ro, req, array, line coordinates, subType:object, range:[0,2], desc:the origin is the upper-left corner of the screen*/
                    {
                        "x":  0.000,
                        /*ro, req, float, x-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, req, float, y-coordinate, range:[0.000,1.000]*/
                    }
                ],
                "ruleLineDirection": {
                /*ro, req, object, detection line positive direction, desc:the origin is the upper-left corner of the screen*/
                    "startPoint": {
                    /*ro, req, object, start point*/
                        "x":  0.000,
                        /*ro, req, float, x-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, req, float, y-coordinate, range:[0.000,1.000]*/
                    },
                    "endPoint": {
                    /*ro, req, object, end point*/
                        "x":  0.000,
                        /*ro, req, float, x-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, req, float, y-coordinate, range:[0.000,1.000]*/
                    }
                }
            },
            "ruleParams": {
            /*ro, opt, object, rule parameters*/
                "statisticalObjectives": ["human", "nonMotor", "vehicle"],
                /*ro, opt, array, statistics target, subType:string, desc:"human", "nonMotor" (non-motor vehicles), "vehicle"*/
                "vehicleStatistical": ["bus", "van", "smallCar", "truck"],
                /*ro, opt, array, subType:string, dep:and,{$.ruleParams.statisticalObjectives[*],eq,vehicle}*/
                "interPupileRegion": {
                /*ro, opt, object, the area of minimum pupil distance, desc:the origin is the upper-left corner of the screen*/
                    "x":  0.120,
                    /*ro, req, float, x-coordinate of the upper-left corner, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*ro, req, float, y-coordinate of the upper-left corner, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*ro, req, float, rectangle width, range:[0.000,1.000]*/
                    "height":  0.120
                    /*ro, req, float, rectangle height, range:[0.000,1.000]*/
                },
                "dataOverlap": {
                /*ro, opt, object, OSD*/
                    "overlapEnabled":  true,
                    /*ro, req, bool, whether to enable the OSD*/
                    "overlapDirection":  "bothway",
                    /*ro, opt, enum, overlay direction, subType:string, desc:"bothway" (bidirectional); "forward", "back" (reverse)*/
                    "forwardOverlapName":  "test",
                    /*ro, opt, string, range:[0,48]*/
                    "backOverlapName":  "test",
                    /*ro, opt, string, range:[0,48]*/
                    "overlapPosition": {
                    /*ro, opt, object*/
                        "x":  0.000,
                        /*ro, req, float, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, req, float, range:[0.000,1.000]*/
                    },
                    "dailyResetTime":  "00:00:00+08:00",
                    /*ro, opt, time, daily reset time*/
                    "fontSize":  "adaptive"
                    /*ro, opt, enum, subType:string*/
                }
            }
        }
    ]
}
```

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19.11.1.3 Set parameters of a single rule of regional target counting for a specified channel

Request URL

PUT /ISAPI/Event/channels/<channelID>/ModifyRegionTargetNumberCountingCfg?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "sceneID":  1,
    /*opt, int, scene ID, range:[1,64], desc:this node is not required for certain types of devices, such as network cameras*/
    "ruleID":  1,
    /*req, int, rule ID, range:[1,4]*/
    "enabled":  true,
    /*req, bool, whether to enable the function*/
    "ruleName":  "test",
    /*opt, string, rule name, range:[0,64]*/
    "ruleDraw": {
    /*opt, object, rule area drawing parameters*/
        "ruleLinelist": [
        /*req, array, line coordinates, subType:object, range:[0,2], desc:the origin is the upper-left corner of the screen*/
            {
                "x":  0.000,
                /*req, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*req, float, y-coordinate, range:[0.000,1.000]*/
            }
        ],
        "ruleLineDirection": {
        /*req, object, detection line positive direction, desc:the origin is the upper-left corner of the screen*/
            "startPoint": {
            /*req, object, start point*/
                "x":  0.000,
                /*req, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*req, float, y-coordinate, range:[0.000,1.000]*/
            },
            "endPoint": {
            /*req, object, end point*/
                "x":  0.000,
                /*req, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*req, float, y-coordinate, range:[0.000,1.000]*/
            }
        }
    },
    "ruleParams": {
    /*opt, object, rule parameters*/
        "statisticalObjectives": ["human", "nonMotor", "vehicle"],
        /*opt, array, statistics target, subType:string, desc:"human", "nonMotor" (non-motor vehicles), "vehicle"*/
        "interPupileRegion": {
        /*ro, opt, object, the area of minimum pupil distance, desc:the origin is the upper-left corner of the screen*/
            "x":  0.120,
            /*req, float, x-coordinate of the upper-left corner, range:[0.000,1.000]*/
            "y":  0.120,
            /*req, float, y-coordinate of the upper-left corner, range:[0.000,1.000]*/
            "width":  0.120,
            /*req, float, rectangle width, range:[0.000,1.000]*/
            "height":  0.120
            /*req, float, rectangle height, range:[0.000,1.000]*/
        },
        "dataOverlap": {
        /*opt, object, OSD*/
            "overlapEnabled":  true,
            /*req, bool, whether to enable the OSD*/
            "overlapDirection":  "bothway",
            /*opt, enum, overlay direction, subType:string, desc:"bothway" (bidirectional); "forward", "back" (reverse)*/
            "dailyResetTime":  "00:00:00+08:00"
            /*opt, time, daily reset time*/
        }
    }
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

19.11.1.4 Get the capability of regional target counting parameters for a specified channel

Request URL

GET /ISAPI/Event/channels/<channelID>/RegionTargetNumberCounting/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "eventRulesCap": {
    /*ro, req, object, event configuration capability*/
        "commonCfgCap": {
        /*ro, req, object, general configuration capability*/
            "enabled": {
            /*ro, req, object, whether to enable the function*/
                "@opt": [true, false],
                /*ro, req, array, options, subType:bool*/
                "@def":  false
                /*ro, opt, bool, default value*/
            },
            "realtimeTriggeUpload": {
            /*ro, opt, object*/
                "enabled": {
                /*ro, req, object*/
                    "@opt": [true, false],
                    /*ro, req, array, subType:bool*/
                    "@def":  false
                    /*ro, opt, bool*/
                }
            },
            "regularUpload": {
            /*ro, opt, object, scheduled uploading*/
                "enabled": {
                /*ro, req, object, whether to enable the function*/
                    "@opt": [true, false],
                    /*ro, req, array, options, subType:bool*/
                    "@def":  false
                    /*ro, opt, bool, default value*/
                },
                "alarmInterval": {
                /*ro, opt, object, scheduled alarm interval*/
                    "@min":  1,
                    /*ro, opt, int, the minimum value, range:[1,60], unit:min*/
                    "@max":  60,
                    /*ro, opt, int, the maximum value*/
                    "@def":  15
                    /*ro, opt, int, the default value*/
                }
            },
            "emailReport": {
            /*ro, opt, object, email report*/
                "dayReport": {
                /*ro, req, object, whether to enable the function*/
                    "@opt": [true, false],
                    /*ro, req, array, options, subType:bool*/
                    "@def":  false
                    /*ro, opt, bool, default value*/
                },
                "weekReport": {
                /*ro, req, object, whether to enable the function*/
                    "@opt": [true, false],
                    /*ro, req, array, options, subType:bool*/
```

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```
                "@def":  false
                /*ro, opt, bool, default value*/
            },
            "monthReport": {
            /*ro, req, object, whether to enable the function*/
                "@opt": [true, false],
                /*ro, req, array, options, subType:bool*/
                "@def":  false
                /*ro, opt, bool, default value*/
            },
            "yearReport": {
            /*ro, req, object, whether to enable the function*/
                "@opt": [true, false],
                /*ro, req, array, options, subType:bool*/
                "@def":  false
                /*ro, opt, bool, default value*/
            },
            "reportFormat": {
            /*ro, opt, object, Format*/
                "@opt": ["excel", "csv", "txt", "xml"],
                /*ro, req, array, options, subType:string, desc:"excel", "csv", "txt", "xml"*/
                "@def": ["excel", "csv", "txt", "xml"]
                /*ro, opt, array, default value, subType:string*/
            }
        }
    },
    "ruleInfoCap": {
    /*ro, opt, object, rule information*/
        "maxSize":  4,
        /*ro, req, int, the maximum number of supported rules*/
        "ruleID": {
        /*ro, req, object, rule ID (unique)*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  4
            /*ro, req, int, the maximum value*/
        },
        "ruleDraw": {
        /*ro, opt, object, rule area drawing parameters*/
            "ruleLineListCap": {
            /*ro, opt, object, rule line capability, desc:the origin is the upper-left corner of the screen*/
                "x": {
                /*ro, req, object, x-coordinate*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, req, object, y-coordinate*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                }
            },
            "ruleLineDirectionCap": {
            /*ro, opt, object, capability of detection line positive direction, desc:the origin is the upper-left corner of the screen*/
                "startPoint": {
                /*ro, req, object, start point, desc:the origin is the upper-left corner of the screen*/
                    "x": {
                    /*ro, req, object, x-coordinate*/
                        "@min":  0.000,
                        /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
                        /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    },
                    "y": {
                    /*ro, req, object, y-coordinate*/
                        "@min":  0.000,
                        /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
                        /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    }
                },
                "endPoint": {
                /*ro, req, object, end point, desc:the origin is the upper-left corner of the screen*/
                    "x": {
                    /*ro, req, object, x-coordinate*/
                        "@min":  0.000,
                        /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
                        /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    },
                    "y": {
                    /*ro, req, object, y-coordinate*/
                        "@min":  0.000,
                        /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                        "@max":  1.000
                        /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    }
                }
            }
        },
```

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```
            "ruleParams": {
            /*ro, opt, object, capability of rule configuration parameters*/
                "statisticalObjectives": {
                /*ro, opt, object, statistics target*/
                    "@opt": ["human", "nonMotor", "vehicle"],
                    /*ro, opt, array, range, subType:string, desc:"human", "nonMotor" (non-motor vehicles), "vehicle"*/
                    "@def": ["human", "nonMotor", "vehicle"]
                    /*ro, opt, array, default value, subType:string*/
                },
                "vehicleStatistical": {
                /*ro, opt, object*/
                    "@opt": ["bus", "van", "smallCar", "truck"],
                    /*ro, opt, array, subType:string*/
                    "@def": ["bus", "van", "smallCar", "truck"]
                    /*ro, opt, array, subType:string*/
                },
                "dataOverlap": {
                /*ro, opt, object, OSD*/
                    "overlapEnabled": {
                    /*ro, req, object, whether to enable the OSD*/
                        "@opt": [true, false],
                        /*ro, req, array, options, subType:bool*/
                        "@def":  false
                        /*ro, opt, bool, default value*/
                    },
                    "overlapDirection": {
                    /*ro, opt, object, overlay direction*/
                        "@opt": ["forward", "back"],
                        /*ro, req, array, options, subType:string, desc:"bothway" (bidirectional), "forward", "back" (reverse)*/
                        "@def":  "bothway"
                        /*ro, opt, string, default value*/
                    },
                    "forwardOverlapName": {
                    /*ro, opt, object*/
                        "@min":  0,
                        /*ro, req, int*/
                        "@max":  48
                        /*ro, req, int*/
                    },
                    "backOverlapName": {
                    /*ro, opt, object*/
                        "@min":  0,
                        /*ro, req, int*/
                        "@max":  48
                        /*ro, req, int*/
                    },
                    "overlapPosition": {
                    /*ro, opt, object*/
                        "x": {
                        /*ro, opt, object*/
                            "@min":  0.000,
                            /*ro, req, float, range:[0.000,1.000]*/
                            "@max":  1.000
                            /*ro, req, float, range:[0.000,1.000]*/
                        },
                        "y": {
                        /*ro, opt, object*/
                            "@min":  0.000,
                            /*ro, req, float, range:[0.000,1.000]*/
                            "@max":  1.000
                            /*ro, req, float, range:[0.000,1.000]*/
                        }
                    },
                    "dailyResetTime":  "00:00:00+08:00",
                    /*ro, opt, time, daily reset time*/
                    "fontSize": {
                    /*ro, opt, object*/
                        "@opt": ["16*16", "32*32", "48*48", "64*64", "96*96", "adaptive"],
                        /*ro, req, array, subType:string*/
                        "@def":  "adaptive"
                        /*ro, opt, string*/
                    }
                }
                },
                "isSupportOSDReset":  true
                /*ro, opt, bool*/
            },
            "isSupportSearchReport":  true,
            /*ro, opt, bool, whether the device supports searching for statistics report, desc:related URL:
/ISAPI/Event/channels/<channelID>/SearchRegionTargetNumberCounting/capabilities?format=json*/
        }
    }
```

19.11.1.5 Set the general parameters of regional target counting for a specified channel

Request URL

PUT /ISAPI/Event/channels/<channelID>/RegionTargetNumberCountingCommonCfg?format=json

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "enabled":  false,
    /*req, bool, whether to enable the function*/
    "realtimeTriggeUpload": {
    /*opt, object, real-time uploading of triggered alarm*/
        "enabled":  true
        /*req, bool, whether to enable the function*/
    },
    "regularUpload": {
    /*opt, object, scheduled uploading*/
        "enabled":  false,
        /*req, bool, whether to enable the function*/
        "alarmInterval":  1
        /*ro, opt, int, scheduled alarm interval, range:[1,60], unit:min*/
    },
    "emailReport": {
    /*opt, object, email report*/
        "dayReport":  true,
        /*opt, bool, whether to enable daily report*/
        "weekReport":  true,
        /*opt, bool, whether to enable weekly report*/
        "monthReport":  true,
        /*opt, bool, whether to enable monthly report*/
        "yearReport":  true,
        /*opt, bool, whether to enable annual report*/
        "reportFormat": ["excel", "csv"]
        /*opt, array, report format, subType:string, desc:"excel", "csv", "txt", "xml"*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

19.11.1.6 Get the general parameters of regional target counting for a specified channel

Request URL

GET /ISAPI/Event/channels/<channelID>/RegionTargetNumberCountingCommonCfg?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dejojin@gmail.com

```
{
    "enabled":  false,
    /*ro, req, bool, whether to enable the function*/
    "realtimeTriggeUpload": {
    /*ro, opt, object, real-time uploading of triggered alarm*/
        "enabled":  true
        /*ro, req, bool, whether to enable the function*/
    },
    "regularUpload": {
    /*ro, opt, object, scheduled uploading*/
        "enabled":  false,
        /*ro, req, bool, whether to enable the function*/
        "alarmInterval":  1
        /*ro, opt, int, scheduled alarm interval, range:[1,60], unit:min*/
    },
    "emailReport": {
    /*ro, opt, object, email report*/
        "dayReport":  true,
        /*ro, opt, bool, whether to enable daily report*/
        "weekReport":  true,
        /*ro, opt, bool, whether to enable weekly report*/
        "monthReport":  true,
        /*ro, opt, bool, whether to enable monthly report*/
        "yearReport":  true,
        /*ro, opt, bool, whether to enable annual report*/
        "reportFormat": ["excel", "csv"]
        /*ro, opt, array, report format, subType:string, desc:"excel", "csv", "txt", "xml"*/
    }
}
```

19.11.1.7 Get the capability of searching for the regional target statistics report of a specified channel

Request URL

GET /ISAPI/Event/channels/<channelID>/SearchRegionTargetNumberCounting/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
{
    "ReportCond": {
    /*ro, opt, object, search condition*/
        "reportType": {
        /*ro, req, object, report type*/
            "@opt": ["daily", "weekly"],
            /*ro, req, array, range, subType:string*/
            "@def":  "daily"
            /*ro, req, enum, search according to report type, subType:string, desc:"daily" (daily report, begins at 00:00 and ends at 24:00), "weekly"
(weekly report, from Monday to Sunday), "monthly" (monthly report), "yearly" (annual report), "custom" (custom report)*/
        },
        "statisticalTime":  "2004-05-03T00:00:00+08:00",
        /*ro, req, string, statistics time, desc:it is accurate to YYYY-MM-DD, the HH:mm:ss can be ignored*/
        "statisticalObjectives": {
        /*ro, req, object, statistics target*/
            "@opt": ["human", "nonMotor", "vehicle"],
            /*ro, req, array, range, subType:string, desc:"human", "nonMotor" (non-motor vehicles), "vehicle"*/
            "@def": ["human", "nonMotor", "vehicle"]
            /*ro, opt, array, default value, subType:string, desc:"human", "nonMotor" (non-motor vehicles), "vehicle"*/
        },
        "vehicleStatistical": {
        /*ro, opt, object*/
            "@opt": ["bus", "van", "smallCar", "truck"],
            /*ro, opt, array, subType:string*/
            "@def": ["bus", "van", "smallCar", "truck"]
            /*ro, opt, array, subType:string*/
        },
        "statisticalDirection": {
        /*ro, opt, object, statistics direction*/
            "@opt": ["forward", "bothway", "back"],
            /*ro, req, array, range, subType:string, desc:"bothway" (bidirectional), "forward", "back" (reverse)*/
            "@def":  "forward"
            /*ro, opt, string, default value*/
        },
        "sceneID": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  64
            /*ro, req, int*/
        },
        "ruleID": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  4
            /*ro, req, int*/
        }
    }
}
```

19.11.1.8 Search for the regional target statistics report of a specified channel

Request URL

POST /ISAPI/Event/channels/<channelID>/SearchRegionTargetNumberCounting?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
{
    "ReportCond": {
    /*opt, object, search condition*/
        "reportType":  "daily",
        /*req, enum, search according to report type, subType:string, desc:"daily" (daily report, begins at 00:00 and ends at 24:00), "weekly" (weekly
report, from Monday to Sunday), "monthly" (monthly report), "yearly" (annual report), "custom" (custom report)*/
        "statisticalTime":  "2004-05-03T00:00:00+08:00",
        /*req, string, statistics time, desc:it is accurate to YYYY-MM-DD, the HH:mm:ss can be ignored*/
        "statisticalObjectives": ["human", "nonMotor", "vehicle"],
        /*req, array, statistics target, subType:string, desc:"human", "nonMotor" (non-motor vehicles), "vehicle", "all"*/
        "vehicleStatistical": ["bus", "van", "smallCar", "truck"],
        /*opt, array, subType:string, dep:and,{$.ReportCond.statisticalObjectives[*],eq,vehicle}*/
        "statisticalDirection":  "bothway",
        /*opt, enum, statistics direction, subType:string, desc:"bothway" (bidirectional), "forward", "back" (reverse)*/
        "sceneID":  1,
        /*opt, int, range:[1,64]*/
        "ruleID":  1
        /*opt, int, range:[1,4]*/
    }
}
```

Response Message

```
{
    "ReportResult": [
    /*ro, opt, array, search result, subType:object*/
        {
            "startTime":  "1970-01-01T00:00:00+08:00",
            /*ro, req, datetime, start time*/
            "endTime":  "1970-01-01T00:00:00+08:00",
            /*ro, req, datetime, end time*/
            "sceneID":  1,
            /*ro, opt, int, range:[1,64]*/
            "ruleID":  1,
            /*ro, req, int, range:[1,4]*/
            "dataList": [
            /*ro, req, array, data list, subType:object*/
                {
                    "statisticalDirection":  "bothway",
                    /*ro, req, enum, statistics direction, subType:string, desc:"bothway" (bidirectional), "forward", "back" (reverse)*/
                    "humanCount":  1,
                    /*ro, opt, int, number of people, range:[0,99999]*/
                    "nonMotorCount":  1,
                    /*ro, opt, int, number of non-motor vehicles, range:[0,99999]*/
                    "vehicleCount":  1,
                    /*ro, opt, int, number of vehicles, range:[0,99999]*/
                    "busCount":  1,
                    /*ro, opt, int, range:[0,99999]*/
                    "vanCount":  1,
                    /*ro, opt, int, range:[0,99999]*/
                    "smallCarCount":  1,
                    /*ro, opt, int, range:[0,99999]*/
                    "truckCount":  1
                    /*ro, opt, int, range:[0,99999]*/
                }
            ]
        }
    ]
}
```

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dejojin@gmail.com

19.11.1.9 Regional target counting

EventType:regionTargetNumberCounting

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm, range:[1,32]*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm, range:[1,128]*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm, range:[1,65535]*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address, range:[1,128]*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:the event triggering frequency. The pulse event is the event which is triggered continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object recognition occurs, and the recognition of each object will trigger the event once, e.g., the facial recognition. When the pulse events are triggered, this node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously according to the device detection frequency; if it changes to another source, then the counting will start again.*/
    "eventType":  "regionTargetNumberCounting",
    /*ro, req, enum, event type, subType:string, desc:"regionTargetNumberCounting" (regional target counting)*/
    "eventState":  "active",
    /*ro, req, enum, durative event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "region Target Number Counting",
    /*ro, req, string, event description, desc:regional target counting*/
    "channelName":  "ABC",
    /*ro, opt, string, channel name (camera name), desc:it should be the same as the channelName in the message of URI (/ISAPI/Streaming/channels/<ID>)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID (PUID), range:[1,64]*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers*/
    "regionTargetNumberCounting": {
    /*ro, opt, object, regional target counting*/
        "statisticalMethod":  "realTime",
        /*ro, opt, enum, counting method, subType:string, desc:"realTime" (real-time counting), "regular" (scheduled)*/
        "realTime":  "1970-01-01T00:00:00+08:00",
        /*ro, opt, datetime, real-time counting time, dep:and,{$.regionTargetNumberCounting[*].statisticalMethod,eq,realtime}*/
        "timeRange": {
        /*ro, opt, object, time period of real-time counting, dep:and,{$.regionTargetNumberCounting[*].statisticalMethod,eq,regular}*/
            "startTime":  "test",
            /*ro, opt, string, counting start time*/
            "endTime":  "test"
            /*ro, opt, string, counting end time*/
        },
        "dataList": [
        /*ro, opt, array, counting data list, subType:object*/
            {
                "sceneID":  1,
                /*ro, opt, int, scene ID, range:[1,64], desc:this node is not required for certain types of devices, such as network cameras*/
                "ruleID":  1,
                /*ro, opt, int, rule ID, range:[1,4]*/
                "dataList": [
                /*ro, req, array, data list, subType:object*/
                    {
                        "statisticalDirection":  "forward",
                        /*ro, req, enum, statistics direction, subType:string, desc:"bothway" (bidirectional), "forward", "back" (reverse)*/
                        "humanCount":  1,
                        /*ro, opt, int, number of people, range:[0,99999]*/
                        "nonMotorCount":  1,
                        /*ro, opt, int, number of non-motor vehicles, range:[0,99999]*/
                        "vehicleCount":  1
                        /*ro, opt, int, number of vehicles, range:[0,99999]*/
                    }
                ]
            }
        ]
    }
}
```

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19.12 Multi-Target-Type Detection

19.12.1 Global Target Detection

19.12.1.1 Get the uploading parameters of recognized target

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/recognitionDataUpload

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RecognitionDataUpload xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <protocolA>
    <!--ro, opt, bool-->true
  </protocolA>
  <protocolB>
    <!--ro, opt, bool-->true
  </protocolB>
</RecognitionDataUpload>
```

19.12.1.2 Set the uploading parameters of recognized target

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/recognitionDataUpload

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RecognitionDataUpload xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <protocolA>
    <!--opt, bool-->true
  </protocolA>
  <protocolB>
    <!--opt, bool-->true
  </protocolB>
</RecognitionDataUpload>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.12.2 Multi-Target-Type Detection

19.12.2.1 Get advanced configuration capability of multi-target-type detection algorithm library

Request URL

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GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/algParam/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetAlgParamCap": {
    /*ro, req, object, algorithm configuration capability of multi-target-type detection*/
        "enabled":  "true,false",
        /*ro, req, string, whether to enable algorithm configuration*/
        "targetSpeed": {
        /*ro, opt, object, target generation speed*/
            "@min":  1,
            /*ro, opt, int, minimum value*/
            "@max":  5,
            /*ro, opt, int, maximum value*/
            "@def":  3
            /*ro, req, int, the default value*/
        },
        "sensitiveLevel": {
        /*ro, opt, object, sensitivity*/
            "@min":  1,
            /*ro, opt, int, minimum value*/
            "@max":  5,
            /*ro, opt, int, maximum value*/
            "@def":  3
            /*ro, req, int, the default value*/
        },
        "snapMode": {
        /*ro, opt, object, Capture Mode*/
            "@opt":  "best, quick",
            /*ro, opt, string*/
            "@def":  "best"
            /*ro, opt, string, the default value*/
        },
        "bestSnapThreshold": {
        /*ro, opt, object, picture number threshold of optimal capture: 0 to 20 frames; it is valid only when snapMode is "best"*/
            "@min":  0,
            /*ro, opt, int, minimum value*/
            "@max":  100,
            /*ro, opt, int, maximum value*/
            "@def":  20
            /*ro, req, int, the default value*/
        },
        "quickSnapThreshold": {
        /*ro, opt, object, picture number threshold of fast capture: 20 to 80 frames; it is valid only when snapMode is "quick"*/
            "@min":  20,
            /*ro, opt, int, minimum value*/
            "@max":  80,
            /*ro, opt, int, maximum value*/
            "@def":  50
            /*ro, opt, int, the default value*/
        },
        "exposureEnabled":  "true,false",
        /*ro, opt, string, whether to enable face picture exposure*/
        "brightRef": {
        /*ro, opt, object, reference brightness*/
            "@min":  0,
            /*ro, opt, int, minimum value*/
            "@max":  100,
            /*ro, opt, int, maximum value*/
            "@def":  50
            /*ro, opt, int, the default value*/
        },
        "expDurationTime": {
        /*ro, opt, object, shortest exposure time*/
            "@min":  0,
            /*ro, opt, int, minimum value*/
            "@max":  3600,
            /*ro, opt, int, maximum value*/
            "@def":  60
            /*ro, req, int, the default value*/
        },
        "faceFilteringTime": {
        /*ro, opt, object, Face Filtering Time*/
            "@min":  0,
            /*ro, opt, int, minimum value*/
            "@max":  100,
            /*ro, opt, int, maximum value*/
            "@def":  5
```

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```
            /*ro, req, int, the default value*/
        },
        "longestSnapTime": {
        /*ro, opt, object, longest capture duration: 0 to 10 seconds. It is valid when snapMode is "quick"*/
            "@min":  1,
            /*ro, opt, int, minimum value*/
            "@max":  10,
            /*ro, opt, int, maximum value*/
            "@def":  5
            /*ro, opt, int, the default value*/
        },
        "UploadDataType": {
        /*ro, opt, object*/
            "faceEnabled":  "true,false",
            /*ro, opt, string*/
            "humanEnabled":  "true,false",
            /*ro, opt, string*/
            "nonMotorEnabled":  "true,false",
            /*ro, opt, string*/
            "vehiceEnabled":  "true,false"
            /*ro, opt, string*/
        },
        "RemoveDuplicateCap": {
        /*ro, opt, object, remove duplicates of captured face pictures; it takes effect when the value of snapMode is "best"*/
            "enabled":  "true,false",
            /*ro, req, string, whether to enable algorithm configuration*/
            "threshold": {
            /*ro, opt, object, threshold of removing duplicates, when the picture similarity is larger than the value, the picture is duplicated*/
                "@min":  70,
                /*ro, opt, int, minimum value*/
                "@max":  100,
                /*ro, opt, int, maximum value*/
                "@def":  88
                /*ro, req, int, the default value*/
            },
            "faceScore": {
            /*ro, opt, object, the face grading threshold for removing duplicates. Only when the face grading is larger than the configured threshold, the
face picture will be compared for removing duplicates, in case that low quality face pictures influence the comparison effect*/
                "@min":  20,
                /*ro, opt, int, minimum value*/
                "@max":  60,
                /*ro, opt, int, maximum value*/
                "@def":  20
                /*ro, opt, int, the default value*/
            },
            "updateTime": {
            /*ro, opt, object, face modal data updating time interval (the time interval between the modal data being added into the library and being
deleted)*/
                "@min":  0,
                /*ro, opt, int, minimum value*/
                "@max":  300,
                /*ro, opt, int, maximum value*/
                "@def":  60
                /*ro, opt, int, the default value*/
            }
        },
        "uploadWithoutLicense":  "true,false",
        /*ro, opt, string*/
        "vehicleDisplacementFilterEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false],
            /*ro, opt, array, subType:bool*/
            "@def":  false
            /*ro, opt, bool*/
        },
        "vehicleDisplacementFilterThreshold": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  100,
            /*ro, opt, int*/
            "@def":  30
            /*ro, opt, int*/
        },
        "framesCaptureEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false],
            /*ro, req, array, subType:bool*/
            "@def":  false
            /*ro, req, bool*/
        },
    }
}
```

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19.12.2.2 Get advanced parameters of multi-target-type detection algorithm library

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/algParam?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetAlgParam": {
    /*ro, req, object, algorithm configuration capability of multi-target-type detection*/
        "enabled": true,
        /*ro, req, bool, whether to enable algorithm configuration*/
        "targetSpeed": 3,
        /*ro, opt, int, target speed*/
        "sensitiveLevel": 3,
        /*ro, opt, int, sensitivity level*/
        "snapMode": "best",
        /*ro, opt, enum, capture mode, subType:string, desc:"best" (optimal capture), "quick" (fast capture)*/
        "bestSnapInterval": 1,
        /*ro, opt, int, it is valid only snapMode is "best", range:[1,255], desc:it is valid only snapMode is "best"*/
        "bestSnapThreshold": 20,
        /*ro, opt, int, picture number threshold of optimal capture, range:[0,100], desc:it is valid only snapMode is "best"*/
        "quickSnapThreshold": 20,
        /*ro, opt, int, picture number threshold of fast capture, range:[20,80], desc:it is valid only snapMode is "quick"*/
        "exposureEnabled": true,
        /*ro, opt, bool, whether to enable face picture exposure*/
        "brightRef": 50,
        /*ro, opt, int, reference brightness, range:[0,100]*/
        "expDurationTime": 60,
        /*ro, req, int, shortest exposure time, range:[0,3600], unit:s*/
        "faceFilteringTime": 5,
        /*ro, req, int, face filtering time, range:[0,100], unit:s*/
        "bestSnapNum": 1,
        /*ro, opt, int, range:[0,10]*/
        "longestSnapTime": 1,
        /*ro, opt, int, longest capture duration, range:[1,10], unit:s, desc:it is valid when snapMode is "quick"*/
        "UploadDataType": {
        /*ro, opt, object*/
            "faceEnabled": true,
            /*ro, opt, bool*/
            "humanEnabled": true,
            /*ro, opt, bool*/
            "nonMotorEnabled": true,
            /*ro, opt, bool*/
            "vehiceEnabled": true
            /*ro, opt, bool*/
        },
        "RemoveDuplicate": {
        /*ro, req, object, remove duplicates of captured face pictures, desc:it is valid when snapMode is "best"*/
            "enabled": true,
            /*ro, req, bool, whether to enable algorithm configuration*/
            "threshold": 88,
            /*ro, req, int, threshold of removing duplicates, desc:when the picture similarity is larger than the value, the picture is duplicated*/
            "faceScore": 20,
            /*ro, opt, int, the face grading threshold for removing duplicate, desc:only when the face grading is larger than the configured threshold, the face picture will be compared for removing duplicates, in case that low quality face pictures influence the comparison effect*/
            "updateTime": 60
            /*ro, opt, int, duplicates removing library update time, range:[0,300], desc:duplicates removing library update time*/
        },
        "uploadWithoutLicense": true,
        /*ro, opt, bool*/
        "panoramaTargetRemoveDuplicate": false,
        /*ro, opt, bool*/
        "miscatchFilterEnabled": false,
        /*ro, opt, bool*/
        "capturingPeopleEnabled": true,
        /*ro, opt, bool*/
        "lineCrossingCaptureSensitivity": 50,
        /*ro, opt, int, range:[0,100]*/
        "relatedMixedTargetUidTime": 30,
        /*ro, opt, int, range:[5,30], step:1, unit:s*/
        "nonMotorPlateCustomText": "test",
        /*ro, opt, string, range:[0,16]*/
        "filterDataType": {
        /*ro, opt, object*/
            "retrogradeEnabled": true,
            /*ro, opt, bool*/
            "mannedEnabled": true,
            /*ro, opt, bool*/
            "unproopedHelmetEnabled": true,
            /*ro, opt, bool*/
            "unblockedBicycleEnabled": true,
            /*ro, opt, bool*/
            "blockedBicycleEnabled": true
```

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```
            /*ro, opt, bool*/
        },
        "nonMotorOverspeedThreshoid":  30,
        /*ro, opt, int, range:[10,50], step:1, unit:km/h*/
        "vehicleOverspeedThreshoid":  40,
        /*ro, opt, int, range:[30,80], step:1, unit:km/h*/
        "vehicleDisplacementFilterEnabled":  true,
        /*ro, opt, bool*/
        "vehicleDisplacementFilterThreshold":  30,
        /*ro, opt, int, range:[1,100]*/
        "framesCaptureEnabled":  false,
        /*ro, opt, bool*/
        "facePosture":  "normal",
        /*ro, opt, enum, subType:string*/
        "snapAlgLevel":  1
        /*ro, opt, int, range:[1,5]*/
    }
}
```

19.12.2.3 Set advanced parameters of multi-target-type detection algorithm library

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/algParam?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
{
    "MixedTargetAlgParam": {
    /*req, object, algorithm configuration capability of multi-target-type detection*/
        "enabled":  true,
        /*req, bool, whether to enable algorithm configuration*/
        "targetSpeed":  3,
        /*opt, int, target generation speed, range:[1,5]*/
        "sensitiveLevel":  3,
        /*opt, int, sensitivity level, range:[1,5]*/
        "snapMode":  "best",
        /*opt, enum, capture mode, subType:string, desc:"best" (optimal capture), "quick" (fast capture)*/
        "bestSnapInterval":  1,
        /*opt, int, it is valid when the snapMode is "best", range:[1,255]*/
        "bestSnapThreshold":  20,
        /*opt, int, picture number threshold of optimal capture: 0 to 20 frames, it is valid when snapMode is "best", range:[0,100]*/
        "quickSnapThreshold":  50,
        /*opt, int, picture number threshold of fast capture: 20 to 80 frames, it is valid when snapMode is "quick", range:[20,80]*/
        "exposureEnabled":  true,
        /*opt, bool, whether to enable face picture exposure*/
        "brightRef":  50,
        /*opt, int, reference brightness, range:[0,100]*/
        "expDurationTime":  60,
        /*opt, int, minimum exposure time, range:[0,3600]*/
        "faceFilteringTime":  5,
        /*opt, int, face filtering time, range:[0,100]*/
        "bestSnapNum":  1,
        /*opt, int, optimal capture times, range:[0,10]*/
        "longestSnapTime":  5,
        /*opt, int, maximum capture duration: 0 to 10 seconds, it is valid when snapMode is "quick", range:[1,10]*/
        "UploadDataType": {
        /*opt, object*/
            "faceEnabled":  true,
            /*opt, bool*/
            "humanEnabled":  true,
            /*opt, bool*/
            "nonMotorEnabled":  true,
            /*opt, bool*/
            "vehiceEnabled":  true
            /*opt, bool*/
        },
        "RemoveDuplicate": {
        /*opt, object, remove duplicates of captured face pictures, it is valid when snapMode is "best"*/
            "enabled":  true,
            /*req, bool, whether to enable algorithm configuration*/
            "threshold":  88,
            /*opt, int, threshold of removing duplicates, when the picture similarity exceeds the threshold, the picture is duplicated, range:[70,100]*/
            "faceScore":  20,
            /*opt, int, the face grading threshold for removing duplicates, only when the face grading exceeds the configured threshold, the face picture
will be compared for removing duplicates, in case that low quality face pictures leads to less-effective comparison, range:[20,60]*/
            "updateTime":  60
            /*opt, int, range:[0,300]*/
        },
        "uploadWithoutLicense":  true,
        /*opt, bool*/
```

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```
        "panoramaTargetRemoveDuplicate": true,
        /*opt, bool*/
        "miscatchFilterEnabled": true,
        /*opt, bool*/
        "capturingPeopleEnabled": true,
        /*opt, bool*/
        "lineCrossingCaptureSensitivity": 50,
        /*opt, int, range:[0,100]*/
        "relatedMixedTargetUidTime": 30,
        /*opt, int, range:[5,30], step:1, unit:s*/
        "nonMotorPlateCustomText": "test",
        /*opt, string, range:[0,16]*/
        "filterDataType": {
        /*opt, object*/
            "retrogradeEnabled": true,
            /*opt, bool*/
            "mannedEnabled": true,
            /*opt, bool*/
            "unproopedHelmetEnabled": true,
            /*opt, bool*/
            "unblockedBicycleEnabled": true,
            /*opt, bool*/
            "blockedBicycleEnabled": true
            /*opt, bool*/
        },
        "nonMotorOverspeedThreshoid": 30,
        /*opt, int, range:[10,50], step:1, unit:km/h*/
        "vehicleOverspeedThreshoid": 40,
        /*opt, int, range:[30,80], step:1, unit:km/h*/
        "vehicleDisplacementFilterEnabled": true,
        /*opt, bool*/
        "vehicleDisplacementFilterThreshold": 30,
        /*opt, int, range:[1,100]*/
        "framesCaptureEnabled": false,
        /*opt, bool*/
        "facePosture": "normal",
        /*opt, enum, subType:string*/
        "snapAlgLevel": 1
        /*opt, int, range:[1,5]*/
    }
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode": "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx": "0x00000000"
    /*ro, opt, string*/
}
```

19.12.2.4 Get the rule parameters capability of frequently appeared person detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetDetectionCap": {
    /*ro, opt, object*/
        "enabled": "true,false",
        /*ro, req, string, whether to enable multi-target-type detection*/
```

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```
"TypesDetectionCap": [
/*ro, opt, array, subType:object*/
    {
        "ruleMode":  "human,face,vehice,nonMotor",
        /*ro, opt, string, rule mode*/
        "enabled":  "true,false",
        /*ro, opt, string, whether to enable the function*/
        "Property": [
        /*ro, opt, array, subType:object*/
            {
                "description":  "playMobilePhone",
                /*ro, opt, string, attributes description*/
                "enabled":  "true,false"
                /*ro, opt, string, whether to enable multi-target-type detection*/
            }
        ]
    }
],
"RuleInfoCap": {
/*ro, opt, object, rule information, desc:rule information,currently,only one rule is supported*/
    "maxSize":  1,
    /*ro, req, int, capability of maximum queue management rules*/
    "ruleID": {
    /*ro, req, object, rule ID*/
        "@min":  1,
        /*ro, req, int, the minimum value*/
        "@max":  1
        /*ro, req, int, the maximum value*/
    },
    "RegionCap": {
    /*ro, req, object, region that configured with rule, desc:polygon with 3 to 10 edges*/
        "minSize":  3,
        /*ro, req, int, capability of minimum edges of rule region*/
        "maxSize":  10,
        /*ro, req, int, capability of maximum queue management rules*/
        "x": {
        /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
        }
    },
    "InterPupileRegionCap": {
    /*ro, req, object, minimum rectangle region to detect pupil distance*/
        "x": {
        /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, req, object, float,Y-coordinate,the value is between 0.000 and 1, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "width": {
        /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "height": {
        /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        }
    },
    "alarmDisplayMode": {
    /*ro, opt, object*/
        "@opt":  "face_human_vehice_nonMotor,human_vehice_nonMotor,face_human",
        /*ro, opt, string*/
        "@def":  "face human vehice nonMotor"
```

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```
                /*ro, opt, string*/
            },
        },
        "isSupportCapturePicture":  true,
        /*ro, opt, bool, whether supports setting capture parameters for multi-target-type detection, desc:whether supports setting capture parameters for
multi-target-type detection,if not supports,this node will not be returned*/
        "isSupportCapturePicOverlap":  true,
        /*ro, opt, bool, whether supports setting parameters of overlaying information on captured pictures, desc:whether supports setting parameters of
overlaying information on captured pictures,if not supports,this node will not be returned*/
        "isSupportAlgParam":  true,
        /*ro, opt, bool, whether supports setting advanced algorithm library parameters for multi-target-type detection, desc:whether supports setting
advanced algorithm library parameters for multi-target-type detection,if not supports,this node will not be returned*/
        "isSupportAlarmInfoGet":  true,
        /*ro, opt, bool, whether supports getting captured multi-target-type information, desc:whether supports getting captured multi-target-type
information,"true"-support,"false"-not support*/
        "plateInfo": {
        /*ro, opt, object*/
            "region": {
            /*ro, opt, object*/
                "@opt": ["AFandAM"],
                /*ro, opt, array, subType:string*/
                "@def":  "APAC"
                /*ro, opt, string*/
            },
            "CRIndex": {
            /*ro, opt, object*/
                "@opt": ["0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12", "13", "14", "15", "16", "17", "18", "19", "20", "21", "22",
"23", "25", "26", "28", "29", "30", "31", "33", "34", "35", "36", "37", "38", "39", "40", "42", "43", "44", "45", "46", "47", "49", "51", "53", "55", "59",
"60", "62", "65", "68", "70", "71", "72", "73", "76", "77", "84", "87", "89", "90", "94", "95", "91", "104", "107", "108", "110", "114", "115", "130",
"137", "139", "140", "144", "145", "146", "151", "156", "157", "169", "174", "175", "177", "180", "181", "188", "206", "210", "215", "216", "217", "218",
"219", "220", "221", "222", "227", "228", "256"]
                /*ro, opt, array, subType:string*/
            },
            "relatedRegionAndCRList": [
            /*ro, opt, array, subType:object*/
                {
                    "region":  "AFandAM",
                    /*ro, req, enum, subType:string*/
                    "CRIndex": {
                    /*ro, opt, object*/
                        "@opt": ["0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12", "13", "14", "15", "16", "17", "18", "19", "20", "21",
"22", "23", "25", "26", "28", "29", "30", "31", "33", "34", "35", "36", "37", "38", "39", "40", "42", "43", "44", "45", "46", "47", "49", "51", "53", "55",
"59", "60", "62", "65", "68", "70", "71", "72", "73", "76", "77", "84", "87", "89", "90", "94", "95", "91", "104", "107", "108", "110", "114", "115", "130",
"137", "139", "140", "144", "145", "146", "151", "156", "157", "169", "174", "175", "177", "180", "181", "188", "206", "210", "215", "216", "217", "218",
"219", "220", "221", "222", "227", "228", "256"]
                        /*ro, opt, array, subType:string*/
                    }
                }
            ]
        },
    },
}
```


19.12.2.5 Get configuration capability of overlaying information on multi-target-type detection pictures

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/capturePicOverlap/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

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```
{
    "MixedTargetCapturePicOverlapCap": {
    /*req, object, configuration capability of overlaying information on multi-target-type detection pictures*/
        "overlapItemNum": {
        /*opt, object*/
            "@max":  1
            /*req, int, the maximum value*/
        },
        "overlapItemId": {
        /*opt, object*/
            "@min":  1,
            /*opt, int, the minimum value*/
            "@max":  "3"
            /*req, string, the maximum value*/
        },
        "itemType": {
        /*opt, object, attribute details of different overlay element types*/
            "@opt":
"cameraNo,captureTime,positionInfo,motionDirection,jacketColor,trousersColor,jacketType,trousersType,bag,things,hat,plateNo,vehicleLogo,vehicleType,GPS,enterLeaveTime,hatStyle,cyclingType"
            /*opt, string, attribute details of different overlay element types*/
        },
        "targetAttribute": {
        /*opt, object*/
            "@opt":
"motionDirection,jacketColor,trousersColor,jacketType,trousersType,bag,things,hat,plateNo,vehicleLogo,vehicleType,enterLeaveTime,hatStyle,cyclingType"
            /*opt, string, attribute details of different overlay element types*/
        }
    }
}
```

Response Message

```
{
    "MixedTargetCapturePicOverlapCap": {
    /*ro, req, object*/
        "overlapItemNum": {
        /*ro, req, object, the maximum number of elements can be overlaid*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "overlapItemId": {
        /*ro, req, object, overlaid element ID*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  3
            /*ro, req, int, the maximum value*/
        },
        "itemType": {
        /*ro, req, object, overlay element types*/
            "@opt":  "cameraNo"
            /*ro, req, string, "cameraNo"-camera No., "captureTime"-capture time, "positionInfo"-camera information, "motionDirection"-motion direction,
"jacketColor"-Tops' color, "trousersColor"-bottoms' color, "jacketType"-Tops' type, "trousersType"-bottoms' type, "bag"-backpack, "things"-handbag, "hat"-
wearing hat, "plateNo"-license plate number, "vehicleLogo"-vehicle brand, "vehicleType"-vehicle type, "GPS"-longitude and latitude information,
"enterLeaveTime"-entering and leaving time, desc:"cameraNo"-camera No., "captureTime"-capture time, "positionInfo"-camera information, "motionDirection"-
motion direction, "jacketColor"-Tops' color, "trousersColor"-bottoms' color, "jacketType"-Tops' type, "trousersType"-bottoms' type, "bag"-backpack,
"things"-handbag, "hat"-wearing hat, "plateNo"-license plate number, "vehicleLogo"-vehicle brand, "vehicleType"-vehicle type, "GPS"-longitude and latitude
information, "enterLeaveTime"-entering and leaving time*/
        },
        "targetAttribute": {
        /*ro, opt, object, attribute details of different overlay element types*/
            "@opt":  "motionDirection"
            /*ro, req, string, attribute details of different overlay element types, desc:attribute details of different overlay element types*/
        }
    }
}
```

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19.12.2.6 Get the captured picture overlay parameters of multi-target-detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/capturePicOverlap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetCapturePicOverlap": {
    /*ro, req, object*/
        "OverlapItemList": [
        /*ro, req, array, subType:object, range:[0,22]*/
            {
                "overlapItem": {
                /*ro, req, object*/
                    "overlapItemId":  1,
                    /*ro, req, int*/
                    "itemType":  "cameraNo"
                    /*ro, req, enum, subType:string*/
                }
            }
        ]
    }
}
```

19.12.2.7 Set the captured picture overlay parameters of multi-target-detection

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/capturePicOverlap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "MixedTargetCapturePicOverlap": {
    /*req, object*/
        "OverlapItemList": [
        /*req, array, subType:object*/
            {
                "overlapItem": {
                /*req, object*/
                    "overlapItemId":  1,
                    /*req, int*/
                    "itemType":  "cameraNo"
                    /*req, enum, subType:string*/
                }
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

19.12.2.8 Set the parameters of multiple scene mode switching of multi-target-detection

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/multiScene/sceneMode?format=json

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "MixedTargetMultiSceneMode": {
    /*req, object*/
        "sceneMode":  "normal"
        /*req, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

19.12.2.9 Get the parameters of multiple scene mode switching of multi-target-detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/multiScene/sceneMode?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetMultiSceneMode": {
    /*ro, req, object*/
        "sceneMode":  "normal"
        /*ro, req, enum, subType:string*/
    }
}
```

19.12.2.10 Get OSD overlay configuration capability for detecting mixed targets

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/overlay/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "MixedTargetOverlayCap": {
    /*ro, req, object, OSD overlay configuration capability for mixed targets*/
        "enabled": {
        /*ro, opt, object, whether to enable OSD overlay capability for detecting mixed targets; OSD calculation data will be cleared if this node is
reset*/
            "@opt":  "true,false",
            /*ro, opt, string, options*/
            "@def":  false
            /*ro, opt, bool, the default value*/
        },
        "x": {
        /*ro, opt, object, x-coordinate; the value ranges from 0.000 to 1; the upper-left corner is set as the origin point*/
            "@min":  0.000,
            /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000
            /*ro, opt, float, the maximum value, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, opt, object, y-coordinate; the value ranges from 0.000 t0 1; the upper-left corner is set as the origin point*/
            "@min":  0.000,
            /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000
            /*ro, opt, float, the maximum value, range:[0.000,1.000]*/
        },
        "fontSize": {
        /*ro, opt, object, font size of overlaid information*/
            "@opt": ["16*16", "32*32", "48*48", "64*64", "96*96", "adaptive"]
            /*ro, opt, array, "adaptive"-size adaptive (by default), subType:string*/
        },
        "targetDynamicMaskEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "targetDynamicMaskLevel": {
        /*ro, opt, object*/
            "@opt": ["high", "low", "medium"]
            /*ro, opt, array, subType:string*/
        },
        "isSupportOSDReset":  true,
        /*ro, opt, bool, if "true" is returned, it indicates that manually clearing OSD calculation data of mixed targets detection is supported (the URI is
available: /ISAPI/Intelligent/channels/<ID>/mixedTargetDetection/overlay/OSDReset?format=json*/
        "isSupportOSDAutoReset":  true
        /*ro, opt, bool, if "true" is returned, it indicates that auto clearing OSD calculation data of mixed targets detection is supported (the URI is
available: /ISAPI/Intelligent/channels/<ID>/mixedTargetDetection/overlay/OSDAutoReset?format=json*/
    }
}
```

19.12.2.11 Clear OSD calculation data of detecting mixed targets manually

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/overlay/OSDReset?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

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19.12.2.12 Set OSD overlay configuration parameters for detecting mixed targets

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/overlay?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
{
    "MixedTargetOverlay": {
    /*req, object, OSD overlay configuration parameters for detecting mixed targets*/
        "enabled": false,
        /*opt, bool, whether to enable OSD overlay capability for detecting mixed targets*/
        "x": 0.120,
        /*opt, float, x-coordinate, the value ranges from 0.000 to 1, the upper-left corner is set as the origin point, range:[0.000,1.000]*/
        "y": 0.120,
        /*opt, float, y-coordinate, the value ranges from 0.000 to 1, the upper-left corner is set as the origin point, range:[0.000,1.000]*/
        "fontSize": "adaptive",
        /*opt, enum, font size of overlaid information, subType:string, desc:"16*16", "32*32", "48*48", "64*64", "96*96", "adaptive" (size adaptive, by
default)*/
        "targetDynamicMaskEnabled": true,
        /*opt, bool*/
        "targetDynamicMaskLevel": "high"
        /*opt, enum, subType:string, dep:or,{$.MixedTargetOverlay.targetDynamicMaskEnabled,eq,true}*/
    }
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode": "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx": "0x00000000"
    /*ro, opt, string*/
}
```

19.12.2.13 Get OSD overlay configuration parameters for detecting mixed targets

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/overlay?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "MixedTargetOverlay": {
    /*ro, req, object, OSD overlay configuration parameters for detecting mixed targets*/
        "enabled":  false,
        /*ro, opt, bool, whether to enable OSD overlay capability for detecting mixed targets. By default, it is set to false*/
        "x":  0.120,
        /*ro, opt, float, X-coordinate, the value ranges from 0.000 to 1. The upper-left corner is set as the origin point, range:[0.000,1.000]*/
        "y":  0.120,
        /*ro, opt, float, Y-coordinate, the value ranges from 0.000 to 1. The upper-left corner is set as the origin point, range:[0.000,1.000]*/
        "fontSize":  "adaptive",
        /*ro, opt, enum, font size of overlaid information, subType:string, desc:"16*16", "32*32", "48*48", "64*64", "96*96", "adaptive" (size adaptive, by
default)*/
        "targetDynamicMaskEnabled":  true,
        /*ro, opt, bool*/
        "targetDynamicMaskLevel":  "high"
        /*ro, opt, enum, subType:string, dep:or,{$.MixedTargetOverlay.targetDynamicMaskEnabled,eq,true}*/
    }
}
```

19.12.2.14 Set single-scene mode parameters of multi-target-type detection

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "MixedTargetDetection": {
    /*opt, object, rule parameters of multi-target-type detection (by default, the first scene)*/
        "enabled":  true,
        /*req, bool, whether to enable globally*/
        "RuleInfo": [
        /*opt, array, rule information, subType:object*/
            {
                "ruleID":  1,
                /*req, int, rule ID*/
                "Region": [
                /*req, array, region configured with rules, subType:object, range:[3,10], desc:the origin is the lower-left corner of the screen*/
                    {
                        "x":  0.120,
                        /*req, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.120
                        /*req, float, Y-coordinate, range:[0.000,1.000]*/
                    }
                ],
                "InterPupileRegion": {
                /*req, object, minimum rectangle region to detect pupil distance, desc:The origin is the lower-left corner of the screen.*/
                    "x":  0.120,
                    /*req, float, X-coordinate of the point in the top left corner, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*req, float, Y-coordinate of the point in the top left corner, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*req, float, rectangle width, range:[0.000,1.000]*/
                    "height":  0.120
                    /*req, float, rectangle height, range:[0.000,1.000]*/
                },
            }
        ],
    }
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

19.12.2.15 Get single-scene mode parameters of multi-target-type detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetDetection": {
    /*ro, opt, object*/
        "enabled":  true,
        /*ro, req, bool, whether to enable multi-target-type detection*/
        "TypesDetection": [
        /*ro, opt, array, subType:object*/
            {
                "ruleMode":  "human",
                /*ro, opt, enum, rule type, subType:string, desc:face + human body + vehicle + non-motor vehicle, human body + vehicle + non-motor vehicle*/
                "enabled":  true,
                /*ro, opt, bool, whether to enable multi-target-type detection*/
                "Property": [
                /*ro, opt, array, Attribute Information, subType:object*/
                    {
                        "description":  "jacketColor",
                        /*ro, opt, enum, subType:string*/
                        "enabled":  false
                        /*ro, opt, bool, whether to enable multi-target-type detection*/
                    }
                ]
            }
        ],
        "RuleInfo": [
        /*ro, opt, array, rule information, subType:object*/
            {
                "ruleID":  1,
                /*ro, req, int, rule ID*/
                "ruleMode":  "face_human_vehice_nonMotor",
                /*ro, opt, enum, rule type, subType:string, desc:rule type: face + human body + vehicle + non-motor vehicle,human body + vehicle + non-motor
vehicle*/
                "Region": [
                /*ro, req, array, rule area, subType:object, range:[3,10], desc:the origin is the lower-left corner of the screen*/
                    {
                        "x":  0.120,
                        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.120
                        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    }
                ],
                "InterPupileRegion": {
                /*ro, req, object, minimum rectangle region to detect pupil distance, desc:the origin is the lower-left corner of the screen*/
                    "x":  0.120,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*ro, req, float, rectangle height, range:[0.000,1.000]*/
                    "height":  0.120
                    /*ro, req, float, rectangle width, range:[0.000,1.000]*/
                },
                "MaxInterPupilRegion": {
                /*ro, opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen*/
                    "x":  0.120,
```

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```
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*ro, req, float, rectangle height, range:[0.000,1.000]*/
                    "height":  0.120
                    /*ro, req, float, rectangle width, range:[0.000,1.000]*/
                },
                "LaneInfo": {
                /*ro, opt, object, lane information, desc:it is valid when ruleMode is "face_human_vehice"*/
                    "LaneList": [
                    /*ro, req, array, lane list, subType:object*/
                        {
                            "LeftLaneLine": [
                            /*ro, req, array, left lane line capability, subType:object, range:[0,2], desc:the origin is the lower-left corner of the
screen*/
                                {
                                    "x":  0.120,
                                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                                    "y":  0.120
                                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                                }
                            ]
                        }
                    ],
                    "RightLaneLine": [
                    /*ro, req, array, right lane line, subType:object, range:[0,2], desc:the origin is the lower-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                },
                "retrogradeDetectionList": [
                /*ro, opt, array, subType:object, range:[0,2]*/
                    {
                        "retrogradeDetectionLine": [
                        /*ro, opt, array, subType:object, range:[2,2]*/
                            {
                                "x":  0.120,
                                /*ro, req, float, range:[0.000,1.000]*/
                                "y":  0.120
                                /*ro, req, float, range:[0.000,1.000]*/
                            }
                        ],
                        "retrogradeDirectionLine": {
                        /*ro, opt, object*/
                            "startPoint": {
                            /*ro, req, object*/
                                "x":  0.120,
                                /*ro, req, float, range:[0.000,1.000]*/
                                "y":  0.120
                                /*ro, req, float, range:[0.000,1.000]*/
                            },
                            "stopPoint": {
                            /*ro, req, object*/
                                "x":  0.120,
                                /*ro, req, float, range:[0.000,1.000]*/
                                "y":  0.120
                                /*ro, req, float, range:[0.000,1.000]*/
                            }
                        }
                    }
                ],
                "nonMotorOverspeedThreshoid":  30,
                /*ro, opt, int, range:[10,50], step:1, unit:km/h*/
                "vehicleOverspeedThreshoid":  40
                /*ro, opt, int, range:[30,80], step:1, unit:km/h*/
            }
        ],
        "KeyRegionInfo": {
        /*ro, opt, object, key region information*/
            "enabled":  true,
            /*ro, req, bool, whether to enable multi-target-type detection*/
            "RegionList": [
            /*ro, opt, array, key region array, at least two key regions are required, subType:object*/
                {
                    "id":  1,
                    /*ro, req, int, detection area ID*/
                    "Region": [
                    /*ro, req, array, Key Area, subType:object, desc:the origin is the lower-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120,
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                            "width":  0.120,
                            /*ro, req, float, rectangle height, range:[0.000,1.000]*/
                            "height":  0.120
                            /*ro, req, float, rectangle width, range:[0.000,1.000]*/
                        }
                    ]
```

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```
                }
            ]
        },
        "province": 1,
        /*ro, opt, enum, reserved, subType:int, desc:reserved*/
        "pictureQuality": "high",
        /*ro, opt, enum, picture quality, subType:string, desc:picture quality*/
        "isSupportBinaryPicUp": true,
        /*ro, opt, bool*/
        "convertBinToBmpEnabled": true,
        /*ro, opt, bool*/
        "MEF": {
        /*ro, opt, object*/
            "enabled": true,
            /*ro, req, bool, whether to enable multi-target-type detection*/
            "offsetTimes": 1,
            /*ro, opt, int, range:[0,10], unit:min*/
            "leftOffsetTimes": 60,
            /*ro, opt, int, range:[0,300], unit:s*/
            "rightOffsetTimes": 30
            /*ro, opt, int, range:[0,30], unit:s*/
        },
        "plateInfo": {
        /*ro, opt, object*/
            "region": "AFandAM",
            /*ro, opt, enum, subType:string*/
            "CRIndex": "0"
            /*ro, opt, enum, subType:string*/
        },
        "nonMotorizedViolationUpload": {
        /*ro, opt, object*/
            "postEnabled": true,
            /*ro, opt, bool*/
            "illegalEnabled": true
            /*ro, opt, bool*/
        }
    }
}
```

19.12.2.16 Set the shielded area parameters of multi-target-detection event

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/Shield/mixedTargetDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Shield xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, enum, subType:string-->true
  </enabled>
  <RegionList>
    <!--opt, array, subType:object-->
    <Region>
      <!--opt, object-->
      <id>
        <!--req, string-->1
      </id>
      <enabled>
        <!--req, enum, subType:string-->true
      </enabled>
      <RegionCoordinatesList>
        <!--opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </RegionList>
</Shield>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.12.2.17 Multi-target-type detection

EventType:mixedTargetDetection

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "dateTimeMillisecond":  999,
    /*ro, opt, int, the milliseconds of the alarm trigger time, range:[0,999], desc:it should be used together with field dateTime*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:the event triggering frequency. The pulse event is the event which is triggered continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object recognition occurs, and the recognition of each object will trigger the event once, e.g., the facial recognition. When the pulse events are triggered, this node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously according to the device detection frequency; if it changes to another source, then the counting will start again*/
    "eventType":  "mixedTargetDetection",
    /*ro, req, enum, event type, subType:string, desc:"mixedTargetDetection" (multi-target-type detection event)*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "Mixed target detection",
    /*ro, req, enum, event description, subType:string, desc:"Mixed target detection" (multi-target-type detection event)*/
    "channelName":  "ABC",
    /*ro, opt, string, channel name (camera name), desc:it can be configured via URI (/ISAPI/Streaming/channels/<ID>)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID, desc:it can be configured via URI (/ISAPI/System/Network/Ehome)*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished, desc:whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers.)*/
    "customChanID":  "test",
    /*ro, opt, string, custom camera channel No., desc:the maximum string size is 64. The value of this node is that of the node <customChanID> configured by /ISAPI/ContentMgmt/InputProxy/channels/<ID>*/
    "faceBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"faceBackgroundImage" (face background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "humanBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"humanBackgroundImage" (human body background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "humanImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"humanImage" (human body thumbnail)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
```

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```
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "nonMotorBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"nonMotorBackgroundImage" (background picture of the non-motor vehicle)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "vehicleBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"vehicleBackgroundImage" (vehicle background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "vehicleImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"vehicleImage" (vehicle captured picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "CaptureResult": [
    /*ro, opt, array, captured results of detected target, subType:object*/
        {
            "targetID":  1,
            /*ro, req, int, captured target ID*/
            "Face": {
            /*ro, opt, object, face attribute*/
                "Rect": {
                /*ro, opt, object, face thumbnail coordinates, desc:rectangular frame coordinates of the face thumbnail, which is based on the background
picture (faceBackgroundImage). The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, face attribute details, subType:object*/
                    {
                        "description":  "age",
                        /*ro, opt, enum, attributes description, subType:string, desc:"age" (age), "ageGroup" (age group), "gender" (gender), "glass"
(whether the person wears glasses), "smile" (whether the person wears a smile), "mask" (whether the person wears a mask), "faceExpression" (face
expression), "hat" (whether the person wears a hat), "score" (face score)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "contentID1":  "faceImage",
                /*ro, opt, enum, face thumbnail, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "nonMotorImage"
(non-motor vehicle thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share
one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "faceBackgroundImage",
                /*ro, opt, enum, "faceBackgroundImage" (background picture of the face), subType:string, desc:"faceBackgroundImage" (face background
picture), "humanBackgroundImage" (human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture), "plateImage" (vehicle
license plate thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and
non-motor vehicle are detected, these two properties share one picture and "nonMotorBackgroundImage" will be returned*/
                "FacePictureRect": {
                /*ro, opt, object, coordinates of face picture containing head and shoulder, desc:it is based on the face background picture matting and
equals to the face thumbnail uploaded by the device. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, face thumbnail (contains head and shoulder) ID, which indicates the uniqueness of the picture data, range:[0,32],
desc:both "contentID" and "pId" are available for the data transmission format Content-Type: multipart/form-data, and implement the relation between picture
data and structured data of corresponding contents. "contentID1" and "contentID2" correspond to "contentID", "pId1" and "pId2" correspond to "name" and
"filename". The advantage of "pId" is that it has better compatibility, it can match data via "name" and indicate the file name and format via "filename".
Both "contentID" and "pId" need to be returned, while "faceImageURL" and "faceBackgroundImageURL" are not required*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, face background picture ID*/
                "faceImageURL":  "test",
                /*ro, opt, string, face thumbnail URL*/
                "faceBackgroundImageURL":  "test",
                /*ro, opt, string, face background picture URL, desc:face background picture URL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, face capture time*/
                "snapTimeMillisecond":  999
                /*ro, opt, int, the milliseconds of face capture time, range:[0,999], desc:it should be used together with the field snapTime*/
            },
            "Human": {
            /*ro, opt, object, human body attributes*/
```

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```
            /*ro, opt, object, human body attributes*/
                "Rect": {
                /*ro, req, object, target frame (based on the background picture), desc:coordinates of the human body target frame, which is based on the background picture matting. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "jacketColor",
                        /*ro, opt, enum, attributes description, subType:string, desc:"jacketColor" (top color), "trousersColor" (bottoms color), "ageGroup" (age group), "gender", "glass" (whether the person wears glasses), "ride" (whether the person is riding a bike), "bag" (whether the person is carrying a bag), "things"(whether the person is carrying things), "hat" (whether the person wears a hat), "trousersType" (bottoms type), "mask" (whether the person wears a mask), "hairStyle" (hairstyle), "cyclingPersonNumber" (number of riders), "direction" (target direction), "score" (human body grading), "hatStyle" (hat style), "cyclingType" (riding type), "jacketType" (top type)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "modeldata":  "test",
                /*ro, opt, string, human body model data (encoded by base64)*/
                "contentID1":  "humanImage",
                /*ro, opt, enum, human body thumbnail, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "nonMotorImage" (non-motor vehicle thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "humanBackgroundImage",
                /*ro, opt, enum, human body background picture, subType:string, desc:"faceBackgroundImage" (face background picture), "humanBackgroundImage" (human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture), "plateImage" (vehicle license plate thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorBackgroundImage" will be returned*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, human body thumbnail ID, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one thumbnail. The value of pId1 is same as that of pId1 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle thumbnail*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, background picture ID of the human body, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one background picture. The value of pId2 is same as that of pId2 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "humanImageURL":  "test",
                /*ro, opt, string, human body thumbnail URL, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanImageURL is same as that of nonMotorImageURL, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle thumbnail*/
                "humanBackgroundImageURL":  "test",
                /*ro, opt, string, background picture URL of the human body, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, human body capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, milliseconds of human body capture time, range:[0,999], desc:it should be used together with the field snapTime*/
                "contentID3":  "humanMergeImage",
                /*ro, opt, string, "humanMergeImage" (picture of human body merging details)*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of the picture of human body merging details, range:[0,32]*/
                "contentID4":  "humanBackgroundImage2",
                /*ro, opt, enum, 2nd background picture of human body, subType:string, desc:"humanBackgroundImage2" (2nd background picture of human body)*/
                "contentID5":  "humanBackgroundImage3",
                /*ro, opt, enum, 3rd background picture of human body, subType:string, desc:"humanBackgroundImage3" (3rd background picture of human body)*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of 2nd background picture of human body, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one background picture. The value of pId4 is same as that of pId2 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "pId5":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of 3rd background picture of human body, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one background picture. The value of pId5 is same as that of pId2 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "humanBackgroundImage2URL":  "test",
                /*ro, opt, string, URL of 2nd human body background picture, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "humanBackgroundImage3URL":  "test",
                /*ro, opt, string, URL of 3rd human body background picture, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "HumanImageRect": {
                /*ro, opt, object, target rectangle frame based on the human body thumbnail, desc:the coordinates of the human body target frame, which is based on the human body thumbnail matting. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "smallImageRectInBackground": {
                /*ro, opt, object, rectangle frame of human body thumbnail (based on the background picture), desc:the rectangle frame is the area of the small human body picture based on humanBackgroundImage. The matted picture is the same as humanImage. The origin is the upper-left corner of the screen*/
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                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "relatedMixedTargetUid": ["test", "1"],
                /*ro, opt, array, UID array of linked multi-target-type detection events, subType:string, desc:corresponding to the "uid" of
mixedTargetDetection (multi-target-type detection event). when a device uploads human body properties, other related targets attributes will be linked, and
the client software can get the information about targets around the captured human body*/
                "contentID6":  "featurePicture",
                /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, close-up picture ID, range:[0,32]*/
                "featurePictureURL":  "test"
                /*ro, opt, string, close-up picture URL*/
            },
            "NonMotor": {
            /*ro, opt, object, non-motor vehicle attributes*/
                "Rect": {
                /*ro, req, object, frame information of non-motor vehicle thumbnail, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "nonMotorType",
                        /*ro, opt, enum, attributes description, subType:string, desc:"detectType" (device type), "nonMotorType" (non-motor vehicle type),
"plateColor" (license plate color of non-motor vehicle), "plateNo" (license plate number of non-motor vehicle), "directionIndex" (vehicle driving
direction), "laneNo" (lane number), "RFID" (linked RFID), "EPlateChipNo" (chip number of electronic license plate), "EPlateCardNo" (card number of
electronic license plate), "EPlateListenAuthCode" (licensing authority code of electronic license plate), "EPlateNo" (electronic license plate number),
"EPlateManufacturingDate" (data of production of electronic license plate), "EPlateInspectionValidity" (inspection validity of electronic license plate),
"EPlateMandaScrapPeriod" (compulsory scrapping period of electronic license plate), "EPlatePassengersCapacity" (authorized passenger capacity),
"EPlateCarryingCapacity" (tractive tonnage), "EPlateVehicleColor" (non-motor vehicle color), "EPlateNonMotorType" (non-motor vehicle type),
"EPlateNonMotorPlateType" (license plate type of non-motor vehicle), "EPlateColor" (license plate color of non-motor vehicle), "EPlateUseType" (function
properties of non-motor vehicle), "EPlateTriggerTime" (detection trigger time), "retrograde" (wrong-way driving of non-motor vehicle), "manned" (carrying
passengers on non-motor vehicle), "hatStyle" (hat style)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "contentID1":  "nonMotorImage",
                /*ro, opt, enum, face picture ID, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "nonMotorImage"
(non-motor vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body
and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "nonMotorBackgroundImage",
                /*ro, opt, enum, background picture of the non-motor vehicle, subType:string, desc:"faceBackgroundImage" (face background picture),
"humanBackgroundImage" (human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture). When there are multiple properties
for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one
picture and "nonMotorBackgroundImage" will be returned*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, ID of the non-motor vehicle thumbnail, range:[0,32]*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, background picture ID of the non-motor vehicle, range:[0,32]*/
                "nonMotorImageURL":  "test",
                /*ro, opt, string, URL of the non-motor vehicle thumbnail*/
                "nonMotorBackgroundImageURL":  "test",
                /*ro, opt, string, background picture URL of the non-motor vehicle*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, non-motor vehicle capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, it should be used together with the field snapTime, range:[0,999], desc:it should be used together with the field snapTime*/
                "contentID3":  "nonMotorBackgroundImage2",
                /*ro, opt, enum, 2nd background picture of the non-motor vehicle, subType:string, desc:"nonMotorBackgroundImage2" (2nd background picture of
the non-motor vehicle)*/
                "contentID4":  "nonMotorBackgroundImage3",
                /*ro, opt, enum, 3rd background picture of the non-motor vehicle, subType:string, desc:"nonMotorBackgroundImage3" (3rd background picture of
the non-motor vehicle)*/
                "contentID5":  "faceImage",
                /*ro, opt, enum, face thumbnail, subType:string, desc:"faceImage" (face thumbnail)*/
                "contentID6":  "plateImage",
                /*ro, opt, enum, license plate thumbnail, subType:string, desc:"plateImage" (license plate thumbnail)*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD2223",
                /*ro, opt, string, 2nd background picture ID of the non-motor vehicle, range:[0,32]*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD2224",
                /*ro, opt, string, 3rd background picture ID of the non-motor vehicle, range:[0,32]*/
                "pId5":  "F4F665D6A18E41308CE9934DCDDD2225",
                /*ro, opt, string, face thumbnail ID, range:[0,32]*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD2226",
                /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                "nonMotorBackgroundImage2URL":  "test",
                /*ro, opt, string, 2nd background picture URL of the non-motor vehicle*/
                "nonMotorBackgroundImage3URL":  "test",
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                /*ro, opt, string, 3rd background picture URL of the non-motor vehicle*/
                "faceImageURL":  "test",
                /*ro, opt, string, face thumbnail URL*/
                "plateImageURL":  "test",
                /*ro, opt, string, license plate thumbnail URL*/
                "illegalCode":  "12313",
                /*ro, opt, string, violation code*/
                "illegalName":  "test",
                /*ro, opt, string, violation name, range:[0,128]*/
                "illegalDescription":  "test",
                /*ro, opt, string, violation description, range:[0,256]*/
                "dwIllegalTime":  100,
                /*ro, opt, int, duration of the violation, unit:ms, desc:it is the difference between the capture time of the last picture and the capture
time of the first picture*/
                "relatedMixedTargetUid": ["test", "1"],
                /*ro, opt, array, UID array of linked multi-target-type detection events, subType:string, desc:corresponding to the "uid" of
mixedTargetDetection (multi-target-type detection event). when a device uploads non-motor vehicle properties, other related targets attributes will be
linked, and the client software can get the information about targets around the captured non-motor vehicle*/
                "contentID7":  "featurePicture",
                /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
                "pId7":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, close-up picture ID, range:[0,32]*/
                "featurePictureURL":  "test",
                /*ro, opt, string, close-up picture URL*/
                "nonMotorSpeed":  23.4
                /*ro, opt, float, driving speed of non-motor vehicle, range:[0,1024], unit:km/h*/
            },
            "Vehicle": {
            /*ro, opt, object, vehicle attributes*/
                "VehicleRect": {
                /*ro, req, object, target frame information of vehicle picture (based on background picture), desc:the origin is the upper-left corner of
the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "PlateRect": {
                /*ro, req, object, frame information of license plate picture, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "plateNo",
                        /*ro, opt, enum, attributes description, subType:string, desc:"plateNo" (license plate No.), "plateType" (license plate type),
"plateColor" (license plate color), "vehicleType"(vehicle type), "vehicleColor" (vehicle color), "vehicleLogo" (vehicle brand), "vehicleSublogo" (vehicle
sub-brand), "vehicleSize" (vehicle size)*/
                        "value":  "23",
                        /*ro, opt, string, it depends on the value of description, desc:it depends on the value of description*/
                        "confidence":  99.00
                        /*ro, opt, float, confidence, range:[0.00,100.00]*/
                    }
                ],
                "contentID1":  "vehicleImage",
                /*ro, opt, enum, vehicle thumbnail (on the basis of the target frame, a certain proportion is expanded), subType:string, desc:"faceImage"
(face thumbnail), "humanImage" (human body thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target,
multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and
"nonMotorImage" will be returned*/
                "contentID2":  "plateImage",
                /*ro, opt, enum, license plate thumbnail, subType:string, desc:"plateImage" (license plate thumbnail)*/
                "contentID3":  "vehicleBackgroundImage",
                /*ro, opt, enum, vehicleBackgroundImage, subType:string, desc:"vehicleBackgroundImage" (vehicle background picture 1)*/
                "contentID4":  "vehicleBackgroundImage2",
                /*ro, opt, enum, vehicleBackgroundImage2, subType:string, desc:"vehicleBackgroundImage2" (vehicle background picture 2)*/
                "contentID5":  "vehicleBackgroundImage3",
                /*ro, opt, enum, vehicleBackgroundImage3, subType:string, desc:"vehicleBackgroundImage3" (vehicle background picture 3)*/
                "contentID6":  "vehicleBackgroundImage4",
                /*ro, opt, enum, vehicleBackgroundImage4, subType:string, desc:"vehicleBackgroundImage4" (vehicle background picture 4)*/
                "contentID8":  "vehicleBackgroundImage5",
                /*ro, opt, enum, vehicleBackgroundImage5, subType:string, desc:"vehicleBackgroundImage5" (vehicle background picture 5)*/
                "contentID9":  "pilotPicture",
                /*ro, opt, enum, pilotPicture, subType:string, desc:"pilotPicture" (driver's picture)*/
                "contentID10":  "copilotPicture",
                /*ro, opt, enum, copilotPicture, subType:string, desc:"copilotPicture" (co-driver's picture)*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, vehicle thumbnail ID, range:[0,32]*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, vehicle background picture ID, range:[0,32]*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 2nd vehicle background picture, range:[0,32]*/
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                "pId5":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 3rd vehicle background picture, range:[0,32]*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 4th vehicle background picture, range:[0,32]*/
                "pId8":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 5th vehicle background picture, range:[0,32]*/
                "pId9":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, driver's picture, range:[0,32]*/
                "pId10":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, co-driver's picture, range:[0,32]*/
                "vehicleImageURL":  "test",
                /*ro, opt, string, vehicle thumbnail URL*/
                "plateImageURL":  "test",
                /*ro, opt, string, license plate thumbnail URL*/
                "vehicleBackgroundImageURL":  "test",
                /*ro, opt, string, vehicle background picture URL*/
                "vehicleBackgroundImage2URL":  "test",
                /*ro, opt, string, URL of the 2nd vehicle background picture*/
                "vehicleBackgroundImage3URL":  "test",
                /*ro, opt, string, URL of the 3rd vehicle background picture*/
                "vehicleBackgroundImage4URL":  "test",
                /*ro, opt, string, URL of the 4th vehicle background picture*/
                "vehicleBackgroundImage5URL":  "test",
                /*ro, opt, string, URL of the 5th vehicle background picture*/
                "pilotPictureURL":  "test",
                /*ro, opt, string, driver's picture URL*/
                "copilotPictureURL":  "test",
                /*ro, opt, string, co-driver's picture URL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, vehicle capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, it should be used together with the field snapTime, range:[0,999], desc:it should be used together with the field snapTime*/
                "humanMergeImageURL":  "test",
                /*ro, opt, string, URL of the picture of human body merging details*/
                "illegalCode":  "12313",
                /*ro, opt, string, violation code, range:[0,64]*/
                "illegalName":  "test",
                /*ro, opt, string, violation name, range:[0,128]*/
                "illegalDescription":  "test",
                /*ro, opt, string, violation description, range:[0,256]*/
                "dwIllegalTime":  100,
                /*ro, opt, int, duration of the violation, unit:ms, desc:it is the difference between the capture time of the last picture and the capture time of the first picture*/
                "VehicleImageRect": {
                /*ro, opt, object, rectangle target frame in the vehicle thumbnail, desc:the frame is the target area based on vehicleImage. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "smallImageRectInBackground": {
                /*ro, opt, object, vehicle thumbnail frame (based on the background picture), desc:the rectangle frame of vehicle thumbnail is based on vehicleBackgroundImage. The matted picture equals to vehicleImage. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "MultiPlateInfo": [
                /*ro, opt, array, information about multiple license plates, subType:object*/
                    {
                    "MultiPlateRect": {
                    /*ro, opt, object, license plate thumbnail frame, desc:based on the vehicle background picture*/
                        "height":  1.000,
                        /*ro, req, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, req, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, req, float, x-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, req, float, y-coordinate, range:[0.000,1.000]*/
                    },
                    "plateNo":  "noPlate",
                    /*ro, req, enum, license plate number, subType:string, desc:"noPlate" (vehicle without license plate), "unknown" (no license plate recognized), XXXXXXXX (license plate number. For motor vehicles, it is a 16-byte string; for non-motor vehicles, it is a 48-byte string*/
                    "plateColor":  "black",
                    /*ro, req, enum, license plate color, subType:string, desc:"black", "blue", "civilAviationBlack" (civil aviation black), "civilAviationGreen" (civil aviation green), "golden", "green", "mixedColor" (mixed color), "newEnergyGreen" (new energy green), "newEnergyYellowGreen" (new energy green and yellow), "orange", "other" (other color), "red", "unknown", "white", "yellow"*/
                    "confidence":  100.0,
                    /*ro, opt, float, confidence, range:[0,100.0]*/
                    "province":  0,
                    /*ro, req, enum, province index, subType:int*/
                    "contentID7":  "HongkongPlateImage",
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                        /*ro, opt, enum, license plate thumbnail, subType:string, desc:license plate thumbnail*/
                        "pId7":  "F4F665D6A18E41308CE9934DCDDD3333",
                        /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                        "multiplateImageURL":  "test"
                        /*ro, opt, string, thumbnail URL of multiple license plates*/
                    }
                ],
                "carDriveDirect":  "forward",
                /*ro, opt, enum, vehicle driving direction, subType:string, desc:"back" (backward), "eastWest" (from east to west), "forward" (forward),
"northSouth" (from north to south), "northeastToSouthwest" (from northeast to southwest), "northwestToSoutheast" (from northwest to southeast), "other"
(other), "southNorth" (from south to north), "southeastToNorthwest" (from southeast to northwest), "southwestToNortheast" (from southwest to northeas),
"unknown" (unknown), "westEast" (from west to east)*/
                "contentID11":  "featurePicture",
                /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
                "pId11":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, close-up picture ID, range:[0,32]*/
                "featurePictureURL":  "test",
                /*ro, opt, string, close-up picture URL*/
                "vehicleSpeed":  23.4
                /*ro, opt, float, vehicle driving speed, range:[0,1024], unit:km/h*/
            },
            "AbsoluteHigh": {
            /*ro, opt, object, PTZ position information*/
                "elevation":  11,
                /*ro, opt, int, tilting angle, desc:tilting angle*/
                "azimuth":  22,
                /*ro, opt, int, azimuth, desc:panning angle*/
                "absoluteZoom":  33
                /*ro, opt, int, zooming ratio, desc:zooming rate,value range: 1 to 1000*/
            },
            "GPS": {
            /*ro, opt, object, device GPS information*/
                "longitude":  -180.000000,
                /*ro, req, float, longitude*/
                "latitude":  -90.000000
                /*ro, req, float, latitude*/
            },
            "FaceContrastResult": [
            /*ro, opt, array, face picture comparison result, one result can contains comparison results of multiple persons, the captured picture with
highest comparison similarity is adopted, subType:object*/
                {
                    "errorCode":  1,
                    /*ro, opt, int, error code, desc:error information returned when analyzing face picture failed*/
                    "errorMsg":  "ok",
                    /*ro, opt, string, the modeling data of base picture in the face picture library, desc:when analyzing face picture failed, the
corresponding error information will be returned*/
                    "modelData":  "test",
                    /*ro, opt, string, the modeling data of base picture in the face picture library, range:[0,2048], desc:base64 encoded*/
                    "faces": [
                    /*ro, opt, array, matched face information, subType:object, desc:the information includes detection results, properties, modeling data,
and matched list. One picture can contains multiple faces*/
                        {
                            "faceId":  1,
                            /*ro, opt, int, face ID, desc:the face ID cannot be duplicated in the same picture*/
                            "identify": [
                            /*ro, opt, array, features, subType:object*/
                                {
                                    "relationId": "9887",
                                    /*ro, opt, string, arming linkage ID, range:[0,64], desc:arming linkage ID*/
                                    "maxsimilarity":  0.204000,
                                    /*ro, opt, float, the maximum similarity, range:[0.000000,1.000000], desc:the maximum similarity in the matched results
the similarity is between 0 and 1, and it corrects to two decimals*/
                                    "candidate": [
                                    /*ro, opt, array, person information list, subType:object*/
                                        {
                                            "blacklist_id":  "2323",
                                            /*ro, opt, string, alarm list ID, range:[0,64], desc:it corresponds to the FDID of face picture library. For
cameras, only the default FDID will be returned, the custom face picture library ID will be returned in customFaceLibID*/
                                            "human_data": [
                                            /*ro, opt, array, face picture information list, subType:object*/
                                                {
                                                    "face_id":  "123445",
                                                    /*ro, opt, string, face ID*/
                                                    "contentID":  "faceLibImage",
                                                    /*ro, opt, string, picture in the face picture library*/
                                                    "similarity":  0.204000,
                                                    /*ro, opt, float, face picture similarity, range:[0.000000,1.000000]*/
                                                    "isNoSaveFDPicture":  true,
                                                    /*ro, opt, bool, whether to not save the picture of face picture library, desc:true (not save). If the
function of not saving original pictures during picture importing or modeling is enabled, true should be returned, otherwise, this field will not be
returned*/
                                                    "pId":  "F4F665D6A18E41308CE9934DCDDD766F",
                                                    /*ro, opt, string, picture ID in face picture library, range:[0,32], desc:picture ID in face picture
library*/
                                                    "face_picurl":  "http://127.0.0.1:6120/face_picurl"
                                                    /*ro, opt, string, face thumbnail URL*/
                                                }
                                            ],
                                            "human_id":  "123",
                                            /*ro, opt, string, corresponds to the FPID in the face picture library, range:[0,64], desc:person ID in the
list, which corresponds to FPID in the face picture library*/
                                            "reserve_field": {
                                            /*ro, req, object, transmitted data*/
                                                "name":  "张三"
```

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```
                                        "name": "张三",
                                        /*ro, opt, string, name, range:[0,96]*/
                                        "gender": "any",
                                        /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition), "unknown", "male", "female"*/
                                        "bornTime": "1970-01-01+08:00",
                                        /*ro, opt, date, date of birth*/
                                        "city": "130100",
                                        /*ro, opt, string, city code of person birth, desc:see the data dictionary for details*/
                                        "certificateType": "officerID",
                                        /*ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport" (passport), "other" (other certificate)*/
                                        "certificateNumber": "test",
                                        /*ro, opt, string, certificate No., range:[0,32]*/
                                        "ageGroup": "unknown",
                                        /*ro, opt, enum, age group, subType:string, desc:“any”, “child”, “infant”, “kid”, “middle”, “middleAged” (the middle age), “old”, “prime”, “teenager”, “unknown”, “young”*/
                                        "phoneNumber": "test",
                                        /*ro, opt, string, phone number, range:[0,64]*/
                                        "cityName": "石家庄市",
                                        /*ro, opt, string, city (corresponding to the field "city"), range:[0,32], desc:city (corresponding to the field "city)*/
                                        "provinceName": "河北省"
                                        /*ro, opt, string, province, range:[0,32], desc:province*/
                                    },
                                    "similarity": 0.780,
                                    /*ro, opt, float, the maximum similarity, range:[0.000,1.000], desc:the maximum similarity of all matched pictures for the person*/
                                    "listType": "blackList",
                                    /*ro, opt, enum, list type, subType:string, desc:"blacklist" (blocklist), "whitelist" (allowlist)*/
                                    "extendData": [
                                    /*ro, opt, array, person extended information, subType:object*/
                                        {
                                            "extendID": 1,
                                            /*ro, opt, int, ID of the extended person information*/
                                            "enable": true,
                                            /*ro, opt, bool, whether to enable the person extended information*/
                                            "name": "test",
                                            /*ro, opt, string, name of person tag extended information, range:[0,96]*/
                                            "value": "test"
                                            /*ro, opt, string, person tag extended information, range:[0,64]*/
                                        }
                                    ],
                                    "FDLibName": "faceLib1",
                                    /*ro, opt, string, detection threshold of face picture library*/
                                    "FDLibThreshold": 50,
                                    /*ro, opt, int, detection threshold of face picture library, range:[0,100]*/
                                    "customFaceLibID": "test",
                                    /*ro, opt, string, custom face picture library ID*/
                                    "customHumanID": "test"
                                    /*ro, opt, string, custom person ID*/
                                }
                            ]
                        }
                    ],
                    "stayDuration": 20
                    /*ro, opt, int, dwell time duration in the image, unit:ms*/
                }
            ]
        }
    ],
    "ModelingResult": {
    /*ro, opt, object, modeling result of captured face picture*/
        "modelingStatus": "succeeded",
        /*ro, req, enum, modeling status, subType:string, desc:"succeeded" (modeling succeeded), "resourceInsuf" (insufficient engine capacity), "faceDetectingFail" (face detection failed), "faceModelingFail" (face modeling failed), "faceGradingFail" (face quality grading), "featureExtractFail" (extracting features failed), "featureAnalyzeFail" (features analysis failed), "otherErrors" (other error of the engine), "DSPInternalError" (DSP internal error)*/
        "facePicQuality": "lower",
        /*ro, opt, enum, quality level of face thumbnail, subType:string, desc:"lower", "medium", "high"*/
        "stayDuration": 20,
        /*ro, opt, int, dwell time duration in the image, unit:ms*/
        "modeldata": "test"
        /*ro, opt, string, model data (encoded by base64)*/
    },
    "sid": "123",
    /*ro, opt, string, scene ID*/
    "sceneName": "场景1",
    /*ro, opt, string, scene name*/
    "uid": "test",
    /*ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits*/
    "behaviorLinkageUid": "test",
    /*ro, opt, string, which can consists of time (accurate to millisecond) and random digits,the maximum value size is 64 bytes, range:[0,64], desc:which can consists of time (accurate to millisecond) and random digits,the maximum value size is 64 bytes*/
    "position": [500, 500],
    /*ro, opt, array, offset position coordinates of target center point, subType:int, desc:an X-coordinate is positive if it is to the right of the Y-axis and negative if it is to the left of the Y-axis. A Y-coordinate is positive if it is above of the X-axis and negative values are not supported. The origin is the center point (the center point of all lane lines) of the horizontal line where the camera is located and which is perpendicular to the lane line*/
    "laneNo": 1,
    /*ro, opt, int, number of the lane where the target is located*/
    "speed": 23.4,
    /*ro, opt, float, vehicle speed*/
    "targetType": "human",
```


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```
            /*ro, opt, enum, vehicle type, subType:string, desc:"vehicle", "human", "nonMotor" (non-motor vehicles)*/
            "horizonSpeed":  33.4,
            /*ro, opt, float, horizontal speed of target*/
            "NormalizationPosition": [500, 500],
            /*ro, opt, array, normalized coordinates, subType:int, desc:normalized position coordinates,the value of x-coordinate is between 0 and 1000,and
the value of y-coordinate is between -500 and 1500,e.g.,[500,500]*/
            "targetMaxY":  340,
            /*ro, opt, int, the maximum Y-coordinate of target, desc:the value should be smaller than 1500*/
            "TargetGPSInfo": [
            /*ro, opt, array, GPS information of the target, subType:object*/
                {
                    "longitude":  -180.000000,
                    /*ro, req, float, longitude information, corrects to 6 decimal places, and the value range is [-180.000000,180.000000]*/
                    "latitude":  -90.000000
                    /*ro, req, float, latitude information, corrects to 6 decimal places, and the value range is [-180.000000,180.000000]*/
                }
            ],
            "perimeterLinkageUid":  "test",
            /*ro, opt, string, UID of linked perimeter events uploading, desc:64-byte unique identifier, which can be composed of time (accurate to the
millisecond) and random digits*/
            "MEFData": [
            /*ro, opt, array, mobile electronic fence data corresponding to the target, subType:object, desc:mobile electronic fence data of multiple users
is supported (the data needs to be encrypted)*/
                {
                    "time":  "2004-05-03T17:30:08+08:00",
                    /*ro, req, datetime, electronic fence data acquired time*/
                    "IMSI":  "460001357924680",
                    /*ro, req, string, SIM card number (international mobile subscriber identity), range:[1,15]*/
                    "IMEI":  "359355041886388",
                    /*ro, opt, string, international mobile equipment identity, range:[0,17]*/
                    "bandIdx":  11
                    /*ro, opt, enum, frequency band type used by the current user, subType:int, desc:0 (China Unicom B3), 1 (China Telecom B3), 2 (China
Mobile B39), 3 (China Mobile B38), 4 (China Mobile B40), 5 (China Mobile B41), 6 (China Unicom B1), 7 (China Telecom B1), 8 (China Mobile 2G), 9 (China
Unicom 2G), 10 (China Unicom WCDMA), 11 (China Telecom CDMA), 12 (China Telecom CDMA2000), 13 (China Mobile TD), 19 (China Mobile B1), 20 (China Mobile B3),
21 (China Mobile B8)*/
                }
            ]
        }
    ],
    "captureResultTargetNum":  10,
    /*ro, opt, int, number of uploaded alarm targets*/
    "URLCertificationType":  "digest",
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no"-no authentication, "digest"-digest authentication (it is used for local
picture URL returned by NVR or DVR)*/
    "triggerSourceType":  "linkedTrackingCapture"
    /*ro, opt, enum, alarm source type, subType:string, desc:"ruleTrigger" (rule parameters triggered alarm), "linkedTrackingCapture" (alarm triggered by
linked tracking capture). If this field is not returned or the returned value is "ruleTrigger", it indicates that the alarm is triggered by multi-target-
type detection rules (corresponding URL for rule configurations: /ISAPI/Intelligent/channels/<ID>/mixedTargetDetection?format=json). If the returned value
is "linkedTrackingCapture", it indicates that the alarm is triggered by linked tracking capture (corresponding URL for configurations:
/ISAPI/MasterSlaveTracking/linkedTracking?format=json)*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name
mixedTargetDetection	[Message content]	application/json	--	--
..value	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg
..value	[Binary picture data]	image/jpeg	faceBackgroundImage	faceBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	humanImage	humanImage.jpg
..value	[Binary picture data]	image/jpeg	humanBackgroundImage	humanBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	humanMergeImage	humanMergeImage.jpg

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..value	[Binary picture data]	image/jpeg	humanBackgroundImage2	humanBackgroundImage2.jpg
..value	[Binary picture data]	image/jpeg	humanBackgroundImage3	humanBackgroundImage3.jpg
..value	[Binary picture data]	image/jpeg	nonMotorImage	nonMotorImage.jpg
..value	[Binary picture data]	image/jpeg	nonMotorBackgroundImage	nonMotorBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	nonMotorBackgroundImage2	nonMotorBackgroundImage2.jpg
..value	[Binary picture data]	image/jpeg	nonMotorBackgroundImage3	nonMotorBackgroundImage3.jpg
..value	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg
..value	[Binary picture data]	image/jpeg	plateImage	plateImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleImage	vehicleImage.jpg
..value	[Binary picture data]	image/jpeg	plateImage	plateImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage	vehicleBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage2	vehicleBackgroundImage2.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage3	vehicleBackgroundImage3.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage4	vehicleBackgroundImage4.jpg
..value	[Binary picture data]	image/jpeg	HongkongPlateImage	HongkongPlateImage.jpg
	[Binary			

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..value	[Binary picture data]	image/jpeg	MacaoPlateImage	MacaoPlateImage.jpg
.....value	[Binary picture data]	image/jpeg	faceLibImage	faceLibImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage5	vehicleBackgroundImage.jpg5
..value	[Binary picture data]	image/jpeg	pilotPicture	pilotPicture.jpg
..value	[Binary picture data]	image/jpeg	copilotPicture	copilotPicture.jpg
..value	[Binary picture data]	image/jpeg	featurePicture	featurePicture
..value	[Binary picture data]	image/jpeg	featurePicture	featurePicture
..value	[Binary picture data]	image/jpeg	featurePicture	featurePicture
..value	[Binary picture data]	image/jpeg	compositePicture	compositePicture.jpg
..value	[Binary picture data]	image/jpeg	compositePicture	compositePicture.jpg
..value	[Binary picture data]	image/jpeg	compositePicture	compositePicture.jpg
..value	[Binary picture data]	image/jpeg	copilotPicture	copilotPicture.jpg

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Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.

- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.13 Algorithm Scheduling

19.13.1 Scheduling and Managing Local Smart Resources

19.13.1.1 Get parameters of intelligent resources switch

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/VCAResource

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VCAResource xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, intelligent resources, attr:version{req, string, protocolVersion}-->
  <type>
    <!--ro, req, enum, “basicBehavior”, subType:string, desc:“basicBehavior”-->basicBehavior
  </type>
</VCAResource>
```

19.13.1.2 Set parameters of intelligent resources switch

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/VCAResource

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VCAResource xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, VCA Resource, attr:version{req, string, protocolVersion}-->
  <type>
    <!--req, enum, type, subType:string,
desc:"basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDetection,smart,judicial,smart264AndRoadDetection,smart264AndFaceDetection,smart264AndHeatMap,smartIntelligentMonitor,smartTrafficDataCollection,roadDetection,humanRecognition,perimeterCapture,vehicleDetection,HVTDetection,mixedTargetDetection,trackingCaptureMode,nonTrackingCaptureMode,close,faceHumanModelingContrast,cityManagement,teacherBehavior,12MPLiveView,personQueueDetection,verticalPeopleCounting,safetyHelmet,faceCounting,personArming,AIOpenPlatform"-->basicBehavior
  </type>
</VCAResource>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
VCAResource	[Message content]	application/xml	--	--	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

19.13.1.3 Get capability of switching intelligent resources

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/VCAResource/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<VCAResource xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, intelligent resources, attr:version{req, string, protocolVersion}-->
  <type
opt="basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDetection,smart,judicial,smart264AndRoadDetection,smart264AndFac
eDetection,smart264AndHeatMap,smartVehicleIllegalParkingDetection,smartIntelligentMonitor,smartTrafficDataCollection, roadDetection,humanRecognition,
perimeterCapture, vehicleDetection,HVTDetection,mixedTargetDetection,     trackingCaptureMode,nonTrackingCaptureMode,close,
faceHumanModelingContrast,cityManagement,teacherBehavior,12MPLiveView,personQueueDetection,verticalPeopleCounting,AIOpenPlatform,mixedTargetFaceHuman,mixedT
argetAll,safetyHelmet,faceContrast,none,faceMonitor,vehicleMonitor,faceCounting,heelPDC,personDensity,behaviorMixedTargetAll,operationMonitor,fielddetection
,personArming,smokeDetectAlarm,smokeDetection,personDensityQueueLeavePosition,bannerDetection,ATMSurround,
ATMPanel,ATMSafetyCabin,intelligentTraffic,heatmap,teacherBehaviorAndFaceContrast,mixedTargetAllCapture,riverwayGaugeReading,MixedTargetAllAuxiliaryCapture,
radarHearNurse,smokeDetectionAndFlameCombustion,PackingSpaceRecognition" def="basicBehavior">
    <!--ro, req, enum, resource type, subType:string, attr:opt{req, string},def{opt, string}, desc:resource type-->basicBehavior
  </type>
  <PromptList>
    <!--ro, opt, array, prompt for switching intelligent resources, subType:object-->
    <Prompt>
      <!--ro, opt, object, list-->
      <resourceType opt="roadDetection">
        <!--ro, opt, string, currently,only "roadDetection" is supported, attr:opt{req, string}, desc:currently, only "roadDetection" is supported-->test
      </resourceType>
      <resourcePrompt opt="smartCode">
        <!--ro, opt, string, prompt contents, attr:opt{req, string}, desc:prompt contents-->test
      </resourcePrompt>
    </Prompt>
  </PromptList>
  <isSupportAPPWorking>
    <!--ro, opt, bool-->true
  </isSupportAPPWorking>
  <isSupportTest>
    <!--ro, opt, bool-->true
  </isSupportTest>
  <VCAResourcePrompt opt="streamingCodingPrompt">
    <!--ro, opt, string, attr:opt{req, string}-->streamingCodingPrompt
  </VCAResourcePrompt>
  <sceneRecommendationTypeList>
    <!--ro, opt, array, subType:object-->
    <sceneRecommendationType>
      <!--ro, opt, object-->
      <scene opt="Y,L,one,concentricBayonet">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->Y
      </scene>
      <recommendationType
opt="basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDetection,smart,judicial,smart264AndRoadDetection,smart264AndFac
eDetection,smart264AndHeatMap,smartVehicleIllegalParkingDetection,smartIntelligentMonitor,smartTrafficDataCollection, roadDetection,humanRecognition,
perimeterCapture, vehicleDetection,HVTDetection,mixedTargetDetection,     trackingCaptureMode,nonTrackingCaptureMode,close,
faceHumanModelingContrast,cityManagement,teacherBehavior,12MPLiveView,personQueueDetection,verticalPeopleCounting,AIOpenPlatform,mixedTargetFaceHuman,mixedT
argetAll,safetyHelmet,faceContrast,none,faceMonitor,vehicleMonitor,faceCounting,heelPDC,personDensity,behaviorMixedTargetAll,operationMonitor,fielddetection
,personArming,smokeDetectAlarm,smokeDetection,personDensityQueueLeavePosition,bannerDetection,ATMSurround,
ATMPanel,ATMSafetyCabin,intelligentTraffic,heatmap" def="basicBehavior">
        <!--ro, req, enum, subType:string, attr:opt{req, string},def{opt, string}-->basicBehavior
      </recommendationType>
    </sceneRecommendationType>
  </sceneRecommendationTypeList>
  <synergyRecommendationTypeList>
    <!--ro, opt, array, subType:object-->
    <synergyRecommendationType>
      <!--ro, opt, object-->
      <synergyMode opt="associationDeDuplication,trackAssociation,targetAssociation">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->associationDeDuplication
      </synergyMode>
      <priority min="1" max="3">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
      </priority>
      <recommendationType
opt="basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDetection,smart,judicial,smart264AndRoadDetection,smart264AndFac
eDetection,smart264AndHeatMap,smartVehicleIllegalParkingDetection,smartIntelligentMonitor,smartTrafficDataCollection, roadDetection,humanRecognition,
perimeterCapture, vehicleDetection,HVTDetection,mixedTargetDetection,     trackingCaptureMode,nonTrackingCaptureMode,close,
faceHumanModelingContrast,cityManagement,teacherBehavior,12MPLiveView,personQueueDetection,verticalPeopleCounting,AIOpenPlatform,mixedTargetFaceHuman,mixedT
argetAll,safetyHelmet,faceContrast,none,faceMonitor,vehicleMonitor,faceCounting,heelPDC,personDensity,behaviorMixedTargetAll,operationMonitor,fielddetection
,personArming,smokeDetectAlarm,smokeDetection,personDensityQueueLeavePosition,bannerDetection,ATMSurround,
ATMPanel,ATMSafetyCabin,intelligentTraffic,heatmap" def="basicBehavior">
        <!--ro, req, enum, subType:string, attr:opt{req, string},def{opt, string}-->basicBehavior
      </recommendationType>
    </synergyRecommendationType>
  </synergyRecommendationTypeList>
</VCAResource>
```

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19.14 Video Recognition

19.14.1 (Professional) Smart Abnormal Event Detection

19.14.1.1 Get the intelligent capability

Request URL

GET /ISAPI/Intelligent/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IntelliCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <isFaceSupport>
    <!--ro, opt, bool, whether it supports face detection, desc:whether it supports face detection-->true
  </isFaceSupport>
  <isBehaviorSupport>
    <!--ro, opt, bool, whether it supports abnormal event detection, desc:whether it supports abnormal event detection-->true
  </isBehaviorSupport>
  <isLineDetectionSupport>
    <!--ro, opt, bool, whether it supports line crossing detection, desc:whether it supports line crossing detection-->true
  </isLineDetectionSupport>
  <isFieldDetectionSupport>
    <!--ro, opt, bool, whether it supports intrusion detection, desc:whether it supports intrusion detection-->true
  </isFieldDetectionSupport>
  <isRegionEntranceSupport>
    <!--ro, opt, bool, whether it supports region entrance detection, desc:whether it supports region entrance detection-->true
  </isRegionEntranceSupport>
  <isRegionExitingSupport>
    <!--ro, opt, bool, whether it supports region exiting detection, desc:whether it supports region exiting detection-->true
  </isRegionExitingSupport>
  <isLoiteringSupport>
    <!--ro, opt, bool, whether it supports loitering detection, desc:whether it supports loitering detection-->true
  </isLoiteringSupport>
  <isGroupSupport>
    <!--ro, opt, bool, whether it supports people gathering detection, desc:whether it supports people gathering detection-->true
  </isGroupSupport>
  <isRapidMoveSupport>
    <!--ro, opt, bool, whether it supports fast moving detection, desc:whether it supports fast moving detection-->true
  </isRapidMoveSupport>
  <isParkingSupport>
    <!--ro, opt, bool, whether it supports parking detection, desc:whether it supports parking detection-->true
  </isParkingSupport>
  <isUnattendedBaggageSupport>
    <!--ro, opt, bool, whether it supports unattended baggage detection, desc:whether it supports unattended baggage detection-->true
  </isUnattendedBaggageSupport>
  <isAttendedBaggageSupport>
    <!--ro, opt, bool, whether it supports object removal detection, desc:whether it supports object removal detection-->true
  </isAttendedBaggageSupport>
  <isTeacherSupport>
    <!--ro, opt, bool, whether it supports teachers' behavior detection, desc:whether it supports teacher's behavior detection-->true
  </isTeacherSupport>
  <isStudentSupport>
    <!--ro, opt, bool, whether it supports students' behavior detection, desc:whether it supports student's behavior detection-->true
  </isStudentSupport>
</IntelliCap>
```

19.14.1.2 Set all abnormal event detection rules of a specific scene

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
SID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BehaviorRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, all abnormal event detection rules of a scene, attr:version{req, string, protocolVersion}-->
  <sid>
    <!--req, string, scene ID-->test
  </sid>
  <RuleInfoList>
    <!--opt, array, subType:object, range:[1,200]-->
```

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```
<RuleInfo>
  <!--opt, object, rule information-->
  <ruleId>
    <!--req, string, rule ID-->test
  </ruleId>
  <ruleName>
    <!--req, string, rule name-->test
  </ruleName>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <eventType>
    <!--req, enum, event type, subType:string, desc:"none"-none, "lineDetection"-line crossing, "fieldDetection"-intrusion, "regionEntrance"-region entrance, "regionExiting"-region exit, "loitering"-loitering detection, "group"-people gathering, "rapidMove"fast moving, "parking"-parking, "unattendedBaggage"-unattended baggage, "attendedBaggage"-attended baggage, "teacher"-teacher, "student"-student, "peopleNumChange"-change of the number of people, "spacingChange"-change of distance, "violentMotion"-violent motion, "leavePosition"-absence, "failDown"-fall down, "running"-running, "retentionParam"-retention, "advReachHight"-climbing, "situationAnalysis"-trend analysis, "toiletTarry"-in toilet overtime, "audioAbnormal"-sudden change of sound intensity, "standUp"-stand up, "getUp"-get up, "playCellphone"-play cellphone-->none
  </eventType>
  <ruleType>
    <!--req, enum, subType:string-->region
  </ruleType>
  <LineDetectionParam>
    <!--opt, object, line crossing detection-->
    <directionSensitivity>
      <!--req, enum, subType:string-->left-right
    </directionSensitivity>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "human_vehicle". The symbol of "_" indicates that multiple options are supported, for example, "human_vehicle" represents human and vehicle-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <sensitivity>
      <!--req, int, sensitivity, range:[0,100]-->1
    </sensitivity>
    <priority>
      <!--opt, enum, priority, subType:string, desc:priority-->high
    </priority>
    <stackTarget>
      <!--opt, bool, whether to overlay the alarm target frame-->true
    </stackTarget>
    <stackRule>
      <!--opt, bool, whether to overlay the alarm rule frame-->true
    </stackRule>
  </LineDetectionParam>
  <FieldDetectionParam>
    <!--opt, object-->
    <durationTime>
      <!--req, int-->1
    </durationTime>
    <sensitivityLevel>
      <!--req, int, range:[0,100]-->1
    </sensitivityLevel>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "human_vehicle". The symbol of "_" indicates that multiple options are supported, for example, "human_vehicle" represents human and vehicle-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <sensitivity>
      <!--req, int, sensitivity, range:[0,100]-->1
    </sensitivity>
    <priority>
      <!--opt, enum, priority, subType:string, desc:priority-->high
    </priority>
    <stackTarget>
      <!--opt, bool, whether to overlay the alarm target frame-->true
    </stackTarget>
    <stackRule>
      <!--opt, bool, whether to overlay the alarm rule frame-->true
    </stackRule>
  </FieldDetectionParam>
  <RegionEntranceParam>
    <!--opt, object-->
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "human_vehicle". The symbol of "_" indicates that multiple options are supported, for example, "human_vehicle" represents human and vehicle-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <sensitivity>
```

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```
        <sensitivity>
          <!--req, int, sensitivity, range:[0,100]-->1
        </sensitivity>
        <priority>
          <!--opt, enum, priority, subType:string, desc:"high", "medium", "low"-->high
        </priority>
        <stackTarget>
          <!--opt, bool, whether to overlay the alarm target frame-->true
        </stackTarget>
        <stackRule>
          <!--opt, bool, whether to overlay the alarm rule frame-->true
        </stackRule>
      </RegionEntranceParam>
      <RegionExitingParam>
        <!--opt, object-->
        <detectionTarget>
          <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "human_vehicle". The symbol of "_" indicates that multiple
options are supported, for example, "human_vehicle" represents human and vehicle-->all
        </detectionTarget>
        <humanMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (human)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <sensitivity>
          <!--req, int, sensitivity, range:[0,100]-->1
        </sensitivity>
        <priority>
          <!--opt, enum, priority, subType:string, desc:priority-->high
        </priority>
        <stackTarget>
          <!--opt, bool, whether to overlay the alarm target frame-->true
        </stackTarget>
        <stackRule>
          <!--opt, bool, whether to overlay the alarm rule frame-->true
        </stackRule>
      </RegionExitingParam>
      <LoiteringParam>
        <!--opt, object, loitering detection-->
        <durationTime>
          <!--req, int, range:[1,100], unit:s-->1
        </durationTime>
        <sensitivity>
          <!--opt, int, sensitivity-->1
        </sensitivity>
        <stackTarget>
          <!--opt, bool, whether to overlay the alarm target frame-->true
        </stackTarget>
        <stackRule>
          <!--opt, bool, whether to overlay the alarm rule frame-->true
        </stackRule>
      </LoiteringParam>
      <GroupParam>
        <!--opt, object-->
        <populDensity>
          <!--opt, int, range:[1,10]-->1
        </populDensity>
        <durationTime>
          <!--opt, int-->1
        </durationTime>
        <interval>
          <!--opt, int-->1
        </interval>
        <threshold>
          <!--opt, int-->1
        </threshold>
        <stackTarget>
          <!--opt, bool, whether to overlay the alarm target frame-->true
        </stackTarget>
        <stackRule>
          <!--opt, bool, whether to overlay the alarm rule frame-->true
        </stackRule>
        <confidence>
          <!--opt, float, range:[0,99.9]-->60
        </confidence>
      </GroupParam>
      <RapidMoveParam>
        <!--opt, object, fast moving-->
        <rapidMoveMode>
          <!--opt, enum, mode of fast moving, subType:string, desc:"pixels"-pixel, "actualSize"-actual size-->pixels
        </rapidMoveMode>
        <distanceThreshold>
          <!--opt, int, if the value of rapidMoveMode is "pixels", the value range of this node is from 1 to 10; if the value of rapidMoveMode is
"actualSize", the value range of this node is from 1 to 20, desc:if the value of rapidMoveMode is "pixels", the value range of this node is from 1 to 10; if
the value of rapidMoveMode is "actualSize", the value range of this node is from 1 to 20-->1
        </distanceThreshold>
      </RapidMoveParam>
      <ParkingParam>
        <!--opt, object, parking detection-->
        <durationTime>
          <!--req, int, range:[5,100]-->5
        </durationTime>
      </ParkingParam>
```

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```
        <UnattendedBaggageParam>
          <!--opt, object-->
          <durationTime>
            <!--req, int, range:[5,100], unit:s-->5
          </durationTime>
          <sensitivity>
            <!--opt, int, sensitivity, desc:sensitivity-->0
          </sensitivity>
        </UnattendedBaggageParam>
        <AttendedBaggageParam>
          <!--opt, object-->
          <durationTime>
            <!--req, int, range:[5,100], unit:s-->5
          </durationTime>
        </AttendedBaggageParam>
        <TeacherParam>
          <!--opt, object-->
          <durationTime>
            <!--req, int-->1
          </durationTime>
        </TeacherParam>
        <StudentParam>
          <!--opt, object-->test
        </StudentParam>
        <GetUpParam>
          <!--opt, object-->
          <durationTime>
            <!--opt, int, range:[1,100], unit:s-->1
          </durationTime>
          <mode>
            <!--opt, enum, subType:string-->overBed
          </mode>
          <sensitivity>
            <!--opt, int-->1
          </sensitivity>
        </GetUpParam>
        <ToiletTarryParam>
          <!--opt, object-->
          <durationTime>
            <!--opt, int, range:[1,3600]-->5
          </durationTime>
        </ToiletTarryParam>
        <PeopleNumChangeParam>
          <!--opt, object-->
          <peopleNumThreshold>
            <!--opt, int-->1
          </peopleNumThreshold>
          <detectMode>
            <!--opt, enum, "greaterthan"-greater than, “lessThan”-less than, “equal”-equal to, “notEqual”-not equal to, subType:string, desc:"greaterthan"-
greater than, “lessThan”-less than, “equal”-equal to, “notEqual”-not equal to-->greaterthan
          </detectMode>
          <noneStateEffectiveEnabled>
            <!--opt, bool-->true
          </noneStateEffectiveEnabled>
          <durationTime>
            <!--opt, int-->1
          </durationTime>
          <stackTarget>
            <!--opt, bool, whether to overlay the alarm target frame-->true
          </stackTarget>
          <stackRule>
            <!--opt, bool, whether to overlay the alarm rule frame-->true
          </stackRule>
        </PeopleNumChangeParam>
        <SpacingChangeParam>
          <!--opt, object-->
          <spacingThreshold>
            <!--opt, int-->1
          </spacingThreshold>
          <detectMode>
            <!--opt, enum, subType:string-->greaterthan
          </detectMode>
          <durationTime>
            <!--opt, int-->1
          </durationTime>
        </SpacingChangeParam>
        <ViolentMotionParam>
          <!--opt, object-->
          <sensitivity>
            <!--opt, int, sensitivity-->1
          </sensitivity>
          <durationTime>
            <!--opt, int-->1
          </durationTime>
          <interval>
            <!--opt, int-->1
          </interval>
          <mode>
            <!--opt, enum, mode, subType:string, desc:mode-->video
          </mode>
          <stackTarget>
            <!--opt, bool, whether to overlay the alarm target frame-->true
          </stackTarget>
```

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```
      <stackRule>
        <!--opt, bool, whether to overlay the alarm rule frame-->true
      </stackRule>
      <detectType>
        <!--opt, enum, subType:string-->fight
      </detectType>
      <confidence>
        <!--opt, float, range:[0,99.9]-->60
      </confidence>
    </ViolentMotionParam>
    <LeavePositionParam>
      <!--opt, object-->
      <leaveDelay>
        <!--opt, int, range:[1,86400], unit:s-->1
      </leaveDelay>
      <onPosition>
        <!--opt, int-->1
      </onPosition>
      <mode>
        <!--opt, enum, mode, subType:string, desc:mode-->leavePosition
      </mode>
      <sleepDelay>
        <!--opt, int, range:[1,1800], unit:s-->120
      </sleepDelay>
      <stackTarget>
        <!--opt, bool, whether to overlay the alarm target frame-->true
      </stackTarget>
      <stackRule>
        <!--opt, bool, whether to overlay the alarm rule frame-->true
      </stackRule>
      <personType>
        <!--opt, enum, subType:string-->single
      </personType>
      <alarmFrames>
        <!--opt, int-->1
      </alarmFrames>
      <stackPersonCountEnabled>
        <!--opt, bool-->true
      </stackPersonCountEnabled>
      <stackPersonPositionX>
        <!--opt, int, X-coordinate of the OSD overlay (the number of people), desc:it depends on the node stackPersonCountEnabled-->1
      </stackPersonPositionX>
      <stackPersonPositionY>
        <!--opt, int, Y-coordinate of the OSD overlay (the number of people), desc:it depends on the node stackPersonCountEnabled-->1
      </stackPersonPositionY>
      <sleepNumber>
        <!--opt, int, range:[1,128]-->120
      </sleepNumber>
    </LeavePositionParam>
    <FailDownParam>
      <!--opt, object-->
      <durationTime>
        <!--opt, int-->1
      </durationTime>
      <sensitivity>
        <!--opt, int, sensitivity-->1
      </sensitivity>
      <heightThreshold>
        <!--opt, int-->1
      </heightThreshold>
      <stackTarget>
        <!--opt, bool, whether to overlay the alarm target frame-->true
      </stackTarget>
      <stackRule>
        <!--opt, bool, whether to overlay the alarm rule frame-->true
      </stackRule>
      <interval>
        <!--opt, int, range:[4,600], unit:s-->4
      </interval>
      <confidence>
        <!--opt, float, range:[0,99.9]-->60
      </confidence>
    </FailDownParam>
    <RunningParam>
      <!--opt, object-->
      <speed>
        <!--opt, int-->1
      </speed>
      <durationTime>
        <!--opt, int-->1
      </durationTime>
      <sensitivity>
        <!--opt, int, sensitivity-->1
      </sensitivity>
      <stackTarget>
        <!--opt, bool, whether to overlay the alarm target frame-->true
      </stackTarget>
      <stackRule>
        <!--opt, bool, whether to overlay the alarm rule frame-->true
      </stackRule>
      <mode>
        <!--opt, string, running mode-->test
      </mode>
      <peopleNum>
```

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```
      <!--opt, int, range:[1,50]-->0
    </peopleNum>
    <confidence>
      <!--opt, float, range:[0,99.9]-->60
    </confidence>
  </RunningParam>
  <RetentionParam>
    <!--opt, object-->
    <durationTime>
      <!--req, int, range:[1,3600], unit:s-->1800
    </durationTime>
    <mode>
      <!--opt, enum, mode, subType:string, desc:mode-->single
    </mode>
    <stackTarget>
      <!--opt, bool, whether to overlay the alarm target frame-->true
    </stackTarget>
    <stackRule>
      <!--opt, bool, whether to overlay the alarm rule frame-->true
    </stackRule>
    <alarmCount>
      <!--opt, int, range:[0,64]-->1
    </alarmCount>
    <alarmInterval>
      <!--opt, int, range:[0,300], unit:s-->5
    </alarmInterval>
  </RetentionParam>
  <AdvReachHightParam>
    <!--opt, object-->
    <crossDirection>
      <!--opt, enum, subType:string-->bothDirection
    </crossDirection>
  </AdvReachHightParam>
  <SituationAnalysisParam>
    <!--opt, object-->
    <peopleNumThreshold>
      <!--opt, int-->1
    </peopleNumThreshold>
    <stackRule>
      <!--opt, bool, whether to overlay the alarm rule frame-->true
    </stackRule>
  </SituationAnalysisParam>
  <SizeFilter>
    <!--opt, object-->
    <enabled>
      <!--req, enum, subType:string-->true
    </enabled>
    <mode>
      <!--req, enum, mode, subType:string, desc:"pixels"-pixel, "actualSize"-actual size-->pixels
    </mode>
    <MaxObjectSize>
      <!--opt, object, the maximum size, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
      <width>
        <!--req, int, width, range:[0,1000]-->1
      </width>
      <height>
        <!--req, int, range:[0,1000]-->1
      </height>
    </MaxObjectSize>
    <MinObjectSize>
      <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
      <width>
        <!--req, int, width, range:[0,1000]-->1
      </width>
      <height>
        <!--req, int, range:[0,1000]-->1
      </height>
    </MinObjectSize>
    <maxObjectSizeHighPrecision>
      <!--opt, object-->
      <positionX>
        <!--req, int, range:[0,10000]-->1
      </positionX>
      <positionY>
        <!--req, int, range:[0,10000]-->1
      </positionY>
      <width>
        <!--req, int, range:[0,10000]-->1
      </width>
      <height>
        <!--req, int, range:[0,10000]-->1
```

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```
      </height>
    </maxObjectSizeHighPrecision>
    <minObjectSizeHighPrecision>
      <!--opt, object-->
      <positionX>
        <!--req, int, range:[0,10000]-->1
      </positionX>
      <positionY>
        <!--req, int, range:[0,10000]-->1
      </positionY>
      <width>
        <!--req, int, range:[0,10000]-->1
      </width>
      <height>
        <!--req, int, range:[0,10000]-->1
      </height>
    </minObjectSizeHighPrecision>
  </SizeFilter>
  <RuleRegion>
    <!--opt, object-->
    <RegionCoordinatesList>
      <!--opt, array, area coordinates list, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <regionCoordinatesHighPrecisionList>
      <!--opt, array, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--opt, object-->
        <positionX>
          <!--req, int, range:[0,10000]-->1
        </positionX>
        <positionY>
          <!--req, int, range:[0,10000]-->1
        </positionY>
      </RegionCoordinates>
    </regionCoordinatesHighPrecisionList>
  </RuleRegion>
  <backgroundSuppression>
    <!--opt, enum, subType:string-->open
  </backgroundSuppression>
  <KeyPersonGetUpParam>
    <!--opt, object-->
    <durationTime>
      <!--opt, int-->1
    </durationTime>
    <sensitivity>
      <!--opt, int, sensitivity-->1
    </sensitivity>
    <mode>
      <!--opt, enum, mode, subType:string, desc:mode-->overBed
    </mode>
  </KeyPersonGetUpParam>
  <YardTarryParam>
    <!--opt, object-->
    <durationTime>
      <!--opt, int-->1
    </durationTime>
  </YardTarryParam>
  <SitQuietly>
    <!--opt, object-->
    <durationTime>
      <!--opt, int-->1
    </durationTime>
    <sensitivity>
      <!--opt, int, range:[0,100]-->1
    </sensitivity>
  </SitQuietly>
  <StandUpParam>
    <!--opt, object-->
    <sensitivity>
      <!--opt, int, sensitivity-->1
    </sensitivity>
    <heightThreshold>
      <!--opt, int, unit:cm-->1
    </heightThreshold>
    <durationTime>
      <!--opt, int-->1
    </durationTime>
  </StandUpParam>
  <PlayCellphoneParam>
    <!--opt, object-->
    <durationTime>
      <!--opt, int, range:[1,3600], unit:s-->1
    </durationTime>
    <stackTarget>
      <!--opt, bool, whether to overlay the alarm target frame-->true
```

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```
        </stackTarget>
        <stackRule>
          <!--opt, bool, whether to overlay the alarm rule frame-->true
        </stackRule>
      </PlayCellphoneParam>
      <AudioAbnormalParam>
        <!--opt, object-->
        <audioMode>
          <!--opt, enum, subType:string-->sensitivity
        </audioMode>
        <sensitivity>
          <!--opt, int, sensitivity-->1
        </sensitivity>
        <decibel>
          <!--opt, int-->1
        </decibel>
      </AudioAbnormalParam>
      <alarmDelay>
        <!--opt, enum, subType:int-->1
      </alarmDelay>
      <TailParam>
        <!--opt, object-->
        <sensitivity>
          <!--opt, int, sensitivity, range:[1,5]-->0
        </sensitivity>
      </TailParam>
      <OperateOverTimeParam>
        <!--opt, object-->
        <durationTime>
          <!--opt, int, range:[4,6000], unit:s-->4
        </durationTime>
      </OperateOverTimeParam>
      <StickUpParam>
        <!--opt, object-->
        <sensitivity>
          <!--opt, int, sensitivity, range:[1,5]-->1
        </sensitivity>
        <durationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </durationTime>
      </StickUpParam>
      <InstallScannerParam>
        <!--opt, object-->
        <sensitivity>
          <!--opt, int, sensitivity, range:[1,5]-->1
        </sensitivity>
        <durationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </durationTime>
      </InstallScannerParam>
      <ObjectPlacedOrTakenParam>
        <!--opt, object-->
        <durationTime>
          <!--opt, int, range:[4,60], unit:s-->4
        </durationTime>
      </ObjectPlacedOrTakenParam>
      <FakeCardDetectionParam>
        <!--opt, object-->
        <sensitivity>
          <!--opt, int, sensitivity, range:[1,5]-->1
        </sensitivity>
      </FakeCardDetectionParam>
      <sceneMode>
        <!--opt, enum, subType:string-->general
      </sceneMode>
      <relatedTriggerRuleList>
        <!--opt, array, subType:object, range:[1,3]-->
        <relatedTriggerRule>
          <!--opt, object-->
          <enabled>
            <!--req, bool-->true
          </enabled>
          <combinationRuleType>
            <!--req, enum, subType:string-->rule
          </combinationRuleType>
          <relatedRuleID>
            <!--opt, int, range:[1,200], dep:and,
{$.BehaviorRule.RuleInfoList[*].RuleInfo.relatedTriggerRuleList[*].relatedTriggerRule.combinationRuleType,eq,rule}-->1
          </relatedRuleID>
          <relatedEventType>
            <!--opt, enum, subType:string, dep:and,
{$.BehaviorRule.RuleInfoList[*].RuleInfo.relatedTriggerRuleList[*].relatedTriggerRule.combinationRuleType,eq,event}-->VMD
          </relatedEventType>
          <relatedAlarmInputID>
            <!--opt, int, range:[1,2], dep:and,
{$.BehaviorRule.RuleInfoList[*].RuleInfo.relatedTriggerRuleList[*].relatedTriggerRule.combinationRuleType,eq,alarmInput}-->1
          </relatedAlarmInputID>
        </relatedTriggerRule>
      </relatedTriggerRuleList>
      <shieldRegionList>
        <!--opt, array, subType:object-->
        <region>
          <!--opt, object-->
```

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```
            <id>
              <!--req, string-->1
            </id>
            <regionCoordinatesList>
              <!--opt, array, subType:object, range:[3,10]-->
              <regionCoordinates>
                <!--req, object-->
                <positionX>
                  <!--req, int, range:[0,1000]-->1
                </positionX>
                <positionY>
                  <!--req, int, range:[0,1000]-->1
                </positionY>
              </regionCoordinates>
            </regionCoordinatesList>
          </region>
        </shieldRegionList>
        <DisplacementFilter>
          <!--opt, object-->
          <enabled>
            <!--req, enum, subType:bool-->true
          </enabled>
        </DisplacementFilter>
        <TrackFilter>
          <!--opt, object-->
          <enabled>
            <!--req, enum, subType:bool-->true
          </enabled>
          <trackDirectionRatio>
            <!--opt, int, range:[0,100]-->1
          </trackDirectionRatio>
          <trackCenterDist>
            <!--opt, int, range:[0,1000]-->1
          </trackCenterDist>
          <trackMinTgtW>
            <!--opt, int, range:[0,100]-->1
          </trackMinTgtW>
        </TrackFilter>
        <AspectRatioFilter>
          <!--opt, object-->
          <enabled>
            <!--req, enum, subType:bool-->true
          </enabled>
          <aspectRatioThresh>
            <!--opt, int, range:[1,10]-->1
          </aspectRatioThresh>
        </AspectRatioFilter>
        <TargetMutationsFilter>
          <!--opt, object-->
          <enabled>
            <!--req, enum, subType:bool-->true
          </enabled>
          <targetMutationsMinTgtWh>
            <!--opt, int, range:[1,10]-->1
          </targetMutationsMinTgtWh>
          <tandemFramesTargetAreaChanges>
            <!--opt, int, range:[1,100]-->1
          </tandemFramesTargetAreaChanges>
        </TargetMutationsFilter>
      </RuleInfo>
    </RuleInfoList>
    <enabledAllRules>
      <!--opt, bool, whether to enable the smart channel, desc:you can configure rules only after enabling the smart channel-->true
    </enabledAllRules>
    <subBehaviorType>
      <!--opt, enum, sub types of abnormal event detection, subType:string, desc:"streetBehavior"-street behavior, "workBehavior"-work behavior--
>streetBehavior
    </subBehaviorType>
    <analysisMode>
      <!--opt, enum, analysis mode, subType:string, desc:"realTimeMode" (real time), "autoSwitchMode" (auto switch). If this node is not returned, the
analysis mode is set to "realTimeMode" by default. The engine also has its analysis mode. The device will be added to the engine with the same analysis mode
via the load balancing method-->realTimeMode
    </analysisMode>
    <WorkBehaviorParam>
      <!--opt, object-->
      <interval>
        <!--opt, int, range:[2,1200]-->1
      </interval>
    </WorkBehaviorParam>
    <sceneName>
      <!--opt, string-->test
    </sceneName>
    <targetAttributeOverlapEnabled>
      <!--opt, bool-->false
    </targetAttributeOverlapEnabled>
</BehaviorRule>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
BehaviorRule	[Message content]	application/xml	--	--	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

19.14.1.3 Get the capability of professional event detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
SID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<RuleInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, rule information, attr:version{req, string, protocolVersion}-->
  <ruleId opt="1,2,3,4,5,6,7,8">
    <!--ro, req, string, rule ID, attr:opt{req, string}-->test
  </ruleId>
  <ruleName min="1" max="10">
    <!--ro, req, string, Rule name., attr:min{req, int},max{req, int}-->test
```

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```
    </ruleName>
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <ruleType opt="region,line,none">
      <!--ro, req, enum, rule type, subType:string, attr:opt{req, string}, desc:"region", "line", "none" (rule area configuration not supported)-->region
    </ruleType>
    <PeopleNumChangeParam>
      <!--ro, opt, object, people amount change-->
      <peopleNumThreshold min="1" max="10" default="null">
        <!--ro, opt, int, threshold number of people, attr:min{req, int},max{req, int},default{opt, string}-->1
      </peopleNumThreshold>
      <detectMode opt="greaterthan,lessThan,equal,notEqual">
        <!--ro, opt, enum, checking mode, subType:string, attr:opt{req, string}, desc:"greaterthan"-greater than, “lessThan”-less than, “equal”-equal to,
“notEqual”-not equal to-->greaterthan
      </detectMode>
      <durationTime min="1" max="10" default="null">
        <!--ro, opt, int, duration, attr:min{req, int},max{req, int},default{opt, string}-->1
      </durationTime>
    </PeopleNumChangeParam>
    <ViolentMotionParam>
      <!--ro, opt, object, violent motion detection-->
      <sensitivity min="1" max="10" default="null">
        <!--ro, opt, int, sensitivity, attr:min{req, int},max{req, int},default{opt, string}-->1
      </sensitivity>
      <durationTime min="1" max="10" def="1">
        <!--ro, opt, int, playing duration, attr:min{req, int},max{req, int},def{req, int}-->1
      </durationTime>
    </ViolentMotionParam>
    <eventType
opt="none,lineDetection,fieldDetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,teacher,student,peop
leNumChange,spacingChange,violentMotion,leavePosition,failDown,running,retentionParam,advReachHight,situationAnalysis,toiletTarry,audioAbnormal,standUp,getU
p,playCellphone,combination,objectPlacedOrTaken,tailing,electricFence">
      <!--ro, req, enum, event type, subType:string, attr:opt{opt, string}, desc:"none"-none, "lineDetection"-line crossing, "fieldDetection"-intrusion,
"regionEntrance"-region entrance, "regionExiting"-region exit, "loitering"-loitering detection, "group"-people gathering, "rapidMove"fast moving, "parking"-
parking, "unattendedBaggage"-unattended baggage, "attendedBaggage"-attended baggage, "teacher"-teacher, "student"-student, "peopleNumChange"-change of the
number of people, "spacingChange"-change of distance, "violentMotion"-violent motion, "leavePosition"-absence, "failDown"-fall down, "running"-running,
"retentionParam"-retention, "advReachHight"-climbing, "situationAnalysis"-trend analysis, "toiletTarry"-in toilet overtime, "audioAbnormal"-sudden change of
sound intensity, "standUp"-stand up, "getUp"-get up, "playCellphone"-play cellphone-->none
    </eventType>
    <GetUpParam>
      <!--ro, opt, object, out of bed detection parameters-->
      <mode opt="overBed,areaMove,sitting">
        <!--ro, opt, enum, getting up detection, subType:string, attr:opt{req, string},
desc:"overBed' (wide bed mode), "areaMove" (
bunk bed mode), "sitting" (sitting and getting up mode)-->overBed
      </mode>
    </GetUpParam>
    <SizeFilter>
      <!--ro, opt, object, size filter-->
      <enabled opt="true,false">
        <!--ro, req, string, whether to enable, attr:opt{req, string}-->test
      </enabled>
      <mode opt="pixels,actualSize">
        <!--ro, req, enum, mode, subType:string, attr:opt{req, string}, desc:"pixels"-pixel, "actualSize"-actual size-->pixels
      </mode>
      <MaxObjectSize>
        <!--ro, opt, object, the maximum size, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, y-coordinate, range:[0,1000]-->1
        </positionY>
        <width>
          <!--ro, req, int, width, range:[0,1000]-->1
        </width>
        <height>
          <!--ro, req, int, high, range:[0,1000]-->1
        </height>
      </MaxObjectSize>
      <MinObjectSize>
        <!--ro, opt, object, the minimum size, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, y-coordinate, range:[0,1000]-->1
        </positionY>
        <width>
          <!--ro, req, int, width, range:[0,1000]-->1
        </width>
        <height>
          <!--ro, req, int, high, range:[0,1000]-->1
        </height>
      </MinObjectSize>
    </SizeFilter>
    <RuleRegion>
      <!--ro, opt, object, rule area-->
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
```

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```
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </RuleRegion>
  <SitQuietly>
    <!--ro, opt, object, sedentary-->
    <durationTime min="1" max="3600" def="300">
      <!--ro, opt, int, playing duration, attr:min{req, int},max{req, int},def{req, int}-->1
    </durationTime>
  </SitQuietly>
</RuleInfo>
```

19.14.1.4 Get the intelligent capability of device

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IntelliCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <isFaceSupport>
    <!--ro, req, bool, whether it supports face detection, desc:whether it supports face detection-->true
  </isFaceSupport>
  <isBehaviorSupport>
    <!--ro, req, bool, whether it supports abnormal event detection, desc:whether it supports abnormal event detection-->true
  </isBehaviorSupport>
  <isLineDetectionSupport>
    <!--ro, req, bool, whether it supports line crossing detection, desc:whether it supports line crossing detection-->true
  </isLineDetectionSupport>
  <isFieldDetectionSupport>
    <!--ro, req, bool, whether it supports intrusion detection, desc:whether it supports intrusion detection-->true
  </isFieldDetectionSupport>
  <isRegionEntranceSupport>
    <!--ro, req, bool, whether it supports region entrance detection, desc:whether it supports region entrance detection-->true
  </isRegionEntranceSupport>
  <isRegionExitingSupport>
    <!--ro, req, bool, whether it supports region exiting detection, desc:whether it supports region exiting detection-->true
  </isRegionExitingSupport>
  <isLoiteringSupport>
    <!--ro, req, bool, whether it supports loitering detection, desc:whether it supports loitering detection-->true
  </isLoiteringSupport>
  <isGroupSupport>
    <!--ro, opt, bool, whether it supports people gathering detection, desc:whether it supports people gathering detection-->true
  </isGroupSupport>
  <isRapidMoveSupport>
    <!--ro, opt, bool, whether it supports fast moving detection, desc:whether it supports fast moving detection-->true
  </isRapidMoveSupport>
  <isParkingSupport>
    <!--ro, opt, bool, whether it supports parking detection, desc:whether it supports parking detection-->true
  </isParkingSupport>
  <isUnattendedBaggageSupport>
    <!--ro, opt, bool, whether it supports unattended baggage detection, desc:whether it supports unattended baggage detection-->true
  </isUnattendedBaggageSupport>
  <isAttendedBaggageSupport>
    <!--ro, opt, bool, whether it supports object removal detection, desc:whether it supports object removal detection-->true
  </isAttendedBaggageSupport>
  <isTeacherSupport>
    <!--ro, opt, bool, whether it supports teacher's behavior detection, desc:whether it supports teacher's behavior detection-->true
  </isTeacherSupport>
  <isStudentSupport>
    <!--ro, opt, bool, whether it supports student's behavior detection, desc:whether it supports student's behavior detection-->true
  </isStudentSupport>
  <isCombinedSupport>
    <!--ro, opt, bool-->true
  </isCombinedSupport>
  <isFaceCaptureStatisticsSupport>
    <!--ro, opt, bool, whether it supports face picture statistics, desc:whether it supports face picture statistics-->true
  </isFaceCaptureStatisticsSupport>
  <isSupportPersonDensityDetection>
```

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```
    <!--ro, opt, bool, whether the device supports people density detection, desc:whether the device supports people density detection-->true
  </isSupportPersonDensityDetection>
  <MultiSceneFaceRule>
    <!--ro, opt, object-->
    <sid min="1" max="8">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </sid>
  </MultiSceneFaceRule>
  <isSupportAlgLibParamKey>
    <!--ro, opt, bool-->true
  </isSupportAlgLibParamKey>
  <isStudentsStoodUpSupport>
    <!--ro, opt, bool-->true
  </isStudentsStoodUpSupport>
  <isFramesPeopleCountingSupport>
    <!--ro, opt, bool-->true
  </isFramesPeopleCountingSupport>
  <isSupportPersonQueueDetection>
    <!--ro, opt, bool, whether it supports queue management, desc:whether it supports queue management-->true
  </isSupportPersonQueueDetection>
  <isSupportModeling>
    <!--ro, opt, bool-->true
  </isSupportModeling>
  <RestoreAlgLibParam>
    <!--ro, opt, object, the capabilities of restoring algorithm library parameters according to algorithm library name, desc:the capabilities of restoring
algorithm library parameters according to algorithm library name-->
    <libName opt="faceSnap,HMS,behaviorAnalysis,shipsDetection">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </libName>
  </RestoreAlgLibParam>
  <isSupportAlarmInfoGet>
    <!--ro, opt, bool-->true
  </isSupportAlarmInfoGet>
  <isFaceSnapPicParamSupport>
    <!--ro, opt, bool-->true
  </isFaceSnapPicParamSupport>
  <isSupportCityManagement>
    <!--ro, opt, bool, whether it supports intelligent city management, desc:whether it supports intelligent city management. Refer to URI:
/ISAPI/Intelligent/channels/<ID>/cityManagement/capabilities?format=json-->true
  </isSupportCityManagement>
  <isSupportMixedTargetDetection>
    <!--ro, opt, bool, whether the device supports multi-target-type detection events, desc:whether the device supports multi-target-type detection events--
>true
  </isSupportMixedTargetDetection>
  <isSupportFaceContrastMode>
    <!--ro, opt, bool-->false
  </isSupportFaceContrastMode>
  <isSupportPictureCaptureComparision>
    <!--ro, opt, bool-->false
  </isSupportPictureCaptureComparision>
  <isFaceAttendanceSupport>
    <!--ro, opt, bool-->true
  </isFaceAttendanceSupport>
  <isTeacherBehaviorDetectSupport>
    <!--ro, opt, bool-->true
  </isTeacherBehaviorDetectSupport>
  <isSupportIntersectionAnalysis>
    <!--ro, opt, bool, whether it supports intersection analysis, desc:whether it supports intersection analysis-->true
  </isSupportIntersectionAnalysis>
  <isSupportFacePrivacyMask>
    <!--ro, opt, bool, whether the device supports privacy mask on faces, desc:whether the device supports privacy mask on faces-->true
  </isSupportFacePrivacyMask>
  <IntelliType>
    <!--ro, opt, object-->
    <type opt="behavior,face,behaviorandface,humanAttribute">
      <!--ro, opt, string, attr:opt{req, string}-->humanAttribute
    </type>
  </IntelliType>
</IntelliCap>
```

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19.14.1.5 Get the capability of picture overlay parameters of smart event

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/picInfoOverlap/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "PicInfoOverlapCap": {
    /*ro, req, object*/
        "enabled": [true, false],
        /*ro, req, array, subType:bool*/
        "overlapItemNum": {
        /*ro, req, object*/
            "@max":  5
            /*ro, opt, int*/
        },
        "overlapItemId": {
        /*ro, req, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  3
            /*ro, req, int*/
        },
        "itemType": ["positionInfo", "cameraNo", "captureTime", "sex", "age", "GPS", "ruleName", "count"],
        /*ro, req, array, subType:string*/
        "eventType": ["faceSnap", "behaviorAnalysis", "judiciary"]
        /*ro, req, array, subType:string*/
    }
}
```

19.14.1.6 Get the intelligent channel list

Request URL

GET /ISAPI/Intelligent/intelliChannelList

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, intelligent channel list, subType:object, attr:version{req, string, protocolVersion}-->
  <IntelliChannel>
    <!--ro, opt, object, intelligent channel-->
    <id>
      <!--ro, req, string, channel ID-->test
    </id>
  </IntelliChannel>
</IntelliChannelList>
```

19.14.1.7 People number change detection

EventType:peopleNumChange

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
```

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```
  <relatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </relatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"peopleNumChange" (people number change detection)-->peopleNumChange
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <personCount>
    <!--ro, opt, int, the number of people-->1
  </personCount>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
```

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```
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

19.14.1.8 Overstay detection

EventType:retention

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform, multiple channel No.s are separated by commas, e.g.1,2,3-->
```

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```
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:retention (overstay detection)-->retention
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
```

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```
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

19.14.1.9 Playing mobile phone detection

EventType:playCellphone

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:times that the same alarm has been uploaded*/
    "eventType":  "playCellphone",
    /*ro, req, enum, event type, subType:string, desc:"playCellphone" (playing mobile phone detection)*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "playCellphone",
    /*ro, req, string, event description*/
    "channelName":  "ABC",
    /*ro, opt, string, channel name (camera name)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID (PUID)*/
    "targetAttrs": {
    /*ro, opt, object, transmitted parameters*/
        "deviceId":  "1",
        /*ro, opt, string, device ID*/
        "deviceChannel":  1,
        /*ro, opt, int, device channel No.*/
        "deviceName":  "shebei"
        /*ro, opt, string, device name, range:[0,128]*/
    },
    "PlayCellphone": {
    /*ro, opt, object, Playing Mobile Phone Detection*/
        "TaskInfo": {
        /*ro, opt, object, task information submitted by data center products*/
            "taskID":  "test",
            /*ro, req, string, task ID, range:[0,64]*/
            "algorithmType":  1,
            /*ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)*/
            "taskName":  "test",
            /*ro, opt, string, task name*/
            "algorithmID":  "test",
            /*ro, opt, string, algorithm package ID*/
            "monitorPointID":  "test",
            /*ro, opt, string, camera No.*/
            "monitorPointName":  "test",
            /*ro, opt, string, camera name*/
            "streamType":  "realtime",
            /*ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)*/
            "ruleID":  "1",
            /*ro, opt, string, rule ID*/
            "ruleCustomName":  "ruleName"
```

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```
            "ruleCustomName":  "ruleName",
            /*ro, req, string, user-defined event name, range:[0,128], desc:It is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed.*/
            "algorithmInfo": {
            /*ro, opt, object, algorithm version information, this field is required for secondary analysis*/
                "algorithmId":  "test",
                /*ro, opt, string, algorithm package ID*/
                "algoVendorId":  "test",
                /*ro, opt, string, manufacturer ID*/
                "algoVersionId":  "test",
                /*ro, opt, string, algorithm version No.*/
                "algoChipType":  "test",
                /*ro, opt, string, chip type*/
                "algoModelVersion":  "test"
                /*ro, opt, string, model version*/
            }
        },
        "BackgroundImage": {
        /*ro, opt, object, background picture when the event occurred*/
            "resourcesContentType":  "base64",
            /*ro, opt, enum, resource transmission type, subType:string, desc:"url", "binary" "base64"*/
            "resourcesContent":  "test"
            /*ro, opt, string, resource ID, desc:When the returned alarm message contains additional resources (e.g., pictures), this field should be
returned. When the value of resourcesContentType is "binary", it must be the same as Content-ID of the resource. When the value of resourcesContentType is
"url", its value is the resource URL. When the value of resourcesContentType is "base64", its value is the data encoded via base64.*/
        },
        "timeStamp":  "2019-12-30T15:29:23+08:00",
        /*ro, opt, datetime, ISO 8601 format*/
        "ruleID":  0,
        /*ro, opt, int, rule No., range:[0,7], desc:rule No.*/
        "ruleName":  "test"
        /*ro, opt, string, rule name, range:[0,32], desc:rule name*/
    }
}
```

19.14.1.10 Getting up or falling down detection

EventType:failDown

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event uploading, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"failDown" (getting up or falling down detection)-->failDown
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
```

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```
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
```

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```
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle, the origin is the upper-left corner of the screen-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:area origin coordinates-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:Rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:Rule Name-->test
  </ruleName>
  <isDataRetransmission>
    <!--ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be
uploaded again after the device recovers-->true
  </isDataRetransmission>
</EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
failDown	[Message content]	application/xml	--	--	--
video	--	video/mp4	随机值	video.mp4	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.14.1.11 Violent motion detection

EventType:violentMotion

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
```

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```
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"violentMotion" (violent motion detection)-->violentMotion
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
```

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```
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
  <isDataRetransmission>
    <!--ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers-->true
  </isDataRetransmission>
</EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
violentMotion	[Message content]	application/xml	--	--	--
video	--	video/mp4	随机值	video.mp4	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.

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- Parameter Value: the body content of form unit.

19.14.1.12 People running detection

EventType:running

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"running" (people running detection)-->running
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for EHome alarms, e.g., test0123 (EHome2.0、EHome4.0、ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission method, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

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19.14.2 Hard Hat Detection

19.14.2.1 Get the parameters of hard hat detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, hard hat detection parameters, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, opt, object, normalized value-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[1,1000]-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height, range:[1,1000]-->1000
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <enabled>
    <!--ro, opt, bool, whether to enable the rule-->true
  </enabled>
  <SafetyHelmetDetectionRegionList>
    <!--ro, opt, array, subType:object-->
    <SafetyHelmetDetectionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, opt, string, region ID-->test
      </id>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </SafetyHelmetDetectionRegion>
  </SafetyHelmetDetectionRegionList>
  <stackTarget>
    <!--ro, req, bool, whether to overlay target frame on the alarm picture-->true
  </stackTarget>
  <stackRule>
    <!--ro, req, bool, whether to overlay rule frame on the alarm picture-->true
  </stackRule>
  <alarmLinkageDuration>
    <!--ro, req, enum, alarm linkage duration, subType:string, desc:alarm linkage duration-->5
  </alarmLinkageDuration>
  <ImageParam>
    <!--ro, opt, object-->
    <Resolution>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, picture ID-->test
      </id>
      <name>
        <!--ro, req, enum, pixels, subType:string, desc:pixels-->1080p
      </name>
      <value>
        <!--ro, req, enum, resolution, subType:string, desc:resolution-->1920*1080
      </value>
    </Resolution>
    <quality>
      <!--ro, req, enum, picture quality, subType:string, desc:picture quality-->best
    </quality>
    <alarmUpload>
      <!--ro, req, bool, whether to upload alarm picture-->true
    </alarmUpload>
    <localStorage>
      <!--ro, opt, bool, whether to store pictures locally-->true
    </localStorage>
  </ImageParam>
  <videoOverlay>
```

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```
    <!--ro, opt, object, video overlay parameters-->
    <stackTarget>
      <!--ro, req, bool, whether to overlay alarm target frame-->true
    </stackTarget>
    <stackRule>
      <!--ro, req, bool, whether to overlay alarm rule frame-->true
    </stackRule>
  </videoOverlay>
  <wearHelmetEnabled>
    <!--ro, opt, bool-->true
  </wearHelmetEnabled>
  <analysisMode>
    <!--ro, opt, enum, analysis mode, subType:string, desc:the engine also has its analysis mode (refer to the <analysisMode> node in
/ISAPI/Intelligent/analysisEngines/<ID>). The device will be added to the engine with the same analysis mode via the load balancing method. If the engine is
not configured correspondingly,this node will be failed to e returned,and the error code of taskNotMatchEngine will be returned; "realTimeMode"-real time
mode,"autoSwitchMode"-auto-switch mode-->realTimeMode
  </analysisMode>
  <helmetFilterEnabled>
    <!--ro, opt, bool-->false
  </helmetFilterEnabled>
  <durationCapEnabled>
    <!--ro, opt, bool-->false
  </durationCapEnabled>
  <durationCapInterval>
    <!--ro, opt, int, range:[0,3600]-->30
  </durationCapInterval>
  <shieldRegionList>
    <!--ro, opt, array, subType:object-->
    <shieldRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, opt, string-->test
      </id>
      <regionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[3,10]-->
        <regionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </regionCoordinates>
      </regionCoordinatesList>
    </shieldRegion>
  </shieldRegionList>
  <sensitivity>
    <!--ro, opt, int, range:[0,100]-->1
  </sensitivity>
</SafetyHelmetDetection>
```

19.14.2.2 Set the parameters of hard hat detection

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, hard hat detection, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--opt, object-->
    <normalizedScreenWidth>
      <!--req, int, normalized width, range:[1,1000]-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height, range:[1,1000]-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <enabled>
    <!--opt, bool, whether to enable the hard hat detection rules-->true
  </enabled>
  <SafetyHelmetDetectionRegionList>
    <!--opt, array, subType:object-->
    <SafetyHelmetDetectionRegion>
      <!--opt, object-->
      <id>
        <!--opt, string, detection area ID-->test
      </id>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin (0, 0) is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </SafetyHelmetDetectionRegion>
  </SafetyHelmetDetectionRegionList>
  <stackTarget>
    <!--req, bool, whether to overlay the alarm target frame-->true
  </stackTarget>
  <stackRule>
    <!--req, bool, whether to overlay the alarm rule frame-->true
  </stackRule>
  <alarmLinkageDuration>
    <!--req, enum, alarm linkage duration time, subType:string, desc:alarm linkage duration time-->5
  </alarmLinkageDuration>
  <ImageParam>
    <!--opt, object, image parameters-->
    <Resolution>
      <!--opt, object-->
      <id>
        <!--req, string, picture ID-->test
      </id>
      <name>
        <!--req, enum, subType:string-->1080p
      </name>
      <value>
        <!--req, enum, resolution, subType:string, desc:resolution-->1920*1080
      </value>
    </Resolution>
    <quality>
      <!--req, enum, picture quality, subType:string, desc:picture quality-->best
    </quality>
    <alarmUpload>
      <!--req, bool-->true
    </alarmUpload>
    <localStorage>
      <!--opt, bool-->true
    </localStorage>
  </ImageParam>
  <videoOverlay>
    <!--opt, object-->
    <stackTarget>
      <!--req, bool, whether to overlay the target frame-->true
    </stackTarget>
    <stackRule>
      <!--req, bool, whether to overlay the rule frame-->true
    </stackRule>
  </videoOverlay>
  <wearHelmetEnabled>
    <!--opt, bool-->true
  </wearHelmetEnabled>
  <analysisMode>
    <!--opt, enum, analysis mode, subType:string, desc:analysis mode-->realTimeMode
  </analysisMode>
</SafetyHelmetDetection>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
SafetyHelmetDetection	[Message content]	application/xml	--	--	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.14.2.3 Set advanced parameters of hard hat detection

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection/advanceConfiguration

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<AdvanceConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <headSensitivity>
    <!--opt, int, head detection sensitivity, range:[1,5], desc:the larger value indicates that the detection is more sensitive-->3
  </headSensitivity>
  <humanSensitivity>
    <!--opt, int, human body detection sensitivity, range:[1,5], desc:the larger value indicates that the detection is more sensitive-->3
  </humanSensitivity>
  <alarmAnalysisFrame>
    <!--opt, int, number of frames required for alarm analysis, range:[1,50], desc:the larger value indicates that more frames need to be analyzed, and the
alarm will be triggered more slowly-->1
  </alarmAnalysisFrame>
  <FDLibList>
    <!--opt, array, channel's linked face picture library list, subType:object-->
    <FDLib>
      <!--opt, object, channel's linked face picture library-->
      <id>
        <!--req, int, No.-->1
      </id>
      <FDID>
        <!--req, int, face picture library ID-->1
      </FDID>
      <thresholdValue>
        <!--opt, int, similarity threshold for comparison, range:[0,100], desc:the higher the threshold is, the more accurate the comparison is. If the
threshold for all libraries is enabled, then threshold for single library will not take effect-->1
      </thresholdValue>
      <customFaceLibID>
        <!--opt, string, custom face picture library ID, desc:custom ID has priority over the FDID-->test
      </customFaceLibID>
    </FDLib>
  </FDLibList>
</AdvanceConfiguration>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.14.2.4 Get advanced parameters of hard hat detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection/advanceConfiguration

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<AdvanceConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <headSensitivity>
    <!--ro, opt, int, head detection sensitivity, and the default value is 3, range:[1,5], desc:the larger value indicates that the detection is more sensitive-->3
  </headSensitivity>
  <humanSensitivity>
    <!--ro, opt, int, human body detection sensitivity, and the default value is 3, range:[1,5], desc:the larger value indicates that the detection is more sensitive-->3
  </humanSensitivity>
  <alarmAnalysisFrame>
    <!--ro, opt, int, number of frames required for alarm analysis, and the default value is 12, range:[1,50], desc:the larger value indicates that more frames need to be analyzed, and the alarm will be triggered more slowly-->1
  </alarmAnalysisFrame>
  <FDLibList>
    <!--ro, opt, array, channel's linked face picture library list, subType:object-->
    <FDLib>
      <!--ro, opt, object, list-->
      <id>
        <!--ro, req, int, No. of items in the list-->1
      </id>
      <FDID>
        <!--ro, req, int, a unique ID of face picture library-->1
      </FDID>
      <thresholdValue>
        <!--ro, opt, int, similarity threshold for comparison, range:[0,100], desc:the higher the threshold is, the more accurate the comparison is. If the threshold for all libraries is enabled, then threshold for single library will not take effect-->1
      </thresholdValue>
      <customFaceLibID>
        <!--ro, opt, string, custom face picture library ID, desc:custom ID has priority over the FDID-->test
      </customFaceLibID>
    </FDLib>
  </FDLibList>
</AdvanceConfiguration>
```

19.14.2.5 Get the configuration capability of hard hat detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetectionCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, opt, object, ro,normalized value-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[1,1000]-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height, range:[1,1000]-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <SafetyHelmetDetectionRegionList size="8">
    <!--ro, opt, array, detection area list, subType:object, attr:size{req, int}-->
    <SafetyHelmetDetectionRegion>
      <!--ro, opt, object, detection area-->
      <id>
        <!--ro, opt, string-->test
      </id>
      <RegionCoordinatesList size="10">
        <!--ro, opt, array, subType:object, range:[3,10], attr:size{req, int}-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:area coordinates-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
```

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```
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </SafetyHelmetDetectionRegion>
    </SafetyHelmetDetectionRegionList>
    <stackTarget opt="true,false">
      <!--ro, req, bool, whether to overlay alarm target frame, attr:opt{req, string}-->true
    </stackTarget>
    <stackRule opt="true,false">
      <!--ro, req, bool, whether to overlay alarm rule frame, attr:opt{req, string}-->true
    </stackRule>
    <alarmLinkageDuration opt="5,10,30,60,120,300,600">
      <!--ro, req, enum, alarm linkage duration, subType:int, attr:opt{req, string}, desc:alarm linkage duration-->5
    </alarmLinkageDuration>
    <ImageParam>
      <!--ro, opt, object, picture parameters-->
      <ResolutionList size="3">
        <!--ro, opt, array, resolution, subType:object, attr:size{req, int}-->
        <Resolution>
          <!--ro, opt, object, resolution-->
          <id>
            <!--ro, req, string, ID-->test
          </id>
          <name>
            <!--ro, req, enum, resolution name, subType:string, desc:resolution name-->1080p
          </name>
          <value>
            <!--ro, req, enum, subType:string-->1920*1080
          </value>
        </Resolution>
      </ResolutionList>
      <quality opt="best,good,general">
        <!--ro, req, enum, picture quality, subType:string, attr:opt{req, string}, desc:"best","good","general"-normal-->best
      </quality>
      <alarmUpload opt="true,false">
        <!--ro, req, bool, whether to upload alarm picture, attr:opt{req, string}-->true
      </alarmUpload>
      <localStorage opt="true,false">
        <!--ro, opt, bool, whether to store pictures locally, attr:opt{req, string}-->true
      </localStorage>
    </ImageParam>
    <AdvanceConfiguration>
      <!--ro, opt, object-->
      <headSensitivity min="1" max="5" def="3">
        <!--ro, opt, int, head detection sensitivity, attr:min{req, int},max{req, int},def{req, int}, desc:larger value indicates more sensitive detection-->1
      </headSensitivity>
      <humanSensitivity min="1" max="5" def="3">
        <!--ro, opt, int, human body detection sensitivity, attr:min{req, int},max{req, int},def{req, int}, desc:larger value indicates more sensitive
detection-->1
      </humanSensitivity>
      <alarmAnalysisFrame min="1" max="50" def="12">
        <!--ro, opt, int, number of frames required for alarm analysis, attr:min{req, int},max{req, int},def{req, int}, desc:larger value indicates that more
frames are needed to be analyzed and more slowly the alarm will be triggered-->1
      </alarmAnalysisFrame>
      <FDLibList size="10">
        <!--ro, opt, array, subType:object, attr:size{req, int}-->
        <FDLib>
          <!--ro, opt, object, face picture library-->
          <id min="1" max="10">
            <!--ro, req, int, No., range:[1,10], attr:min{req, int},max{req, int}-->1
          </id>
          <FDID min="1" max="10">
            <!--ro, req, string, attr:min{req, int},max{req, int}-->test
          </FDID>
          <thresholdValue min="1" max="10">
            <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
          </thresholdValue>
          <customFaceLibID min="1" max="10">
            <!--ro, opt, string, custom face picture library ID, attr:min{req, int},max{req, int}, desc:custom face picture library ID-->test
          </customFaceLibID>
        </FDLib>
      </FDLibList>
    </AdvanceConfiguration>
    <SearchDescriptionCap>
      <!--ro, opt, object, ability of searching for safety helmet event, desc:ability of searching for safety helmet event-->
      <timeSpanMaxNum>
        <!--ro, req, enum, the maximum number of time periods in which search can be conducted simultaneously, subType:string, desc:the maximum number of time
periods in which search can be conducted simultaneously-->all
      </timeSpanMaxNum>
      <resultMaxNum>
        <!--ro, req, int, the maximum number of items that can be searched for at a time-->1
      </resultMaxNum>
      <faceContrastResult opt="all,success,failed">
        <!--ro, req, string, comparison result, attr:opt{req, string}-->test
      </faceContrastResult>
      <FDLibList size="10">
        <!--ro, opt, array, subType:object, attr:size{req, int}-->
        <FDLib>
          <!--ro, opt, object, face picture library-->
          <FDID min="1" max="10">
            <!--ro, req, string, attr:min{req, int},max{req, int}-->test
          </FDID>
```

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```
        </FDLib>
      </FDLibList>
      <name min="1" max="10">
        <!--ro, opt, string, (search condition) name, attr:min{req, int},max{req, int}, desc:720p,D1"-->test
      </name>
    </SearchDescriptionCap>
    <videoOverlay>
      <!--ro, opt, object, video overlay parameters-->
      <stackTarget opt="true,false">
        <!--ro, req, bool, whether to overlay alarm target frame, attr:opt{req, string}-->true
      </stackTarget>
      <stackRule opt="true,false">
        <!--ro, req, bool, whether to overlay alarm rule frame, attr:opt{req, string}-->true
      </stackRule>
    </videoOverlay>
    <wearHelmetEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </wearHelmetEnabled>
    <analysisMode opt="realTimeMode,autoSwitchMode">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </analysisMode>
    <helmetFilterEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->false
    </helmetFilterEnabled>
    <durationCapEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->false
    </durationCapEnabled>
    <durationCapInterval min="0" max="3600">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->30
    </durationCapInterval>
    <shieldRegionList size="8">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <shieldRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, opt, string-->test
        </id>
        <regionCoordinatesList size="10">
          <!--ro, opt, array, subType:object, range:[3,10], attr:size{req, int}-->
          <regionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->1
            </positionY>
          </regionCoordinates>
        </regionCoordinatesList>
      </shieldRegion>
    </shieldRegionList>
    <sensitivity min="0" max="100">
      <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
  </SafetyHelmetDetectionCap>
```

19.14.2.6 Set hard hat detection regions in a batch

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetectionRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <SafetyHelmetDetectionRegion>
    <!--req, object-->
    <id>
      <!--opt, string, detection area ID-->test
    </id>
    <RegionCoordinatesList size="10">
      <!--opt, array, area coordinate list, subType:object, range:[3,10], attr:size{req, int}-->
      <RegionCoordinates>
        <!--opt, object, coordinates of the area, desc:the origin is the upper-left corner of the screen(0, 0)-->
        <positionX>
          <!--req, int, x-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--req, int, y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </SafetyHelmetDetectionRegion>
</SafetyHelmetDetectionRegionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.14.2.7 Get the batch configure of the hard hat detection region

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetectionRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <SafetyHelmetDetectionRegion>
    <!--ro, req, object-->
    <id>
      <!--ro, opt, string, area ID-->test
    </id>
    <RegionCoordinatesList size="10">
      <!--ro, opt, array, area coordinate list, subType:object, range:[3,10], attr:size{req, int}-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen(0, 0)-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </SafetyHelmetDetectionRegion>
</SafetyHelmetDetectionRegionList>
```

19.14.2.8 Get a specific configuration of hard hat detection area

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, string, area ID-->test
  </id>
  <RegionCoordinatesList size="10">
    <!--ro, opt, array, area coordinate list, subType:object, range:[3,10], attr:size{req, int}-->
    <RegionCoordinates>
      <!--ro, opt, object, coordinates of the area, desc:the origin is the upper-left corner of the screen(0, 0)-->
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</SafetyHelmetDetectionRegion>
```

19.14.2.9 Set a hard hat detection region of a specific video input channel

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

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Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, string, ID-->test
  </id>
  <RegionCoordinatesList size="10">
    <!--opt, array, subType:object, range:[3,10], attr:size{req, int}-->
    <RegionCoordinates>
      <!--opt, object-->
      <positionX>
        <!--req, int, x-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</SafetyHelmetDetectionRegion>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.14.2.10 Hard hat detection results

EventType:safetyHelmetDetection

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The
value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video
channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI
protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:times that the same alarm has been uploaded*/
    "eventType":  "safetyHelmetDetection",
    /*ro, req, string, event type, desc:"safetyHelmetDetection" (hard hat detection event)*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event
ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "Safety Helmet Detection",
    /*ro, req, string, event description, desc:Hard Hat Detection*/
    "channelName":  "ABC",
    /*ro, req, string, channel name (camera name)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID which is also the PUID, desc:this field must be returned when ISAPI event information is transmitted via ISUP*/
    "safetyHelmetDetectionBackgroundImageResolution": {
    /*ro, opt, object*/
```

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```
        "height":  1,
        /*ro, req, int, range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, range:[1,65535], unit:px*/
    },
    "Target": [
    /*ro, opt, array, target, subType:object, desc:multiple targets can be output after being recognized once*/
        {
            "recognitionType":  "safetyHelmet",
            /*ro, req, enum, recognition type, subType:string, desc:"safetyHelmet" (hard hat detection)*/
            "TargetInfo": {
            /*ro, opt, object, target information*/
                "recognition":  "safetyHelmet",
                /*ro, req, enum, recognition type, subType:string, desc:"safetyHelmet" (hard hat detection)*/
                "dataSource":  "timing"
                /*ro, req, enum, detection data source, subType:string, desc:"timing" (scheduled), "trigger" (when exceeded the configured level limit)*/
            },
            "rect": [
            /*ro, opt, array, rectangle frame of the hard hat, subType:object, desc:only one element will be returned. The origin is the upper-left corner
of the screen*/
                {
                    "height":  1.000,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                }
            ],
            "contentID":  "safetyHelmet1_image",
            /*ro, opt, string, thumbnail content ID, range:[0,256]*/
            "pId":  "F4F665D6A18E41308CE9934DCDDD2222",
            /*ro, opt, string, thumbnail ID of hard hat detection alarm, range:[0,64]*/
            "uid":  "test",
            /*ro, opt, string, unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits*/
            "safetyHelmetURL":  "test",
            /*ro, opt, string, thumbnail URL of hard hat detection alarm, range:[0,256]*/
            "faceContrastResult":  "success",
            /*ro, opt, enum, face picture comparison result, subType:string, desc:this field is returned only when the linkage between hard hat detection
and face picture (face comparison linkage) has been configured. "success" (comparison succeeded), "failed" (comparison failed)*/
            "Face": {
            /*ro, opt, object, face picture information*/
                "FDID":  "1123",
                /*ro, req, string, face picture library ID, range:[0,63]*/
                "FPID":  "test",
                /*ro, opt, string, face record ID, range:[0,63], desc:unique ID consisting of letters and digits*/
                "contentID":  "faceLibPic",
                /*ro, opt, string, contentID of picture, which matches the captured face picture, range:[0,256]*/
                "pId":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of picture, which matches the captured face picture, range:[0,64]*/
                "libFacePicURL":  "test"
                /*ro, opt, string, URL of picture, which matches the captured face picture, range:[0,256]*/
            },
            "FaceList": [
            /*ro, opt, array, face picture information list, subType:object, desc:face picture information list*/
                {
                    "FDID":  "1123",
                    /*ro, req, string, face picture library ID, range:[0,63]*/
                    "FPID":  "test",
                    /*ro, opt, string, face record ID, range:[0,63], desc:unique ID consisting of letters and digits*/
                    "contentID":  "faceLibPic",
                    /*ro, opt, string, contentID of picture, which matches the captured face picture, range:[0,256]*/
                    "pId":  "F4F665D6A18E41308CE9934DCDDD3333",
                    /*ro, opt, string, ID of picture, which matches the captured face picture, range:[0,64]*/
                    "libFacePicURL":  "test",
                    /*ro, opt, string, URL of picture, which matches the captured face picture, range:[0,256]*/
                    "name":  "test",
                    /*ro, opt, string, person name, range:[1,32], desc:the sensitive information should be encrypted*/
                    "certificateType":  "passportID",
                    /*ro, opt, enum, certificate type, subType:string, desc:the sensitive information should be encrypted. "officerID" (officer ID), "ID"
(ID card), "passportID" (passport), "other" (other type)*/
                    "certificateNumber":  "test"
                    /*ro, opt, string, certificate number, range:[8,32], desc:the sensitive information should be encrypted*/
                }
            ],
            "status":  "wear",
            /*ro, opt, enum, whether the person wears a hard hat, subType:string, desc:"unwear" (not wear, default value), "wear" (wear)*/
            "safetyHelmetColour":  "yellow",
            /*ro, opt, enum, subType:string, dep:and,{$.Target[*].status,eq,wear}*/
            "reflectiveVestStatus":  "wear",
            /*ro, opt, enum, subType:string*/
            "presetID":  0
            /*ro, opt, int*/
        }
    ],
    "contentID":  "background_image",
    /*ro, opt, string, thumbnail content ID, range:[0,256]*/
    "pId":  "F4F665D6A18E41308CE9934DCDDD1111",
    /*ro, opt, string, thumbnail ID of hard hat detection alarm, range:[0,64]*/
    "imageURL":  "test",
    /*ro, opt, string, background picture URL, range:[0,256]*/
    "URLCertificationType":  "digest"
```

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```
    urlCertificationType : "digest"
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (it is for the cloud storage protocol), "digest" (digest authentication,
it is for the device local storage)*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
safetyHelmetDetection	[Message content]	application/json	--	--	--
.value	[Binary picture data]	image/jpeg	safetyHelmet1_image	safetyHelmet1_image.jpg	--
.value	[Binary picture data]	image/jpeg	faceLibPic	faceLibPic.jpg	--
.value	[Binary picture data]	image/jpeg	background_image	background_image.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.15 AIOP

19.15.1 AIOP Algorithm Model Management

19.15.1.1 Get the status information of a specified engine

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engine/<engineID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
engineID	string	--

Request Message

None

Response Message

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```
{
    "LoadingInfo": {
    /*ro, req, object, engine status information*/
        "status":  1,
        /*ro, req, enum, engine status information, subType:int, desc:0 (model package is not loaded), 1 (model package is loaded and operates well), 2 (model package operates abnormally), 10 (engine is initializing), 11 (engine is abnormal), 12 (current engine is used for facial application, cannot be used for AI intelligent application), 13 (hard hat detection), 14 (smoke detection), 15 (perimeter protection, including  line crossing detection, region entrance, region exit, intrusion, leaving items, picking up items, wandering), 16 (person management, including absence and abnormal number), 17 (abnormal event detection, including people-falling-down, people gathering, and rapid movement), 18 (mixed mode, supports loading multiple traditional intelligent algorithms and multiple AI algorithms on the engine at the same time)*/
        "MPID":  "example",
        /*ro, req, string, model package ID, range:[0,64], desc:model package ID*/
        "MPName":  "example",
        /*ro, req, string, model package name, range:[0,128]*/
        "MPVersion":  "V1.1.0",
        /*ro, opt, string, range:[0,64]*/
        "description": [
        /*ro, opt, array, model description, subType:object, desc:when the model is downloaded, the packed model URL and a description file in JSON format will be returned, and this node is a whole interception of file*/
            {}
        ],
        "algSortingParams": {},
        /*ro, opt, object, added transparent transmission node of algorithm rule arrangement platform parameter, desc:the arrangement platform combination applies AIOP and other algorithm dynamic libraries, adding a node to transparently transmit the algorithm parameters of the arrangement platform, and notifies the web how to display the interface, such as supporting the display of face library and version information. The field value is not required in the algorithm engine*/
        "MPType":  "reasonModel",
        /*ro, opt, enum, subType:string*/
        "relVideoChannelNum":  4,
        /*ro, opt, int, the number of video channels bound to the engine, range:[0,4], desc:this field is returned when calculating the multiple models loading supported by the engine used for displaying the bound video channel number*/
        "surplusVideoChannelNum":  1,
        /*ro, opt, int, the number of remaining video channels bound to the engine, desc:this field is returned when calculating the multiple models loading supported by the engine this field is used for determining whether the engine can be bound to the new video channel*/
        "relPictureChannelNum":  4,
        /*ro, opt, int, the number of picture channels bound to the engine, range:[0,4], desc:this field is returned when calculating the multiple models loading supported by the engine used for displaying the bound video channel number*/
        "surplusPictureChannelNum":  1,
        /*ro, opt, int, the number of remaining picture channels bound to the engine, desc:this field is returned when calculating the multiple models loading supported by the engine used for displaying the bound picture channel number*/
        "maxRelChannelNum":  2,
        /*ro, opt, int, the maximum number of channel that can be bound to the device*/
        "supportModelType":  [1, 2, 11, 12],
        /*ro, opt, enumarray, supported (AIOP) model type, subType:int, desc:this field is not returned indicates that all AIOP model types are supported, 1(detection), 2 (classification), 3 (OCR frame detection), 4 (OCR character recognition), 11 (vdeo abnormal event detection sub-detection algorithm), 12 (sub classification algorithm for video abnormal event detection)*/
        "supportSortingType":  [2001, 2002],
        /*ro, opt, enumarray, subType:int*/
        "sharingEnabled":  true,
        /*ro, opt, bool, whether to enable then engine sharing mode, desc:default value is false, sharing mode run a task in multiple engines, which are required to be bound to the same model inclusive mode: run a task in the specified engine*/
        "relChannelList": [1]
        /*ro, opt, array, channel list bound to the engine, subType:int, desc:used for displaying the bound channel number*/
    }
}
```

19.15.1.2 Get the status of all engines

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/engines?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "LoadingInfoList": [
    /*ro, req, array, engine status information list, subType:object*/
        {
            "engine":  1,
            /*ro, req, int, which is used to distinguish different engine No.*/
            "status":  1,
            /*ro, req, enum, engine status information list, subType:int, desc:0 (model package is not loaded), 1 (model package is loaded and operates
well), 2 (model package operates abnormally), 10 (engine is initializing), 11 (engine is abnormal), 12 (current engine is used for facial application,
cannot be used for AI intelligent application), 13 (hard hat detection), 14 (smoke detection), 15 (perimeter protection, including  line crossing detection,
region entrance, region exit, intrusion, leaving items, picking up items, wandering), 16 (person management, including absence and abnormal number), 17
(abnormal event detection, including people-falling-down, people gathering, and rapid movement), 18 (mixed mode, supports loading multiple traditional
intelligent algorithms and multiple AI algorithms on the engine at the same time)*/
            "MPID":  "example",
            /*ro, req, string, model package ID, range:[0,64], desc:model package ID*/
            "MPName":  "example",
            /*ro, req, string, model package name, range:[0,128]*/
            "MPVersion":  "V1.1.0",
            /*ro, opt, string, range:[0,64]*/
            "description": [
            /*ro, opt, array, model description information, subType:object, desc:description*/
                {}
            ],
            "algSortingParams": {},
            /*ro, opt, object, added transparent transmission node of algorithm rule arrangement platform parameter, desc:the arrangement platform
combination applies AIOP and other algorithm dynamic libraries, adding a node to transparently transmit the algorithm parameters of the arrangement
platform, and notifies the web how to display the interface, such as supporting the display of face library and version information. The field value is not
required in the algorithm engine*/
            "algRelationPicUrl":  "test",
            /*ro, opt, string, range:[0,512]*/
            "algRelationPicParams": {},
            /*ro, opt, object*/
            "MPType":  "reasonModel",
            /*ro, opt, enum, subType:string*/
            "relVideoChannelNum":  4,
            /*ro, opt, int, the number of video channels bound to the engine, range:[0,4], desc:this field is returned when calculating the multiple models
loading supported by the engine used for displaying the bound video channel number*/
            "surplusVideoChannelNum":  1,
            /*ro, opt, int, the number of remaining video channels bound to the engine, desc:this field is returned when calculating the multiple models
loading supported by the engine used for determining whether the new video channel can be bound by the field*/
            "relPictureChannelNum":  4,
            /*ro, opt, int, the number of picture channels bound to the engine, range:[0,4], desc:this field is returned when calculating the multiple
models loading supported by the engine used for displaying the bound picture channel number*/
            "surplusPictureChannelNum":  1,
            /*ro, opt, int, the number of remaining picture channels bound to the engine, desc:this field is returned when calculating the multiple models
loading supported by the engine used for determining whether the new video channel can be bound by the field*/
            "supportModelType":  [1, 2, 11, 12],
            /*ro, opt, enumarray, supported (AIOP) model type, subType:int, desc:this field is not returned indicates that all AIOP model types are
supported, 1(detection), 2 (classification), 3 (OCR frame detection), 4 (OCR character recognition), 11 (vdeo abnormal event detection sub-detection
algorithm), 12 (sub classification algorithm for video abnormal event detection)*/
            "supportSortingType":  [2001, 2002],
            /*ro, opt, enumarray, subType:int*/
            "sharingEnabled":  true,
            /*ro, opt, bool, whether to enable the engine sharing mode, desc:default value is false, sharing mode run a task in multiple engines, which are
required to be bound to the same model inclusive mode: run a task in the specified engine*/
            "relChannelList": [1]
            /*ro, opt, array, channel list bound to the engine, subType:int, desc:used for displaying the bound picture channel number*/
        }
    ],
    "specialEngineList": [
    /*ro, opt, array, subType:object*/
        {
            "engine":  1,
            /*ro, req, int*/
            "status":  1,
            /*ro, req, enum, subType:int*/
            "MPID":  "example",
            /*ro, req, string, range:[0,64]*/
            "MPName": "example",
            /*ro, req, string, range:[0,128]*/
            "MPVersion":  "V1.1.0",
            /*ro, opt, string, range:[0,64]*/
            "description": [
            /*ro, opt, array, subType:object*/
                {}
            ]
        }
    ]
}
```

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19.15.1.3 Search for the model package storage space of the device

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/management/space?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "totalCapacity":  1024,
    /*ro, req, int, total storage space of model packages supported by device, unit:MB*/
    "surplusCapacity":  1024,
    /*ro, req, int, remaining storage space of model packages supported by device, unit:MB*/
    "MPNumbers":  1
    /*ro, req, int, total number of supported model packages*/
}
```

19.15.1.4 Search for locally stored algorithm model package

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/management?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AlgorithmModels": [
    /*ro, req, array, algorithm package information list, subType:object*/
        {
            "MPID":  "example",
            /*ro, req, string, model package ID, range:[0,64], desc:used for distinguishing the model information*/
            "MPName":  "example",
            /*ro, req, string, model package name, range:[0,128]*/
            "MPVersion":  "V1.1.0",
            /*ro, opt, string, model package version, range:[0,64]*/
            "MPRuleVersion":  "V1.1.0",
            /*ro, opt, string, range:[0,64]*/
            "trainModeID":  "12930",
            /*ro, opt, string, range:[0,64]*/
            "status":  1,
            /*ro, req, enum, status information, subType:int, desc:loaded engine No., refer to the node "engine": 0 (not loaded), 1 (loaded), 2 (decrypting
failed)*/
            "engine": [1, 2],
            /*ro, opt, array, one model package can be loaded by multiple engines, subType:int, desc:one model package can be loaded by multiple engines,
and multiple engines are separated by commas; if this node is not returned, it indicates the model package of current MPID is not loaded*/
            "description": [
            /*ro, opt, array, when the model is downloaded, the packed model URL and a description file in JSON format will be returned, and this node is a
whole interception of file, subType:object*/
                {}
            ],
            "algSortingParams": {},
            /*ro, opt, object, added transparent transmission node of algorithm rule arrangement platform parameter, desc:the arrangement platform
combination applies AIOP and other algorithm dynamic libraries, adding a node to transparently transmit the algorithm parameters of the arrangement
platform, and notifies the web how to display the interface, such as supporting the display of face library and version information. The field value is not
required in the algorithm engine*/
            "algRelationPicUrl":  "test",
            /*ro, opt, string, Algorithm App Relationship Diagram, range:[0,512], desc:model package is valid when arranging the model. The picture is used
to describe the relationship between the arrangement  model application nodes. The picture is packaged in model package*/
            "algRelationPicParams": {},
            /*ro, opt, object, algorithm application relationship diagram for explaining transparent node, desc:valid when the model package is
orchestration model. The content passed through this node is the JSON file of the algorithm application relationship diagram (i.e., the orchestration
diagram) mentioned above. This JSON file is packed into the model package*/
            "isDefaultFile":  true,
            /*ro, opt, bool, whether it is the model package file carried by the device during factory delivery, desc:“false" (not carried by the device
during factory delivery), "true" (carried by the device during factory delivery), related URI: DELETE
/ISAPI/Intelligent/AIOpenPlatform/algorithmModel/management/space/<MPID>?format=json*/
            "MPType":  "reasonModel",
            /*ro, opt, enum, model type, subType:string, desc:if this filed does not exist, it is inference model by default. The inference model, also
known as the application model, is used for classification and detection; the training model is used for self-learning training of AI algorithm models.
After training is completed and the target performance value is reached, it is converted into an inference model "trainMode", "reasonModel"*/
            "videoTaskAnalysisCap":  4,
            /*ro, opt, int, number of video task analysis channels, desc:The device will calculate and load the number of video and image analysis channels
supported for each orchestration model based on the current resources*/
            "pictureTaskAnalysisCap":  16,
            /*ro, opt, int, number of picture task analysis channels, desc:different model packages occupy different resources, and the number of video
picture channels supporting analysis is different*/
            "authorisationStatus":  "expired",
            /*ro, opt, enum, authorization status, subType:string, desc:"free", "authorized", "expired", "notAuthorized"*/
            "authorisationDate":  "2023-08-18",
```

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```
            /*ro, opt, date, authorization date*/
            "residualValidity": 10,
            /*ro, opt, int, remaining validity period, range:[0,240000], step:1, unit:h*/
            "modelPrecision": 0,
            /*ro, opt, enum, model accuracy, subType:int, desc:consistent with the definitions in model description file. Common model: small model size, short training time, short prediction time, and average model accuracy; High-precision model: higher model accuracy, but longer training time, larger model size, and longer prediction time; Ultra-high-precision model: highest model accuracy, but long training time, large model size, and long prediction time. 0 (common model), 1 (high-precision model), 2 (ultra-high-precision model)*/
            "isSupportRelatedMPID": true,
            /*ro, opt, bool, dep:and,{$.AlgorithmModels[*].modelPrecision,eq,2}*/
            "isSupportIndependentAnalysis": true,
            /*ro, opt, bool, dep:and,{$.AlgorithmModels[*].modelPrecision,eq,2}*/
            "loadingResources": 25,
            /*ro, opt, int, range:[0,100]*/
            "defaultIconURL": "test",
            /*ro, opt, string*/
            "importTime": "1970-01-01T08:00:00.000+08:00",
            /*ro, opt, datetime*/
            "loadingTime": "1970-01-01T08:00:00.000+08:00",
            /*ro, opt, datetime*/
            "platform": 1,
            /*ro, opt, enum, subType:int*/
            "platformName": "K81",
            /*ro, opt, string*/
            "isSupportModelUpdate": true,
            /*ro, opt, bool*/
            "isSupportModelRestore": true,
            /*ro, opt, bool*/
            "taskCap": ["pic", "video"],
            /*ro, opt, array, subType:string*/
            "packageURL": "http://10.66.109.30:20080/ISAPI/System/GetFile?/home/ssd/algorithm/package/dc_custom_KT2card_l64_baseline_build230426165935_a403a734e21f00db853cdbf1f5609c4a.app",
            /*ro, opt, string*/
            "MD5": "test",
            /*ro, opt, string*/
            "FunctionDescription": "test",
            /*ro, opt, string*/
            "updateLog": "test",
            /*ro, opt, string, range:[0,512]*/
            "abilityInfo": [
            /*ro, opt, array, subType:object*/
                {
                    "Resolution": {
                    /*ro, opt, object*/
                        "width": 0,
                        /*ro, req, int*/
                        "height": 0
                        /*ro, req, int*/
                    },
                    "ability": 0,
                    /*ro, opt, int*/
                    "picAbility": 0.0
                    /*ro, opt, float*/
                }
            ],
            "algorithm": [
            /*ro, opt, array, subType:object*/
                {
                    "targetType": "1",
                    /*ro, req, enum, subType:string*/
                    "analysisType": "1",
                    /*ro, opt, enum, subType:string*/
                    "analysisSourceType": "video,picture",
                    /*ro, req, string*/
                    "algorithmID": "test",
                    /*ro, opt, string*/
                    "classificationID": "test",
                    /*ro, opt, string*/
                    "version": "test",
                    /*ro, opt, string*/
                    "vendor": "test"
                    /*ro, opt, string*/
                }
            ],
            "algorithmType": "analysis",
            /*ro, opt, enum, subType:string*/
            "AppIDList": ["test"]
            /*ro, opt, array, subType:string*/
        }
    ],
    "coordinationModels": [
    /*ro, opt, array, subType:object*/
        {
            "MPID": "example",
            /*ro, req, string, range:[0,64]*/
            "MPName": "example",
            /*ro, opt, string, range:[0,128]*/
            "MPVersion": "V1.1.0",
            /*ro, opt, string, range:[0,64]*/
            "modelAssociation": {
            /*ro, opt, object*/
                "performanceMPID": "undefined",
```


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```
                /*ro, req, string, range:[0,64]*/
                "performanceMPEngine": [1, 2],
                /*ro, opt, array, subType:int*/
                "superPrecisionMPID":  "undefined",
                /*ro, req, string, range:[0,64]*/
                "superPrecisionMPEngine": [1, 2]
                /*ro, opt, array, subType:int*/
            },
            "applicationMode":  3
            /*ro, opt, enum, subType:int*/
        }
    ]
}
```

19.15.1.5 Get the download progress of preset model package

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/algorithmModel/progress/<MPID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
MPID	string	--

Request Message

None

Response Message

```
{
    "percent":  100
    /*ro, req, int, algorithm download progress, range:[0,100], desc:0 (download is not started), 100 (download completed); the device should be rebooted
when the download progress has reached 100%*/
}
```

19.15.1.6 Preset the algorithm model in binary form format

Request URL

POST /ISAPI/Intelligent/AIOpenPlatform/algorithmModelBinaryPackets?format=json

Query Parameter

None

Request Message

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```
{
    "engine": [1, 2, 3, 4],
    /*opt, array, engine ID, subType:int, desc:if this node is not configured,it indicates that store the model in the device without binding to engine; if this node is configured,the model will be bond to the engine*/
    "MPID":  "test",
    /*req, string, model ID, range:[1,64], desc:which is managed by platform*/
    "MPName":  "test",
    /*opt, string, model package name, range:[1,128]*/
    "MPVersion":  "V1.1.0",
    /*opt, string, range:[0,64]*/
    "trainModeID":  "12930",
    /*opt, string, range:[0,64]*/
    "description": [
    /*opt, array, description file, subType:object, desc:when the model is downloaded, the packed model URL and a description file in JSON format will be returned, this node is a whole interception of file*/
        {}
    ],
    "platform":  1,
    /*opt, enum, subType:int*/
    "platformName":  "K81",
    /*opt, string*/
    "algSortingParams": {},
    /*opt, object, added transparent transmission node of algorithm rule arrangement platform parameter, desc:the arrangement platform combination applies AIOP and other algorithm dynamic libraries, adding a node to transparently transmit the algorithm parameters of the arrangement platform, and notifies the web how to display the interface, such as supporting the display of face library and version information. The field value is not required in the algorithm engine*/
    "authorized":  true,
    /*opt, bool, whether the model is authorized*/
    "appID":  "test",
    /*opt, string, application ID, range:[1,64], desc:after the user submits the training task, the background executes the model training operation (this step is the same as the original step), and SaaS saves the training task ID as appid*/
    "MPType":  "reasonModel",
    /*opt, enum, subType:string*/
    "secretKey":  "test",
    /*opt, string, range:[8,16]*/
    "chunckInfo": {
    /*opt, object*/
        "fileType":  "modelPackage",
        /*req, enum, subType:string*/
        "totalSize":  "32132131",
        /*req, string, range:[1,16]*/
        "packNum":  4,
        /*req, int, range:[1,32]*/
        "chunkSize":  2048,
        /*req, int, range:[1,2147483647], unit:Byte*/
        "chunck":  1
        /*req, int, range:[1,32]*/
    },
    "packageAbility": {},
    /*opt, object*/
    "md5":  "test"
    /*opt, string*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
Algorithm_Model	[Message content]	application/json	--	--	--
Model_Package_Size	[Stream data]	application/octet-stream		Model_Package.bin	--
LicenseKey_Package_Size	[Stream data]	application/octet-stream		LicenseKey_Package.bin	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.

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- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded), it is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded), it is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded), it is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

19.15.1.7 Get the AI Open Platform capability supported by the device

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json&engine=<engineID>

Query Parameter

Parameter Name	Parameter Type	Description
engineID	string	--

Request Message

None

Response Message

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```
{
    "isSupportVideoTask": true,
    /*ro, opt, bool, whether the device supports video analysis task*/
    "isSupportPicturePollingAnalysisTask": true,
    /*ro, opt, bool, whether the device supports captured picture polling task*/
    "isSupportUploadModelwithURL": true,
    /*ro, opt, bool, whether the device supports importing algorithm model in URL format*/
    "isSupportUploadModelwithPushBinaryData": true,
    /*ro, opt, bool, whether the device supports importing algorithm model in binary format*/
    "isSupportAIRuleConfig": true,
    /*ro, opt, bool, whether the device supports AI rule configuration*/
    "isSupportAuthorizationModel": true,
    /*ro, opt, bool, whether the device supports model authorization*/
    "isSupportEventTrigger": true,
    /*ro, opt, bool, whether the device supports linkage configuration*/
    "isSupportOSDConfig": true,
    /*ro, opt, bool, whether the device supports OSD overlay configuration*/
    "isSupportConfidenceConfig": true,
    /*ro, opt, bool, whether the device supports confidence configuration in AI rule configuration, dep:and,{当isSupportAIRuleConfig,eq,true}, desc:this
node is valid when the value of isSupportAIRuleConfig is "true"*/
    "isSupportAIRuleCount": true,
    /*ro, opt, bool, whether the device supports manually counting AI rule for line crossing*/
    "isSupportAIRuleCapabilities": true,
    /*ro, opt, bool, whether the device supports getting AI rule capability, desc:if supported, get supported parameters (not including the transmitted
algorithm parameters) via /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<ID>/channels/<ID>/capabilities?format=json*/
    "isSupportContrast": true,
    /*ro, opt, bool, whether the device supports importing model, desc:whether it supports importing the comparison model package of target detection*/
    "isSupportAIRuleCountManualReset": true,
    /*ro, opt, bool, whether the device supports manually resetting the result of counting targets crossing the line, desc:whether it supports manually
resetting the result of counting targets crossing the line,related URI: /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/channels/<channelID>/manualReset?
format=json*/
    "isSupportAITargetLibrary": true,
    /*ro, opt, bool, whether the device supports AI target comparison library, desc:/ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json*/
    "isSupportAITargetBP": true,
    /*ro, opt, bool, whether the device supports AI target comparison library, desc:the default value is used when the capability is not returned*/
    "isSupportAIRuleTrigger": true,
    /*ro, opt, bool, whether the device supports AI rule linkage capability, desc:/ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/channels/<
ID>//manualReset?format=json*/
    "isSupportAIRuleSchedules": true,
    /*ro, opt, bool*/
    "isSupportAIRuleDefaultParams": true,
    /*ro, opt, bool, whether it supports getting the default parameters of special AI intelligent rule,
desc:/ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/defaultParams?format=json*/
    "isSupportDefaultParamsTaskEnabled": true,
    /*ro, opt, bool, whether it supports enabling default rule task, desc:Corresponding URI: defaultParamsTaskEnabled in
/ISAPI/Intelligent/AIOpenPlatform/algorithmModel/loading/<engineID>?format=json*/
}
```

19.15.1.8 Get the AIOP computing power supported by device

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "algorithmVersion": "AV1.5.0",
    /*ro, req, string, HCNN algorithm version, desc:its Data Dictionary is provided by research institute for matching the algorithm model*/
    "firmware": "FW006",
    /*ro, req, string, firmware version, desc:its Data Dictionary is provided by research institute for matching the algorithm model*/
    "engine": [1, 2, 3, 4],
    /*ro, req, array, engine ID, subType:int, desc:device computing power*/
    "platform": 1,
    /*ro, req, enum, the platform where the algorithm runs, subType:int, desc:the field is used for evaluating the adaptation of algorithm model*/
    "platformName": "K81",
    /*ro, req, string, the platform where the algorithm runs, desc:the platform where the algorithm runs*/
    "loadMPMaxNum": 4,
    /*ro, opt, int, the number of supported algorithm package for a single calculating engine, desc:this field is returned when multiple models uploading is
supported by calculating engine*/
    "maxRelChannelNum": 64
    /*ro, opt, int, the maximum number of channel that can be linked to the device, desc:this field is returned when multiple models uploading is supported
by calculating engine*/
}
```

19.15.2 AIOP Analysis Task Rule Configuration

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19.15.2.1 Get the capability of AI rule linkage configuration

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/channels/<channelID>/capabilities?format=json&channelType=<channelType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.
channelType	enum	When channelType is cameraIndexCode, the value of cameraIndexCode in is in string type; when channelType does not exist, the value in is video channel No. of integer type.

Request Message

None

Response Message

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```
{
    "AIRuleConfigCap": {
    /*ro, req, object, the capability of configuring AI rules*/
        "LineCrossingTargetStatisticsMode": {
        /*ro, opt, object, counting mode for line crossing target counting, desc:line crossing target counting mode; if this node is returned,it indicates
the mode is supported; otherwise not supported*/
            "mode": {
            /*ro, req, object, line crossing target counting mode*/
                "@opt": ["ScheduledCounting", "BatchCounting", "ManualCounting", "fixedQuantity"]
                /*ro, opt, array, "ScheduledCounting", "BatchCounting", "ManualCounting", subType:string, desc:"ScheduledCounting", "BatchCounting",
"ManualCounting"*/
            },
            "BatchCounting": {
            /*ro, req, object, parameters of counting in batches*/
                "timeInterval": {
                /*ro, req, object, time interval of batch counting*/
                    "@min": 1,
                    /*ro, opt, int, the minimum time interval of batch counting, unit:min*/
                    "@max": 60,
                    /*ro, opt, int, the maximum time interval of batch counting, unit:min*/
                    "@def": 5
                    /*ro, opt, int, default time interval of batch counting, unit:min*/
                }
            },
            "ScheduledCounting": {
            /*ro, req, object, schedule counting parameters*/
                "@size": 4
                /*ro, req, int, the maximum number of supported time ranges for scheduled counting*/
            },
        },
        "RuleNameInfo": {
        /*ro, opt, object, rule name configuration parameters*/
            "ruleName": {
            /*ro, req, object*/
                "@min": 0,
                /*ro, req, int, the minimum length*/
                "@max": 32
                /*ro, req, int, the maximum length*/
            },
            "ruleNameSpecCap": [
            /*ro, opt, array, subType:object*/
                {
                    "modelId": "000120220624010081ajidetect00000",
                    /*ro, opt, string, range:[1,64]*/
                    "ruleNameSpec": ["暴露垃圾", "打包垃圾", "垃圾桶满", "建筑垃圾", "烟火检测"]
                    /*ro, req, array, subType:string, range:[0,32]*/
                }
            ],
        },
        "isSupportManualTargetComparison": true,
        /*ro, opt, bool, this node is not returned if it is not supported, desc:this node is not returned if it is not supported*/
        "ruleID": {
        /*ro, opt, object*/
            "@min": 1,
            /*ro, req, int, range:[1,16]*/
            "@max": 16
            /*ro, req, int, range:[1,16]*/
        },
        "alarmRuleConfig": {
        /*ro, opt, object*/
            "ruleID": {
            /*ro, req, object*/
                "@min": 1,
                /*ro, req, int, unit:min*/
                "@max": 16
                /*ro, req, int, unit:min*/
            },
            "alarmPictureUploadEnabled": {
            /*ro, opt, object*/
            "@opt": [true, false],
            /*ro, req, array, subType:bool*/
            "@def": true
                /*ro, opt, bool*/
            }
        },
    }
}
```

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19.15.2.2 Get the capability of AI linkage rule configuration

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/channels/<channelID>/trigger/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.

Request Message

None

Response Message

```
{
    "AIRuleTriggerListCap": {
    /*ro, opt, object*/
        "size":  1024,
        /*ro, req, int, the maximum number of rule linkages supported by each channel*/
        "ruleType": {
        /*ro, opt, object*/
            "@opt": ["single", "multiple"]
            /*ro, opt, array, notification method, subType:string, desc:notification method: IO,syslog (system log linkage),beep,record,center (uploading to
center),audio,AIAudio (AI custom autio file)*/
        },
        "notificationMethod": {
        /*ro, req, object*/
            "@opt": ["email", "IM", "IO", "syslog", "HTTP", "FTP", "beep", "ptz", "record", "monitorAlarm", "center", "monitorAlarm", "LightAudioAlarm",
"focus", "trace", "whiteLight", "audio", "AIAudio", "modbus"]
            /*ro, opt, array, notification method, subType:string, desc:"IO", "syslog" (system log linkage), "beep", "record", "center" (uploading to
center), "audio", "AIAudio" (AI custom autio file)*/
        },
        "AudioAction": {
        /*ro, opt, object*/
            "childDevID": {
            /*ro, opt, object*/
                "@min":  9,
                /*ro, opt, int*/
                "@max":  32,
                /*ro, opt, int*/
                "@def":  "1"
                /*ro, opt, string*/
            },
            "audioContent": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, opt, int*/
                "@max":  256,
                /*ro, opt, int*/
                "@def":  "test"
                /*ro, opt, string*/
            },
            "isSupportAudioID":  true,
            /*ro, opt, bool*/
            "defAudioID":  1
            /*ro, opt, int*/
        },
        "AIAudioID": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int*/
            "@max":  100
            /*ro, req, int*/
        }
    }
}
```

19.15.2.3 Get AI rule linkage parameter

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>/trigger?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--
channelID	string	--

Request Message

None

Response Message

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```
{
    "AIRuleTriggerCfg": [
    /*ro, req, array, AI rule linkage configuration list, subType:object*/
        {
            "ruleID": 1,
            /*ro, opt, int, rule ID*/
            "ruleType": "single",
            /*ro, opt, enum, rule type, subType:string, desc:default value: single; “single” (single rule), "multiple" (multiple rules)*/
            "notificationMethod": "AIAudio",
            /*ro, req, enum, linkage method, subType:string, desc:"IO", "syslog" (system log linkage), "beep", "record", "center" (uploading to center),
"audio", "AIAudio" (AI custom autio file)*/
            "outputIOPortID": "1",
            /*ro, opt, string, linkage alarm output*/
            "dynOutputIOPortID": "1",
            /*ro, opt, string, dynamic linkage alarm output*/
            "videoInputID": "1",
            /*ro, opt, string, linked analog channel No., desc:which indicates the specified for operations*/
            "dynVideoInputID": "1",
            /*ro, opt, string, digital channel No., desc:which indicates the specified for operations*/
            "AIAudioID": 1
            /*ro, opt, int, linked audio file ID, dep:and,{$.AIRuleTriggerCfg[*].notificationMethod,eq,AIAudio}*/
        }
    ]
}
```

19.15.2.4 Set AI rule linkage configuration parameters

Request URL

PUT /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>/trigger?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--
channelID	string	--

Request Message

```
{
    "AIRuleTriggerCfg": [
    /*req, array, AI rule linkage configuration list, subType:object*/
        {
            "ruleID": 1,
            /*opt, int, rule ID*/
            "ruleType": "single",
            /*opt, enum, rule type, subType:string, desc:default value: single; single "single rule", multiple "multiple rules"*/
            "notificationMethod": "AIAudio",
            /*req, enum, notification method, subType:string, desc:"IO", "syslog" (system log linkage), "beep", "record", "center" (uploading to center),
"audio", "AIAudio" (AI custom autio file), "email", "FTP", "whiteLight"*/
            "outputIOPortID": "1",
            /*opt, string, alarm output IO*/
            "dynOutputIOPortID": "1",
            /*opt, string, linked digital channel ID*/
            "videoInputID": "1",
            /*opt, string, linked analog channel ID, desc:linked analog channel ID*/
            "dynVideoInputID": "1",
            /*opt, string, digital channel No., desc:which indicates the specified for operations*/
            "AIAudioID": 1
            /*opt, int, audio file name, dep:and,{$.AIRuleTriggerCfg[*].notificationMethod,eq,AIAudio}*/
        }
    ]
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

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19.15.3 AIOP Capture Auto-Switch Task Management

19.15.3.1 Get capability of getting capture polling task information

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/picturePollingAnalysisTask/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "tasksNum":  1,
    /*ro, req, int, number of concurrent tasks supported by the device*/
    "taskIDLen": {
    /*ro, req, object, the length of the task ID*/
        "@min":  0,
        /*ro, req, int, the minimum value, unit:Byte*/
        "@max":  64
        /*ro, req, int, the maximum value, unit:Byte*/
    },
    "taskName": {
    /*ro, opt, object, task name*/
        "@min":  0,
        /*ro, opt, int, minimum string length*/
        "@max":  64
        /*ro, opt, int, maximum string length*/
    },
    "taskEnabled": {
    /*ro, opt, object*/
        "@opt": [true, false],
        /*ro, opt, array, subType:bool*/
        "@def":  true
        /*ro, opt, bool*/
    },
    "engine": {
    /*ro, req, object, engine No.*/
        "@opt": [1, 2, 3, 4]
        /*ro, req, array, engine No. array, subType:int*/
    },
    "channel": {
    /*ro, opt, object, channel No., desc:device channel No. which is the video analysis channel No. of the device*/
        "@opt": [1, 2]
        /*ro, req, array, channel No. array, subType:int*/
    },
    "analysisType": {
    /*ro, opt, object, Analysis type., desc:"local" (local analysis), "camera" (remote analysis).The "local" refers to the capture polling task. After
streaming from the linked channel, the capture analysis will be performed locally. The "camera" refers to the camera analysis task, and the alarm pictures
from intelligent cameras will be received locally for secondary analysis. If this filed does not exist, "local" is the default mode and valid when the
streaming method is channel (channel field should exist). Application: If the camera is intelligent camera, camera analysis mode can save back-end engine
analysis computing power and increase analysis channels or other intelligent services.*/
        "@opt": ["local", "camera"],
        /*ro, req, array, the optional value, subType:string*/
        "@def":  "local"
        /*ro, req, string, the default value*/
    },
    "cameraAnalysisType": {
    /*ro, opt, object, event type supported to be analyzed by camera, dep:and,{$.analysisType,eq,camera}, desc:"safetyHelmetDetection"*/
        "@opt": ["safetyHelmetDetection"]
        /*ro, req, array, the optional value, subType:string*/
    },
    "Stream": {
    /*ro, opt, object, real-time stream information, desc:either Stream or channel is required*/
        "streamType": {
        /*ro, req, object, stream type, desc:"realtime" (streaming remotely and in real time), "historyvideo" (streaming remotely from the recorded data),
the default value is "realtime"*/
            "@opt": ["realTime", "historyVideo"]
            /*ro, req, array, the optional value, subType:string*/
        },
        "streamSourceType": {
        /*ro, req, object, streaming mode, desc:"URL" (streaming via URL generated by the platform), "Camera" (streaming via streaming protocol of the
camera); in the "Camera" mode, the channel streams via the URL of streaming RTSP based on the access protocol of the camera*/
            "@opt": ["URL", "Camera"]
            /*ro, opt, array, the optional value, subType:string*/
        },
        "StreamSourceList": [
        /*ro, opt, array, streaming information list, subType:object*/
            {
                "cameraIndexCode": {
                /*ro, req, object, custom device local channel No. (the actual device does not access the analysis device), desc:custom device local channel
No. (the actual device does not access the analysis device)*/
                    "@min":  0,
                    /*ro, req, int, the minimum value*/
```

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```
                    /*ro, req, int, the minimum value*/
                    "@max":  64
                    /*ro, req, int, the maximum value*/
                },
                "AITaskType": {
                /*ro, opt, object, AI picture comparison task type, desc:"AITargetBPTask" (AI reference comparison task). If it is not configured, it is
normal task by default*/
                    "@opt": ["AITargetBPTask"]
                    /*ro, req, array, the optional value, subType:string*/
                },
                "isSupportTargetAttrs":  true,
                /*ro, opt, bool, whether the device supports getting the transmitted parameters, desc:it is used for uploading events*/
                "URL": {
                /*ro, opt, object, platform streaming information, dep:and,{$.Stream.streamSourceType,eq,URL}, desc:it is valid when the value of
streamSourceType is URL*/
                    "URLMode": {
                    /*ro, opt, object, streaming URL type, desc:"RTSP" (default value), "DAC"*/
                        "@opt": ["DAC", "RTSP"]
                        /*ro, opt, array, the optional value, subType:string*/
                    },
                    "URL": {
                    /*ro, req, object, streaming via RTSP URL*/
                        "@min":  0,
                        /*ro, req, int, the minimum value*/
                        "@max":  1024
                        /*ro, req, int, the maximum value*/
                    },
                    "userName": {
                    /*ro, opt, object, user name*/
                        "@min":  0,
                        /*ro, req, int, the minimum value*/
                        "@max":  128
                        /*ro, req, int, the maximum value*/
                    },
                    "password": {
                    /*ro, opt, object, password*/
                        "@min":  0,
                        /*ro, req, int, the minimum value*/
                        "@max":  128
                        /*ro, req, int, the maximum value*/
                    }
                },
                "Camera": {
                /*ro, opt, object, information on streaming via local direct connection, dep:and,{$.Stream.streamSourceType,eq,Camera}, desc:it is valid
when the value of streamSourceType is camera*/
                    "protocol": {
                    /*ro, opt, object, the protocol that is used for devices to communicate with the platform which assigns tasks, desc:"Private" (private
protocol), "Onvif" (ONVIF), get streaming URL of RTSP according to the protocol type*/
                        "@opt": ["Private", "Onvif", "ISUP"],
                        /*ro, req, array, the optional value, subType:string*/
                        "@def":  "Private"
                        /*ro, req, string, the default value*/
                    },
                    "StreamingViaIP": {
                    /*ro, opt, object, streaming via IP, desc:it is valid when the value of proxyProtocol is Private or ONVIF*/
                        "ipV4Address": {
                        /*ro, opt, object, IPv4 address*/
                            "@min":  0,
                            /*ro, req, int, minimum string length*/
                            "@max":  32
                            /*ro, req, int, maximum string length*/
                        },
                        "port": {
                        /*ro, req, object, port No.*/
                            "@min":  0,
                            /*ro, req, int, the minimum value*/
                            "@max":  65535
                            /*ro, req, int, the maximum value*/
                        },
                        "userName": {
                        /*ro, req, object, user name*/
                            "@min":  0,
                            /*ro, req, int, the minimum value*/
                            "@max":  128
                            /*ro, req, int, the maximum value*/
                        },
                        "password": {
                        /*ro, req, object, password*/
                            "@min":  0,
                            /*ro, req, int, the minimum value*/
                            "@max":  128
                            /*ro, req, int, the maximum value*/
                        },
                        "channel": {
                        /*ro, opt, object, device channel No., desc:device channel No. which is the video analysis channel No. of the device*/
                            "@min":  0,
                            /*ro, req, int, the minimum value*/
                            "@max":  256
                            /*ro, req, int, the maximum value*/
                        }
                    },
                    "StreamingViaDeviceID": {
                    /*ro, opt, object, information on streaming via local direct connection, desc:it is valid when the value of proxyProtocol is ISUP*/
                        "deviceID": {
```

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```
                        /*ro, req, object, device ID, desc:deviceID; get deviceID by the URL: /ISAPI/System/Network/Ehome*/
                            "@min":  0,
                            /*ro, req, int, minimum string length*/
                            "@max":  256
                            /*ro, req, int, maximum string length*/
                        },
                        "channel": {
                        /*ro, opt, object, device channel No., desc:device channel No. which is the video analysis channel No. of the device*/
                            "@min":  0,
                            /*ro, req, int, the minimum value*/
                            "@max":  256
                            /*ro, req, int, the maximum value*/
                        }
                    }
                },
                "RecordTime": {
                /*ro, opt, object, streaming time of history streams, dep:and,{$.Stream.streamType,eq,historyVideo}, desc:it is valid when the value of
streamType is historyVideo*/
                    "startTime":  "1970-01-01T00:00:00+08:00",
                    /*ro, req, datetime, streaming start time, desc:in ISO 8601 format*/
                    "endTime":  "1970-01-01T00:00:00+08:00"
                    /*ro, req, datetime, streaming end time, desc:in ISO 8601 format*/
                }
            }
        ]
    },
    "pollingTime": {
    /*ro, req, object, polling time range*/
        "@min":  1,
        /*ro, req, int, minimum video polling duration, step:1, unit:s*/
        "@max":  86400,
        /*ro, req, int, maximum video polling duration, step:1, unit:s*/
        "@def":  1
        /*ro, req, int, the default value, step:1, unit:s*/
    },
    "status": {
    /*ro, opt, object, status information, desc:0 (unexecuted), 1 (executing), 2 (abnormal), 3 (queueing)*/
        "@opt": [0, 1, 2, 3]
        /*ro, opt, array, enumeration value, subType:int*/
    },
    "isSupportTimeRange":  true,
    /*ro, opt, bool, whether the daily arming schedules of tasks (by week) is supported, desc:by default there are 8 time periods per day*/
    "alarmTargetOverlay": {
    /*ro, opt, object, whether to enable overlaying alarm target frame*/
        "@opt": ["true", "false"]
        /*ro, opt, array, whether to enable this function, subType:string*/
    },
    "alarmRuleOverlay": {
    /*ro, opt, object, whether to enable overlaying alarm rule information*/
        "@opt": [true, false]
        /*ro, opt, array, whether to enable this function, subType:bool*/
    },
    "alarmRuleOverlayCap": {
    /*ro, opt, object, whether to enable overlaying items of alarm rule, desc:it is used to overlay rule ID, rule name, etc.*/
        "@opt": ["ruleID", "ruleName"]
        /*ro, opt, array, the optional value, subType:string*/
    },
    "EventTrigger": {
    /*ro, opt, object, task linkage configuration capability, desc:if this node is not returned, it indicates that linkage configurations are not
supported*/
        "notificationMethod": {
        /*ro, opt, object, linkage action*/
            "@opt": ["email", "IM", "IO", "syslog", "HTTP", "FTP", "beep", "ptz", "record", "monitorAlarm", "center", "LightAudioAlarm", "focus", "trace",
"cloud", "SMS", "whiteLight"],
            /*ro, req, array, linkage action supported by the device, subType:string*/
            "@def":  "center"
            /*ro, req, string, default linkage action*/
        }
    },
    "maxRelChannelNum":  2,
    /*ro, opt, int, the maximum number of linked channels supported by one real-time video task*/
    "Destination": [
    /*ro, opt, array, receiving destination information, subType:object, desc:it is used for uploading events*/
        {
            "destinationType": {
            /*ro, opt, object, receiving destination type, desc:for details, refer to the Dictionaries*/
                "@opt": ["AlarmRMQ", "ServerClient", "ServerHCS", "ActiveMQ"]
                /*ro, opt, array, the optional value, subType:string*/
            },
            "destinationURL": {
            /*ro, req, object, URL for receiving results, desc:related to destinationType*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  1024
                /*ro, req, int, the maximum value*/
            },
            "userName": {
            /*ro, opt, object, user name of the recipient*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  128
                /*ro, req, int, the maximum value*/
            },
            "password": {
```

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```
            "password": {
            /*ro, opt, object, password of the recipient*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  128
                /*ro, req, int, the maximum value*/
            },
            "SSLEnabled": {
            /*ro, opt, object, whether to enable SSL encryption, desc:true (enable), false (disable, default value)*/
                "@opt": [true, false],
                /*ro, req, array, the optional value, subType:bool*/
                "@def":  false
                /*ro, opt, bool, the default value*/
            }
        }
    ],
    "alarmDataUploadType": {
    /*ro, opt, object, alarm-data upload type*/
        "@opt": ["all", "target"],
        /*ro, opt, array, range, subType:string, desc:"all" (report all regardless of whether the target is recognized or not), "target" (report only when
the target supported by current algorithm model is recognized)*/
        "@def":  "all"
        /*ro, opt, string, the default value*/
    },
    "savePollingPicEnabled": {
    /*ro, opt, object, whether to save original picture of capture polling task*/
        "@opt": ["true", "false"],
        /*ro, opt, array, range, subType:string, desc:true (yes), false (no)*/
        "@def":  "true"
        /*ro, opt, string, the default value*/
    },
    "synchronizationWithSleepCaptureInterval":  true,
    /*ro, opt, bool, whether it supports synchronizing polling interval with sleeping capture interval, desc:returns false by default. The AI image analysis
polling interval follows the image polling task configuration when it is not in sleep mode, and follows the image capture interval in sleeping
(/ISAPI/System/consumptionMode?format=json) when in sleep mode.*/
    "AIOP_Polling_SnapEnabledType": {
    /*ro, opt, object, AIOP algorithm event type, desc:"safetyHelmetDetection", "takingElevatorDetection", "fireEscapeDetection", "playCellphone",
"peopleNumChange", "leavePosition", "retention"*/
        "@opt": ["safetyHelmetDetection", "    takingElevatorDetection", "fireEscapeDetection", "playCellphone", "peopleNumChange", "leavePosition",
"retention"]
        /*ro, opt, array, whether to enable this function, subType:string*/
    },
    "heightThreshold": {
    /*ro, opt, object, height threshold, desc:when deploying a triple-lens camera on-site, it is necessary to configure a threshold of the height difference
between power transmission lines and passing vehicles when configuring tasks. An alarm will be triggered when the height exceeds this threshold*/
        "@min":  1,
        /*ro, opt, int, the minimum value, unit:m*/
        "@max":  255,
        /*ro, opt, int, the maximum value, unit:m*/
        "@def":  10
        /*ro, opt, int, the default value, unit:m*/
    },
    "taskExecutionMode": {
    /*ro, opt, object*/
        "@opt": ["timing", "trigger"],
        /*ro, req, array, subType:string*/
        "@def":  "timing"
        /*ro, req, string*/
    },
    "isSupportTaskTrigger":  true,
    /*ro, opt, bool*/
    "triggerSource": {
    /*ro, opt, object, dep:and,{$.triggerSource,eq,trigger}*/
        "@maxsize":  64,
        /*ro, opt, int*/
        "channelID": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, opt, int*/
            "@max":  64
            /*ro, opt, int*/
        },
        "inputIOPortIDList": {
        /*ro, opt, object*/
            "@size":  1,
            /*ro, opt, int*/
            "inputIOPortID": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, opt, int*/
                "@max":  2
                /*ro, opt, int*/
            }
        }
    },
    "debugEnabled": {
    /*ro, opt, object*/
        "@opt": [true, false],
        /*ro, opt, array, subType:bool*/
        "@def":  false
        /*ro, opt, bool*/
    },
    "backgroundPicQualtity": {
    /*ro, opt, object*/
```

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```
        "@opt":  "high,middle,low",
        /*ro, req, string*/
        "@def":  "middle"
        /*ro, opt, string*/
    }
}
```

19.15.3.2 Get captured picture polling analysis task

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/picturePollingAnalysisTask?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "Tasks": [
    /*ro, opt, array, list of task information, subType:object*/
        {
            "taskID":  "example",
            /*ro, req, string, task ID, range:[0,64], desc:task ID,the maximum size is 64 bytes*/
            "taskName":  "test",
            /*ro, opt, string, task name, range:[0,64]*/
            "taskCreateTime":  "1970-01-01T00:00:00+08:00",
            /*ro, opt, datetime*/
            "taskEnabled":  true,
            /*ro, opt, bool*/
            "engine":  1,
            /*ro, opt, int, engine ID, desc:it is used to distinguish models running in different engines*/
            "analysisUnitID":  "test",
            /*ro, opt, string*/
            "MPID":  "example",
            /*ro, opt, string, range:[0,64]*/
            "channel": [1, 2],
            /*ro, opt, array, device's local channel No., subType:int, desc:device channel No. which is the video analysis channel No. of the device*/
            "analysisType":  "local",
            /*ro, opt, enum, Analysis type., subType:string, desc:"local" (local analysis), "camera" (remote analysis).The "local" refers to the capture
polling task. After streaming from the linked channel, the capture analysis will be performed locally by deepinMind devices. The "camera" refers to the
analysis task of camera, and the alarm pictures of intelligent camera will be received by deepinMind devices for secondary analysis. If this filed does not
exist, "local" is the default mode and valid when the streaming method is channel (channel field should exist). Application: If the camera is intelligent
camera, camera analysis mode can save back-end engine analysis computing power and increase analysis channels or other intelligent services.*/
            "cameraAnalysisType":  "safetyHelmetDetection",
            /*ro, opt, enum, event type supported to be analyzed by camera, subType:string, dep:and,{$.analysisType,eq,camera}, desc:"safetyHelmetDetection"
(hard hat detection event)*/
            "Stream": {
            /*ro, opt, object, stream information, desc:stream information*/
                "streamType":  "realTime",
                /*ro, req, enum, stream type, subType:string, desc:stream type*/
                "streamSourceType":  "URL",
                /*ro, req, enum, streaming mode, subType:string, desc:streaming mode*/
                "StreamSourceList": [
                /*ro, opt, array, streaming information list, subType:object*/
                    {
                        "cameraIndexCode":  "UUIDTest",
                        /*ro, req, string, camera No., range:[0,64], desc:camera No.*/
                        "AITaskType":  "AITargetBPTask",
                        /*ro, opt, enum, AI task analysis type, subType:string, desc:"AITargetBPTask"(AI comparison task based on the reference picture); if
this node does not exist, by default the task is a normal task; if the value of AITaskType is "AIictTargetBPTask", the cameraIndexCode indicates the
reference picture set*/
                        "TargetAttrs": {},
                        /*ro, opt, object, transmitted parameters, desc:transmitted parameters*/
                        "URL": {
                        /*ro, opt, object, streaming via RTSP URL, dep:and,{$.Stream.streamSourceType,eq,URL}, desc:it is valid when the value of
streamSourceType is URL*/
                            "URL":  "dac://rtsp://10.5.160.103:6304/EUrl/qFD77n9",
                            /*ro, req, string, streaming URL, range:[0,1024], desc:streaming URL*/
                            "userName":  "admin"
                            /*ro, opt, string, streaming user name, range:[0,128], desc:the sensitive information should be encrypted*/
                        },
                        "Camera": {
                        /*ro, opt, object, information on streaming via local direct connection, dep:and,{$.Stream.streamSourceType,eq,Camera}, desc:it is
valid when the value of streamSourceType is camera*/
                            "protocol":  "Private",
                            /*ro, opt, string, the protocol that is used for devices to communicate with the platform which assigns tasks, desc:"Private"
(private protocol), "Onvif" (ONVIF), "ISUP"; get streaming URL of RTSP according to the protocol type*/
                            "StreamingViaIP": {
                            /*ro, opt, object, streaming via IP, dep:or,{$.Stream.StreamSourceList[*].Camera.protocol,eq,Onvif},
{$.Stream.StreamSourceList[*].Camera.protocol,eq,Private}, desc:it is valid when the value of protocol is Private or ONVIF*/
                                "ipV4Address":  "10.21.42.34",
                                /*ro, opt, string, IPv4 address, range:[0,32]*/
                                "port":  8080
```

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```
                                "port": 8080,
                                /*ro, opt, int, port No., range:[0,65535]*/
                                "userName": "admin",
                                /*ro, opt, string, streaming user name, range:[0,128], desc:the sensitive information should be encrypted*/
                                "channel": 1
                                /*ro, opt, int, device channel No. which is the video analysis channel No. of the device, range:[0,256], desc:device channel No. which is the video analysis channel No. of the device*/
                            },
                            "StreamingViaDeviceID": {
                            /*ro, opt, object, streaming via device ID, dep:and,{$.Stream.StreamSourceList[*].Camera.protocol,eq,ISUP}, desc:it is valid when the value of protocol is ISUP*/
                                "deviceID": "test",
                                /*ro, opt, string, device ID, range:[0,256], desc:device ID*/
                                "channel": 1
                                /*ro, opt, int, device channel No. which is the video analysis channel No. of the device, range:[0,256]*/
                            }
                        },
                        "RecordTime": {
                        /*ro, opt, object, streaming time of history streams, dep:and,{$.Stream.streamType,eq,historyVideo}, desc:it is valid when the value of streamType is historyVideo*/
                            "startTime": "1970-01-01T00:00:00+08:00",
                            /*ro, req, datetime, streaming start time, desc:in ISO 8601 format*/
                            "endTime": "1970-01-01T00:00:00+08:00"
                            /*ro, req, datetime, streaming end time, desc:in ISO 8601 format*/
                        }
                    }
                ]
            },
            "pollingTime": 60,
            /*ro, req, int, polling time, range:[1,65535], step:1, unit:s*/
            "status": 1,
            /*ro, req, int, status information*/
            "statusList": [
            /*ro, opt, array, subType:object*/
                {
                    "channelID": 1,
                    /*ro, opt, int*/
                    "cameraIndexCode": "UUIDTest",
                    /*ro, opt, string, range:[0,64]*/
                    "status": 1,
                    /*ro, req, enum, subType:int*/
                    "subStatusCode": "channelOffline",
                    /*ro, opt, string, dep:and,{$.Tasks[*].statusList[*].status,eq,2}*/
                    "errorCode": 1073746358
                    /*ro, opt, int*/
                }
            ],
            "TimeRange": [
            /*ro, opt, array, arming the task by day each week, subType:object, desc:daily arming schedules of tasks (by week), 8 time periods per day by default*/
                {
                    "week": 1,
                    /*ro, req, int, day of the week: 1 (Monday),2 (Tuesday),...*/
                    "beginTime": "00:00",
                    /*ro, req, string, start time*/
                    "endTime": "24:00"
                    /*ro, req, string, end time*/
                }
            ],
            "alarmTargetOverlay": true,
            /*ro, opt, bool, whether to enable overlaying alarm target frame*/
            "alarmRuleOverlay": true,
            /*ro, opt, bool, whether to enable overlaying alarm rule information, desc:if it is enabled, overlaying the alarm rule ID and rule name on the capture alarm pictures is supported*/
            "EventTrigger": [
            /*ro, opt, array, linkage action settings, subType:object, desc:linkage action of tasks; if this node is not configured, the configured linkage action will not be edited; if this node is null, it indicates that the linkage action will be deleted*/
                {
                    "notificationMethod": "center",
                    /*ro, req, string, linkage action*/
                    "outputIOPortID": "1",
                    /*ro, opt, string, linked alarm output port ID*/
                    "videoInputID": "1",
                    /*ro, opt, string, analog channel No., desc:digital channel No.*/
                    "dynOutputIOPortID": "1"
                    /*ro, opt, string, digital channel No., desc:digital channel No.*/
                }
            ],
            "Destination": [
            /*ro, opt, array, receiving destination information, subType:object, desc:it is used for uploading events*/
                {
                    "destinationType": "AlarmRMQ",
                    /*ro, req, enum, receiving destination type, subType:string, desc:"AlarmRMQ" (RabbitMQ), "ServerClient" (uploading to the third-party by ISAPI), "ServerHCS" (micro video cloud), "ActiveMQ"*/
                    "destinationURL": "test",
                    /*ro, opt, string, URL for receiving results, range:[0,1024], desc:URL for receiving results*/
                    "userName": "test",
                    /*ro, opt, string, user name of the recipient, range:[0,128], desc:the sensitive information should be encrypted*/
                    "SSLEnabled": true
                    /*ro, opt, bool, whether to enable SSL encryption, desc:true (enable), false (disable, default value)*/
                }
            ],
            "alarmDataUploadType": "all",
            /*ro, opt, enum, alarm-data upload type, subType:string, desc:"all" (report all regardless of whether the target is recognized or not), "target"
```

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```
(report only when the target supported by current algorithm model is recognized)*/
            "savePollingPicEnabled":  true,
            /*ro, opt, bool, whether to save original picture of capture polling task, desc:After the reported alarm is analyzed by engine, the original
captured picture turns to the processed picture. If this filed is not configured, original pictures will not be saved*/
            "AIOP_Polling_SnapEnabled":  true,
            /*ro, opt, bool, report AI capture polling task detection event,
            desc:if the node is not configured, the value is true by default. If the value is false, it means events are reported by AI algorithms. The
supported events can be found via /ISAPI/Intelligent/AIOpenPlatform/picturePollingAnalysisTask/capabilities?format=json.
            Application example: for deepinMind devices, support detecting hard hat events using the algorithm model, applying video analysis tasks using AI
rules, and uploading eventType:safetyHelmetDetection*/
            "heightThreshold":  1,
            /*ro, opt, int, height threshold, range:[1,255], unit:m, desc:when deploying a triple-lens camera on-site, it is necessary to configure a
threshold of the height difference between power transmission lines and passing vehicles when configuring tasks. An alarm will be triggered when the height
exceeds this threshold*/
            "taskExecutionMode":  "timing",
            /*ro, opt, enum, subType:string*/
            "triggerSource": [
            /*ro, opt, array, subType:object, dep:and,{$.taskExecutionMode,eq,trigger}*/
                {
                    "channelID":  1,
                    /*ro, opt, int*/
                    "inputIOPortIDList": [1, 2]
                    /*ro, req, array, subType:int*/
                }
            ],
            "debugEnabled":  true,
            /*ro, opt, bool*/
            "backgroundPicQualtity":  "middle",
            /*ro, opt, enum, subType:string*/
            "smartStreamList": [
            /*ro, opt, array, subType:object*/
                {
                    "channel":  1,
                    /*ro, opt, int, range:[0,256]*/
                    "cameraIndexCode":  "UUIDTest",
                    /*ro, opt, string, range:[0,64]*/
                    "streamID":  "test"
                    /*ro, req, string, range:[1,32]*/
                }
            ]
        }
    ]
}
```

19.15.4 AIOP Video Analysis Task Management

19.15.4.1 Get the configuration capability of the history video analysis tasks

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/historyVideoTask/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "tasksNum":  1,
    /*ro, req, int, number of concurrent tasks supported by the device*/
    "taskIDLen": {
    /*ro, req, object, the length of the task ID*/
        "@min":  0,
        /*ro, req, int, unit:Byte*/
        "@max":  64
        /*ro, req, int, unit:Byte*/
    },
    "engine": {
    /*ro, req, object, supported engine ID*/
        "@opt": [1, 2, 3, 4]
        /*ro, req, array, subType:int*/
    },
    "channel": {
    /*ro, req, object, device channel No.*/
        "@opt": [1, 2]
        /*ro, req, array, subType:int*/
    },
    "frameRate": {
    /*ro, req, object, detection frame rate*/
        "@opt": [2500, 3000]
        /*ro, req, array, subType:int*/
    },
    "alertInterval": {
    /*ro, opt, object, alarm interval, desc:when it is 1, it means once each second; when it is 0.5, it means twice each second*/
        "@opt": [1, 0.5],
        /*ro, opt, array, subType:float*/
        "@def":  1
        /*ro, opt, int*/
    },
    "status": {
    /*ro, req, object, status information*/
        "@opt": [0, 1, 2, 3]
        /*ro, opt, array, subType:int*/
    },
    "maxRelChannelNum":  2,
    /*ro, opt, int, the maximum number of linked channels supported by one history video task*/
    "drawFrame": {
    /*ro, opt, object, the ratio of extracted frames of the server sending the history stream*/
        "@min":  0,
        /*ro, req, int*/
        "@max":  100,
        /*ro, req, int*/
        "@def":  0
        /*ro, req, int*/
    }
}
```

19.15.4.2 Get the history video analysis task

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/historyVideoTask?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "Tasks": [
    /*ro, opt, array, list of task information, subType:object*/
        {
            "taskID": "test",
            /*ro, req, string, task ID, range:[1,64]*/
            "engine": 1,
            /*ro, req, int, engine ID, desc:it is used to distinguish models running in different engines*/
            "channel": [1, 2],
            /*ro, req, array, device channel No., subType:int*/
            "frameRate": 2500,
            /*ro, req, int, detection frame rate, desc:detection frame rate*/
            "alertInterval": 0.5,
            /*ro, opt, float, alarm interval, desc:1 (1 time per second), 0.5 (2 times per second)*/
            "pollingTime": 60,
            /*ro, req, int, polling time, range:[1,65535], step:1, unit:s*/
            "status": 1,
            /*ro, req, enum, status information, subType:int, desc:0 (unexecuted), 1 (executing), 2 (abnormal), 3 (queueing), 4 (task and model type
mismatch), 5 (initializing)*/
            "recordTime": {
            /*ro, req, object, recording time period, desc:it is the time range of recording (e.g. one month)*/
                "beginTime": "2020-02-17 10:00:00",
                /*ro, req, string, start time*/
                "endTime": "2020-03-18 10:00:00"
                /*ro, req, string, end time*/
            },
            "TimeRange": [
            /*ro, req, array, arming schedules, subType:object*/
                {
                    "week": 1,
                    /*ro, req, int, day of a week*/
                    "beginTime": "7:00:00",
                    /*ro, req, string, start time, desc:ISO 8601 time*/
                    "endTime": "9:00:00"
                    /*ro, req, string, end time, desc:ISO 8601 time*/
                }
            ],
            "drawFrame": 0
            /*ro, opt, int, ratio of the frames extracted by the server sending the history stream, range:[0,100], step:1, desc:0 indicates extracting no
frame*/
        }
    ]
}
```

19.15.4.3 Edit a video analysis task

Request URL

PUT /ISAPI/Intelligent/AIOpenPlatform/videoTask/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	task ID

Request Message

```
{
    "channel": [1, 2],
    /*opt, array, device's local channel No., subType:int, desc:device channel No.,specified device channel No. for video analysis task*/
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

19.15.4.4 Get the capability of getting video analysis task information

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Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/videoTask/capabilities?format=json&engine=<engineID>

Query Parameter

Parameter Name	Parameter Type	Description
engineID	string	--

Request Message

None

Response Message

```
{
    "tasksNum":  1,
    /*ro, req, int, concurrent tasks supported by the device*/
    "taskIDLen": {
    /*ro, req, object, the length of the task ID*/
        "@min":  0,
        /*ro, req, int, the minimum value*/
        "@max":  64
        /*ro, req, int, the maximum value*/
    },
    "engine": {
    /*ro, req, object, supported engine No.*/
        "@opt": [1, 2, 3, 4]
        /*ro, req, array, engine No. array, subType:int*/
    },
    "channel": {
    /*ro, opt, object, device channel No., desc:device channel No. (optional); either Stream or channel is required*/
        "@opt": [1, 2]
        /*ro, req, array, channel No. array, subType:int*/
    },
    "frameRate": {
    /*ro, opt, object, detection frame rate*/
        "@opt": [2500, 3000]
        /*ro, req, array, detection frame rate array, subType:int*/
    },
    "isSupportTimeRange":  true,
    /*ro, opt, bool, whether the daily arming schedules of tasks (by week) is supported, desc:by default there are 8 time periods per day*/
    "alarmTargetOverlay": {
    /*ro, opt, object, whether to enable alarm target frame overlay*/
        "@opt": ["true", "false"]
        /*ro, opt, array, whether to enable this function, subType:string*/
    },
    "alarmRuleOverlay": {
    /*ro, opt, object, whether to enable overlaying alarm rule information*/
        "@opt": [true, false]
        /*ro, opt, array, whether to enable this function, subType:bool*/
    },
    "alarmRuleOverlayCap": {
    /*ro, opt, object, whether to enable overlaying items of alarm rule, desc:it is used to overlay rule ID, rule name, etc.*/
        "@opt": ["ruleID", "ruleName"]
        /*ro, opt, array, options, subType:string*/
    },
    "EventTrigger": {
    /*ro, opt, object, linkage configuration capability, desc:if this node is not returned, it indicates that linkage configurations are not supported*/
        "notificationMethod": {
        /*ro, opt, object, linkage mode*/
            "@opt": ["email", "IM", "IO", "syslog", "HTTP", "FTP", "beep", "ptz", "record", "monitorAlarm", "center", "LightAudioAlarm", "focus", "trace",
"cloud", "SMS", "whiteLight", "door"],
            /*ro, req, array, linkage action supported by the device, subType:string*/
            "@def":  "center"
            /*ro, req, string, default linkage mode*/
        },
    },
    "debugEnabled": {
    /*ro, opt, object*/
        "@opt": [true, false],
        /*ro, opt, array, subType:bool*/
        "@def":  false
        /*ro, opt, bool*/
    },
}
```

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19.15.4.5 Search for the executing video analysis tasks

Request URL

GET /ISAPI/Intelligent/AIOpenPlatform/videoTask?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "Tasks": [
    /*ro, req, array, list of task information, subType:object*/
        {
            "taskID":  "example",
            /*ro, req, string, task ID, range:[0,64], desc:managed and maintained by service platform, transparent transmission by the device*/
            "taskName":  "test",
            /*ro, opt, string, task name, range:[0,64]*/
            "taskSource":  "8200",
            /*ro, opt, string*/
            "taskCreateTime":  "1970-01-01T00:00:00+08:00",
            /*ro, opt, datetime, task creation time*/
            "taskEnabled":  true,
            /*ro, opt, bool*/
            "engineMode":  "engineSharing",
            /*ro, opt, enum, engine mode, subType:string, desc:"engineSharing" (engine sharing mode, which indicates that the current tasks can be analyzed by multiple engines according to devices), "engineMonopolizing" (engine inclusive mode, which indicates that the current tasks can only be analyzed by specified engine); it is engine inclusive mode when this node is not configured*/
            "engine":  1,
            /*ro, opt, int, engine ID, desc:it is used to distinguish models running in different engines*/
            "analysisUnitID":  "test",
            /*ro, opt, string*/
            "taskAnalysisMode":  "queue",
            /*ro, opt, enum, task analysis mode, subType:string, desc:"queue" (queueing analysis), "concurrency" (concurrent analysis); queue analysis is used when the number of streams to be analyzed exceeds the analysis capacity of the engine, the streams are queued and wait in line. After analyzing the designated duration of the current stream analysis task, the queued streams will be continued to be analyzed. Concurrency analysis means that the streams are analyzed simultaneously. The number of streams being analyzed is less than or equal to the analysis capacity of the engine. If it exceeds the analysis capacity of the engine, the task will fail to be assigned*/
            "MPID":  "example",
            /*ro, opt, string, model package ID, range:[0,64], dep:and,{$.engineMode,eq,engineSharing}, desc:it is used to identify the model information; the engine bound with the same model ID can analyze tasks in the load balancing mode*/
            "channel": [1, 2],
            /*ro, opt, array, device channel No., subType:int, desc:device channel No. (optional); either Stream or channel is required*/
            "Stream": {
            /*ro, opt, object, stream information, desc:stream source of video analysis task, either Stream or channel is required*/
                "streamType":  "realTime",
                /*ro, req, enum, stream type, subType:string, desc:"realtime" (stream remotely and in real time, default value), "historyvideo" (stream remotely from the recorded data)*/
                "streamSourceType":  "URL",
                /*ro, req, enum, streaming mode, subType:string, desc:"URL" (specify RTSP URL to stream), "Camera" (specify a camera to stream); normally the device acquiring the stream and the device analyze the stream should be on the same LAN*/
                "StreamSourceList": [
                /*ro, opt, array, list of streaming information, subType:object, desc:if the value of taskAnalysisMode is "queue", up to 1 list is allowed*/
                    {
                        "cameraIndexCode":  "UUIDTest",
                        /*ro, req, string, camera No., range:[0,64], desc:it indicates the uniqueness of the picture data; it is used for settings the rule of channel: /ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>?format=json&channelType=<channelType>; /ISAPI/Intelligent/AIOpenPlatform/AITargetBP/channels/<videoID>?format=json&channelType=<channelType>*/
                        "AITaskType":  "AITargetBPTask",
                        /*ro, opt, enum, AI task analysis type, subType:string, desc:"AITargetBPTask"(AI comparison task based on the reference picture); if this node does not exist, by default the task is a normal task; if the value of AITaskType is "AIictTargetBPTask", the cameraIndexCode indicates the reference picture set*/
                        "TargetAttrs": {},
                        /*ro, opt, object, transmitted parameters, desc:it is used for uploading events*/
                        "URL": {
                        /*ro, opt, object, streaming via RTSP URL, dep:and,{$.Stream.streamSourceType,eq,URL}, desc:it is valid when the value of streamSourceType is URL it means that the platform can generate URL by itself*/
                            "URL":  "dac://rtsp://10.5.160.103:6304/EUrl/qFD77n9",
                            /*ro, req, string, streaming URL, range:[0,1024], desc:notes about the value of Stream.StreamSourceList.URL.URLMode.@opt in the URL /ISAPI/Intelligent/AIOpenPlatform/videoTask/capabilities?format=json; 1. if the value includes DAC, the streaming format: dac://+streaming URL+&dactoken=XXX, eg:dac://rtsp://10.5.160.103:6304/EUrl/qFD77n9; dac://rtsp://10.33.43.159:655/dac/realplay/30219e1e924648e68f3fab59b838bfec/MAIN/TCP; 2. if the value includes RTSP, the streaming format: device://rtsp://xxx, eg:device://rtsp://10.66.108.224:554/Streaming/Channels/101?transportmode=unicast*/
                            "userName":  "admin"
                            /*ro, opt, string, user name of the streaming URL, range:[0,128], desc:the sensitive information should be encrypted*/
                        },
                        "Camera": {
                        /*ro, opt, object, streaming via direct connection, dep:and,{$.Stream.streamSourceType,eq,Camera}, desc:it is valid when the value of streamSourceType is camera it is used for generating URLs from devices; the platform which devices access consist of multiple modules*/
                            "protocol":  "Private",
                            /*ro, opt, string, the protocol that is used for devices to communicate with the platform which assigns tasks, desc:"Private" (private protocol), "Onvif" (ONVIF), "ISUP"; get streaming URL of RTSP according to the protocol type*/
                            "StreamingViaIP": {
                            /*ro, opt, object, stream via IP, dep:or,{$.Stream.StreamSourceList[*].Camera.protocol,eq,Onvif}, {$.Stream.StreamSourceList[*].Camera.protocol,eq,Private}, desc:it is valid when the value of protocol is Private or ONVIF*/
                                "ipV4Address":  "10.21.42.34",
                                /*ro, opt, string, IPv4 address, range:[0,32]*/
                                "port":  8080,
                                /*ro, opt, int, port No., range:[0,65535]*/
                                "userName":  "admin",
                                /*ro, opt, string, user name of the streaming URL, range:[0,128], desc:the sensitive information should be encrypted*/
                                "channel":  1
                                /*ro, opt, int, device channel No., which is the video analysis channel No. of the device, range:[0,256], desc:device channel No., which is the video analysis channel No. of the device*/
```

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```
                        },
                        "StreamingViaDeviceID": {
                        /*ro, opt, object, stream via device ID, dep:and,{$.Stream.StreamSourceList[*].Camera.protocol,eq,ISUP}, desc:it is valid when
the value of protocol is ISUP*/
                            "deviceID":  "test",
                            /*ro, opt, string, device ID; get deviceID by the URL: /ISAPI/System/Network/Ehome, range:[0,256], desc:deviceID; get
deviceID by the URL: /ISAPI/System/Network/Ehome*/
                            "channel":  1
                            /*ro, opt, int, device channel No., which is the video analysis channel No. of the device, range:[0,256]*/
                        }
                    },
                    "RecordTime": {
                    /*ro, opt, object, streaming time of history stream, dep:and,{$.Stream.streamType,eq,historyVideo}, desc:it is valid when the value
of streamType is historyVideo*/
                        "startTime":  "1970-01-01T00:00:00+08:00",
                        /*ro, req, datetime, streaming start time, desc:in ISO 8601 format*/
                        "endTime":  "1970-01-01T00:00:00+08:00"
                        /*ro, req, datetime, streaming end time, desc:in ISO 8601 format*/
                    },
                    "analysisDuration":  10,
                    /*ro, opt, int, analysis duration, range:[10,86400], unit:s, desc:it is valid when <streamType> is "realTime" ad it represents the
analysis time for real-time stream. After the analysis time is up, the stream analysis will be stopped*/
                    "maxSplitCount":  1,
                    /*ro, opt, int, range:[1,160]*/
                    "analyzedTime":  10,
                    /*ro, opt, int, performed analysis duration, range:[10,86400], unit:s, dep:and,
{$.Tasks[*].Stream.StreamSourceList[*].taskAnalysisStatus,eq,analyzing}*/
                    "engine":  1,
                    /*ro, opt, int, which is used to distinguish different engine No., dep:and,
{$.Tasks[*].Stream.StreamSourceList[*].taskAnalysisStatus,eq,analyzing}*/
                    "channelID":  1
                    /*ro, opt, int, linked device channel No., dep:and,{$.Tasks[*].Stream.StreamSourceList[*].taskAnalysisStatus,eq,analyzing},
desc:N/A*/
                }
            ]
        },
        "frameRate":  2500,
        /*ro, req, int, detection frame rate, desc:detection frame rate (frame rate*100)*/
        "alertInterval":  0.5,
        /*ro, opt, float, alarm interval*/
        "status":  0,
        /*ro, req, enum, status information, subType:int, desc:0 (not performed), 1 (performing), 2 (exception), 3 (waiting in queue), 4 (task
mismatched with model), 5 (initializing)*/
        "statusList": [
        /*ro, opt, array, subType:object*/
            {
                "channelID":  1,
                /*ro, opt, int*/
                "cameraIndexCode":  "UUIDTest",
                /*ro, opt, string, range:[0,64]*/
                "status":  1,
                /*ro, req, enum, subType:int*/
                "subStatusCode":  "channelOffline",
                /*ro, opt, string, dep:and,{$.Tasks[*].statusList[*].status,eq,2}*/
                "errorCode":  1073746358
                /*ro, opt, int*/
            }
        ],
        "TimeRange": [
        /*ro, opt, array, arming time, subType:object, desc:daily arming schedules of tasks (by week), 8 time periods per day by default*/
            {
                "week":  1,
                /*ro, req, enum, days of the week, subType:int, desc:1(Monday), 2(Tuesday), 3(Wednesday), 4(Thursday), 5(Friday), 6(Saturday),
7(Sunday)*/
                "beginTime":  "00:00",
                /*ro, req, string, start time*/
                "endTime":  "24:00"
                /*ro, req, string, end time*/
            }
        ],
        "alarmTargetOverlay":  true,
        /*ro, opt, bool, whether to enable overlaying alarm target frame*/
        "alarmRuleOverlay":  true,
        /*ro, opt, bool, whether to enable rule overlay, desc:if it is enabled, overlaying the alarm rule ID and rule name on the capture alarm pictures
is supported*/
        "alarmRuleFrameOverlay":  true,
        /*ro, opt, bool, whether to enable the alarm rule frame overlay, desc:if it is enabled, the alarm rule frame overlay will display on the capture
alarm pictures. When this node is not configured, alarm rule frame overlay will not display by default*/
        "EventTrigger": [
        /*ro, opt, array, configuration for linkage action, subType:object, desc:linkage configurations for task,not sending this filed means not
performing the configured linkage operations,sending null filed means deleting the linkage*/
            {
                "notificationMethod":  "center",
                /*ro, req, enum, linkage action, subType:string, desc:"email", "IM" (instant messaging), "IO" (I/O output), "syslog" (system log),
"HTTP" (uploading via HTTP), "FTP" (uploading via FTP), "beep" (buzzer), "ptz" (PTZ), "record" (recording), "monitorAlarm" (alarm on monitor), "center"
(uploading to the center), "LightAudioAlarm" (light and audio alarm), "focus" (focus), "trace" (tracking), "whiteLight" (white light), "audio" (audio)*/
                "outputIOPortID":  "1",
                /*ro, opt, string, linkage alarm output port No.*/
                "videoInputID":  "1",
                /*ro, opt, string, analog channel No., desc:it is used to indicate the channel for linkage operations*/
                "dynOutputIOPortID":  "1",
                /*ro, opt, string, digital channel No., desc:it is used to indicate the channel for linkage operations*/
                "audioAction": {
                /*ro, opt, object*/
```

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```
                        "audioID": 1,
                        /*ro, req, int*/
                        "alarmTimes": 1,
                        /*ro, opt, int, range:[0,9]*/
                        "childDevID": 3434645611
                        /*ro, opt, float*/
                    }
                }
            ],
            "Destination": [
            /*ro, opt, array, result reception list, subType:object, desc:it is used for uploading events*/
                {
                    "destinationType": "AlarmRMQ",
                    /*ro, req, enum, result reception type, subType:string, desc:"AlarmRMQ" (RabbitMQ), "ServerClient" (uploading to the third-party by
ISAPI), "ServerHCS" (micro video cloud), "ActiveMQ"*/
                    "destinationURL": "test",
                    /*ro, opt, string, URL for receiving results, range:[0,1024], desc:related to destinationType*/
                    "userName": "test",
                    /*ro, opt, string, user name of the recipient, range:[0,128], desc:the sensitive information should be encrypted*/
                    "SSLEnabled": true
                    /*ro, opt, bool, whether to enable SSL encryption, desc:true (enable), false (disable, default value)*/
                }
            ],
            "targetAttributeOverlay": true,
            /*ro, opt, bool, whether to enable target attribute overlay*/
            "AIOP_VideoEnabled": true,
            /*ro, opt, bool, whether to upload AI video task detection events,
            desc:if the node is not configured, the value is true by default. If the value is false, it means events are reported by AI algorithms. The
supported events can be found via /ISAPI/Intelligent/AIOpenPlatform/videoTask/capabilities?format=json._x000D_
            Application example: for deepinMind devices, support detecting hard hat events using the algorithm model, applying video analysis tasks using AI
rules, and uploading eventType:safetyHelmetDetection*/
            "heightThreshold": 1,
            /*ro, opt, int, height threshold, range:[1,255], unit:m, desc:when deploying a triple-lens camera on-site, it is necessary to configure a
threshold of the height difference between power transmission lines and passing vehicles when configuring tasks. An alarm will be triggered when the height
exceeds this threshold*/
            "taskExecutionMode": "continue",
            /*ro, opt, enum, subType:string*/
            "triggerSource": [
            /*ro, opt, array, subType:object, dep:and,{$.taskExecutionMode,eq,trigger}*/
                {
                    "channelID": 1,
                    /*ro, opt, int*/
                    "inputIOPortIDList": [1, 2]
                    /*ro, req, array, subType:int*/
                }
            ],
            "debugEnabled": true,
            /*ro, opt, bool*/
            "backgroundPicQualtity": "middle",
            /*ro, opt, enum, subType:string*/
            "smartStreamList": [
            /*ro, opt, array, subType:object*/
                {
                    "channel": 1,
                    /*ro, opt, int, range:[0,256]*/
                    "cameraIndexCode": "UUIDTest",
                    /*ro, opt, string, range:[0,64]*/
                    "streamID": "test"
                    /*ro, req, string, range:[1,32]*/
                }
            ]
        }
    ]
}
```

19.15.4.6 Apply video tasks

Request URL

POST /ISAPI/Intelligent/AIOpenPlatform/videoTask?format=json

Query Parameter

None

Request Message

```
{
    "taskID": "example",
    /*req, string, task ID, range:[0,64], desc:it is managed and maintained by the service platform for device transparent transmission*/
    "taskName": "test",
    /*opt, string, task name, range:[0,64]*/
    "taskSource": "8200",
    /*opt, string*/
    "taskEnabled": true,
    /*opt, bool*/
    "engineMode": "engineSharing",
    /*opt, enum, engine mode, subType:string,
    desc:when this node is not configured, it indicates engine exclusive mode. Engine exclusive mode means that the current task can only run on the
```

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```
specified engine. Engine shared mode means that the current task can run on multiple engines, and the task can be assigned by the device itself.
    "engineSharing" (engine shared mod), "engineMonopolizing" ( engine exclusive mode)*/
    "engine":  1,
    /*opt, int, it is used to distinguish models running in different engines, desc:the node is required when there is no engineMode or the value of
engineMode is engineMonopolizing*/
    "taskAnalysisMode":  "queue",
    /*opt, enum, task analysis mode, subType:string, desc:"queue" (queueing analysis; when the number of streams exceed the limit of engine, the analysis
will continue after certain queueing time), "concurrency" (concurrent analysis, which means that streams are analyzed at the same time; if the number of
streams exceeds the limit of the engine, the task will be failed)*/
    "MPID":  "example",
    /*ro, opt, string, model package ID, range:[0,64], dep:and,{$.engineMode,eq,engineSharing}, desc:it is used to identify the model information; the
engine bound with the same model ID can analyze tasks in the load balancing mode*/
    "channel": [1, 2],
    /*opt, array, device's local channel No., subType:int, desc:device channel No. (optional); either stream or channel is required*/
    "Stream": {
    /*opt, object, stream information, desc:stream source of video analysis task, and either stream or channel is required*/
        "streamType":  "realTime",
        /*req, enum, stream type, subType:string, desc:"realtime" (streaming remotely and in real time), "historyvideo" (streaming remotely from the
recorded data), the default value is "realtime"*/
        "streamSourceType":  "URL",
        /*req, enum, streaming mode, subType:string, desc:"URL" (specify RTSP URL to stream), "Camera" (specify a camera to stream). Normally the device
acquiring the stream and the device analyze the stream should be on the same LAN*/
        "StreamSourceList": [
        /*opt, array, streaming information, subType:object, desc:if the value of taskAnalysisMode is "queue", up to 1 list is allowed*/
            {
                "cameraIndexCode":  "UUIDTest",
                /*req, string, camera No., range:[0,64],
                desc:the ID should be unique the cameras in the list can reuse the rules configured based on <channel>
                for example, (/ISAPI/Intelligent/AIOpenPlatform/AIRuleConfig/task/<taskID>/channels/<channelID>?format=json&channelType=<channelType>, and
/ISAPI/Intelligent/AIOpenPlatform/AITargetBP/channels/<videoID>?format=json&channelType=<channelType>)*/
                "AITaskType":  "AITargetBPTask",
                /*opt, enum, AI task analysis type, subType:string, desc:when this node is not configured, the task type is regular task by default. If the
value of AITaskType is AITargetBPTask, cameraIndexCode means that you still need to manage the corresponding benchmark dataset. "AITargetBPTask" (AI
benchmark dataset comparison task)*/
                "TargetAttrs": {},
                /*opt, object, transmitted parameters, desc:currently the parameter is only used for the products from the data center to report events*/
                "URL": {
                /*opt, object, streaming URL for RTSP, dep:and,{$.Stream.streamSourceType,eq,URL}, desc:it is valid when the value of streamSourceType is
URL it is used when the platform can construct a URL*/
                    "URL":  "dac://rtsp://10.5.160.103:6304/EUrl/qFD77n9",
                    /*req, string, streaming URL, range:[0,1024], desc:notes about the value of Stream.StreamSourceList.URL.URLMode.@opt in the URL
/ISAPI/Intelligent/AIOpenPlatform/VideoTask/capabilities?format=json; 1. if the value includes DAC, the streaming format: dac://+streaming
URL+&dactoken=XXX, eg:dac://rtsp://10.5.160.103:6304/EUrl/qFD77n9; dac://rtsp://10.33.43.159:655/dac/realplay/30219e1e924648e68f3fab59b838bfec/MAIN/TCP; 2.
if the value includes RTSP, the streaming format: device://rtsp://xxx, eg:device://rtsp://10.66.108.224:554/Streaming/Channels/101?transportmode=unicast*/
                    "userName":  "admin",
                    /*opt, string, streaming user name, range:[0,128], desc:the sensitive information should be encrypted*/
                    "password":  "test"
                    /*opt, string, streaming password, range:[0,128], desc:the sensitive information should be encrypted*/
                },
                "Camera": {
                /*opt, object, stream from local devices, dep:and,{$.Stream.streamSourceType,eq,Camera}, desc:it is valid when the value of streamSourceType
is camera*/
                    "protocol":  "Private",
                    /*opt, string, the protocol that is used for devices to communicate with the platform which assigns tasks, desc:"Private" (private
protocol), "Onvif" (ONVIF), "ISUP" (ISUP) get streaming URL of RTSP according to the protocol type*/
                    "StreamingViaIP": {
                    /*opt, object, stream via IP address, dep:or,{$.Stream.StreamSourceList[*].Camera.protocol,eq,Onvif},
{$.Stream.StreamSourceList[*].Camera.protocol,eq,Private}, desc:it is valid when the value of Protocol is Private or ONVIF*/
                        "ipV4Address":  "10.21.42.34",
                        /*opt, string, IPv4 address, range:[0,32]*/
                        "port":  8080,
                        /*opt, int, port No., range:[0,65535]*/
                        "userName":  "admin",
                        /*opt, string, streaming user name, range:[0,128], desc:the sensitive information should be encrypted*/
                        "password":  "test",
                        /*opt, string, streaming password, range:[0,128], desc:the sensitive information should be encrypted*/
                        "channel":  1
                        /*opt, int, device channel No. which is the video analysis channel No. of the device, range:[0,256], desc:the channel number refers
to the channel number of the device corresponding to the IP address*/
                    },
                    "StreamingViaDeviceID": {
                    /*opt, object, get stream information by device ID, dep:and,{$.Stream.StreamSourceList[*].Camera.protocol,eq,ISUP}, desc:it is valid
when the value of Protocol is ISUP*/
                        "deviceID":  "test",
                        /*opt, string, device ID, range:[0,256], desc:the device ID of ISUP, which is returned by the API /ISAPI/System/Network/Ehome*/
                        "channel":  1
                        /*opt, int, device channel No. which is the video analysis channel No. of the device, range:[0,256]*/
                    }
                },
                "RecordTime": {
                /*opt, object, streaming time of history stream, dep:and,{$.Stream.streamType,eq,historyVideo}, desc:it is valid when the value of
streamType is historyVideo*/
                    "startTime":  "1970-01-01T00:00:00+08:00",
                    /*req, datetime, streaming start time, desc:in ISO 8601 format*/
                    "endTime":  "1970-01-01T00:00:00+08:00"
                    /*req, datetime, streaming end time, desc:in ISO 8601 format*/
                },
                "analysisDuration":  10,
                /*opt, int, analysis duration, range:[10,86400], unit:s, desc:it is valid when <streamType> is "realTime" ad it represents the analysis time
for real-time stream. After the analysis time is up, the stream analysis will be stopped*/
                "maxSplitCount":  1
                /*opt, int, range:[1,160]*/
            }
```

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```
        ]
    },
    "frameRate":  2500,
    /*req, int, (frame rate)*100 = detection frame rate*/
    "alertInterval":  0.5,
    /*opt, float, alarm interval, range:[0,2000], desc:1 (1 time per second), 0.5 (2 times per second)*/
    "TimeRange": [
    /*opt, array, arming schedule list, subType:object, desc:daily arming schedules of tasks (by week), 8 time periods per day by default*/
        {
            "week":  1,
            /*req, enum, the day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)*/
            "beginTime":  "00:00",
            /*req, string, start time*/
            "endTime":  "24:00"
            /*req, string, end time*/
        }
    ],
    "alarmTargetOverlay":  true,
    /*opt, bool, whether to enable overlaying alarm target frame*/
    "alarmRuleOverlay":  true,
    /*opt, bool, whether to enable alarm rule overlay, desc:when enabled, the alarm rule ID (No.) and rule name will be overlaid in the captured alarm
pictures*/
    "alarmRuleFrameOverlay":  true,
    /*opt, bool, whether to enable alarm rule frame overlay, desc:when enabled, the alarm rule frame will be overlaid in the captured alarm pictures. When
the node is not configured, the rule frames will not be overlaid by default*/
    "EventTrigger": [
    /*opt, array, linkage action settings, subType:object, desc:linkage action of tasks; if this node is not configured, the configured linkage action will
not be edited; if this node is null, it indicates that the linkage action will be deleted*/
        {
            "notificationMethod":  "center",
            /*req, enum, linkage action, subType:string, desc:"email" (email), "IM" (instant messaging), "IO" (I/O output), "syslog" (system log), "HTTP"
(uploading via HTTP), "FTP" (uploading via FTP), "beep" (buzzer), "ptz" (PTZ), "record" (recording), "center" (uploading to the center), "LightAudioAlarm"
(light and audio alarm), "focus" (focus), "trace" (tracking), "whiteLight" (white light), "audio" (audio), "door" (door)*/
            "outputIOPortID":  "1",
            /*opt, string, linked alarm output port ID*/
            "videoInputID":  "1",
            /*opt, string, simulated channel No.*/
            "dynOutputIOPortID":  "1",
            /*opt, string, digital channel No.*/
            "audioAction": {
            /*opt, object, dep:and,{$.EventTrigger[*].notificationMethod,eq,audio}*/
                "audioID":  1,
                /*req, int, range:[0,65535]*/
                "alarmTimes":  1,
                /*opt, int, range:[0,9]*/
                "childDevID":  "3434645611"
                /*opt, string, range:[9,32]*/
            }
        }
    ],
    "Destination": [
    /*opt, array, receiving destination information, subType:object, desc:currently the parameter is only used for the products from the data center. When
the node is returned, the address will be the address of reporting alarms*/
        {
            "destinationType":  "AlarmRMQ",
            /*req, enum, receiving destination type, subType:string, desc:for details, refer to the Dictionaries. "AlarmRMQ" (RabbitMQ), "ServerClient"
(ISAPI third-party address), "ServerHCS" (micro video cloud), "ActiveMQ" (ActiveMQ)*/
            "destinationURL":  "test",
            /*opt, string, URL for receiving results, range:[0,1024], desc:related to destinationType*/
            "userName":  "test",
            /*opt, string, user name of the recipient, range:[0,128], desc:the sensitive information should be encrypted*/
            "password":  "test",
            /*opt, string, password of the recipient, range:[0,128], desc:the sensitive information should be encrypted*/
            "SSLEnabled":  true
            /*opt, bool, whether to enable SSL encryption, desc:"true" (enable), "false" (disable, default value)*/
        }
    ],
    "targetAttributeOverlay":  true,
    /*opt, bool, whether to enable target attribute overlay*/
    "AIOP_VideoEnabled":  true,
    /*opt, bool, upload events of AI video task detection,
    desc:if the node is not configured, the value is true by default. If the value is false, it means events are reported by AI algorithms. The supported
events can be found via /ISAPI/Intelligent/AIOpenPlatform/videoTask/capabilities?format=json.
    Application example: for deepinMind devices, support detecting hard hat events using the algorithm model, applying video analysis tasks using AI rules,
and uploading eventType:safetyHelmetDetection*/
    "heightThreshold":  1,
    /*opt, int, height threshold, range:[1,255], unit:m, desc:when deploying a triple-lens camera on-site, it is necessary to configure a threshold of the
height difference between power transmission lines and passing vehicles when configuring tasks. An alarm will be triggered when the height exceeds this
threshold*/
    "taskExecutionMode":  "continue",
    /*opt, enum, subType:string*/
    "triggerSource": [
    /*opt, array, subType:object, dep:and,{$.taskExecutionMode,eq,trigger}*/
        {
            "channelID":  1,
            /*opt, int*/
            "inputIOPortIDList": [1, 2]
            /*req, array, subType:int*/
        }
    ],
    "debugEnabled":  true,
    /*opt, bool*/
    "backgroundPicQualtity":  "middle"
    /*opt, enum, subType:string*/
```

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```
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:this field is required when the status code is not 1 (success)*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded), other value (error occurred). This field is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error message, desc:this node is required when the value of statusCode is not 1*/
    "channelID": [1, 2, 3]
    /*ro, opt, array, error channel No., subType:int, desc:for multi-channel tasks, errors may occur when applying to some of the channels because of resolution. An error code is returned if there are errors, and this node is not returned when applying to channels successfully*/
}
```

19.15.4.7 AI video task detection event

EventType:AIOP_Video

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers an alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers an alarm*/
    "portNo":  80,
    /*ro, opt, int, port No. of the device that triggers an alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, communication protocol types, subType:string, desc:communication protocols include HTTP, HTTPS, and Ehome (when transmitting the ISAPI URL in EZVIZ SDK, the protocol will be HTTP; when transmitting the ISAPI URL in ISUP SDK, the protocol will be EHome); the previous field "protocolType" is still compatible with the platform; "HTTP"(Hypertext Transfer Protocol for communication between clients and servers over the internet), "HTTPS"(Hypertext Transfer Protocol Secure that adds an extra layer of security by using encryption protocols), "EHome"(EHome protocol is a push-mode protocol that can realize the communication between the system and Hikvision mobile devices)*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers an alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, the uploading times of the same alarm, desc:the event triggering frequency. The pulse event is the event which is triggered continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object recognition occurs, and the recognition of each object will trigger the event once, e.g., the face capture. When the pulse events are triggered, this node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously according to the device detection frequency; if it changes to another source, then the counting will start again*/
    "eventType":  "AIOP_Video",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for continuous event: "active" (a valid event or event starts), "inactive" (an invalid event or the event ends); in the heartbeat mode, the field indicates the heartbeat data that will be uploaded every 10 seconds*/
    "eventDescription":  "AI Open Platform Video Detection",
    /*ro, req, string, event description, range:[0,128]*/
    "channelName":  "test",
    /*ro, opt, string, channel name*/
    "taskID":  "test",
    /*ro, opt, string, video analysis task ID*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID, desc:it should be returned for ISUP alarms*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers*/
    "childDevID":  "33346456112",
    /*ro, opt, string, (sub-device local ID) device serial No., range:[9,32]*/
    "AIOPData": {},
    /*ro, opt, object, the value of this node is provided by the algorithm library*/
    "algorithmInfo": {
    /*ro, opt, object, algorithm version information, desc:this field is required for secondary analysis*/
        "algorithmId":  "test",
        /*ro, opt, string, algorithm package ID*/
        "algoVendorId":  "test",
        /*ro, opt, string, manufacturer ID*/
        "algoVersionId":  "test",
        /*ro, opt, string, algorithm version No.*/
        "algoChipType":  "test",
        /*ro, opt, string, chip type*/
        "algoModelVersion":  "test"
        /*ro, opt, string, model version*/
    },
    "url":  "test",
    /*ro, opt, string, AIOPdata picture, desc:if the device supports EZ access, the pictures should be uploaded to the EZ server, and the returned URL
```

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should be entered in this node; if the device supports cloud storage, the pictures should be uploaded to the cloud storage server, and the returned URL
should be entered in this node*/
    "PID":  "test",
    /*ro, opt, string, AIOPdata picture*/
    "targetInfoList": [
    /*ro, opt, array, target information list, subType:object, range:[0,1], desc:only the target information which triggered events is uploaded*/
        {
            "targetImage": {
            /*ro, opt, object, most optimal captured thumbnail of the target*/
                "pictureURL":  "test",
                /*ro, opt, string, picture URL, range:[0,512]*/
                "PID":  "test"
                /*ro, opt, string, picture ID, range:[0,64]*/
            },
            "targetBackgroundImage": {
            /*ro, opt, object, most optimal captured background picture of the target*/
                "pictureURL":  "test",
                /*ro, opt, string, picture URL, range:[0,512]*/
                "PID":  "test"
                /*ro, opt, string, picture ID, range:[0,64]*/
            },
            "targetLibImage": {
            /*ro, opt, object, original picture in the library*/
                "pictureURL":  "test",
                /*ro, opt, string, picture URL, range:[0,512]*/
                "PID":  "test"
                /*ro, opt, string, picture ID, range:[0,64]*/
            }
        }
    ],
    "presetNo":  0,
    /*ro, opt, int, linked preset No.*/
    "MPID":  "example",
    /*ro, opt, string, model package ID, range:[0,64], desc:it is used to identify the model information*/
    "MPName":  "test",
    /*ro, opt, string, model package name*/
    "isSecondaryVerificationAlarm":  false,
    /*ro, opt, bool, whether it is the alarm of secondary analysis, desc:true (the alarm of the secondary model verification), false (the alarm of the
independent model analysis for once); if this node does not exist, the default value is "false"*/
    "addInfolUrl":  "test",
    /*ro, opt, string, picture URL of the additional information, range:[1,1024], desc:picture of non-alarm channel, which is the additional information of
"AIOPData"*/
    "Pictures": [
    /*ro, opt, array, picture array of events, subType:object, desc:it is used for storing multiple pictures*/
        {
            "pictureType":  "test",
            /*ro, req, string, picture type, desc:related captured picture type of the specified event type*/
            "URL":  "test"
            /*ro, req, string, picture URL*/
        }
    ],
    "timeStamp":  "2019-12-30T15:29:23+08:00",
    /*ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format*/
    "targetAttrs": {
    /*ro, opt, object, transmitted parameters*/
        "deviceId":  "1",
        /*ro, opt, string, device ID*/
        "deviceChannel":  1,
        /*ro, opt, int, device channel No.*/
        "deviceName":  "shebei",
        /*ro, opt, string, device name, range:[0,128]*/
        "cameraIndexCode":  "UUIDTest",
        /*ro, opt, string, camera No., desc:when task applying, this node can be returned only when it also exists in the platform streaming mode, i.e.
Stream.StreamSourceList.TargetAttrs.cameraIndexCode exists in the URL /ISAPI/Intelligent/AIOpenPlatform/videoTask?format=json*/
    },
    "AITargetBP": [
    /*ro, opt, array, reference picture comparison information, subType:object, range:[0,1], desc:only a background picture with great differences will be
returned*/
        {
            "PID":  "test",
            /*ro, opt, string, picture ID, range:[0,64]*/
            "pictureURL":  "test"
            /*ro, opt, string, picture URL, range:[0,512]*/
        }
    ],
    "dotMatrixInfo": {
    /*ro, opt, object, dot matrix data, which is generated in semantic segmentation algorithm, desc:it includes mask dot matrix data and dot matrix data of
probit probability, and the two types of data are stored in the same file*/
        "compressedUrl":  "test",
        /*ro, opt, string, URL of the compressed dot matrix chart information, desc:it is the returned URL of uploading device information to the cloud
storage (if configured); since the dot matrix chart information has large size, it can be compressed by the lzma algorithm*/
        "compressedContentID":  "compresseDotMatrixInfo"
        /*ro, opt, string, binary data ID of the compressed dot matrix chart information*/
    },
    "sensorList": [
    /*ro, opt, array, linked sensor information, subType:object*/
        {
            "ruleID":  1,
            /*ro, opt, int, linked alarm rule ID of the sensor*/
            "nodeID":  1,
            /*ro, opt, int, computing Node ID, desc:this node is required when isSupportAlgSorting is returned and the value is "true" in the URL:
/ISAPI/Intelligent/AIOpenPlatform/capabilities?format=json*/
            "sensorInfoList": [
            /*ro, opt, array, sensor information, subType:object*/
```

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```
                {
                    "sensorID":  "test",
                    /*ro, req, string, sensor ID*/
                    "sensorType":  "fireSense",
                    /*ro, req, enum, sensor type, subType:string, desc:"fireSense" (flame sensor)*/
                    "fireSenseProperties": {
                    /*ro, opt, object, flame sensor attribute, desc:it is valid when the value of sensorType is "fireSense"*/
                        "flameProperties": {
                        /*ro, opt, object, flame attribute*/
                            "enabled":  true,
                            /*ro, opt, bool, whether to enable this function*/
                            "value":  "fire"
                            /*ro, opt, enum, attribute value, subType:string, desc:this node is valid when the sensorEnabled is true and sensorType is
fireSense: “fire”, "noFire"*/
                        }
                    }
                }
            ]
        }
    ],
    "angleView": {
    /*ro, opt, object, FOV*/
        "horizontalValue":  0,
        /*ro, req, int, current horizontal field of view, range:[0,360], unit:度*/
        "verticalValue":  0,
        /*ro, req, int, current vertical field of view, range:[0,360], unit:度*/
        "visibleRadius":  0,
        /*ro, req, int, current view radius, range:[0,1000000], desc:unit: m*/
        "viewRadiusRange": {
        /*ro, opt, object, view radius range*/
            "maxViewRadius":  0
            /*ro, req, int, max. view radius, range:[0,100000], desc:unit: m*/
        }
    },
    "centerAngle": {
    /*ro, opt, object, angle of the picture center, desc:the platform can calculate the parameters of FoV according to this node*/
        "rotateAngelCalibratation":  30.000001,
        /*ro, opt, float, rotation calibration value, range:[-180,180], desc:supports getting, setting, and saving the rotation calibration value via SSH
command; the value should be accurate to six decimal place*/
        "rotateAngel":  30.000001,
        /*ro, opt, float, current rotation angle, range:[-180,180], desc:supports getting, setting, and saving the rotation calibration value via SSH
command; the value should be accurate to six decimal place*/
        "verticalCalibratation":  30.000001,
        /*ro, opt, float, tilt calibration value, range:[-180,180], desc:supports getting, setting, and saving the rotation calibration value via SSH
command; the value should be accurate to six decimal place*/
        "verticalAngle":  30.000001
        /*ro, opt, float, current tilt angle, range:[-180,180], desc:supports getting, setting, and saving the rotation calibration value via SSH command;
the value should be accurate to six decimal place*/
    }
}
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name
AIOP_Video	[Message content]	application/json	--	--
compressedContentID	[Binary picture data]	image/jpeg	compresseDotMatrixInfo	compresseDotMatrixInfo.jpg
targetImage.PID	[Binary picture data]	image/jpeg	targetImage.PID	targetImage.jpg
targetBackgroundImage.PID	[Binary picture data]	image/jpeg	targetBackgroundImage.PID	targetBackgroundImage.jpg
targetLibImage.PID	[Binary picture data]	image/jpeg	targetLibImage.PID	targetLibImage.jpg
AITargetBP.PID	[Binary picture data]	image/jpeg	AITargetBP.PID	AITargetBP.jpg
PID	[Binary picture data]	image/jpeg	PID	AIOP_Video_image.jpg

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

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19.16 Two-Way Audio

19.16.1 Two-Way Audio

19.16.1.1 Get the parameters of all two-way audio channels

Request URL

GET /ISAPI/System/TwoWayAudio/channels

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, parameters configuration of the two-way audio channel, subType:object, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--ro, opt, object, configuration of the two-way audio channel-->
    <id>
      <!--ro, req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable or not-->true
    </enabled>
    <audioCompressionType>
      <!--ro, req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM",
"MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType>
      <!--ro, opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM",
"MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume>
      <!--ro, opt, int, input volume, range:[1,100]-->100
    </speakerVolume>
    <microphoneVolume>
      <!--ro, opt, int, output volume, range:[1,100]-->100
    </microphoneVolume>
    <noisereduce>
      <!--ro, opt, bool, whether to enable the environmental noise filter or not-->true
    </noisereduce>
    <audioBitRate>
      <!--ro, opt, int, audio frame rate, unit:kbs-->100
    </audioBitRate>
    <audioInputType>
      <!--ro, opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-
adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--ro, opt, object, linked video channel-->
      <enabled>
        <!--ro, req, bool, whether to enable the linked video input channel or not-->true
      </enabled>
      <videoInputChannelList>
        <!--ro, req, array, list of linked video channels, subType:object-->
        <videoInputChannelID>
          <!--ro, opt, string, linked video channel-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate>
      <!--ro, opt, float, audio sampling rate, unit:kHz-->48.00
    </audioSamplingRate>
    <lineOutForbidden>
      <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
    </lineOutForbidden>
    <muteDuringPanTilt>
      <!--ro, opt, bool, whether to mute during motion-->true
    </muteDuringPanTilt>
    <audioOutputType>
      <!--ro, opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <matrixPickUp>
      <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it
is valid only when the audio input type is "Linein")-->true
    </matrixPickUp>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

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19.16.1.2 Set audio parameters for all two-way audio channels

Request URL

PUT /ISAPI/System/TwoWayAudio/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameters configuration of the two-way audio channel, subType:object, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--opt, object, configuration of the two-way audio channel-->
    <id>
      <!--req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function or not-->true
    </enabled>
    <audioCompressionType>
      <!--req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType>
      <!--req, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume>
      <!--opt, int, input volume-->100
    </speakerVolume>
    <microphoneVolume>
      <!--opt, int, output volume-->100
    </microphoneVolume>
    <noisereduce>
      <!--opt, bool, whether to enable the environmental noise filter or not-->true
    </noisereduce>
    <audioBitRate>
      <!--opt, int, audio frame rate, unit:kbs-->100
    </audioBitRate>
    <audioInputType>
      <!--req, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--opt, object, linked video channel-->
      <enabled>
        <!--req, bool, whether to enable the linked video input channel or not-->true
      </enabled>
      <videoInputChannelList>
        <!--req, array, list of linked video channels, subType:object-->
        <videoInputChannelID>
          <!--opt, string, linked video channel-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate>
      <!--opt, float, audio sampling rate, unit:kHz-->48.00
    </audioSamplingRate>
    <lineOutForbidden>
      <!--opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
    </lineOutForbidden>
    <muteDuringPanTilt>
      <!--opt, bool, whether to mute during motion-->true
    </muteDuringPanTilt>
    <audioOutputType>
      <!--opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <matrixPickUp>
      <!--opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
    </matrixPickUp>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.16.1.3 Set audio parameters of a specified two-way audio channel

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, configuration of the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType>
    <!--req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume>
    <!--opt, int, input volume, range:[1,100]-->100
  </speakerVolume>
  <microphoneVolume>
    <!--opt, int, output volume, range:[1,100]-->100
  </microphoneVolume>
  <noisereduce>
    <!--opt, bool, whether to enable the environmental noise filter or not-->true
  </noisereduce>
  <audioBitRate>
    <!--opt, int, audio frame rate, unit:kbs-->100
  </audioBitRate>
  <audioInputType>
    <!--opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--opt, object, linked video channel-->
    <enabled>
      <!--req, bool, whether to enable the linked video input channel or not-->true
    </enabled>
    <videoInputChannelList>
      <!--req, array, list of linked video channels, subType:object-->
      <videoInputChannelID>
        <!--opt, string, linked video channel-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate>
    <!--opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <lineOutForbidden>
    <!--opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the value
is "true", audio output is not supported-->true
  </lineOutForbidden>
  <micInForbidden>
    <!--opt, bool, whether the audio input is not supported. If this node is not returned or its value is "false", audio input is supported;  if the value
is "true", audio input is not supported-->true
  </micInForbidden>
  <muteDuringPanTilt>
    <!--opt, bool, whether to mute during motion-->true
  </muteDuringPanTilt>
  <audioOutputType>
    <!--opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <timeOut>
    <!--opt, int, timeout during two-way audio, range:[10,600], unit:s, desc:when the Web Client initiates the two-way audio actively towards the radar
device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some noise.
When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client stops
speaking or not.-->10
  </timeOut>
  <matrixPickUp>
    <!--opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
  </matrixPickUp>
</TwoWayAudioChannel>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

19.16.1.4 Get the capability of configuring audio parameters for a specified two-way audio channel

Request URL

GET /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/capabilities?audioInputType=<audioInputType>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
audioInputType	string	Audio input type, corresponding to the "audioInputType" in the message. This parameter is mainly used to obtain the parameters corresponding to the specified audio input type.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, capability of configuring audios for the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2,Opus,G.722.1">
    <!--ro, req, enum, encoding type of the audio output, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711uLaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,G.722.1">
    <!--ro, opt, enum, encoding type of the audio input, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume min="0" max="100">
    <!--ro, opt, int, input volume, attr:min{opt, int},max{opt, int}-->100
  </speakerVolume>
  <microphoneVolume min="0" max="100">
    <!--ro, opt, int, output volume, attr:min{opt, int},max{opt, int}-->100
  </microphoneVolume>
  <noisereduce opt="true,false">
    <!--ro, opt, bool, whether to enable the environmental noise filter or not, attr:opt{opt, string}-->true
  </noisereduce>
  <audioBitRate min="1" max="100">
    <!--ro, opt, int, audio frame rate, unit:kbs, attr:min{opt, int},max{opt, int}-->100
  </audioBitRate>
  <audioInputType opt="MicIn,LineIn,selfAdaptive,wirelessPickUp,doubleMic,networkPickUp">
    <!--ro, opt, enum, audio input type, subType:string, attr:opt{opt, string}, desc:"MicIn" (microphone-level input), "LineIn" (line-level input),
"selfAdaptive" (self-adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--ro, opt, object, linked video channel-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the linked video input channel or not, attr:opt{opt, string}-->true
    </enabled>
    <videoInputChannelList>
      <!--ro, req, array, list of linked video channels, subType:object-->
      <videoInputChannelID min="1" max="2048">
        <!--ro, opt, string, linked video channel, attr:min{opt, string},max{opt, string}-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate min="1.00" max="48.00">
    <!--ro, opt, float, audio sampling rate, unit:kHz, attr:min{opt, float},max{opt, float}-->48.00
  </audioSamplingRate>
  <lineOutForbidden opt="true,false">
    <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported, attr:opt{opt, string}-->true
  </lineOutForbidden>
  <muteDuringPanTilt opt="true,false">
    <!--ro, opt, bool, whether to mute during motion, attr:opt{opt, string}-->true
  </muteDuringPanTilt>
  <audioOutputType opt="Close,LineOut,Speaker,selfAdaptive,LineOut_Speaker,HDMI" def="LineOut">
    <!--ro, opt, enum, audio output type, subType:string, attr:opt{opt, string},def{opt, string}, desc:"Close" (output closed), "LineOut" (line-level
output), "Speaker" (speaker output), "selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <isSupportEchoCancellation>
    <!--ro, opt, bool, whether the device supports echo cancellation-->true
  </isSupportEchoCancellation>
  <timeOut min="10" max="600">
    <!--ro, opt, int, timeout during two-way audio, attr:min{req, int},max{req, int}, desc:when the Web Client initiates the two-way audio actively towards
the radar device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some
noise. When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client
stops speaking or not.-->300
  </timeOut>
  <matrixPickUp opt="true,false">
    <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it is
valid only when the audio input type is "Linein"), attr:opt{opt, string}-->false
  </matrixPickUp>
  <isSupportRestore>
    <!--ro, opt, bool, whether the device supports restoring audio parameters of the two-way audio channel, desc:corresponding API:
/ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/restore?format=json-->true
  </isSupportRestore>
  <networkPickUpStatus opt="online,offline">
    <!--ro, opt, enum, network pick-up status, subType:string, dep:and,{$.TwoWayAudioChannel.audioInputType,eq,networkPickUp}, attr:opt{opt, string},
desc:network status of the network pickup linked with the audio channel: "online", "offline"-->online
  </networkPickUpStatus>
  <noisereduceType opt="normal,smart">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->normal
  </noisereduceType>
</TwoWayAudioChannel>
```

19.16.1.5 Stop two-way audio

Request URL

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PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/close?sessionId=<audioSessionID>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-Way Audio Channel ID
audioSessionID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, Custom Error Information Description, range:[0,1024], desc:Custom detailed error information returned by device application, used
for quick positioning and diagnosis.-->badXmlFormat
  </description>
</ResponseStatus>
```

19.16.1.6 Start two-way audio

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/open?type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
type	enum	If this filed does not exist, it indicates two-way audio via channel, and the audioID is the channel ID. When the type is childDevID, it indicates two-way audio with a single sub device, and the audioID is the gateway intercom resource index (obtained from /ISAPI/System/DynamicResource/SearchResources?format=json); when the type is childDevList, it indicates broadcasting to multiple sub devices, with multiple childDevID values specified in the message, and the audioID in the URL is invalid.

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudio xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters for this two-way audio, attr:version{req, string, protocolVersion}-->
  <audioLevel>
    <!--opt, int, audio level, range:[0,15], desc:range: [0,15]-->0
  </audioLevel>
  <microphoneVolume>
    <!--opt, int, output volume-->100
  </microphoneVolume>
  <audioCompressionType>
    <!--opt, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioSamplingRate>
    <!--opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <isBroadcast>
    <!--opt, bool-->true
  </isBroadcast>
  <broadcastType>
    <!--opt, enum, subType:string-->alarm
  </broadcastType>
  <childList>
    <!--opt, array, subType:object, range:[0,8], dep:and,{$.TwoWayAudio.isBroadCast,eq,true}-->
    <childDevID>
      <!--opt, string, range:[9,32]-->test
    </childDevID>
    <voiceTalkIndex>
      <!--opt, int, range:[1,256]-->test
    </voiceTalkIndex>
  </childList>
</TwoWayAudio>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioSession xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, result returned when starting two-way audio succeeds, attr:version{opt, string, protocolVersion}-->
  <sessionId>
    <!--ro, req, string, two-way audio session ID-->1
  </sessionId>
</TwoWayAudioSession>
```

19.16.1.7 Restore audio parameters of the two-way audio channel

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/restore?format=json

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	--

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

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19.16.1.8 Get the audio parameters of a specified two-way audio channel

Request URL

GET /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>?audioInputType=<audioInputType>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
audioInputType	string	Audio input type, corresponding to the "audioInputType" in the message. This parameter is mainly used to obtain the parameters corresponding to the specified audio input type.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, configuration of the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType>
    <!--ro, req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--ro, opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume>
    <!--ro, opt, int, input volume, range:[1,100]-->100
  </speakerVolume>
  <microphoneVolume>
    <!--ro, opt, int, output volume, range:[1,100]-->100
  </microphoneVolume>
  <noisereduce>
    <!--ro, opt, bool, whether to enable the environmental noise filter or not-->true
  </noisereduce>
  <audioBitRate>
    <!--ro, opt, int, audio frame rate, unit:kbs-->100
  </audioBitRate>
  <audioInputType>
    <!--ro, opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--ro, opt, object, linked video channel-->
    <enabled>
      <!--ro, req, bool, whether to enable the linked video input channel or not-->true
    </enabled>
    <videoInputChannelList>
      <!--ro, req, array, list of linked video channels, subType:object-->
      <videoInputChannelID>
        <!--ro, opt, string, linked video channel-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate>
    <!--ro, opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <lineOutForbidden>
    <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
  </lineOutForbidden>
  <micInForbidden>
    <!--ro, opt, bool, whether the audio input is not supported. If this node is not returned or its value is "false", audio input is supported;  if the
value is "true", audio input is not supported-->true
  </micInForbidden>
  <muteDuringPanTilt>
    <!--ro, opt, bool, whether to mute during motion-->true
  </muteDuringPanTilt>
  <audioOutputType>
    <!--ro, opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <timeOut>
    <!--ro, opt, int, timeout during two-way audio, range:[10,600], unit:s, desc:when the Web Client initiates the two-way audio actively towards the radar
device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some noise.
When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client stops
speaking or not.-->10
  </timeOut>
  <matrixPickUp>
    <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
  </matrixPickUp>
  <networkPickUpStatus>
    <!--ro, opt, enum, subType:string, dep:and,{$.TwoWayAudioChannel.audioInputType,eq,networkPickUp}-->online
  </networkPickUpStatus>
  <noisereduceType>
    <!--ro, opt, enum, subType:string-->normal
  </noisereduceType>
</TwoWayAudioChannel>
```

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19.16.1.9 Get the capability of configuring parameters for all two-way audio channels

Request URL

GET /ISAPI/System/TwoWayAudio/channels/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, capability of configuring audios for all two-way audio channels, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--ro, opt, object, capability of configuring audios for the two-way audio channel-->
    <id>
      <!--ro, req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable or not-->true
    </enabled>
    <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2,Opus">
      <!--ro, req, enum, encoding type of the audio output, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729",
"G.729a", "G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM">
      <!--ro, opt, enum, encoding type of the audio input, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume min="0" max="100">
      <!--ro, opt, int, input volume, attr:min{opt, int},max{opt, int}-->100
    </speakerVolume>
    <microphoneVolume min="0" max="100">
      <!--ro, opt, int, output volume, attr:min{opt, int},max{opt, int}-->100
    </microphoneVolume>
    <noisereduce opt="true,false">
      <!--ro, opt, bool, whether to enable the environmental noise filter or not, attr:opt{opt, string}-->true
    </noisereduce>
    <audioBitRate min="1" max="100">
      <!--ro, opt, int, audio frame rate, unit:kbs, attr:min{opt, string},max{opt, string}-->100
    </audioBitRate>
    <audioInputType opt="MicIn,LineIn,selfAdaptive,wirelessPickUp">
      <!--ro, opt, enum, audio input type, subType:string, attr:opt{opt, string}, desc:"MicIn" (microphone-level input), "LineIn" (line-level input),
"selfAdaptive" (self-adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--ro, opt, object, linked video channel-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the linked video input channel or not, attr:opt{opt, string}-->true
      </enabled>
      <videoInputChannelList>
        <!--ro, req, array, list of linked video channels, subType:object-->
        <videoInputChannelID min="1" max="2048">
          <!--ro, opt, string, linked video channel, attr:min{opt, string},max{opt, string}-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate min="1.00" max="48.00">
      <!--ro, opt, float, audio sampling rate, unit:kHz, attr:min{opt, float},max{opt, float}-->48.00
    </audioSamplingRate>
    <lineOutForbidden opt="true,false">
      <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported, attr:opt{opt, string}-->true
    </lineOutForbidden>
    <muteDuringPanTilt opt="true,false">
      <!--ro, opt, bool, whether to mute during motion, attr:opt{opt, string}-->true
    </muteDuringPanTilt>
    <audioOutputType opt="Close,LineOut,Speaker,selfAdaptive,LineOut_Speaker" def="LineOut">
      <!--ro, opt, enum, audio output type, subType:string, attr:opt{opt, string},def{opt, string}, desc:"Close" (output closed), "LineOut" (line-level
output), "Speaker" (speaker output), "selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <isSupportEchoCancellation>
      <!--ro, opt, bool, whether the device supports echo cancellation-->true
    </isSupportEchoCancellation>
    <matrixPickUp opt="true,false">
      <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it
is valid only when the audio input type is "Linein"), attr:opt{opt, string}-->false
    </matrixPickUp>
    <isSupportRestore>
      <!--ro, opt, bool, whether the device supports restoring audio parameters of the two-way audio channel, desc:corresponding API:
/ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/restore?format=json-->true
    </isSupportRestore>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

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19.17 Audio Recognition

19.17.1 Audio Detection

19.17.1.1 Get the arming schedule of audio detection for all channels

Request URL

GET /ISAPI/Event/schedules/audioDetections

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->audioDetections-1
    </id>
    <eventType>
      <!--ro, opt, string-->audioDetections
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</AudioDetectionScheduleList>
```

19.17.1.2 Set the arming schedule of audio detection for all channels

Request URL

PUT /ISAPI/Event/schedules/audioDetections

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<AudioDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->audioDetections-1
    </id>
    <eventType>
      <!--opt, string-->audioDetections
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</AudioDetectionScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.17.1.3 Set the arming schedule of audio detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/audioDetections/audioDetection-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->audioDetections-1
  </id>
  <eventType>
    <!--opt, string-->audioDetections
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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19.17.1.4 Get the arming schedule of audio detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/audioDetections/audioDetection-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->audioDetections-1
  </id>
  <eventType>
    <!--ro, opt, string-->audioDetections
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.17.1.5 Set the audio detection parameters of all channels

Request URL

PUT /ISAPI/Smart/AudioDetection/channels

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<AudioDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, list of audio detection parameters, subType:object, attr:version{req, string, protocolVersion}-->
  <AudioDetection>
    <!--opt, object-->
    <id>
      <!--req, string, ID-->test
    </id>
    <audioInputException>
      <!--opt, object, audio input exception detection-->
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
    </audioInputException>
    <soundIntensityMutation>
      <!--opt, object, sudden change of sound intensity detection-->
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <sensitivityLevel>
        <!--req, int, sensitivity level, range:[1,100]-->0
      </sensitivityLevel>
      <mutationThreshold>
        <!--req, int, 100], range:[1,100]-->0
      </mutationThreshold>
    </soundIntensityMutation>
    <SteepFall>
      <!--opt, object, sudden decrease of sound intensity-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <sensitivityLevel>
        <!--req, int, sensitivity level, range:[1,100]-->0
      </sensitivityLevel>
    </SteepFall>
    <AudioLoss>
      <!--opt, object, audio loss detection-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <sensitivityLevel>
        <!--opt, int, sensitivity level, range:[0,100]-->0
      </sensitivityLevel>
    </AudioLoss>
    <audioMode>
      <!--opt, enum, subType:int-->0
    </audioMode>
  </AudioDetection>
</AudioDetectionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

19.17.1.6 Get the audio detection parameters of all channels

Request URL

GET /ISAPI/Smart/AudioDetection/channels

Query Parameter

None

Request Message

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None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, list of audio detection parameters, subType:object, attr:version{req, string, protocolVersion}-->
  <AudioDetection>
    <!--ro, opt, object, audio detection parameters-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <audioInputException>
      <!--ro, opt, object, audio input exception detection-->
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
    </audioInputException>
    <soundIntensityMutation>
      <!--ro, opt, object, sudden change of sound intensity detection-->
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity level, range:[1,100]-->0
      </sensitivityLevel>
      <mutationThreshold>
        <!--ro, req, int, 100], range:[1,100]-->0
      </mutationThreshold>
    </soundIntensityMutation>
    <SteepFall>
      <!--ro, opt, object, sudden decrease of sound intensity-->
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity level, range:[1,100]-->0
      </sensitivityLevel>
    </SteepFall>
    <AudioLoss>
      <!--ro, opt, object, audio loss detection-->
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity level, range:[0,100]-->0
      </sensitivityLevel>
    </AudioLoss>
    <audioMode>
      <!--ro, opt, enum, audio detection mode, subType:int, desc:audio detection mode-->0
    </audioMode>
  </AudioDetection>
</AudioDetectionList>
```

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19.17.1.7 Get audio detection parameters of a specified channel

Request URL

GET /ISAPI/Smart/AudioDetection/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, audio detection parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <audioInputException>
    <!--ro, opt, object, audio input exception detection-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <minVolumeThreshold>
      <!--ro, opt, int, minimum volume threshold, range:[0,99], unit:dB, desc:Check whether this parameter is supported by devices via the field
minVolumeThreshold (related API: /ISAPI/Smart/AudioDetection/channels/<channelID>/capabilities). 1. When devices support this parameter, they can check
whether the inputted audio is normal by comparing the inputted audio decibel value and the threshold. 2. When devices do not support this parameter, they
can check whether the inputted audio is normal by the background noise change.-->1
    </minVolumeThreshold>
  </audioInputException>
  <soundIntensityMutation>
    <!--ro, opt, object, sudden change of sound intensity detection-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, req, int, sensitivity level, range:[1,100]-->0
    </sensitivityLevel>
    <mutationThreshold>
      <!--ro, req, int, intensity threshold, range:[1,100]-->0
    </mutationThreshold>
    <decibel>
      <!--ro, opt, int, sound intensity, range:[0,65535], unit:dB-->1
    </decibel>
  </soundIntensityMutation>
  <SteepFall>
    <!--ro, opt, object, sudden decrease of sound intensity-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, req, int, sensitivity level, range:[1,100]-->0
    </sensitivityLevel>
  </SteepFall>
  <AudioLoss>
    <!--ro, opt, object, audio loss detection-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, opt, int, sensitivity level, range:[0,100]-->0
    </sensitivityLevel>
  </AudioLoss>
  <audioMode>
    <!--ro, opt, enum, audio detection mode, subType:int, desc:0 (sensitivity detection mode), 1 (threshold detection mode), 2 (sensitivity detection and
threshold detection modes), 3 (both sensitivity and threshold detection are disabled)-->0
  </audioMode>
  <decibelThreshold>
    <!--ro, opt, int, decibel threshold, range:[-200,10000], step:1, unit:dB-->0
  </decibelThreshold>
  <decibelThresholdDuration>
    <!--ro, opt, float, decibel threshold duration, range:[0,10000], step:0.01, unit:dB-->0
  </decibelThresholdDuration>
  <frequencyThresholdDetectionParams>
    <!--ro, opt, object, frequency threshold detection parameters-->
    <enabled>
      <!--ro, req, bool, whether to enable frequency threshold detection-->true
    </enabled>
    <frequencyThreshold>
      <!--ro, opt, int, frequency threshold, range:[0,96000], step:1, unit:dB-->0
    </frequencyThreshold>
    <frequencyThresholdDuration>
      <!--ro, opt, float, frequency threshold duration, range:[0,10000], step:0.01, unit:dB-->0
    </frequencyThresholdDuration>
  </frequencyThresholdDetectionParams>
</AudioDetection>
```

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19.17.1.8 Set audio detection parameters of a channel

Request URL

PUT /ISAPI/Smart/AudioDetection/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, audio detection parameter, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->test
  </id>
  <audioInputException>
    <!--opt, object, audio input exception detection-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <minVolumeThreshold>
      <!--opt, int, minimum volume threshold, range:[0,99], unit:dB, desc:Check whether this parameter is supported by devices via the field
minVolumeThreshold (related API: /ISAPI/Smart/AudioDetection/channels/<channelID>/capabilities). 1. When devices support this parameter, they can check
whether the inputted audio is normal by comparing the inputted audio decibel value and the threshold. 2. When devices do not support this parameter, they
can check whether the inputted audio is normal by the background noise change.-->1
    </minVolumeThreshold>
  </audioInputException>
  <soundIntensityMutation>
    <!--opt, object, sudden change of sound intensity detection-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, sensitivity level, range:[1,100]-->0
    </sensitivityLevel>
    <mutationThreshold>
      <!--req, int, intensity threshold, range:[1,100]-->0
    </mutationThreshold>
    <decibel>
      <!--opt, int, sound intensity, range:[0,65535], unit:dB-->1
    </decibel>
  </soundIntensityMutation>
  <SteepFall>
    <!--opt, object, sudden decrease of sound intensity-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, sensitivity level, range:[1,100]-->0
    </sensitivityLevel>
  </SteepFall>
  <AudioLoss>
    <!--opt, object, audio loss detection-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <sensitivityLevel>
      <!--opt, int, sensitivity level, range:[0,100]-->0
    </sensitivityLevel>
  </AudioLoss>
  <audioMode>
    <!--opt, enum, sound detection mode, subType:int, desc:sound detection mode-->0
  </audioMode>
  <decibelThreshold>
    <!--opt, int, decibel threshold, range:[-200,10000], step:1, unit:dB-->0
  </decibelThreshold>
  <decibelThresholdDuration>
    <!--opt, float, decibel threshold duration, range:[0,10000], step:0.01, unit:dB-->0
  </decibelThresholdDuration>
  <frequencyThresholdDetectionParams>
    <!--opt, object, frequency threshold detection parameters-->
    <enabled>
      <!--req, bool, whether to enable frequency threshold detection-->true
    </enabled>
    <frequencyThreshold>
      <!--opt, int, frequency threshold, range:[0,96000], step:1, unit:dB-->0
    </frequencyThreshold>
    <frequencyThresholdDuration>
      <!--opt, float, frequency threshold duration, range:[0,10000], step:0.01, unit:dB-->0
    </frequencyThresholdDuration>
  </frequencyThresholdDetectionParams>
</AudioDetection>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

19.17.1.9 Get the capability of audio detection parameters of a specified channel

Request URL

GET /ISAPI/Smart/AudioDetection/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AudioDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, audio detection parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <audioInputException>
    <!--ro, opt, object, audio input exception detection-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <minVolumeThreshold min="0" max="99">
      <!--ro, opt, int, minimum volume threshold, range:[0,99], unit:dB, attr:min{req, int},max{req, int}, desc:1. When devices support this parameter, they
can check whether the inputted audio is normal by comparing the inputted audio decibel value and the threshold. 2. When devices do not support this
parameter, they can check whether the inputted audio is normal by the background noise change.-->1
    </minVolumeThreshold>
  </audioInputException>
  <soundIntensityMutation>
    <!--ro, opt, object, sudden change of sound intensity detection-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <sensitivityLevel min="1" max="100">
      <!--ro, req, int, sensitivity level, range:[1,100], attr:min{req, int},max{req, int}-->0
    </sensitivityLevel>
    <mutationThreshold min="1" max="100">
      <!--ro, opt, int, sound intensity threshold, range:[1,100], attr:min{req, int},max{req, int}-->0
    </mutationThreshold>
    <decibel min="0" max="65535">
      <!--ro, opt, int, sound intensity, range:[0,65535], unit:dB, attr:min{req, int},max{req, int}-->1
    </decibel>
  </soundIntensityMutation>
  <SteepFall>
    <!--ro, opt, object, sudden decrease of sound intensity-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
```

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```
      </enabled>
      <sensitivityLevel min="1" max="100">
        <!--ro, req, int, sensitivity level, range:[1,100], attr:min{req, int},max{req, int}-->0
      </sensitivityLevel>
    </SteepFall>
    <mutexAbility opt="PDC">
      <!--ro, opt, enum, mutual exclusion ability, subType:string, attr:opt{req, string}, desc:"PDC" (people counting)-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--ro, opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
    <AudioLoss>
      <!--ro, opt, object, audio loss detection-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
      </enabled>
      <sensitivityLevel min="0" max="100" def="50">
        <!--ro, opt, int, sensitivity level, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->0
      </sensitivityLevel>
    </AudioLoss>
    <audioMode opt="1">
      <!--ro, opt, enum, audio detection mode, subType:int, attr:opt{req, int}, desc:0 (sensitivity detection mode), 1 (threshold detection mode), 2
  (sensitivity detection and threshold detection modes), 3 (both sensitivity and threshold detection are disabled)-->0
    </audioMode>
    <decibelThreshold min="-200" max="10000">
      <!--ro, opt, int, decibel threshold, range:[-200,10000], step:1, unit:dB, attr:min{req, int},max{req, int}-->0
    </decibelThreshold>
    <decibelThresholdDuration min="0" max="10000">
      <!--ro, opt, float, decibel threshold duration, range:[0,10000], step:0.01, unit:dB, attr:min{req, int},max{req, int}-->0
    </decibelThresholdDuration>
    <frequencyThresholdDetectionParams>
      <!--ro, opt, object, frequency threshold detection parameters-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable frequency threshold detection, attr:opt{req, string}-->true
      </enabled>
      <frequencyThreshold min="0" max="96000">
        <!--ro, opt, int, frequency threshold, range:[0,96000], step:1, unit:dB, attr:min{req, int},max{req, int}-->0
      </frequencyThreshold>
      <frequencyThresholdDuration min="0" max="10000">
        <!--ro, opt, float, frequency threshold duration, range:[0,10000], step:0.01, unit:dB, attr:min{req, int},max{req, int}-->0
      </frequencyThresholdDuration>
    </frequencyThresholdDetectionParams>
    <cryDetectionParams>
      <!--ro, opt, object, cry detection parameters-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable cry detection, attr:opt{req, string}-->false
      </enabled>
      <sensitivityLevel min="0" max="100" def="50">
        <!--ro, opt, int, sensitivity level, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->50
      </sensitivityLevel>
    </cryDetectionParams>
    <audioDetectionMode opt="lowFalsePositive,highDetection">
      <!--ro, opt, enum, audio detection mode, subType:string, attr:opt{req, string}, desc:"lowFalsePositive" (low false alarm mode, the default value),
  "highDetection" (high detection mode)-->lowFalsePositive
    </audioDetectionMode>
  </AudioDetection>
```

19.17.1.10 Get the audio strength

Request URL

GET /ISAPI/Smart/AudioDetection/channels/<channelID>/status

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<AudioStrengthStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index-->test
  </id>
  <audioStrength>
    <!--ro, req, int, sound intensity-->0
  </audioStrength>
</AudioStrengthStatus>
```

19.17.1.11 Audio exception

EventType:audioexception

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event uploading, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers an alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers an alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, transmission communication protocol type, range:[0,64], desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <audioInID>
    <!--ro, opt, int-->1
  </audioInID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"audioexception" (audio exception)-->audioexception
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <AudioExceptionDetection>
    <!--ro, opt, object, information about audio exception-->
    <alarmType>
      <!--ro, req, enum, alarm type, subType:string, desc:"audioLost" (audio loss), "decrease" (sudden decrease of sound intensity), "increase" (sudden
increase of sound intensity), "audioDecibel" (decibel is higher than the threshold), "frequency" (frequency is higher than the threshold),
"SoundSourceLocation" (sound source location), "realAudio" (real-time audio), "gasLeakageDetection" (gas leakage detection event), "dischargeDetection"
(discharge detection event)-->audioLost
    </alarmType>
```

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```
    <audioDecibel>
      <!--ro, opt, int, audio decibel, range:[0,100], desc:it is valid when alarmType is "decrease" or "increase"-->0
    </audioDecibel>
    <frequency>
      <!--ro, opt, int, audio frequency, range:[0,100], step:0.001, unit:kHz-->0
    </frequency>
    <gasLeakageDetectionInfo>
      <!--ro, opt, object, information about gas leakage detection event, dep:and,
{$.EventNotificationAlert.AudioExceptionDetection.alarmType,eq,gasLeakageDetection}-->
      <leakageLevel>
        <!--ro, opt, int, leakage level, range:[1,10], desc:the higher the grade the more severe the leakage-->1
      </leakageLevel>
      <leakMode>
        <!--ro, opt, enum, leakage mode, subType:string, desc:"filling" (filling gas), "compressed" (compressed gas)-->filling
      </leakMode>
      <gasLeakage>
        <!--ro, opt, float, gas leakage rate, range:[0,9999999], desc:the gas leakage rate unit is configured via the field volumUnit in the URL of
configuring gas leakage rules (PUT /ISAPI/System/AcousticLeakDetection/AudioIn/<AudioInID>/GasLeakageRuleParams?format=json)-->0.000
      </gasLeakage>
      <leakageLoss>
        <!--ro, opt, float, leakage loss, range:[0,9999999], desc:Leakage loss = leakage rate * unit price * leakage time duration. The configurations for
leakage rate, unit price, and leakage time unit correspond to fields: "volumUnit" under "leakageUnit", "gasUnitPrice" under "fillingParam", and "timeUnit"
under "leakageUnit". The relative URL is (PUT /ISAPI/System/AcousticLeakDetection/AudioIn/<AudioInID>/GasLeakageRuleParams?format=json)-->0.000
      </leakageLoss>
      <CO2Emissions>
        <!--ro, opt, float, carbon dioxide emission, range:[0.0001,99.9999], desc:unit: kg/leakage time. The configuration for leakage time unit corresponds
to the field "timeUnit" under "leakageUnit" in the URL (PUT /ISAPI/System/AcousticLeakDetection/AudioIn/<AudioInID>/GasLeakageRuleParams?format=json)--
>0.0001
      </CO2Emissions>
    </gasLeakageDetectionInfo>
    <dischargeDetectionInfo>
      <!--ro, opt, object, discharge detection event information, dep:and,
{$.EventNotificationAlert.AudioExceptionDetection.alarmType,eq,dischargeDetection}-->
      <dischargeType>
        <!--ro, opt, enum, discharge detection type, subType:string, desc:"freeMetalParticle" (free metal particle discharge), "suspendedPotentiometer"
(suspended potential discharge), "internalAirGapOfInsulatingParts" (insulating part internal air-gap discharge), "metalTip" (metal tip discharge), "noise"
(noise)-->freeMetalParticle
      </dischargeType>
      <dischargeProbability>
        <!--ro, opt, float, discharge probability, range:[0,1], desc:it is analyzed by device's algorithm-->1
      </dischargeProbability>
    </dischargeDetectionInfo>
  </AudioExceptionDetection>
  <ResourcesName>
    <!--ro, opt, object, resource transmission module, desc:when the returned alarm message contains extra resource (e.g., picture), this node should be
returned. ResourcesName is the specific name of the resource, which has no limitation. For example, it is FaceCaptureBackGroudImage for face capture
background picture. When this node is absent, it defaults to uploading pictures via form submission-->
    <resourcesContentType>
      <!--ro, opt, enum, resource transmission type, subType:string, desc:when the returned alarm message contains additional resource (e.g., pictures),
this field should be returned. Authentication is required when the URL is used to download files from device, and the method is the same with that in device
login. Direct request is allowed if the URL parameter contains token(?token=). For other ways, the authentication method is the same with that in device
login. If the returned URL contains the IP and port, the device is only accessible under a direct network connection, the platform should replace the IP and
port in the URL based on the device's network configuration. If the prefix of the returned URL is "/pic", it represents cloud storage. The custom URL of
device does not contain this prefix. "url" (the resource is transmitted via URL), "binary" (the resource is transmitted in binary format)-->url
    </resourcesContentType>
    <resourcesContent>
      <!--ro, opt, string, resource ID, desc:when the returned alarm message contains extra resource (e.g., picture), this node should be returned. If
resourcesContentType==url, this node is the specific picture URL-->test
    </resourcesContent>
    <resourcesFormatType>
      <!--ro, opt, enum, resource file format, subType:string, desc:"JPG", "PNG". When the returned alarm message contains binary or URL resources, this
field should be returned to specify the resource file format type-->JPG
    </resourcesFormatType>
    <pictureResolution>
      <!--ro, opt, object, picture resource resolution, desc:it is valid only when the resource type is picture-->
      <height>
        <!--ro, req, int, resolution height, range:[1,65535], unit:px-->1
      </height>
      <width>
        <!--ro, req, int, resolution (width), range:[1,65535], unit:px-->1
      </width>
    </pictureResolution>
  </ResourcesName>
</EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
AudioException.xml	[Message content]	application/xml	--	--	--
audioExceptionPic	[Binary picture data]	image/jpeg			--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

VISION AND SECURITY SYSTEM CO., LTD. dg.jg.110@gmail.com

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.18 Entrance & Exit Management and Control

19.18.1 Vehicle List Comparison

19.18.1.1 Get the arming schedule of vehicle allowlist detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/blackList/blackList-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->blackList-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->blackList
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.18.1.2 Set the arming schedule of vehicle allowlist detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/blackList/blackList-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->blackList-1
  </id>
  <eventType>
    <!--opt, string-->blackList
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

19.18.1.3 Set the arming schedule of vehicle allowlist detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/whiteList/whiteList-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->whiteList-1
  </id>
  <eventType>
    <!--opt, string-->whiteList
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

19.18.1.4 Get the arming schedule of vehicle allowlist detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/whiteList/whiteList-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->whiteList-1
  </id>
  <eventType>
    <!--ro, opt, string-->whiteList
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

19.18.1.5 Get the linkage parameters of vehicle blocklist detection

Request URL

GET /ISAPI/Event/triggers/blackList-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

VISION AND SECURITY SYSTEM CO.,LTD. dojojin@gmail.com

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->blackList
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <direction>
    <!--ro, opt, enum, subType:string-->both
  </direction>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,IO}-->1
      </outputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <wiegandOutputID>
        <!--ro, opt, int, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,wiegand}-->1
      </wiegandOutputID>
      <triggerDirection>
        <!--ro, opt, enum, subType:string-->forward
      </triggerDirection>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

19.18.1.6 Set the linkage parameters of vehicle blocklist detection

Request URL

PUT /ISAPI/Event/triggers/blackList-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->blackList
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <direction>
    <!--opt, enum, subType:string-->both
  </direction>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->IO-1
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,IO}-->1
      </outputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <wiegandOutputID>
        <!--opt, int, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,wiegand}-->1
      </wiegandOutputID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.18.1.7 Get the linkage parameters of vehicle allowlist

Request URL

GET /ISAPI/Event/triggers/whitelist-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->whitelist
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <direction>
    <!--ro, opt, enum, subType:string-->both
  </direction>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,IO}-->1
      </outputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <wiegandOutputID>
        <!--ro, opt, int, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,wiegand}-->1
      </wiegandOutputID>
      <triggerDirection>
        <!--ro, opt, enum, subType:string-->forward
      </triggerDirection>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

dojojin@gmail.com VISION AND SECURITY SYSTEM CO.,LTD.

19.18.1.8 Set the linkage parameters of vehicle allowlist

Request URL

PUT /ISAPI/Event/triggers/whitelist-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->whitelist
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <direction>
    <!--opt, enum, subType:string-->both
  </direction>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,IO}-->1
      </outputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <wiegandOutputID>
        <!--opt, int, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,wiegand}-->1
      </wiegandOutputID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

19.18.1.9 Get the capability of deleting the license plate information in the blocklist or allowlist

Request URL

GET /ISAPI/Traffic/channels/<channelID>/DelLicensePlateAuditData/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

dojojin@gmail.com
VISION AND SECURITY SYSTEM CO.,LTD.

```
{
    "id": {
    /*ro, req, object, ID list*/
        "@size":  10000
        /*ro, req, int, the maximum number of members in the array (the maximum number of IDs in the list)*/
    },
    "licensePlate": {
    /*ro, opt, object*/
        "@size":  1
        /*ro, req, int, the maximum number of members in the array (the maximum number of IDs in the list)*/
    },
    "cardNo": {
    /*ro, opt, object*/
        "@size":  1
        /*ro, req, int, the maximum number of members in the array (the maximum number of IDs in the list)*/
    },
    "deleteAllEnabled": {
    /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, req, array, subType:bool*/
    },
    "plateColor": {
    /*ro, opt, object*/
        "@size":  1,
        /*ro, req, int*/
        "@opt": ["black", "blue", "civilAviationBlack", "civilAviationGreen", "golden", "green", "mixedColor", "newEnergyGreen", "newEnergyYellowGreen",
"orange", "other", "red", "unknown", "white", "yellow"]
        /*ro, opt, array, options, subType:string*/
    },
    "plateType": {
    /*ro, opt, object*/
        "@size":  1,
        /*ro, req, int*/
        "@opt": ["02TypePersonalized", "04NewMilitay", "92FarmVehicle", "92TypeArm", "92TypeCivil", "arm", "civilAviation", "coach", "consulate", "embassy",
"emergency", "green1325FarmVehicle", "hongKongMacao", "leftRightMilitay", "motorola", "newEnergy", "oneLineArm", "oneLineArmHeadquarters", "tempEntry",
"tempTravl", "trailer", "twoLineArm", "twoLineArmHeadquarters", "unknown", "upDownMilitay", "yellow1225FarmVehicle", "yellow1325FarmVehicle",
"yellowTwoLine"]
        /*ro, opt, array, options, subType:string*/
    },
    "name": {
    /*ro, opt, object*/
        "@size":  1
        /*ro, req, int*/
    },
    "certificateType": {
    /*ro, opt, object*/
        "@size":  1,
        /*ro, req, int*/
        "@opt": ["officerID", "ID", "passportID", "other"]
        /*ro, opt, array, subType:string*/
    },
    "certificateNumber": {
    /*ro, opt, object*/
        "@size":  1
        /*ro, req, int*/
    },
    "virtualParkingNum": {
    /*ro, opt, object*/
        "@size":  1
        /*ro, req, int*/
    },
    "CompoundCond": {
    /*ro, opt, object*/
        "@size":  1,
        /*ro, req, int*/
        "licensePlate": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  32
            /*ro, opt, int*/
        },
        "plateColor": {
        /*ro, opt, object*/
            "@opt": ["black", "blue", "civilAviationBlack", "civilAviationGreen", "golden", "green", "mixedColor", "newEnergyGreen", "newEnergyYellowGreen",
"orange", "other", "red", "unknown", "white", "yellow"]
            /*ro, opt, array, options, subType:string*/
        }
    }
}
```

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19.18.1.10 Delete the license plate information in the blocklist or allowlist

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/DelLicensePlateAuditData?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "id": ["1", "2", "3"],
    /*opt, array, subType:string*/
    "licensePlate": ["浙A12345", "浙A12346", "浙A12347"],
    /*opt, array, subType:string, range:[1,32]*/
    "cardNo": ["abcd1234", "abcd12345", "abcd123456"],
    /*opt, array, subType:string, range:[1,64]*/
    "deleteAllEnabled": true,
    /*opt, bool*/
    "plateColor": ["black", "blue", "civilAviationBlack", "civilAviationGreen", "golden", "green", "mixedColor", "newEnergyGreen", "newEnergyYellowGreen",
"orange", "other", "red", "unknown", "white", "yellow"],
    /*opt, enumarray, license plate color, subType:string, desc:"black", "blue", "civilAviationBlack" (civil aviation black), "civilAviationGreen" (civil
aviation green), "golden", "green", "mixedColor" (mixed color), "newEnergyGreen" (new energy green), "newEnergyYellowGreen" (new energy green and yellow),
"orange", "other" (other color), "red", "unknown", "white", "yellow"*/
    "plateType": ["02TypePersonalized", "04NewMilitay", "92FarmVehicle", "92TypeArm", "92TypeCivil", "arm", "civilAviation", "coach", "consulate",
"embassy", "emergency", "green1325FarmVehicle", "hongKongMacao", "leftRightMilitay", "motorola", "newEnergy", "oneLineArm", "oneLineArmHeadquarters",
"tempEntry", "tempTravl", "trailer", "twoLineArm", "twoLineArmHeadquarters", "unknown", "upDownMilitay", "yellow1225FarmVehicle", "yellow1325FarmVehicle",
"yellowTwoLine"],
    /*opt, enumarray, license plate type, subType:string, desc:"02TypePersonalized" (type 02 custom vehicle), "92FarmVehicle" (two-line license plate civil
vehicle), "92TypeArm" (type 92 armed police vehicle), "92TypeCivil" (type 92 civil vehicle), "arm" (police vehicle), "civilAviation" (civil aviation license
plate), "coach" (driver-training vehicle), "consulate" (consular vehicle), "embassy" (embassy vehicle), "emergency" (emergency license plate),
"green1325FarmVehicle" (agricultural vehicle (green 1325)), "hongKongMacao" (Hong Kong/Macao entrance vehicle), "motorola" (motorcycle), "newEnergy" (new
energy vehicle license plate), "oneLineArm" (new armed police vehicle (one-line)), "oneLineArmHeadquarters" (armed police headquarter license plate (one-
line)), "tempEntry" (temporary entry car), "tempTravl" (temporary license plate car), "trailer", "twoLineArm" (new armed police vehicle (two-line)),
"twoLineArmHeadquarters" (armed police headquarter license plate (two-line)), "unknown", "yellow1225FarmVehicle" (agricultural vehicle (yellow 1225)),
"yellow1325FarmVehicle" (agricultural vehicle (yellow 1325)), "yellowTwoLine" (yellow two-line rear license plate)*/
    "name": ["张三", "李四"],
    /*opt, array, subType:string, range:[1,64]*/
    "certificateType": ["officerID", "ID", "passportID", "other"],
    /*ro, opt, enumarray, subType:string*/
    "certificateNumber": ["1234", "5678"],
    /*ro, opt, array, subType:string*/
    "virtualParkingNum": ["123", "456"],
    /*ro, opt, array, subType:string*/
    "CompoundCond": [
    /*opt, array, subType:object*/
        {
            "licensePlate": "浙A12345",
            /*opt, string, range:[1,32]*/
            "plateColor": "black"
            /*opt, enum, license plate color, subType:string, desc:"black", "blue", "civilAviationBlack" (civil aviation black), "civilAviationGreen" (civil
aviation green), "golden", "green", "mixedColor" (mixed color), "newEnergyGreen" (new energy green), "newEnergyYellowGreen" (new energy green and yellow),
"orange", "other" (other color), "red", "unknown", "white", "yellow"*/
        }
    ]
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

19.18.1.11 Get the capability of adding license plate information to the blocklist or allowlist, or editing the license plate information in the blocklist or allowlist

Request URL

GET /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData/record/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

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Request Message

None

Response Message

```
{
    "LicensePlateInfoCap": {
    /*ro, req, object, list of license plates to be added to the list*/
        "id": {
        /*ro, req, object, index, desc:the value of this node is the ID of the license plate information, and the current ID can be obtained by calling the URI /ISAPI/Traffic/channels/<channelID>/searchLPListAudit*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        },
        "listType": {
        /*ro, opt, object, list type, desc:"blockList", "allowList"*/
            "@opt": ["blockList", "allowList"]
            /*ro, opt, array, subType:string*/
        },
        "licensePlate": {
        /*ro, req, object, license plate number*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  32
            /*ro, opt, int*/
        },
        "cardID": {
        /*ro, opt, object, card ID (Wiegand protocol), desc:object,card ID (Wiegand protocol),which consists of 8 decimal digits*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  9
            /*ro, req, int*/
        },
        "createTime":  "1970-01-01T00:00:00+08:00",
        /*ro, opt, datetime, created time*/
        "effectiveTime":  "1970-01-01T00:00:00+08:00",
        /*ro, opt, datetime, effective time*/
        "effectiveStartDate":  "1970-01-01",
        /*ro, opt, date, start date of the validity period*/
        "cardNo": {
        /*ro, opt, object, card No.*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  64
            /*ro, req, int*/
        },
        "plateColor": {
        /*ro, req, object*/
            "@opt": ["black", "blue", "civilAviationBlack", "civilAviationGreen", "golden", "green", "mixedColor", "newEnergyGreen", "newEnergyYellowGreen", "orange", "other", "red", "unknown", "white", "yellow"]
            /*ro, opt, array, subType:string*/
        },
        "plateType": {
        /*ro, req, object*/
            "@opt": ["02TypePersonalized", "04NewMilitay", "92FarmVehicle", "92TypeArm", "92TypeCivil", "arm", "civilAviation", "coach", "consulate", "embassy", "emergency", "green1325FarmVehicle", "hongKongMacao", "leftRightMilitay", "motorola", "newEnergy", "oneLineArm", "oneLineArmHeadquarters", "tempEntry", "tempTravl", "trailer", "twoLineArm", "twoLineArmHeadquarters", "unknown", "upDownMilitay", "yellow1225FarmVehicle", "yellow1325FarmVehicle", "yellowTwoLine"]
            /*ro, opt, array, subType:string*/
        },
        "operationType": {
        /*ro, opt, object*/
            "@opt": ["add", "modify"]
            /*ro, opt, array, subType:string*/
        },
        "name": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  32
            /*ro, req, int*/
        },
        "certificateType": {
        /*ro, req, object*/
            "@opt": ["officerID", "ID", "passportID", "other"]
            /*ro, opt, array, subType:string*/
        },
        "certificateNumber": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  32
            /*ro, req, int*/
        },
        "virtualParkingNum": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  16
```

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```
            "@max":  16
            /*ro, req, int*/
        },
        "plateDescription": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int*/
            "@max":  128
            /*ro, req, int*/
        },
        "groupName": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  64
            /*ro, req, int*/
        }
    }
}
```


19.18.1.12 Add license plate information to the blocklist or allowlist, or edit the license plate information in the blocklist or allowlist

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData/record?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message


```
{
    "LicensePlateInfoList": [
    /*req, array, subType:object*/
        {
            "id":  "1",
            /*req, string, ID, range:[1,16], desc:index,the maximum string size is 16 bytes. The value of this node is the ID of the license plate
information,and the current ID can be obtained by calling the URI /ISAPI/Traffic/channels/<channelID>/searchLPListAudit*/
            "listType":  "blockList",
            /*req, enum, list type, subType:string, desc:list type: "blockList","allowList"*/
            "LicensePlate":  "test",
            /*req, string, license plate No., range:[1,32]*/
            "createTime":  "1970-01-01T00:00:00+08:00",
            /*opt, datetime, creation time*/
            "effectiveTime":  "1970-01-01T00:00:00+08:00",
            /*opt, datetime, end time of the validity period*/
            "cardID":  "test",
            /*opt, string, card ID (Wiegand protocol), range:[1,9], desc:card ID by Wiegand rule,which is a number consisting of 8 digits in decimal format.
The maximum string size is 9 bytes*/
            "effectiveStartDate":  "1970-01-01",
            /*opt, string, start date of the validity period*/
            "cardNo":  "abcd1234",
            /*opt, string, card No., range:[1,64]*/
            "plateColor":  "blue",
            /*req, enum, license plate color, subType:string, desc:license plate color*/
            "plateType":  "92TypeCivil",
            /*req, enum, license plate type, subType:string, desc:license plate type*/
            "operationType":  "add",
            /*opt, string*/
            "name":  "张三",
            /*ro, opt, string, range:[0,96]*/
            "certificateType":  "officerID",
            /*ro, opt, enum, subType:string*/
            "certificateNumber":  "test",
            /*ro, opt, string, range:[0,32]*/
            "virtualParkingNum":  "123",
            /*opt, string, range:[1,16]*/
            "plateDescription":  "abc",
            /*opt, string, range:[0,128]*/
            "groupName":  "group1"
            /*opt, string, range:[1,64]*/
        }
    ]
}
```


Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

19.18.1.13 Update the vehicle blocklist and allowlist file

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData?fileType=<fileType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
fileType	enum	When "FileType" is xml, it is an XML file. If no "fileType" field or "fileType" is excel, it is an Excel file of XLS format. When "fileType" is csv, it is a Excel file of CSV format.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <AttachInfo>
    <!--ro, opt, object-->
    <StatusList>
      <!--ro, req, object-->
      <Status>
        <!--ro, req, object-->
        <index>
          <!--ro, req, int-->0
        </index>
        <statusCode>
          <!--ro, req, enum, subType:int-->0
        </statusCode>
        <statusString>
          <!--ro, req, enum, subType:string-->OK
        </statusString>
        <subStatusCode>
          <!--ro, req, string, range:[1,64]-->ok
        </subStatusCode>
        <description>
          <!--ro, opt, string, range:[0,1024]-->test
        </description>
        <plateNo>
          <!--ro, opt, string, range:[1,64]-->test
        </plateNo>
        <licensePlateListType>
          <!--ro, opt, enum, subType:string-->allowlist
        </licensePlateListType>
      </Status>
    </StatusList>
  </AttachInfo>
  <successNum>
    <!--ro, opt, int-->100
  </successNum>
</ResponseStatus>
```

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19.18.1.14 Export the file of vehicle blocklist and allowlist data

Request URL

GET /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData?fileType=<fileType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
fileType	enum	When "FileType" is xml, it is an XML file. If no "fileType" field or "fileType" is excel, it is an Excel file of XLS format. When "fileType" is csv, it is a Excel file of CSV format.

Request Message

None

Response Message

None

19.18.1.15 Get the capability set of searching for license plate information in the allowlist and blocklist

Request URL

GET /ISAPI/Traffic/channels/<channelID>/searchLPListAudit/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LPListAuditSearchCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the capability set of the allowlist and blocklist of license plate numbers, attr:version{req, string}-->
  <LPListAuditSearchResultCap>
    <!--ro, opt, object, the capability of searching for license plate numbers in the allowlist or blocklist-->
    <isSupportId>
      <!--ro, req, bool, whether it supports searching by ID-->true
    </isSupportId>
    <isSupportLicensePlate>
      <!--ro, opt, bool, whether it supports searching by license plate number-->true
    </isSupportLicensePlate>
    <isSupportType>
      <!--ro, opt, bool, whether it supports searching by list type-->true
    </isSupportType>
    <isSupportCreateTime>
      <!--ro, opt, bool, whether it supports searching by the time that the list was created-->true
    </isSupportCreateTime>
    <isSupportPlateCategory>
      <!--ro, opt, bool, whether it supports searching by plate type-->true
    </isSupportPlateCategory>
    <isSupportEffectiveTime>
      <!--ro, opt, bool, whether it supports searching by the effective time of the list-->true
    </isSupportEffectiveTime>
    <supportType opt="blackList,whitelist">
      <!--ro, opt, enum, supported list type, subType:string, attr:opt{req, string}, desc:blocklist, allowlist-->blackList
    </supportType>
    <isSupportCardNo>
      <!--ro, opt, bool, whether it supports searching by license plate number-->true
    </isSupportCardNo>
    <supportPlateColor
opt="black,blue,civilAviationBlack,civilAviationGreen,golden,green,mixedColor,newEnergyGreen,newEnergyYellowGreen,orange,other,red,unknown,white,yellow">
      <!--ro, opt, enum, supported plate color, subType:string, attr:opt{req, string}, desc:black, blue, civilAviationBlack (civil aviation black),
civilAviationGreen (civil aviation green), golden, green, mixedColor (mixed color), newEnergyGreen (new energy green), newEnergyYellowGreen (new energy
green and yellow), orange, other (other color), red, unknown, white, yellow-->unknown
    </supportPlateColor>
    <supportPlateCategory
opt="02TypePersonalized,04NewMilitay,92FarmVehicle,92TypeArm,92TypeCivil,arm,civilAviation,coach,consulate,embassy,emergency,green1325FarmVehicle,hongKongMa
cao,leftRightMilitay,motorola,newEnergy,oneLineArm,oneLineArmHeadquarters,tempEntry,tempTravl,trailer,twoLineArm,twoLineArmHeadquarters,unknown,upDownMilita
y,yellow1225FarmVehicle,yellow1325FarmVehicle,yellowTwoLine">
      <!--ro, opt, enum, supported plate type, subType:string, attr:opt{req, string}, desc:02TypePersonalized (type 02 custom vehicle), 92FarmVehicle (two-
line license plate civil vehicle), 92TypeArm (type 92 armed police vehicle), 92TypeCivil (type 92 civil vehicle), arm (police vehicle), civilAviation (civil
aviation license plate), coach (driver-training vehicle), consulate (consular vehicle), embassy (embassy vehicle), emergency (emergency license plate),
green1325FarmVehicle (agricultural vehicle (green 1325)), hongKongMacao (Hong Kong China/Macao China entrance vehicle), motorola (motorcycle), newEnergy
(new energy vehicle license plate), oneLineArm (new armed police vehicle (one-line)), oneLineArmHeadquarters (armed police headquarter license plate (one-
line)), tempEntry (temporary entry car), tempTravl (temporary license plate car), trailer, twoLineArm (new armed police vehicle (two-line)),
twoLineArmHeadquarters (armed police headquarter license plate (two-line)), unknown, yellow1225FarmVehicle (agricultural vehicle (yellow 1225)),
yellow1325FarmVehicle (agricultural vehicle (yellow 1325)), yellowTwoLine (yellow two-line rear license plate)-->unknown
    </supportPlateCategory>
    <isSupportCardID>
      <!--ro, opt, bool, whether it supports searching by card ID, desc:If the device supports searching by Wiegand protocol (related URI:
/ISAPI/System/Wiegand/<ID>?format=json), the Wiegand protocol ID will be returned during search-->true
    </isSupportCardID>
    <isSupportEffectiveStartDate>
      <!--ro, opt, bool, whether it supports searching by the effective time of the list-->true
    </isSupportEffectiveStartDate>
    <plateDescription min="0" max="128">
      <!--ro, opt, string, range:[0,128], attr:min{req, int},max{req, int}-->abc
    </plateDescription>
  </LPListAuditSearchResultCap>
  <LPListAuditSearchDescriptionCap>
    <!--ro, opt, object, the capability of search description-->
    <supportType opt="blackList,whitelist">
      <!--ro, opt, enum, supported list type, subType:string, attr:opt{req, string}, desc:blocklist, allowlist-->blackList
    </supportType>
    <isSupportLicensePlate>
      <!--ro, opt, bool, whether it supports searching by license plate number-->true
    </isSupportLicensePlate>
    <isSupportCardNo>
      <!--ro, opt, bool, whether it supports searching by card No.-->true
    </isSupportCardNo>
    <isSupportCardID>
      <!--ro, opt, bool, whether it supports searching by card ID-->true
```

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```
    </isSupportCardID>
    <isSupportEffectiveTime>
      <!--ro, opt, bool, whether it supports searching by the effective time of the list-->true
    </isSupportEffectiveTime>
    <plateColor
opt="black,blue,civilAviationBlack,civilAviationGreen,golden,green,mixedColor,newEnergyGreen,newEnergyYellowGreen,orange,other,red,unknown,white,yellow">
      <!--ro, opt, enum, supported plate color, subType:string, attr:opt{req, string}, desc:black, blue, civilAviationBlack (civil aviation black),
civilAviationGreen (civil aviation green), golden, green, mixedColor (mixed color), newEnergyGreen (new energy green), newEnergyYellowGreen (new energy
green and yellow), orange, other (other color), red, unknown, white, yellow-->unknown
    </plateColor>
    <plateType
opt="02TypePersonalized,04NewMilitay,92FarmVehicle,92TypeArm,92TypeCivil,arm,civilAviation,coach,consulate,embassy,emergency,green1325FarmVehicle,hongKongMa
cao,leftRightMilitay,motorola,newEnergy,oneLineArm,oneLineArmHeadquarters,tempEntry,tempTravl,trailer,twoLineArm,twoLineArmHeadquarters,unknown,upDownMilita
y,yellow1225FarmVehicle,yellow1325FarmVehicle,yellowTwoLine">
      <!--ro, opt, enum, supported plate type, subType:string, attr:opt{req, string}, desc:02TypePersonalized (type 02 custom vehicle), 92FarmVehicle (two-
line license plate civil vehicle), 92TypeArm (type 92 armed police vehicle), 92TypeCivil (type 92 civil vehicle), arm (police vehicle), civilAviation (civil
aviation license plate), coach (driver-training vehicle), consulate (consular vehicle), embassy (embassy vehicle), emergency (emergency license plate),
green1325FarmVehicle (agricultural vehicle (green 1325)), hongKongMacao (Hong Kong/Macao entrance vehicle), motorola (motorcycle), newEnergy (new energy
vehicle license plate), oneLineArm (new armed police vehicle (one-line)), oneLineArmHeadquarters (armed police headquarter license plate (one-line)),
tempEntry (temporary entry car), tempTravl (temporary license plate car), trailer, twoLineArm (new armed police vehicle (two-line)), twoLineArmHeadquarters
(armed police headquarter license plate (two-line)), yellow1225FarmVehicle (agricultural vehicle (yellow 1225)), yellow1325FarmVehicle (agricultural vehicle
(yellow 1325)), yellowTwoLine (yellow two-line rear license plate)-->unknown
    </plateType>
    <name min="1" max="32">
      <!--ro, opt, string, range:[1,32], attr:min{req, int},max{req, int}-->张三
    </name>
    <certificateType opt="officerID,ID,passportID,other">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->officerID
    </certificateType>
    <certificateNumber min="1" max="32">
      <!--ro, opt, string, range:[1,32], attr:min{req, int},max{req, int}-->123456
    </certificateNumber>
    <virtualParkingNum min="1" max="16">
      <!--ro, opt, string, range:[1,16], attr:min{req, int},max{req, int}-->123
    </virtualParkingNum>
    <isSupportCardNoRestriction>
      <!--ro, opt, bool-->true
    </isSupportCardNoRestriction>
    <groupName min="1" max="64">
      <!--ro, opt, string, range:[1,64], attr:min{req, int},max{req, int}-->test
    </groupName>
  </LPListAuditSearchDescriptionCap>
</LPListAuditSearchCap>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
LPListAuditSearchCap	[Message content]	application/xml	--	--	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

19.18.1.16 Search the vehicle blocklist and allowlist

Request URL

POST /ISAPI/Traffic/channels/<channelID>/searchLPListAudit

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LPListAuditSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, whether the device supports searching the license plate blocklist and allowlist, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the number of search results this time, desc:no more than 2000 result items for one time-->1
  </maxResults>
  <type>
    <!--opt, enum, list type, subType:string, desc:"blackList" (license plates in blocklist), "whitelist" (license plates in allowlist), "allVehicleList" (license plates in all lists), "otherVehicleList" (license plates in other lists)-->blackList
  </type>
  <LicensePlate>
    <!--opt, string, license plate number, fuzzy search is supported, desc:license plate number,fuzzy search is supported-->test
  </LicensePlate>
  <cardNo>
    <!--opt, string, card No.-->test
  </cardNo>
  <cardID>
    <!--opt, string, card ID (Wiegand protocol), range:[1,9]-->test
  </cardID>
  <plateColor>
    <!--opt, enum, license plate color, subType:string, desc:black, blue, civilAviationBlack (civil aviation black), civilAviationGreen (civil aviation green), golden, green, mixedColor (mixed color), newEnergyGreen (new energy green), newEnergyYellowGreen (new energy green and yellow), orange, other (other color), red, unknown, white, yellow-->black
  </plateColor>
  <plateType>
    <!--opt, enum, license plate type, subType:string, desc:02TypePersonalized (type 02 custom vehicle), 92FarmVehicle (two-line license plate civil vehicle), 92TypeArm (type 92 armed police vehicle), 92TypeCivil (type 92 civil vehicle), arm (police vehicle), civilAviation (civil aviation license plate), coach (driver-training vehicle), consulate (consular vehicle), embassy (embassy vehicle), emergency (emergency license plate), green1325FarmVehicle (agricultural vehicle (green 1325)), hongKongMacao (Hong Kong China / Macao China entrance vehicle), motorola (motorcycle), newEnergy (new energy vehicle license plate), oneLineArm (new armed police vehicle (one-line)), oneLineArmHeadquarters (armed police headquarter license plate (one-line)), tempEntry (temporary entry car), tempTravl (temporary license plate car), trailer, twoLineArm (new armed police vehicle (two-line)), twoLineArmHeadquarters (armed police headquarter license plate (two-line)), unknown, yellow1225FarmVehicle (agricultural vehicle (yellow 1225)), yellow1325FarmVehicle (agricultural vehicle (yellow 1325)), yellowTwoLine (yellow two-line rear license plate)-->02TypePersonalized
  </plateType>
  <name>
    <!--opt, string, range:[1,32]-->张三
  </name>
  <certificateType>
    <!--opt, enum, subType:string-->officerID
  </certificateType>
  <certificateNumber>
    <!--opt, string, range:[1,32]-->123456
  </certificateNumber>
  <virtualParkingNum>
    <!--opt, string, range:[1,16]-->1
  </virtualParkingNum>
  <groupName>
    <!--opt, string, range:[1,64]-->test
  </groupName>
</LPListAuditSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LPListAuditSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search result of vehicle blocklist and allowlist, attr:version{req, string}-->
  <searchID>
    <!--ro, req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <responseStatus>
    <!--ro, req, bool, status search-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO_MATCHES" (no matched data)-->OK
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of returned records-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
```

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```
    </totalMatches>
    <LicensePlateInfoList>
      <!--ro, opt, object, license plate information list-->
      <LicensePlateInfo>
        <!--ro, opt, object, license plate information-->
        <id>
          <!--ro, req, string, ID-->test
        </id>
        <LicensePlate>
          <!--ro, opt, string, license plate number-->test
        </LicensePlate>
        <type>
          <!--ro, opt, enum, list type, subType:string, desc:"blackList" (license plates in blocklist), "whitelist" (license plates in allowlist),
"allVehicleList" (license plates in all lists), "otherVehicleList" (license plates in other lists)-->blackList
        </type>
        <createTime>
          <!--ro, opt, datetime, creation time-->1970-01-01T00:00:00+08:00
        </createTime>
        <direction>
          <!--ro, opt, enum, direction, subType:string, desc:"forward,reverse,unknown"-->forward
        </direction>
        <laneNo>
          <!--ro, opt, int, lane No.-->1
        </laneNo>
        <plateCategory>
          <!--ro, opt, string, license plate type-->test
        </plateCategory>
        <country>
          <!--ro, opt, string, country or region-->BLR
        </country>
        <area>
          <!--ro, opt, int, region-->1
        </area>
        <effectiveTime>
          <!--ro, opt, date, validity period (date) of the blocklist and allowlist-->1970-01-01+08:00
        </effectiveTime>
        <countryIndex>
          <!--ro, opt, int, country/region index-->1
        </countryIndex>
        <cardID>
          <!--ro, opt, string, card ID (Wiegand protocol), range:[1,9]-->test
        </cardID>
        <effectiveDateTime>
          <!--ro, opt, datetime, validity period of the blocklist and allowlist-->1970-01-01T00:00:00+08:00
        </effectiveDateTime>
        <effectiveStartDate>
          <!--ro, opt, string, start date of the validity period-->1970-01-01
        </effectiveStartDate>
        <cardNo>
          <!--ro, opt, string, card No.-->test
        </cardNo>
        <plateColor>
          <!--ro, opt, string, license plate color-->test
        </plateColor>
        <name>
          <!--ro, opt, string, range:[1,32]-->张三
        </name>
        <certificateType>
          <!--ro, opt, enum-->officerID
        </certificateType>
        <certificateNumber>
          <!--ro, opt, string, range:[1,32]-->123456
        </certificateNumber>
        <virtualParkingNum>
          <!--ro, opt, string, range:[1,16]-->1
        </virtualParkingNum>
        <plateDescription>
          <!--ro, opt, string, range:[0,128]-->abc
        </plateDescription>
        <groupName>
          <!--ro, opt, string, range:[1,64]-->test
        </groupName>
      </LicensePlateInfo>
    </LicensePlateInfoList>
  </LPListAuditSearchResult>
```

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20 How-To Video Guidance

If you need access to corresponding video guidance for device integration, please register on https://tpp.hikvision.com and visit our Training Center: https://tpp.hikvision.com/tpp/Training. The Training Center is specifically designed to provide technical training and guidance resources for our partners. On this platform, you can find integration video tutorials for various devices, enabling better understanding and learning of the integration process. To offer more personalized service, our Training Center also supports filtering by integration protocols, devices, and applications.